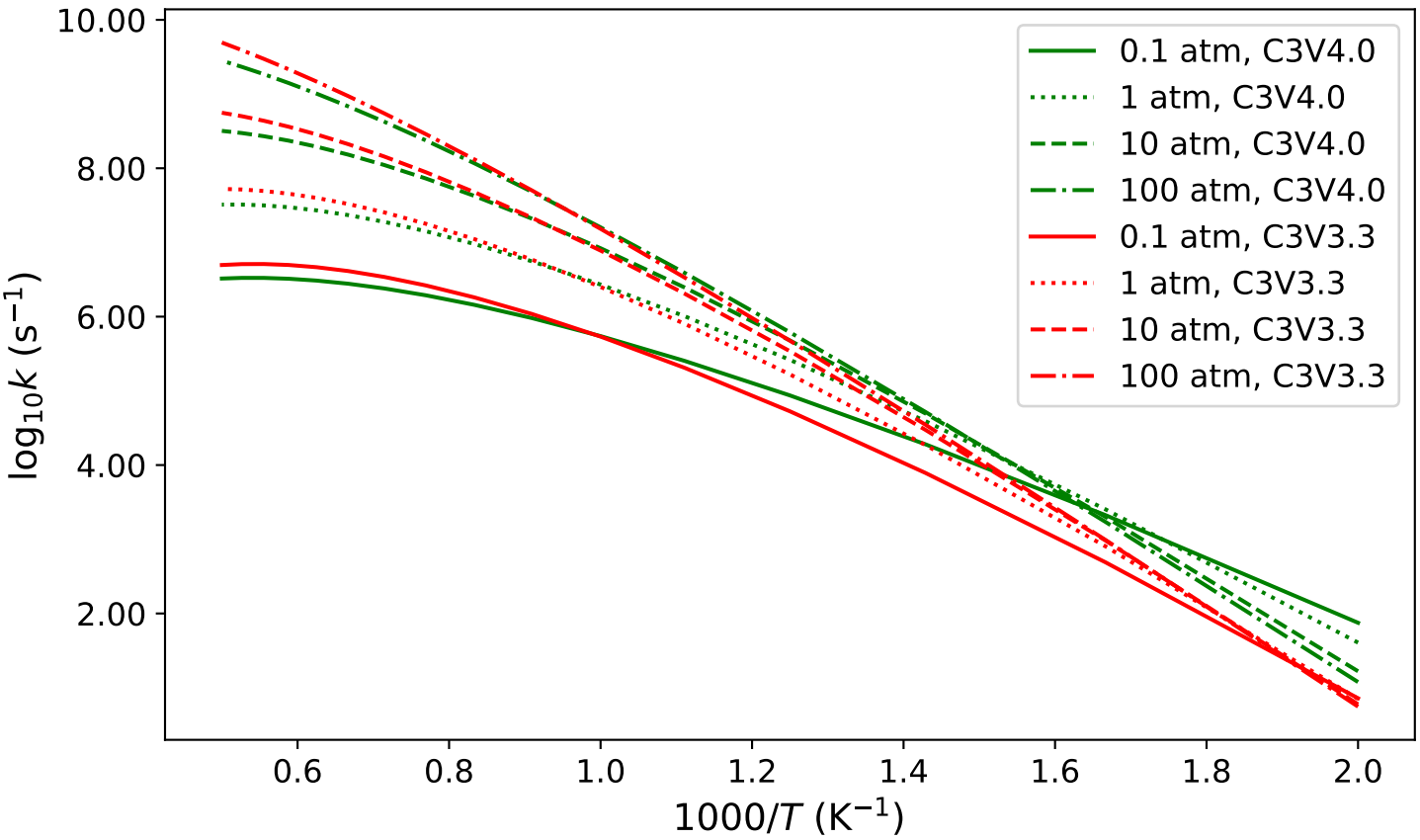
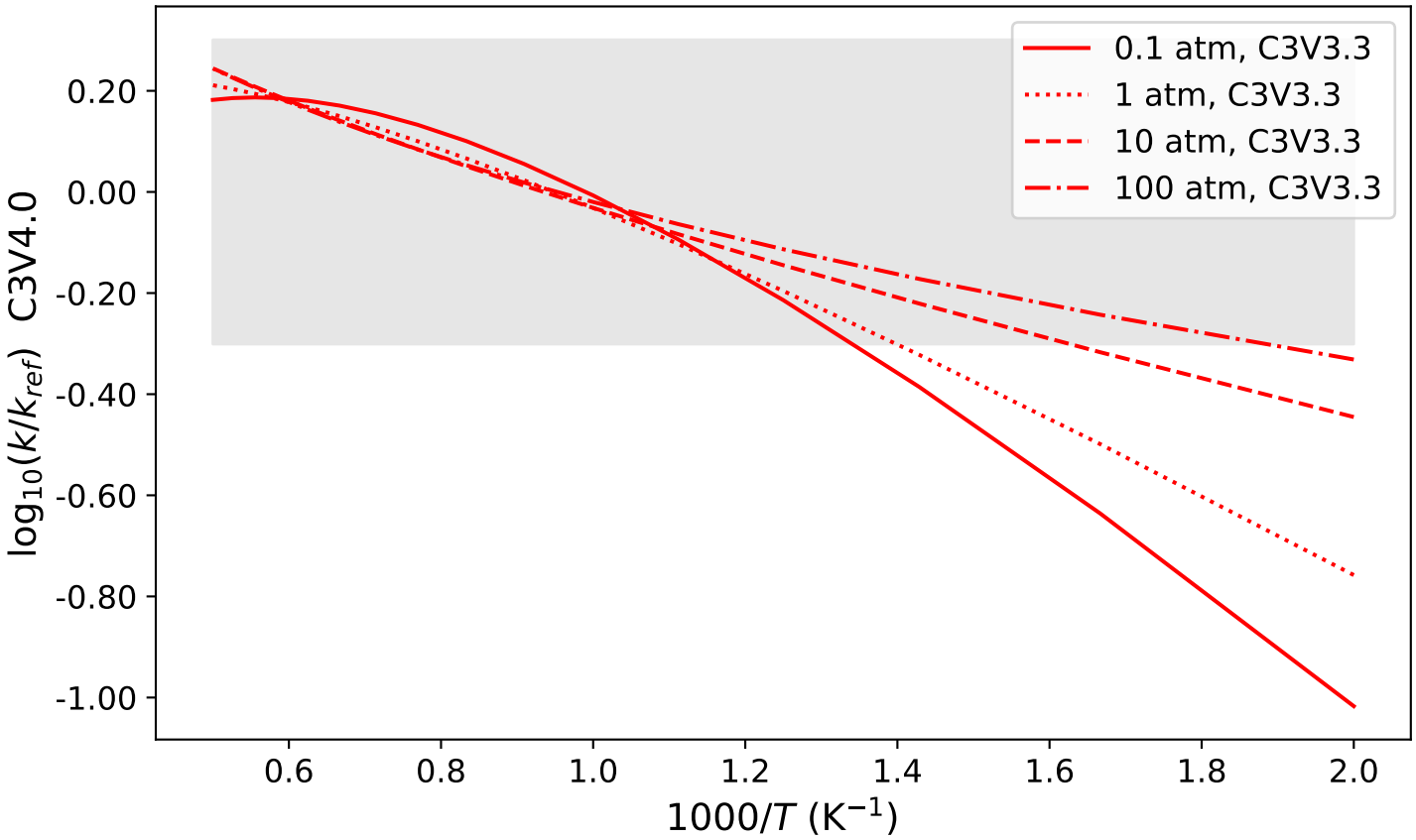
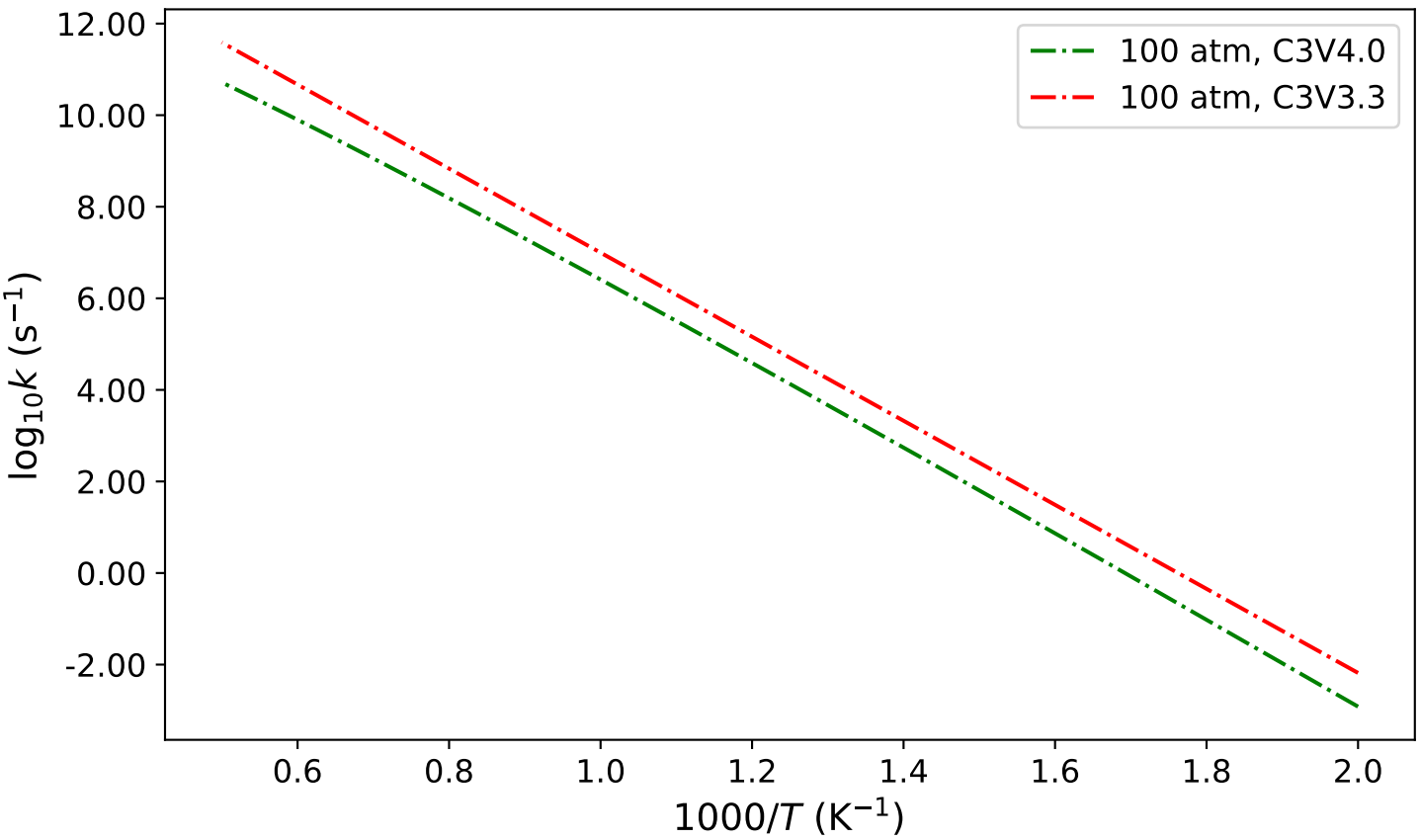
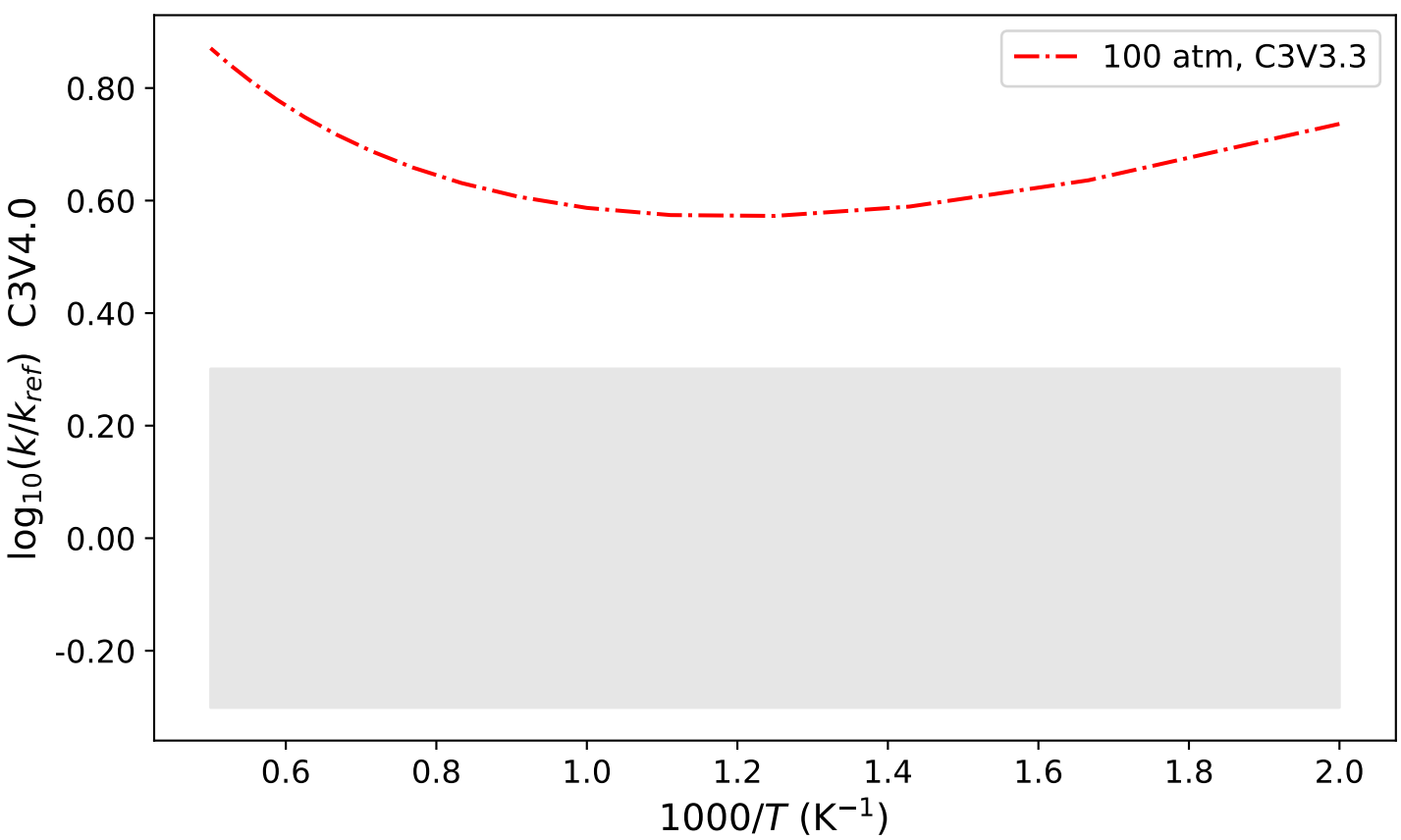
PC4H9 = C2H5 + C2H4



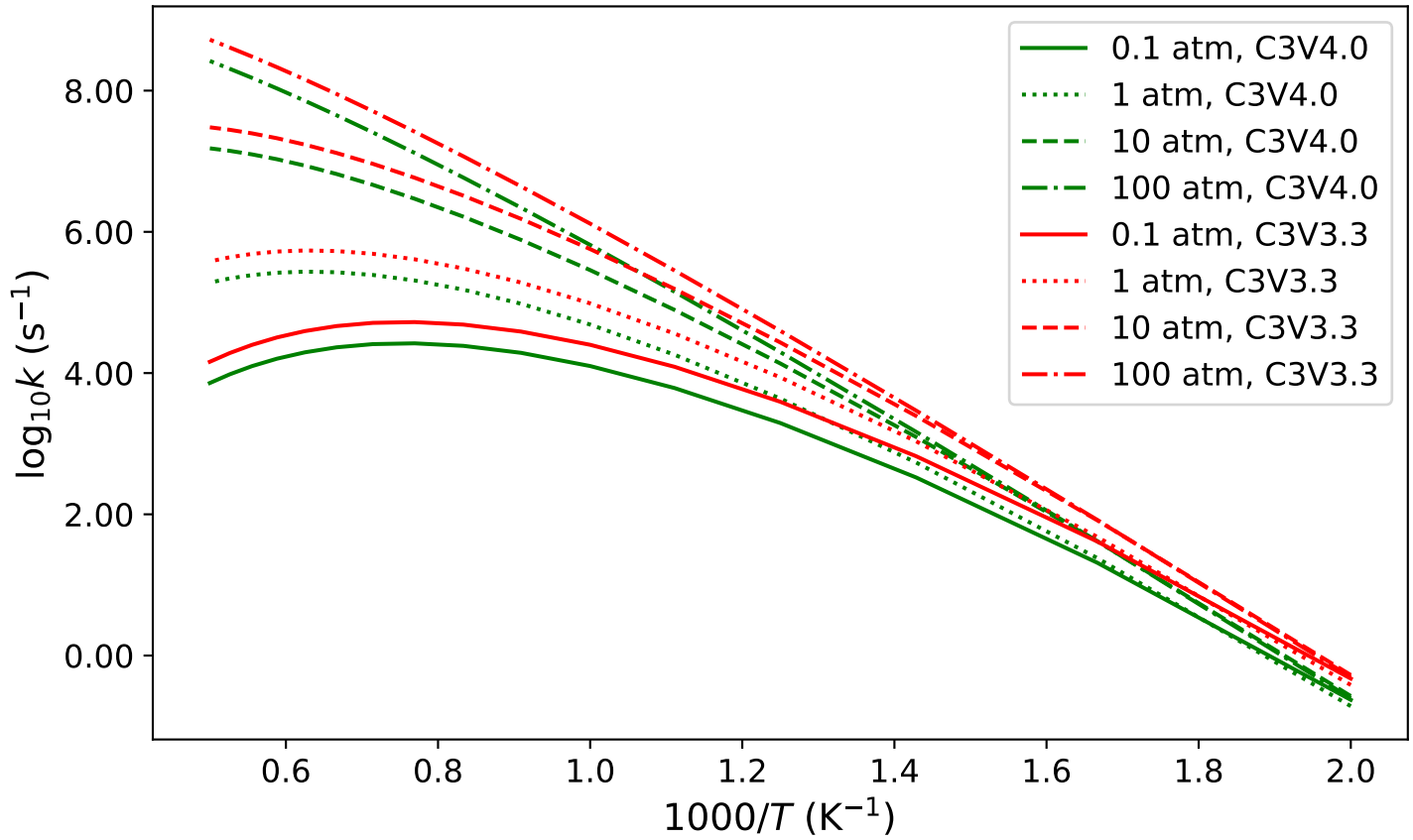



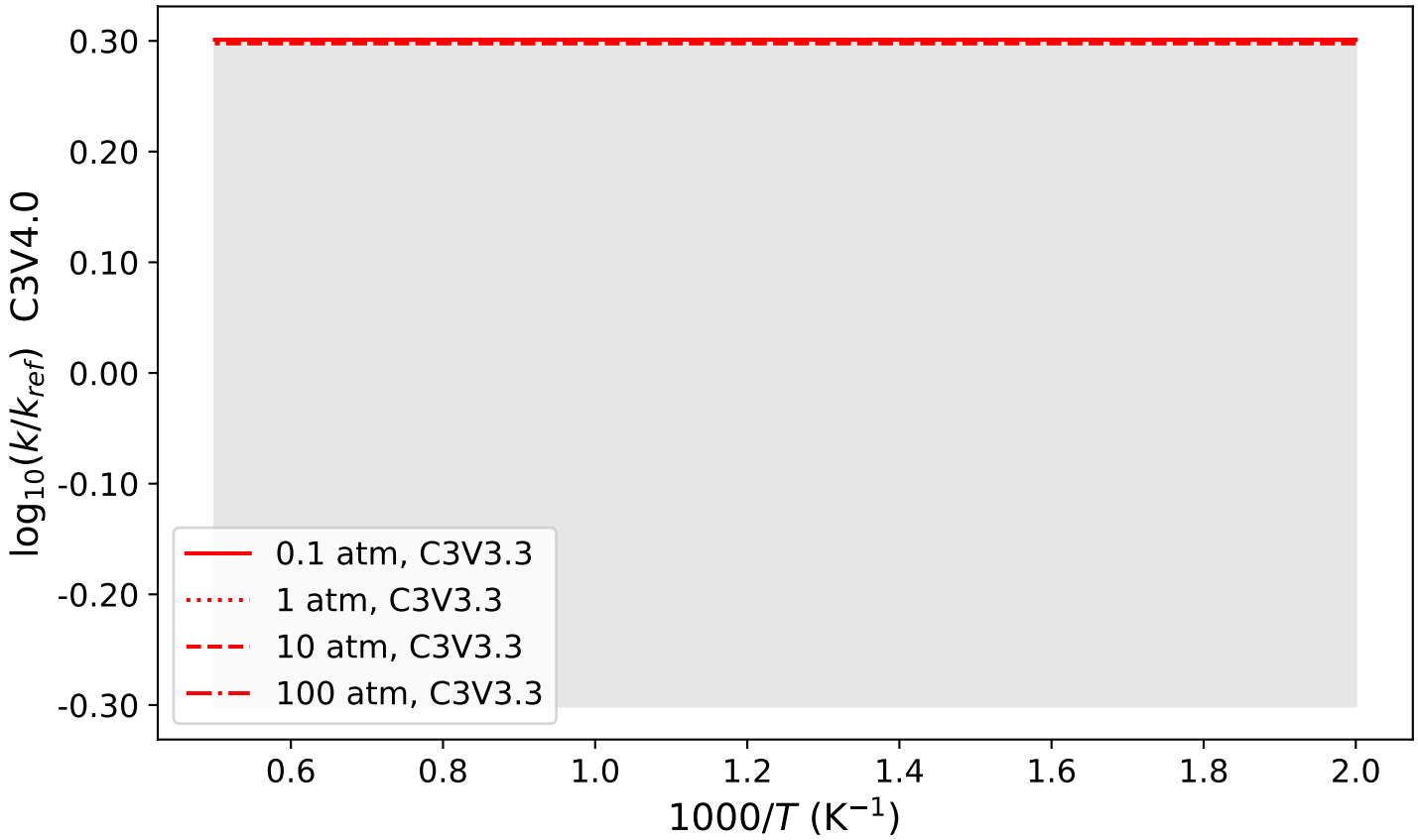
NC4KET21 => CH2O + OH + C2H5CO



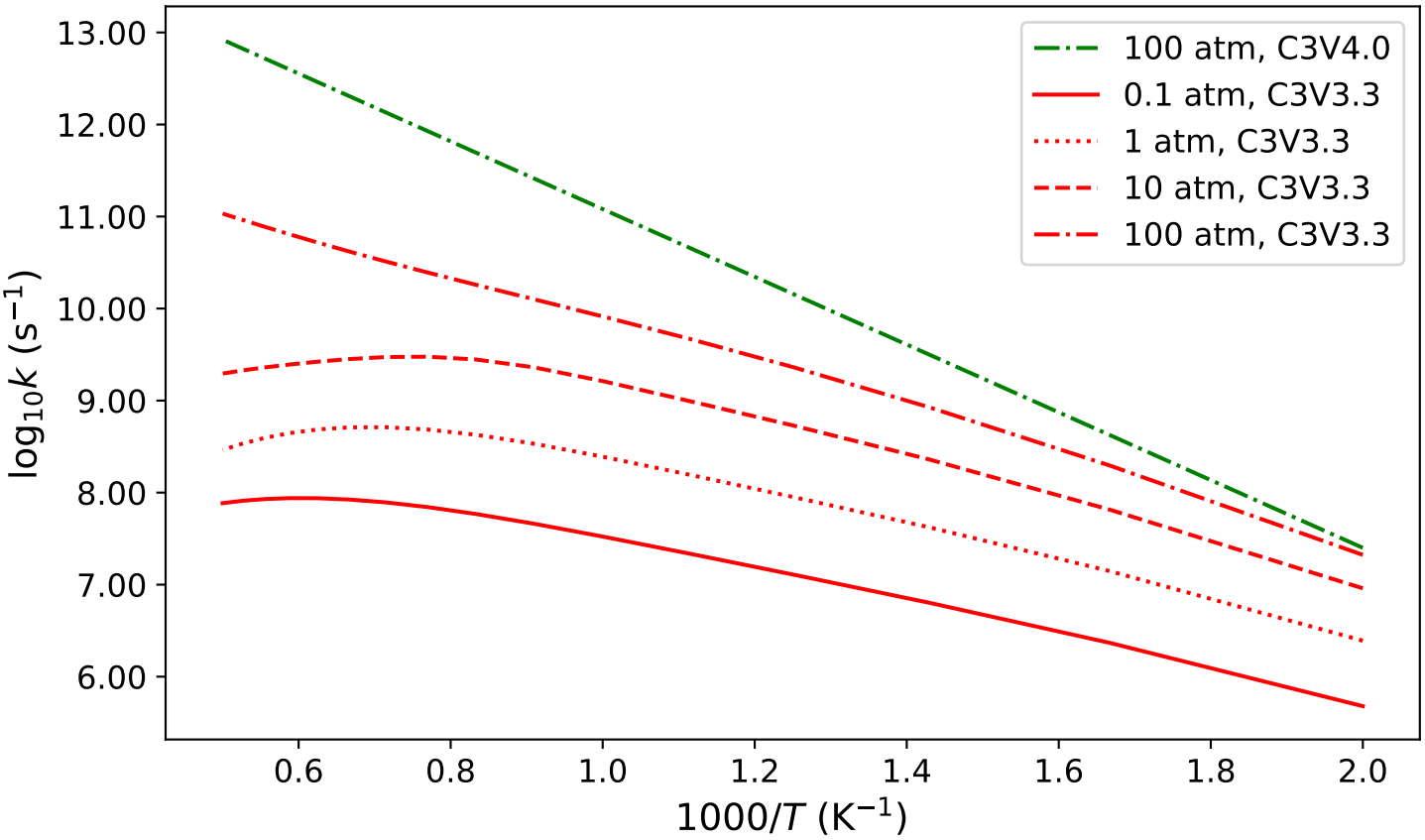



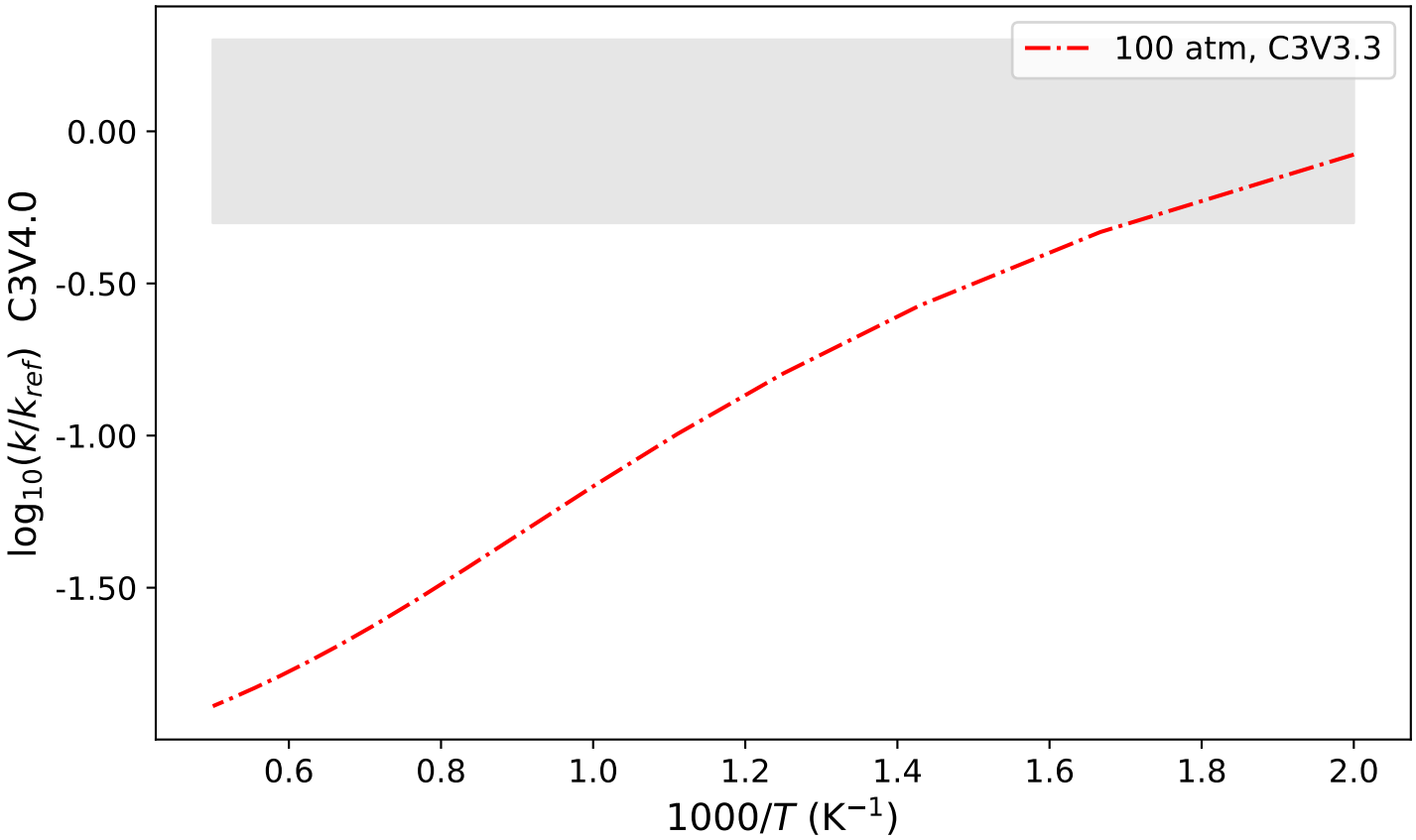
IC3H7O2 = HO2 + C3H6



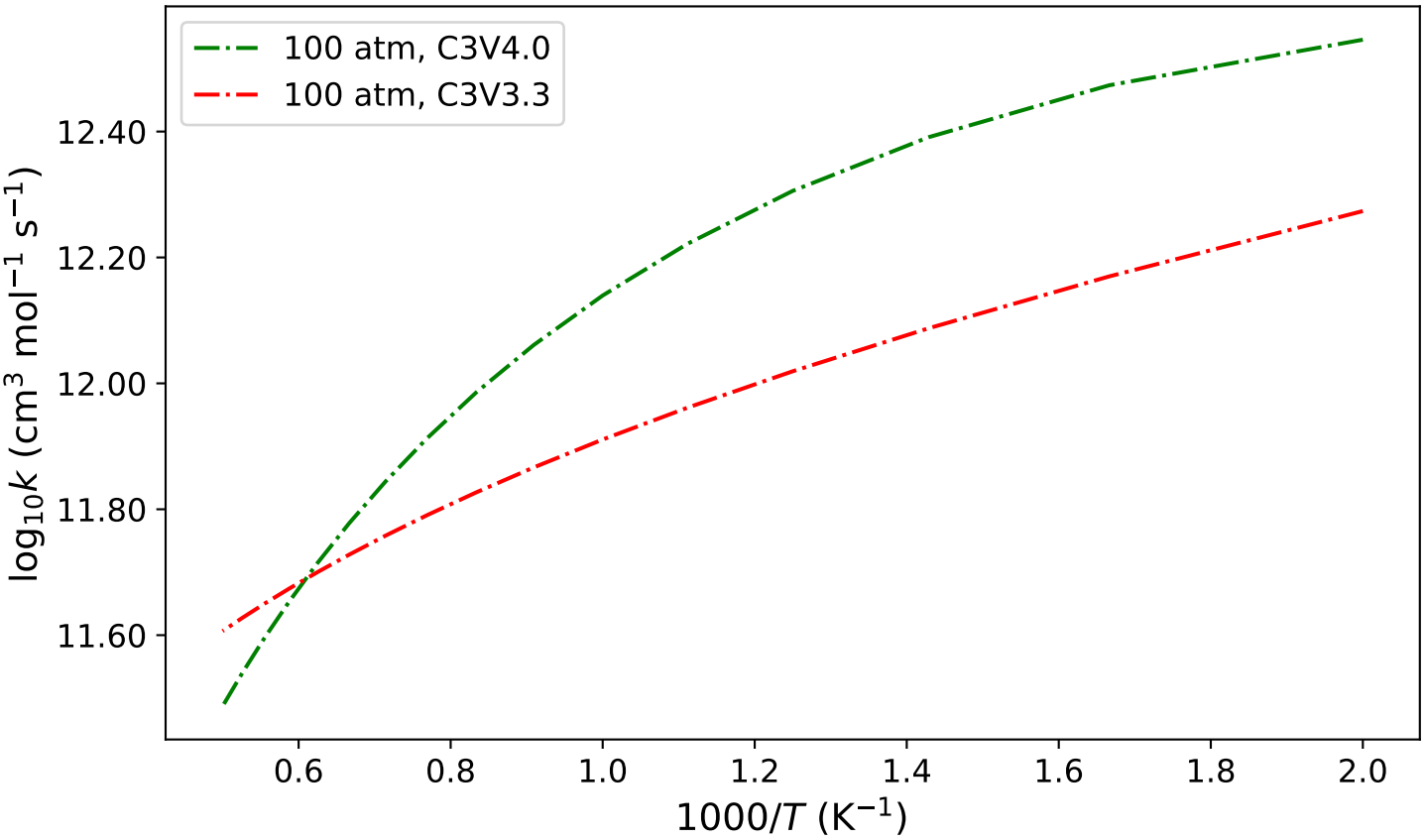



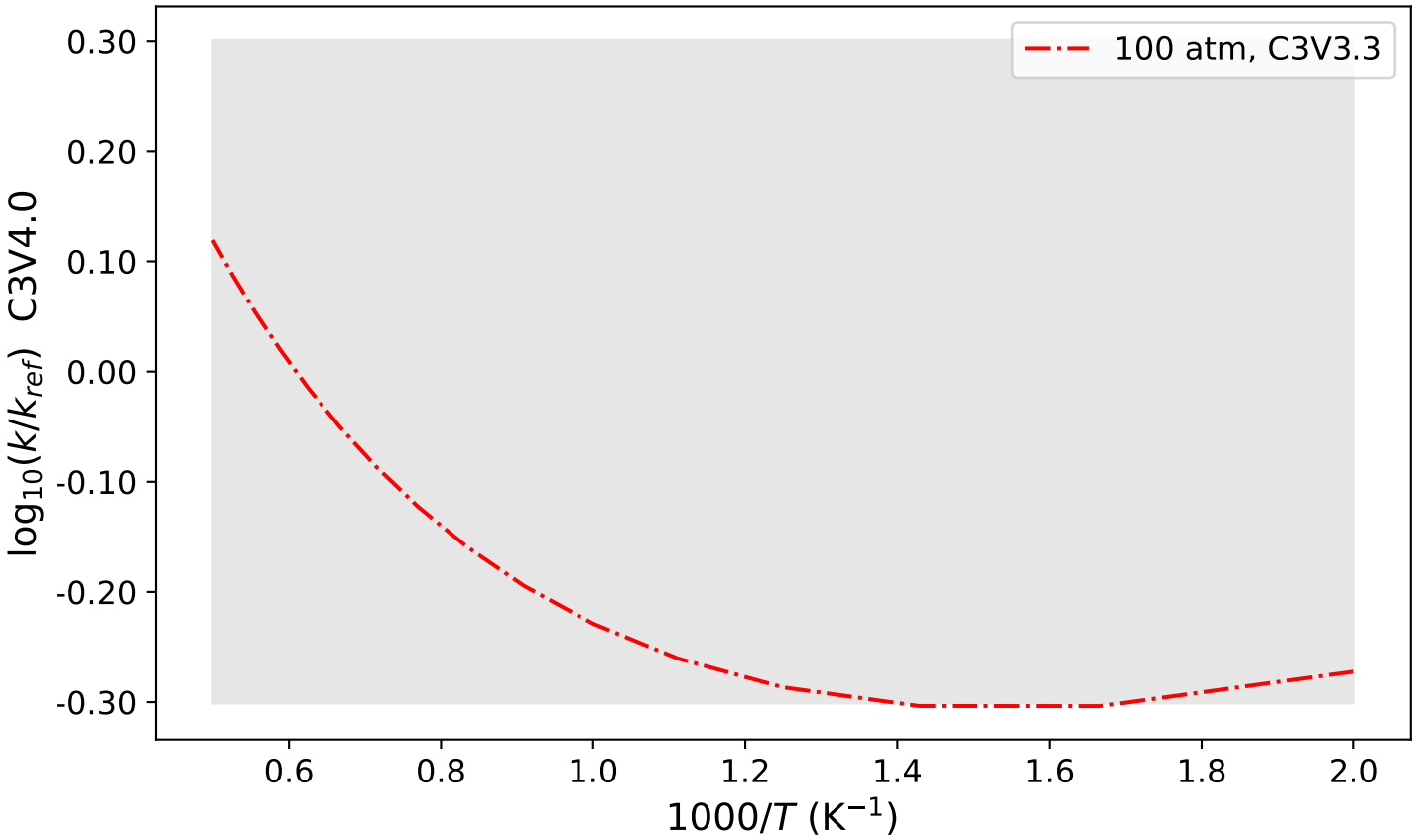
C2H5CO = C2H5 + CO



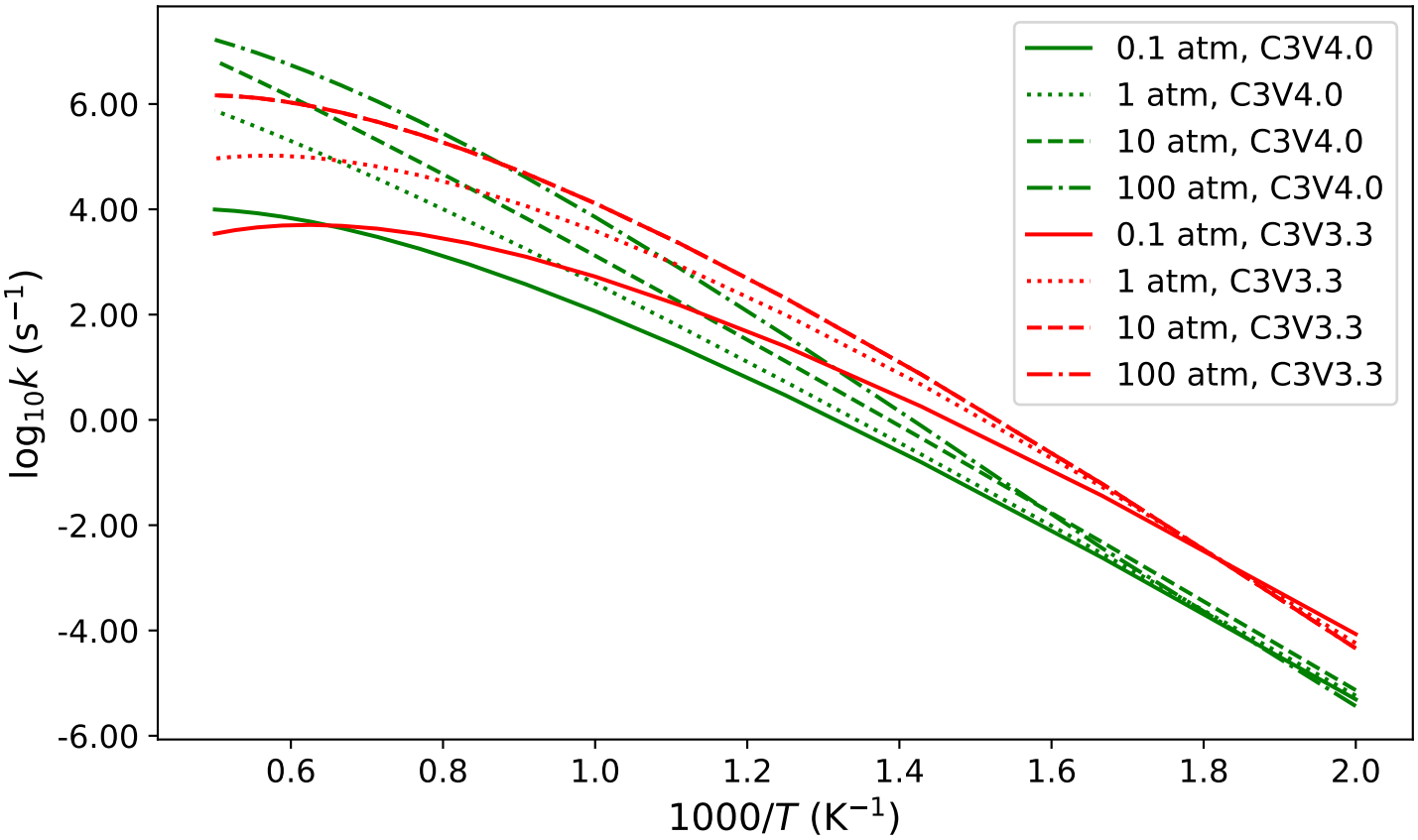



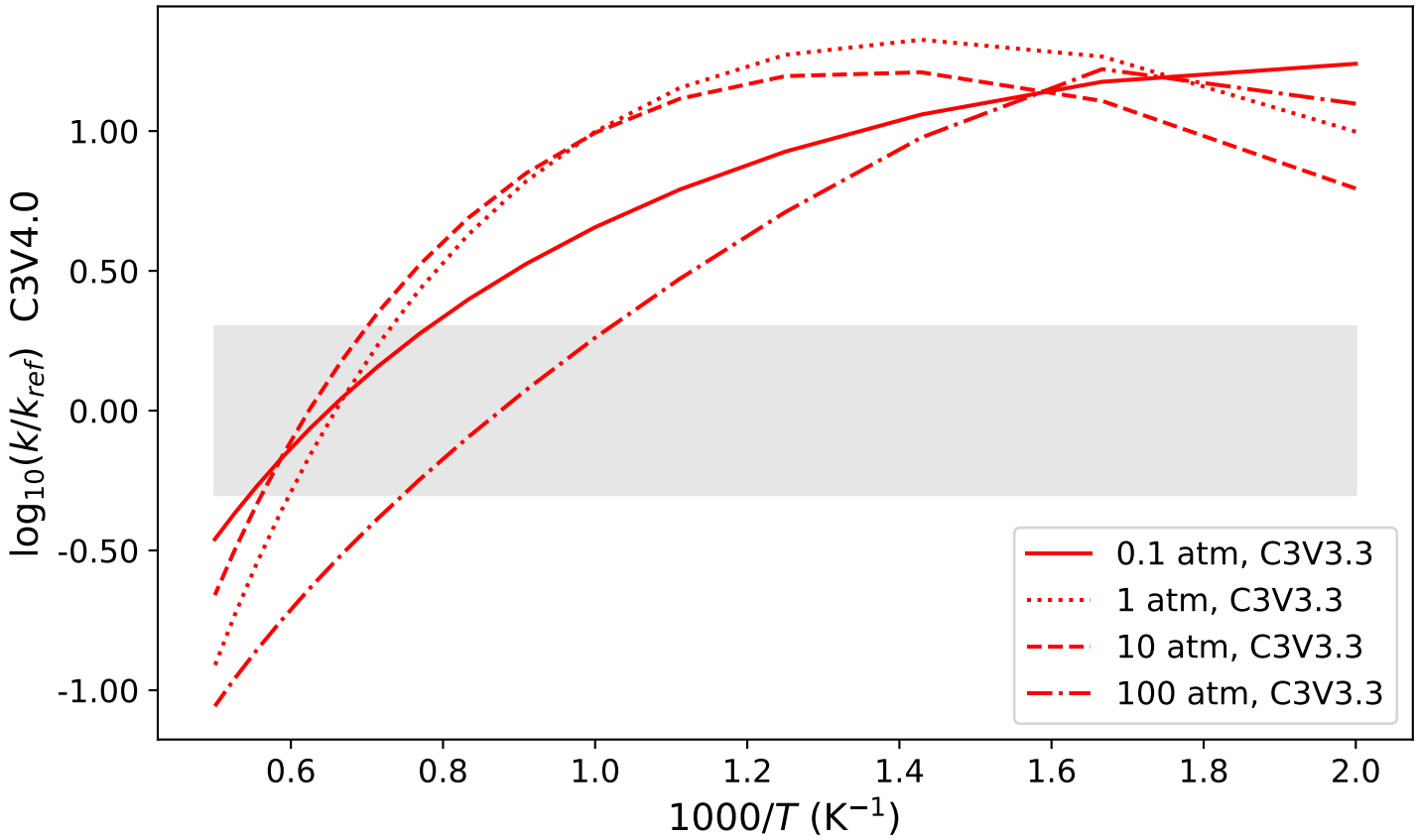
O2 + C4H8OOH1-2 = C4H8OOH1-2O2



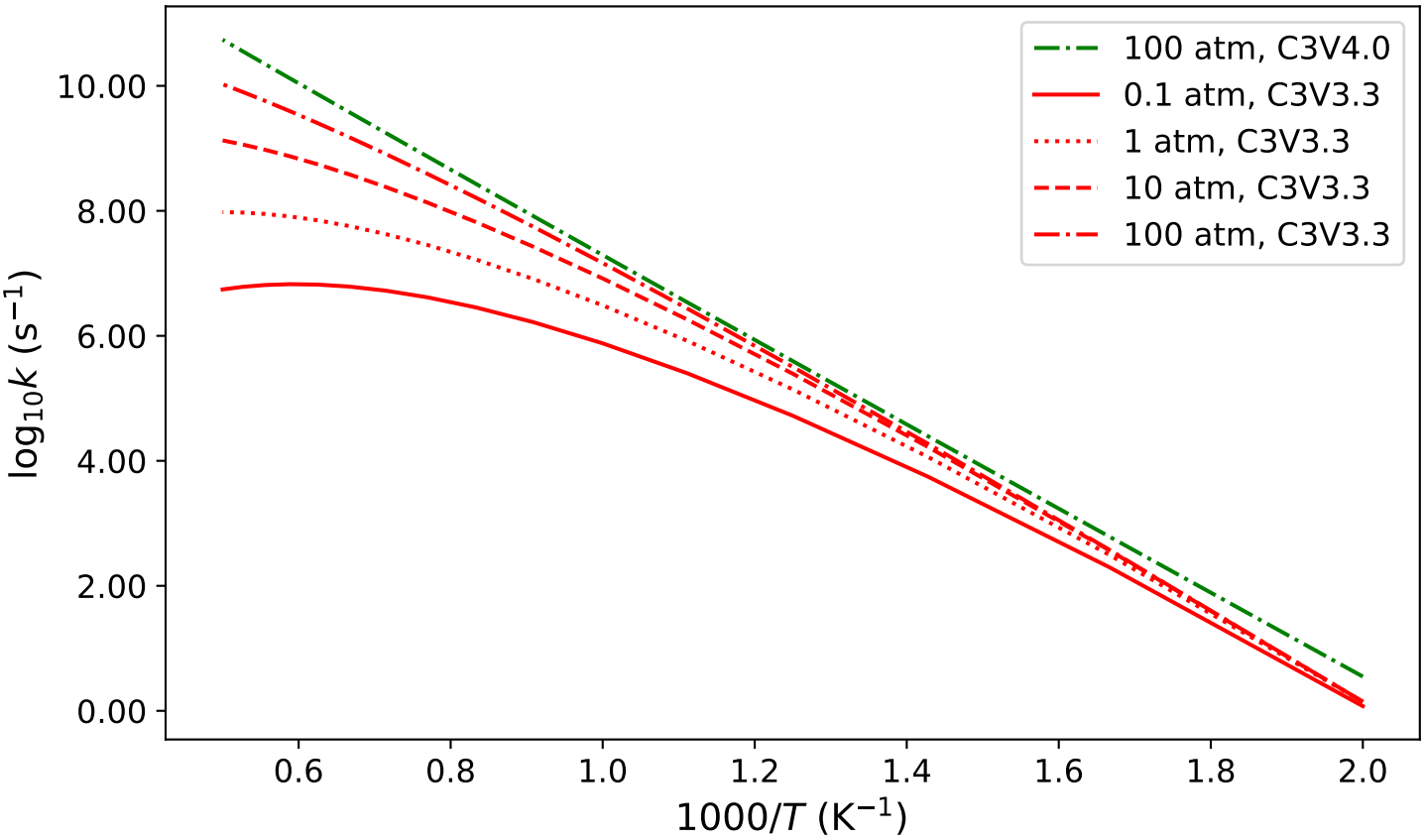



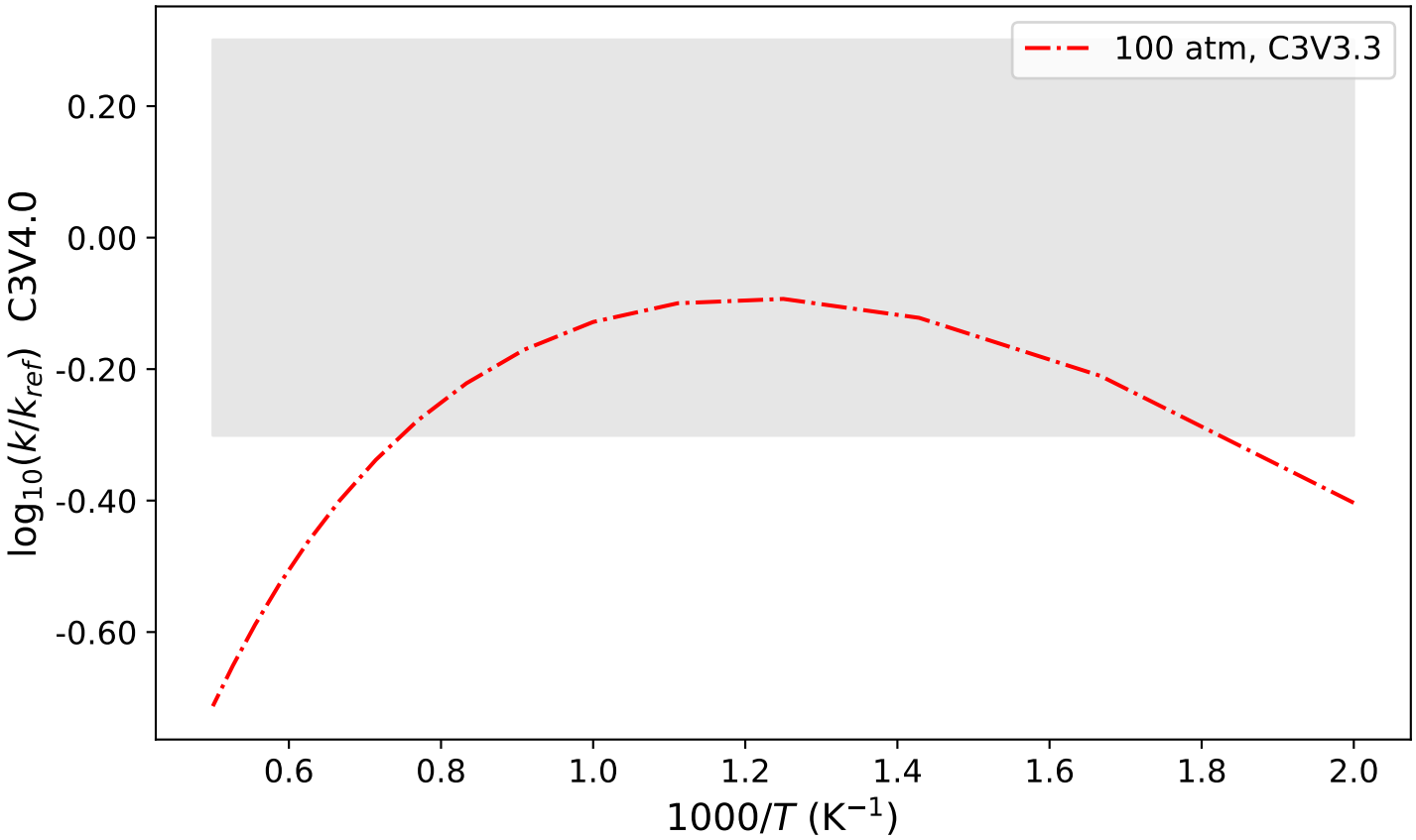
C4H5-N = C4H5-I


0.1 atm, C3V4.0
1 atm, C3V4.0
10 atm, C3V4.0
100 atm, C3V4.0
0.1 atm, C3V3.3
1 atm, C3V3.3
10 atm, C3V3.3
100 atm, C3V3.3
6.00
4.00
2.00
0.00
-2.00
-4.00
-6.00
$\log_{10}k$ ($s^{-1}$)
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
$1000/T$ ($K^{-1}$)

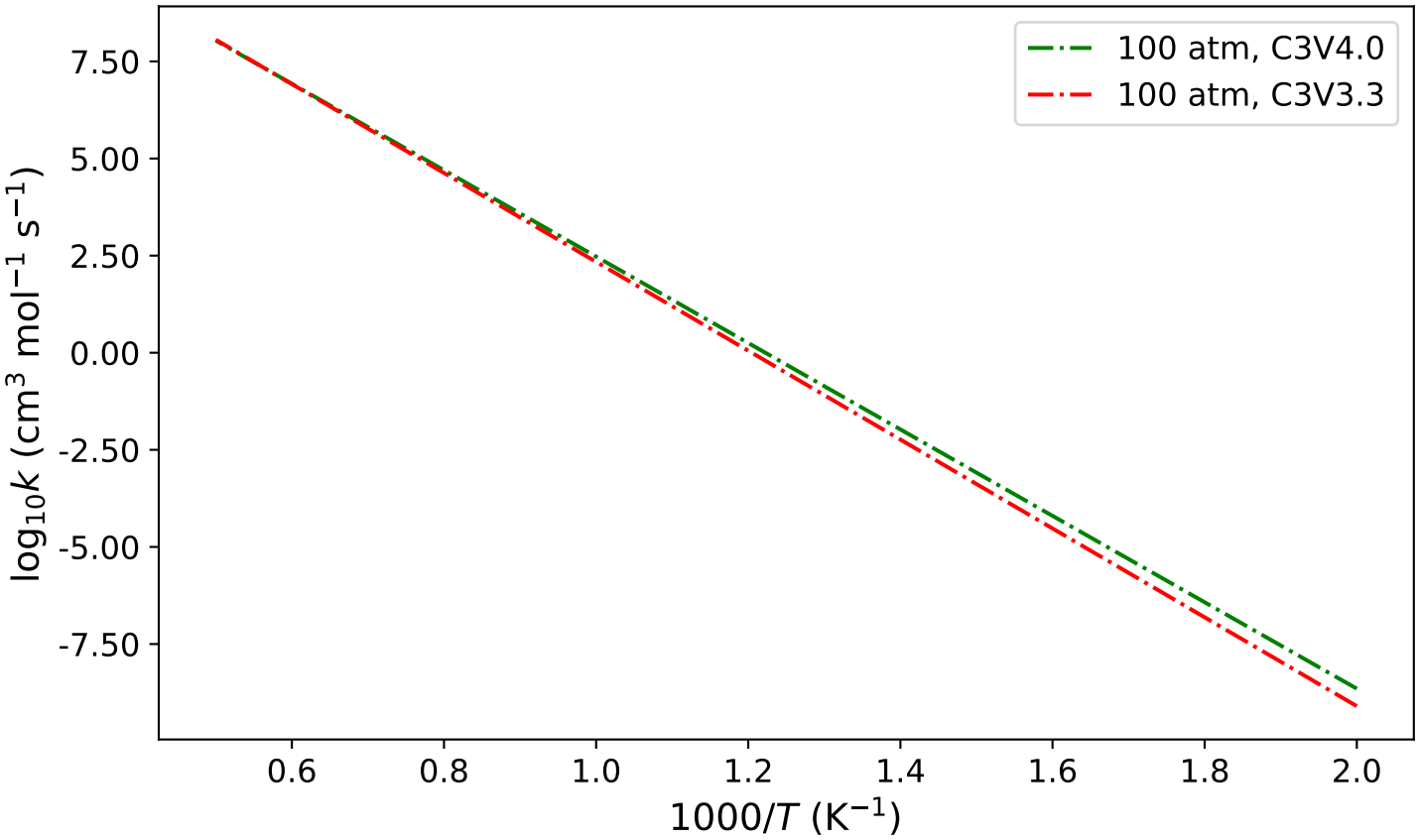

1.00
0.50
0.00
-0.50
-1.00
$\log_{10}(k/k_{ref})$ C3V4.0
0.1 atm, C3V3.3
1 atm, C3V3.3
10 atm, C3V3.3
100 atm, C3V3.3
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
$1000/T$ ($K^{-1}$)

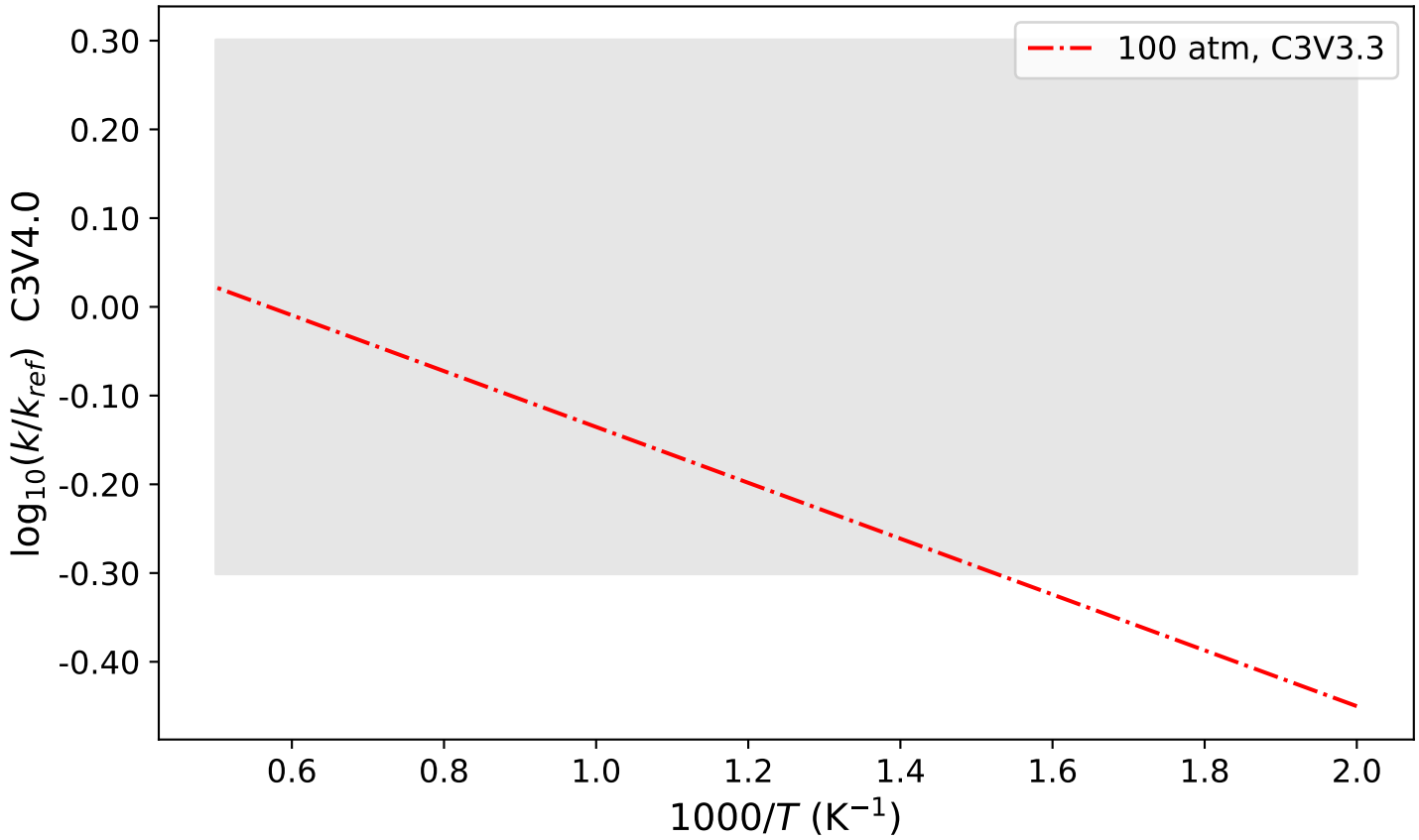
IC4H9 = CH3 + C3H6



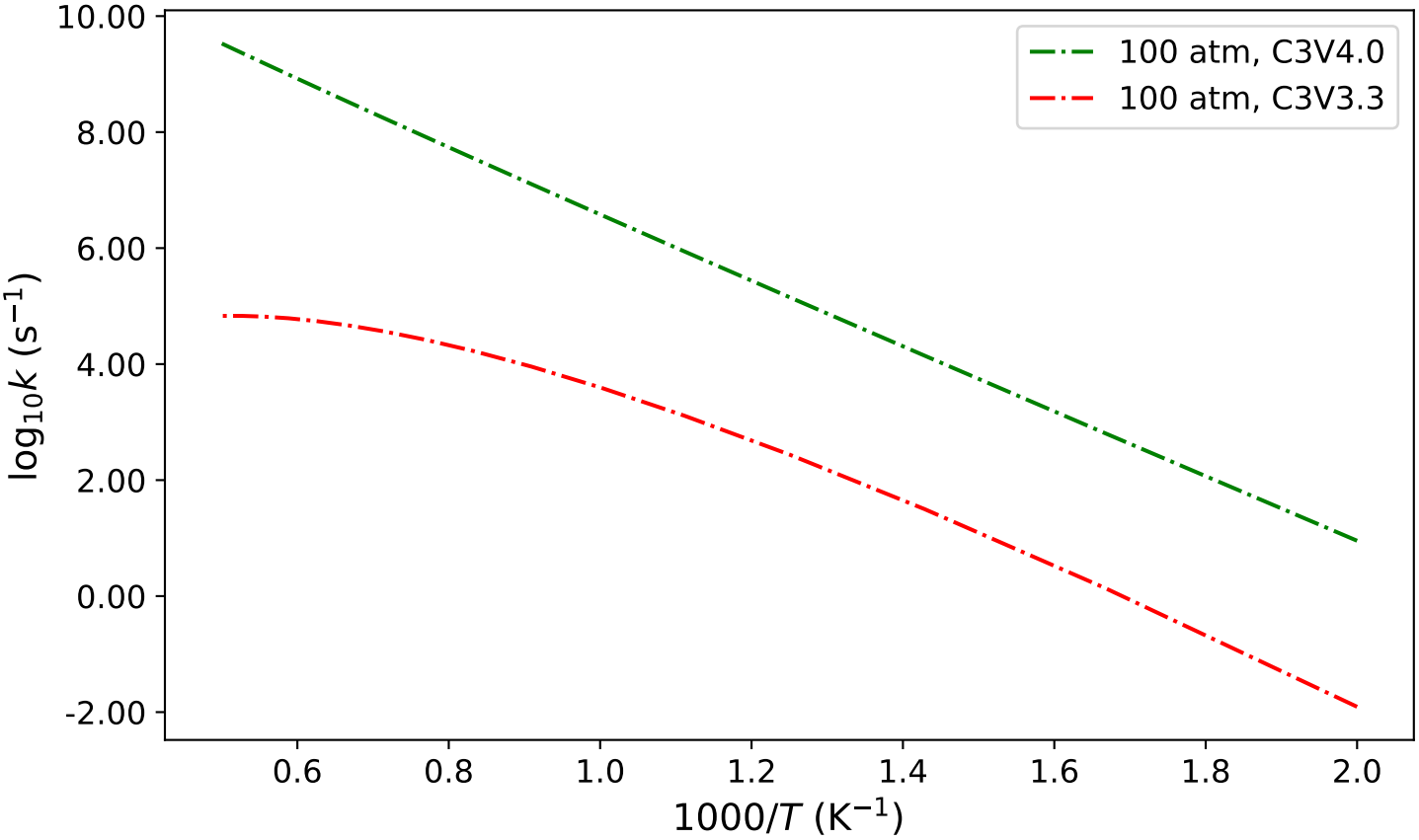



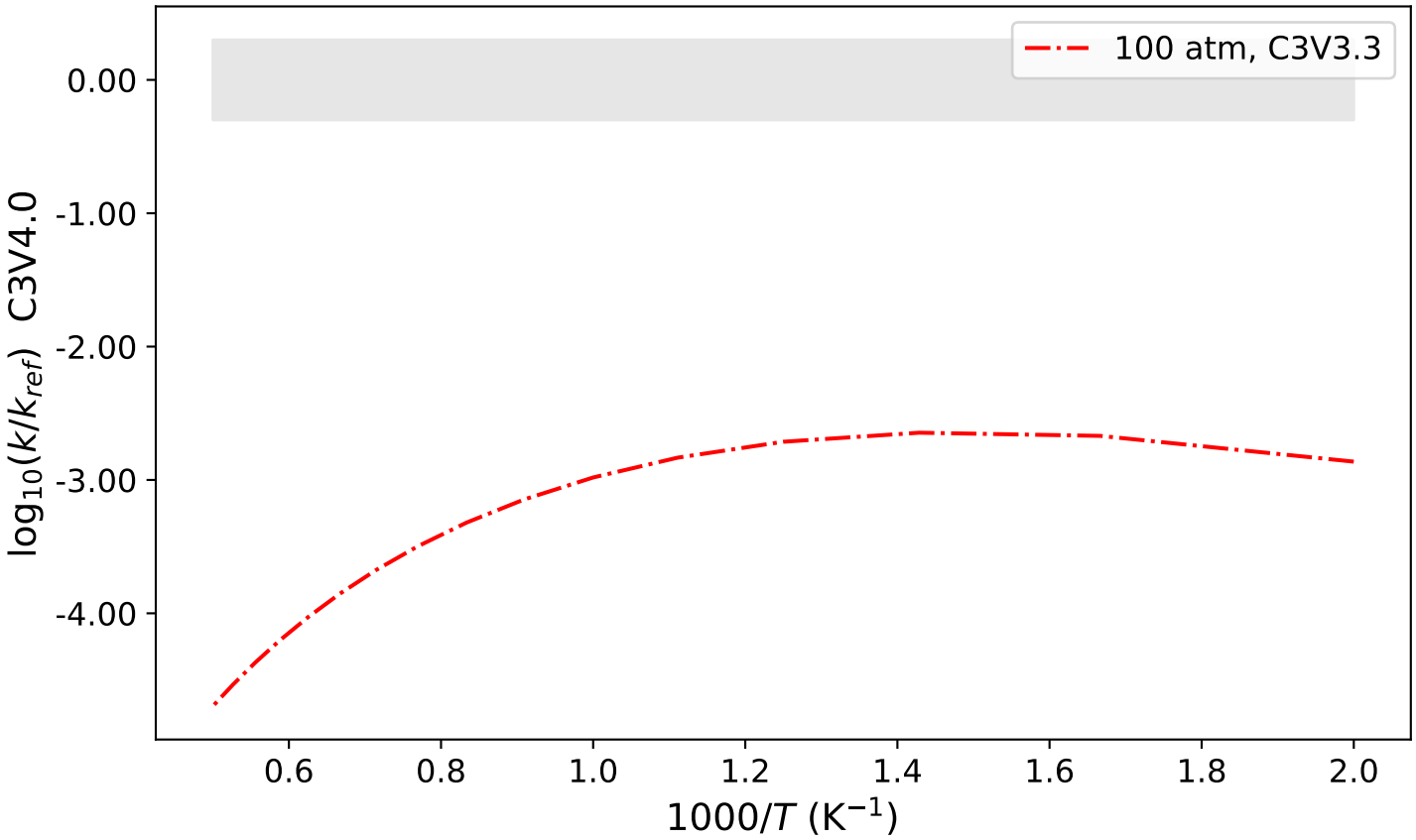
O2 + C4H10 = HO2 + PC4H9



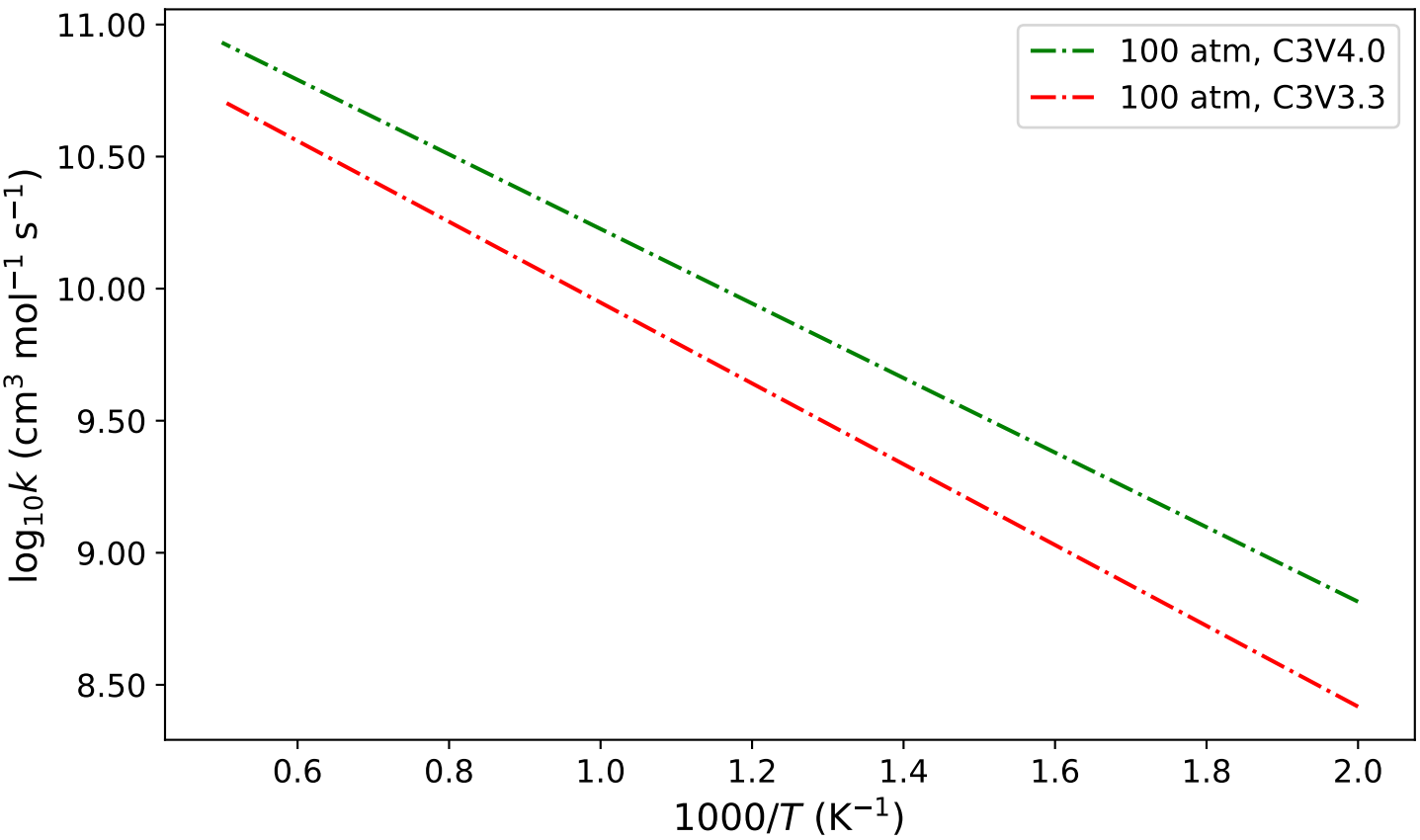



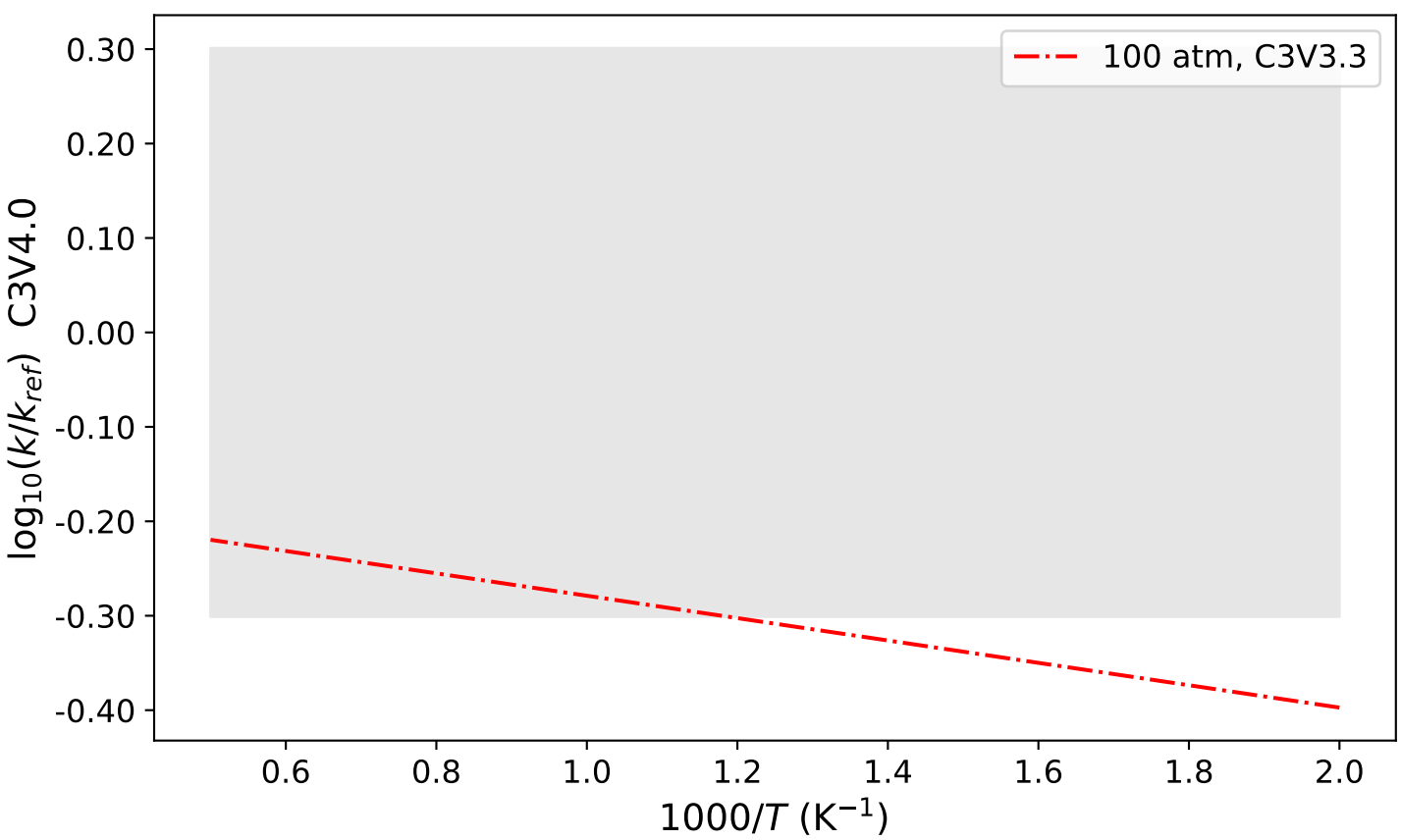
O2CH2CH2CHO = QHO2CH2CHCHO



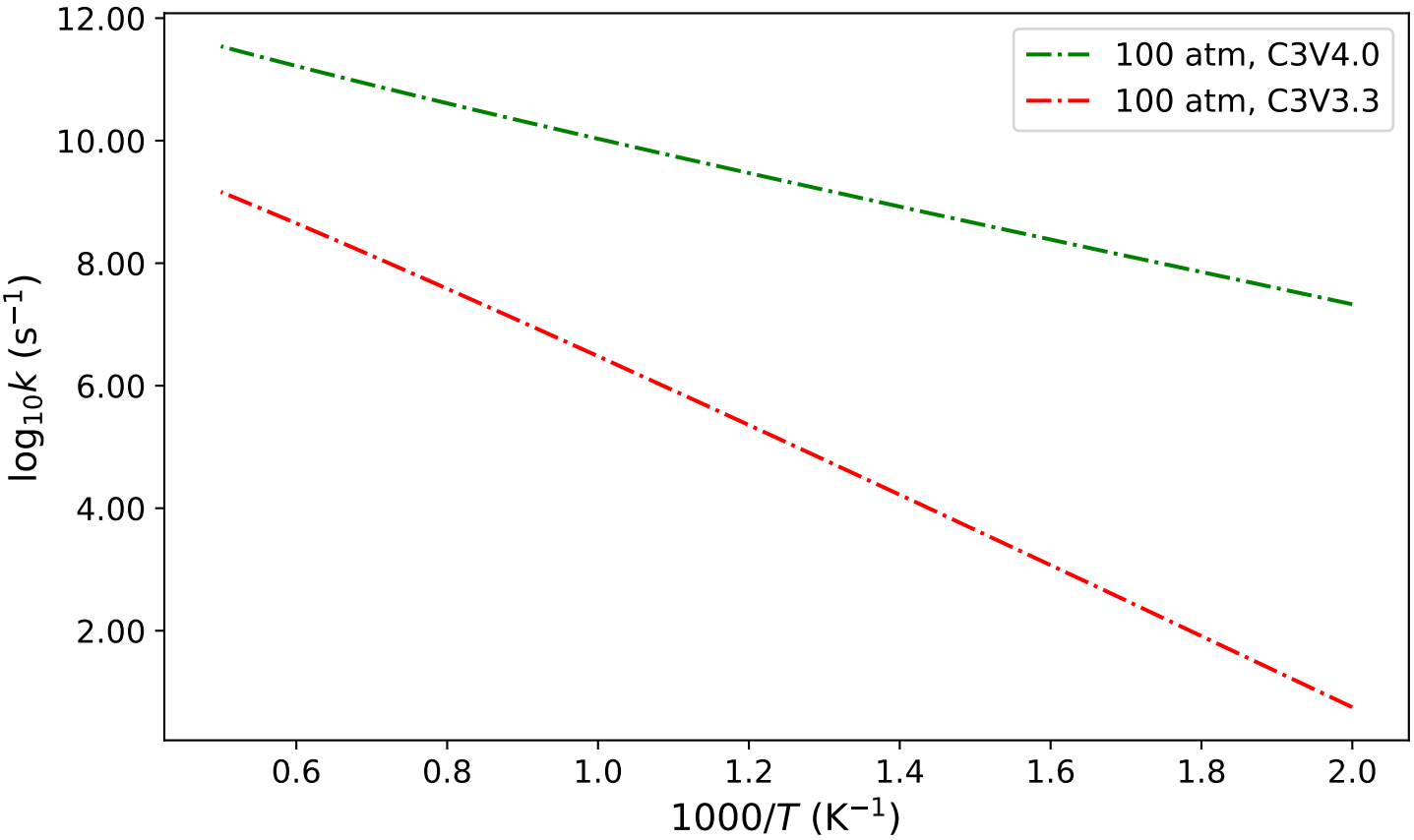



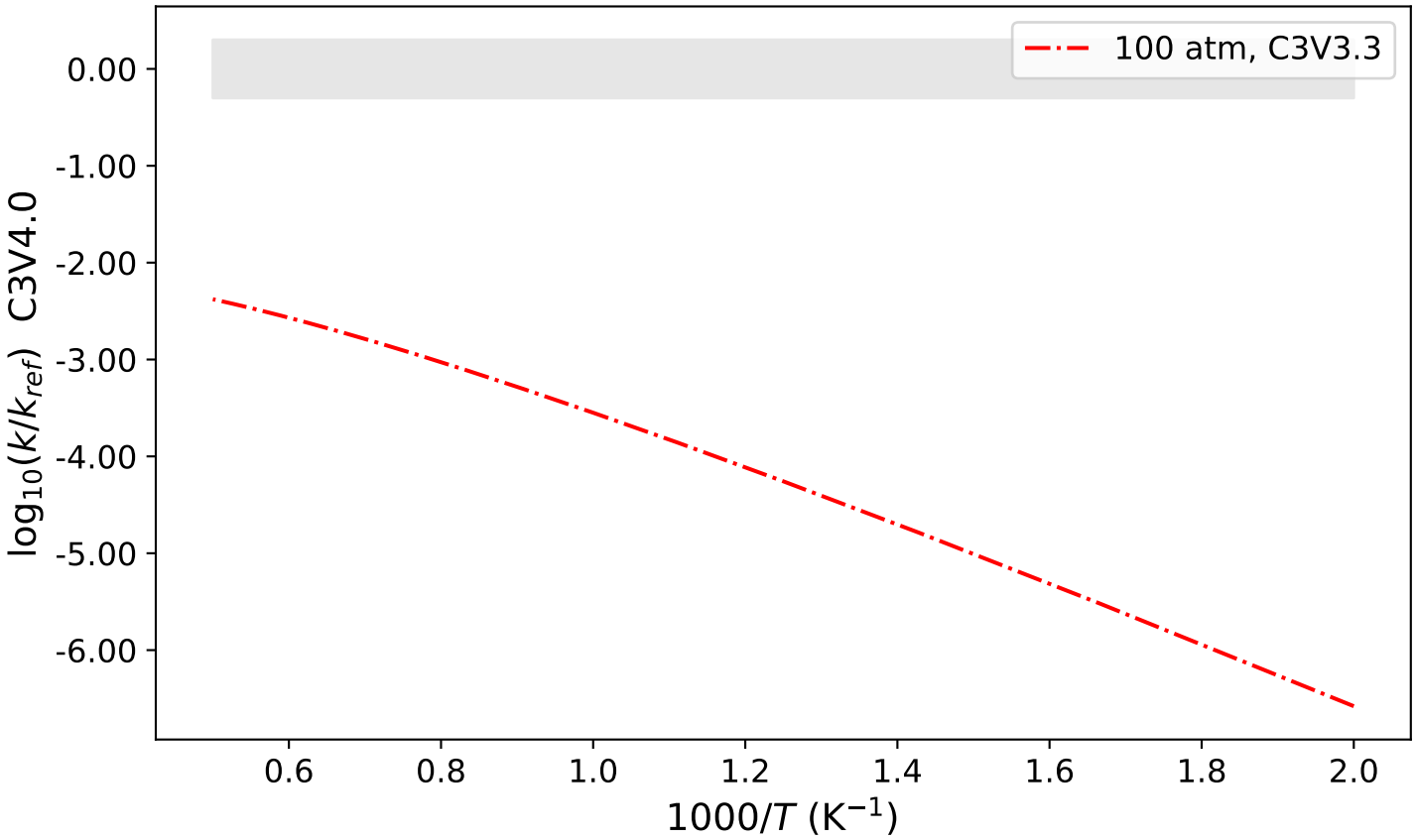
C3H8 + CH3O = NC3H7 + CH3OH



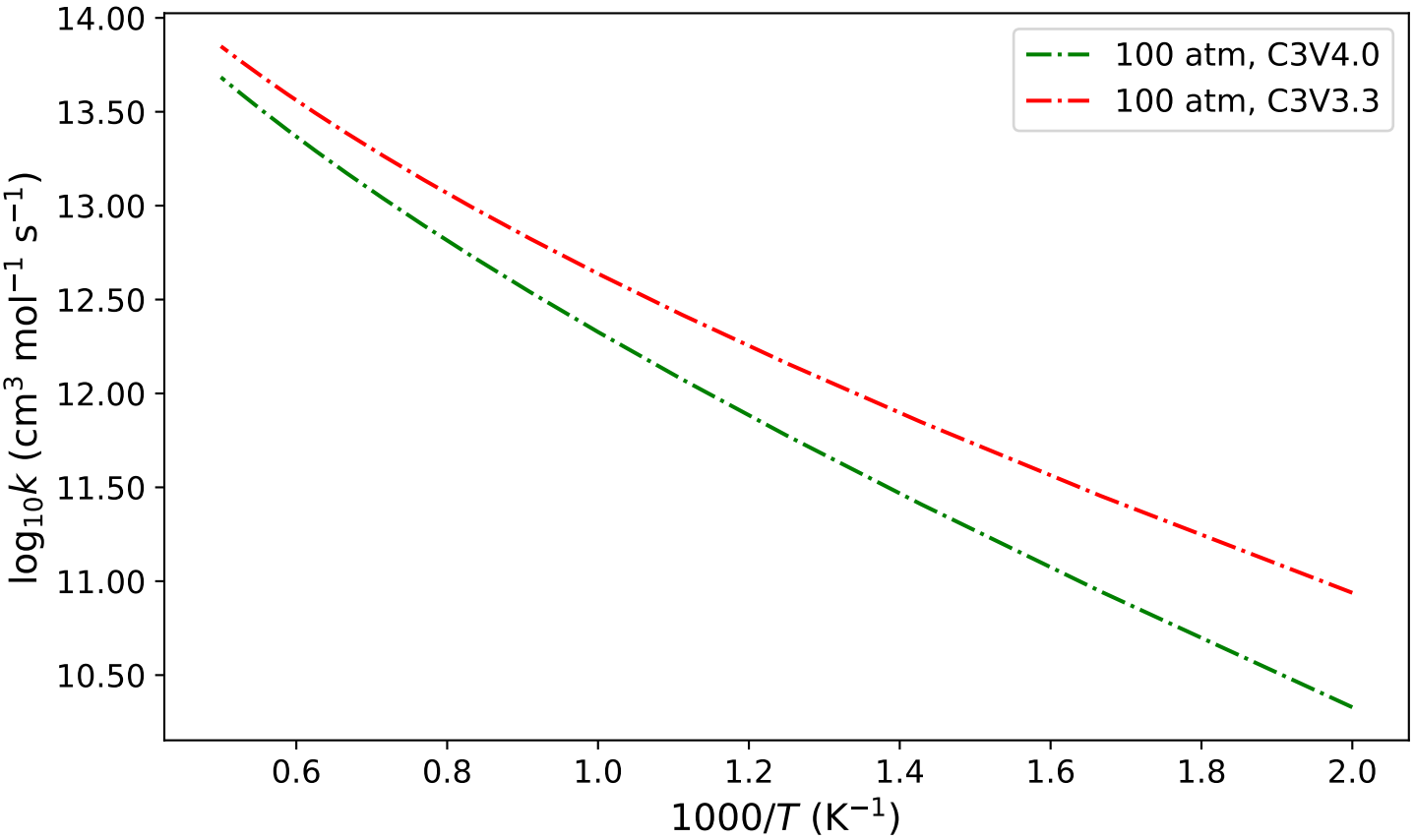



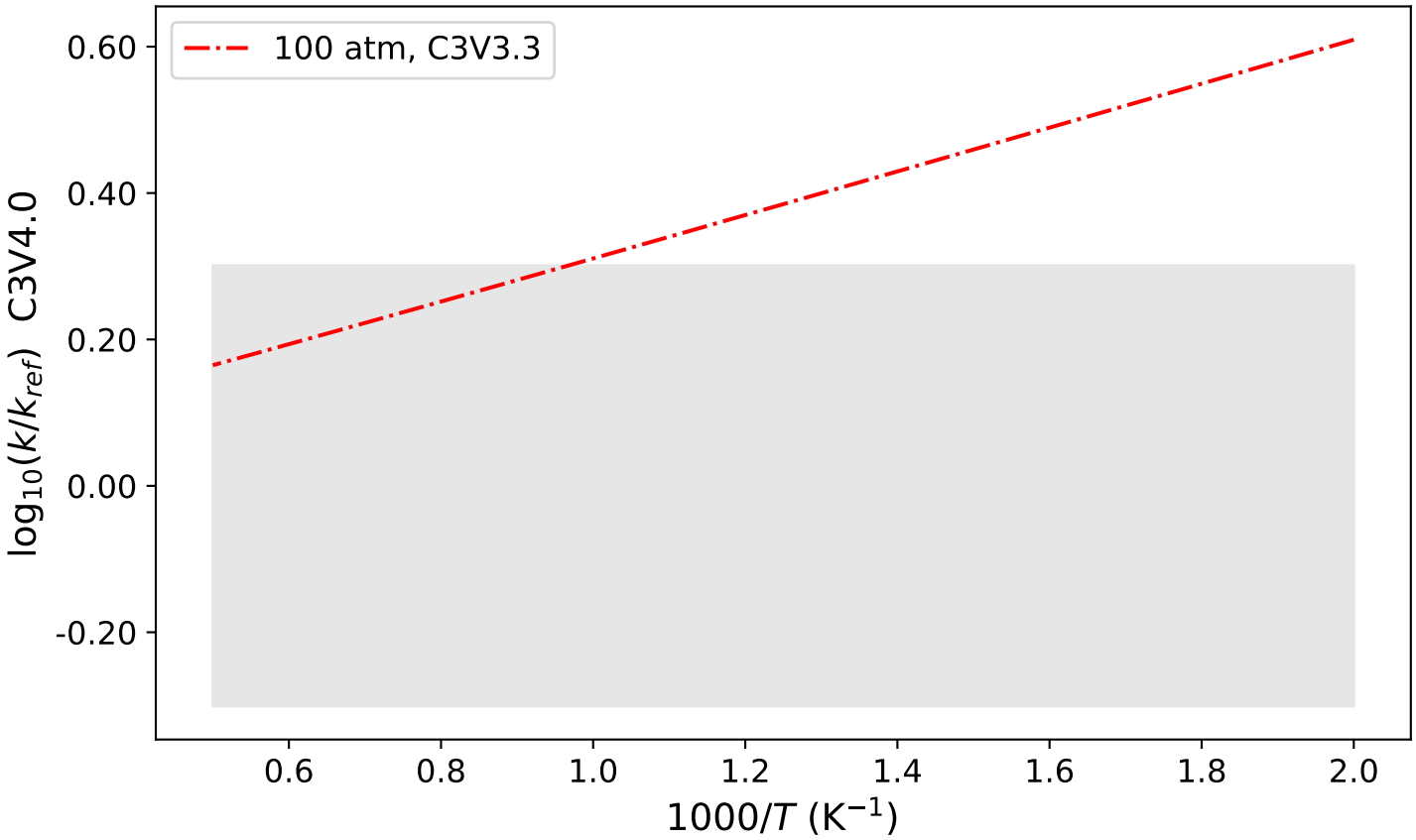
C4H61-3OOH4 = OH + C2H3CHOCH2



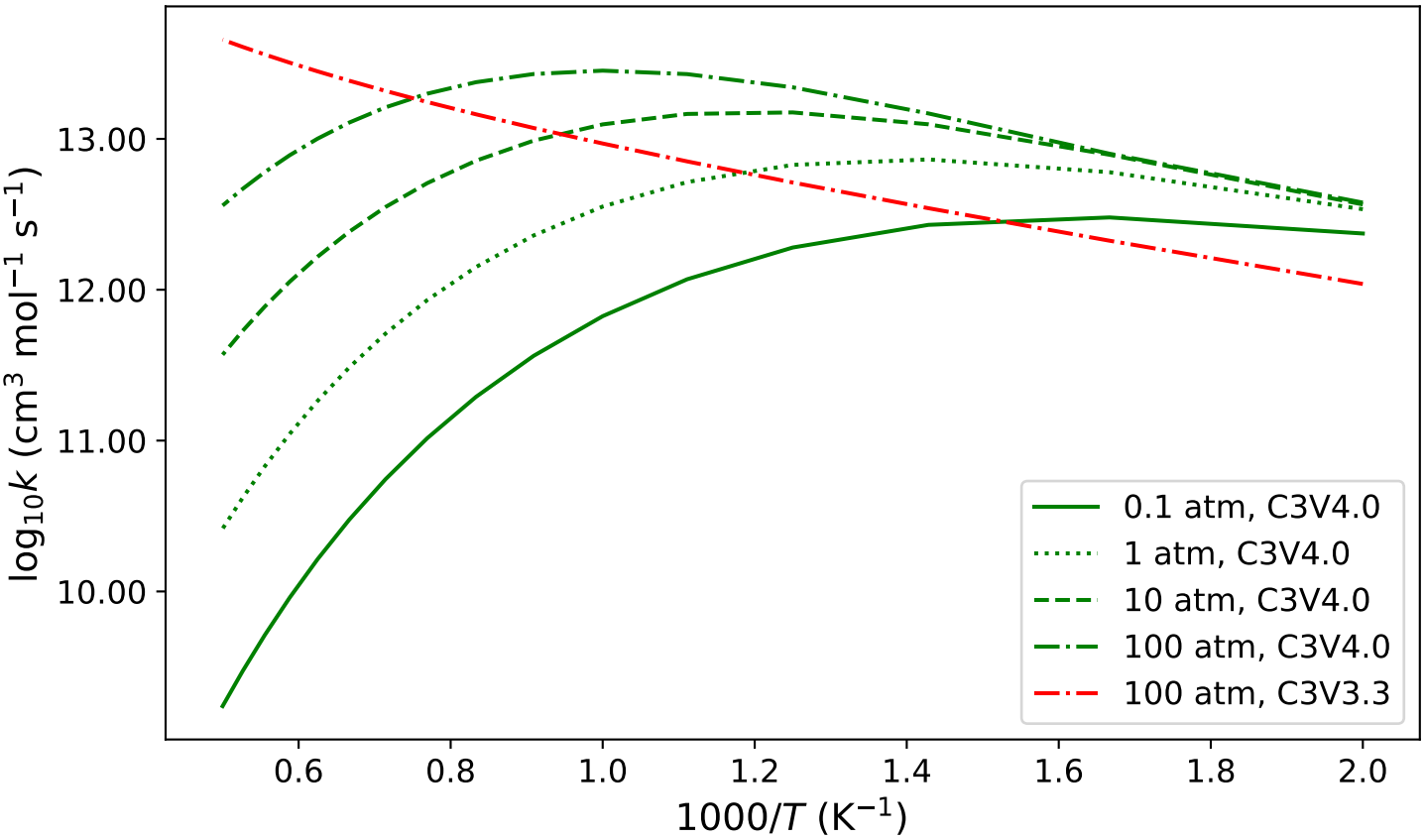



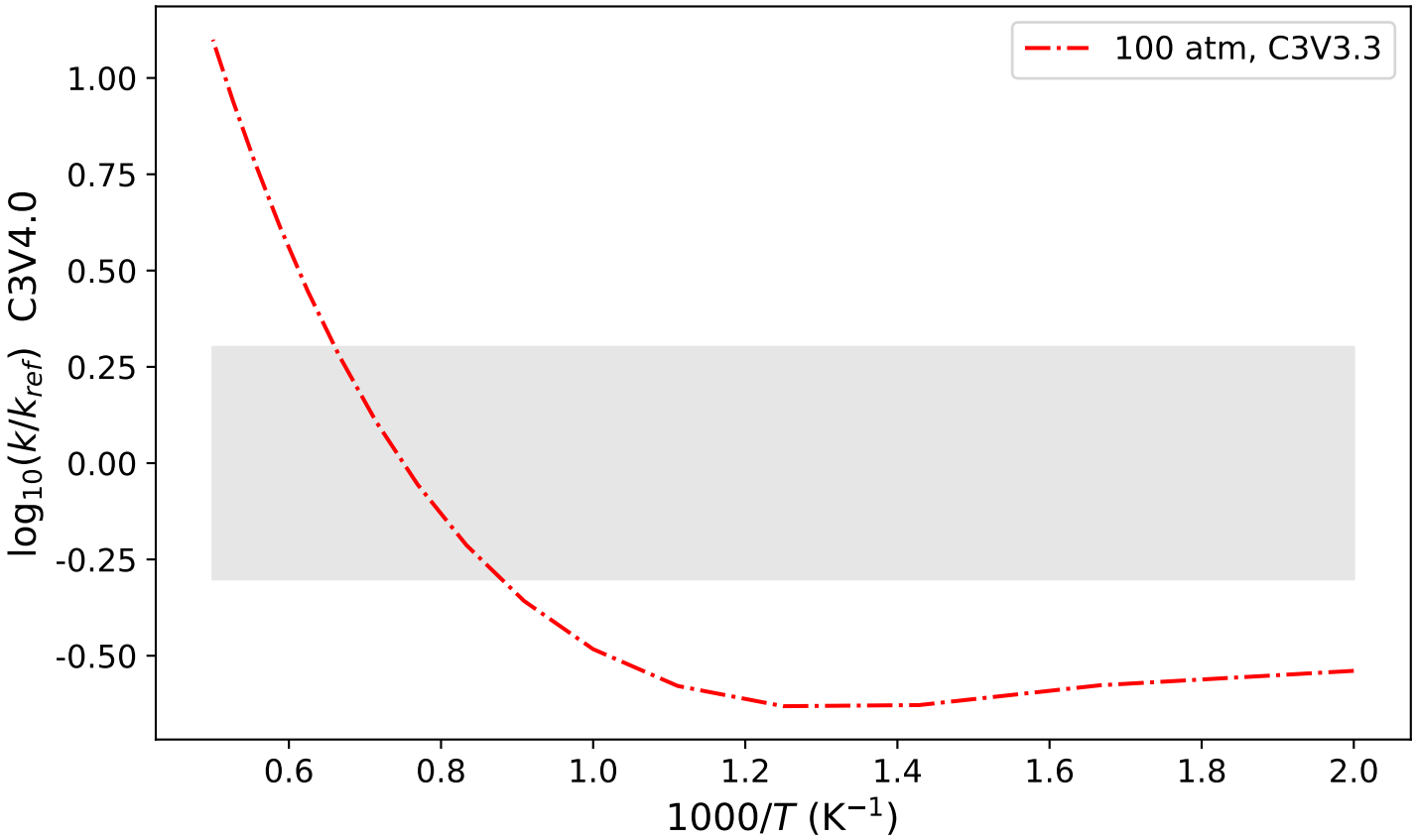
H + C4H10 = H2 + SC4H9



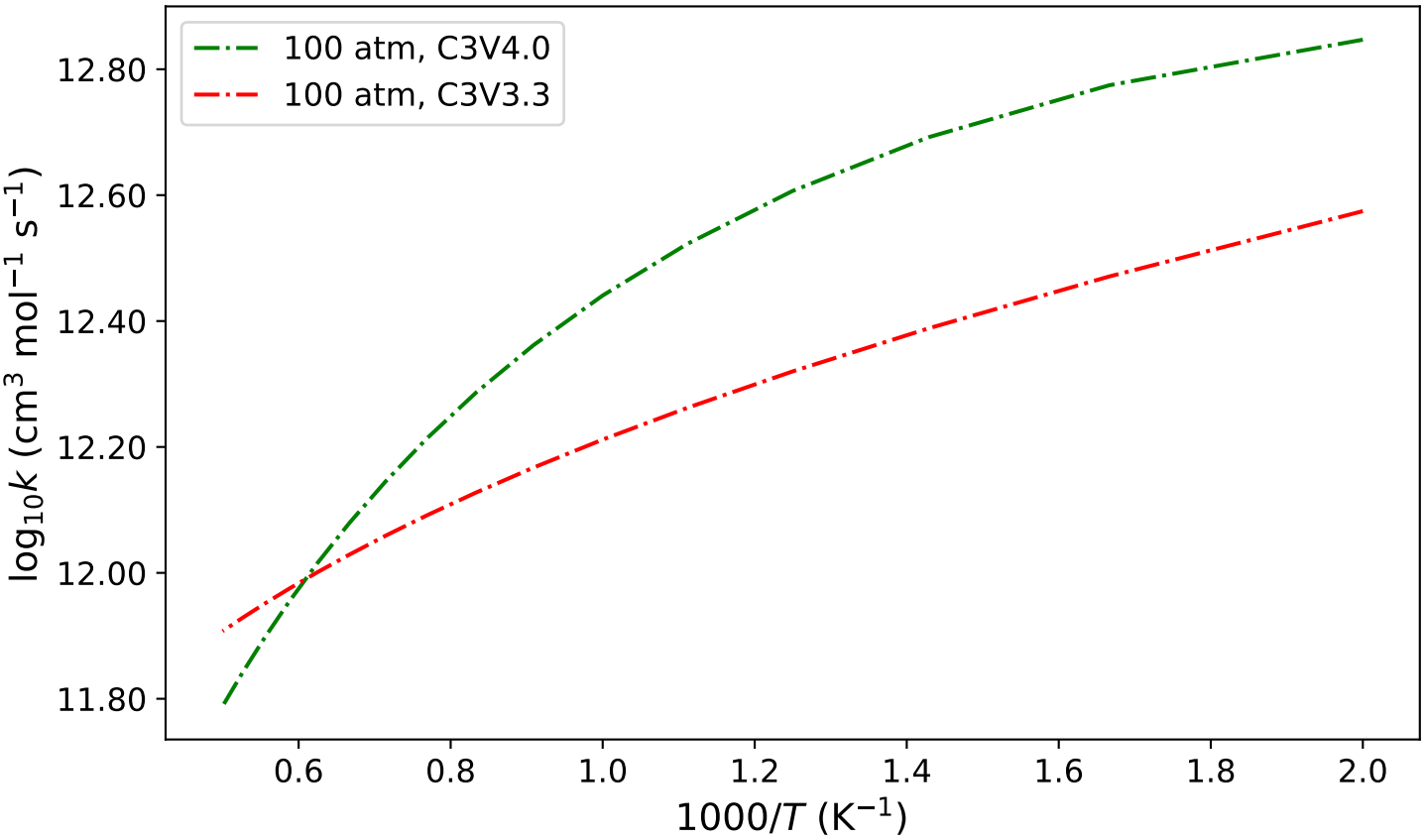



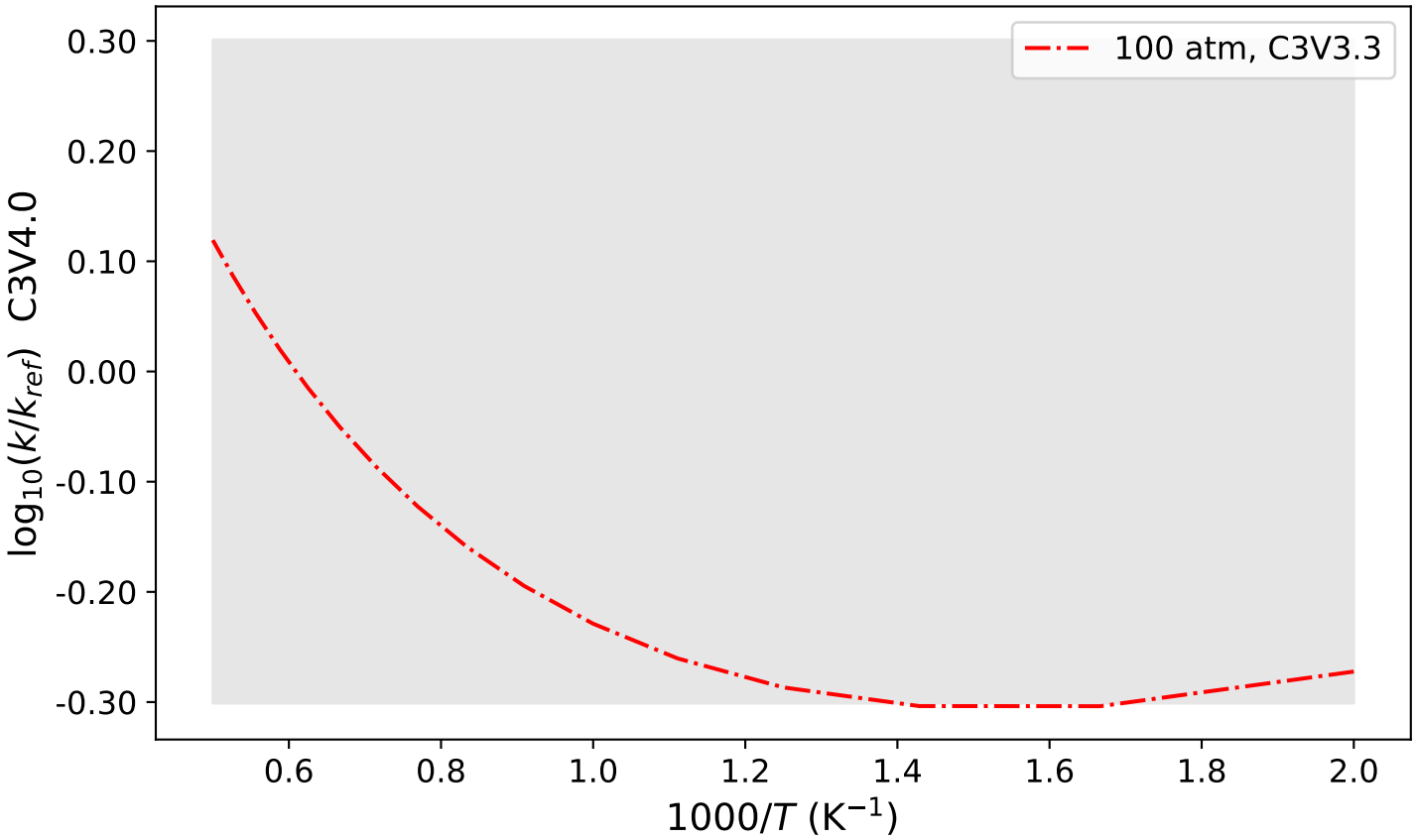
H + C4H6-2 = C4H72-2



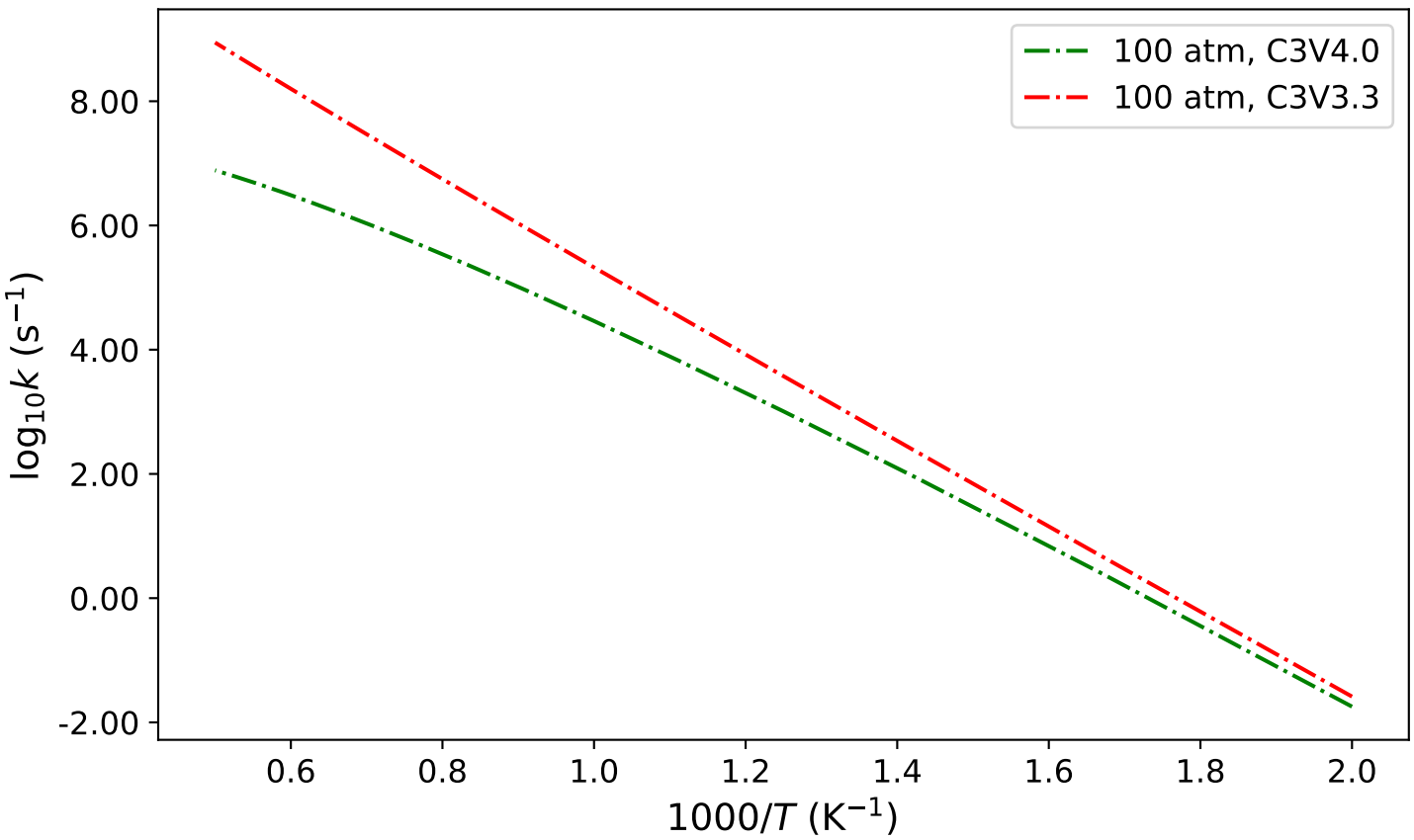



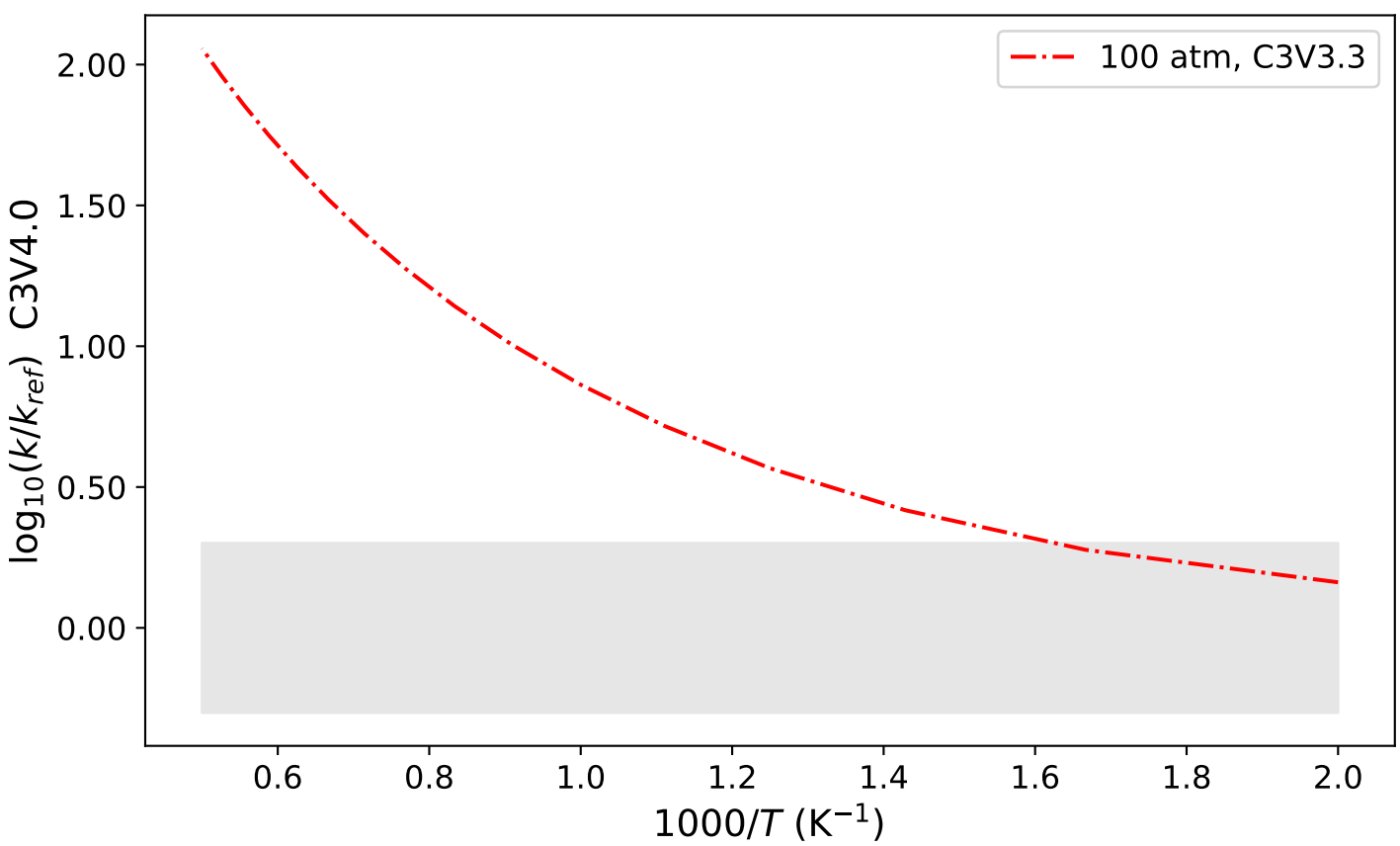
O2 + SC4H9 = SC4H9O2



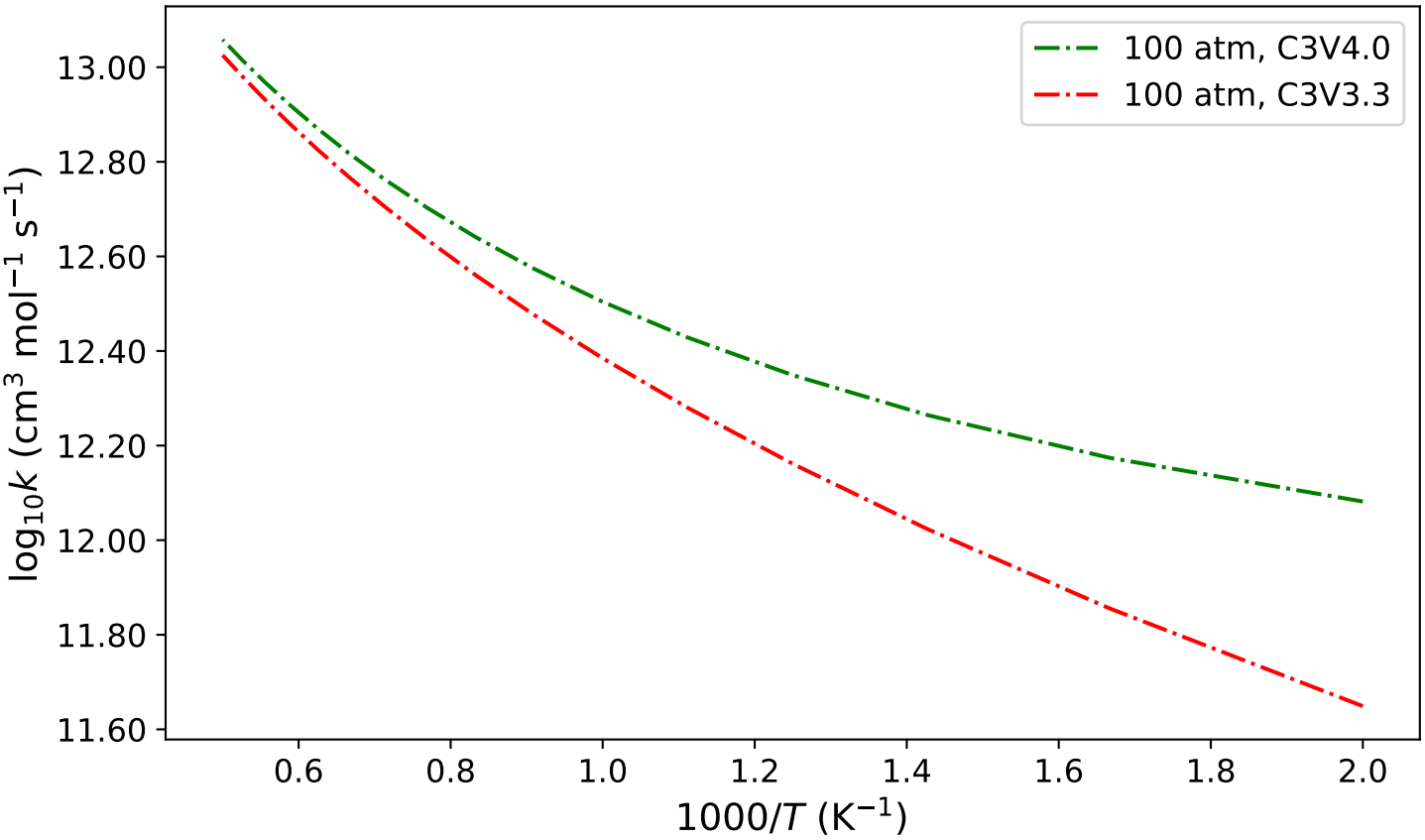



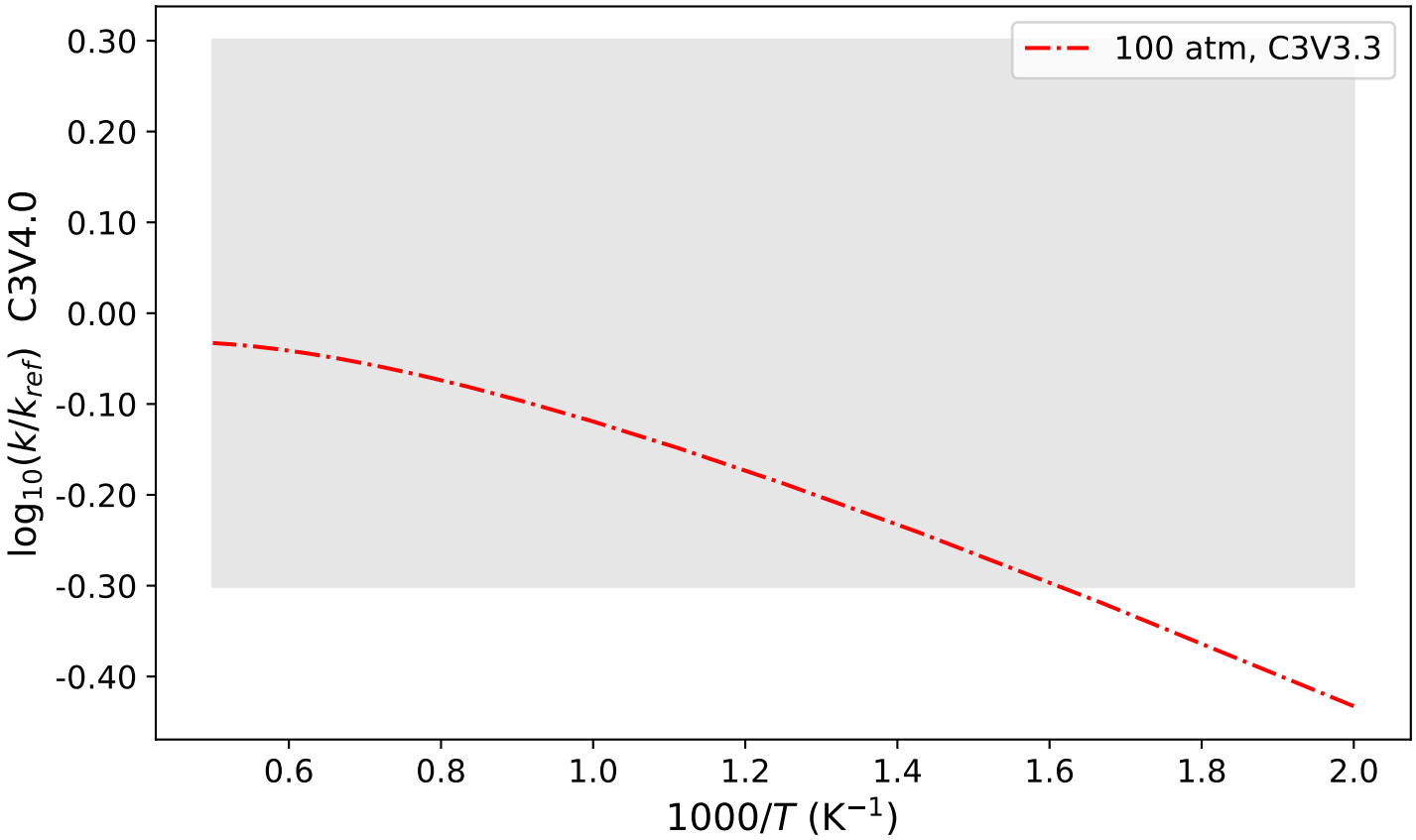
PC4H9O2 = C4H8OOH1-2



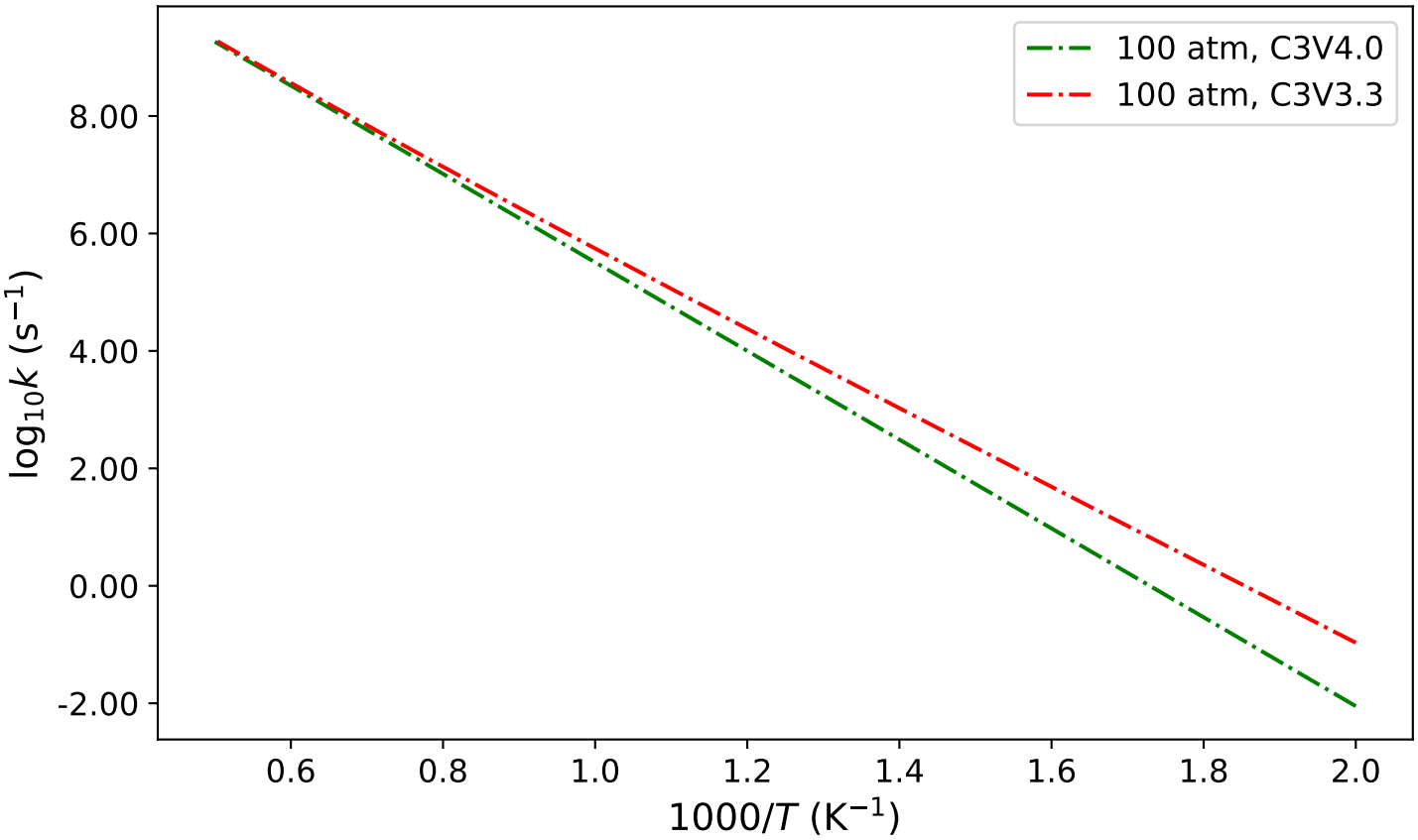



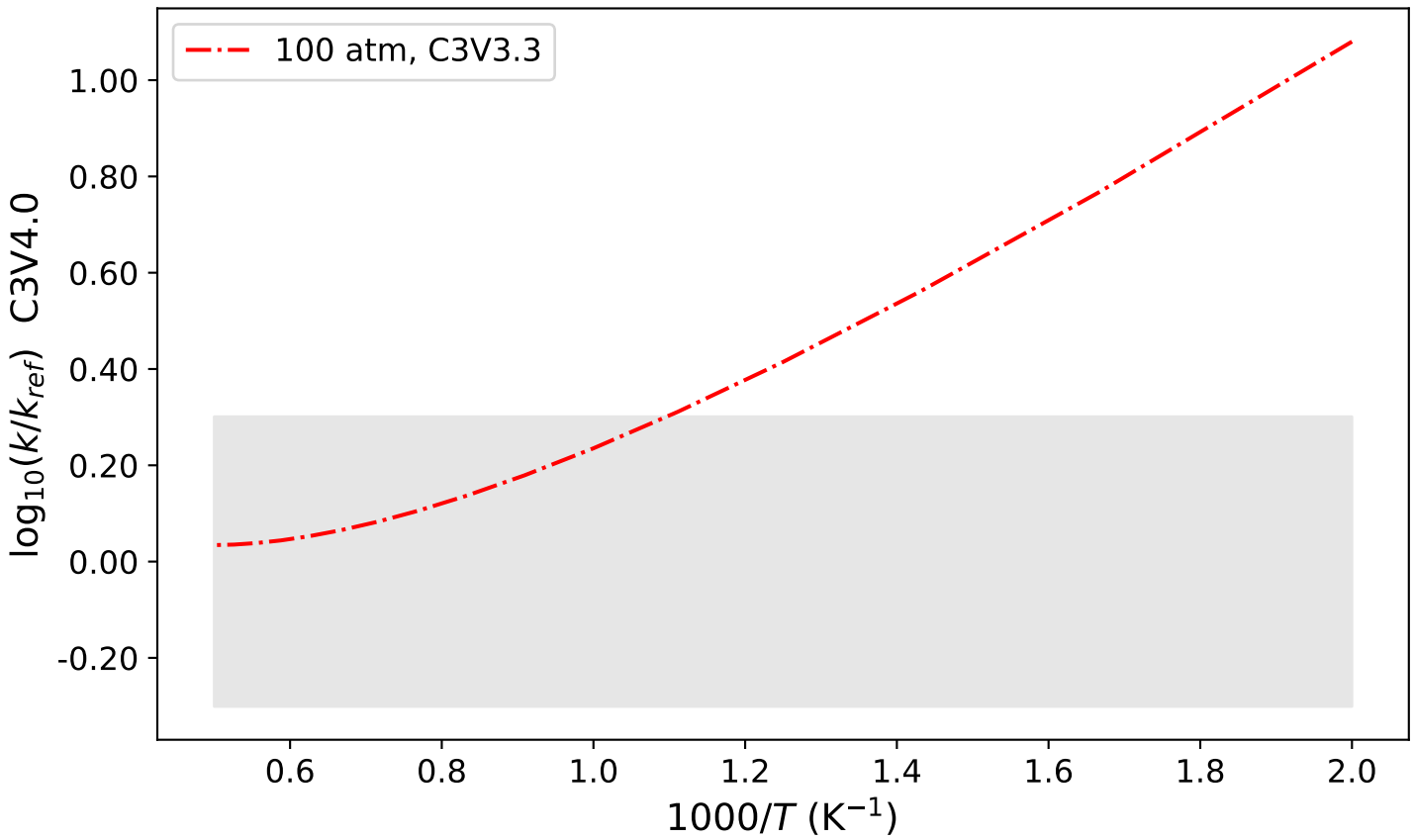
OH + C4H8-1 = H2O + C4H71-4



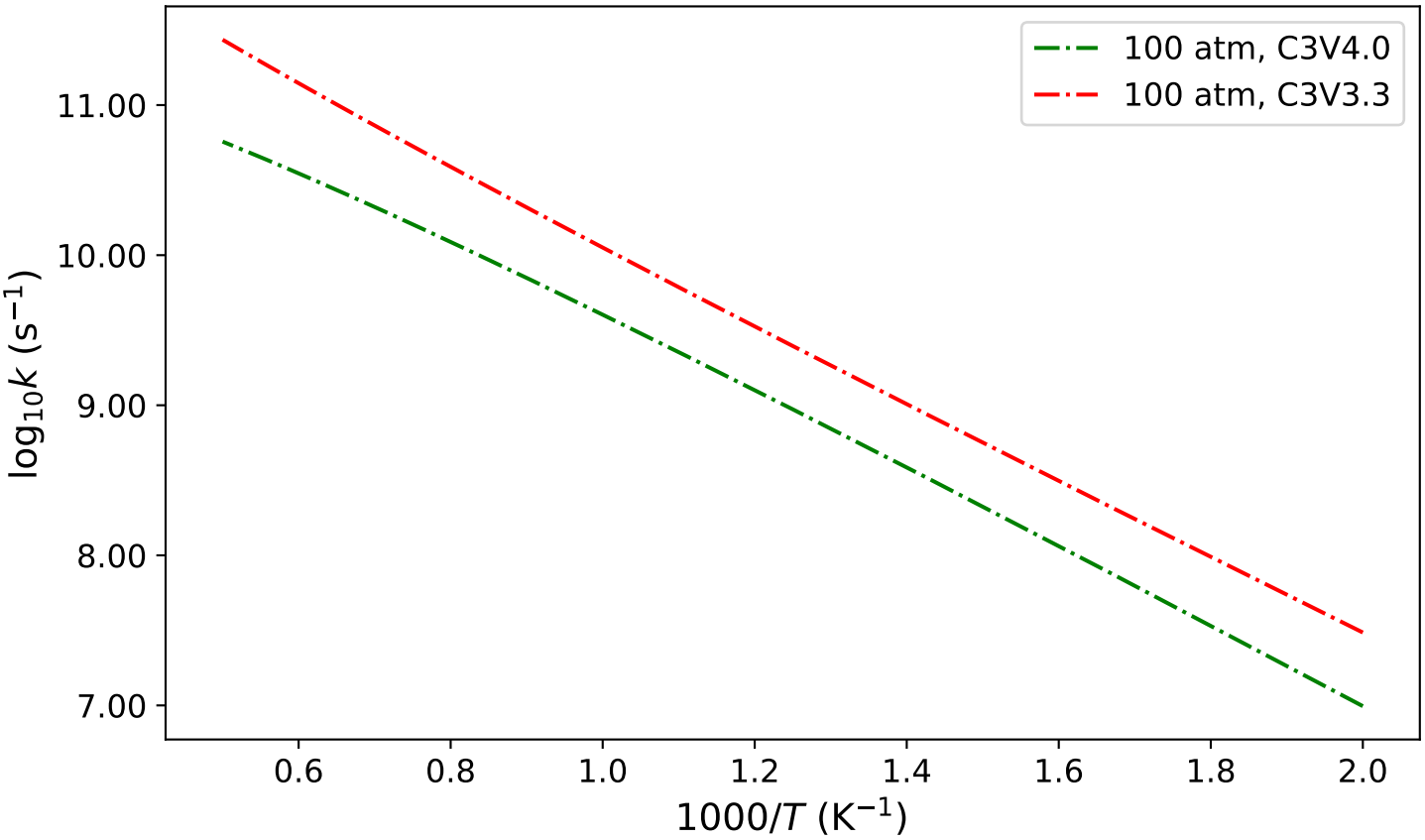



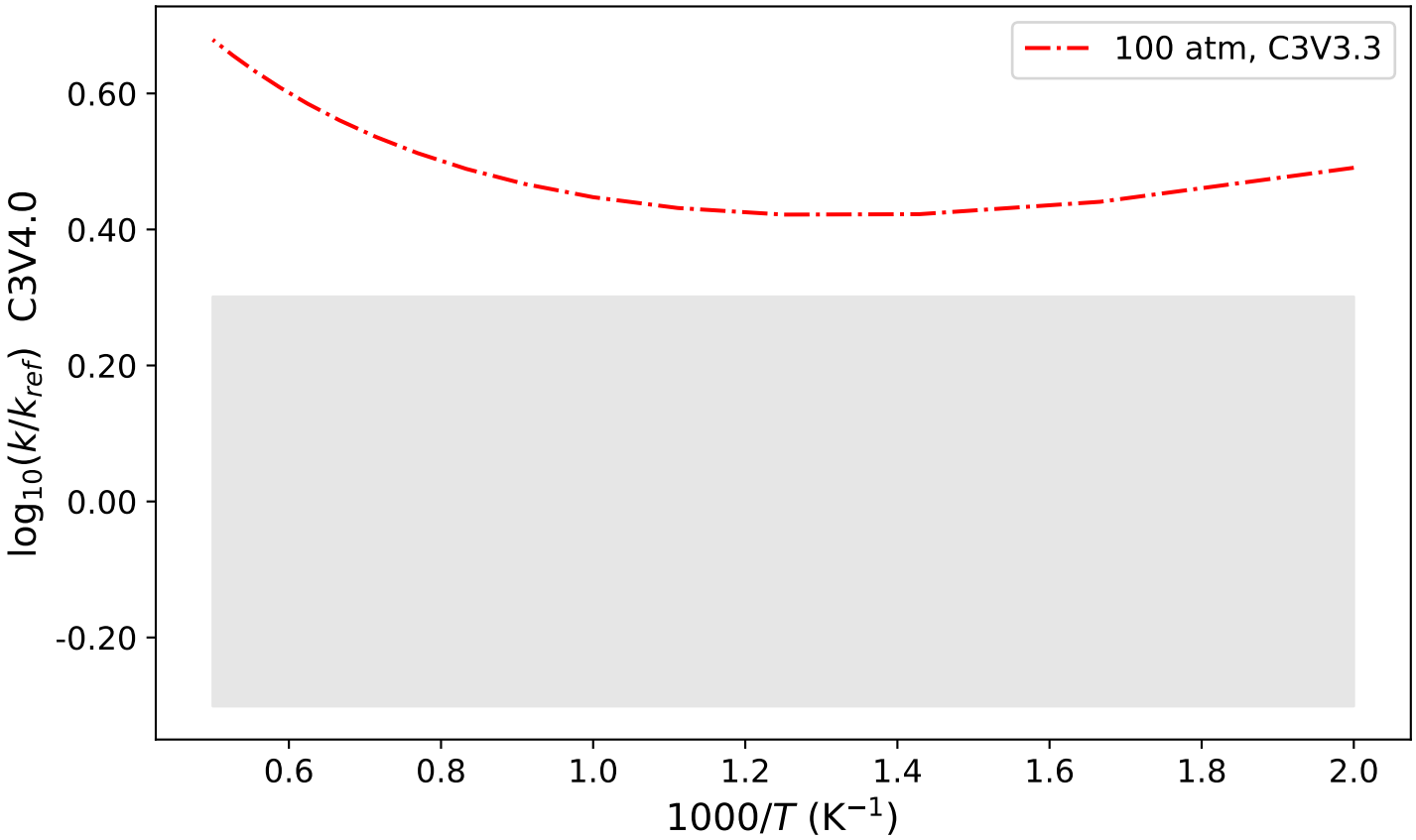
C2H3CHOHCH2OO = HO2 + C4H51,3OH2



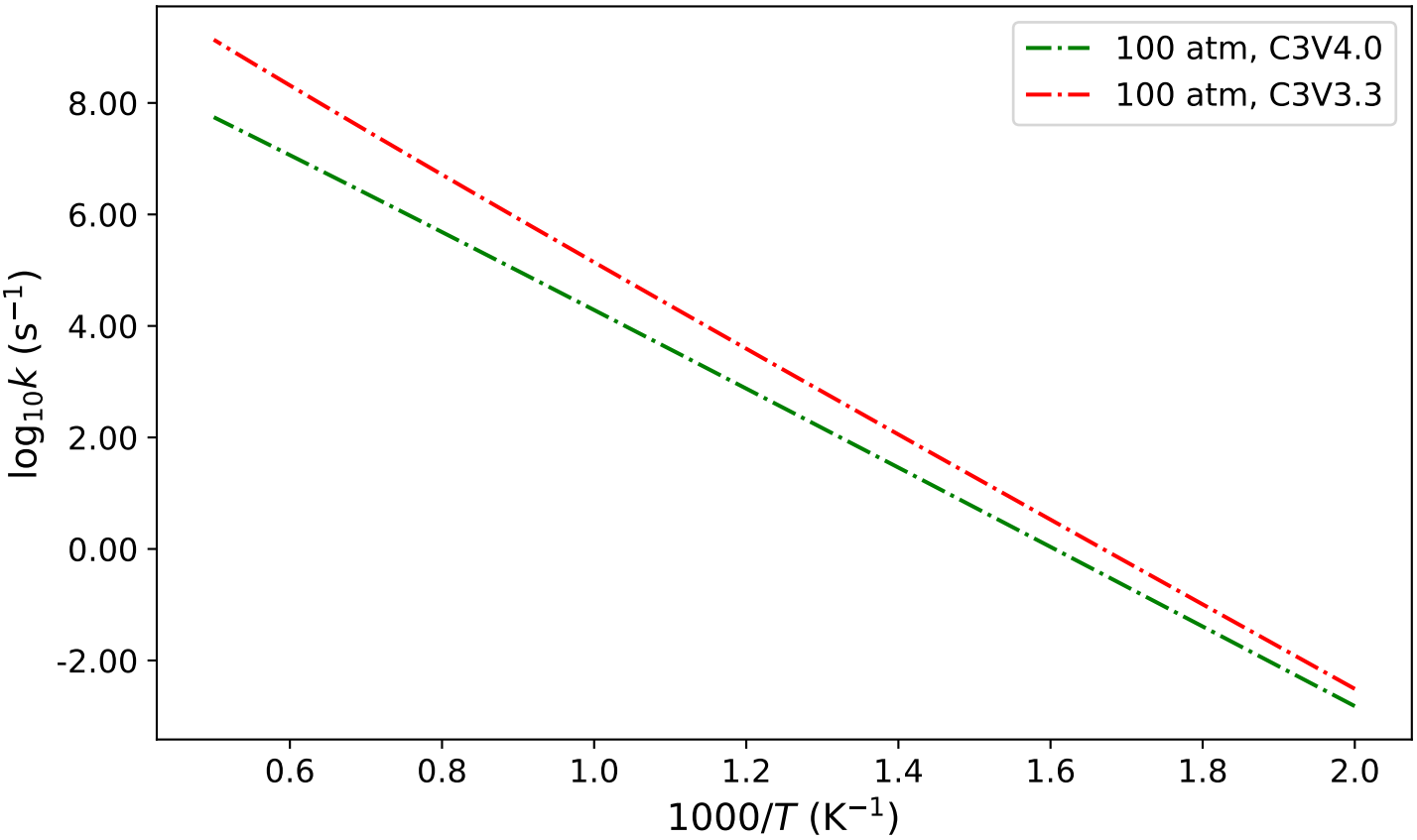



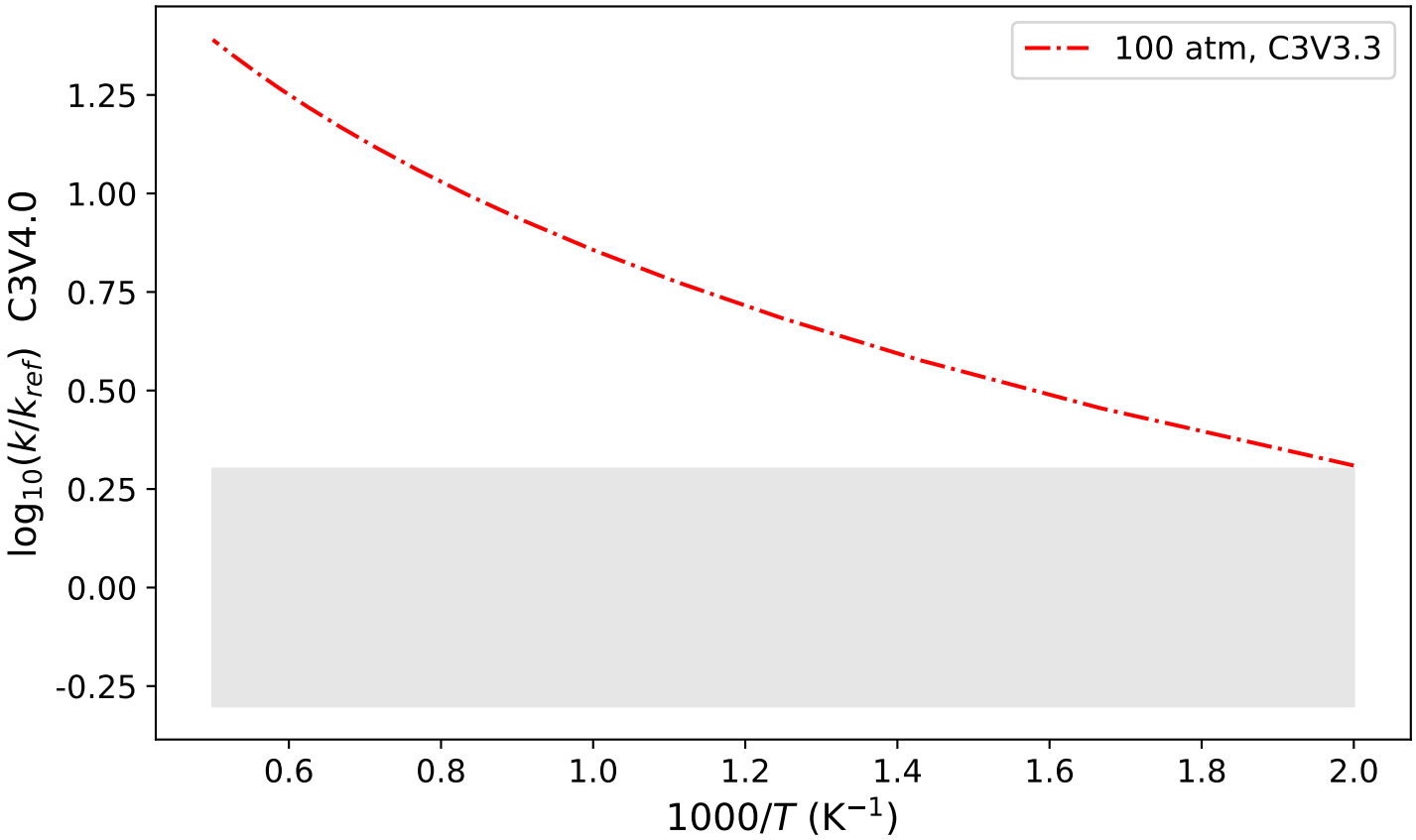
C4H8OOH2-1 = OH + C4H8O1-2



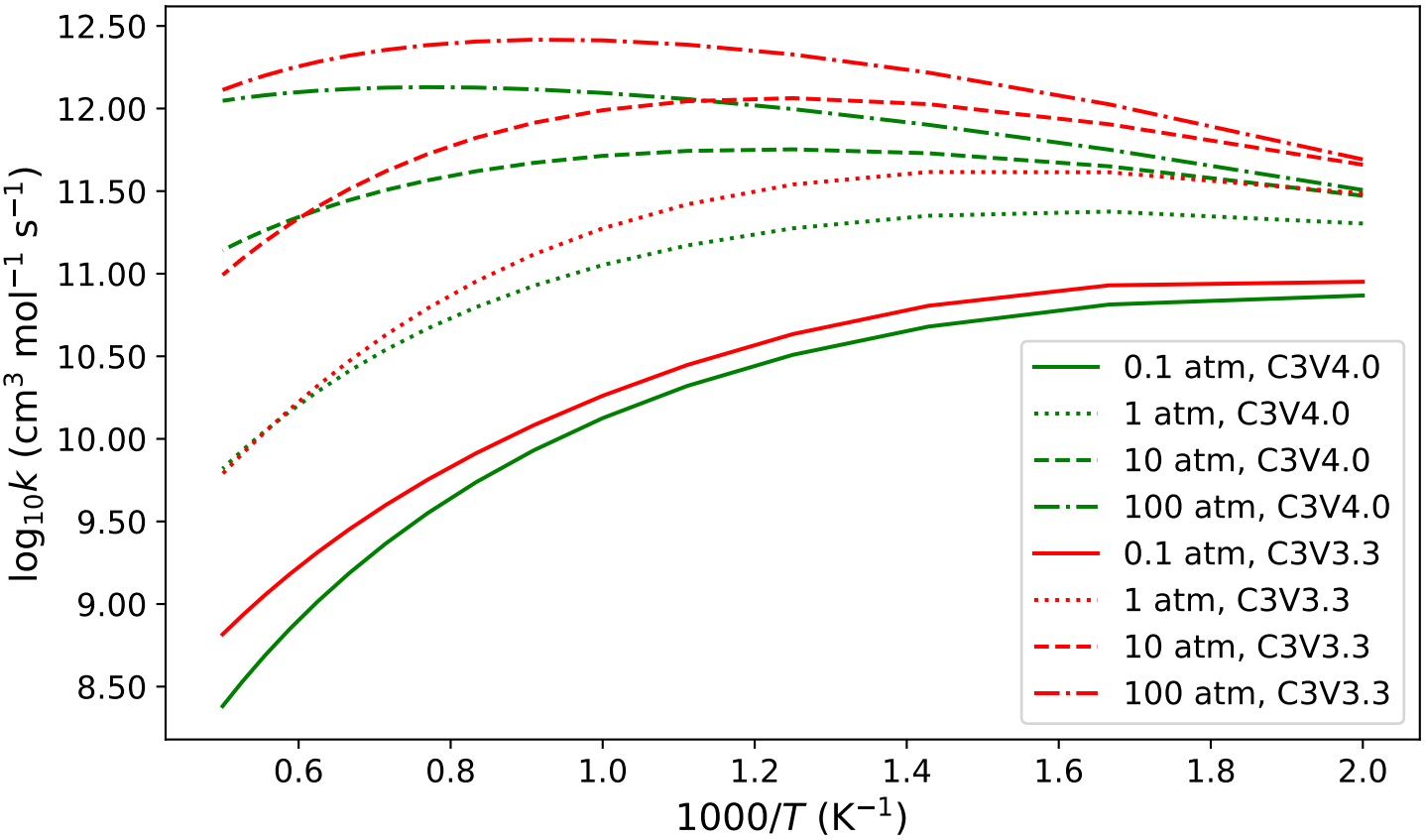



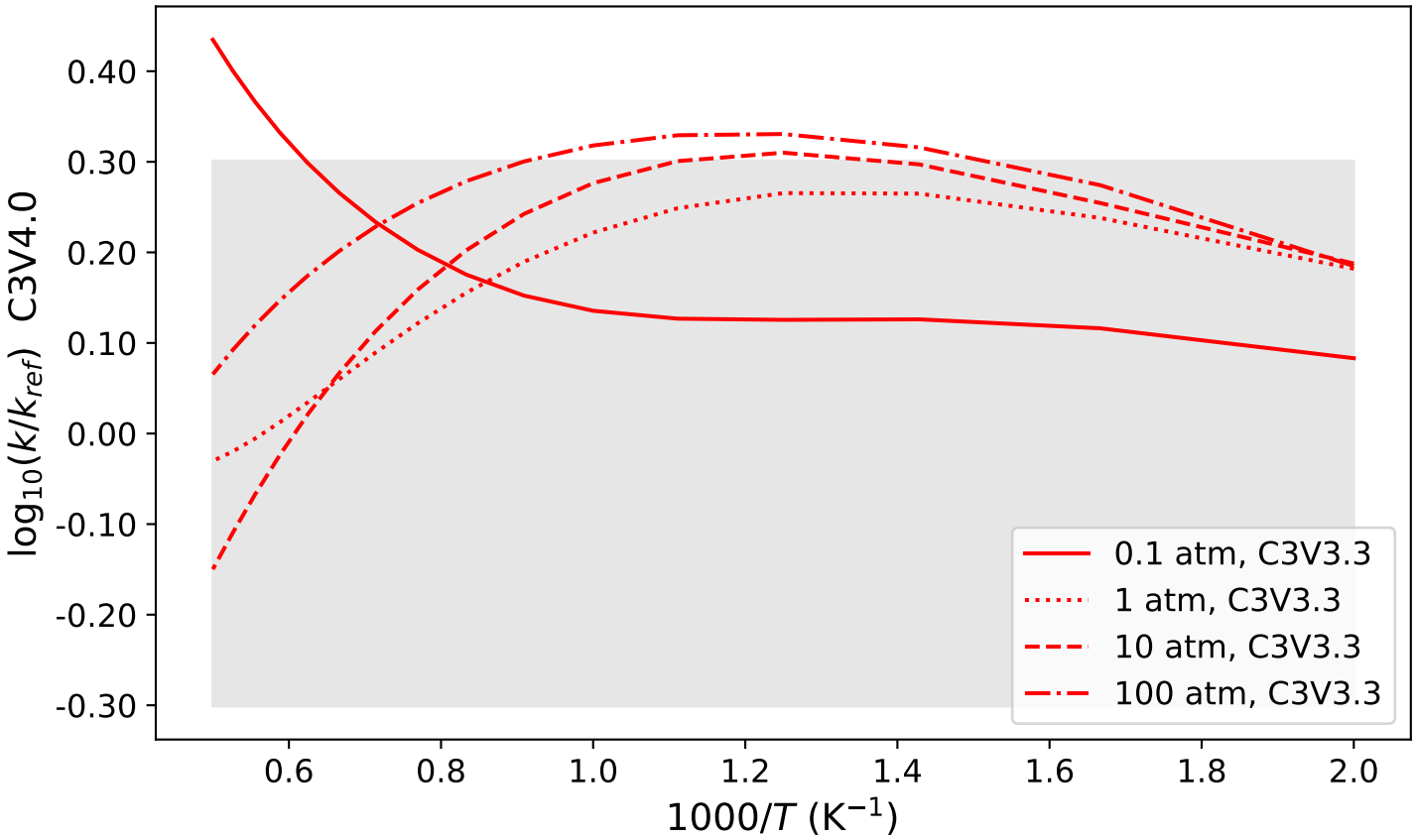
SC4H9O2 = C4H8OOH2-1



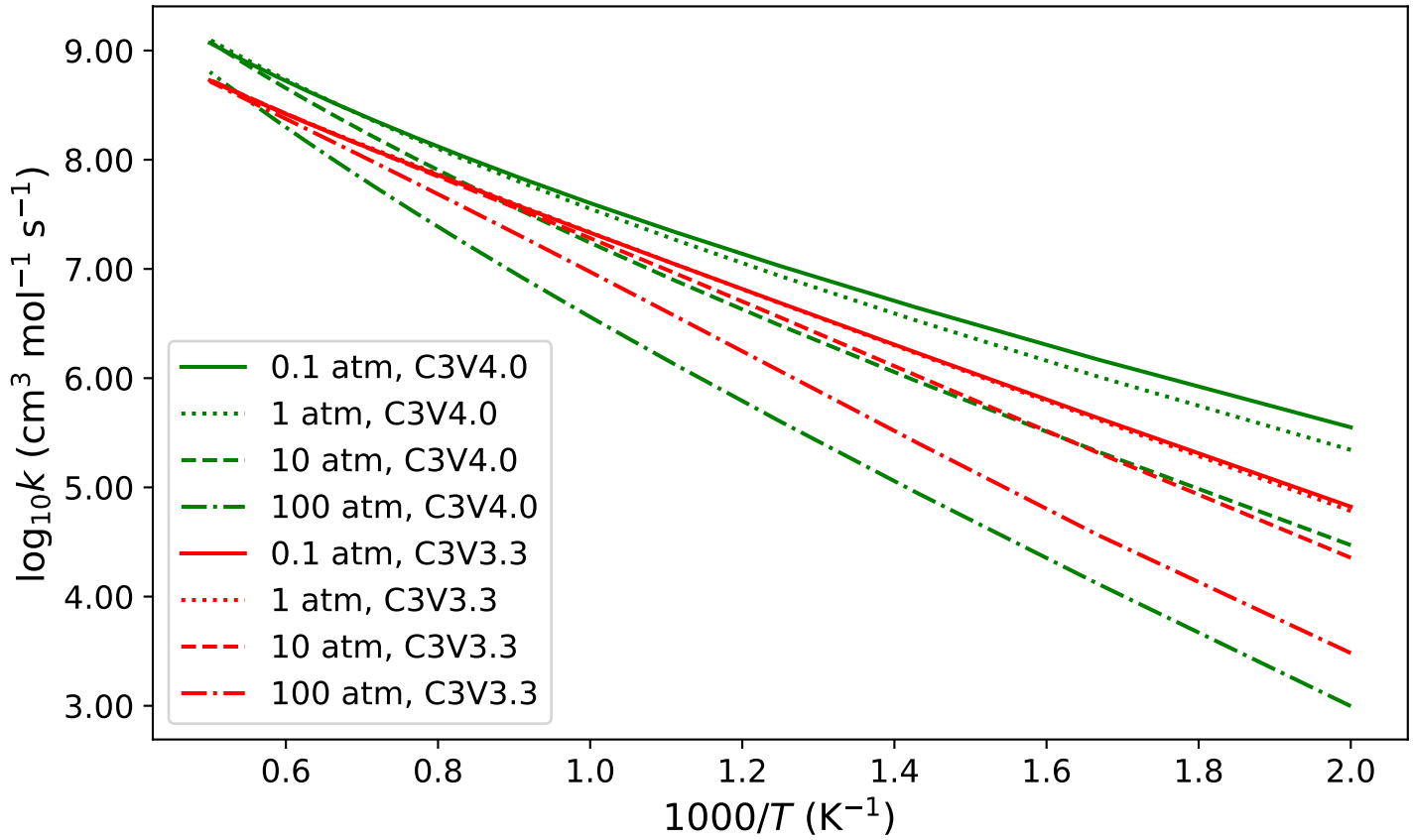



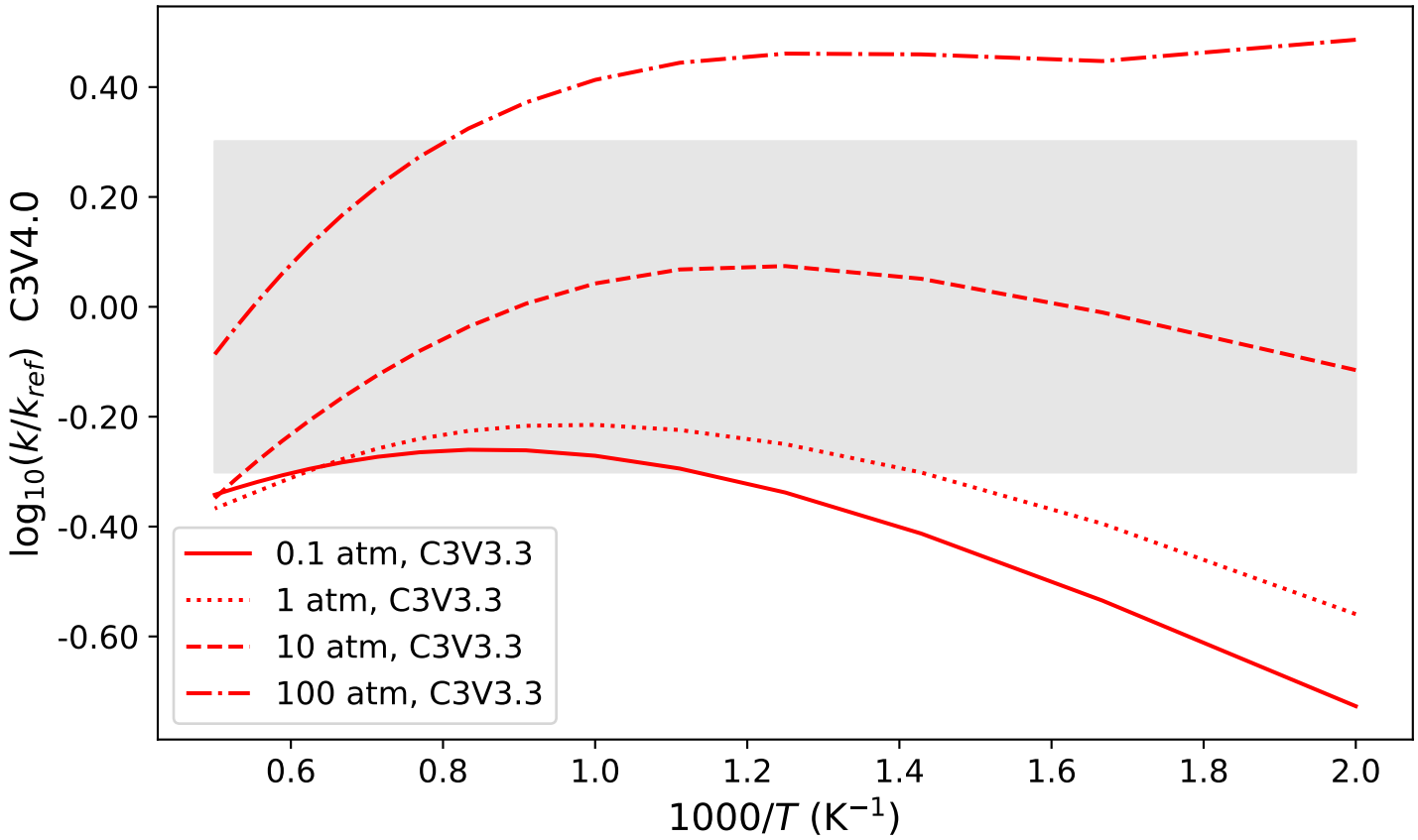
H + C3H6 = NC3H7


12.50
12.00
11.50
11.00
10.50
10.00
9.50
9.00
8.50
$\log_{10}k$ (cm$^3$ mol$^{-1}$ s$^{-1}$)
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ (K$^{-1}$)
0.1 atm, C3V4.0
1 atm, C3V4.0
10 atm, C3V4.0
100 atm, C3V4.0
0.1 atm, C3V3.3
1 atm, C3V3.3
10 atm, C3V3.3
100 atm, C3V3.3

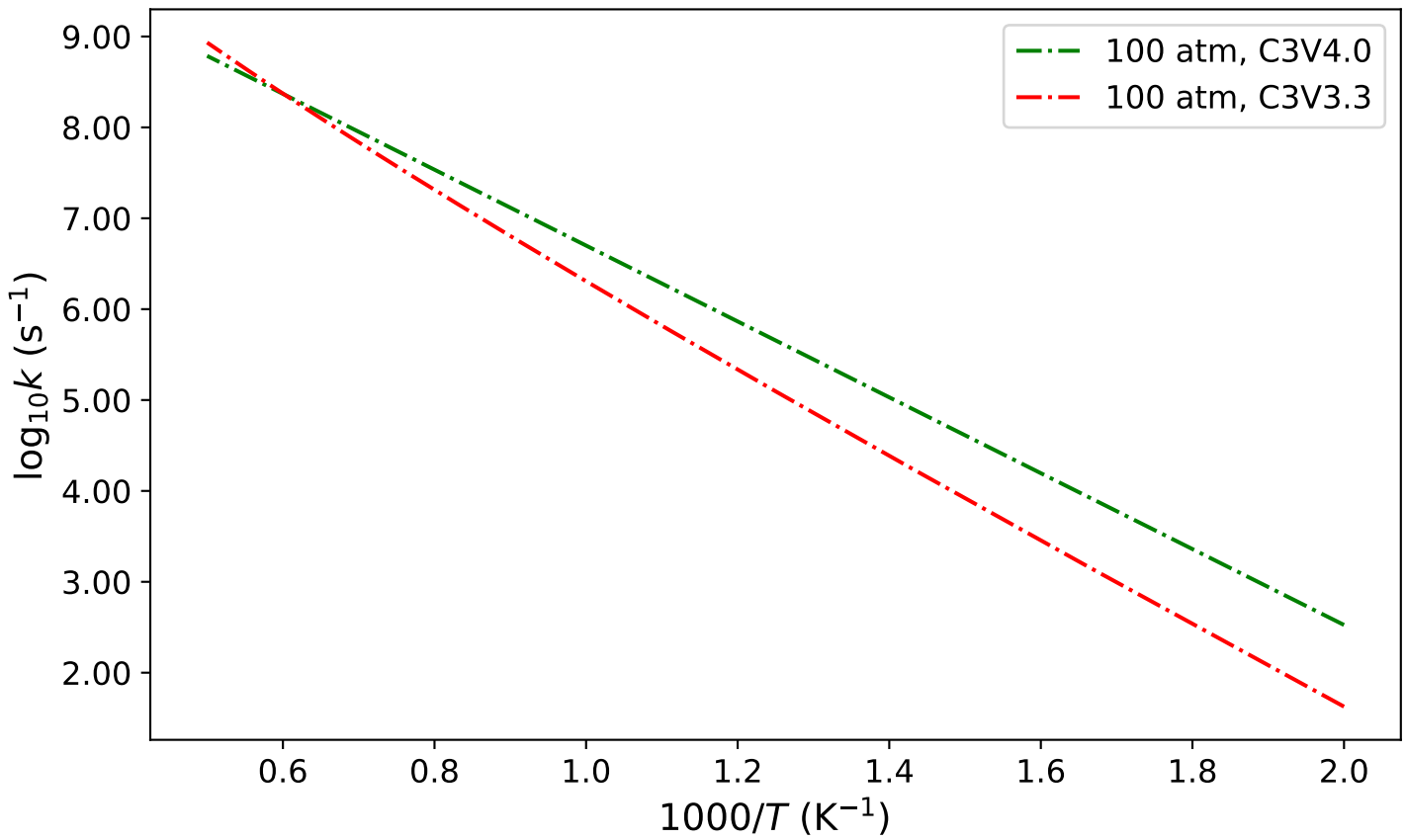

0.40
0.30
0.20
0.10
0.00
-0.10
-0.20
-0.30
$\log_{10}(k/k_{ref})$ C3V4.0
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ (K$^{-1}$)
0.1 atm, C3V3.3
1 atm, C3V3.3
10 atm, C3V3.3
100 atm, C3V3.3

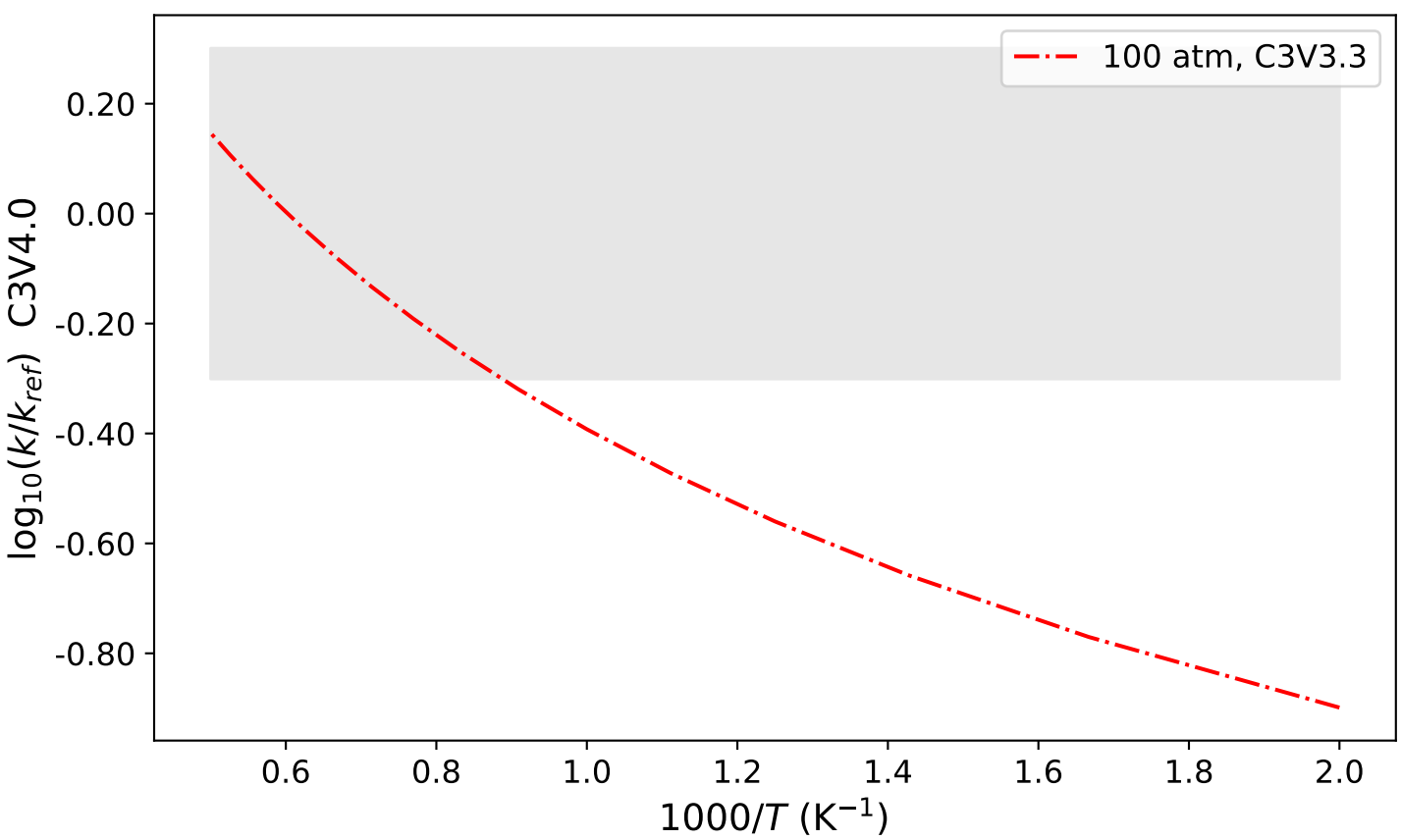
HO2 + C3H4-A = CH2O + CH3CO



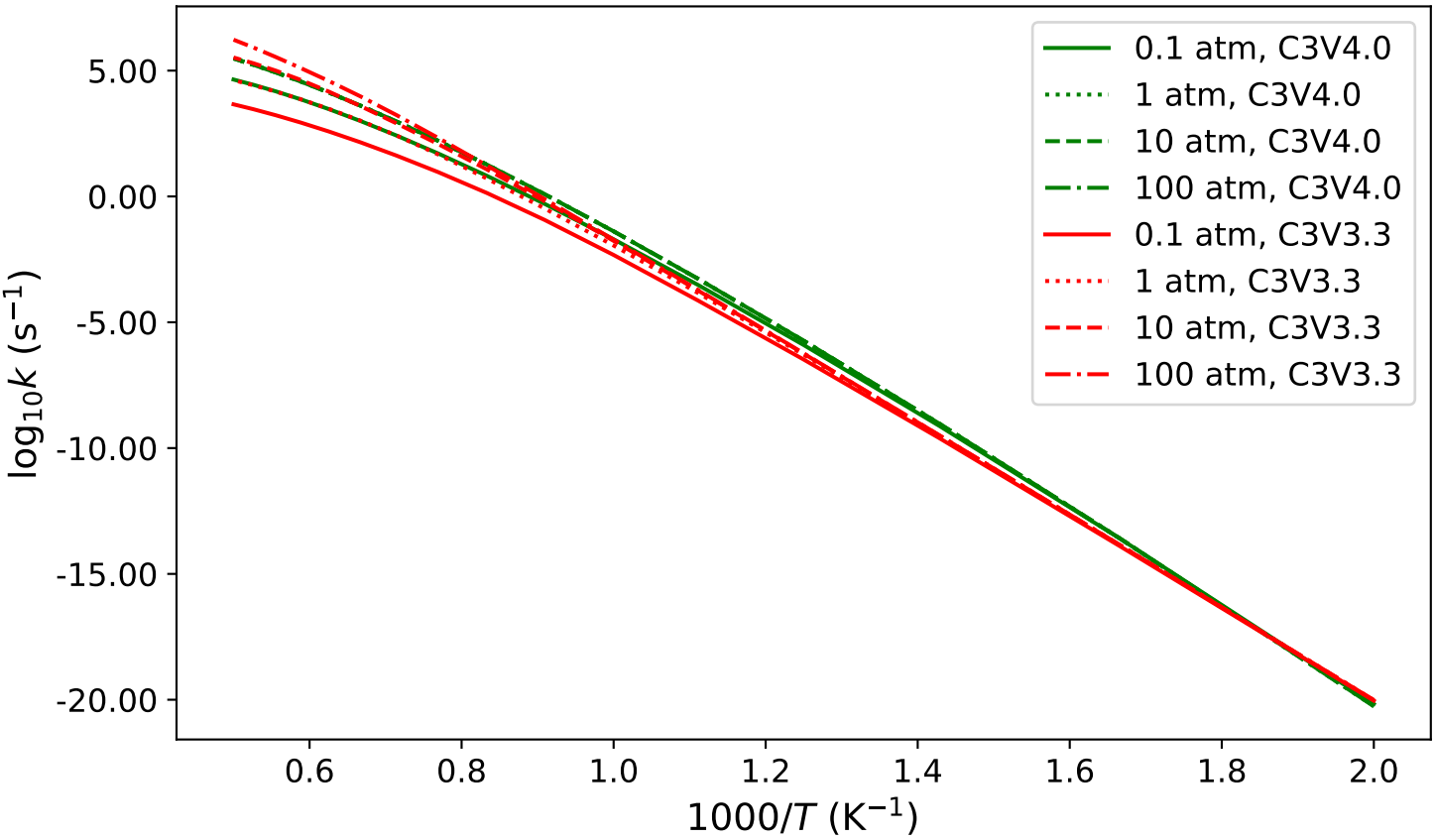



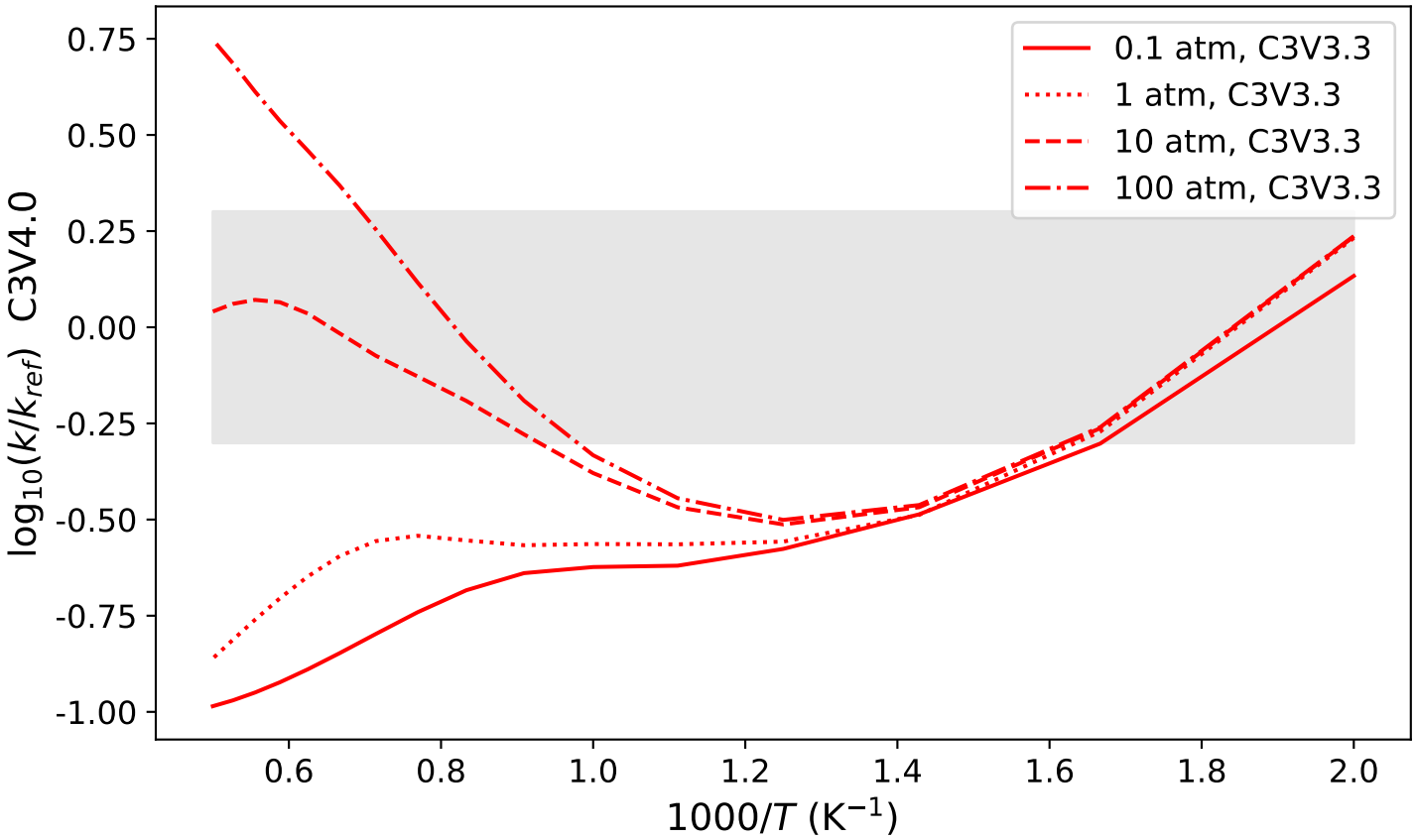
IC4H8OOH-IO2 = OH + IC4KETII



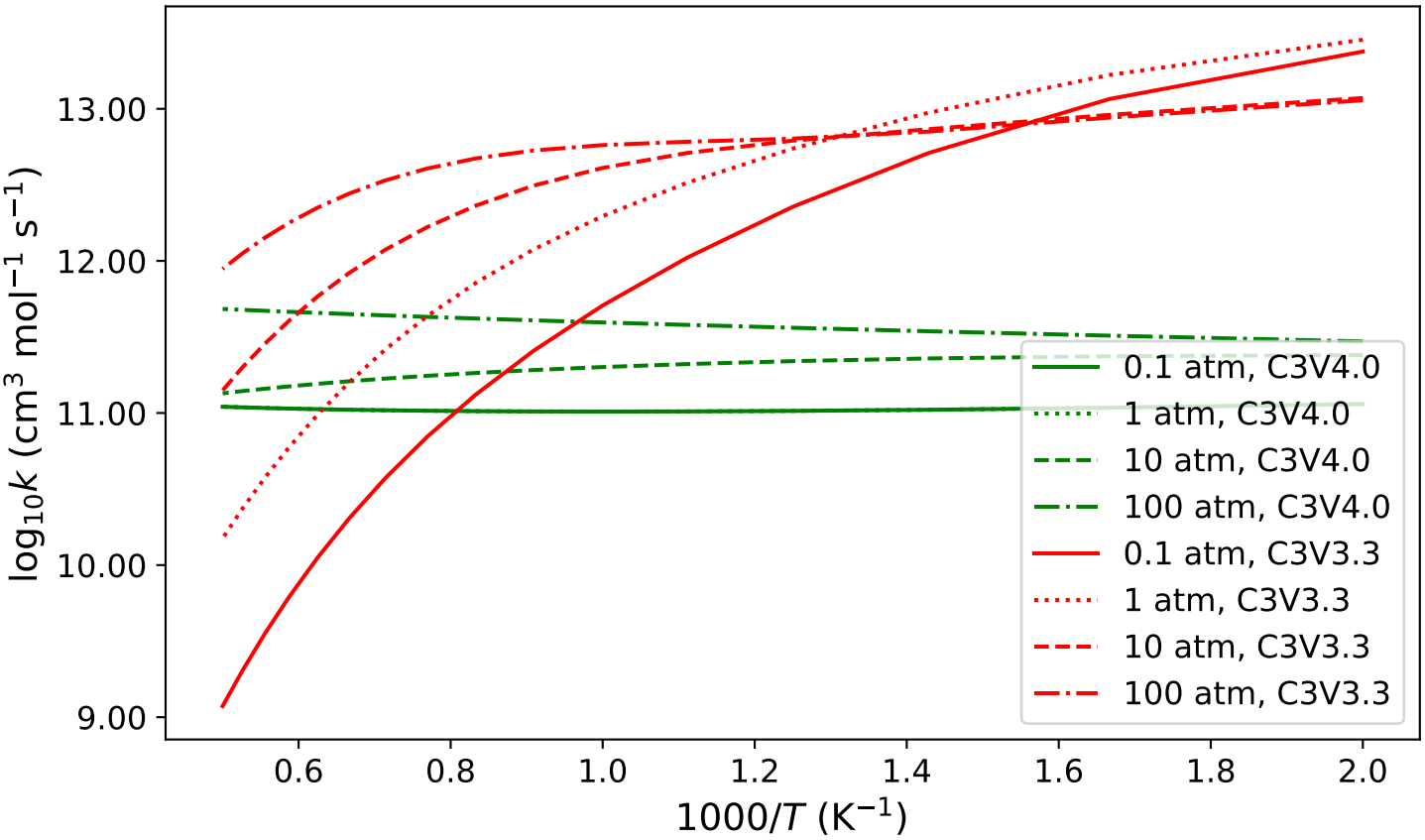



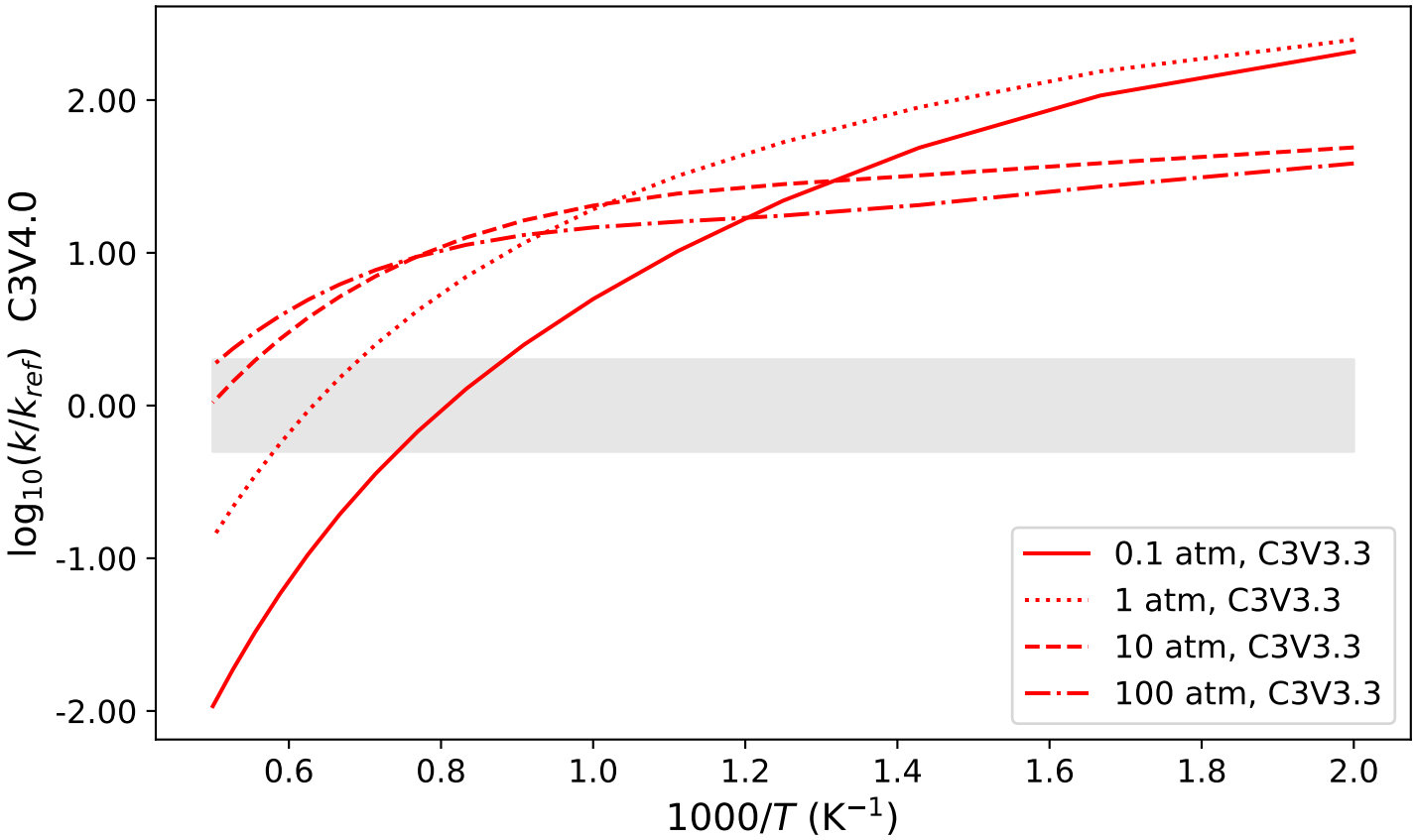
IC3H7OH = CH3 + SC2H4OH



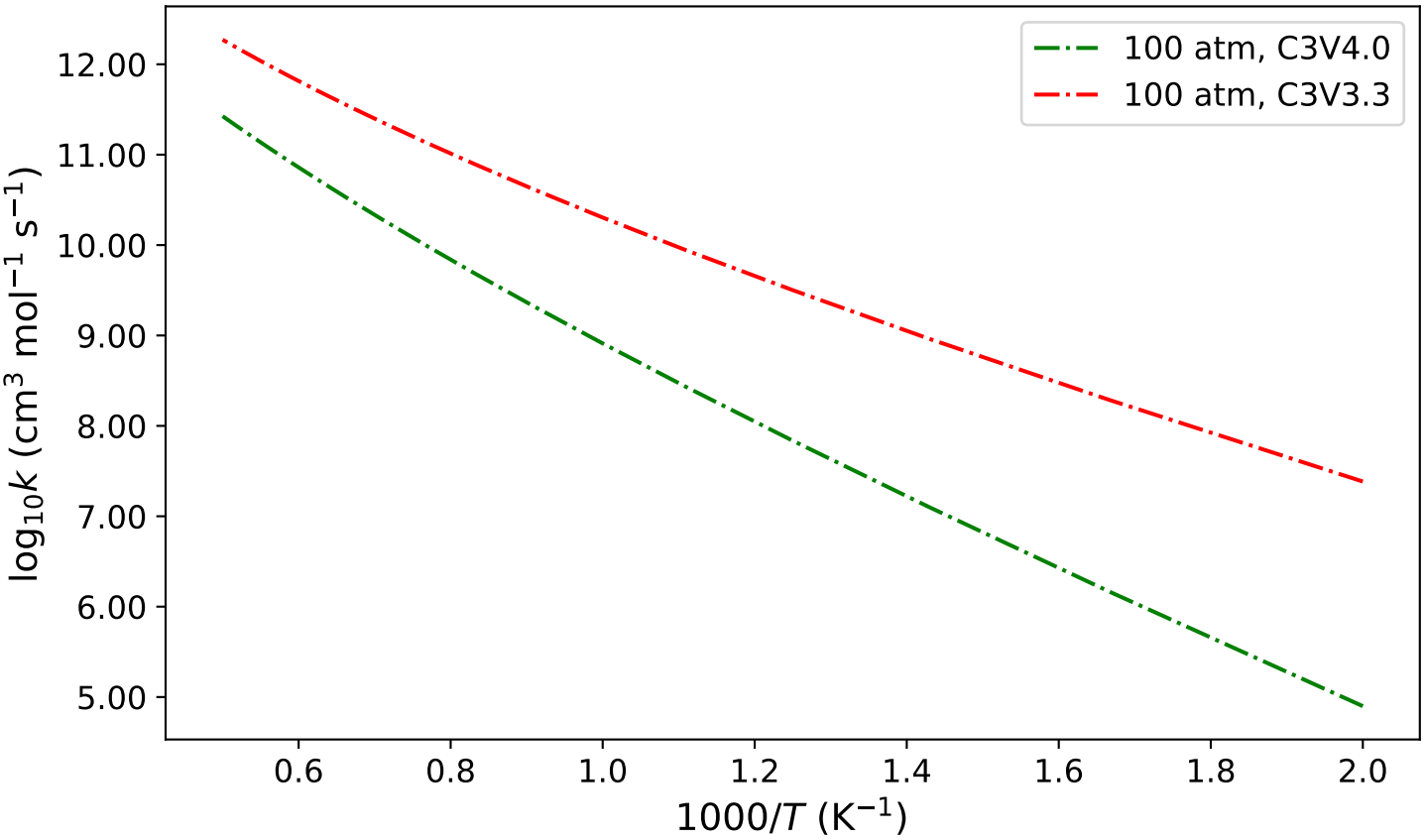



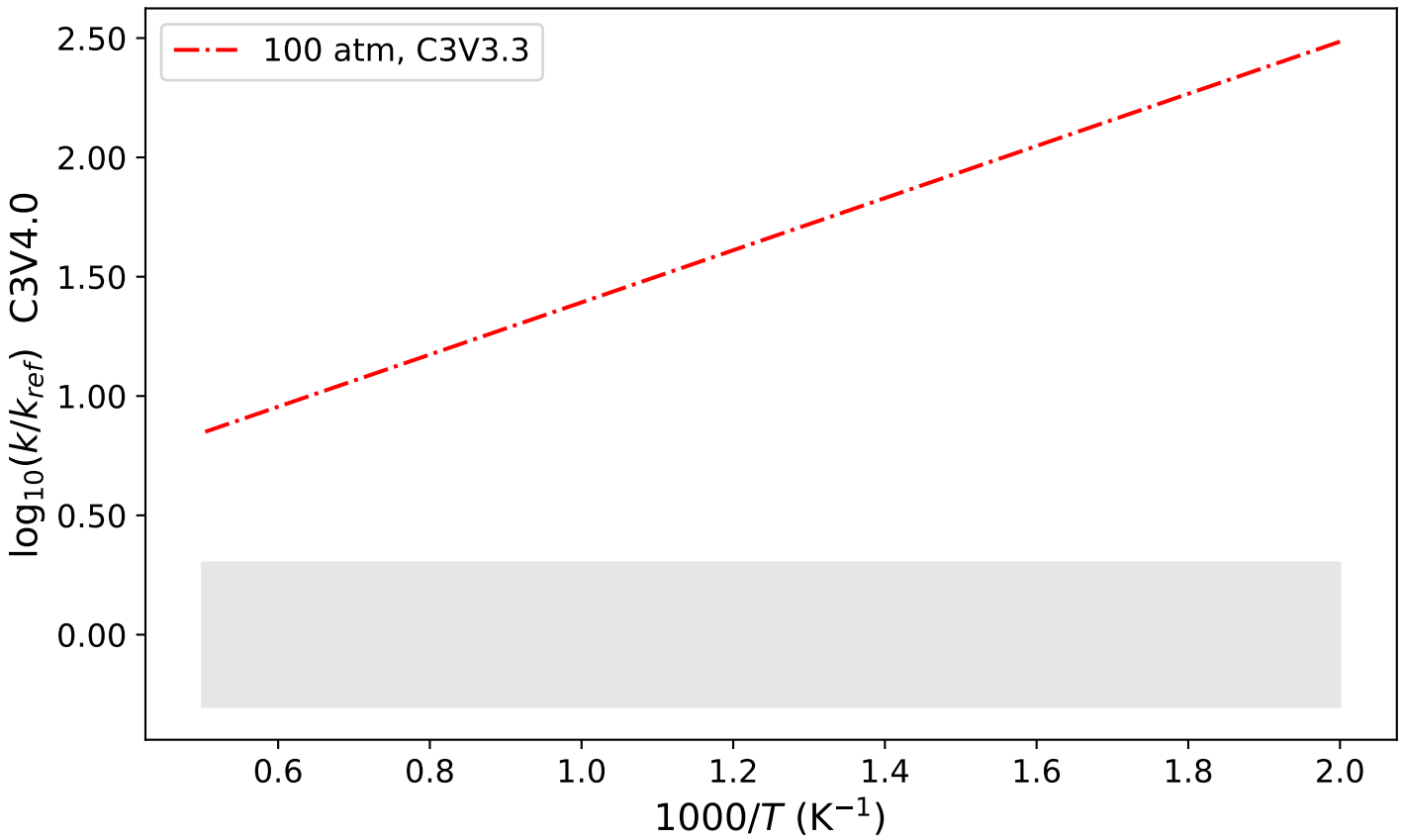
OH + C4H6 = C2H3CHOHCH2



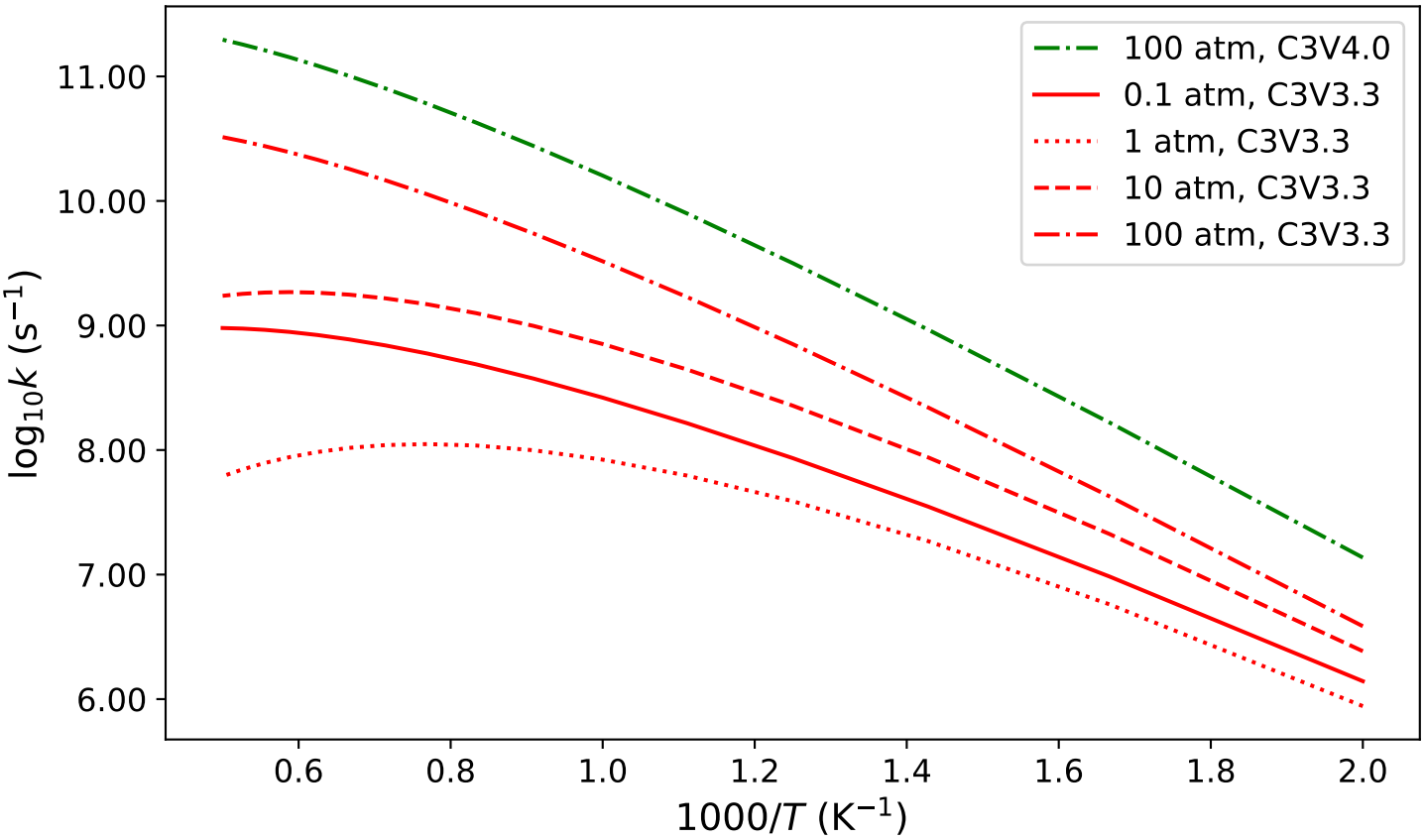



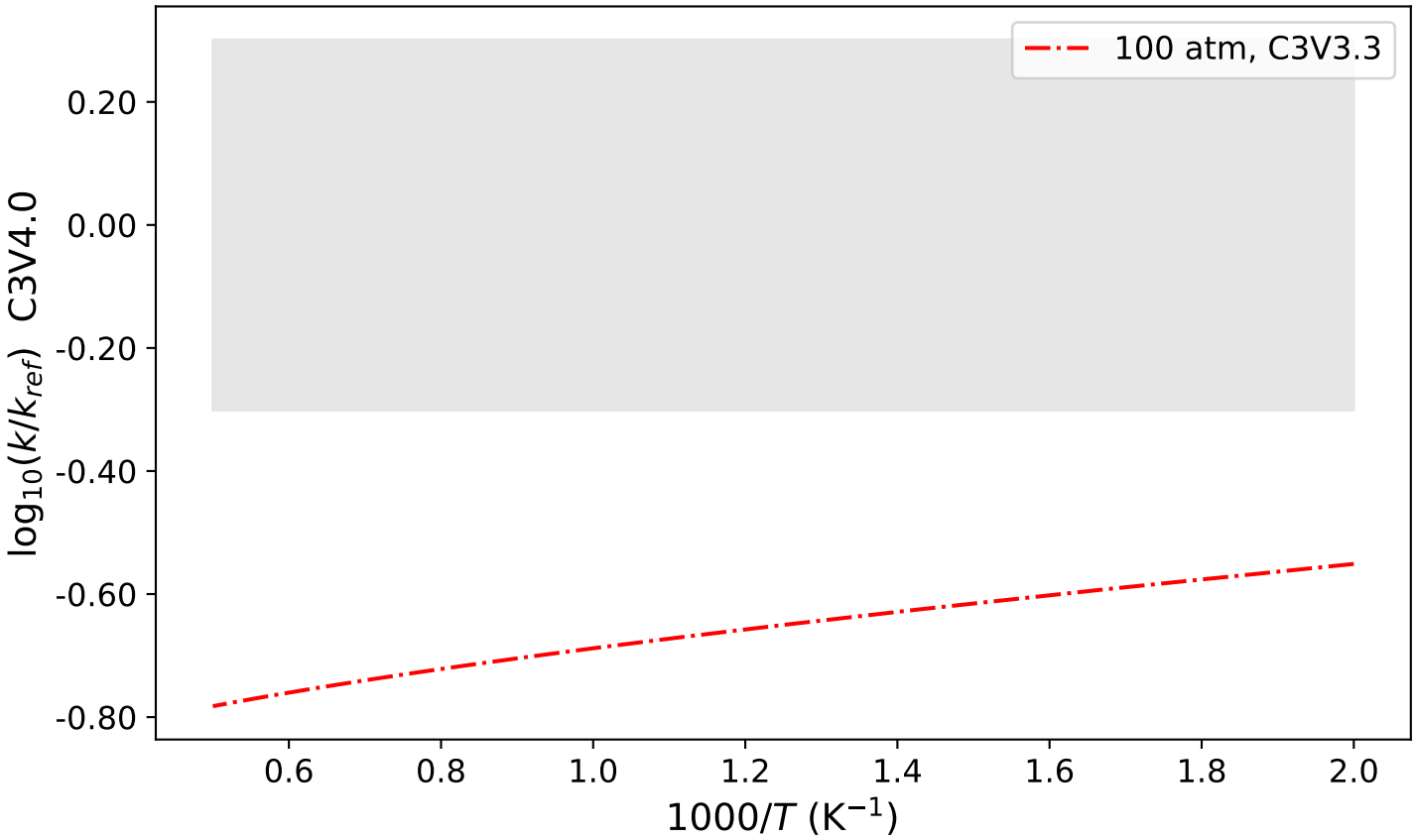
O2CHO + C4H10 = HO2CHO + SC4H9



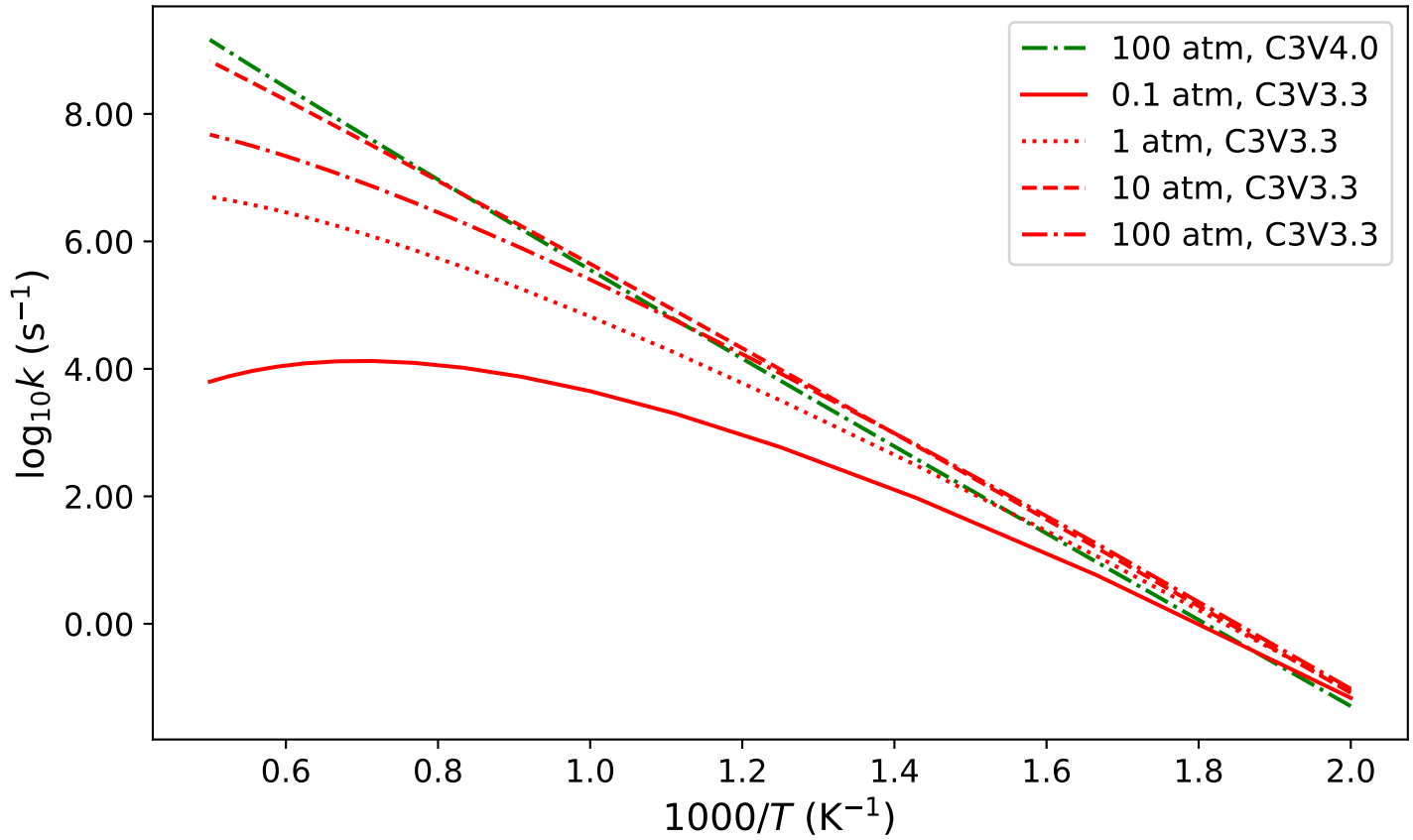



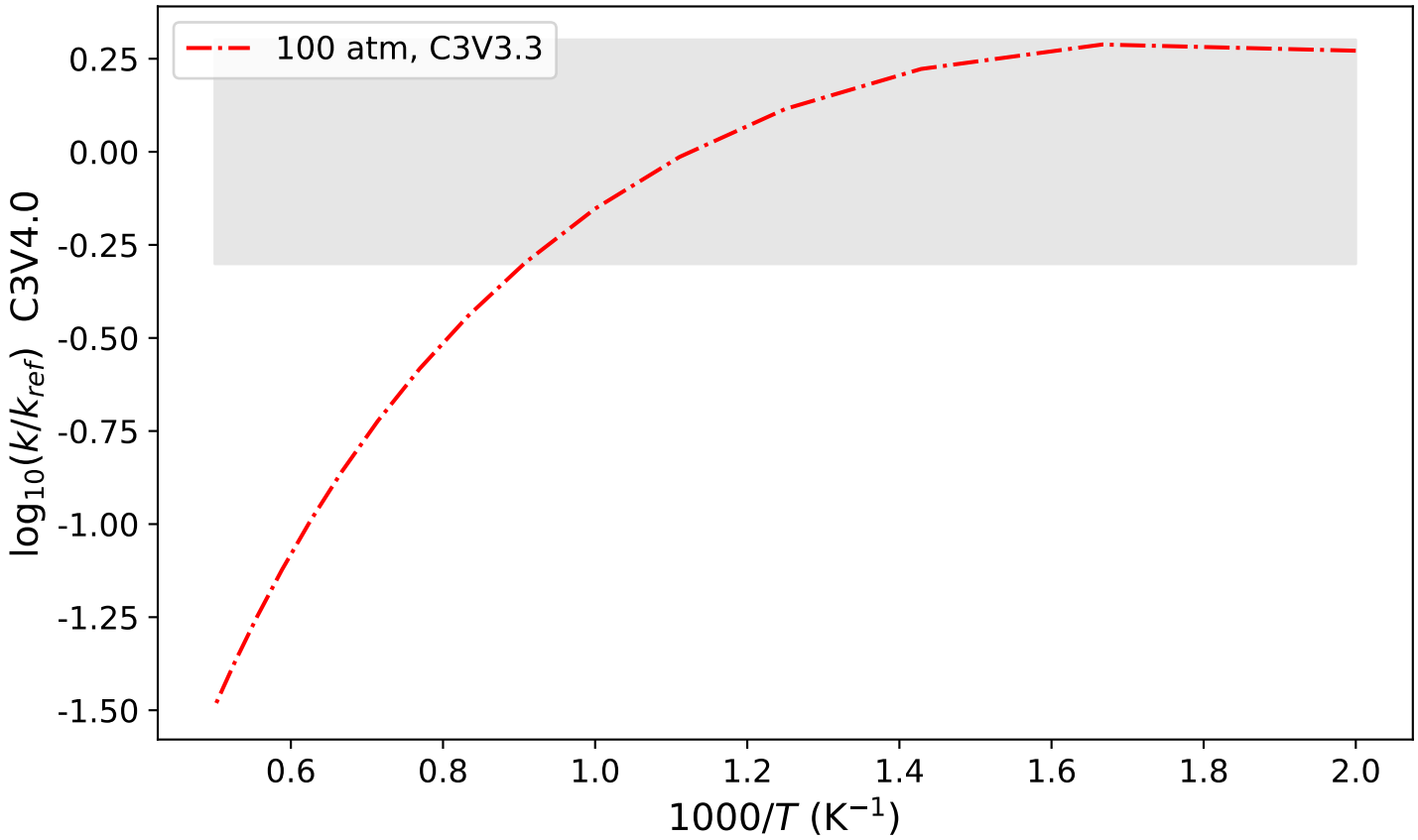
C4H71-O = CH3CHO + C2H3



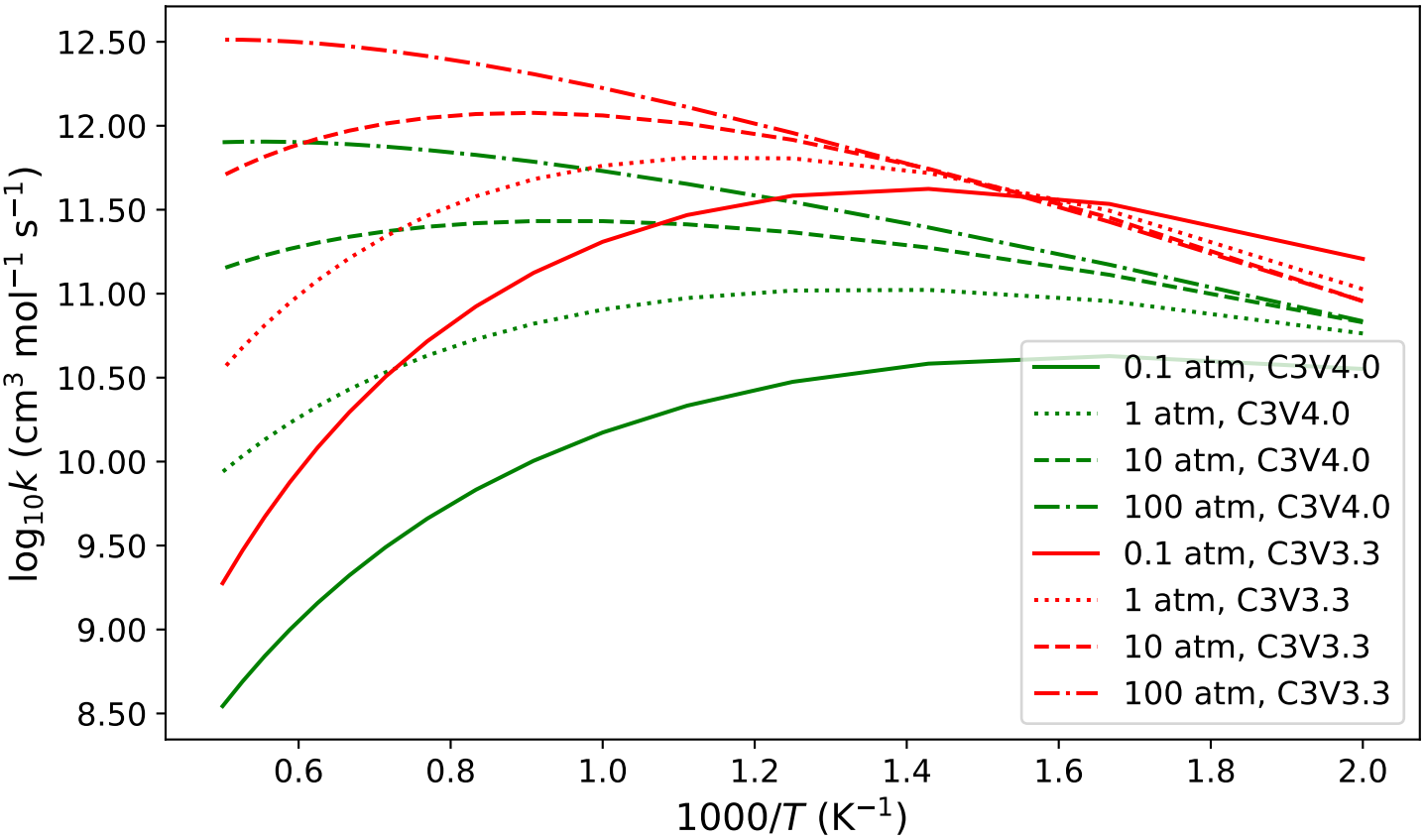



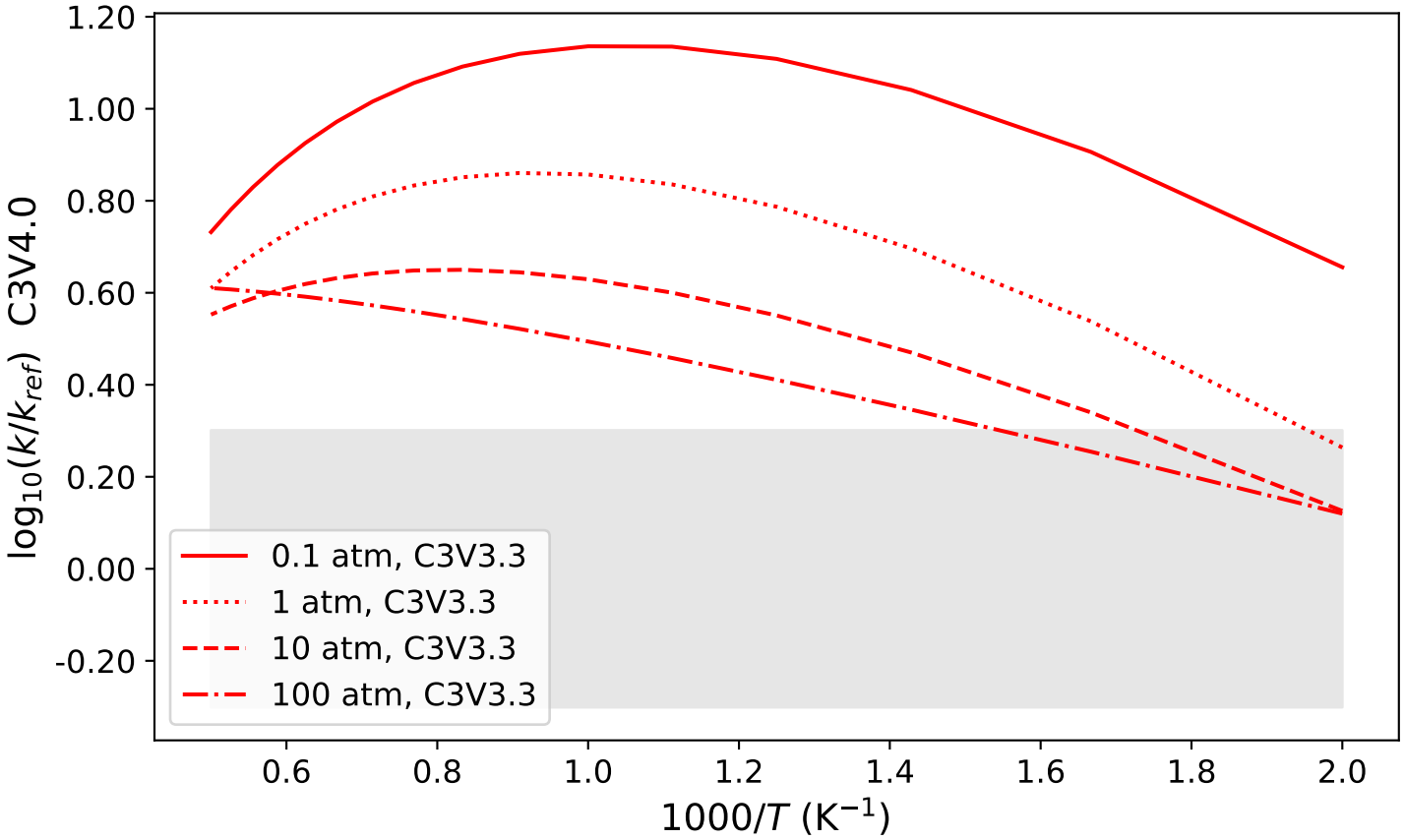
NC3H7O2 = HO2 + C3H6



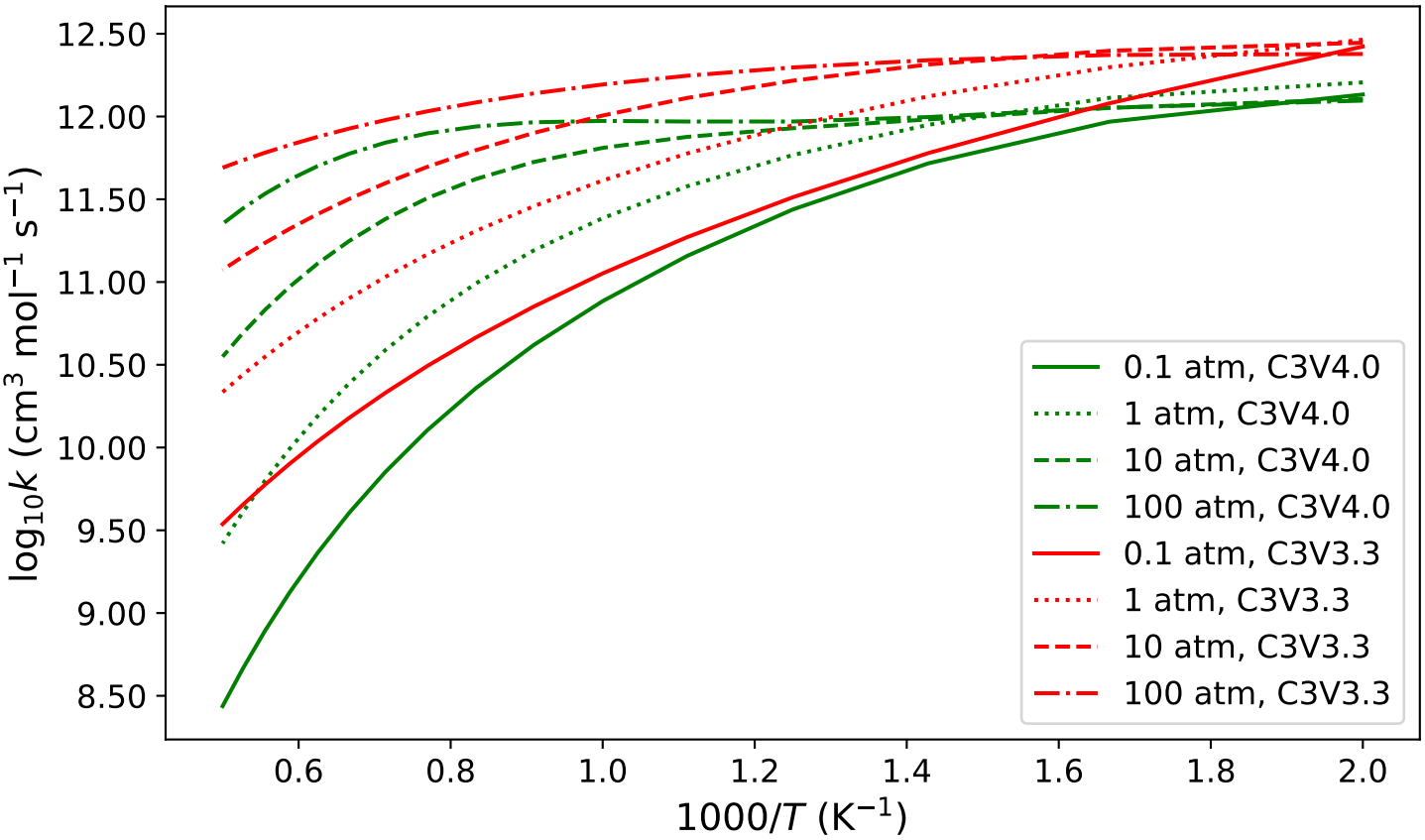



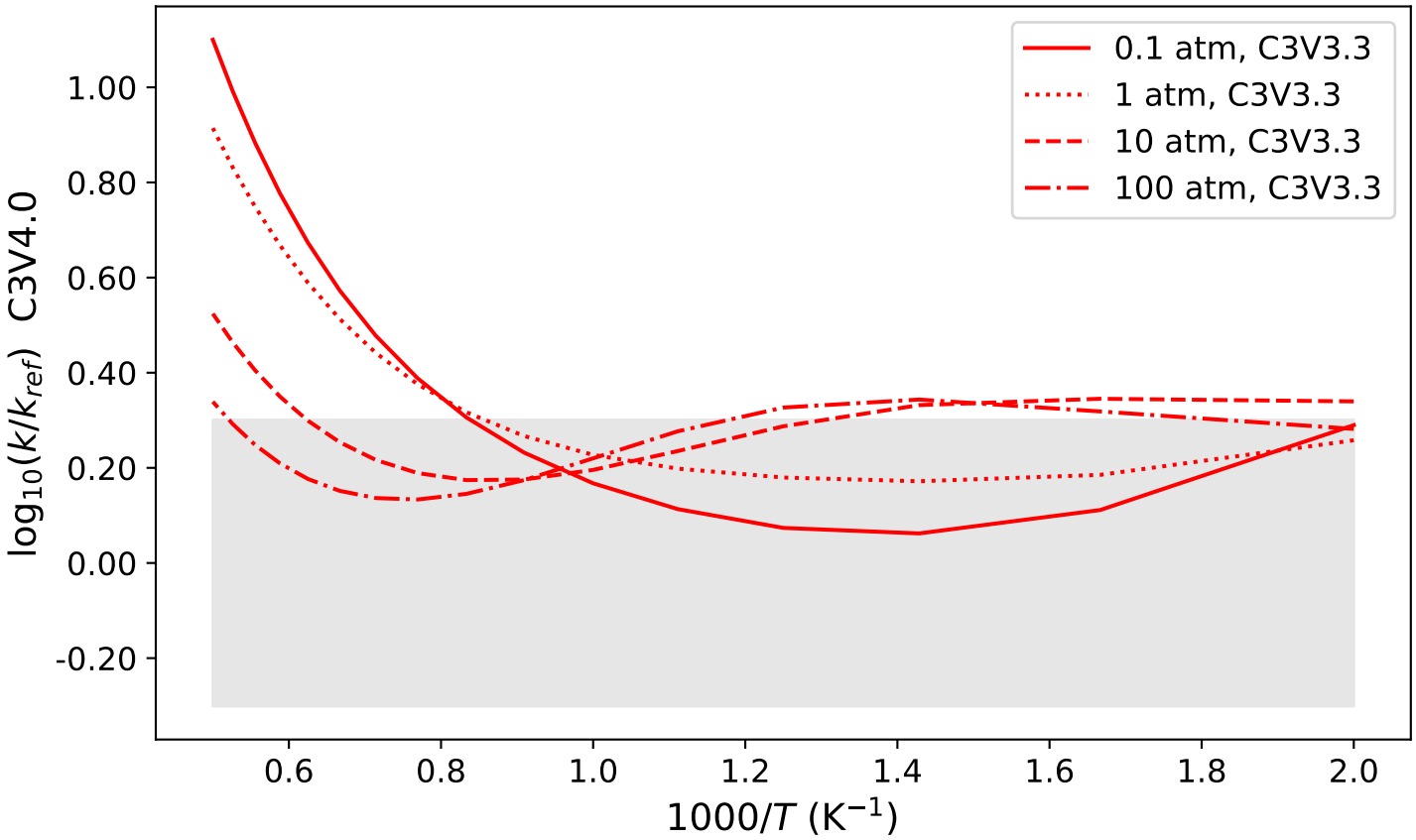
H + IC4H8 = IC4H9



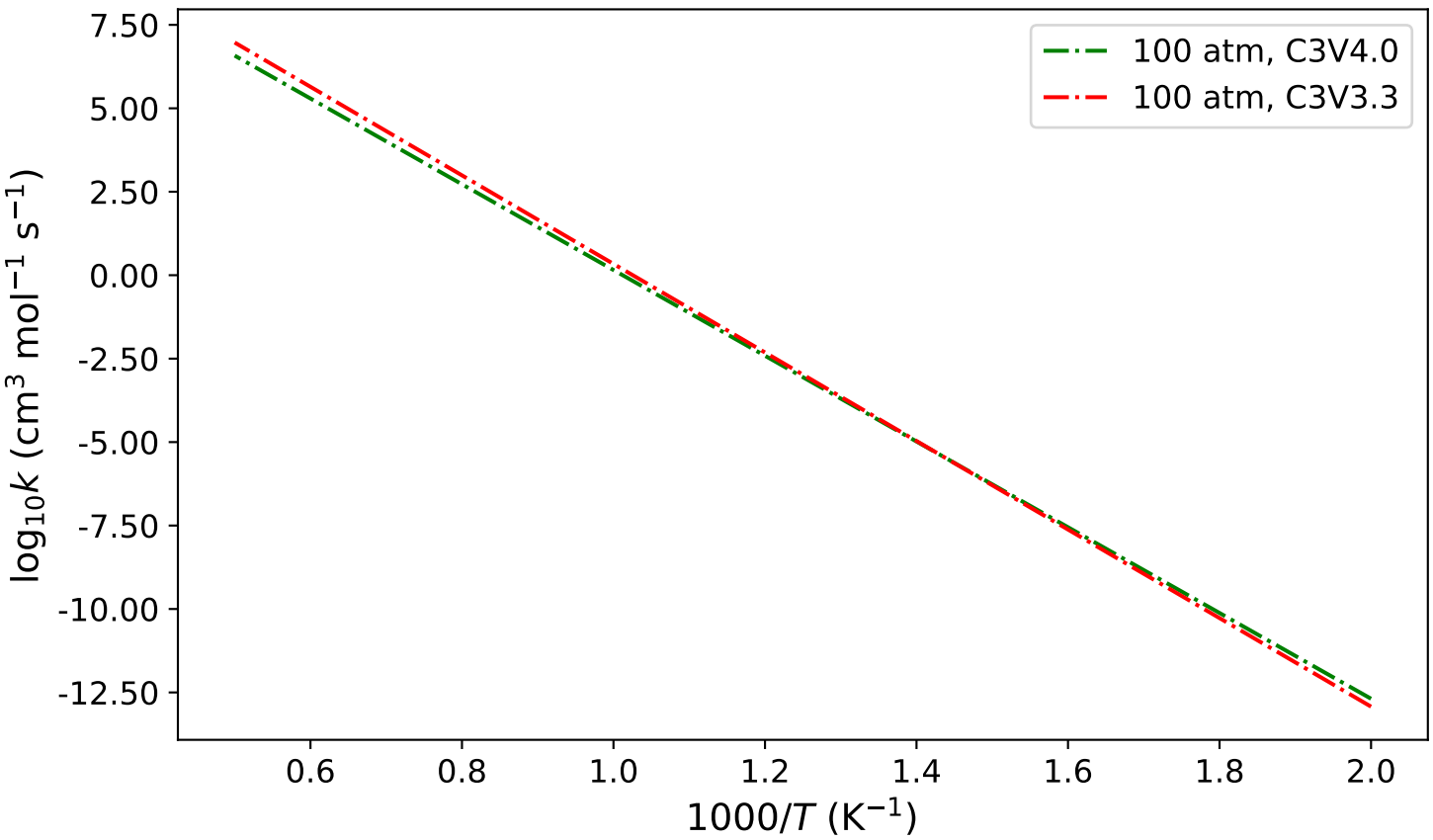



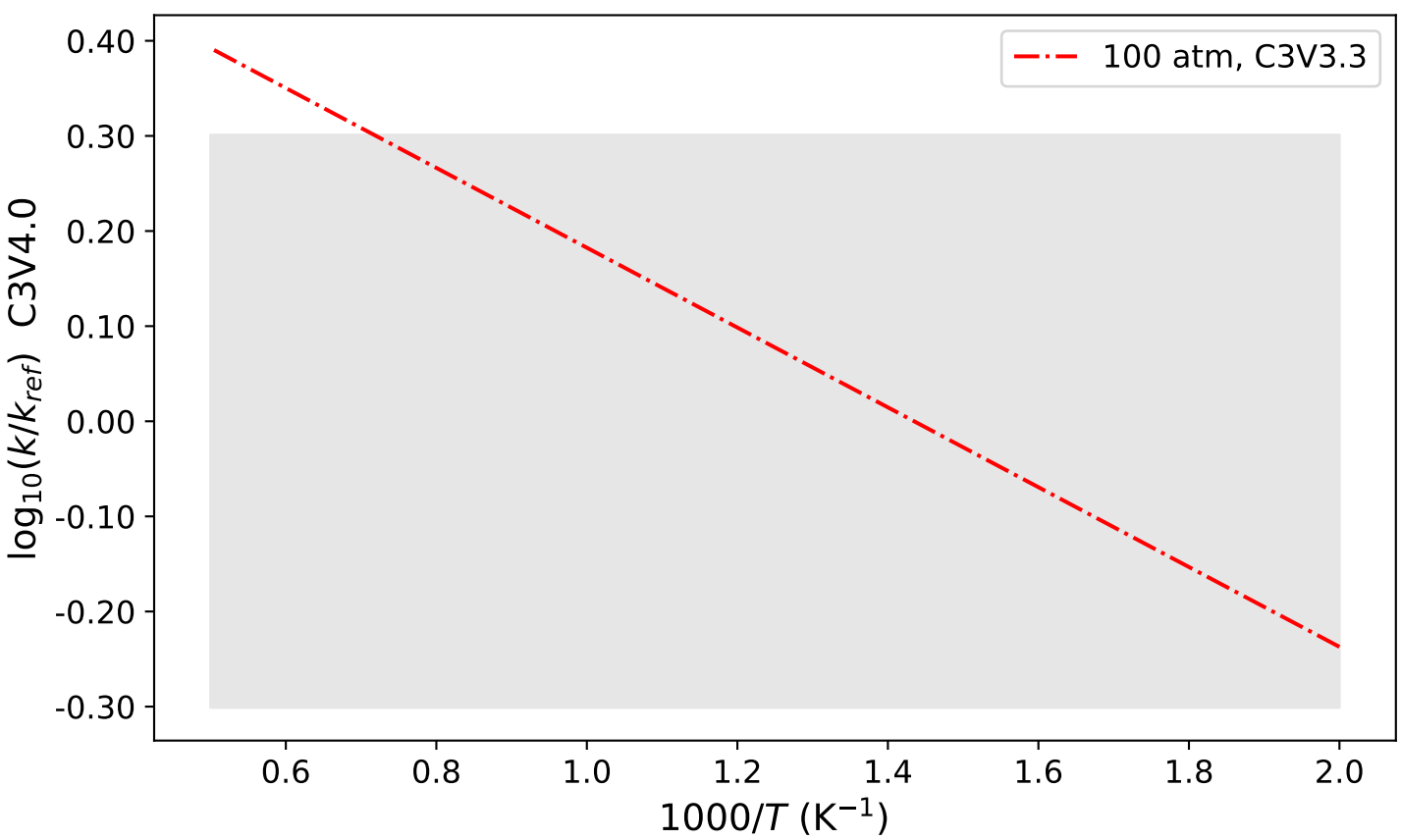
OH + C3H6 = C3H6OH2-1



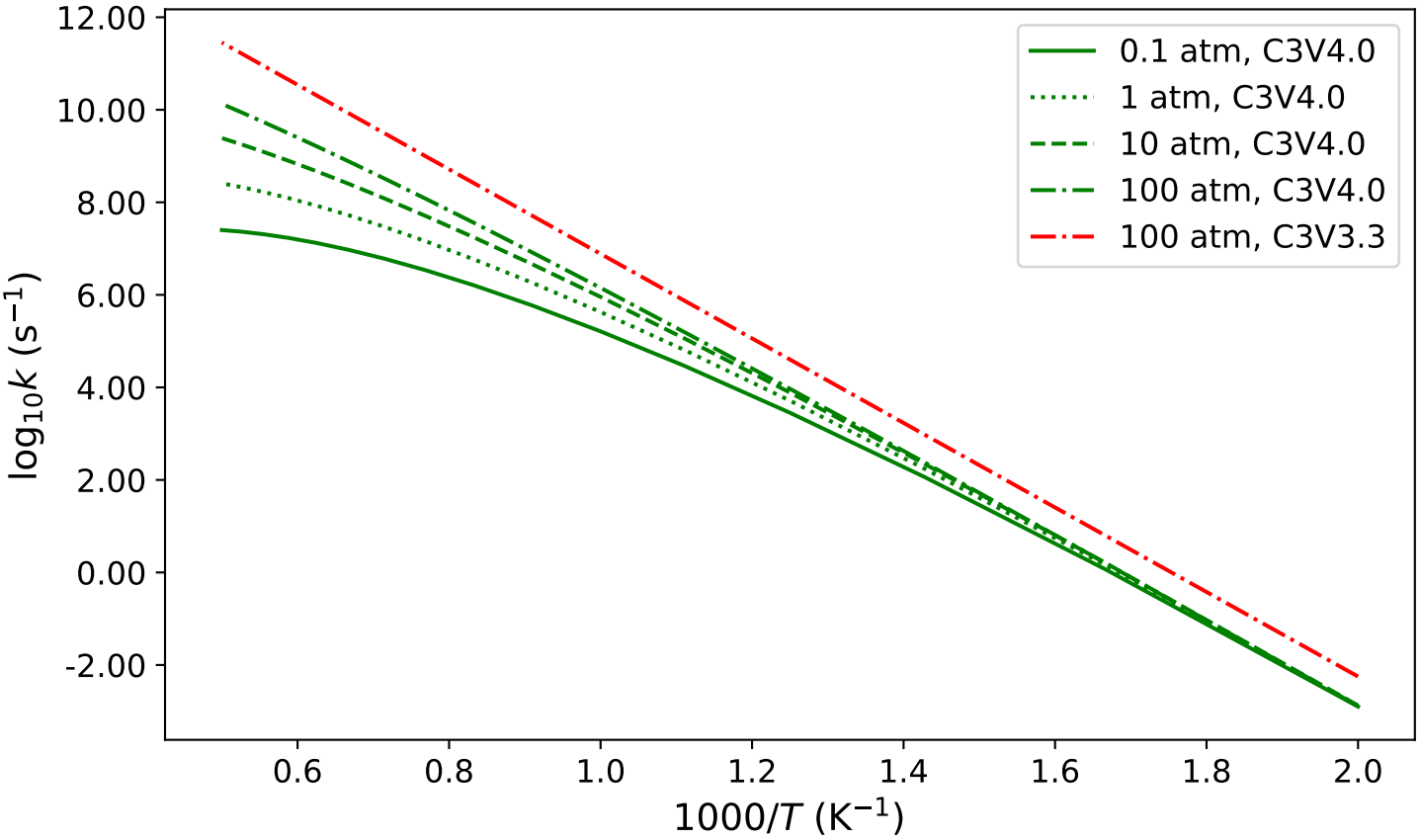



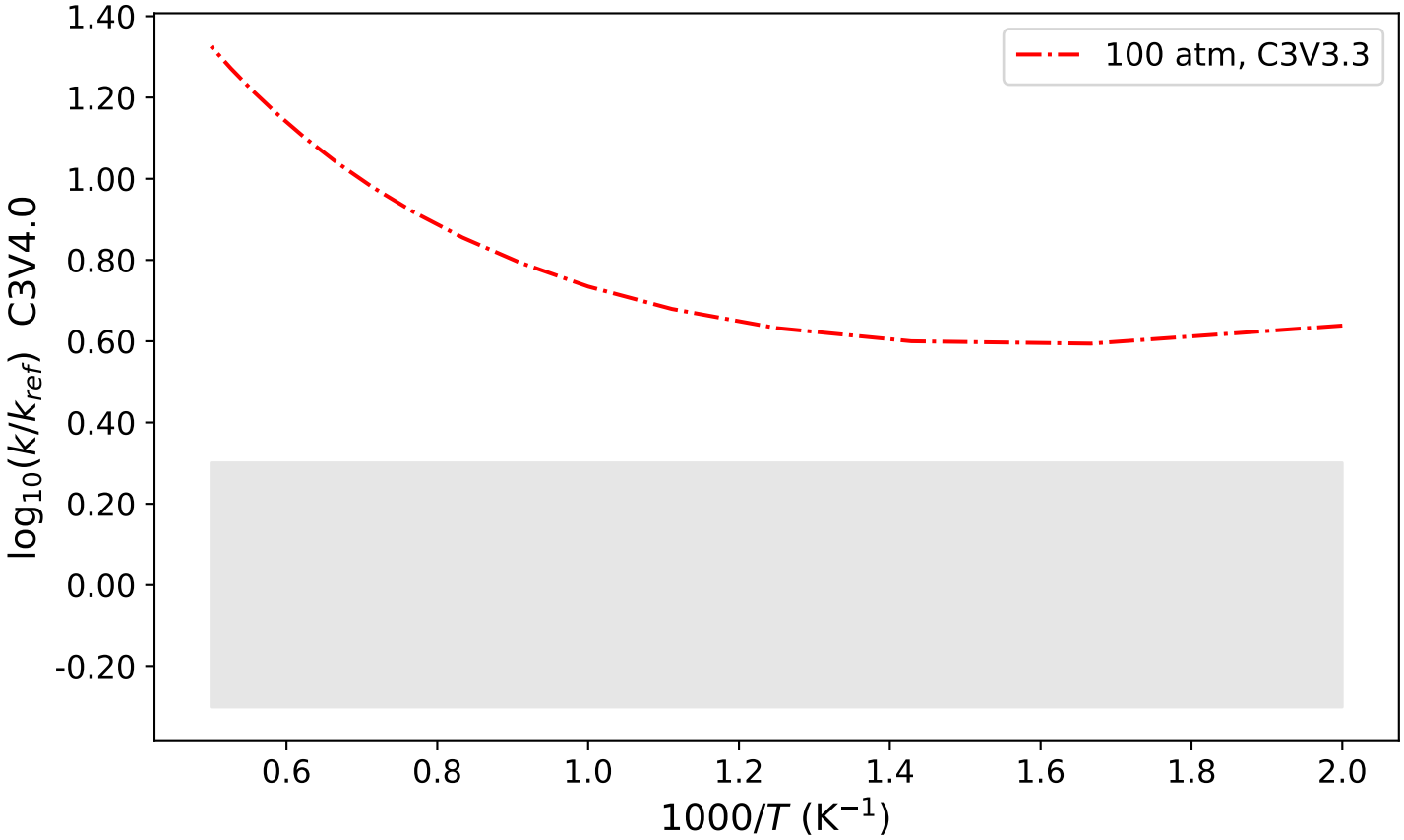
O2 + C3H5OH = HO2 + CH2CCH2OH



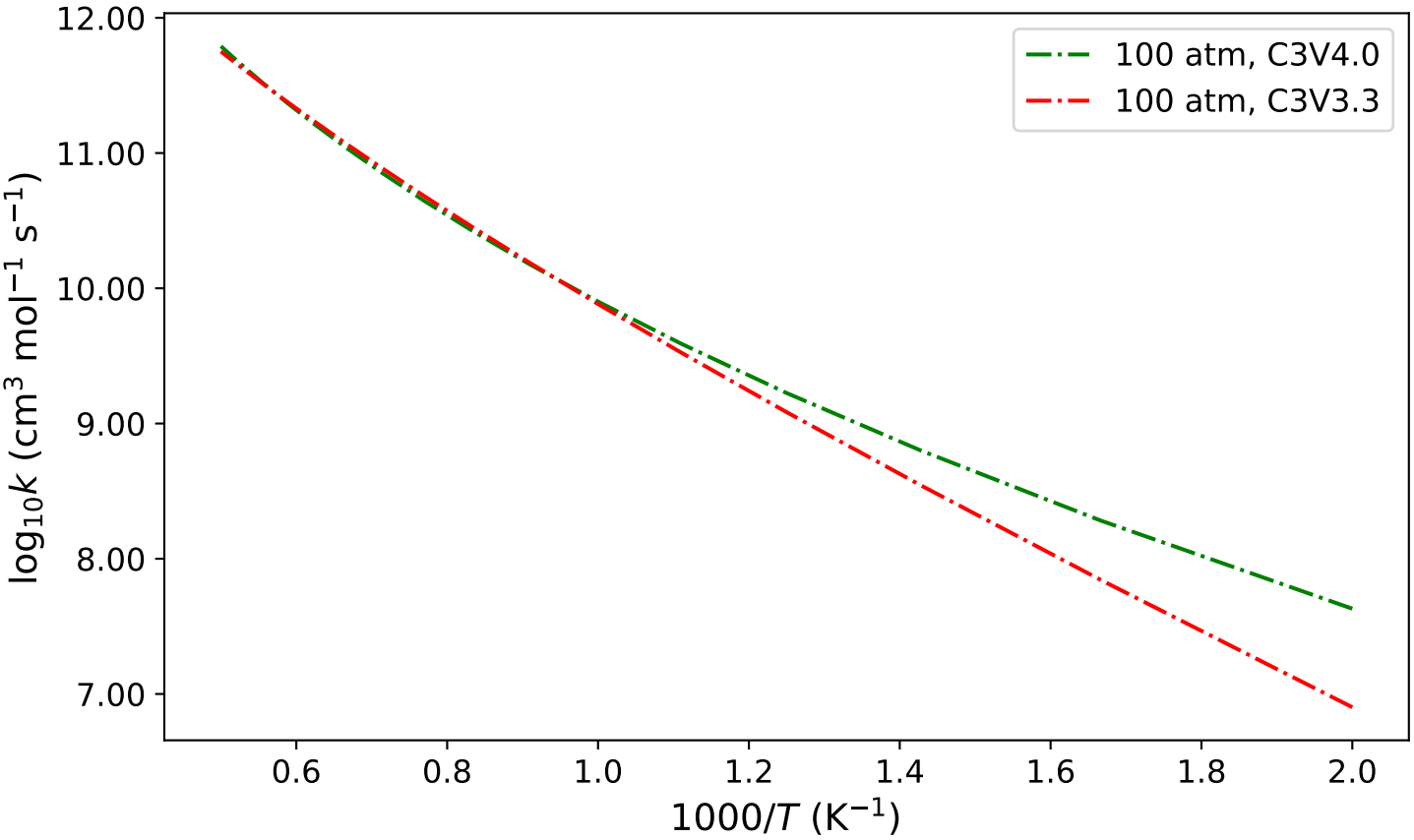



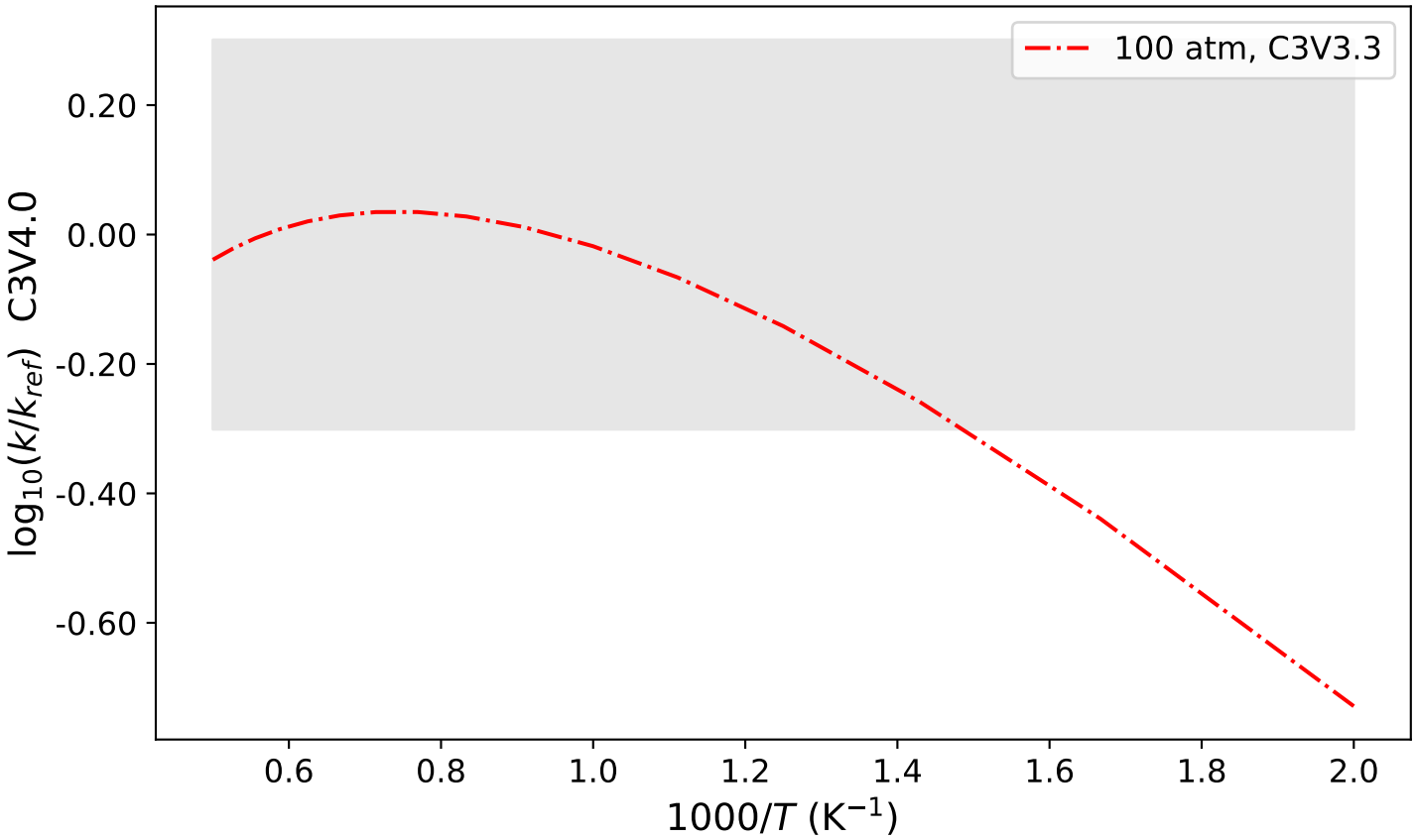
C4H61KET43 = OH + C4H61KET43O



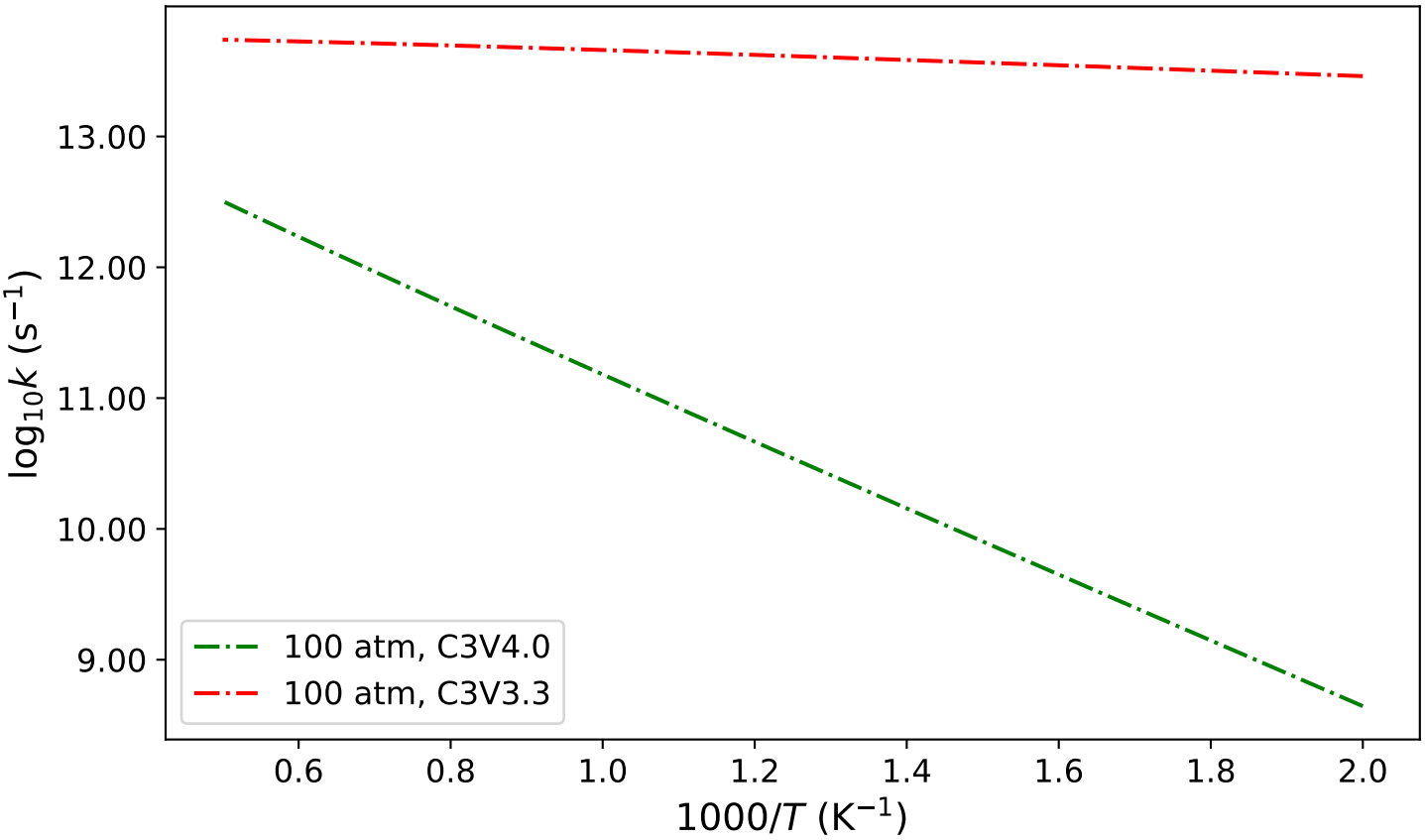



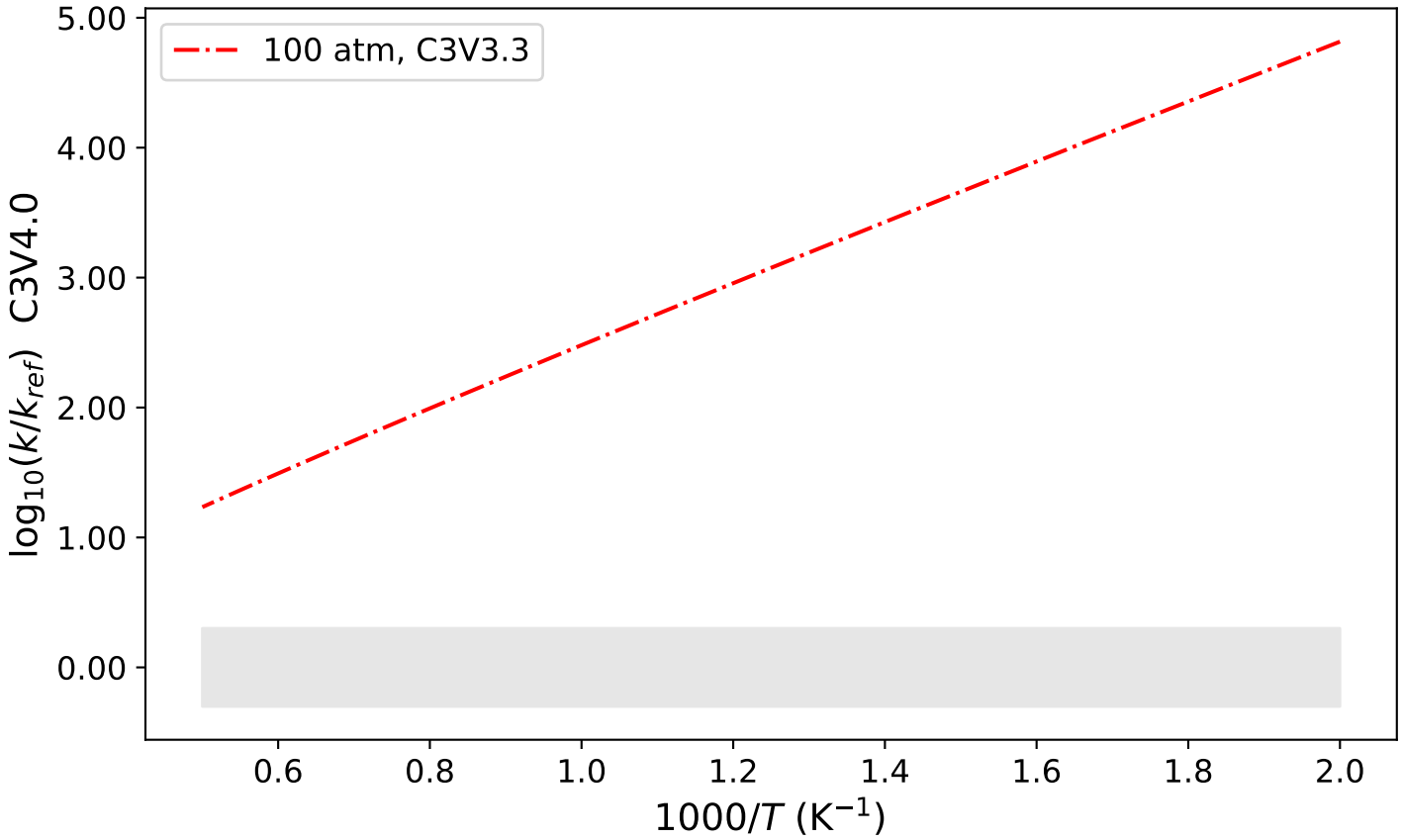
HO2 + C2H5CHO = H2O2 + C2H5CO

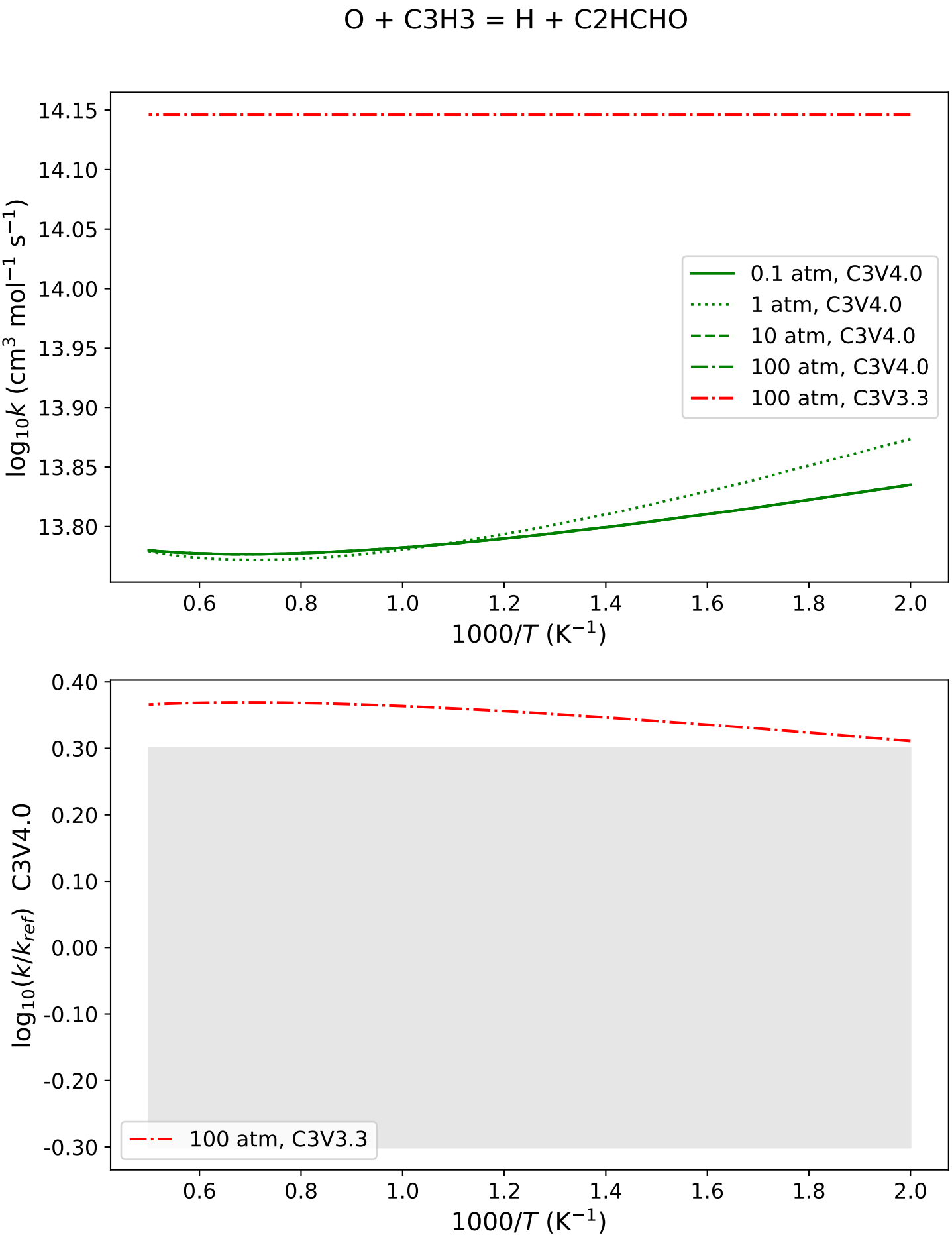


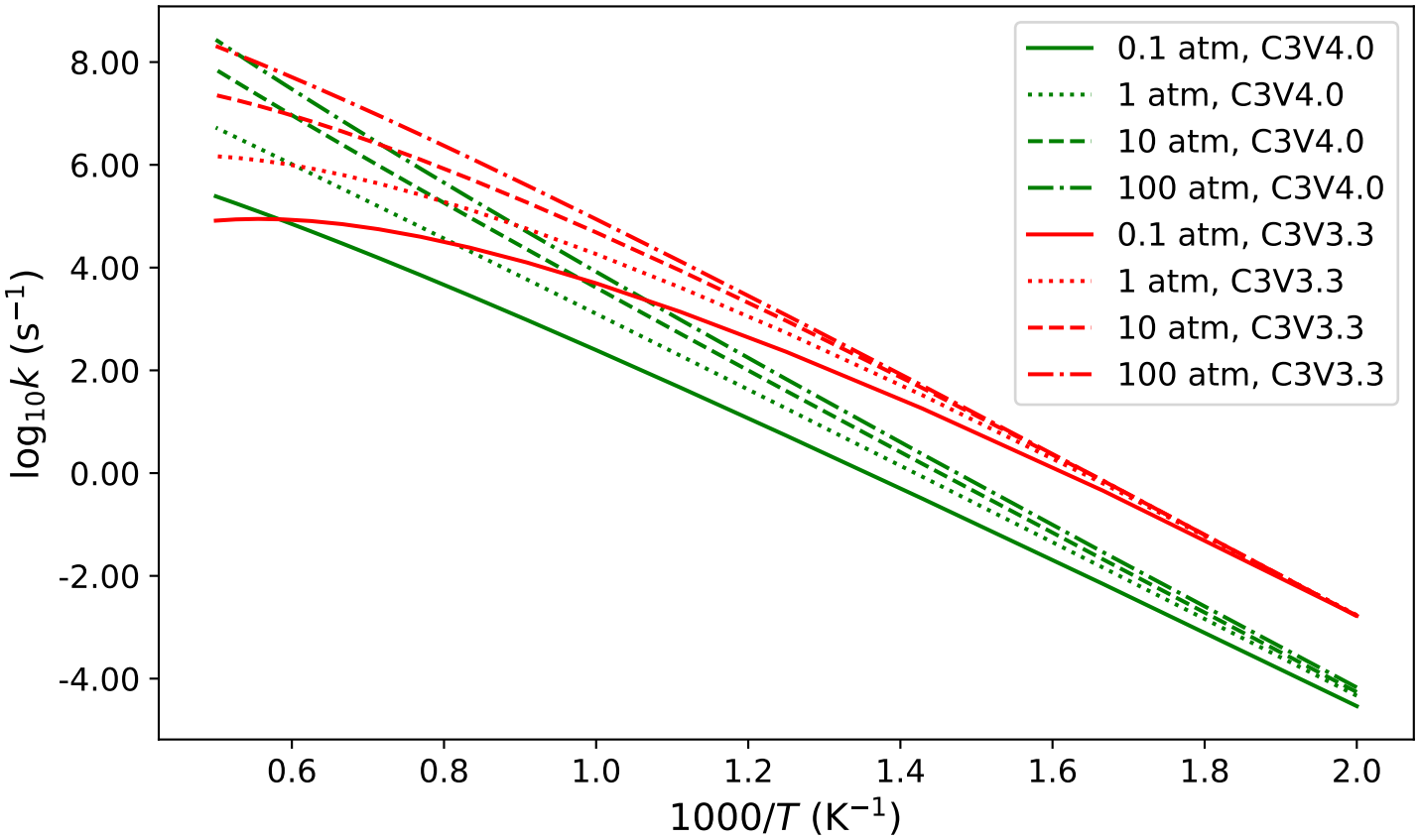
CH2COCHO = HCO + CH2CO

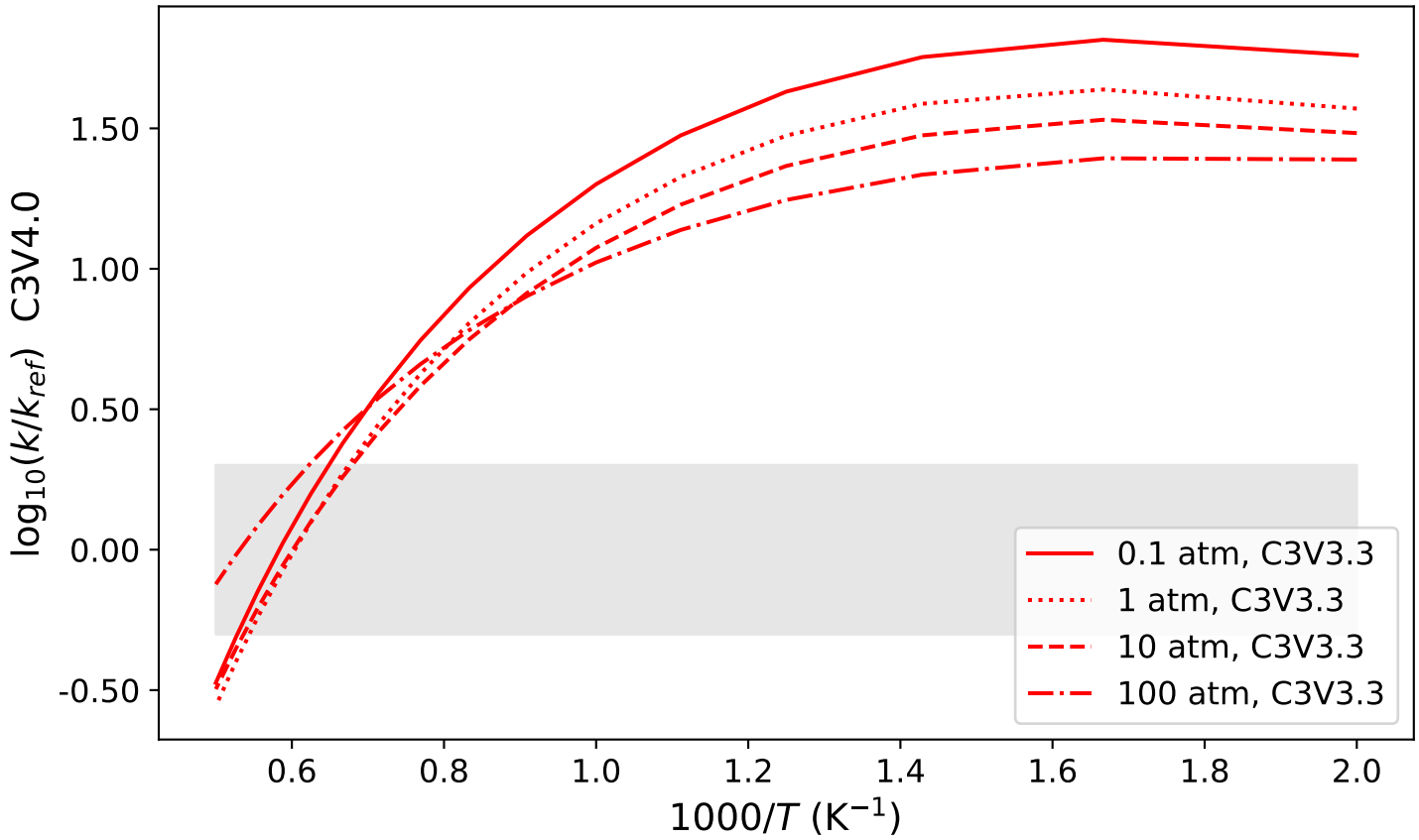


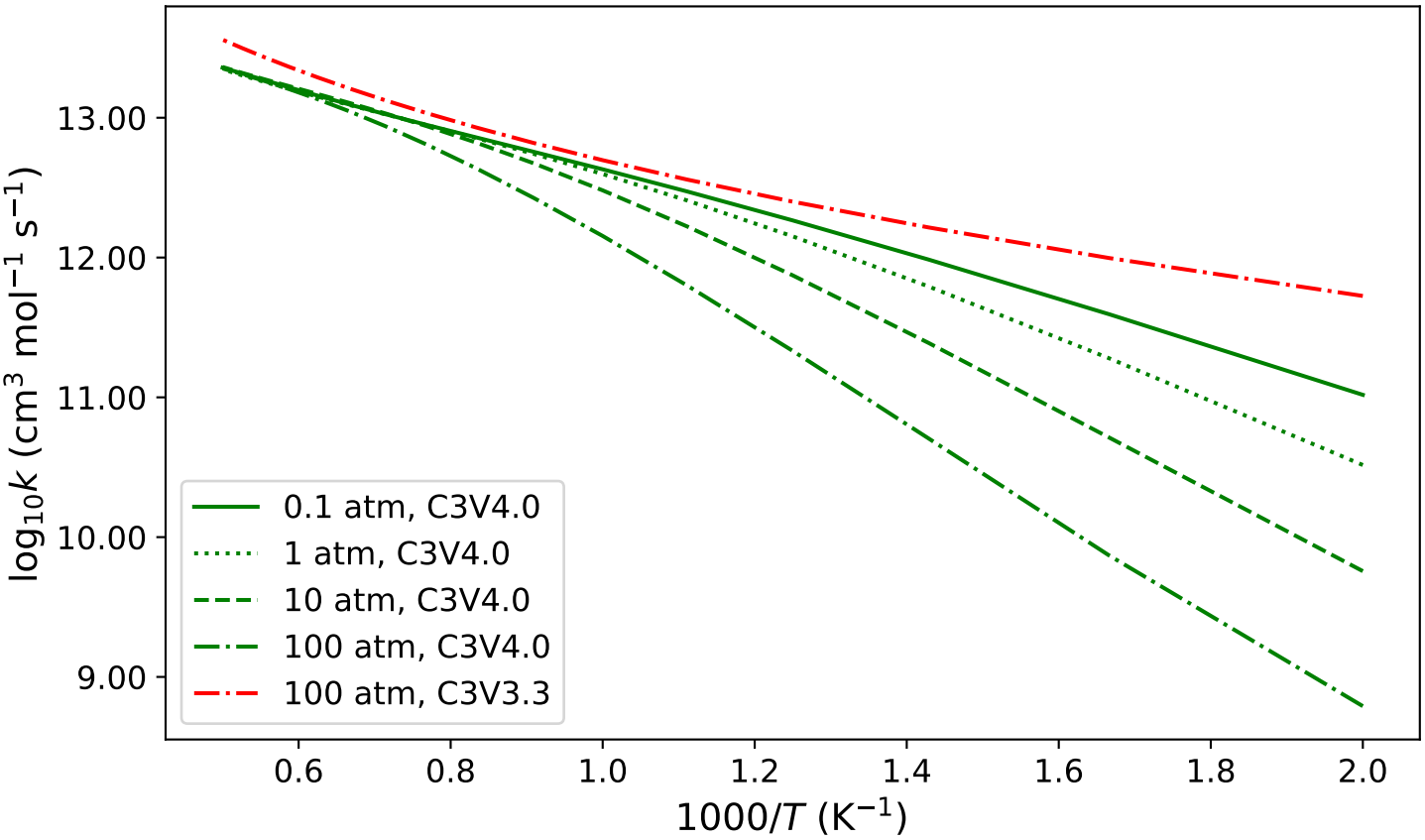

O + C3H3 = H + C2HCHO
14.15
14.10
14.05
14.00
13.95
13.90
13.85
13.80
$\log_{10}k$ (cm$^3$ mol$^{-1}$ s$^{-1}$)
0.1 atm, C3V4.0
1 atm, C3V4.0
10 atm, C3V4.0
100 atm, C3V4.0
100 atm, C3V3.3
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ (K$^{-1}$)
0.40
0.30
0.20
0.10
0.00
-0.10
-0.20
-0.30
$\log_{10}(k/k_{ref})$ C3V4.0
100 atm, C3V3.3
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ (K$^{-1}$)

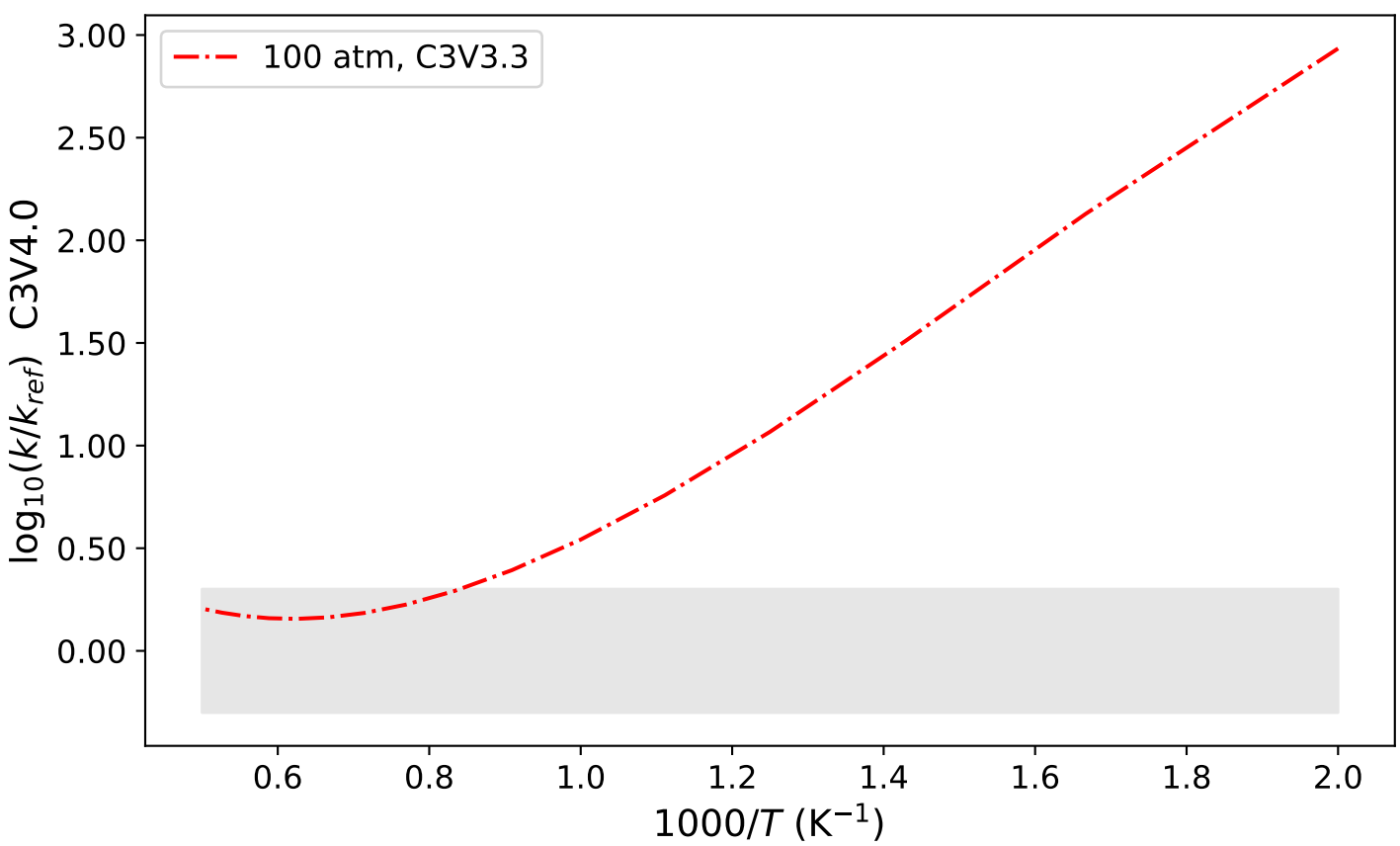
TC4H9 = IC4H9


0.1 atm, C3V4.0
1 atm, C3V4.0
10 atm, C3V4.0
100 atm, C3V4.0
0.1 atm, C3V3.3
1 atm, C3V3.3
10 atm, C3V3.3
100 atm, C3V3.3
8.00
6.00
4.00
2.00
0.00
-2.00
-4.00
$\log_{10}k$ ($s^{-1}$)
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
$1000/T$ ($K^{-1}$)

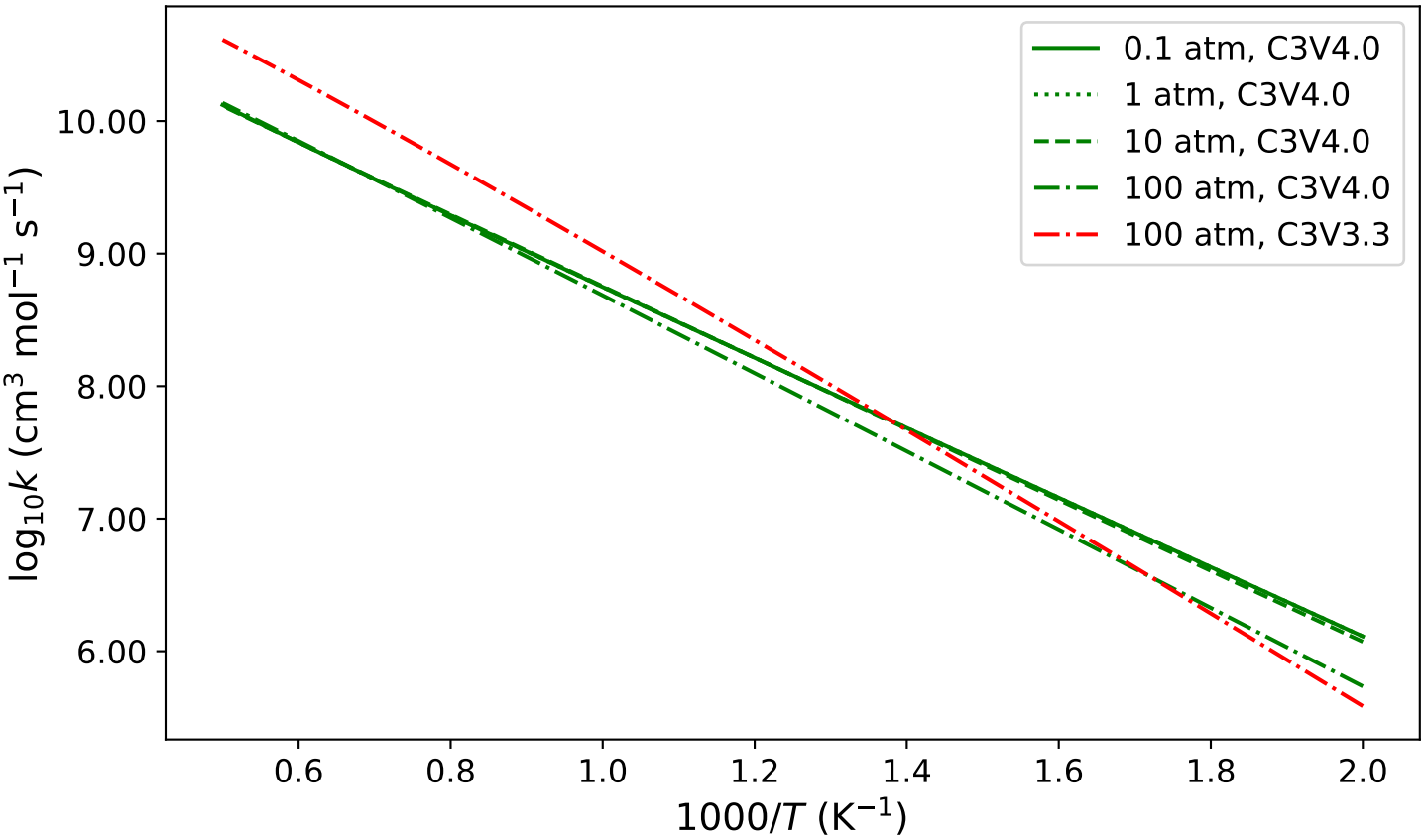

0.1 atm, C3V3.3
1 atm, C3V3.3
10 atm, C3V3.3
100 atm, C3V3.3
1.50
1.00
0.50
0.00
-0.50
$\log_{10}(k/k_{ref})$ C3V4.0
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
$1000/T$ ($K^{-1}$)

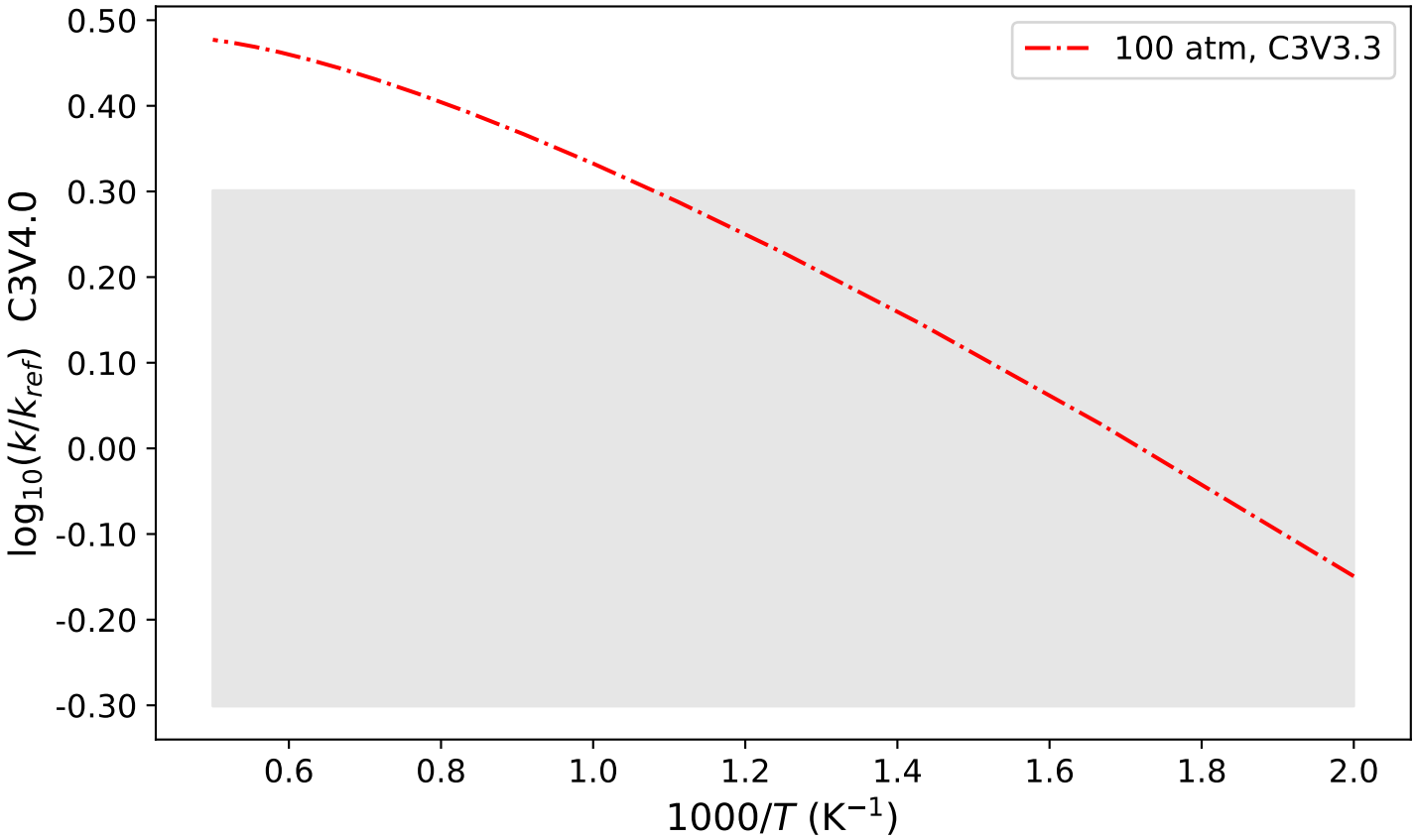
H + C4H6-2 = CH3 + C3H4-P



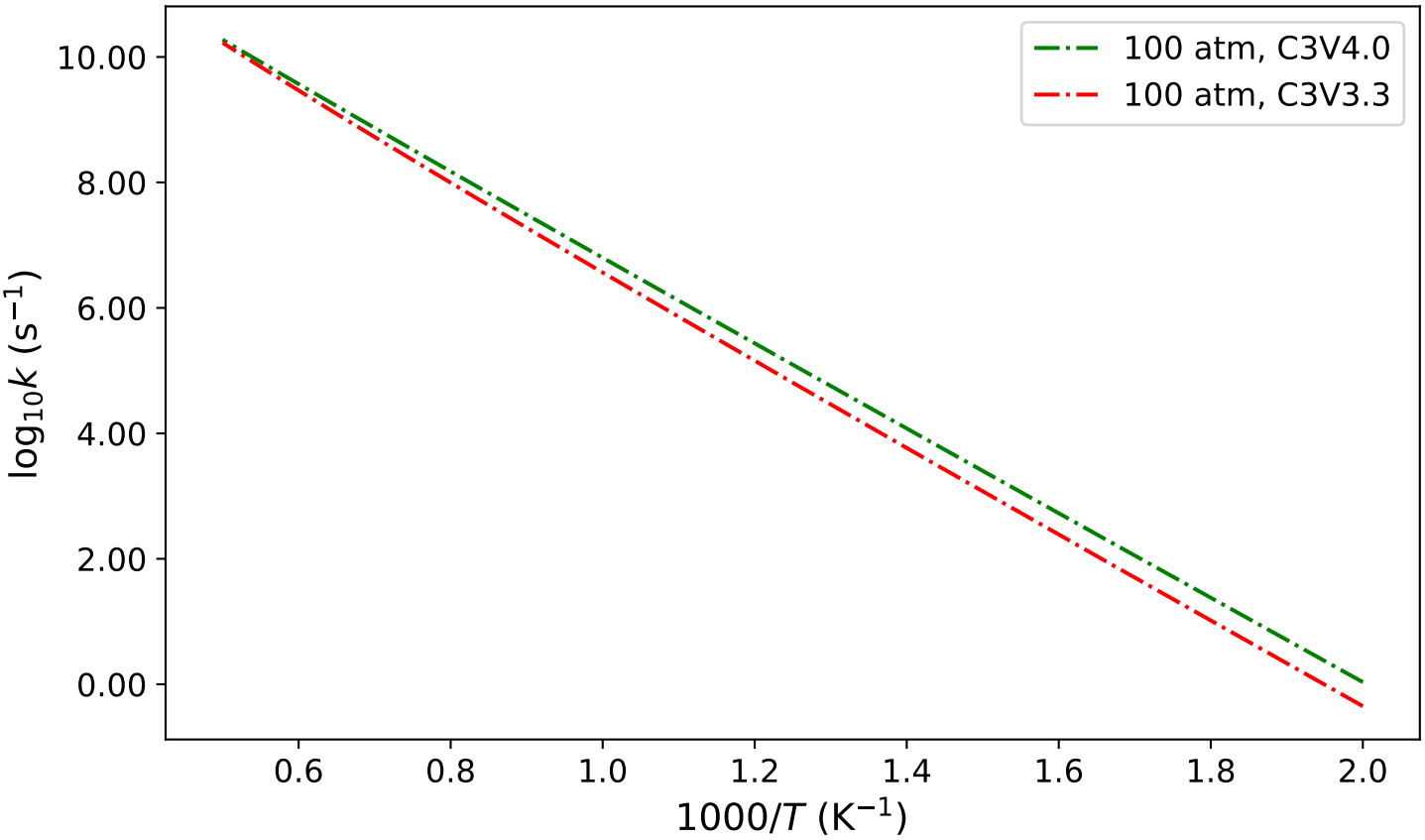



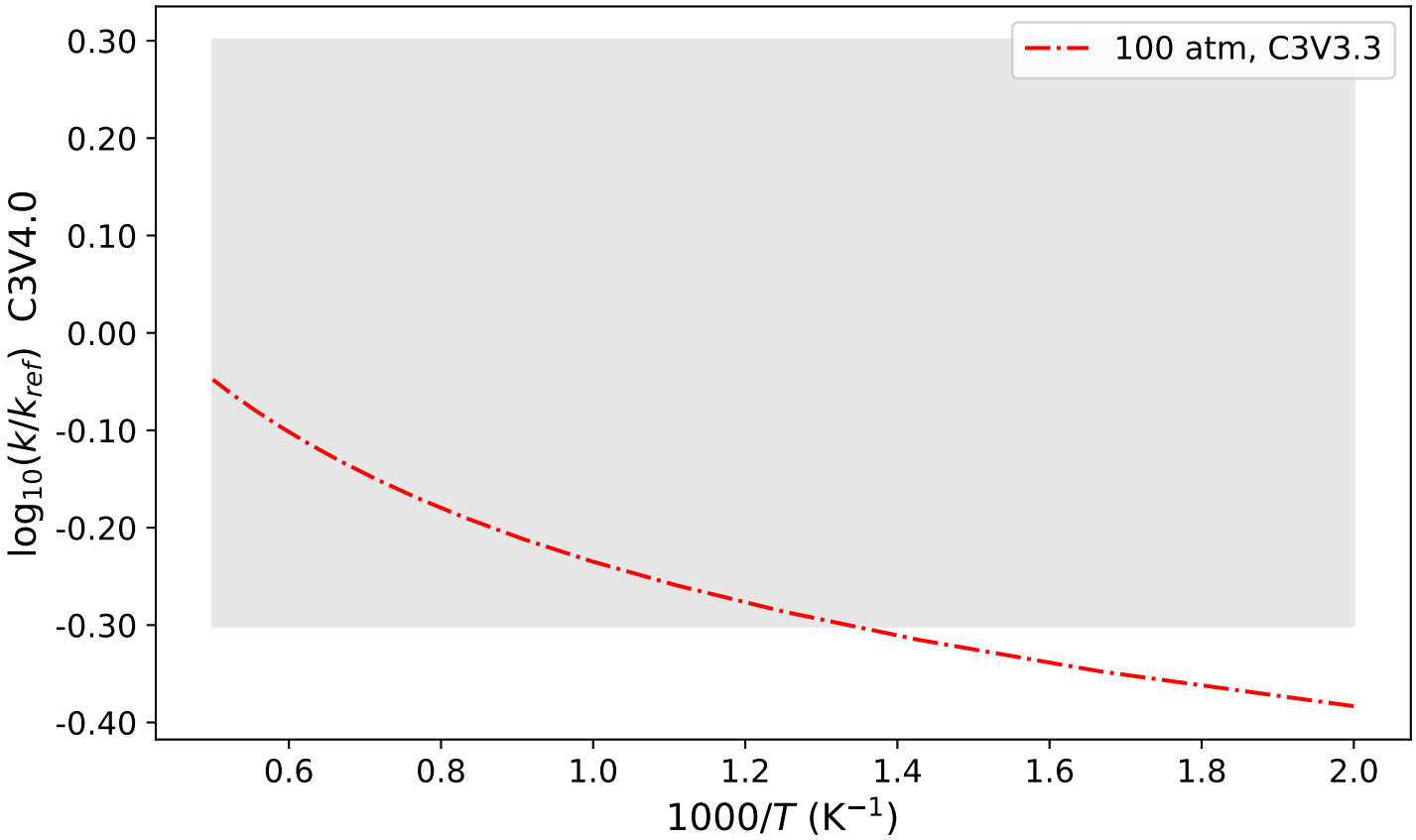
HO2 + C2H3CHO = O2 + CH2CHCHOH



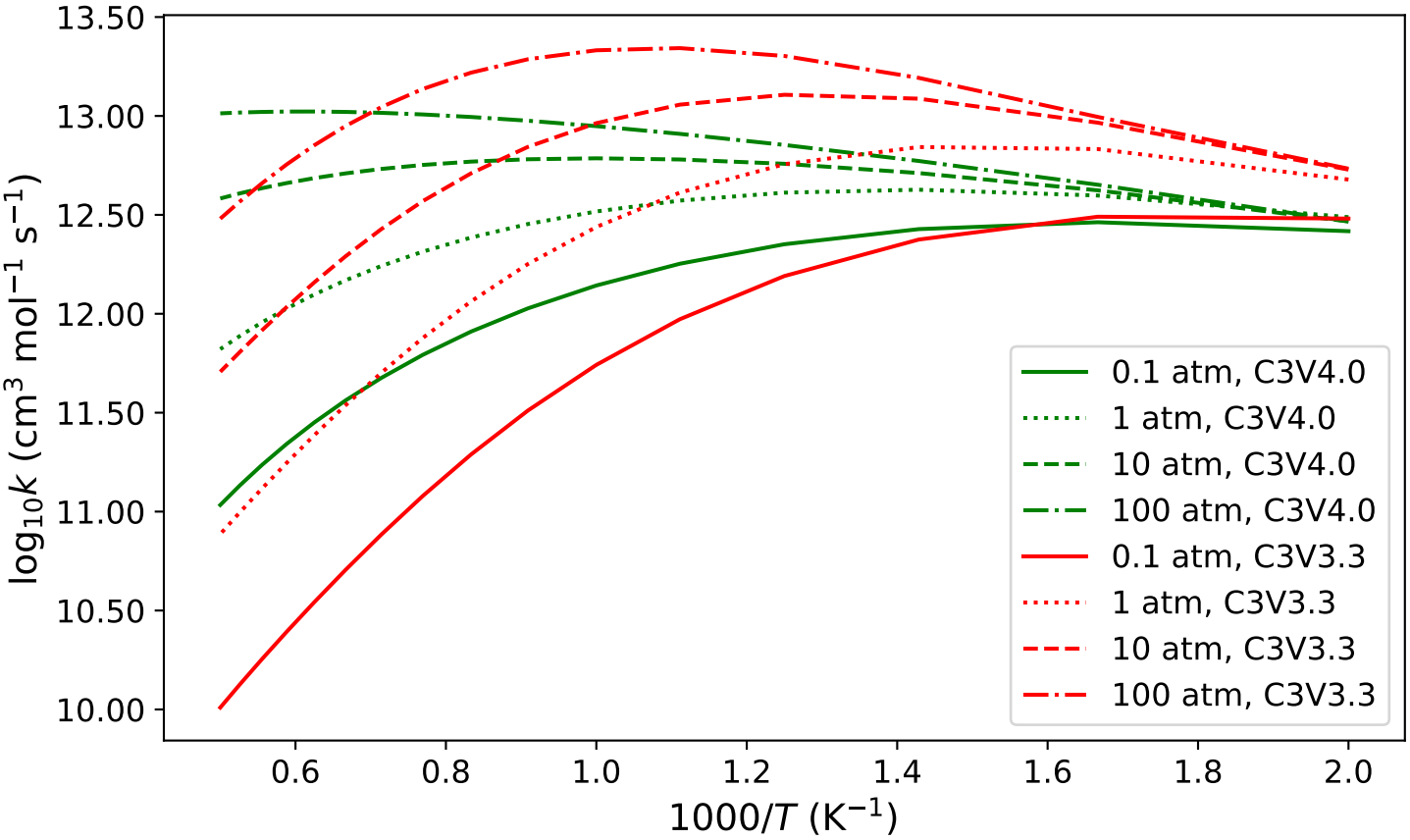



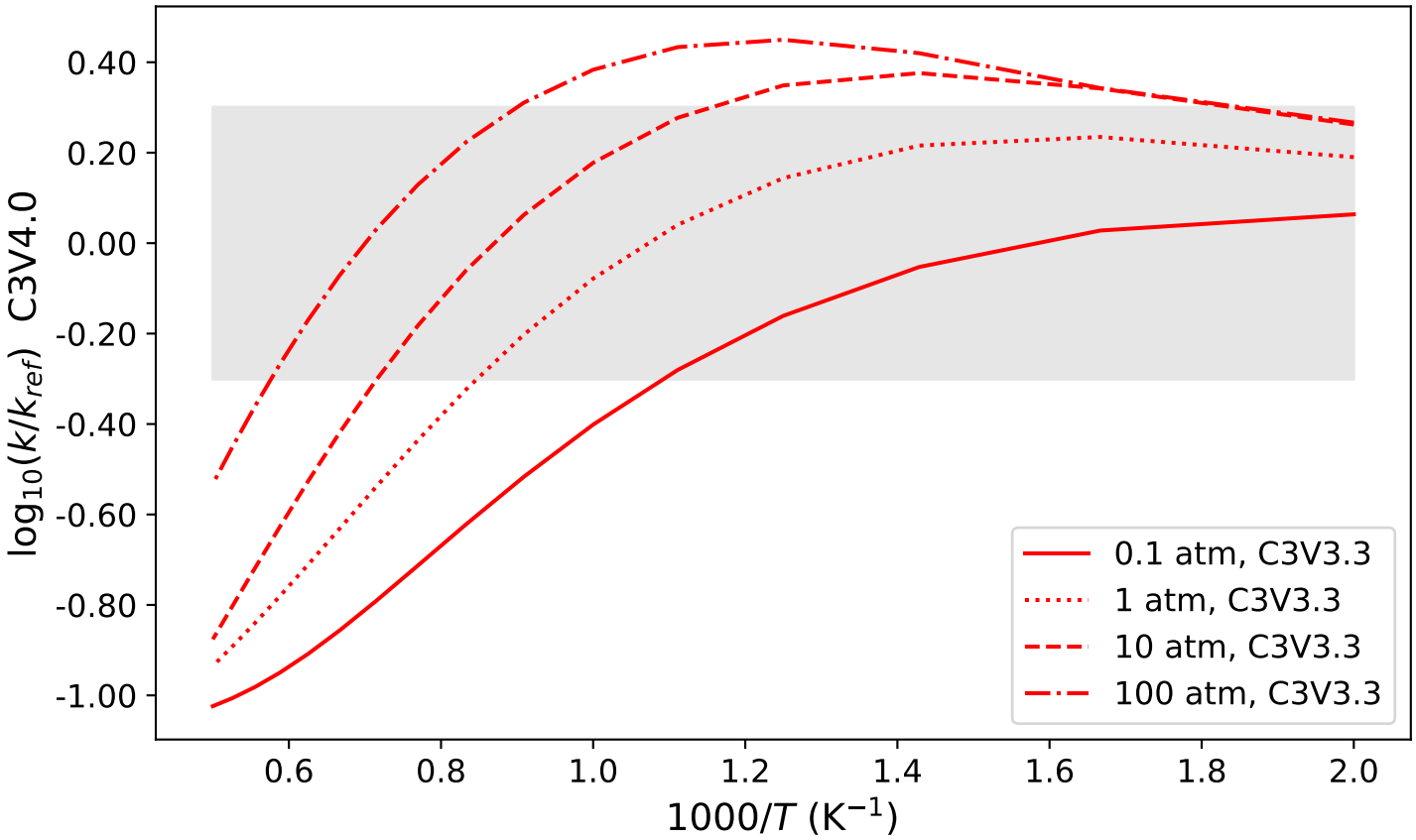
TC4H9O2 = HO2 + IC4H8



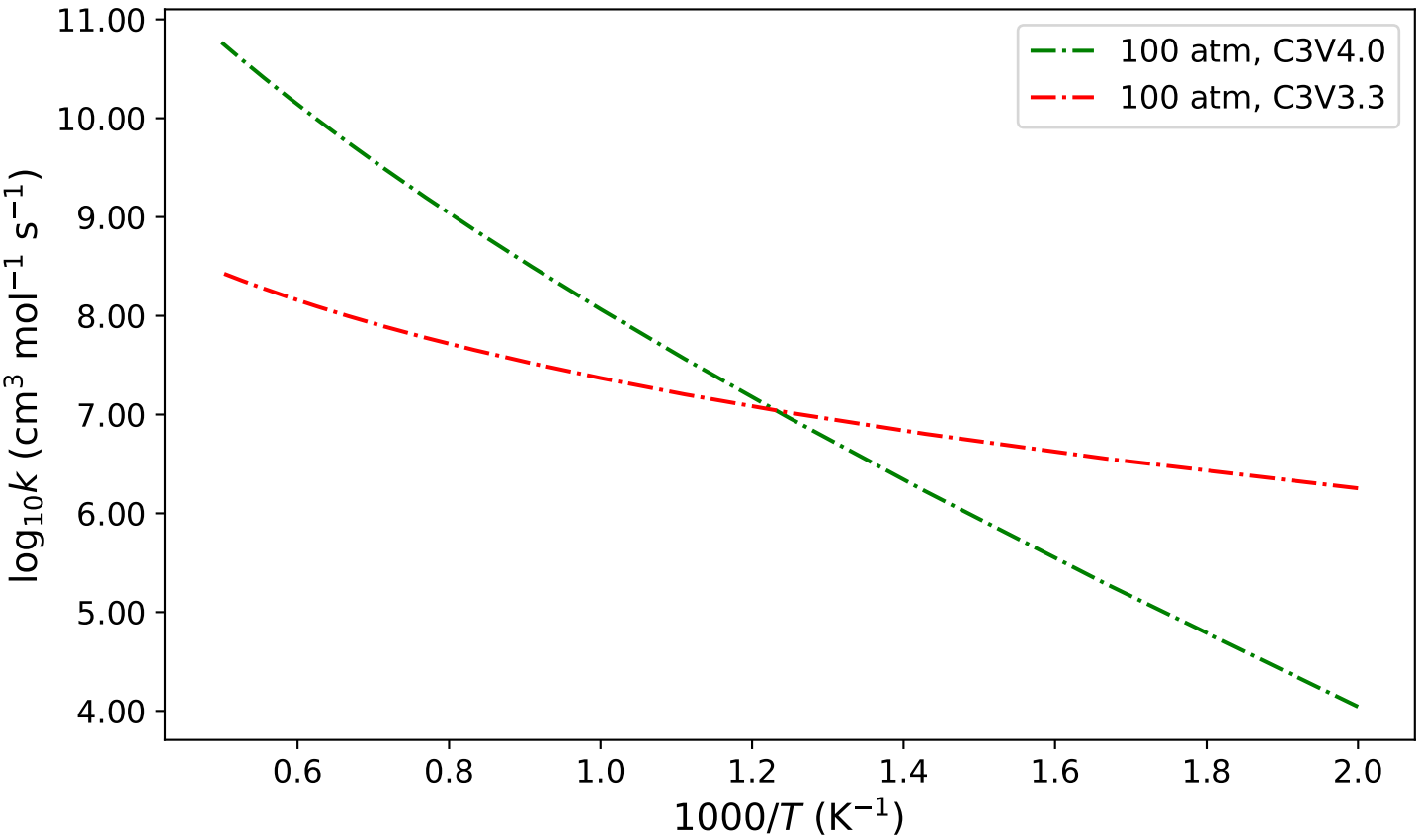



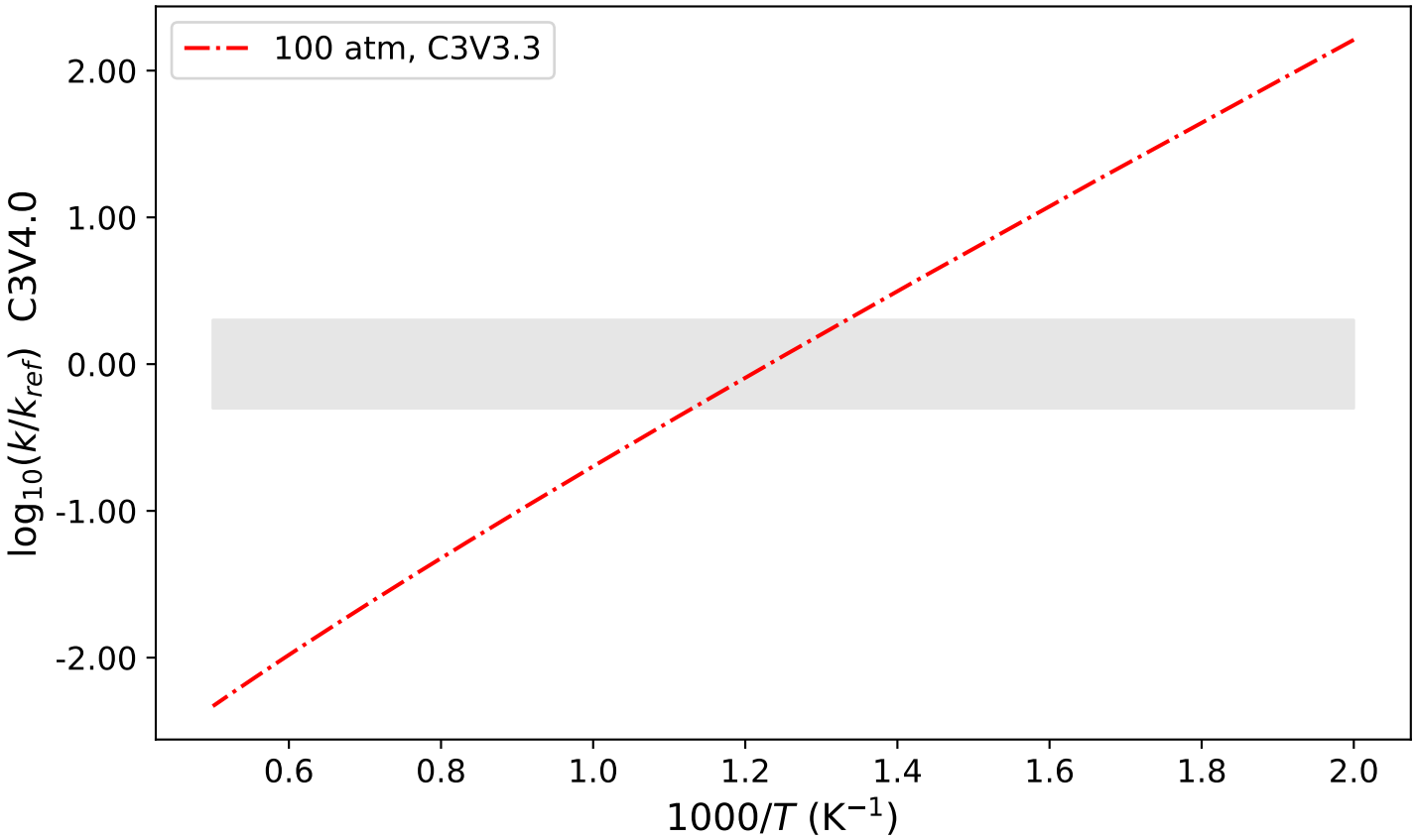
H + C3H6 = IC3H7


13.50
13.00
12.50
12.00
11.50
11.00
10.50
10.00
$\log_{10}k$ (cm$^3$ mol$^{-1}$ s$^{-1}$)
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ (K$^{-1}$)
0.1 atm, C3V4.0
1 atm, C3V4.0
10 atm, C3V4.0
100 atm, C3V4.0
0.1 atm, C3V3.3
1 atm, C3V3.3
10 atm, C3V3.3
100 atm, C3V3.3

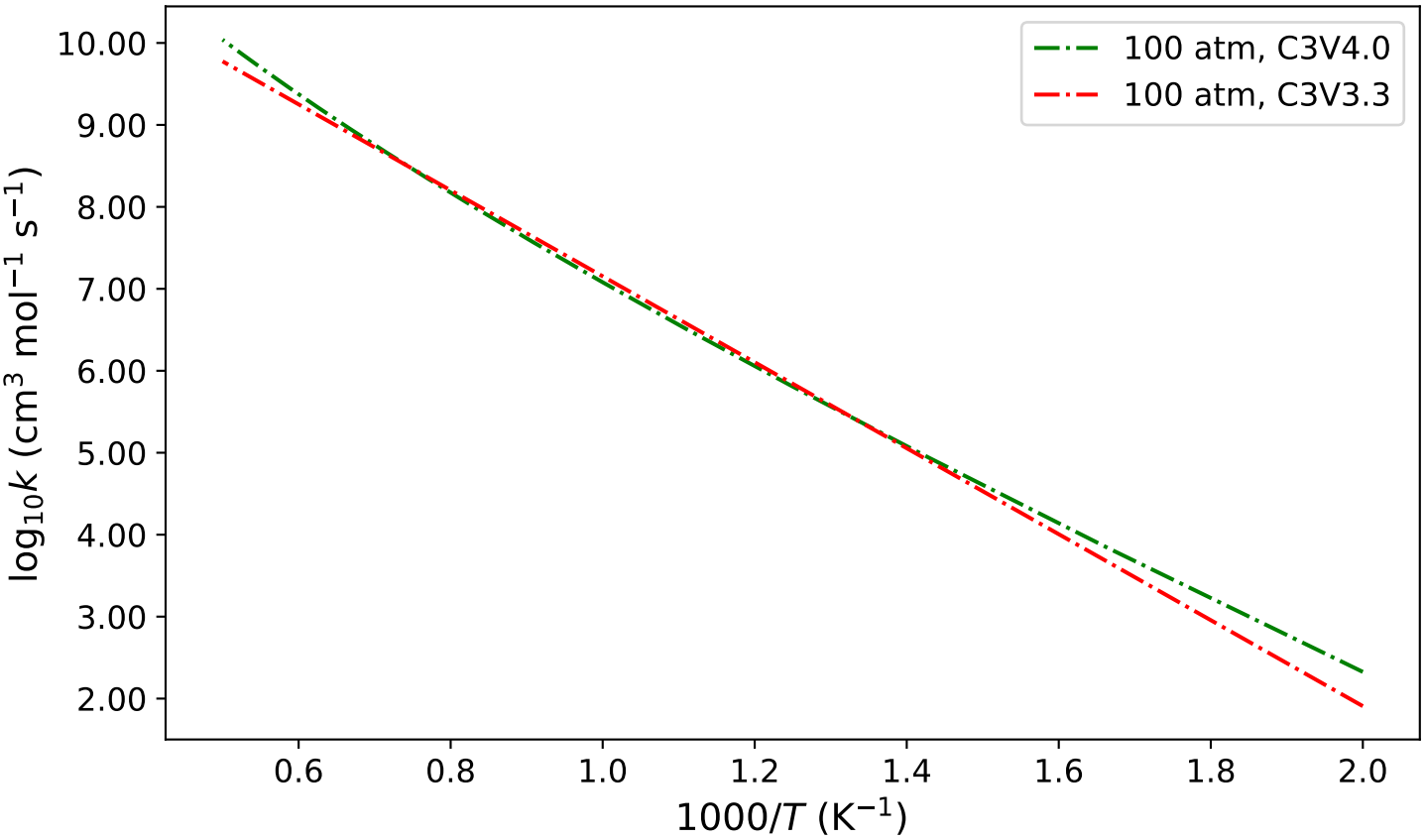

0.40
0.20
0.00
-0.20
-0.40
-0.60
-0.80
-1.00
$\log_{10}(k/k_{ref})$ C3V4.0
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ (K$^{-1}$)
0.1 atm, C3V3.3
1 atm, C3V3.3
10 atm, C3V3.3
100 atm, C3V3.3

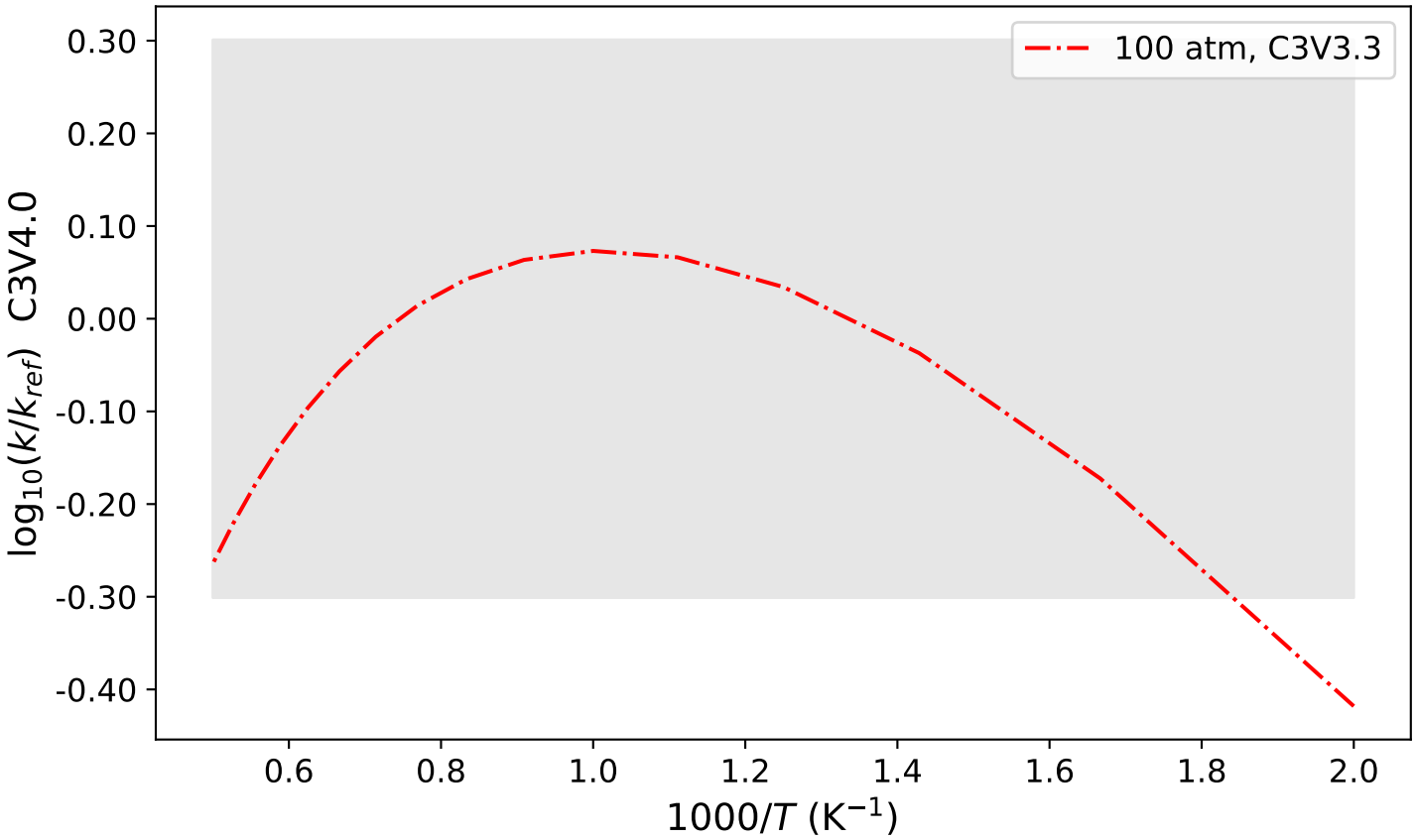
NC3H7OH + HO2 = H2O2 + C3H6OH1-2



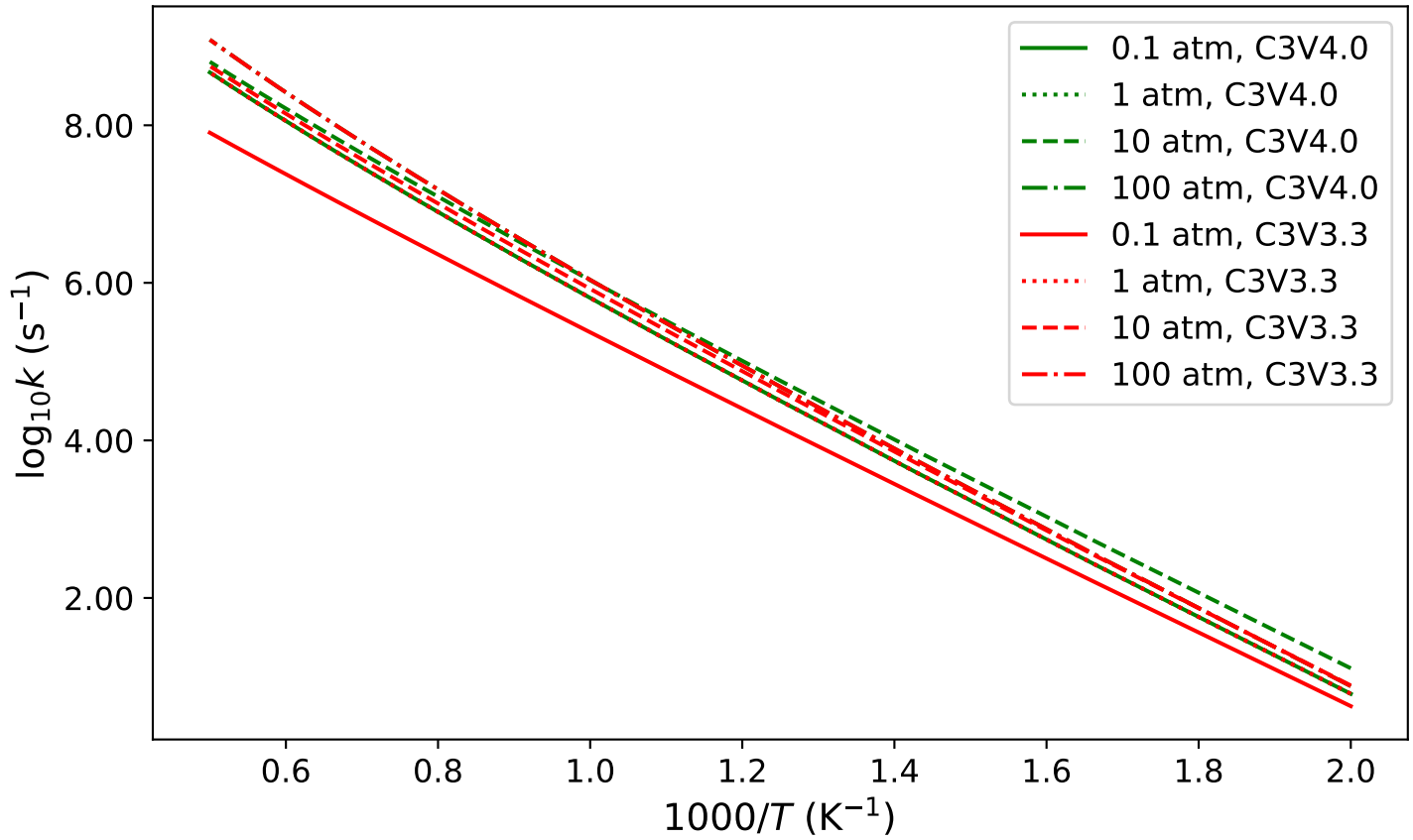



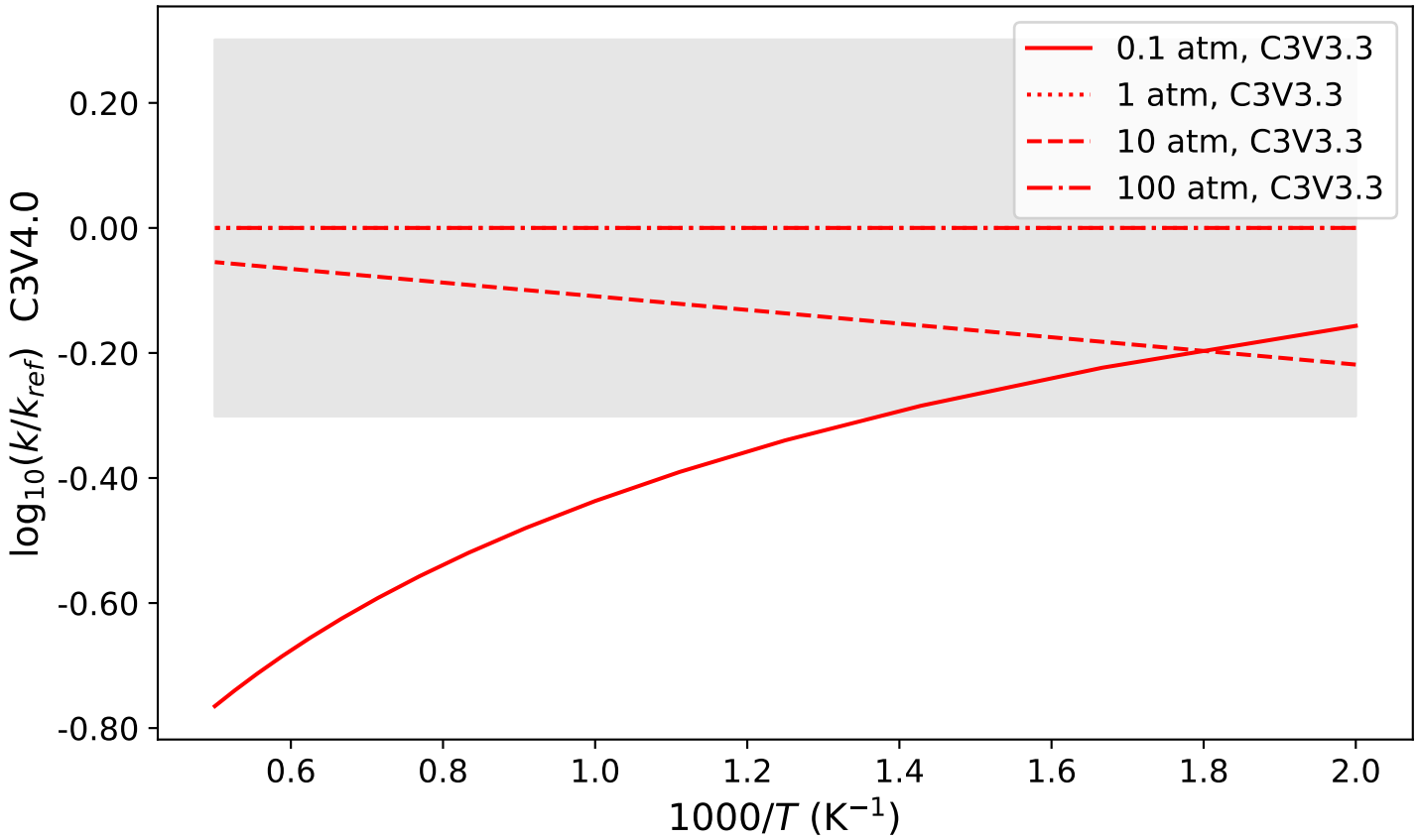
NC3H7OH + HO2 = NC3H7O + H2O2



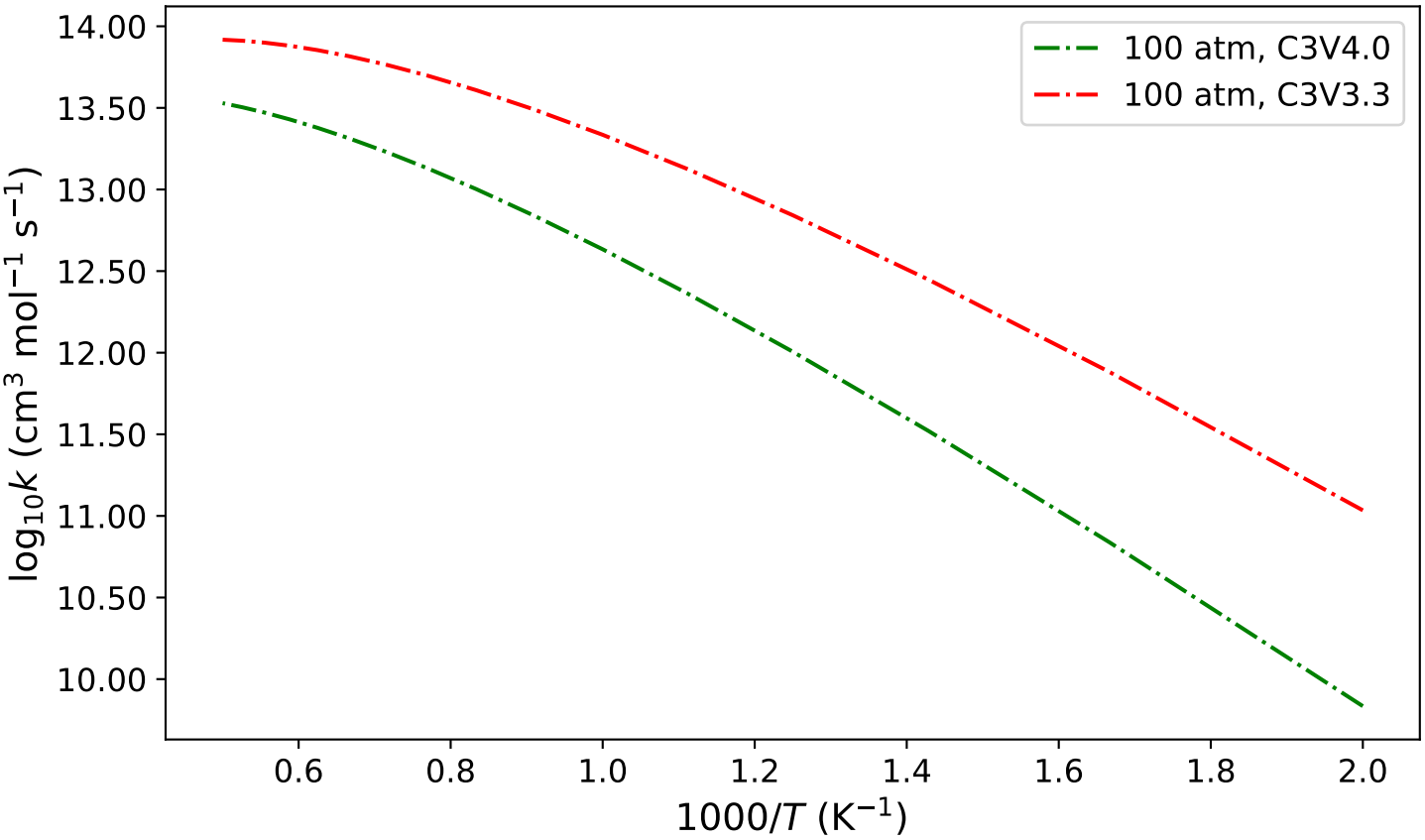



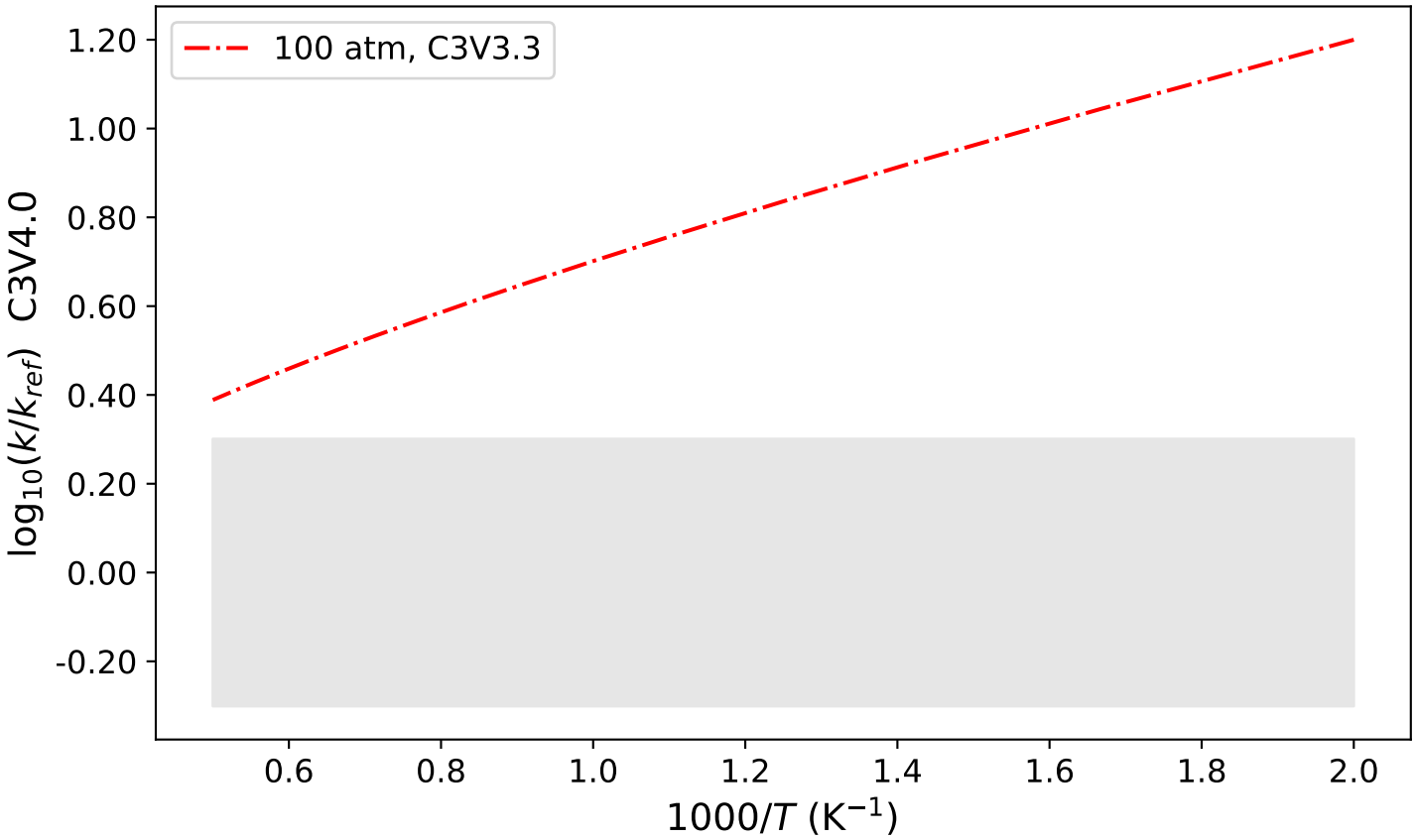
NC3H7O2 = C3H6OOH1-3



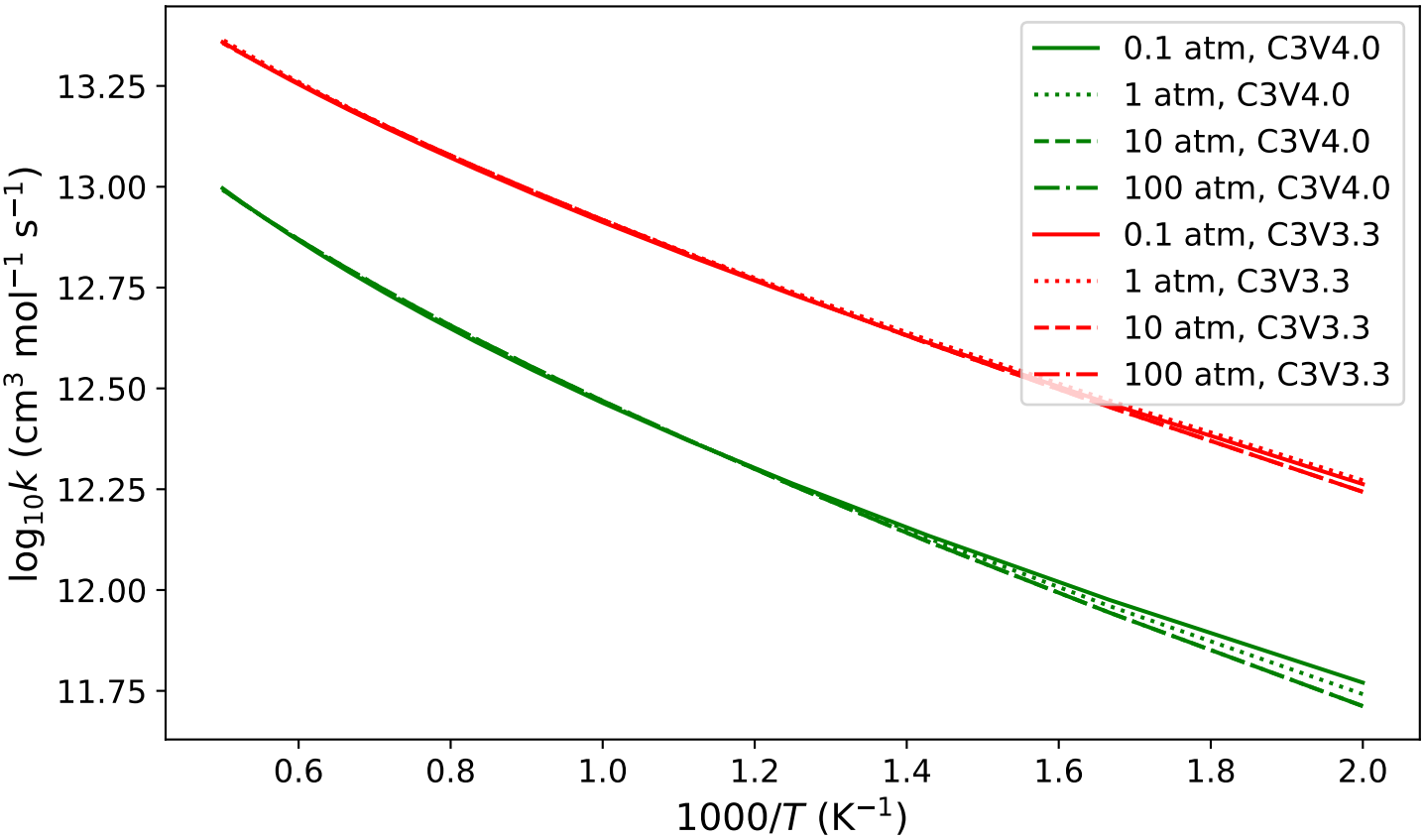



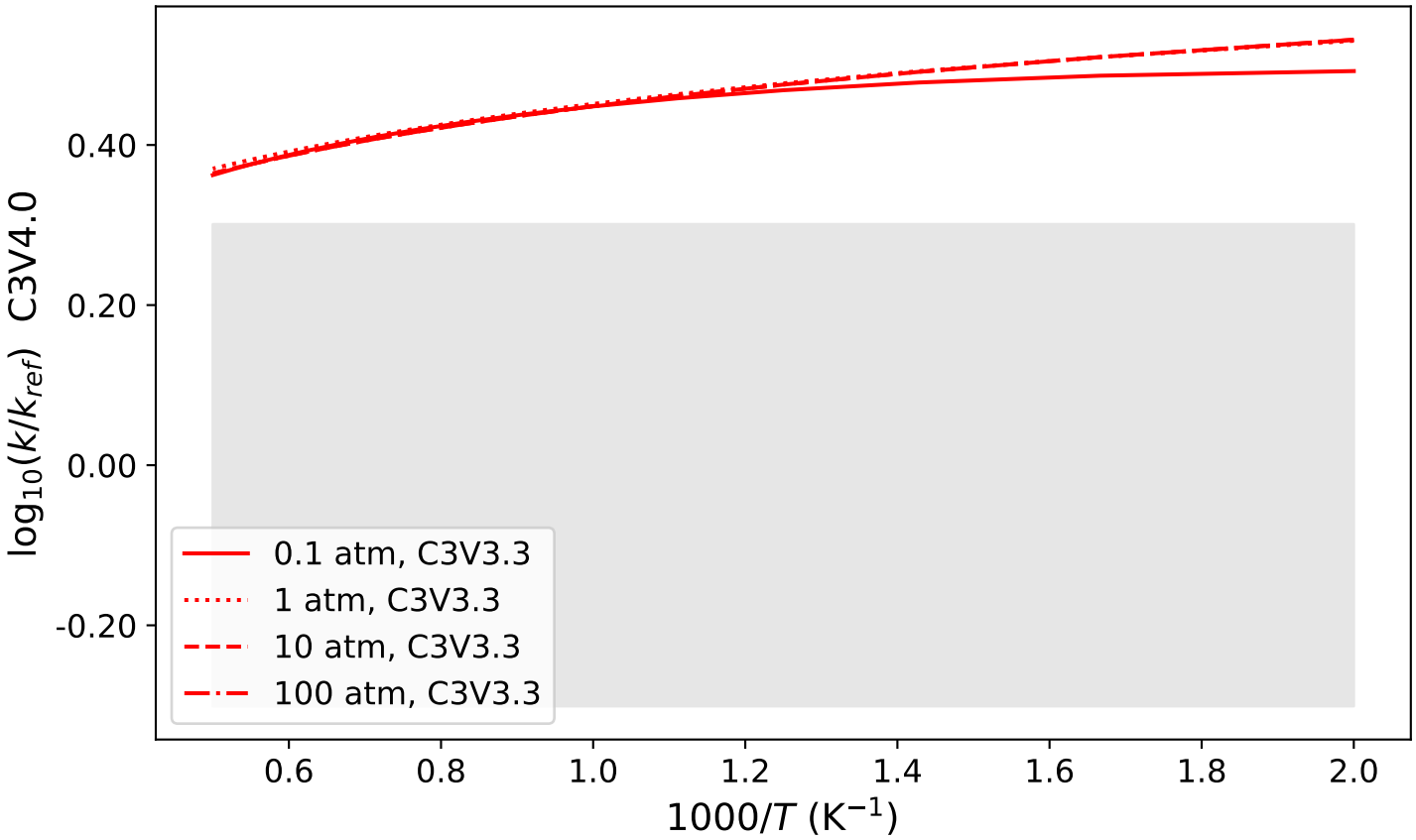
H + C4H5-I = H + C4H5-2



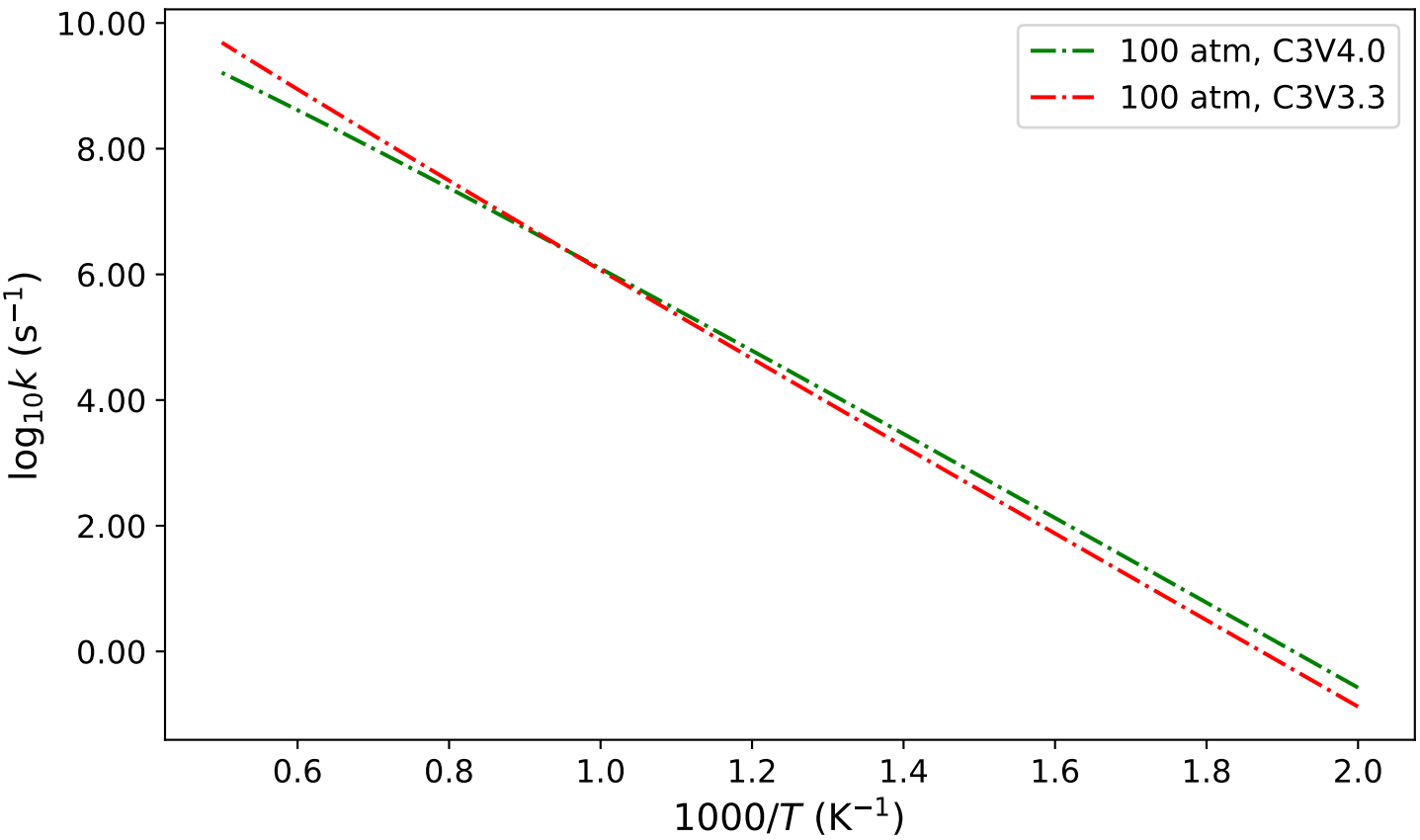



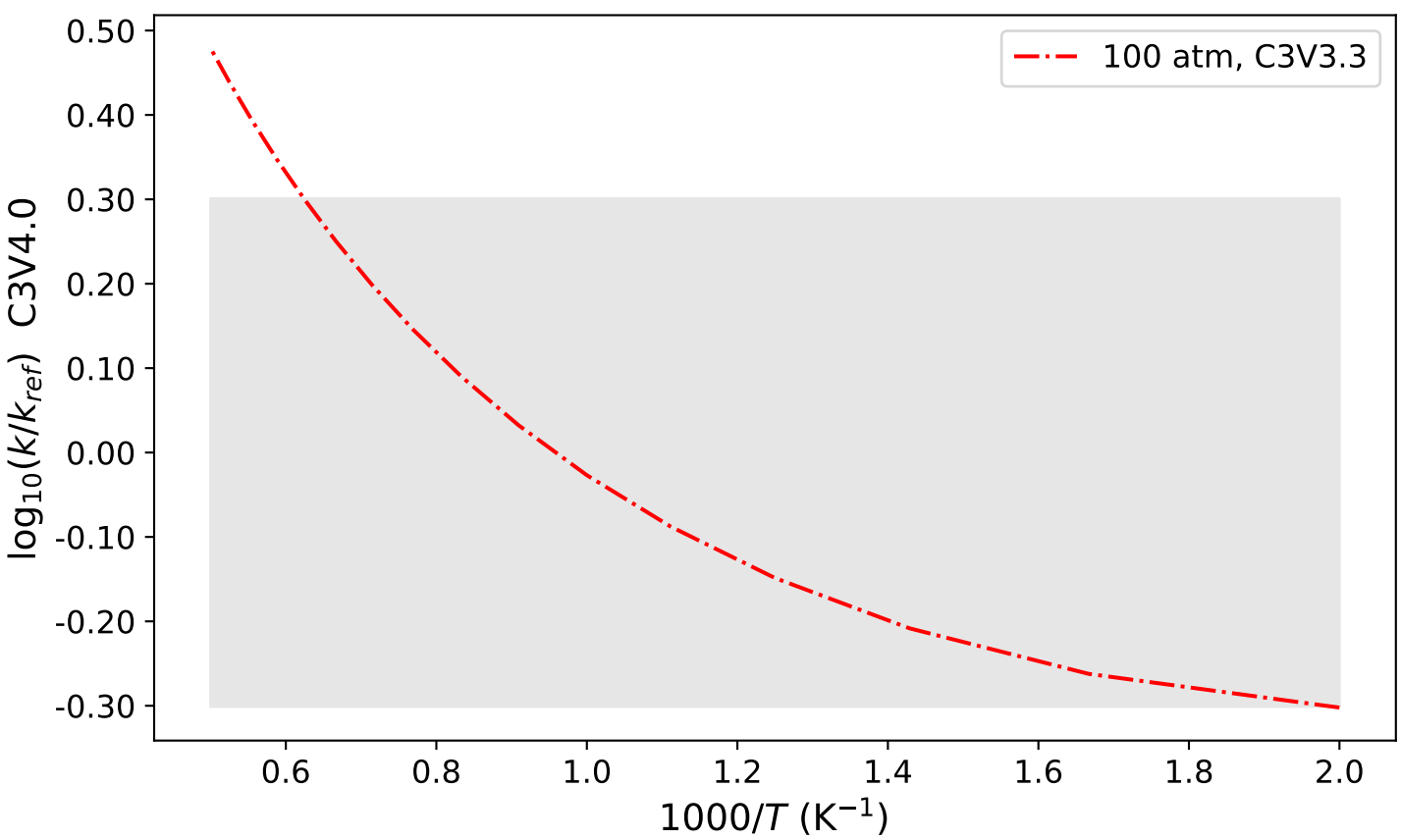
O + C4H6 = CH2O + C3H4-A



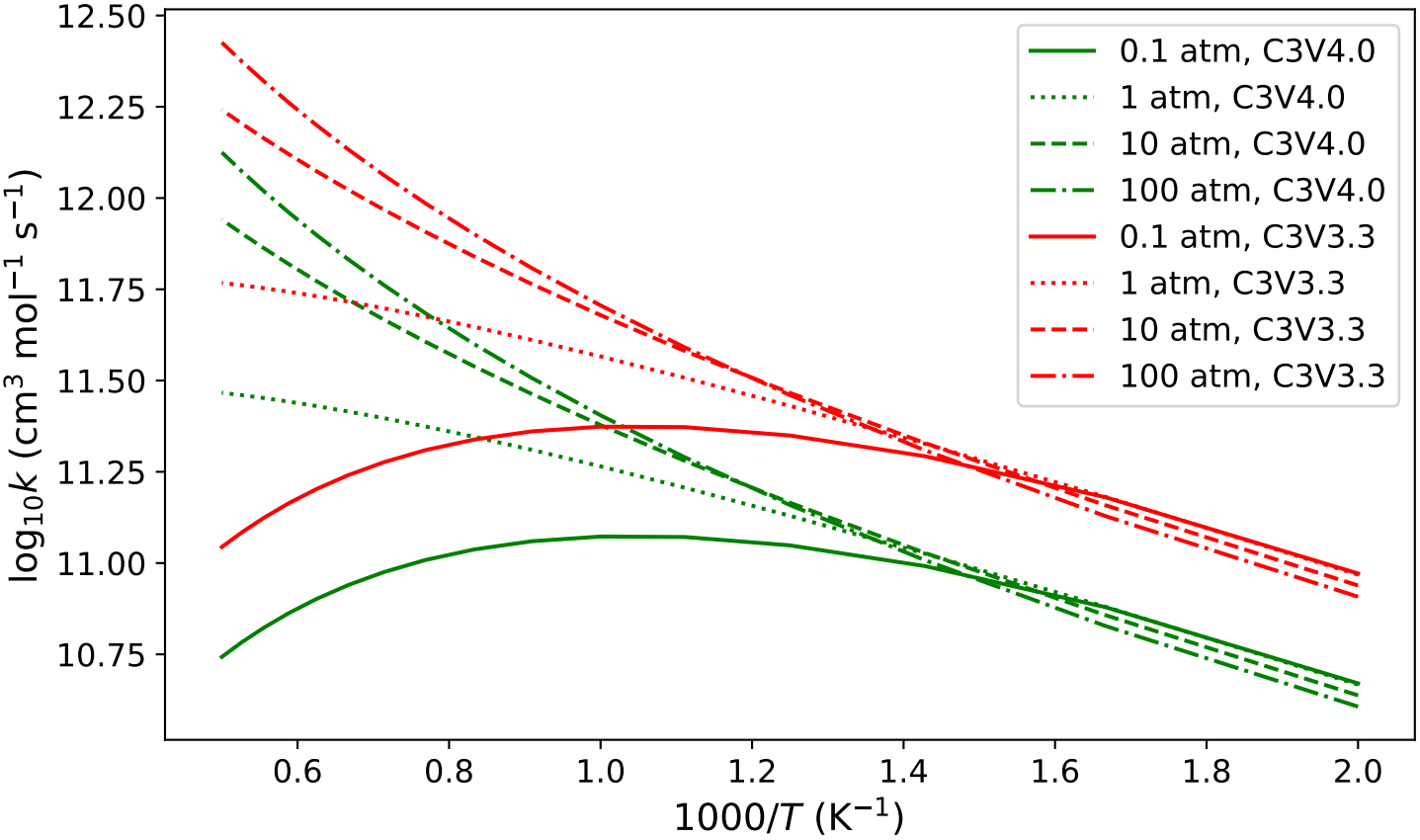



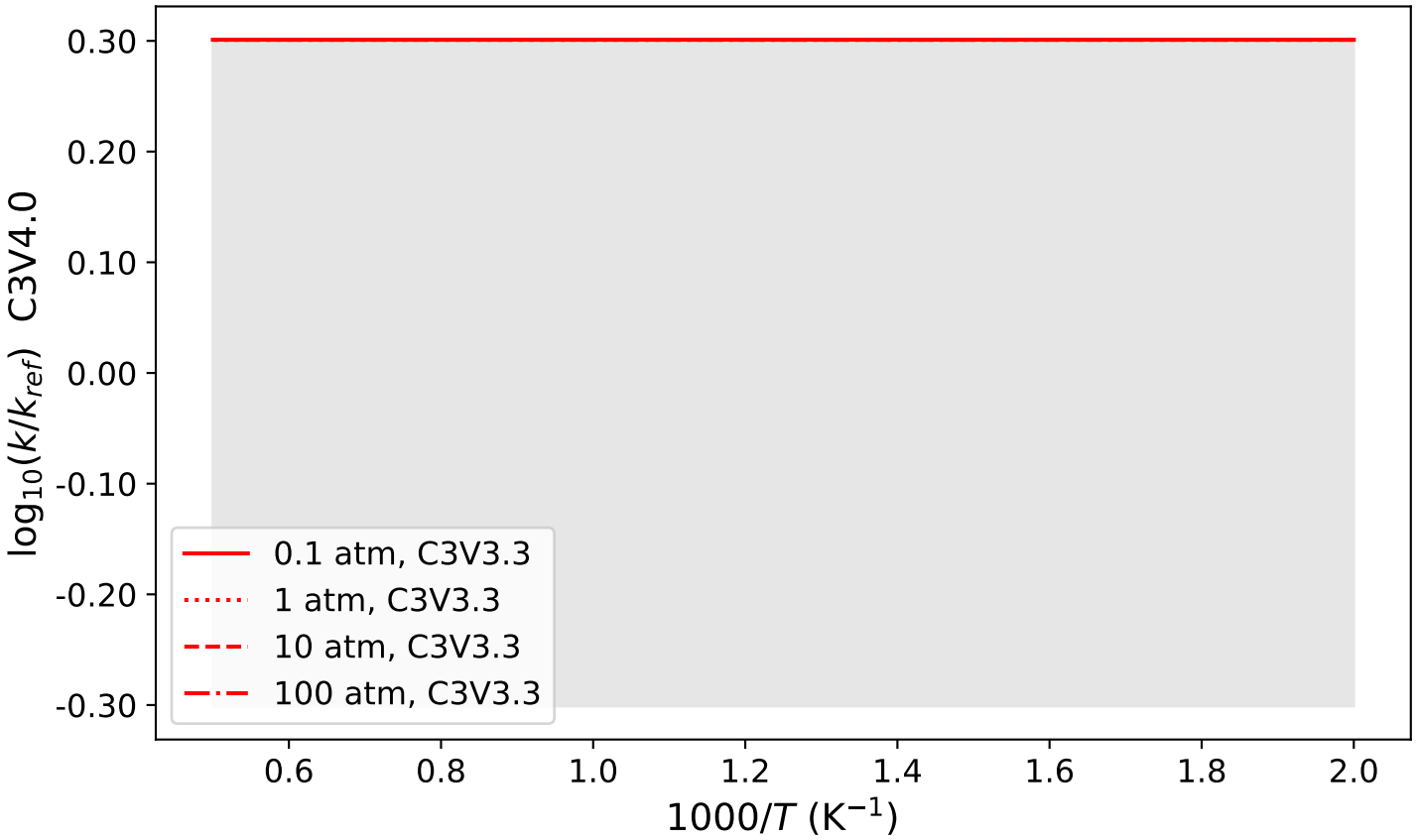
SC4H9O2 = HO2 + C4H8-1



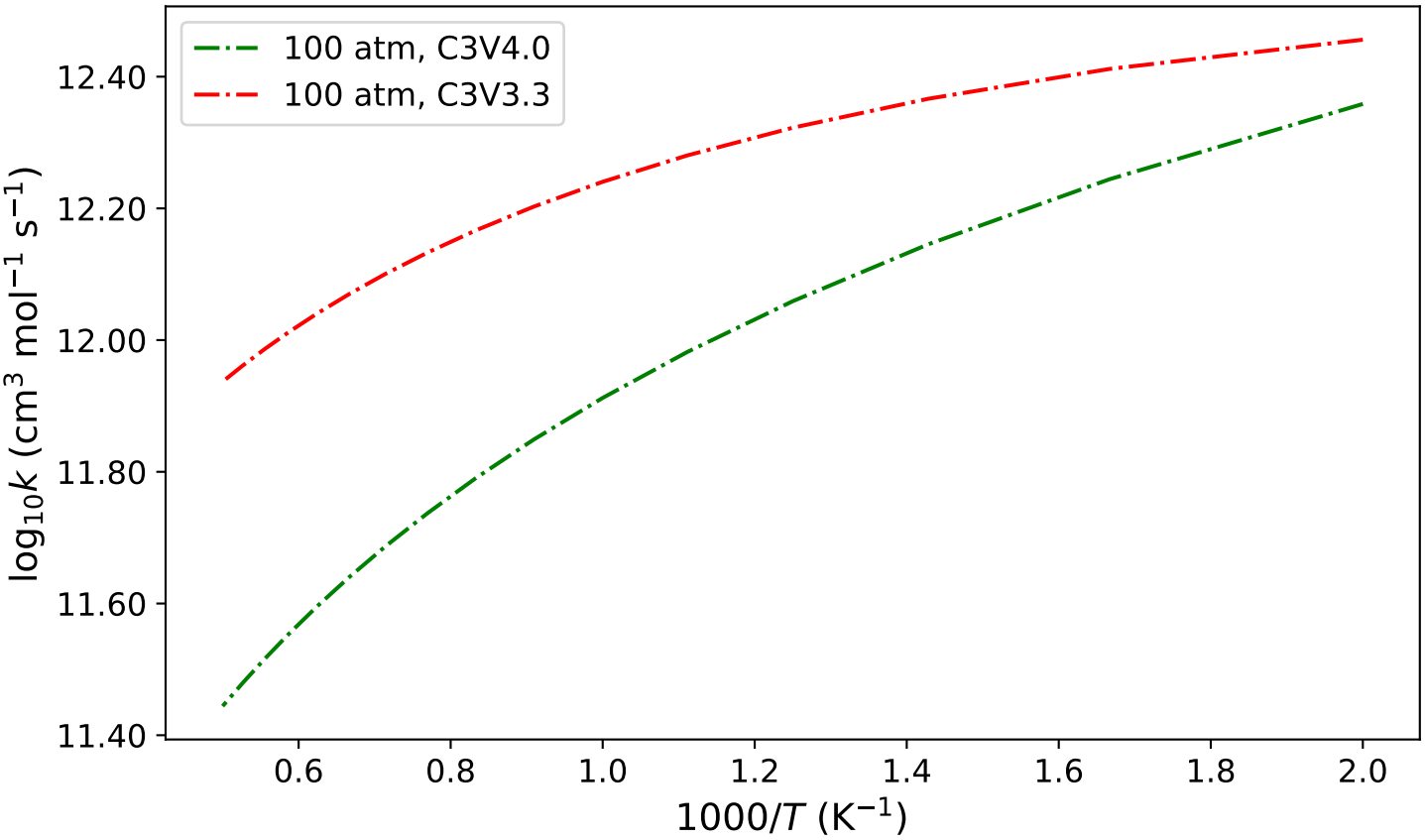



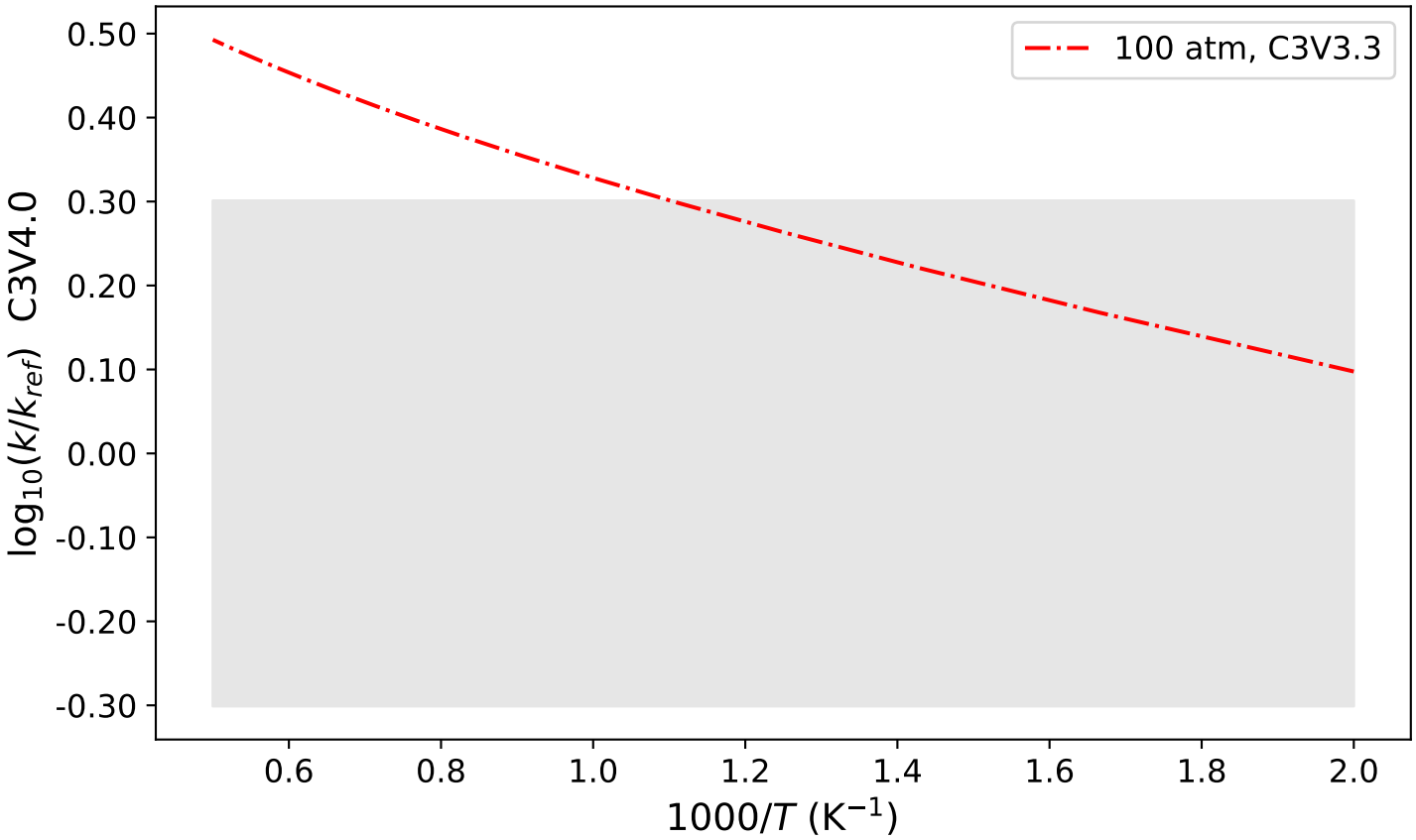
C2H3 + C4H6 = C6H9-A



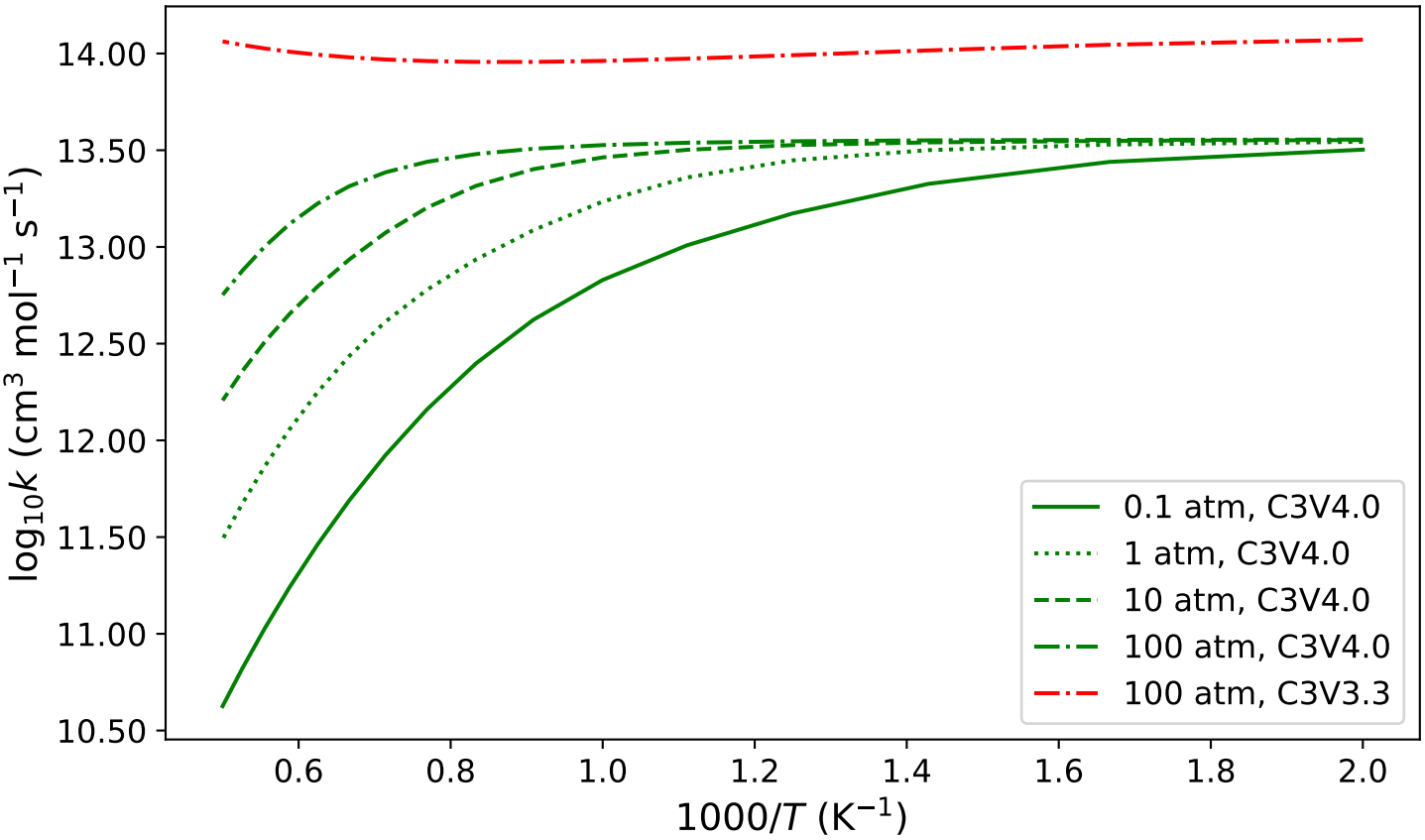



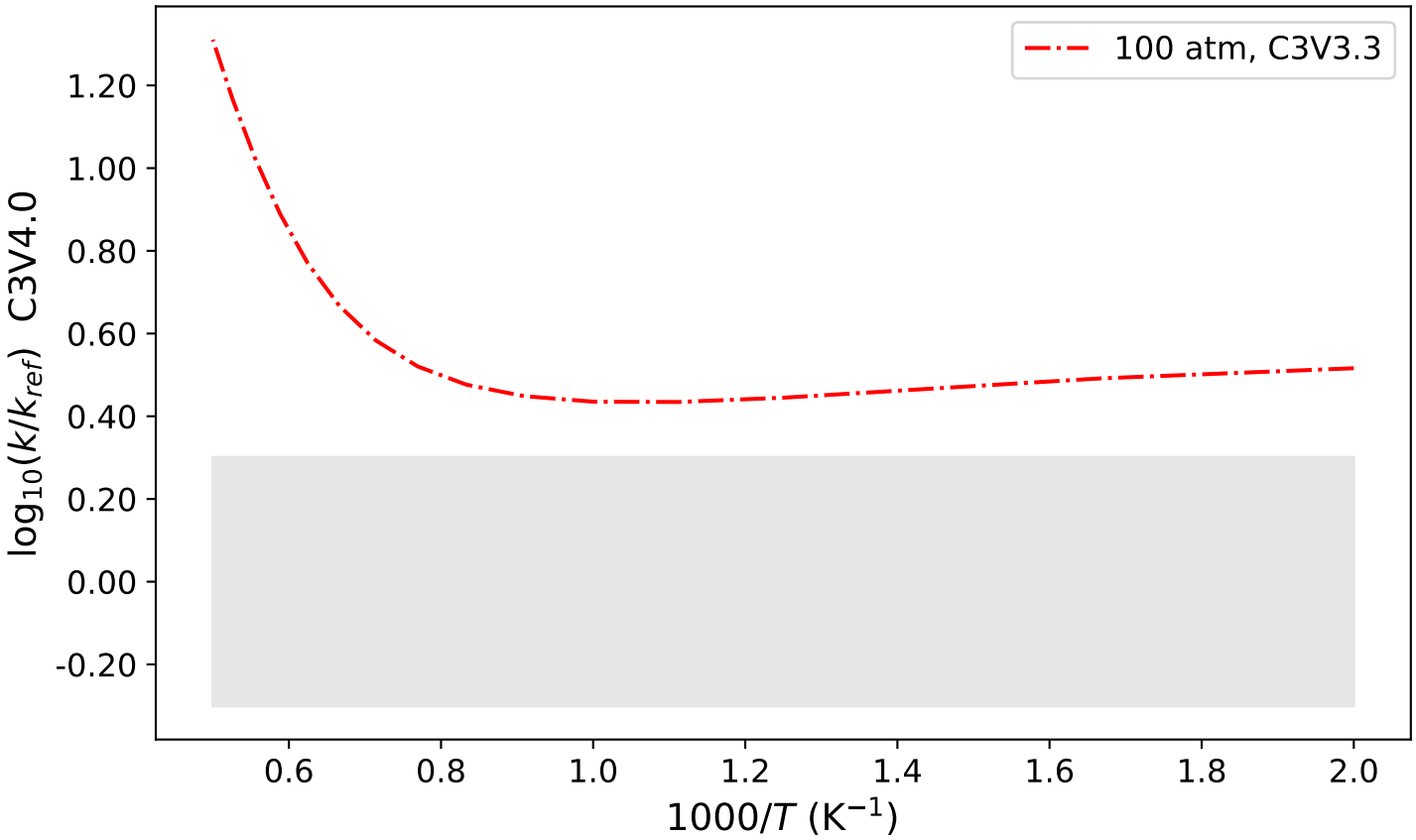
O2 + CH2CH2CHO = O2CH2CH2CHO



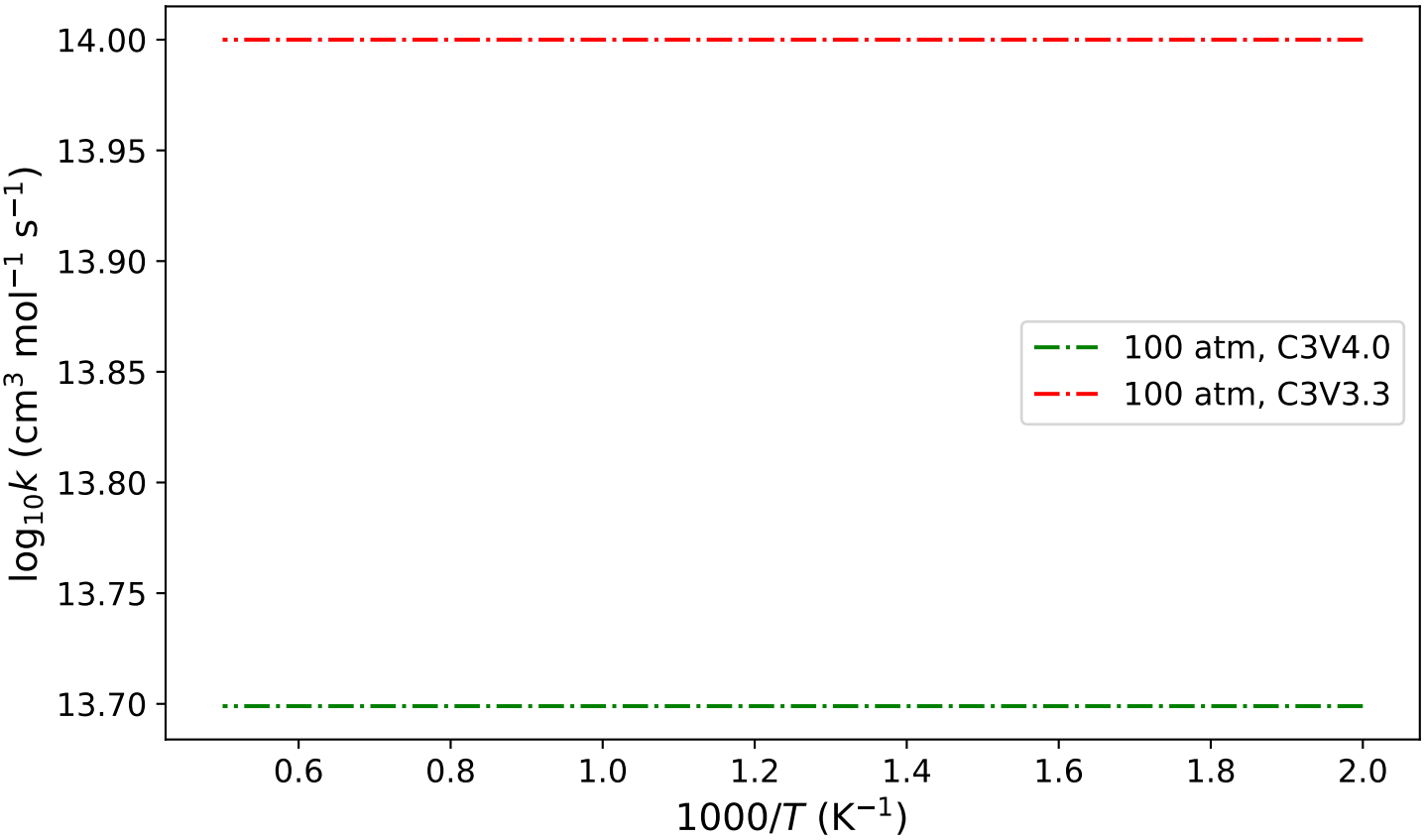



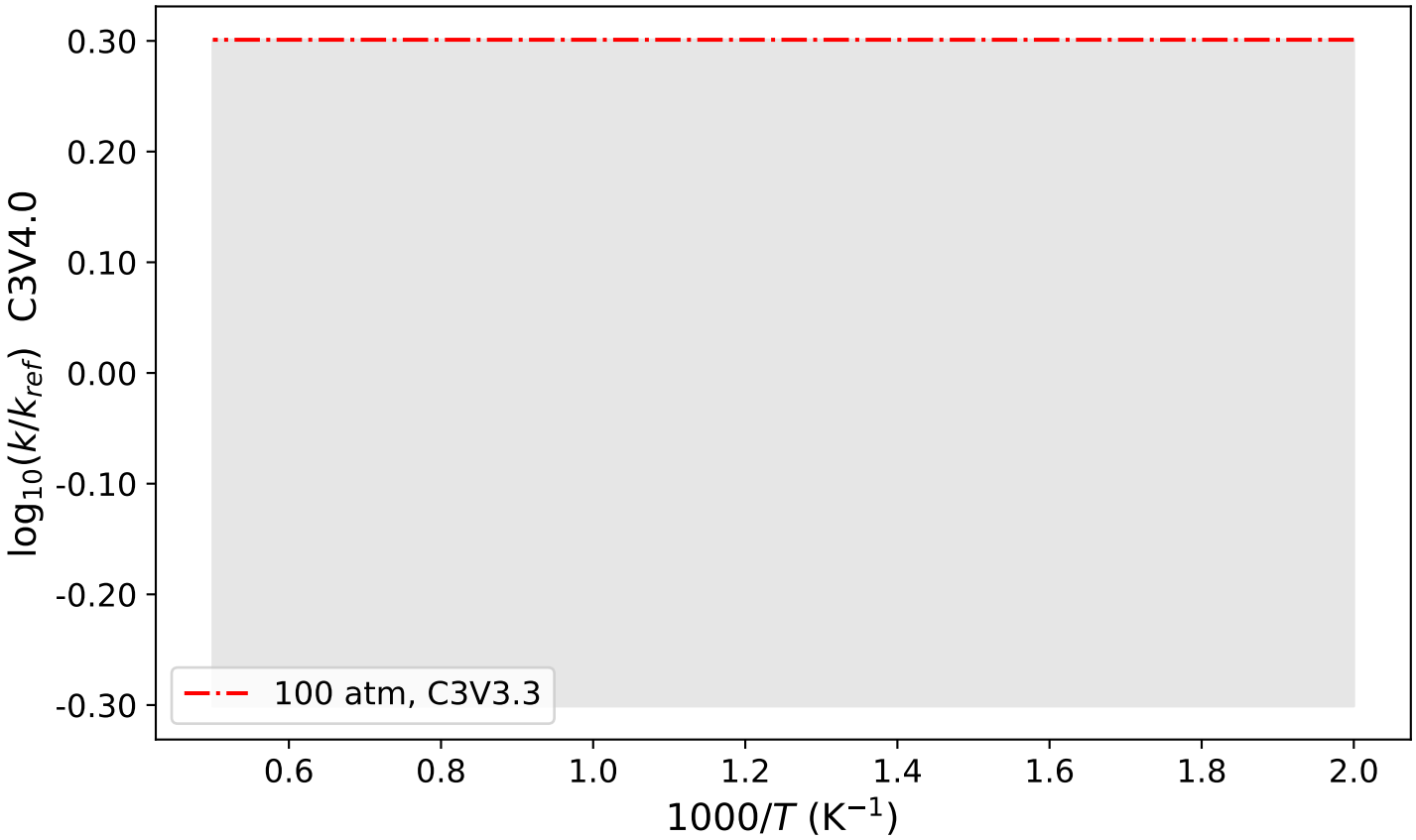
H + C4H5-4 (+M) = C4H6-1 (+M)



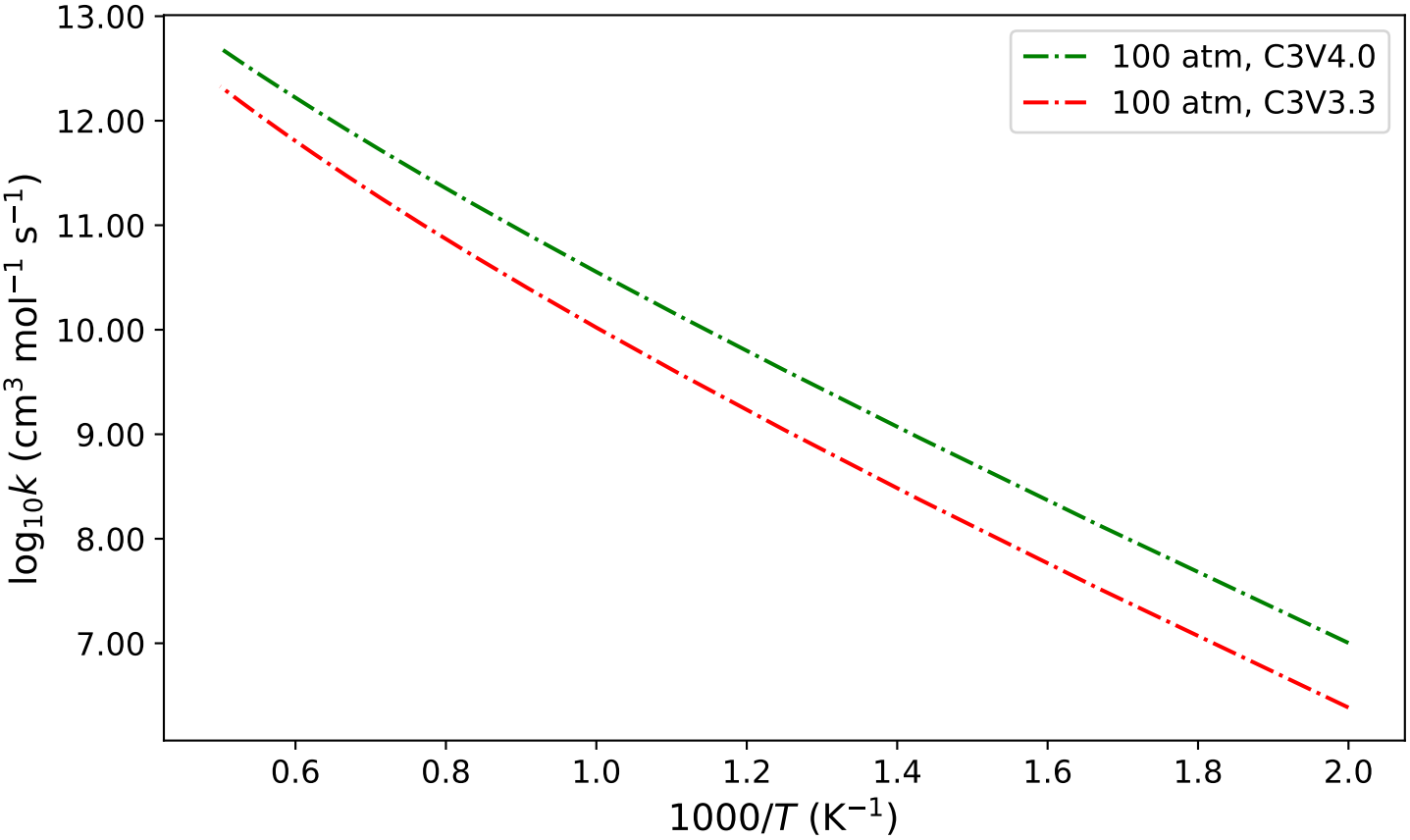



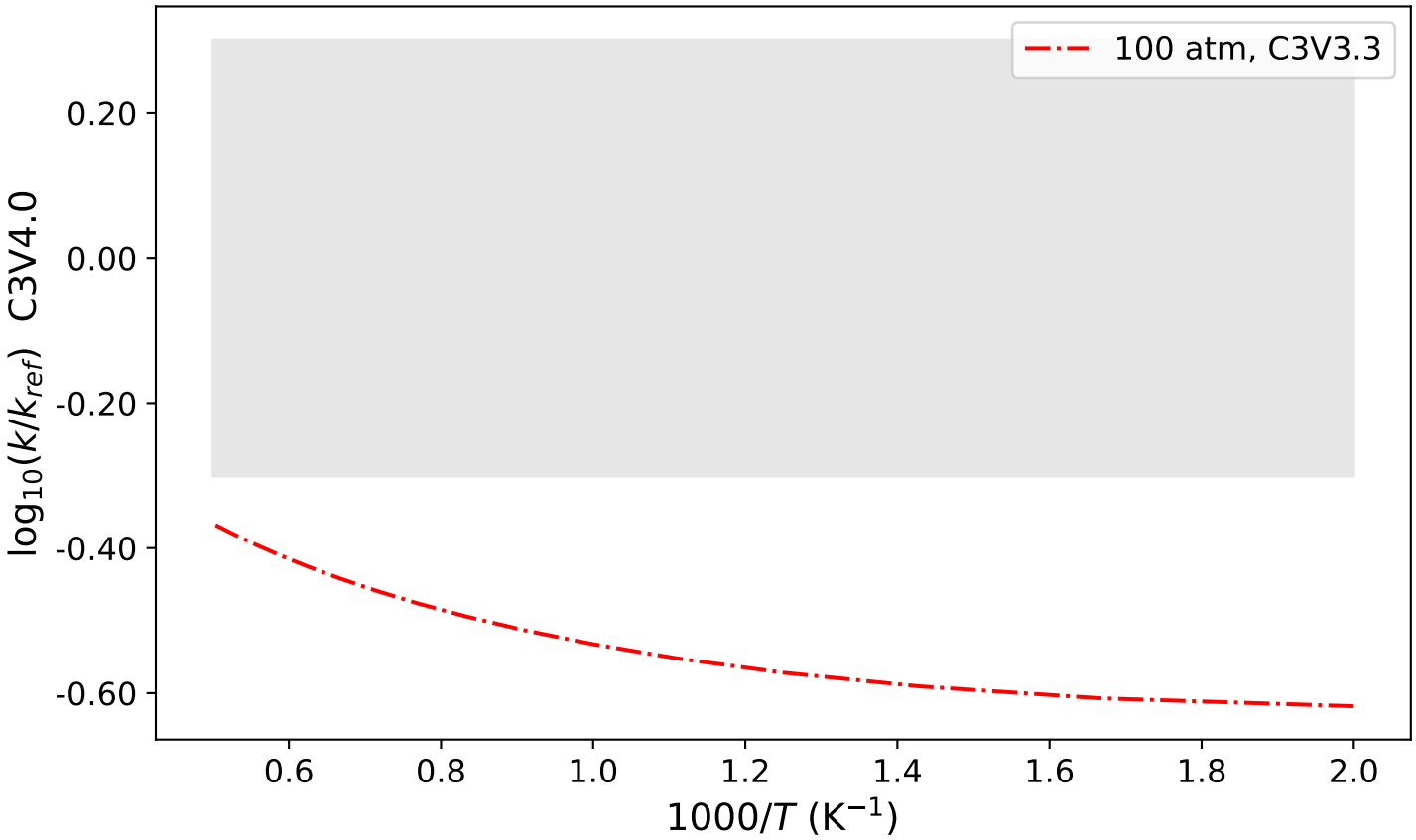
H + CH2CHO = CH3CHO



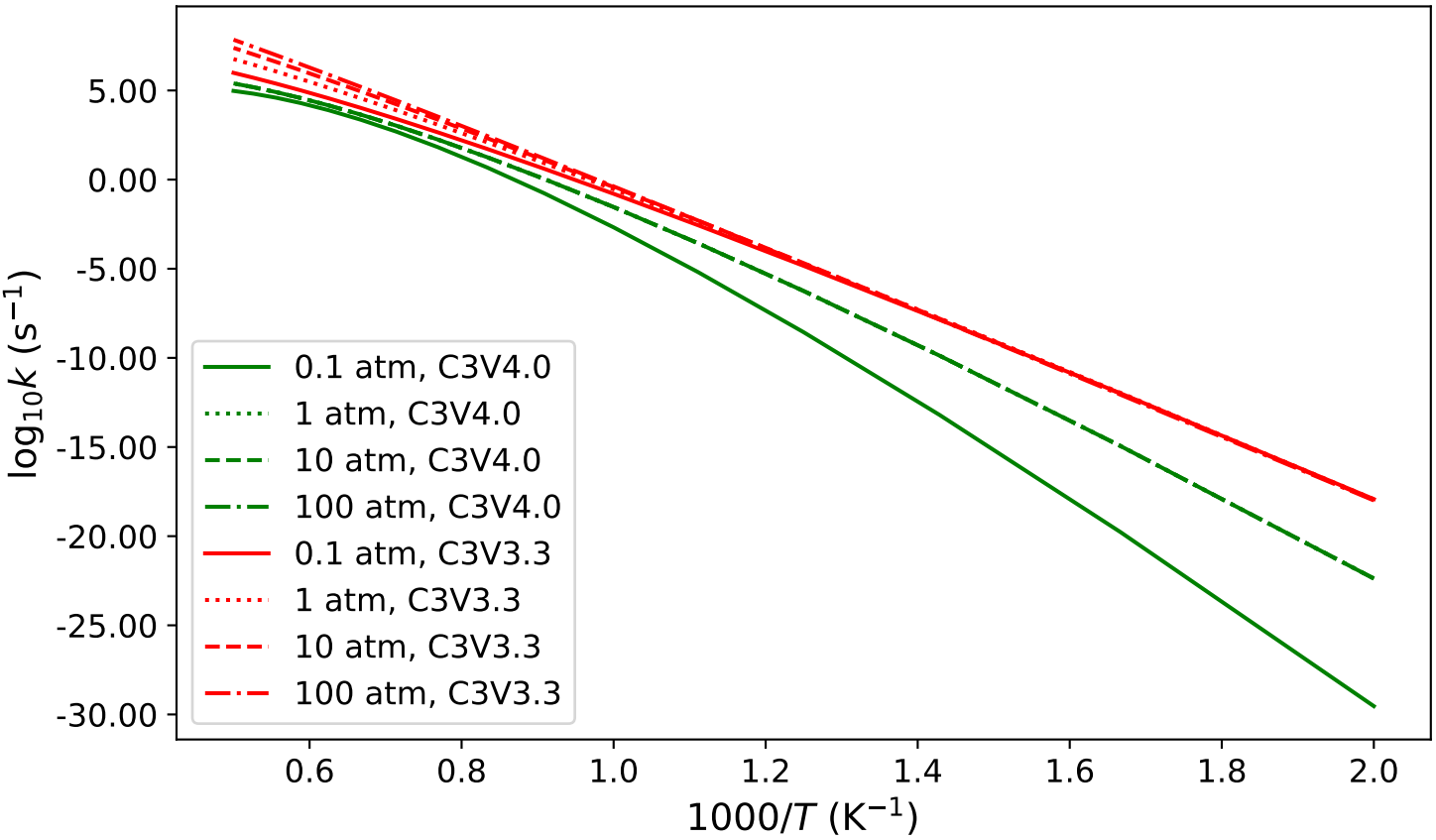



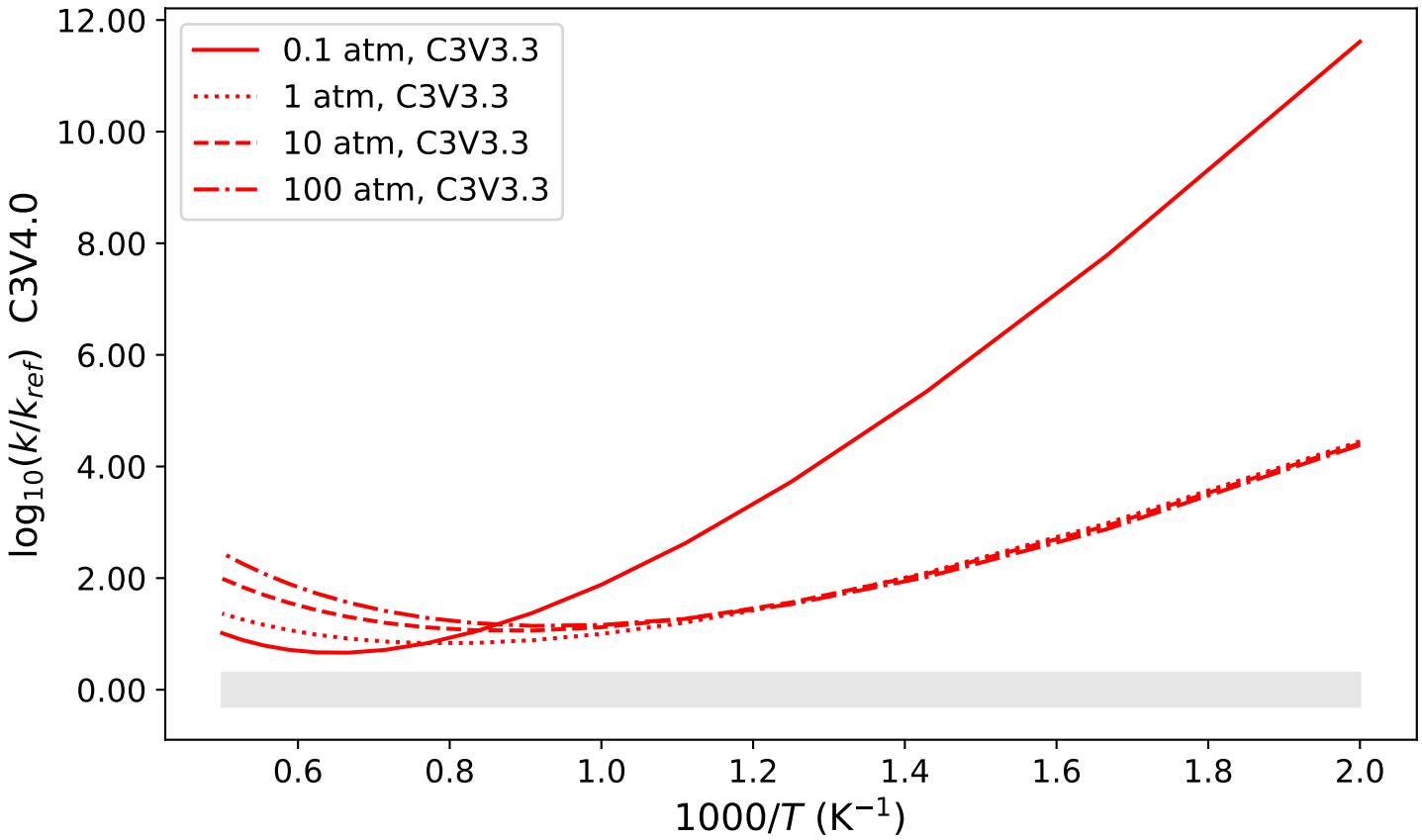
H + C3H6 = H2 + C3H5-S



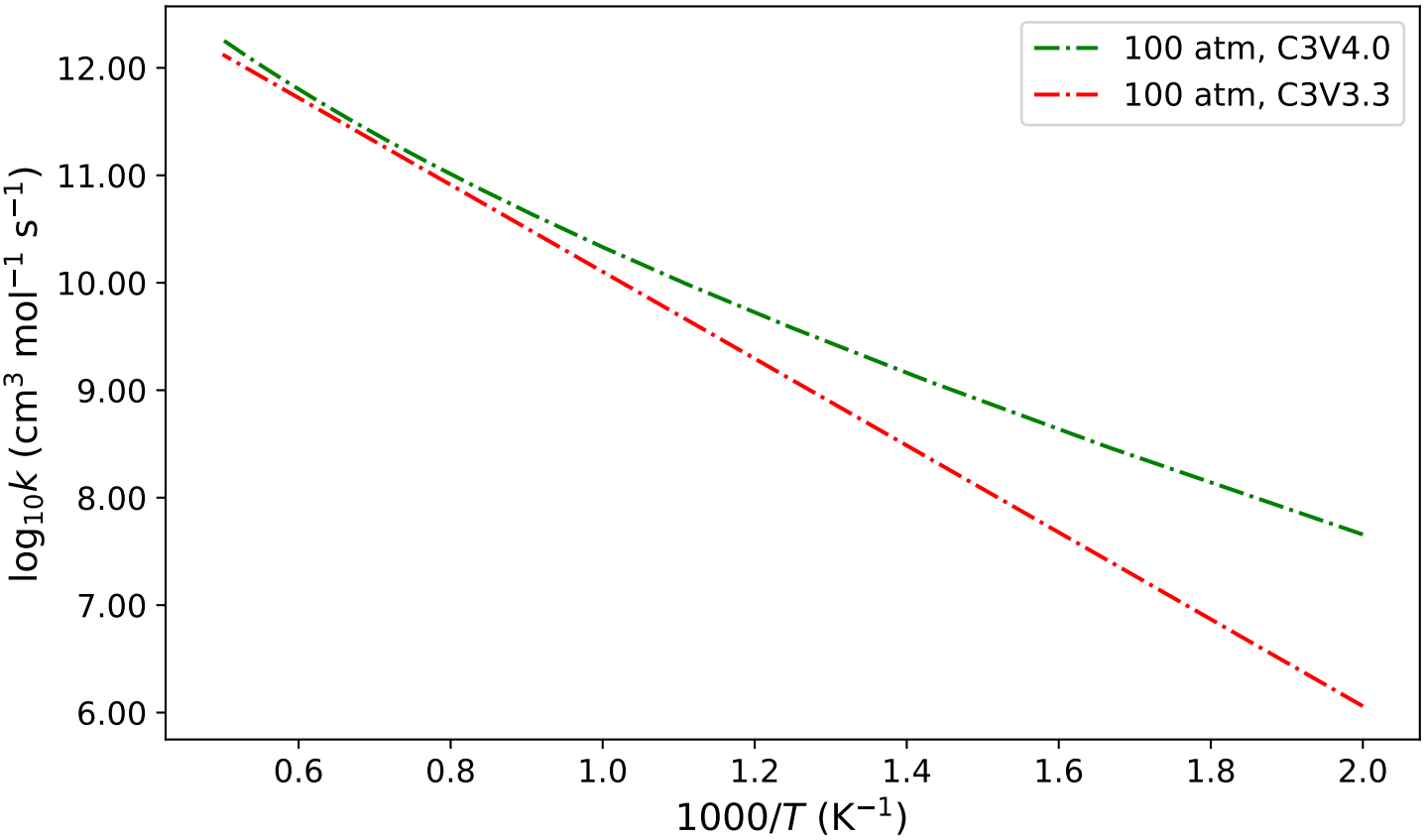



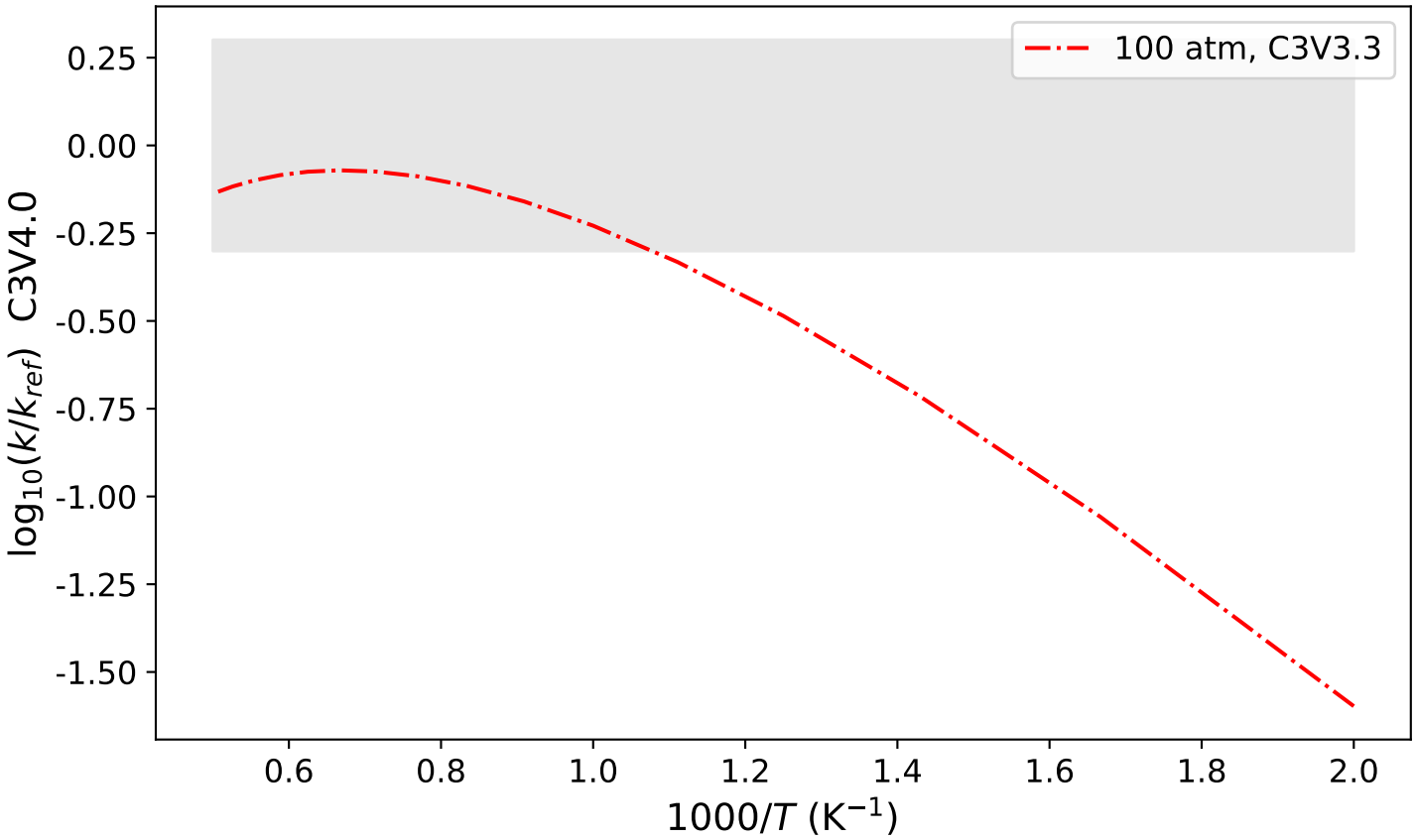
C4H612 = CH3 + C3H3



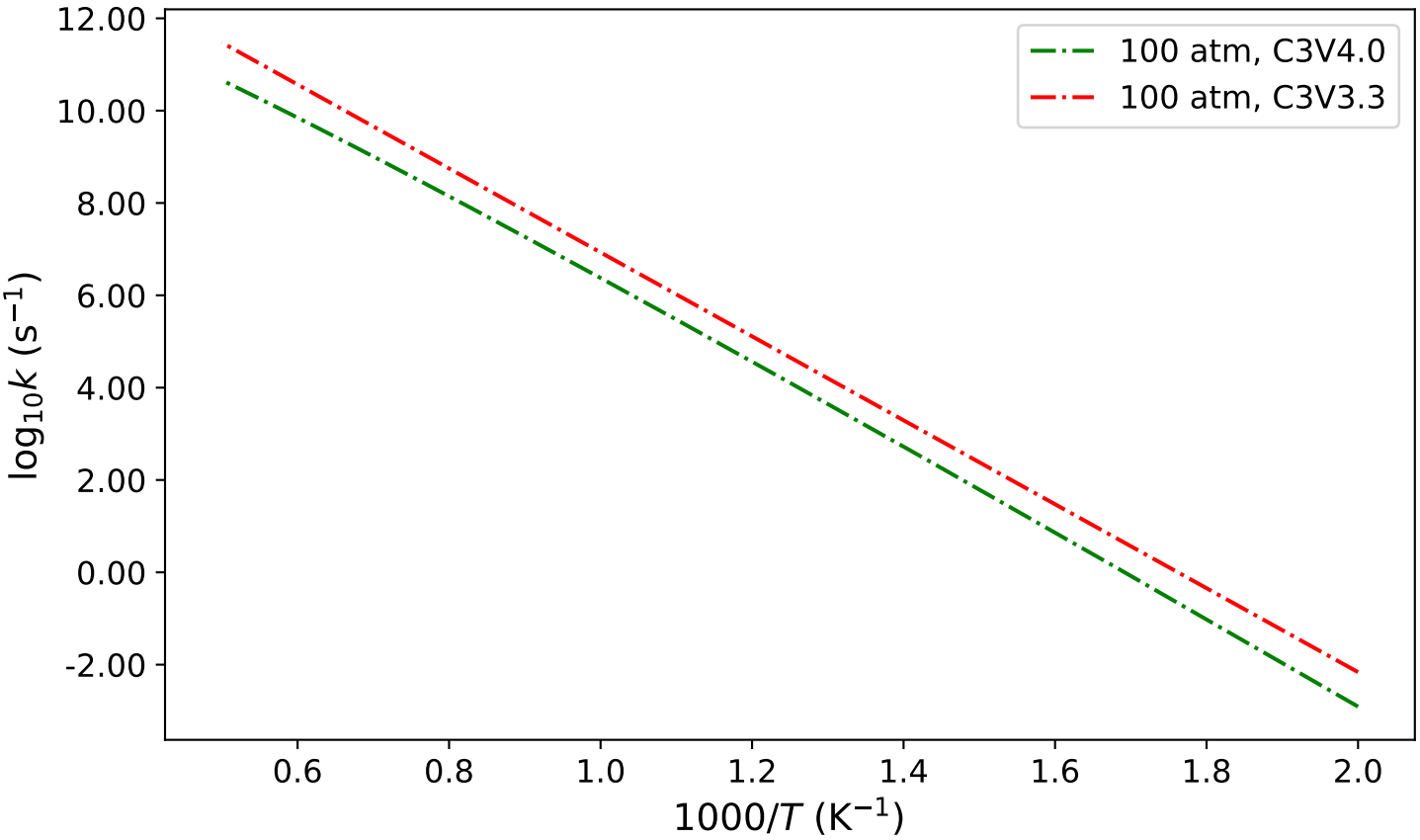



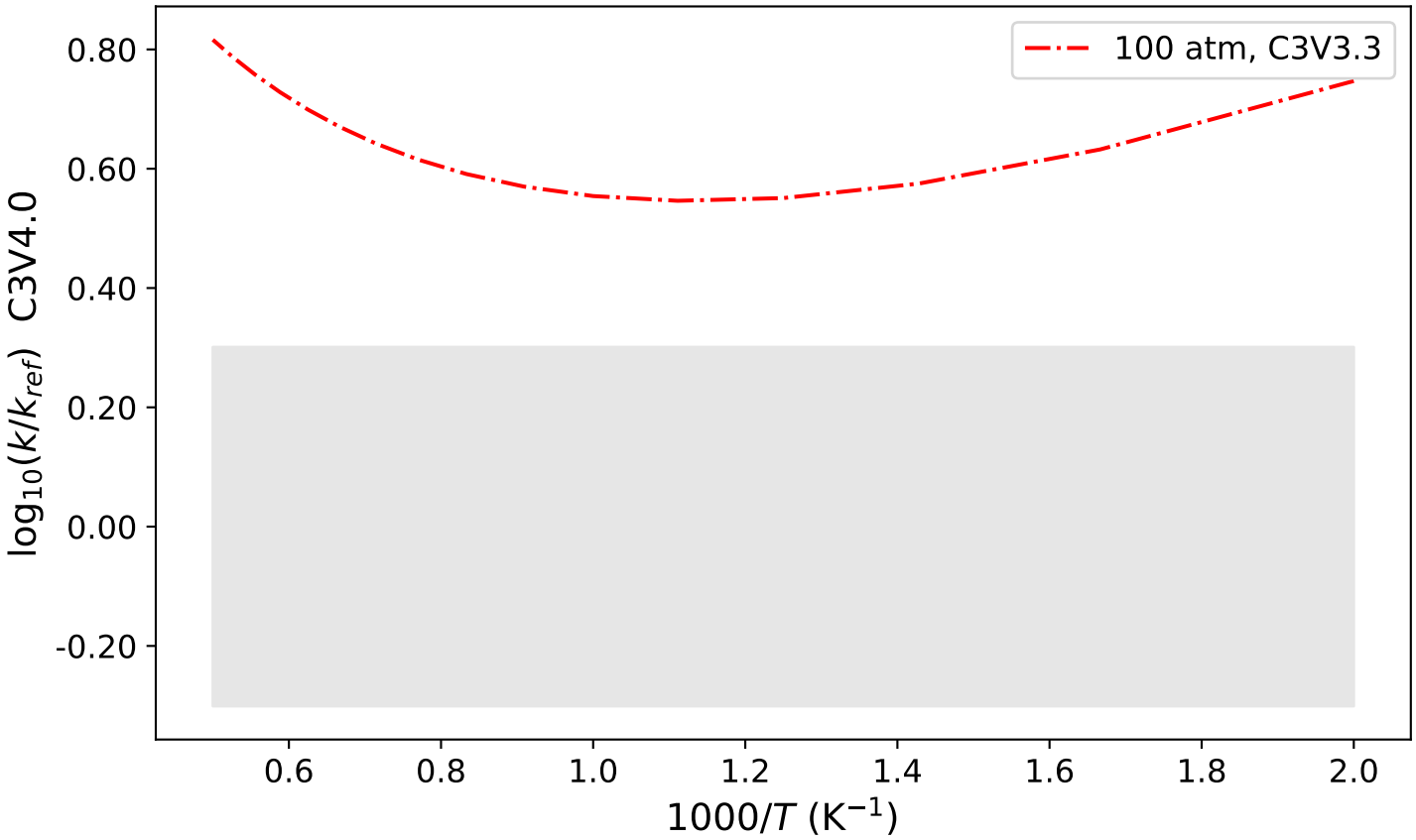
CH3 + C4H6-2 = CH4 + C4H5-2



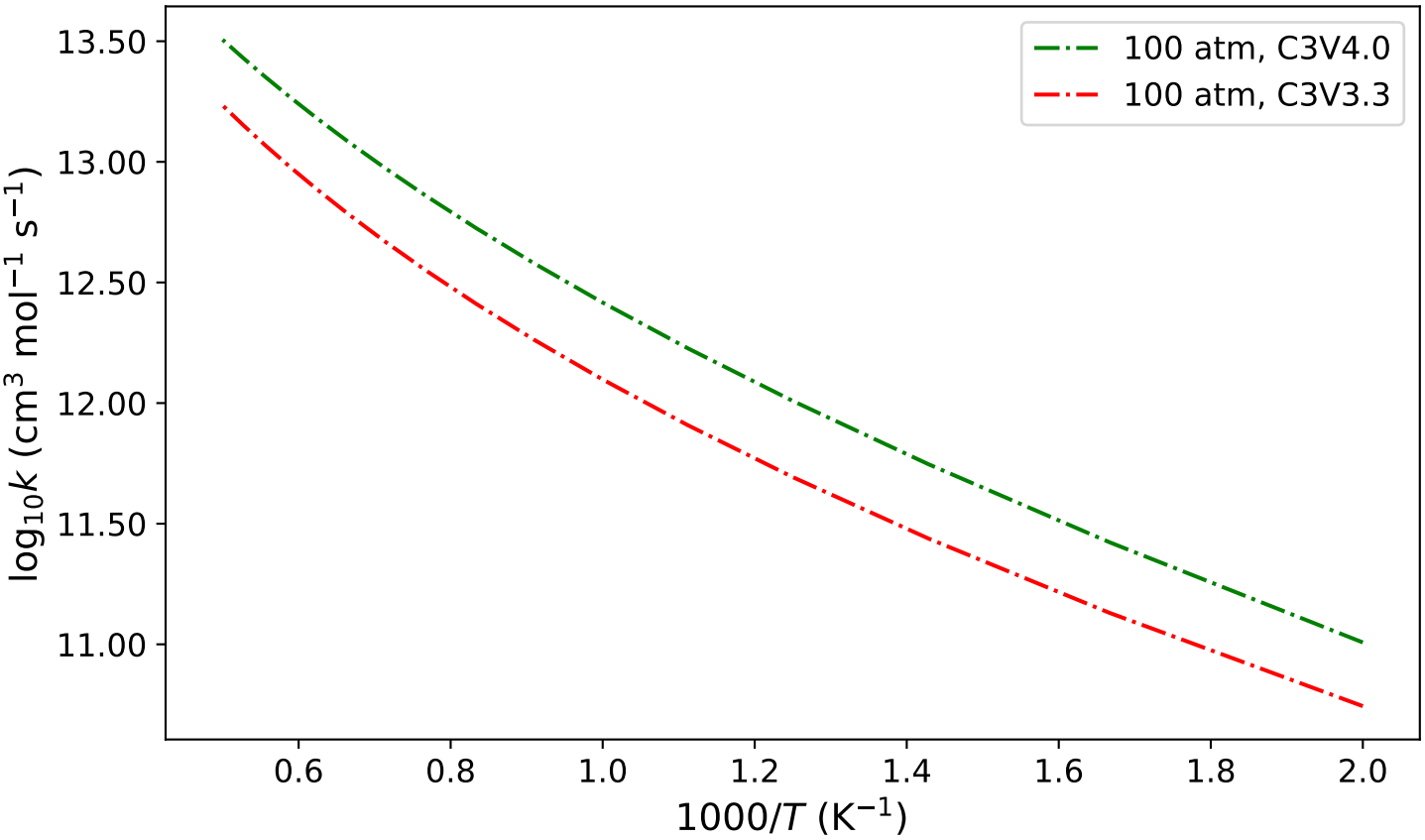



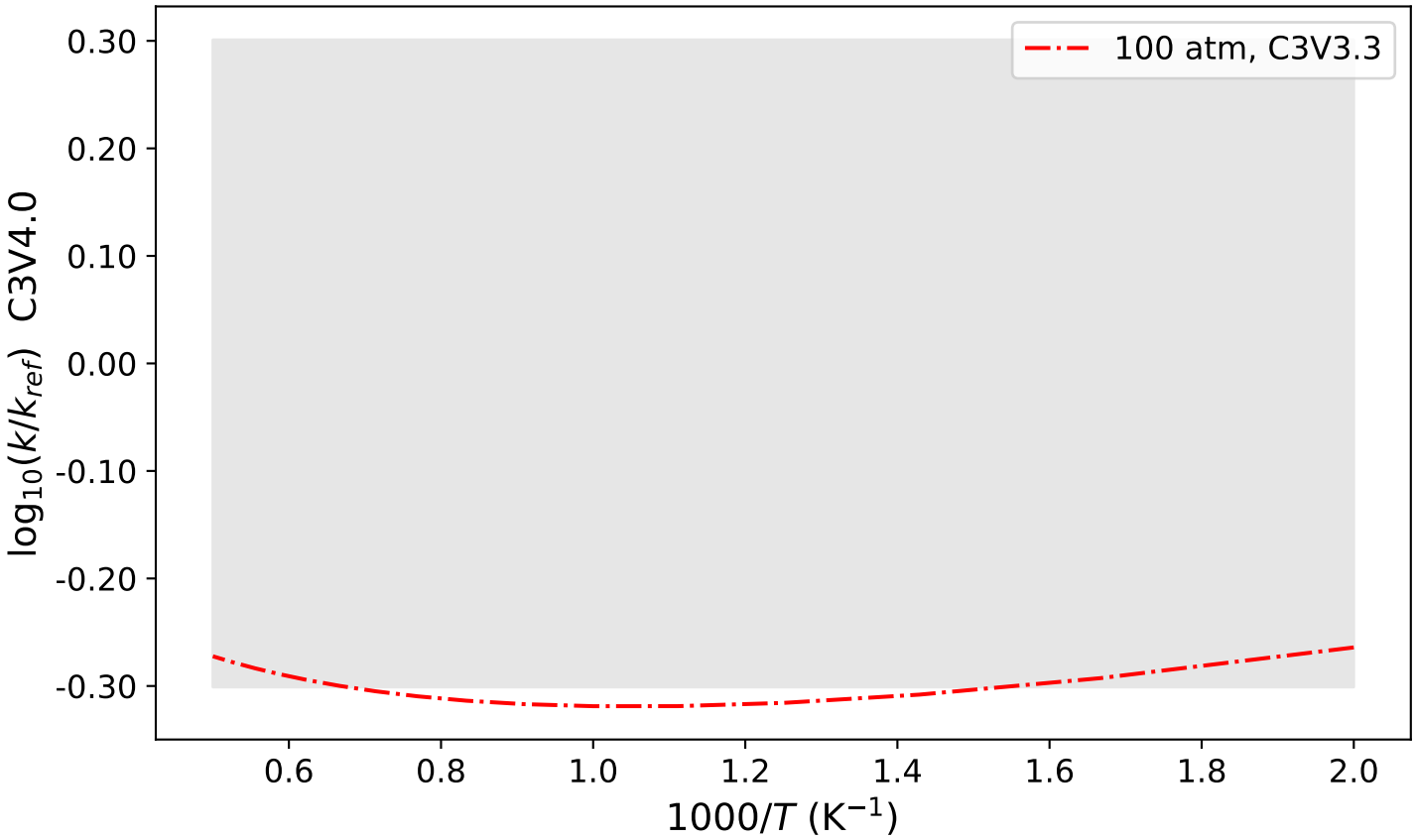
NC4KET23 => CH3CHO + OH + CH3CO



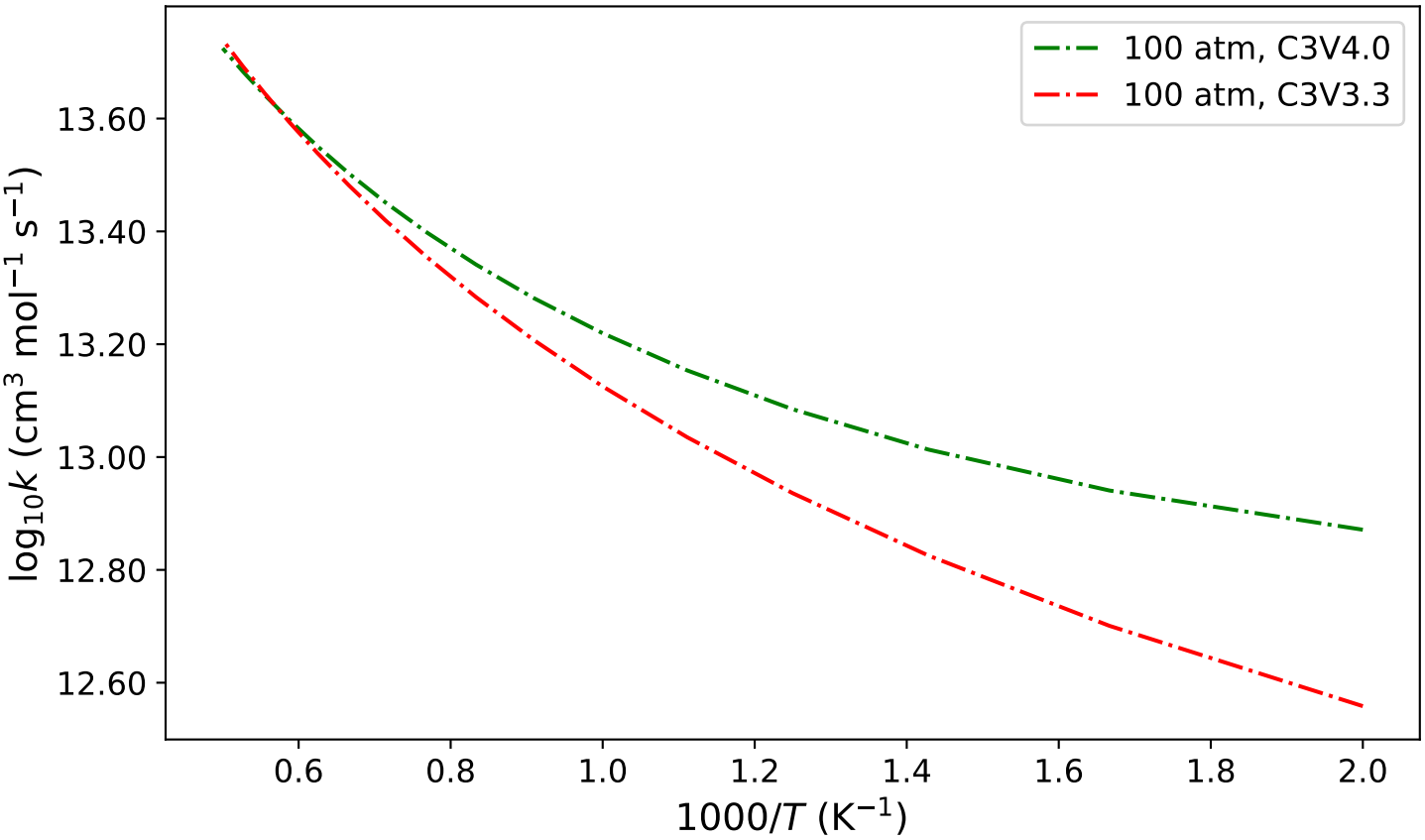



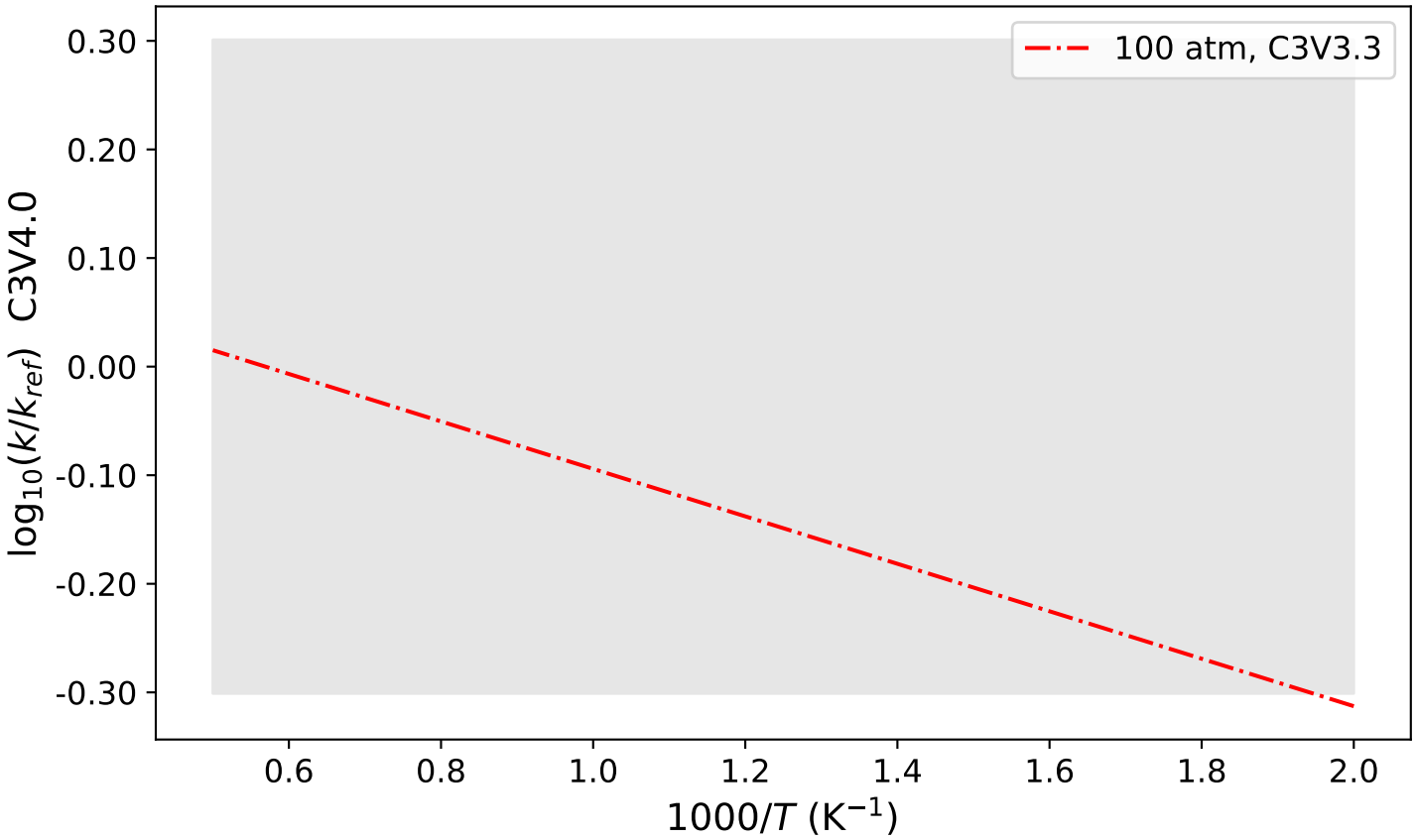
H + C4H8-1 = H2 + C4H71-3



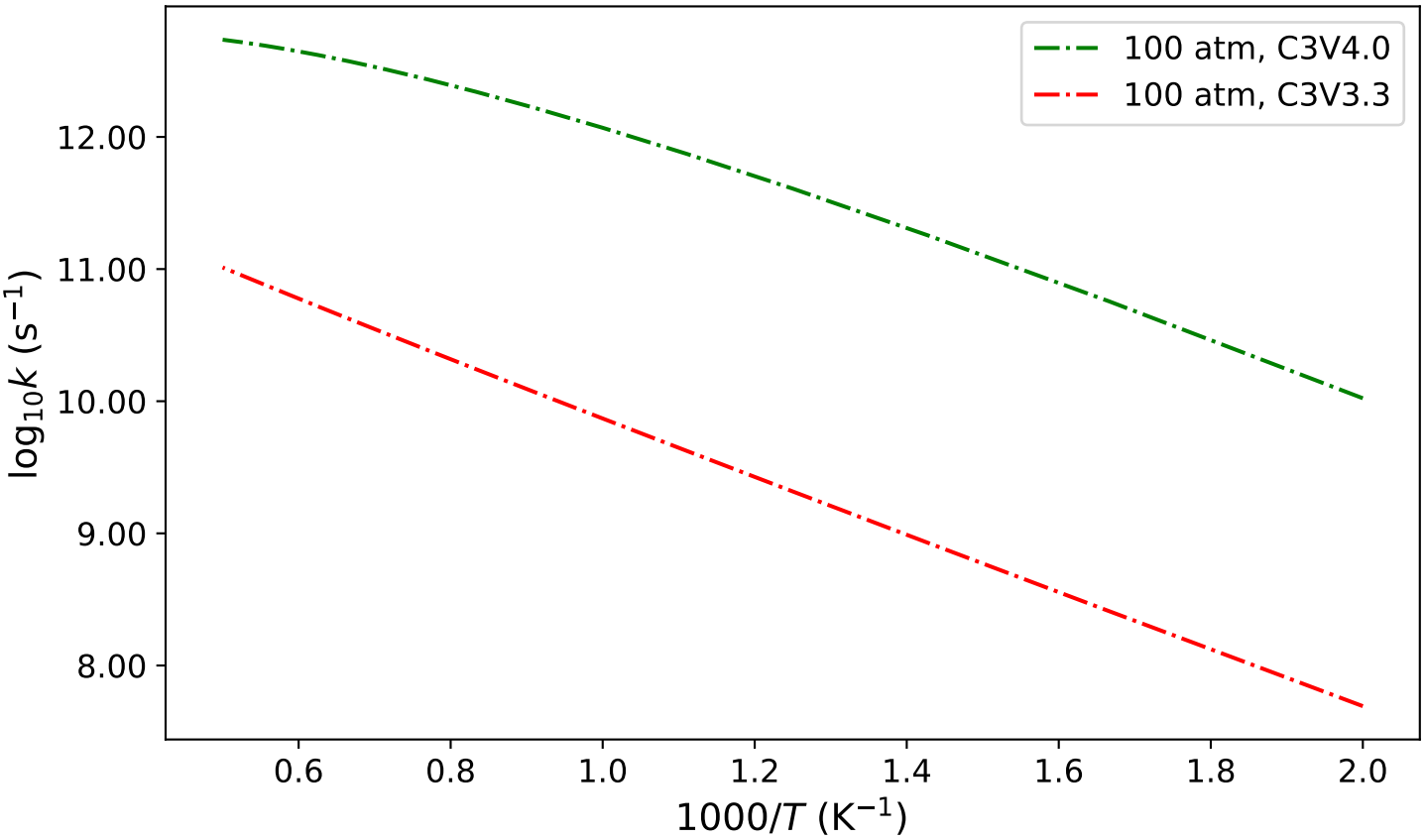



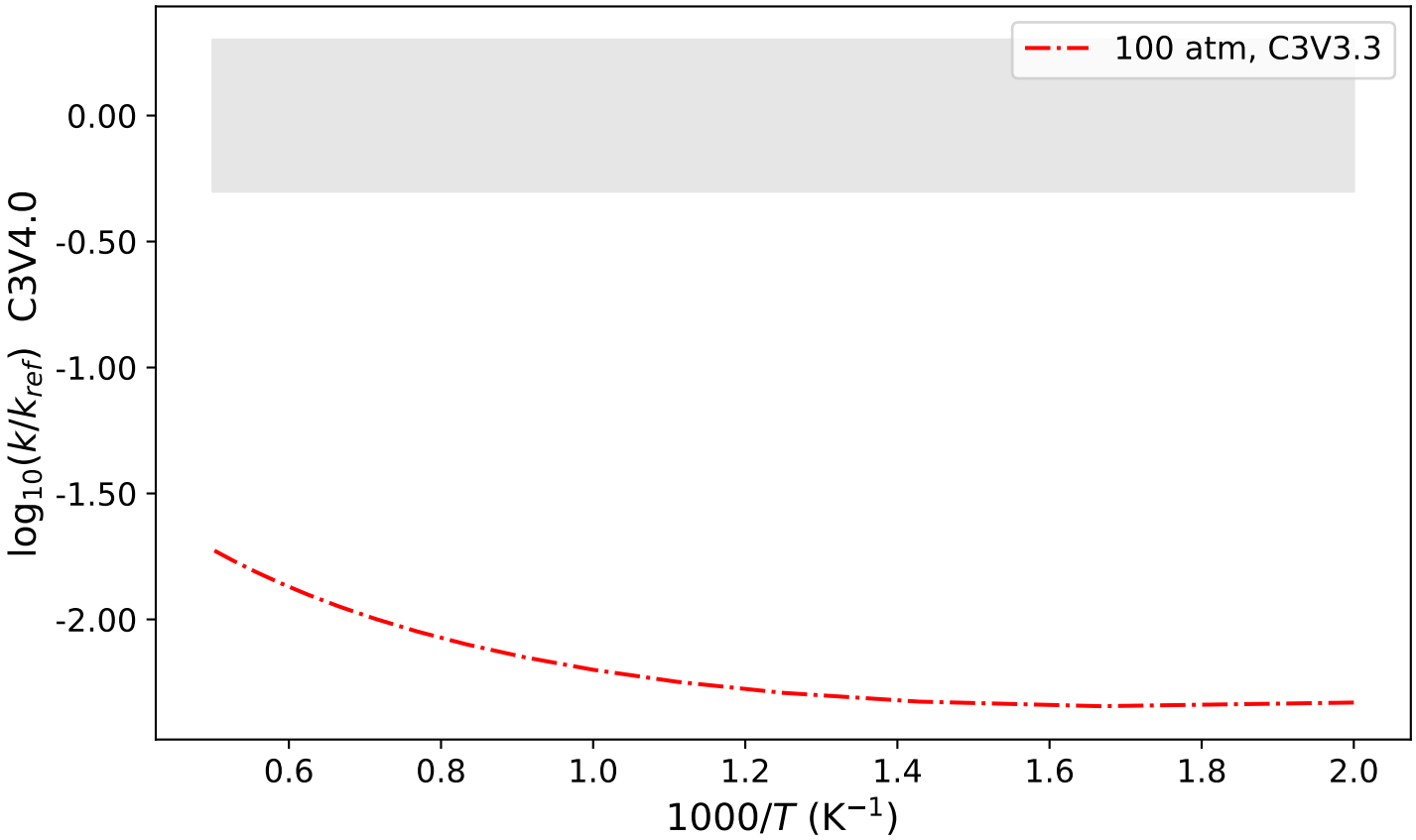
OH + C4H72-1OH = H2O + C4H63,1-1OH



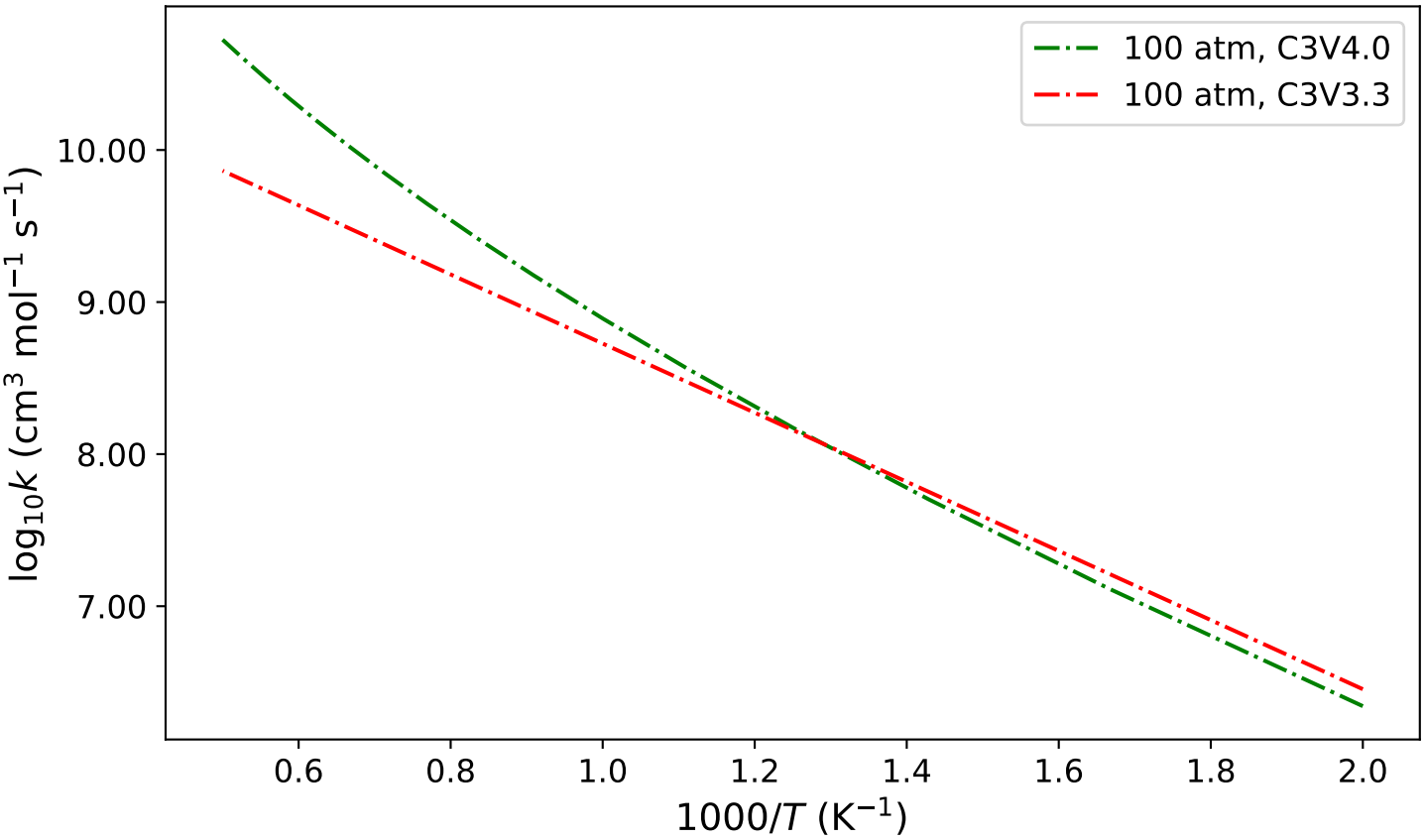



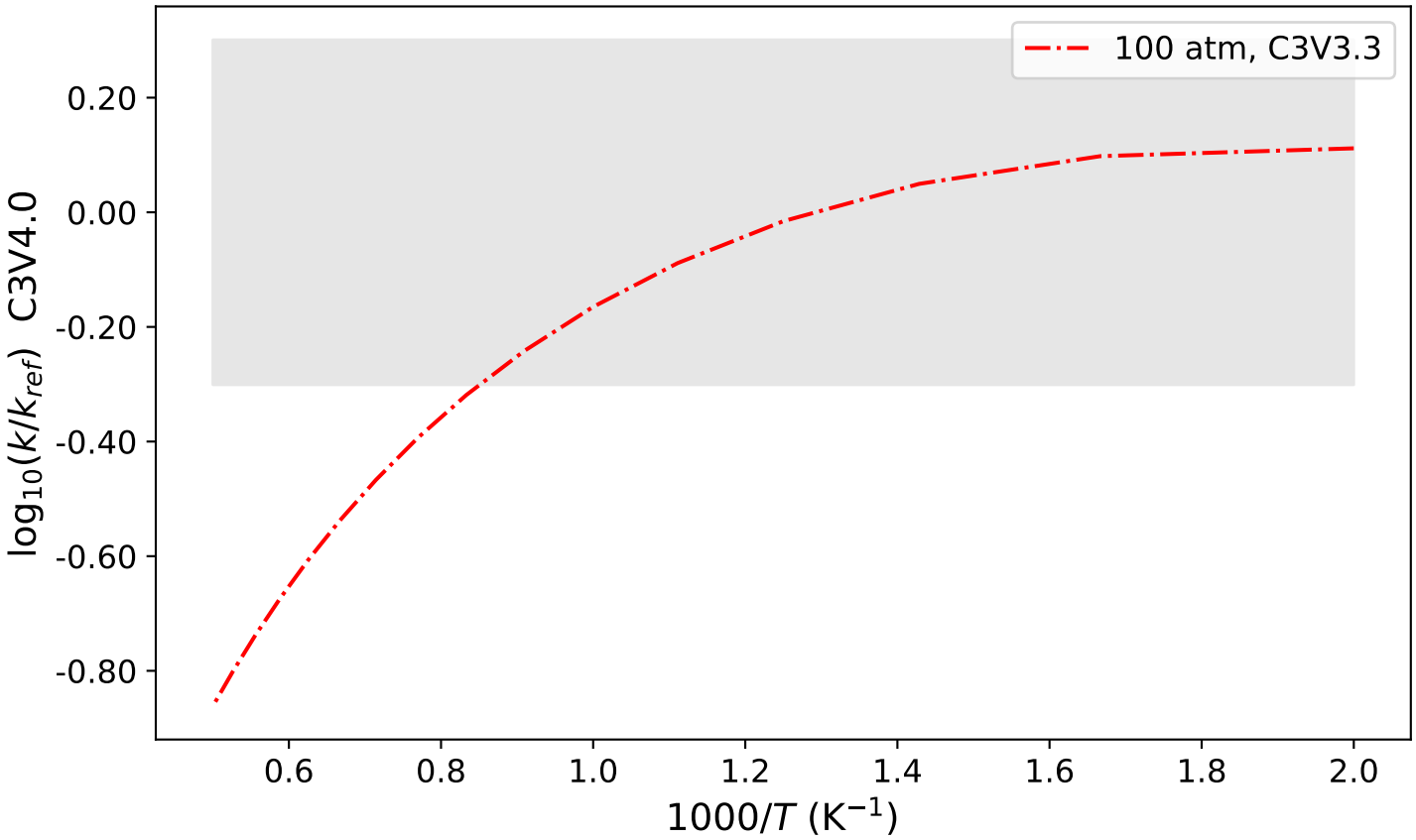
C4H6O1-3OOH4 => CH2O + OH + C2H3CHO



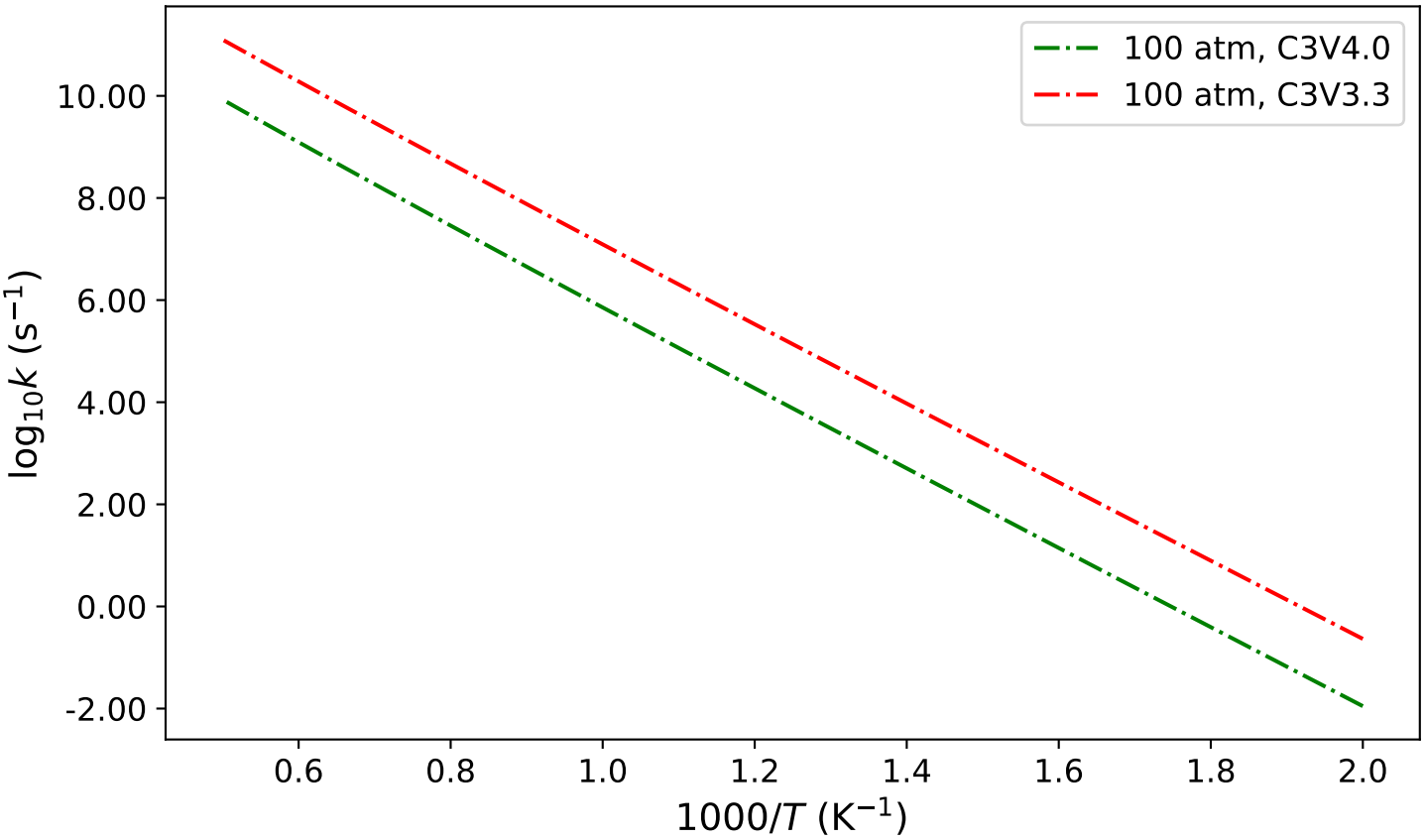



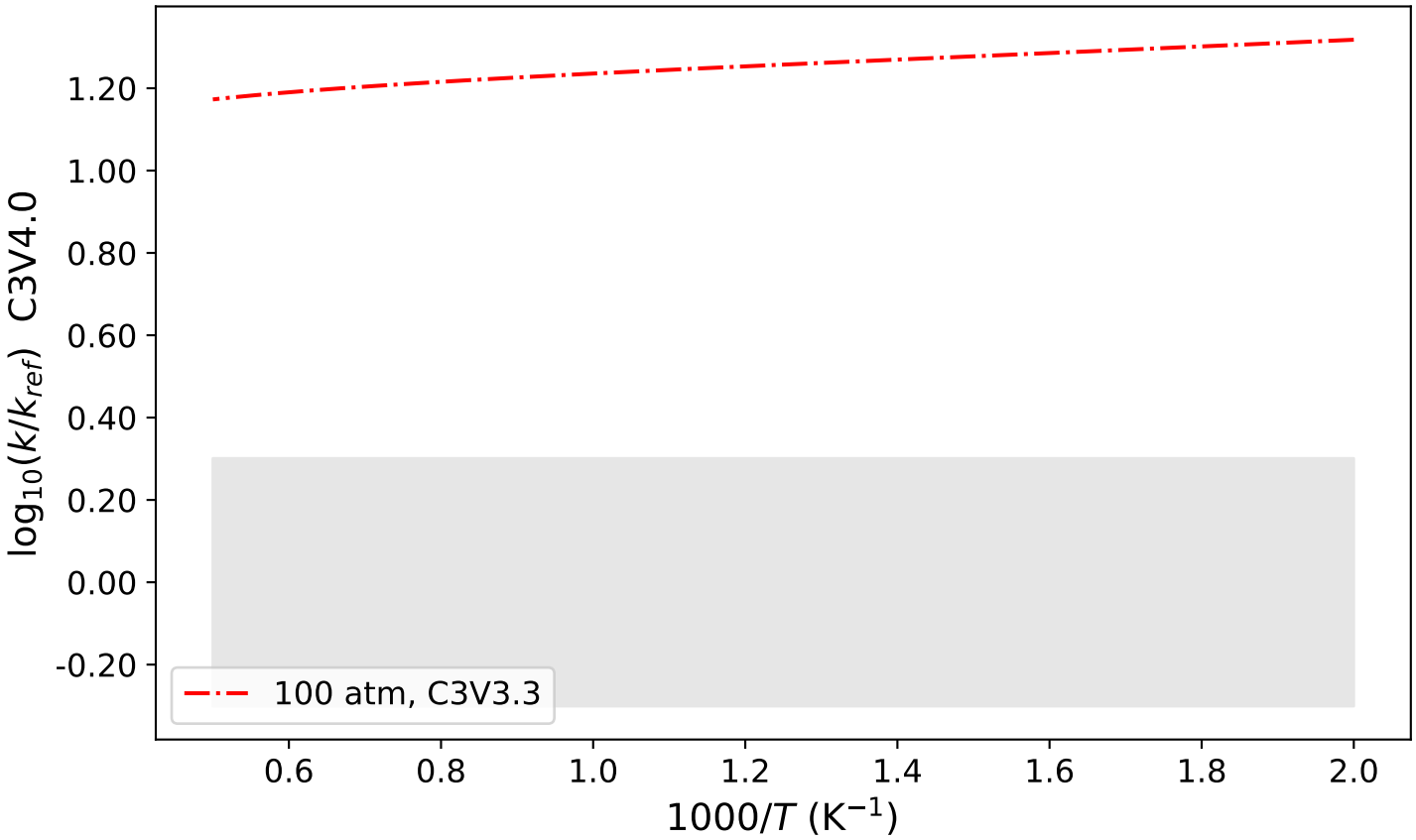
C2H5 + C4H10 = C2H6 + SC4H9



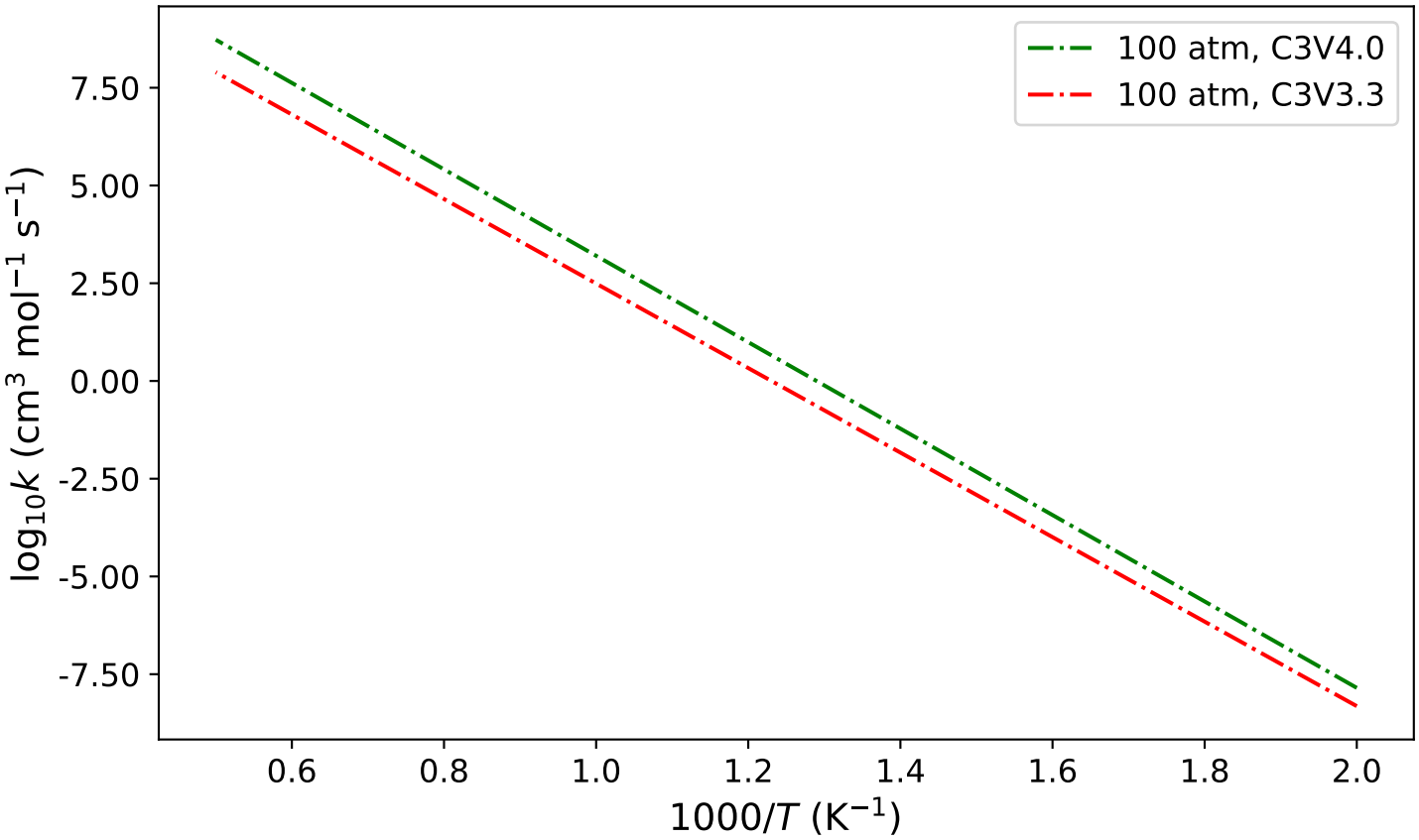



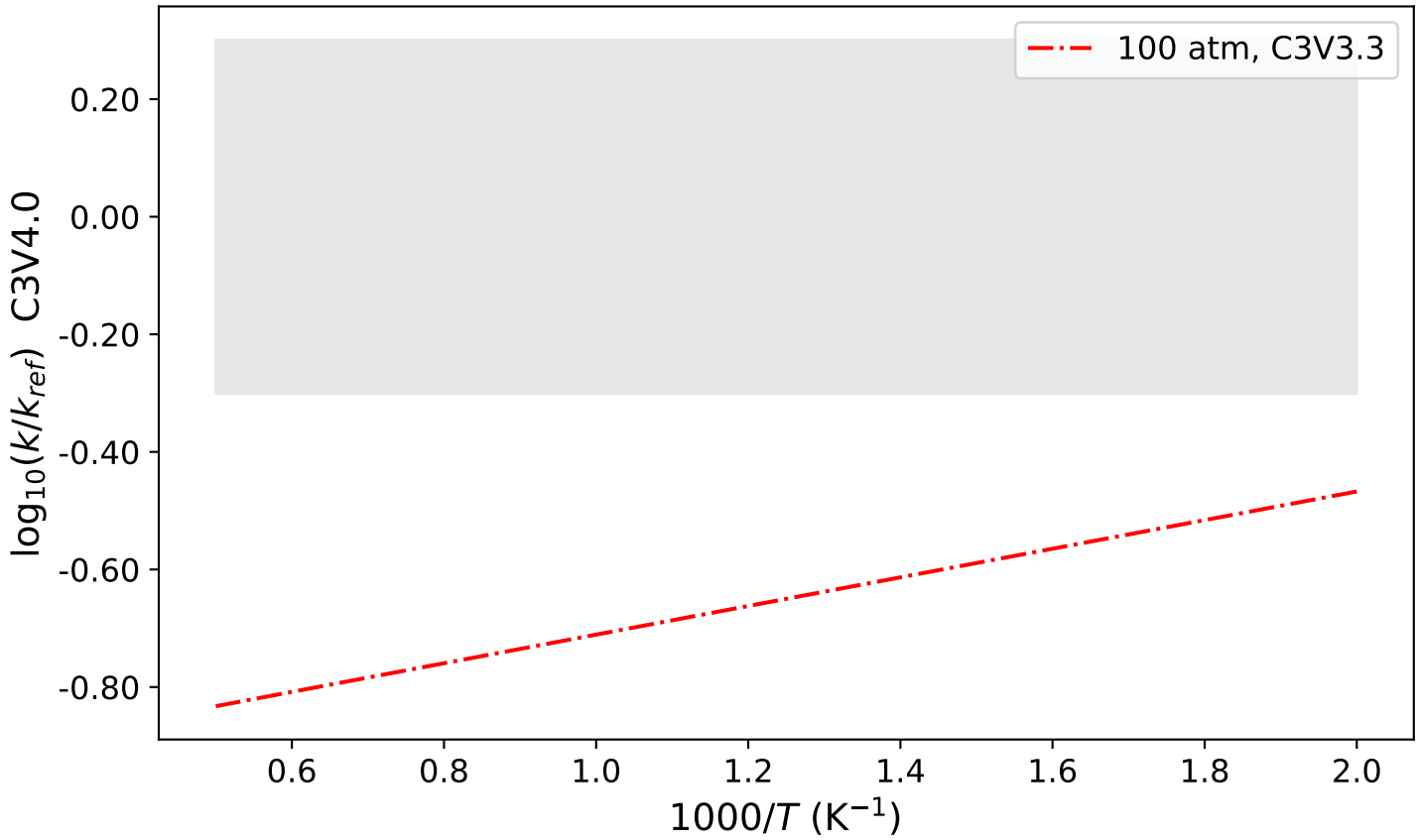
PC4H8OH-1 = H + NC3H7CHO



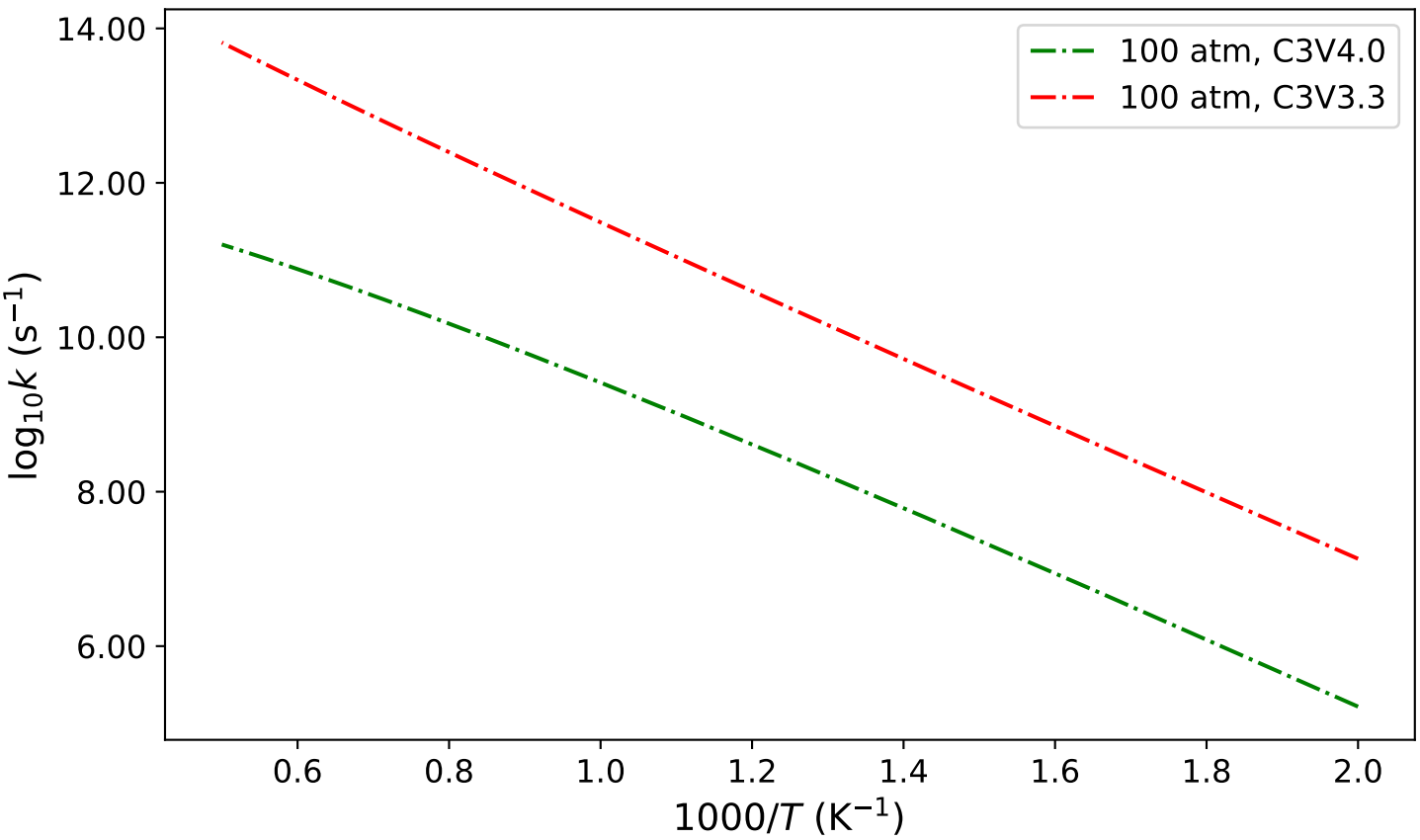



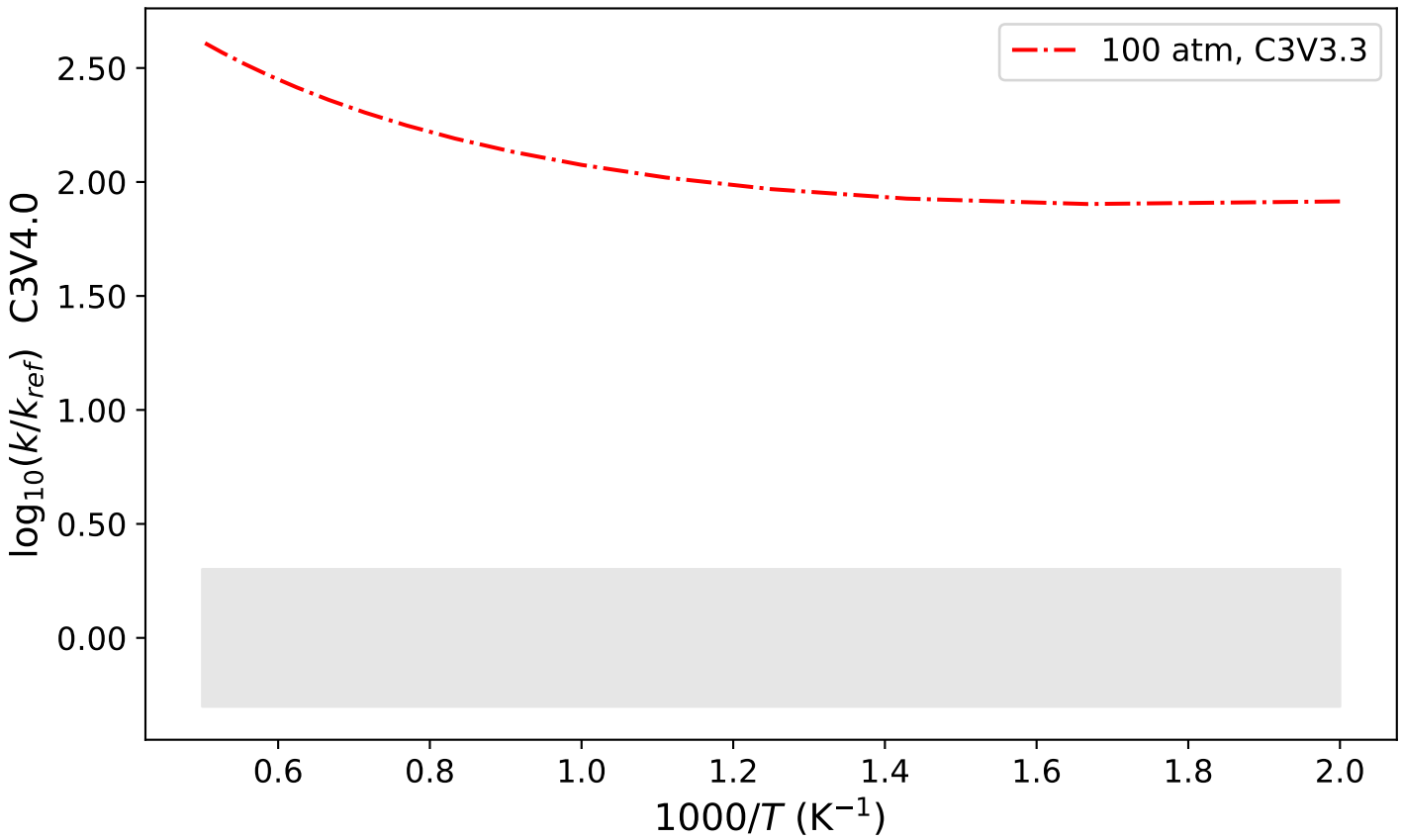
O2 + NC3H7OH = HO2 + C3H6OH1-1



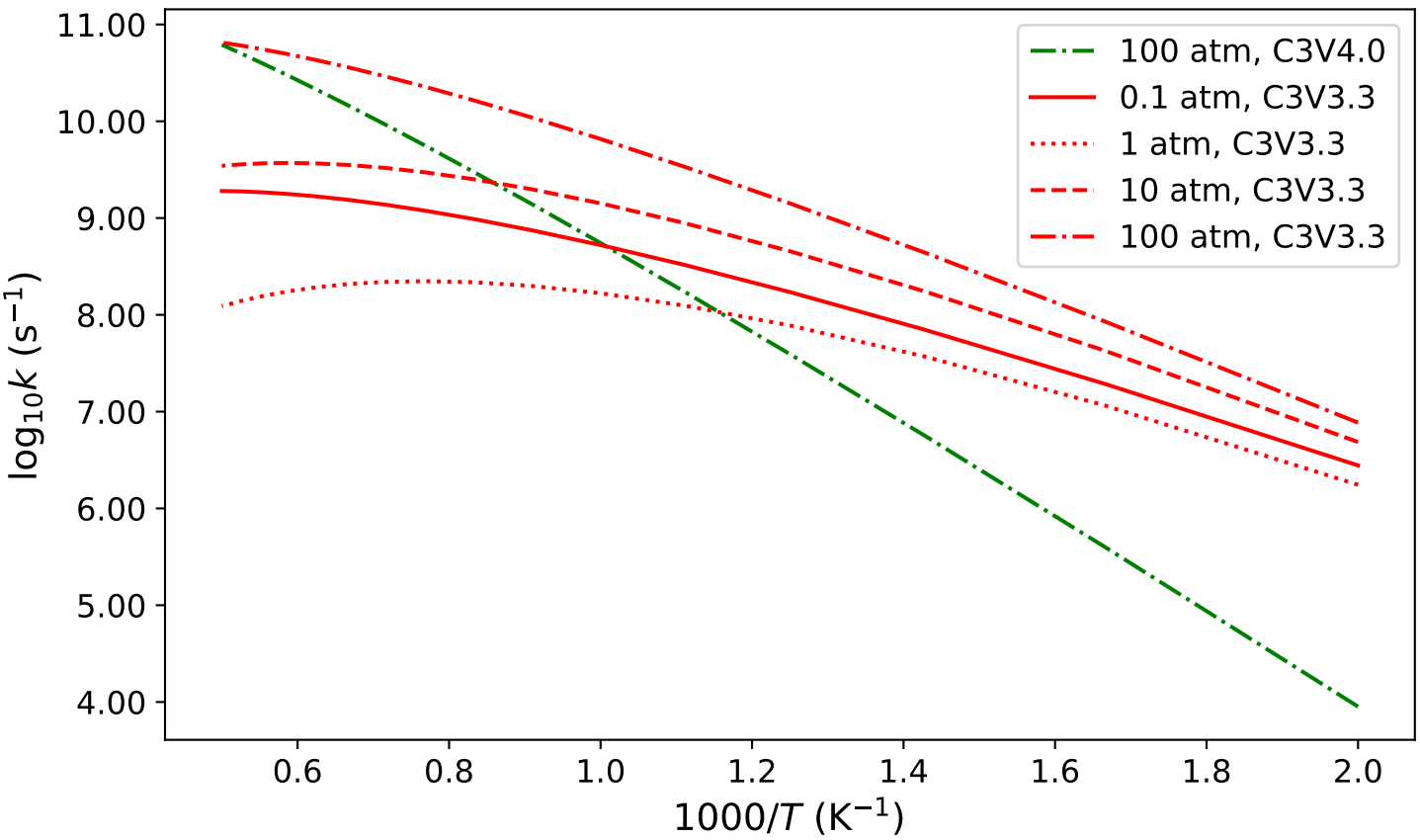



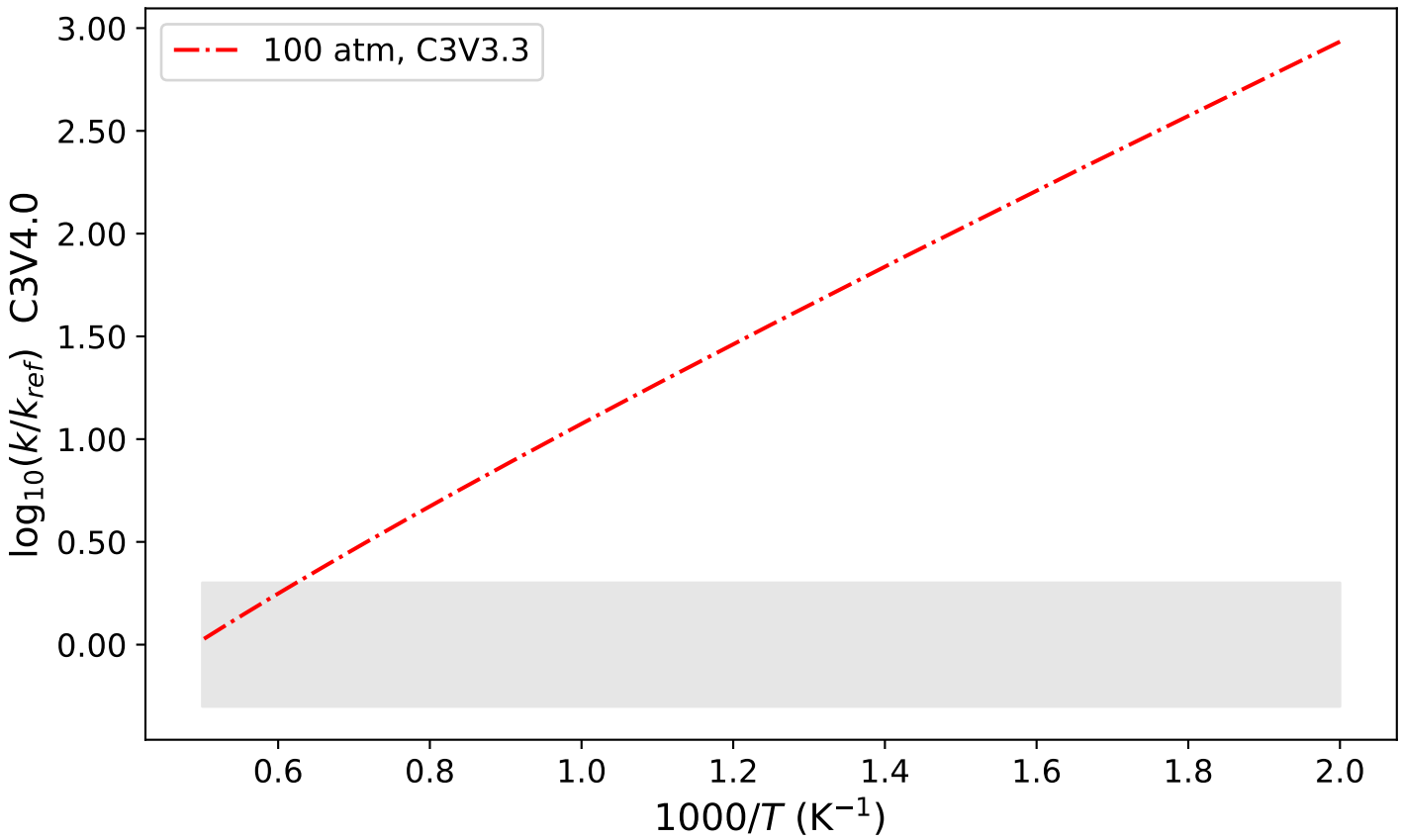
C3H4CH2OH-1O = H + CHOC3H4OH



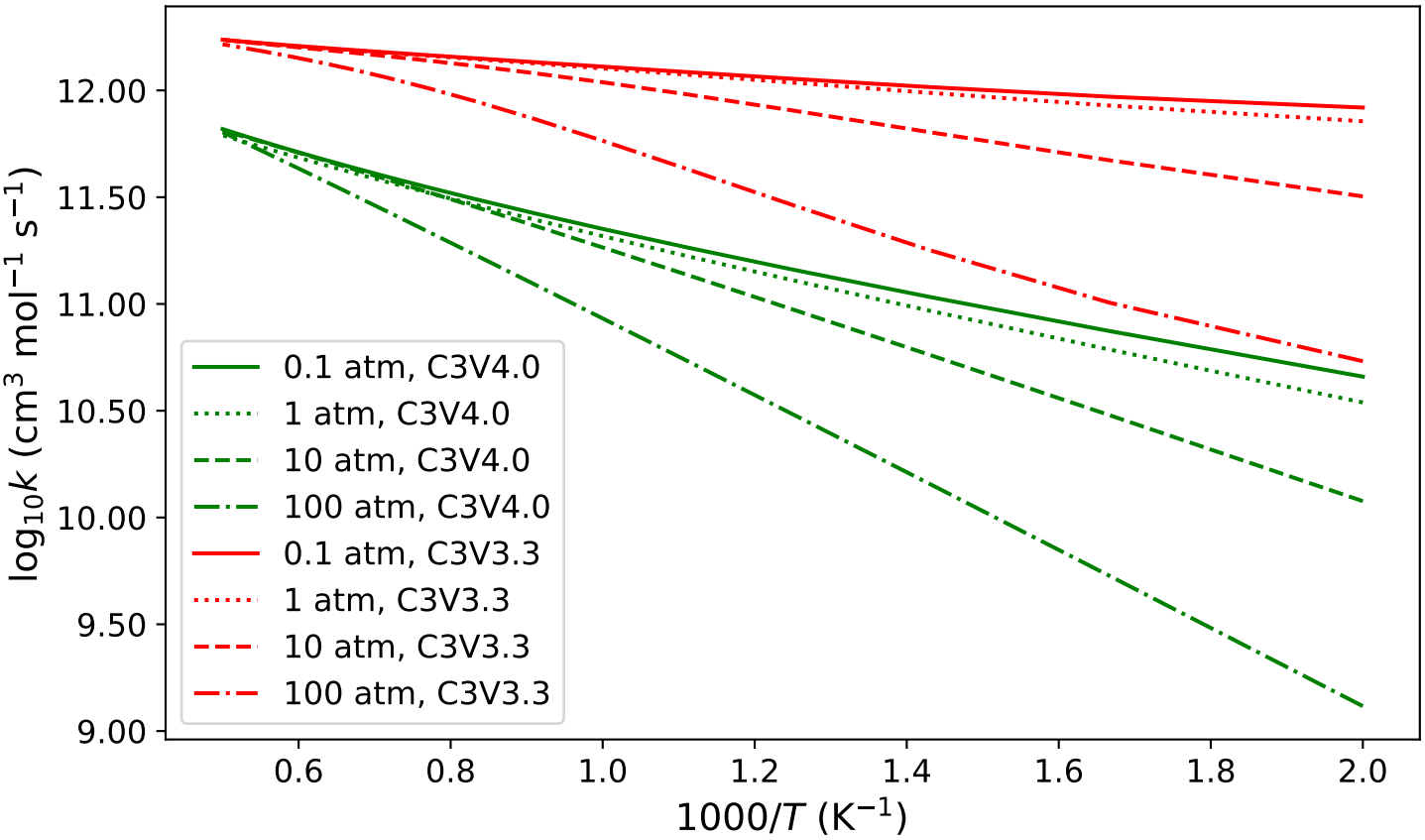



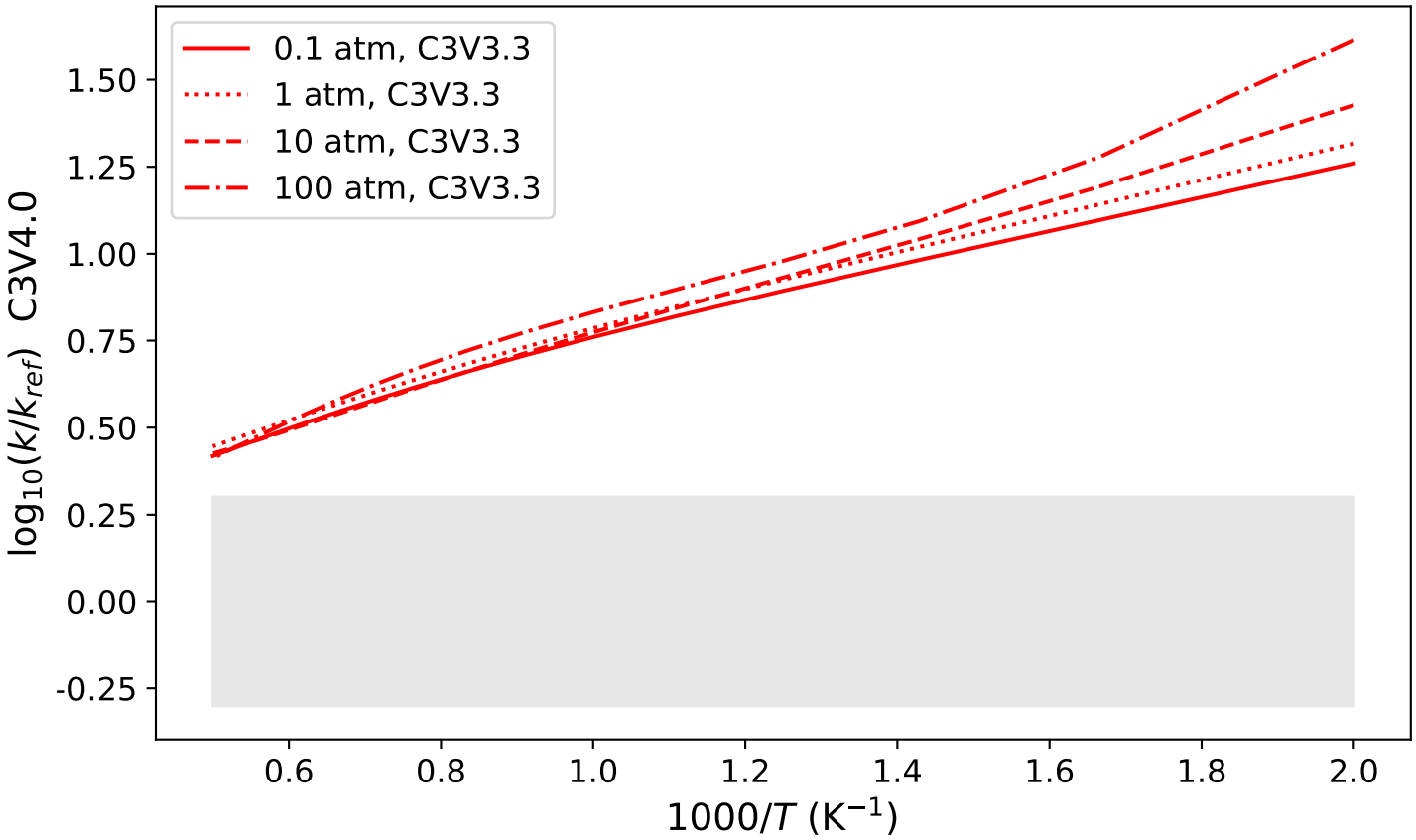
C4H7O1-4 = CH2O + C3H5-A



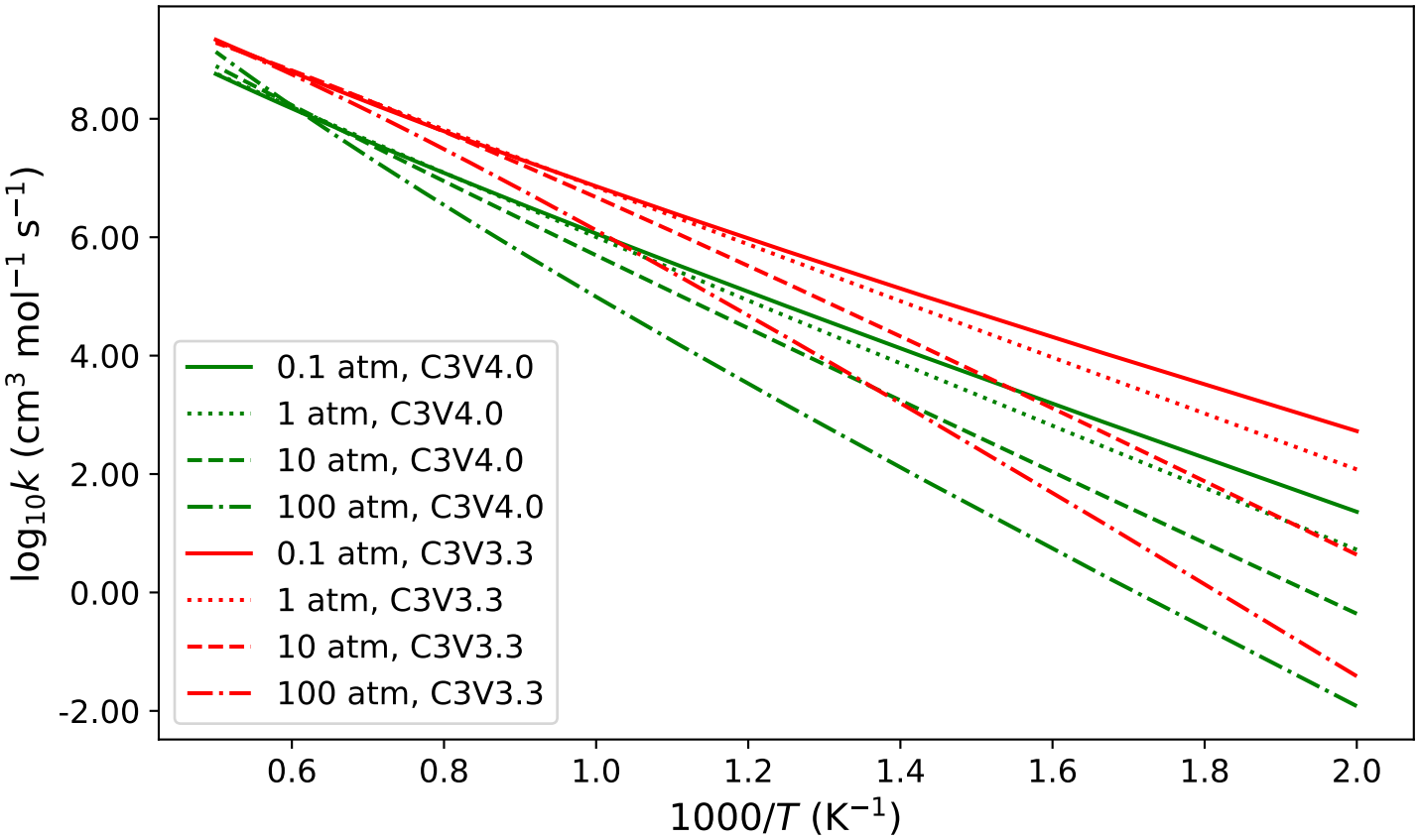



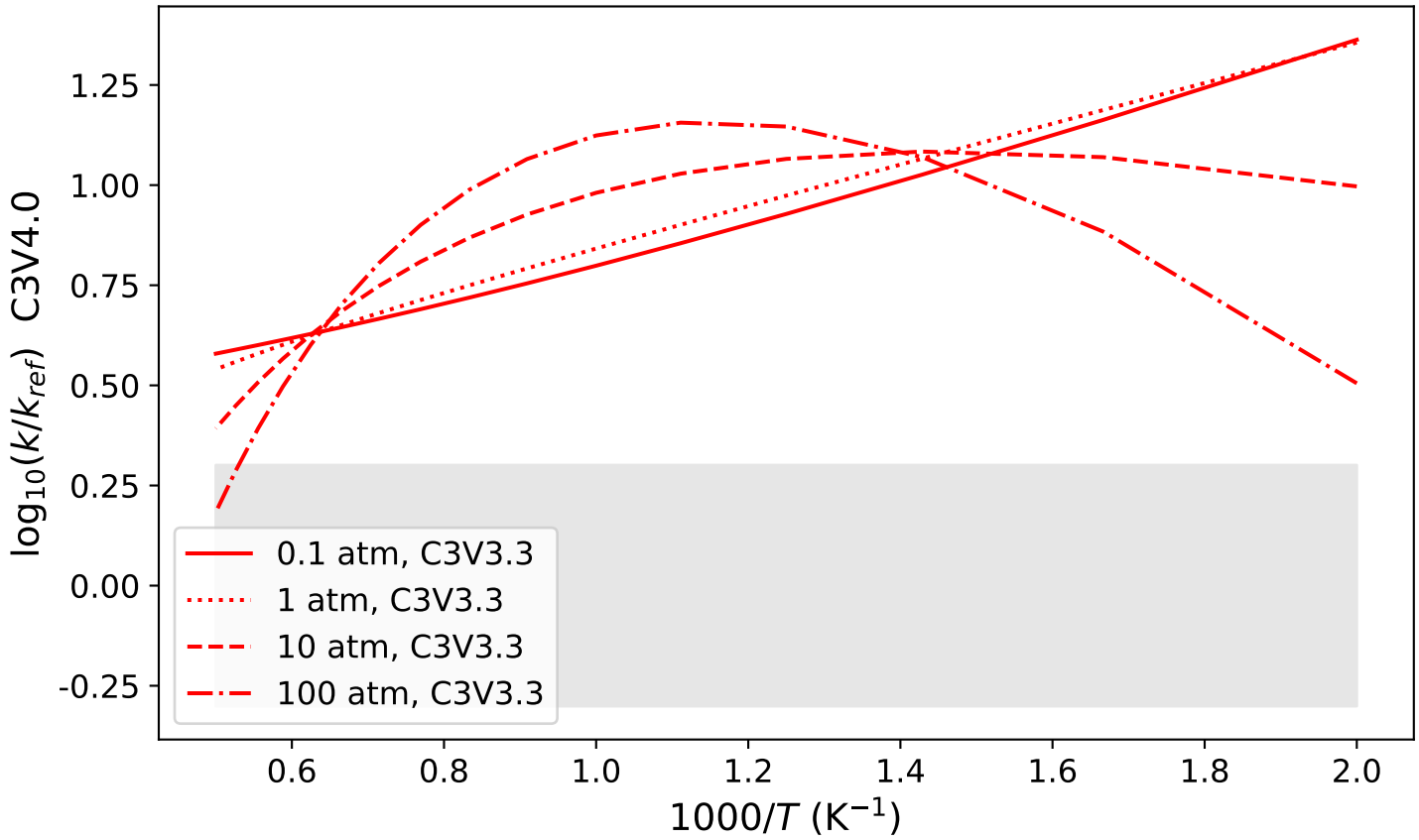
O2 + C4H71-1 = O + C3H6CHO-1

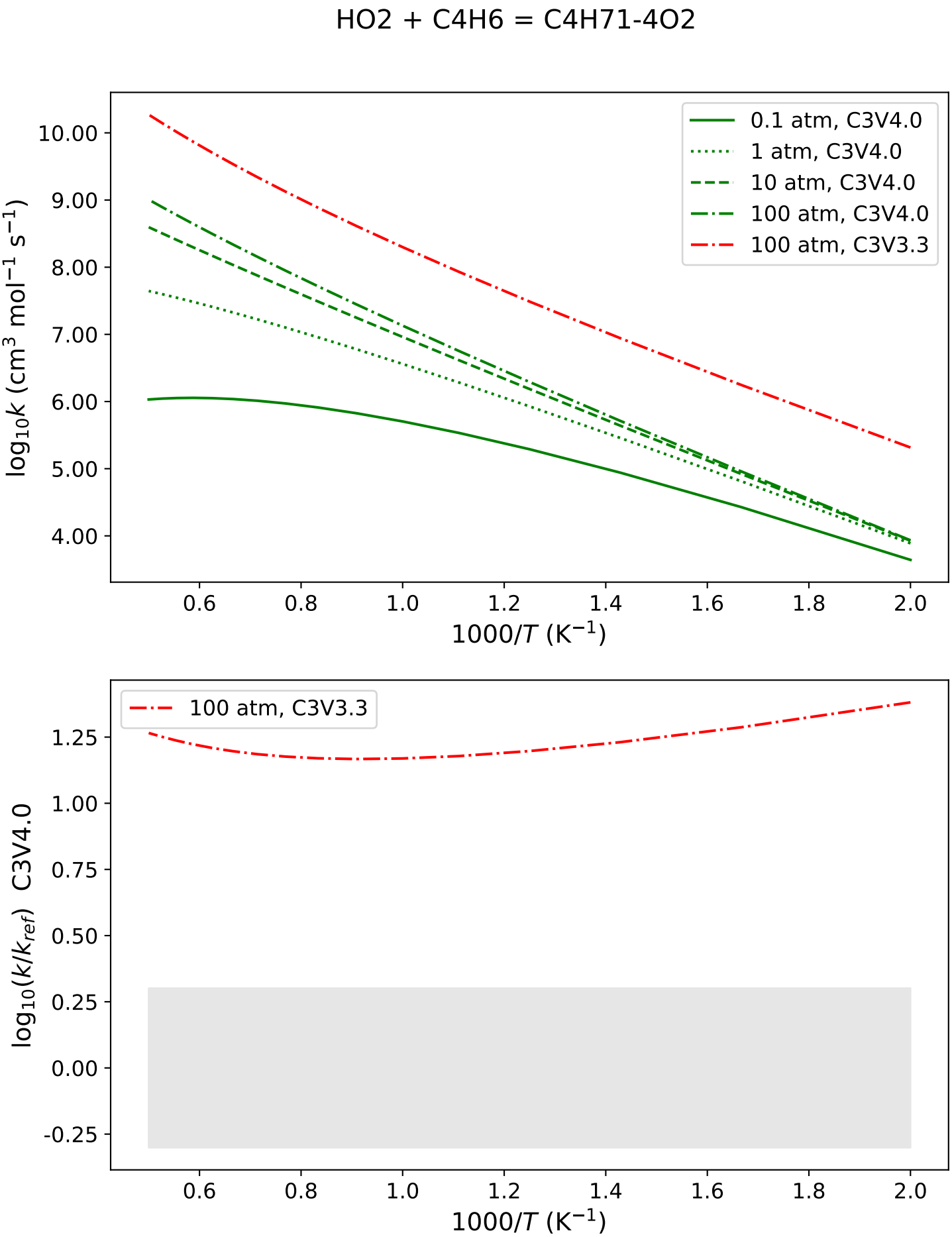


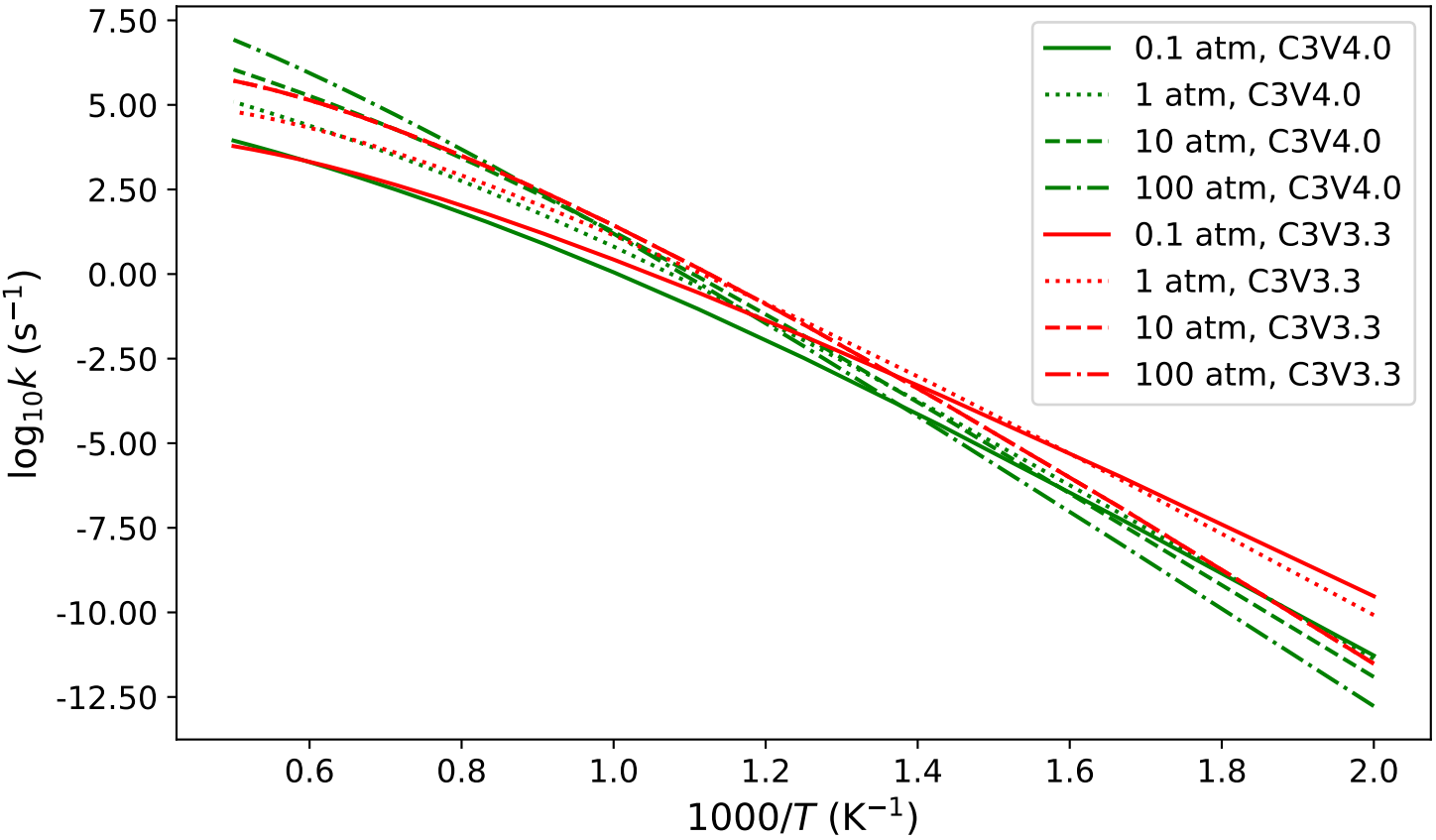
C2H5 + C2H4 = H + C4H8-2

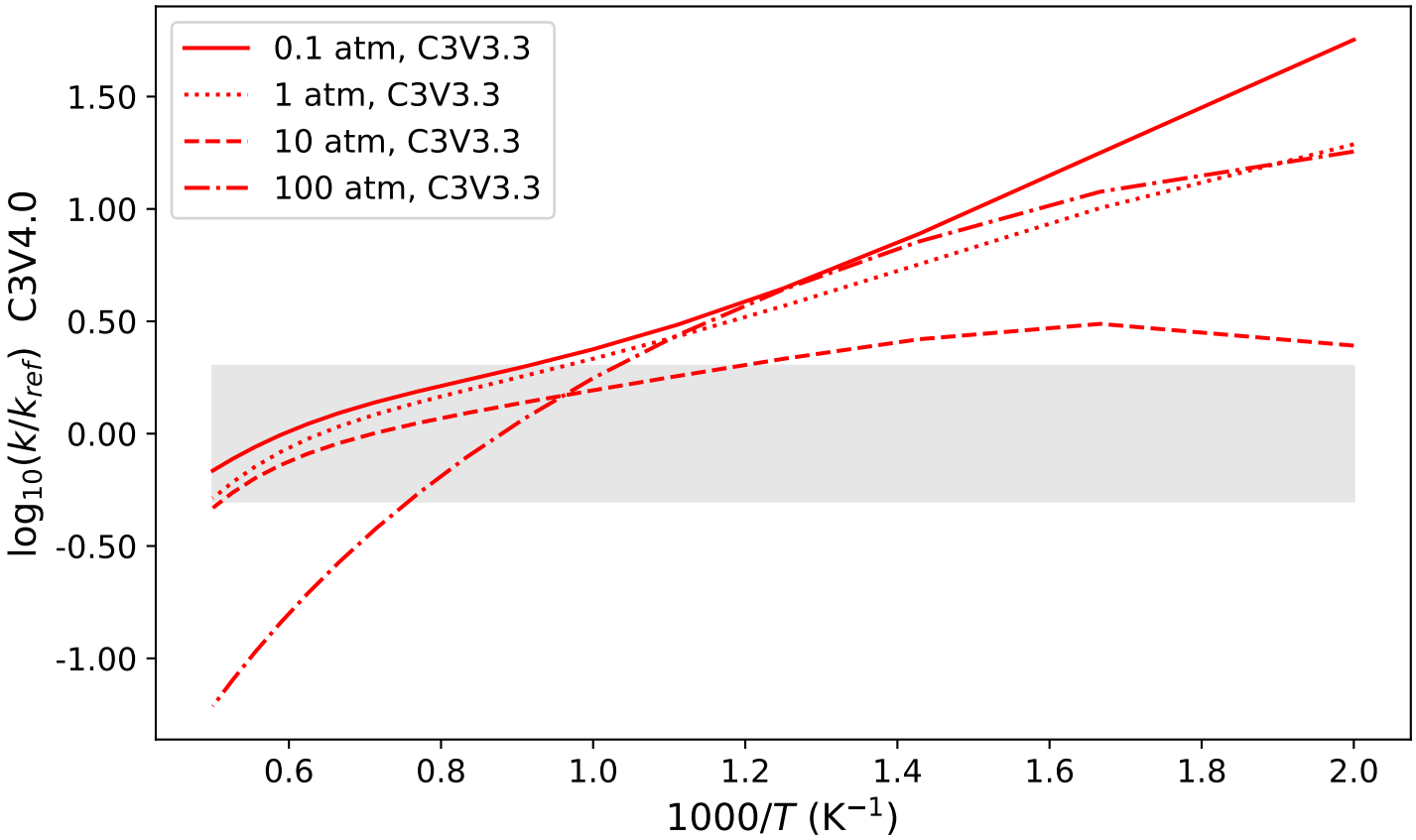


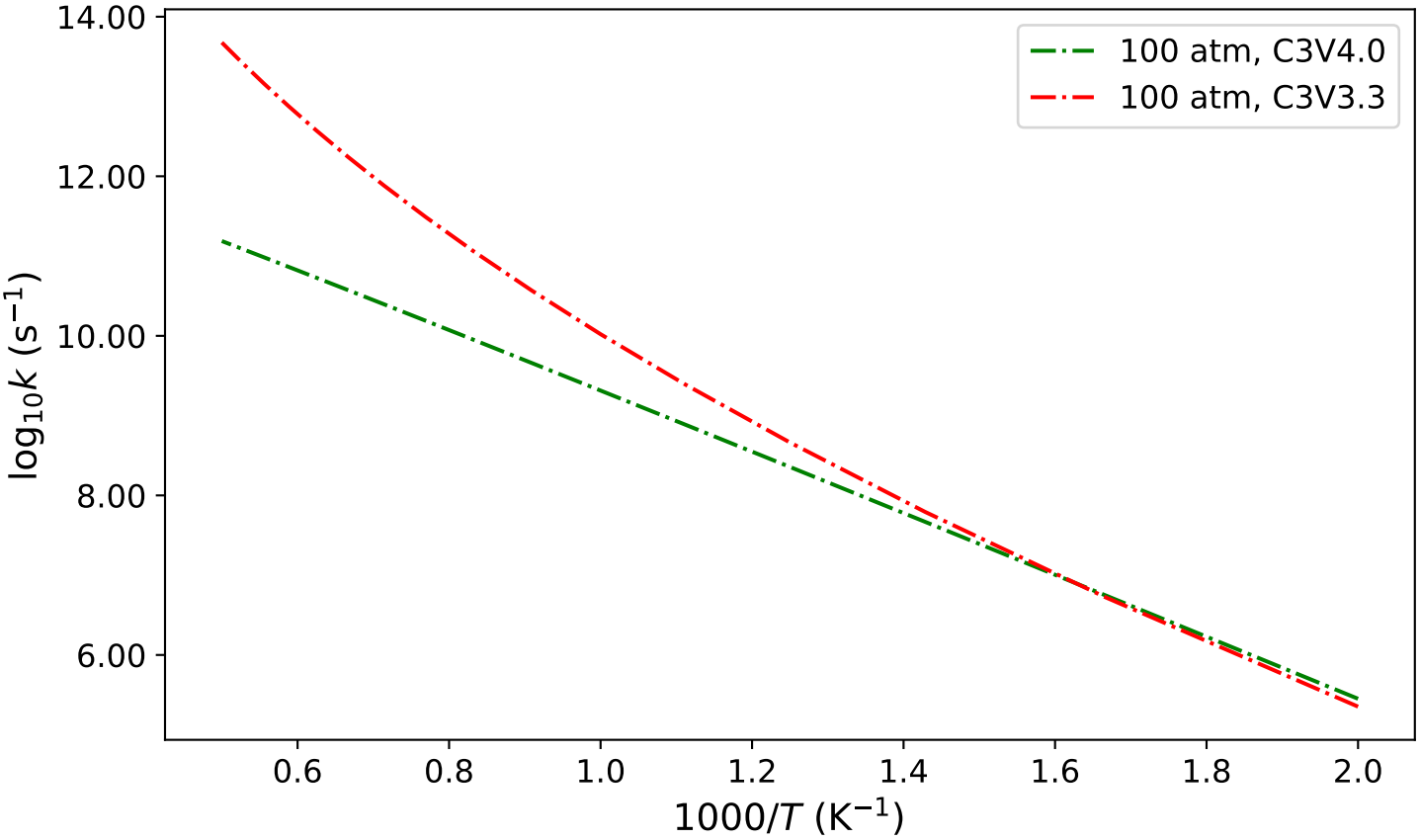

HO2 + C4H6 = C4H71-4O2
0.1 atm, C3V4.0
1 atm, C3V4.0
10 atm, C3V4.0
100 atm, C3V4.0
100 atm, C3V3.3
log10k (cm3 mol−1 s−1)
10.00
9.00
8.00
7.00
6.00
5.00
4.00
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/T (K−1)
100 atm, C3V3.3
log10(k/kref) C3V4.0
1.25
1.00
0.75
0.50
0.25
0.00
-0.25
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/T (K−1)

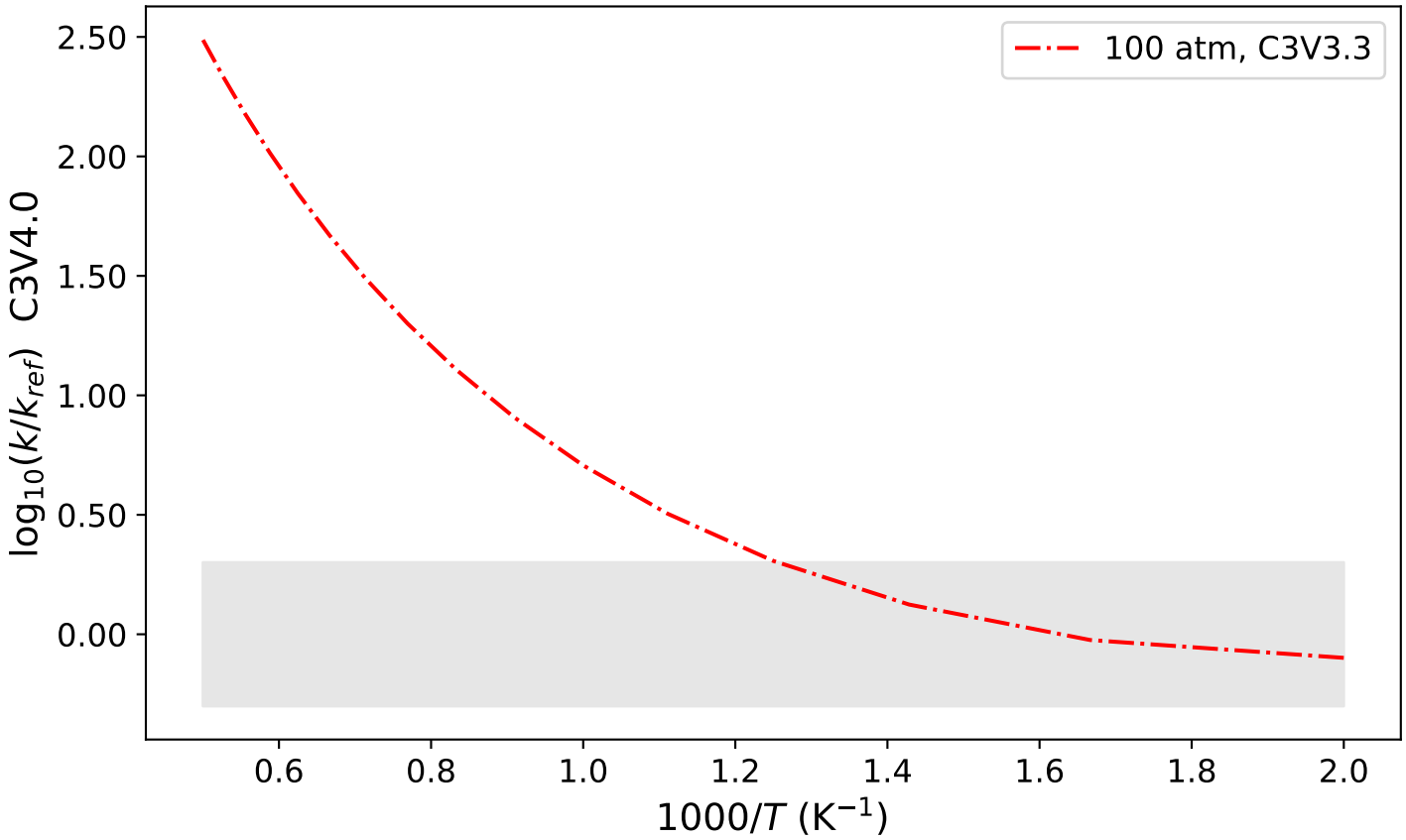
C4H5-I = C2H3 + C2H2



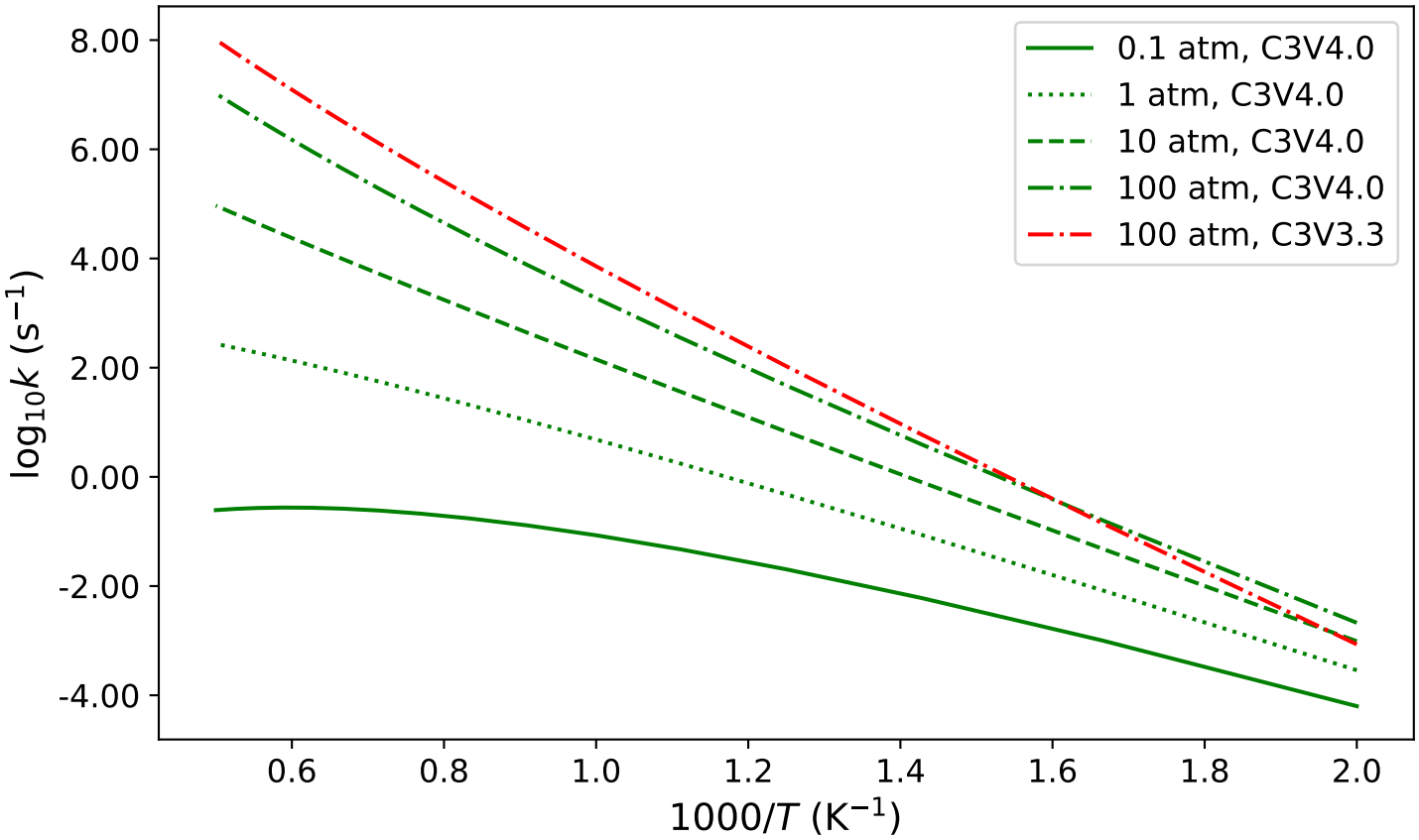



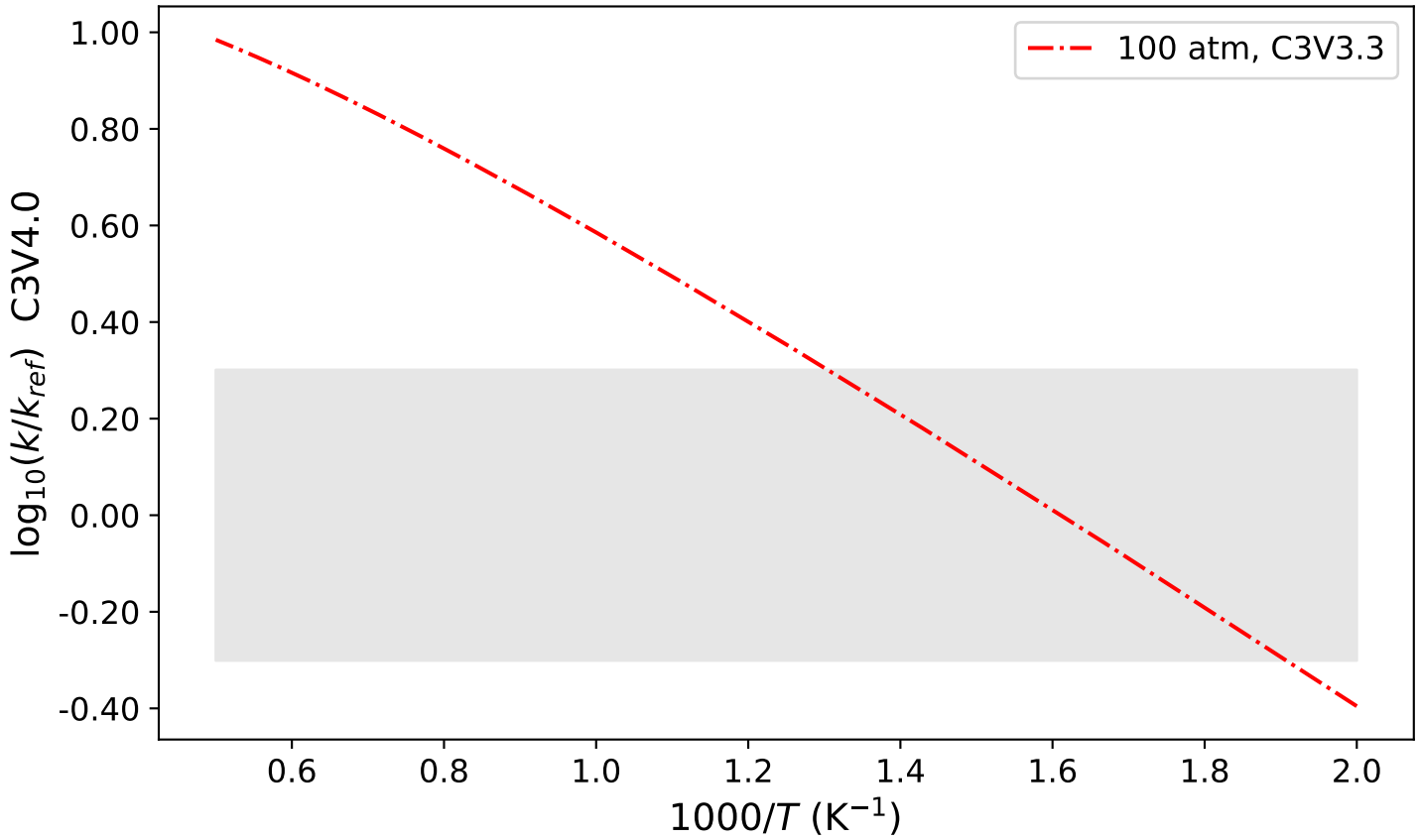
CH2CCH2OOH = HO2 + C3H4-A



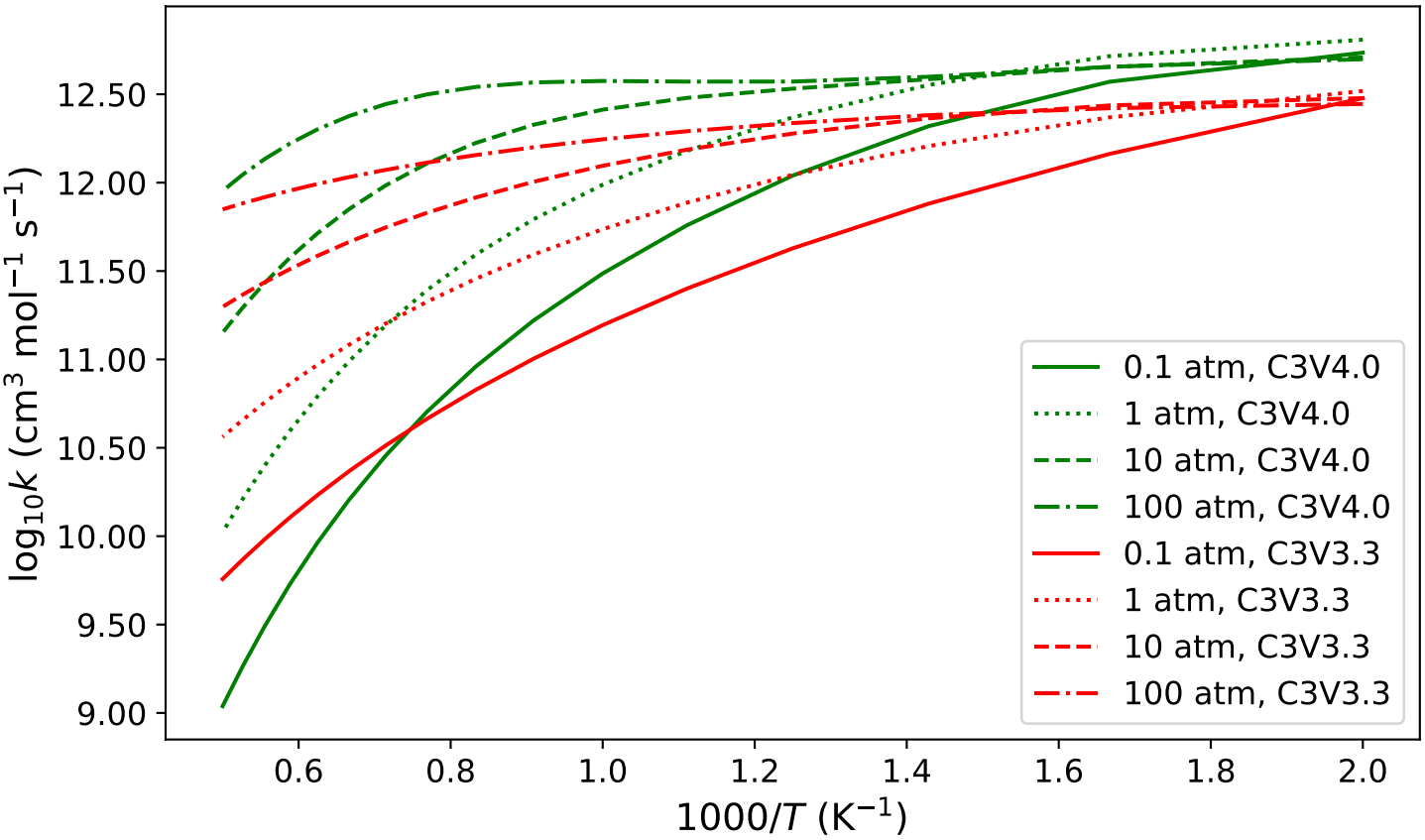



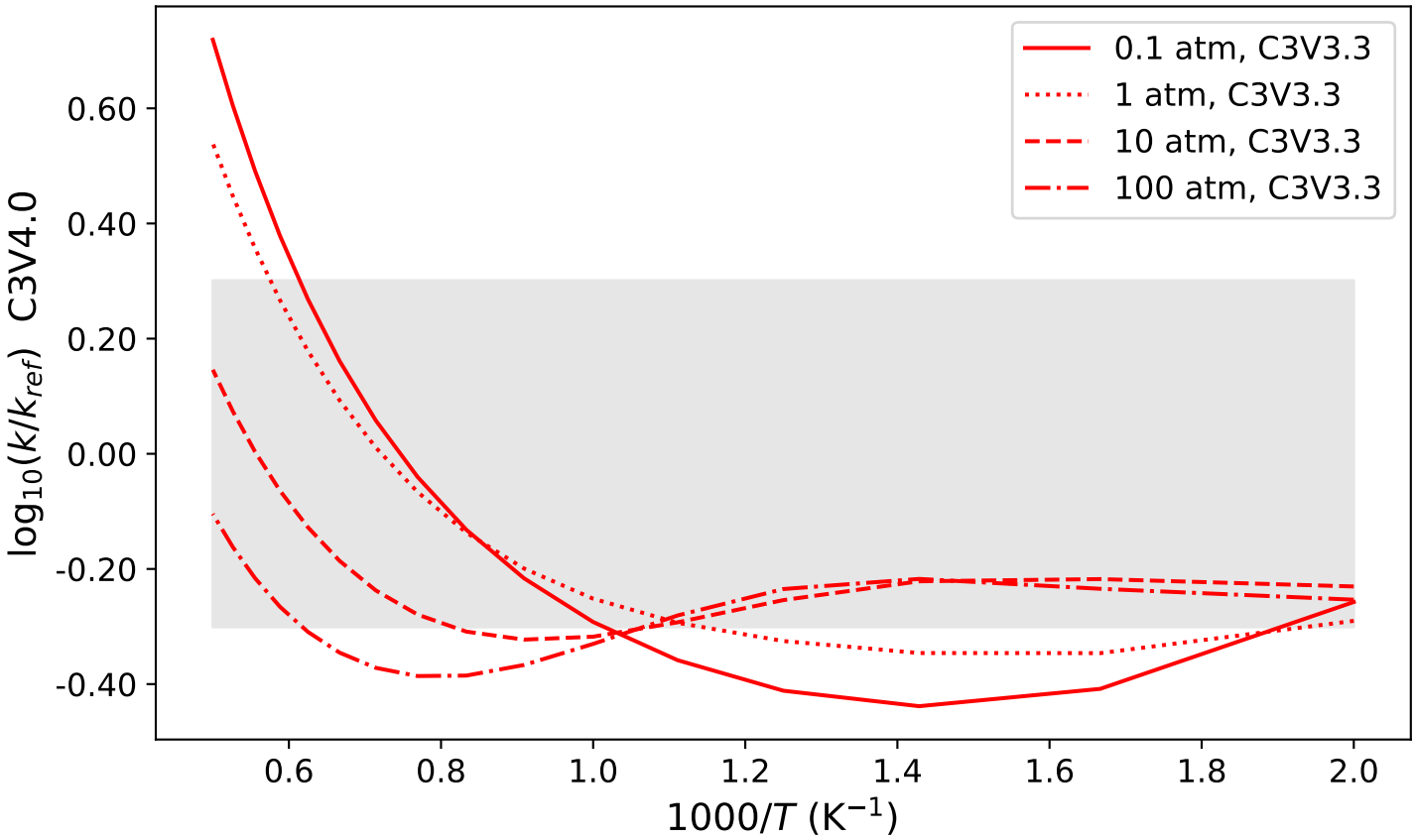
C4H71-3O2 = C4H61OOH3-4



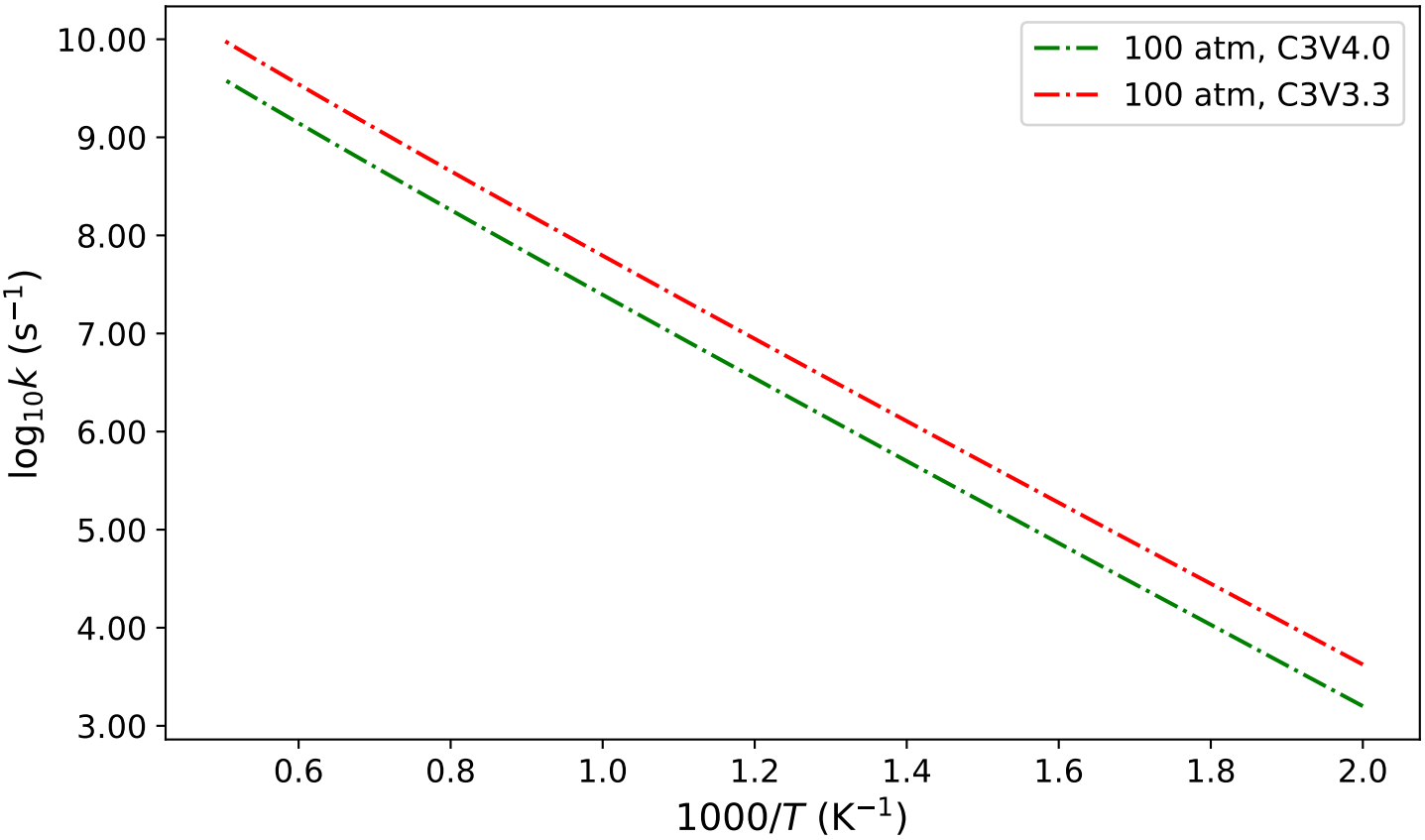



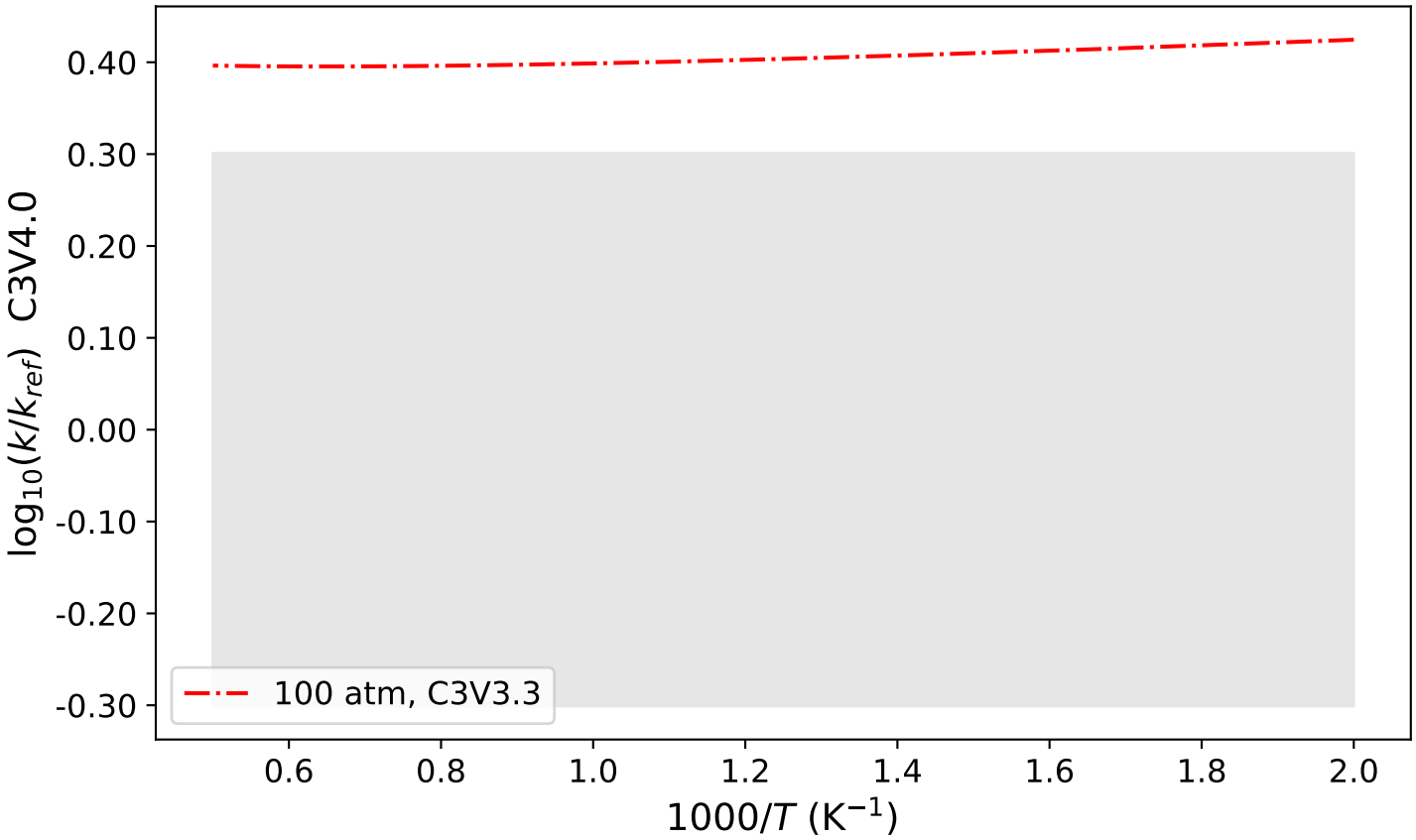
OH + C3H6 = C3H6OH1-2



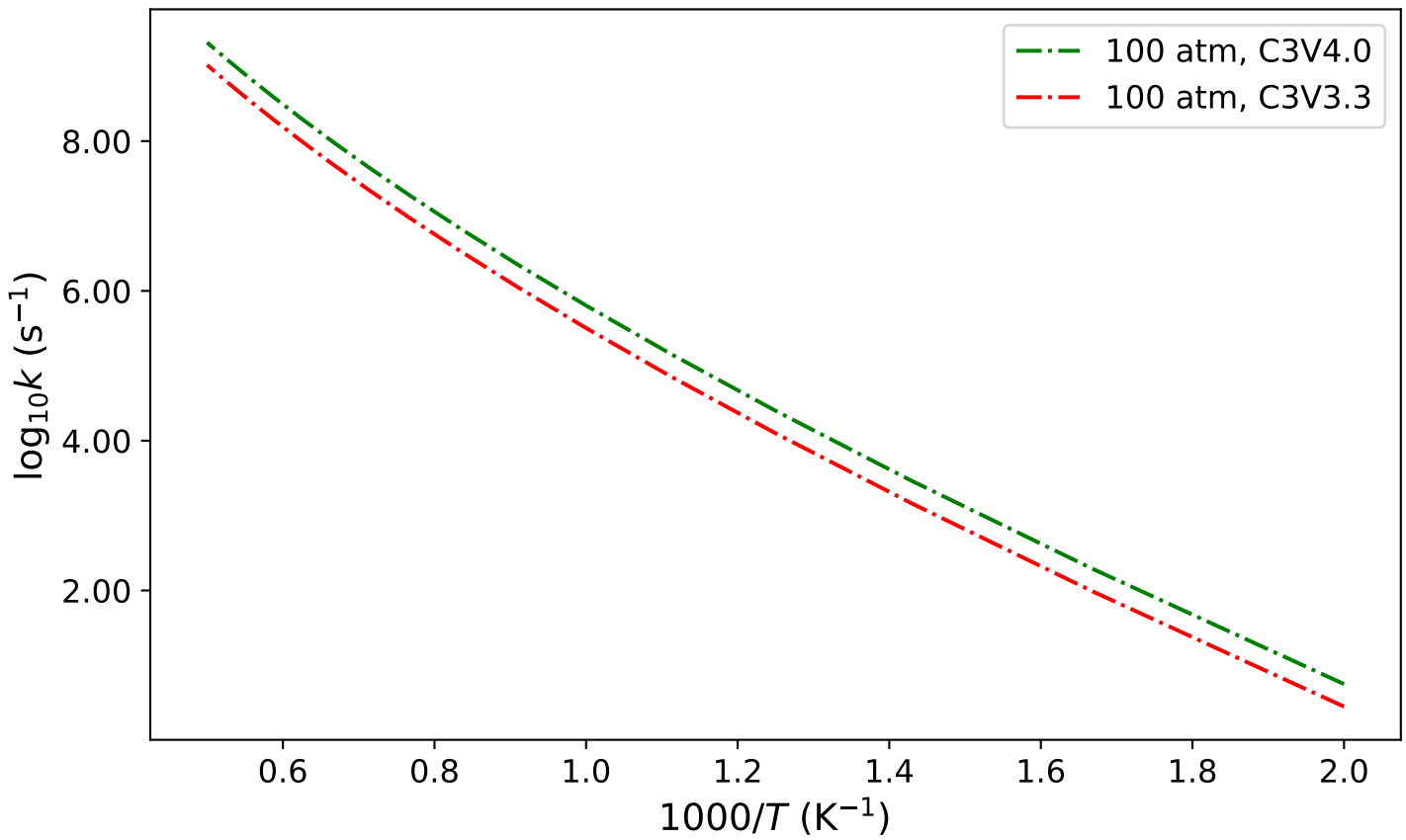



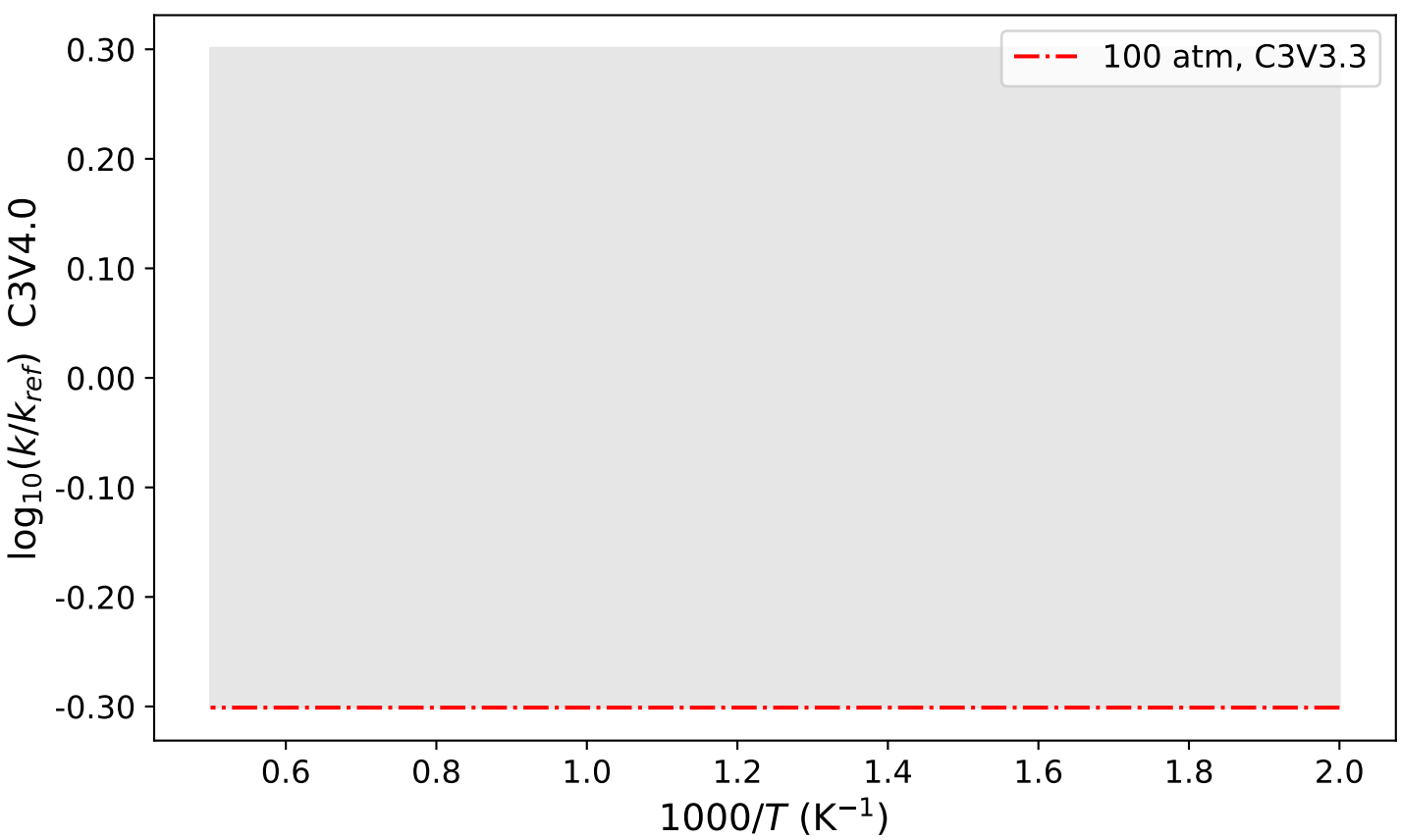
C4H8OOH2-4 = OH + C4H8O1-3



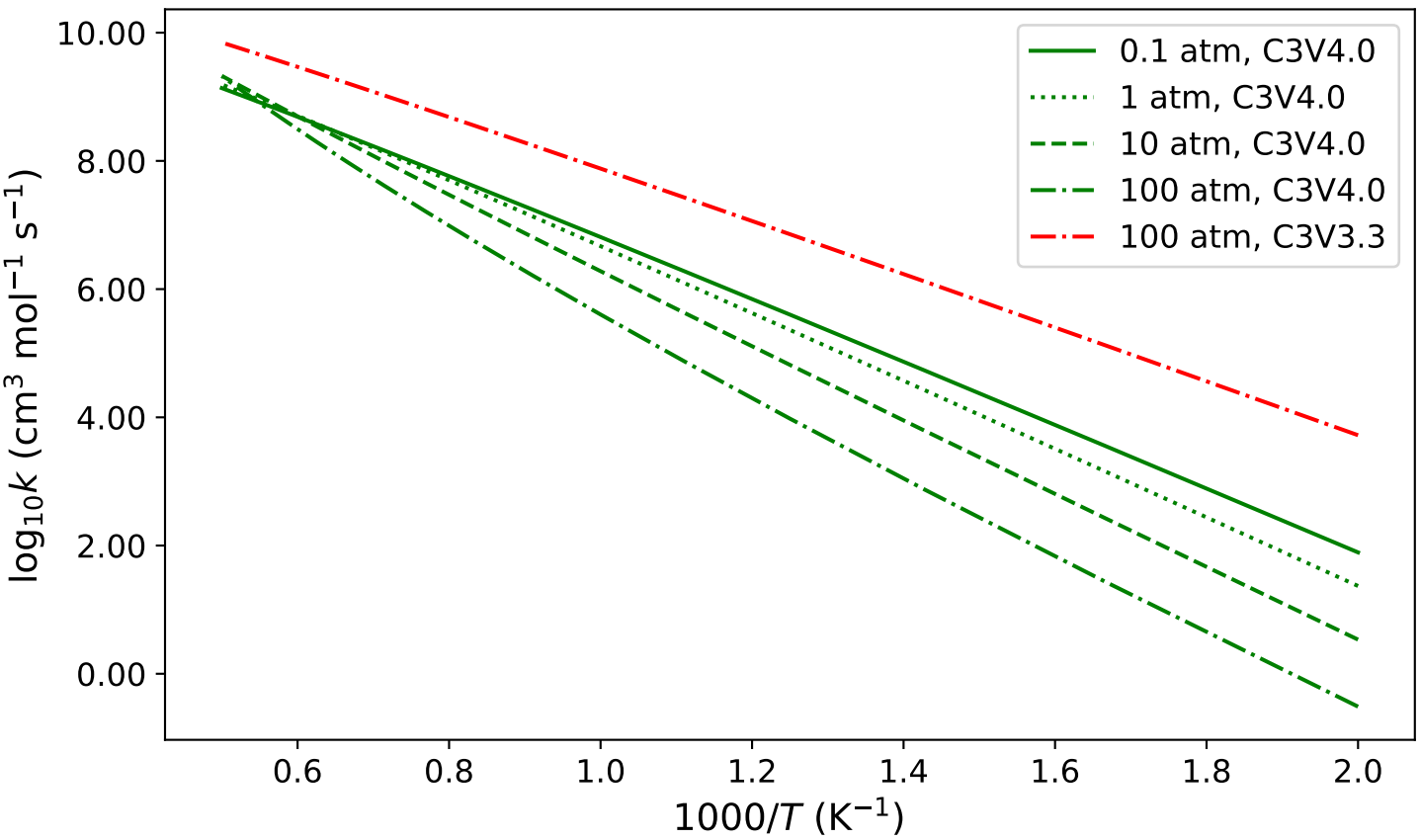



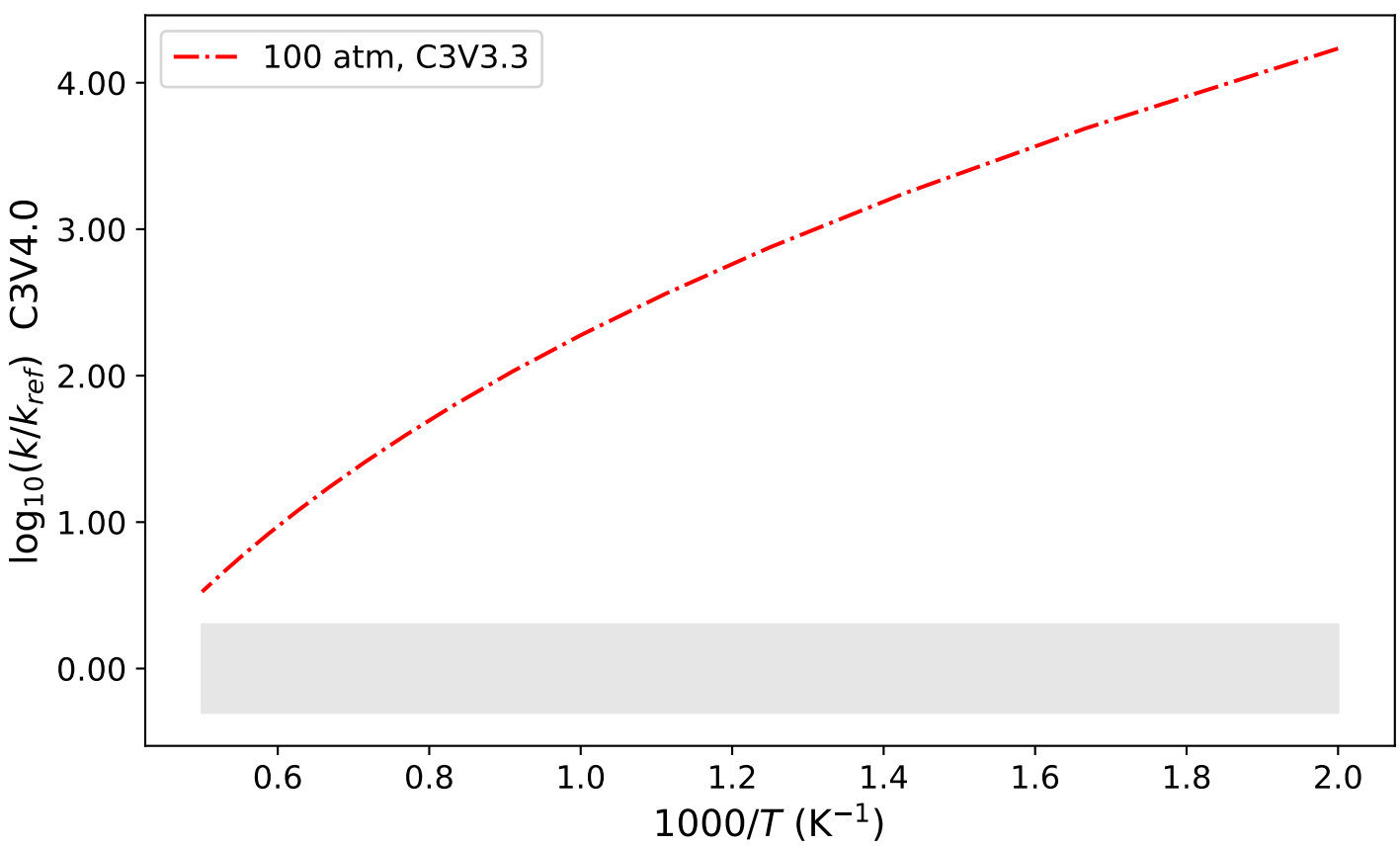
C4H71-4O2 = C4H61-3OOH4



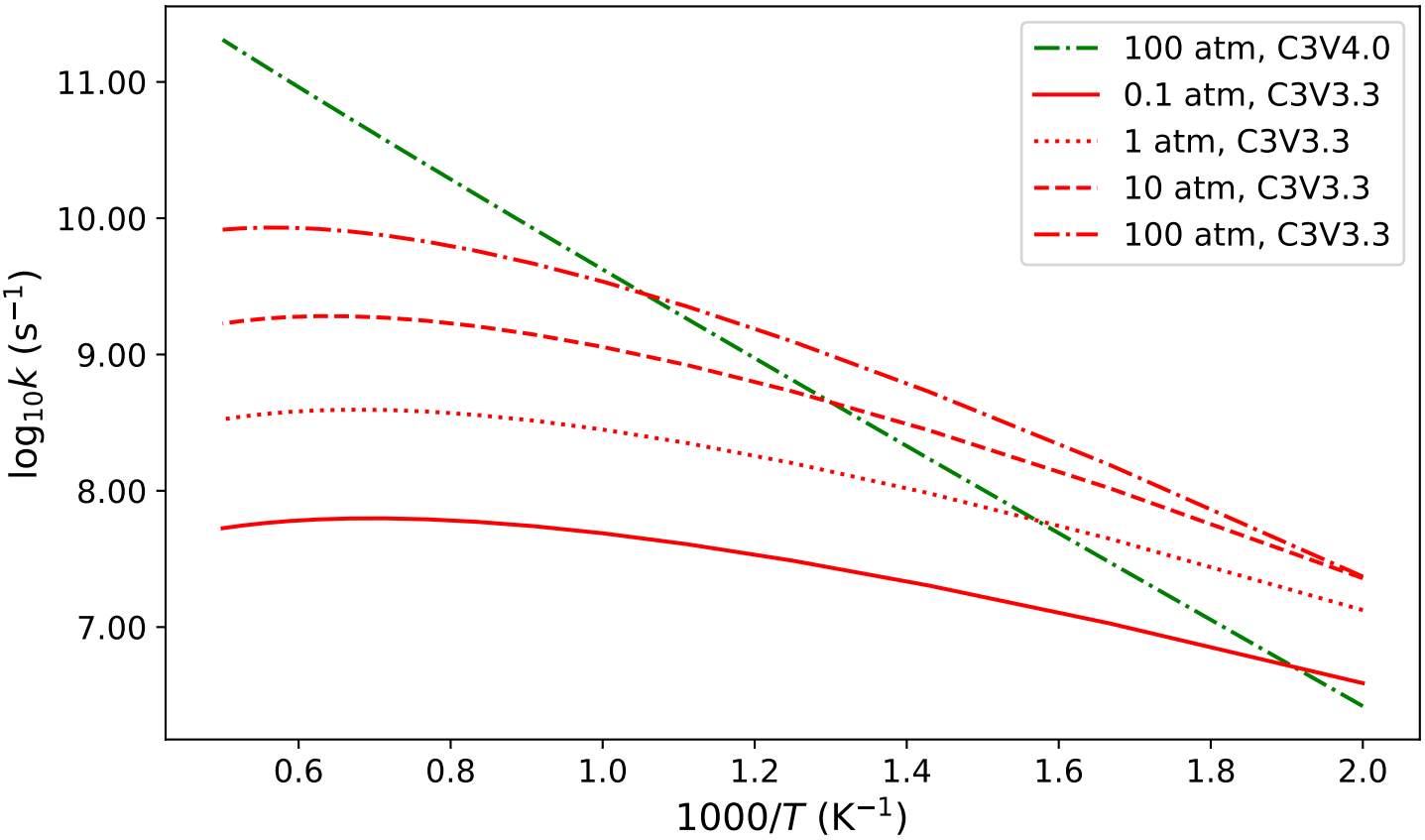



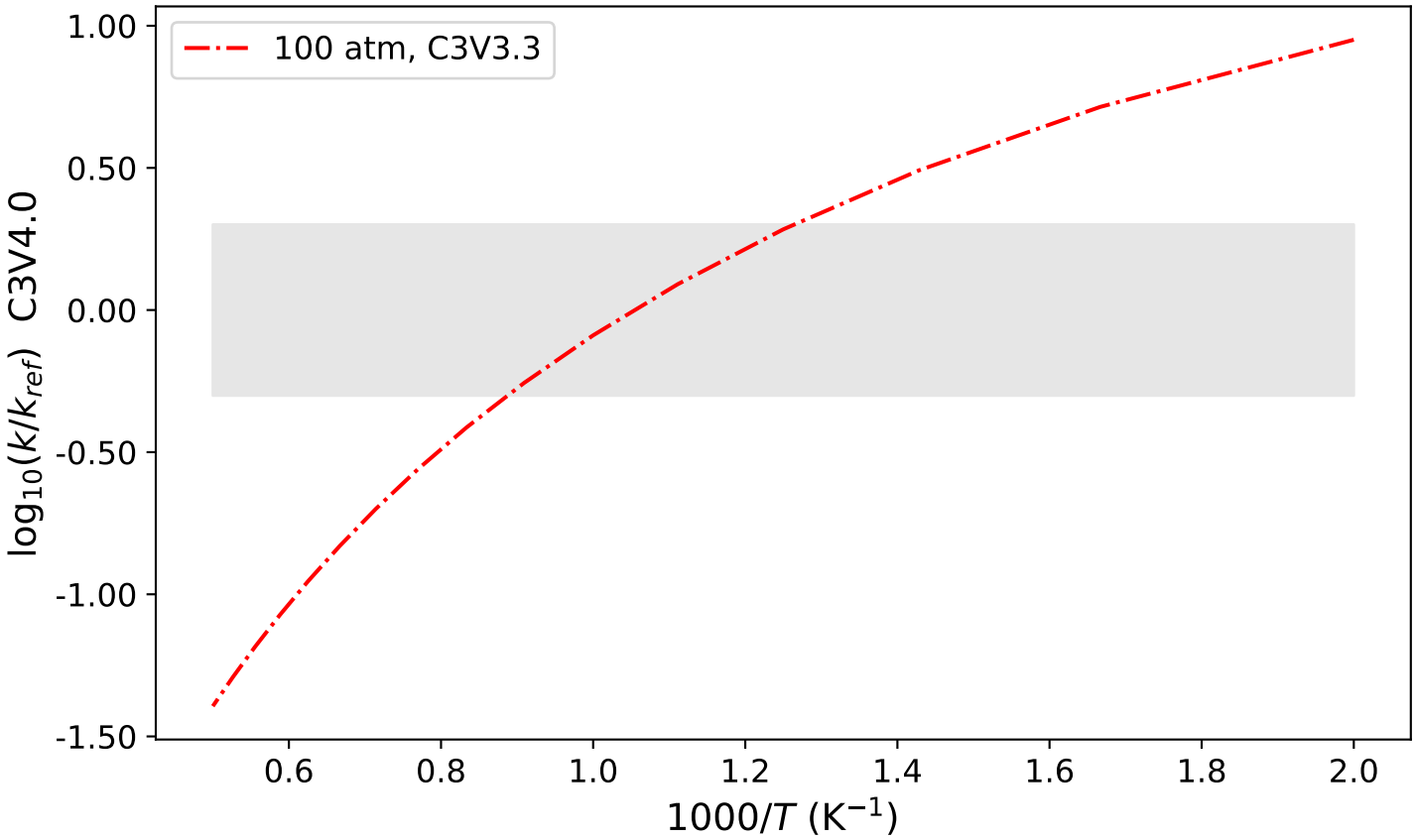
HO2 + C4H6 = O2 + C4H71-4



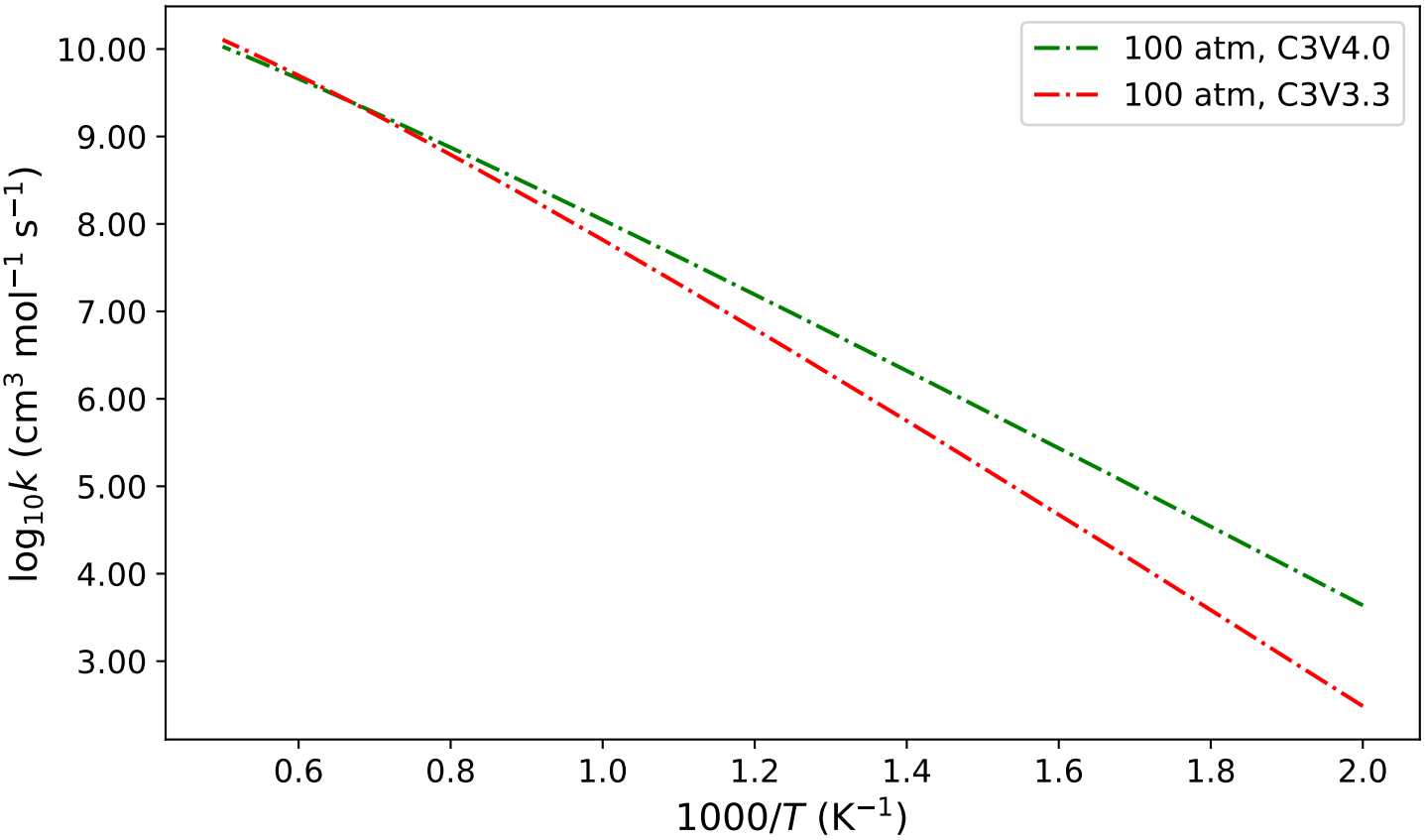



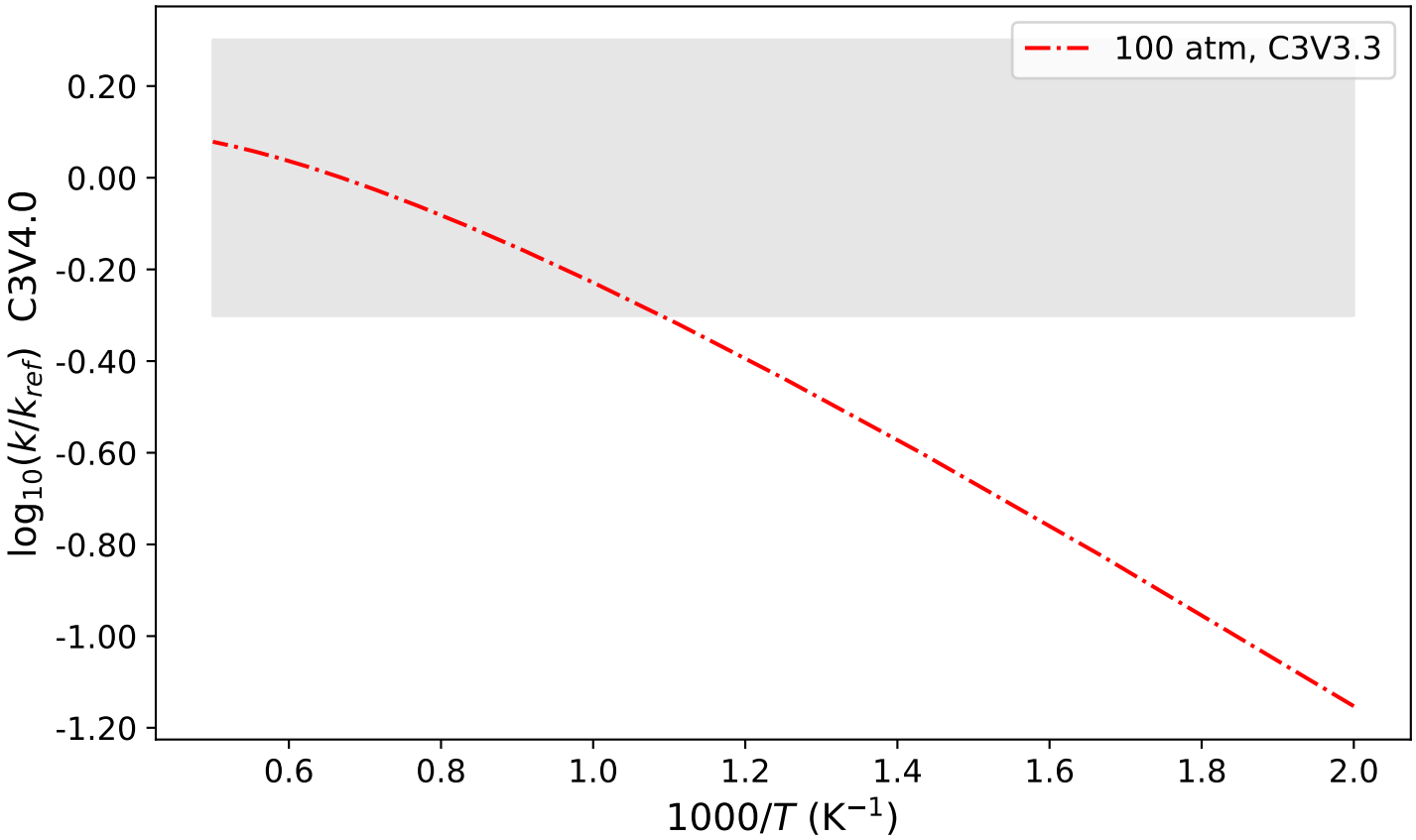
TC4H8O2H-I = OH + IC4H8O



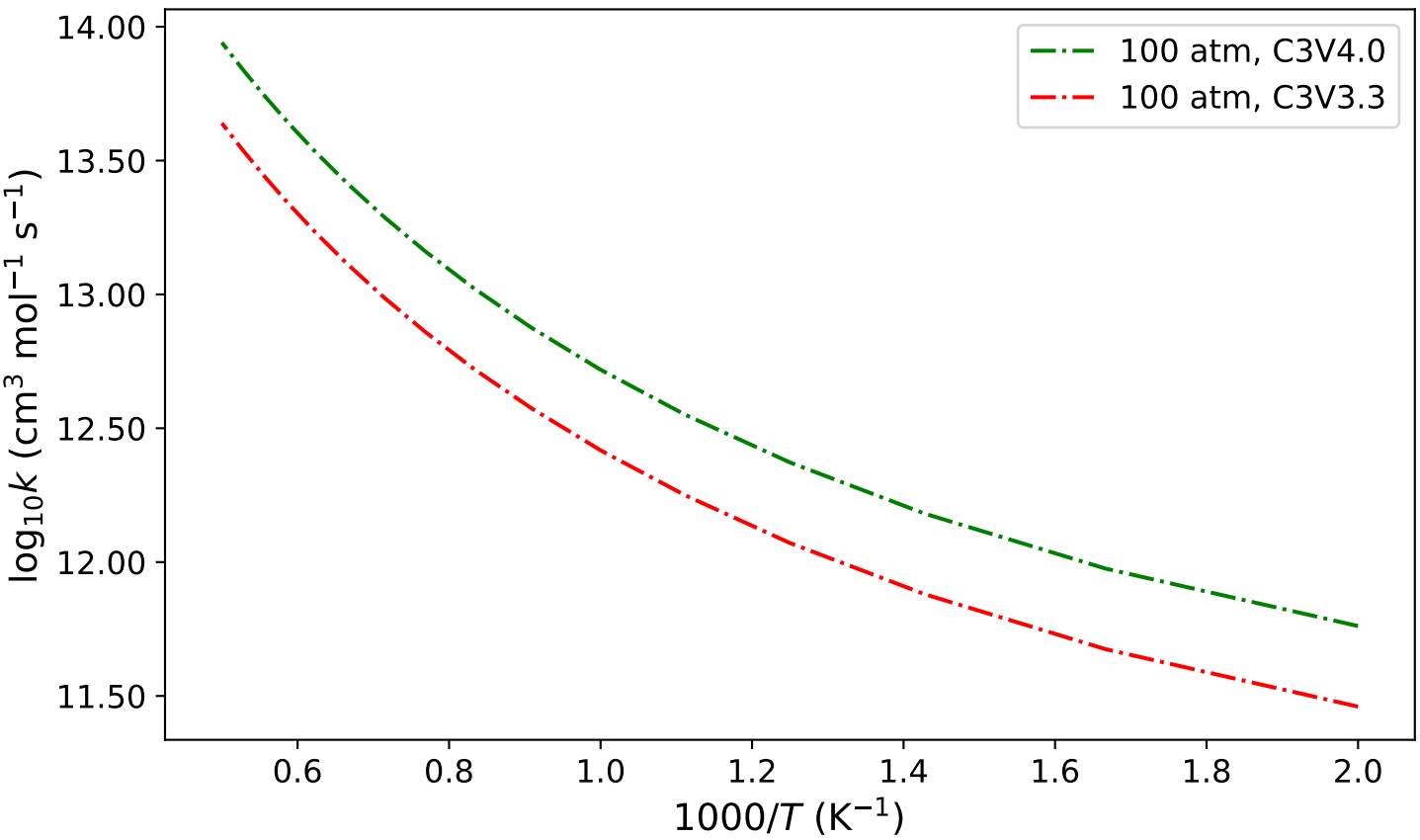



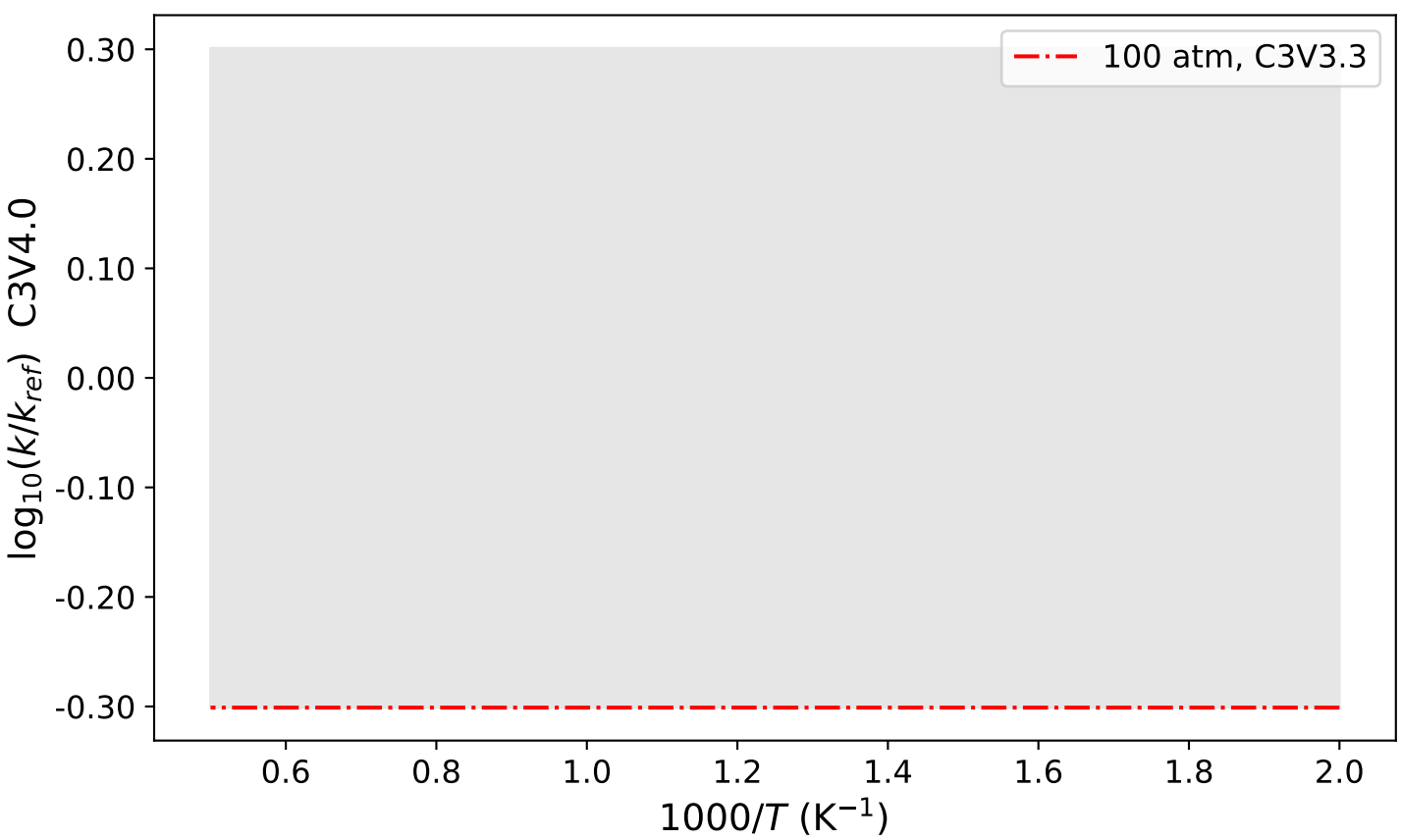
HO2 + C4H8-1 = OH + C4H8O1-2



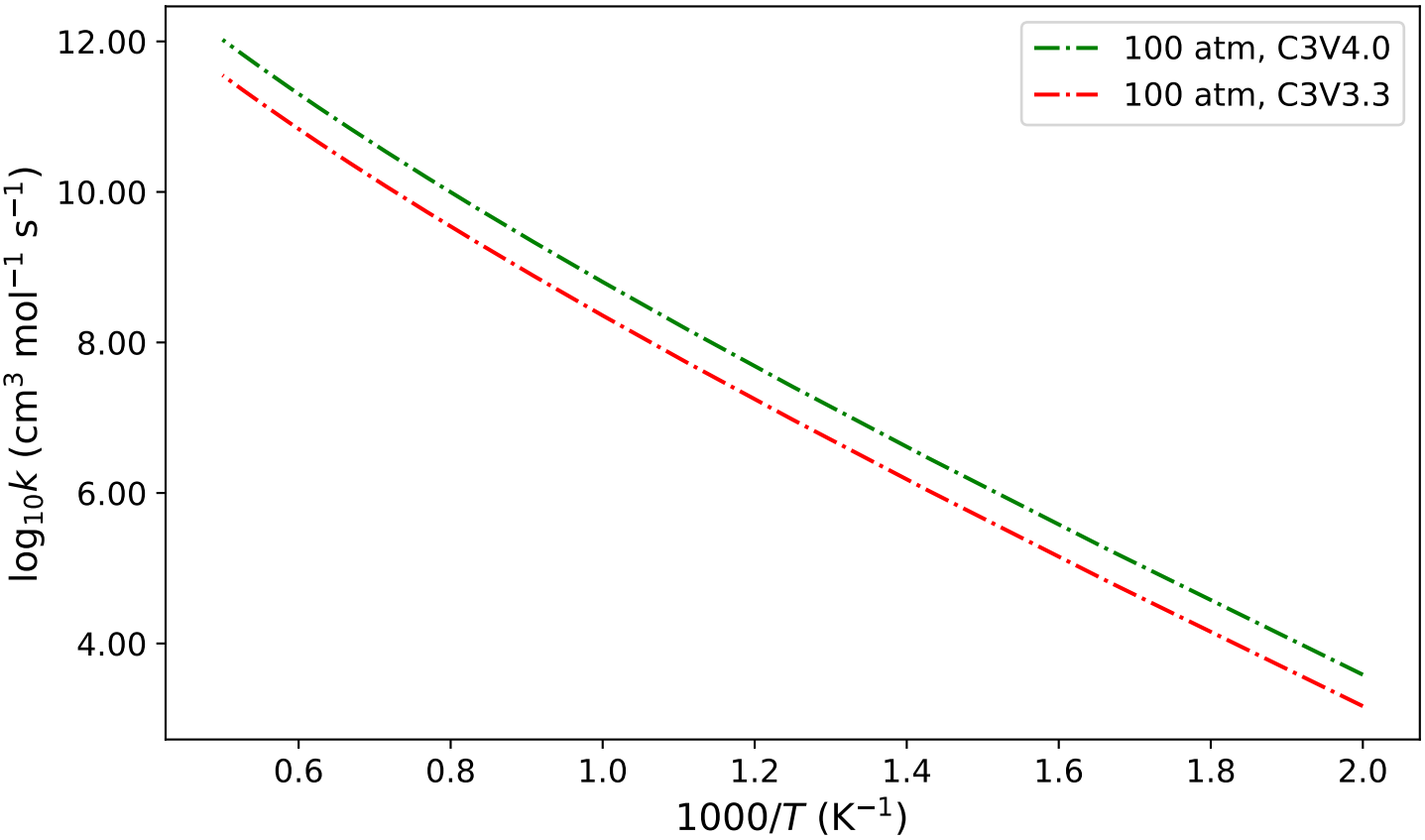



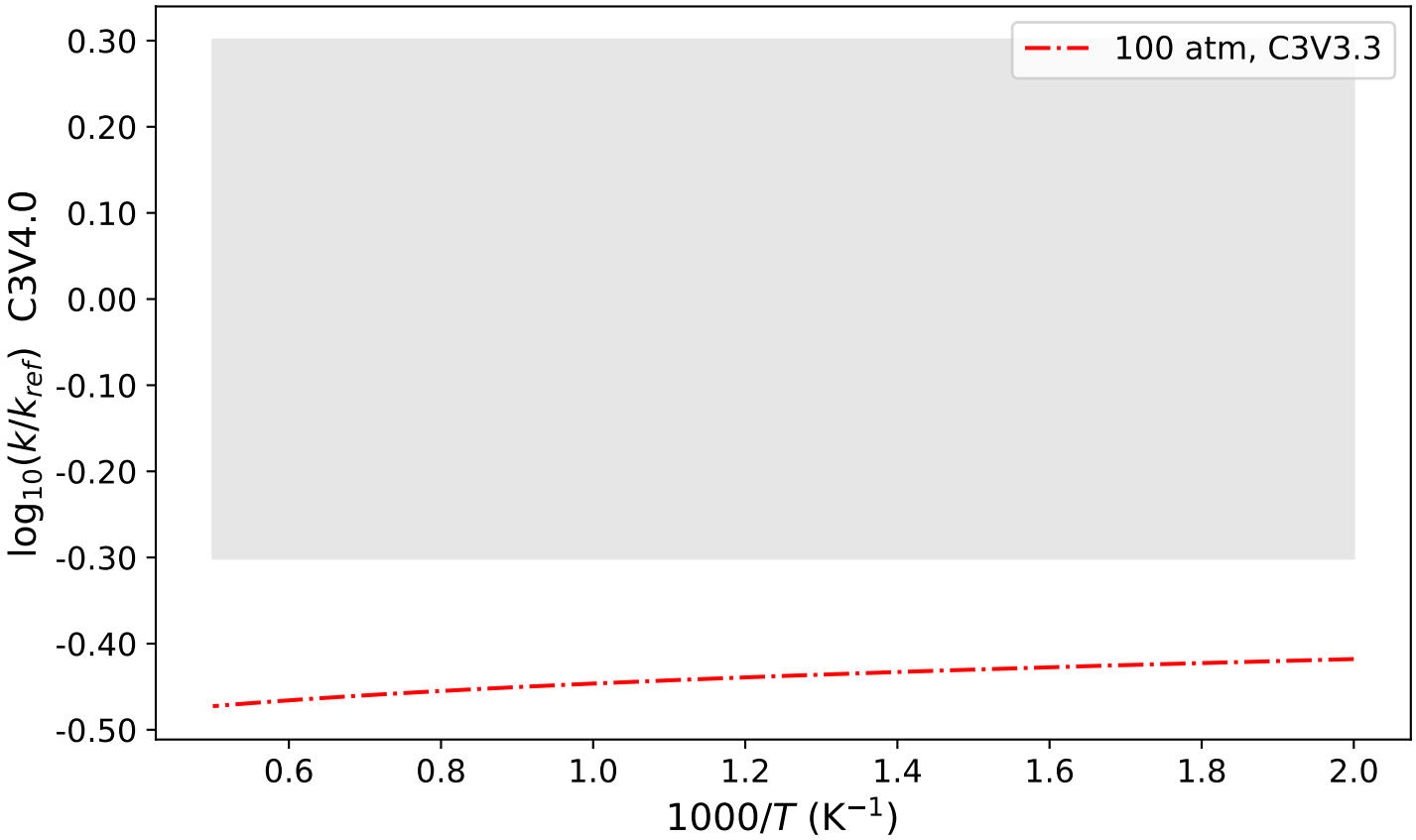
O + C3H6 = H + CH3CHCHO



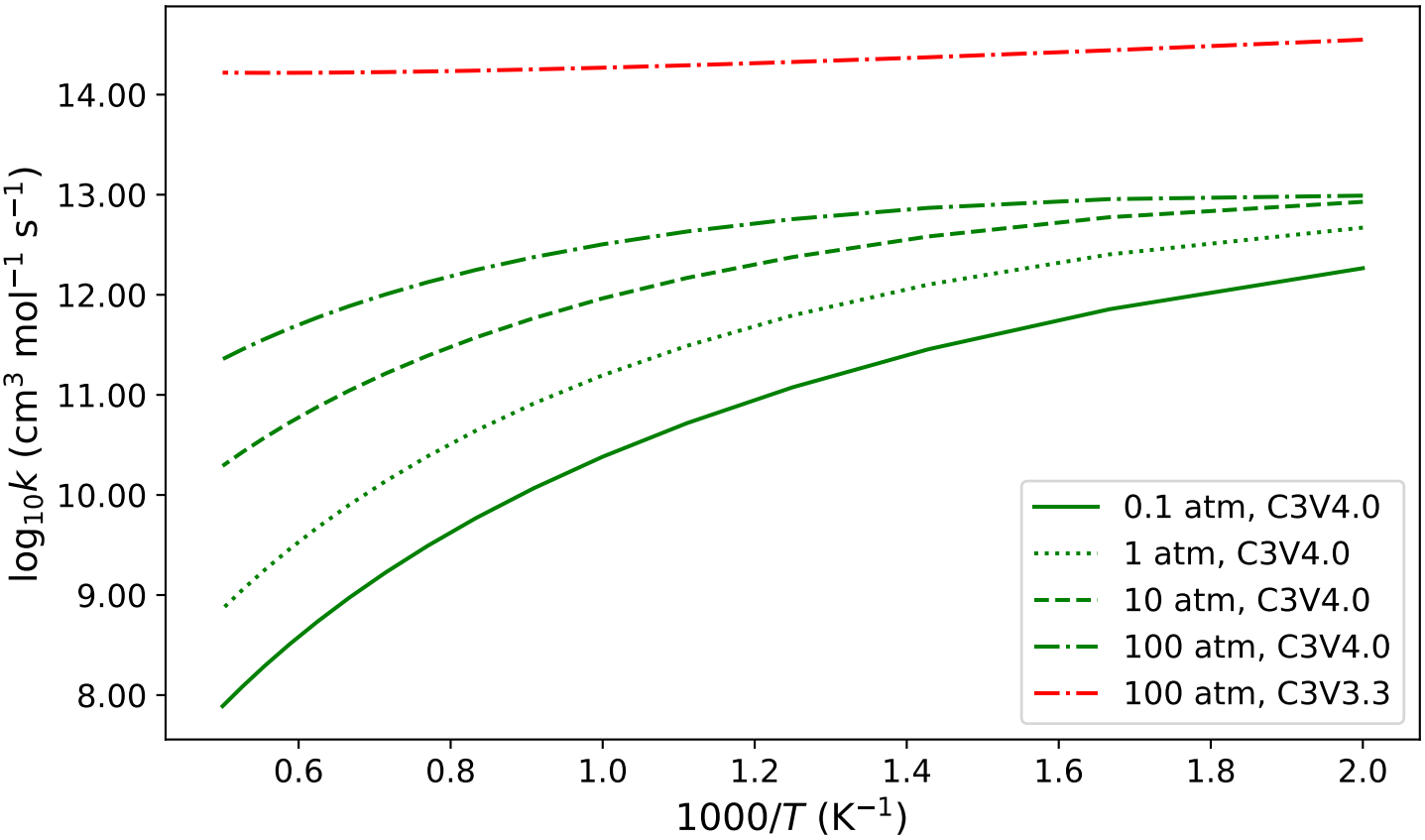



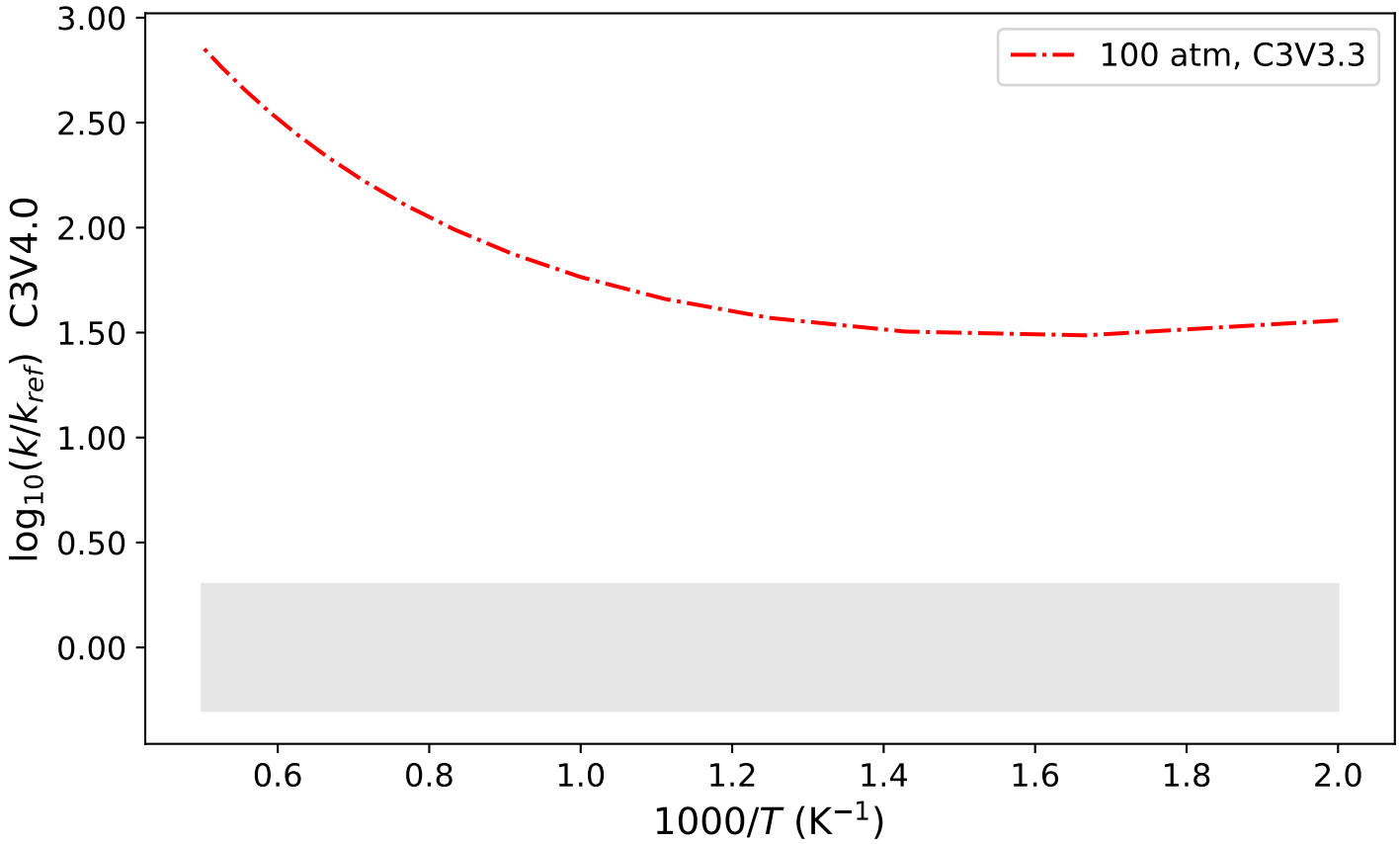
CH3O2 + IC4H10 = CH3O2H + IC4H9



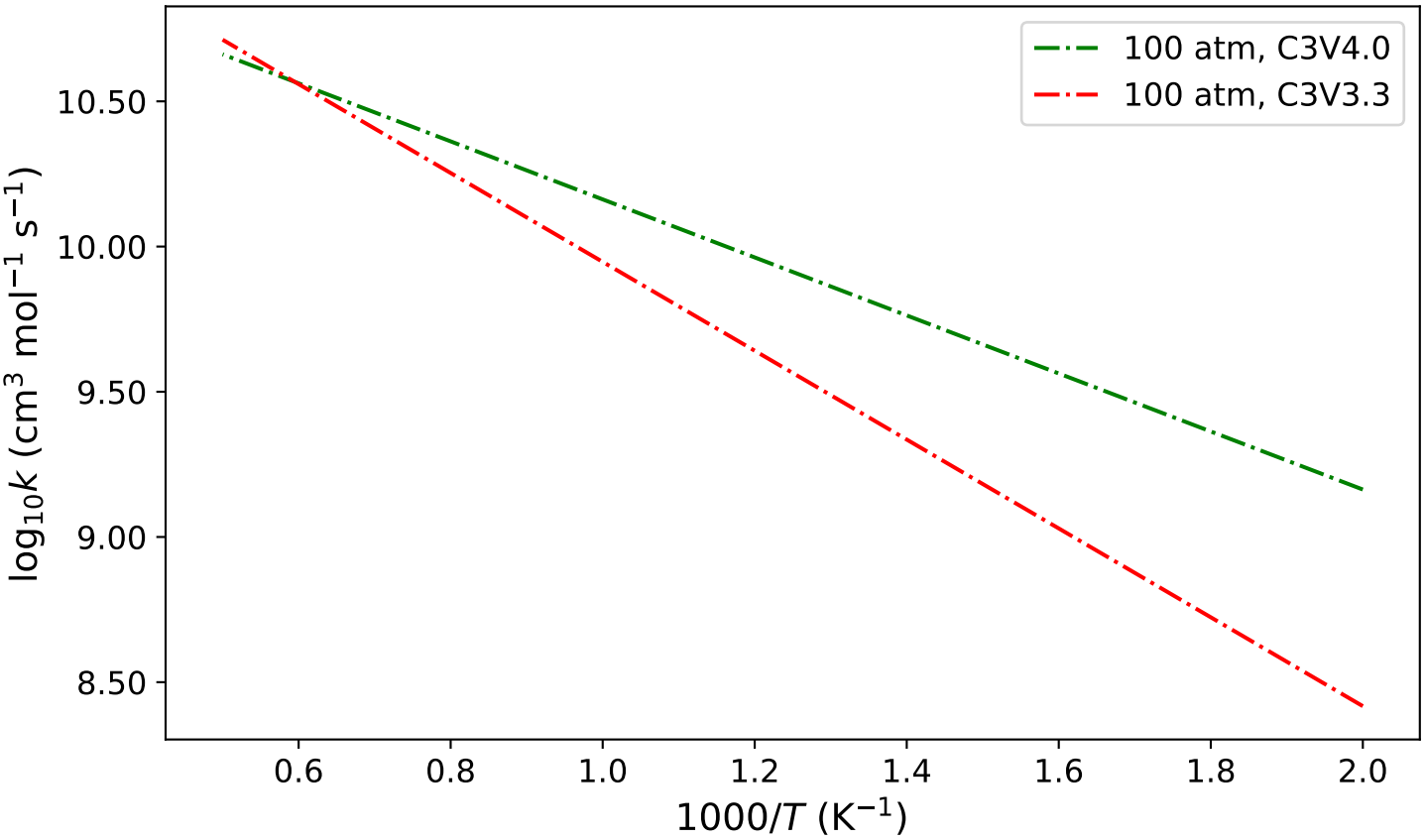



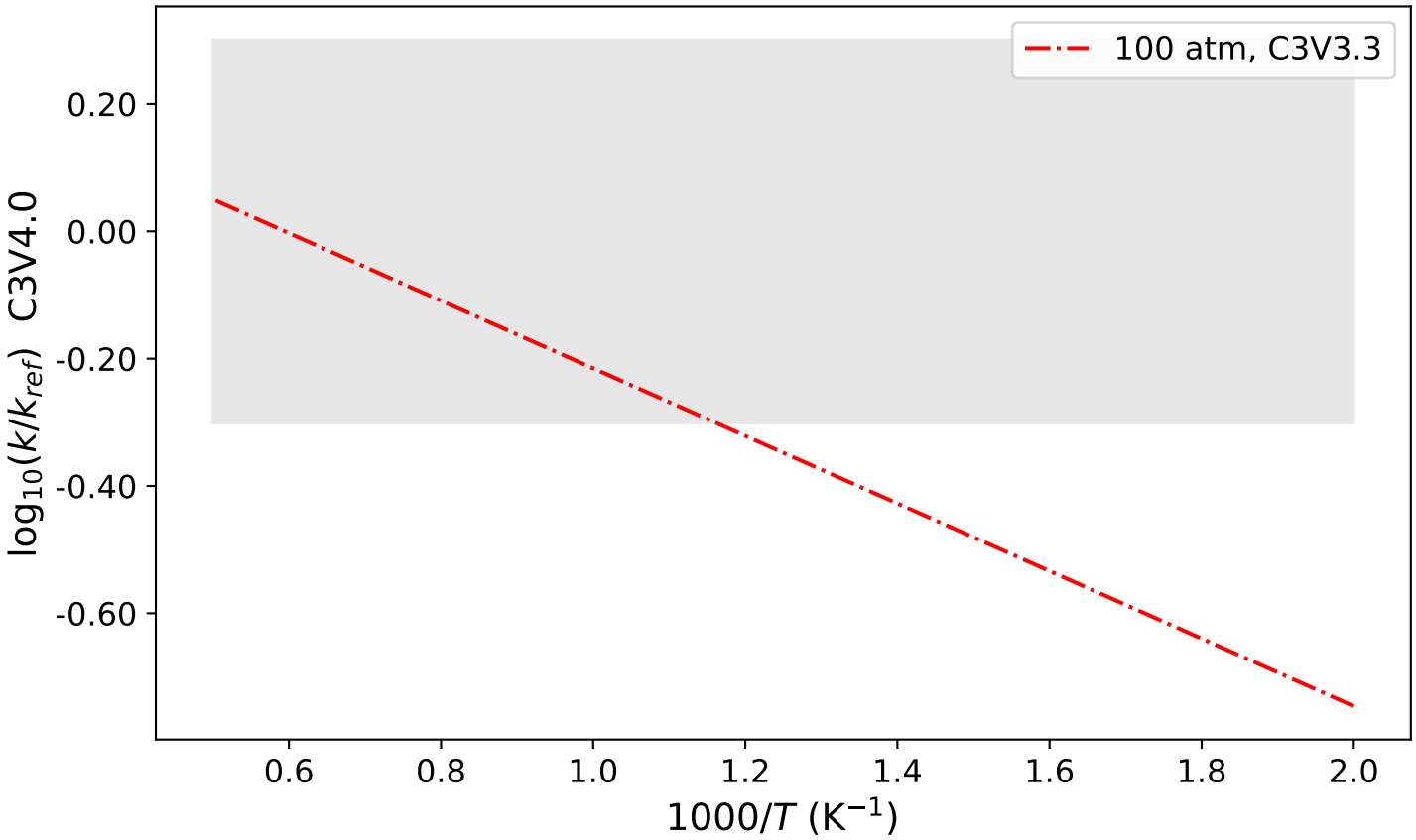
HO2 + C4H64,2-1OH = C3H4CH2OH-1OOH



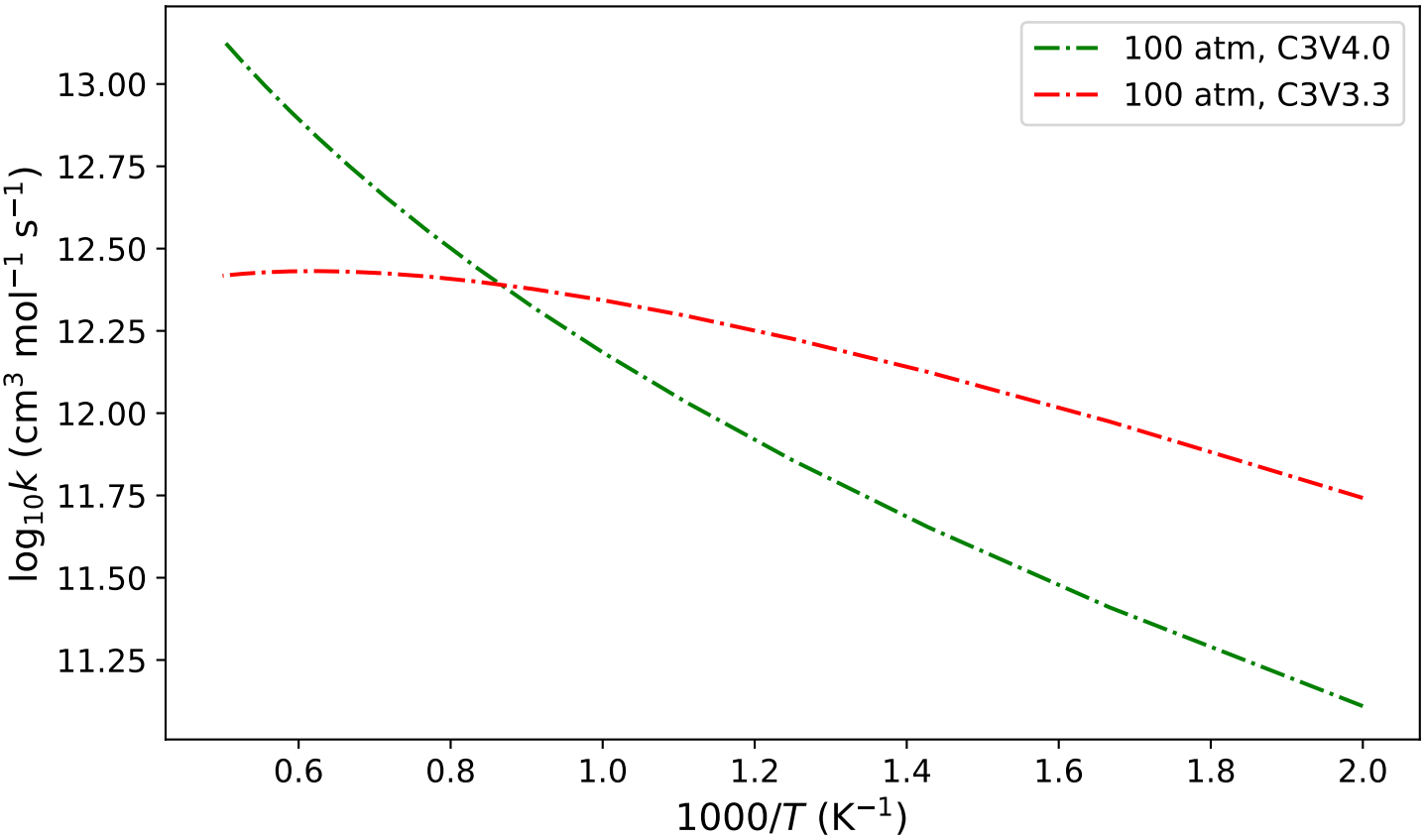



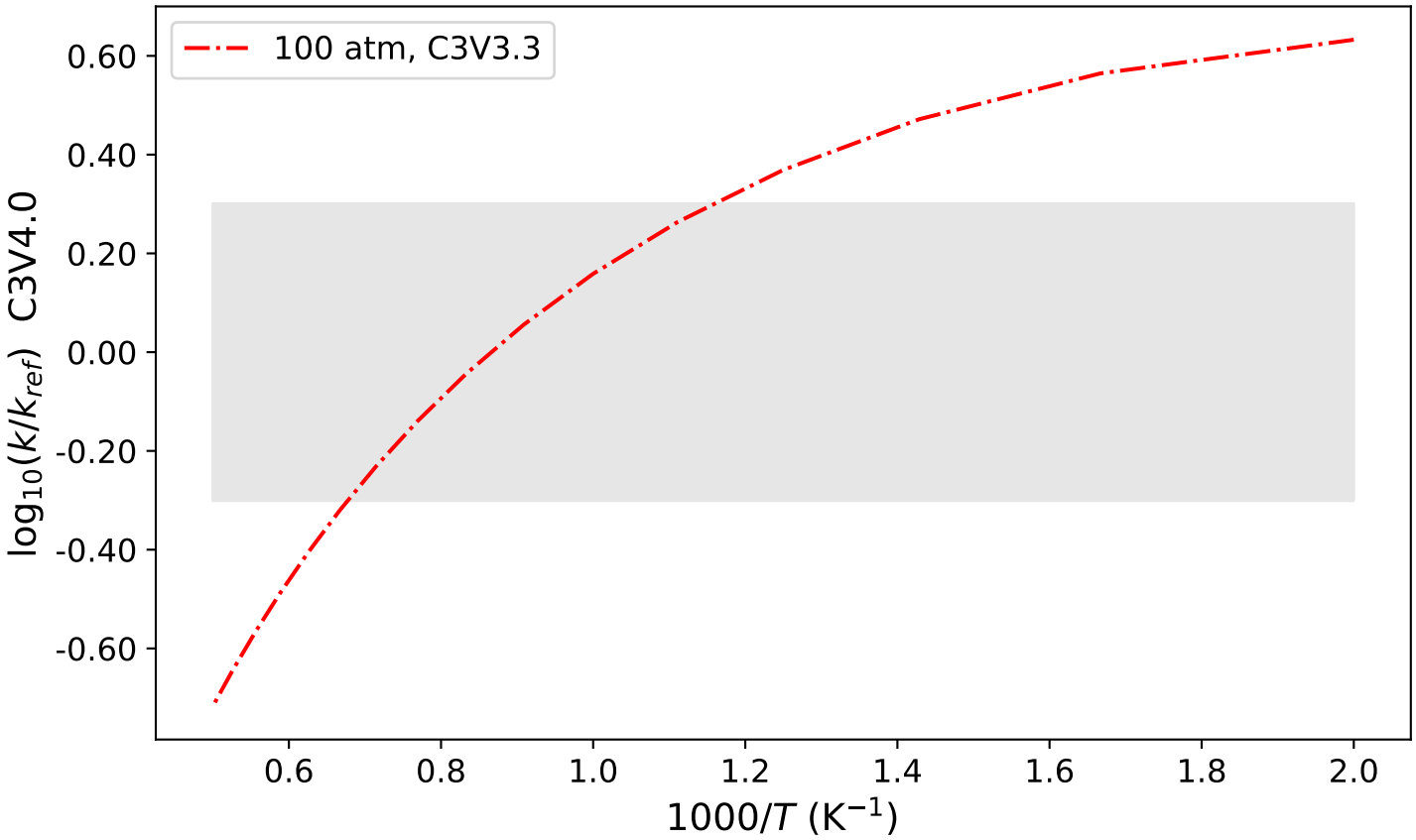
C3H8 + CH3O = IC3H7 + CH3OH



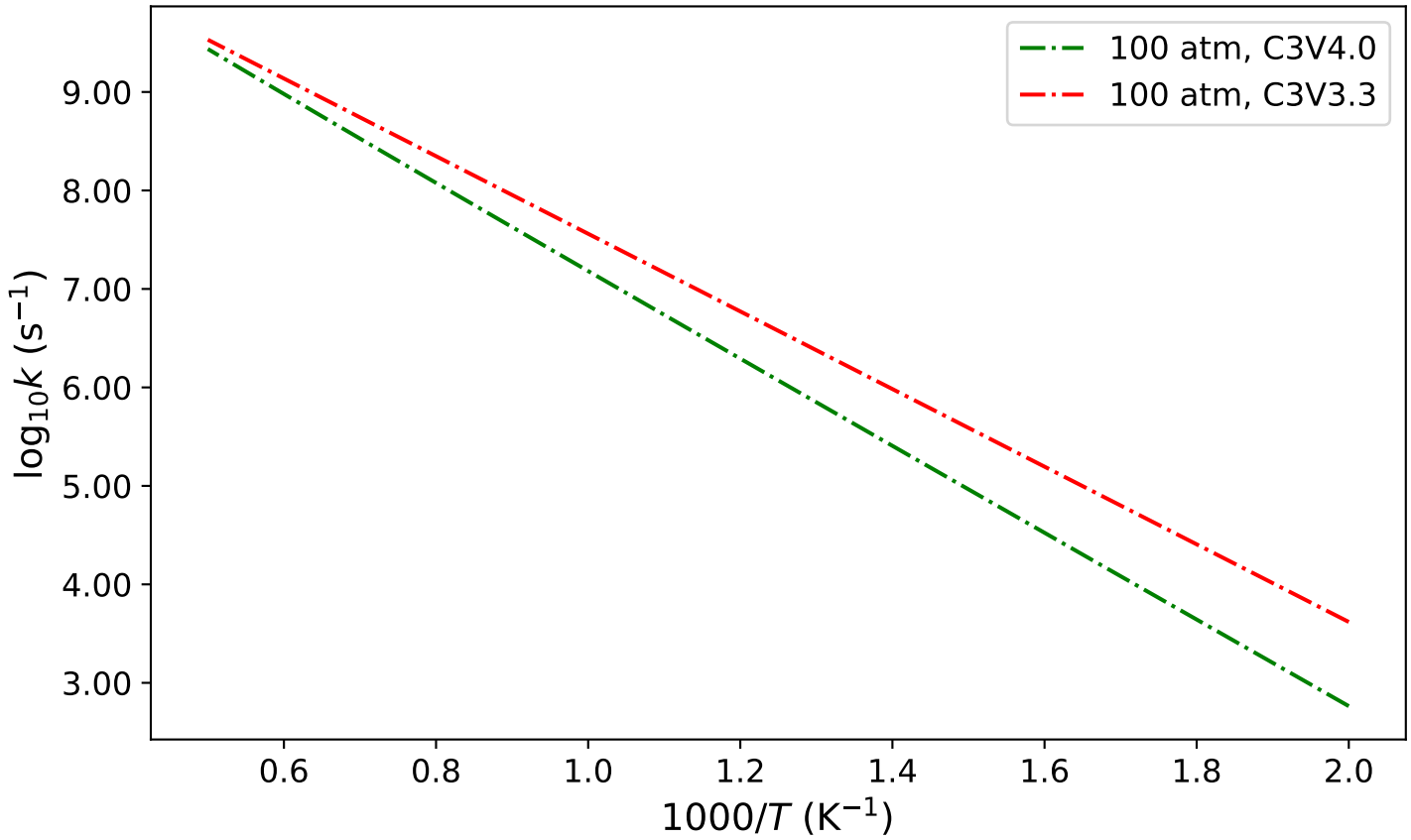



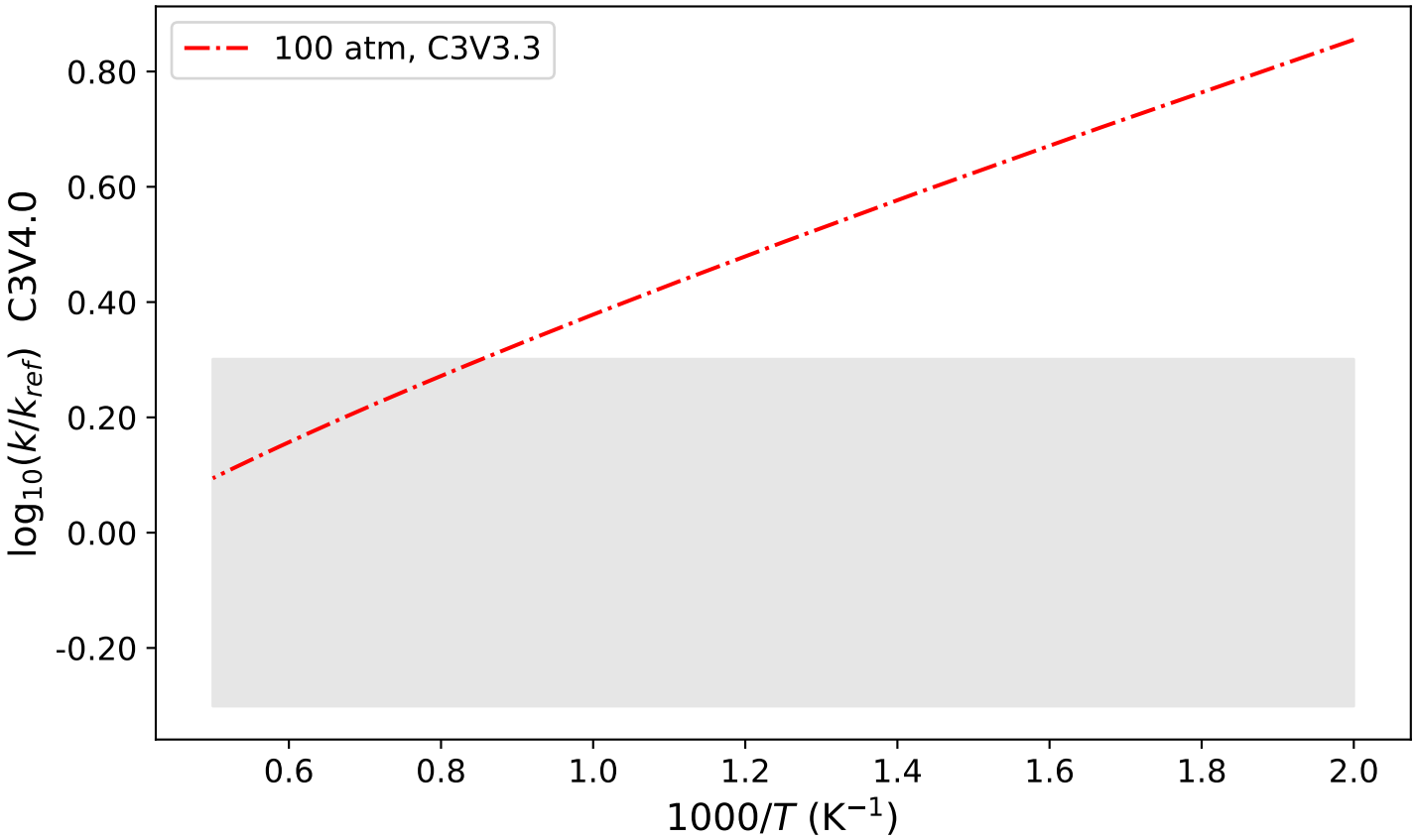
HO2 + CH3COCH2 = OH + CH3COCH2O



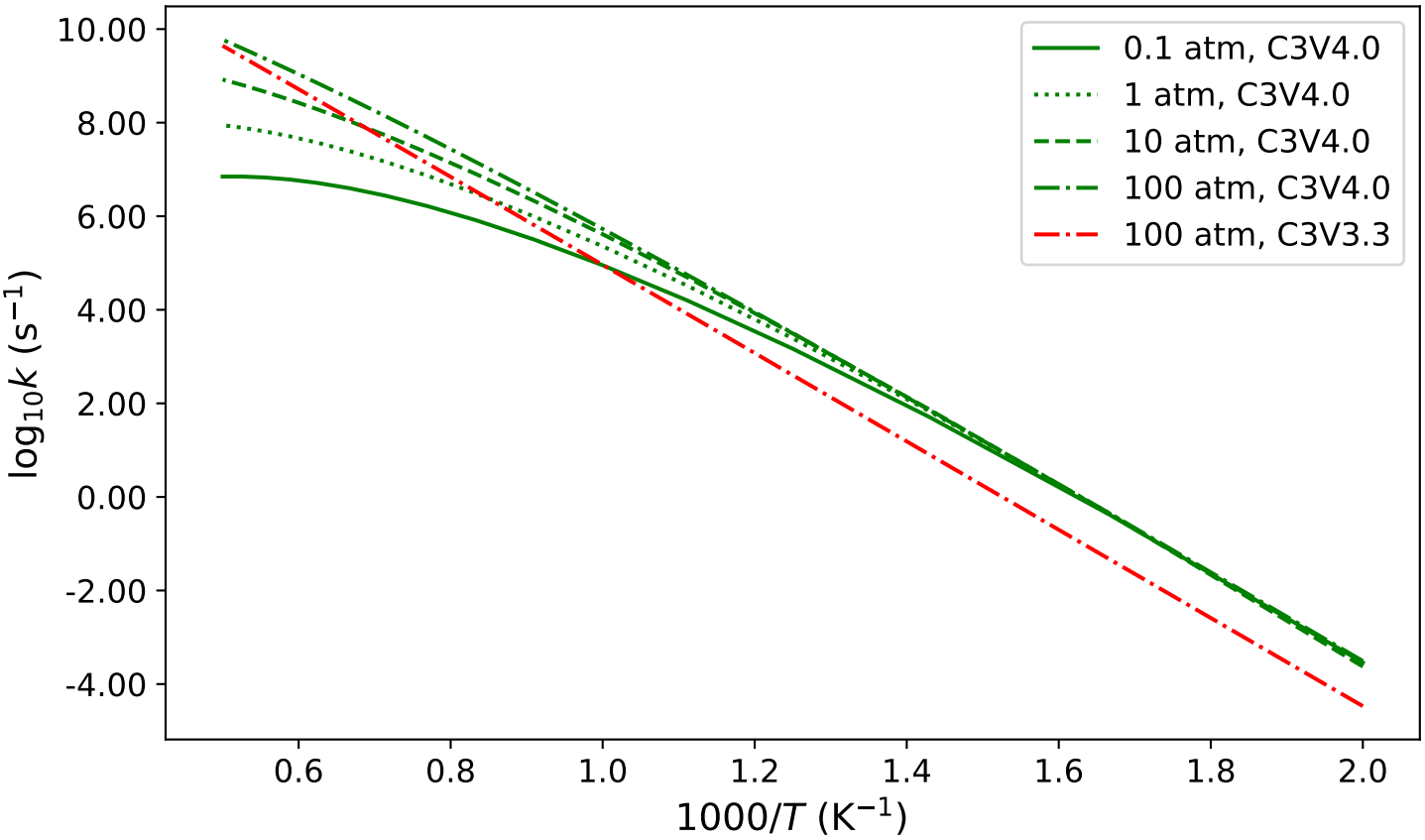



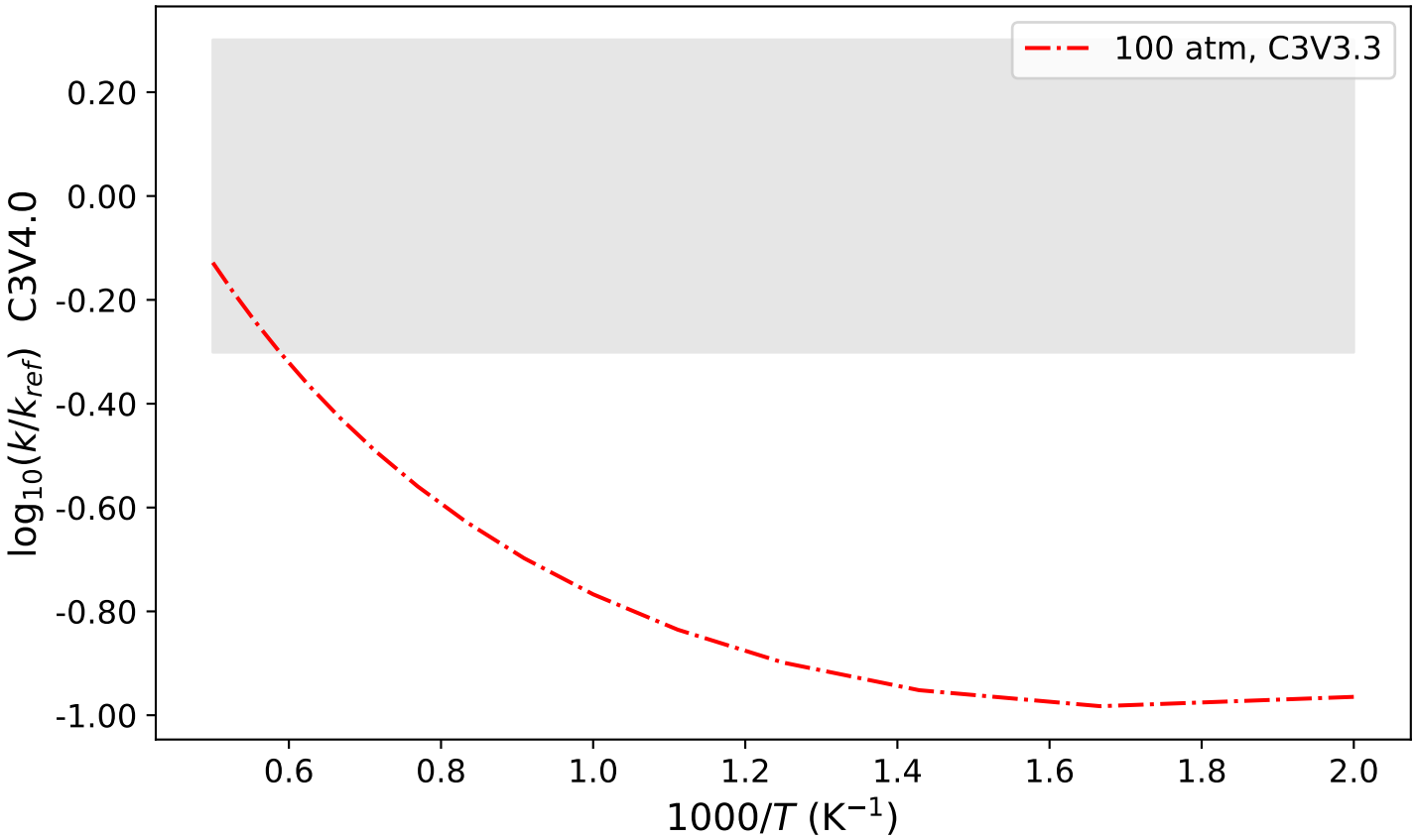
IC4H8O2H-I = OH + CC4H8O



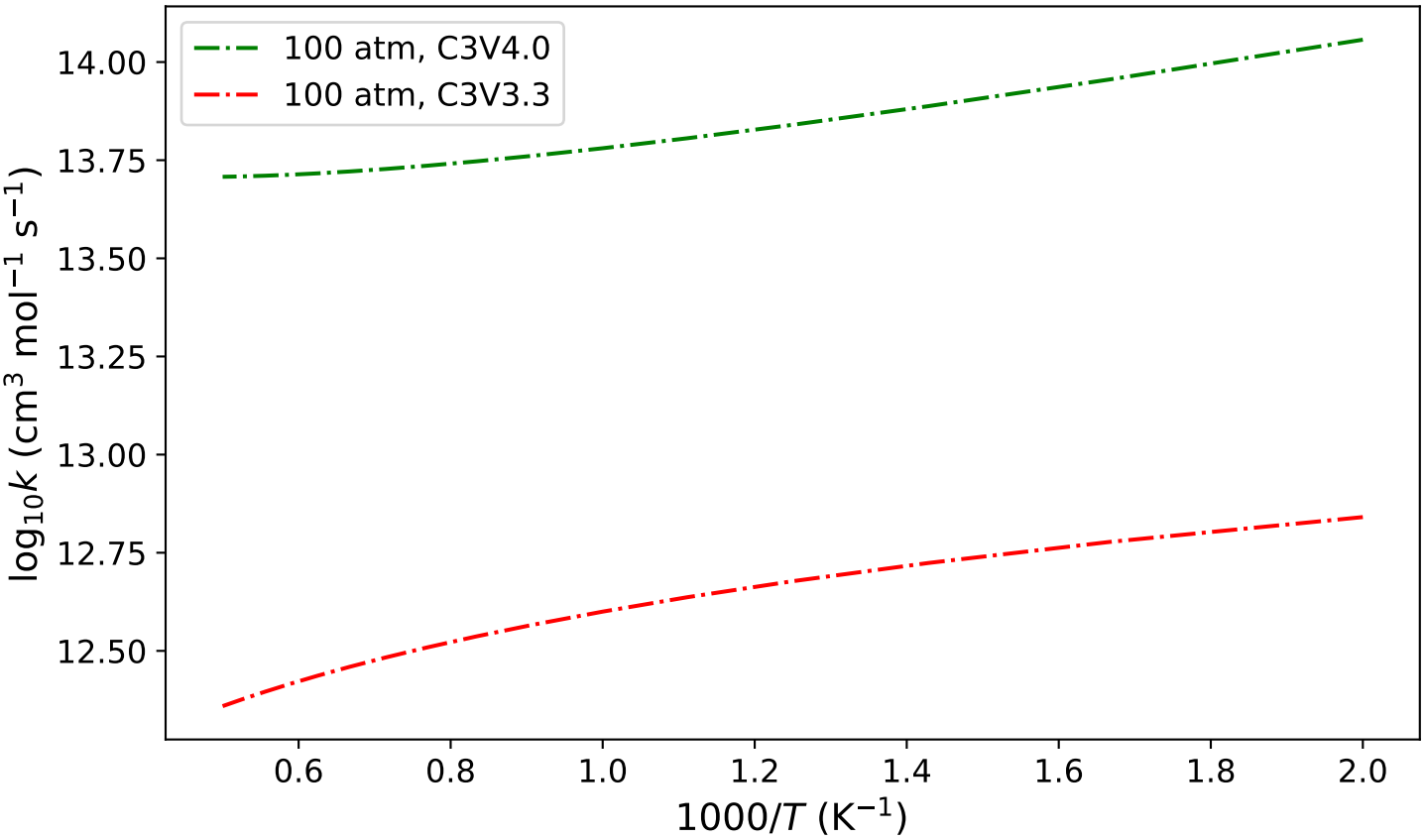



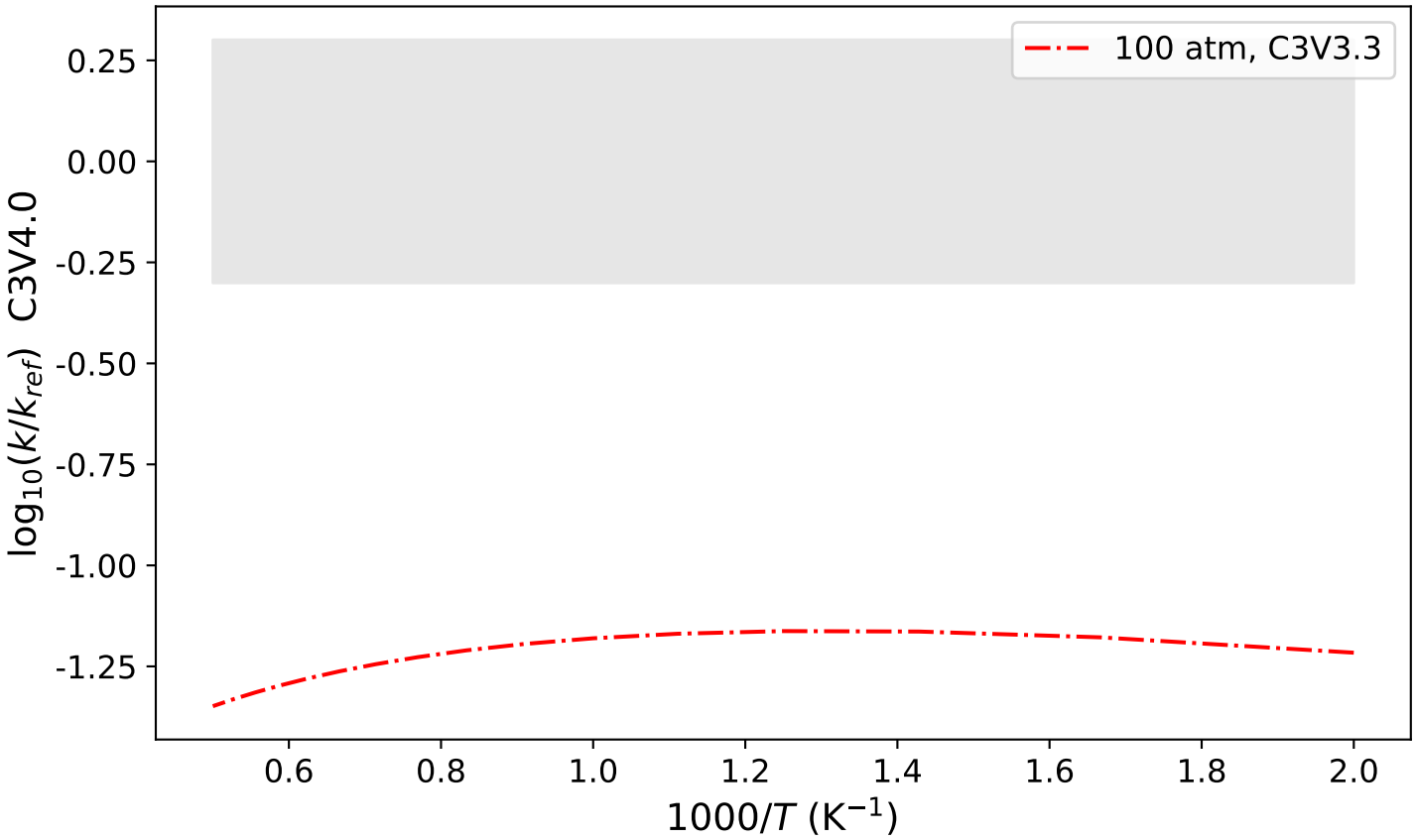
C3H6CHO-1 = CH3 + C2H3CHO



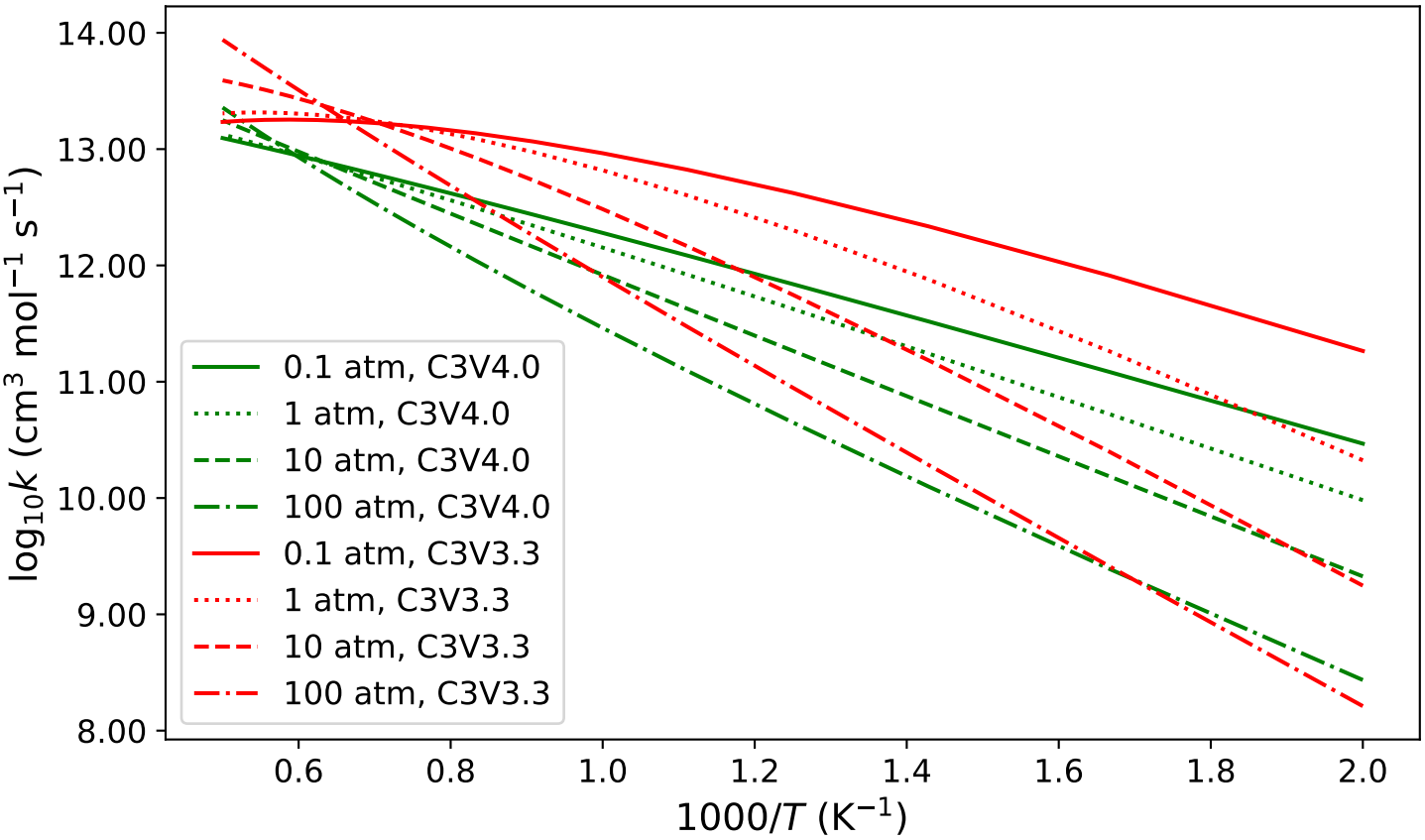



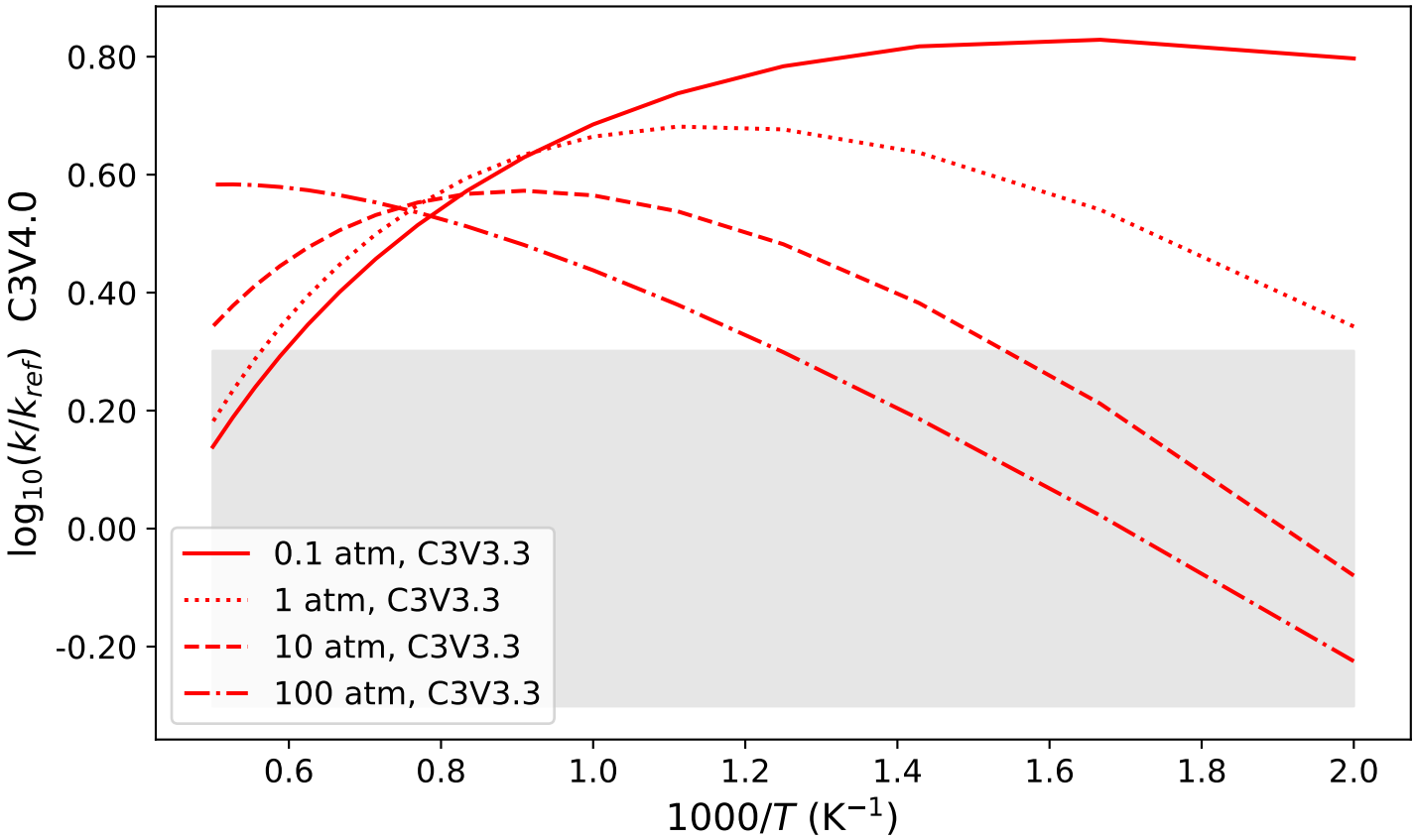
OH + SC4H9O = SC4H9O2H



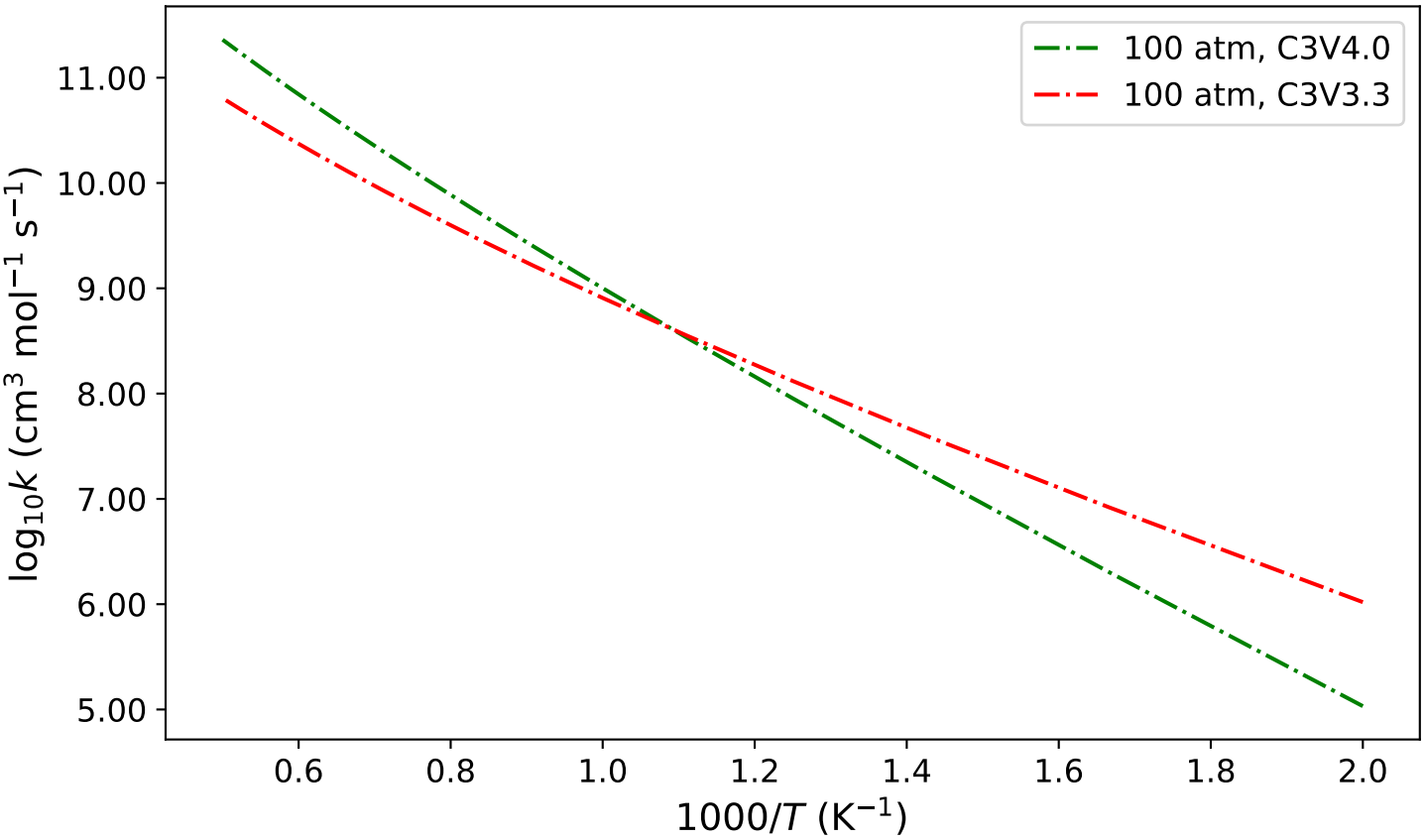



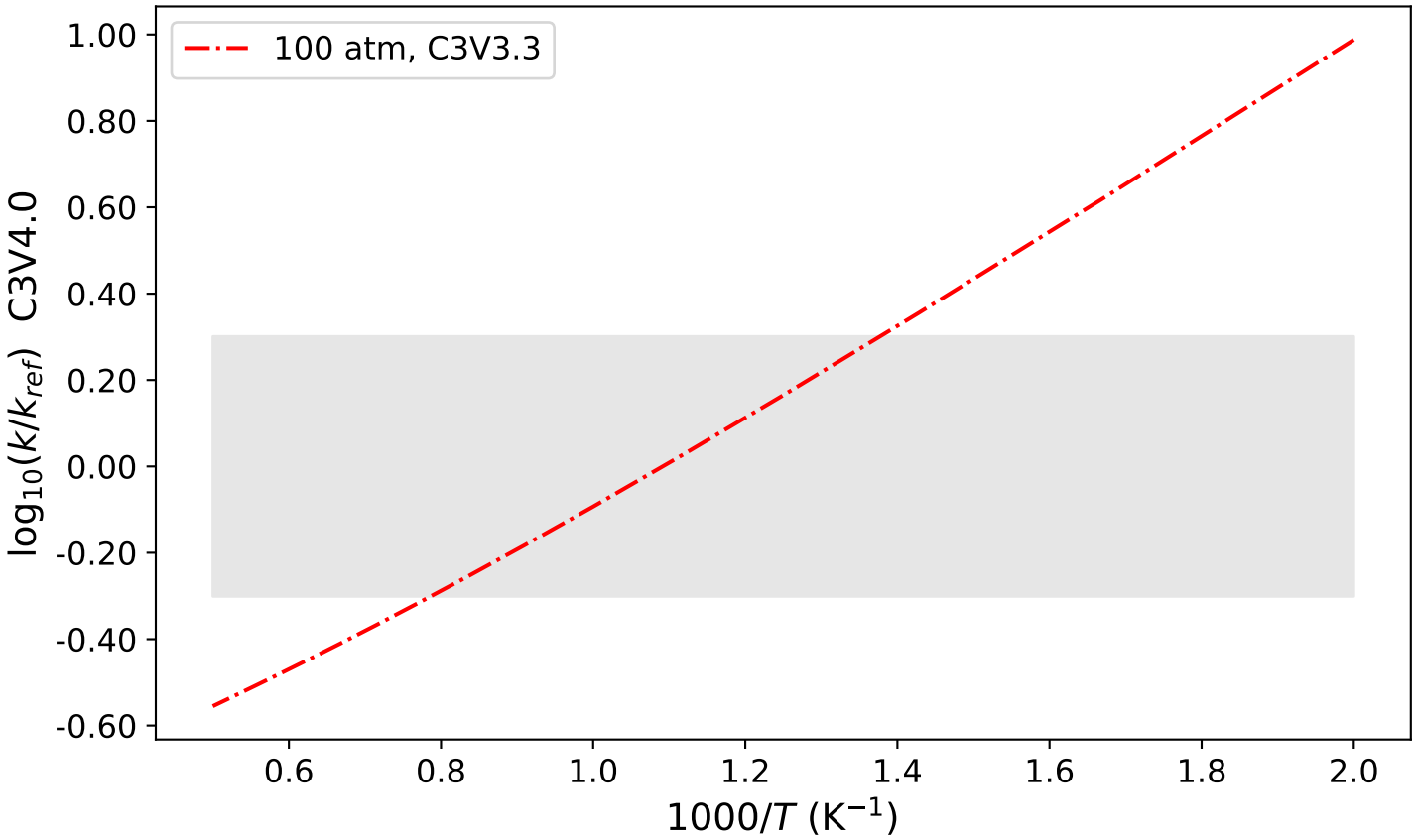
H + IC4H8 = CH3 + C3H6



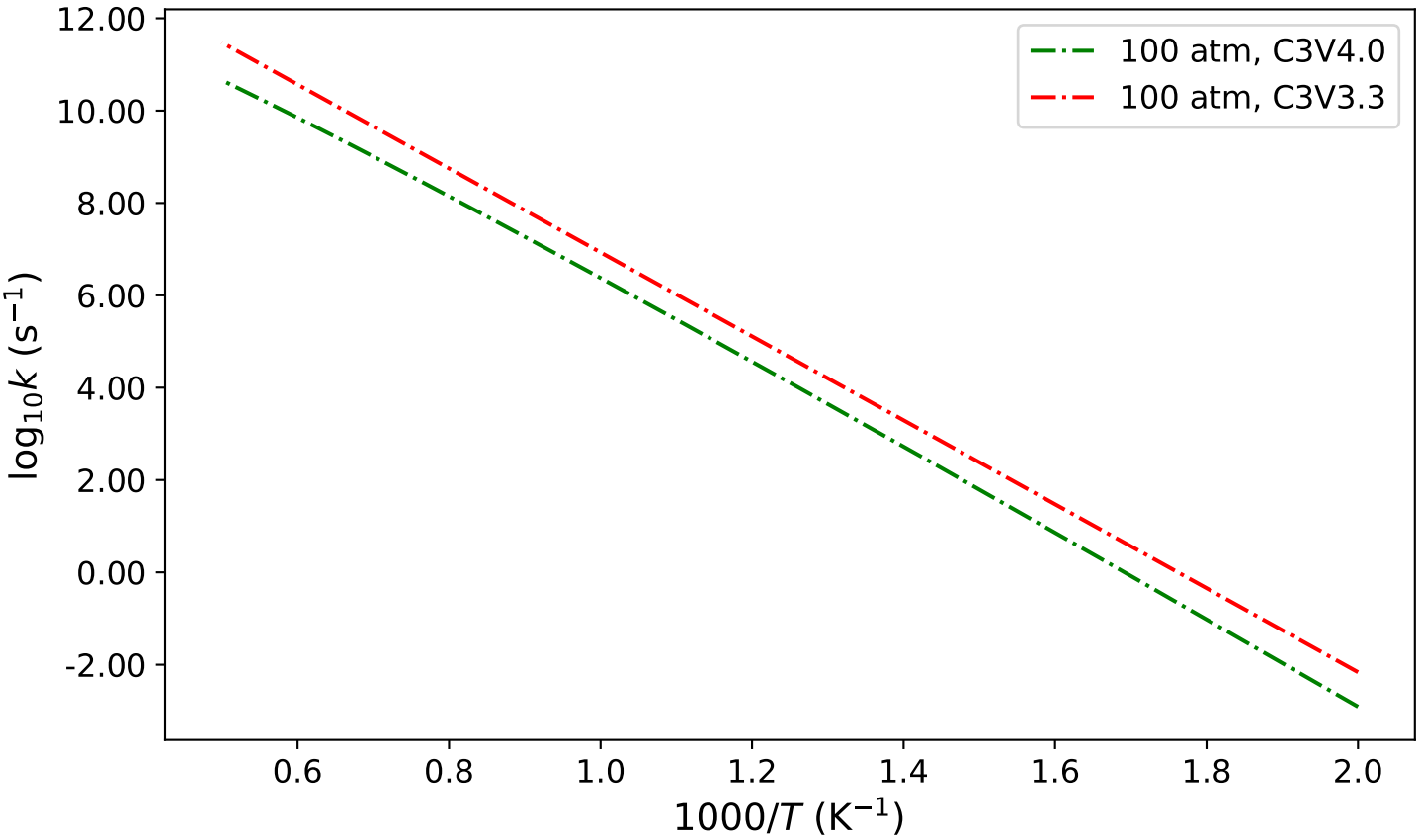



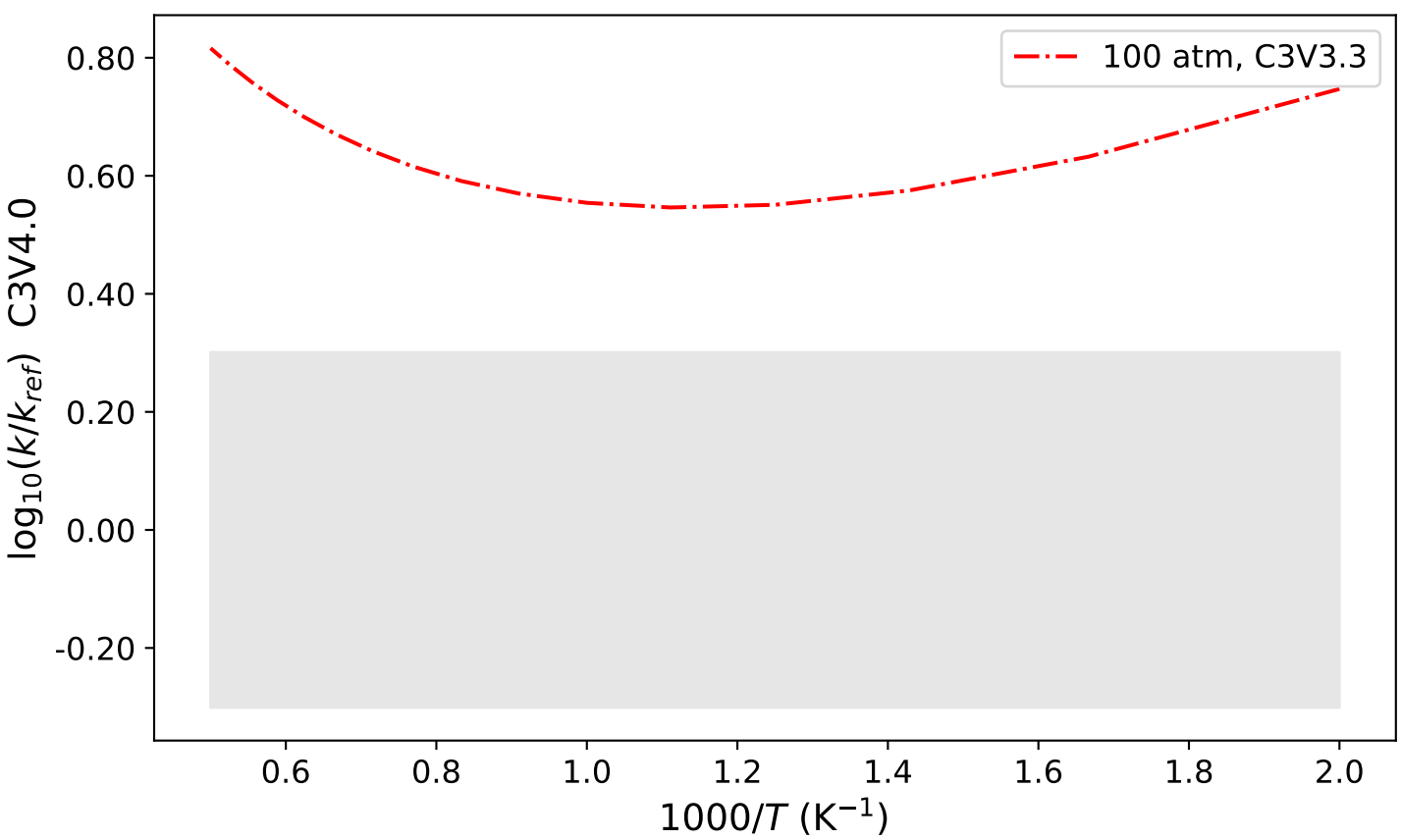
HO2 + C4H8-1 = C4H8OOH1-2



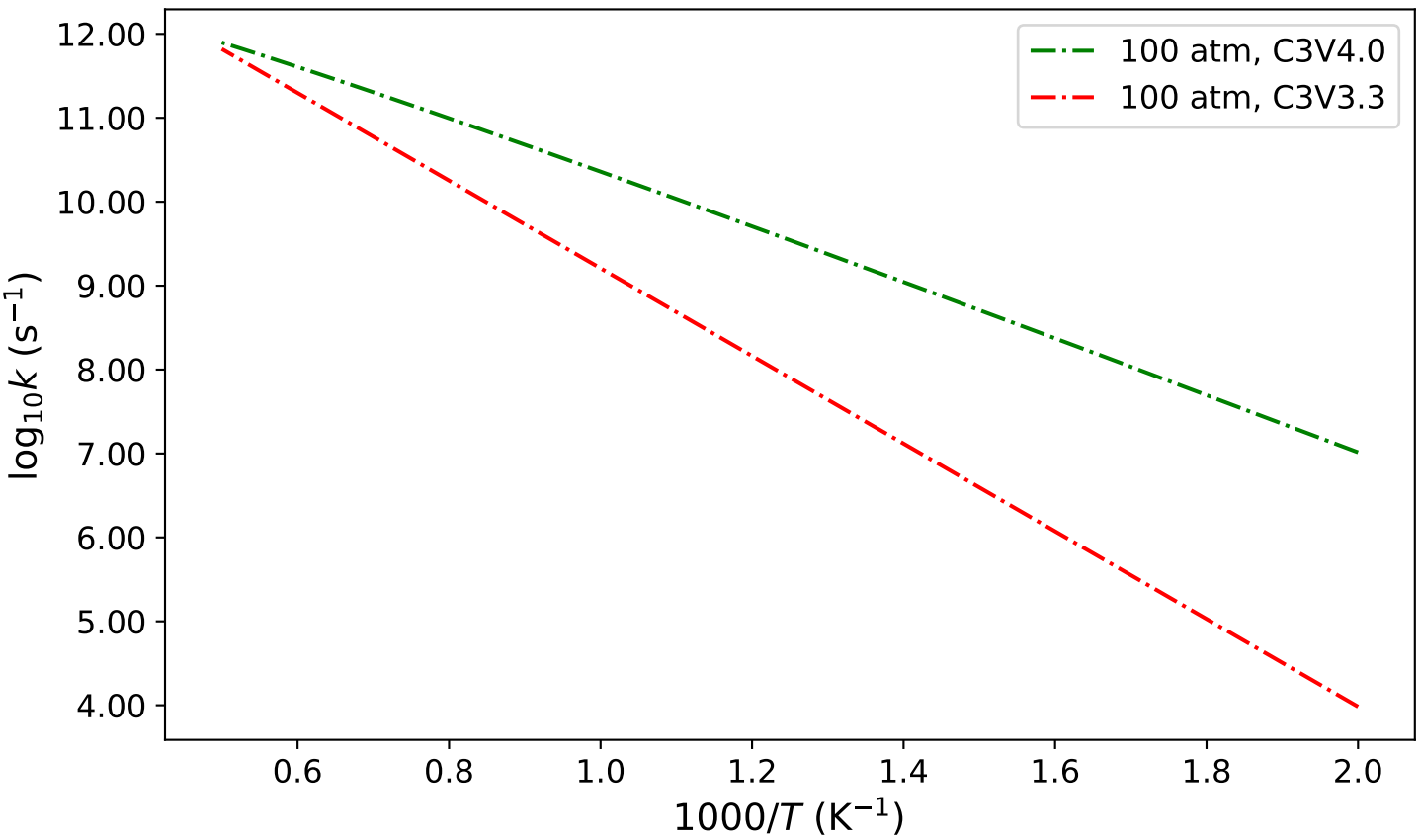



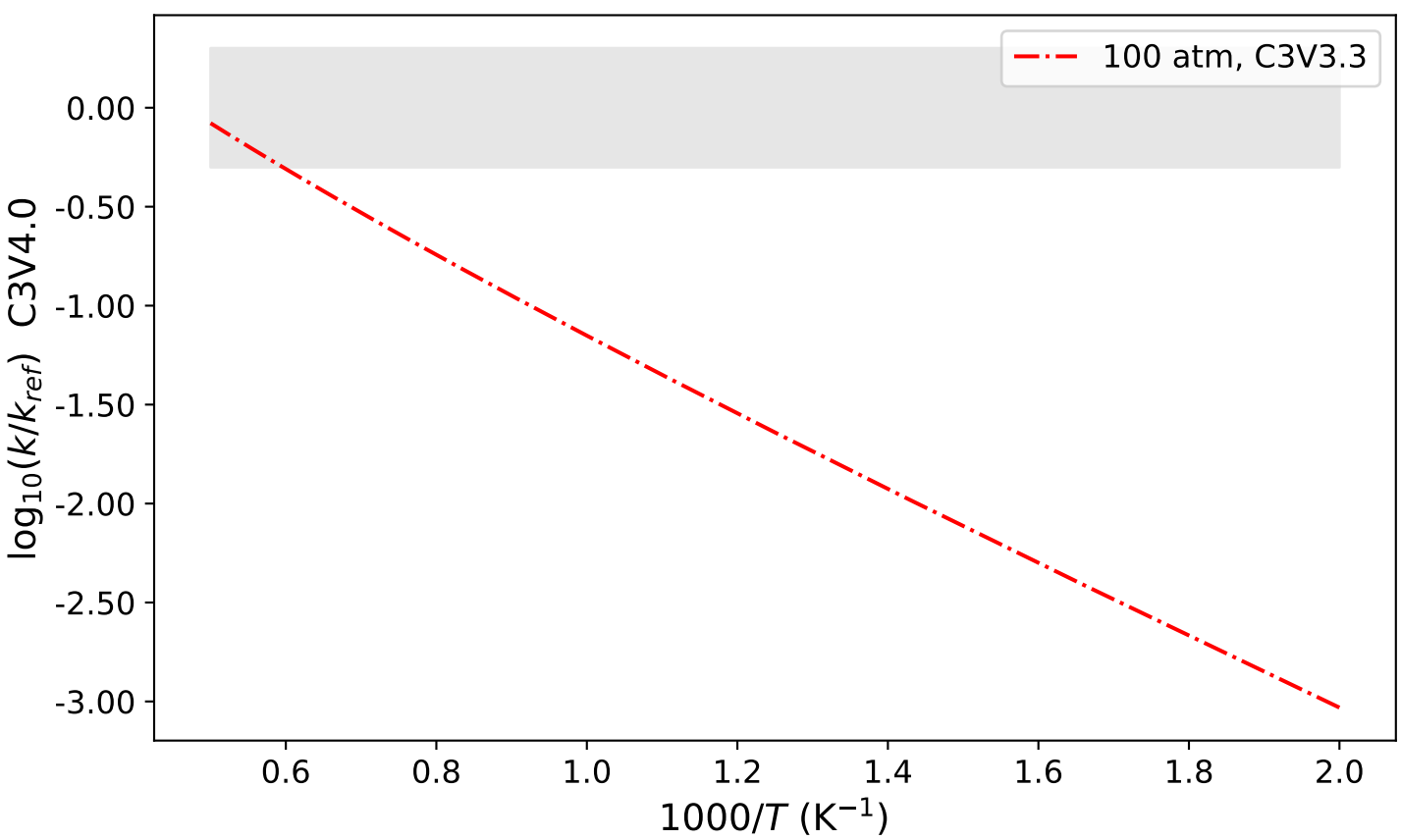
NC4KET12 => OH + C2H5CHO + HCO



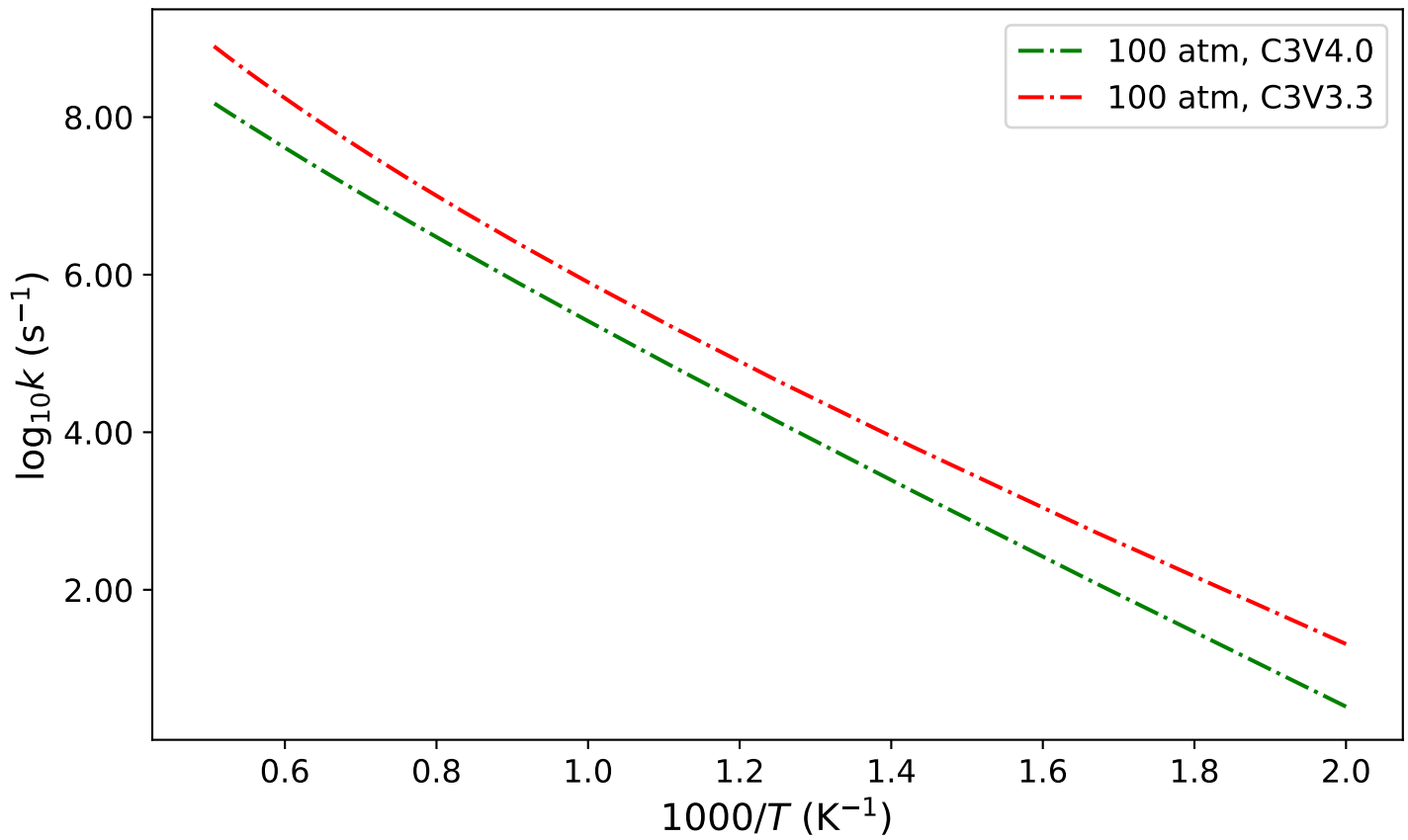



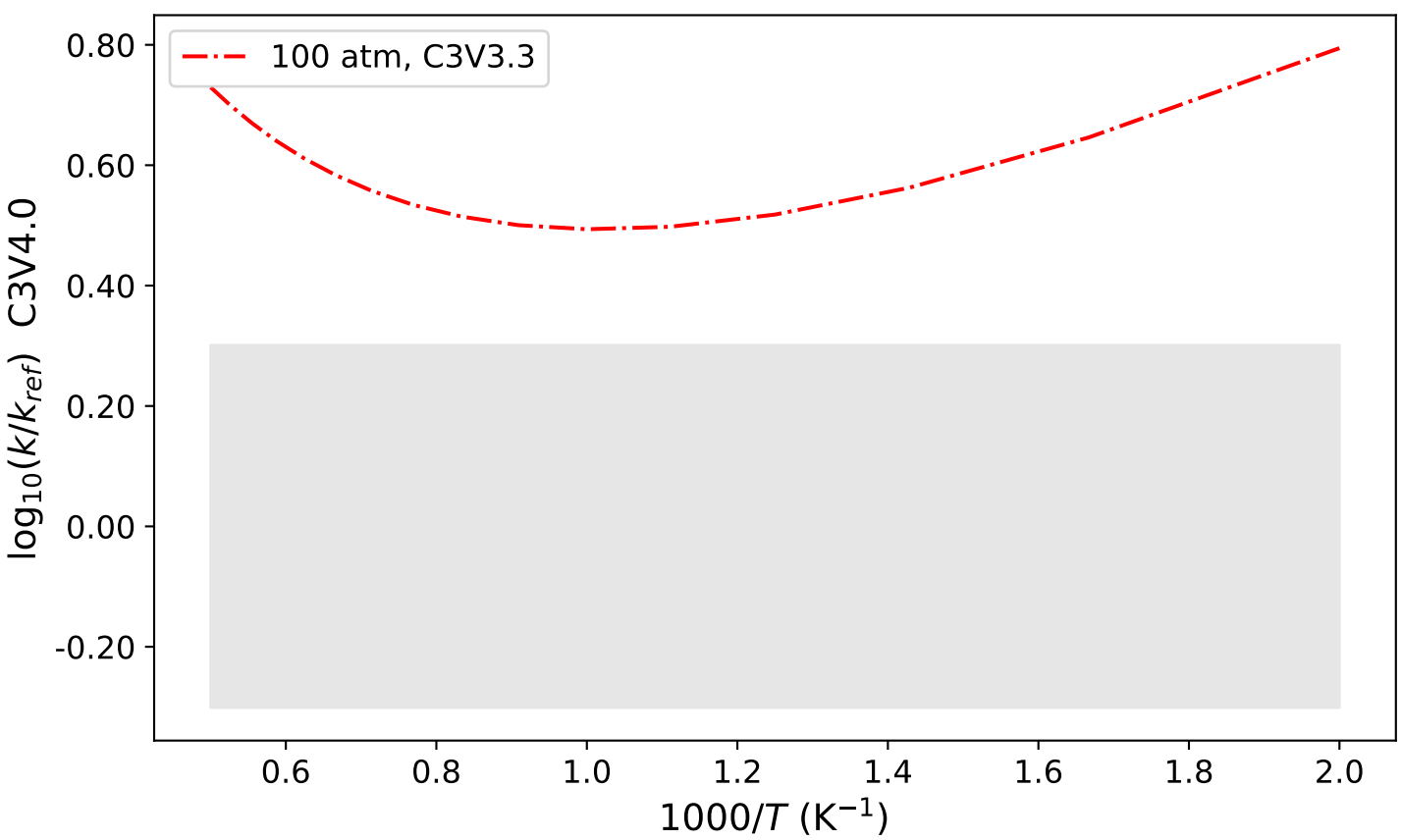
C3H4CH2OH-3O = C2H3CHO + CH2OH



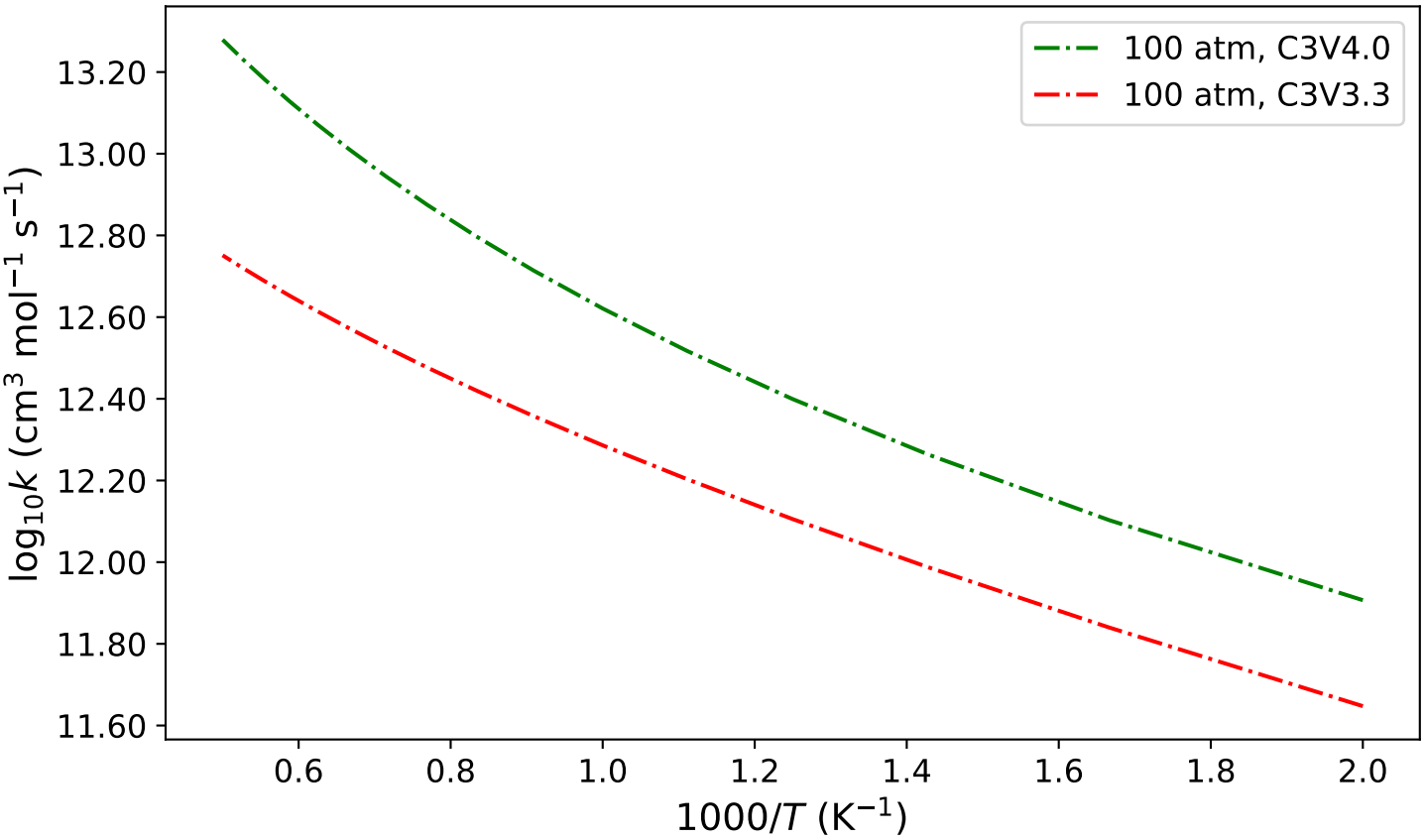



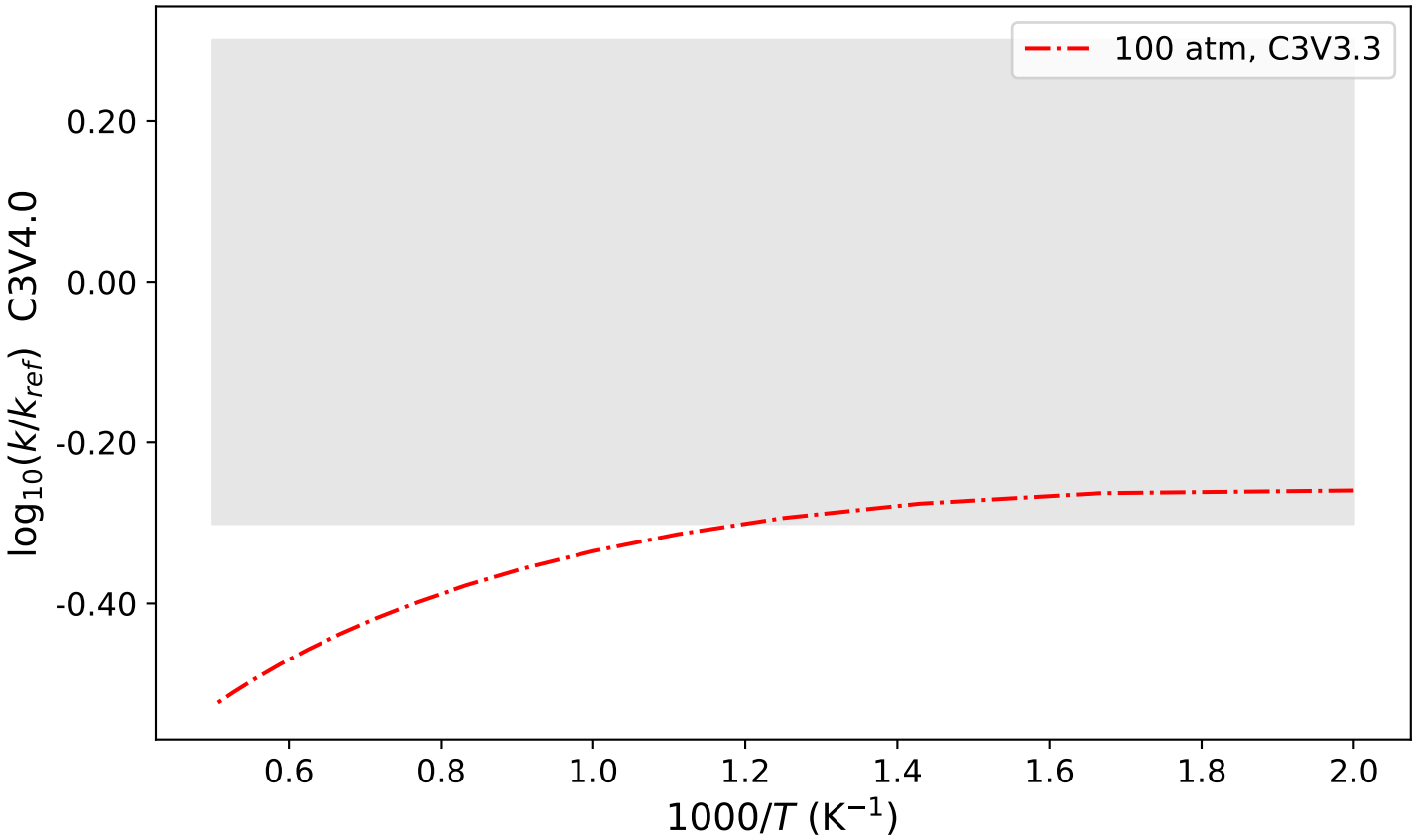
C3H6OH1-3 = NC3H7O



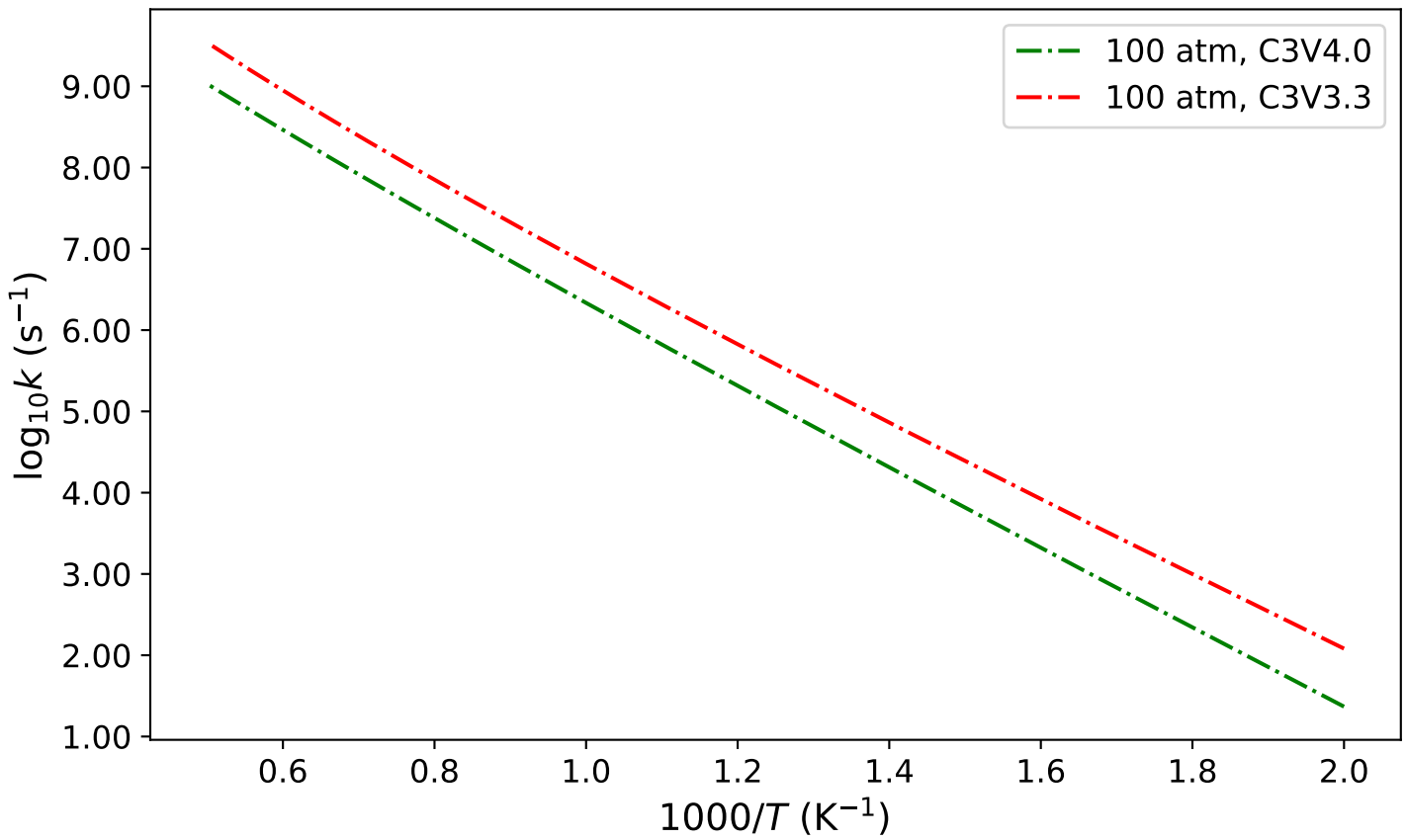



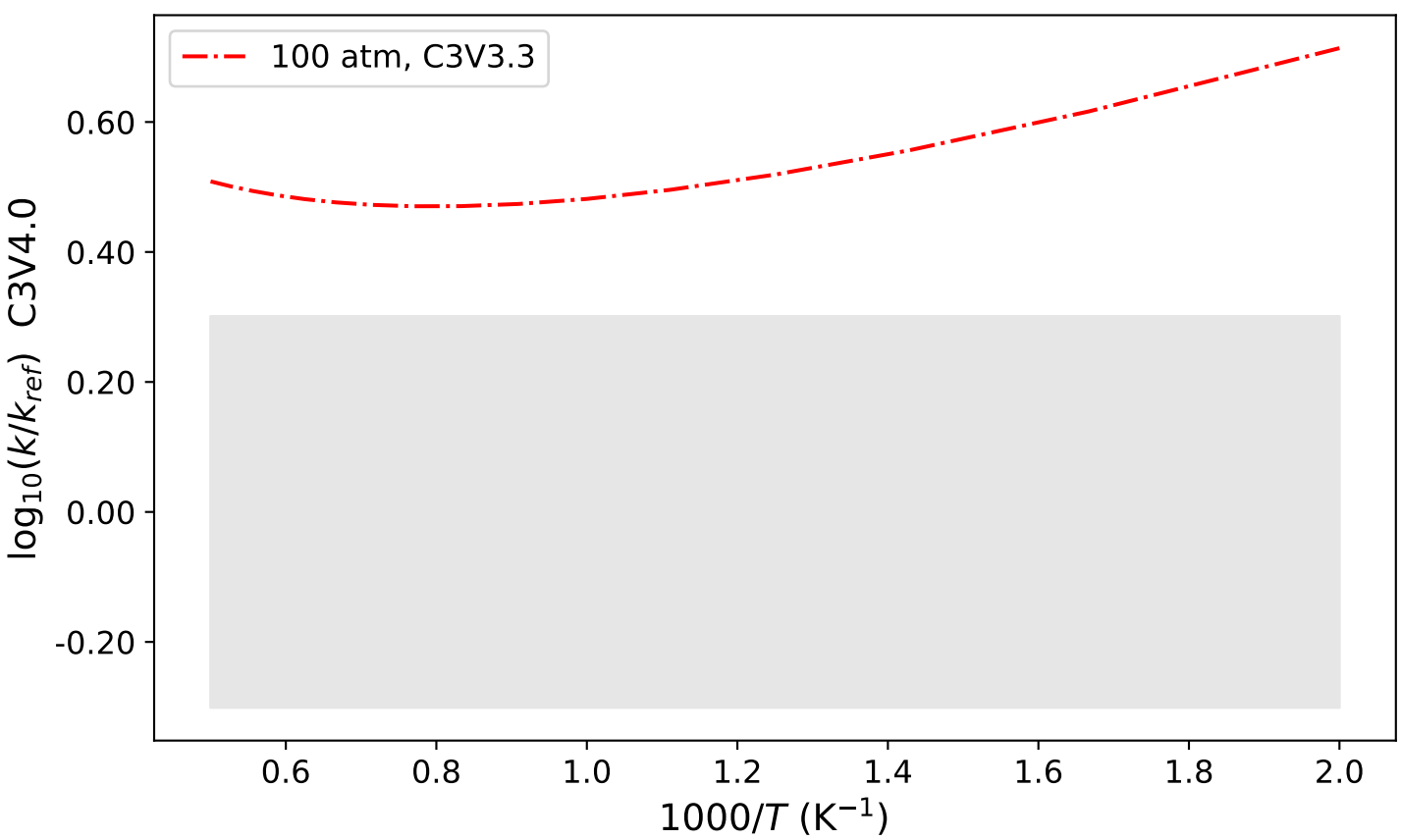
NC3H7OH + OH = H2O + C3H6OH1-3

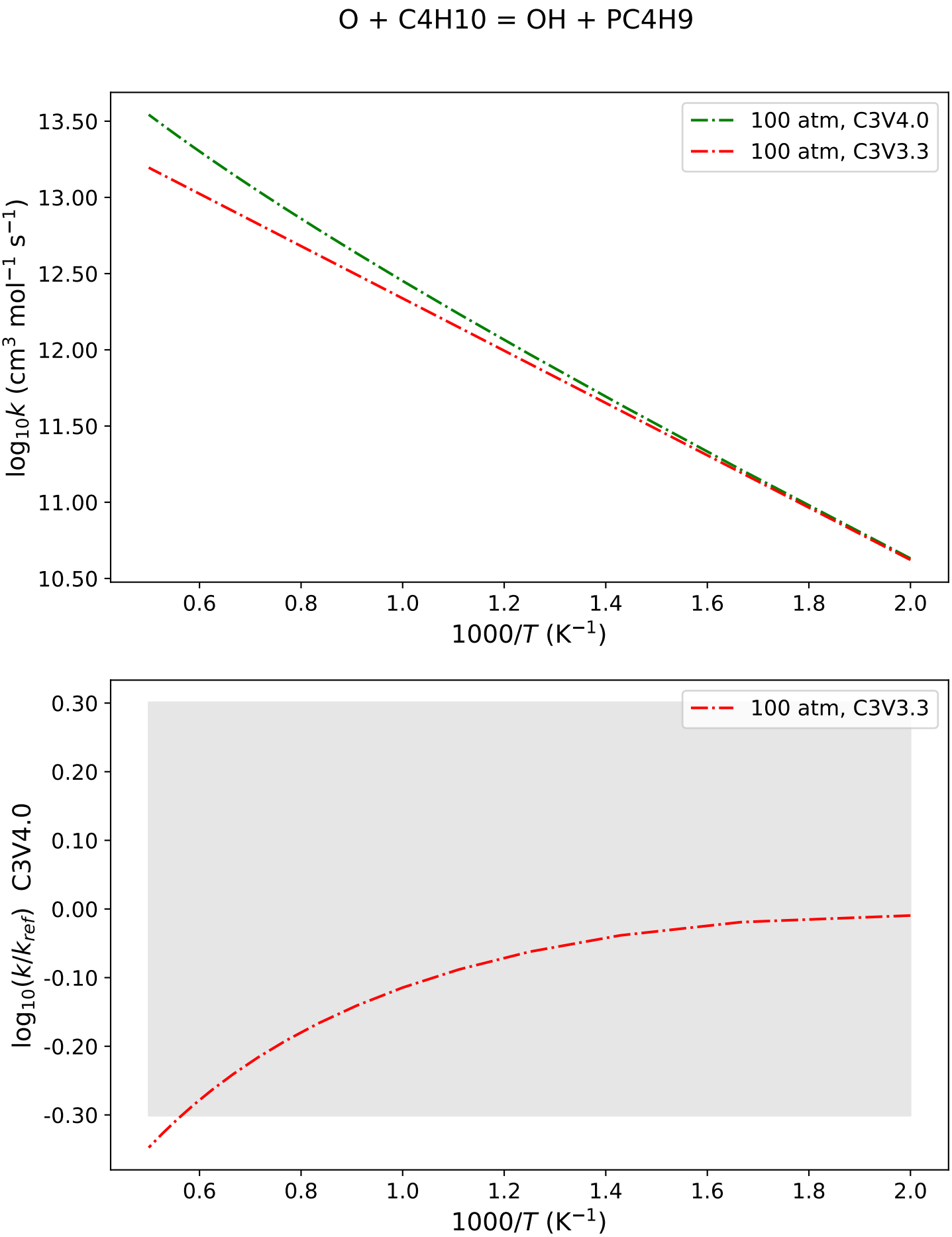


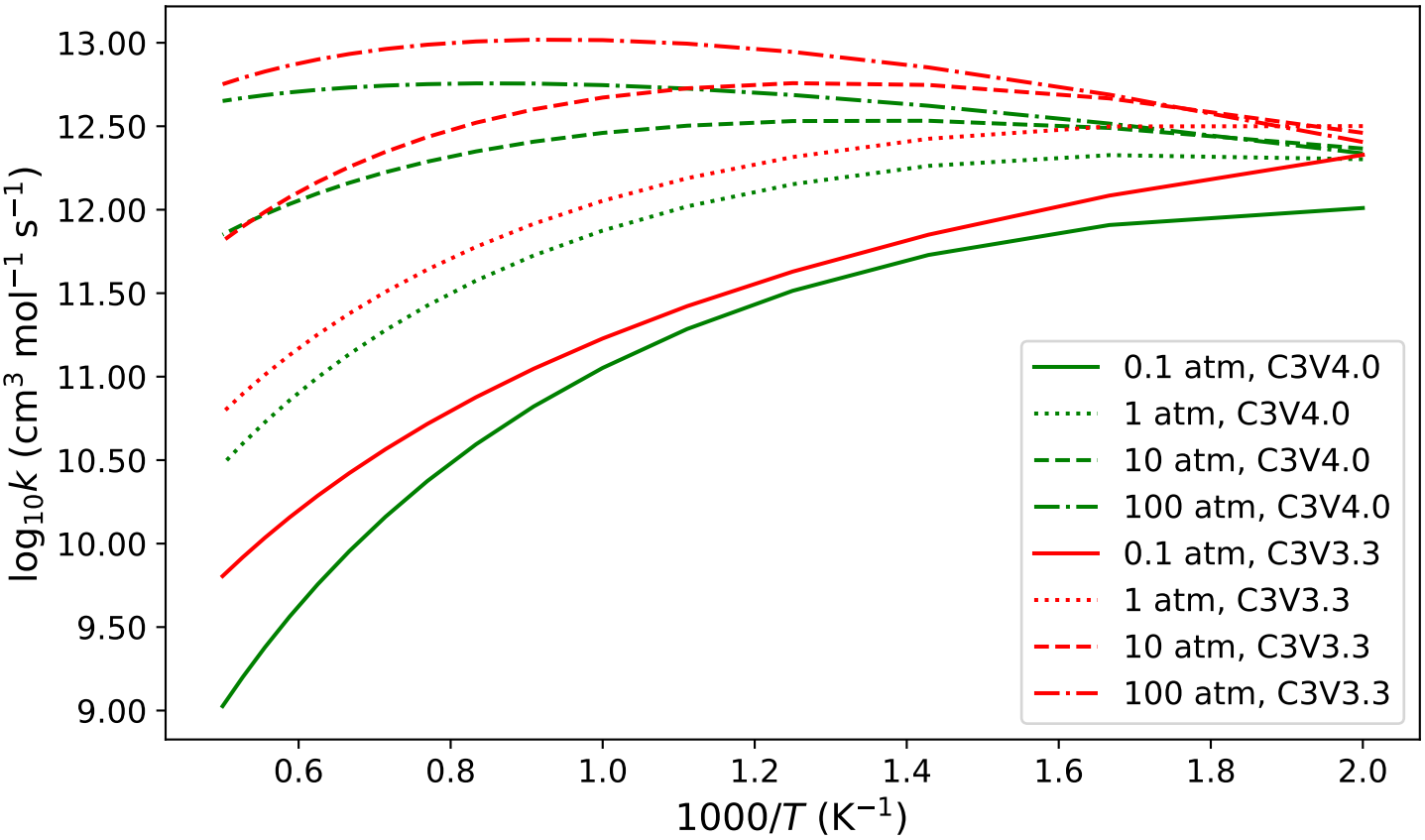
PC4H8OH-2O2 = SQC4H7OHP-4

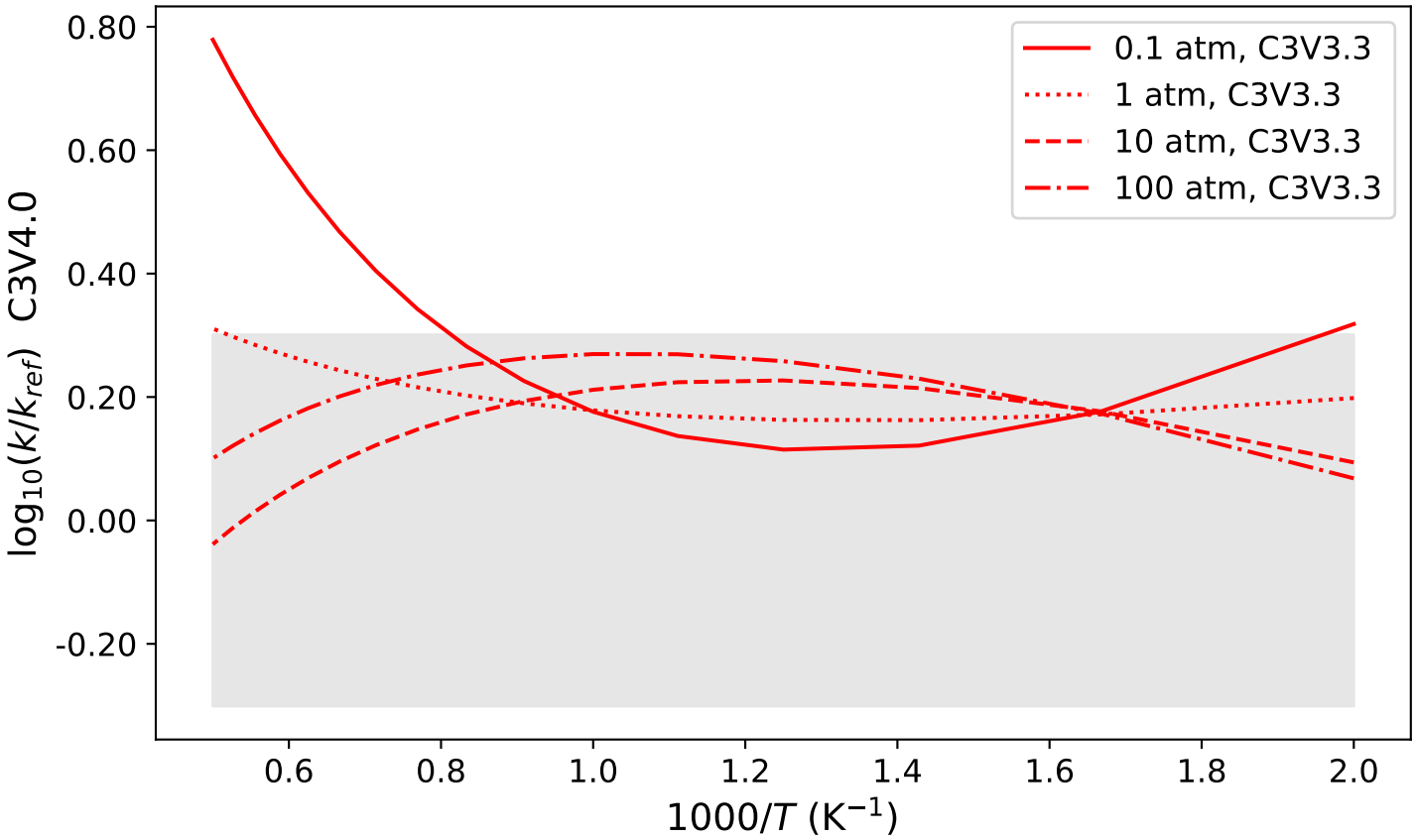


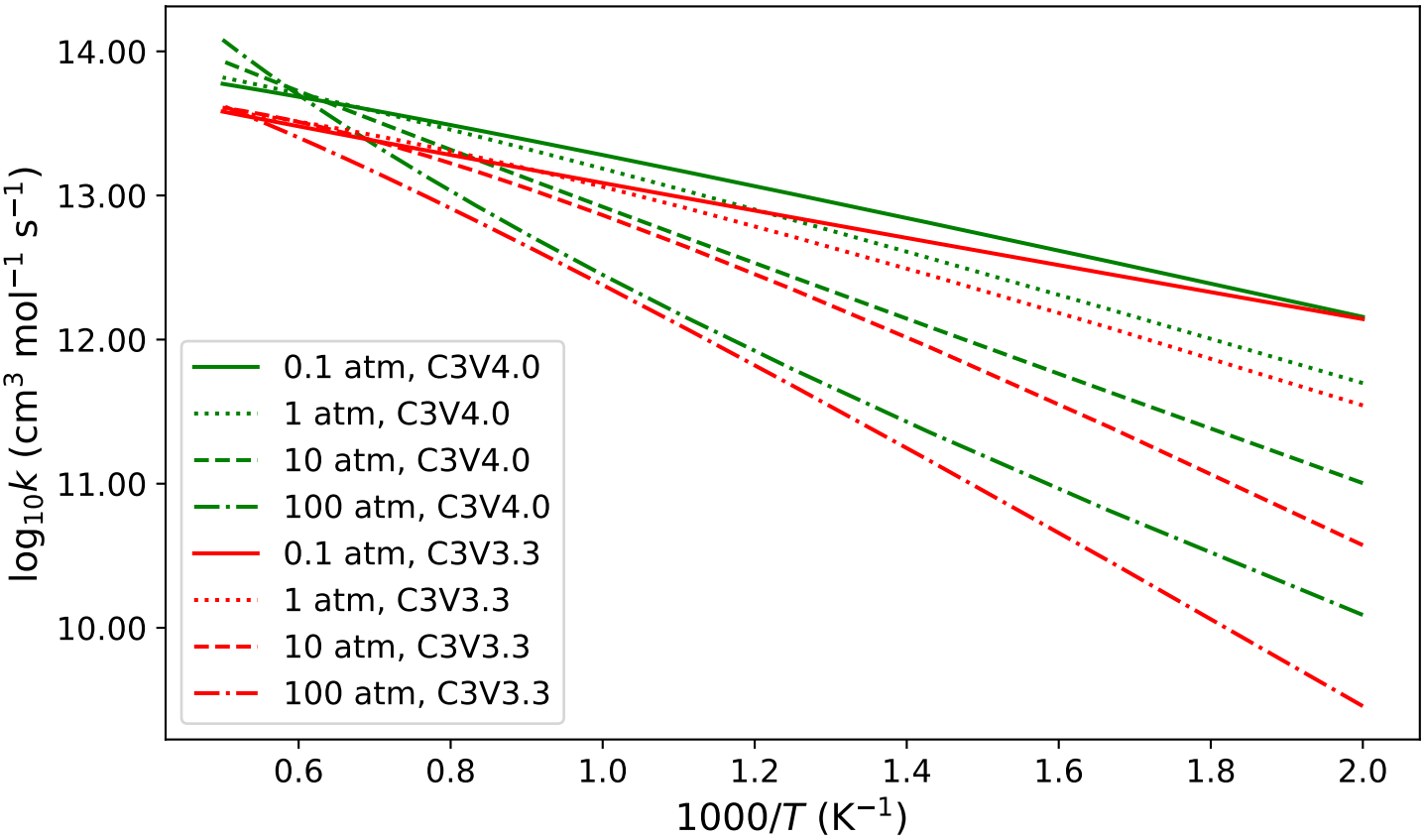

O + C4H10 = OH + PC4H9
100 atm, C3V4.0
100 atm, C3V3.3
$\log_{10}k$ (cm$^3$ mol$^{-1}$ s$^{-1}$)
13.50
13.00
12.50
12.00
11.50
11.00
10.50
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ (K$^{-1}$)
100 atm, C3V3.3
$\log_{10}(k/k_{ref})$ C3V4.0
0.30
0.20
0.10
0.00
-0.10
-0.20
-0.30
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ (K$^{-1}$)

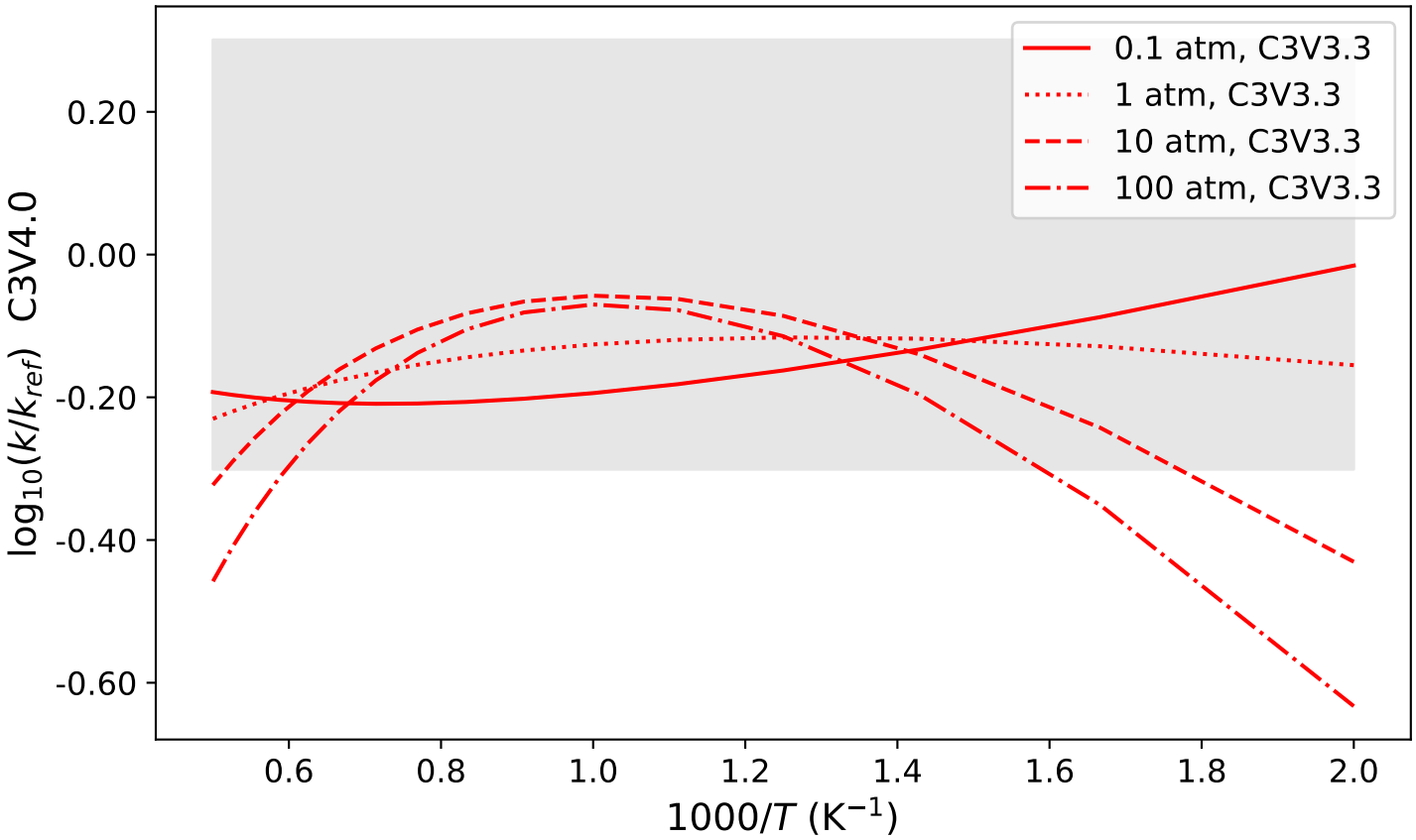
H + C4H8-1 = SC4H9



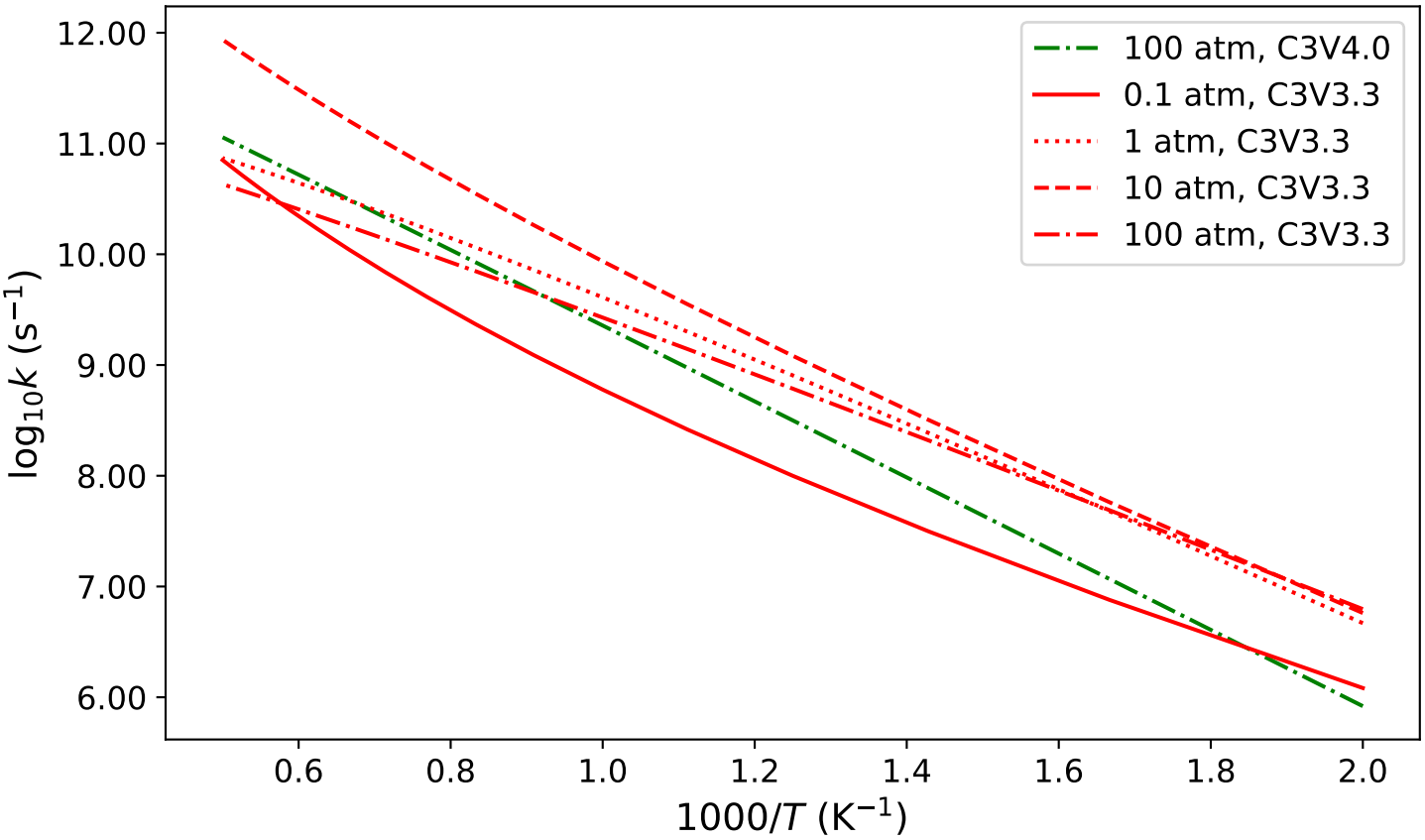



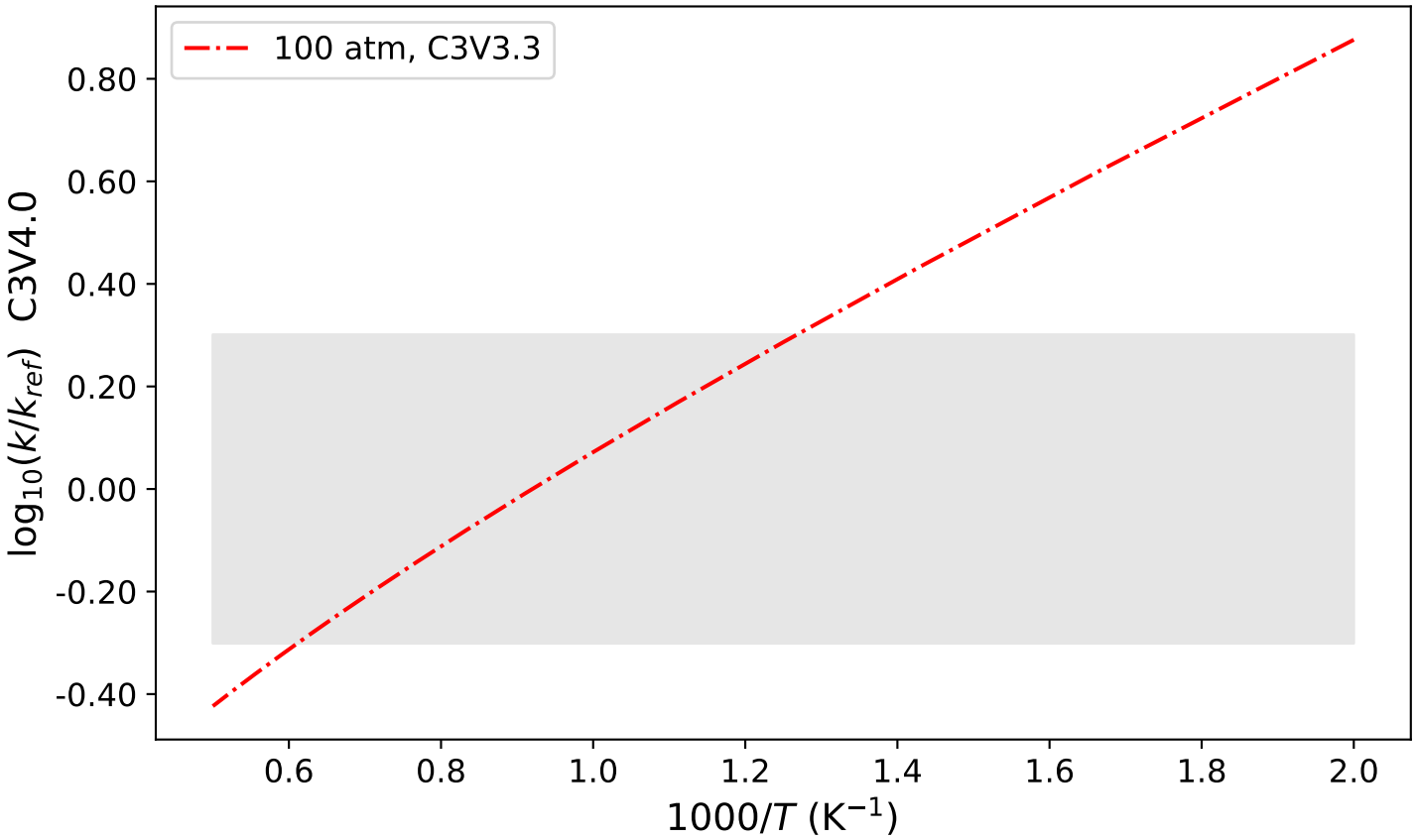
H + C4H8-1 = CH3 + C3H6



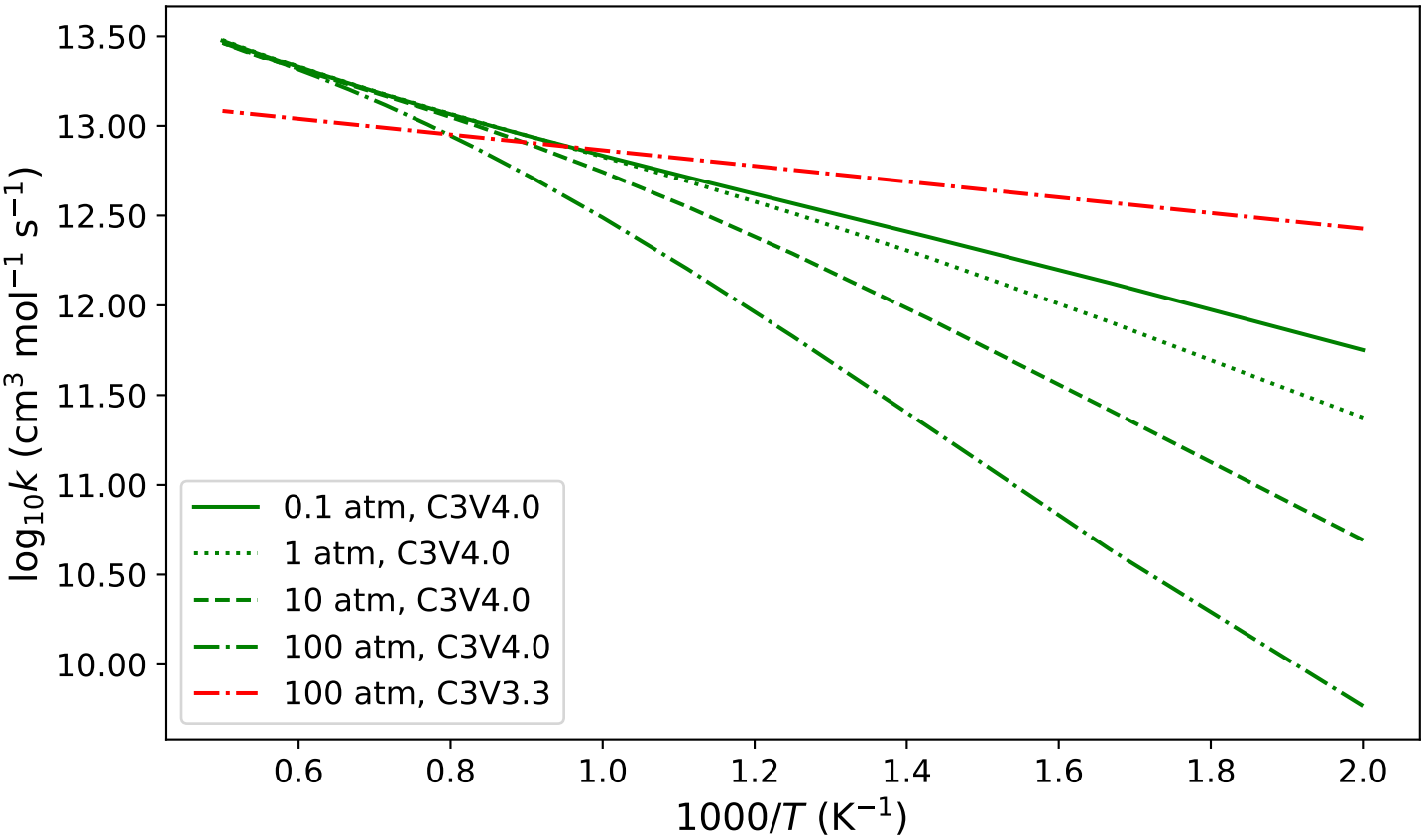



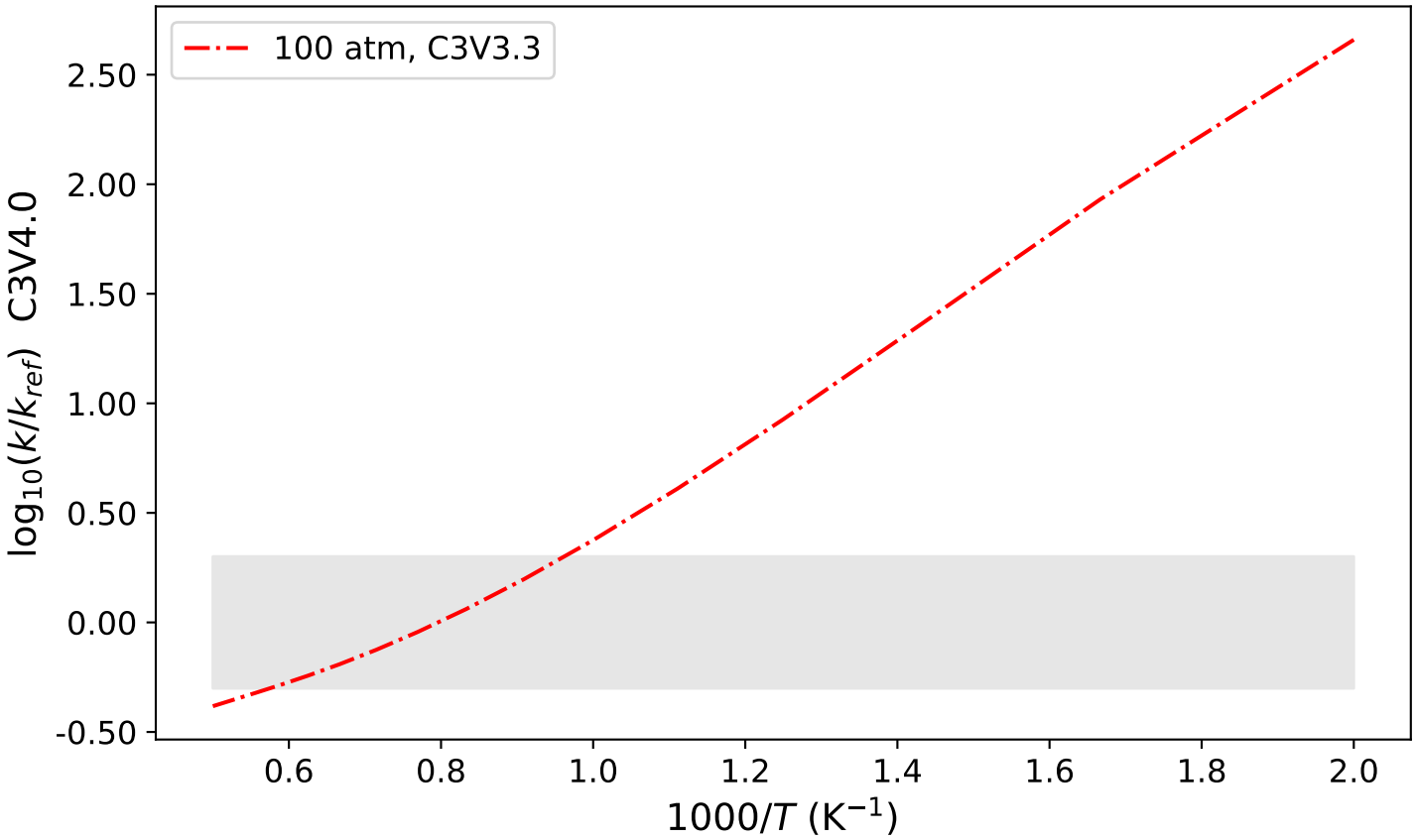
CH2CCH2OOH = OH + C3H4O1-2



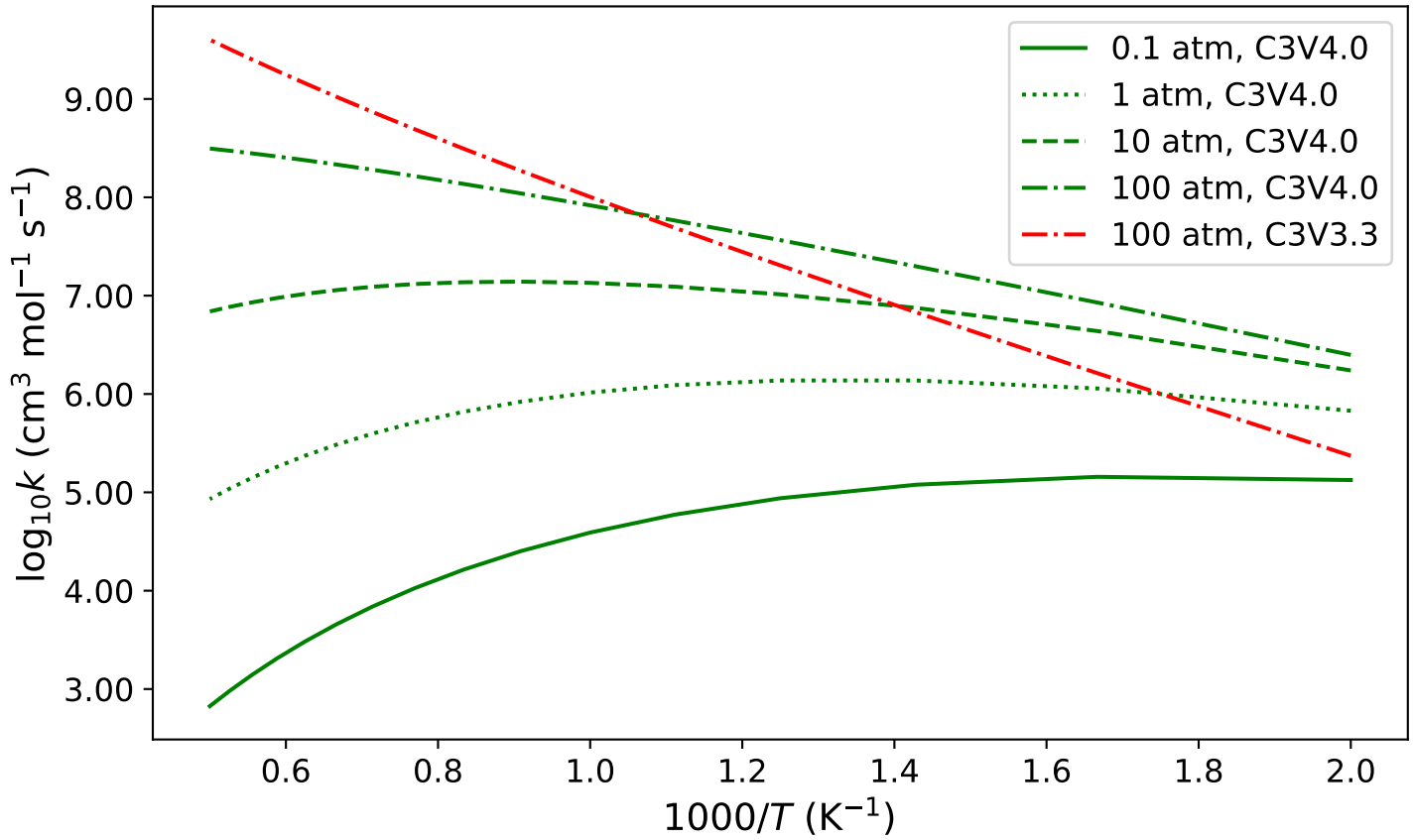



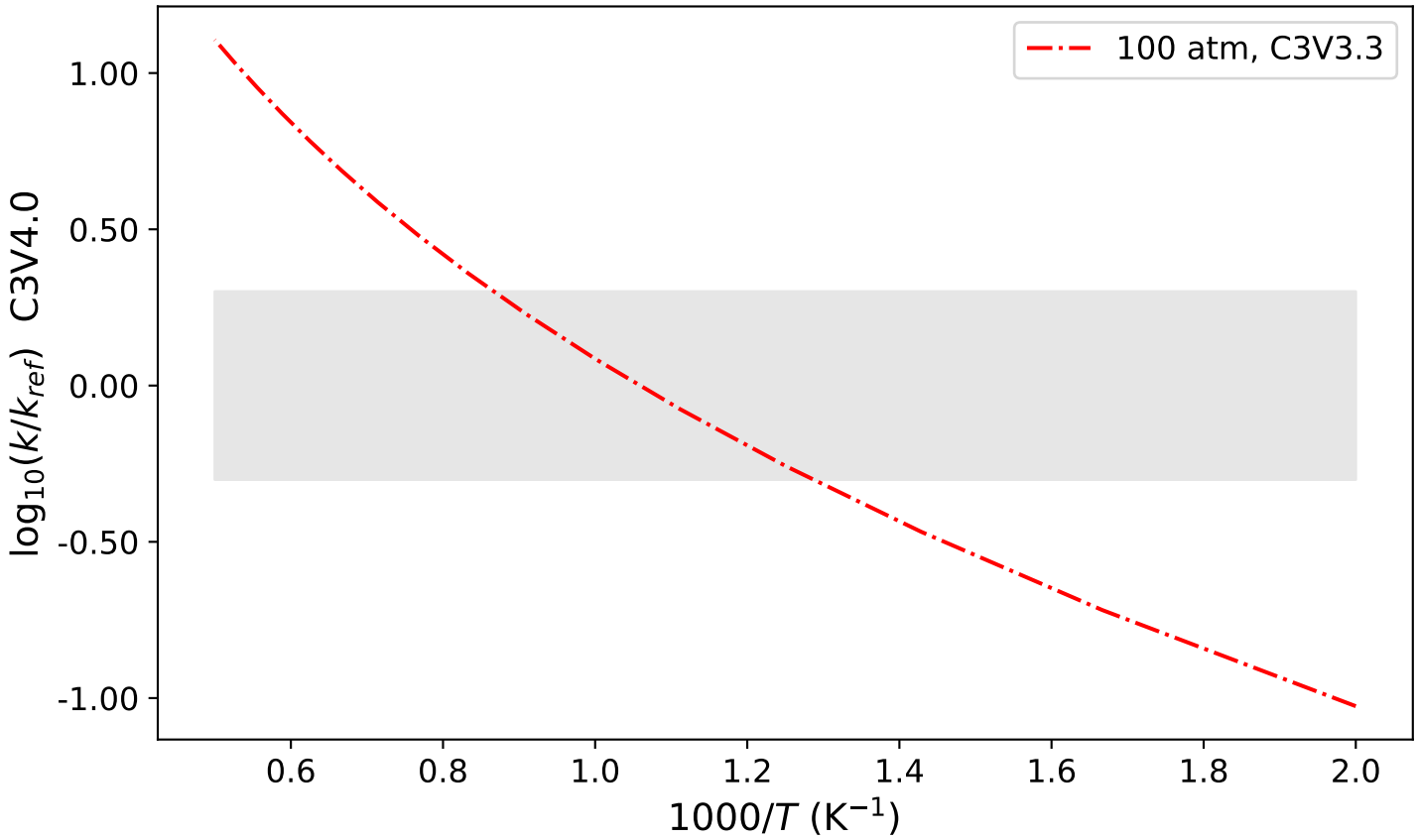
H + C4H612 = CH3 + C3H4-P



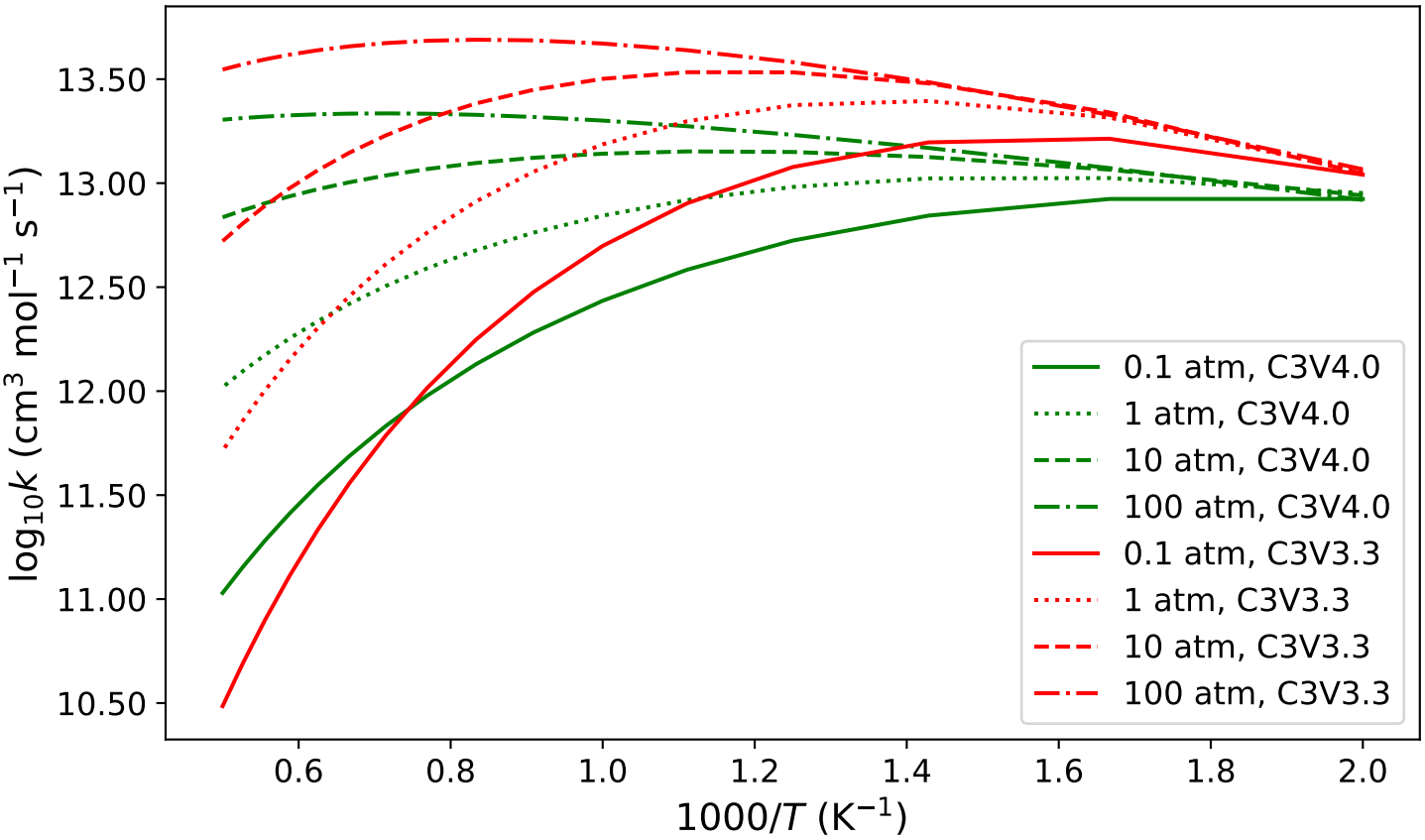



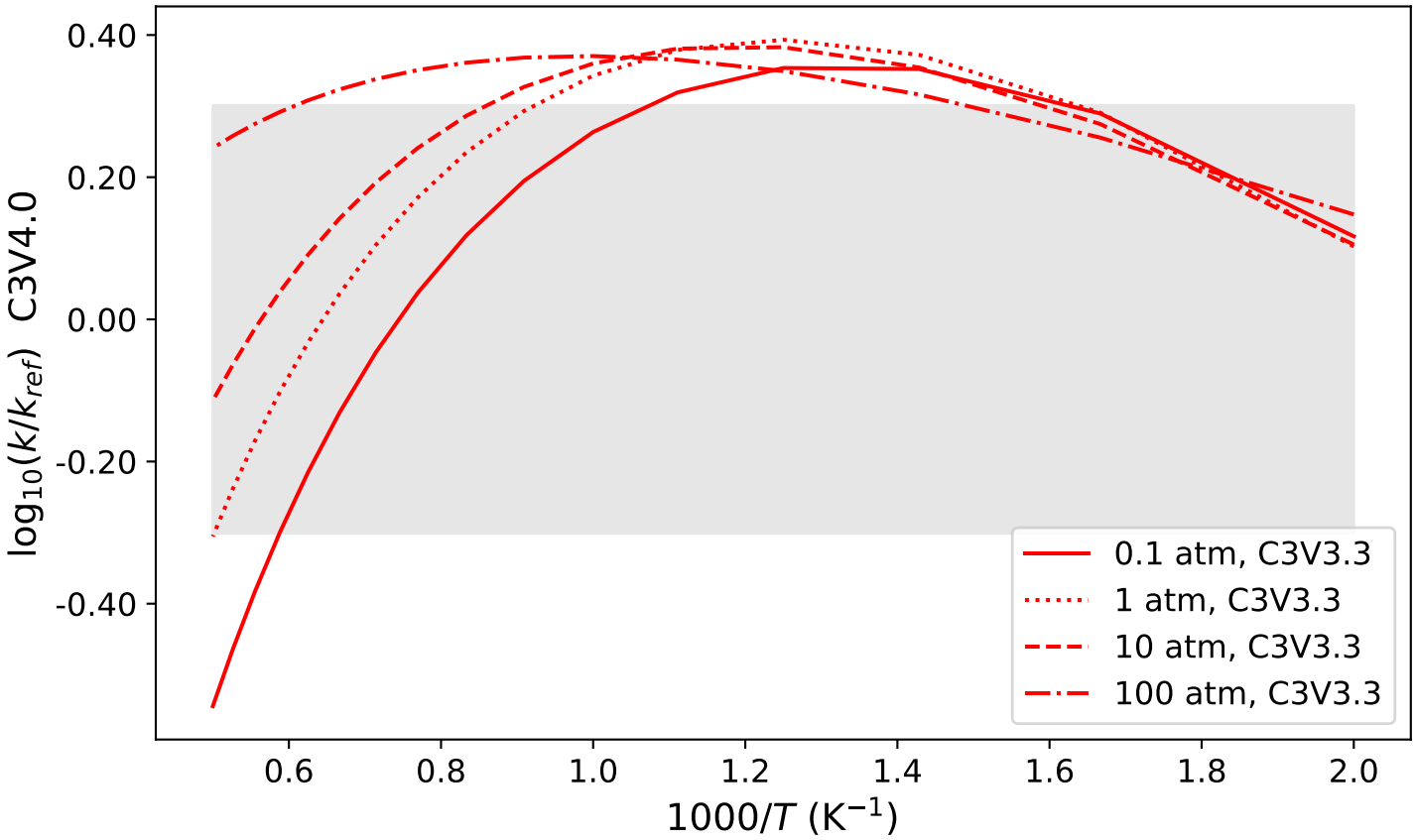
HO2 + C4H6 = C4H71-3O2



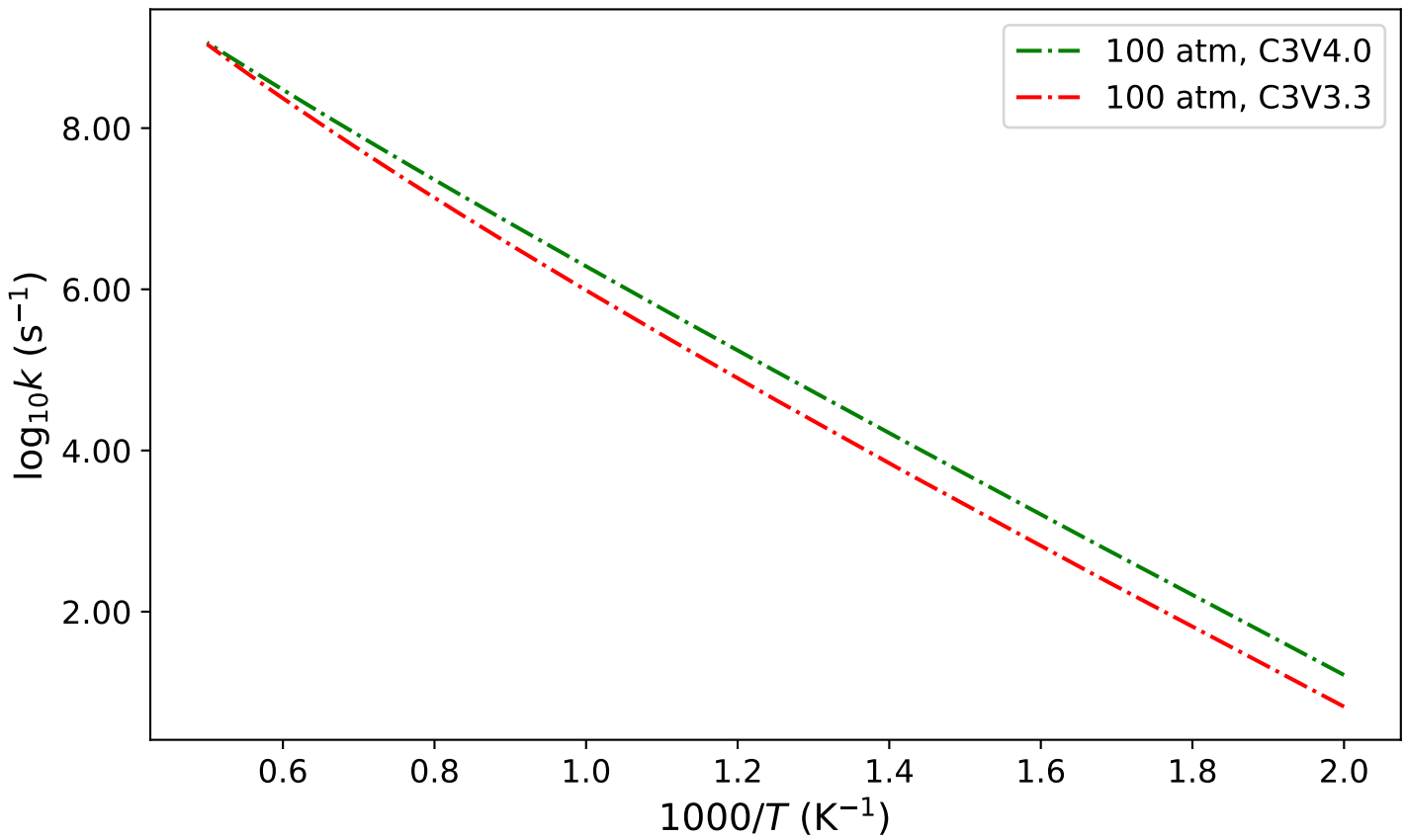



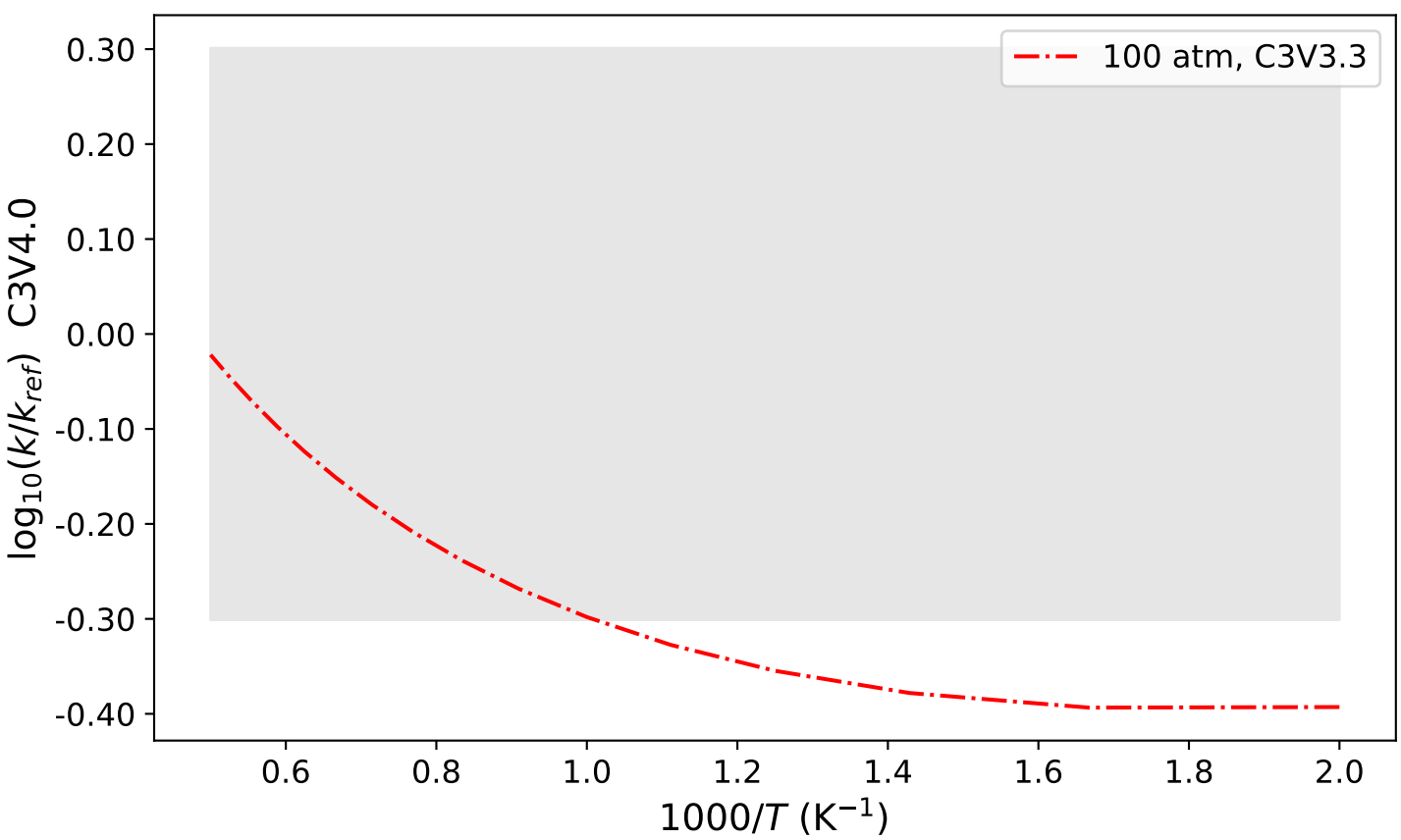
H + IC4H8 = TC4H9



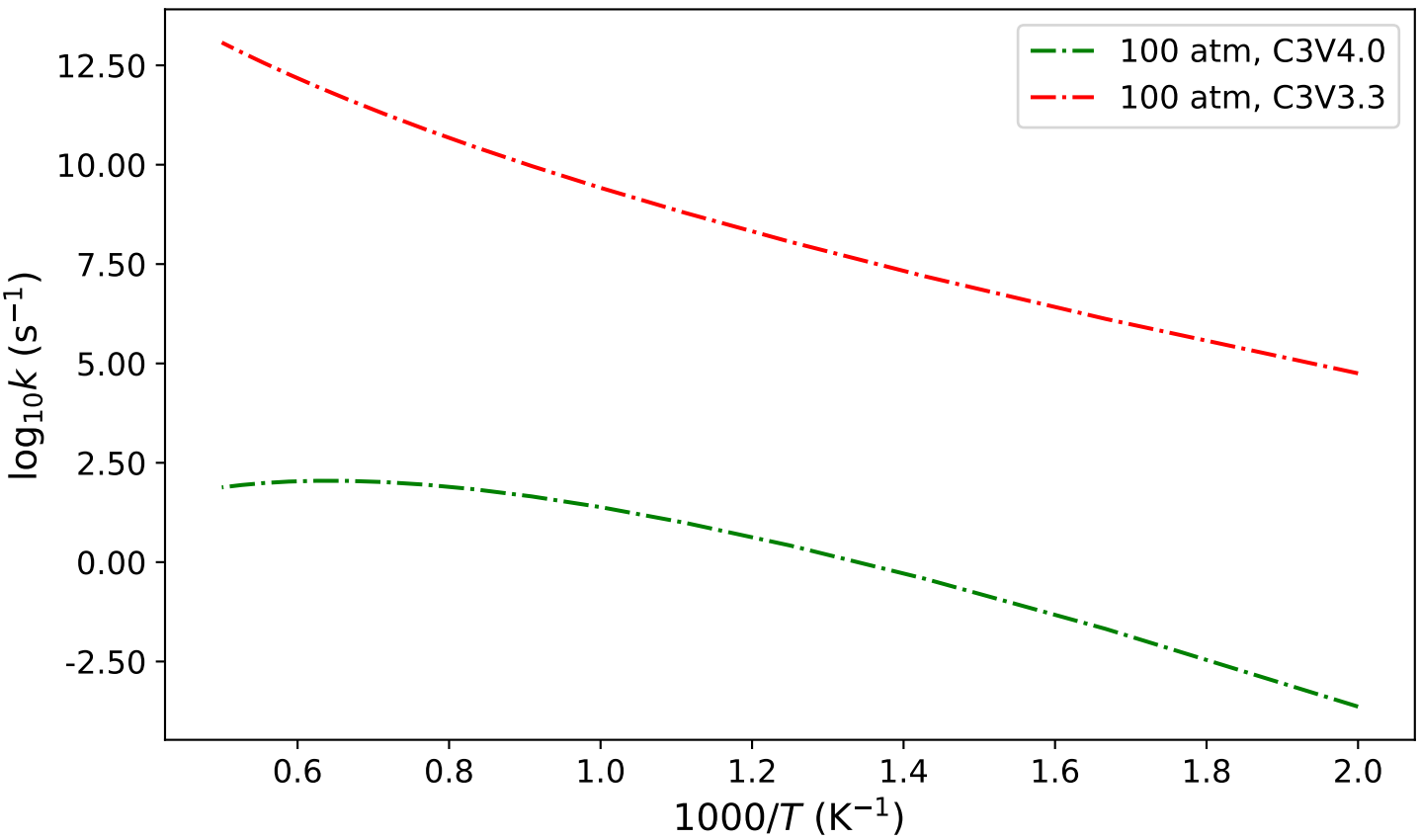



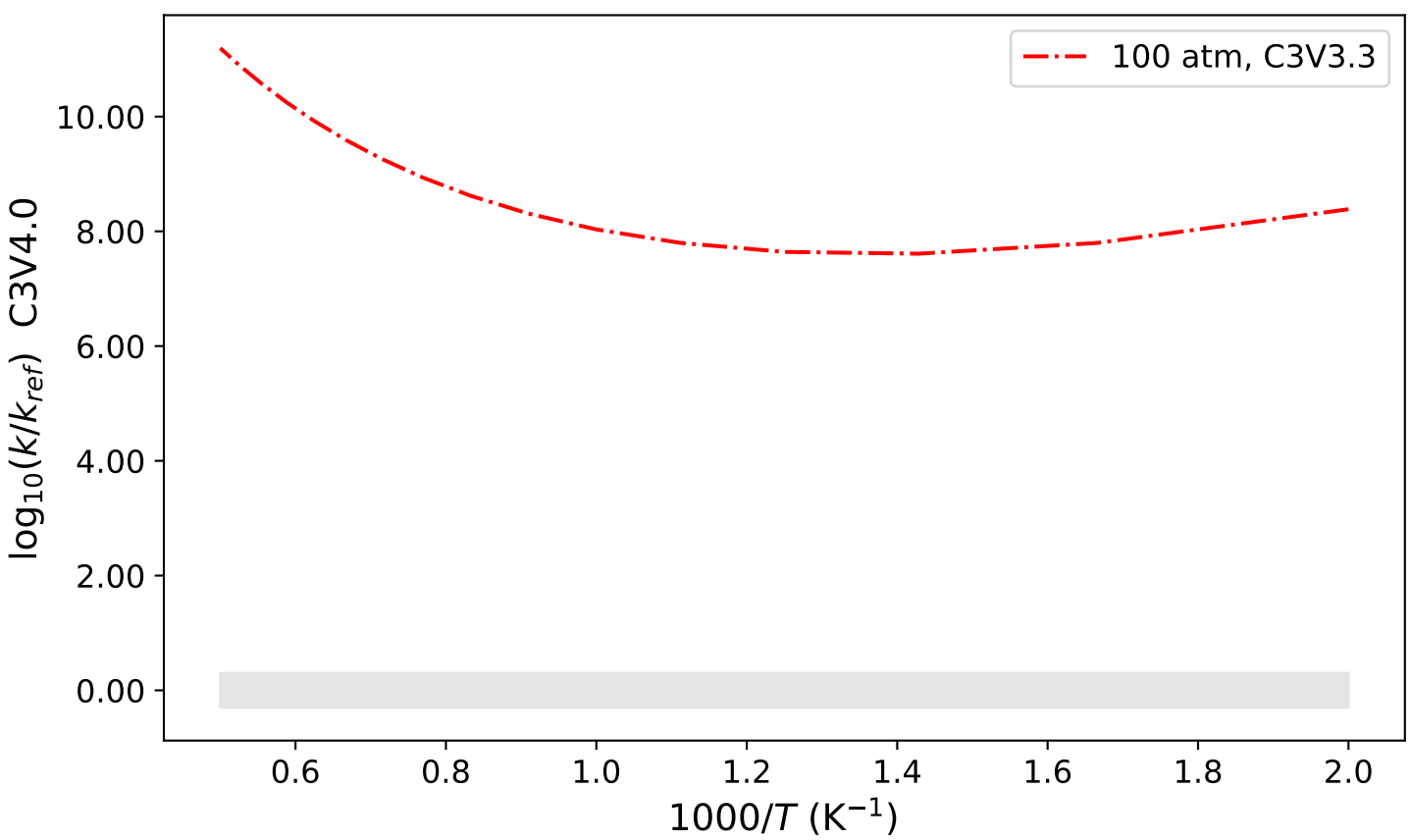
C2H5CO3 = QCH2CH2CO3H



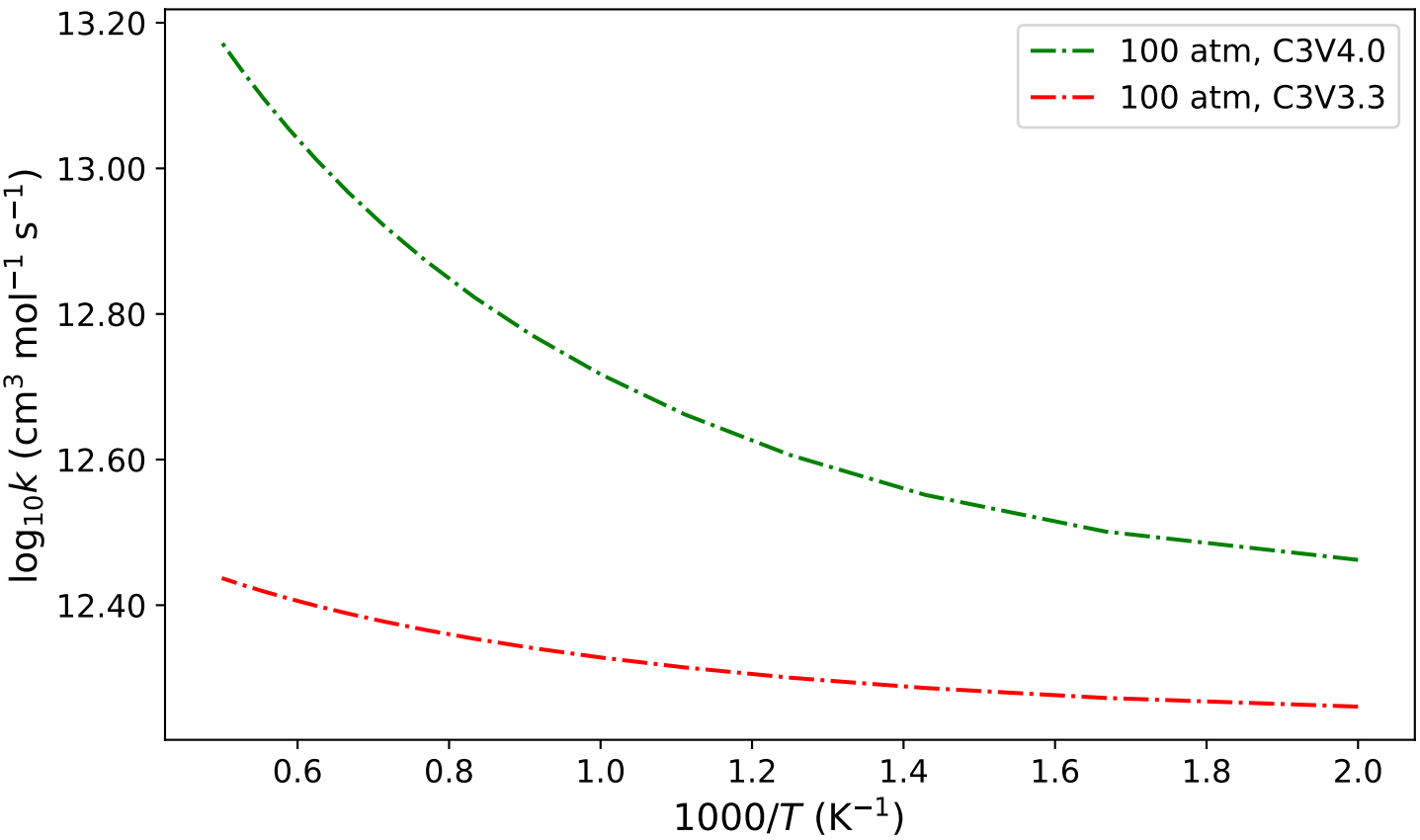



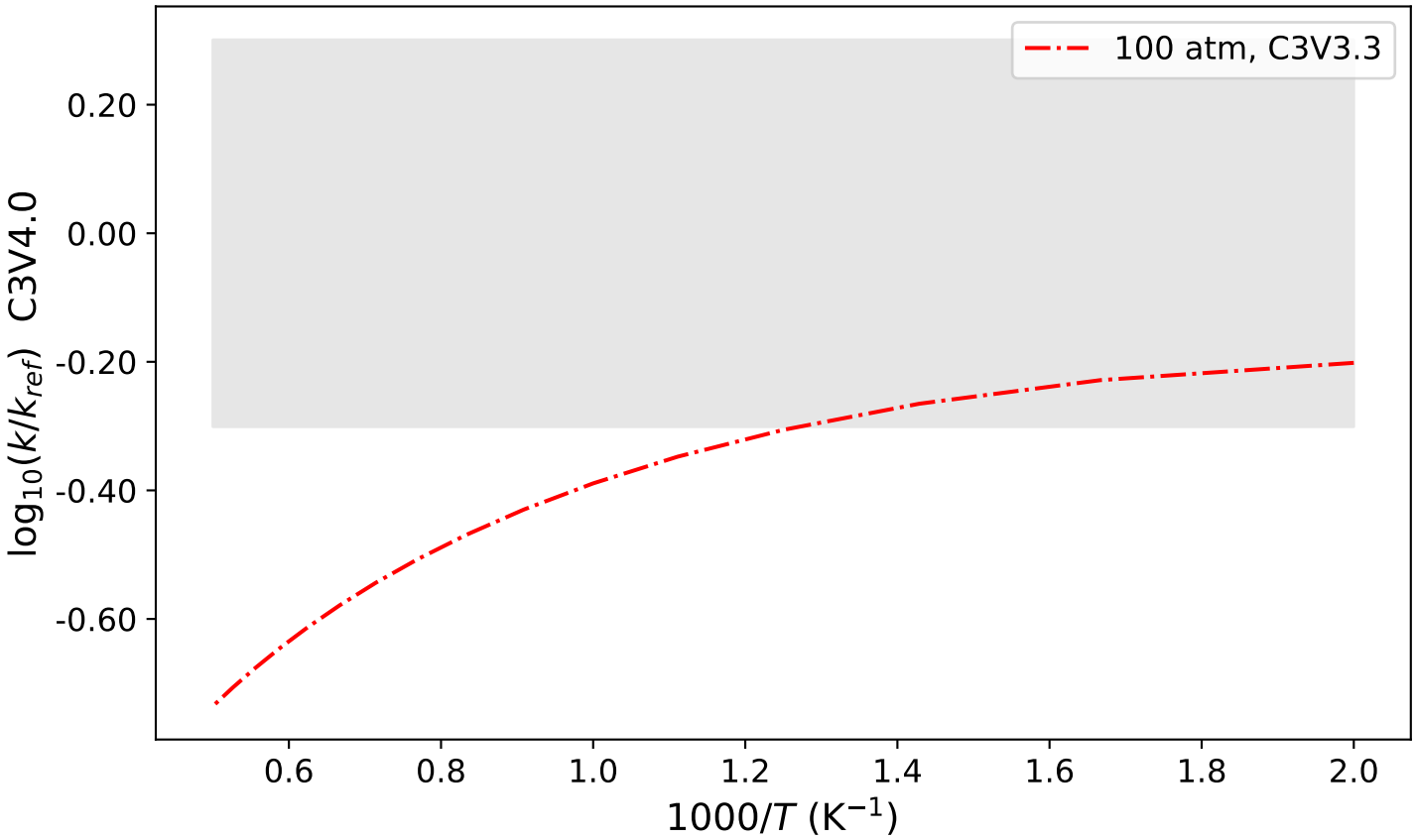
QCH3CHCO3H = HO2 + CH3CHCO



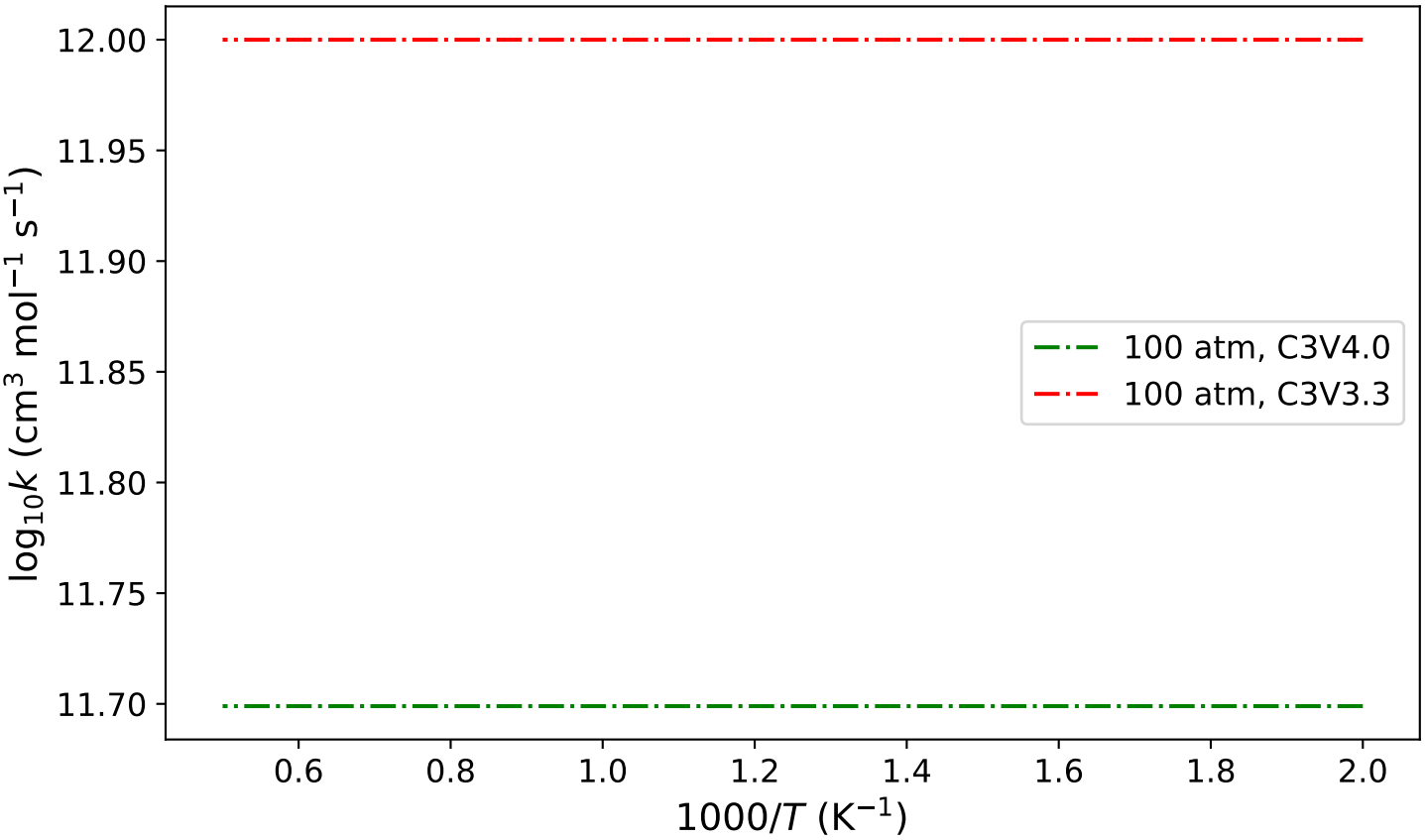



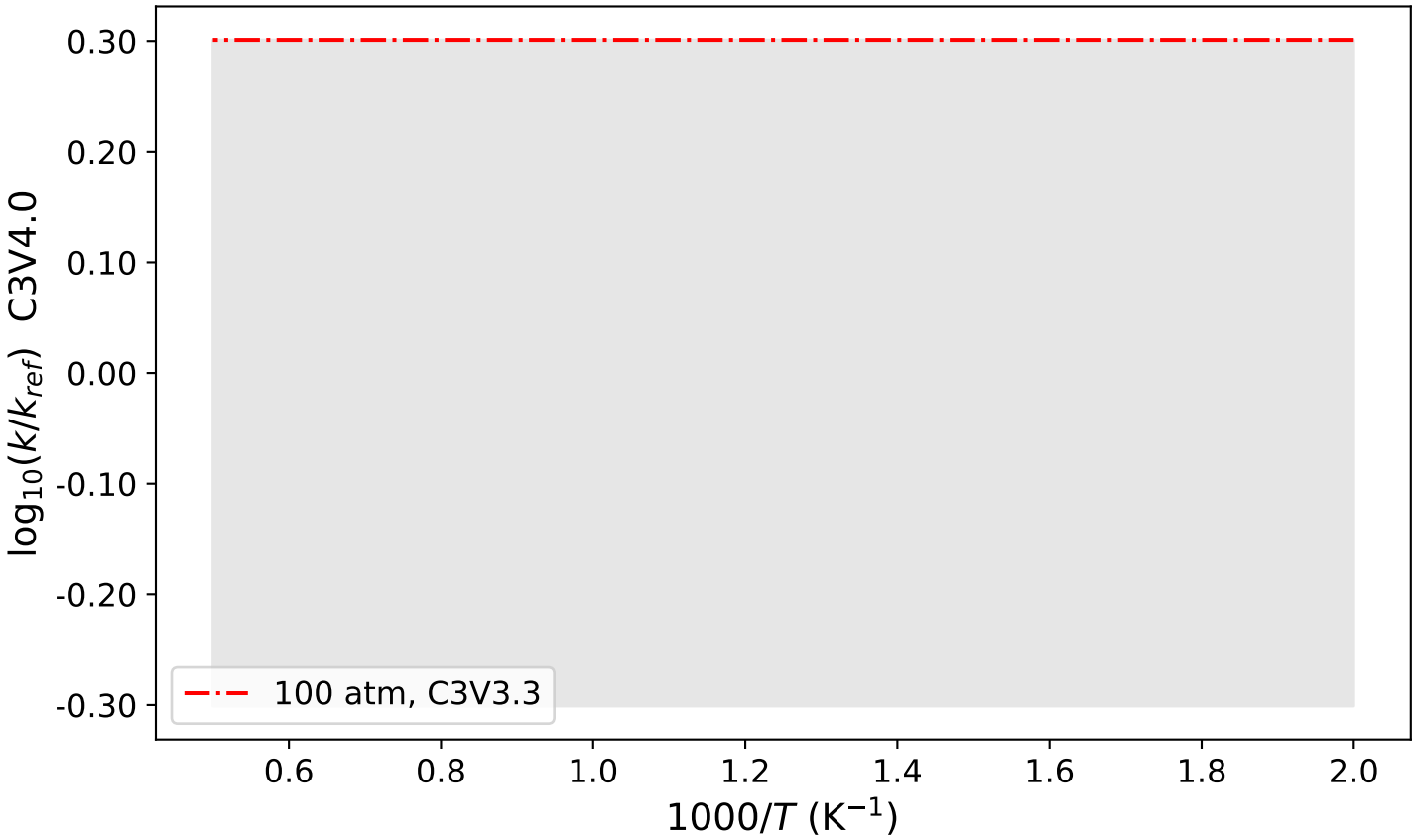
NC3H7OH + OH = H2O + C3H6OH1-1



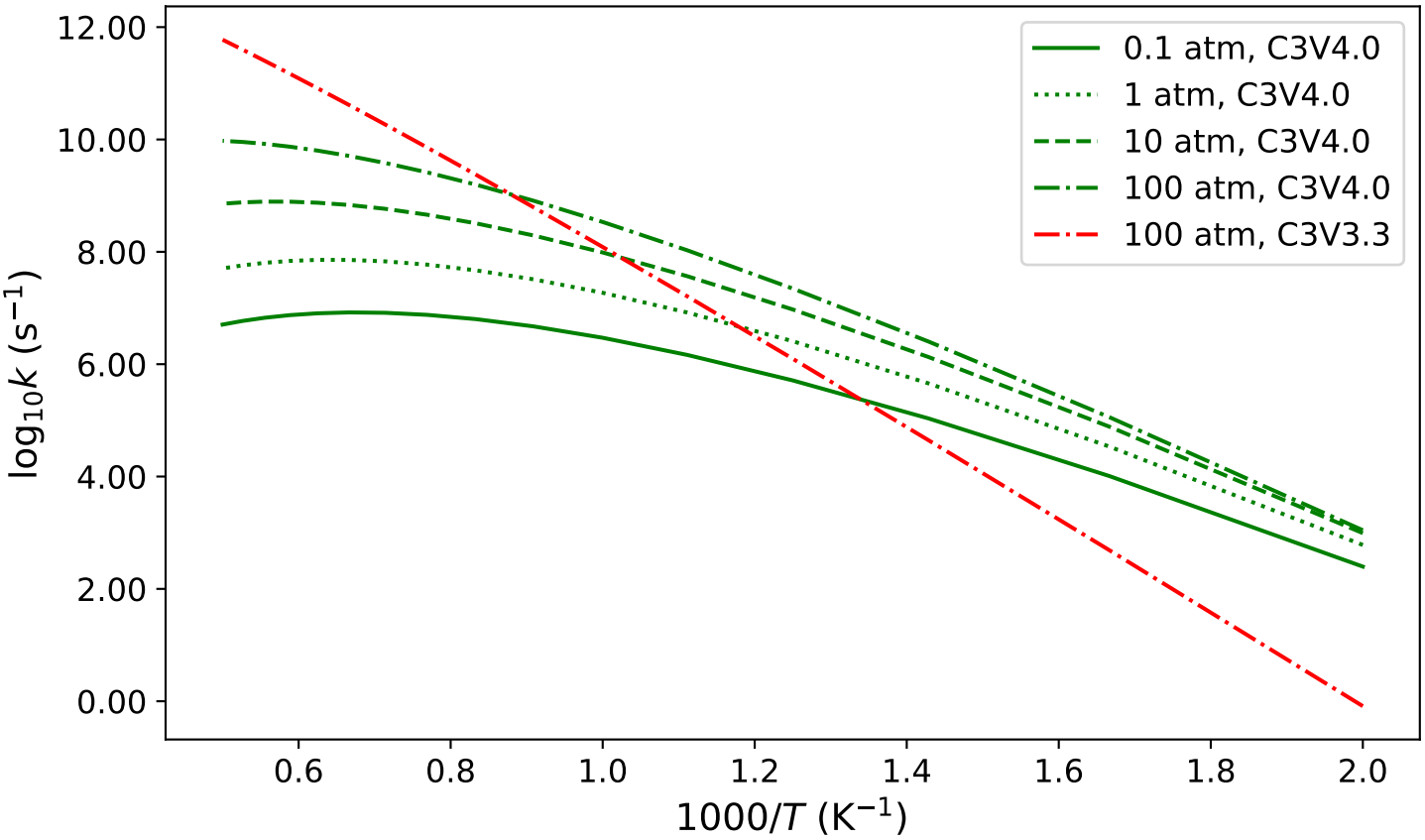



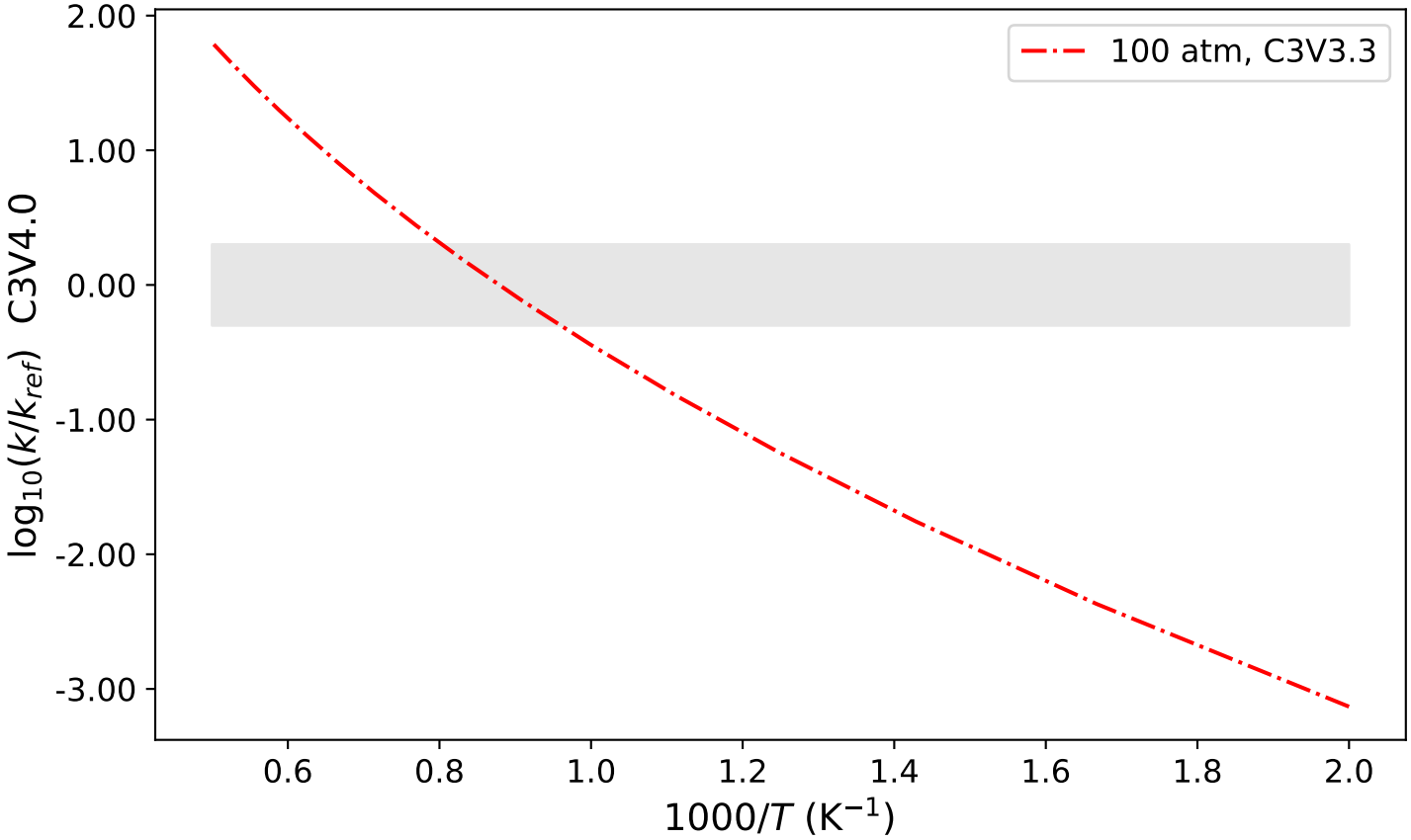
HO2 + CH3COCH2O2 = O2 + C3KET21



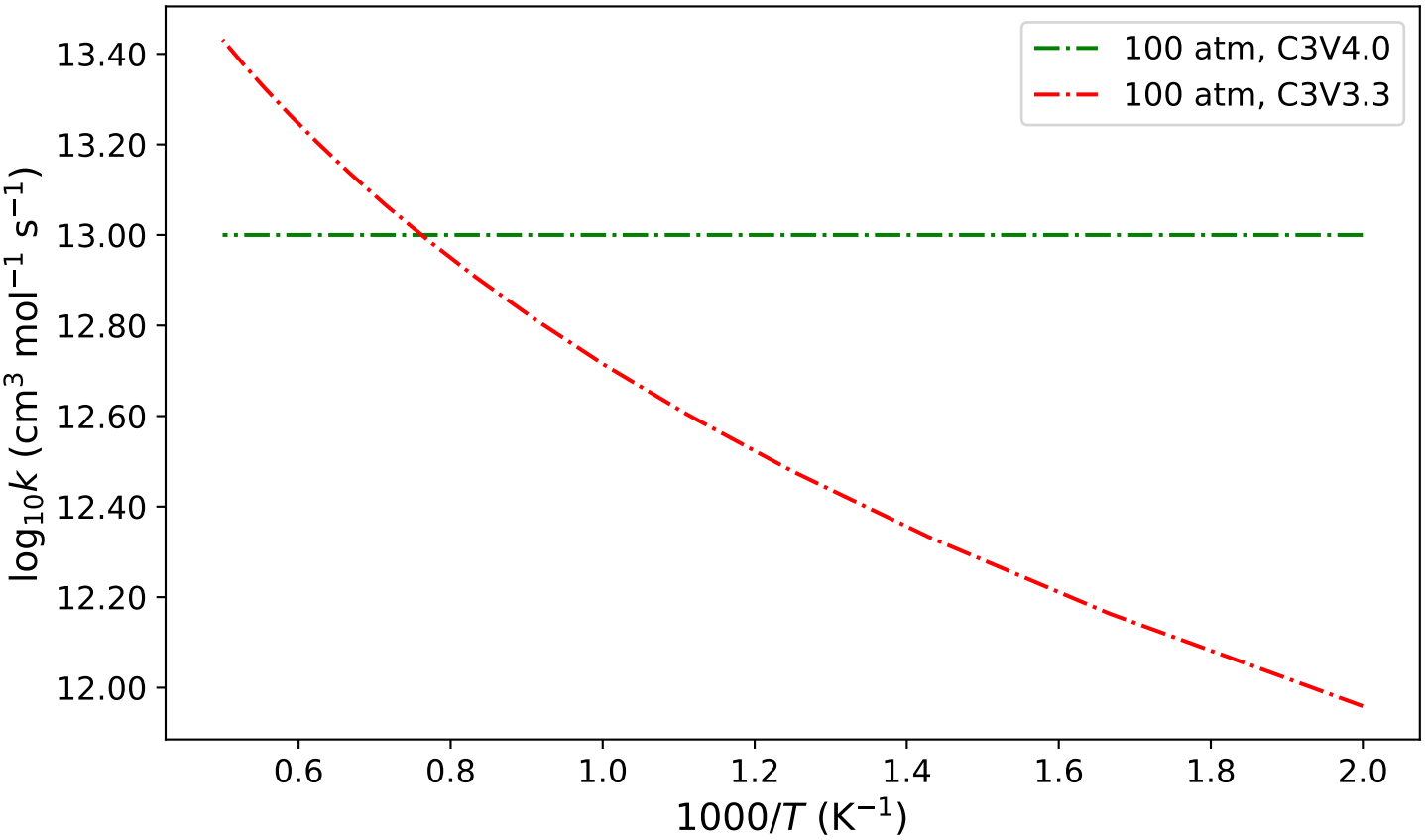



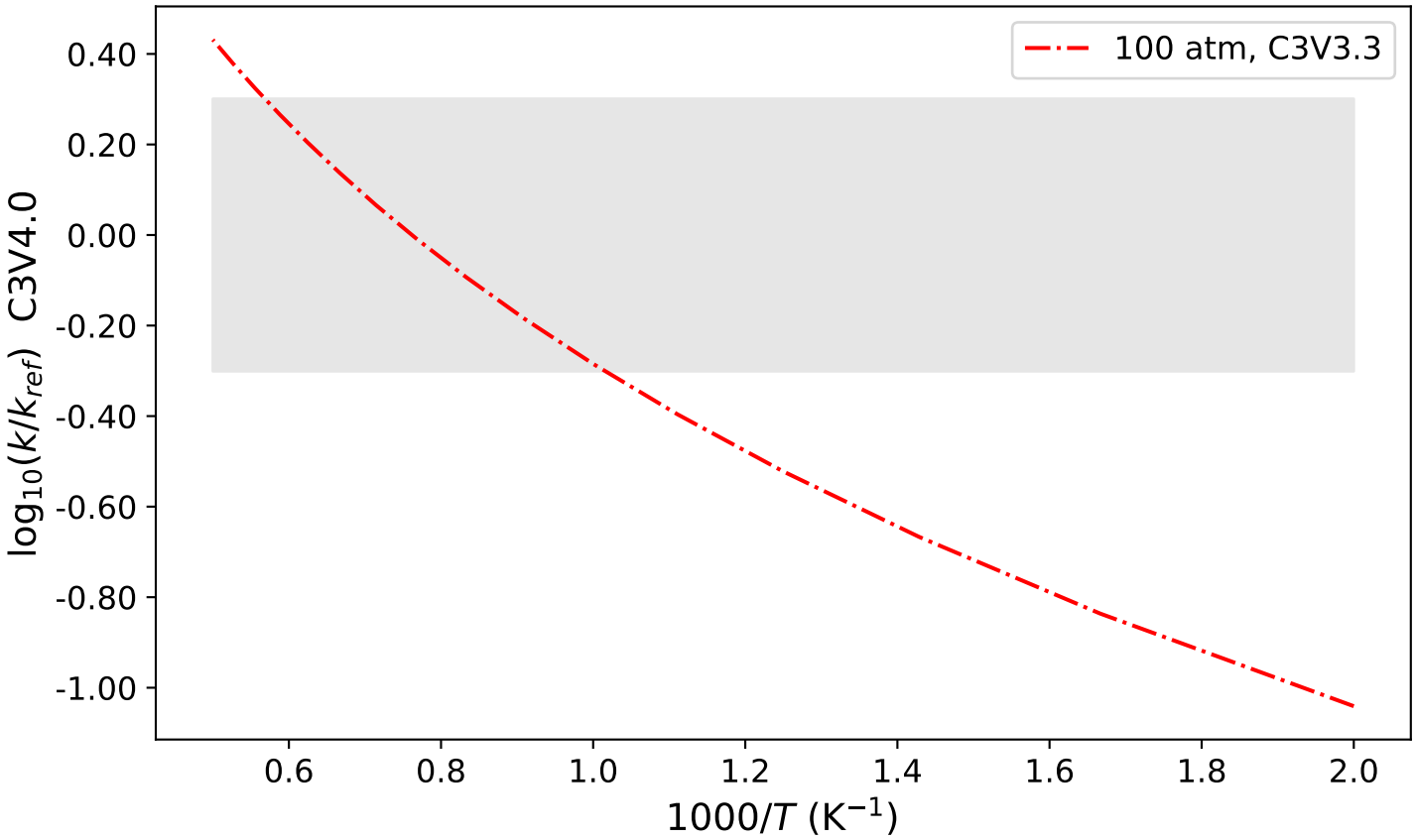
SC3H5CO = CO + C3H5-S



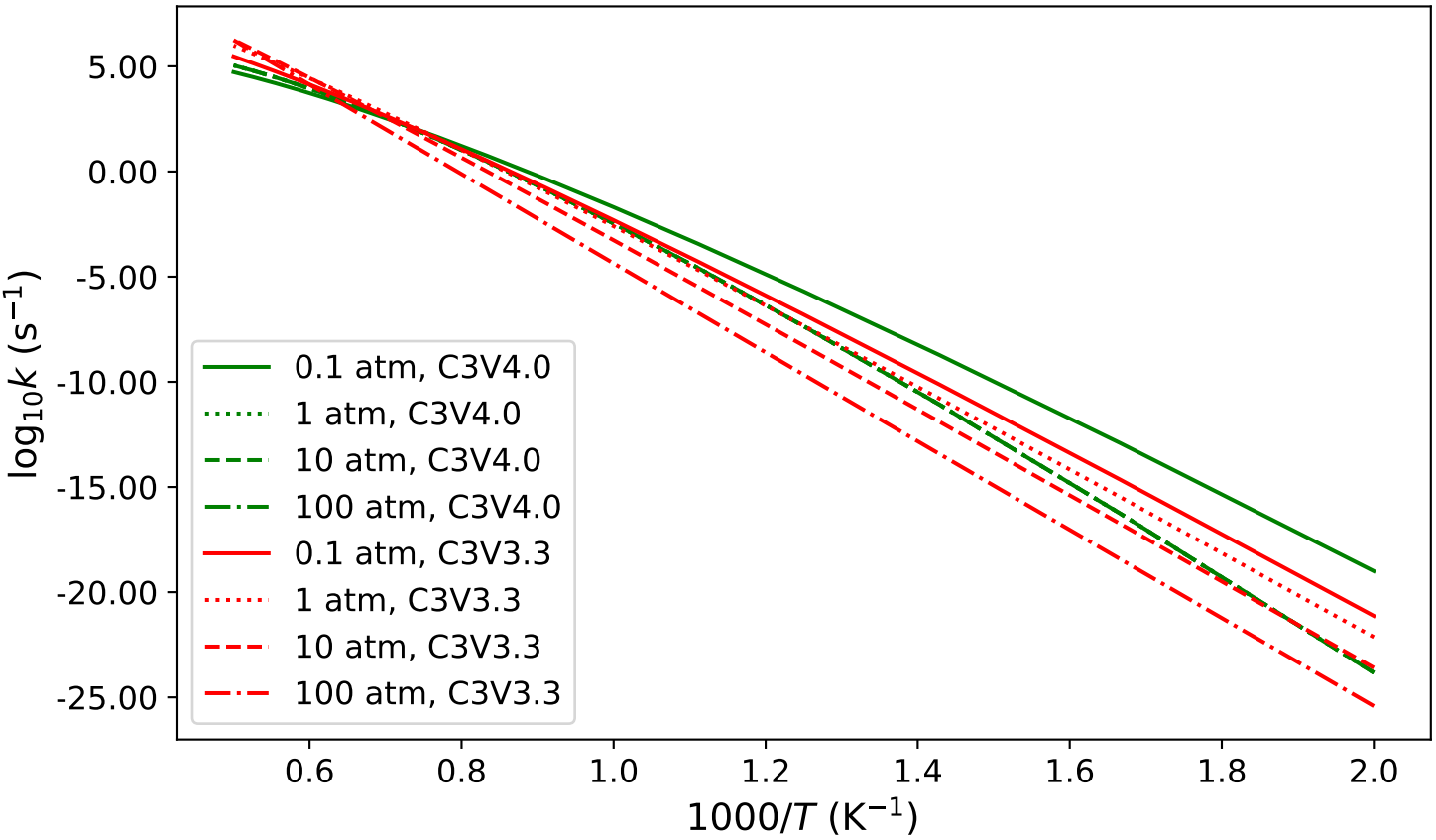



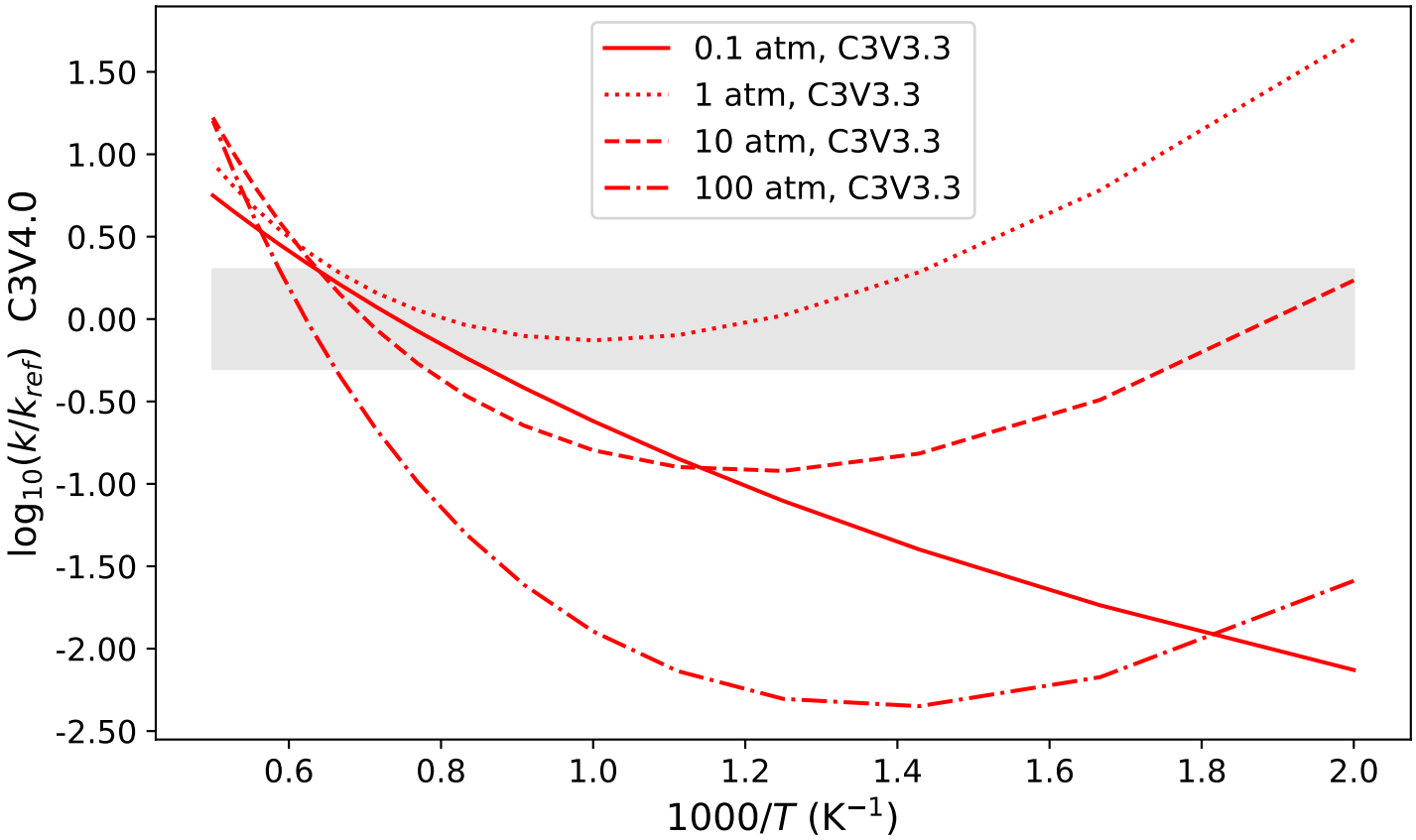
O2 + C3H2 = CO2 + C2H2



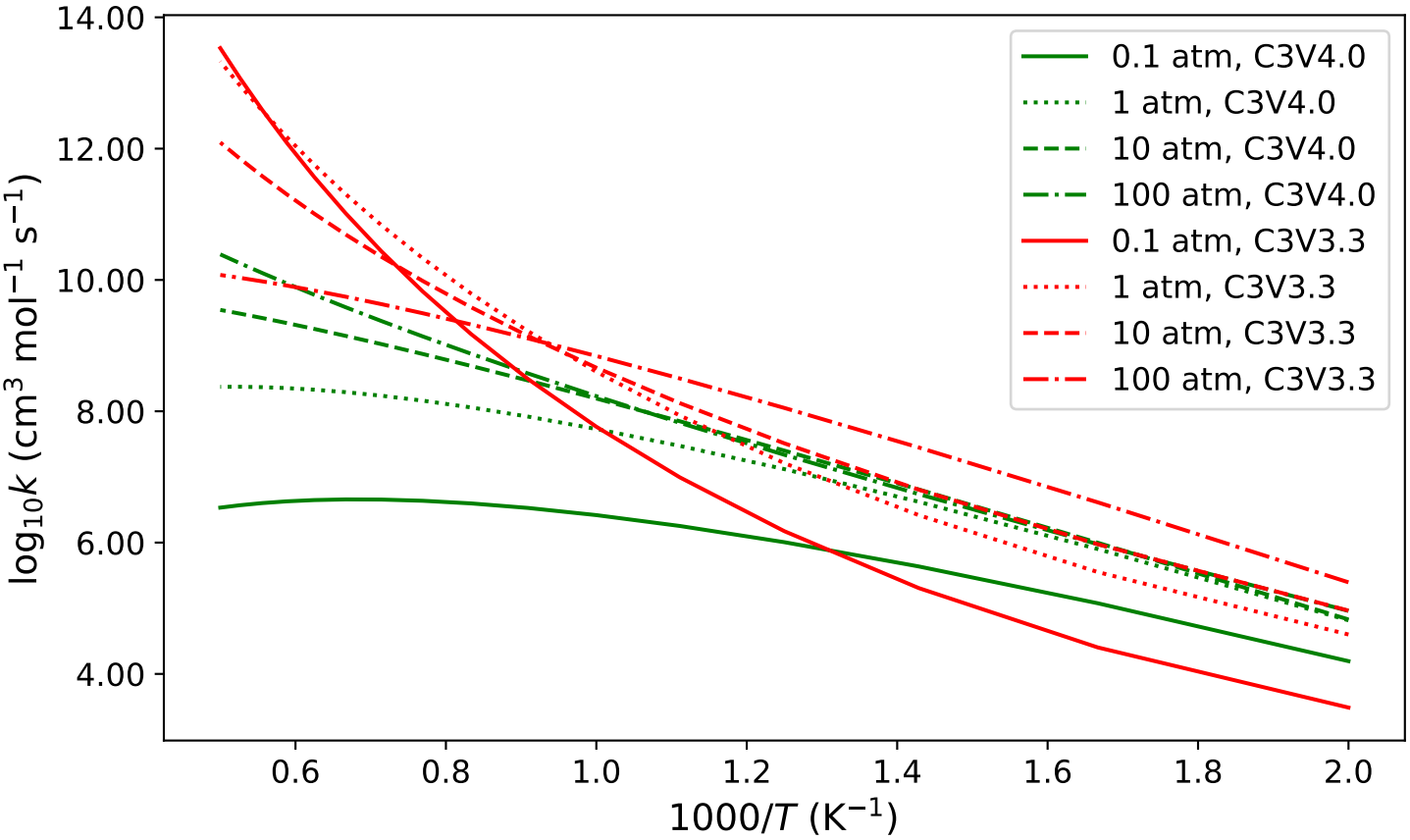



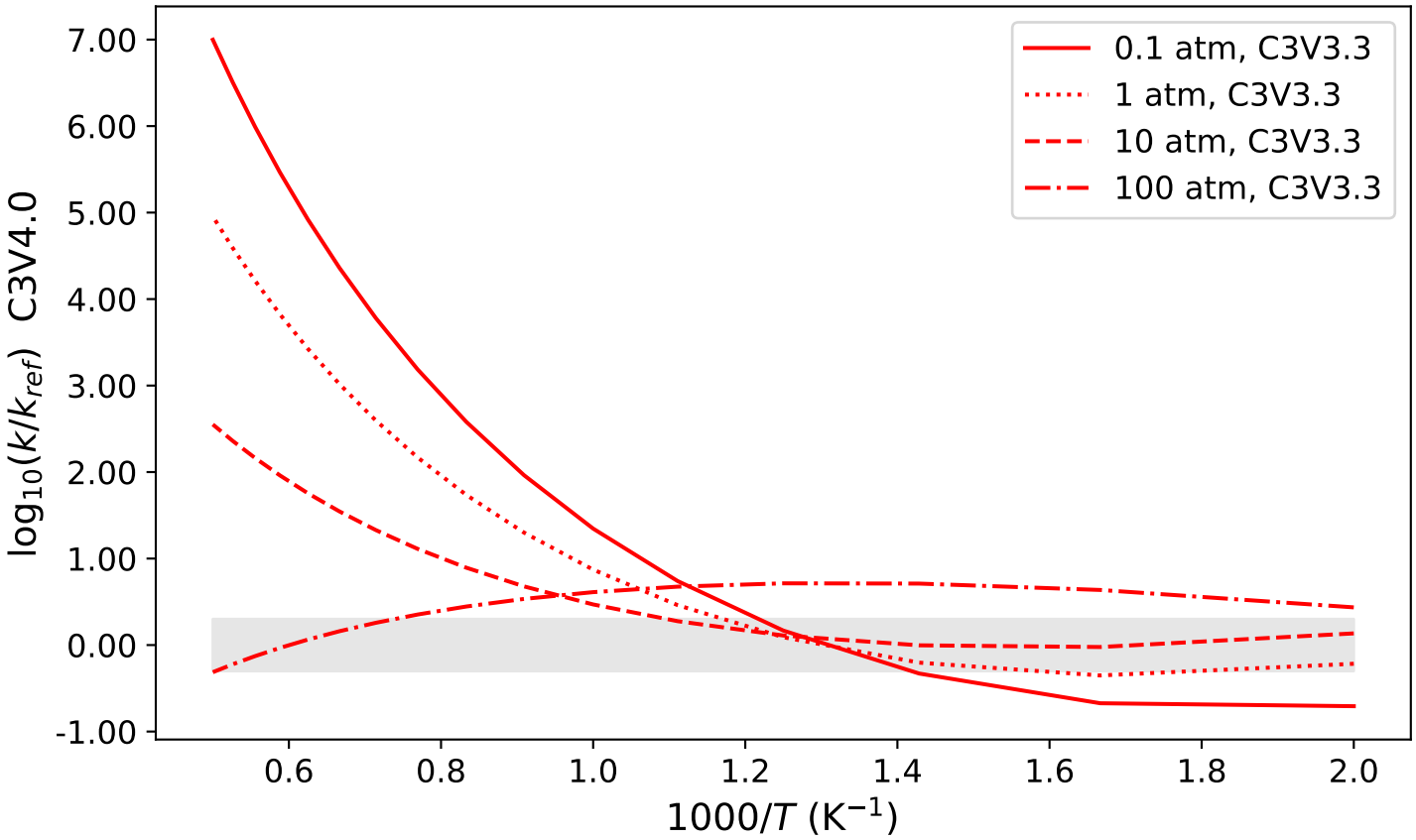
C4H6-2 = CH3 + C3H3



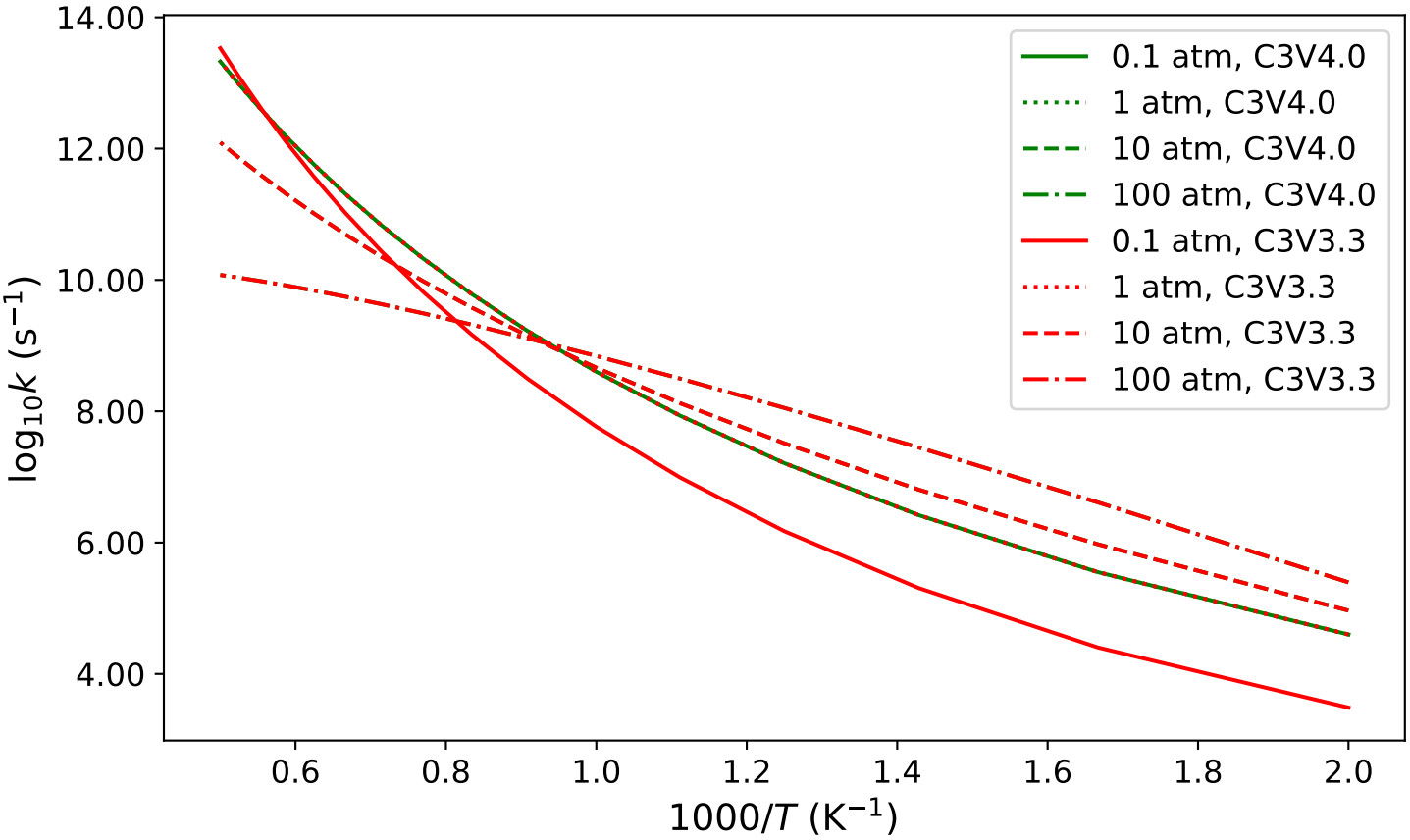



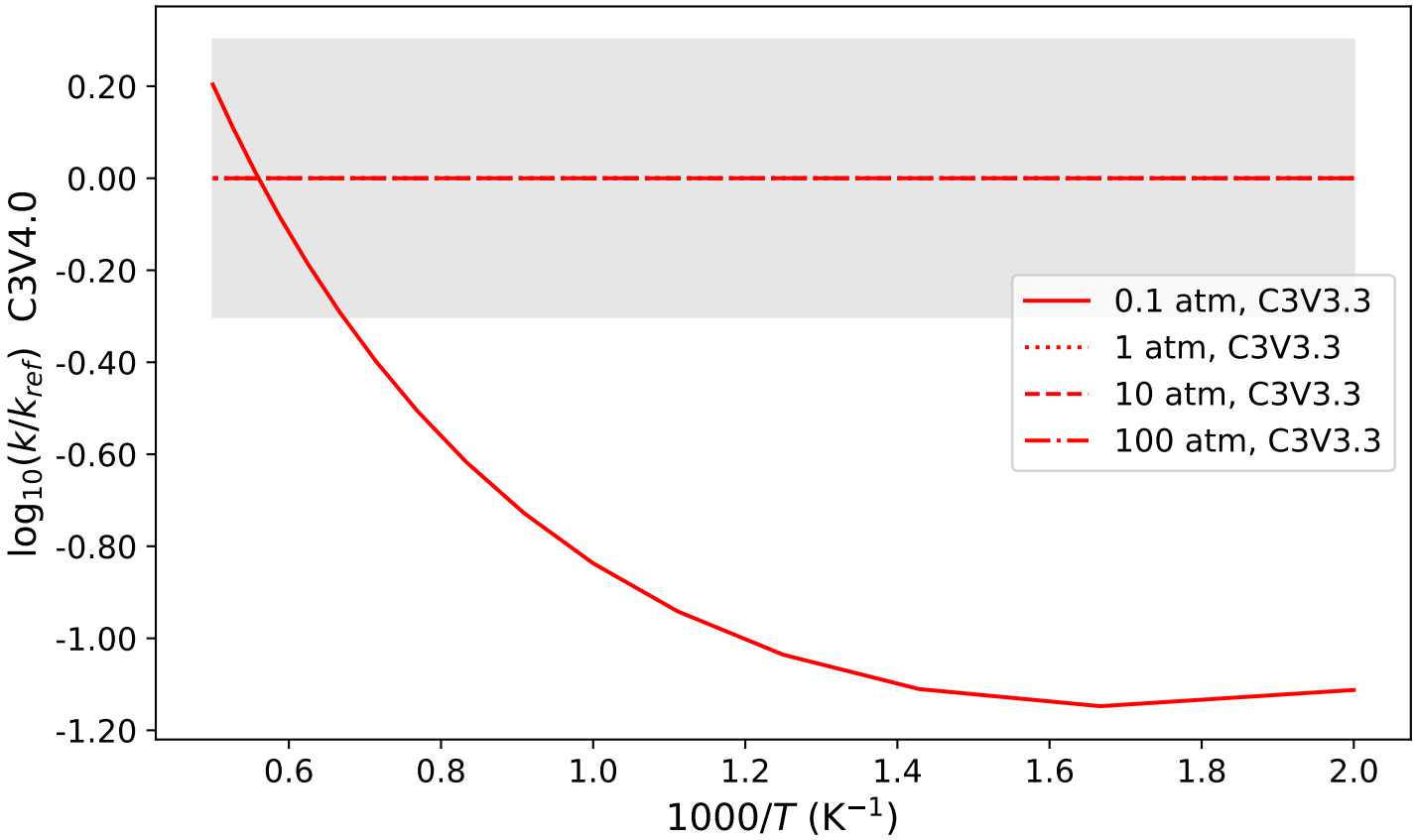
HO2 + C3H4-A = CH2COOHCH2



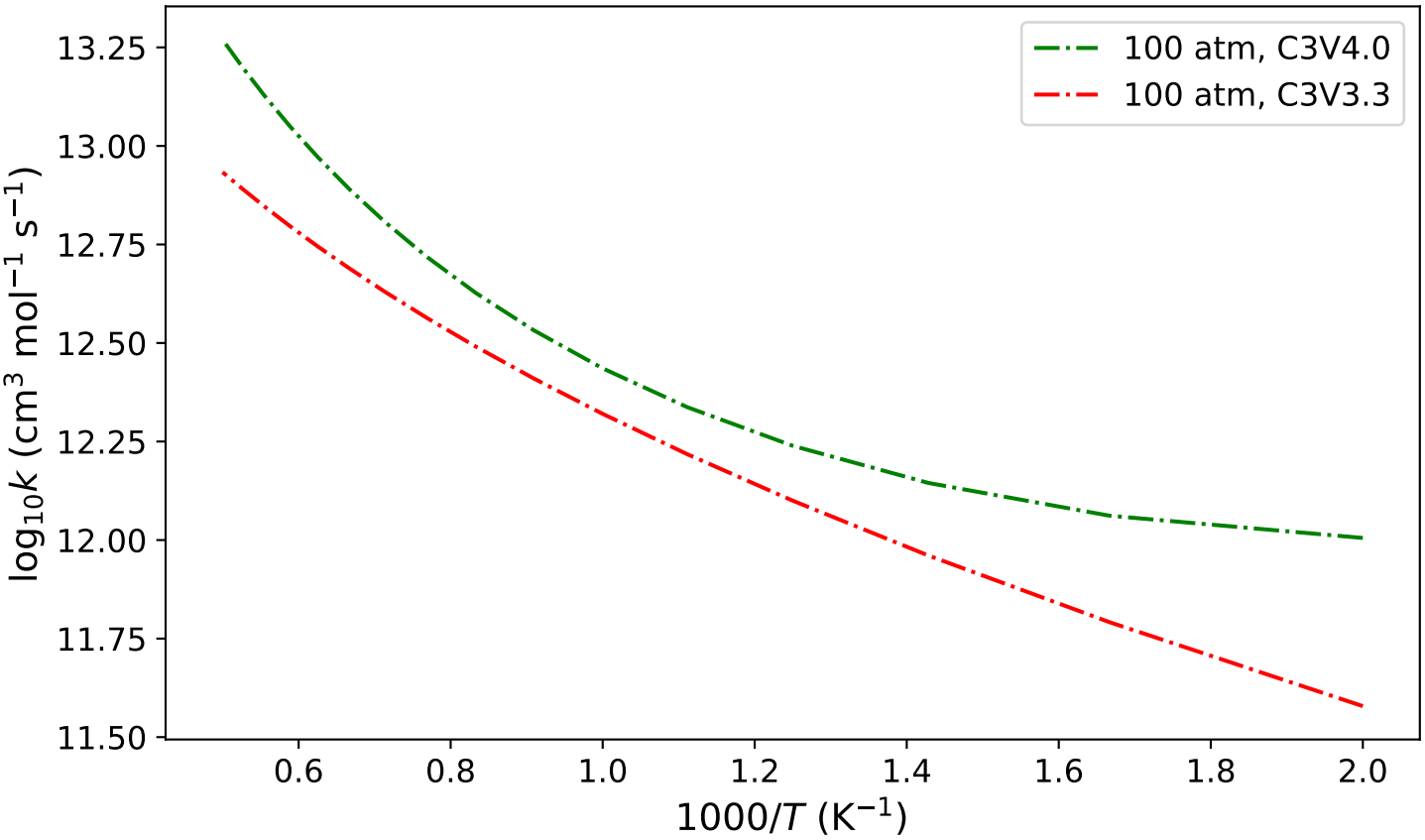



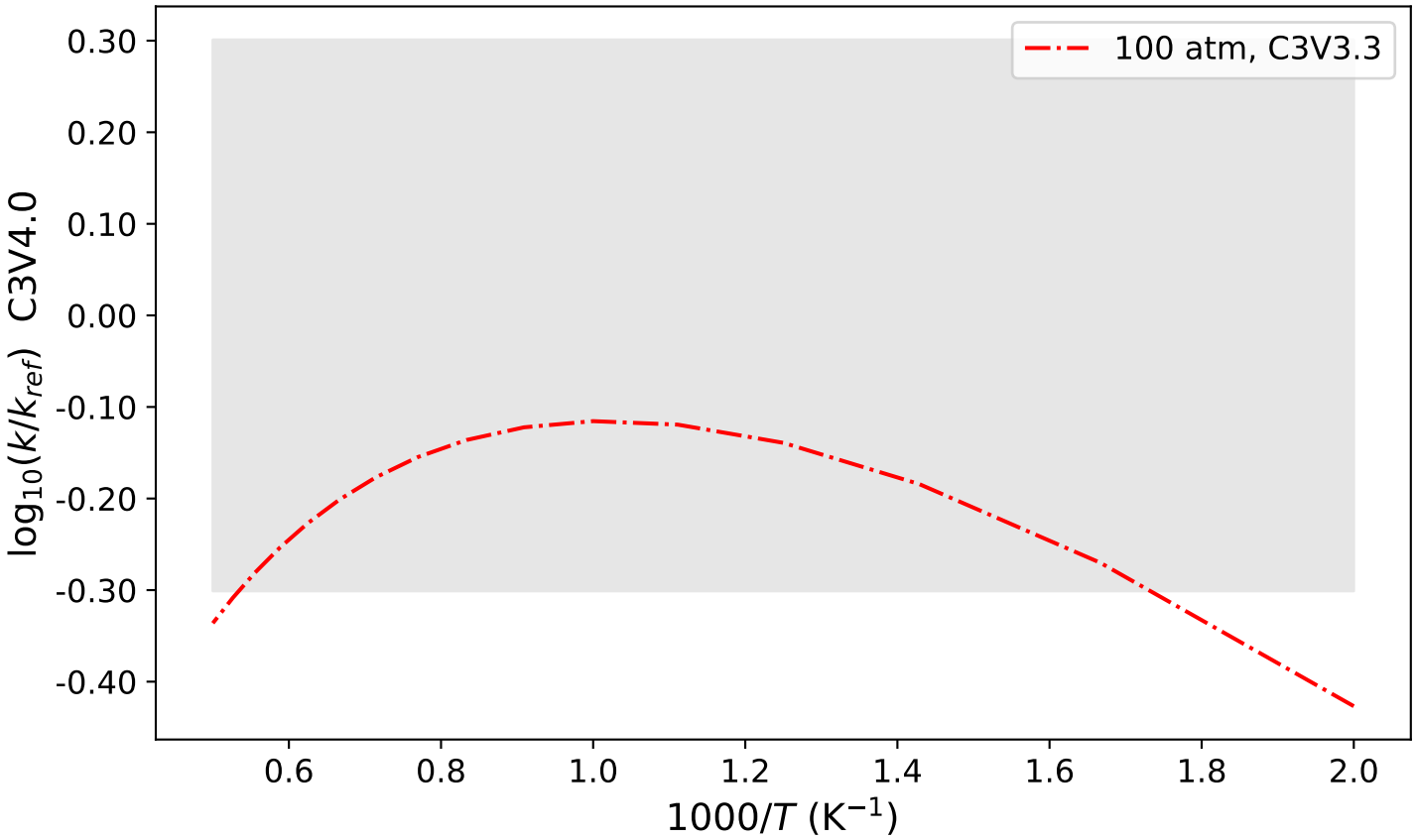
C3H6OOH2-1 = HO2 + C3H6



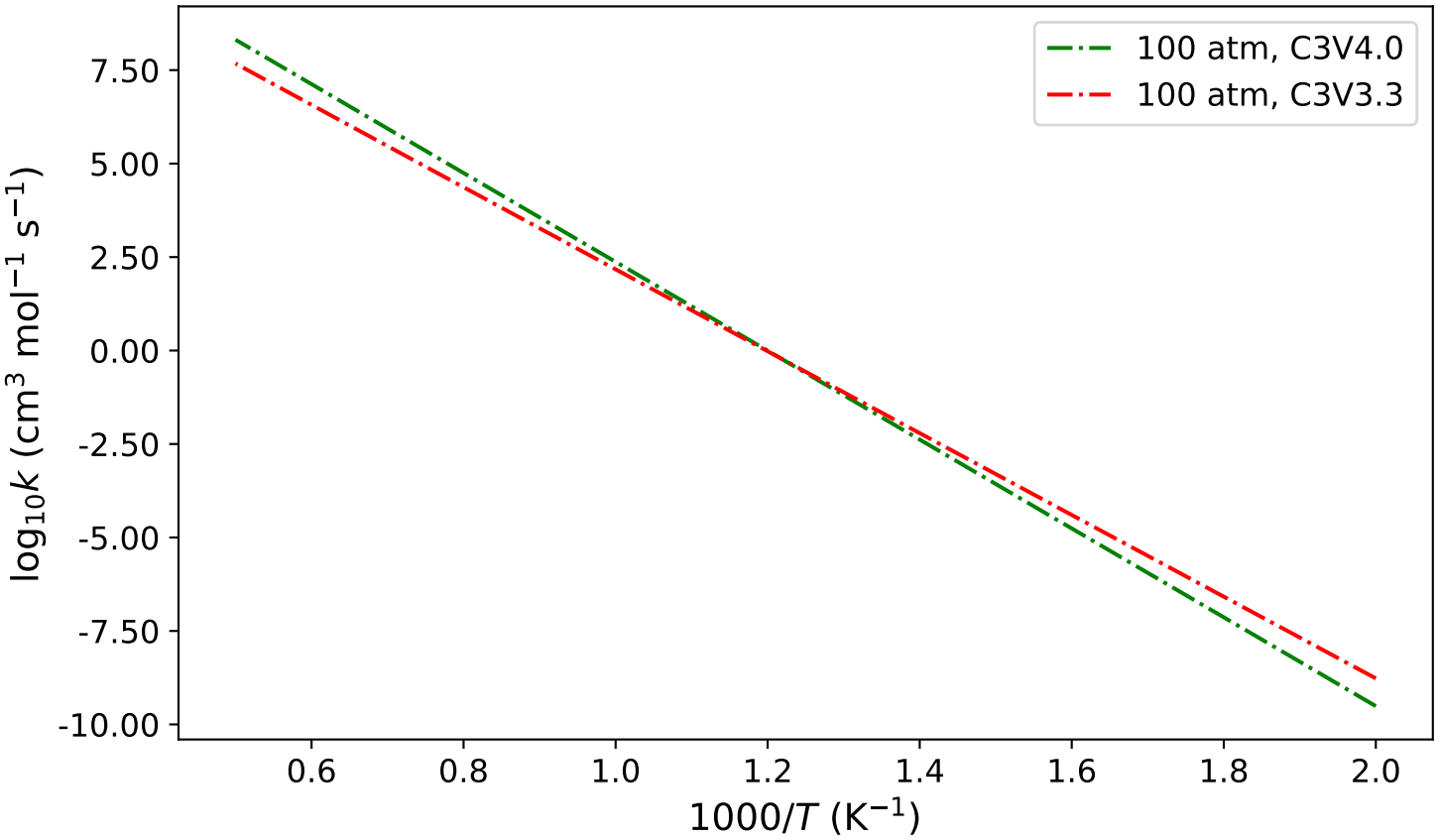



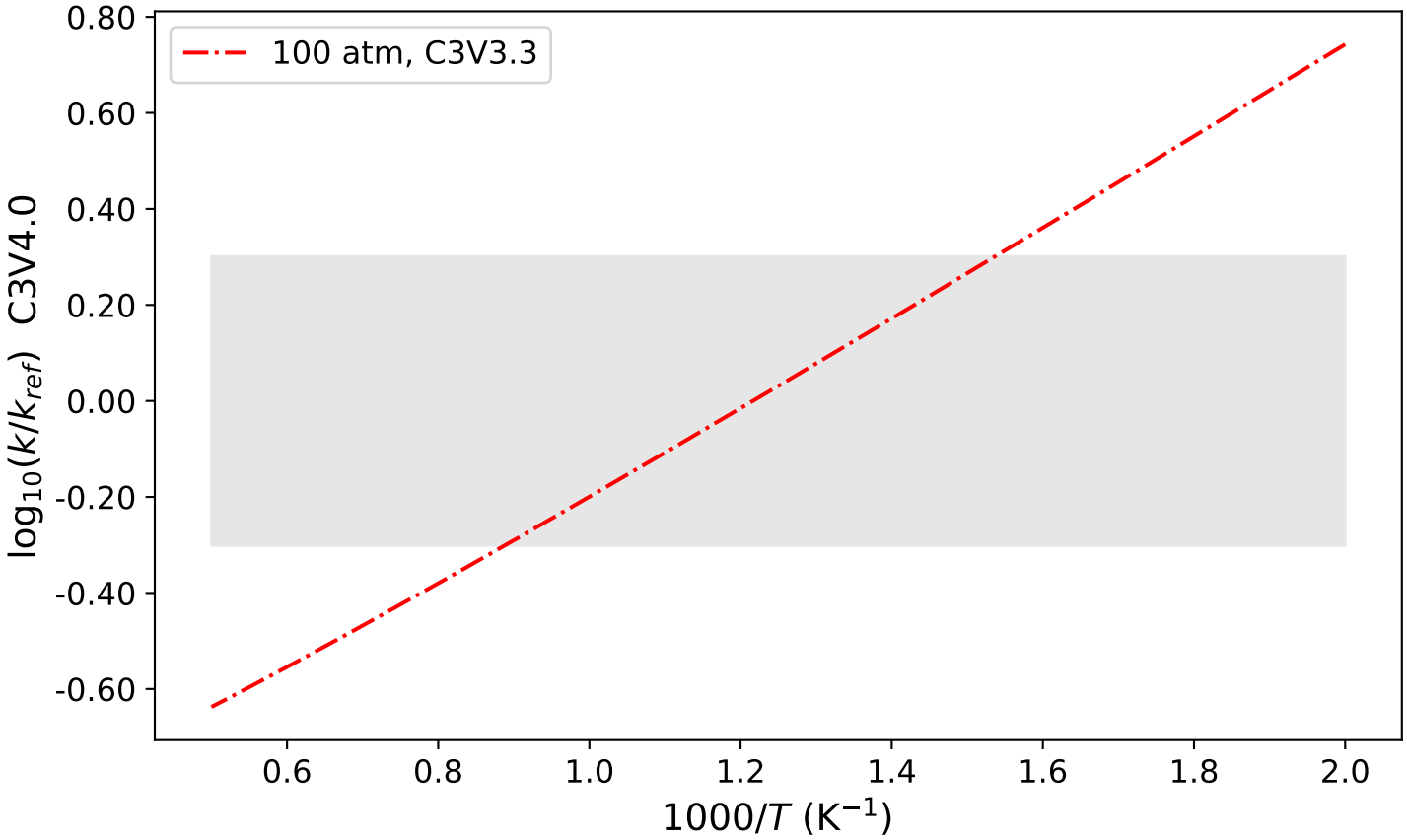
NC3H7OH + OH = H2O + C3H6OH1-2



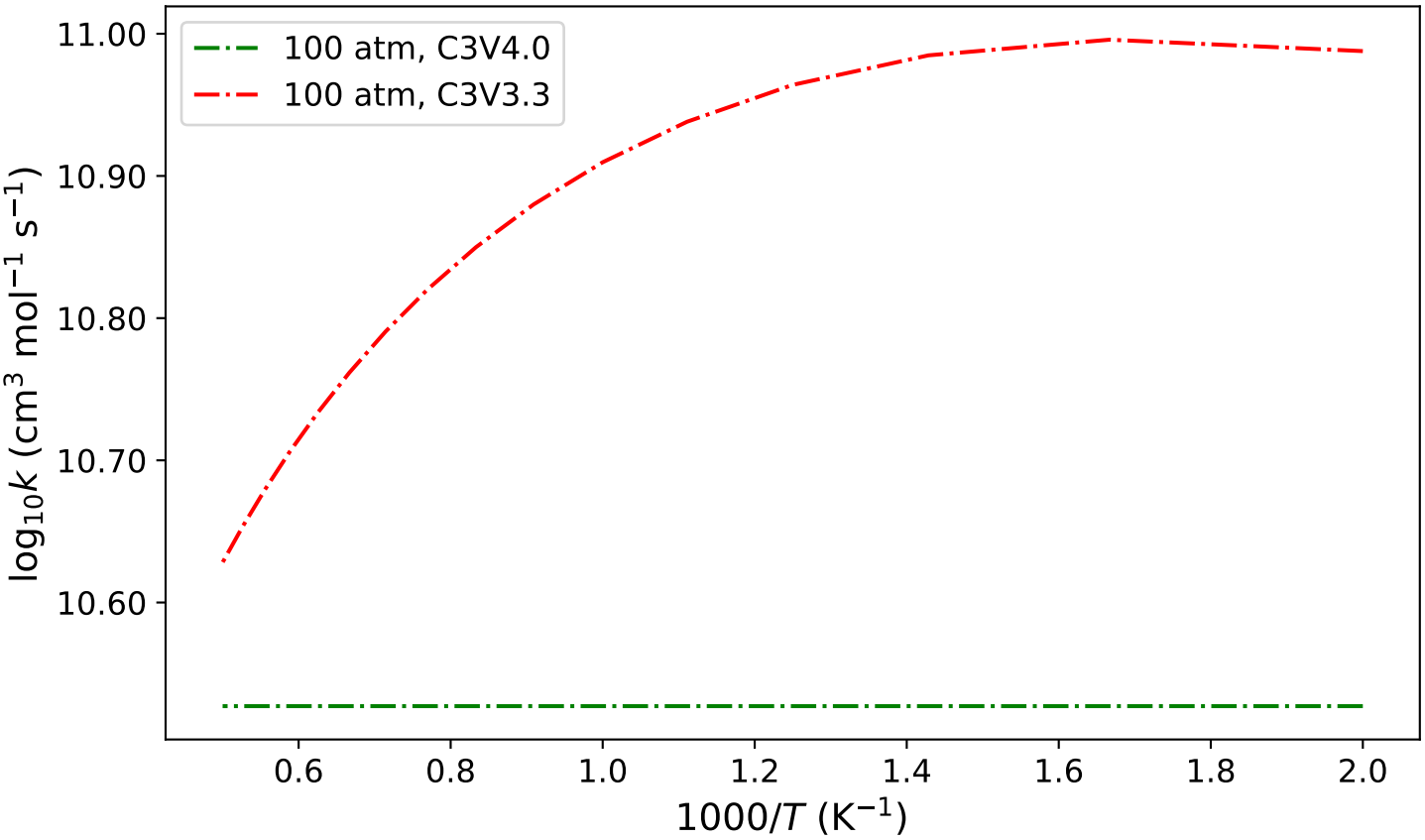



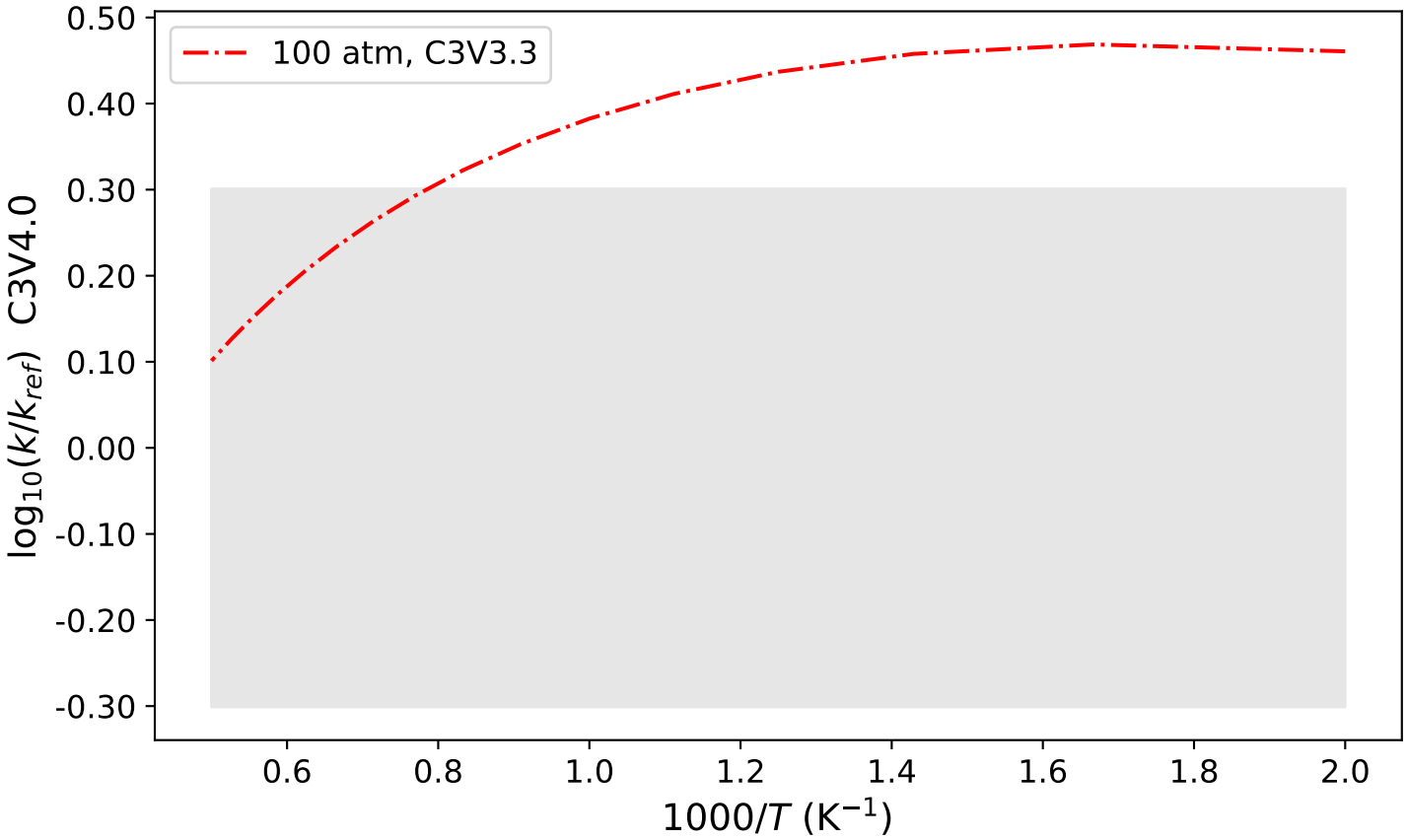
O2 + NC3H7OH = HO2 + C3H6OH1-2



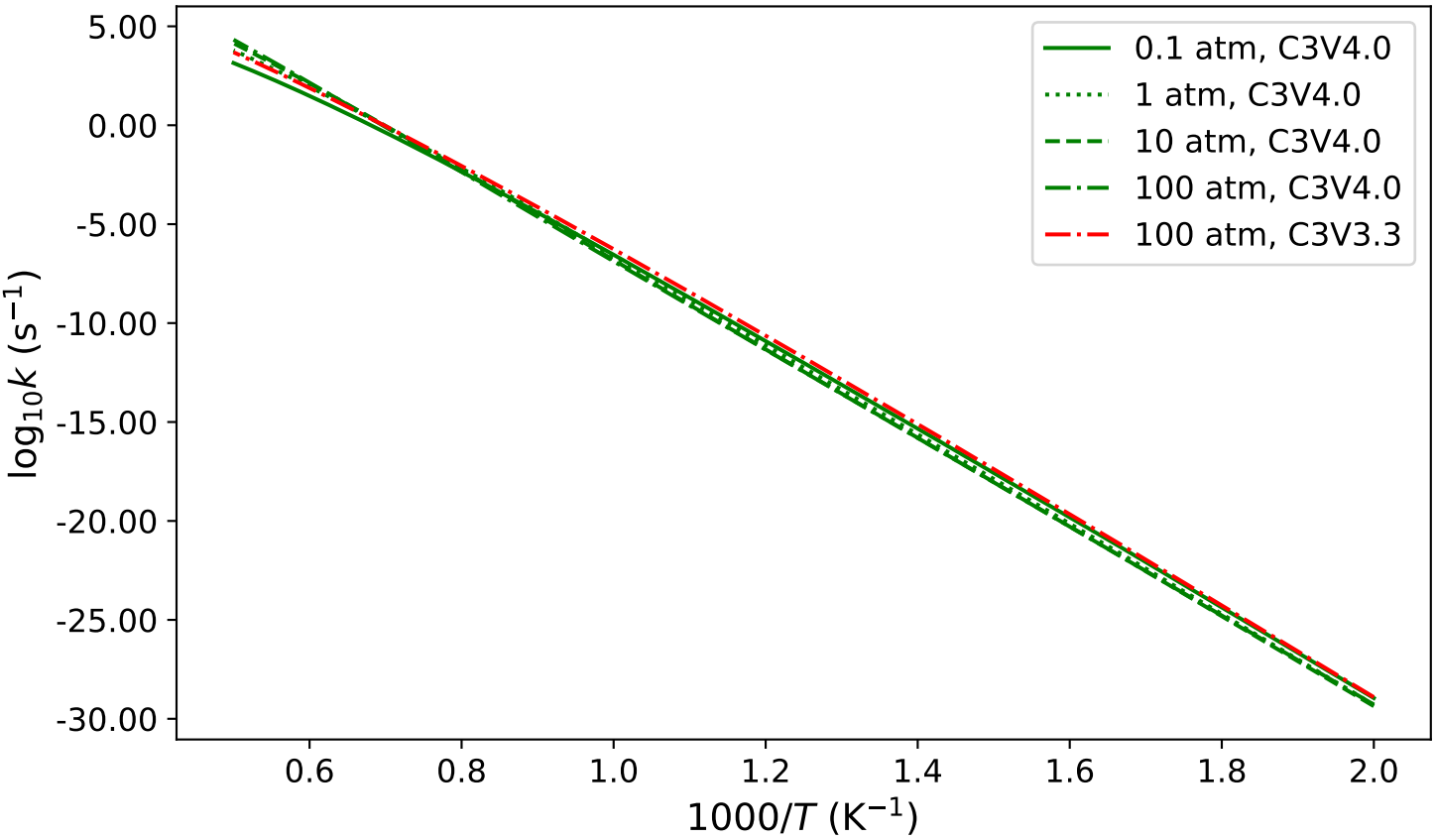



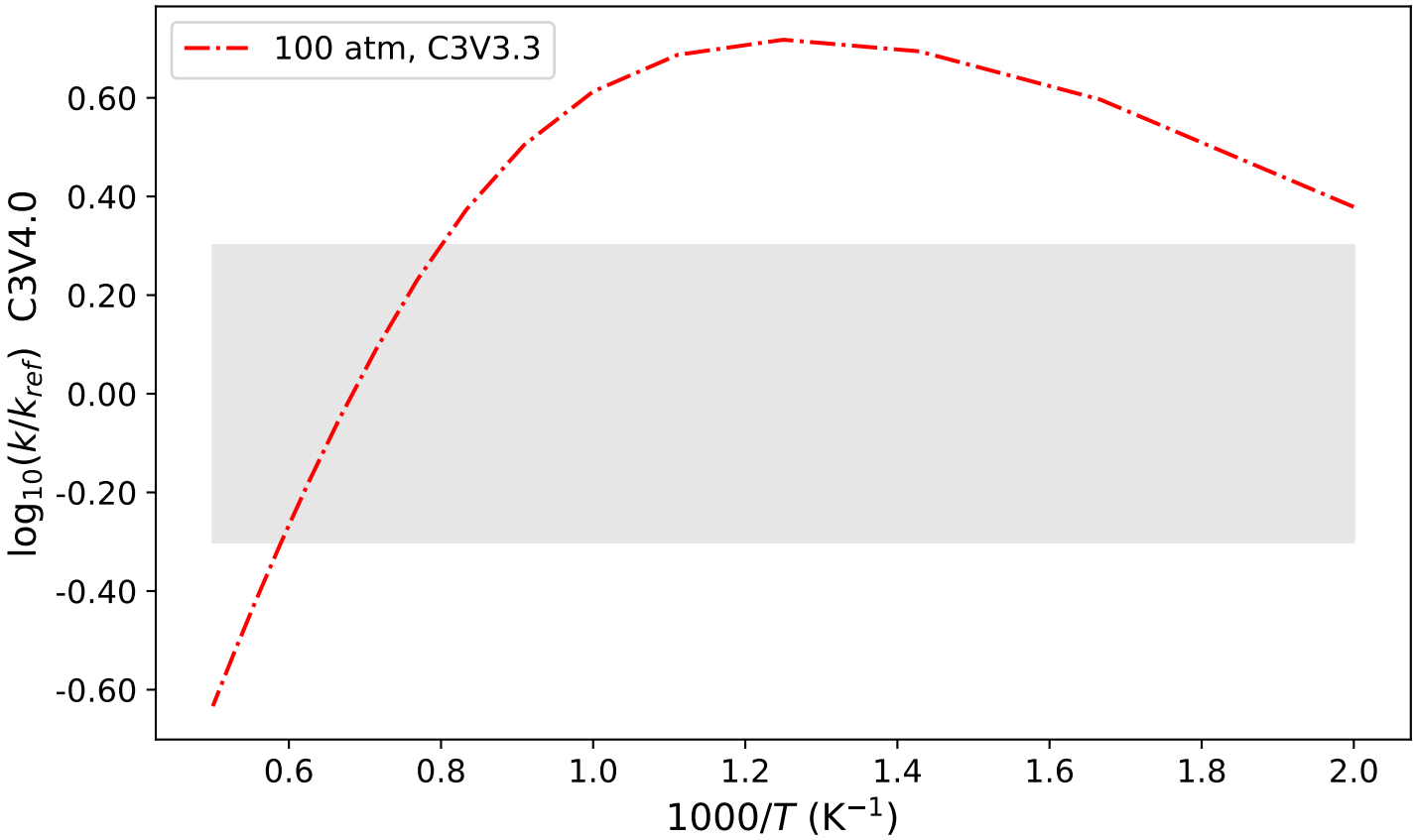
NC3H7O2 + C2H5O2 => NC3H7O + O2 + C2H5O



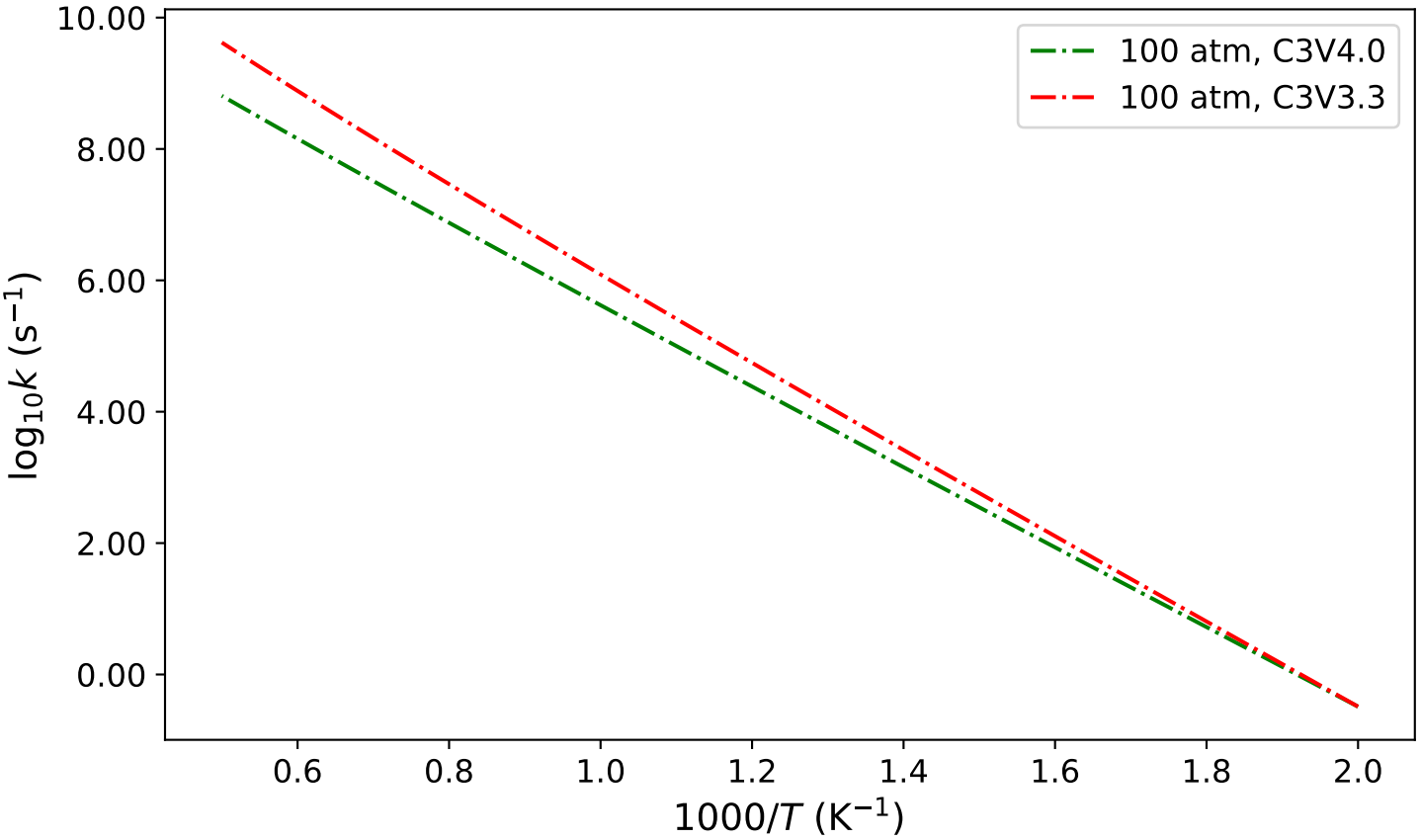



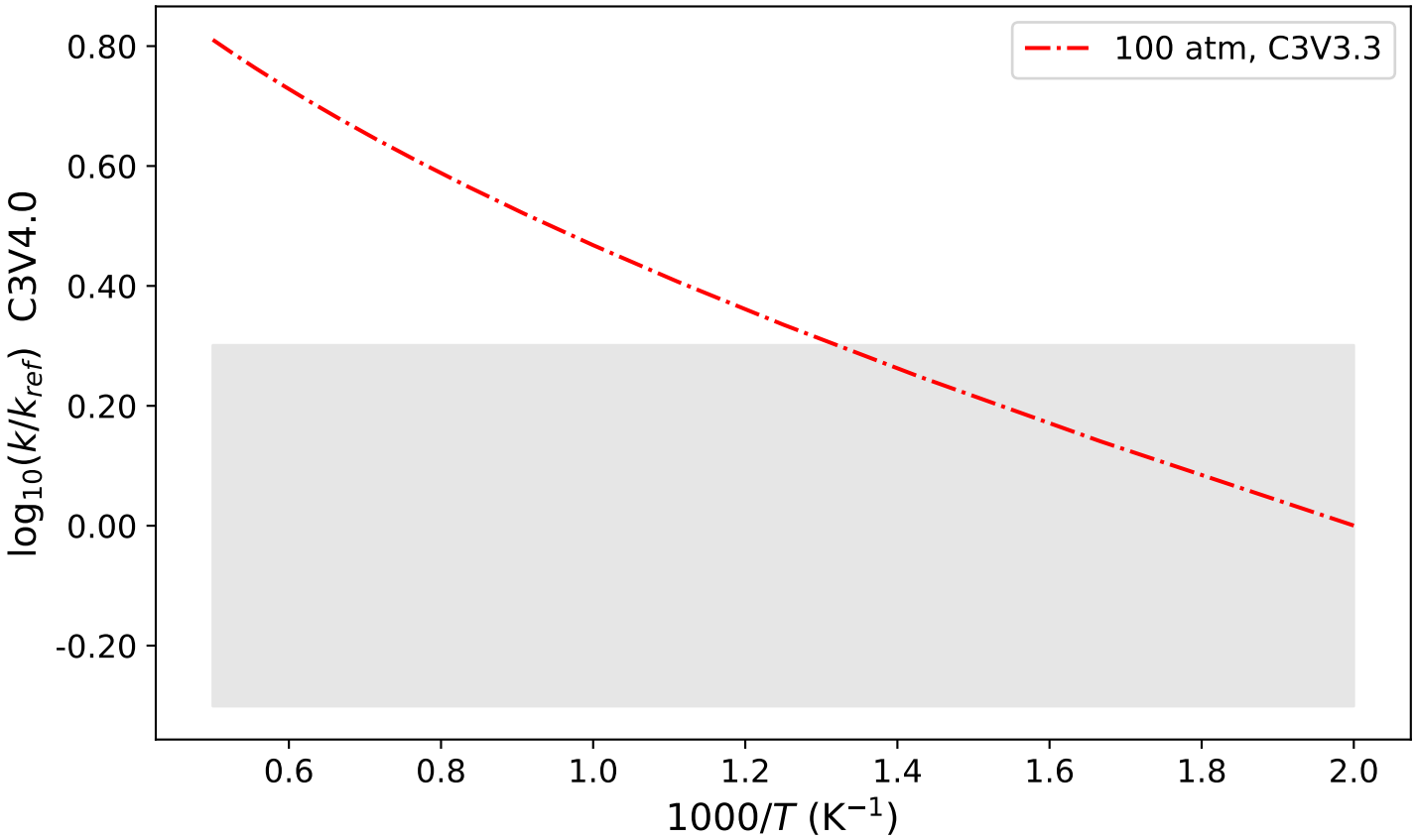
C4H4 = H + C4H3-I


0.1 atm, C3V4.0
1 atm, C3V4.0
10 atm, C3V4.0
100 atm, C3V4.0
100 atm, C3V3.3
5.00
0.00
-5.00
-10.00
-15.00
-20.00
-25.00
-30.00
$\log_{10}k$ ($s^{-1}$)
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
$1000/T$ ($K^{-1}$)

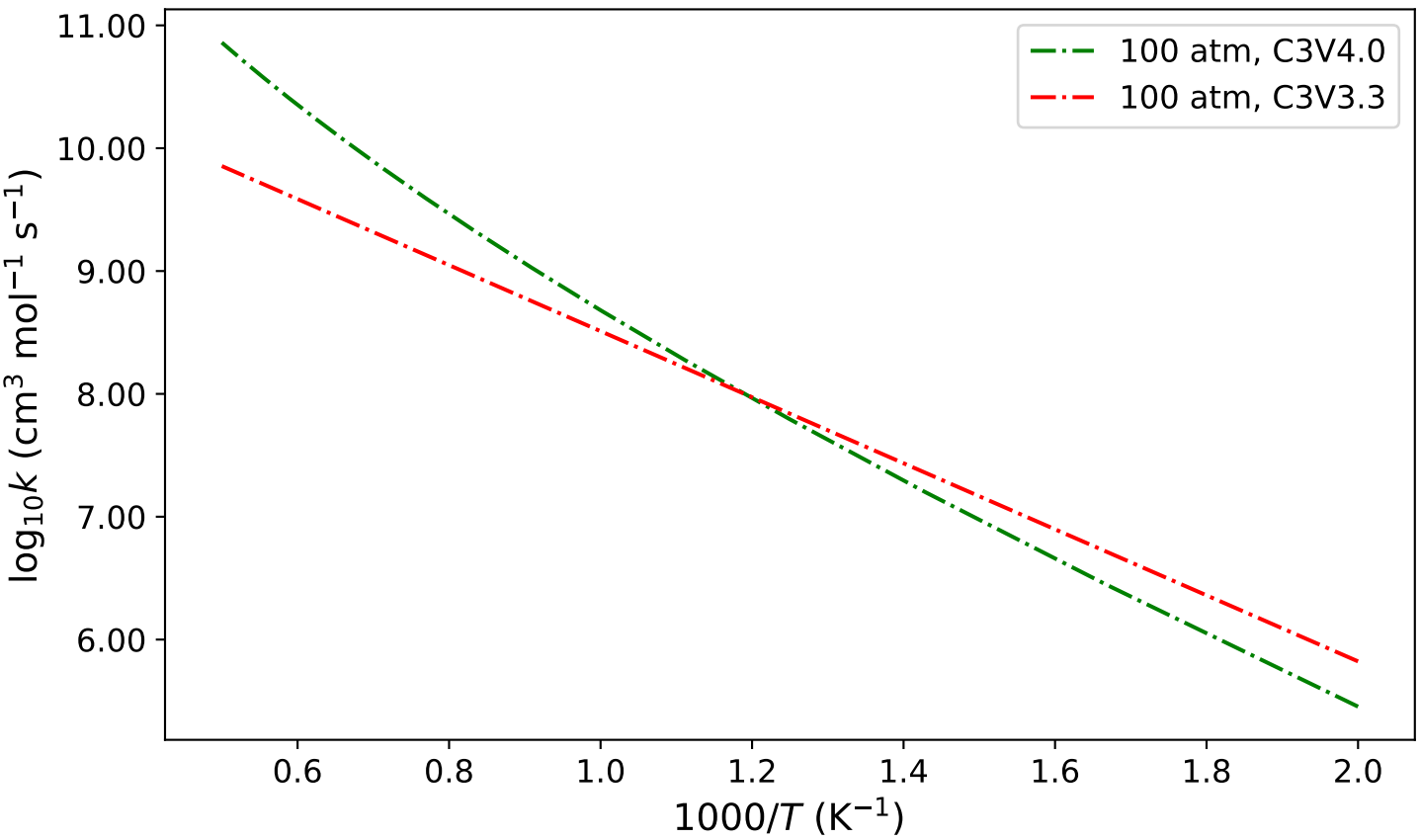

100 atm, C3V3.3
0.60
0.40
0.20
0.00
-0.20
-0.40
-0.60
$\log_{10}(k/k_{ref})$ C3V4.0
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
$1000/T$ ($K^{-1}$)

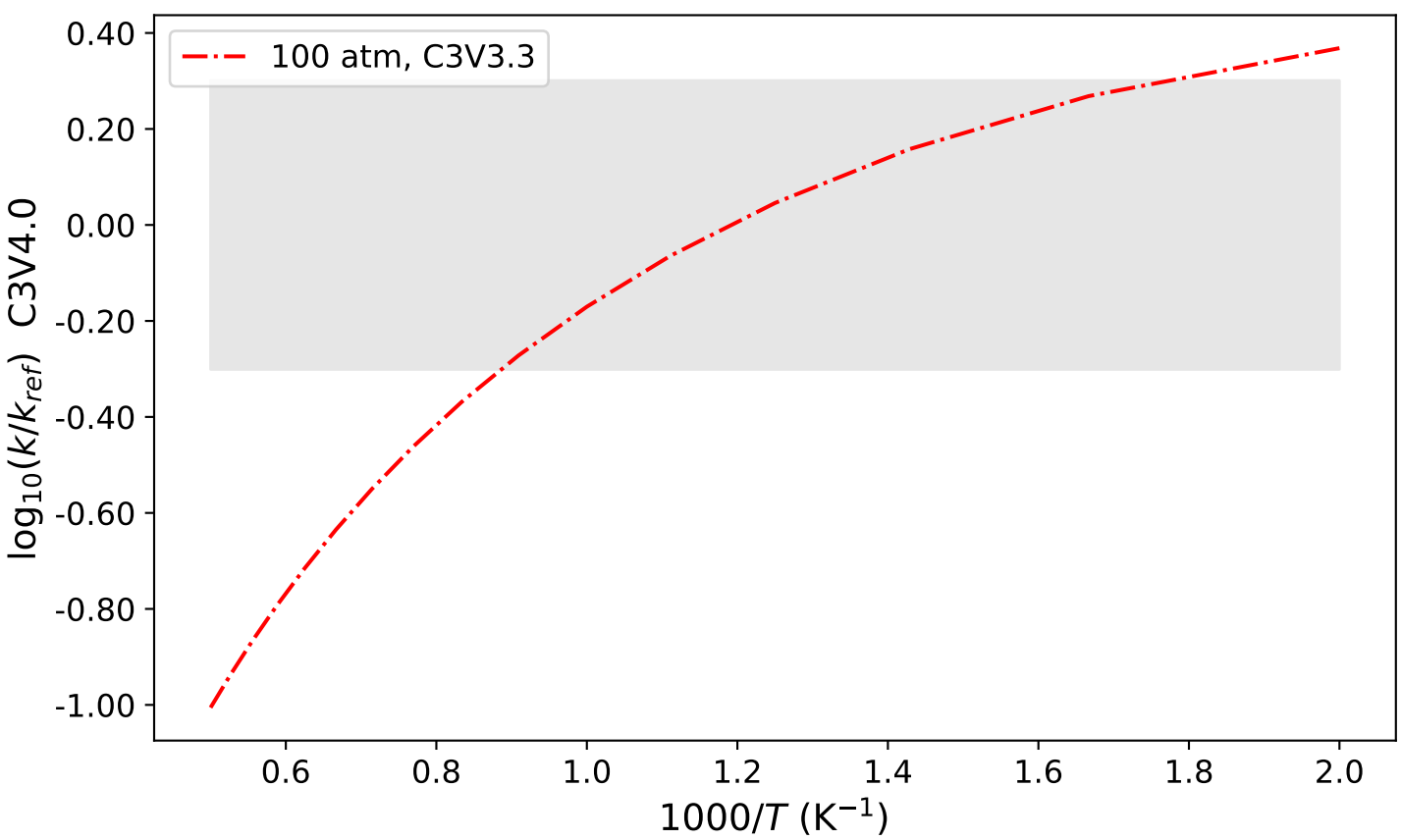
C4H8OOH1-2O2 = OH + NC4KET12



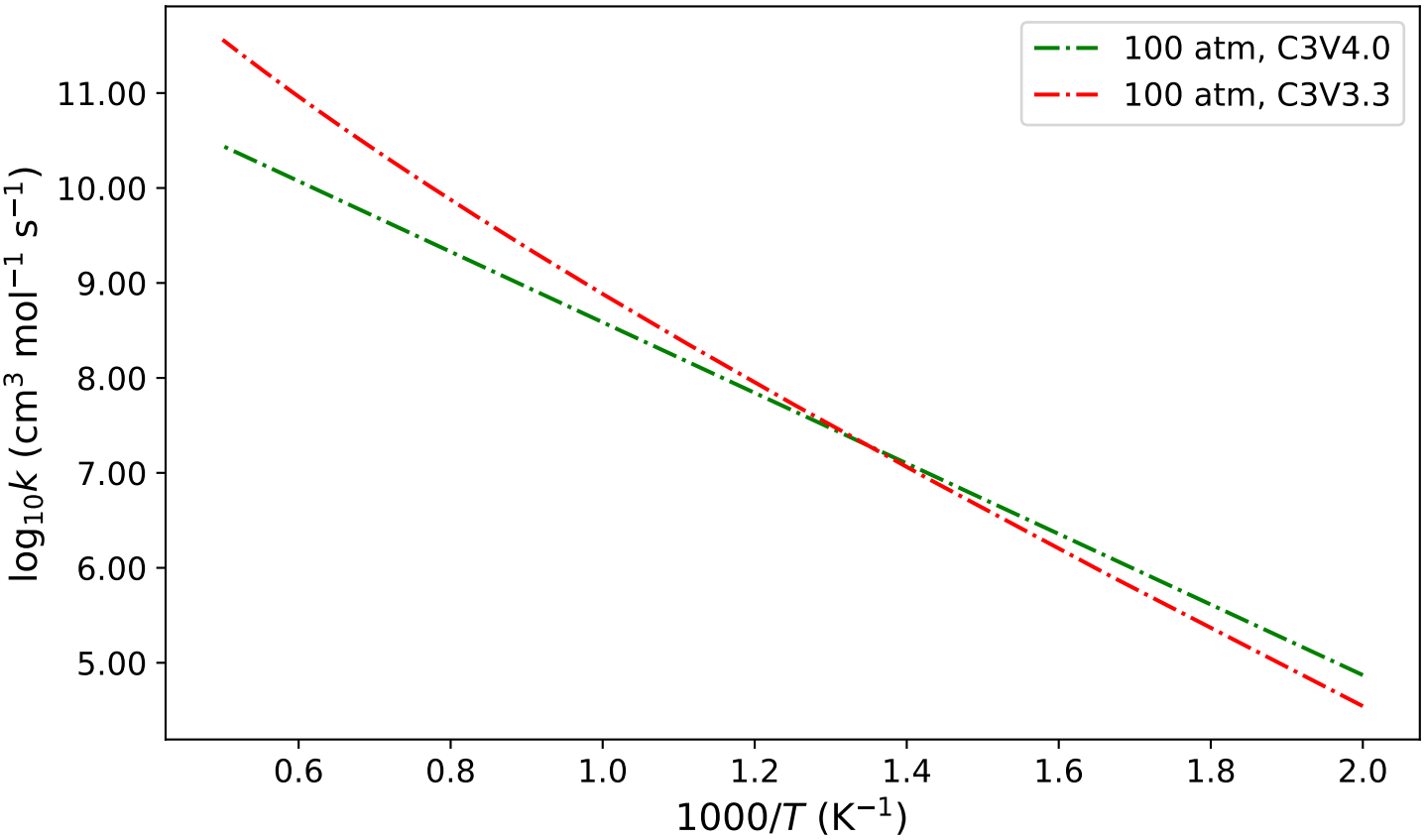



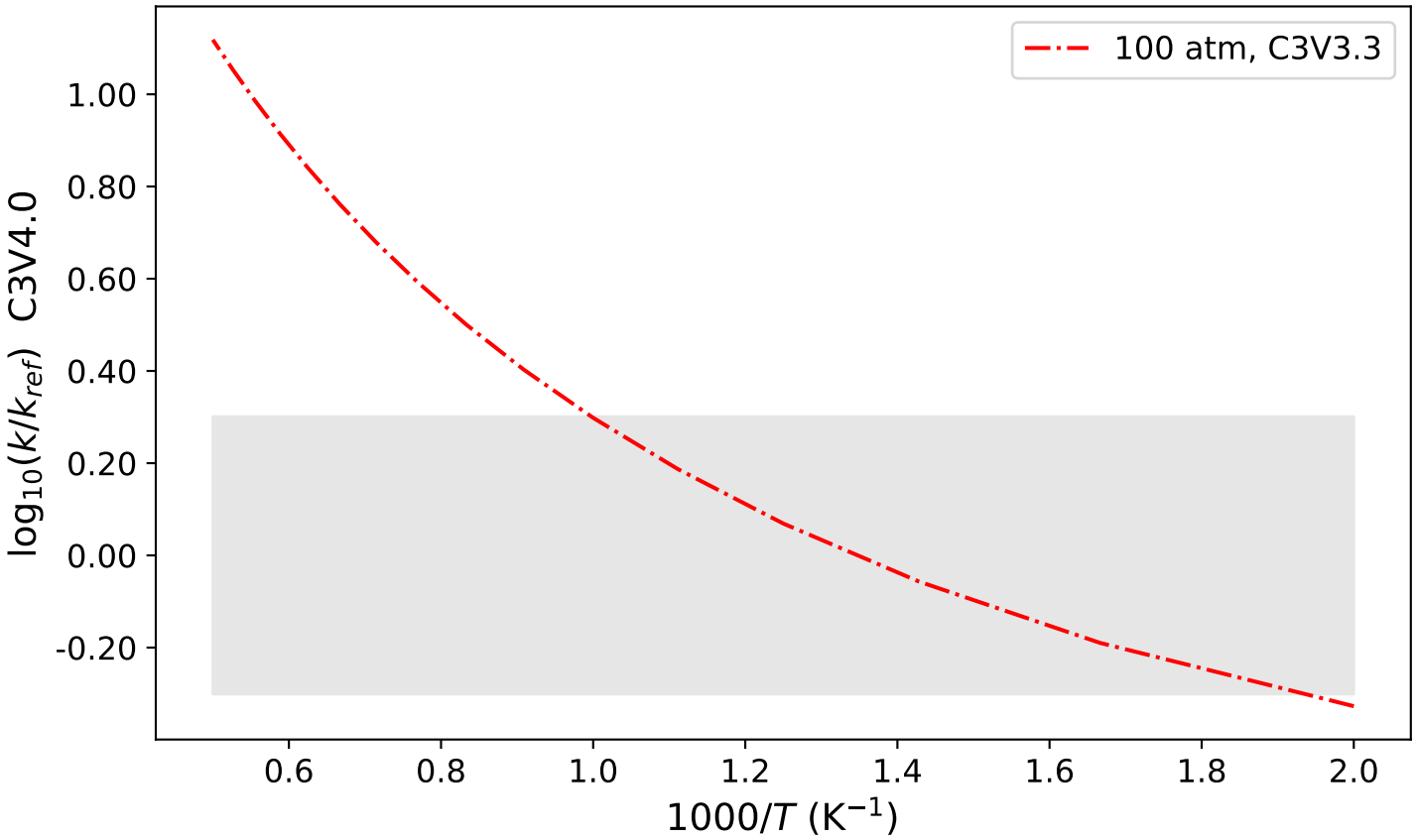
C2H5 + C4H10 = C2H6 + PC4H9



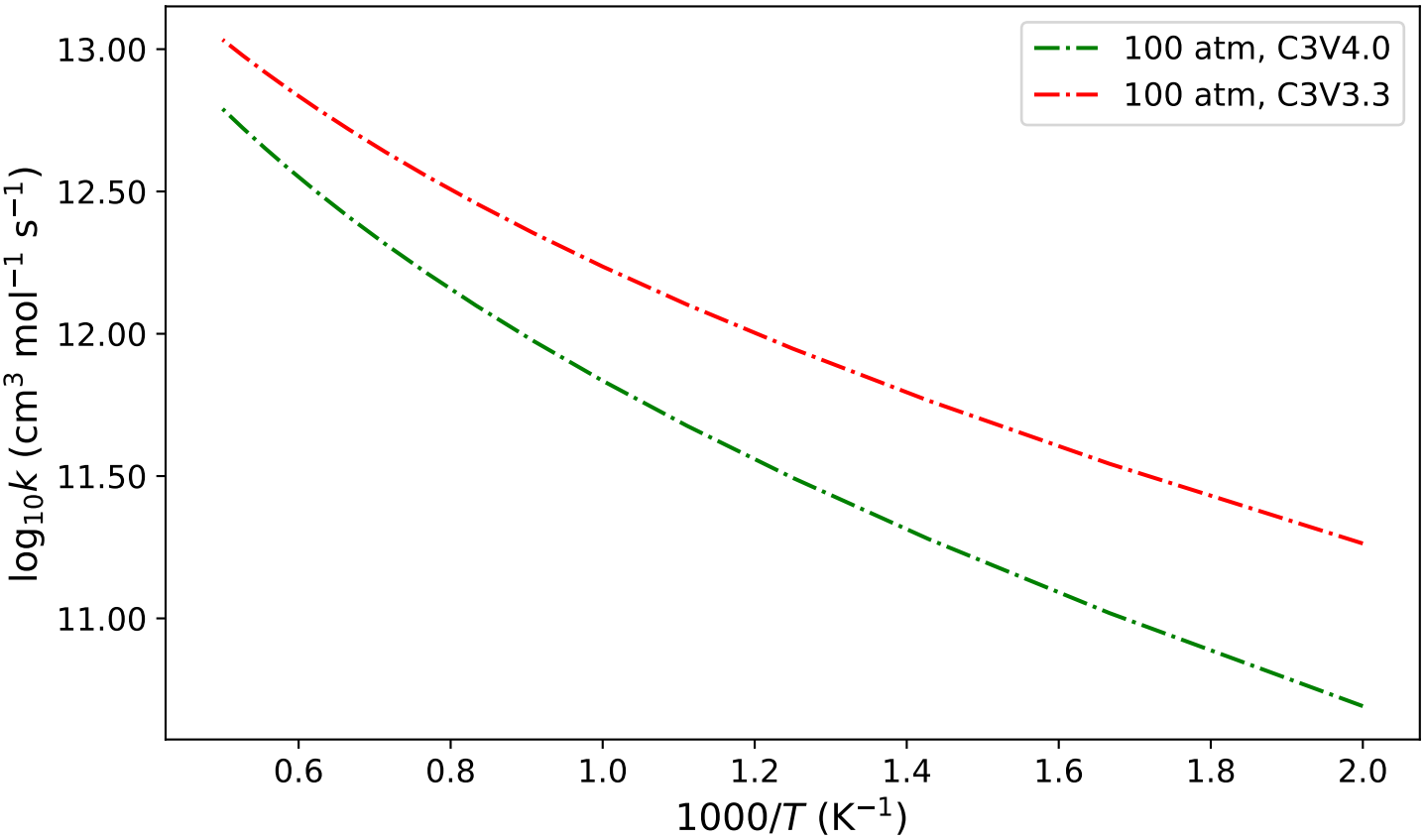



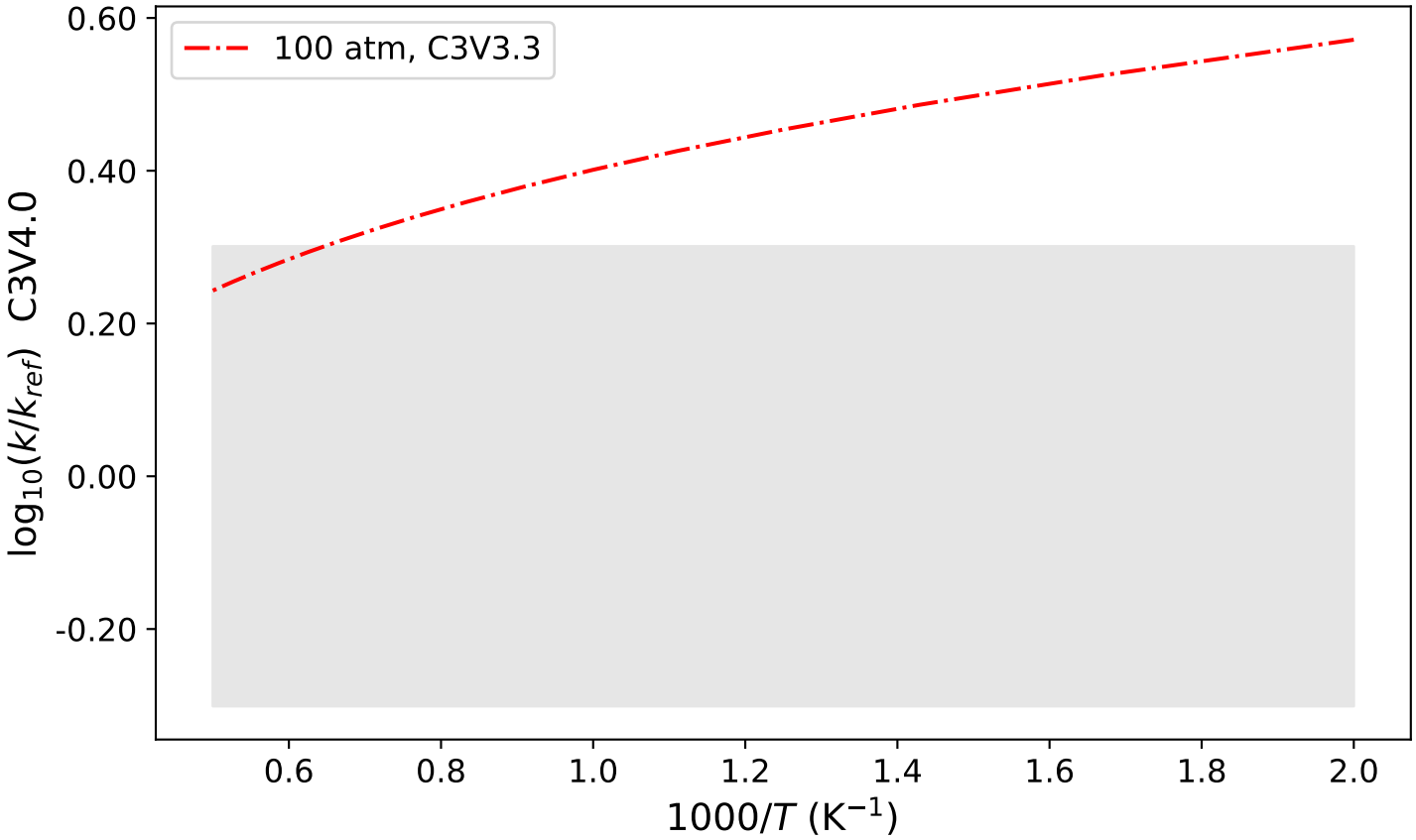
CH3O2 + C4H10 = CH3O2H + SC4H9



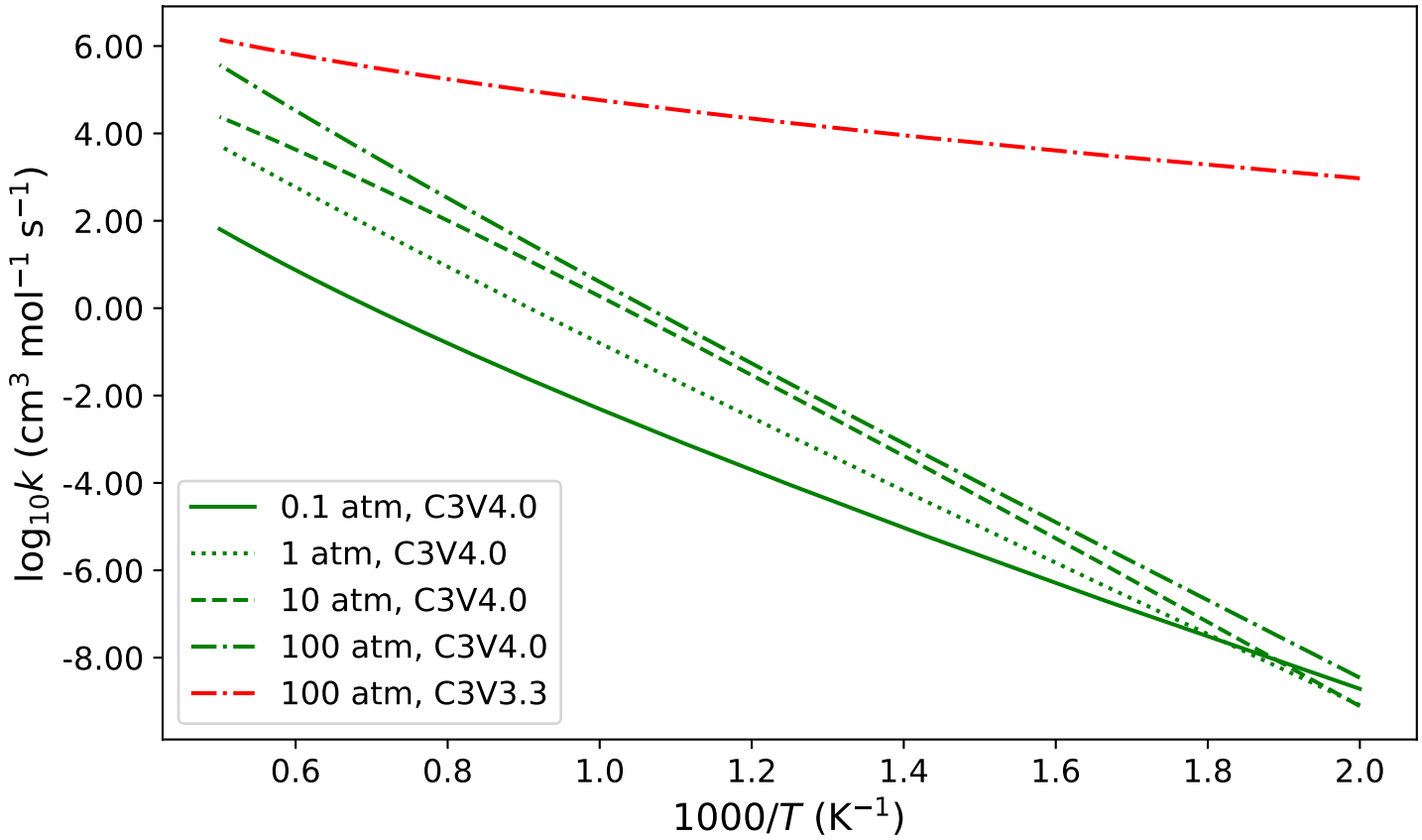



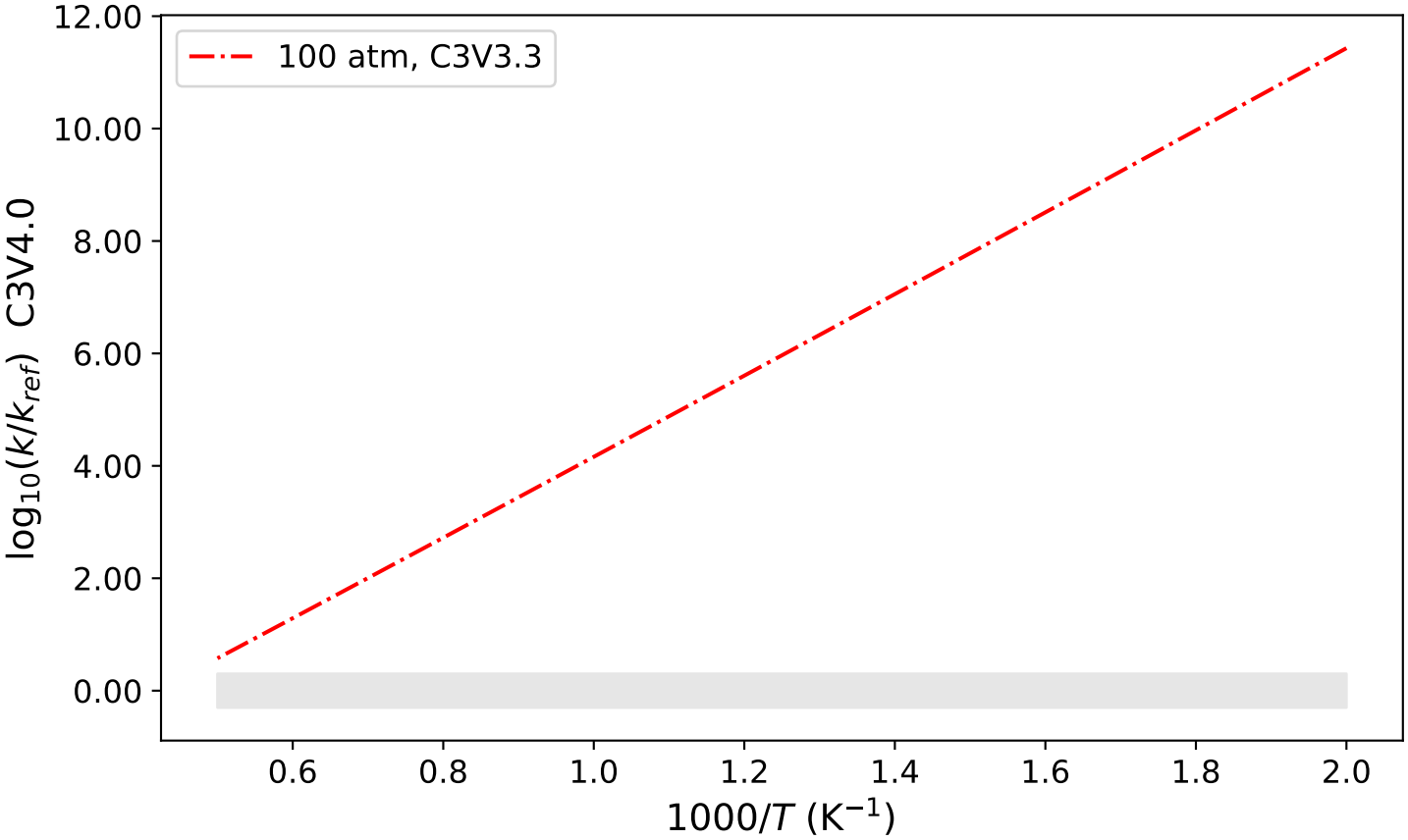
OH + C4H6 = H2O + C4H5-I



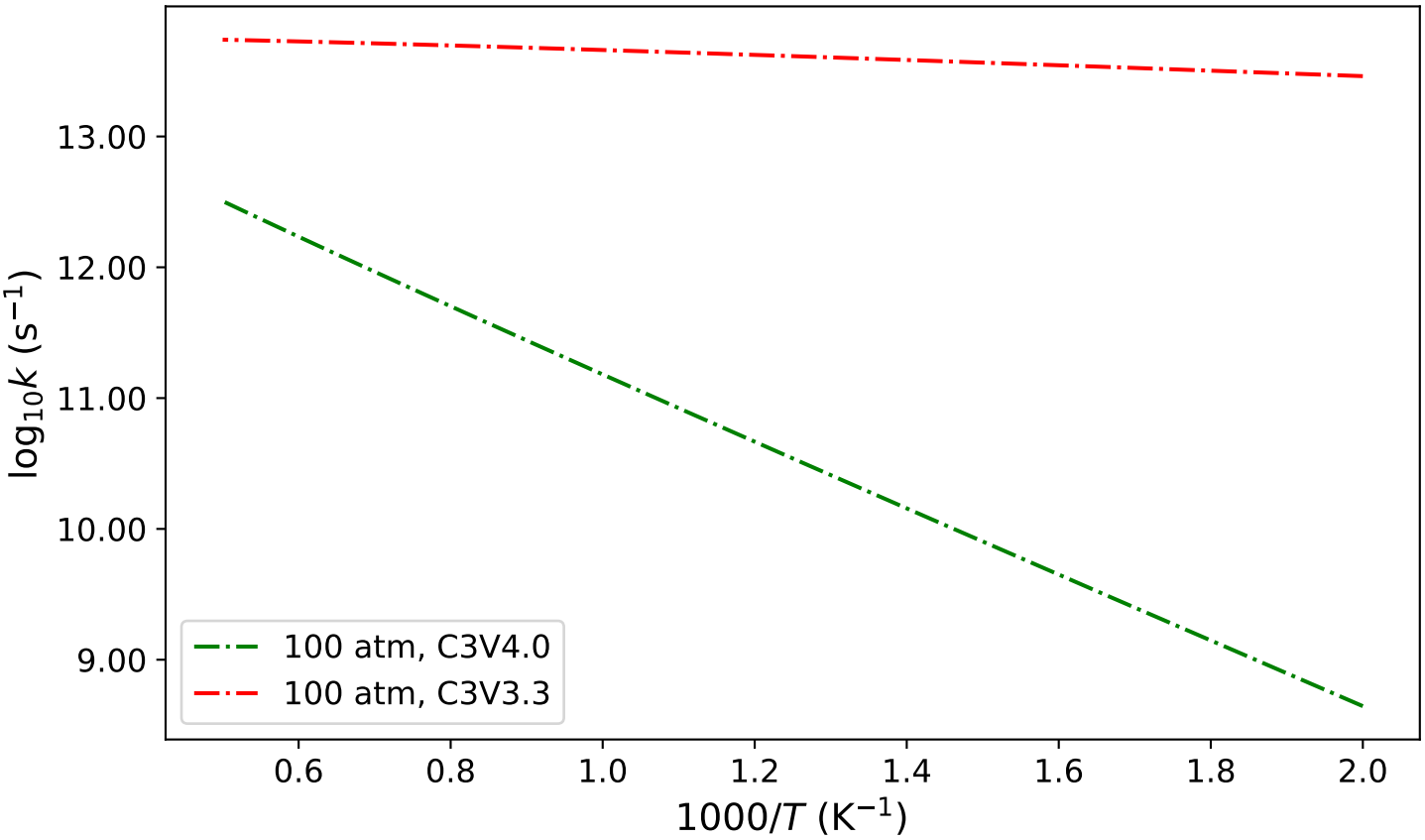



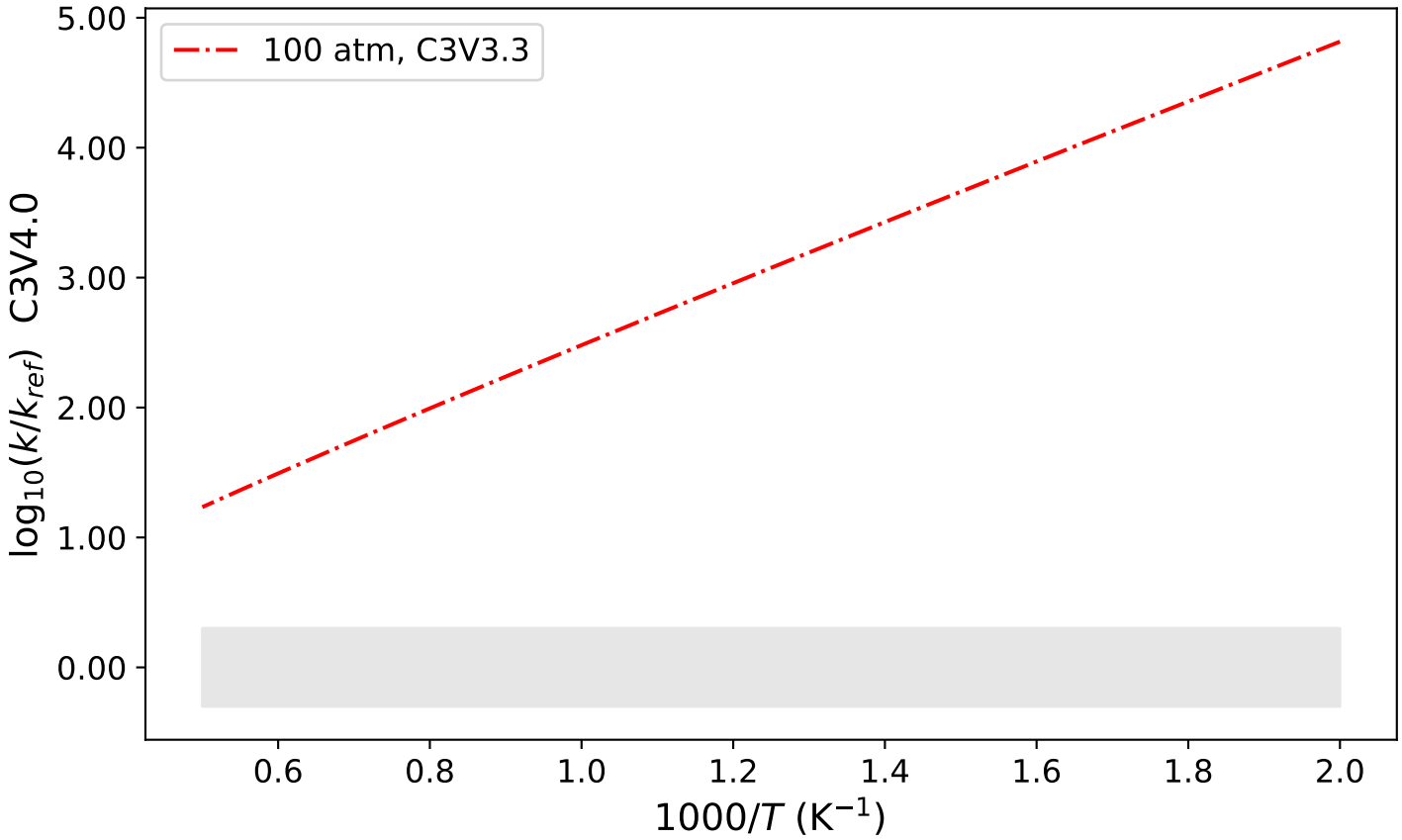
C3H5-A + CO = CH2CHCHCHO



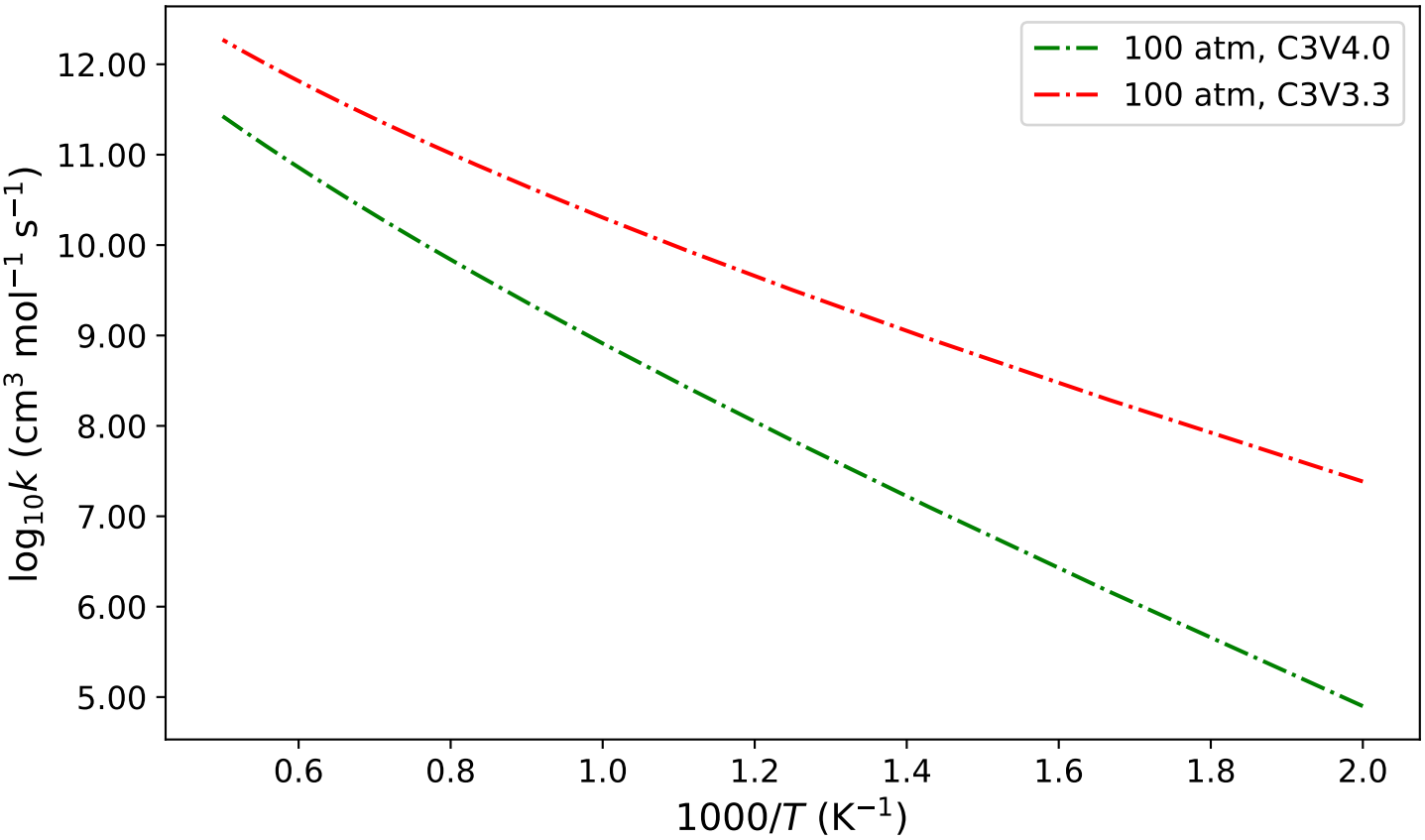



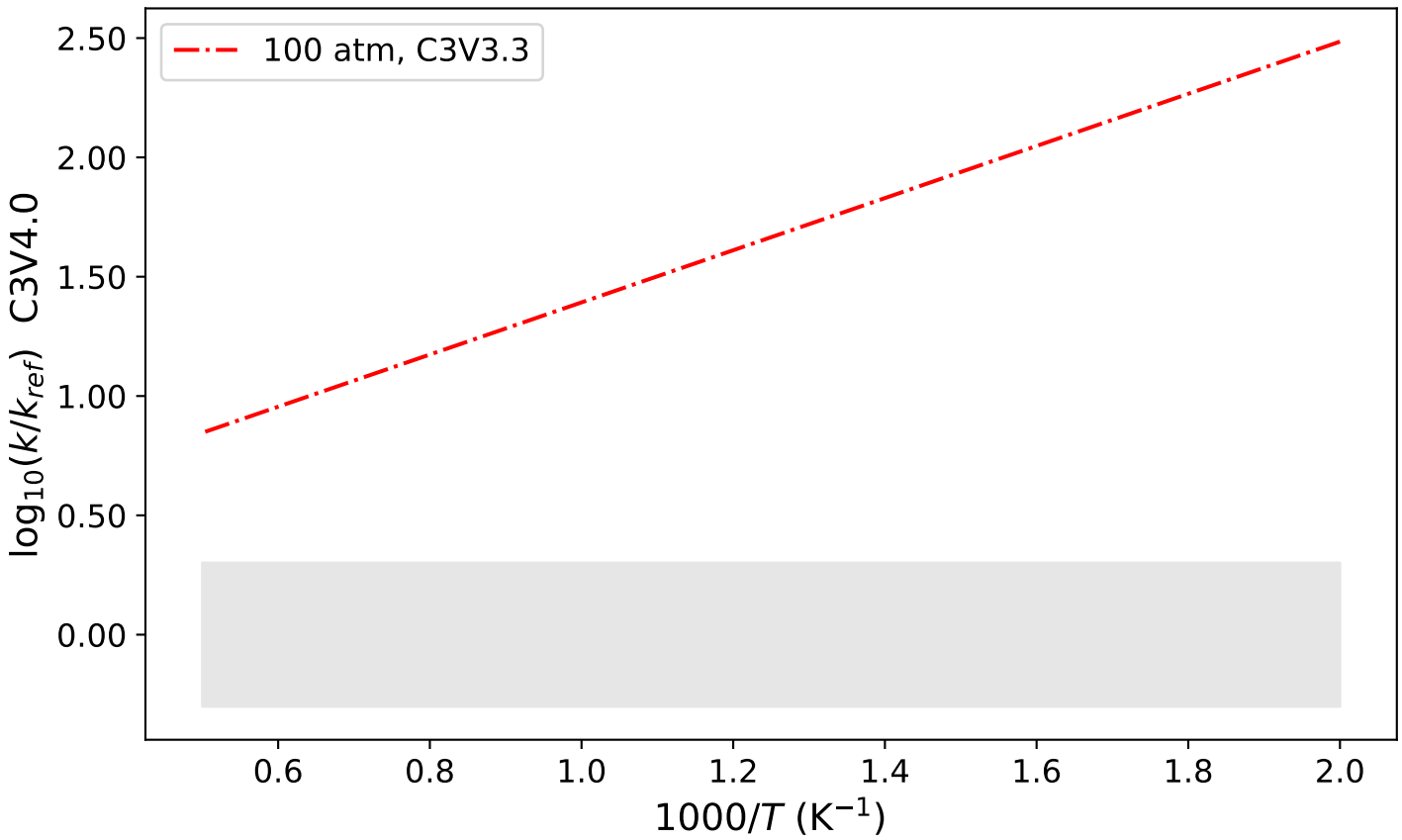
C2H4OCHO = CH3CHO + HCO



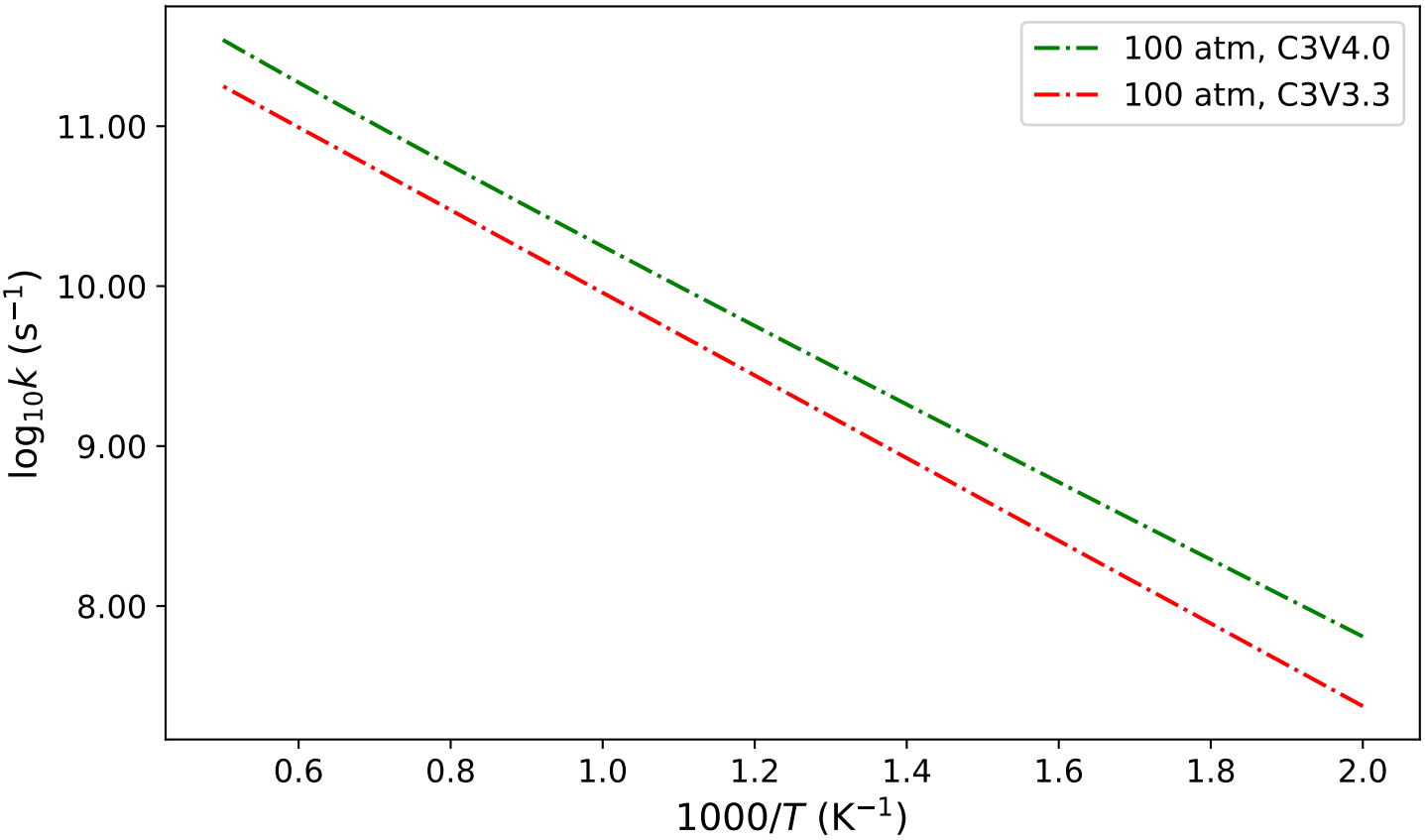



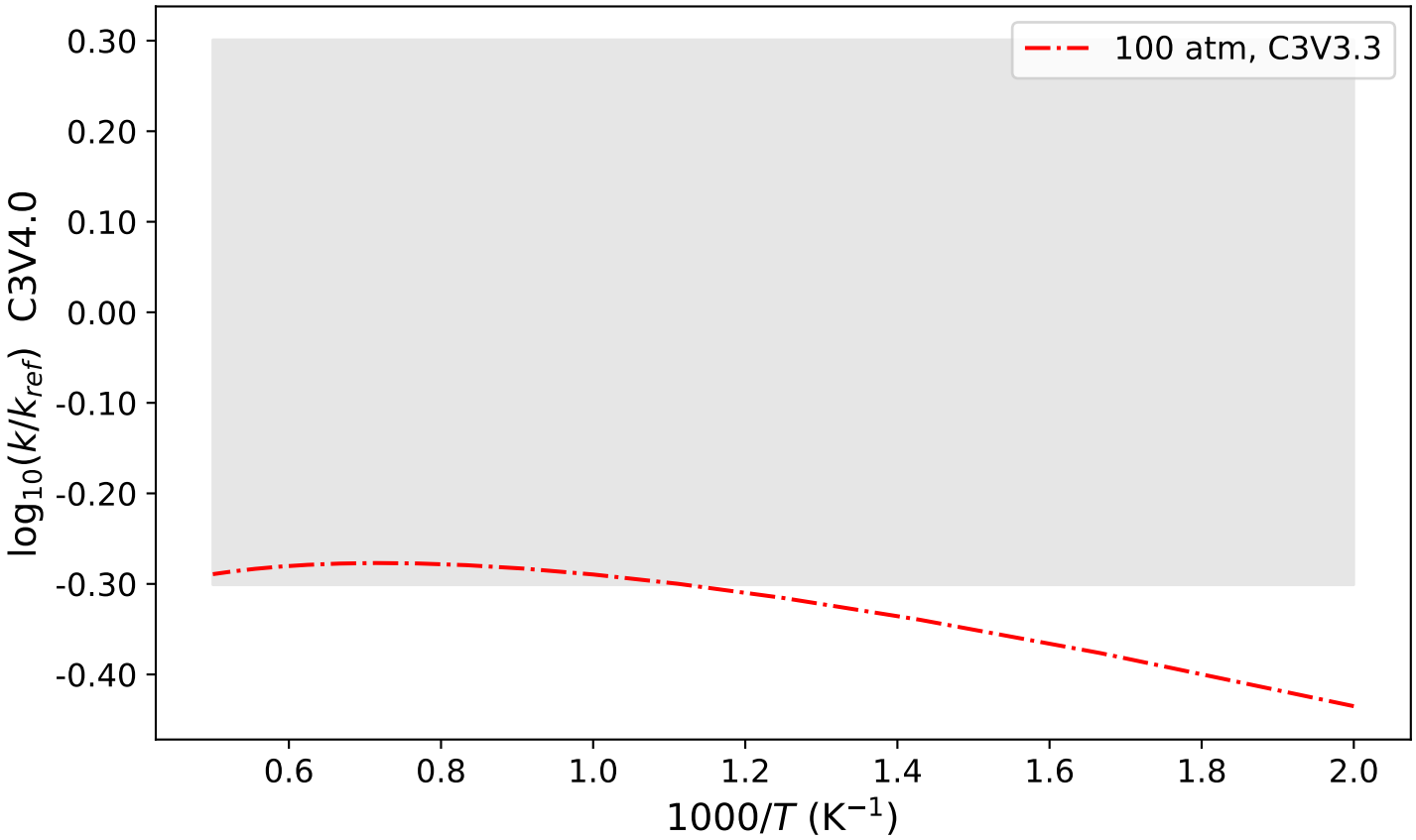
CH3CO3 + C4H10 = CH3CO3H + SC4H9



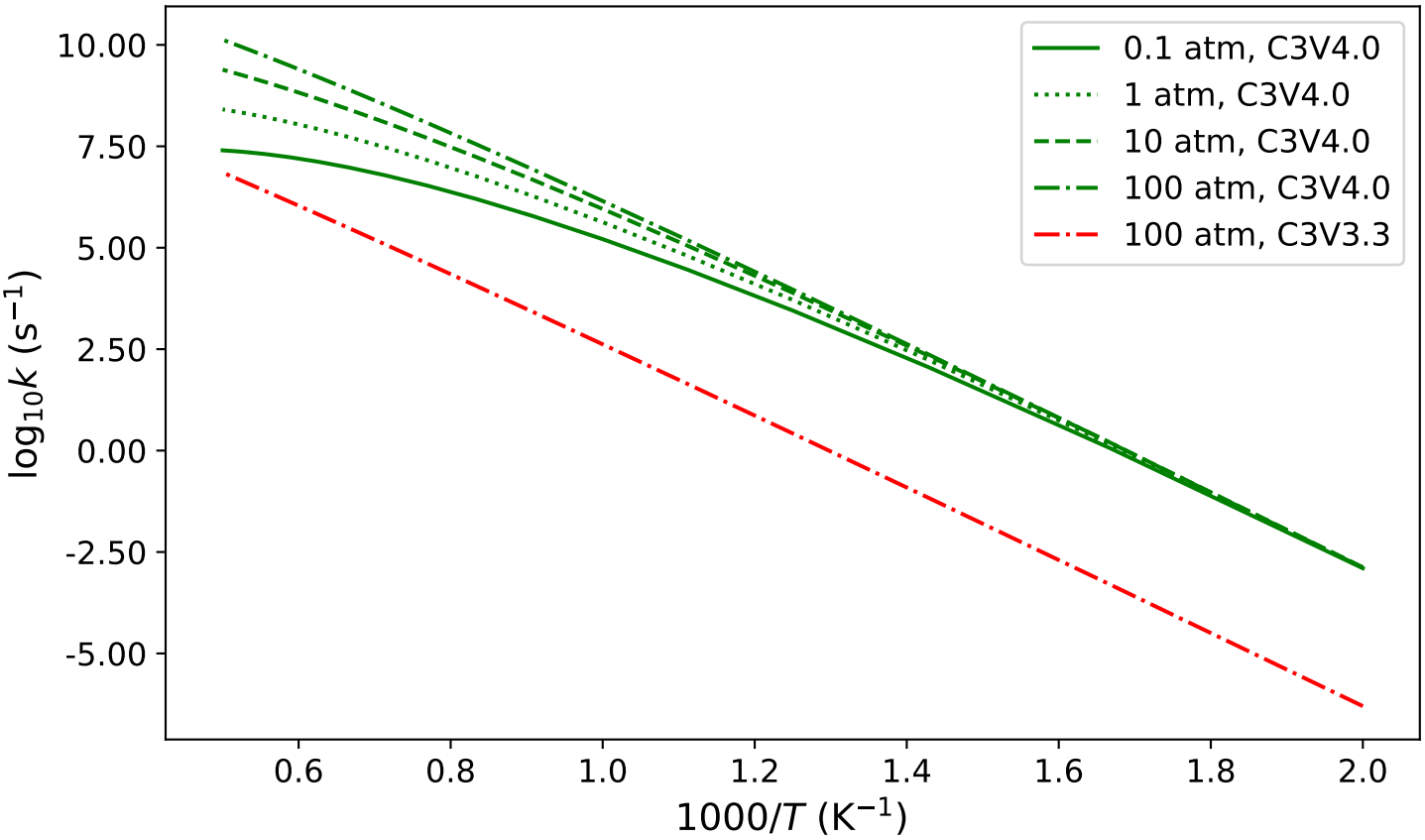



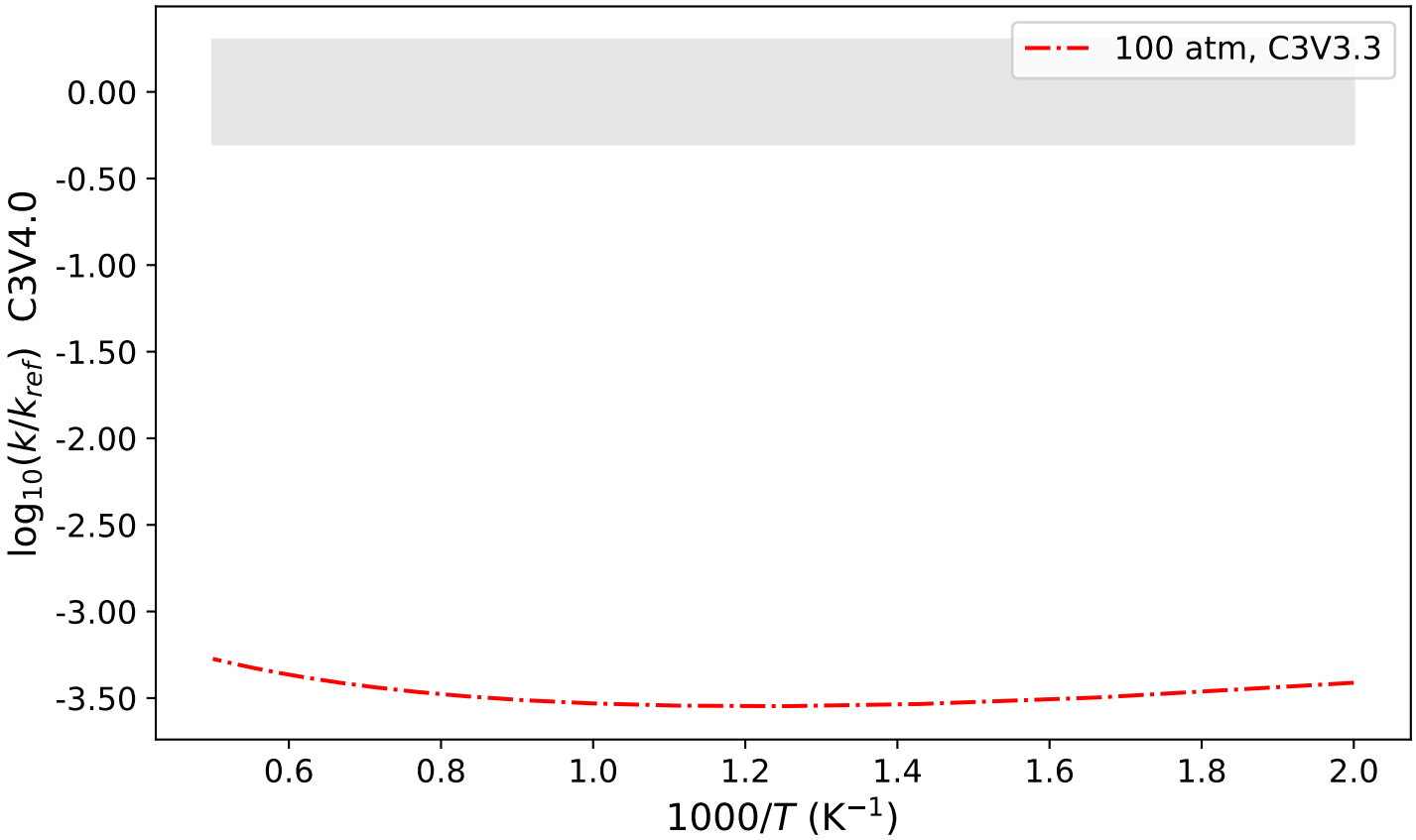
IC4H8O2H-T = OH + IC4H8O



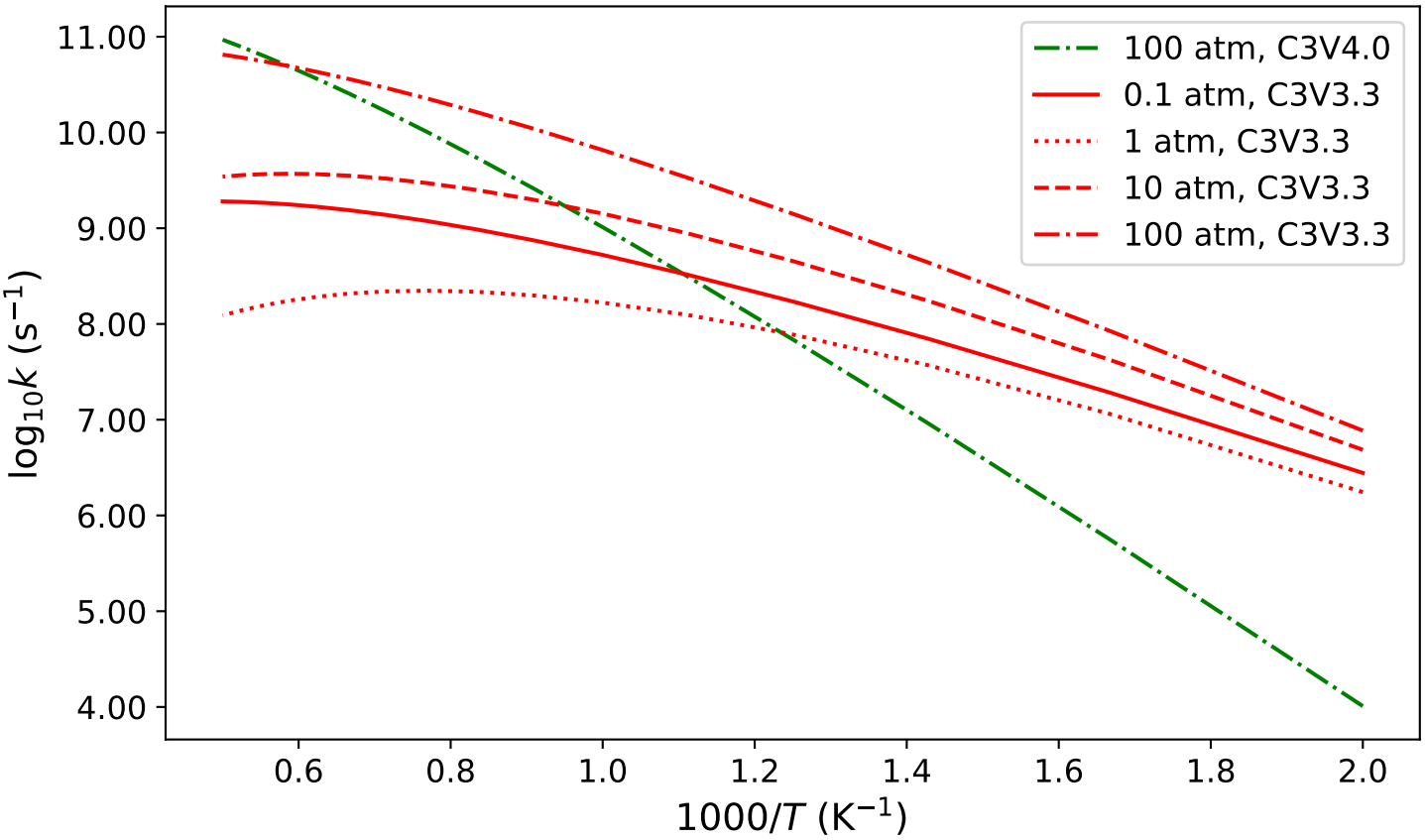



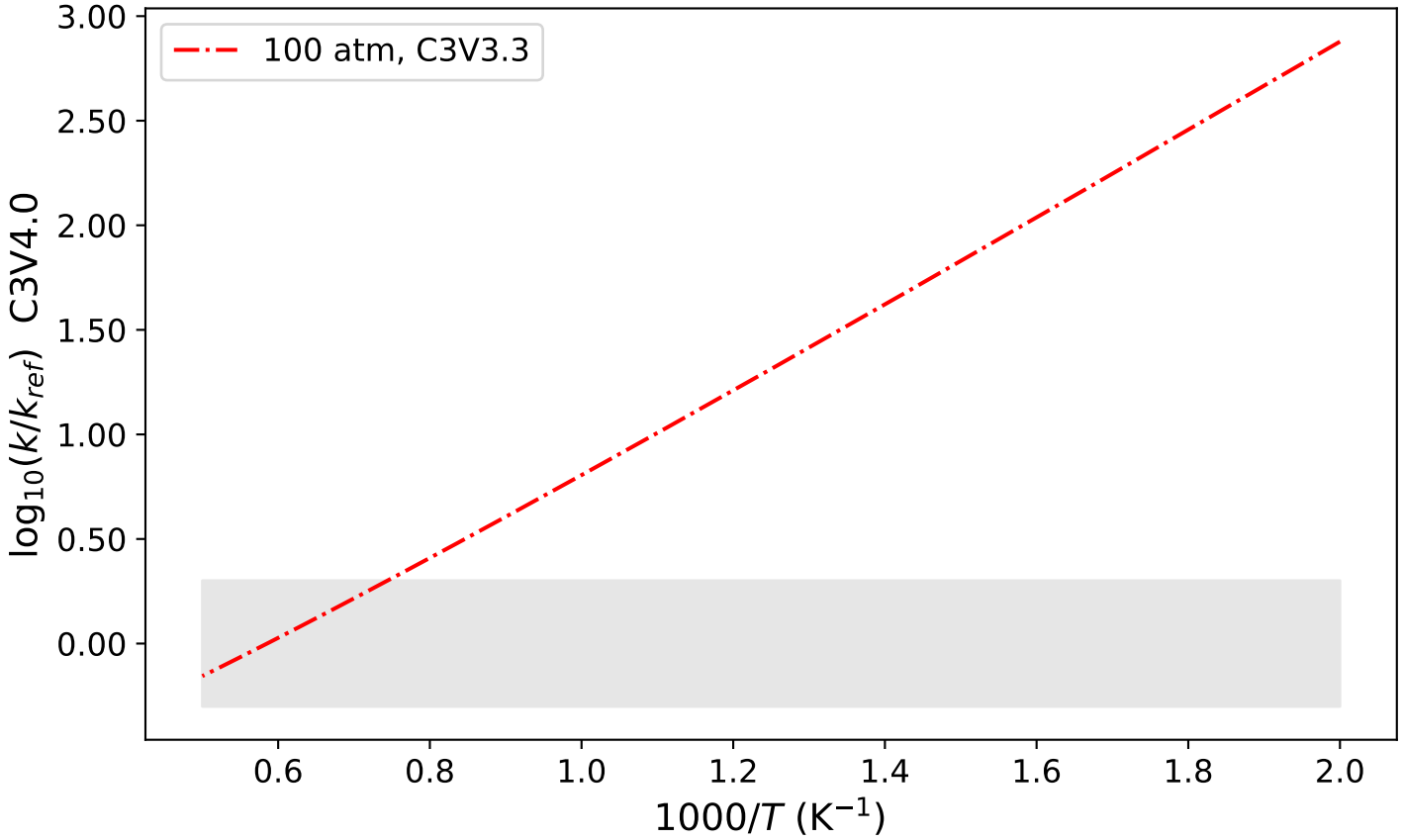
C3H4CH2OH-1OOH = OH + C3H4CH2OH-1O



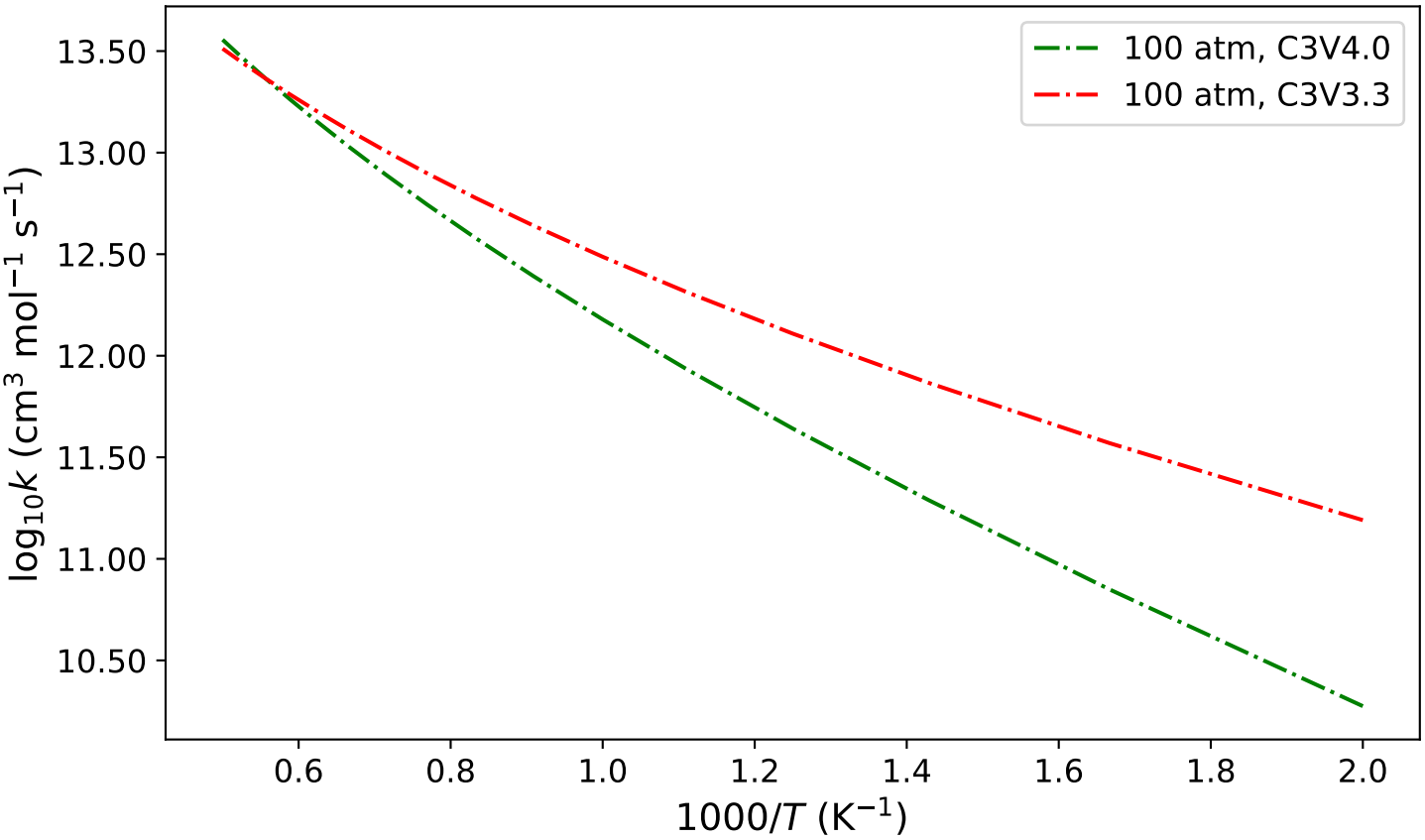



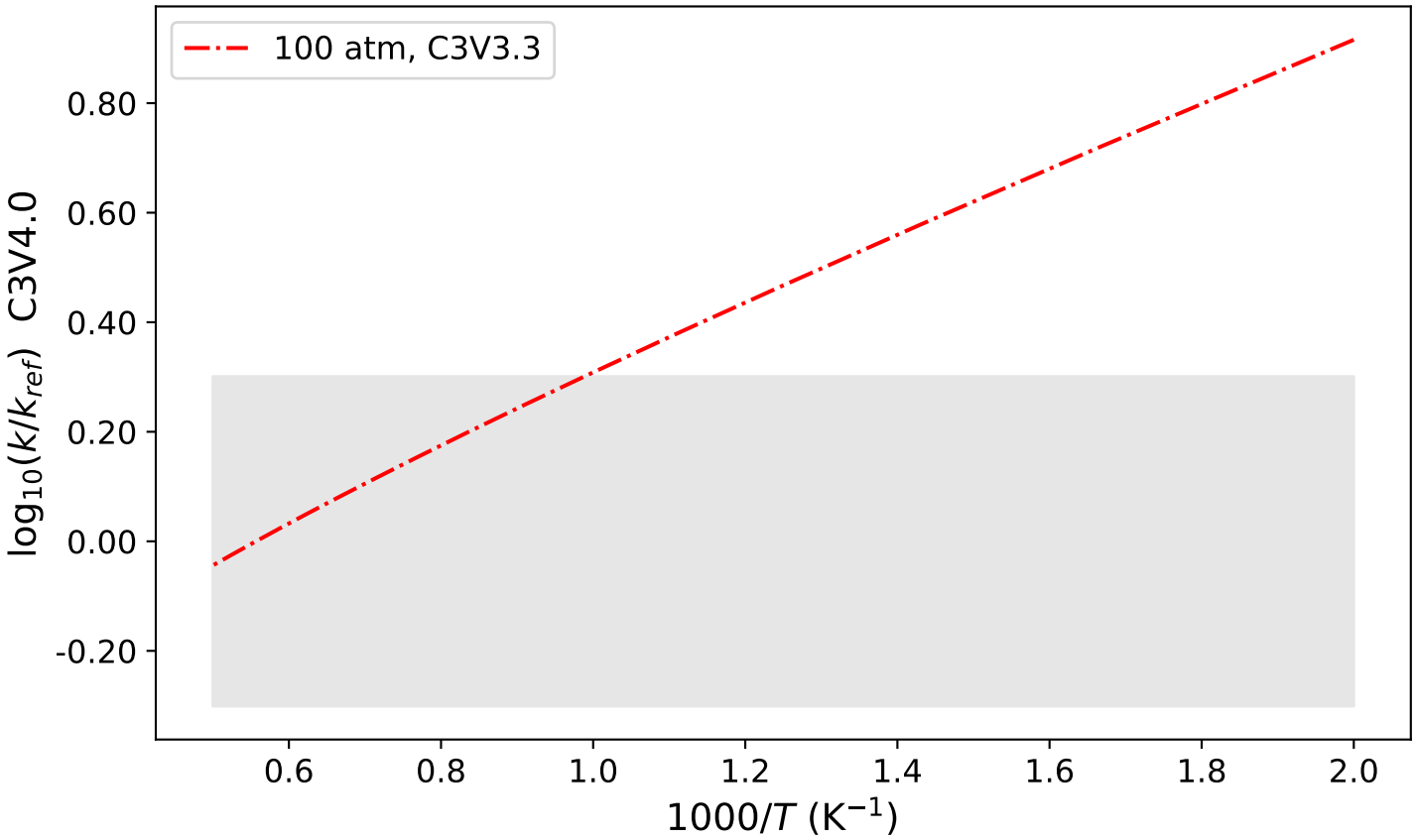
C4H7O2-1 = CH2O + C3H5-S



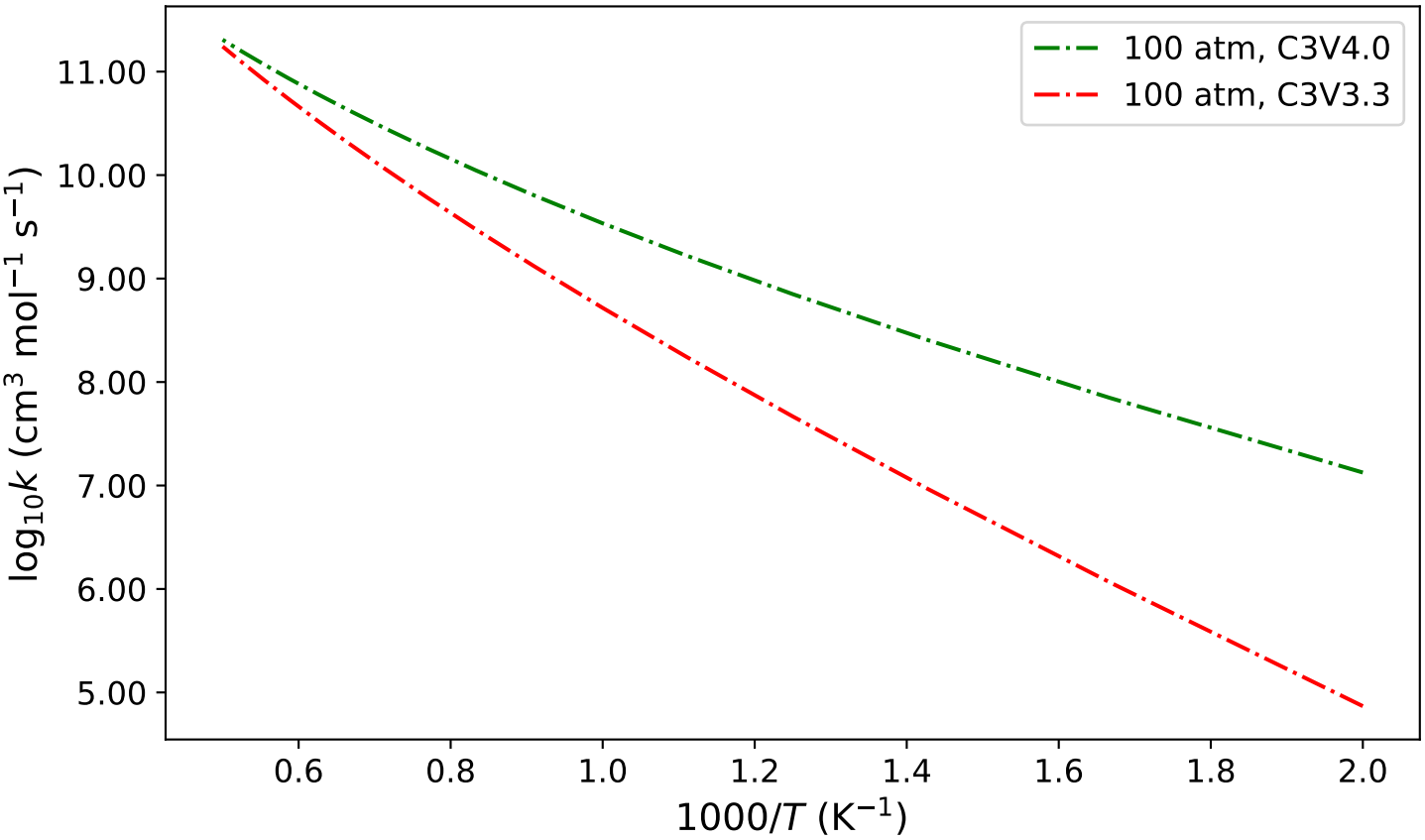



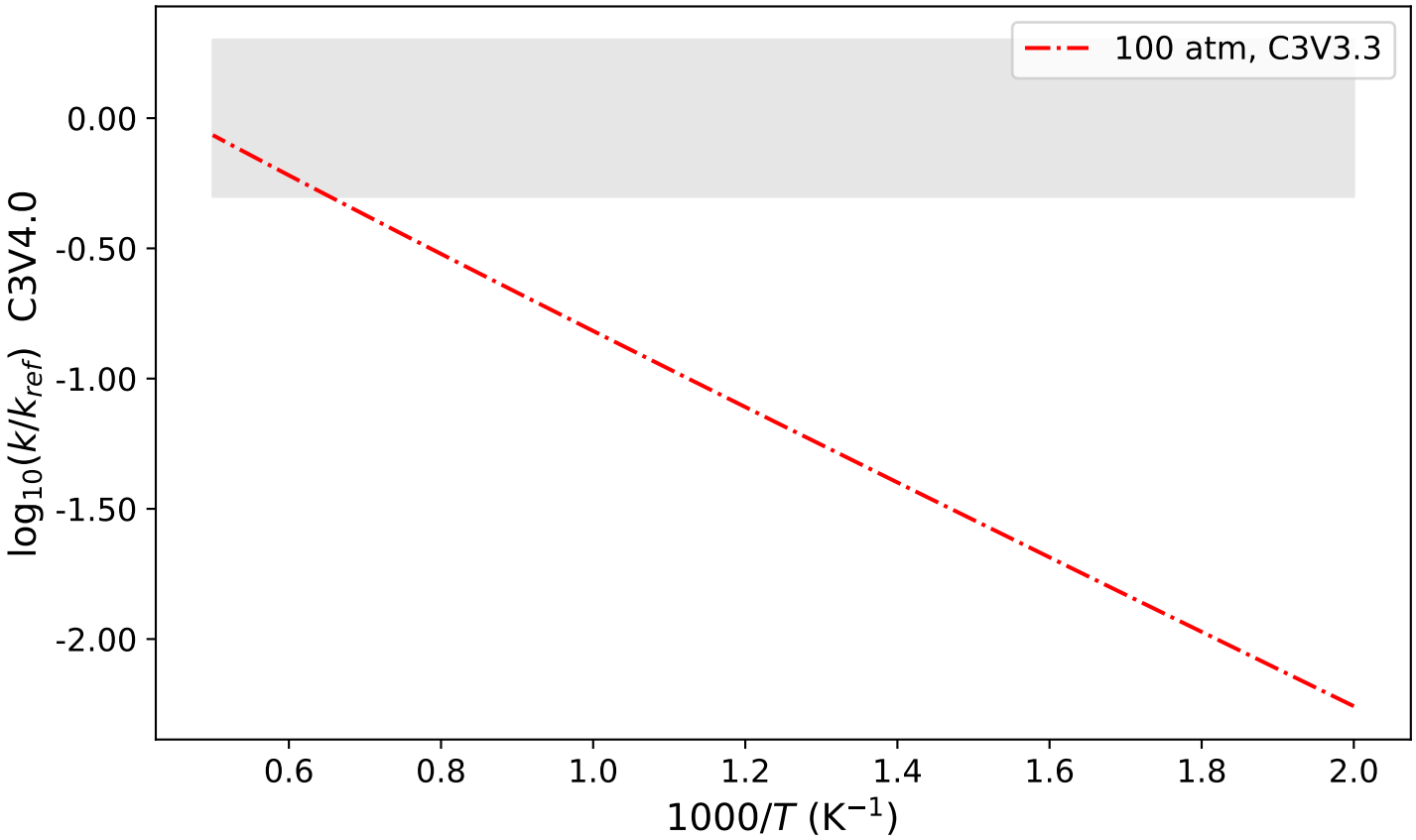
H + C4H6-2 = H2 + C4H5-2



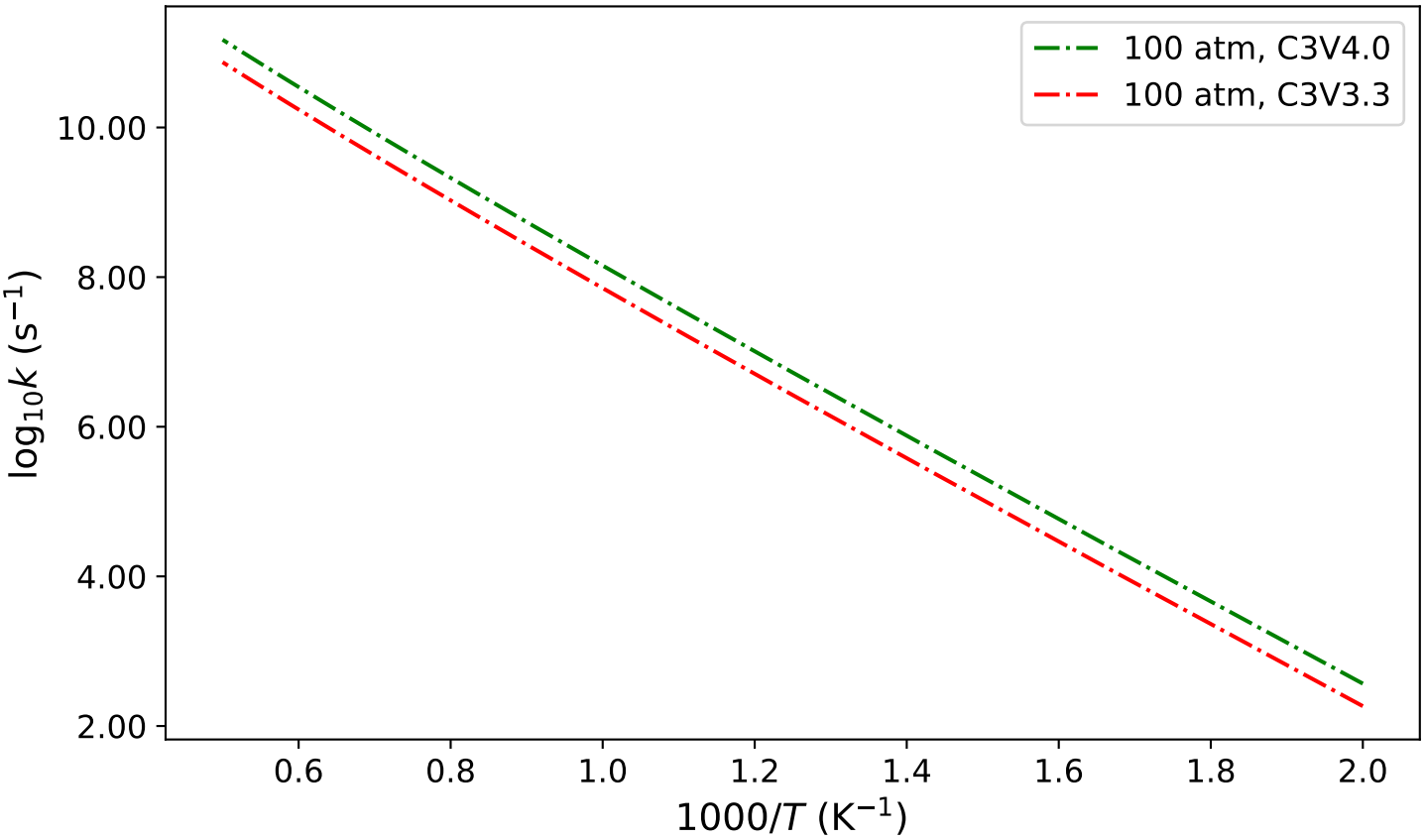



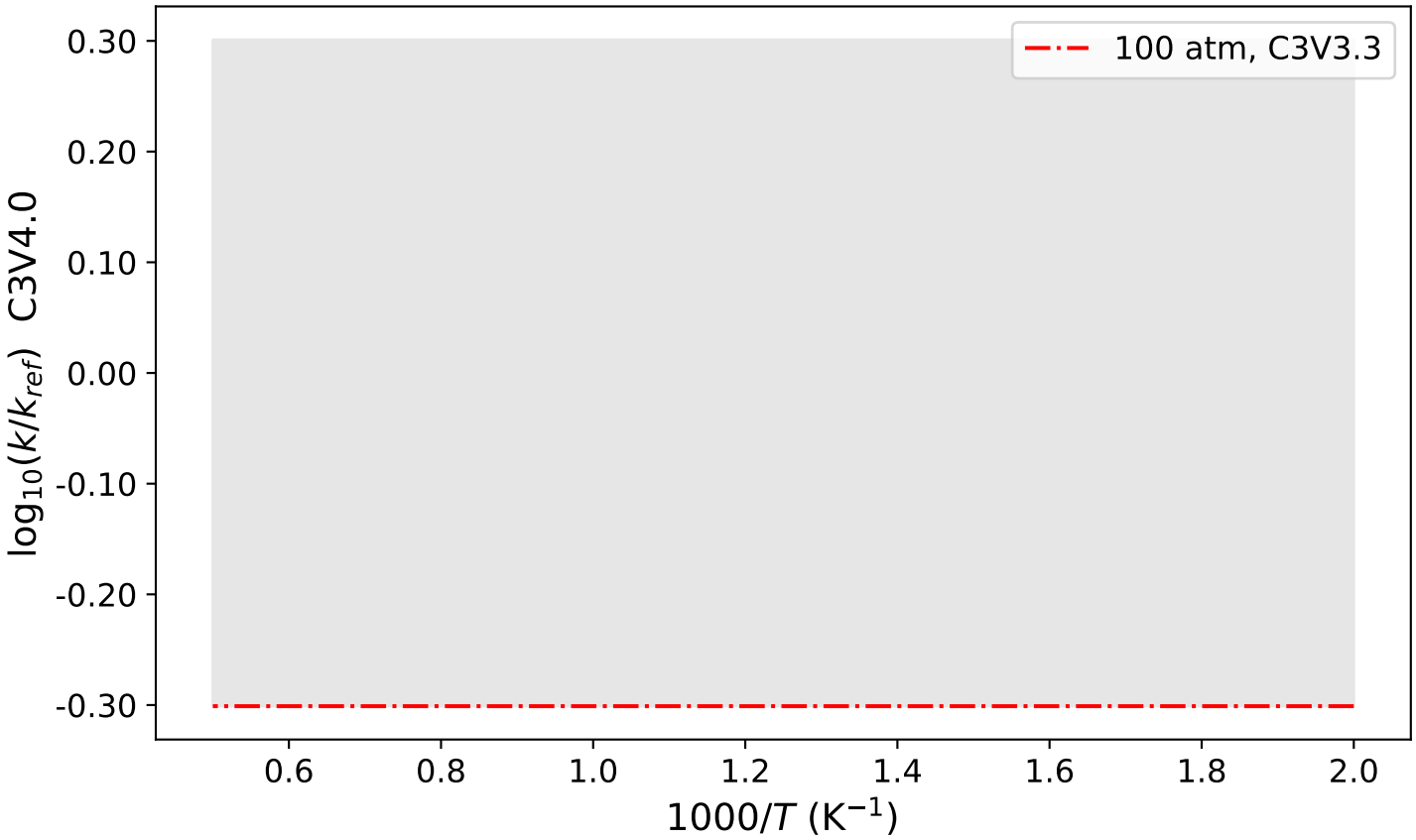
HO2 + C4H72-1OH = H2O2 + C4H63,1-1OH



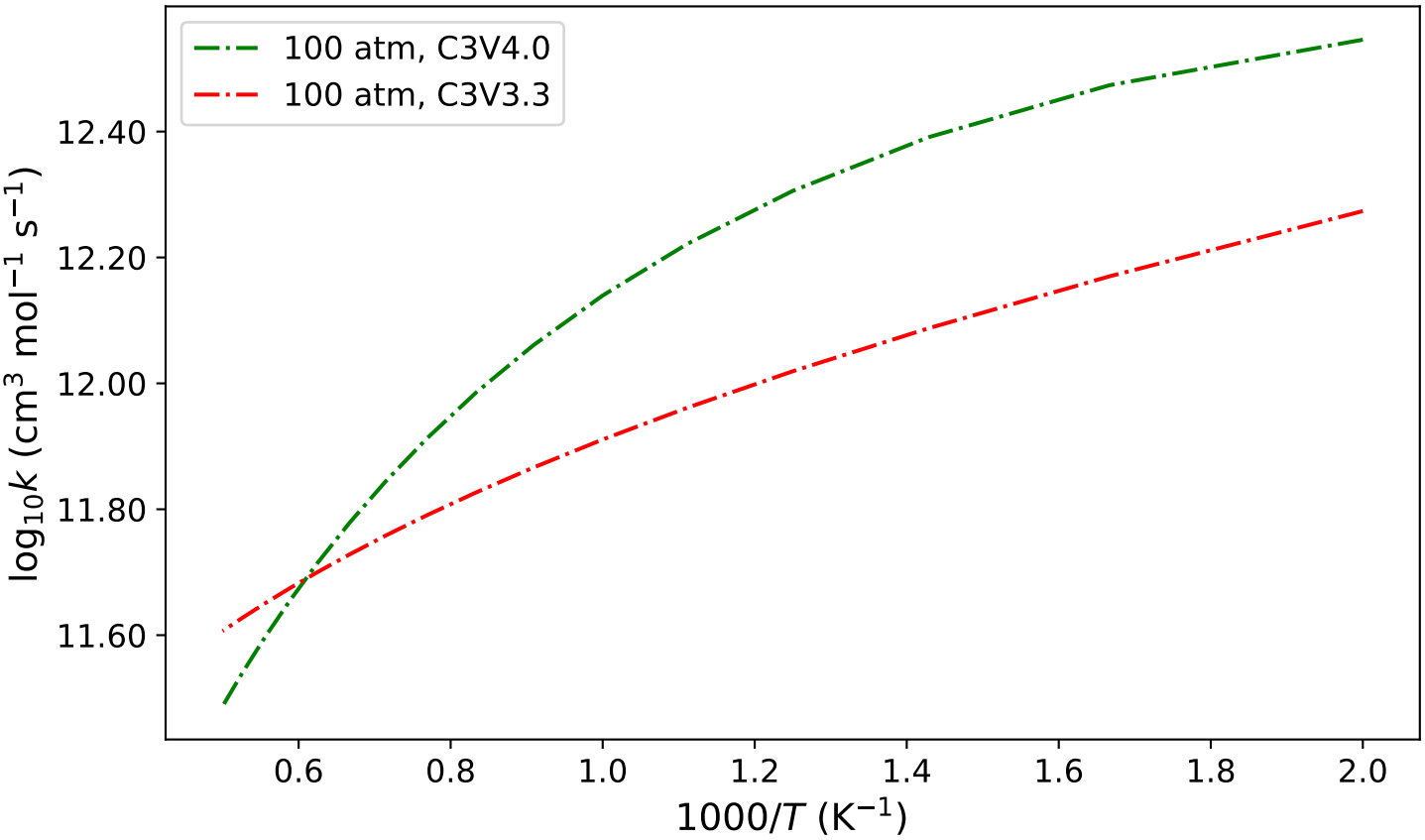



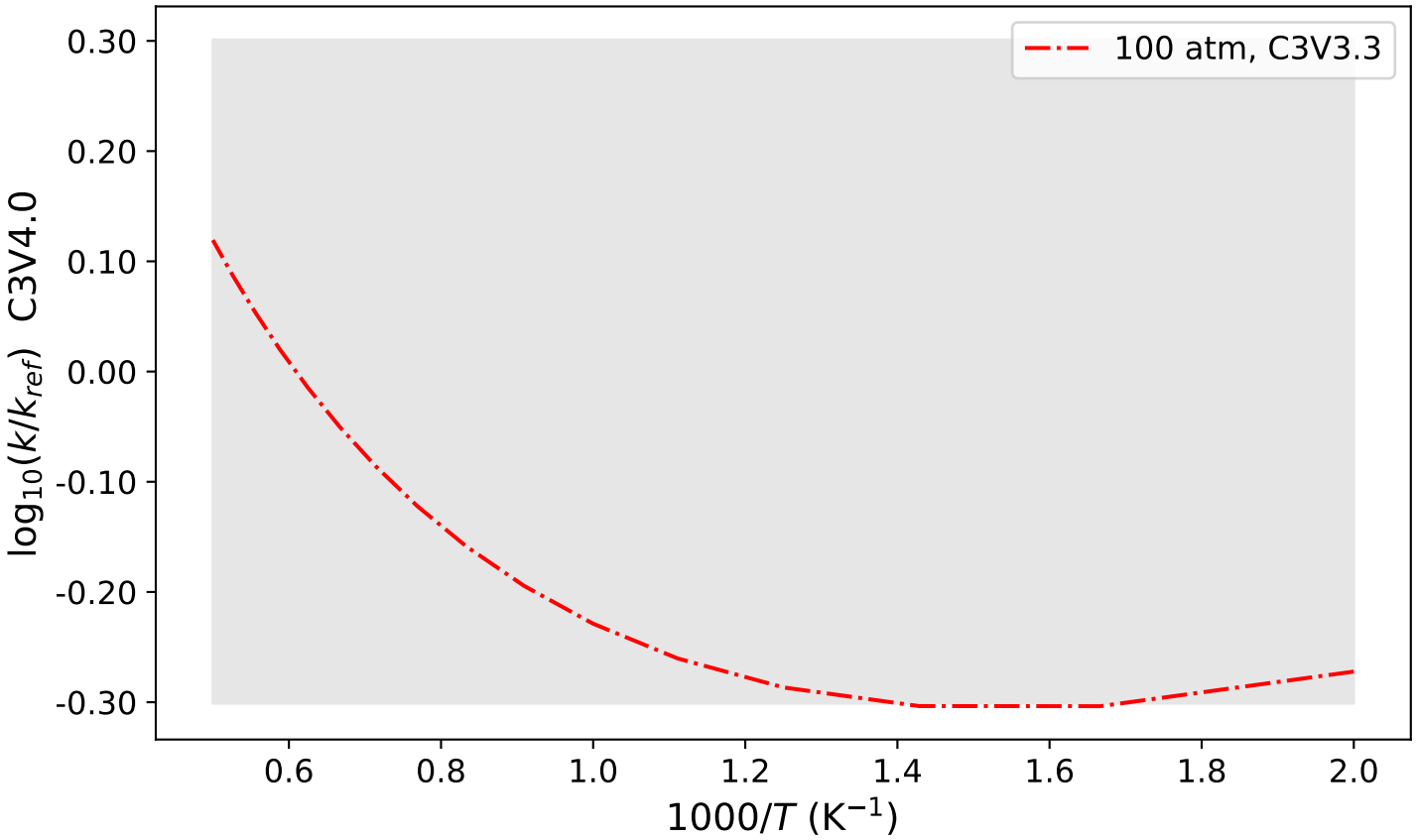
C4H8OOH2-4 => CH3CHO + OH + C2H4



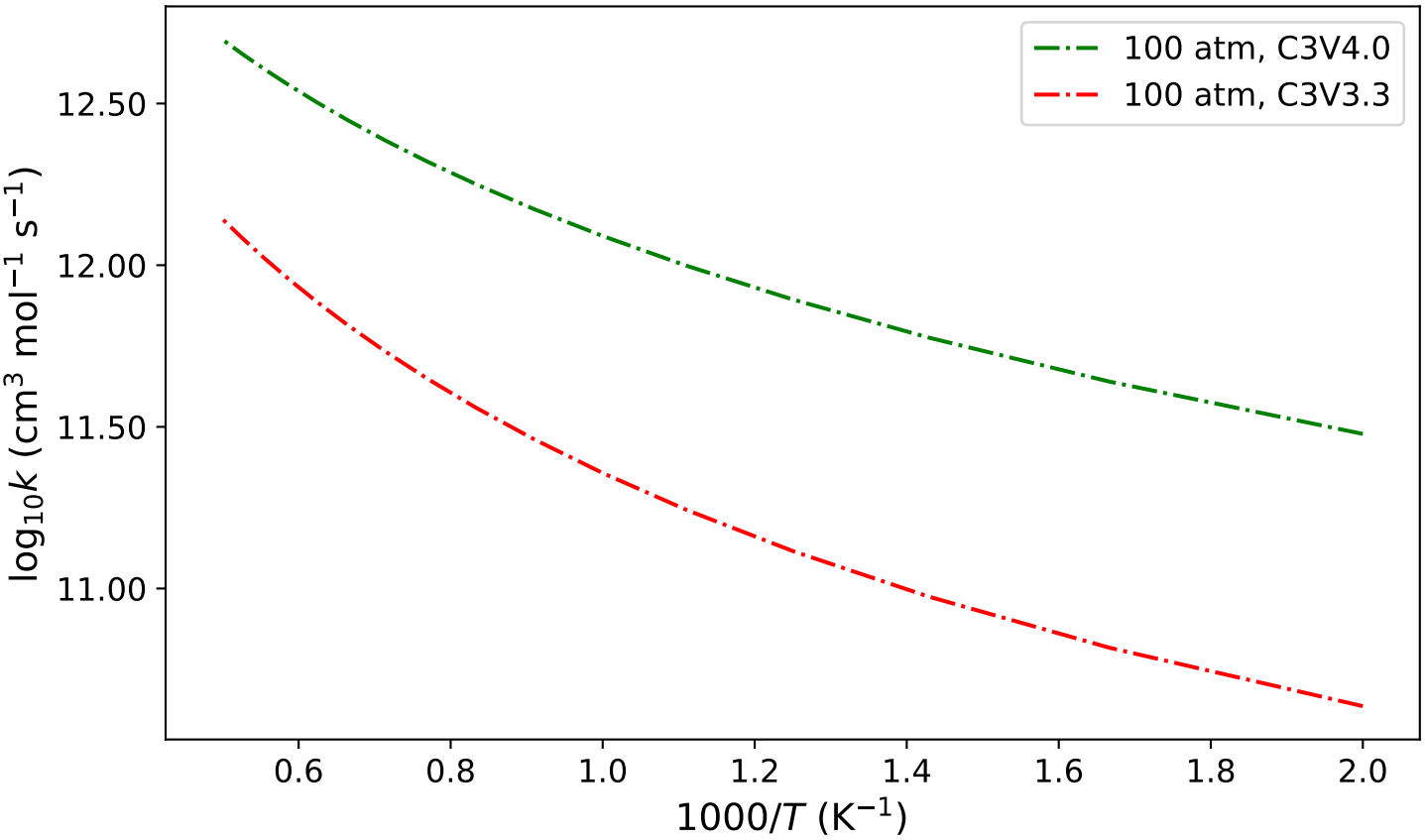



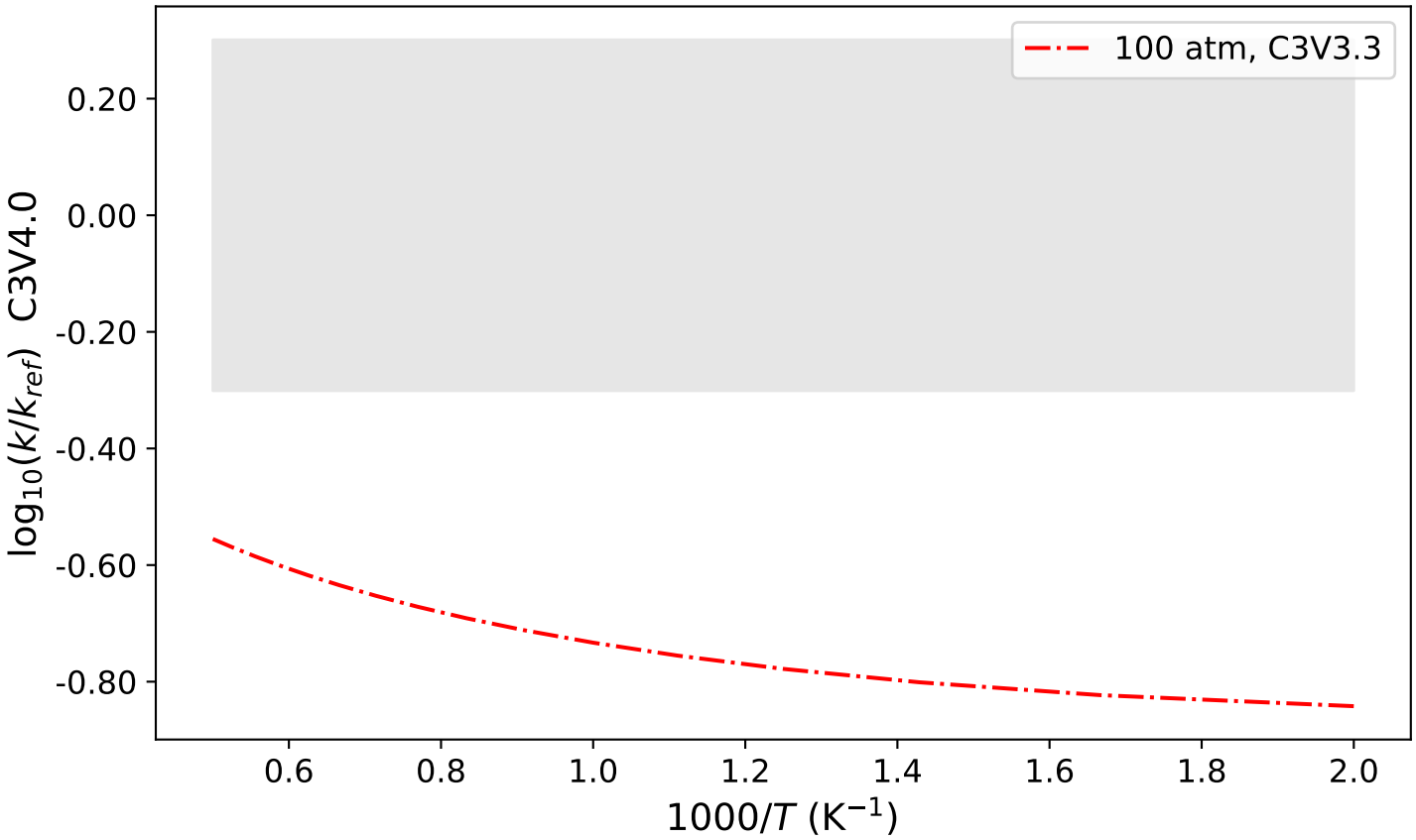
O2 + C4H8OOH1-3 = C4H8OOH1-3O2



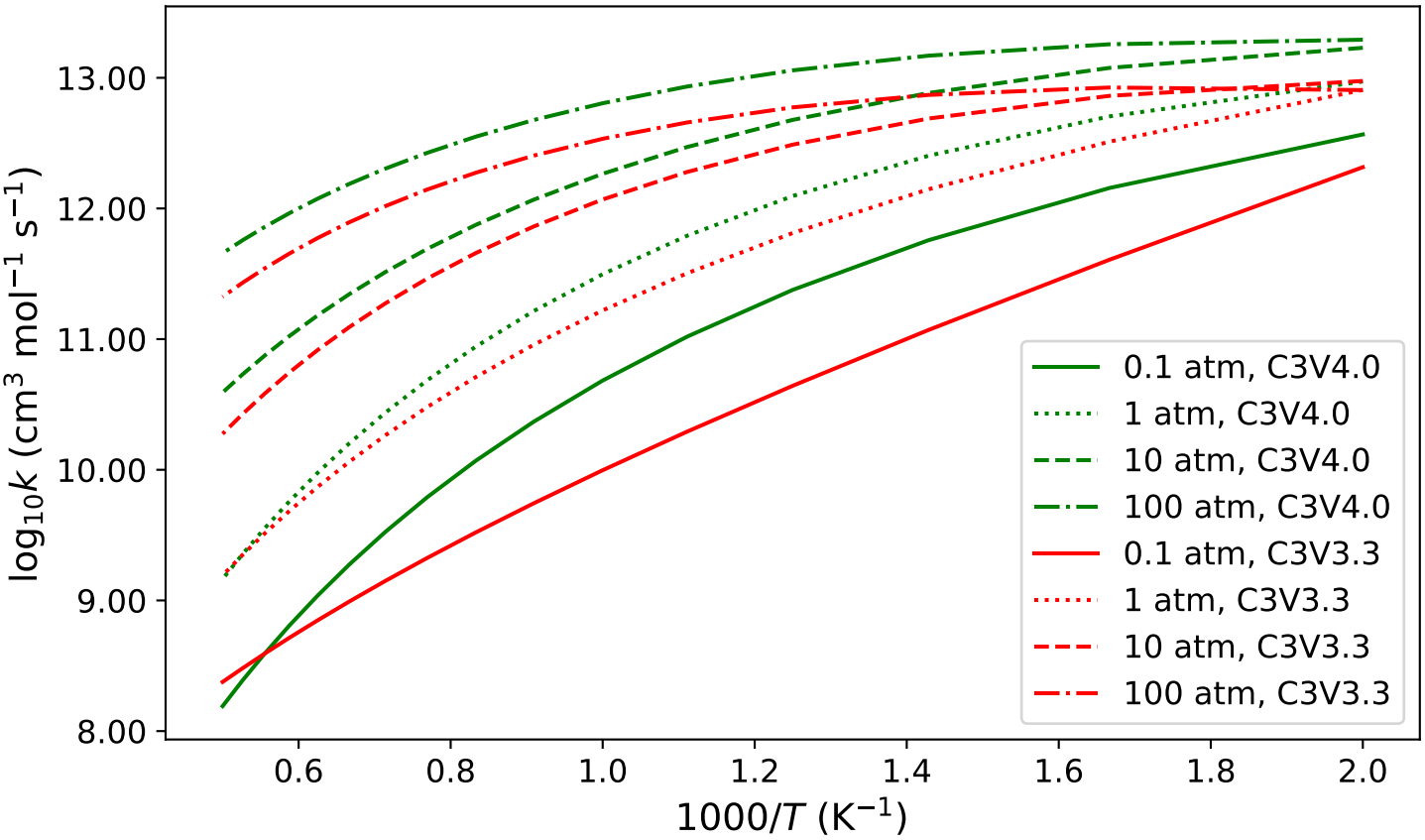



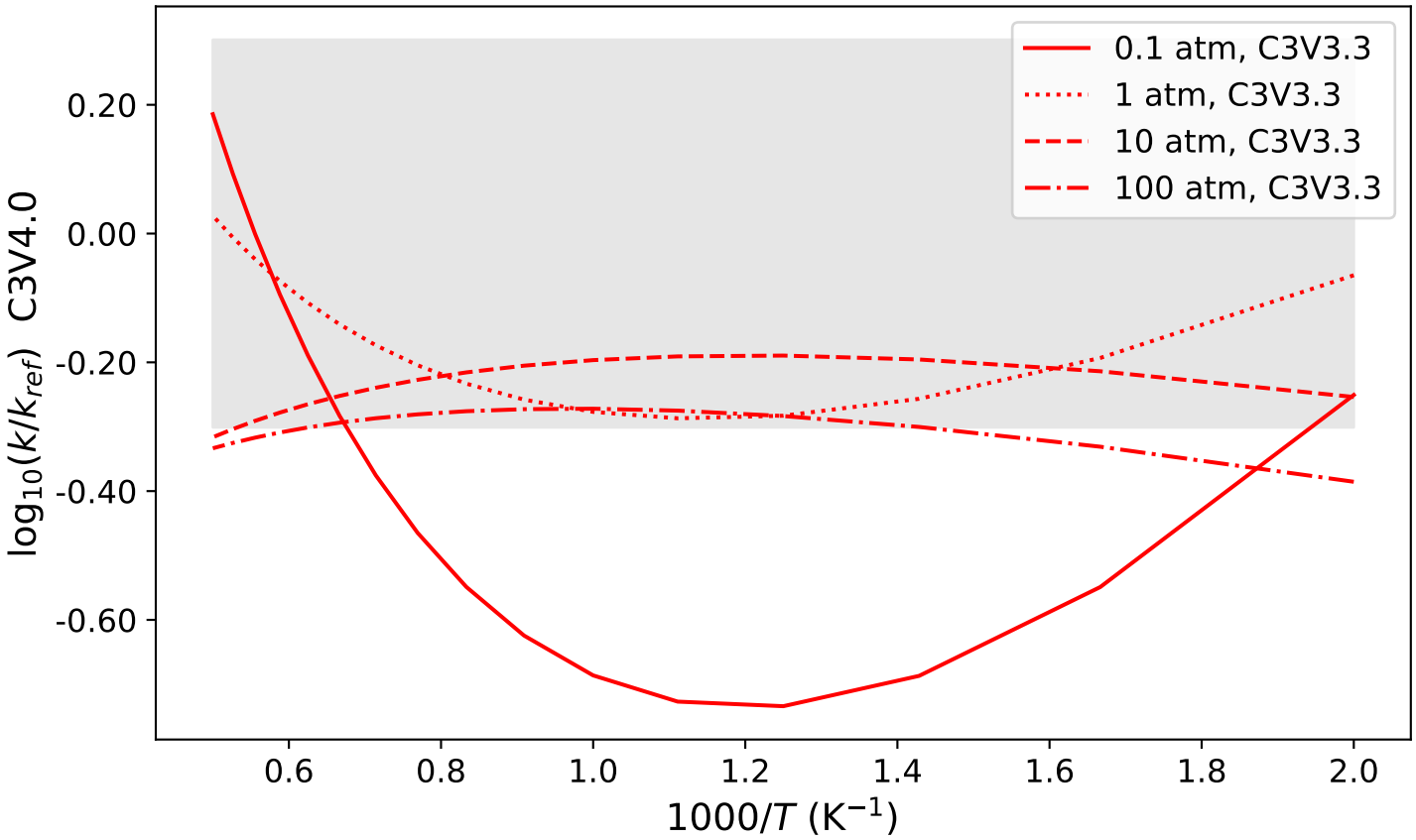
OH + IC3H7OH = IC3H7O + H2O



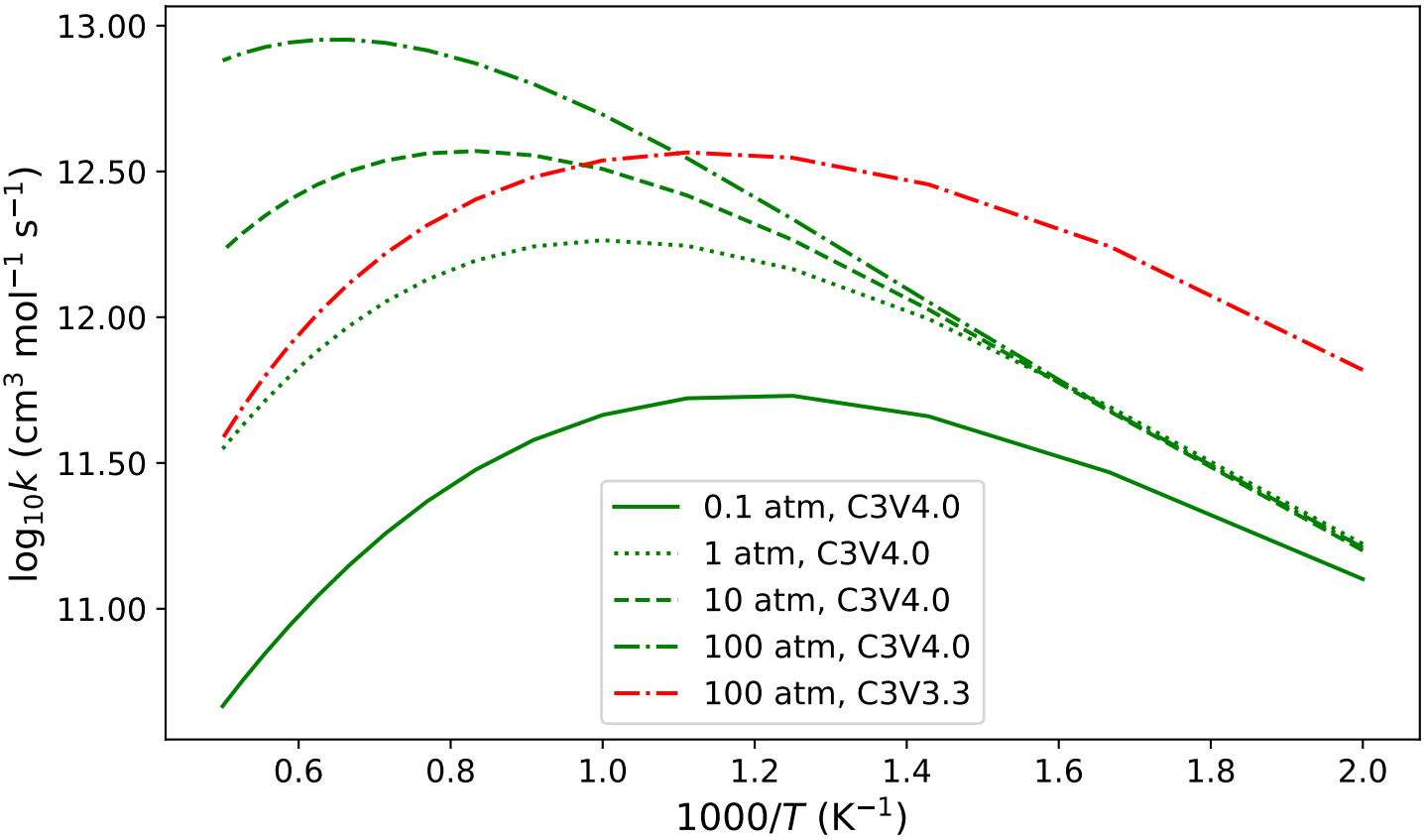



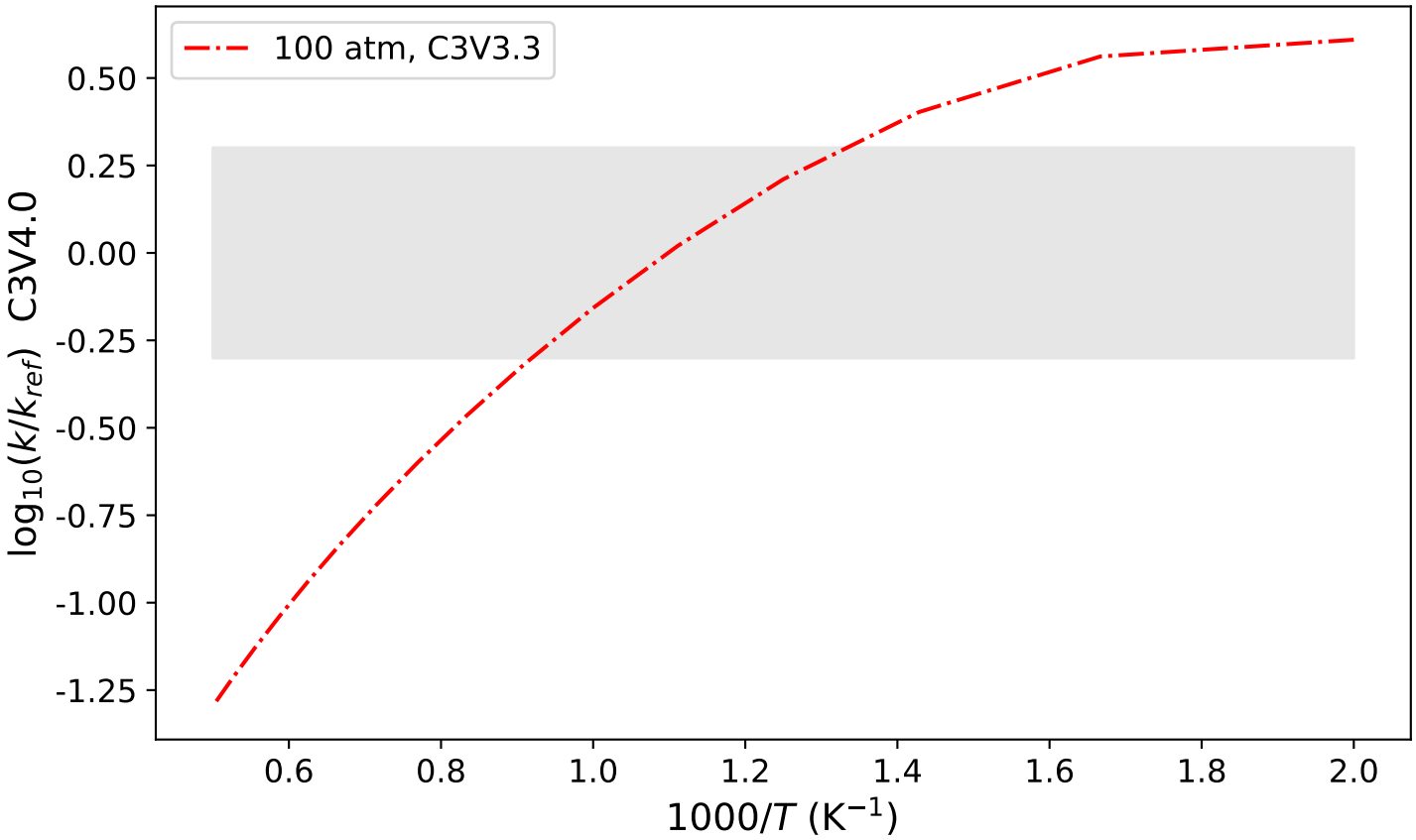
HO2 + IC4H7 = IC4H7OOH



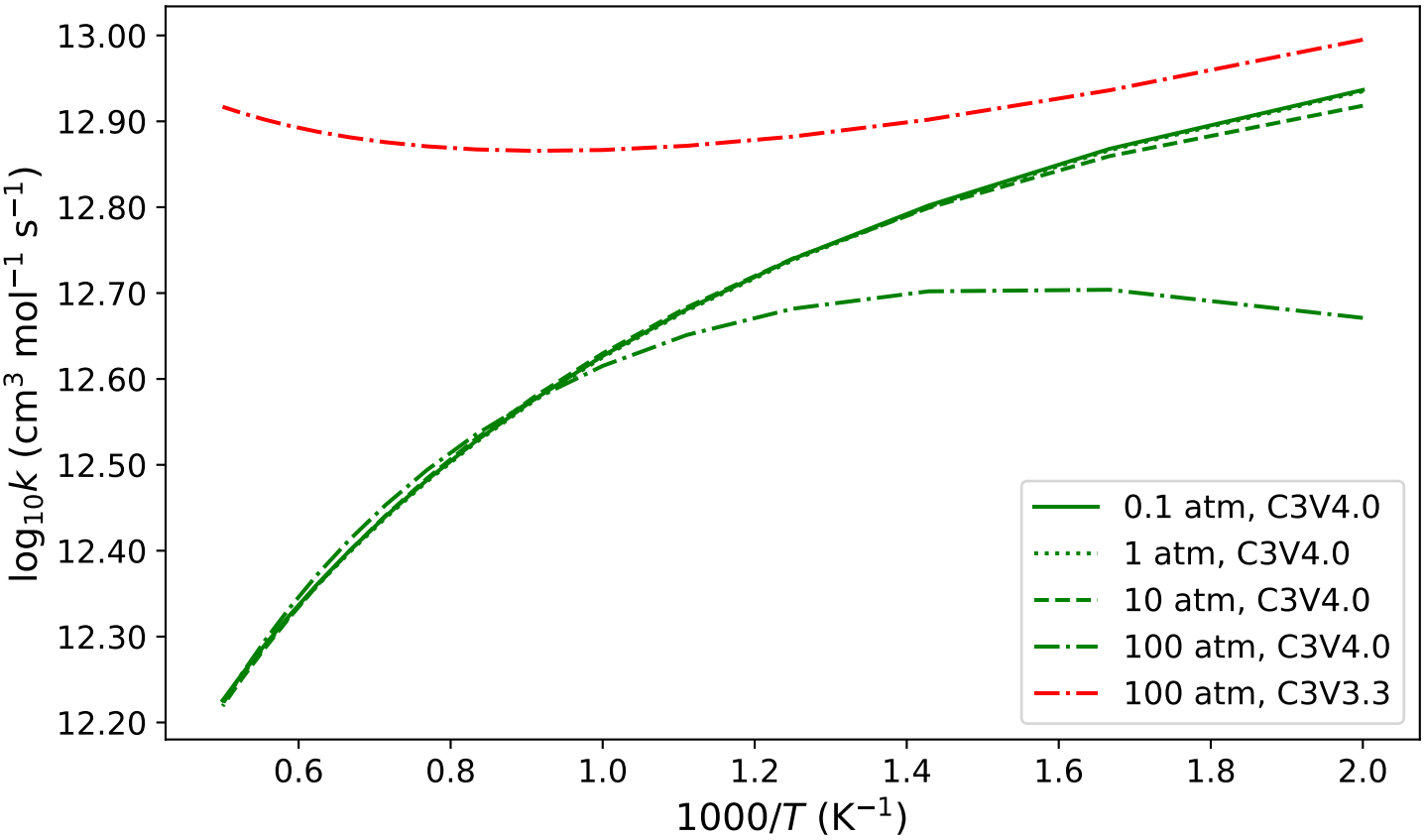



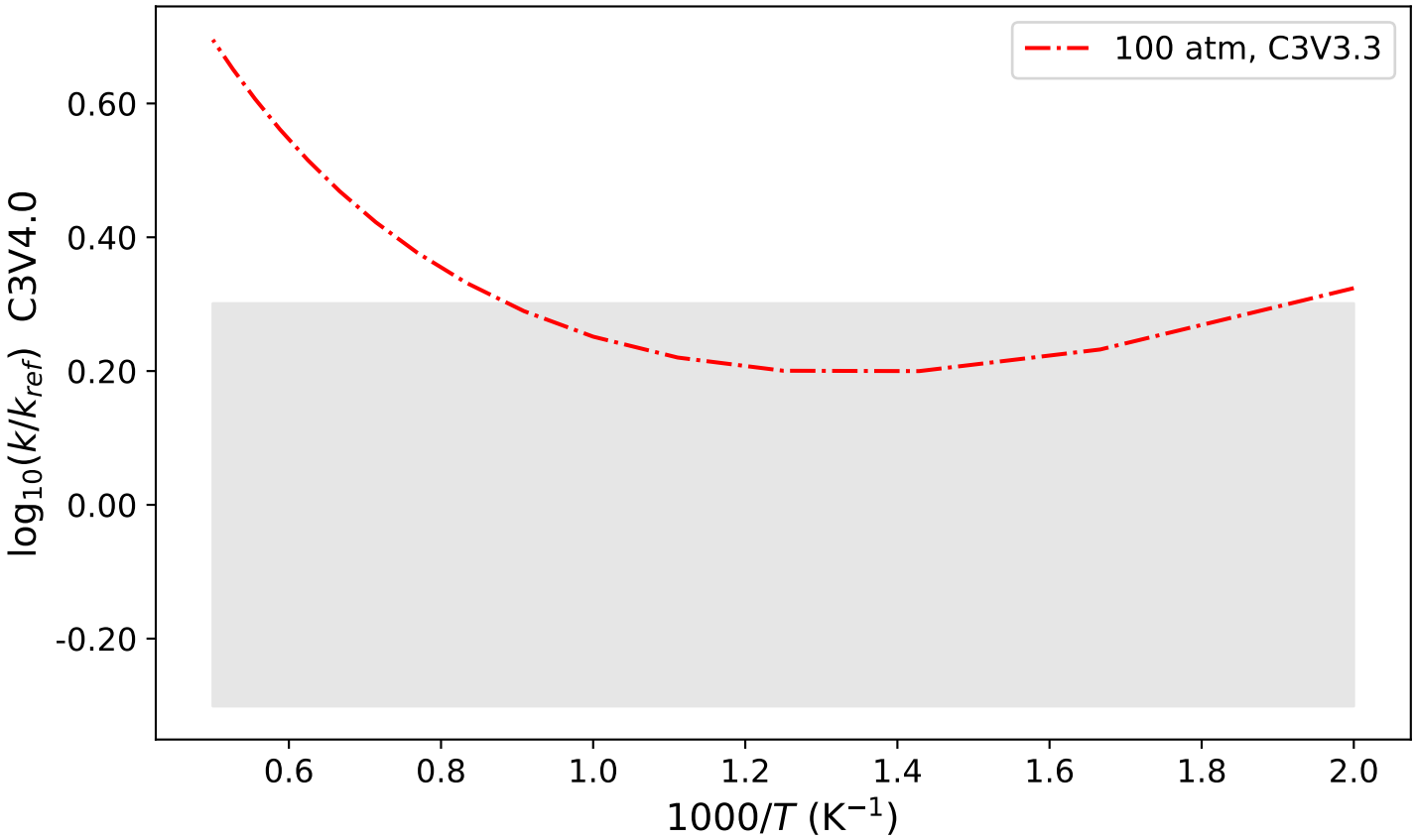
H + C4H2 = C4H3-N


13.00
12.50
12.00
11.50
11.00
$\log_{10}k$ (cm$^3$ mol$^{-1}$ s$^{-1}$)
0.1 atm, C3V4.0
1 atm, C3V4.0
10 atm, C3V4.0
100 atm, C3V4.0
100 atm, C3V3.3
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ (K$^{-1}$)

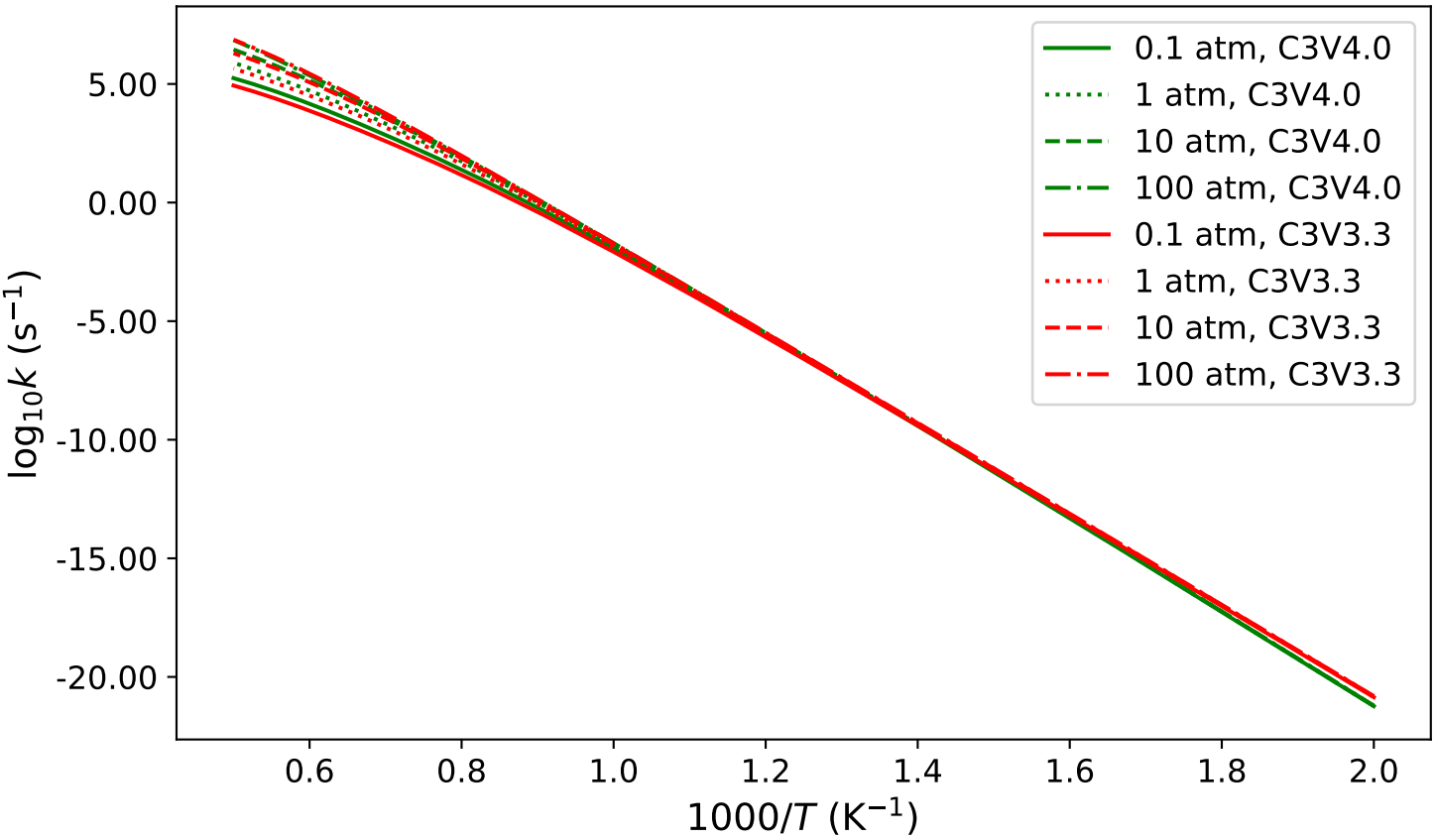

100 atm, C3V3.3
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00
-1.25
$\log_{10}(k/k_{ref})$ C3V4.0
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ (K$^{-1}$)

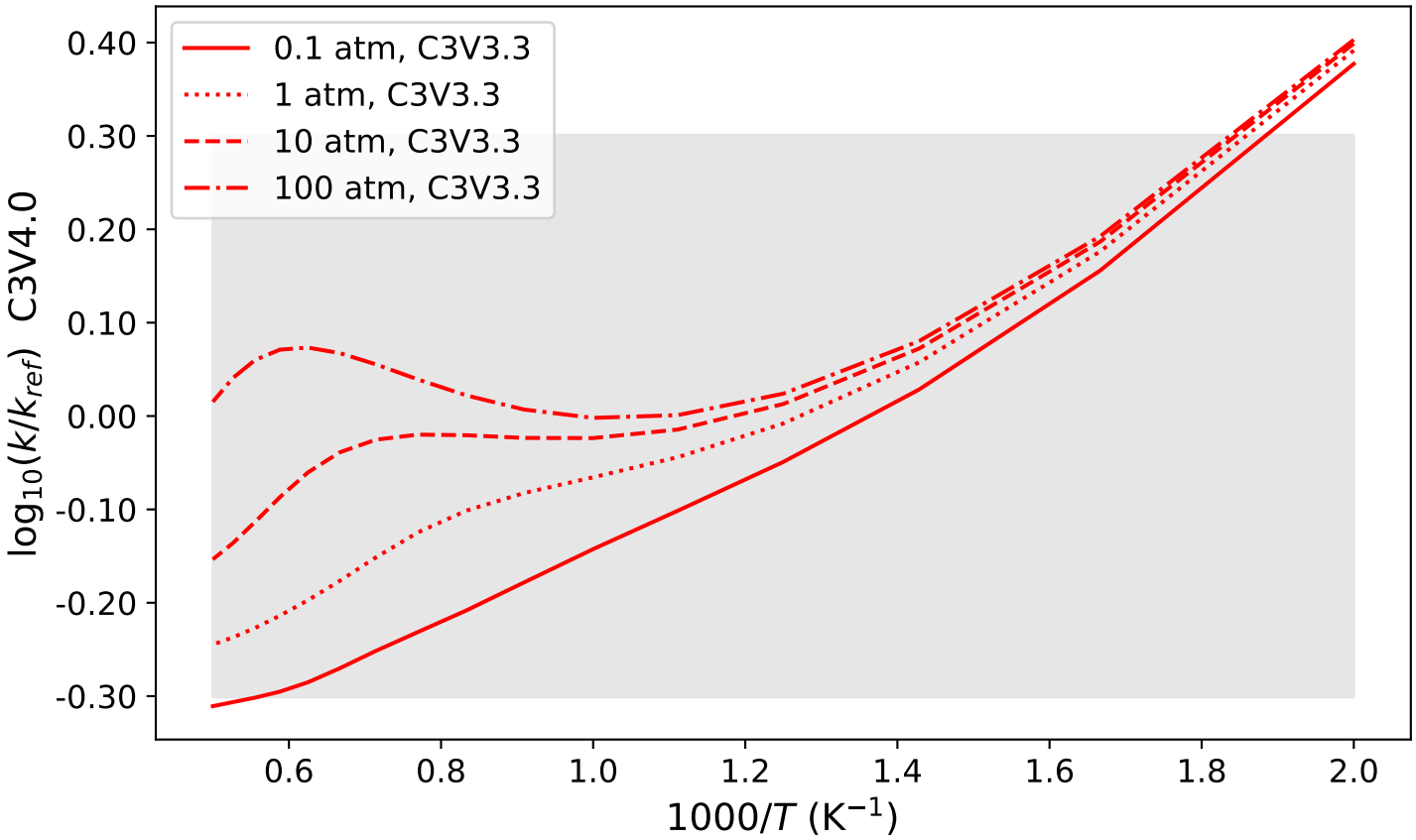
O2 + SQC4H7OHP-1 = HO2 + NC4KET12



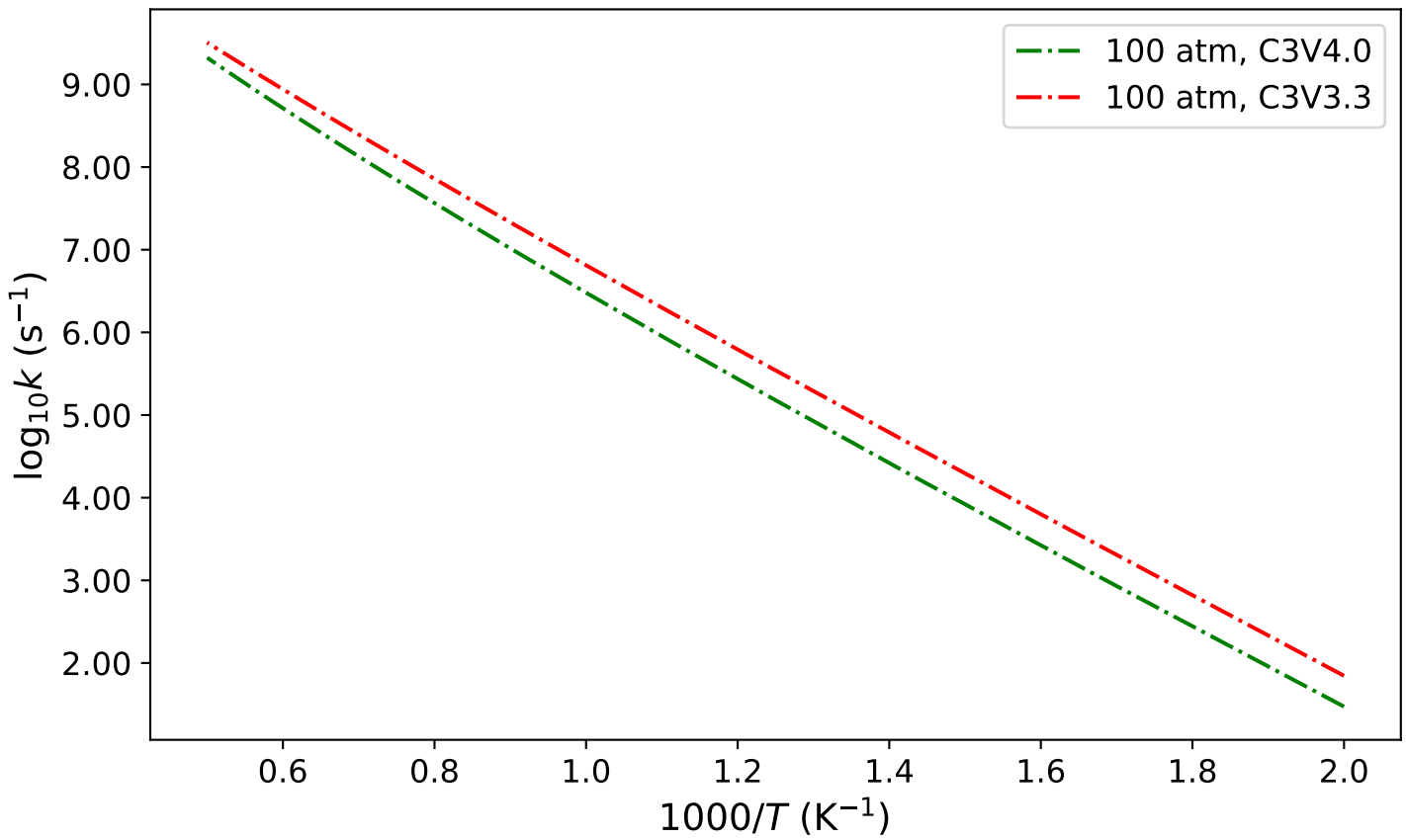



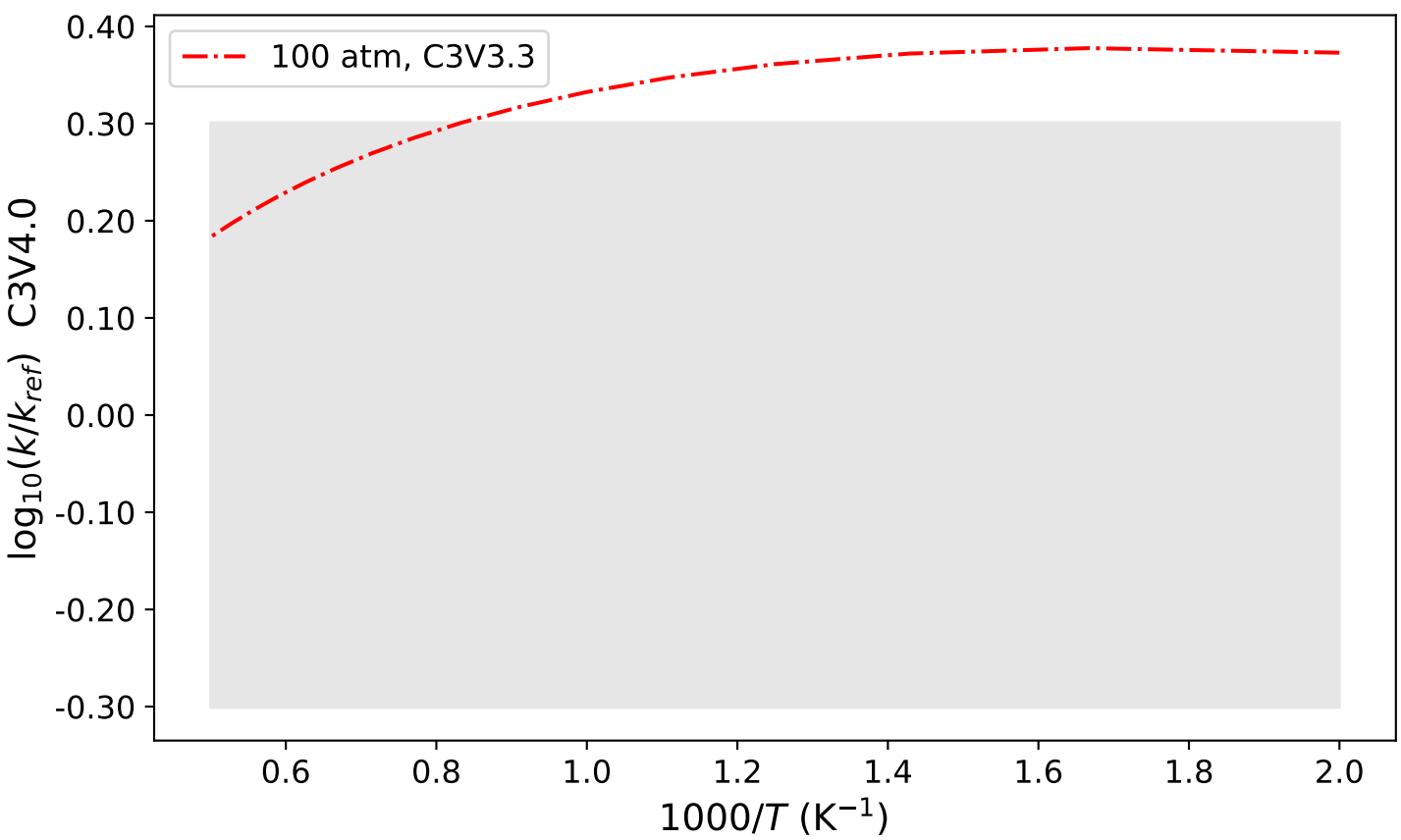
C3H8 (+M) = CH3 + C2H5 (+M)



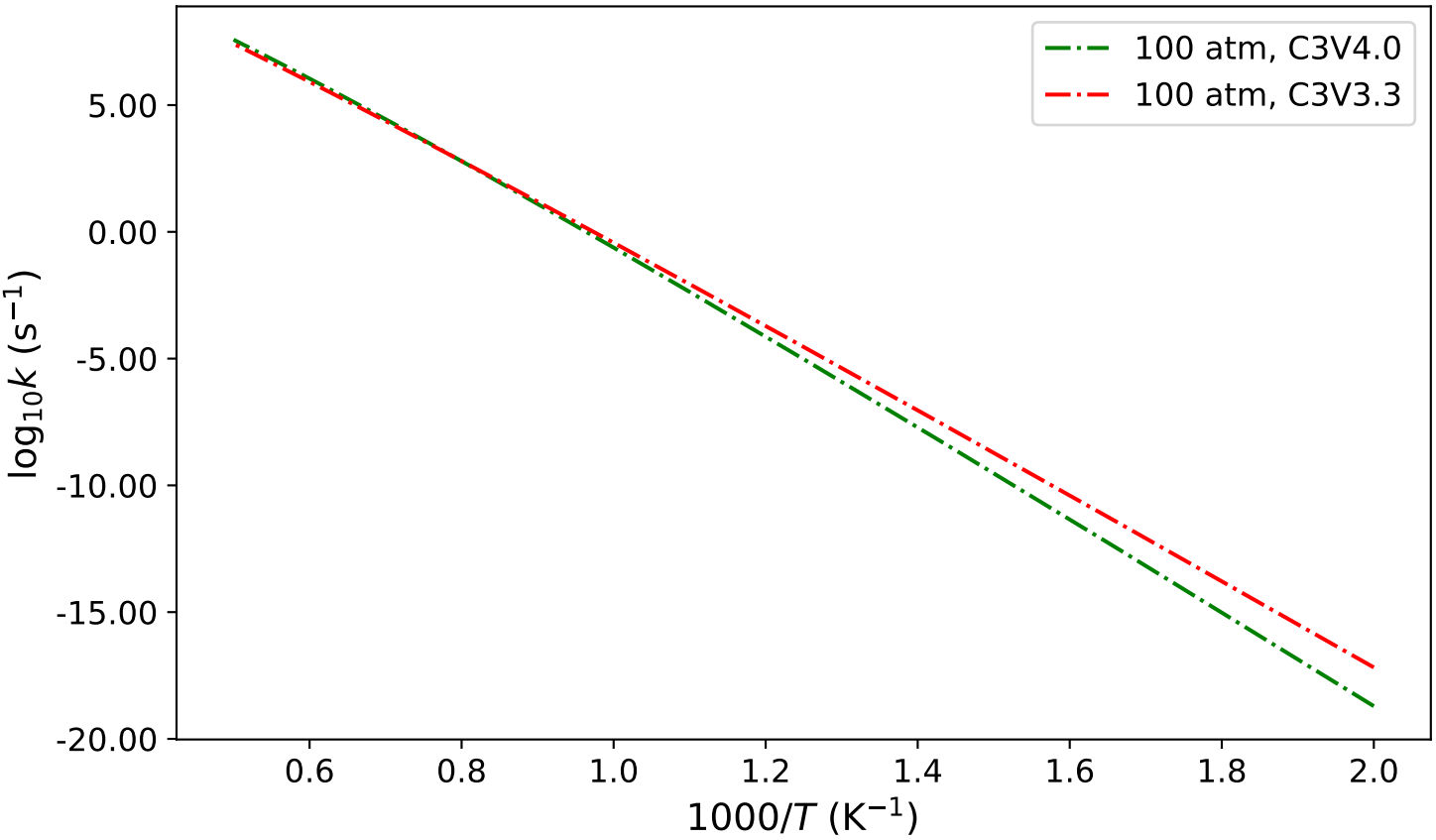



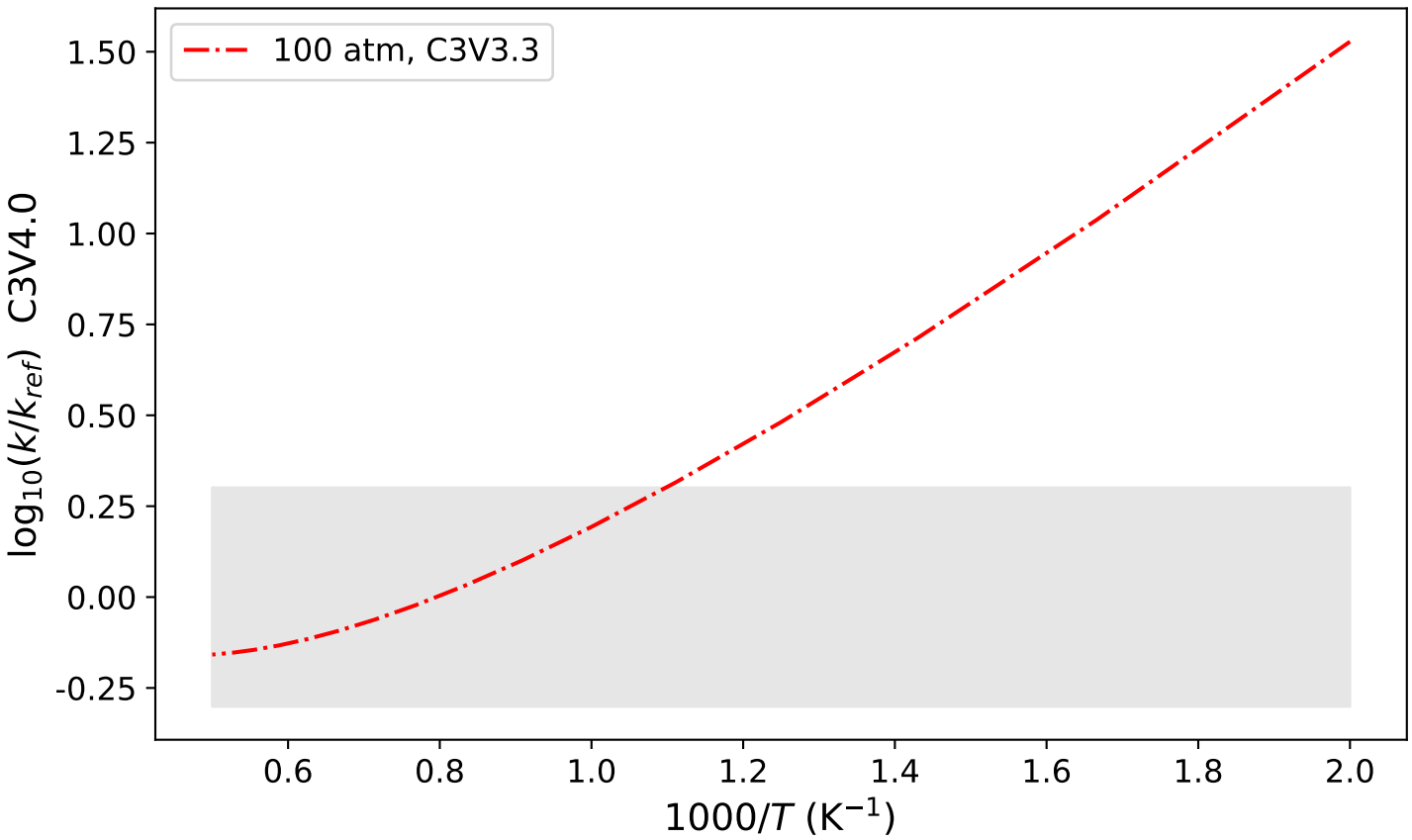
SC4H9O2 = C4H8OOH2-4



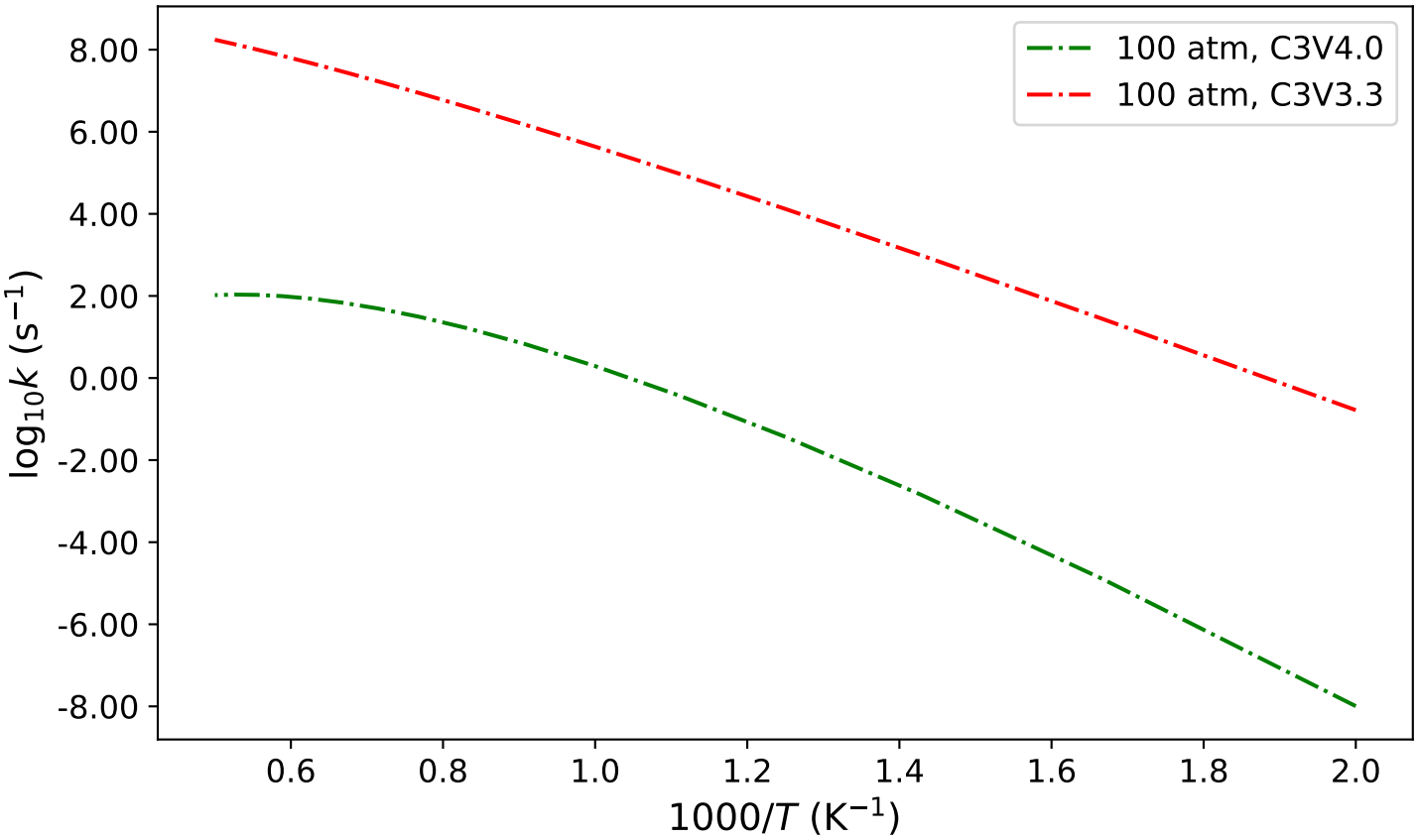



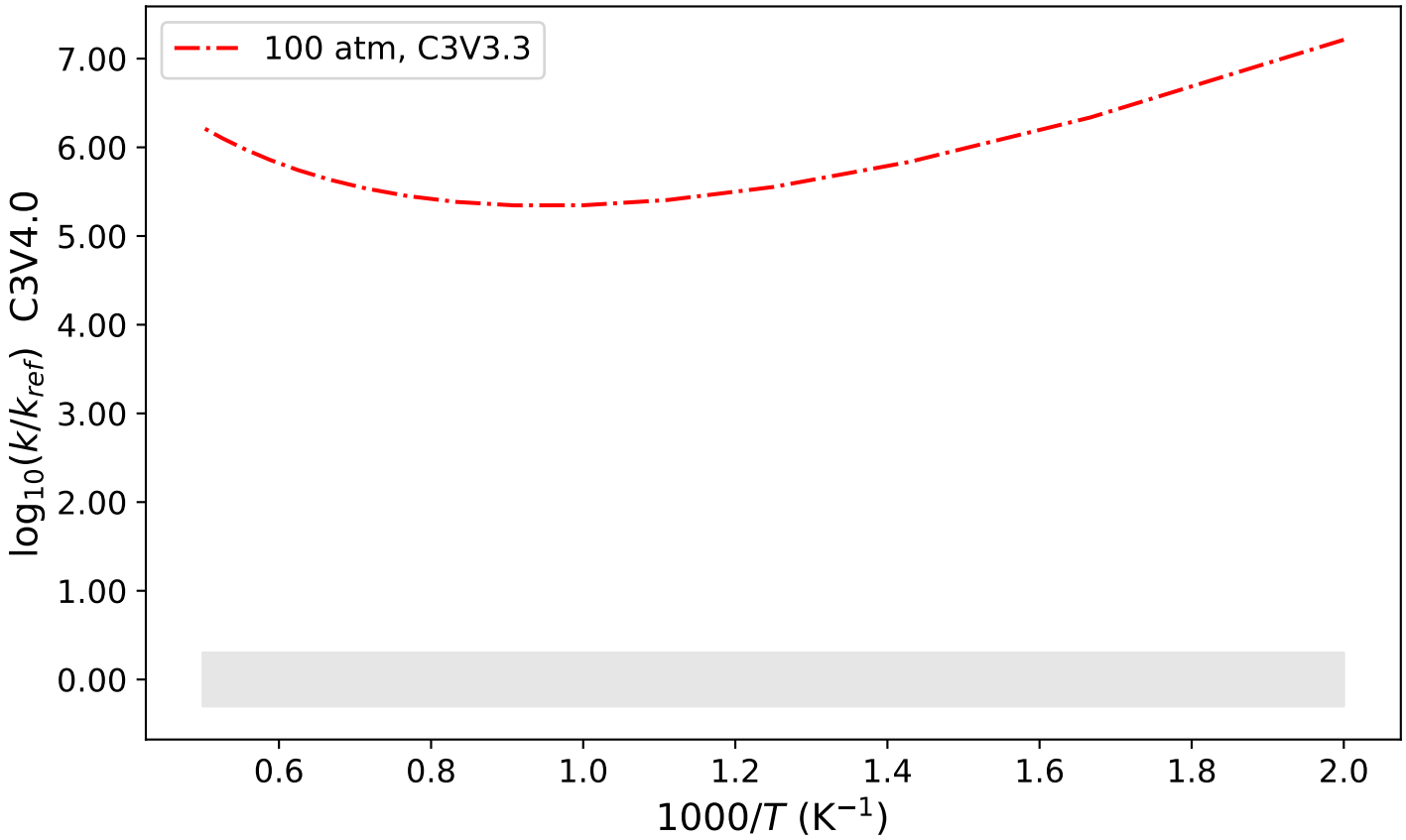
NC3H7CHO = C2H5 + CH2CHO



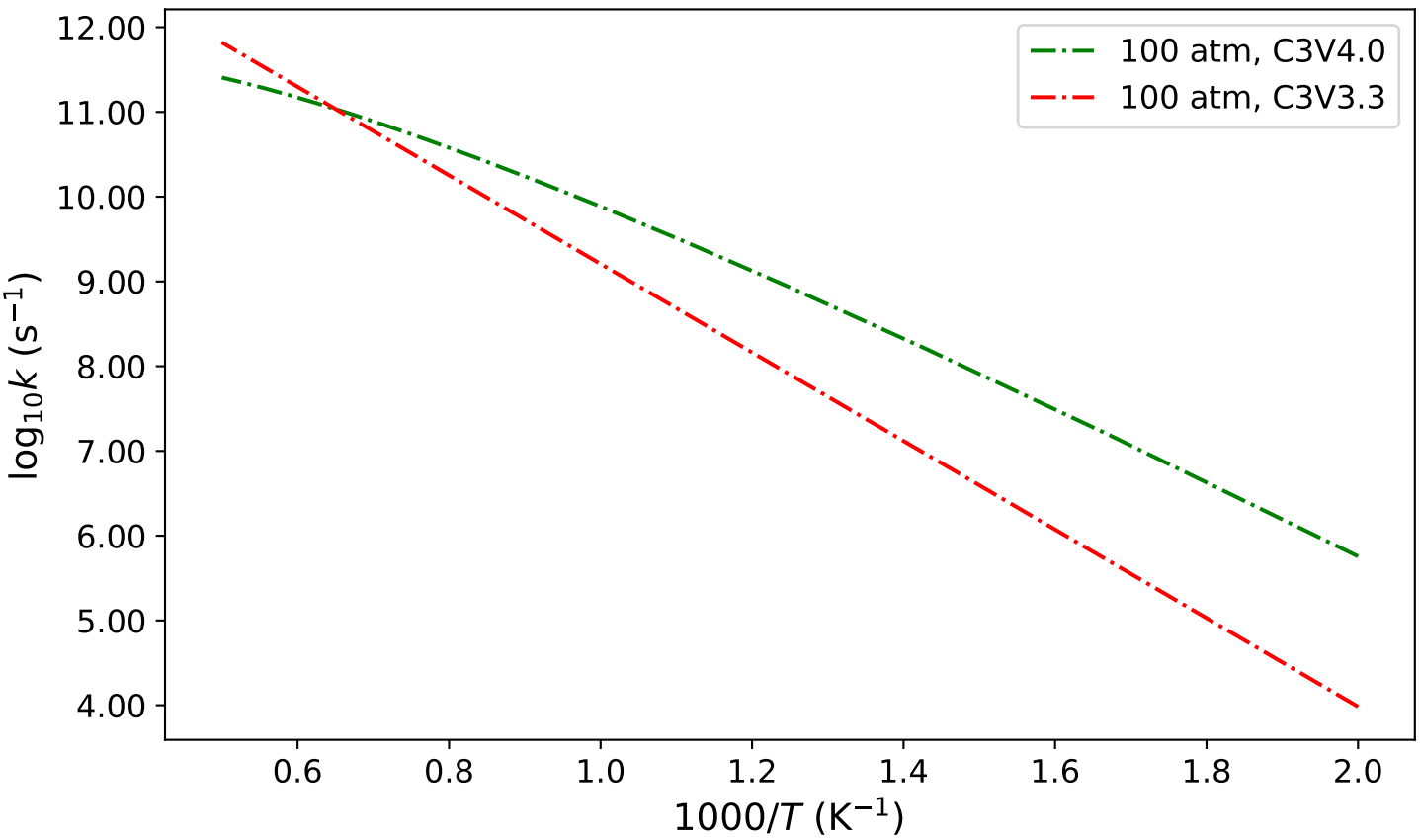



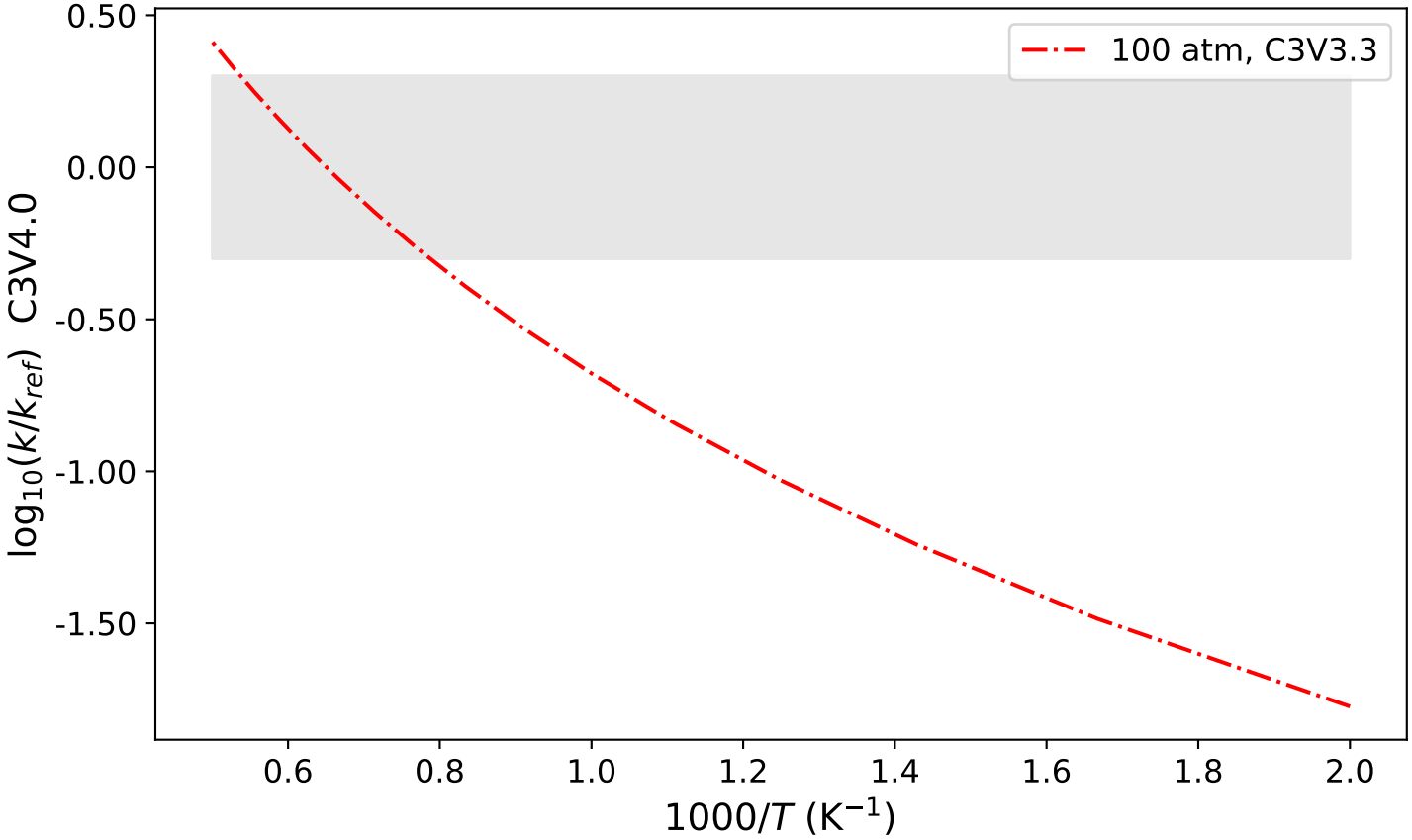
CH3CH(OO)CHO = HO2 + CH3CHCO



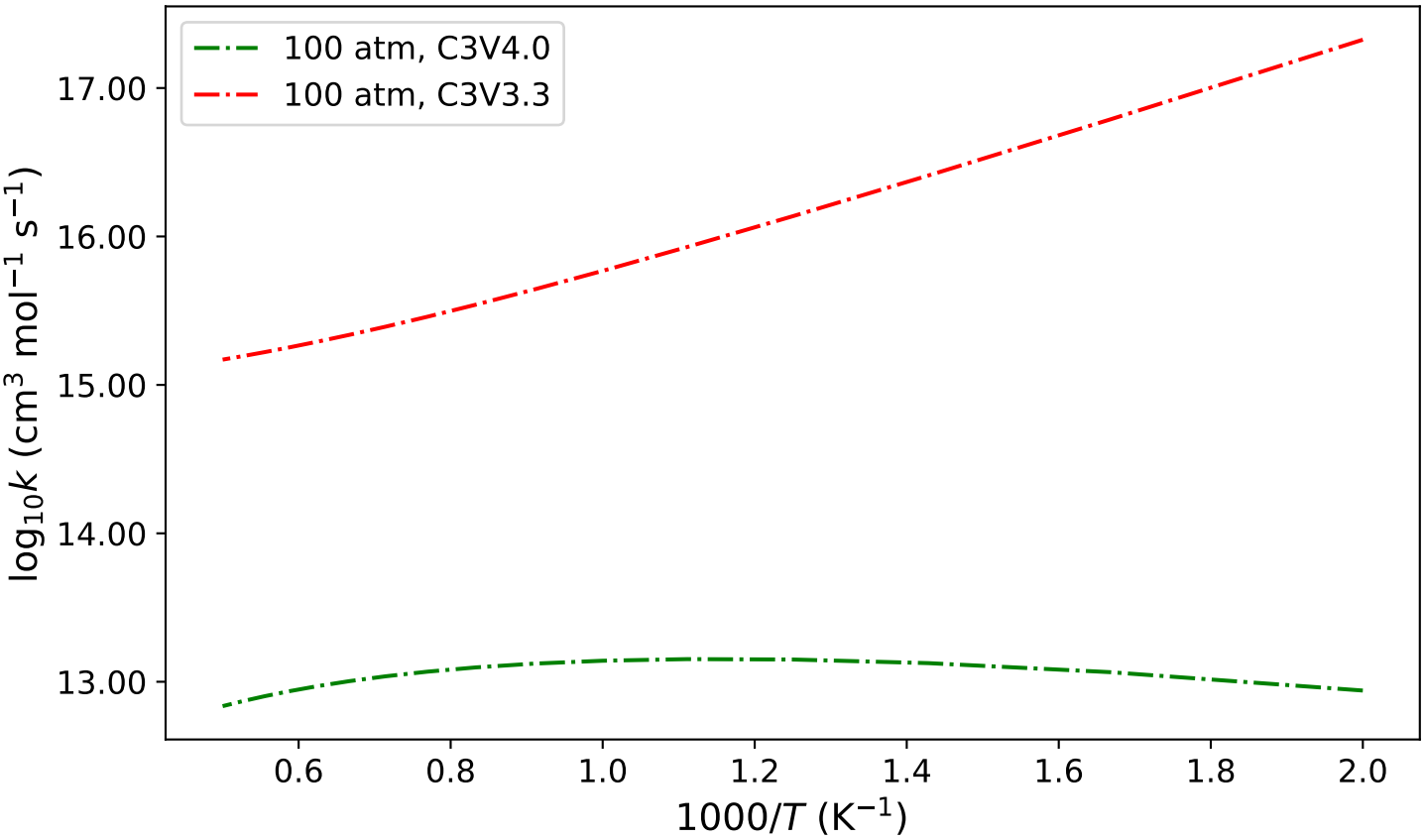



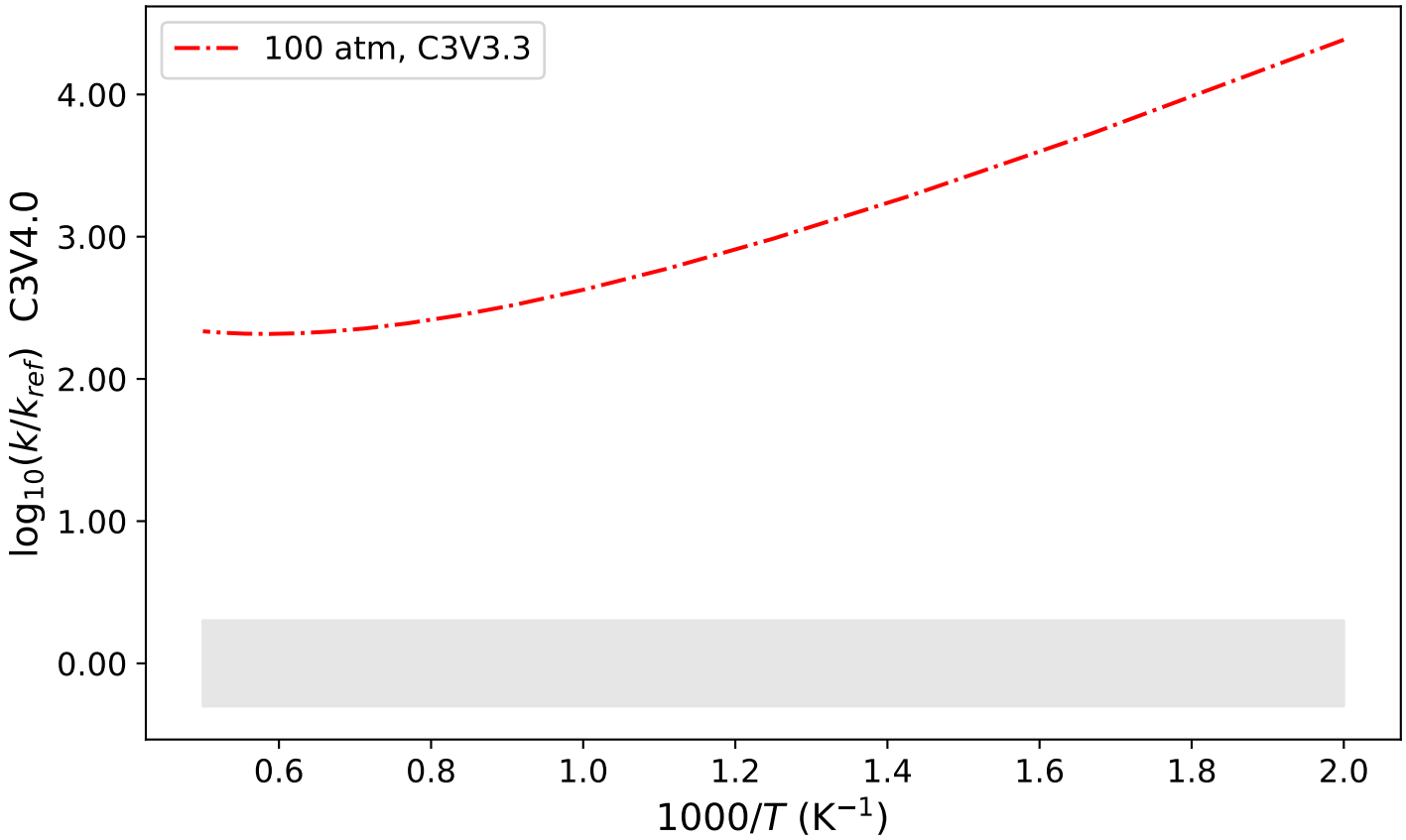
C3H4CH2OH-3O = C2H3 + HOCH2CHO



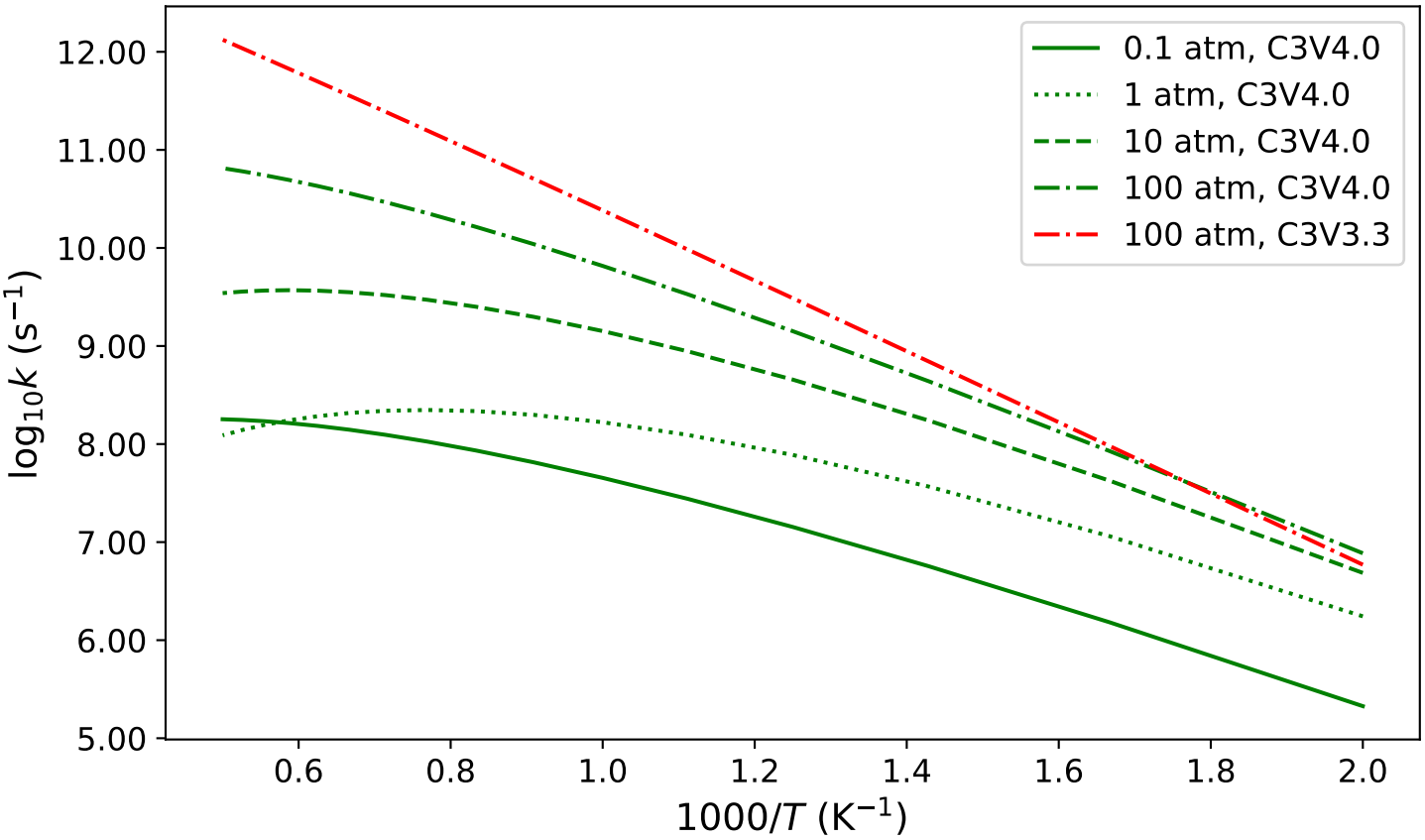



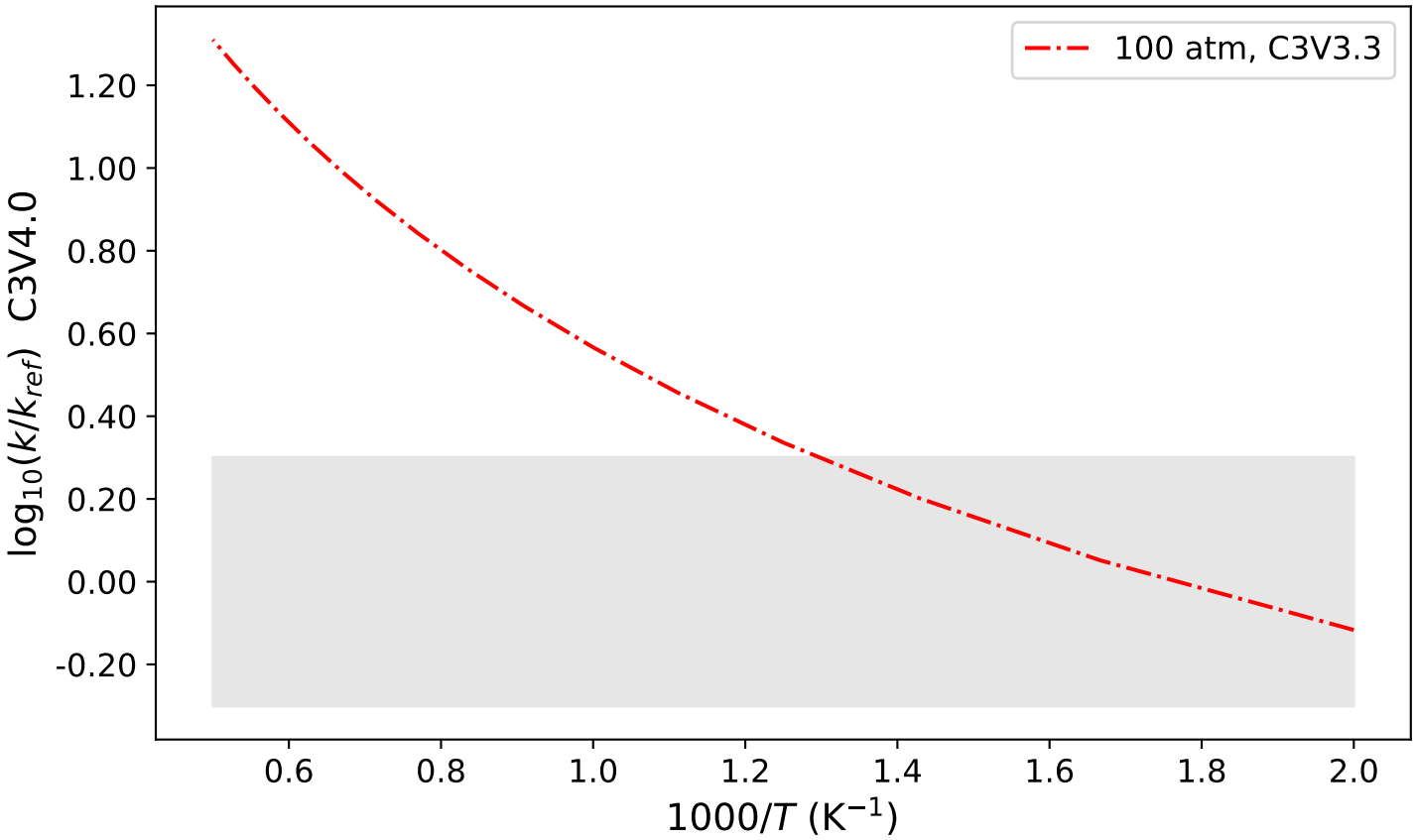
H + IC4H7OOH = IC4H8O2H-T



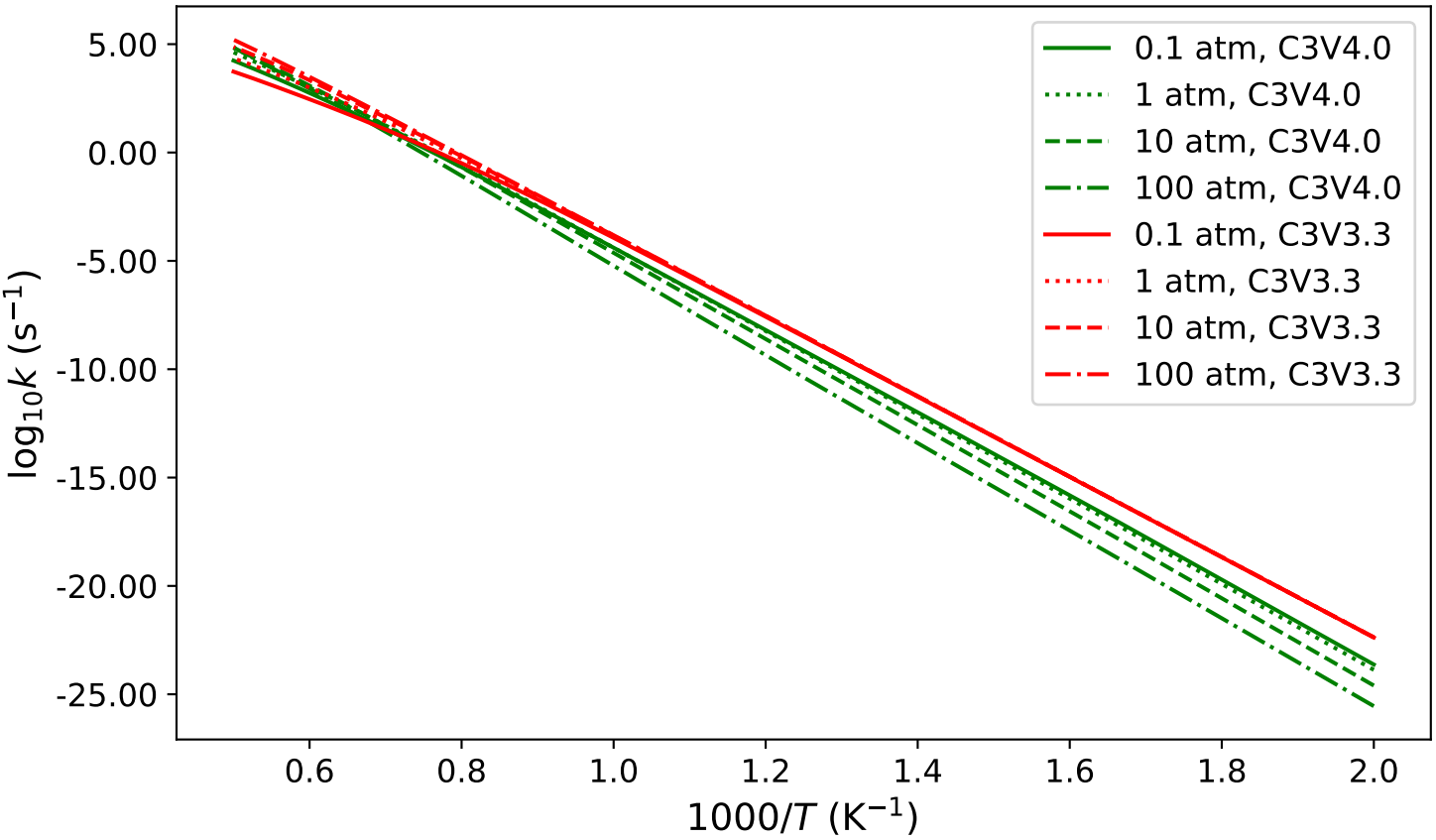



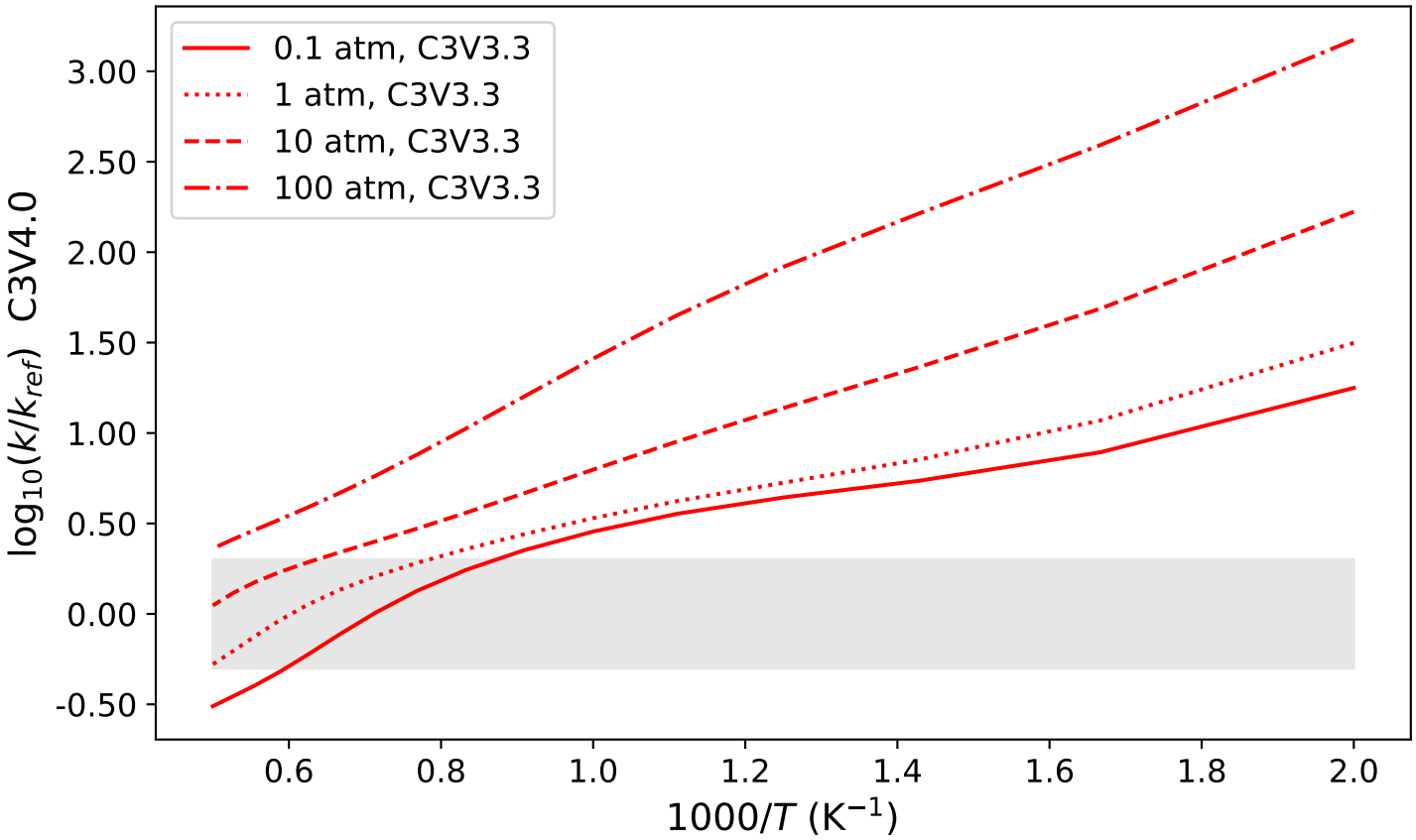
PC4H9O = NC3H7 + CH2O



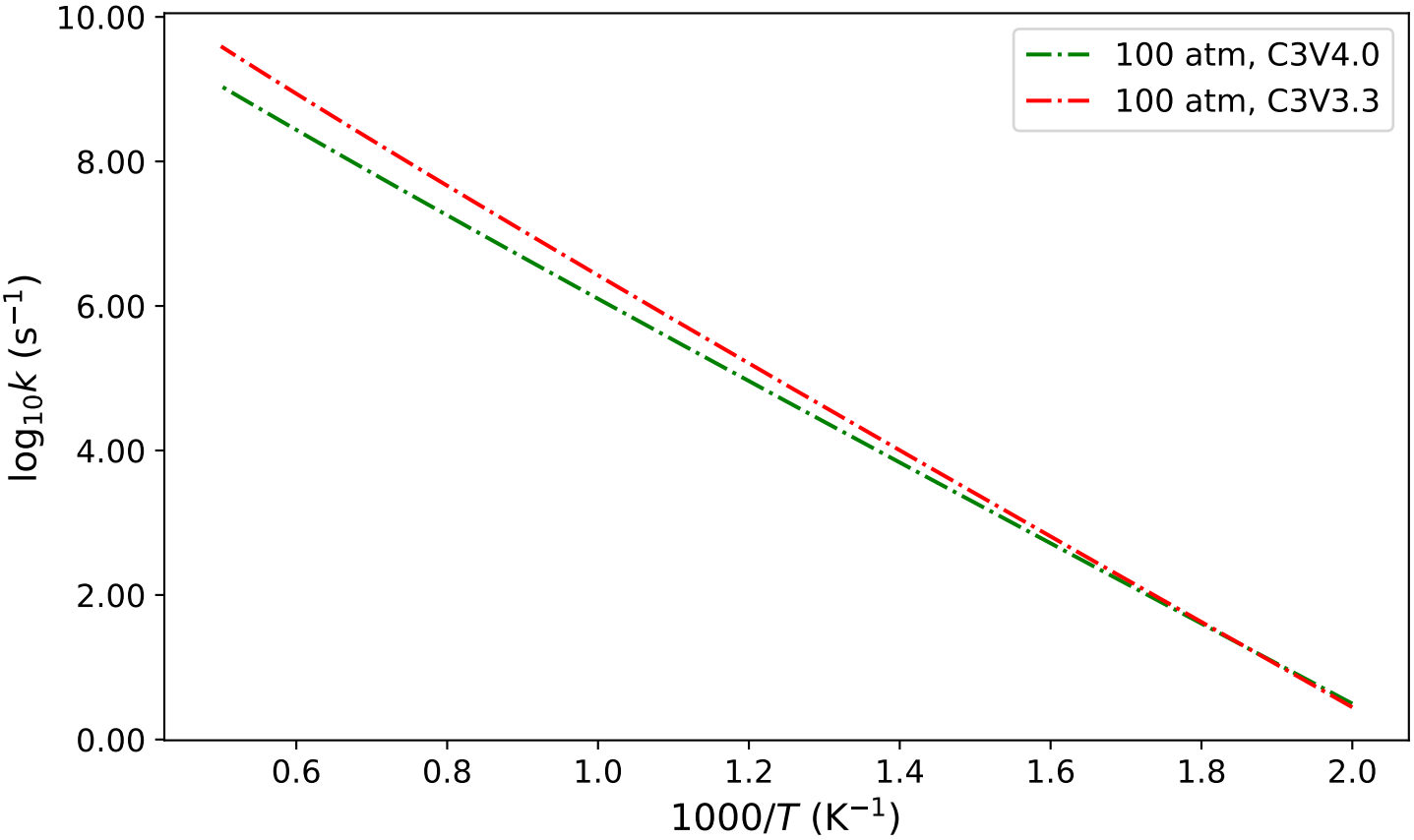



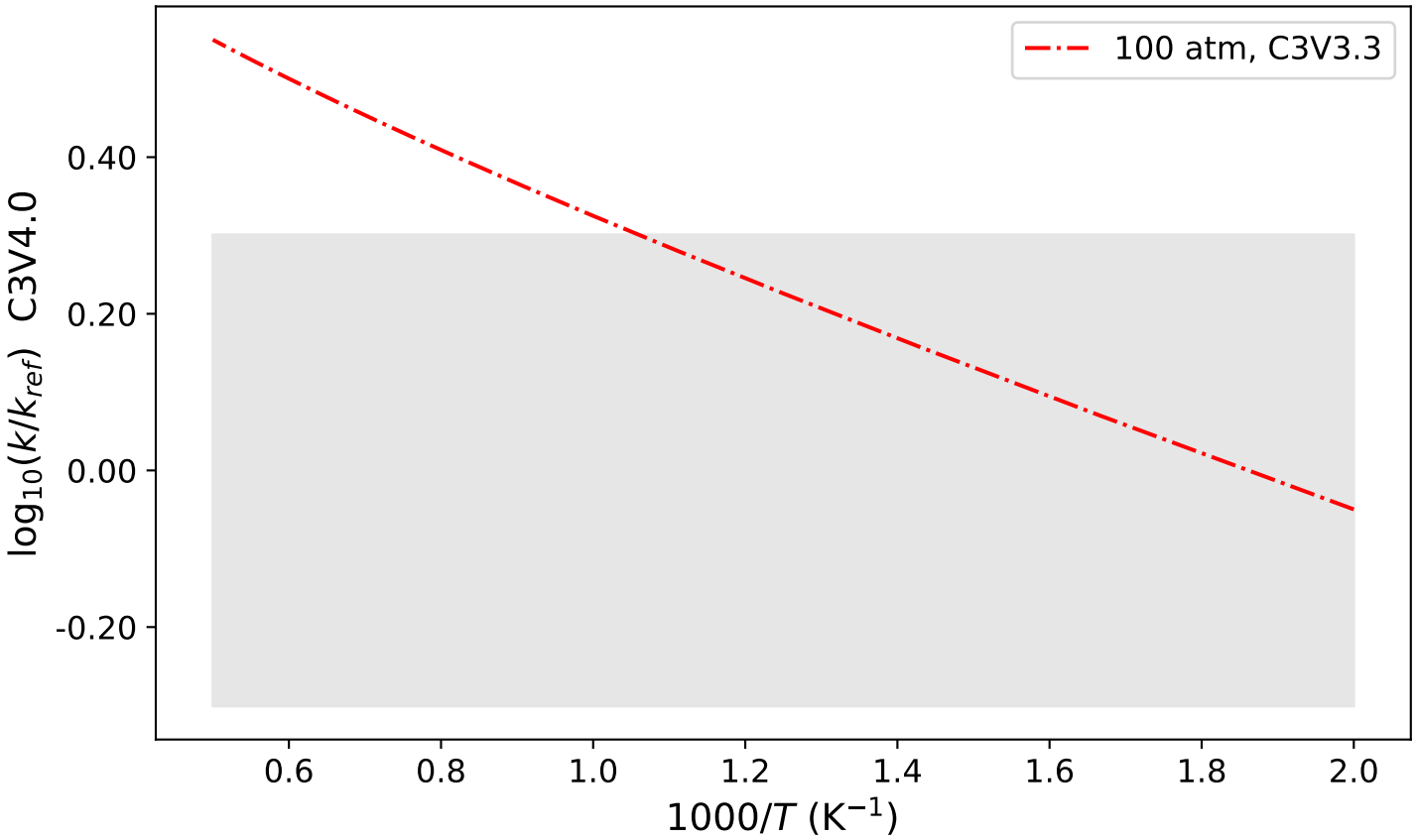
C4H4 = C2H2 + H2CC



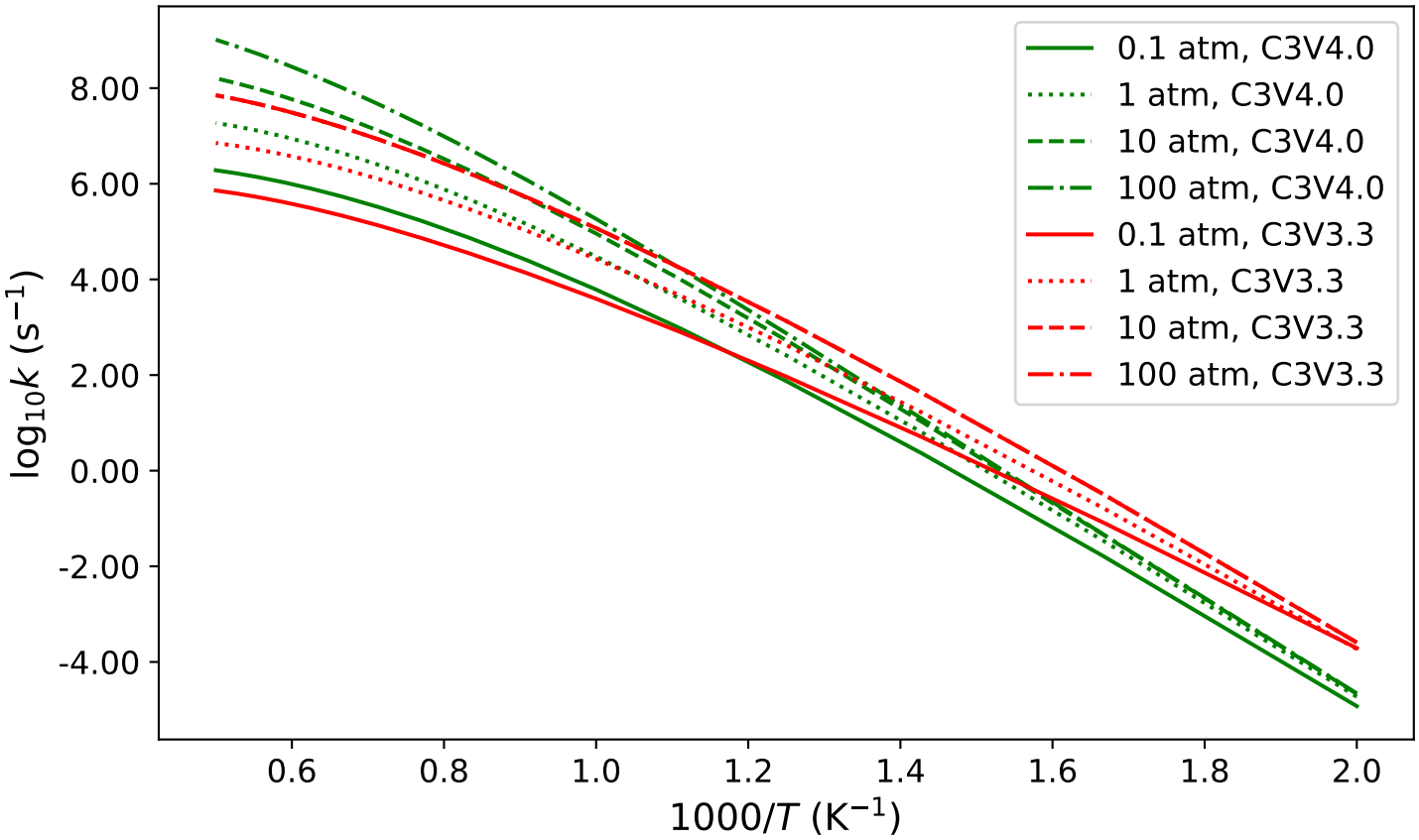



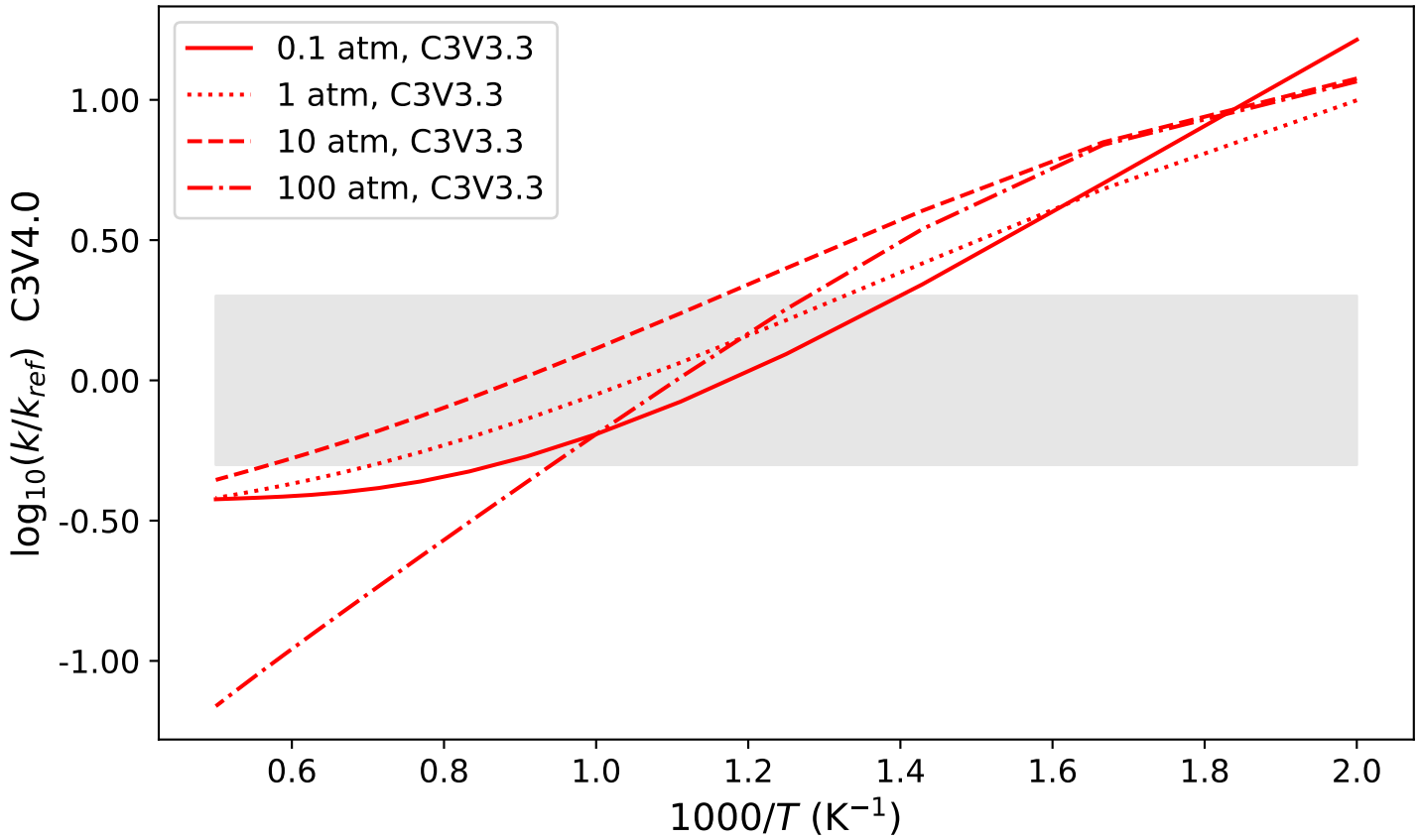
C4H8OOH2-1O2 = OH + NC4KET21



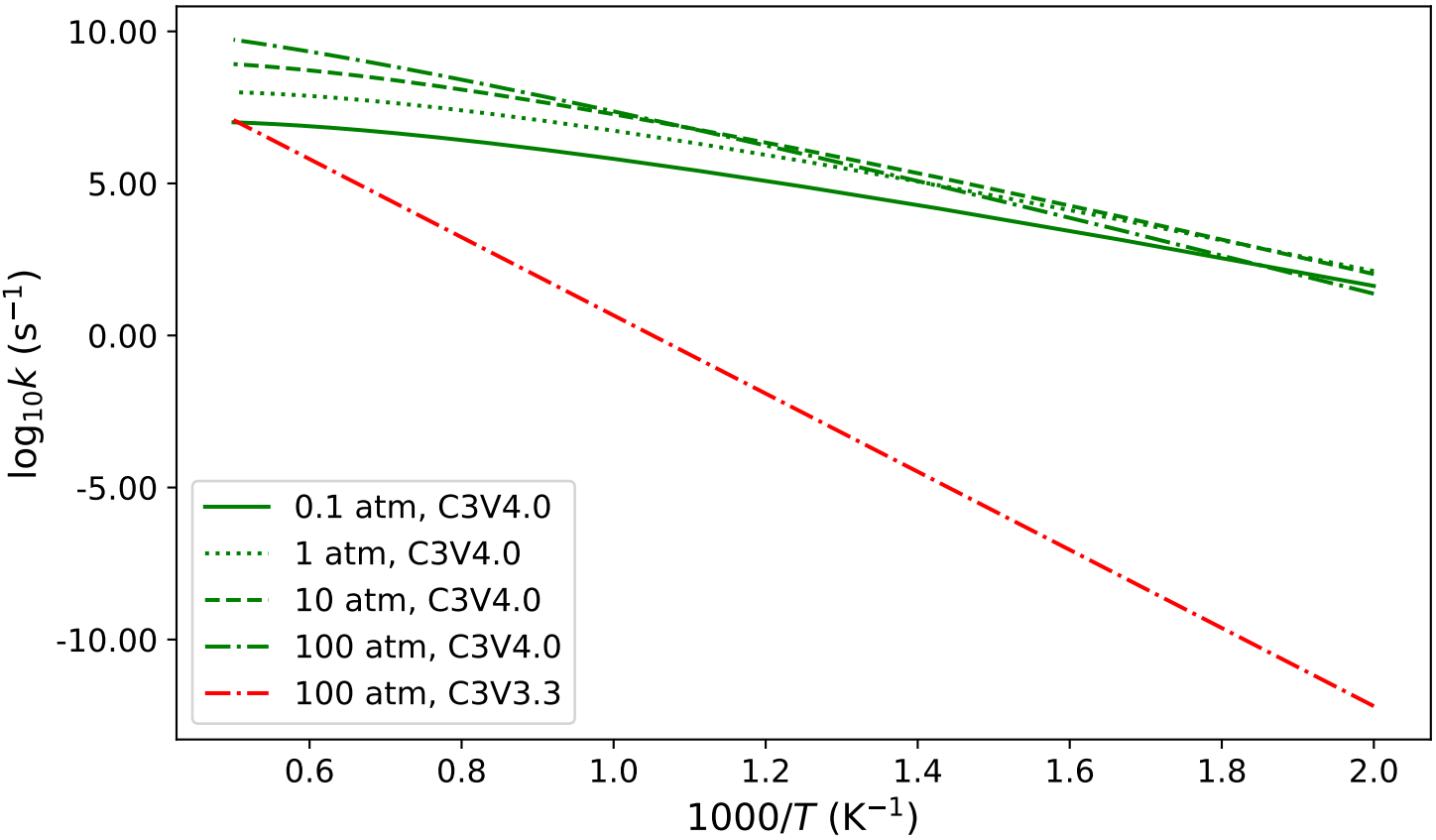



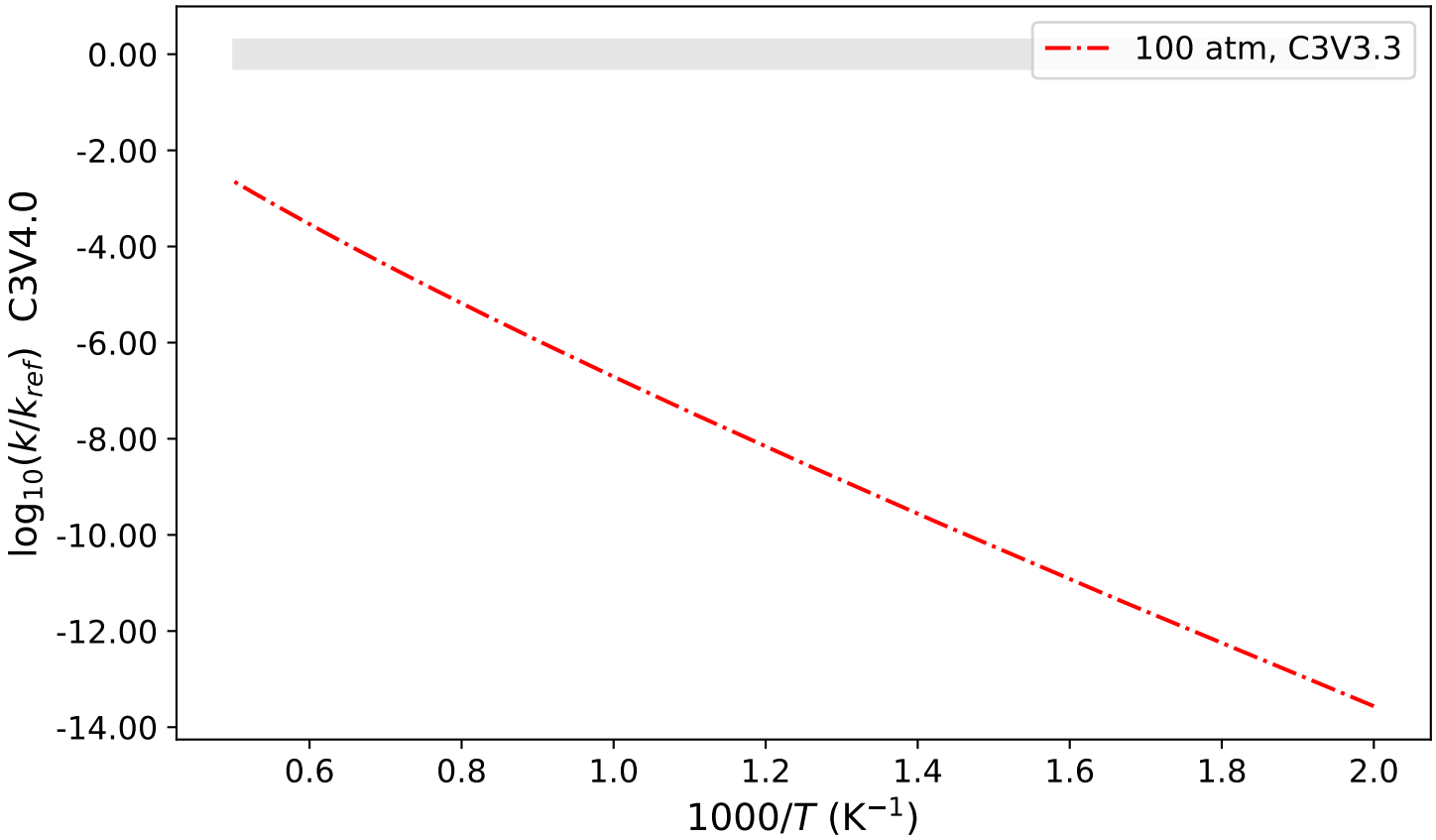
C4H5-N = C2H3 + C2H2



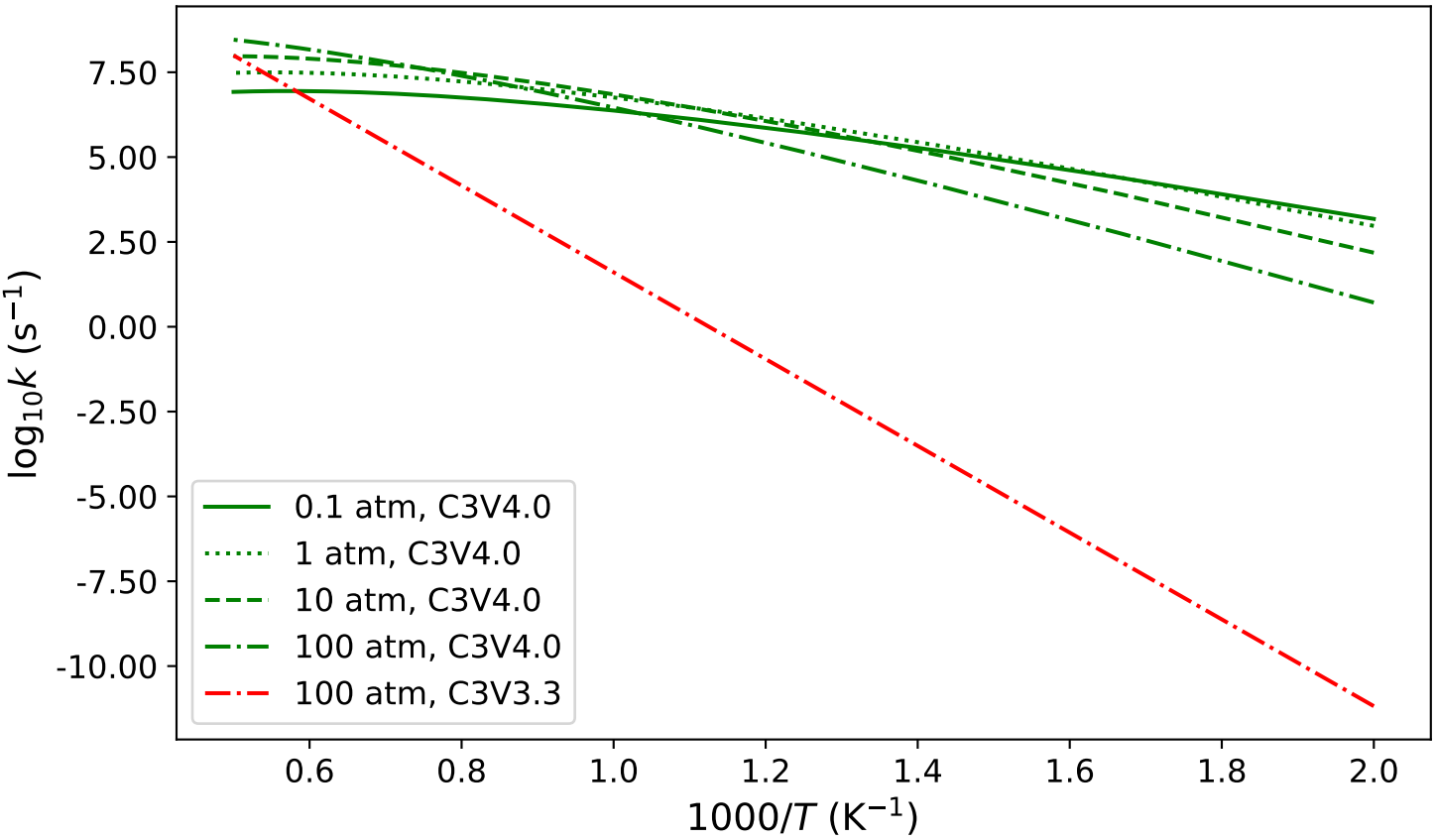



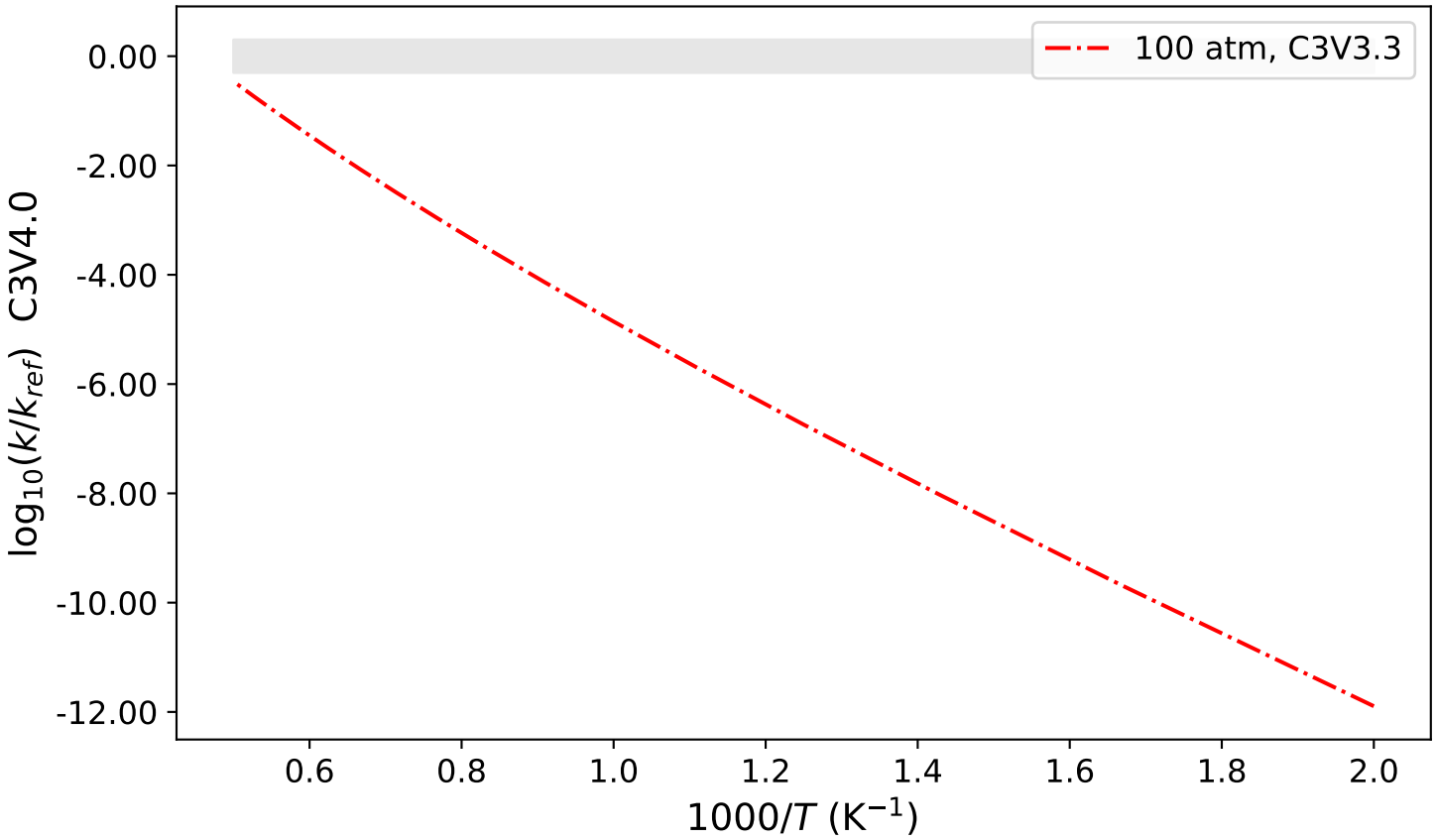
CJCYCCO = CH2O + C2H3



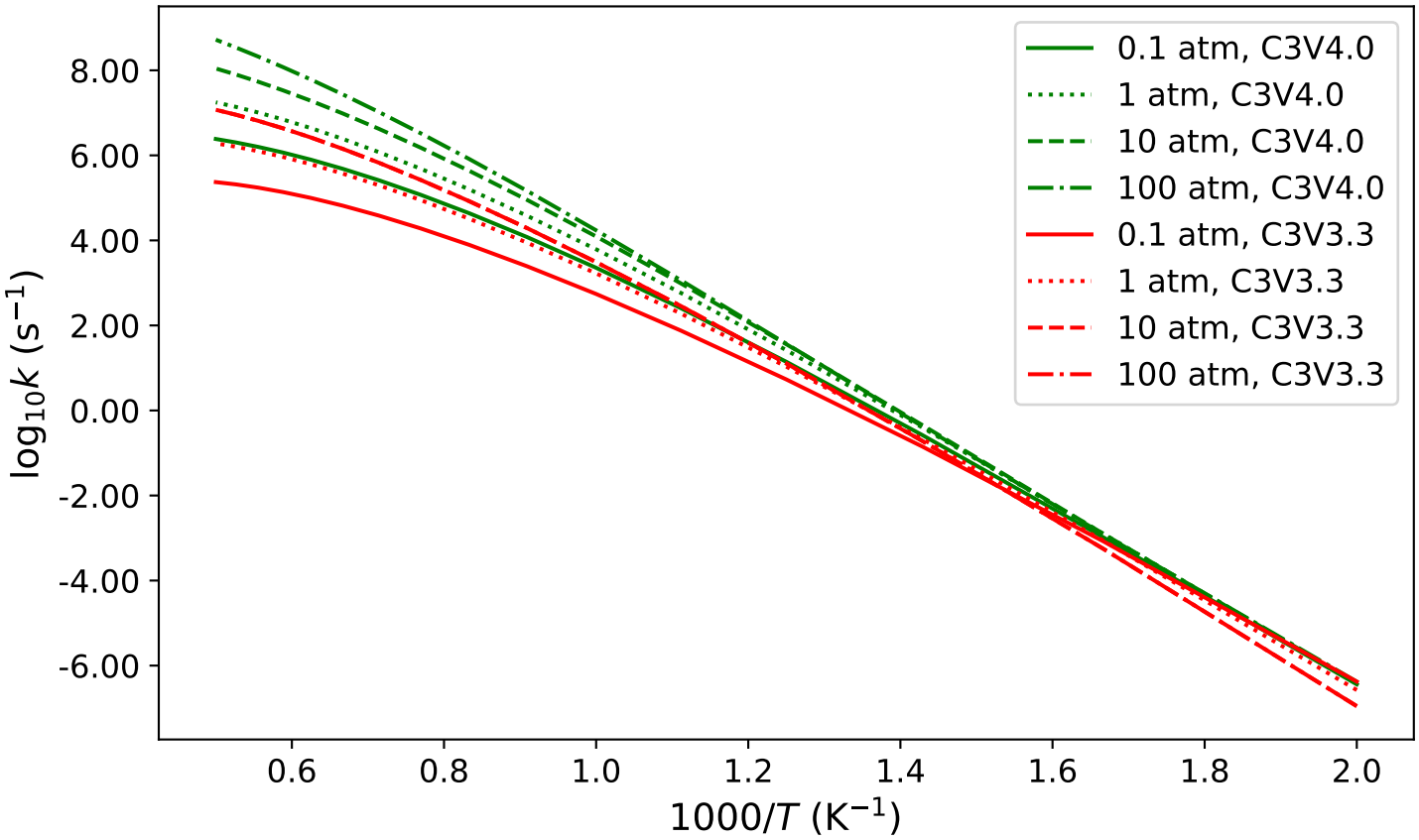



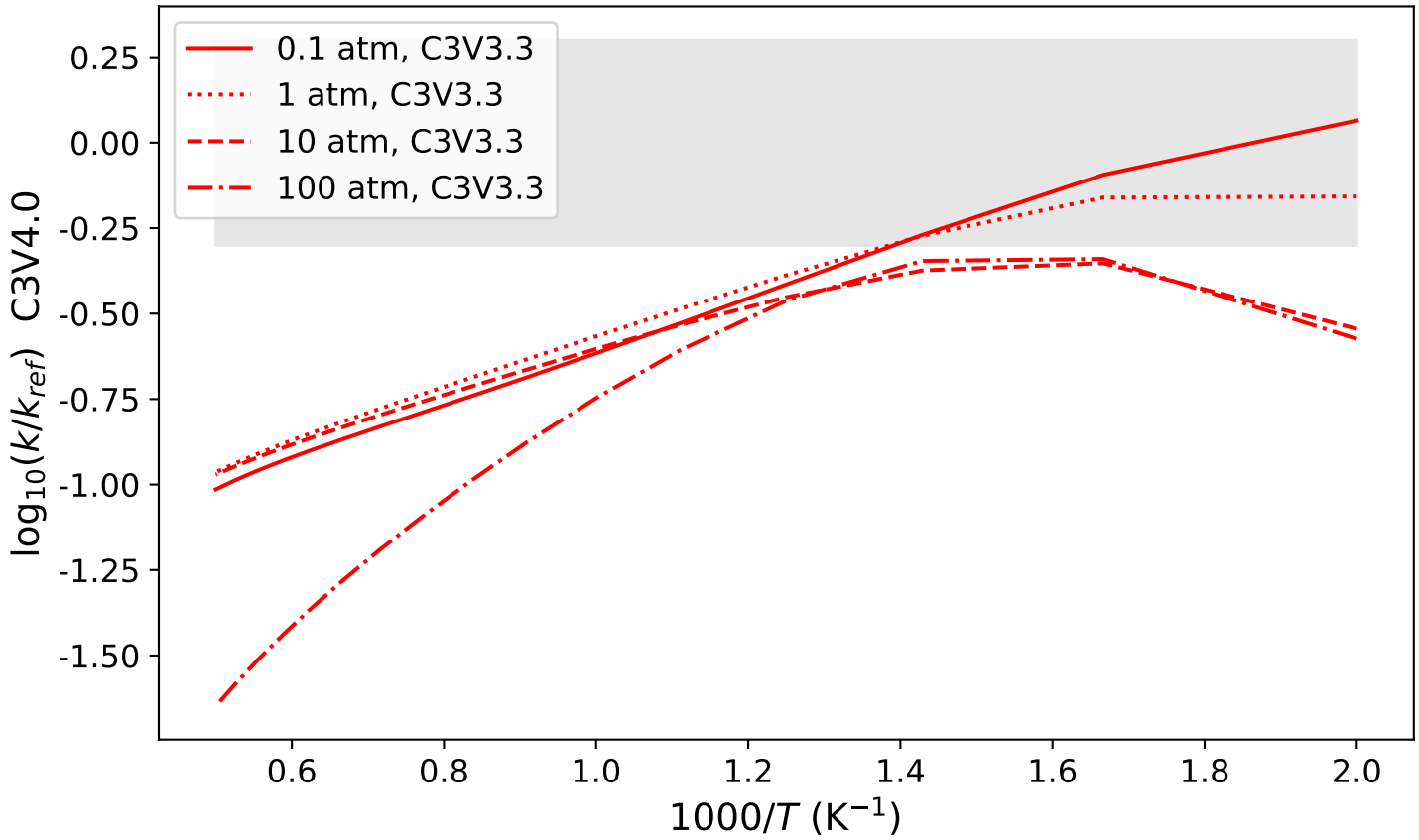
CJCYCCO = C2H4 + HCO



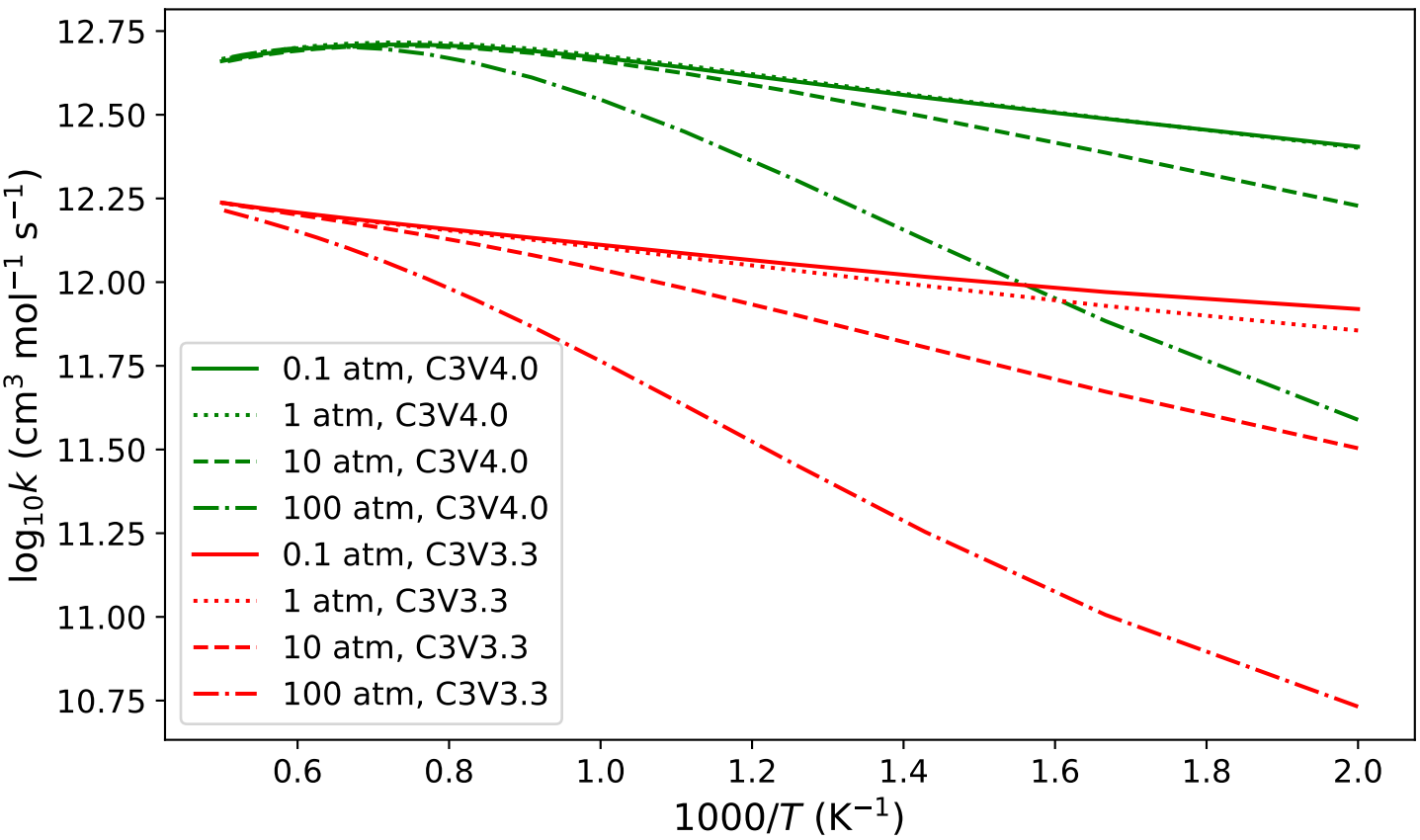



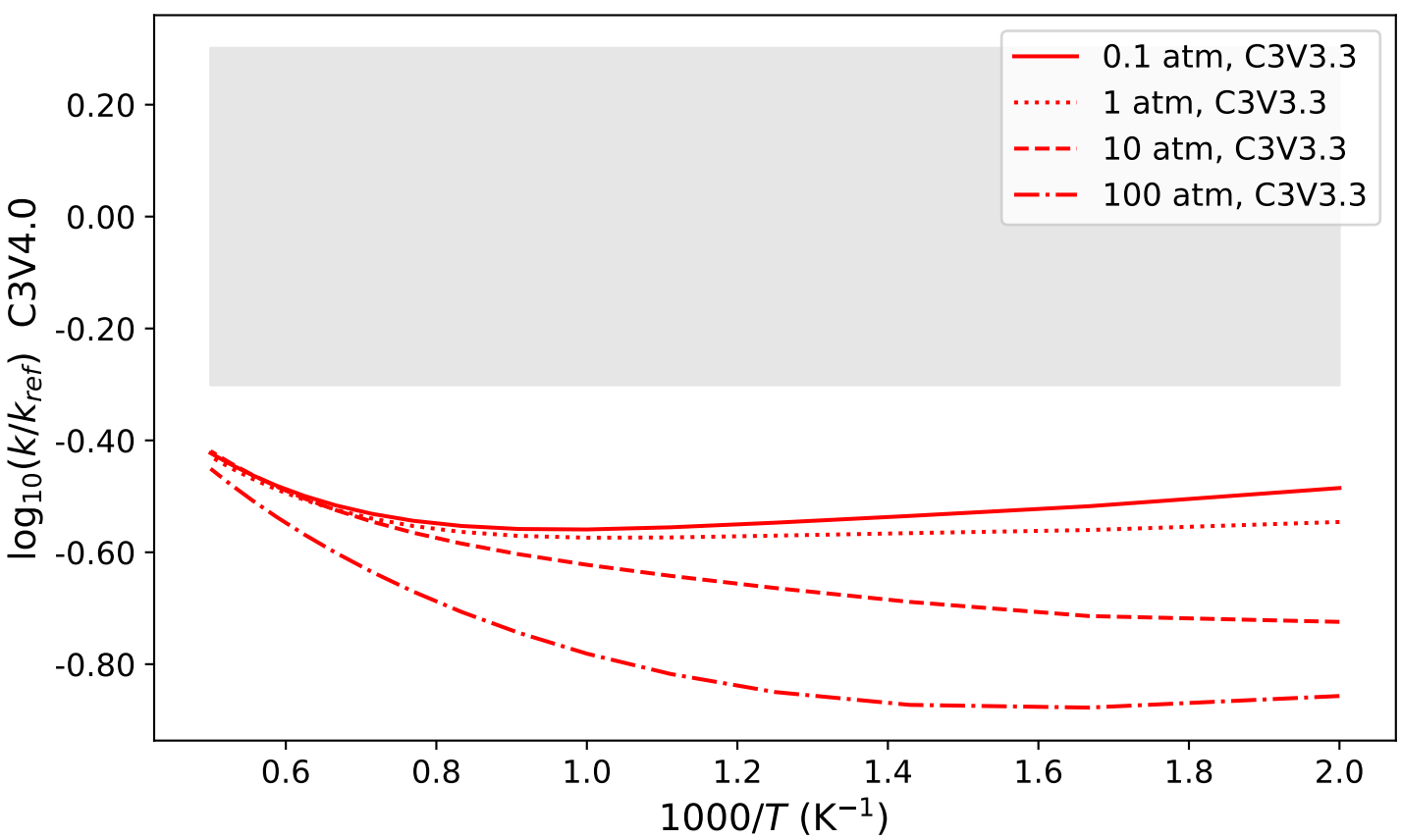
C4H5-I = H + C4H4


0.1 atm, C3V4.0
1 atm, C3V4.0
10 atm, C3V4.0
100 atm, C3V4.0
0.1 atm, C3V3.3
1 atm, C3V3.3
10 atm, C3V3.3
100 atm, C3V3.3
8.00
6.00
4.00
2.00
0.00
-2.00
-4.00
-6.00
$\log_{10}k$ ($s^{-1}$)
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ ($K^{-1}$)

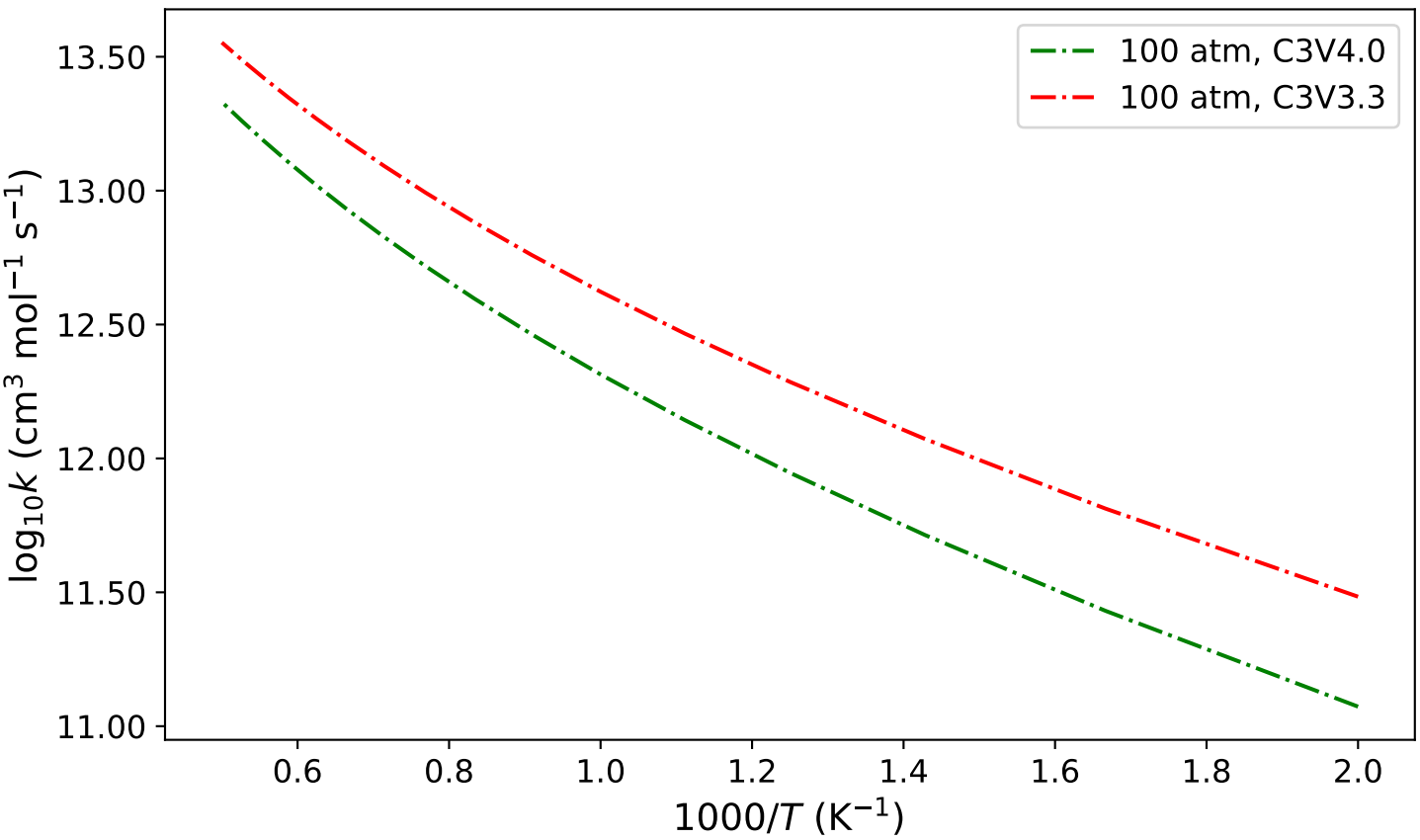

0.1 atm, C3V3.3
1 atm, C3V3.3
10 atm, C3V3.3
100 atm, C3V3.3
0.25
0.00
-0.25
-0.50
-0.75
-1.00
-1.25
-1.50
$\log_{10}(k/k_{ref})$ C3V4.0
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ ($K^{-1}$)

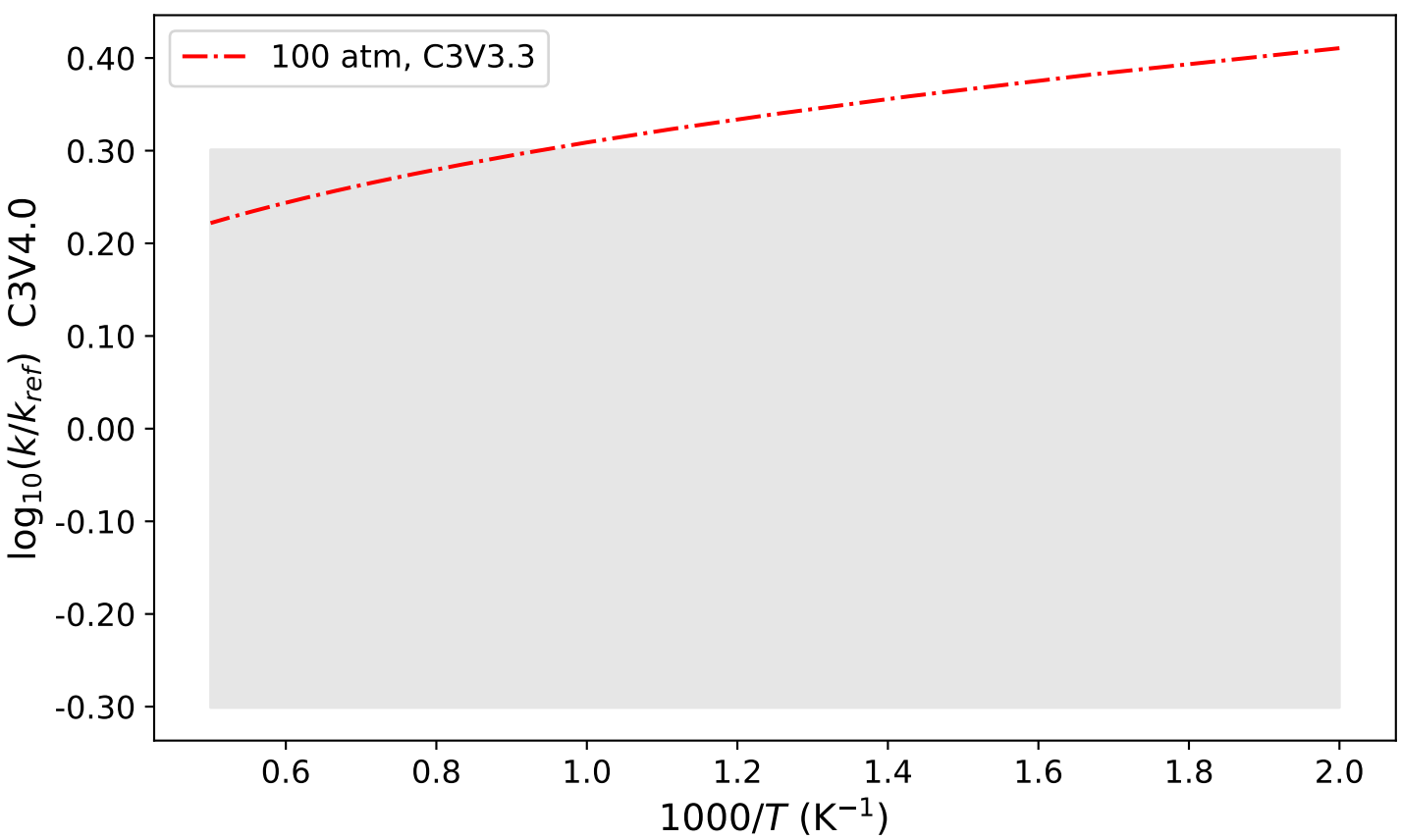
O2 + C4H5-N = O + CH2CHCHCHO



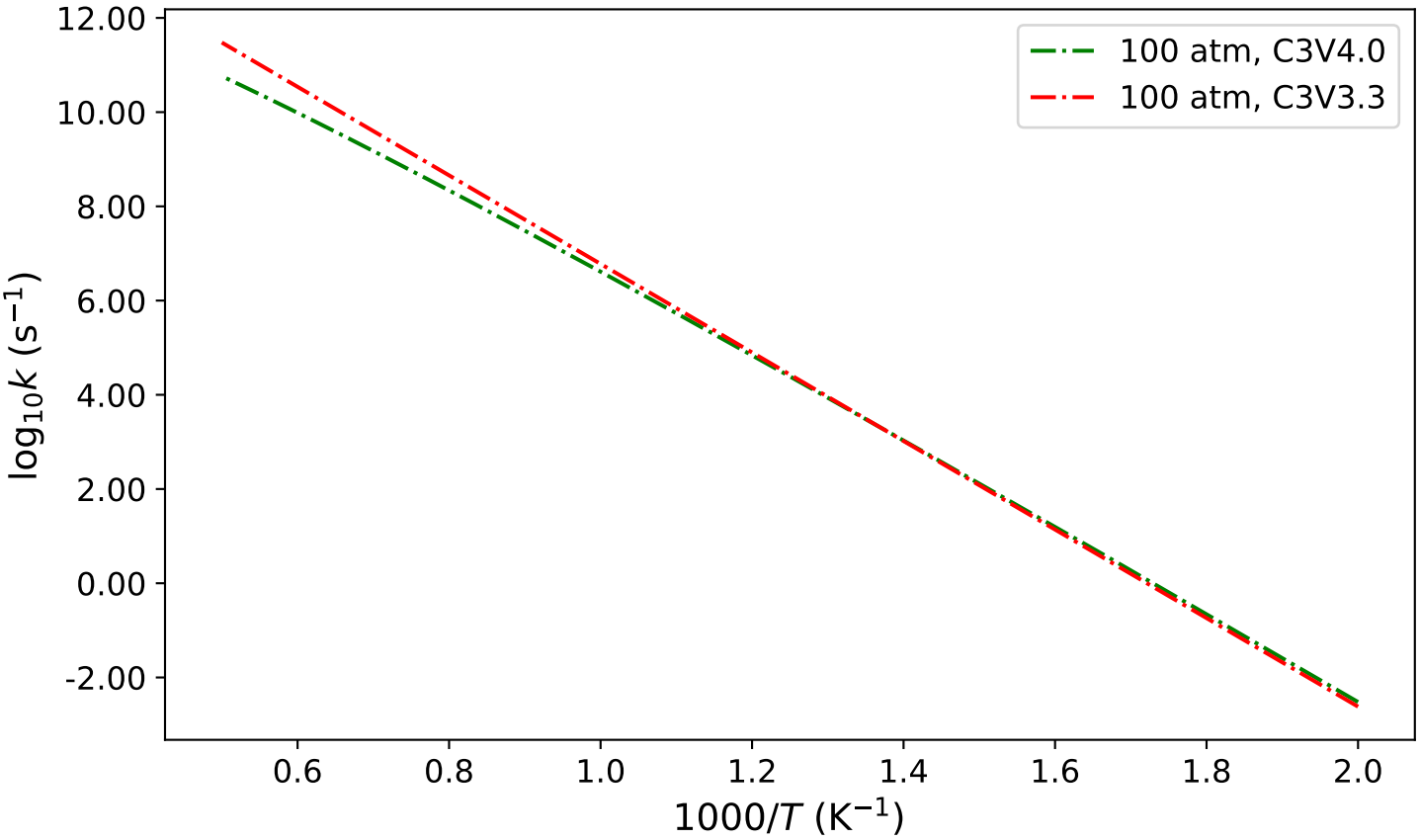



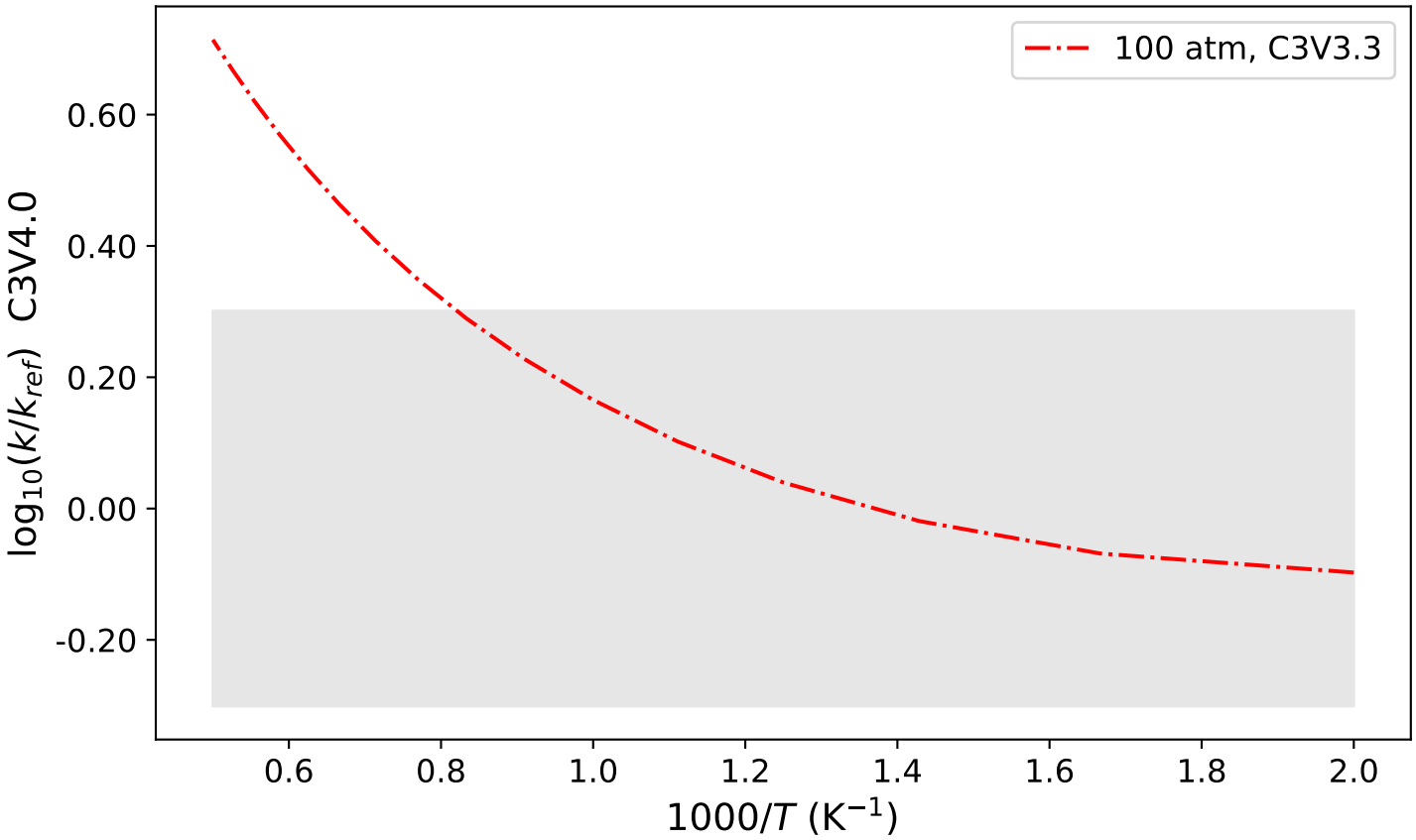
H + C2H3CHO = H2 + C2H3CO



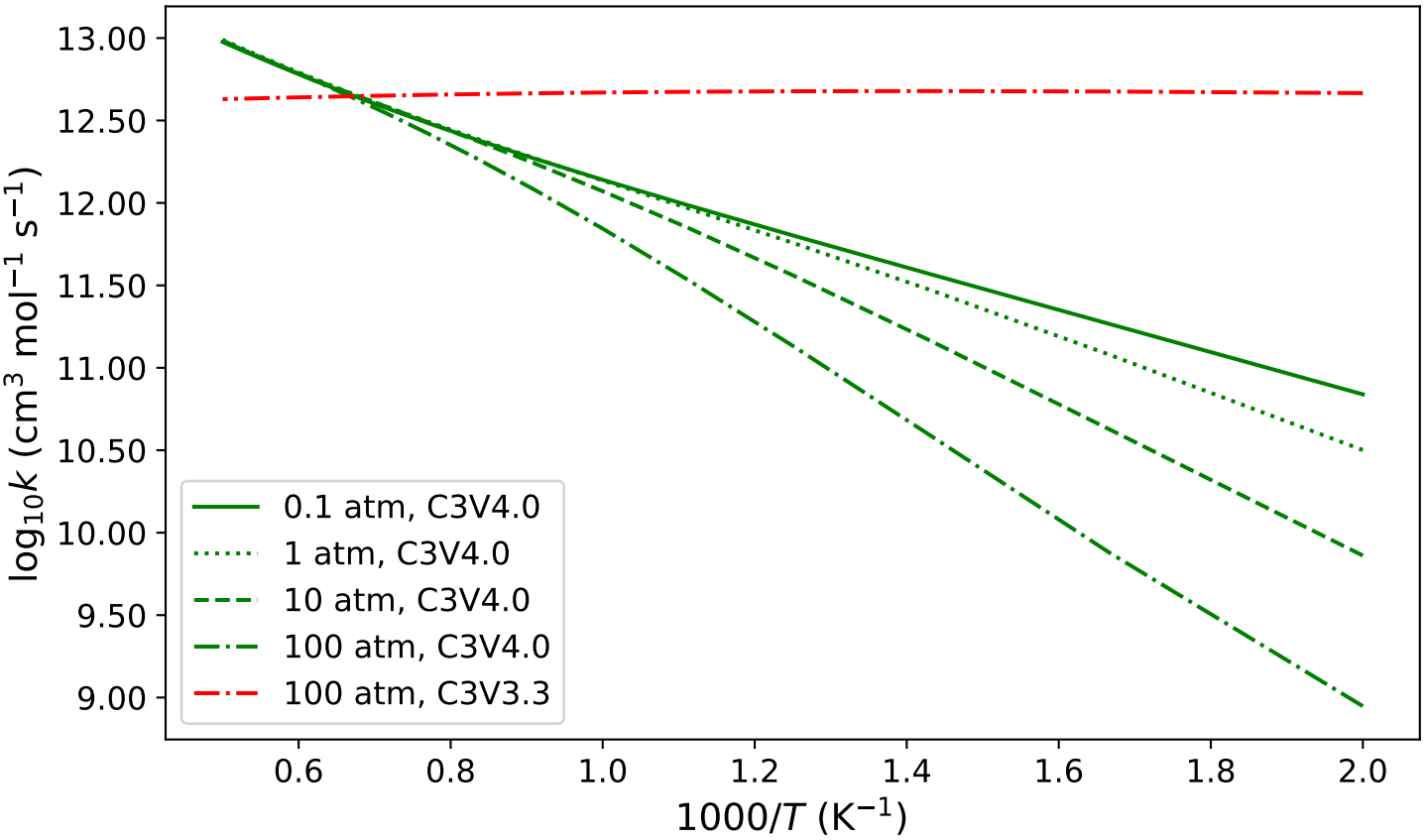



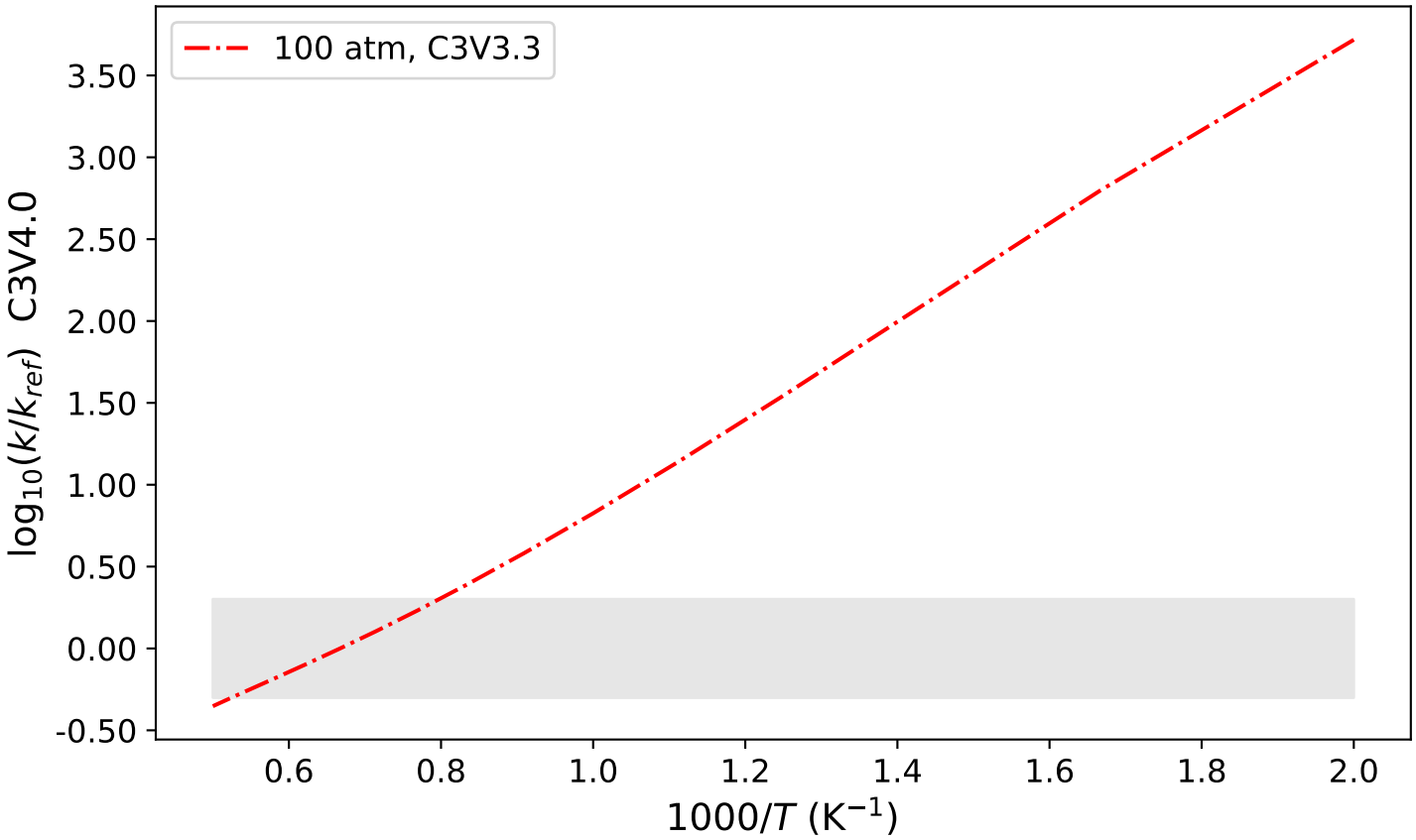
NC4KET24 => CH2O + OH + CH3COCH2



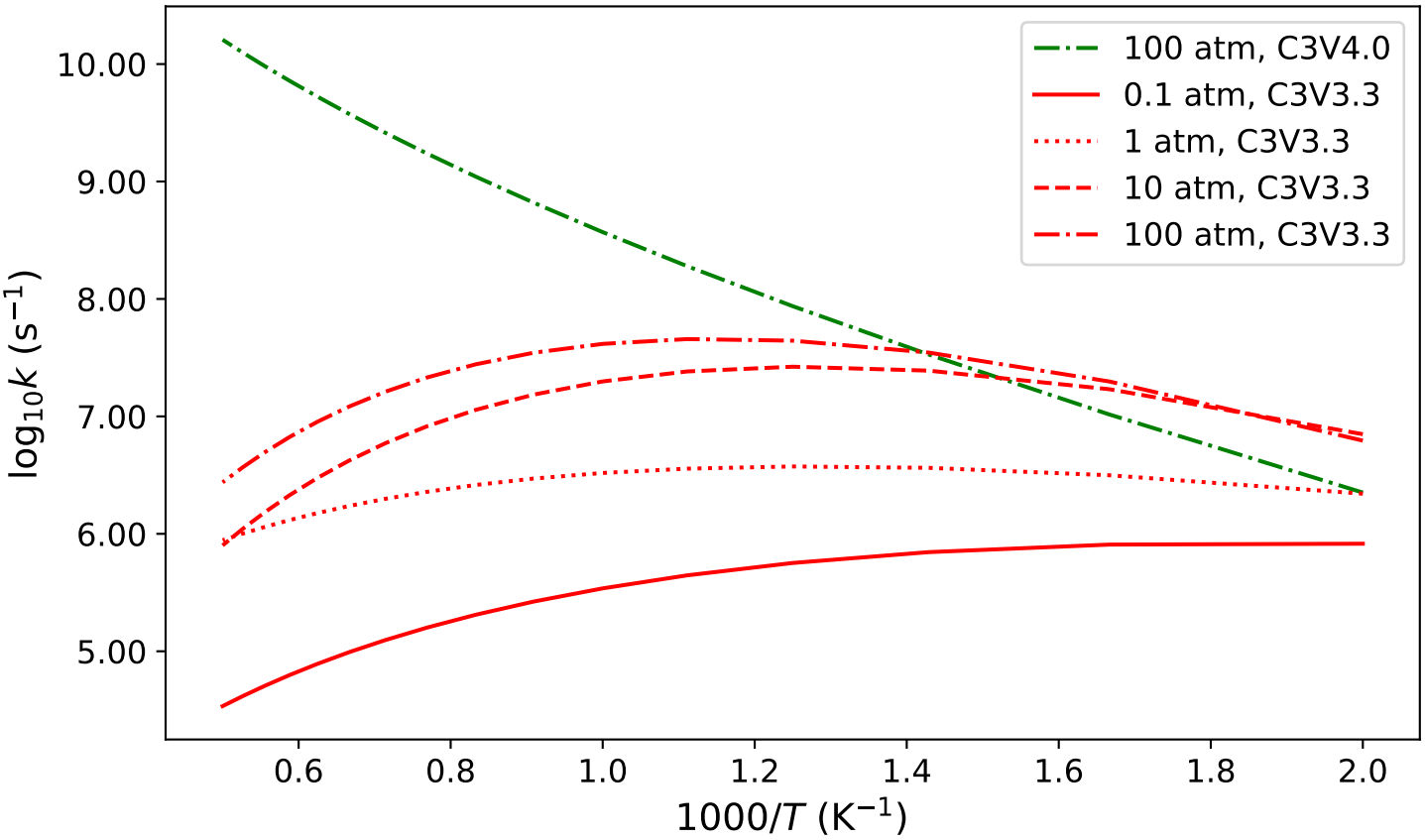



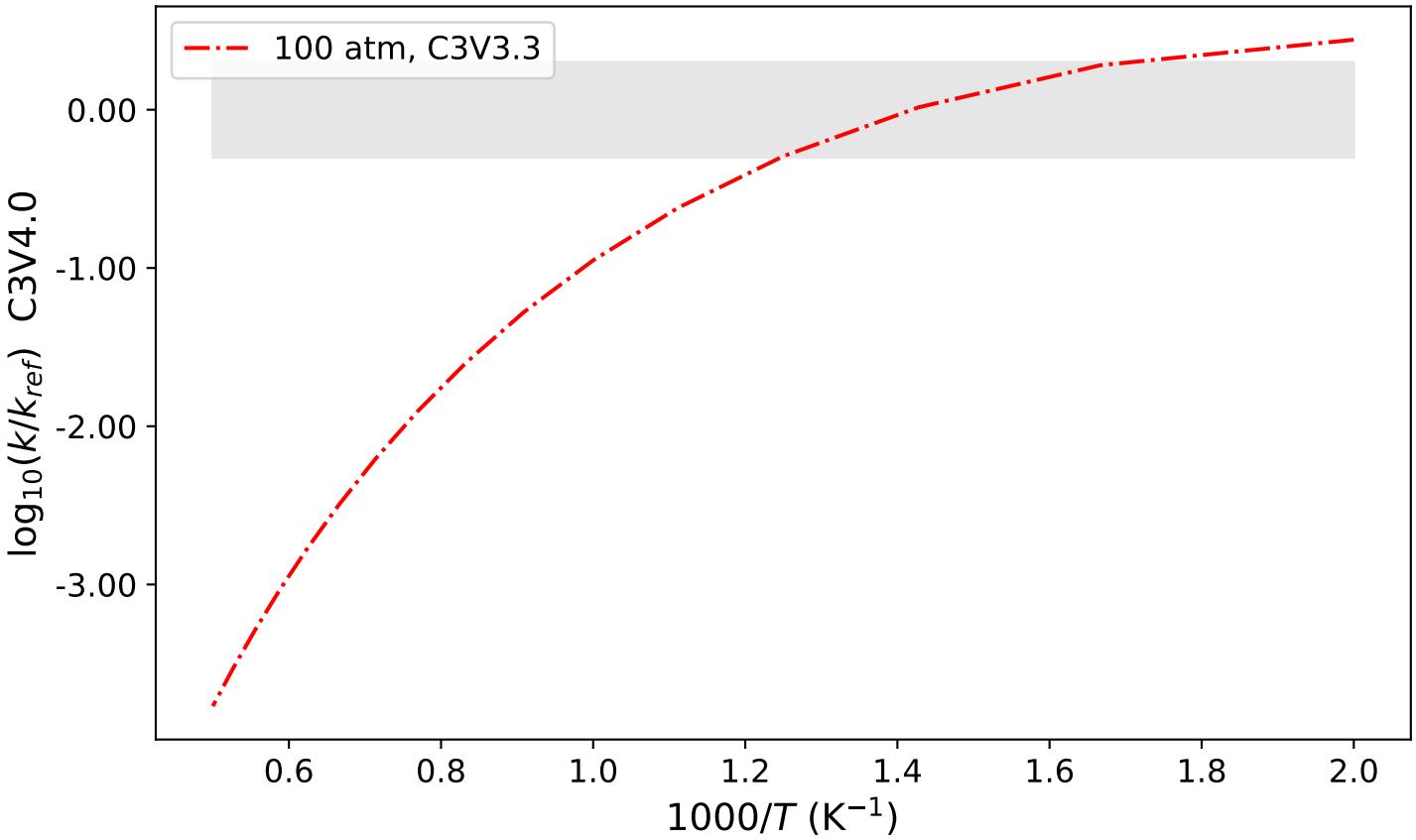
H + C4H612 = H + C4H6-2



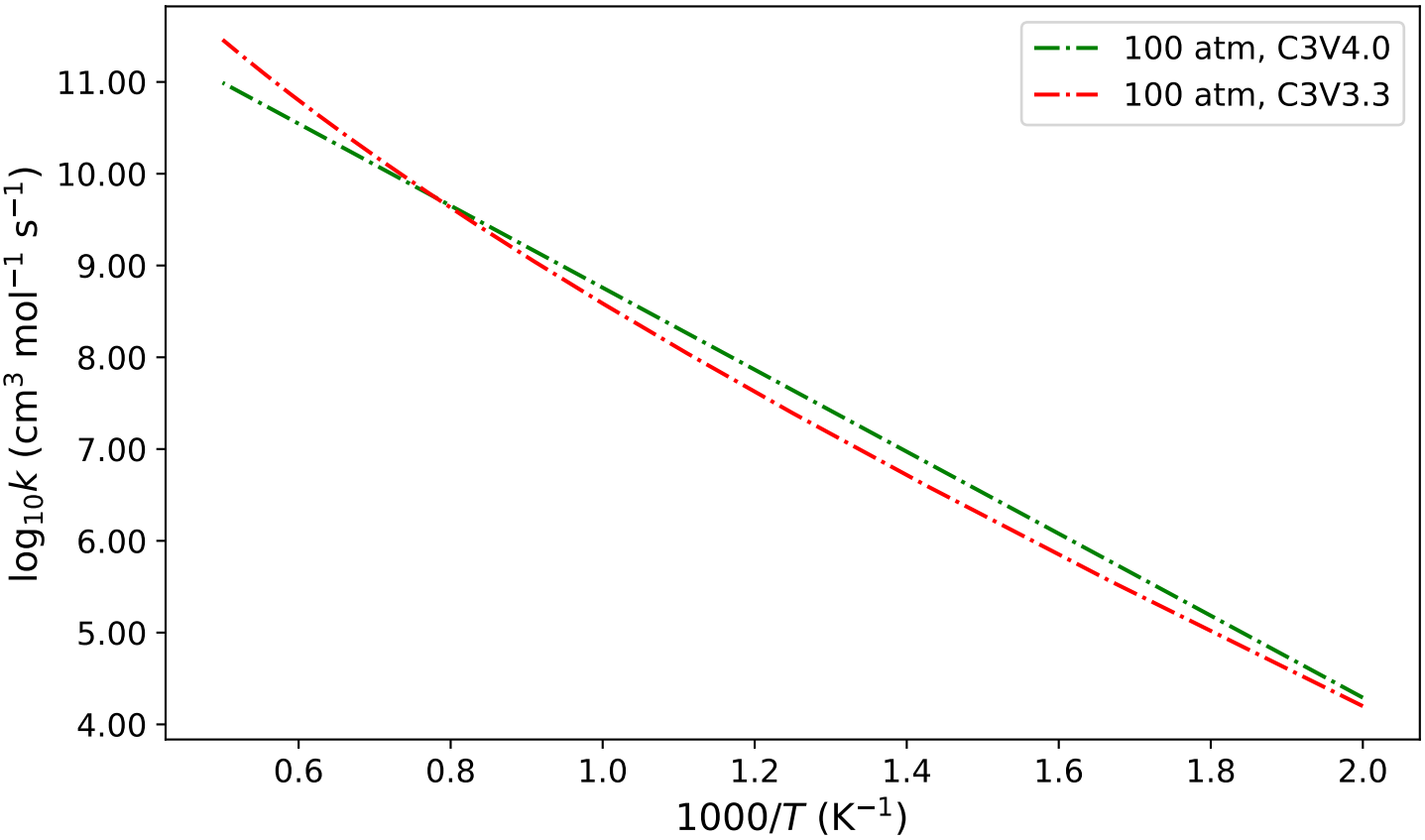



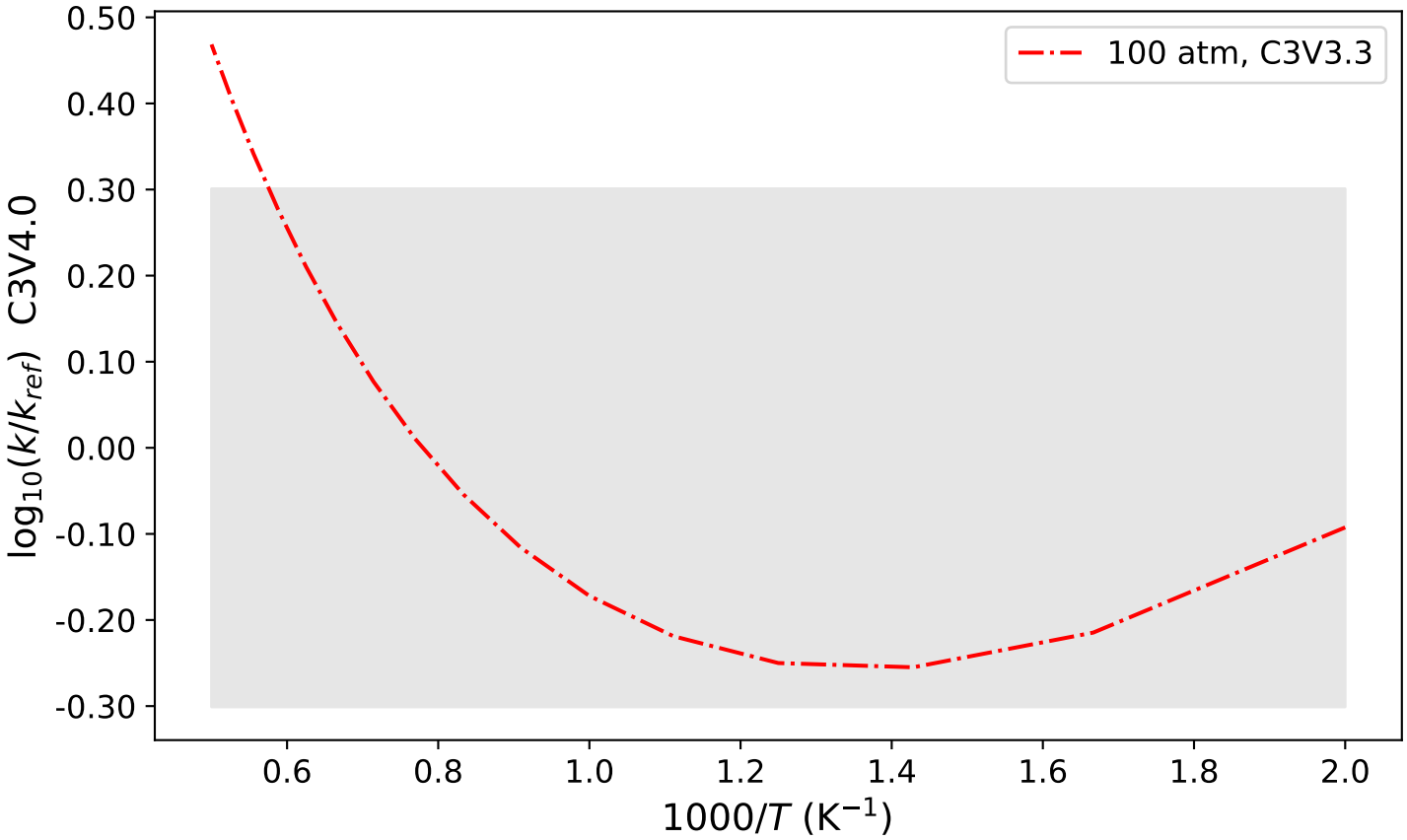
CH2CCH2OOH = CH2CHCH2OO



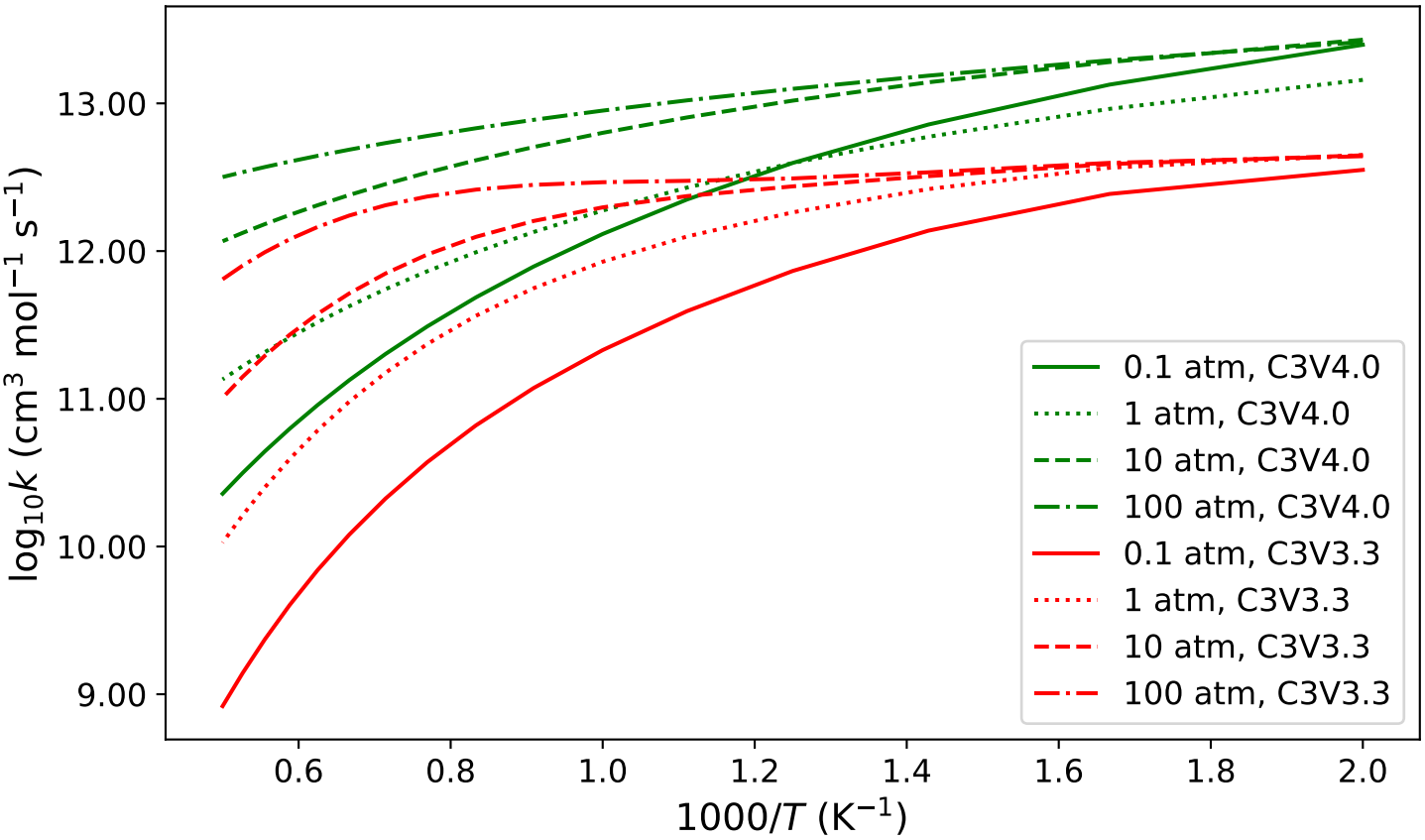



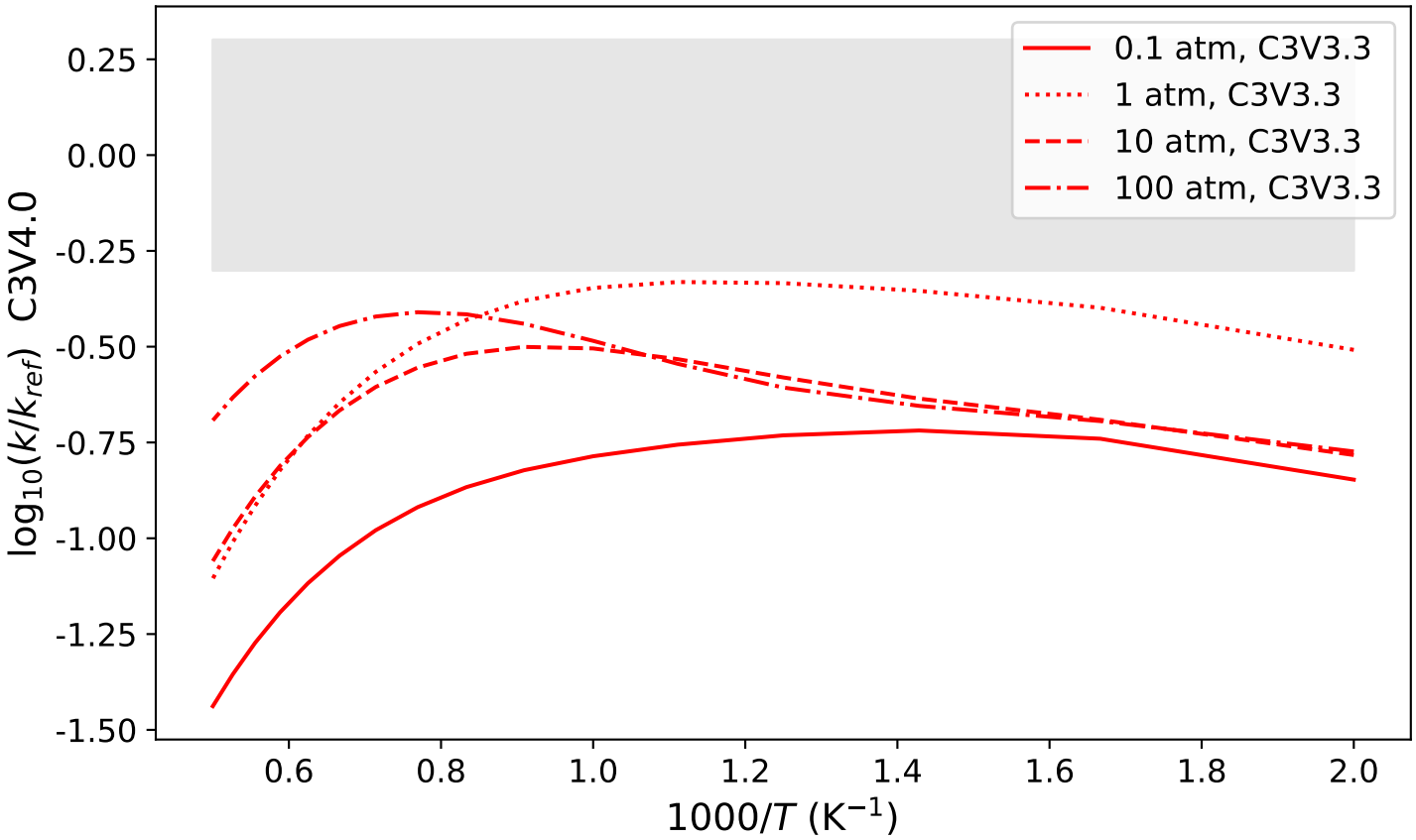
HO2 + IC3H7OH = H2O2 + C3H6OH2-1



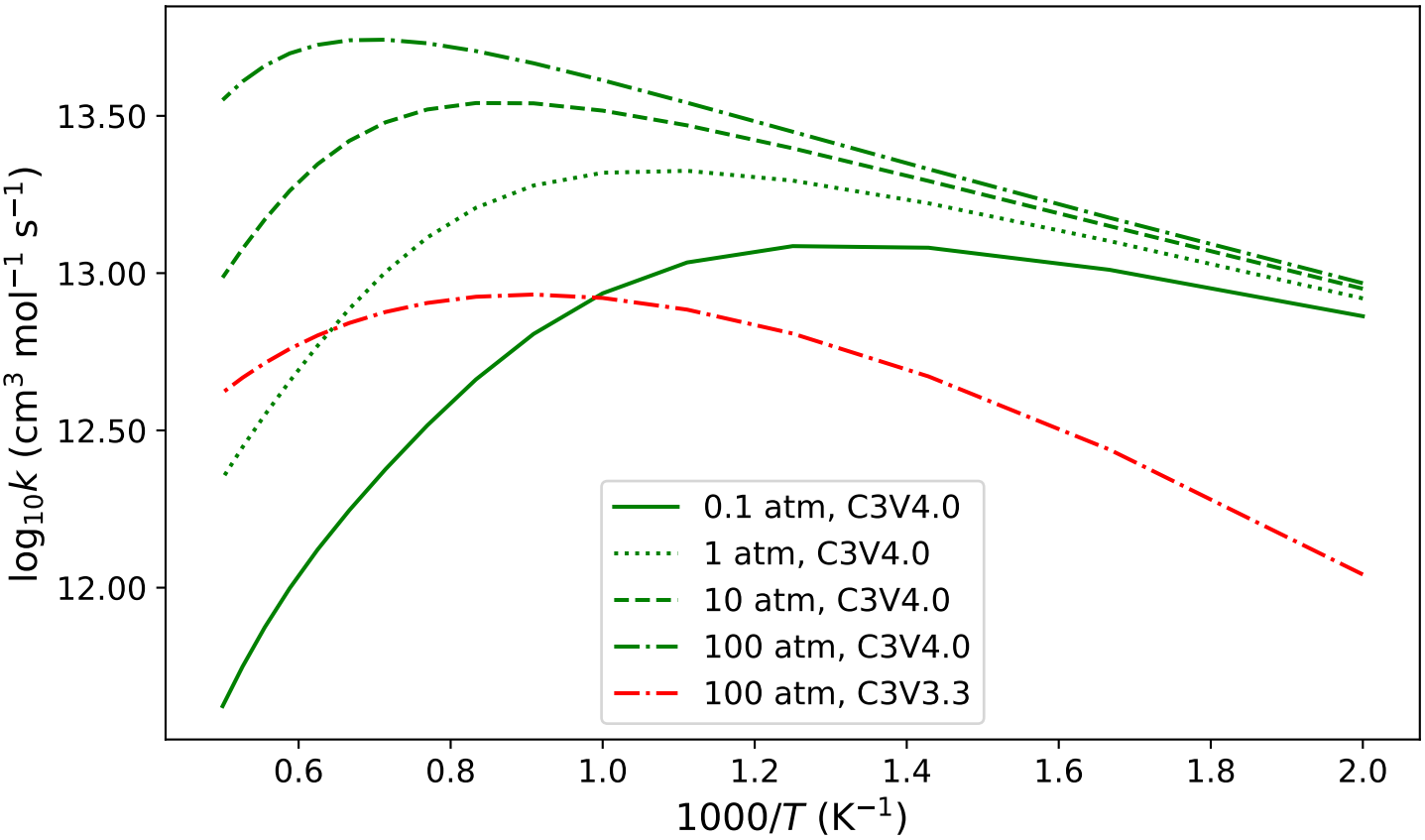



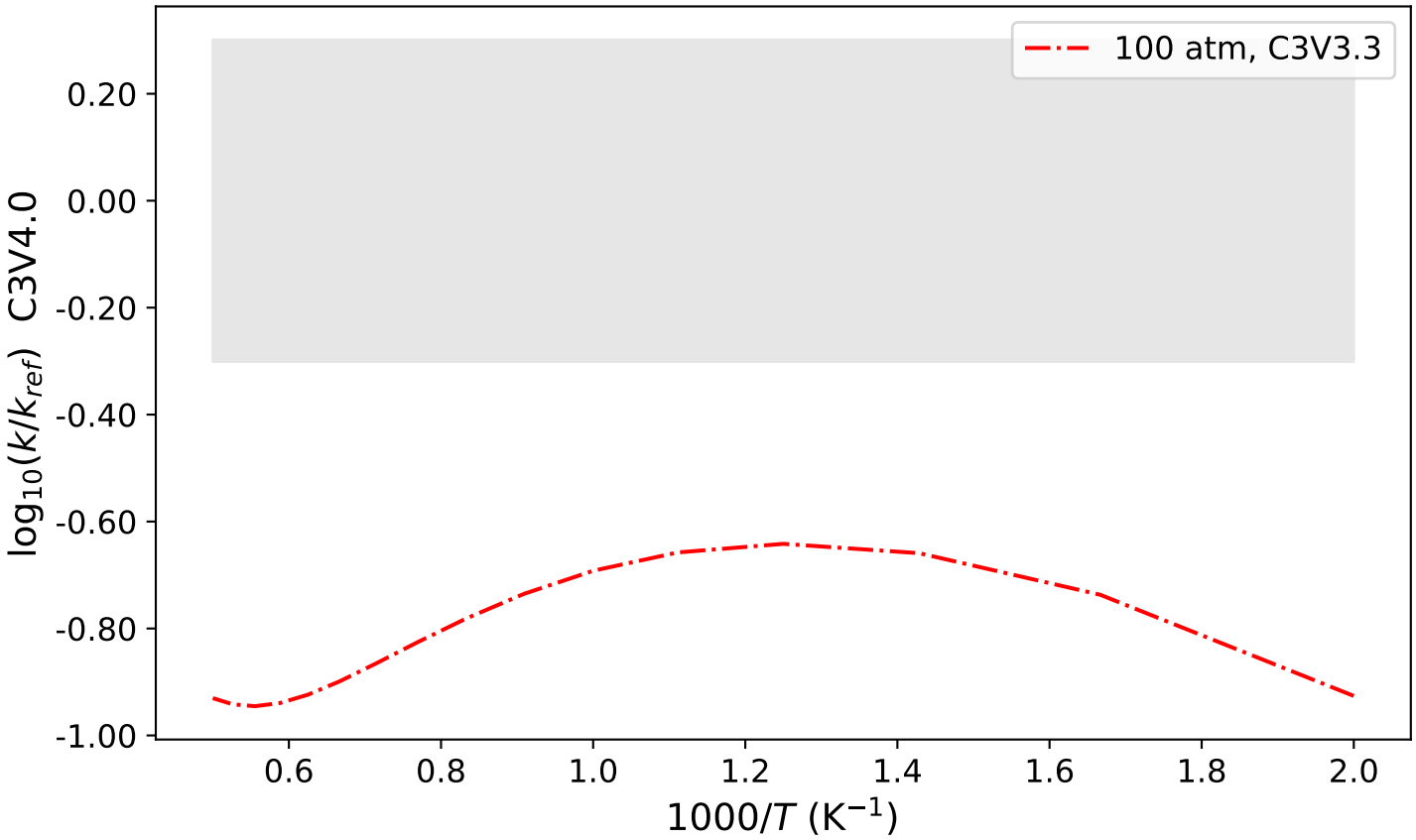
OH + C4H6 = C4H64,2-1OH



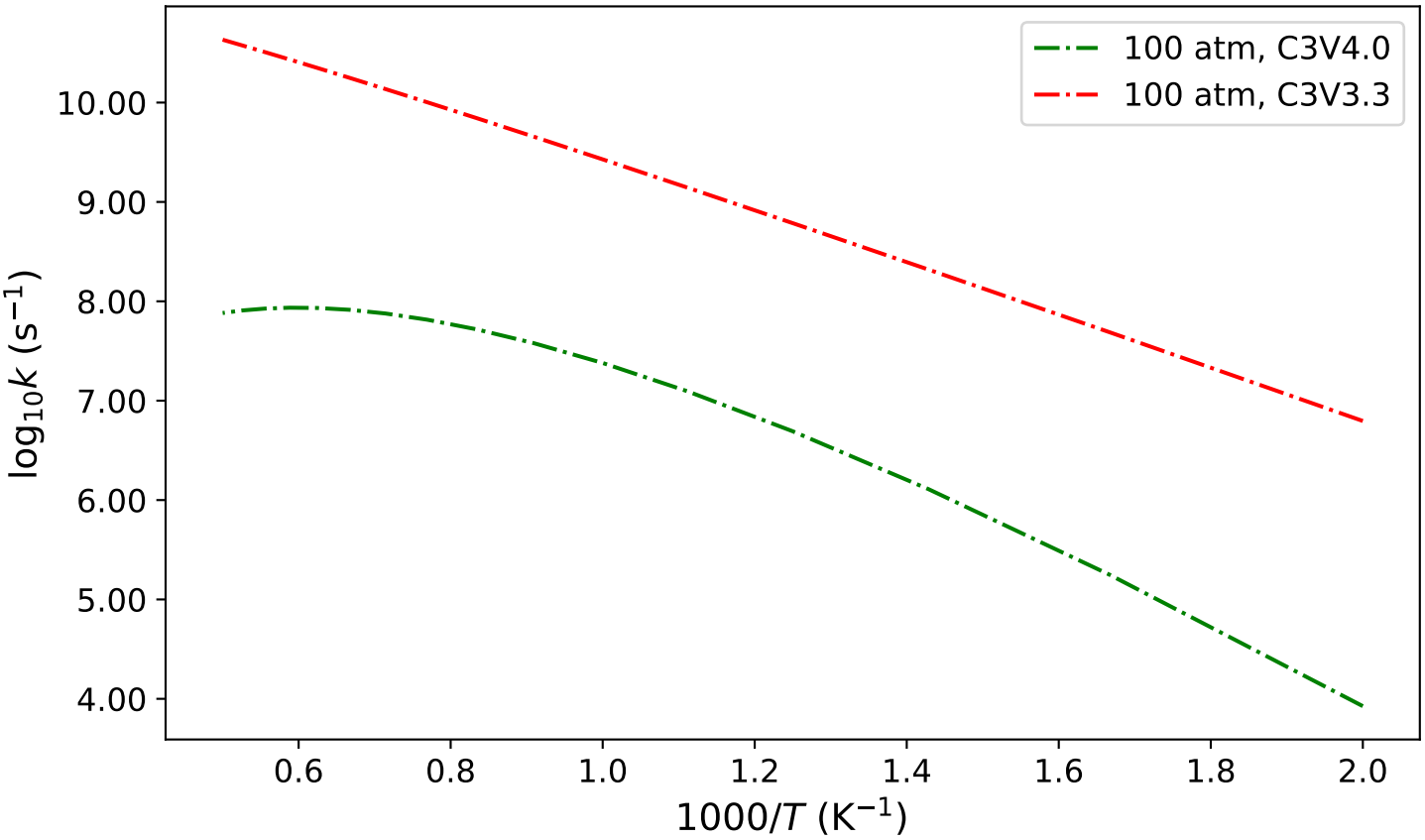



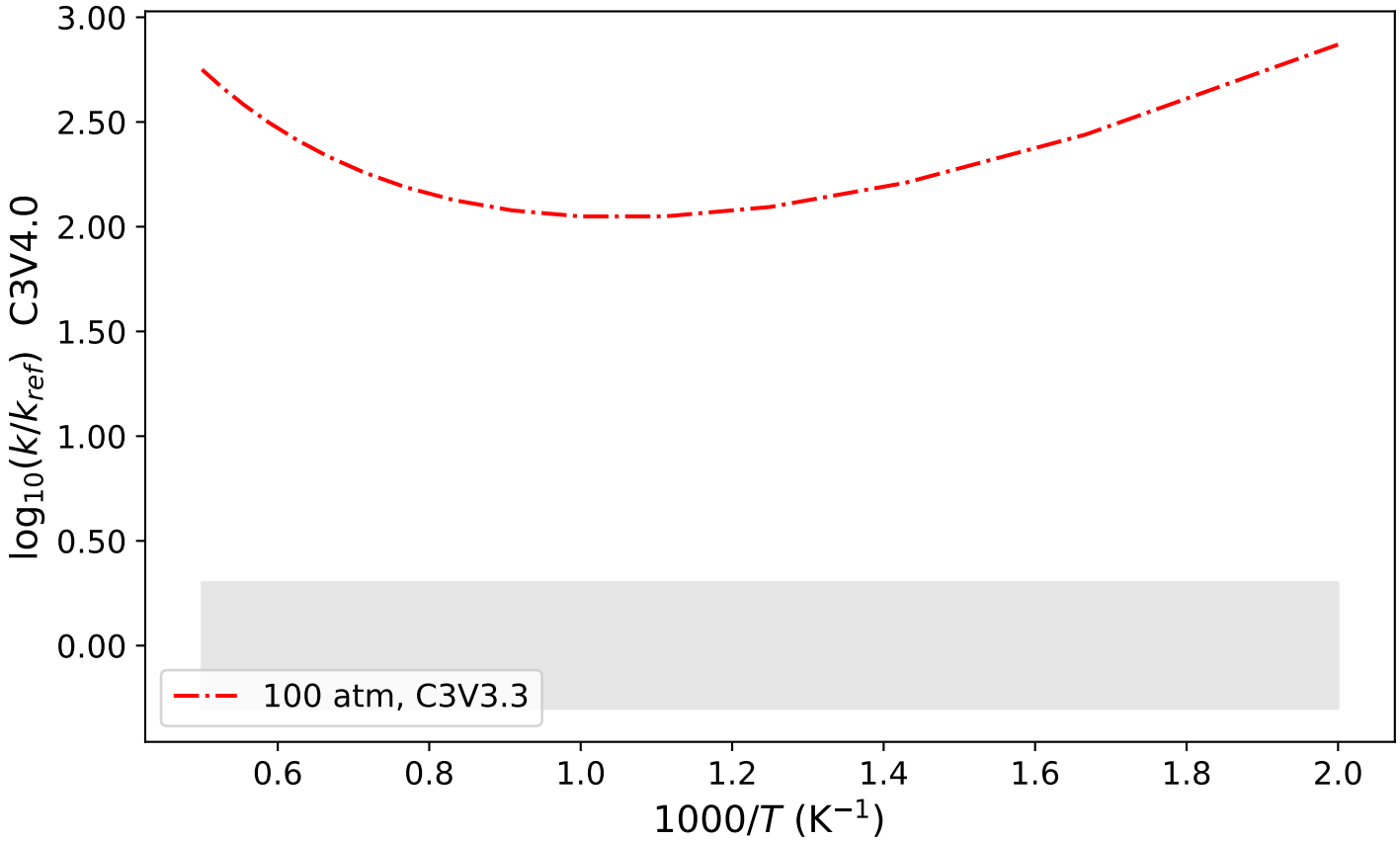
H + C4H2 (+M) = C4H3-I (+M)



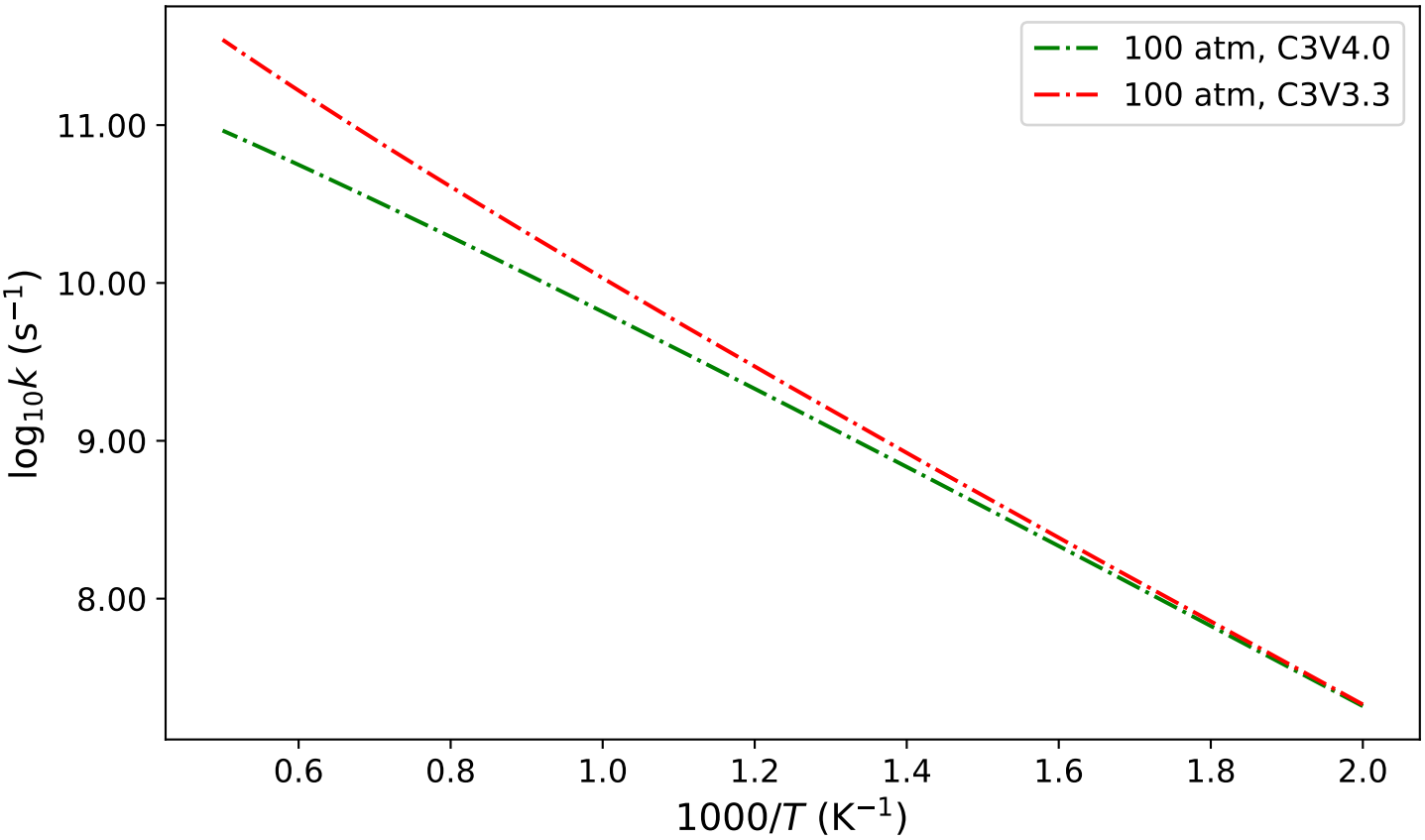



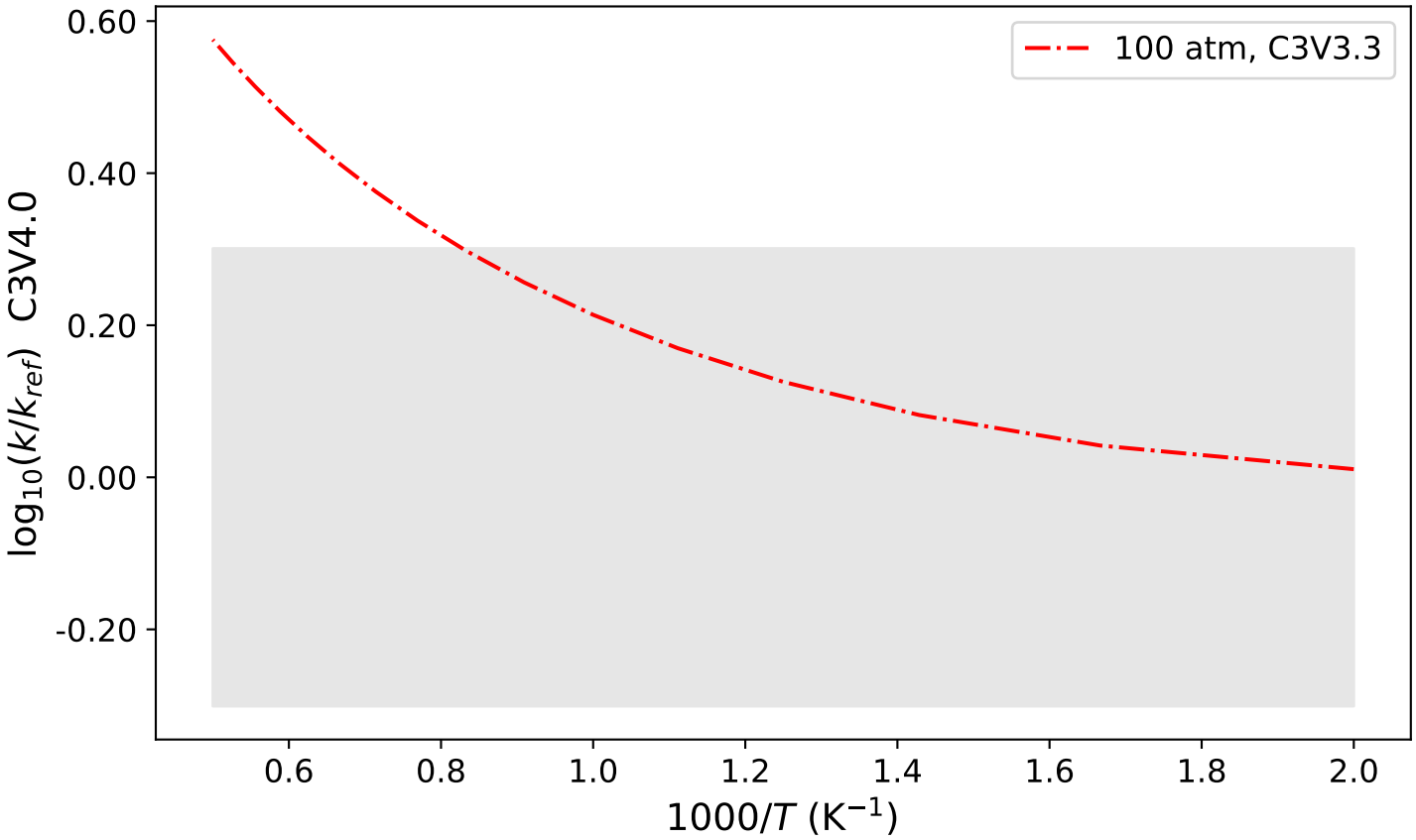
QCH3CHCO3H = OH + CH3#CHCOO#



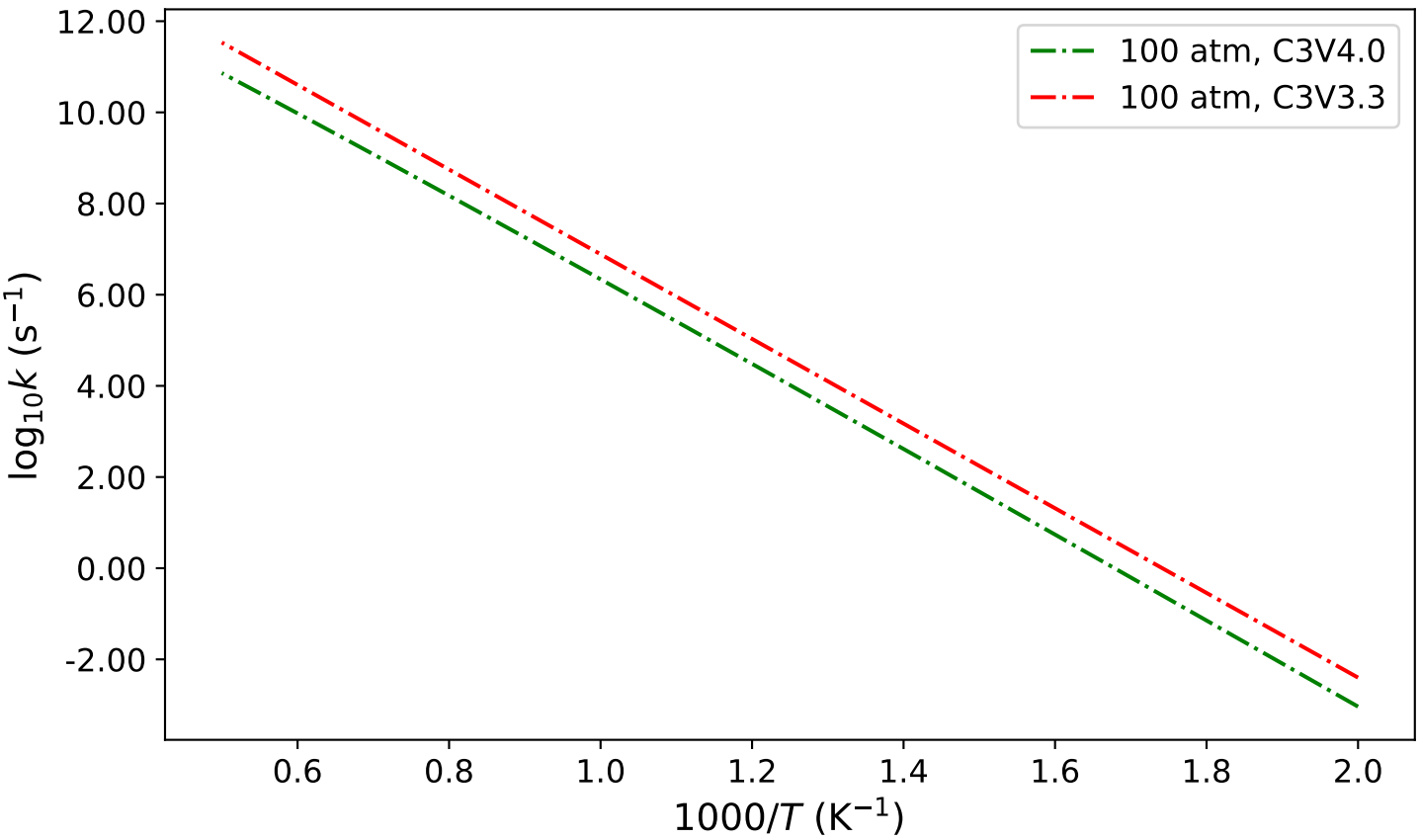



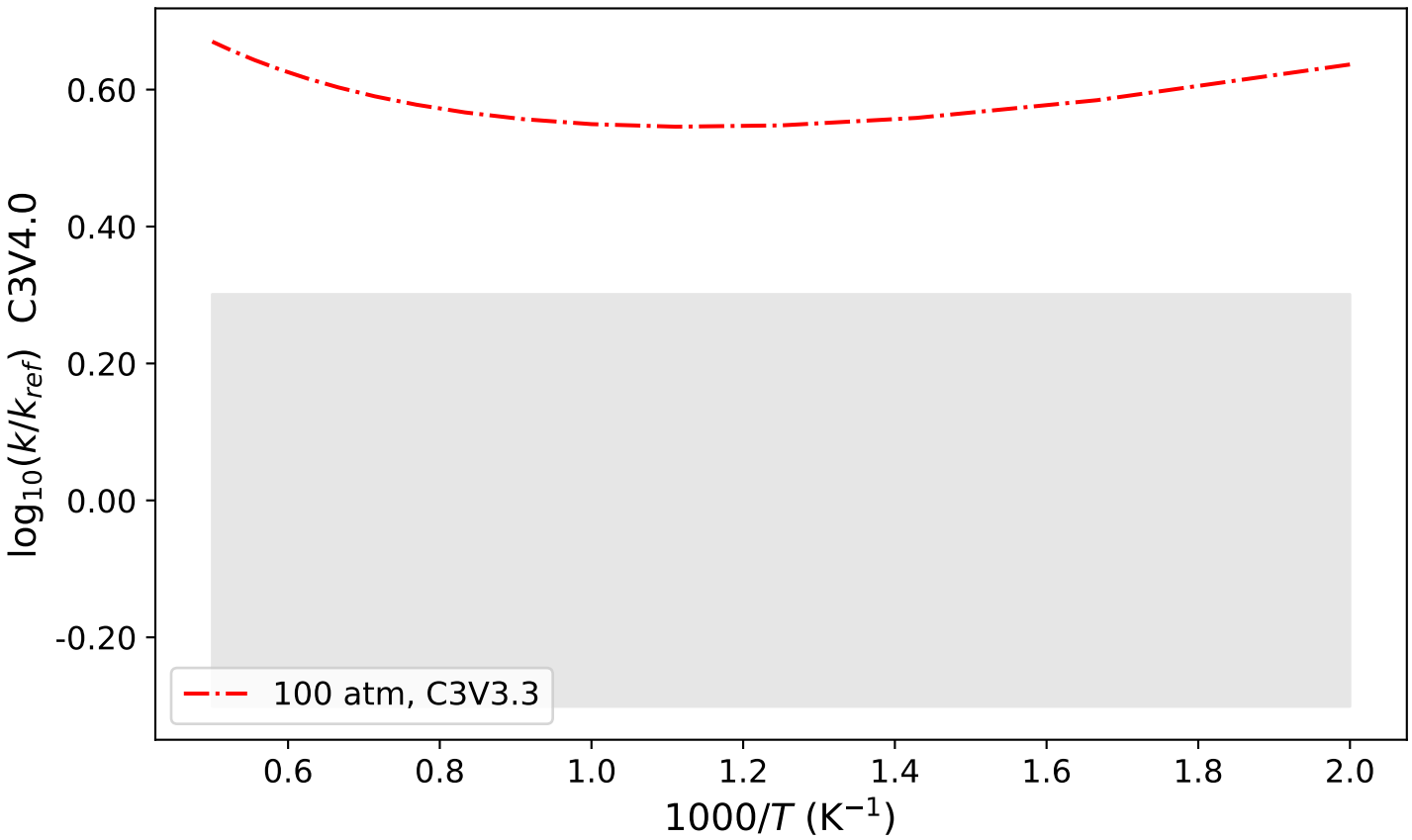
C4H8OOH1-2 = OH + C4H8O1-2



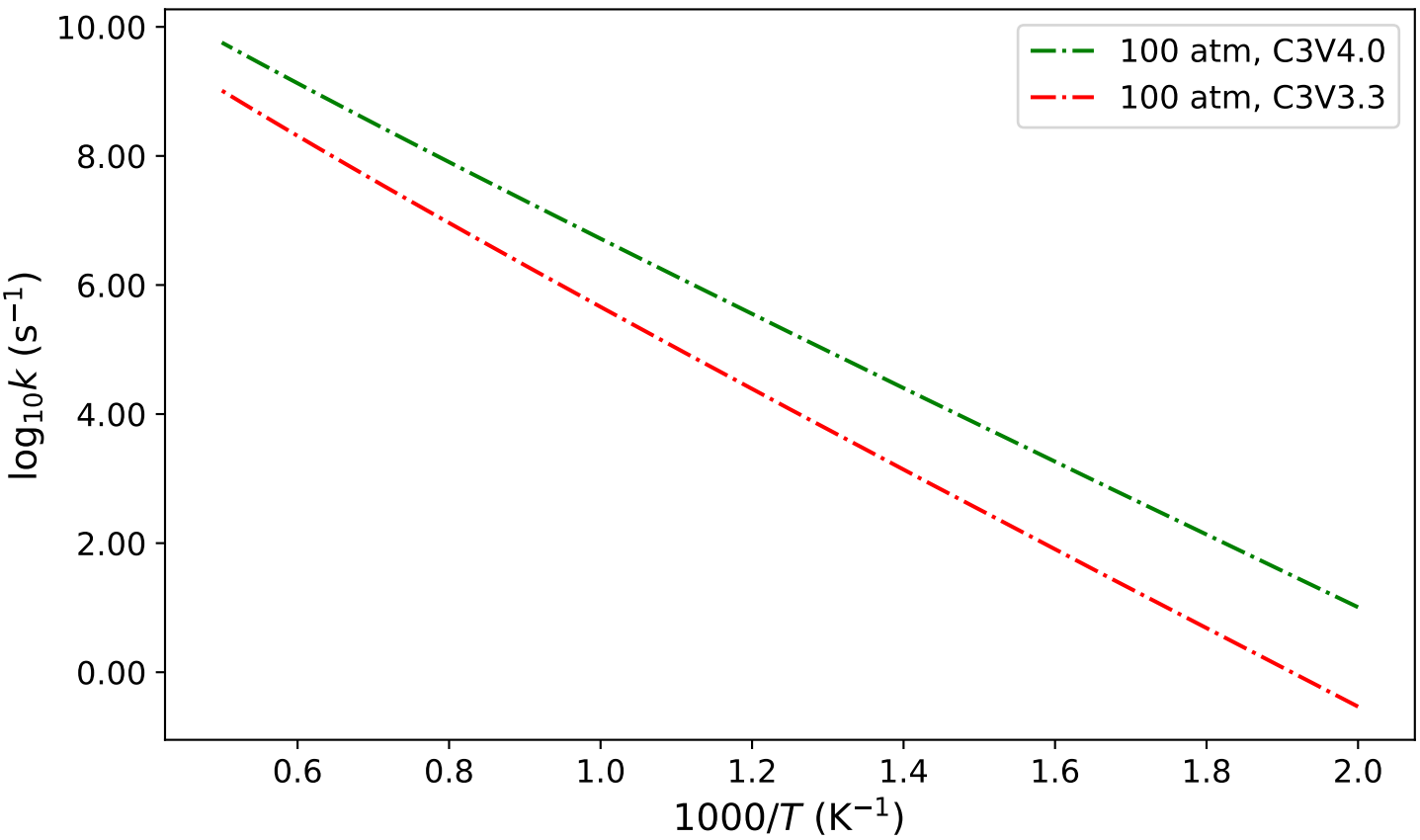



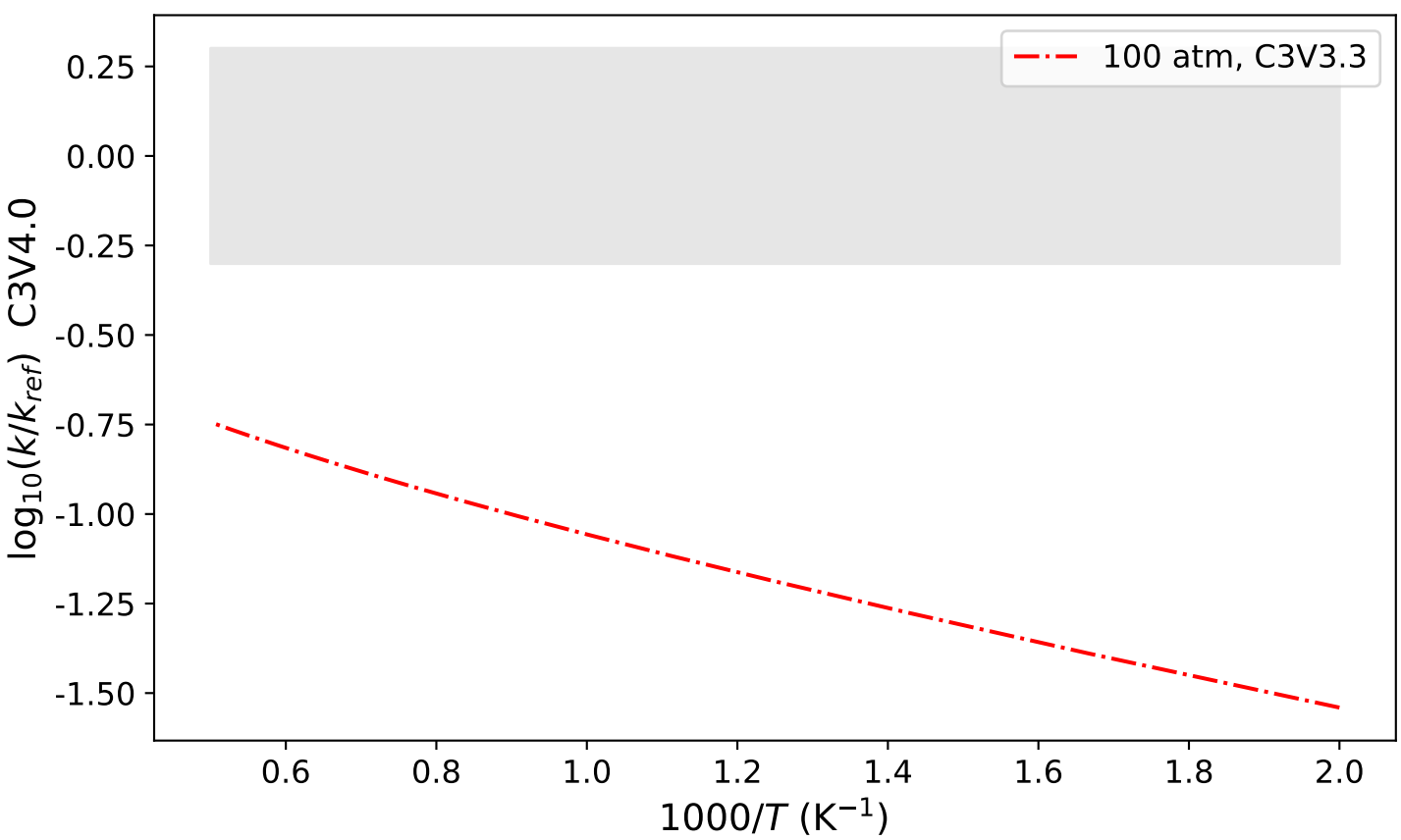
PC4H9O2H = OH + PC4H9O



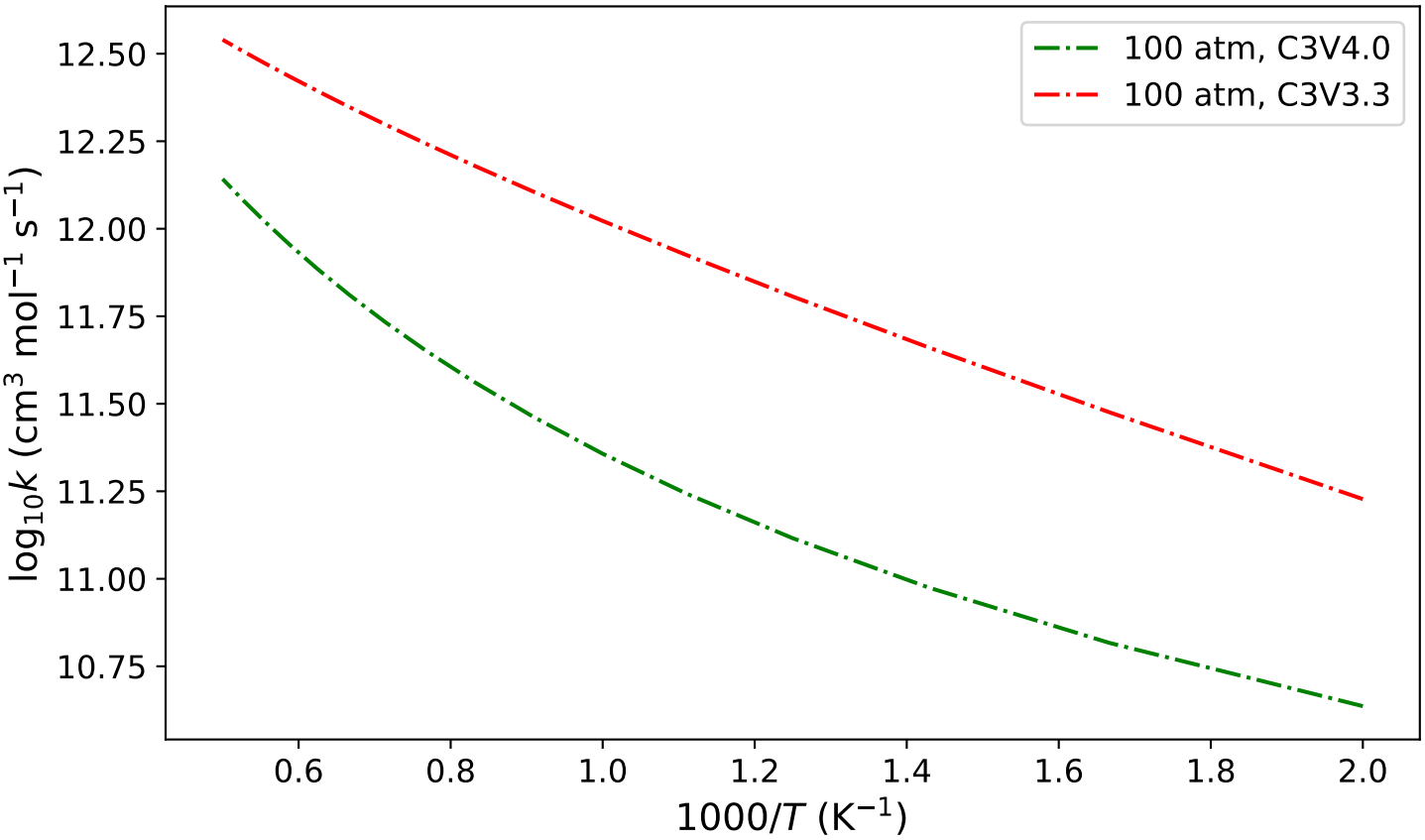



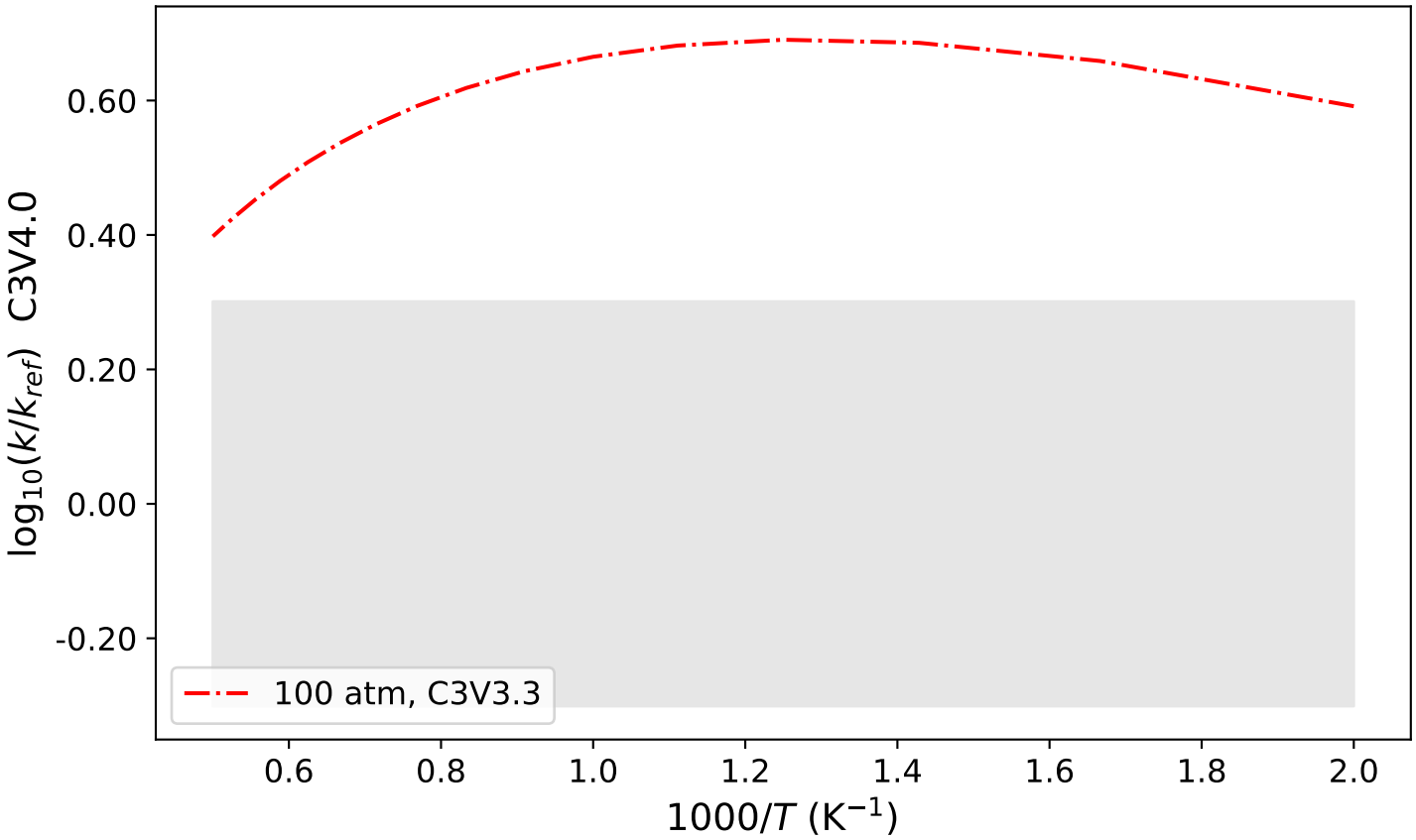
C4H8OOH2-3O2 = OH + NC4KET23



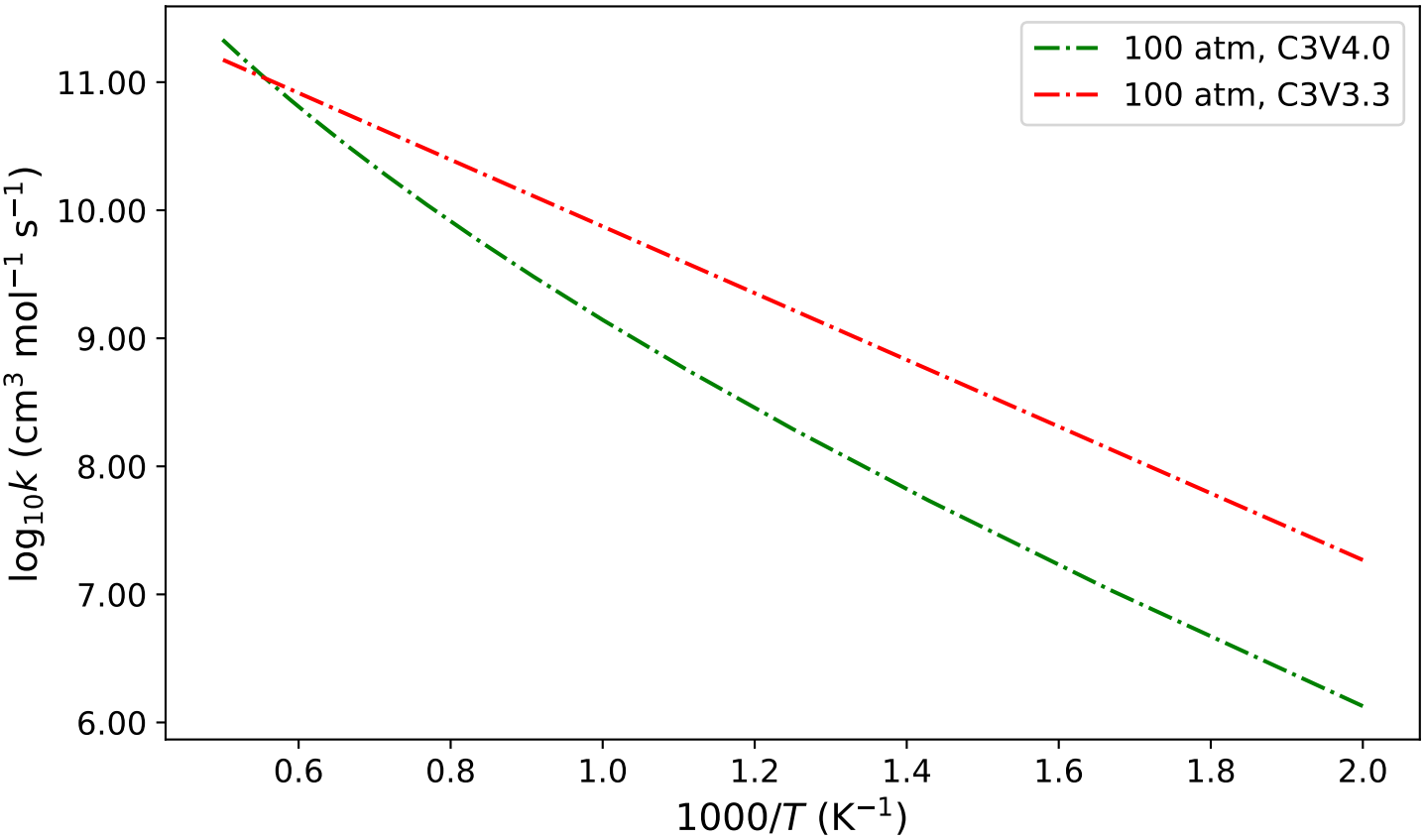



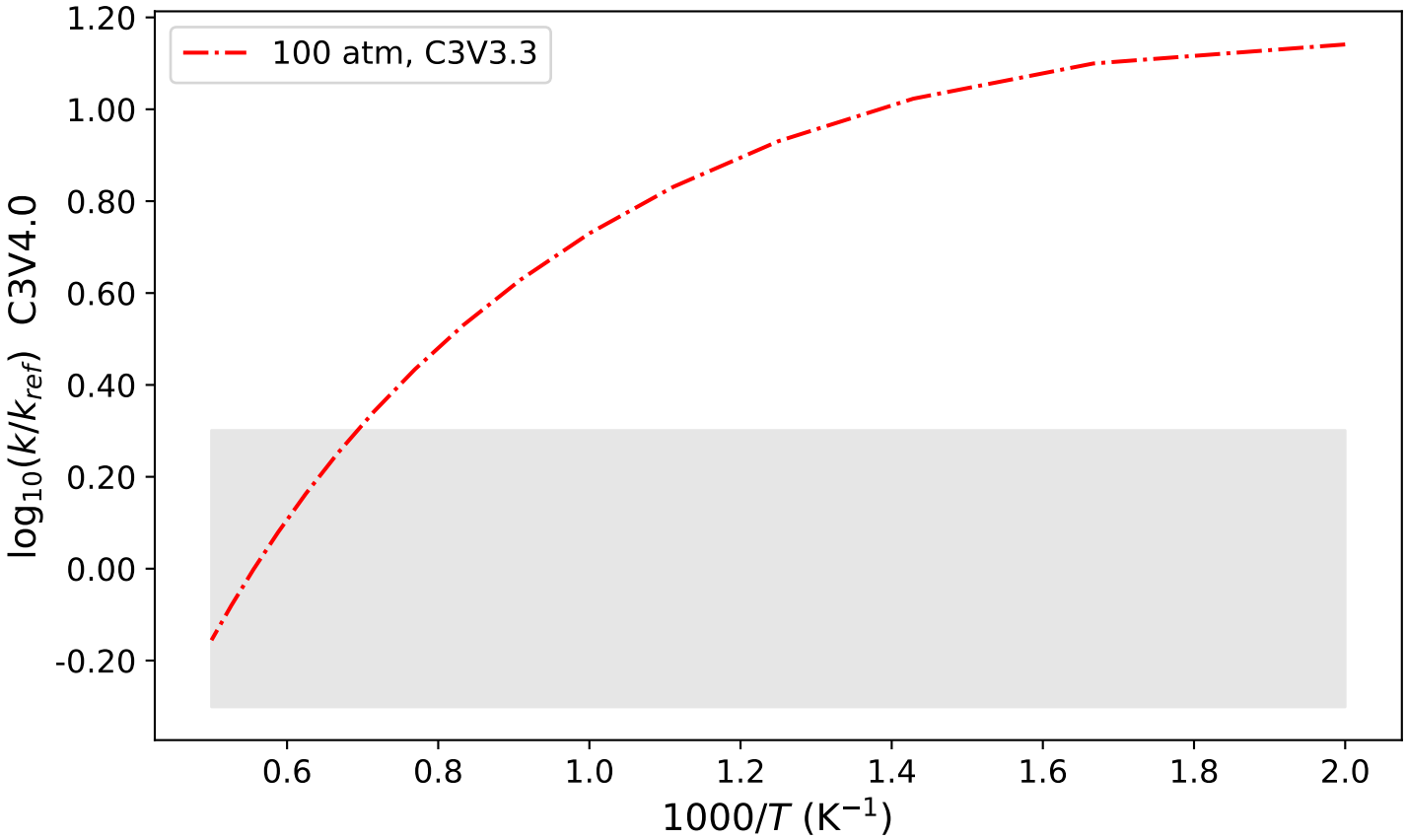
NC3H7OH + OH = NC3H7O + H2O



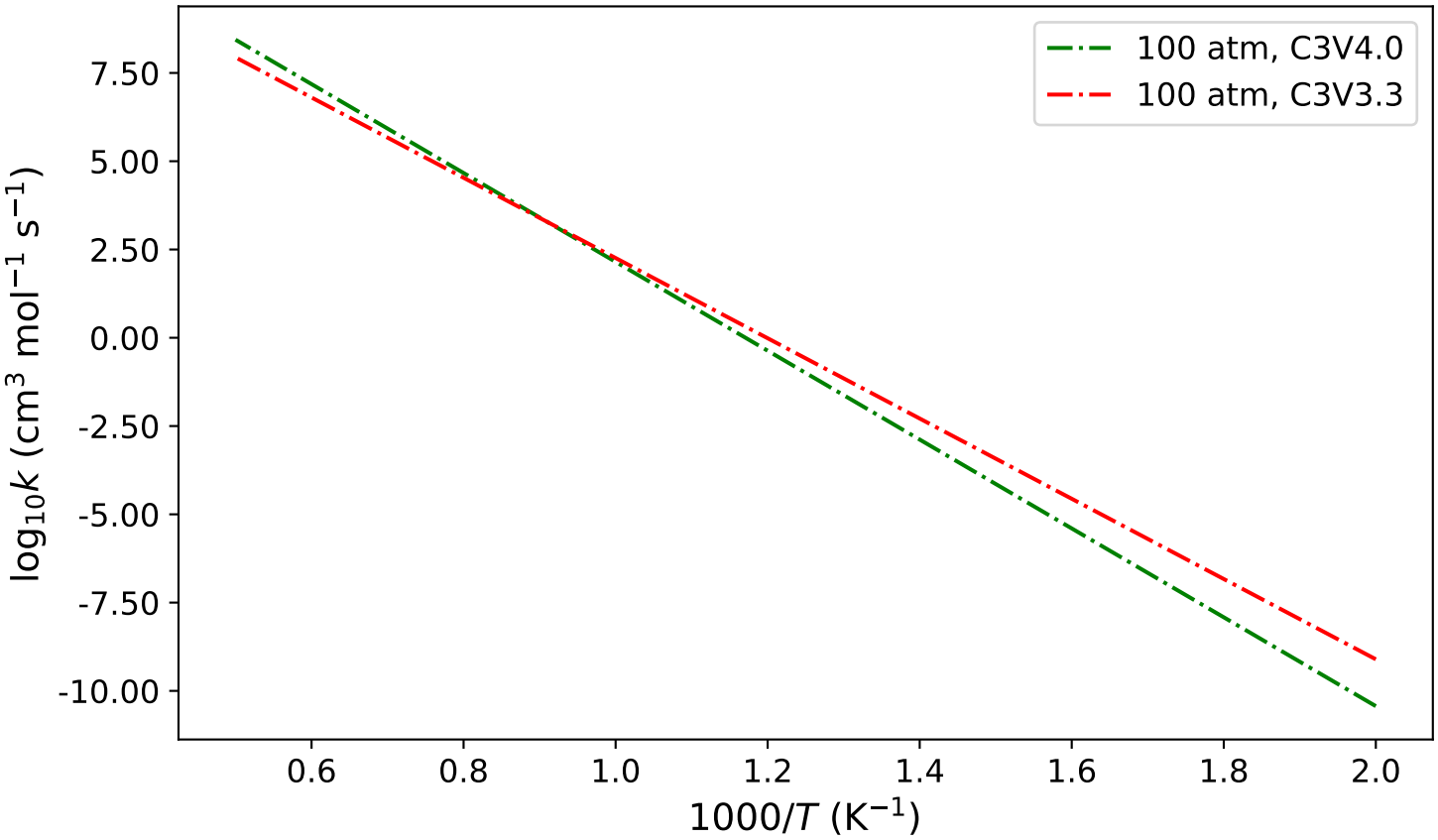



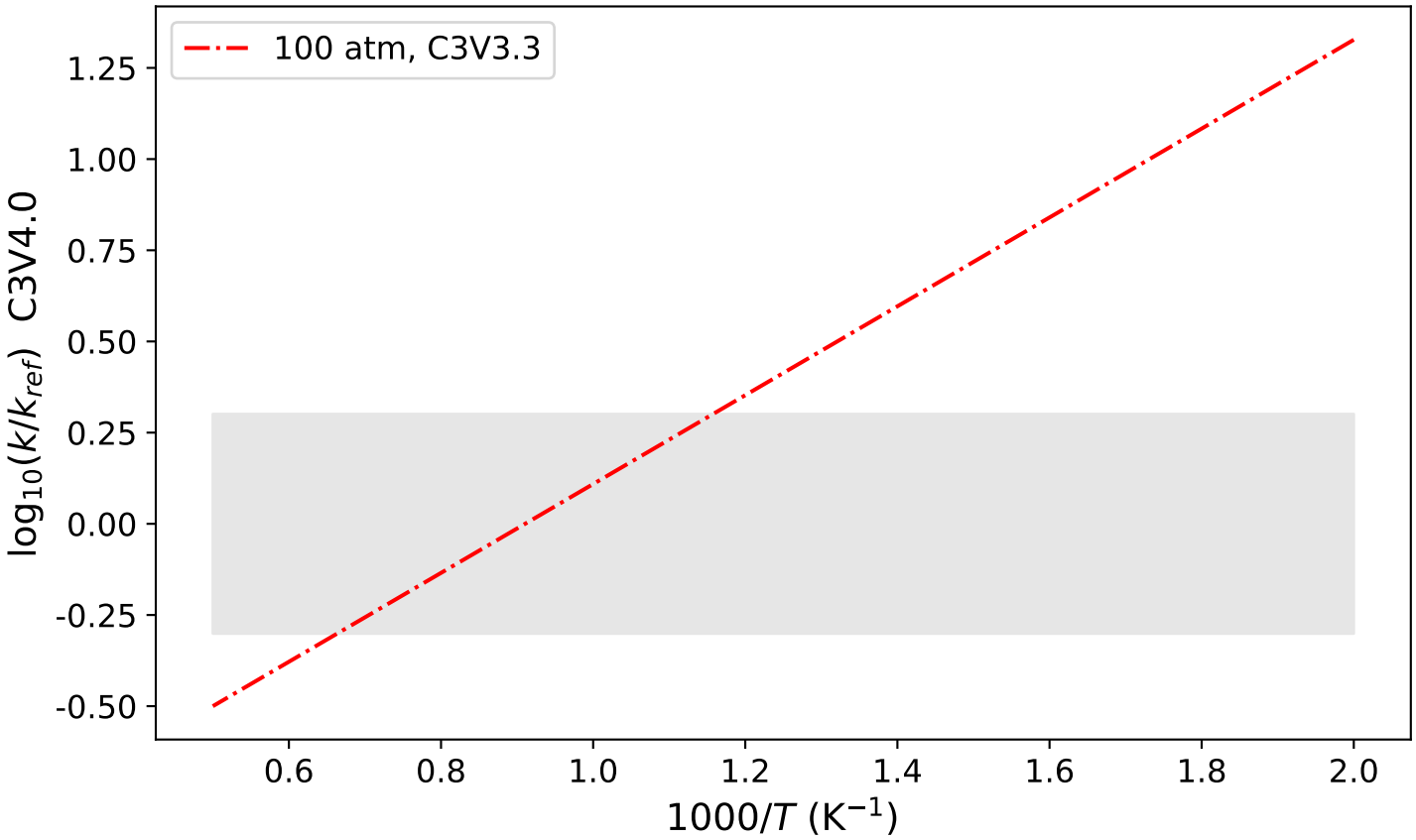
HO2 + C2H3CHO = H2O2 + C2H3CO



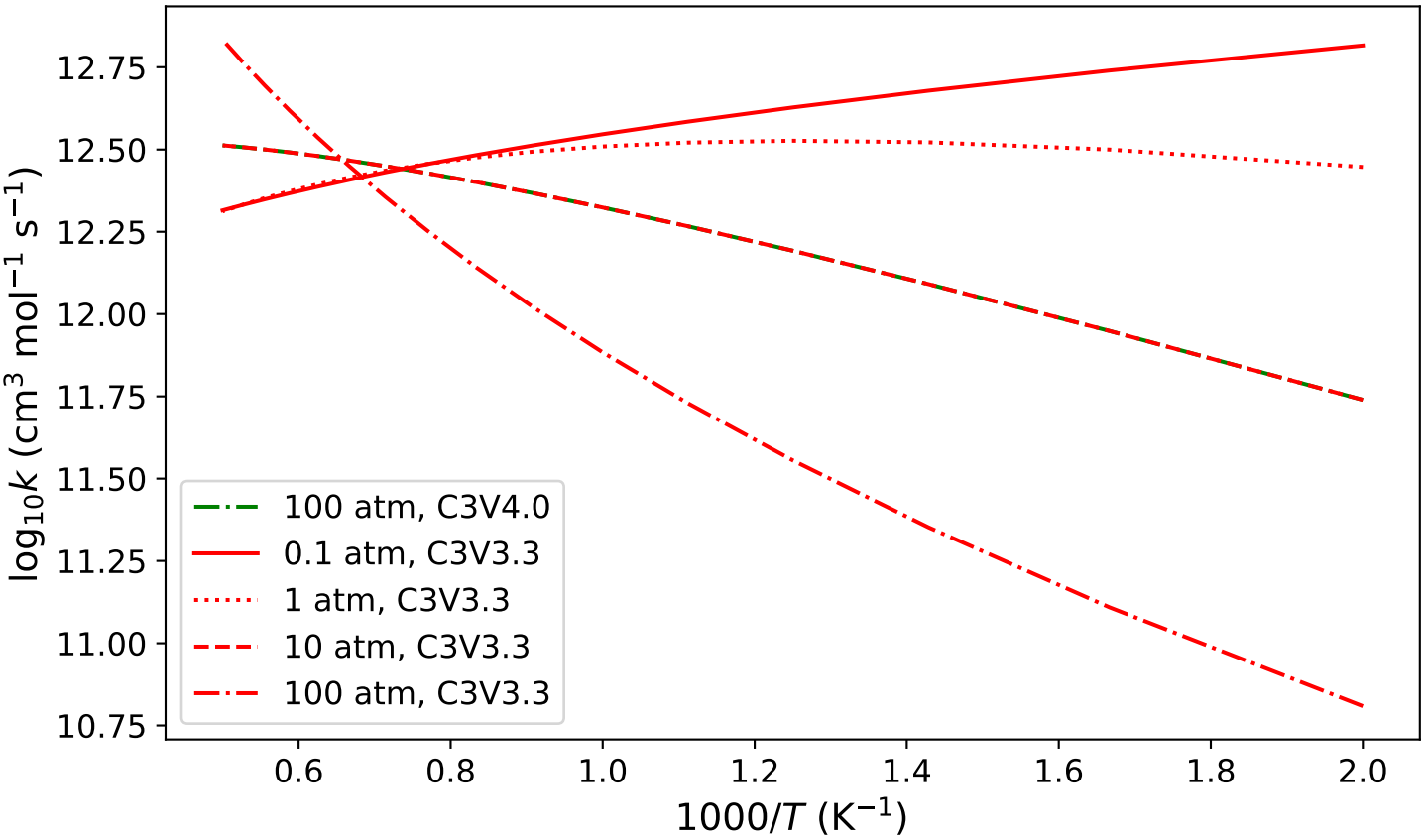



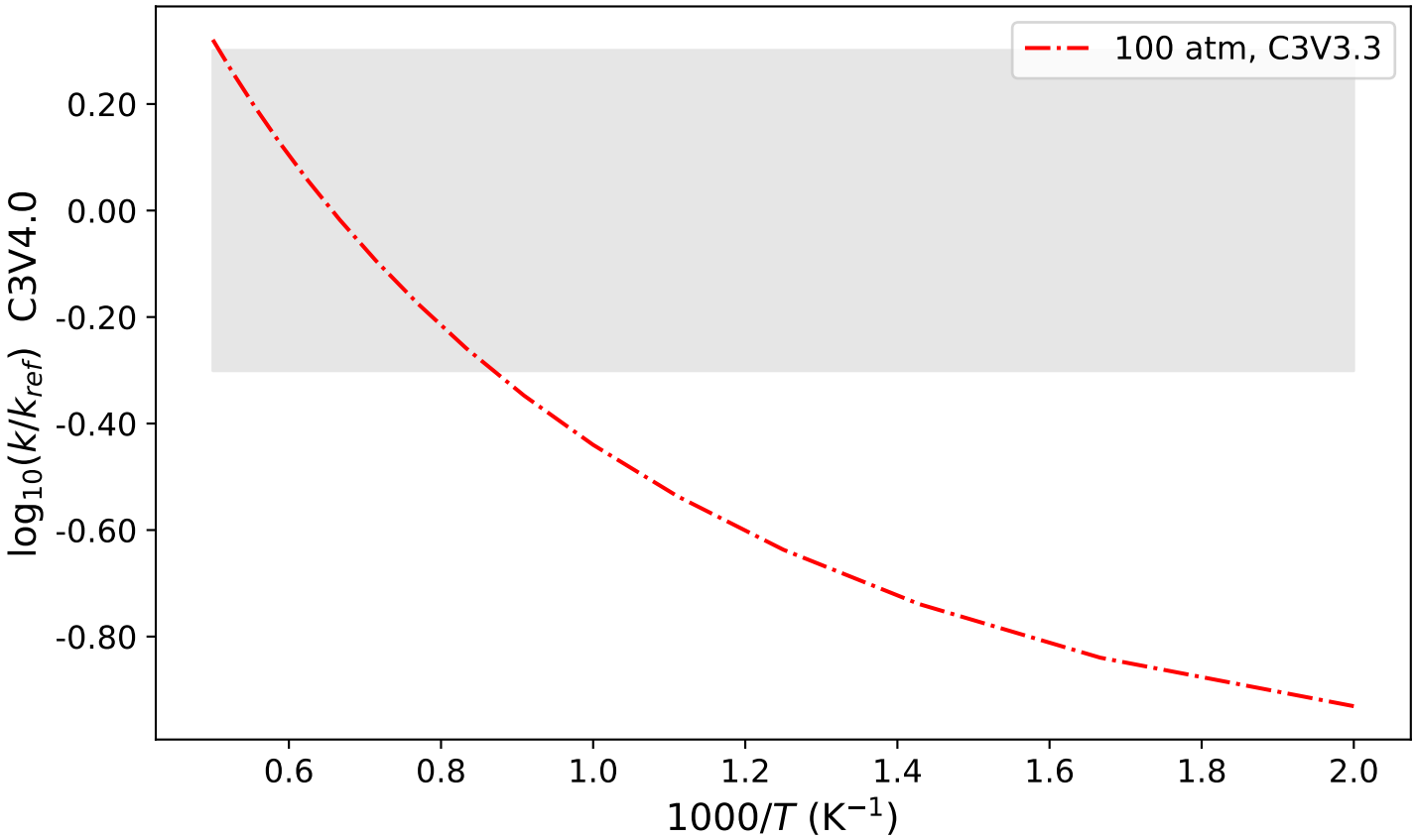
O2 + IC3H7OH = HO2 + C3H6OH2-1



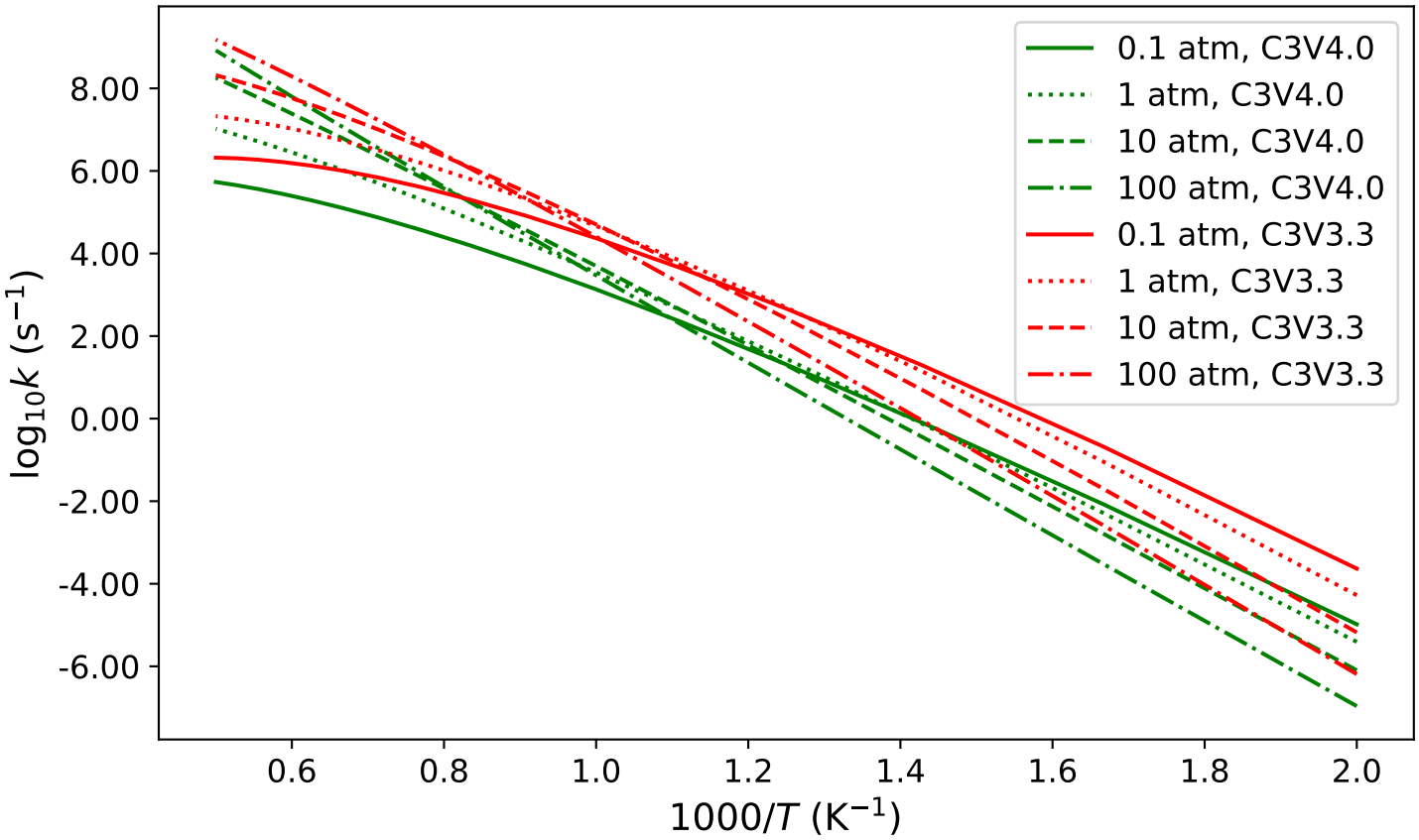



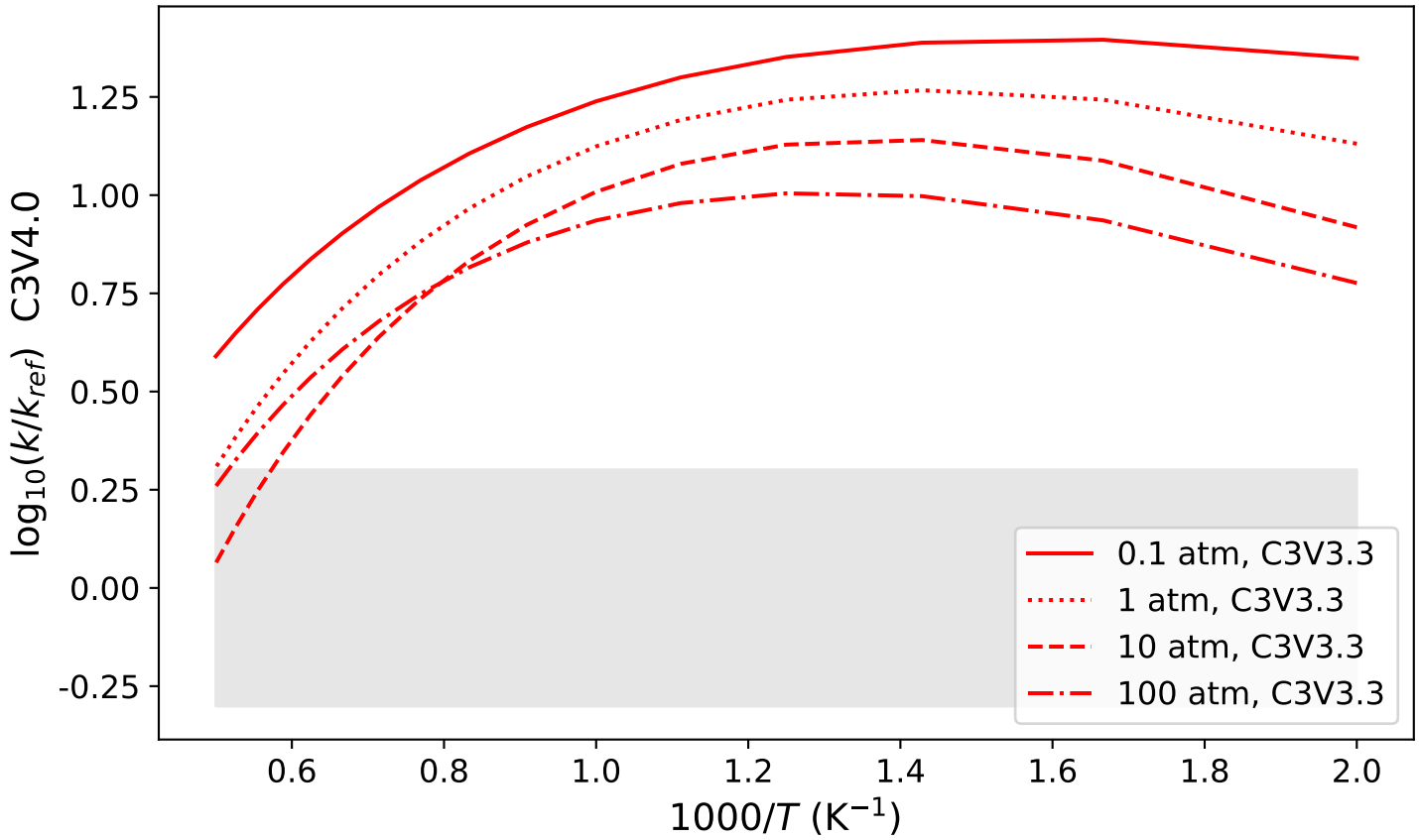
HO2 + C4H71-3 = OH + C4H7O2-1



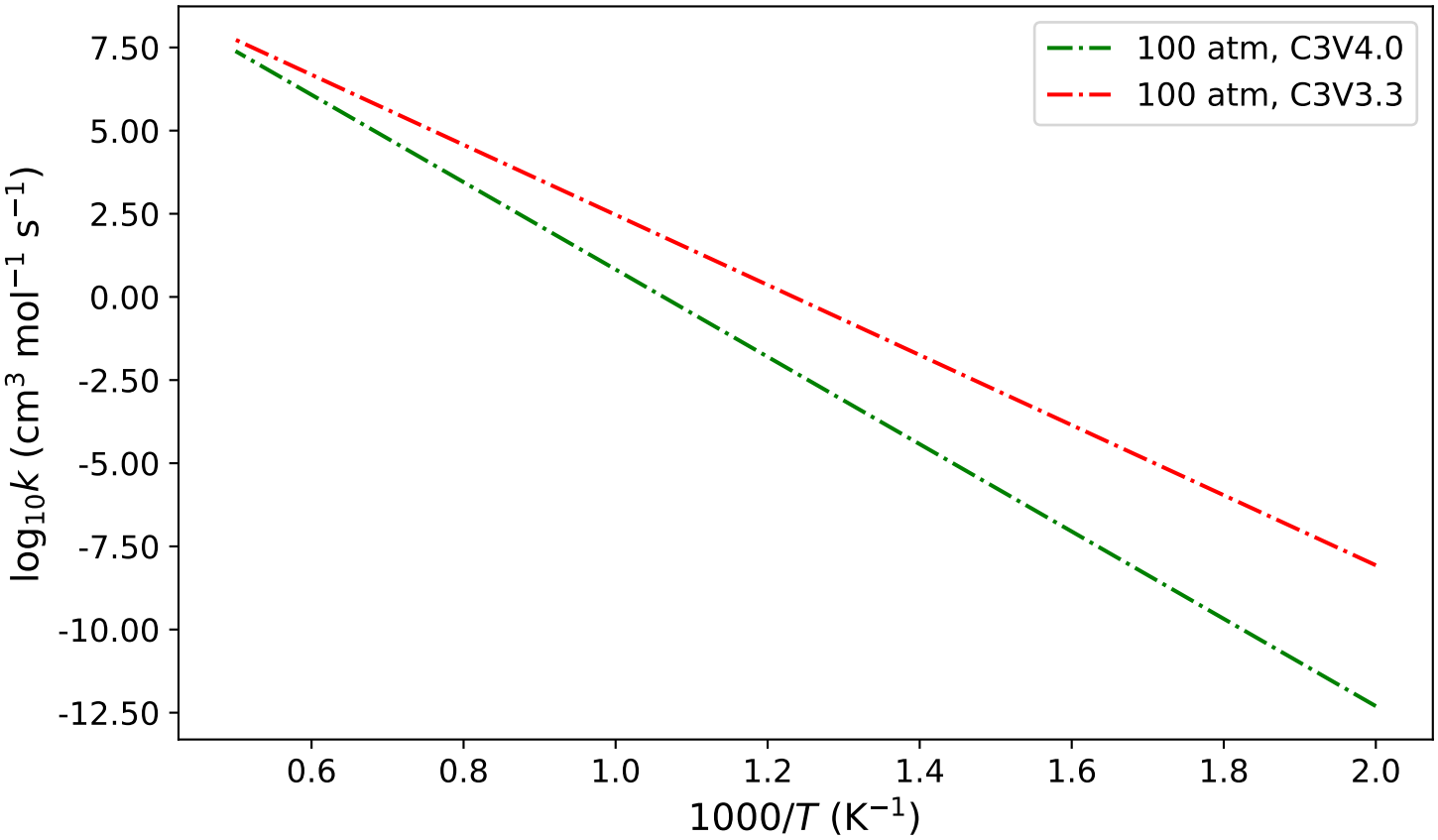



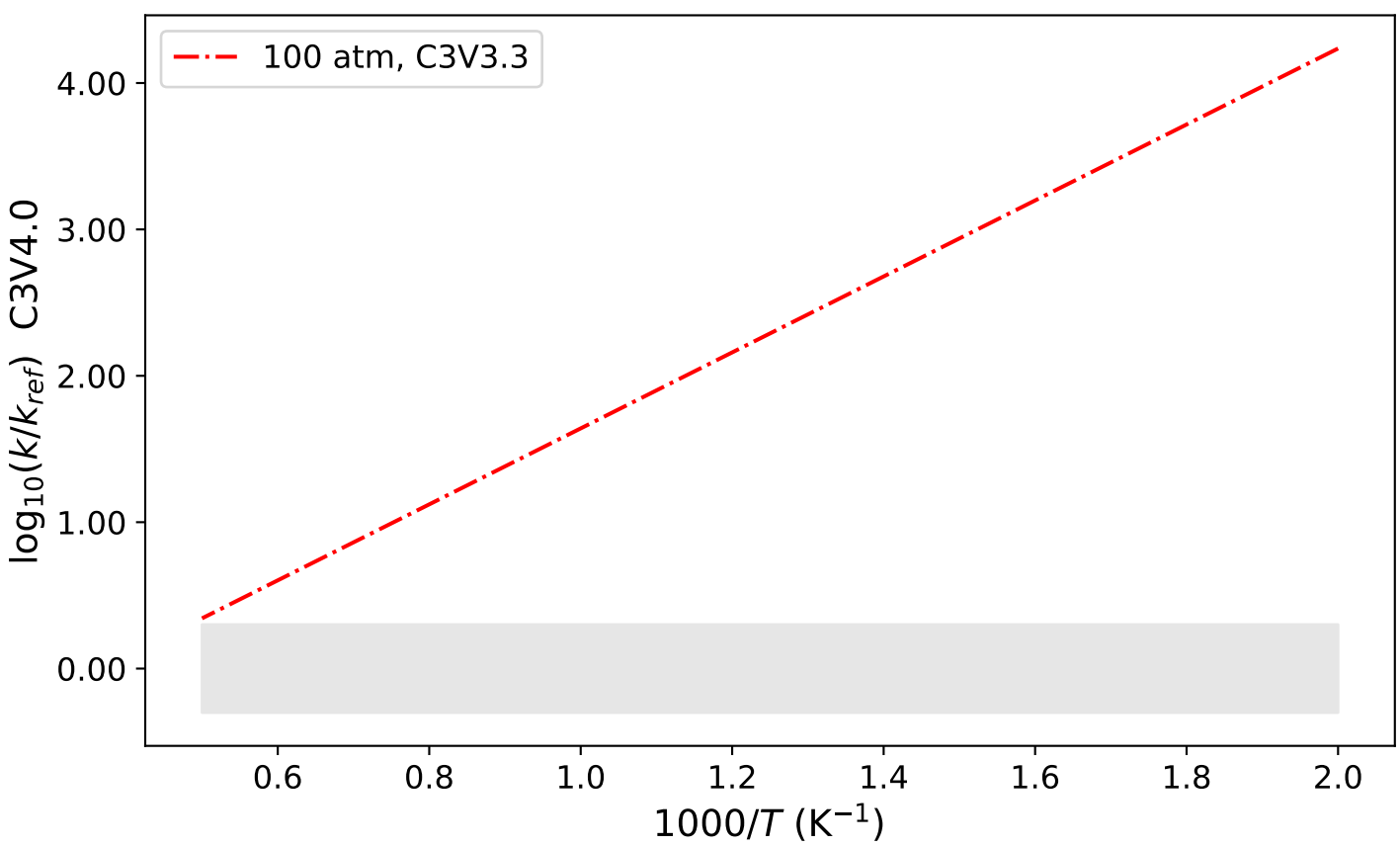
TC4H9 = CH3 + C3H6



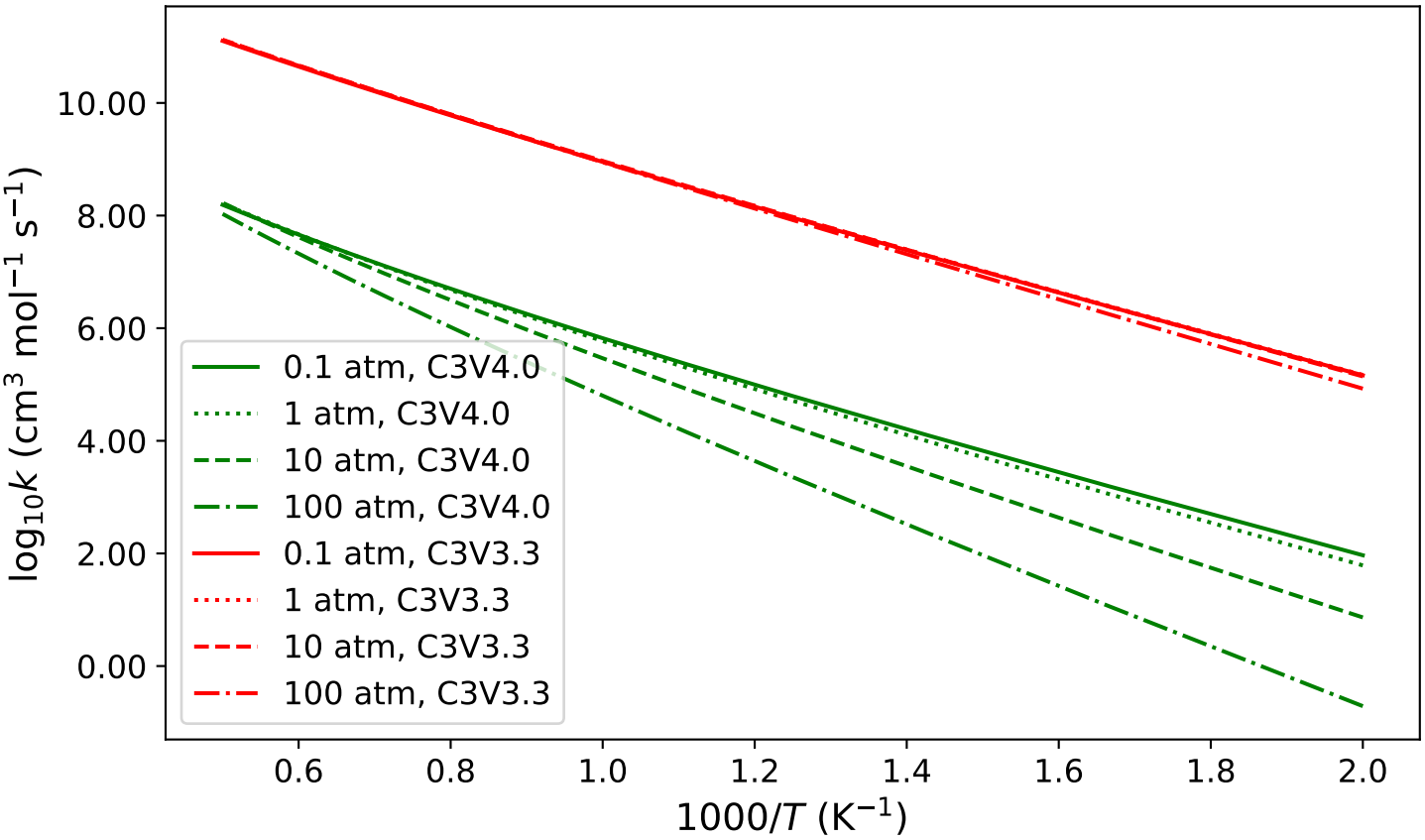



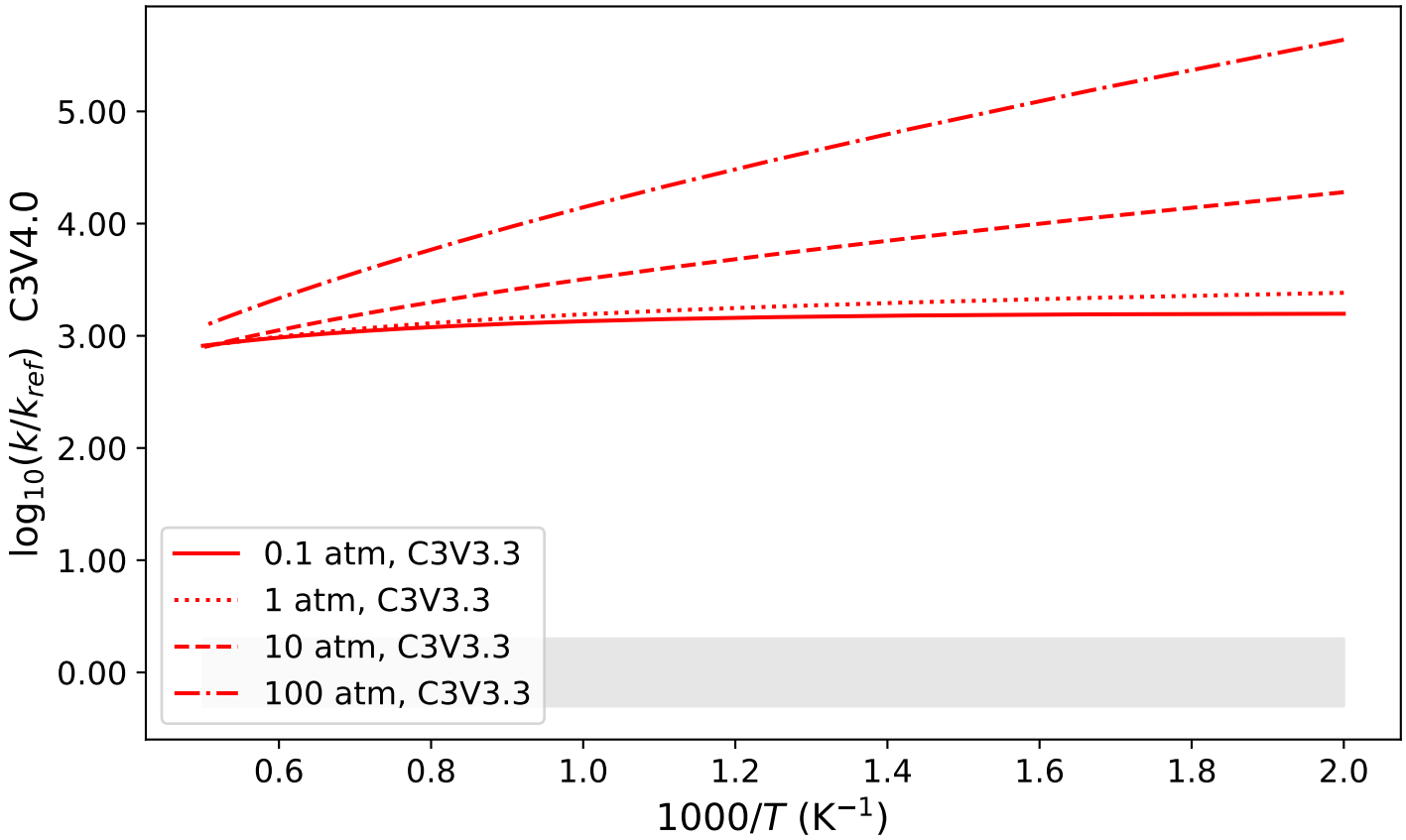
O2 + NC3H7OH = NC3H7O + HO2



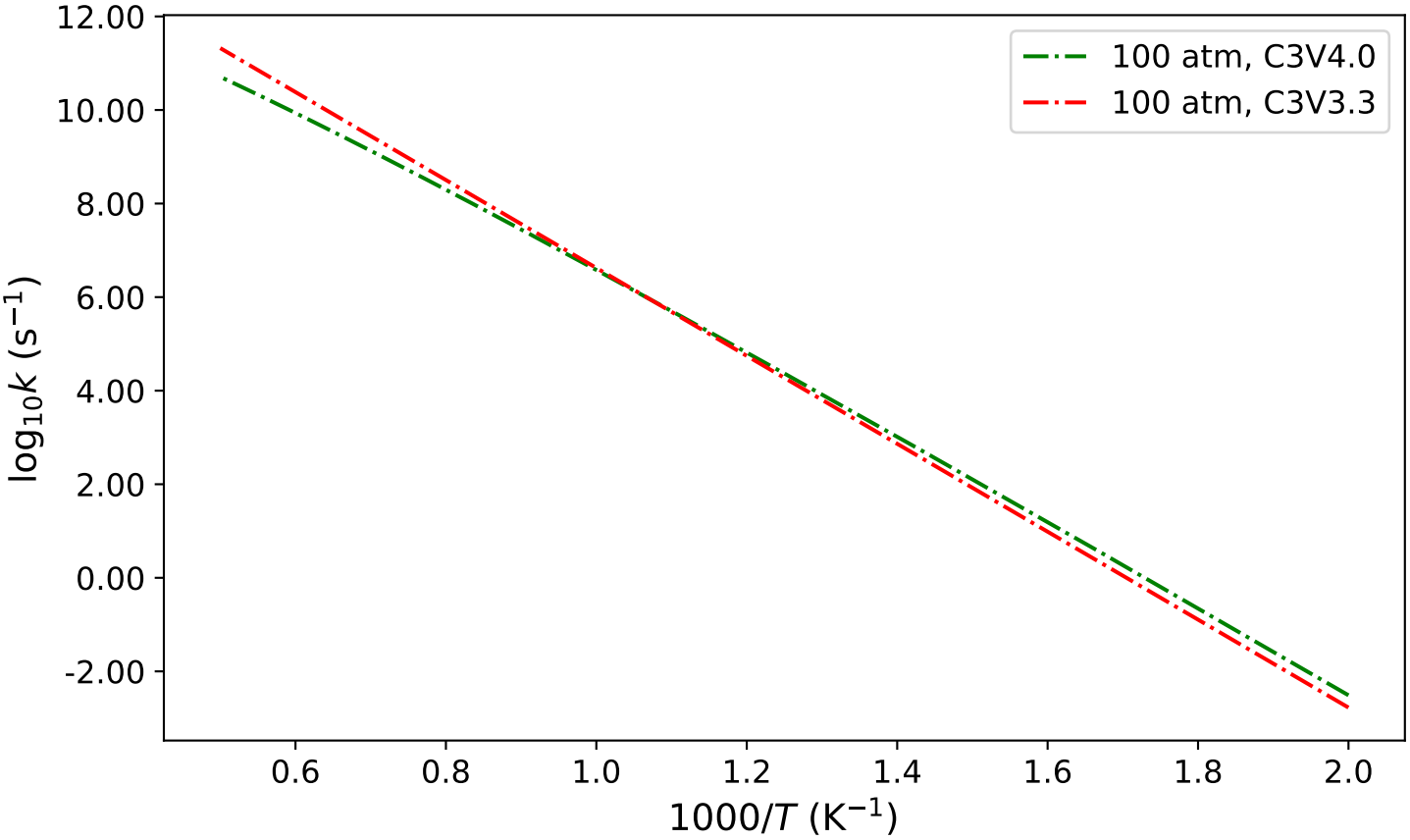



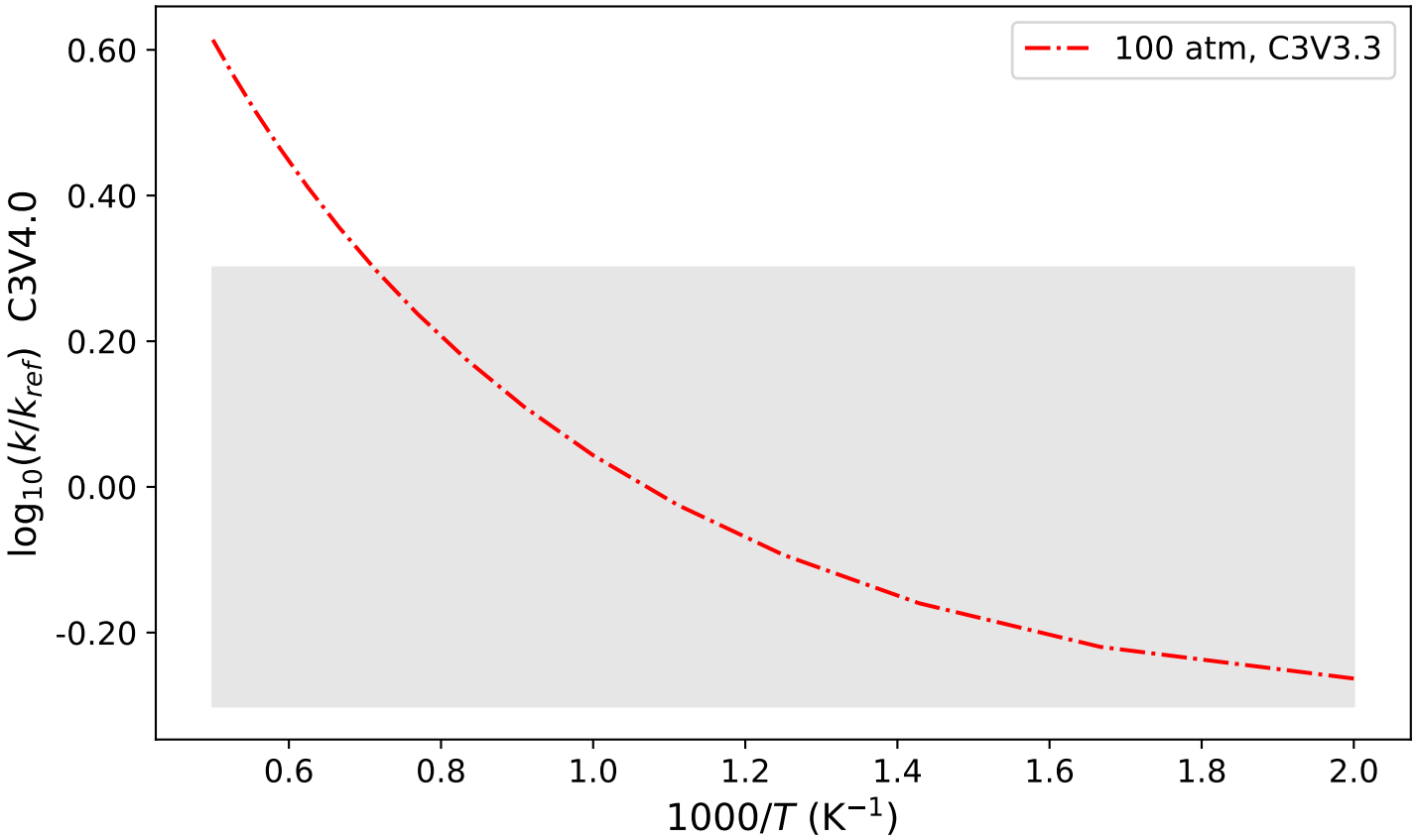
HO2 + C3H4-A = OH + CH3COCH



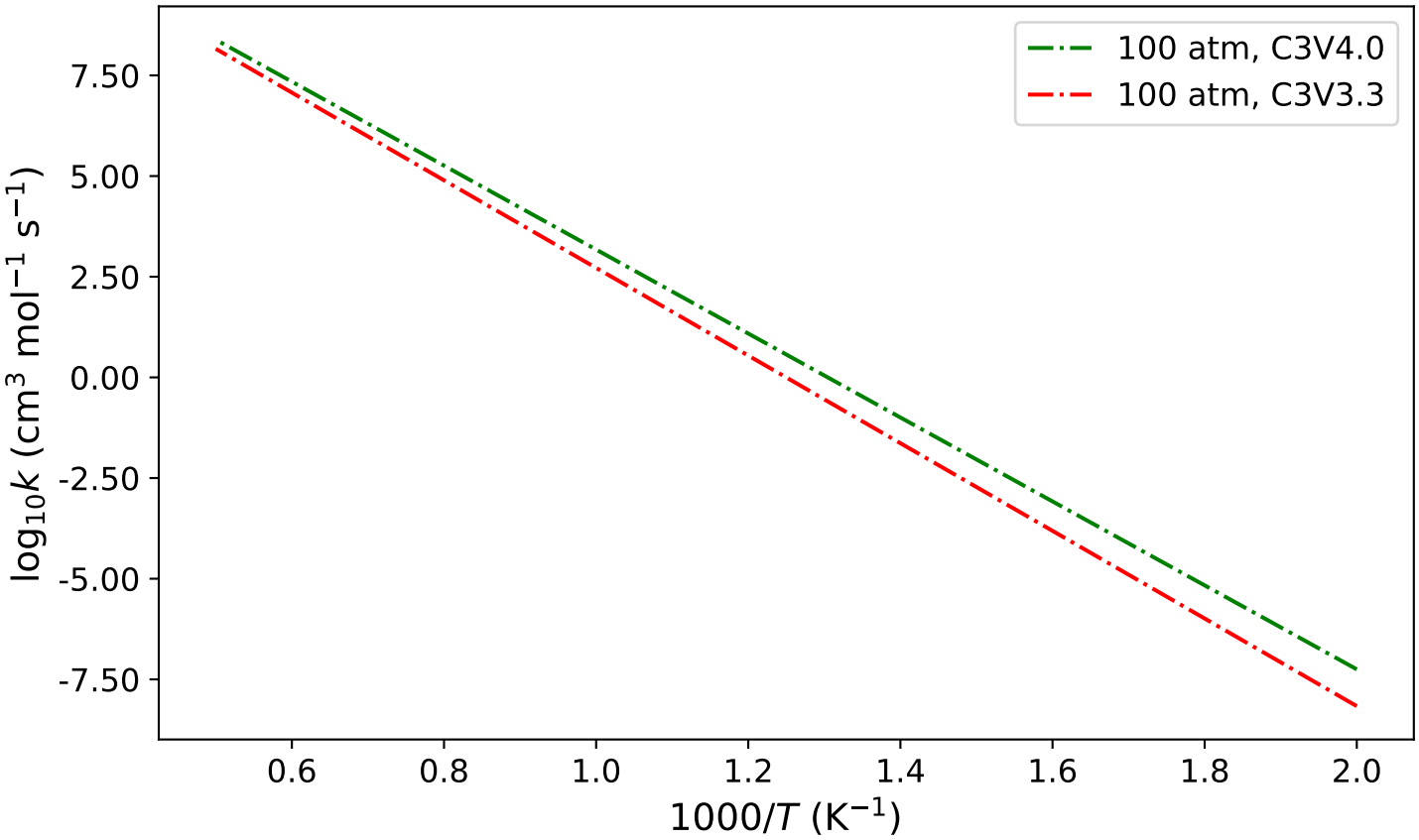



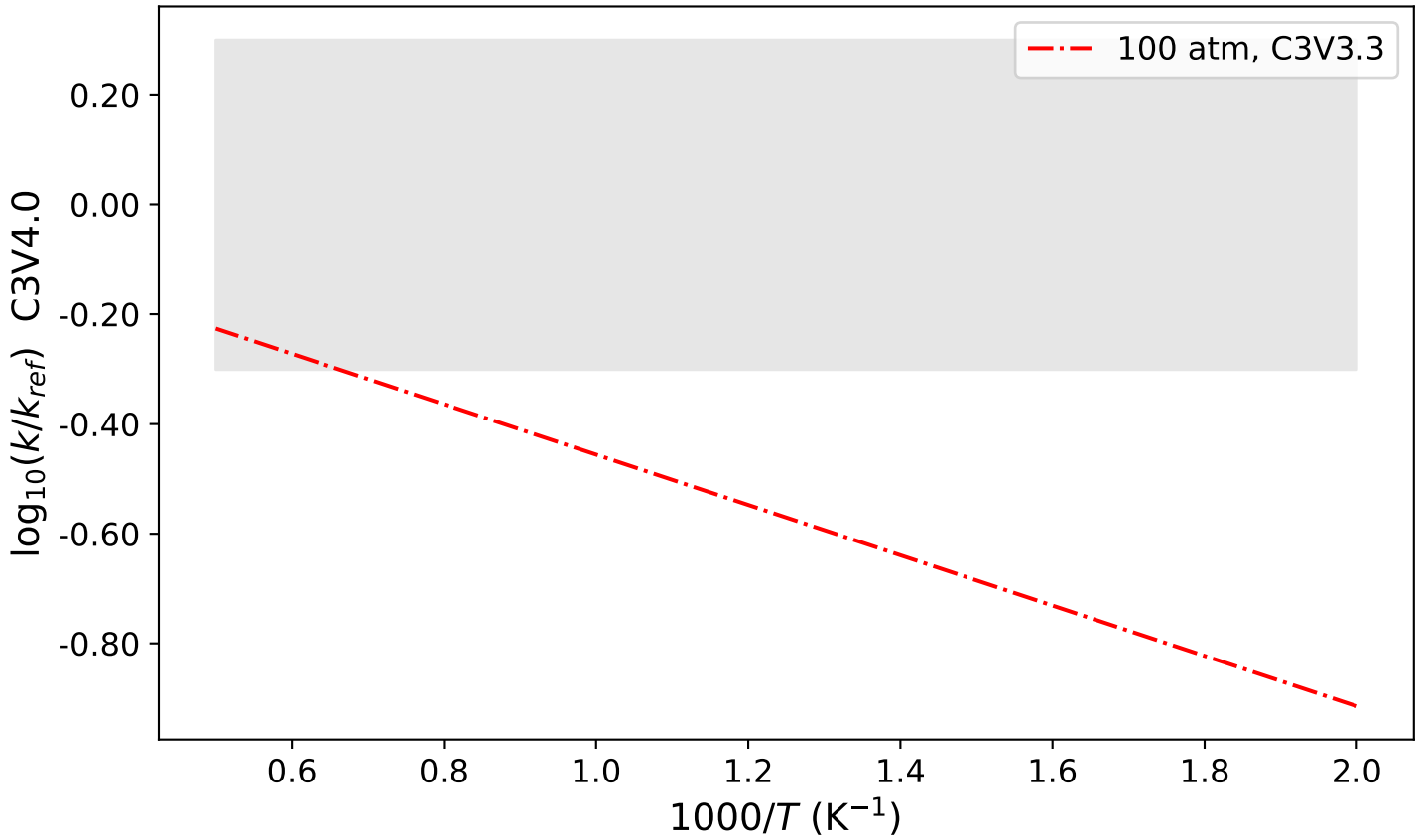
NC4KET13 => CH3CHO + OH + CH2CHO



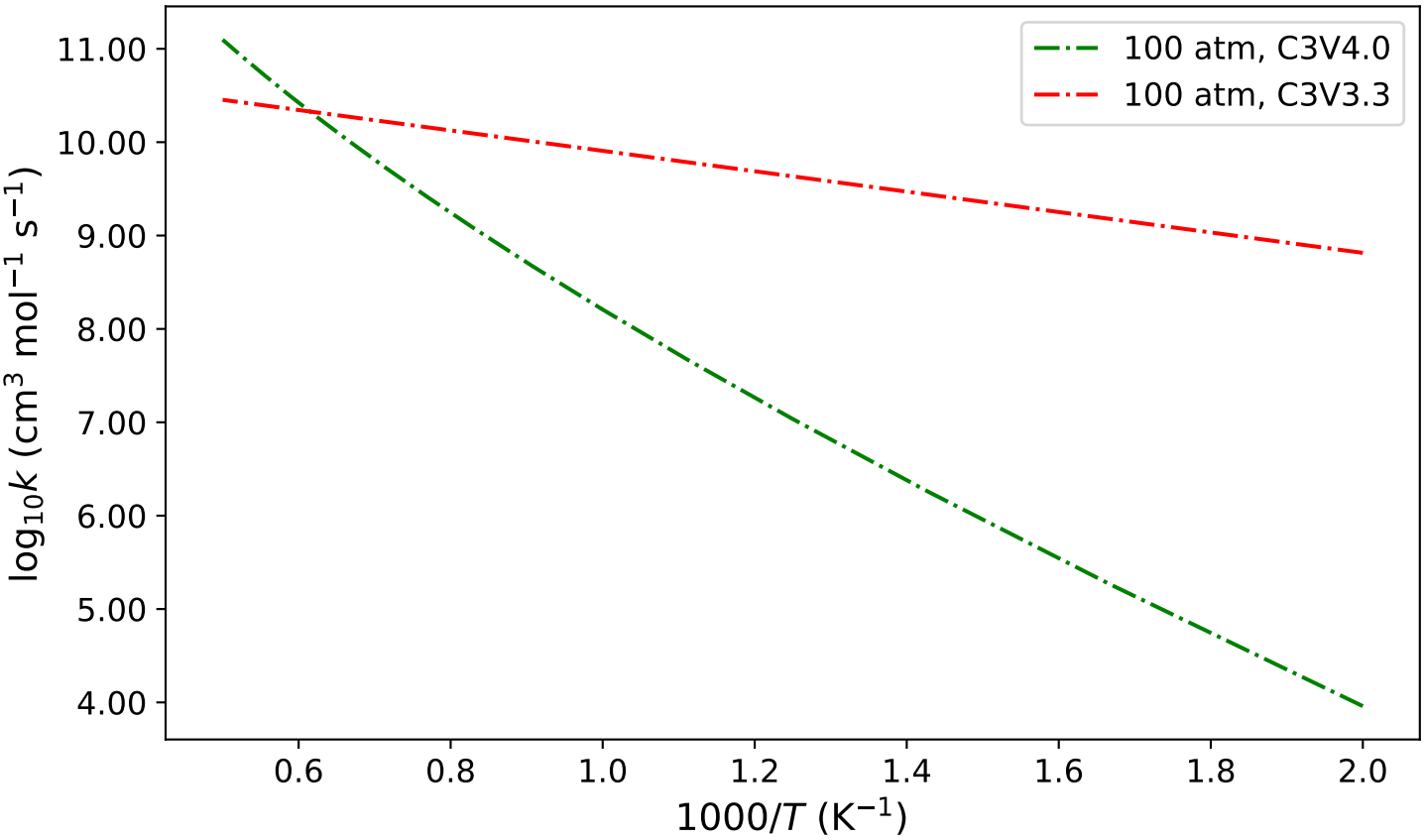



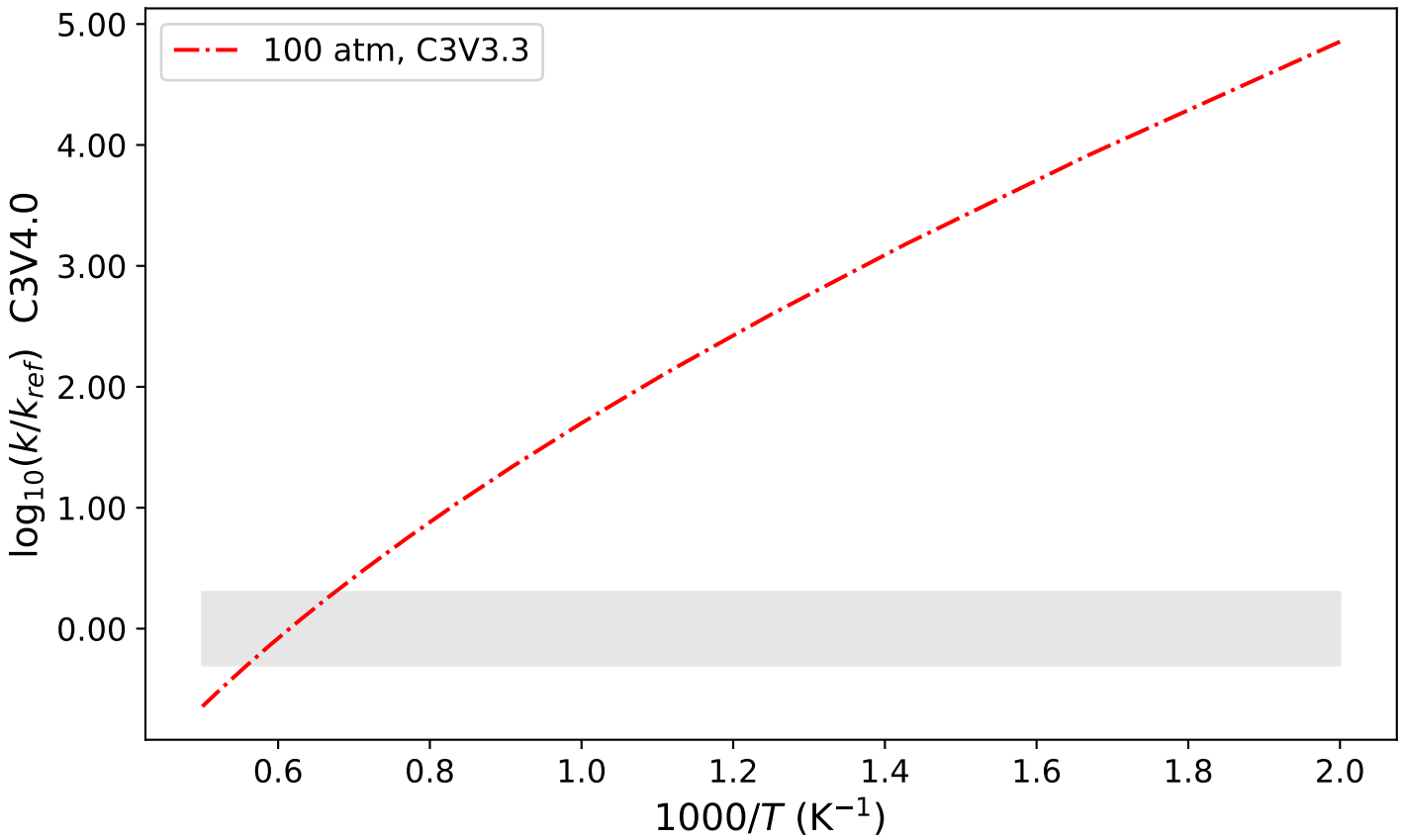
O2 + C4H10 = HO2 + SC4H9



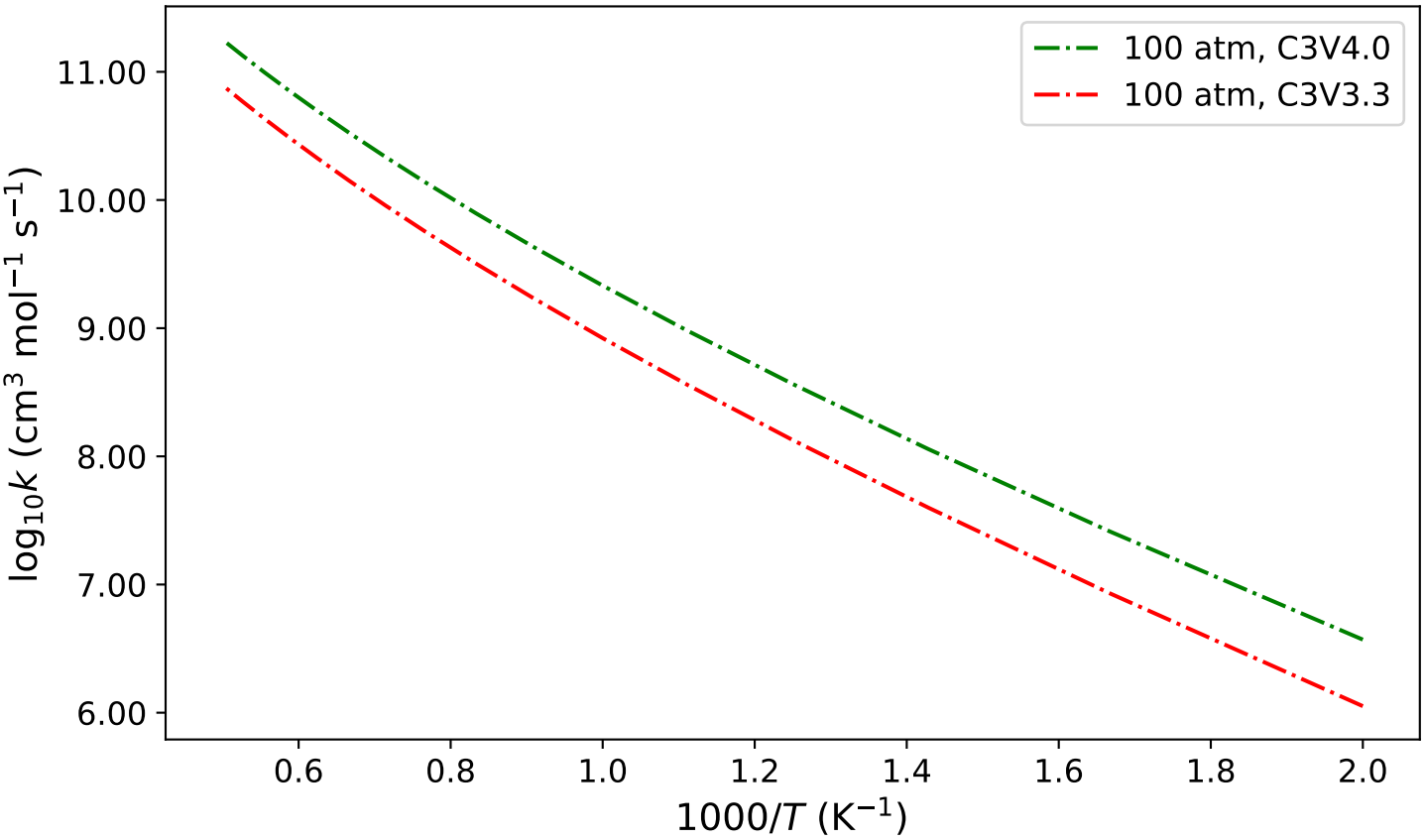



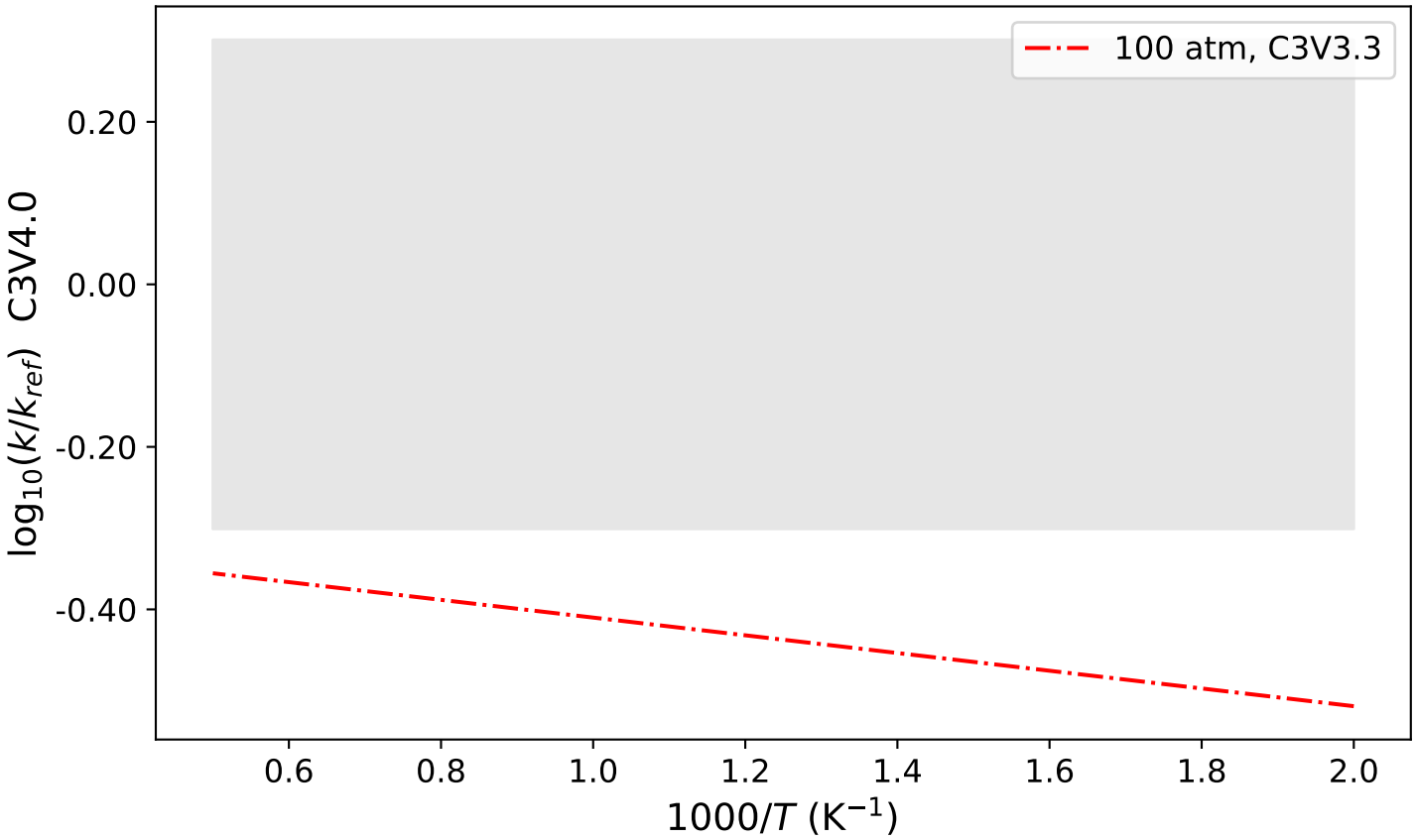
CH3COCH3 + CH3COCH2O2 = C3KET21 + CH3COCH2



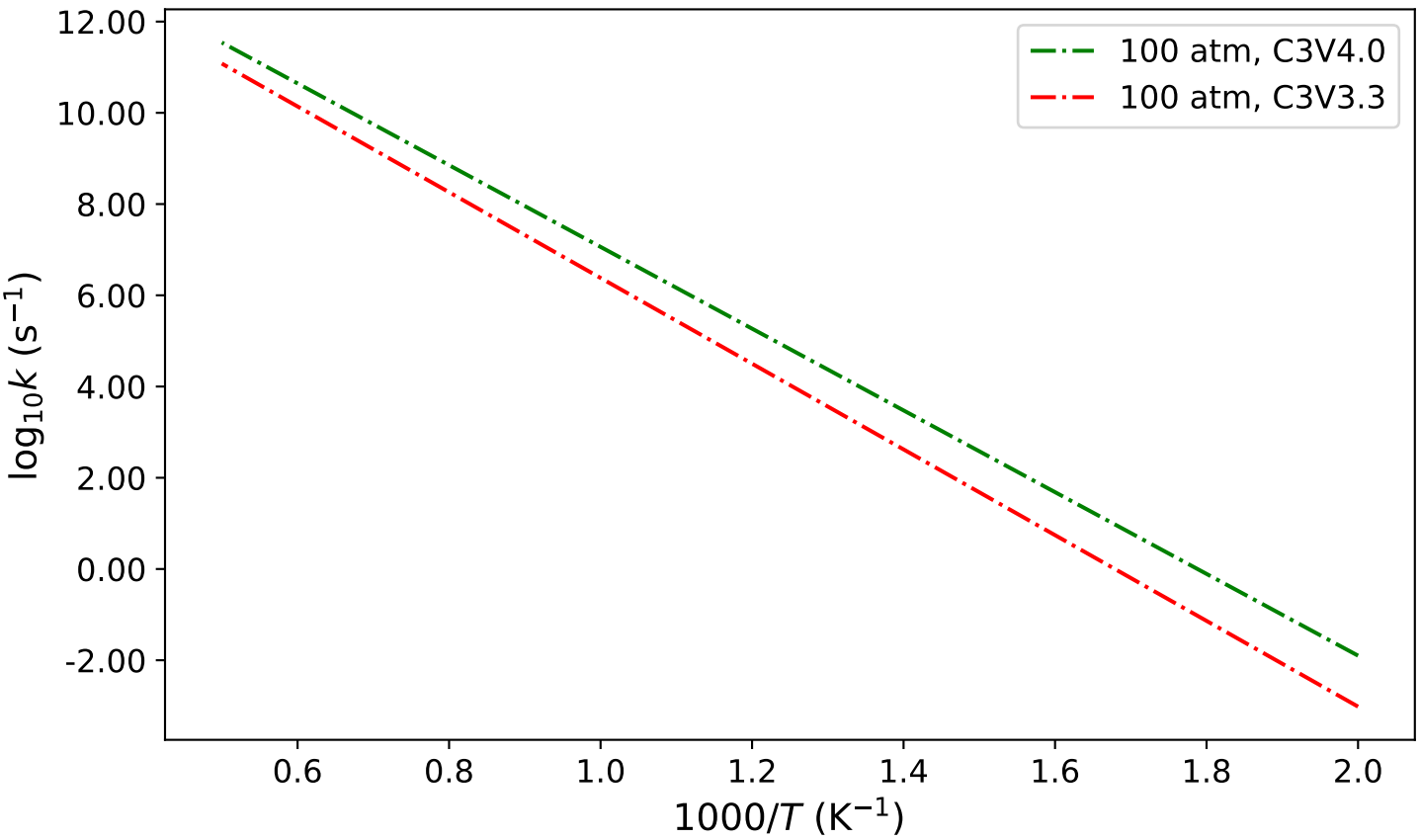



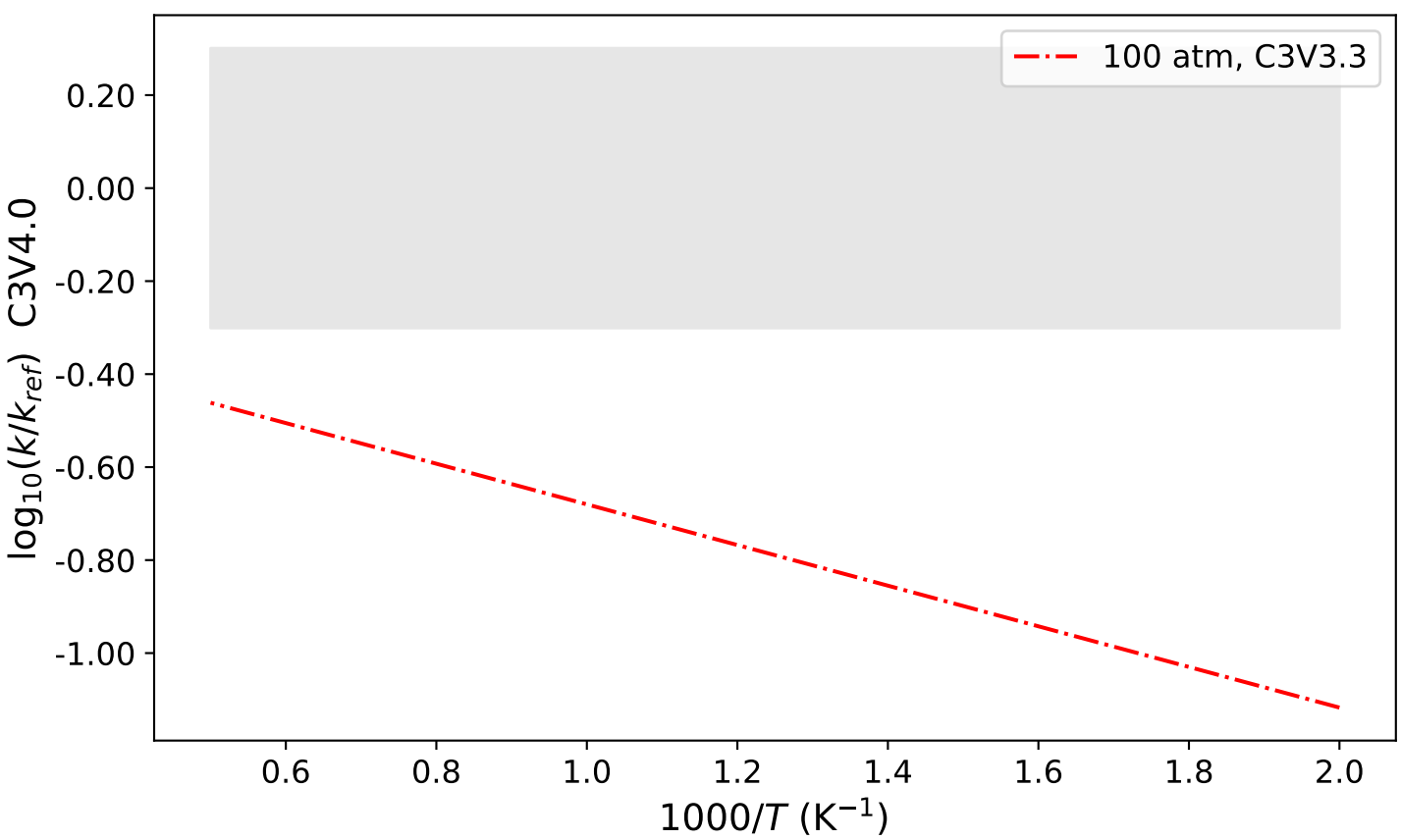
CH3 + IC3H7OH = IC3H7O + CH4



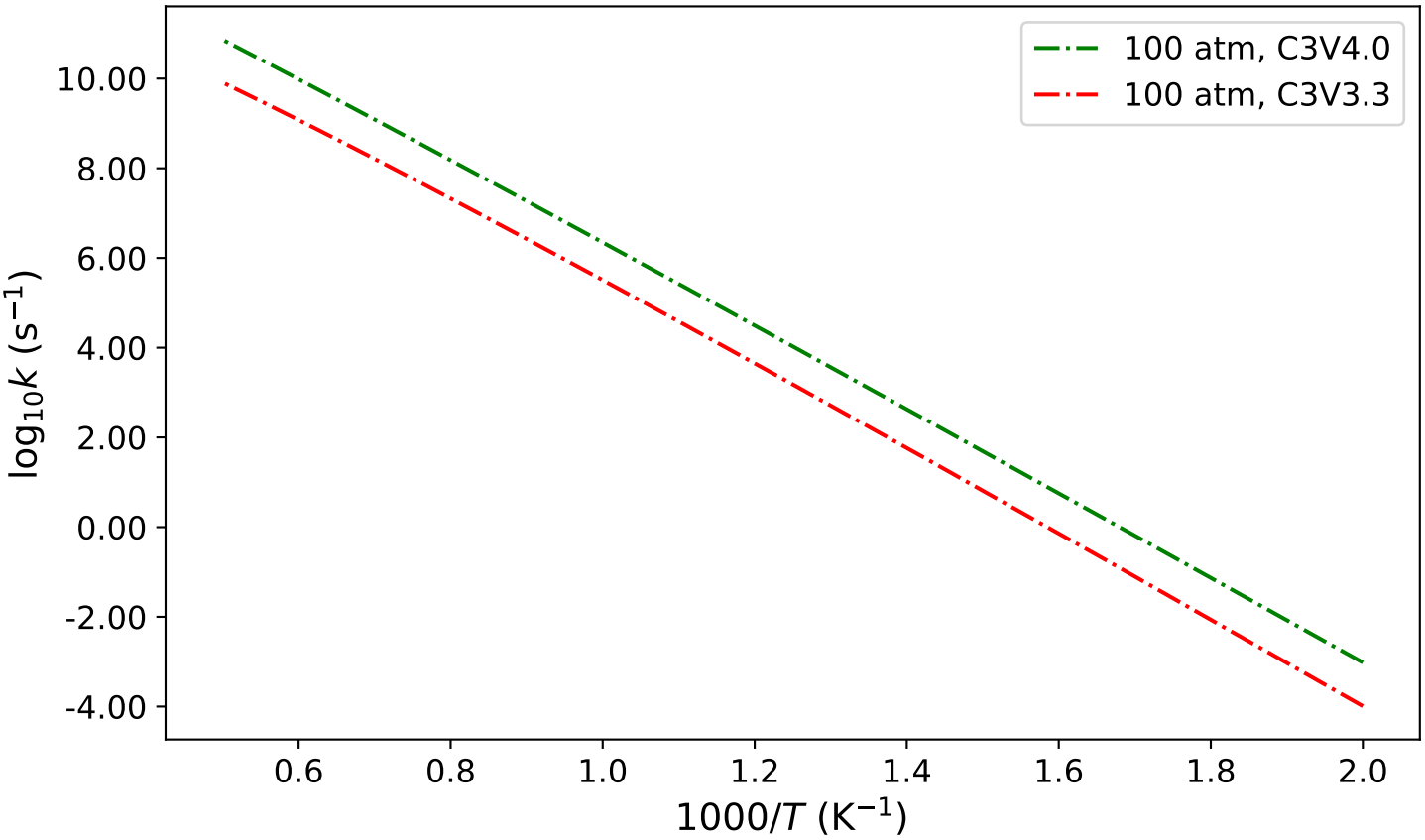



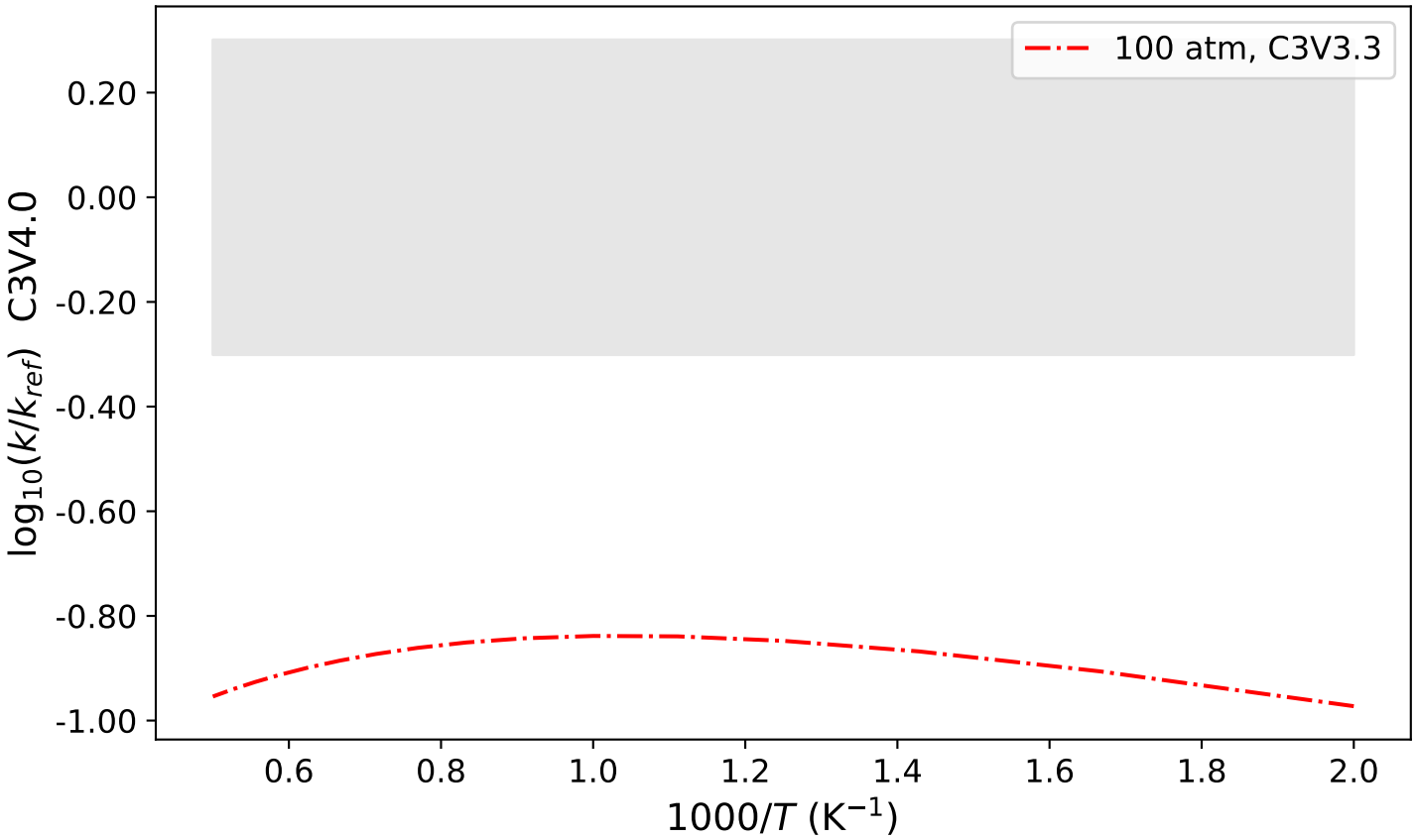
IC4KETIT => OH + CH3COCH3 + HCO



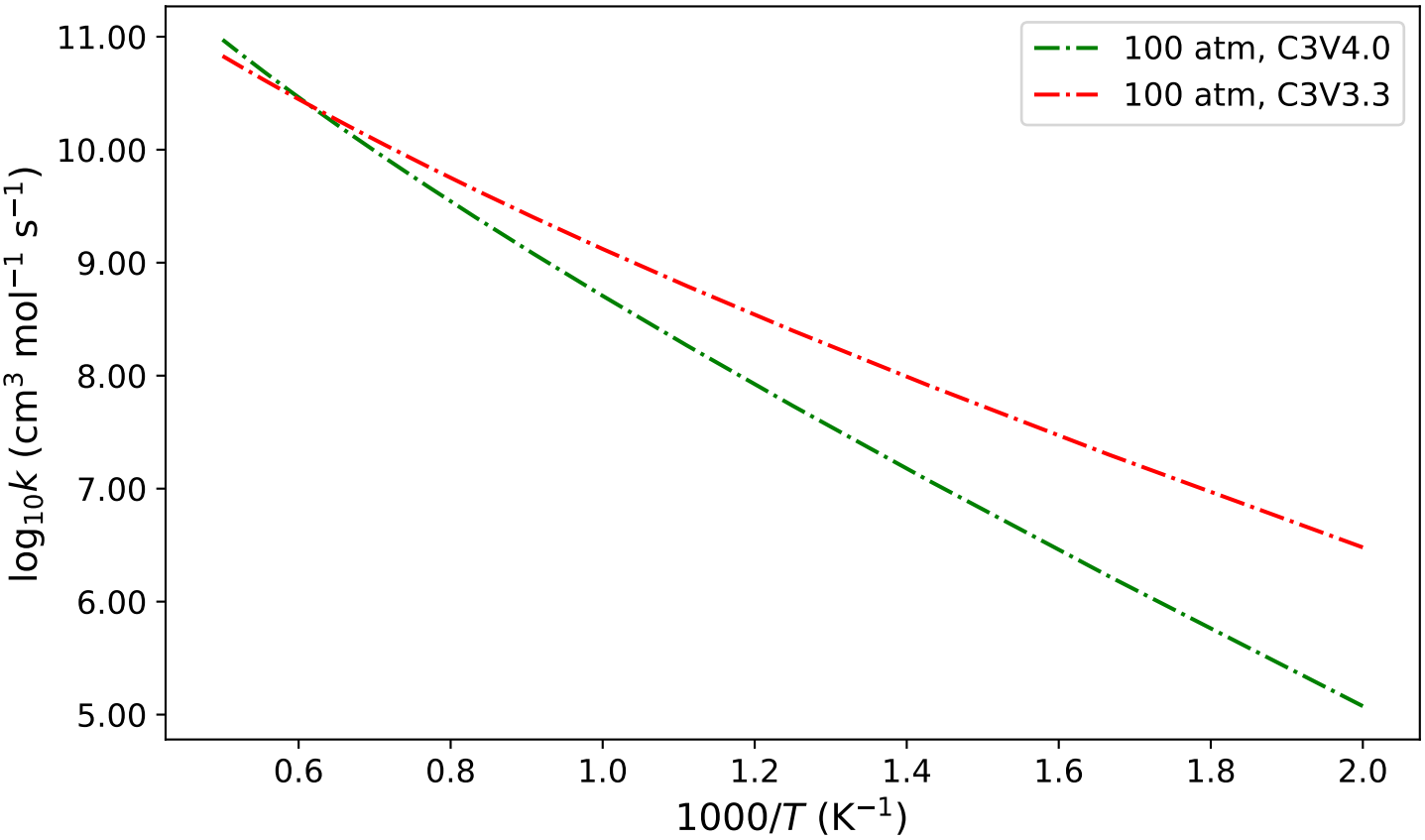



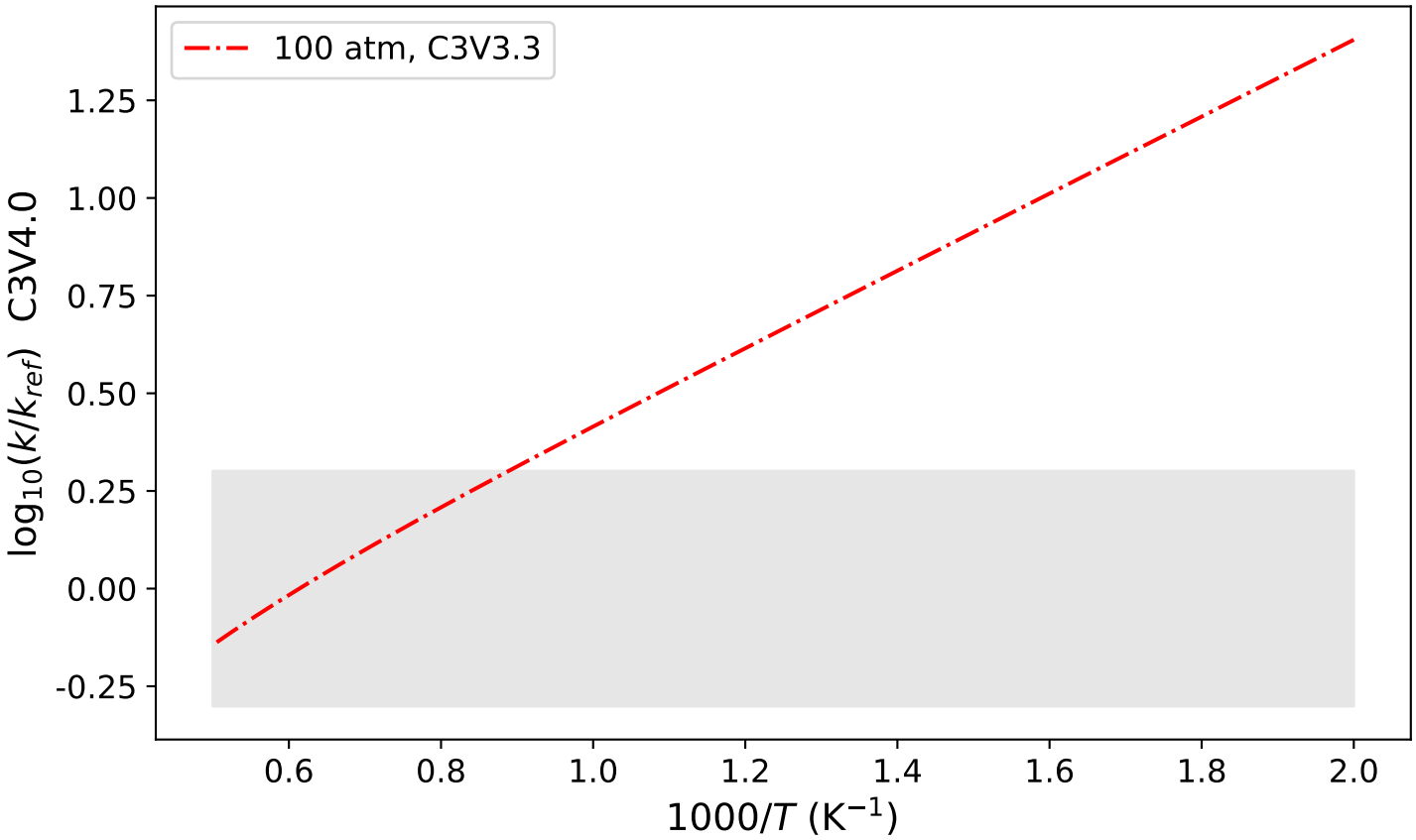
IC3H7O2H = IC3H7O + OH



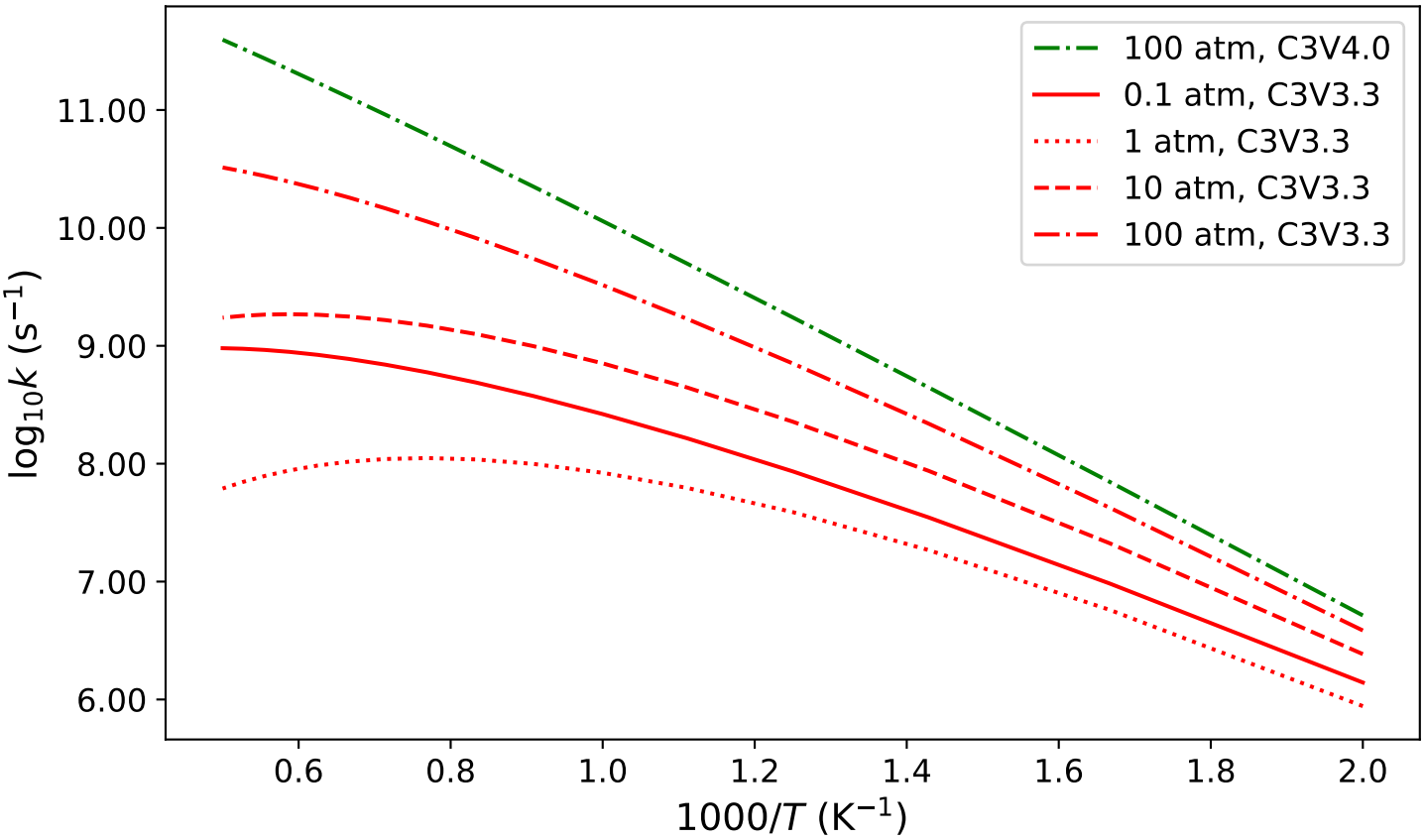



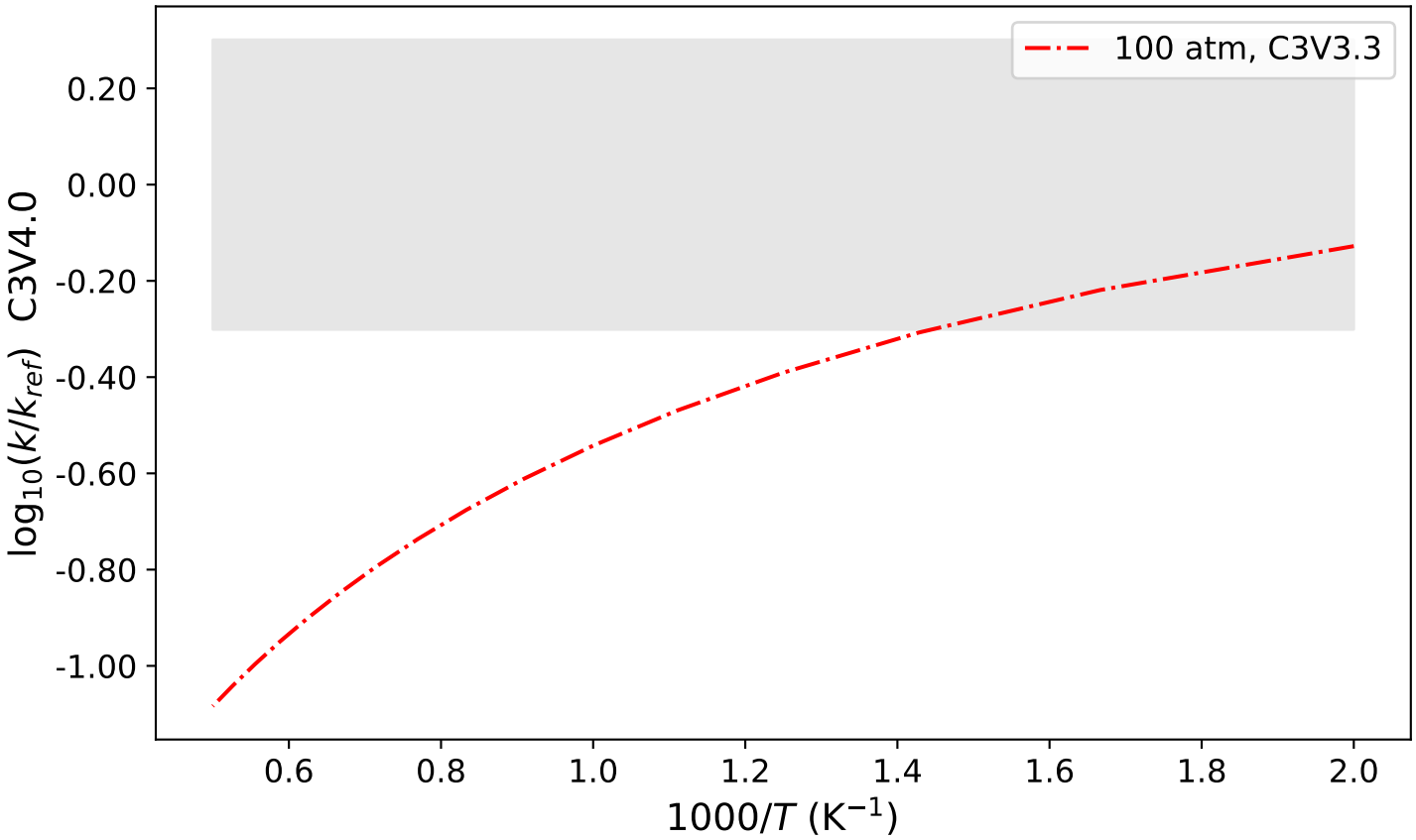
HO2 + C4H8-1 = C4H8OOH2-1



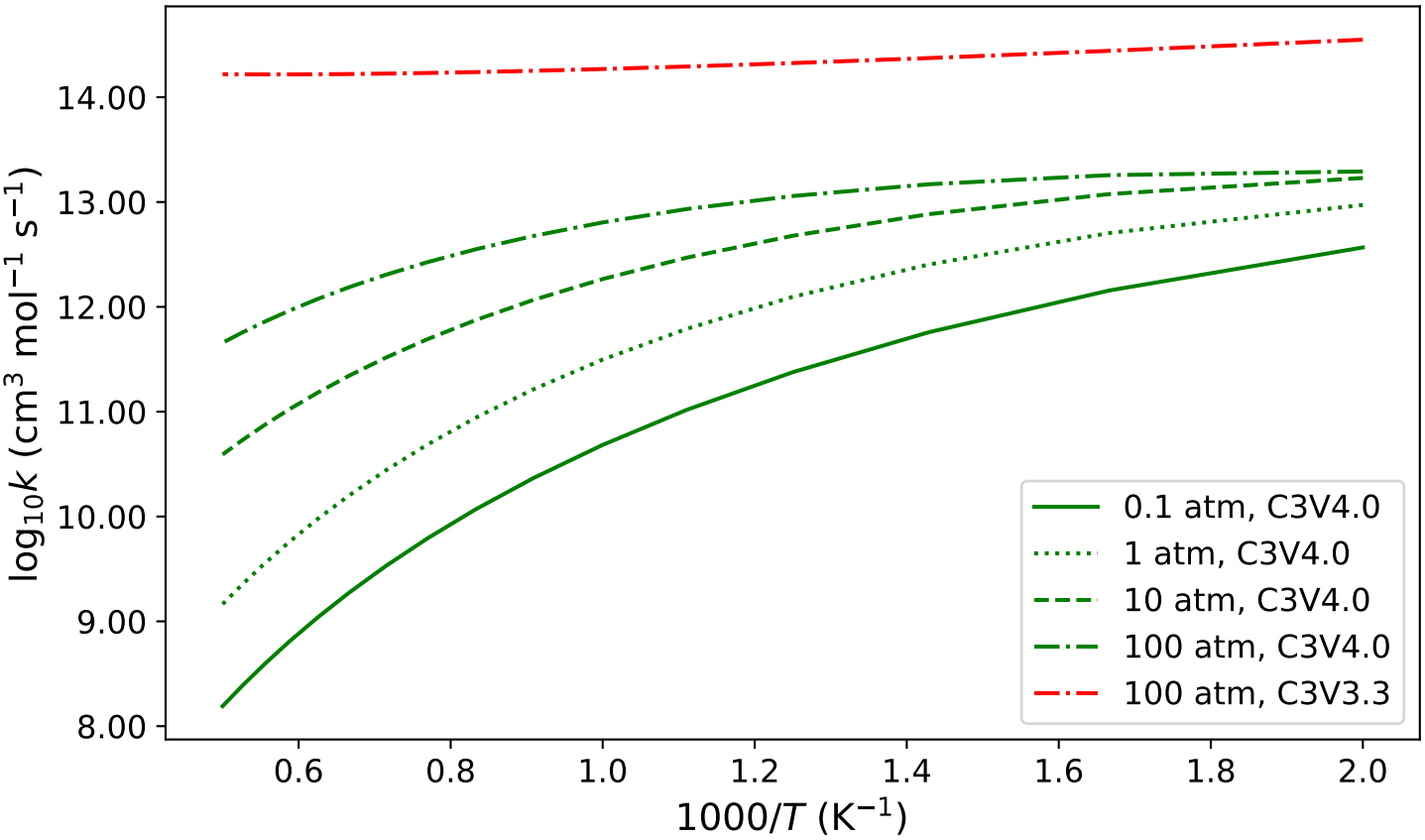



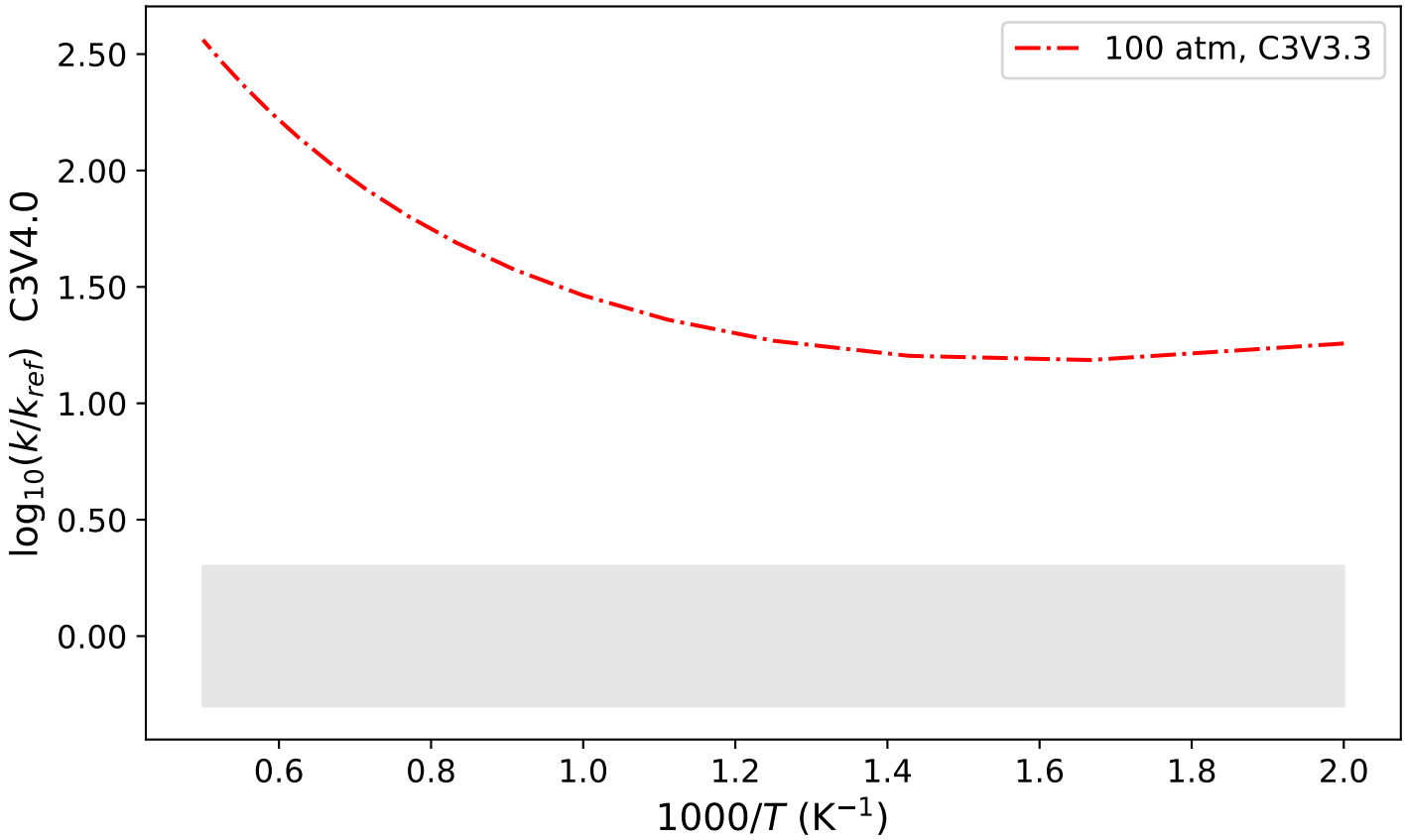
C4H71-O = CH3 + C2H3CHO



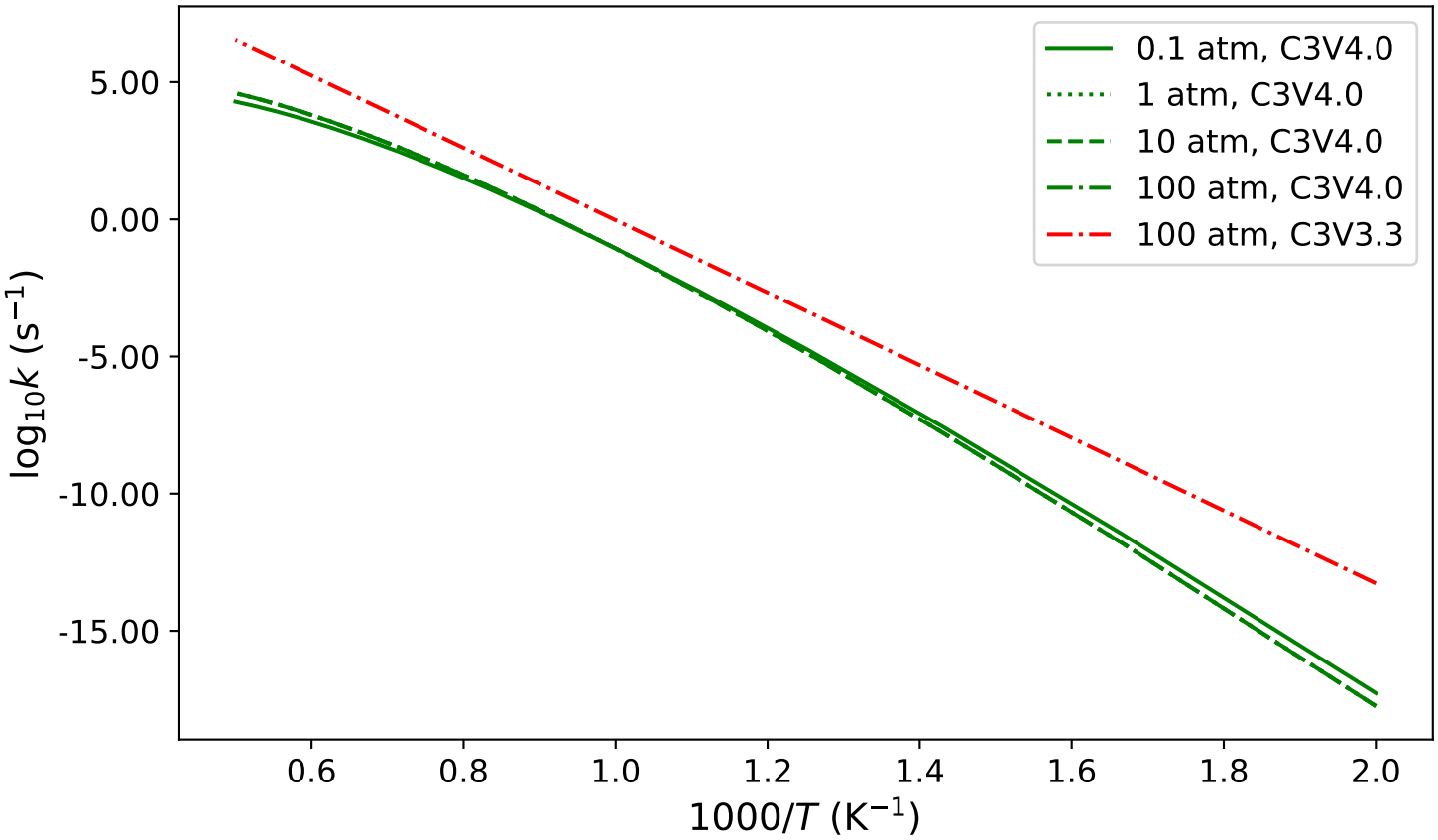



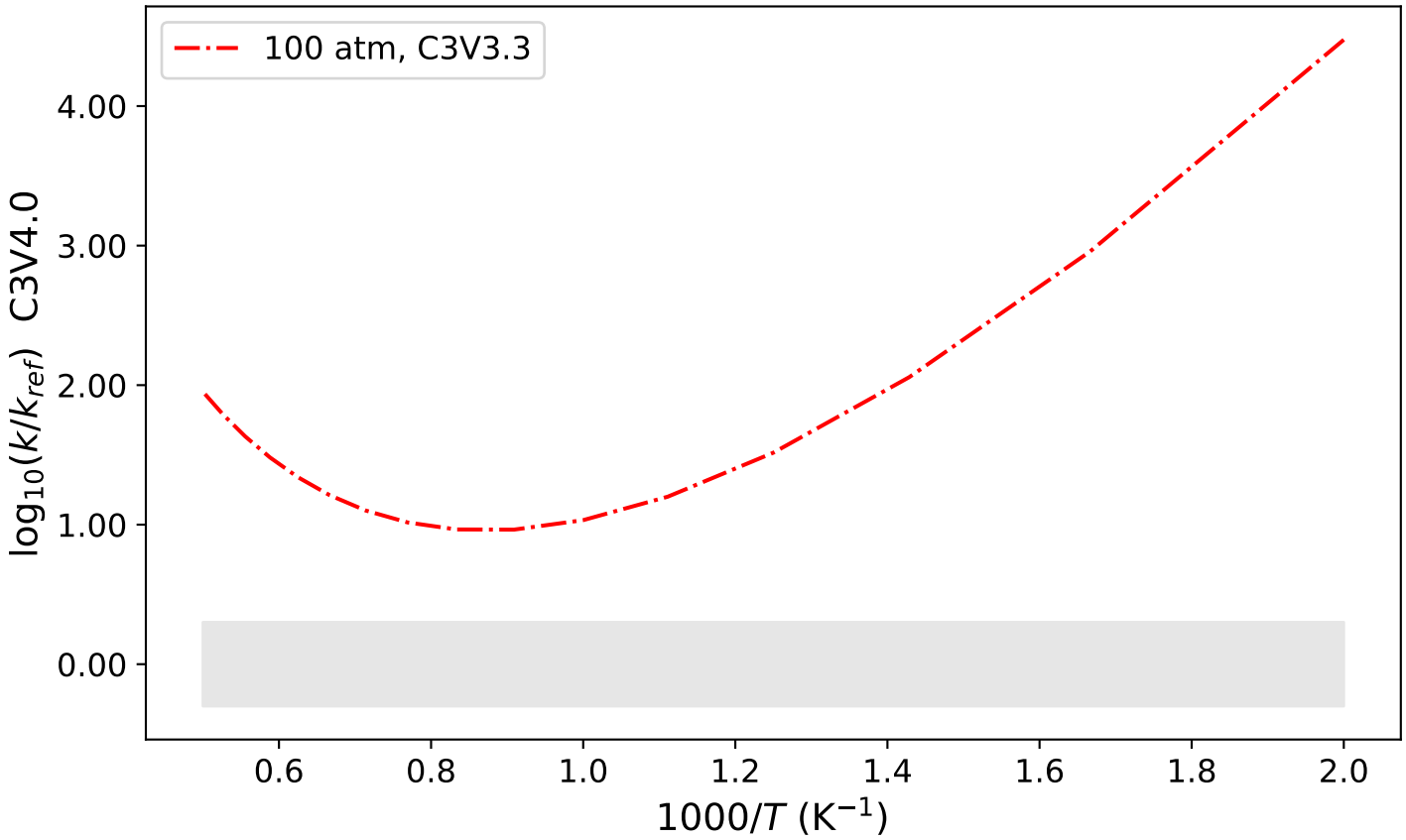
HO2 + C4H64,2-1OH = C4H6OHOOH1-4-3



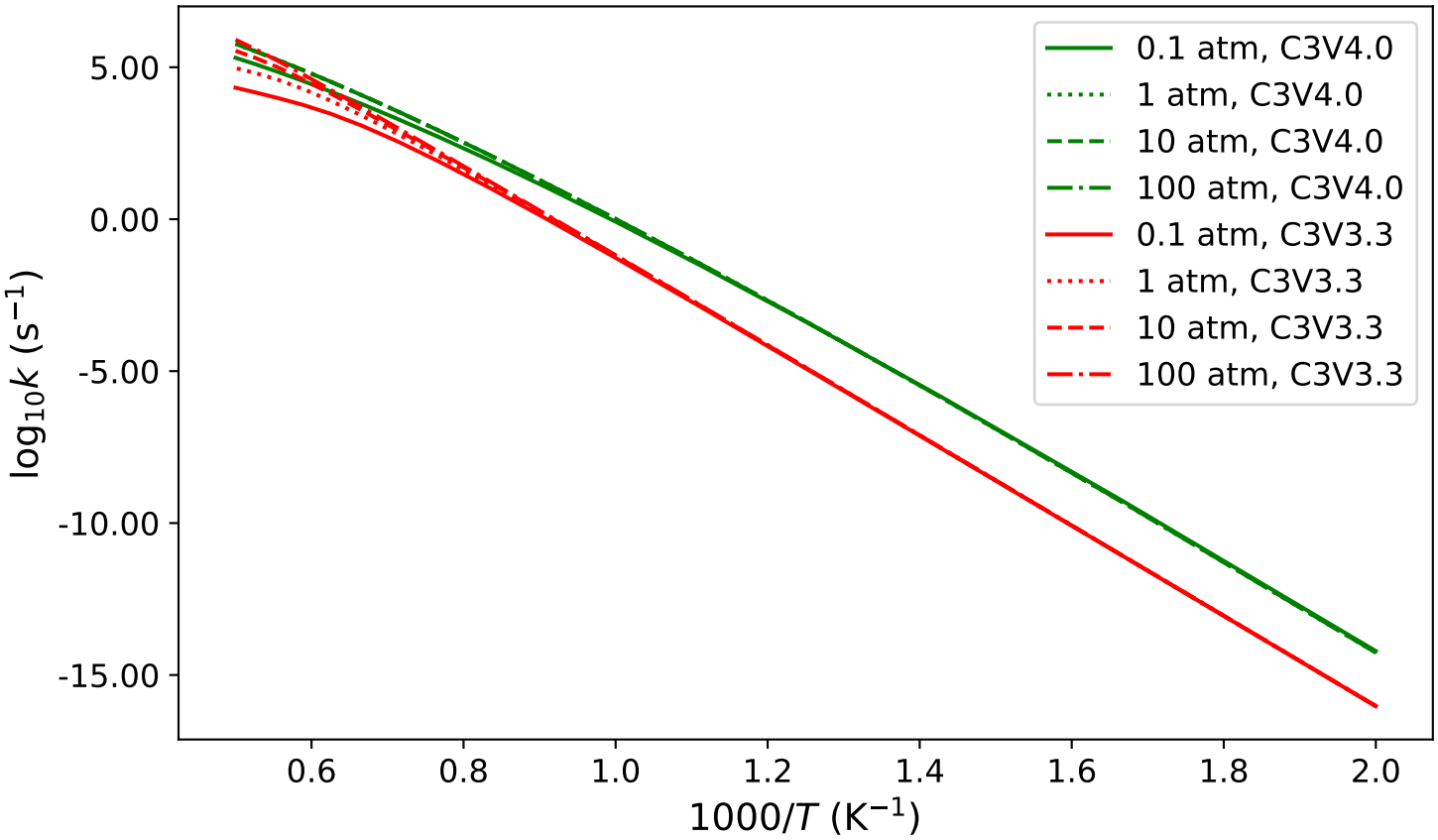



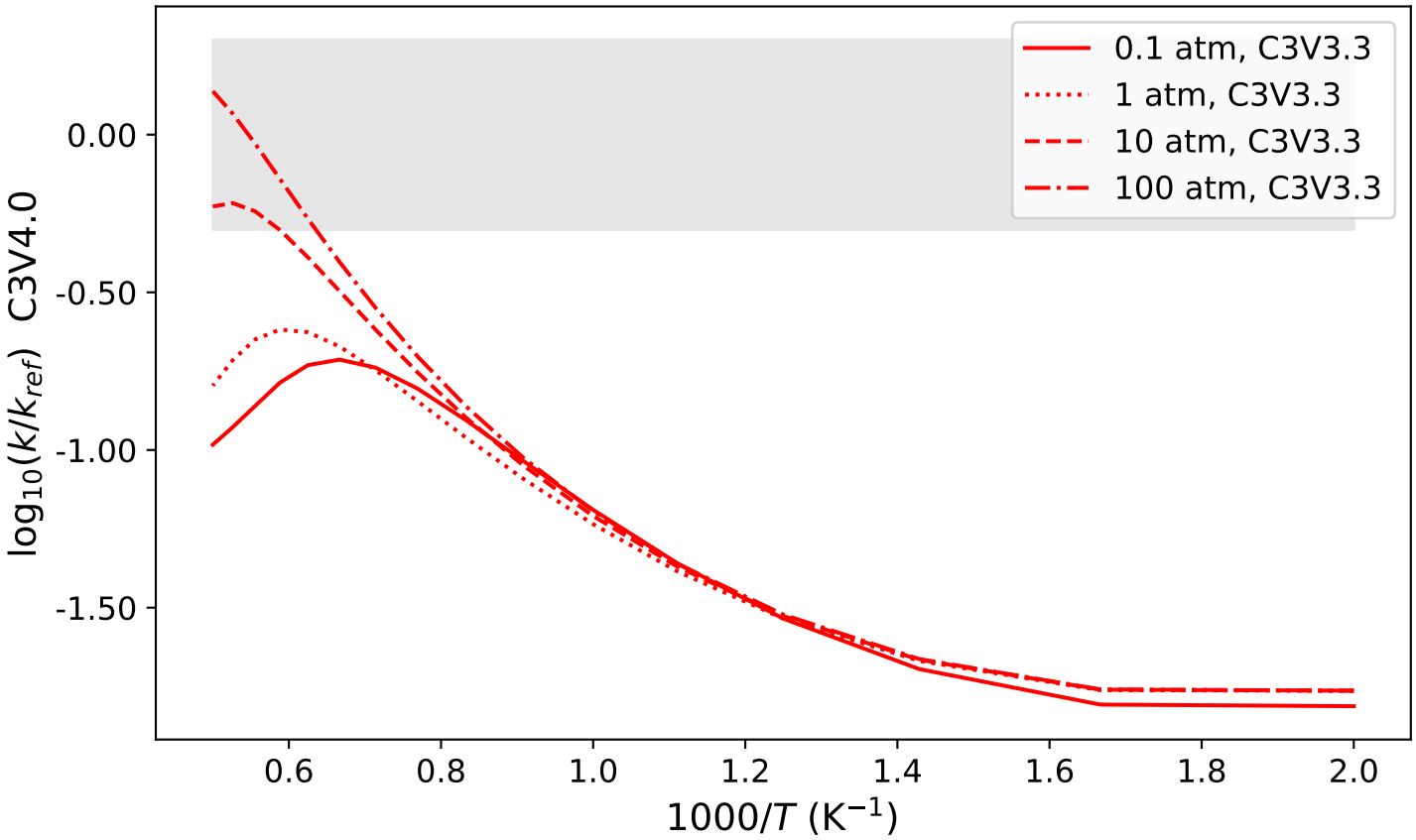
C4H612 = C4H6-2



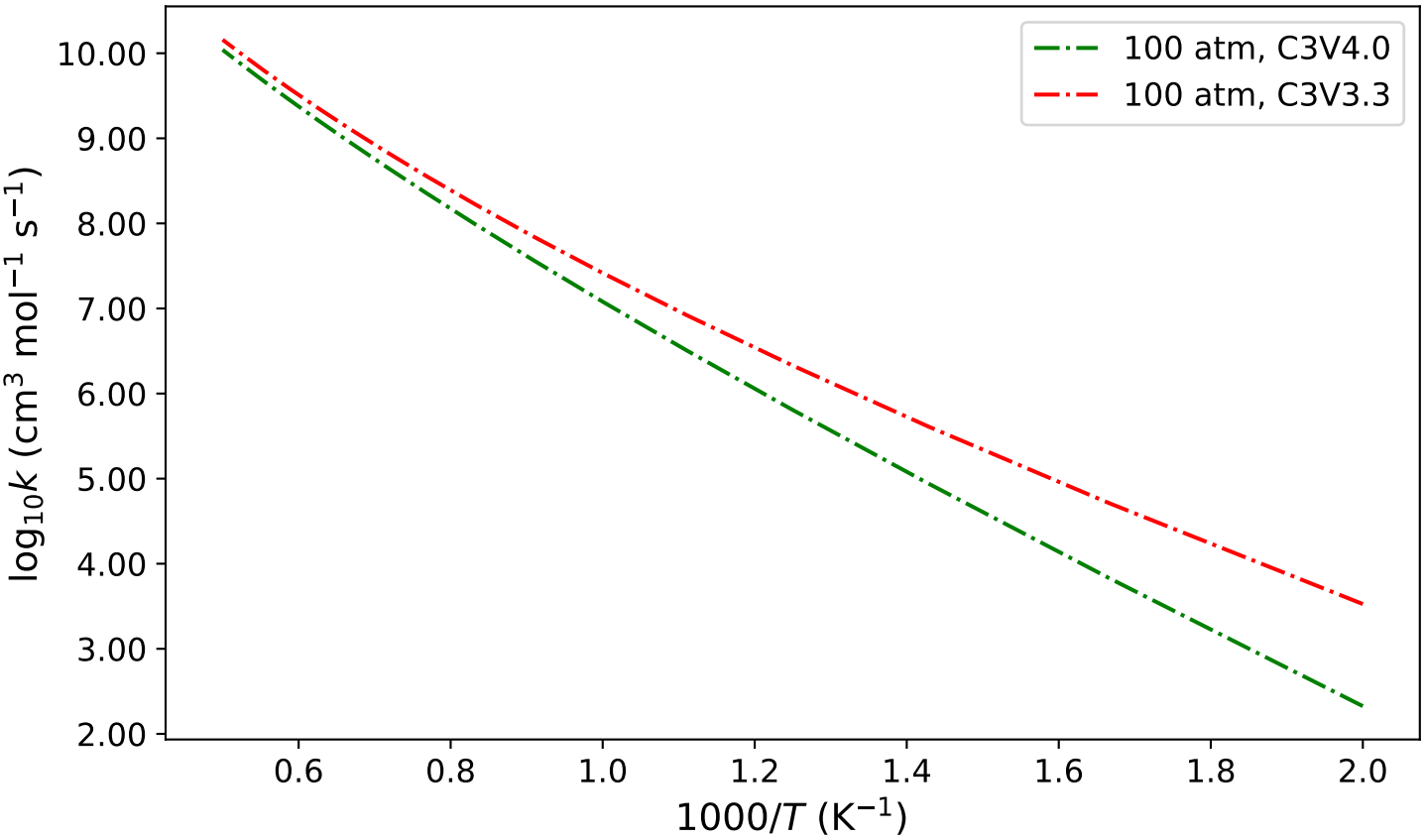



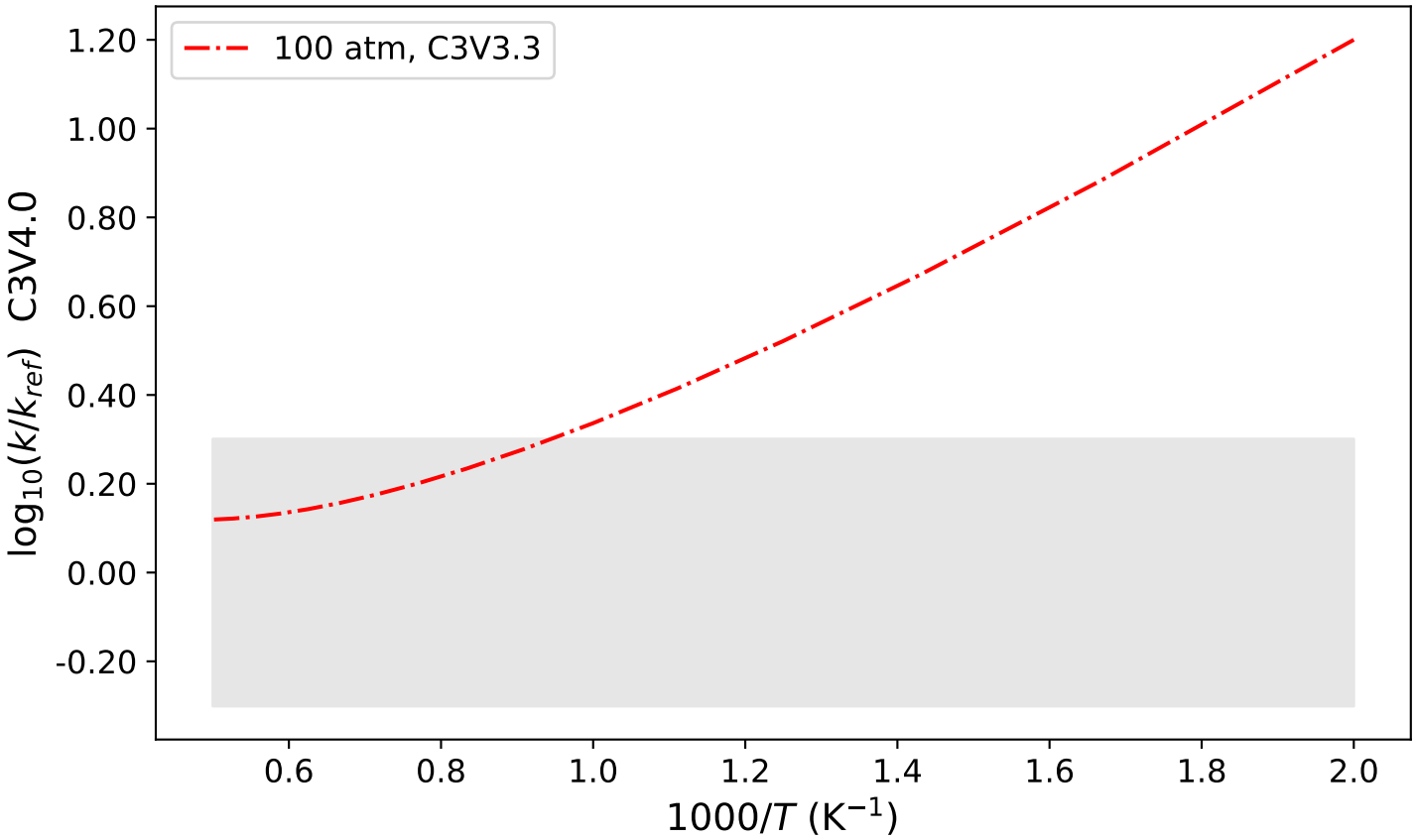
IC3H7OH = H2O + C3H6



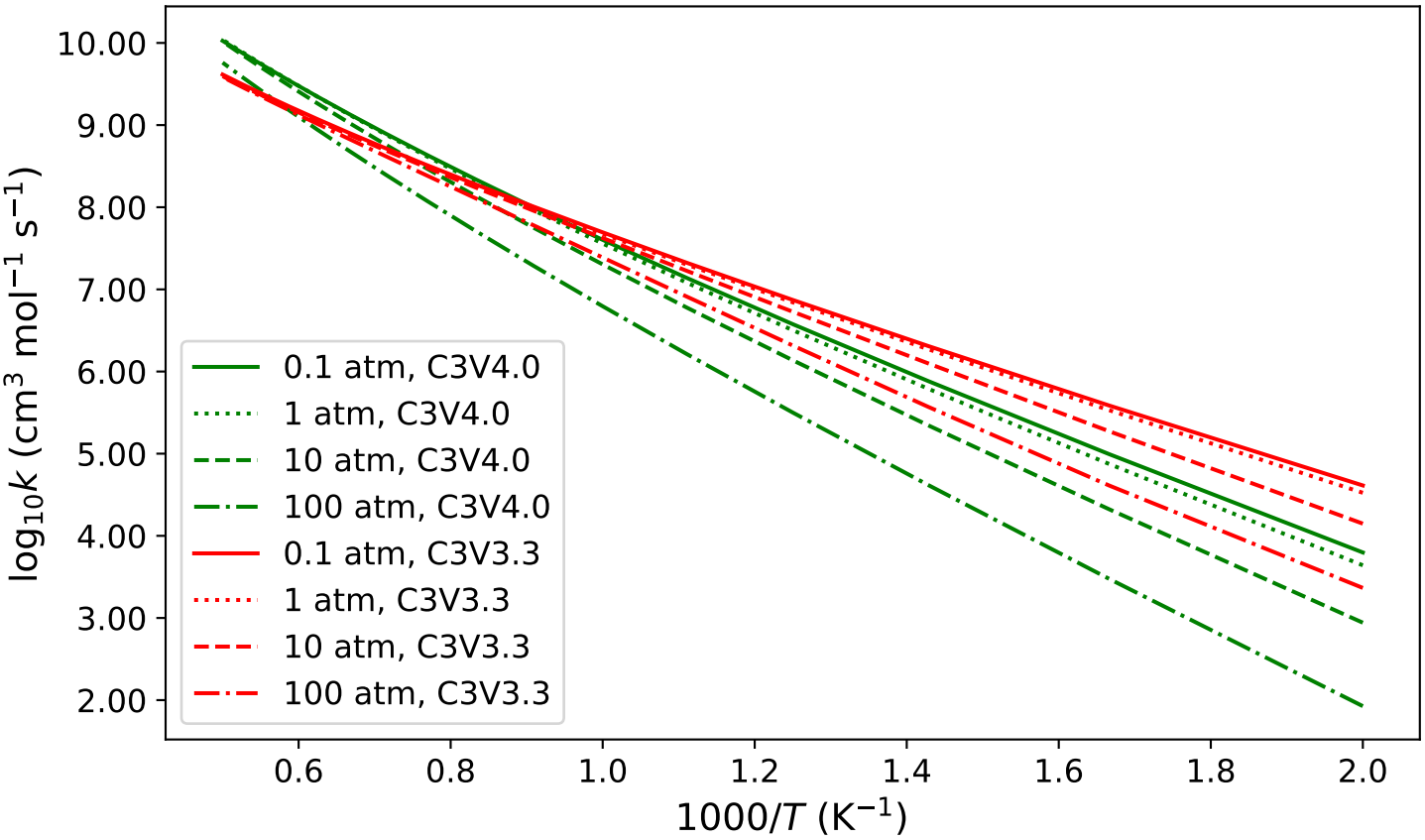



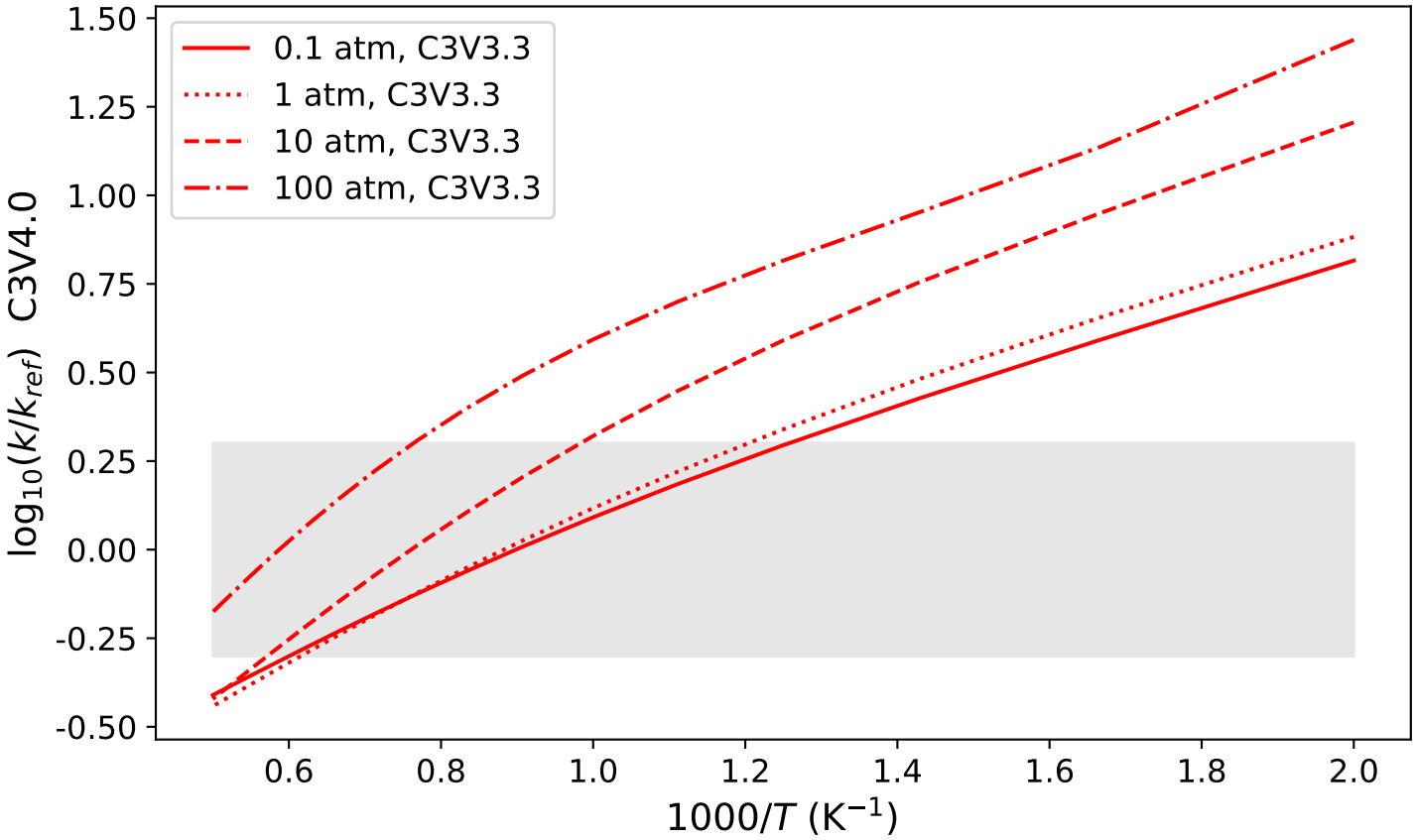
HO2 + IC3H7OH = IC3H7O + H2O2



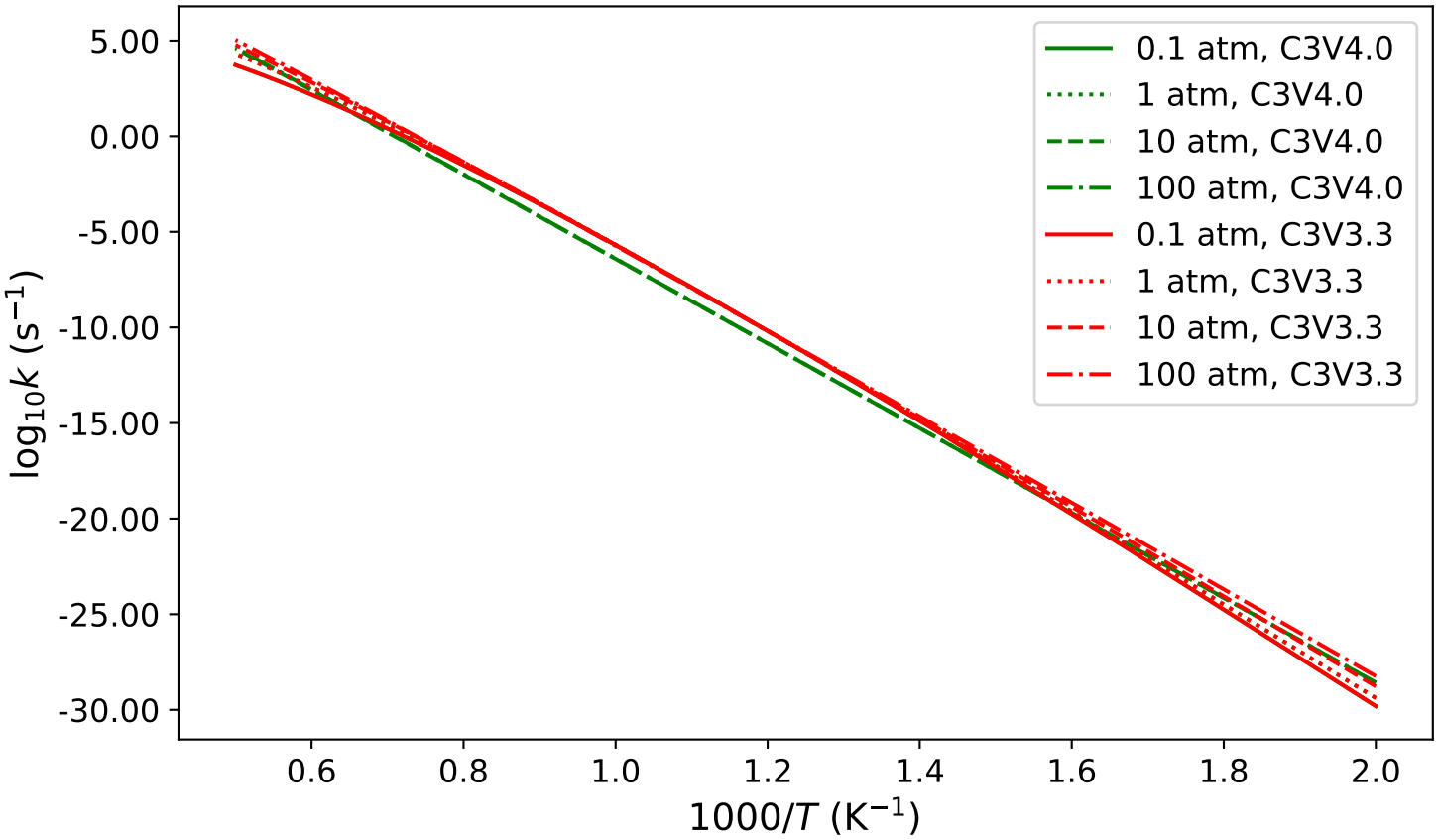



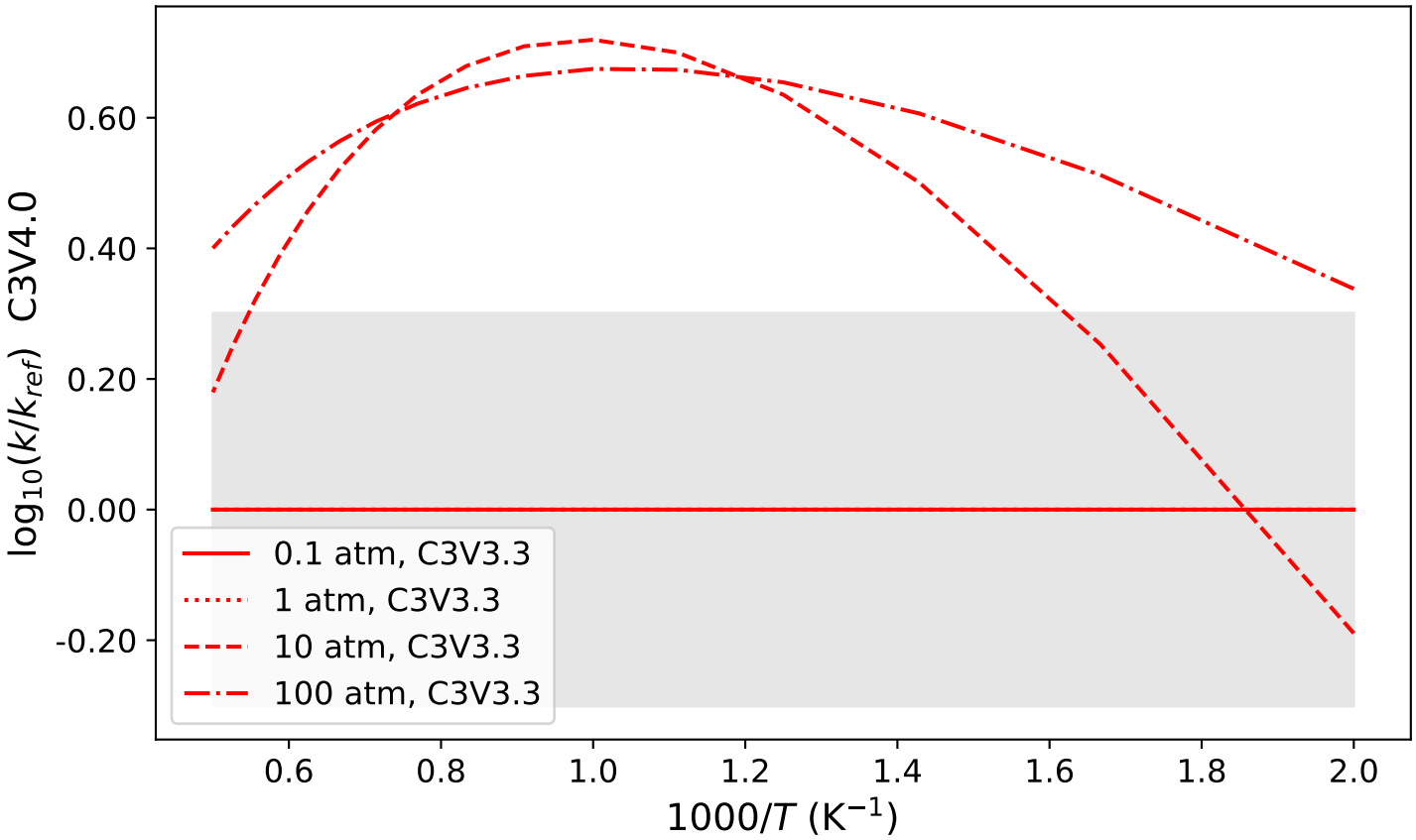
HO2 + C3H4-A = O + CH3COCH2



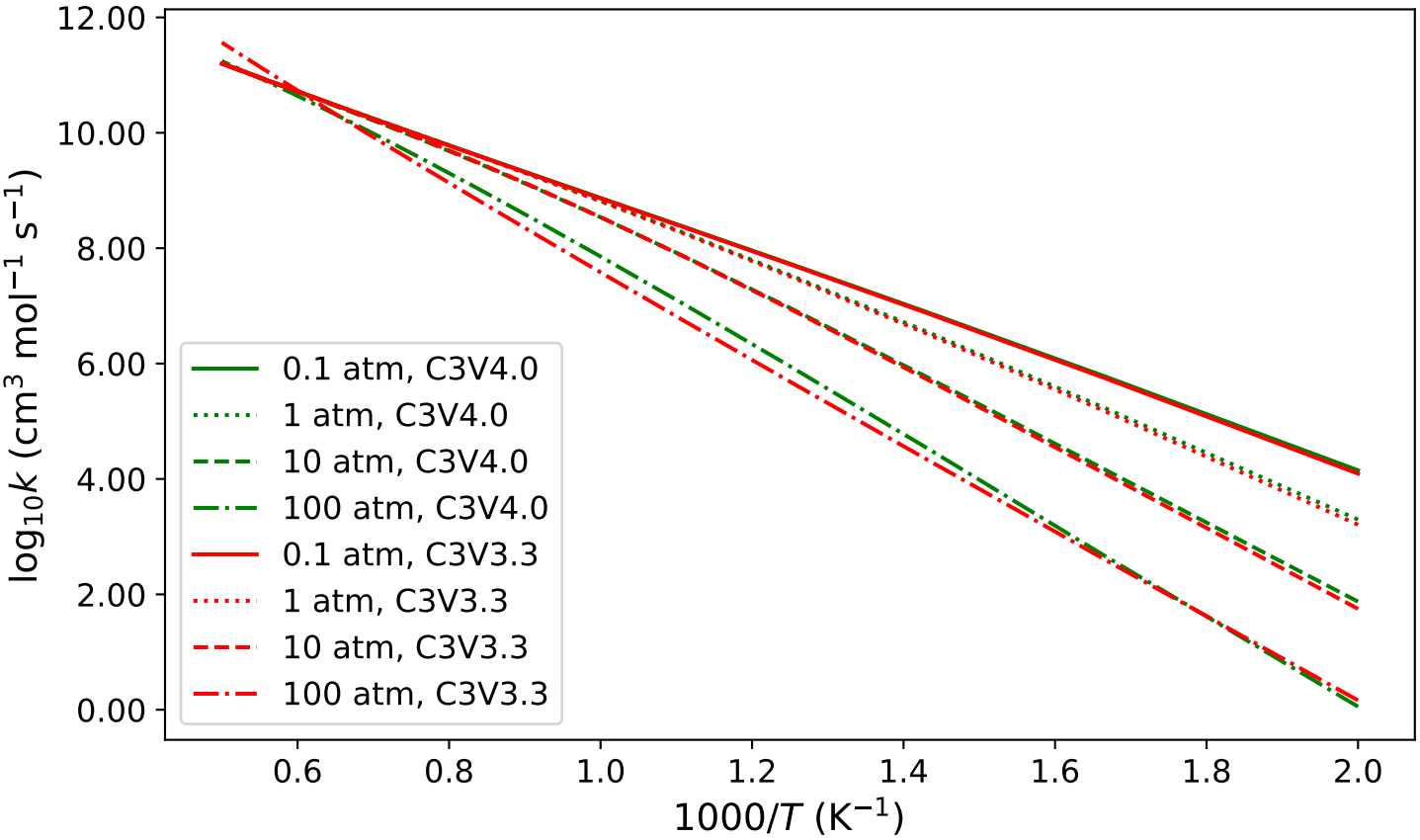



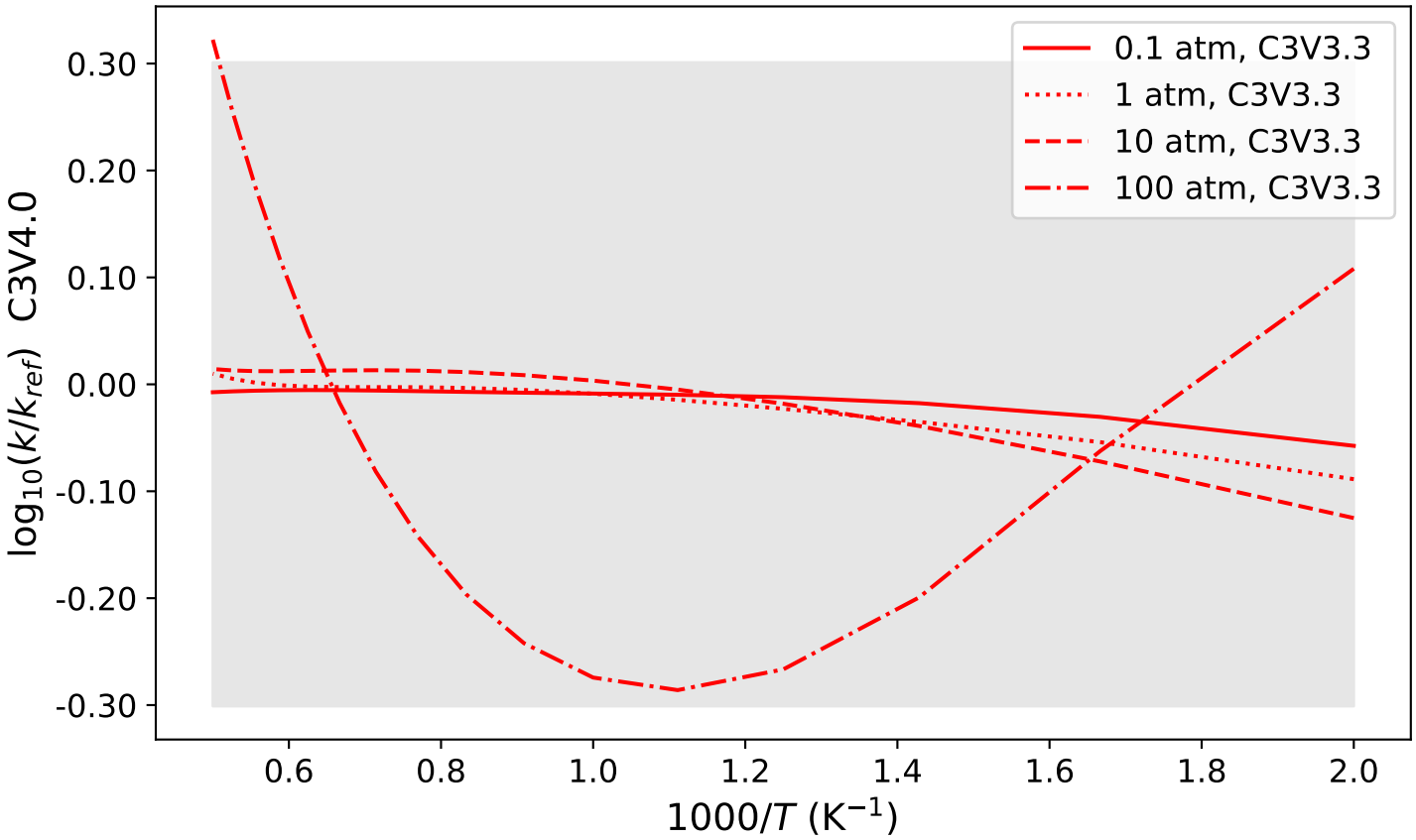
C4H10 = H + PC4H9



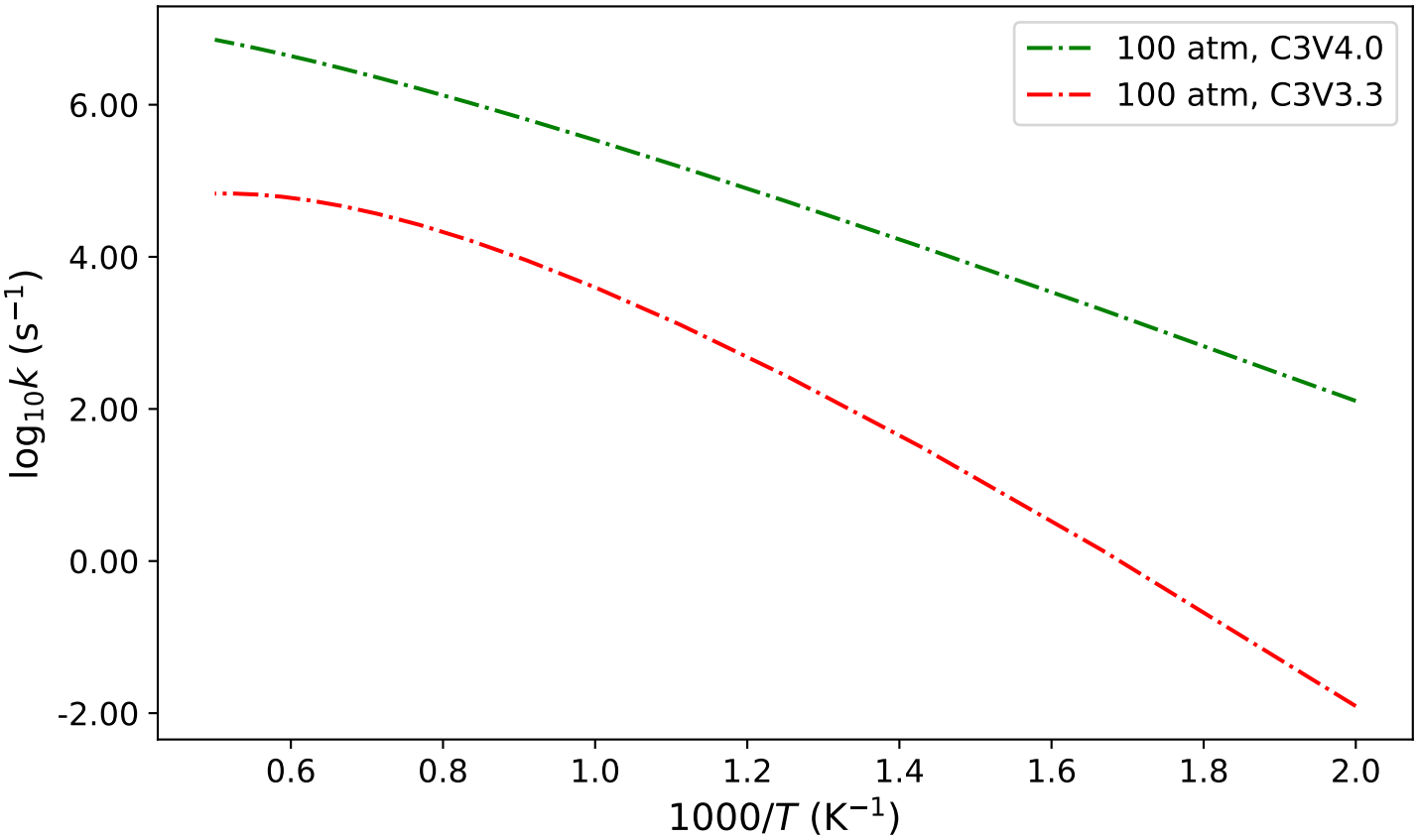



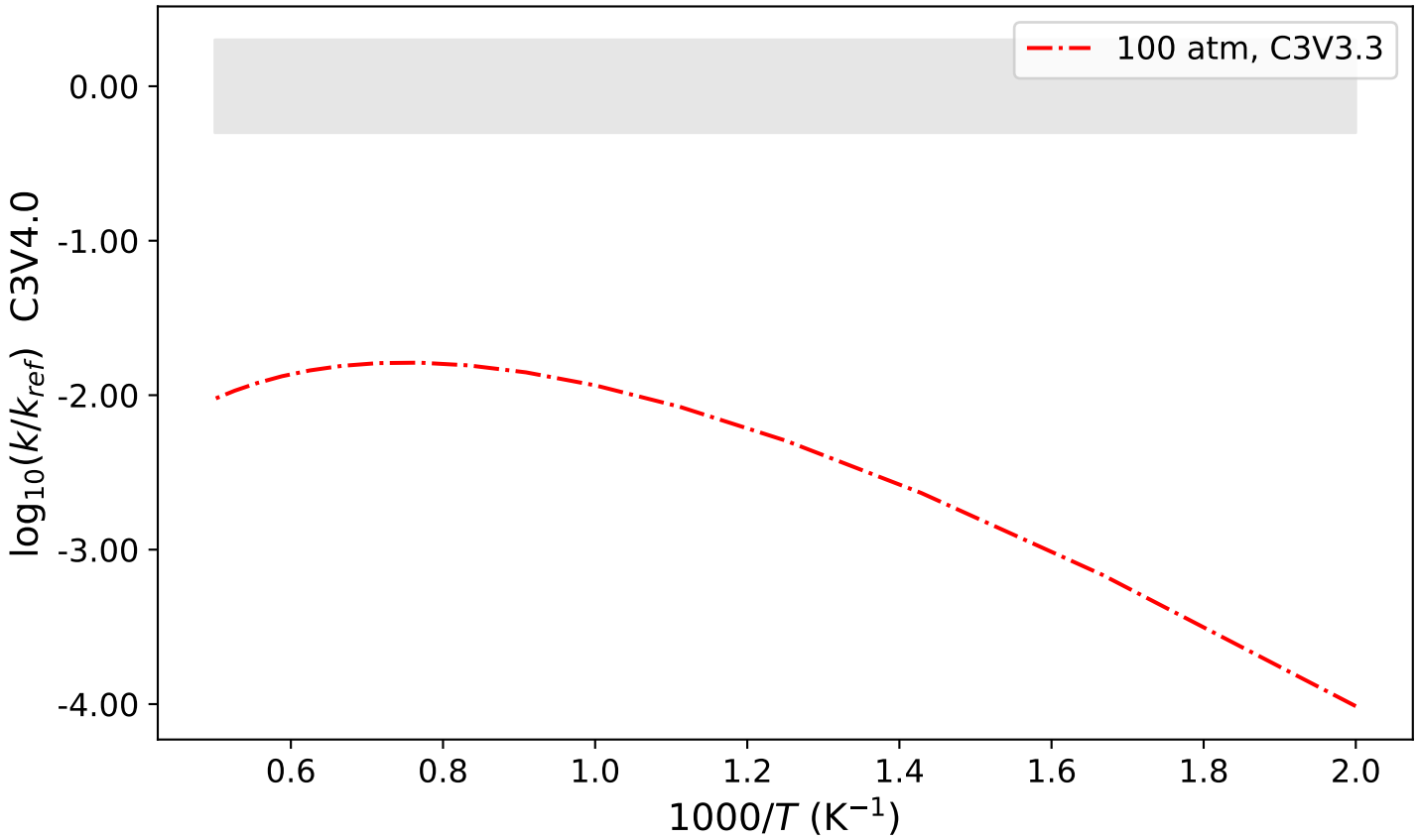
H + C4H6 = H + C4H6-1



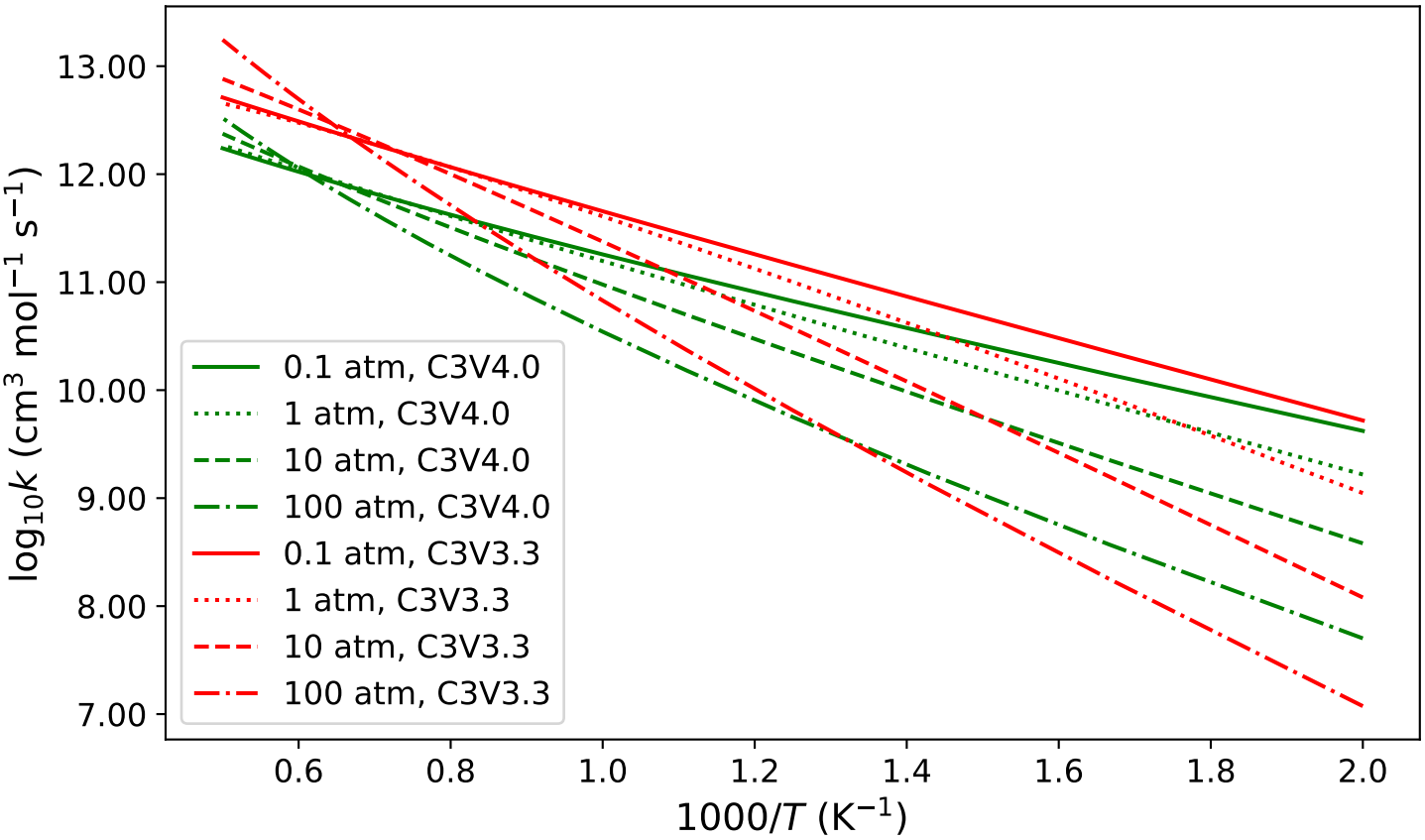



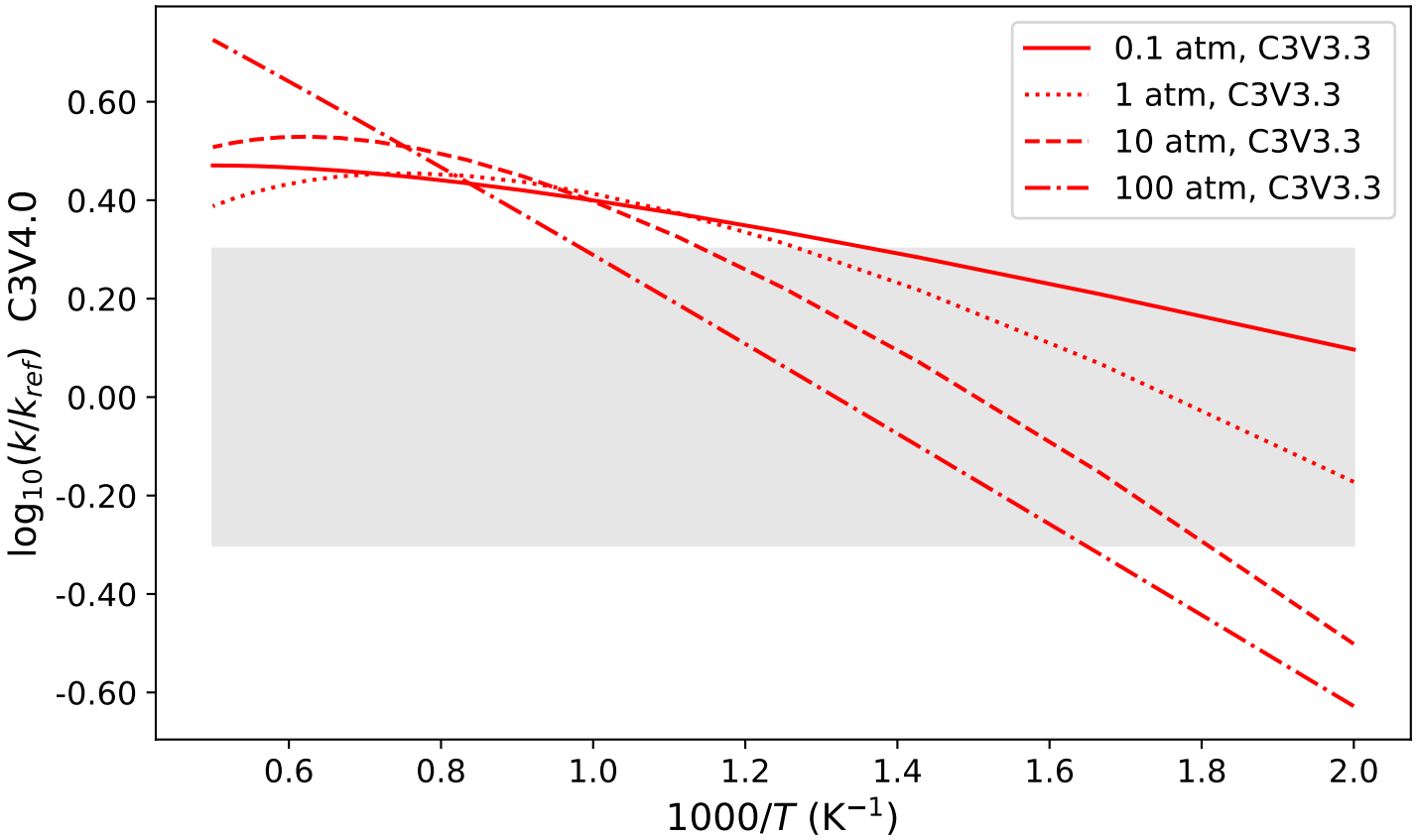
C2H5CO3 = QCH3CHCO3H



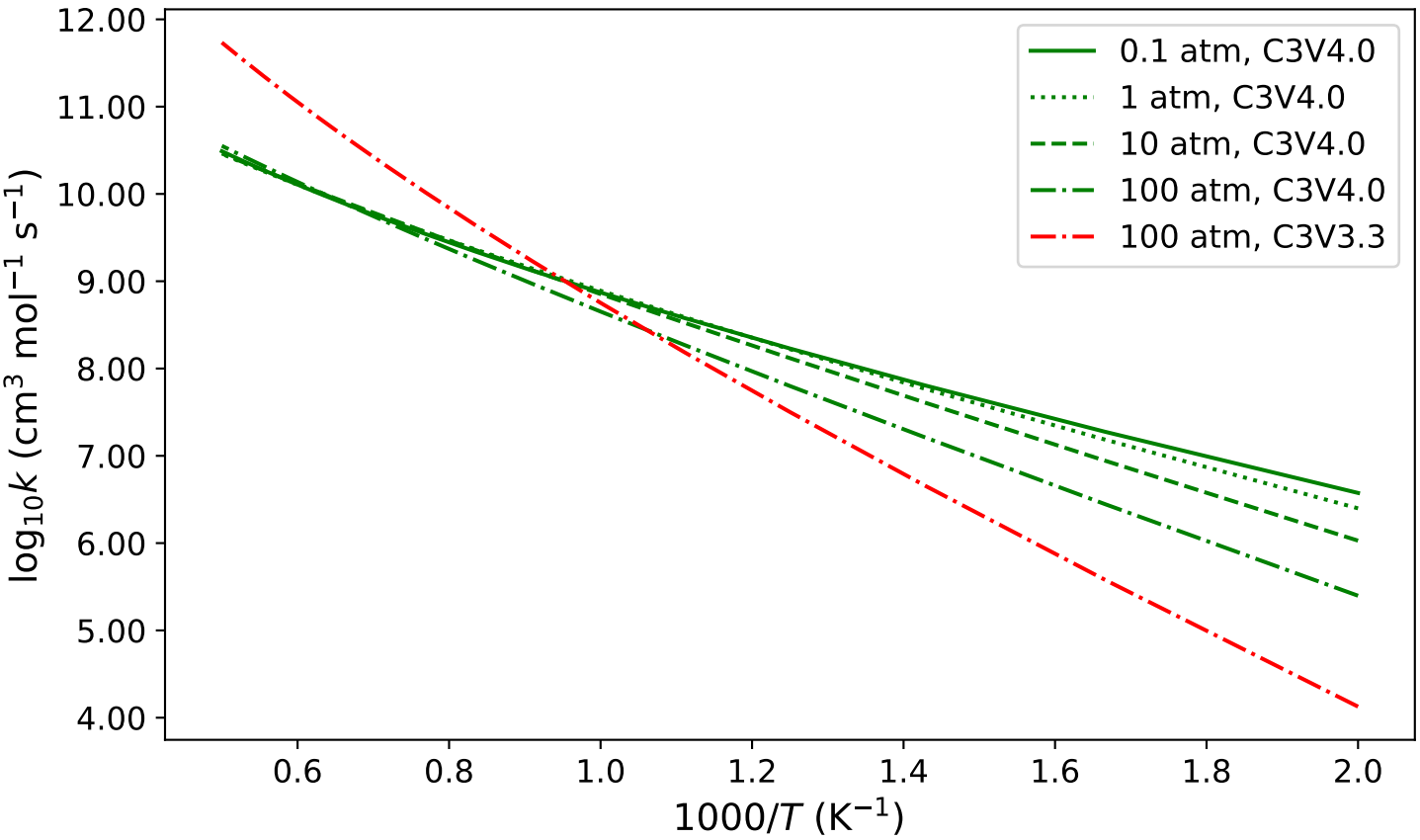



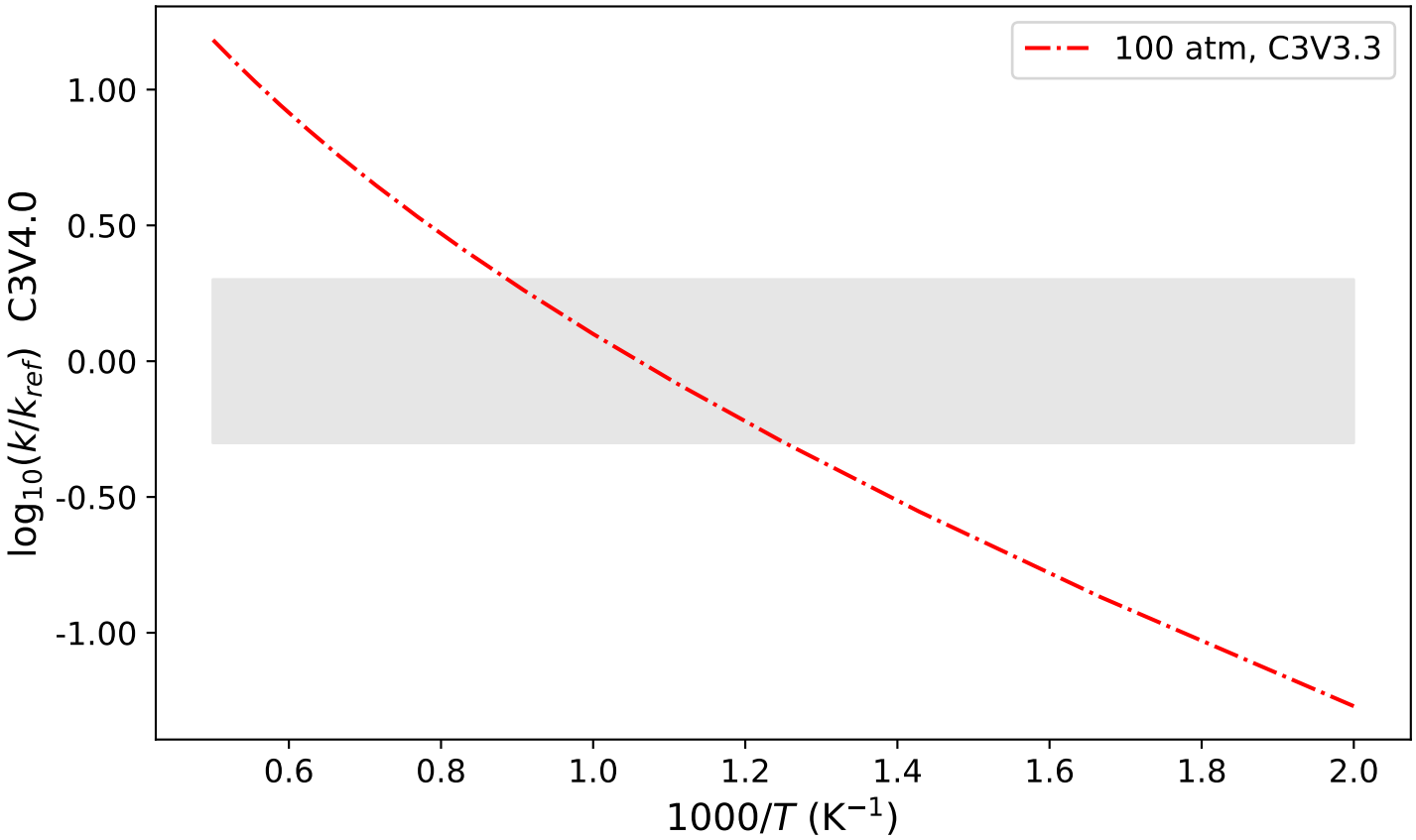
H + C4H8-1 = H + C4H8-2



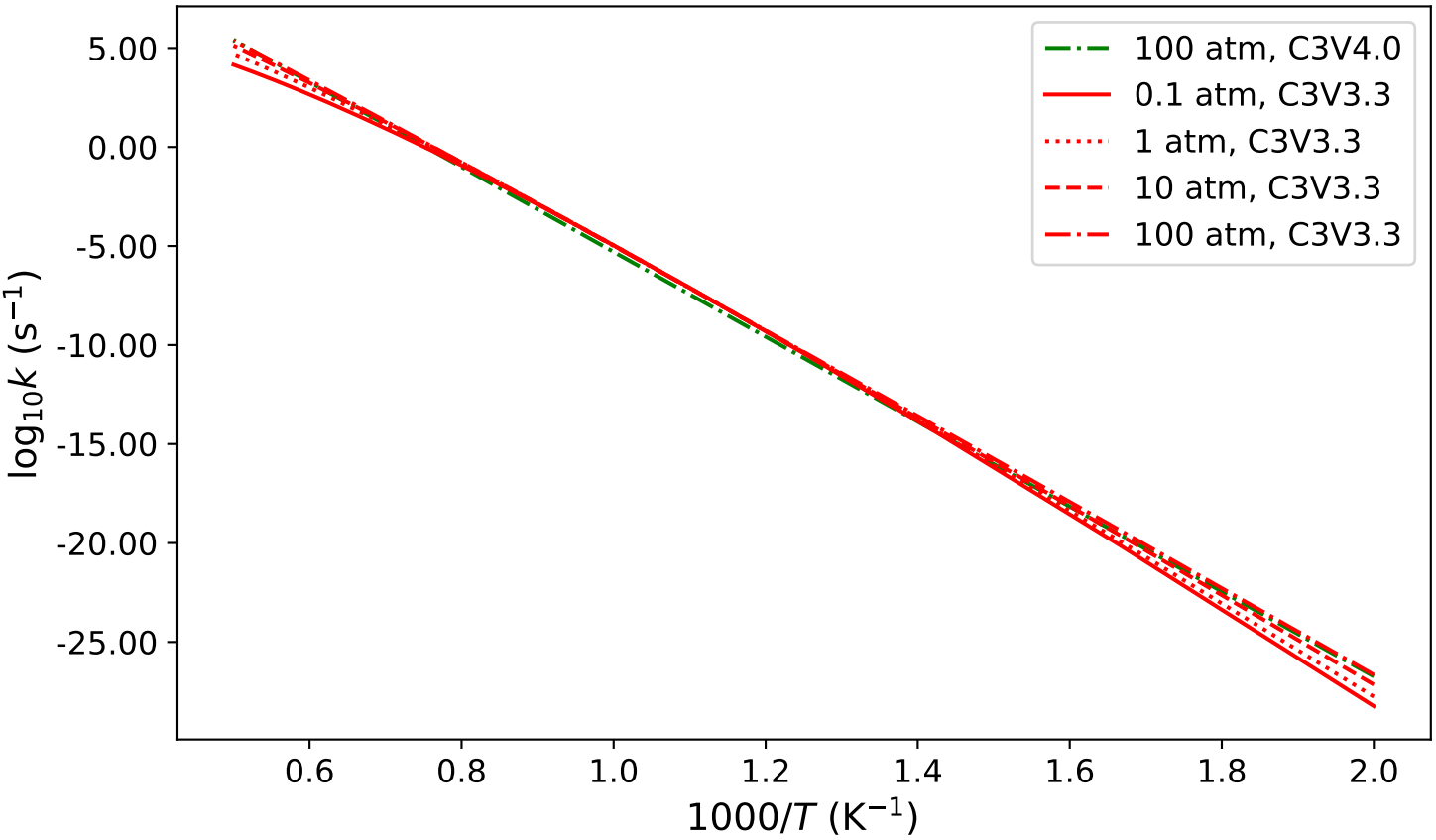



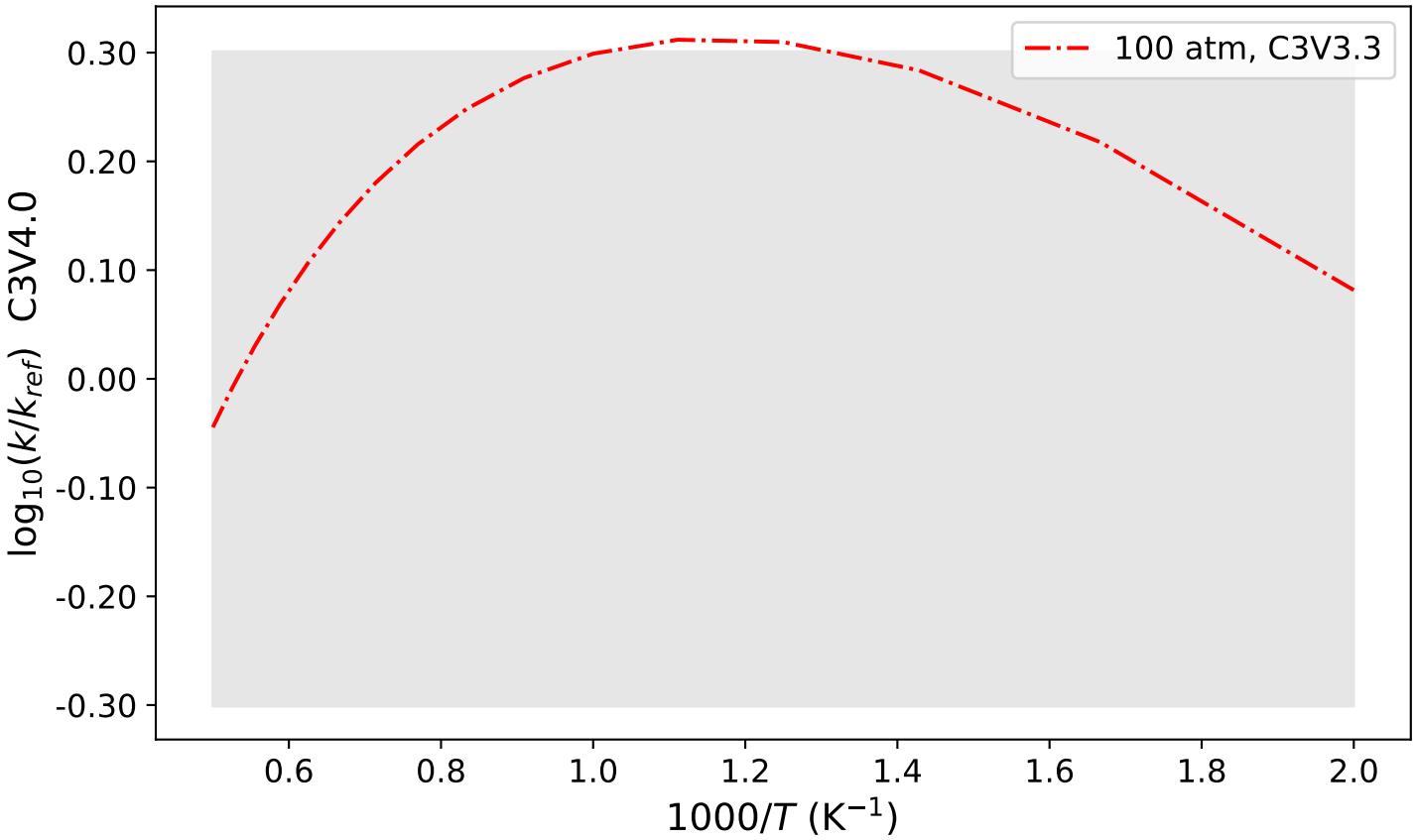
HO2 + C4H6 = O2 + C4H71-3



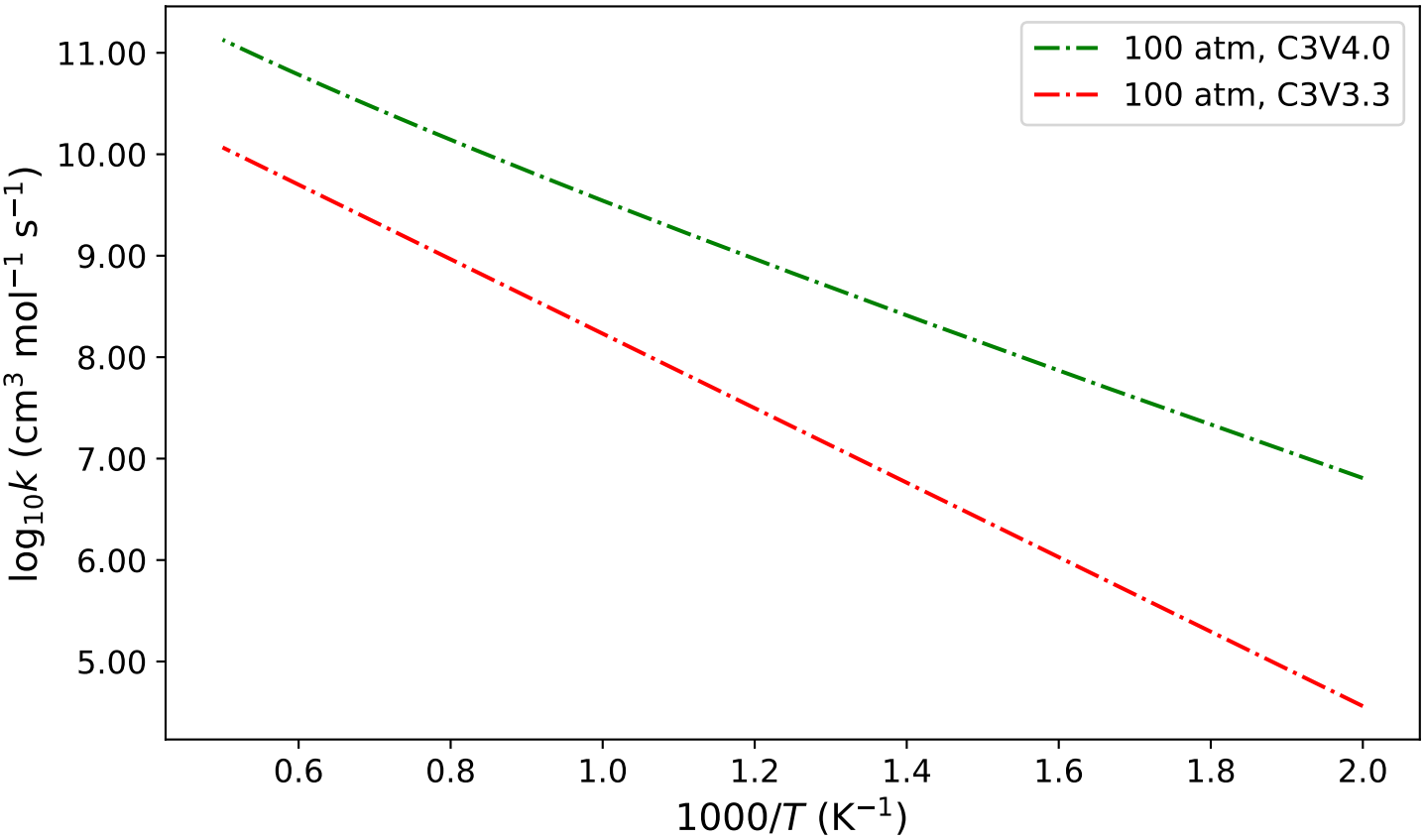



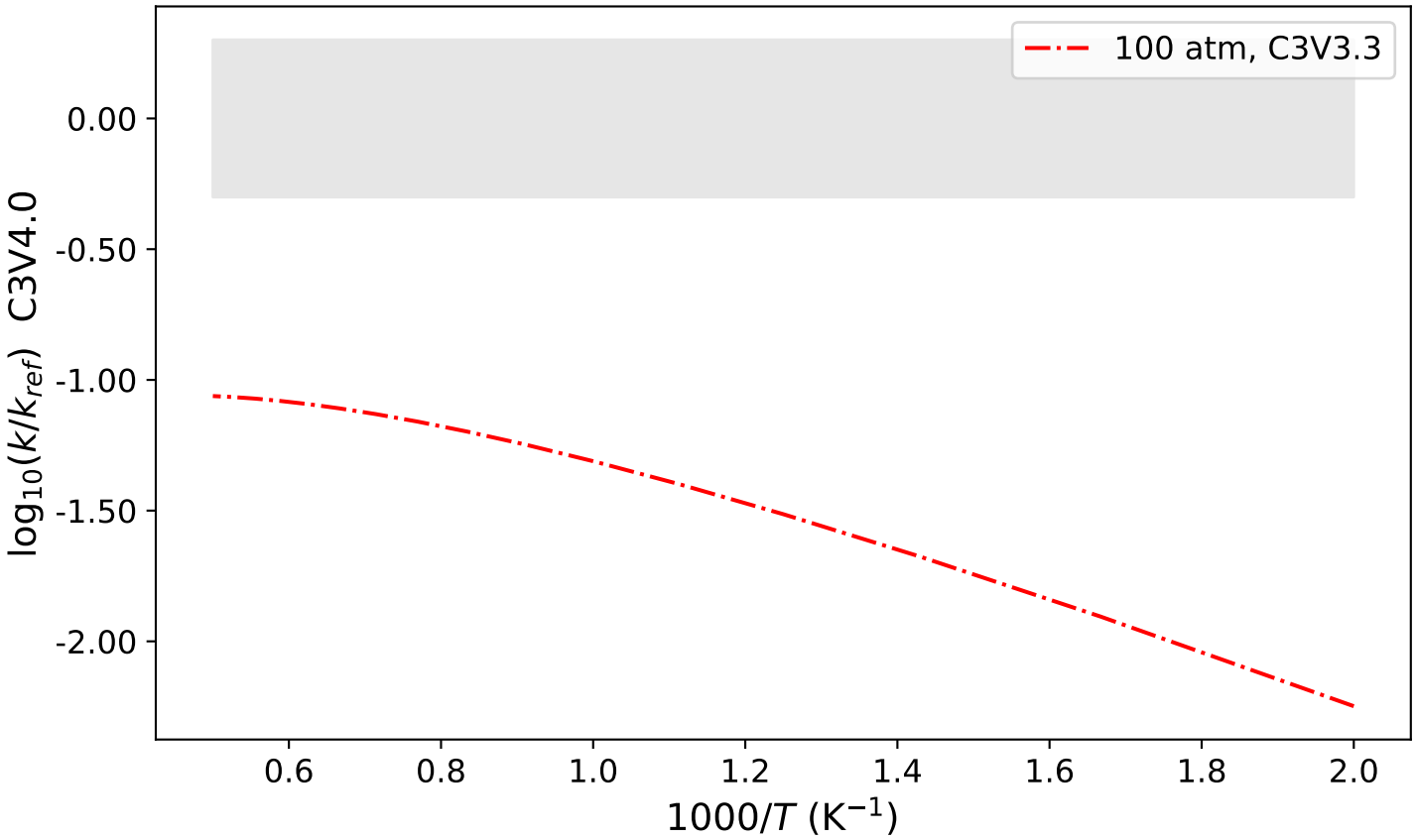
C4H10 = H + SC4H9



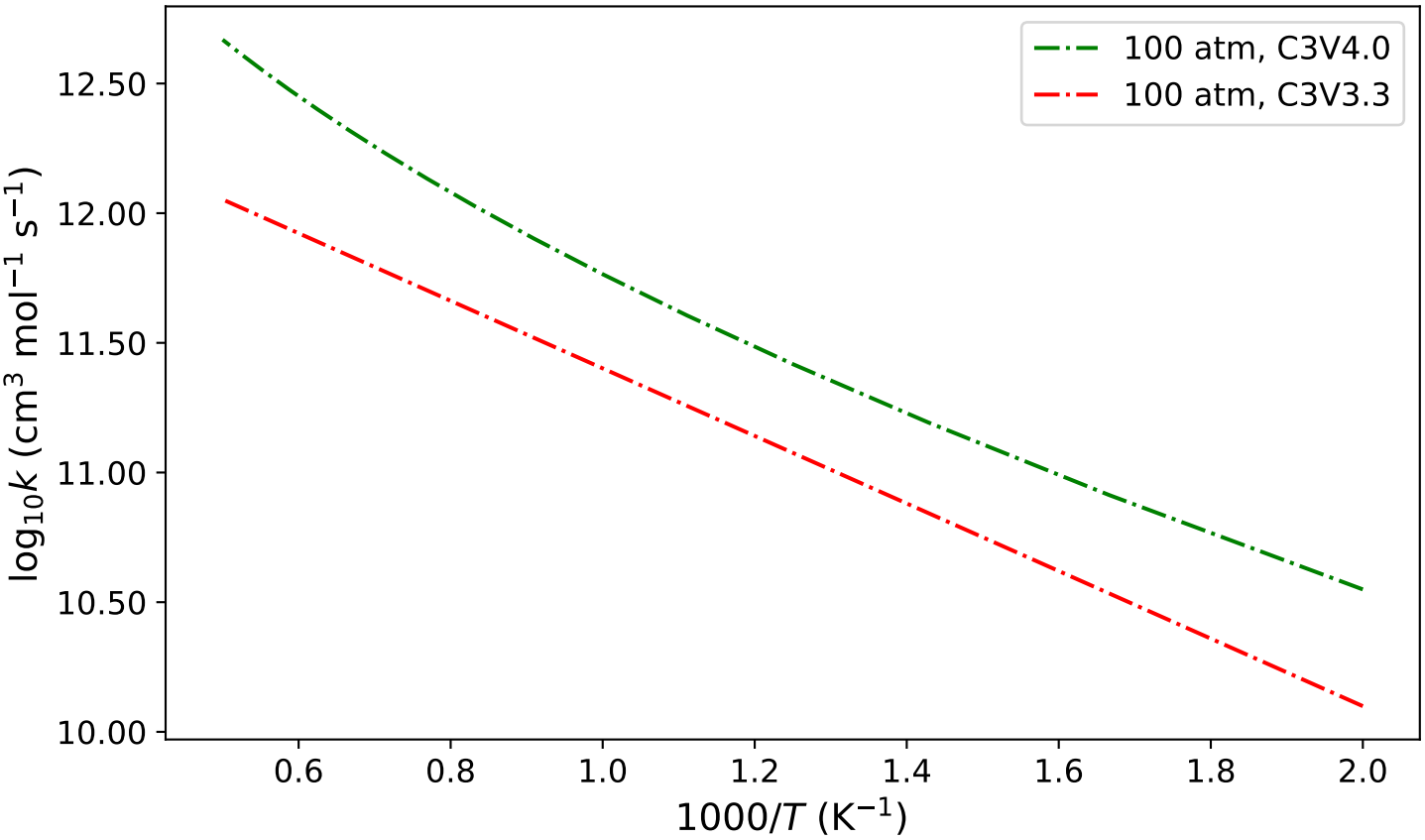



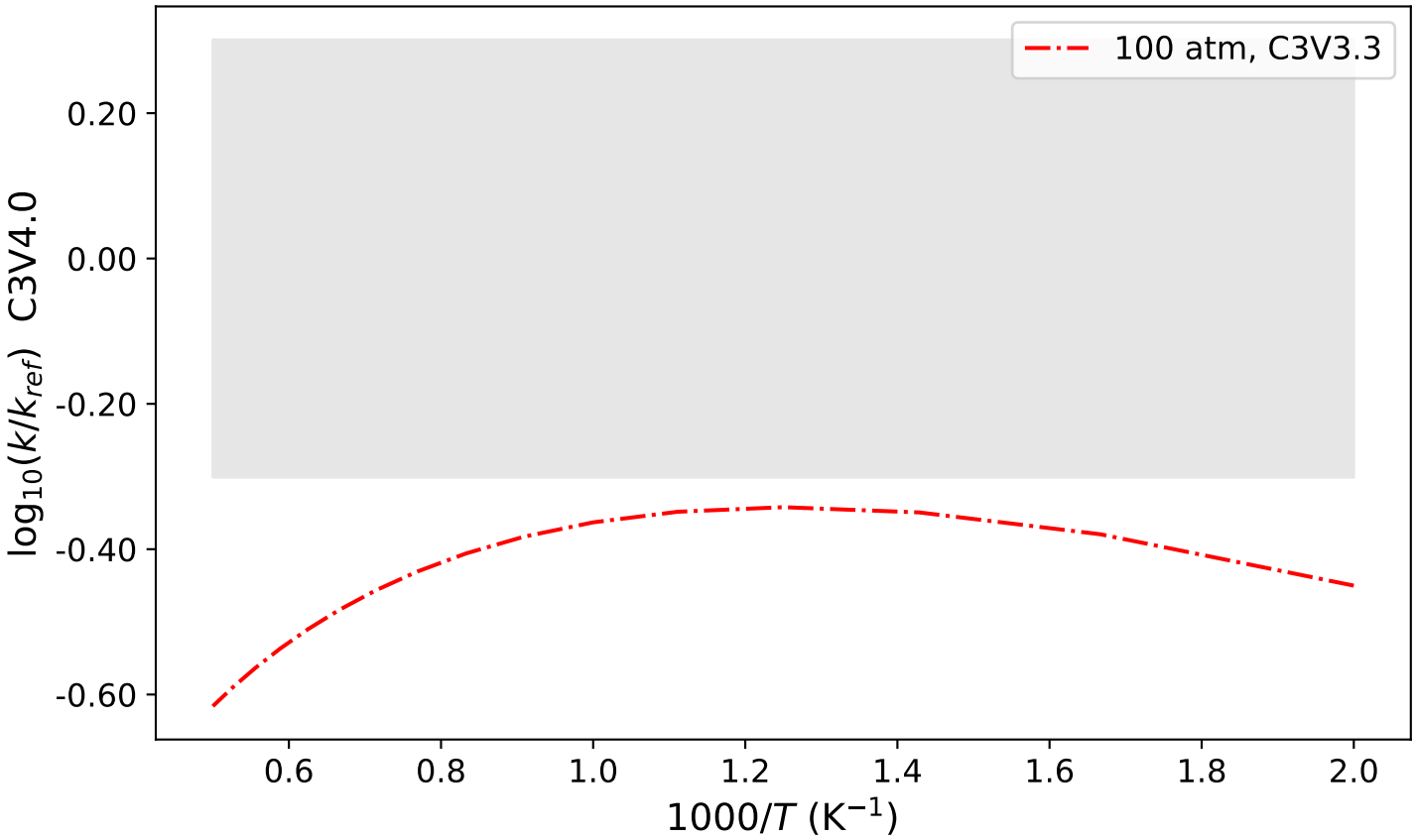
C2H3 + C4H10 = C2H4 + SC4H9



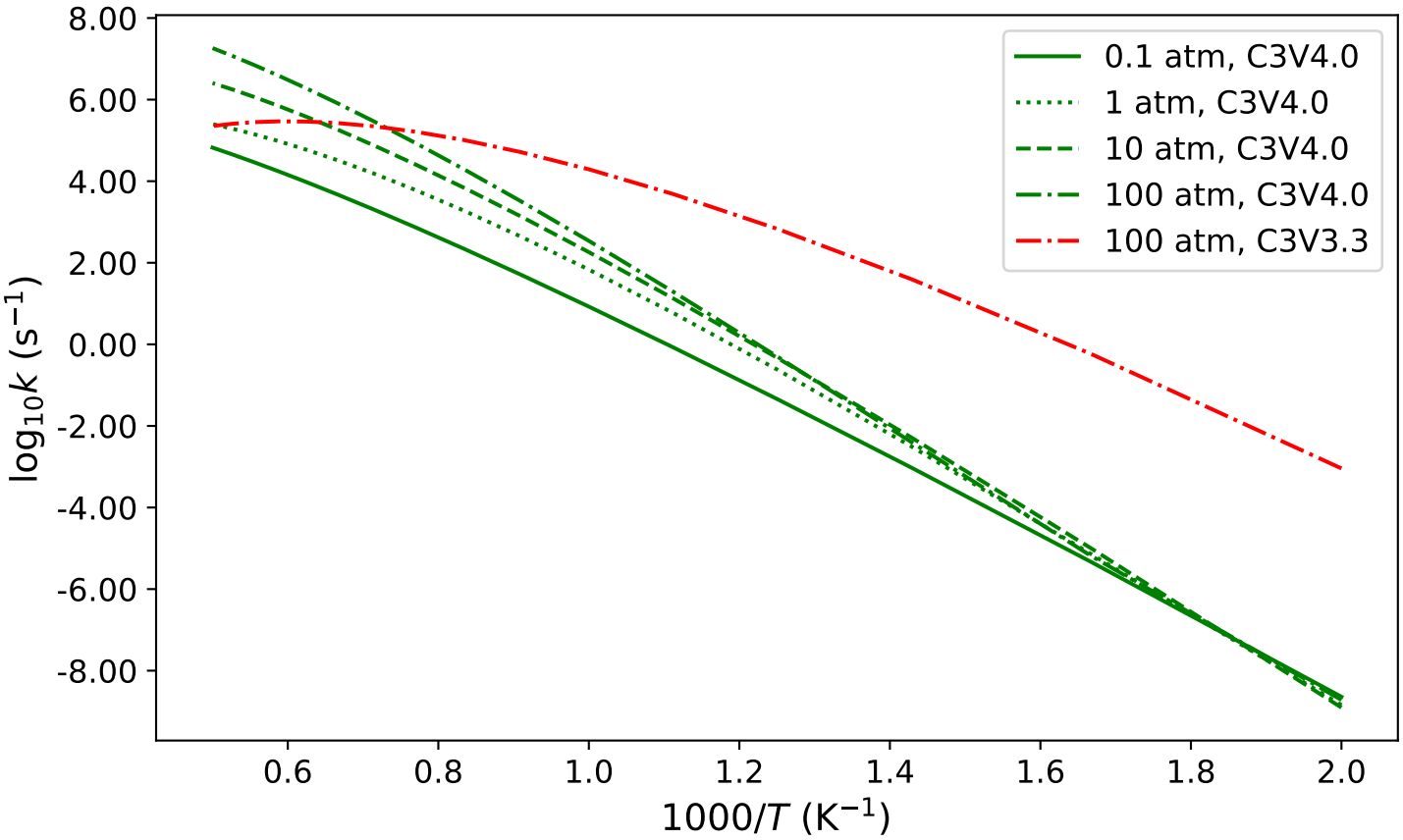



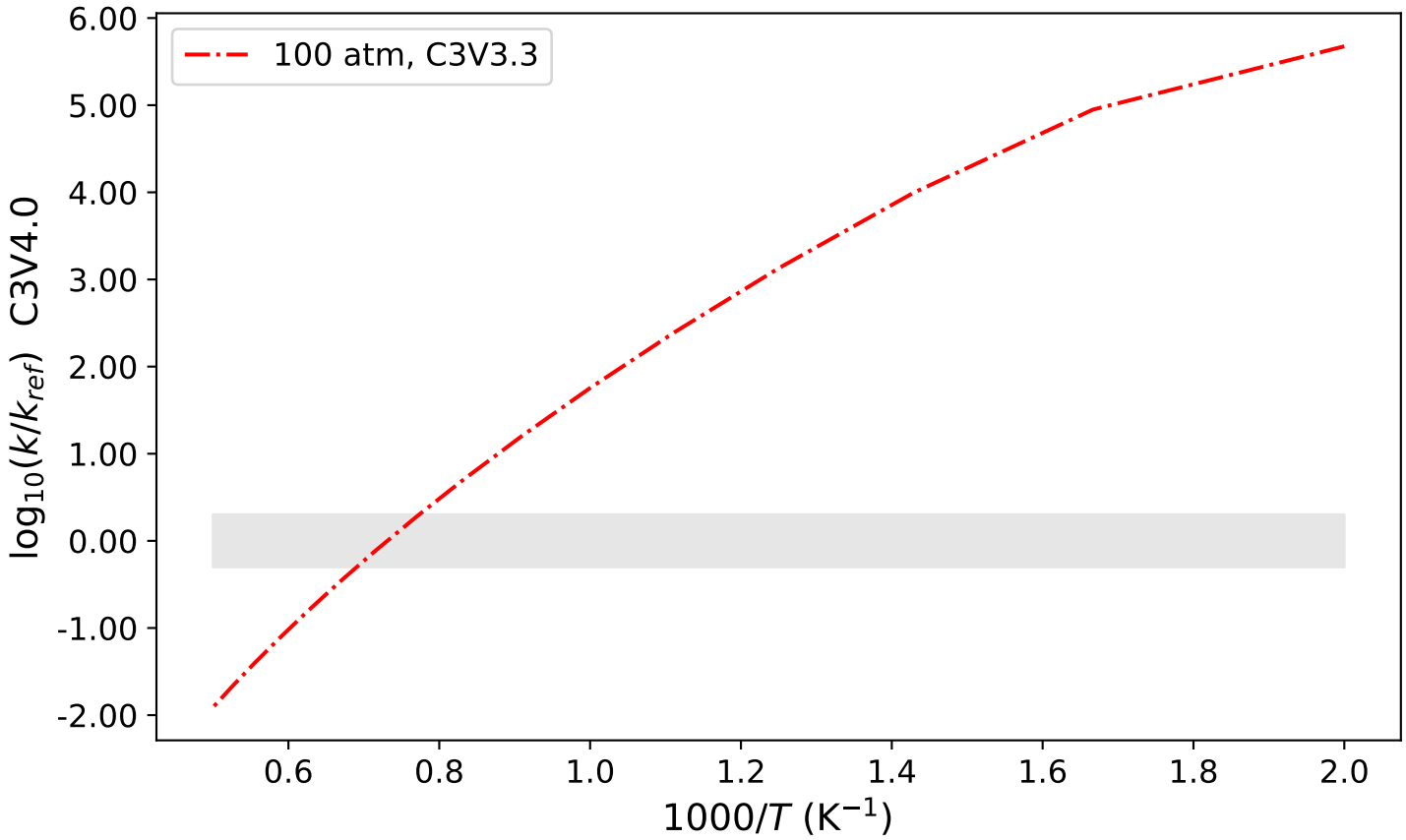
OH + C3H5OH = H2O + CH2CCH2OH



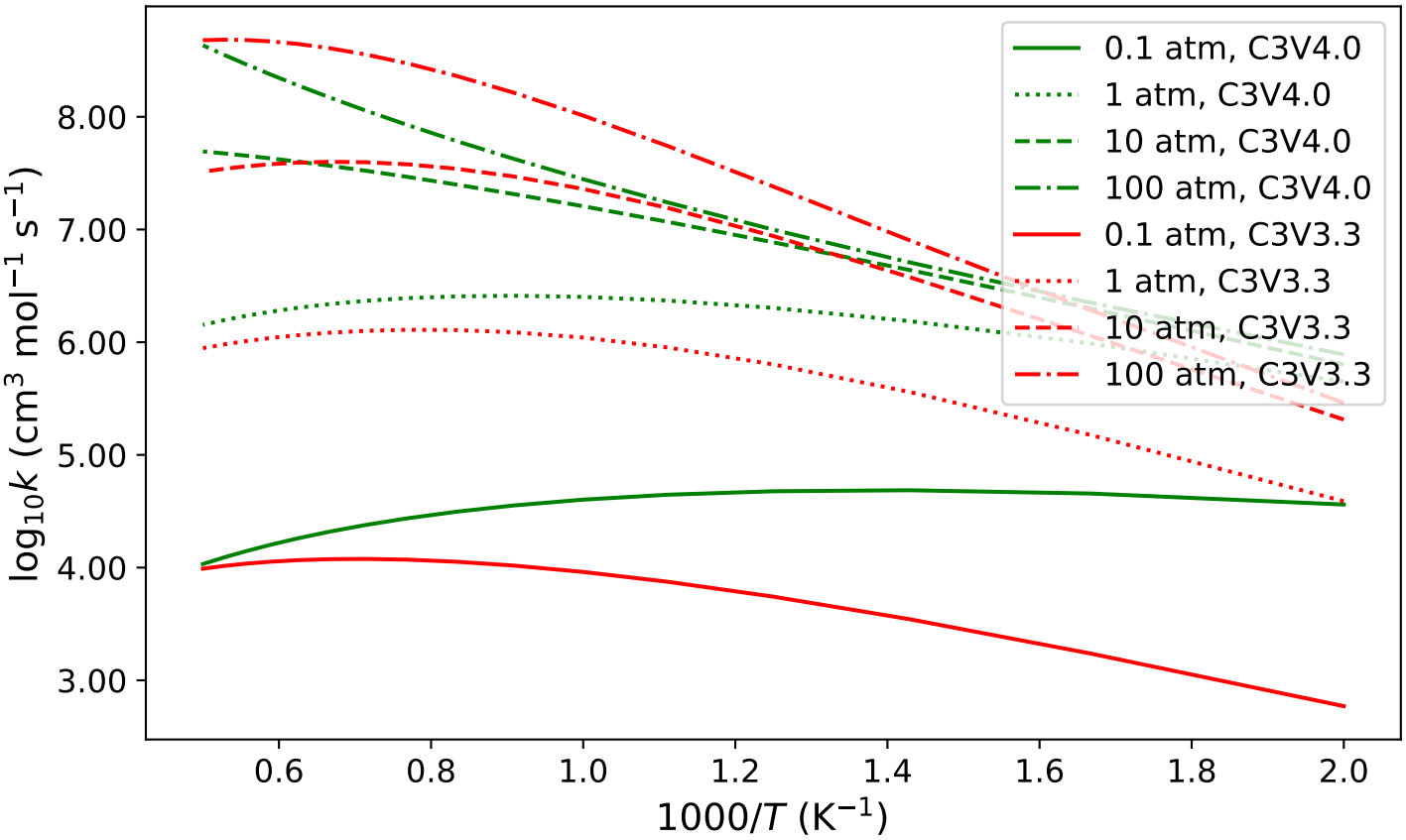



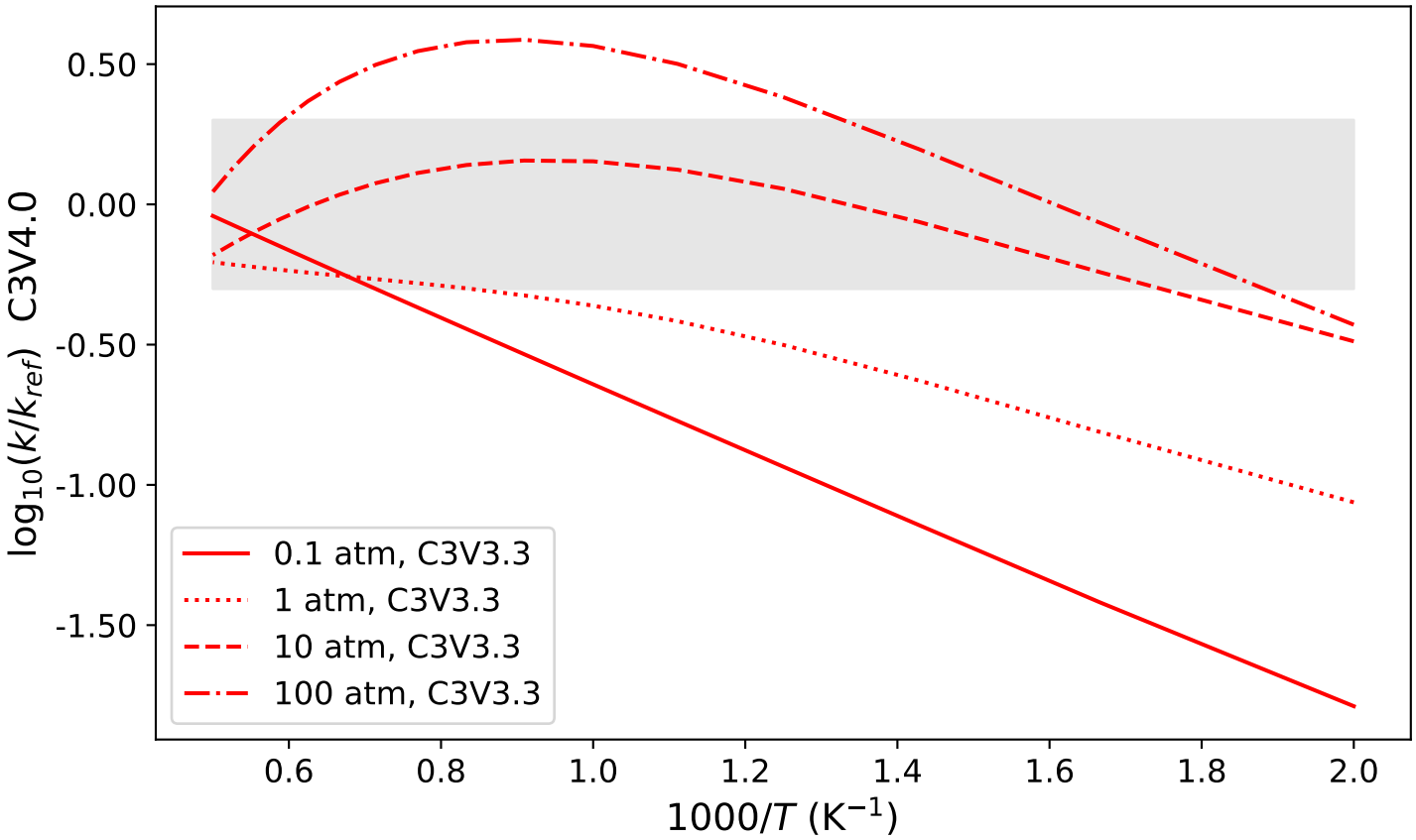
C4H5-I = C4H5-2


0.1 atm, C3V4.0
1 atm, C3V4.0
10 atm, C3V4.0
100 atm, C3V4.0
100 atm, C3V3.3
8.00
6.00
4.00
2.00
0.00
-2.00
-4.00
-6.00
-8.00
$\log_{10}k$ ($s^{-1}$)
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ ($K^{-1}$)

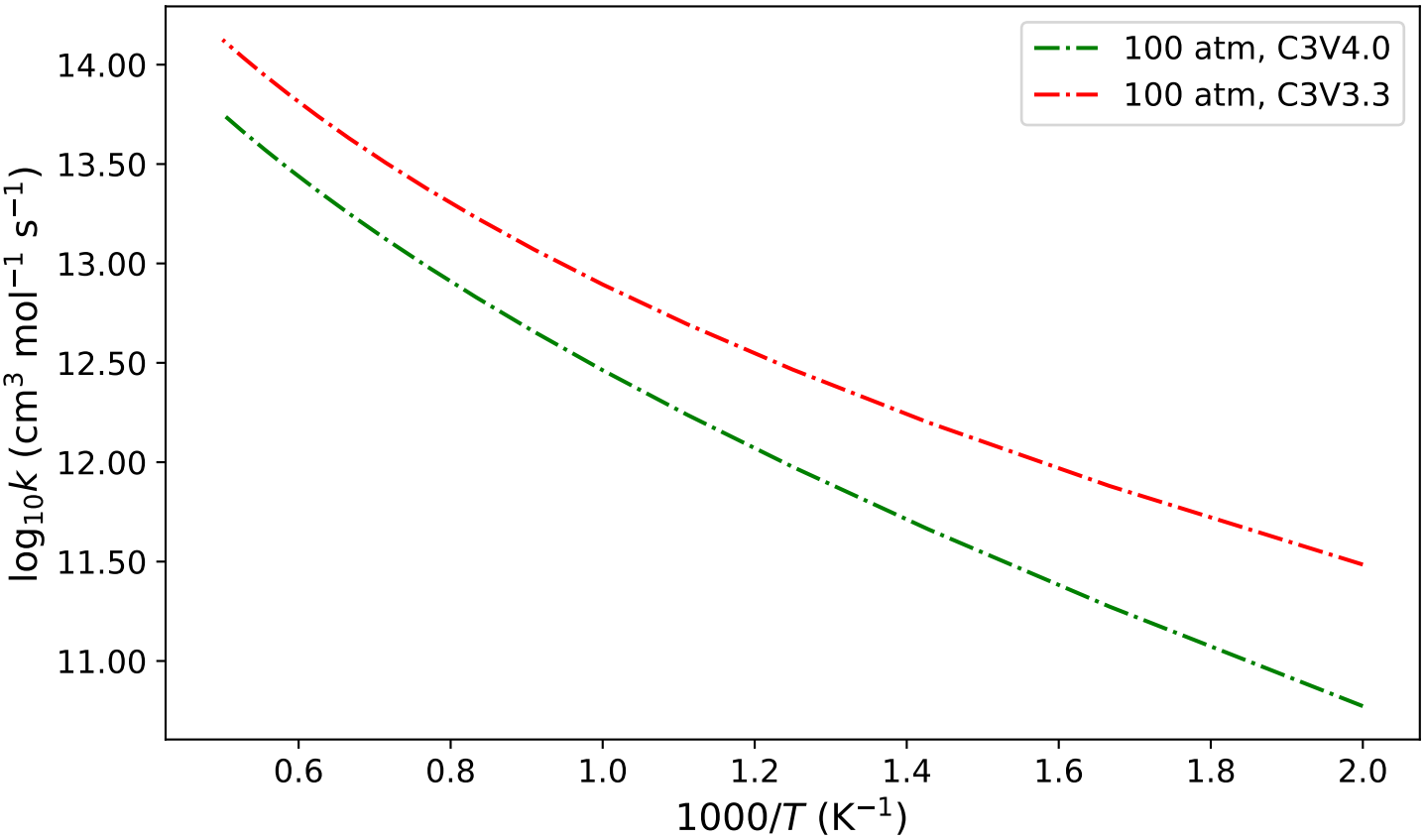

100 atm, C3V3.3
6.00
5.00
4.00
3.00
2.00
1.00
0.00
-1.00
-2.00
$\log_{10}(k/k_{ref})$ C3V4.0
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ ($K^{-1}$)

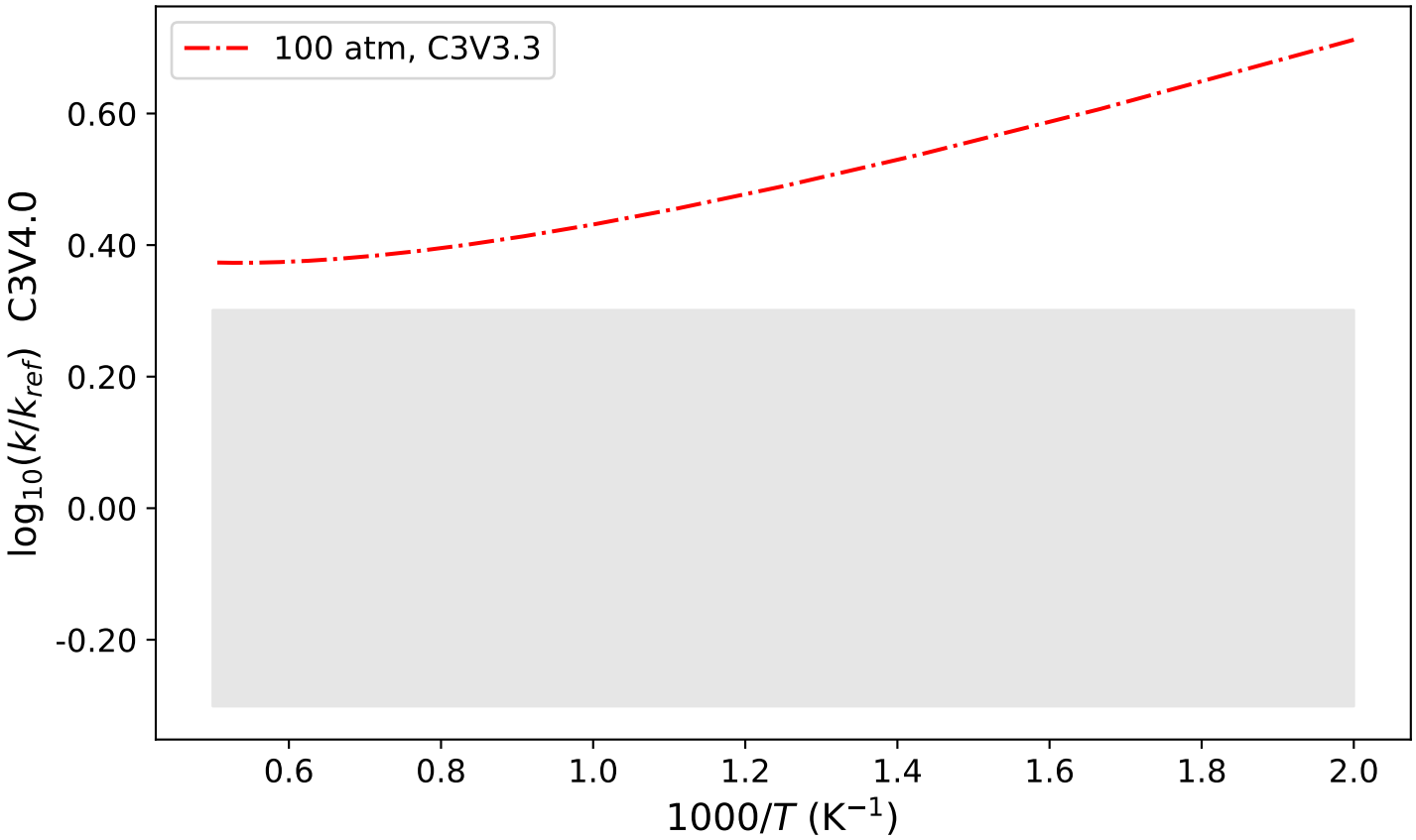
HO2 + C3H4-A = TC3H5OO



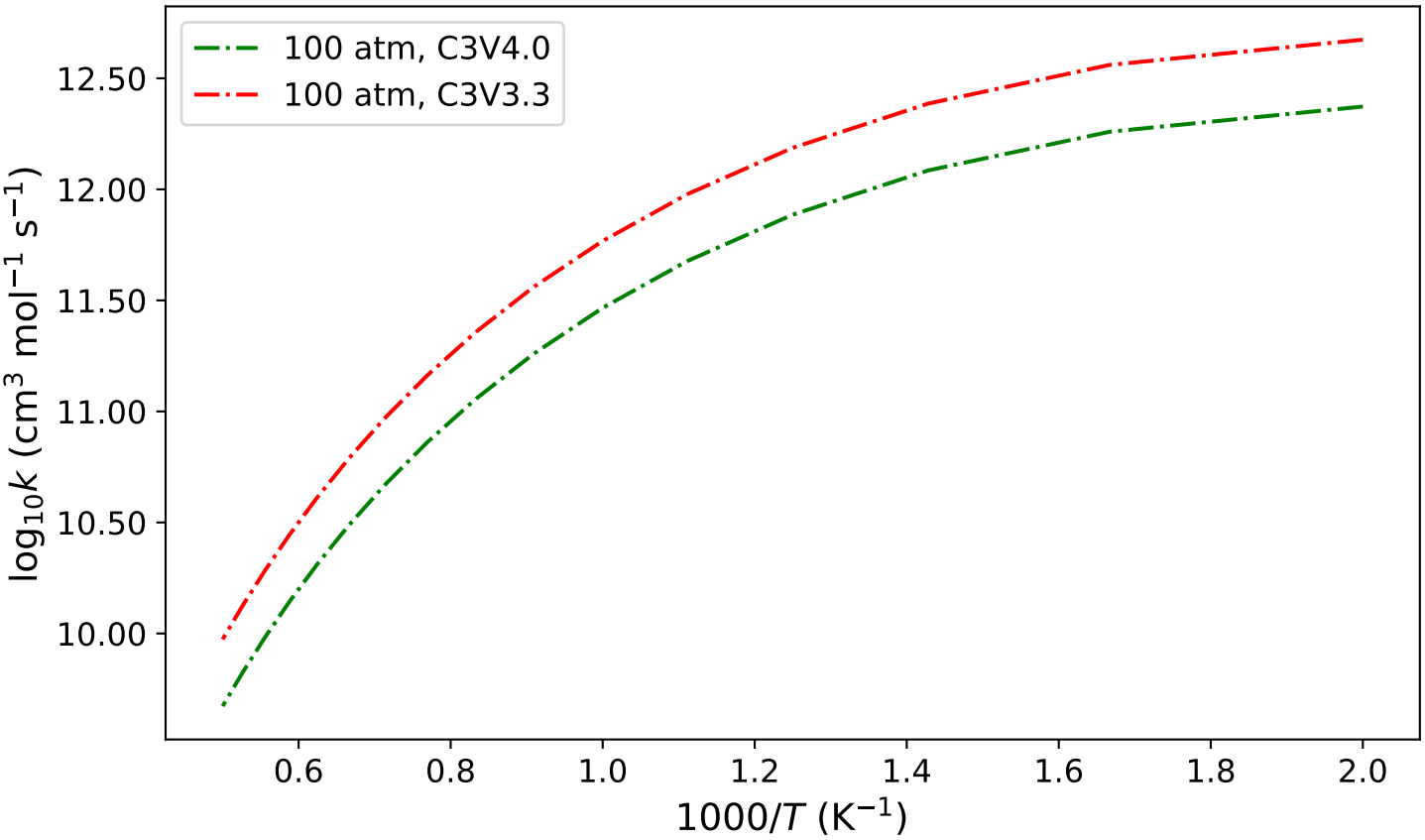



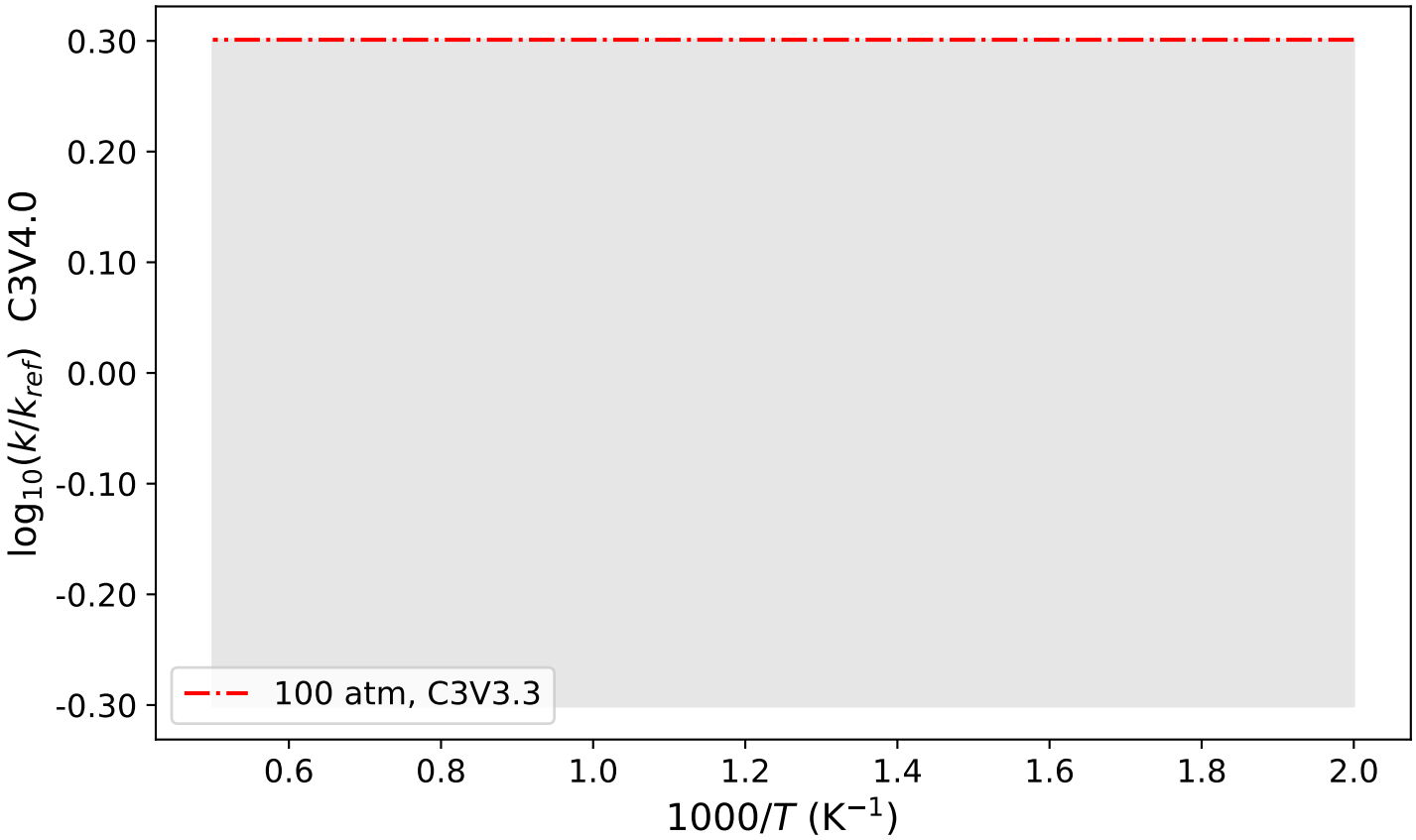
H + C4H8-2 = H2 + C4H71-3



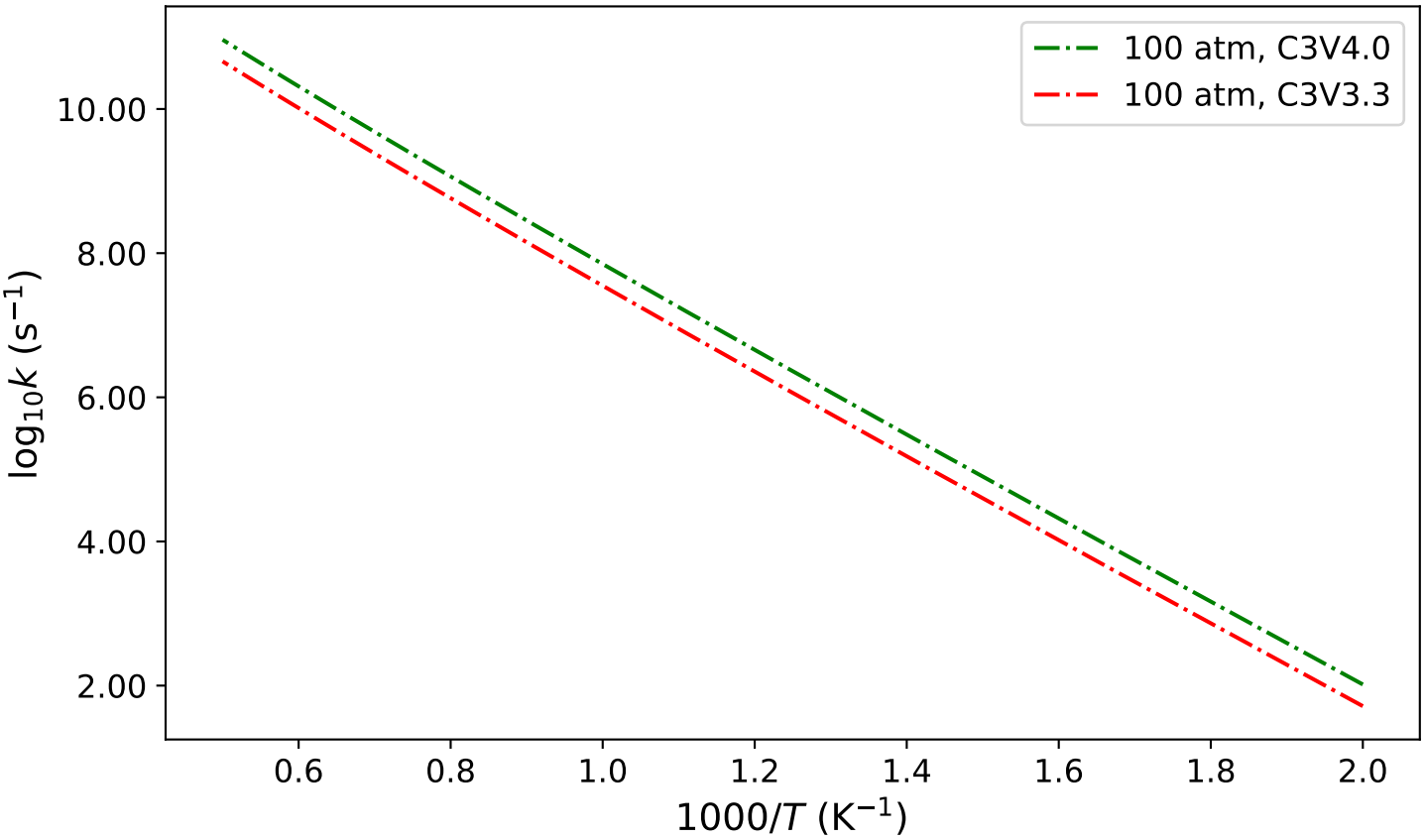



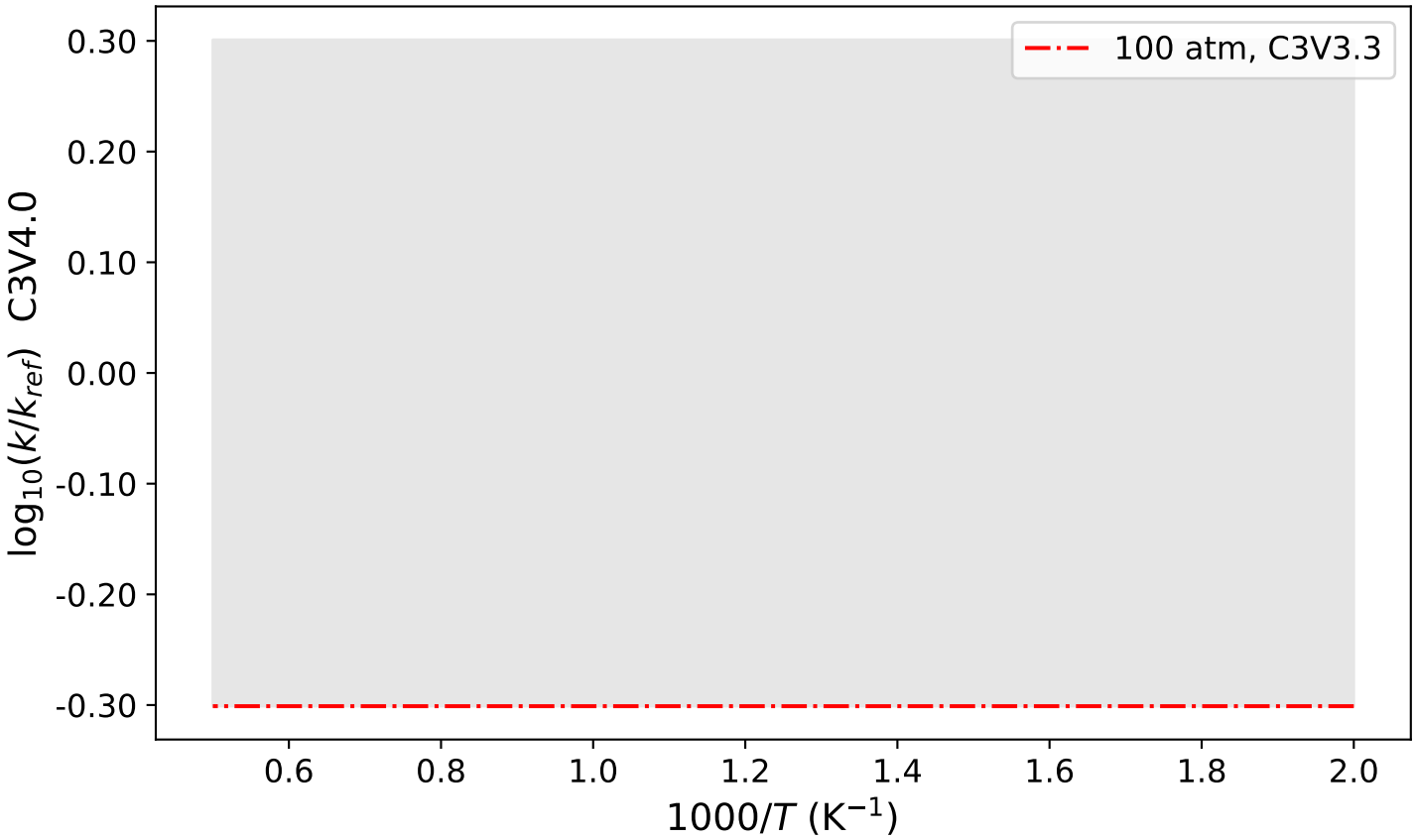
HO2 + CH3COCH2 = C3KET21



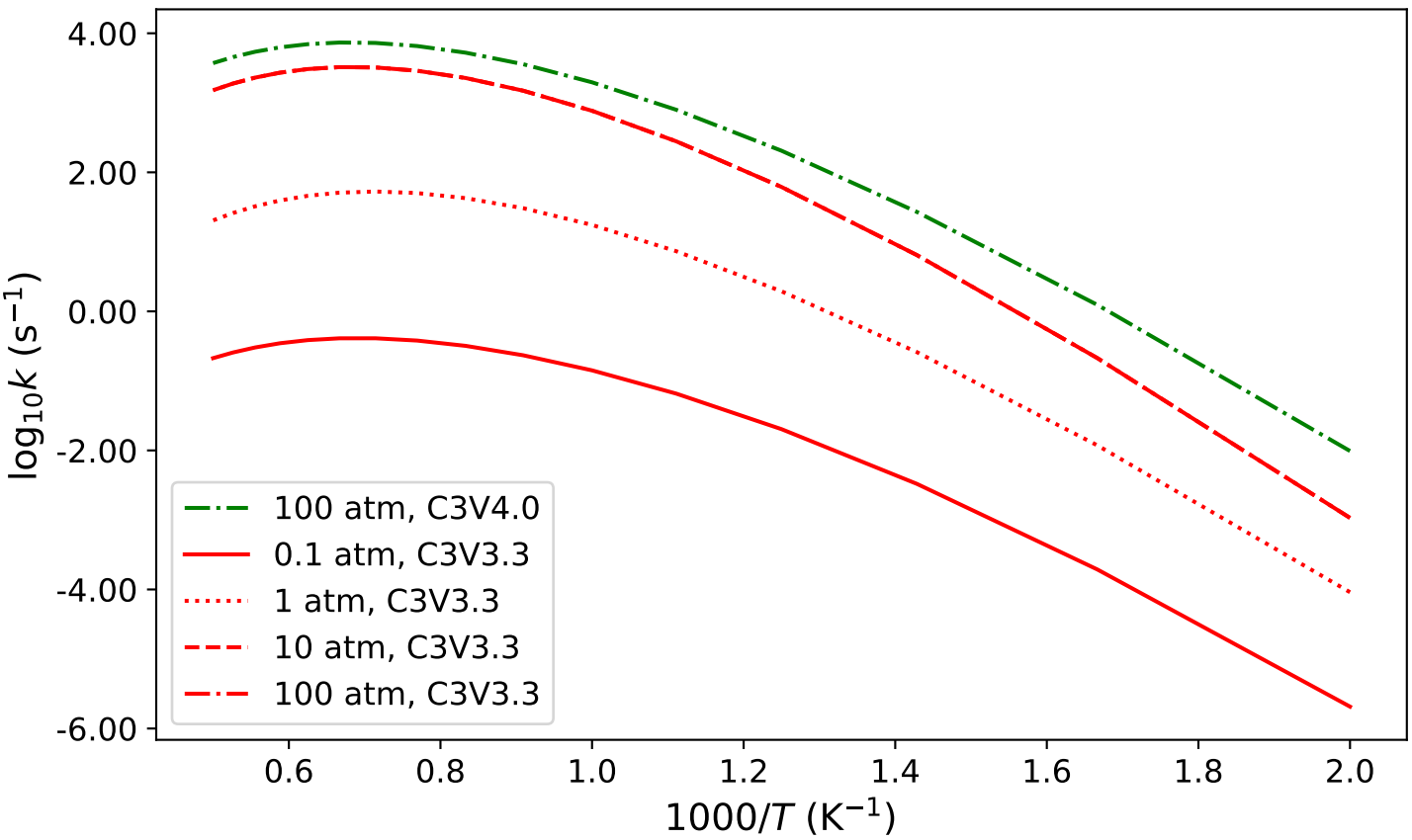



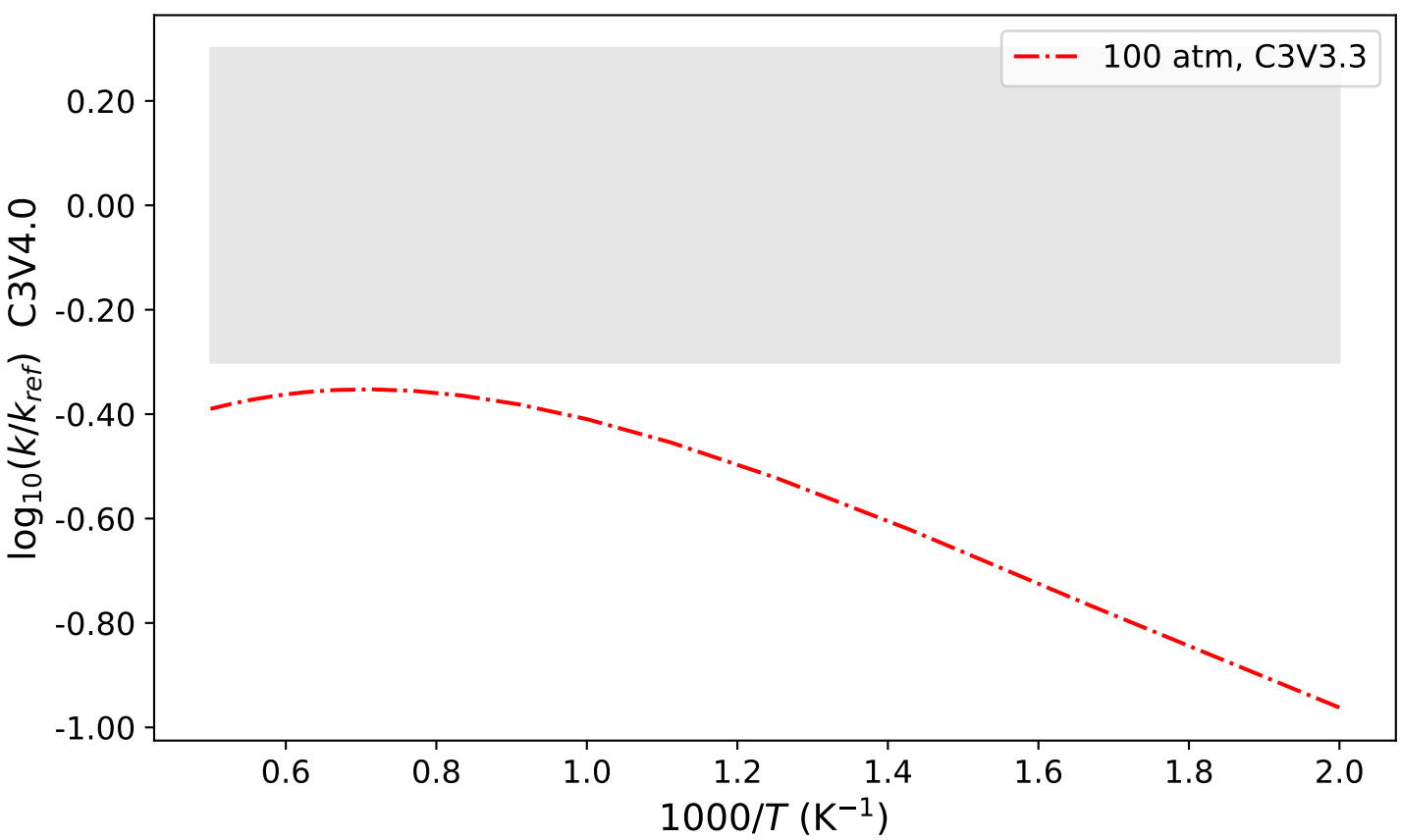
C4H8OOH1-3 => CH2O + OH + C3H6



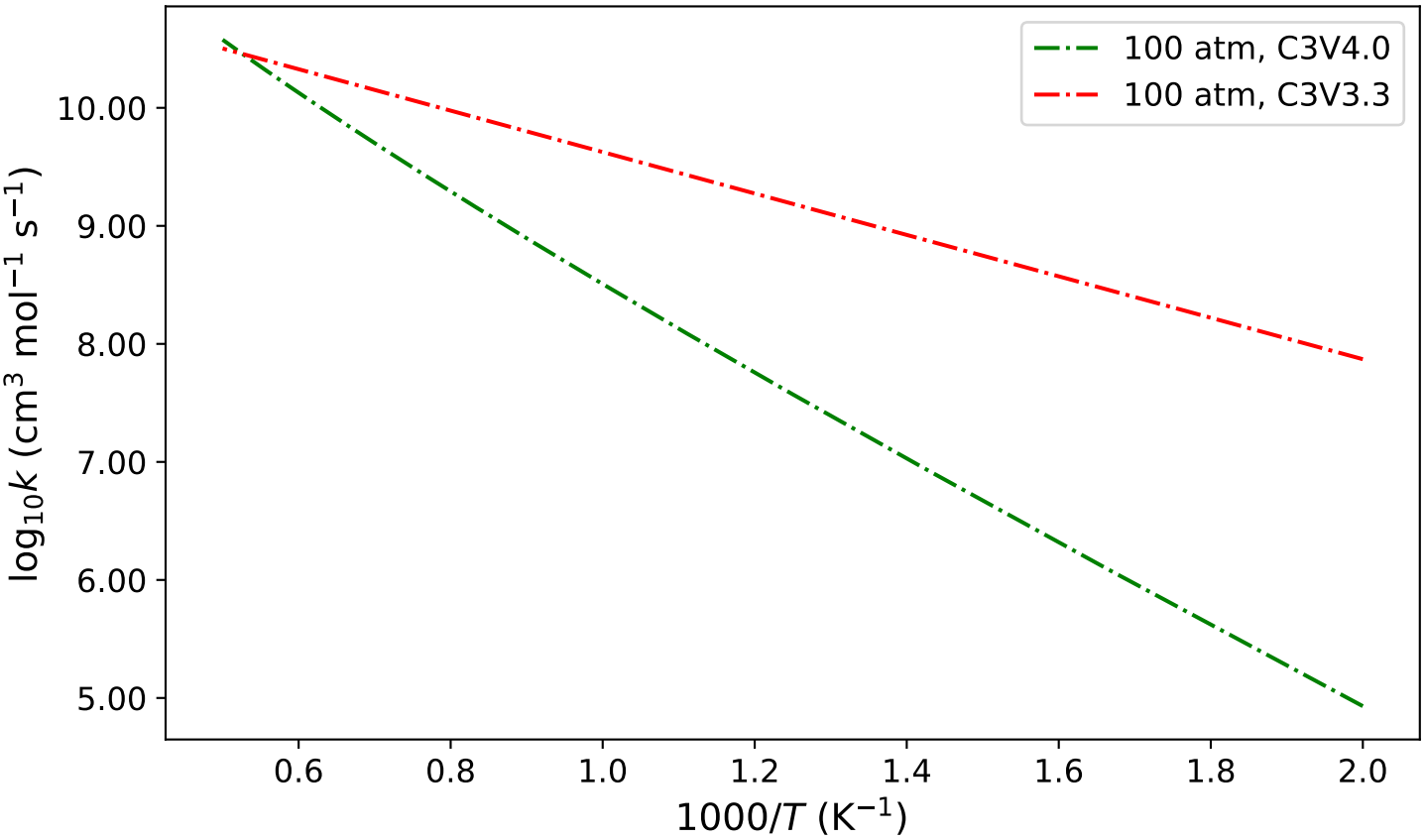



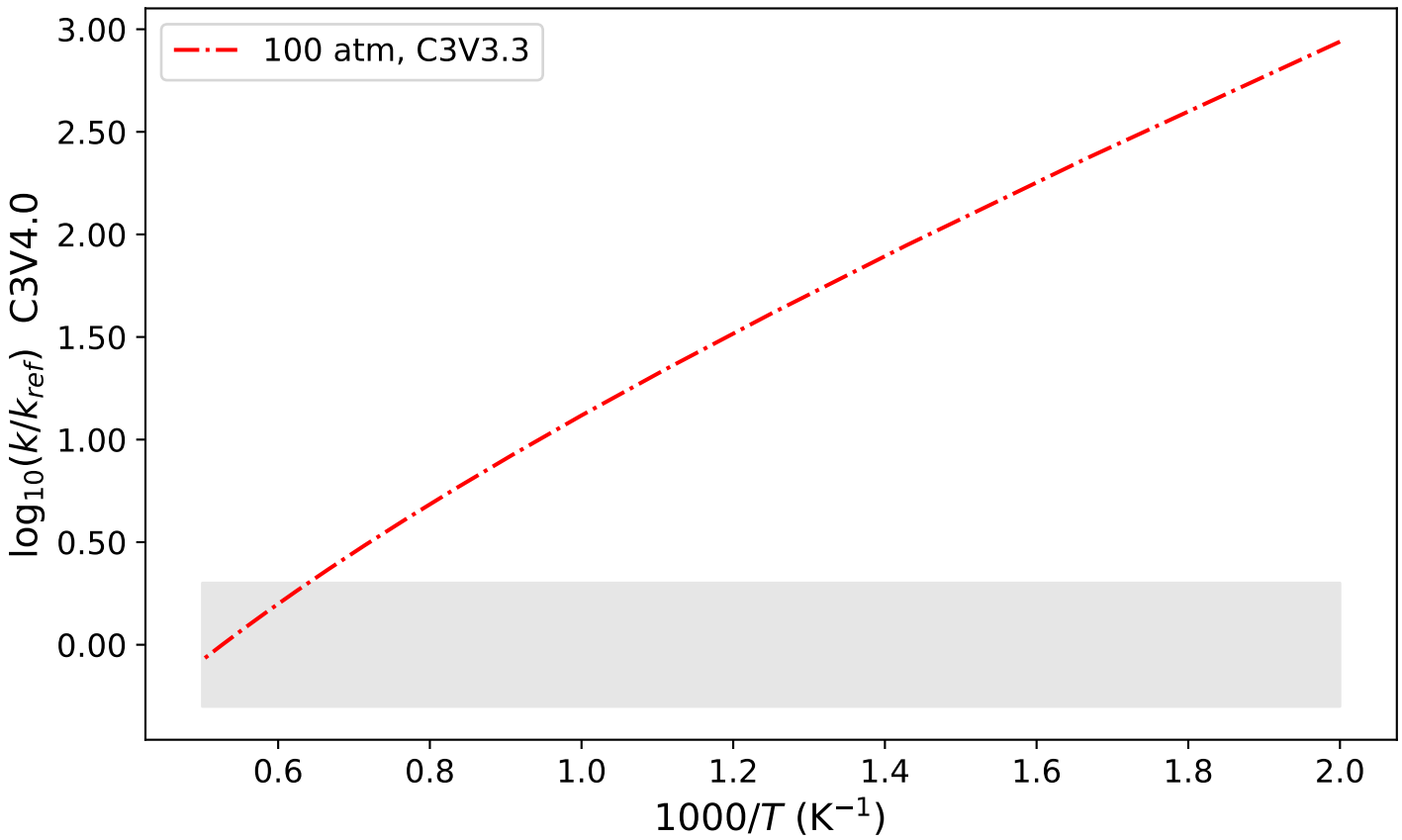
C2H5CO3 = HO2 + CH3CHCO



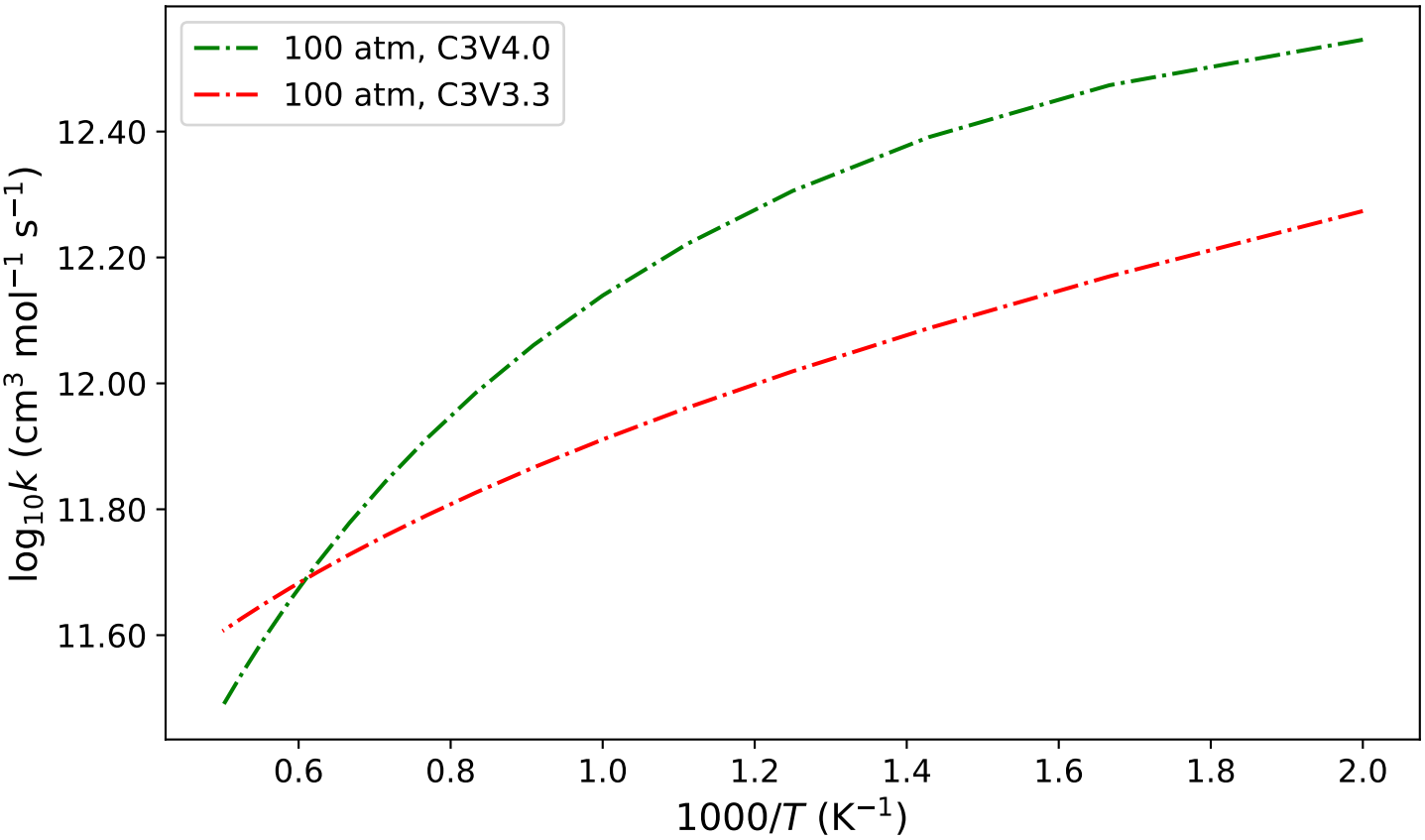



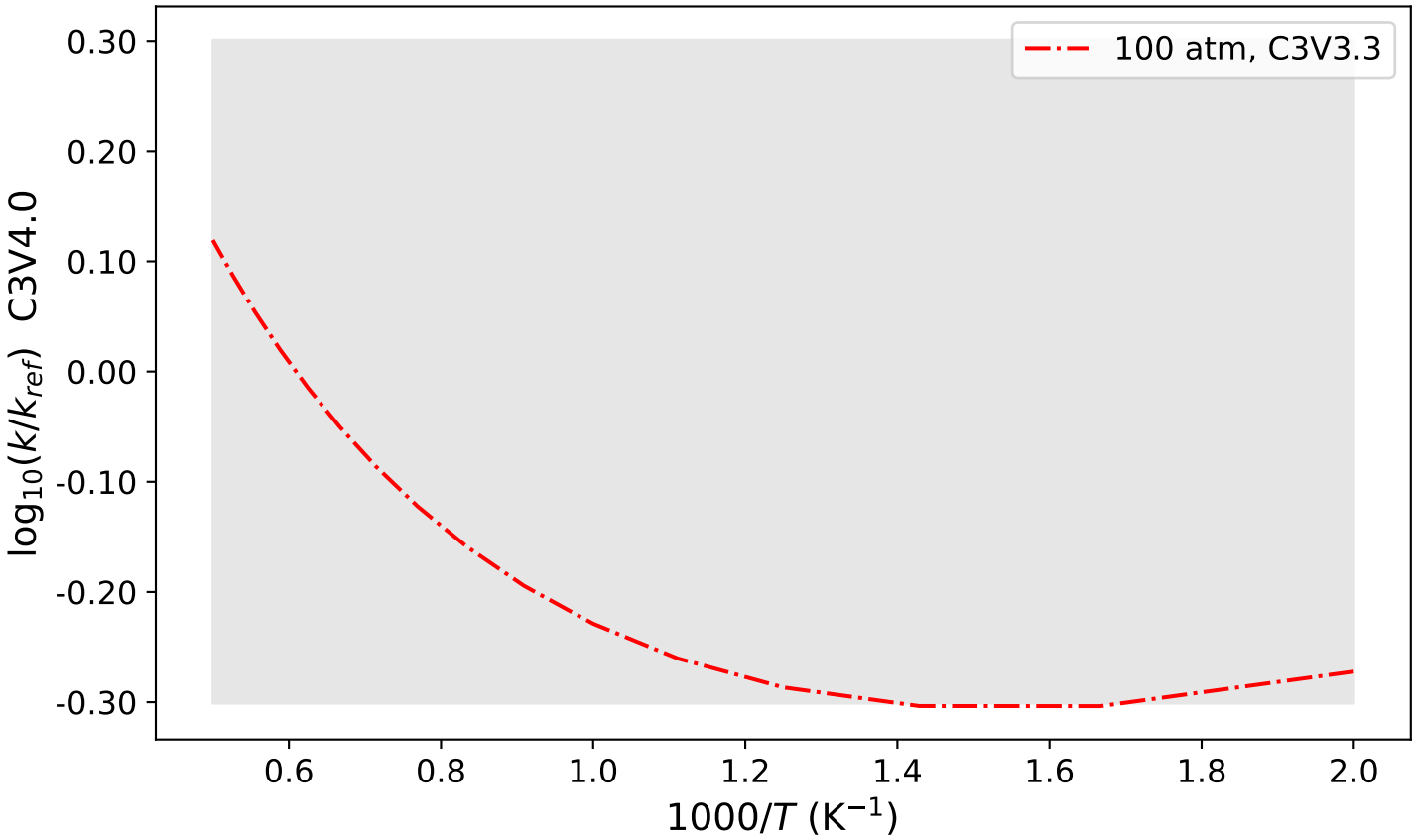
CH3 + C3H5OH = CH4 + CH2CCH2OH



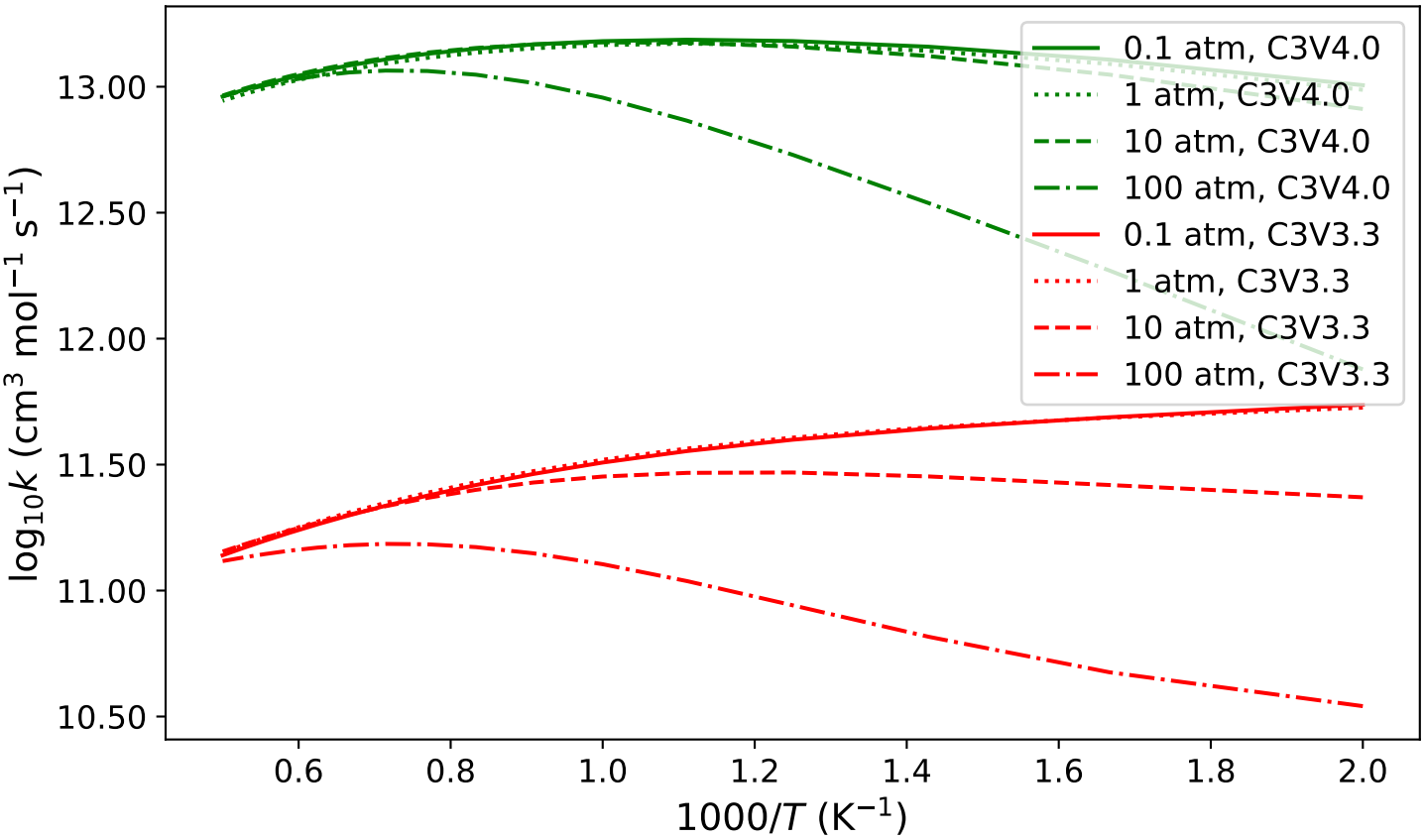



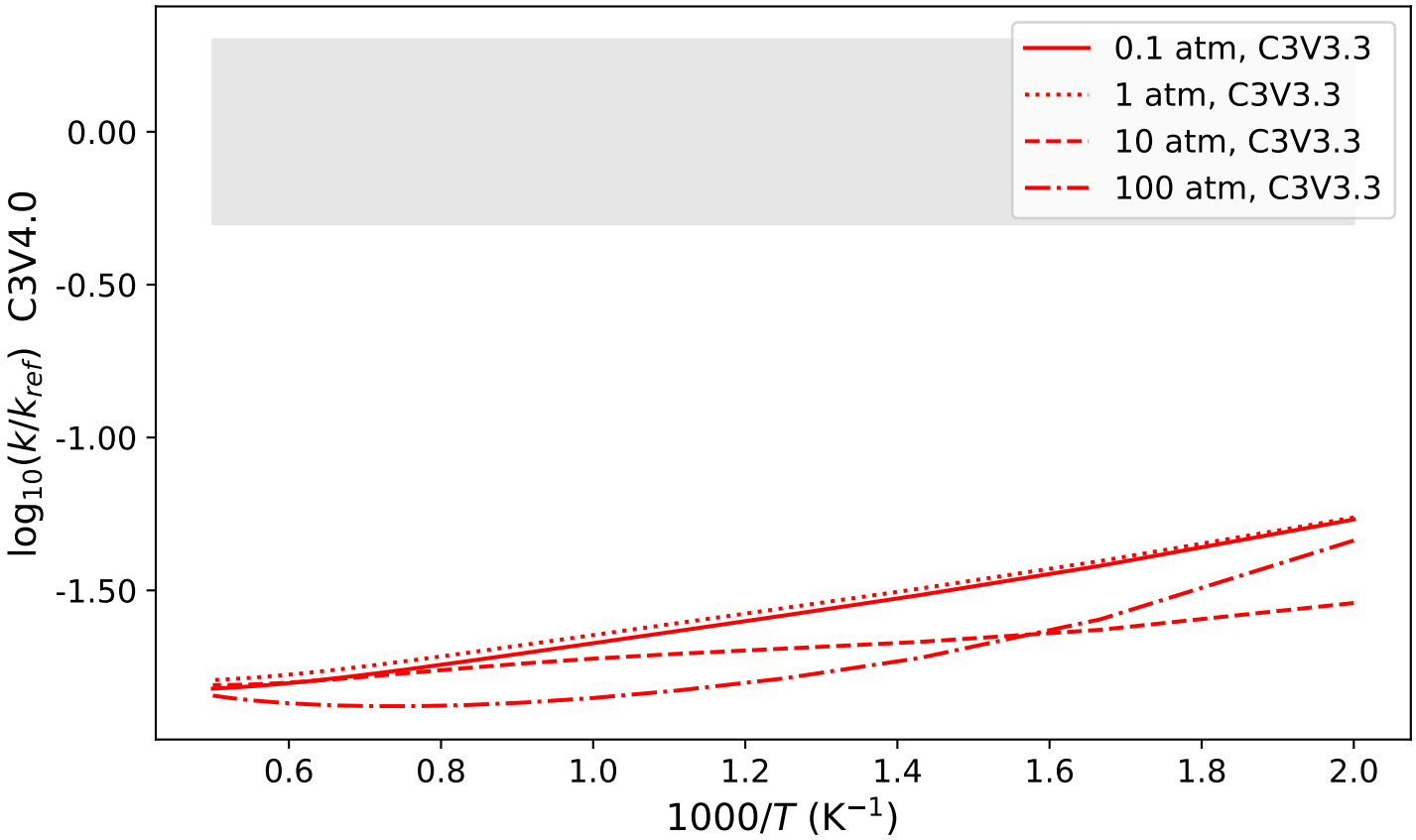
O2 + C4H8OOH2-3 = C4H8OOH2-3O2



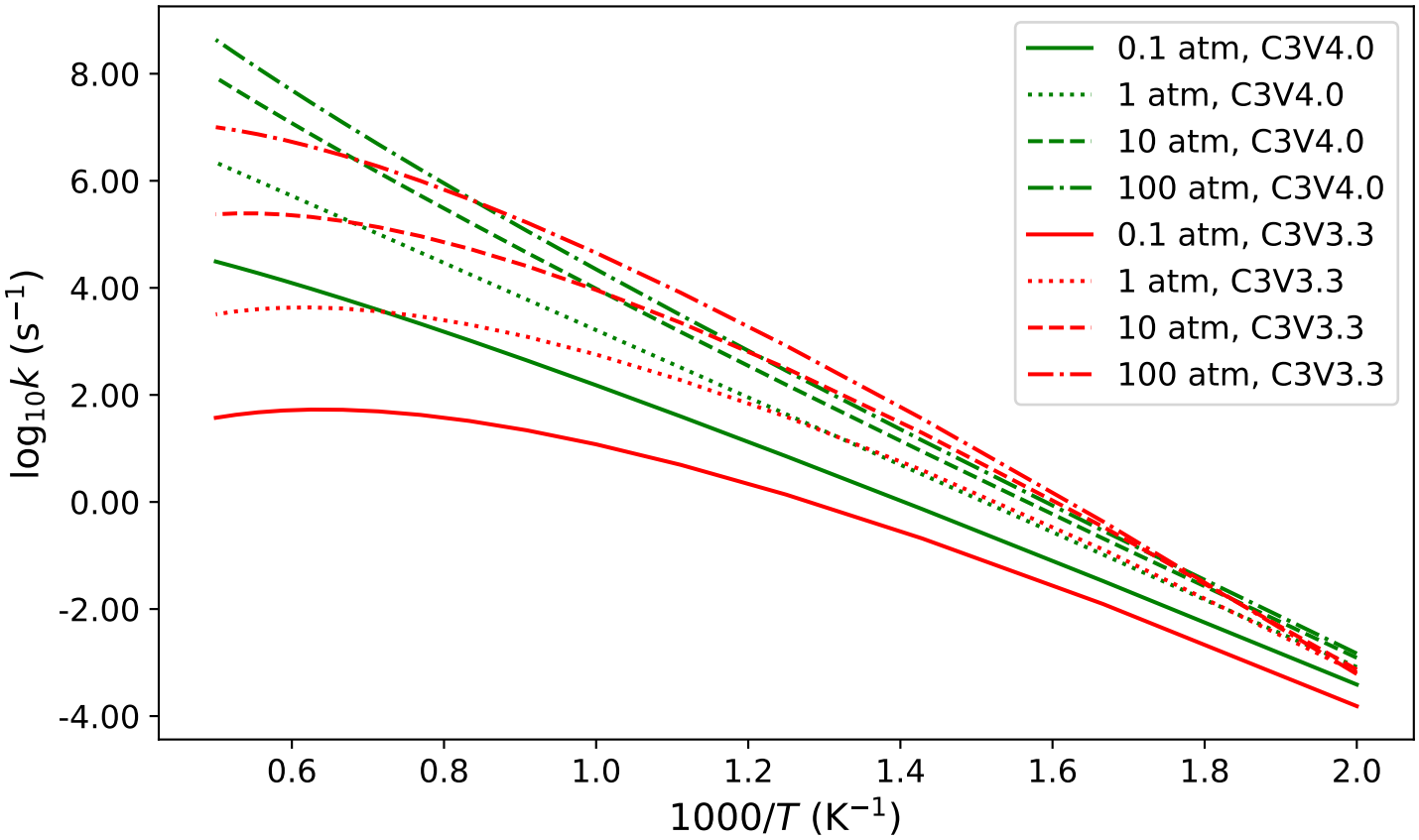



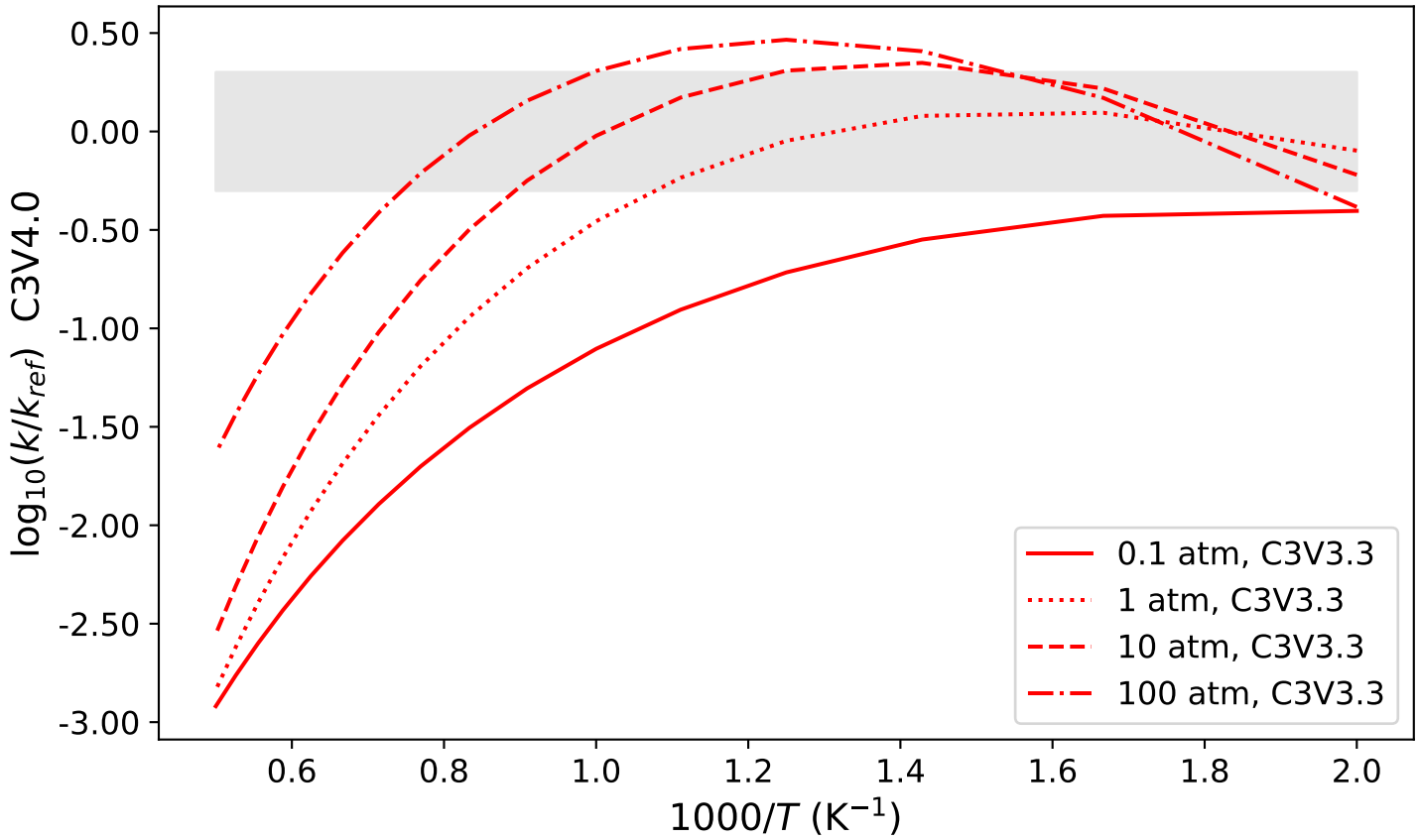
O2 + C4H5-N = HCO + C2H3CHO



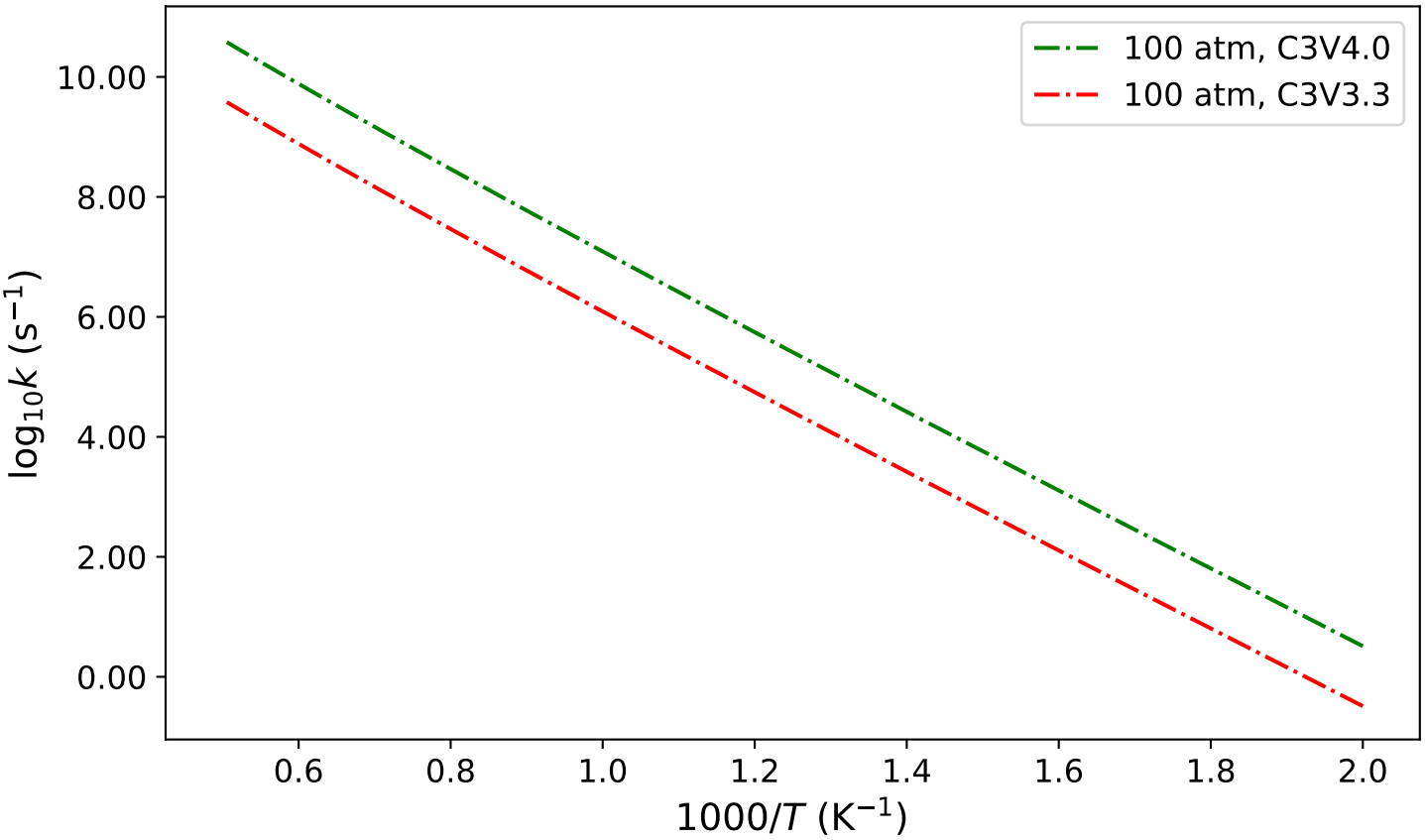



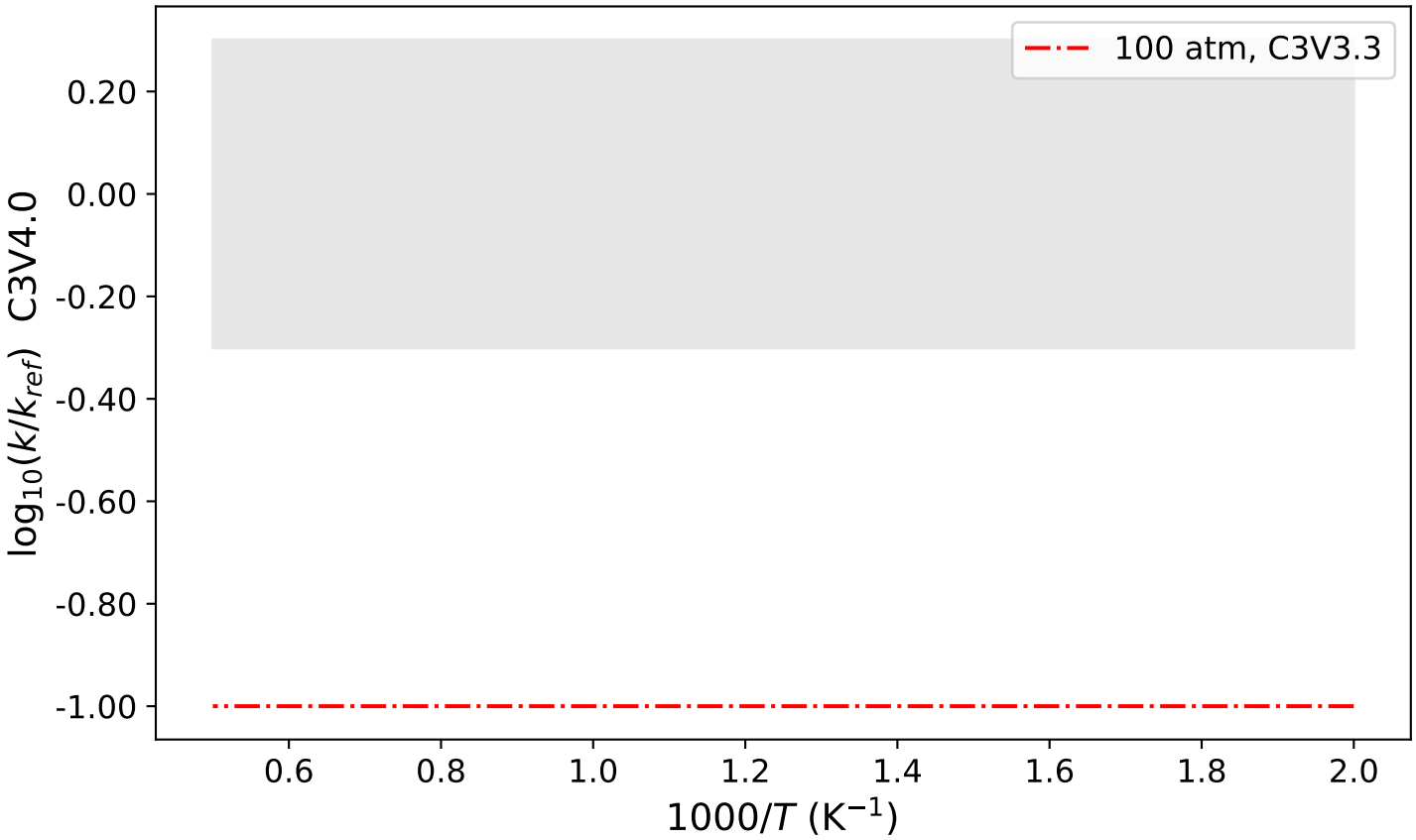
PC4H9 = SC4H9


0.1 atm, C3V4.0
1 atm, C3V4.0
10 atm, C3V4.0
100 atm, C3V4.0
0.1 atm, C3V3.3
1 atm, C3V3.3
10 atm, C3V3.3
100 atm, C3V3.3
8.00
6.00
4.00
2.00
0.00
-2.00
-4.00
$\log_{10}k$ ($s^{-1}$)
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
$1000/T$ ($K^{-1}$)

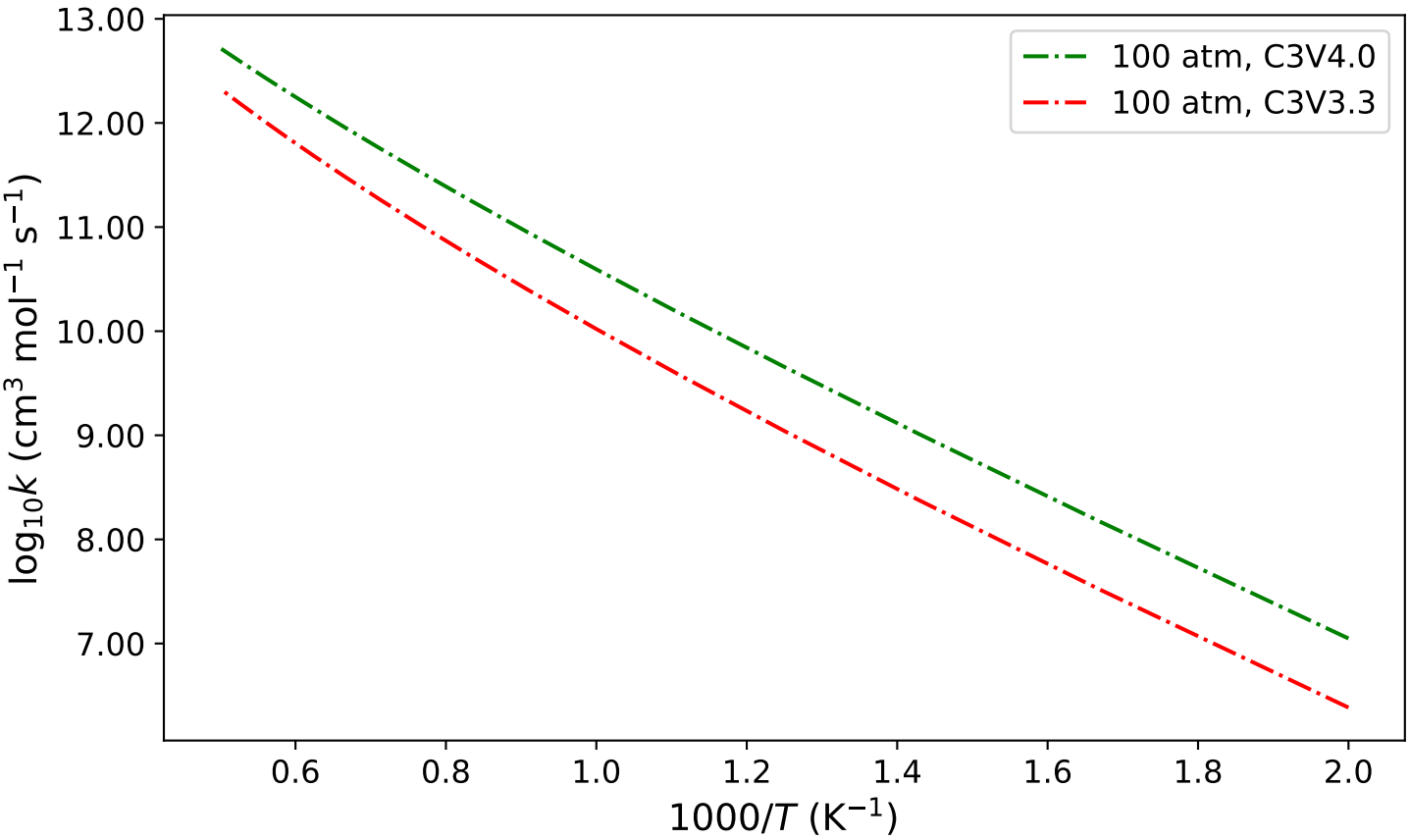

0.50
0.00
-0.50
-1.00
-1.50
-2.00
-2.50
-3.00
$\log_{10}(k/k_{ref})$ C3V4.0
0.1 atm, C3V3.3
1 atm, C3V3.3
10 atm, C3V3.3
100 atm, C3V3.3
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
$1000/T$ ($K^{-1}$)

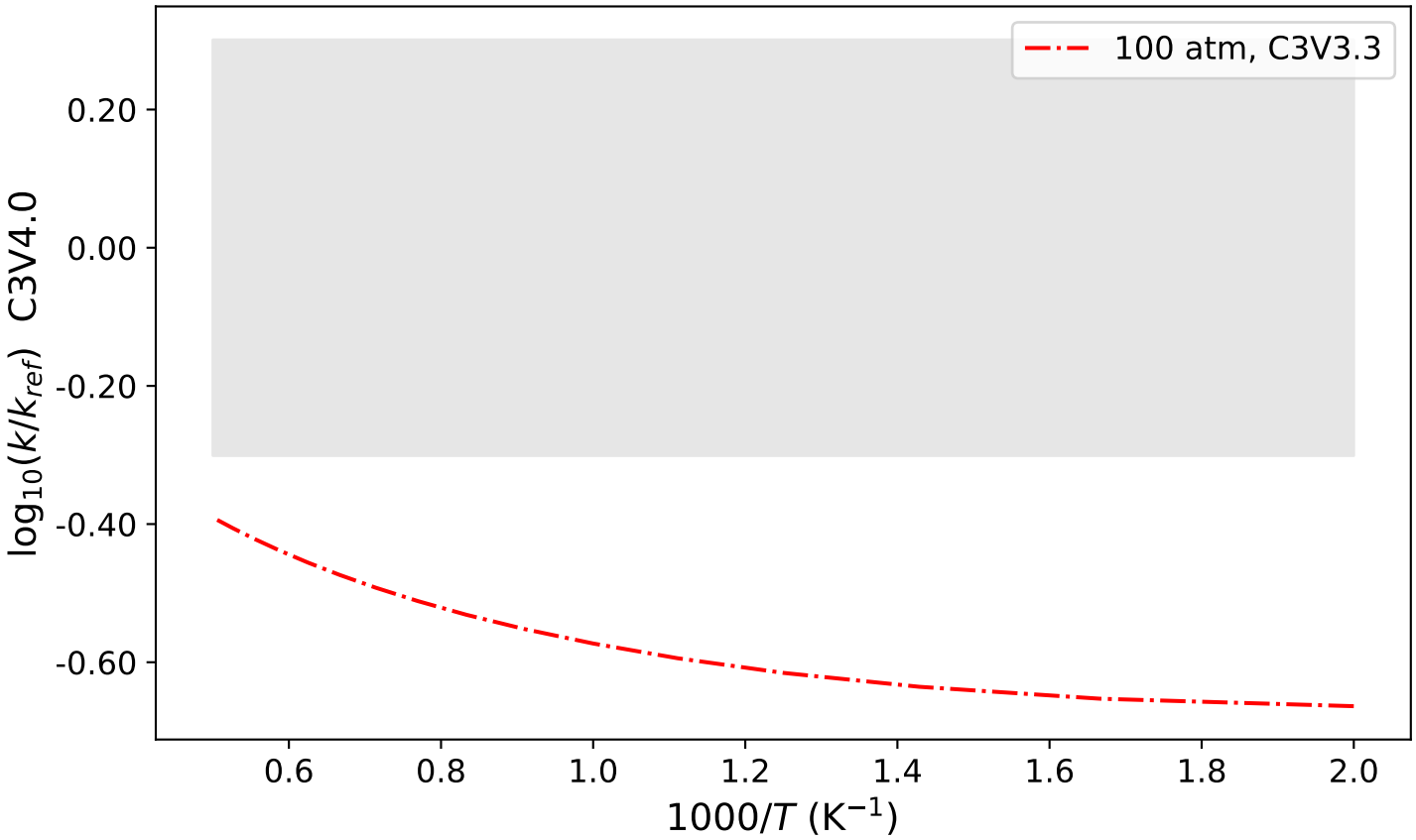
C4H61OOH4-3O2 = OH + C4H61KET43



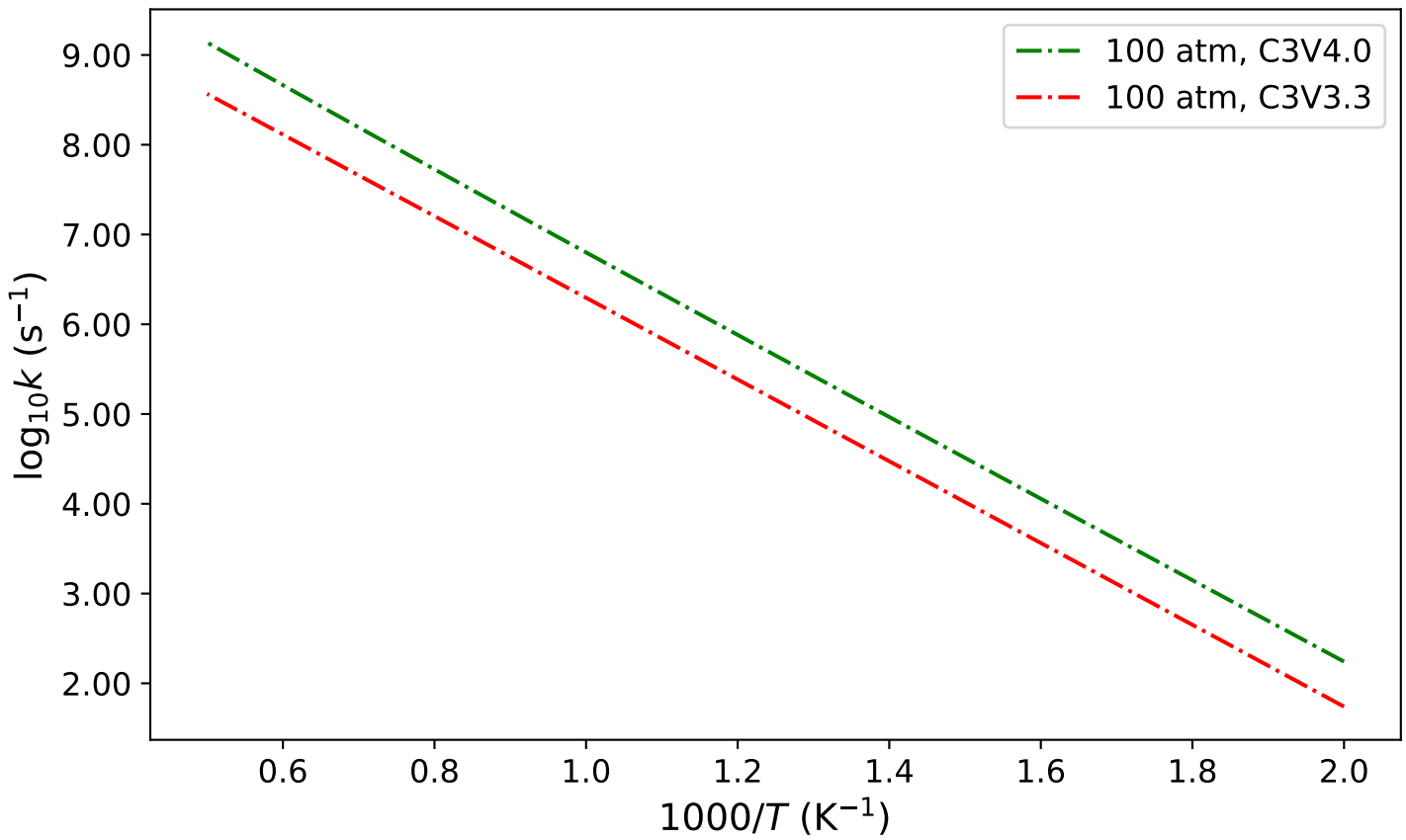



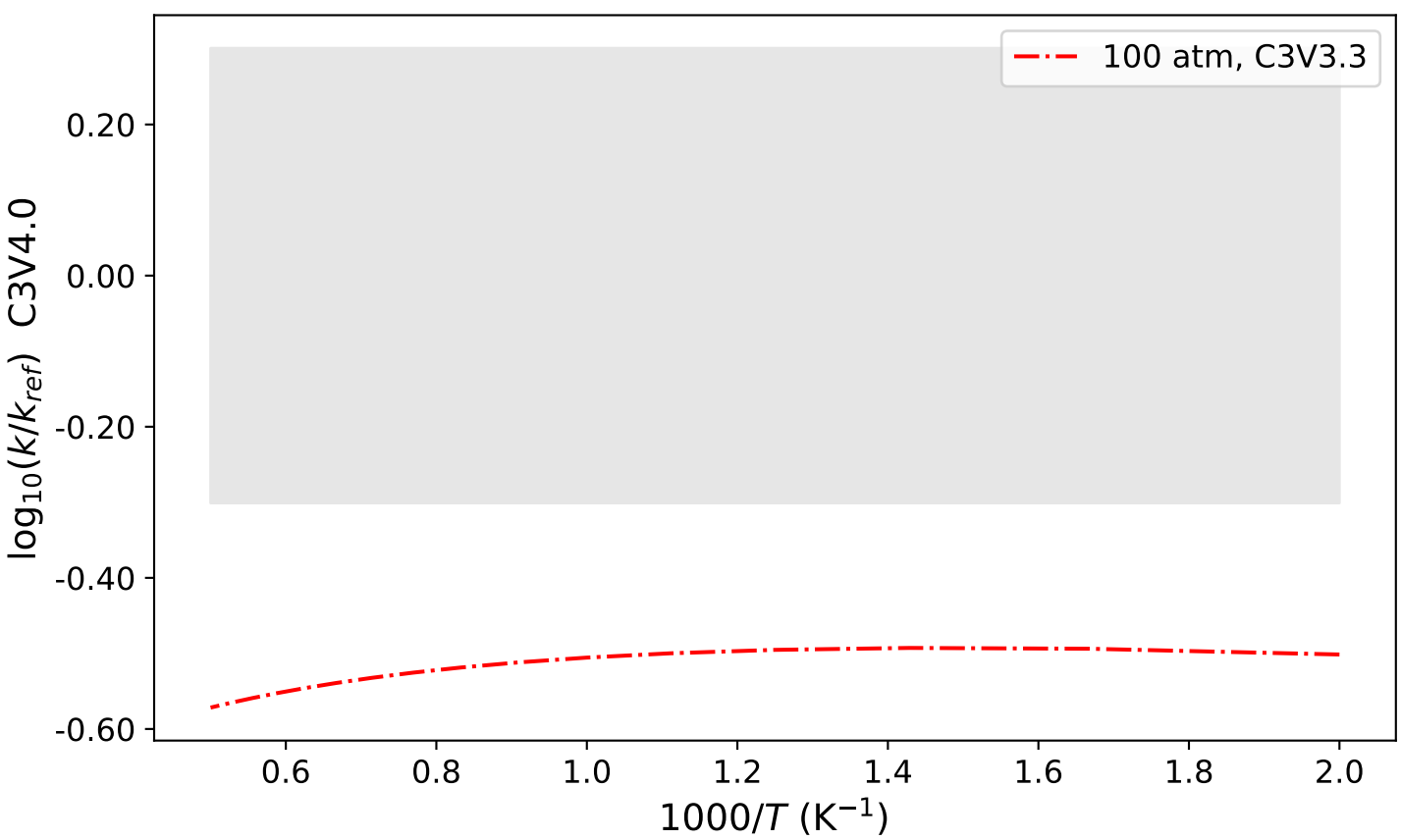
H + IC4H8 = H2 + IC4H7-I1



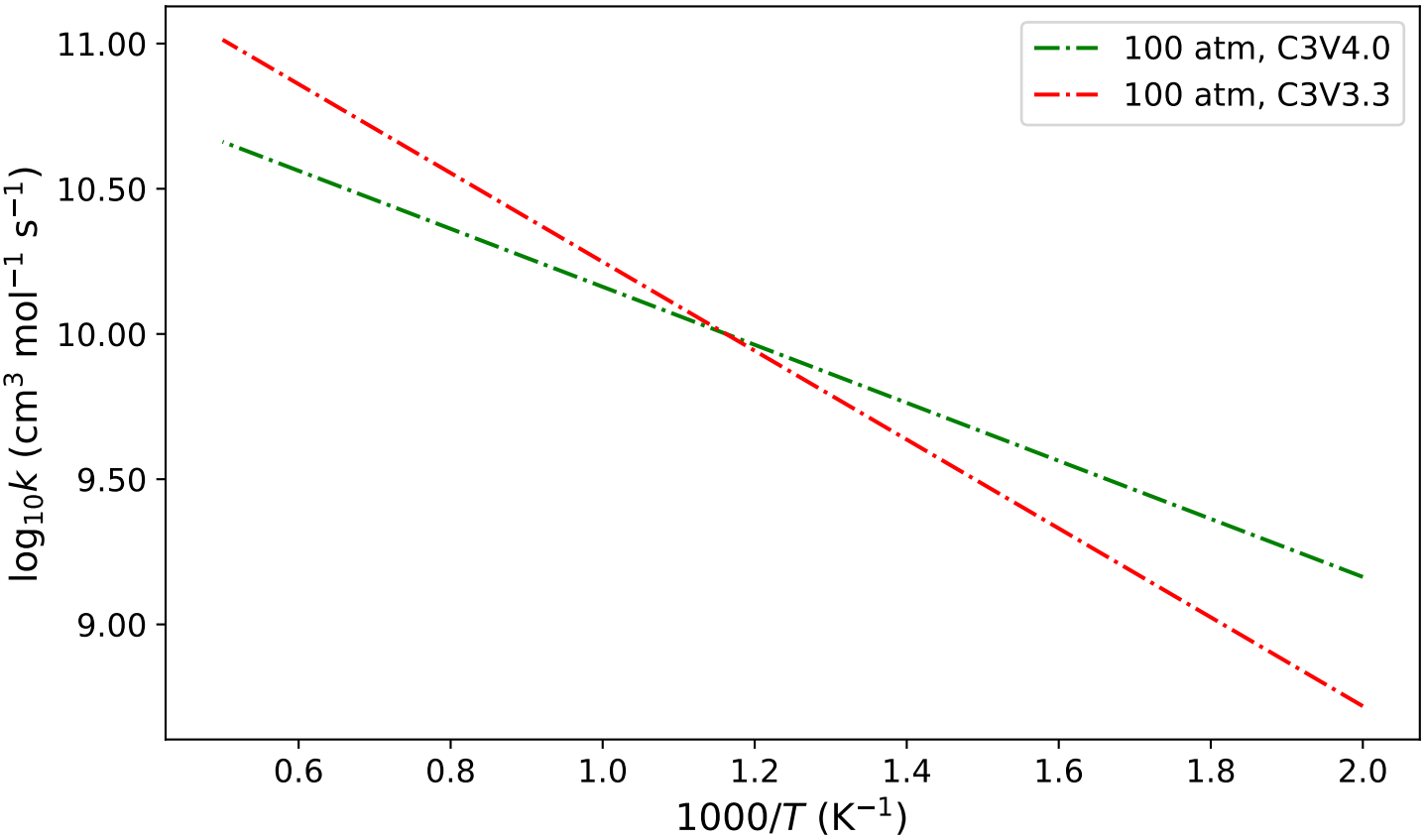



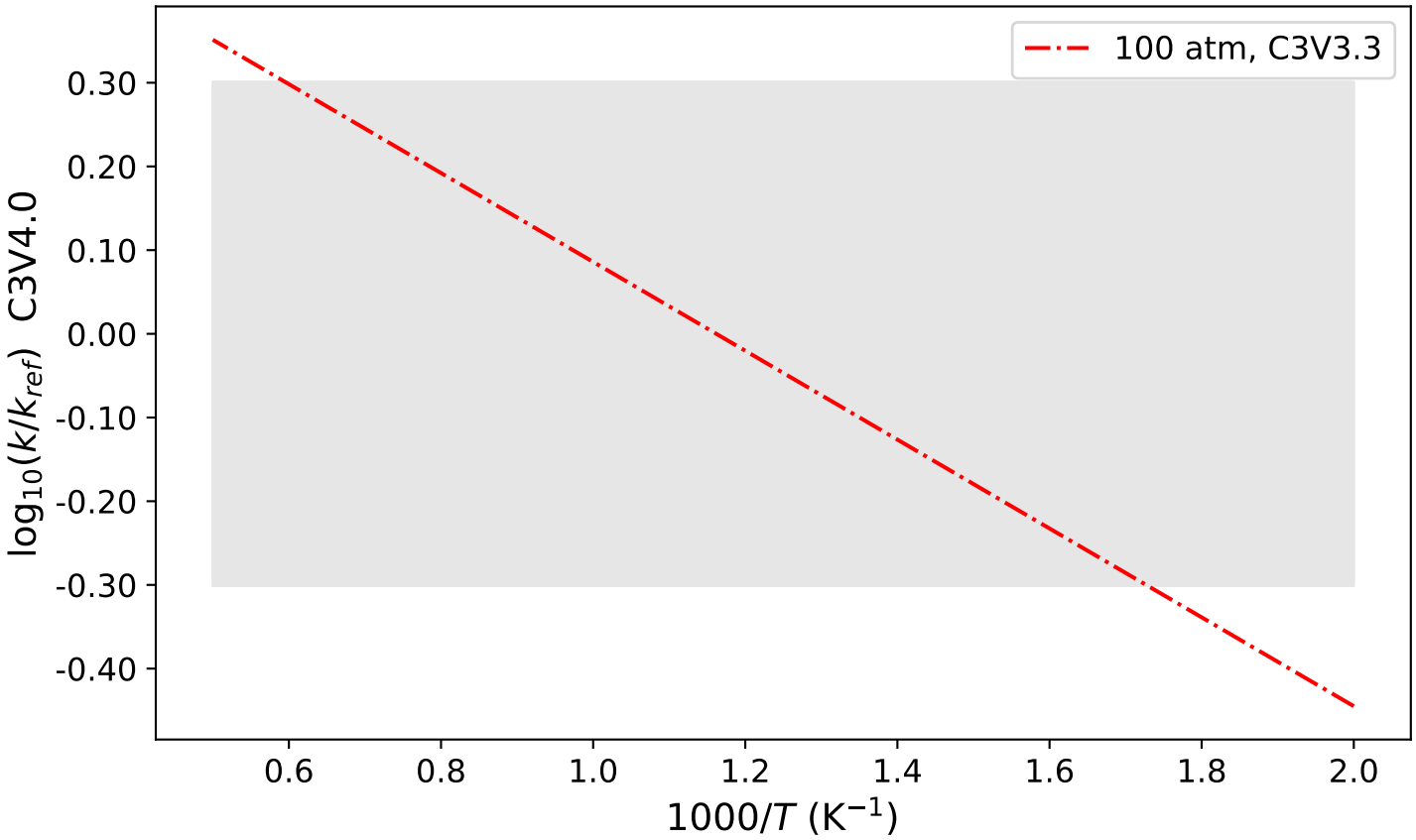
C2H3CHOHCH2OO = C4H6O1-3OOH4



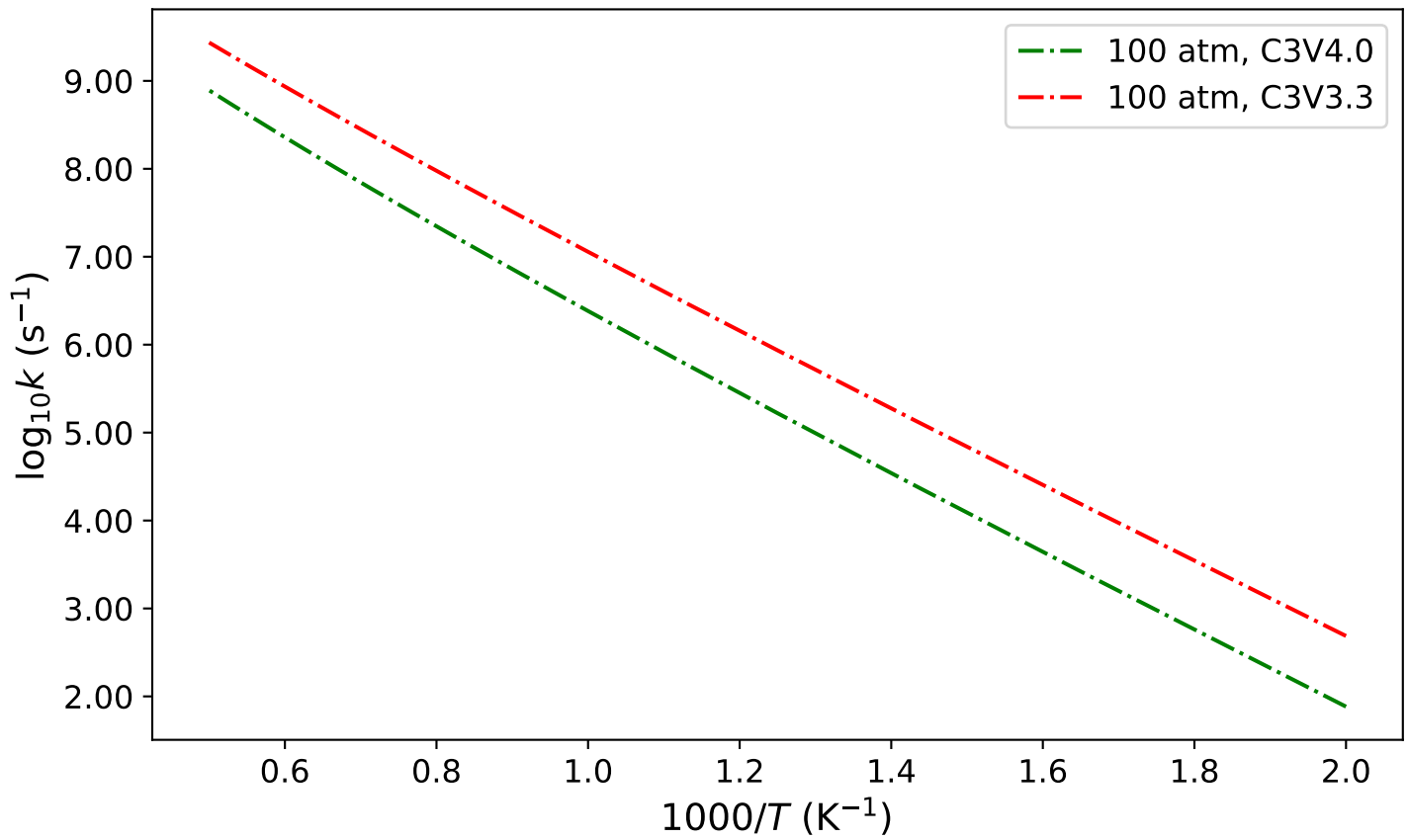



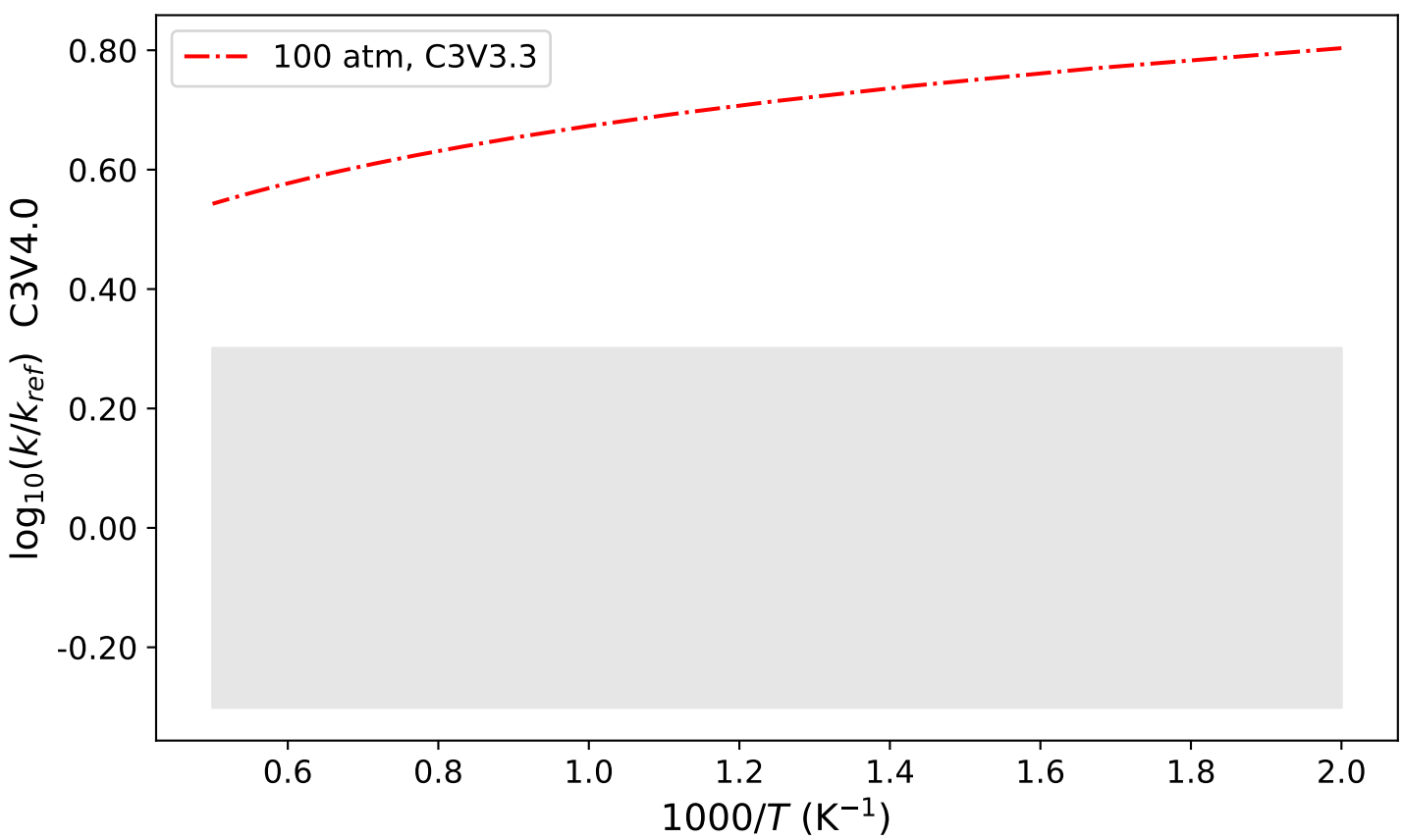
CH3O + C4H10 = CH3OH + SC4H9



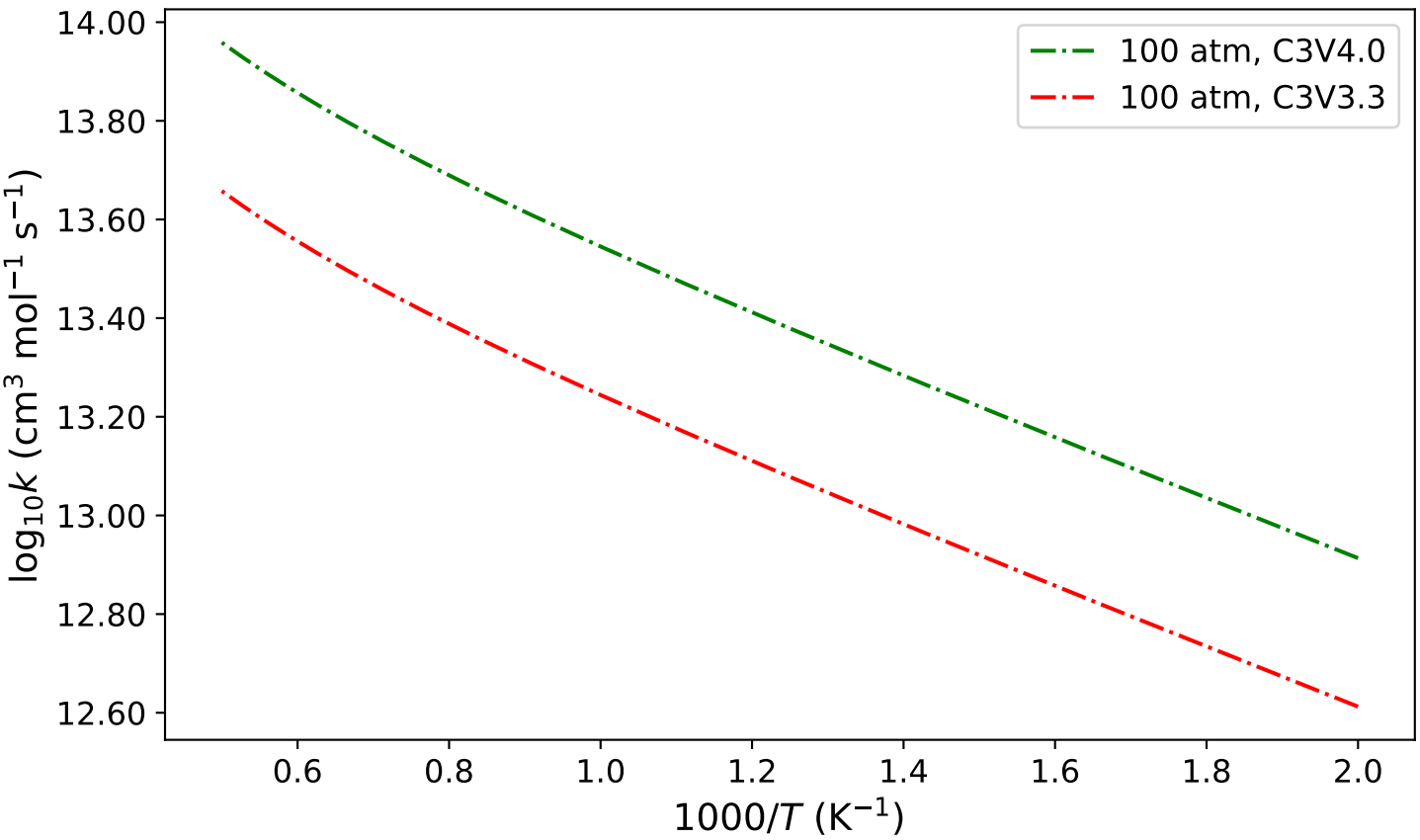



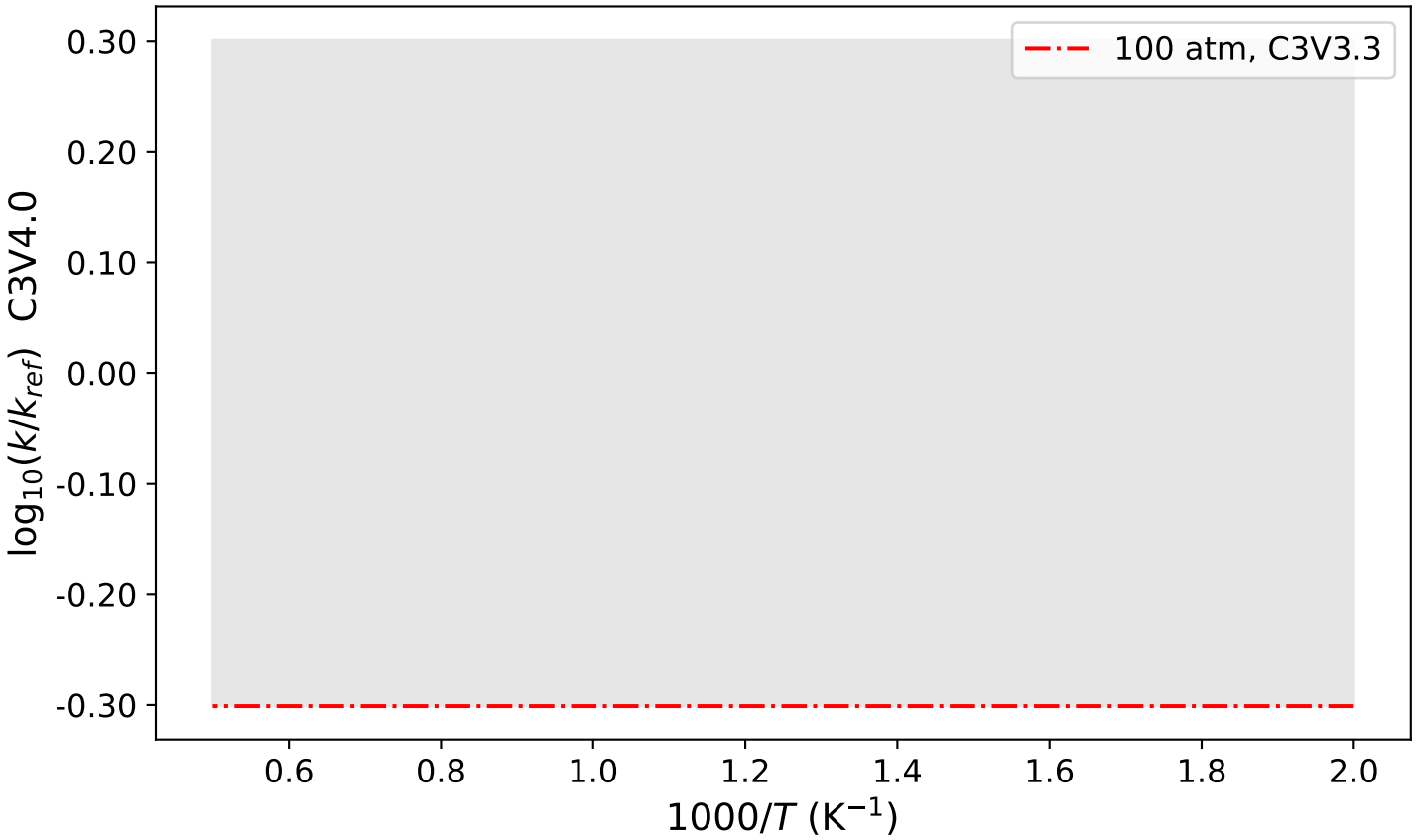
PC4H9O2 = C4H8OOH1-3



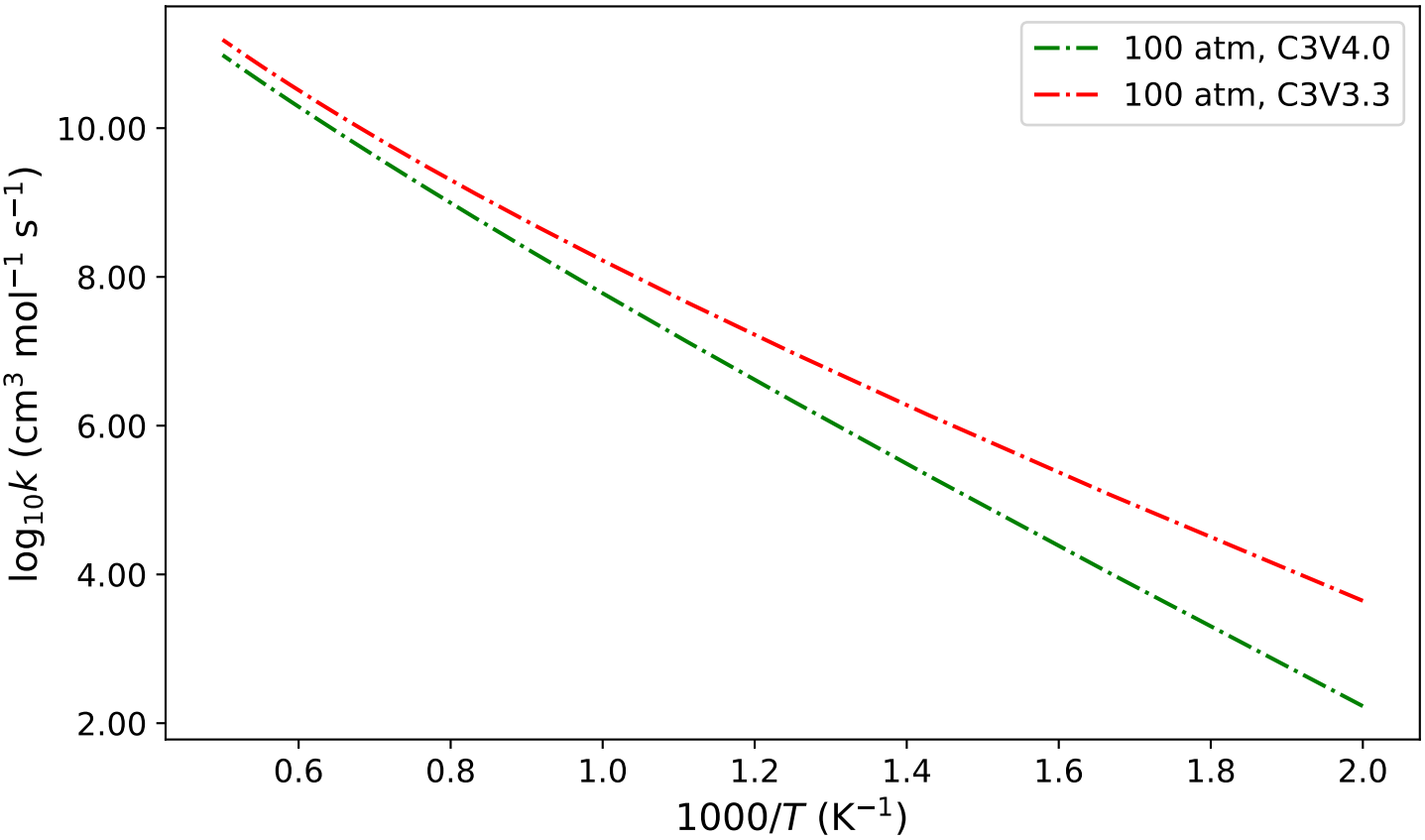



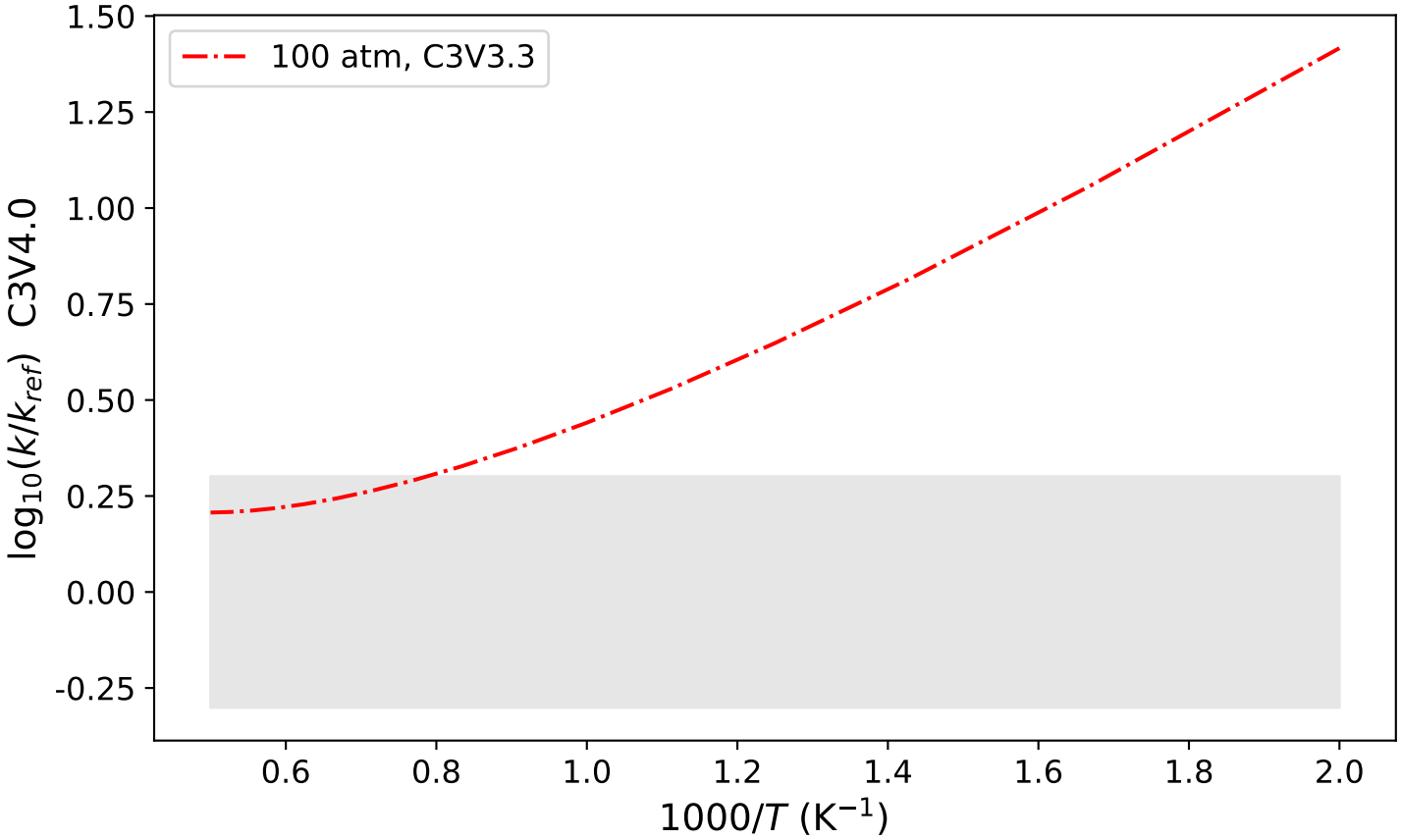
H + C4H4O = C3H5-A + CO



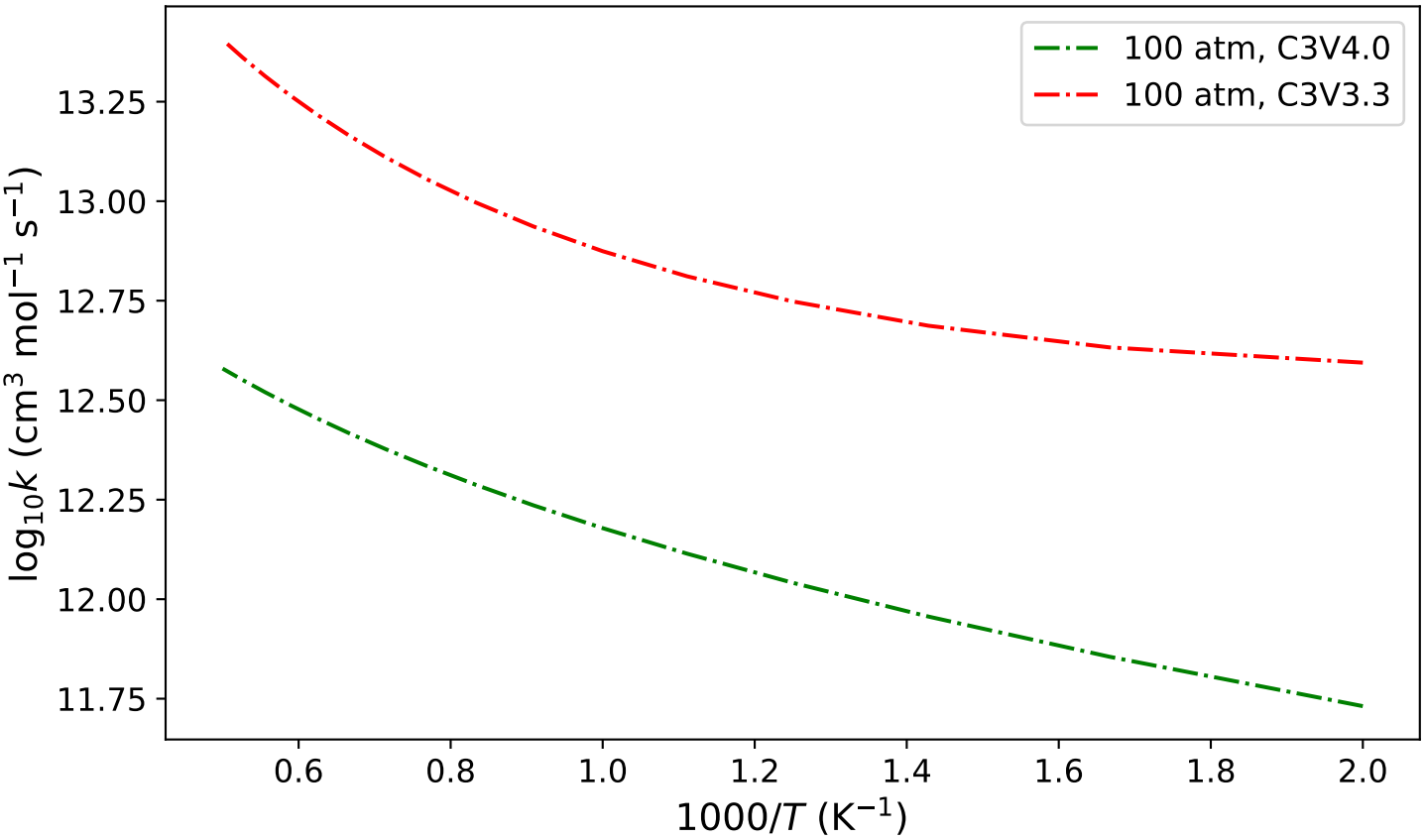



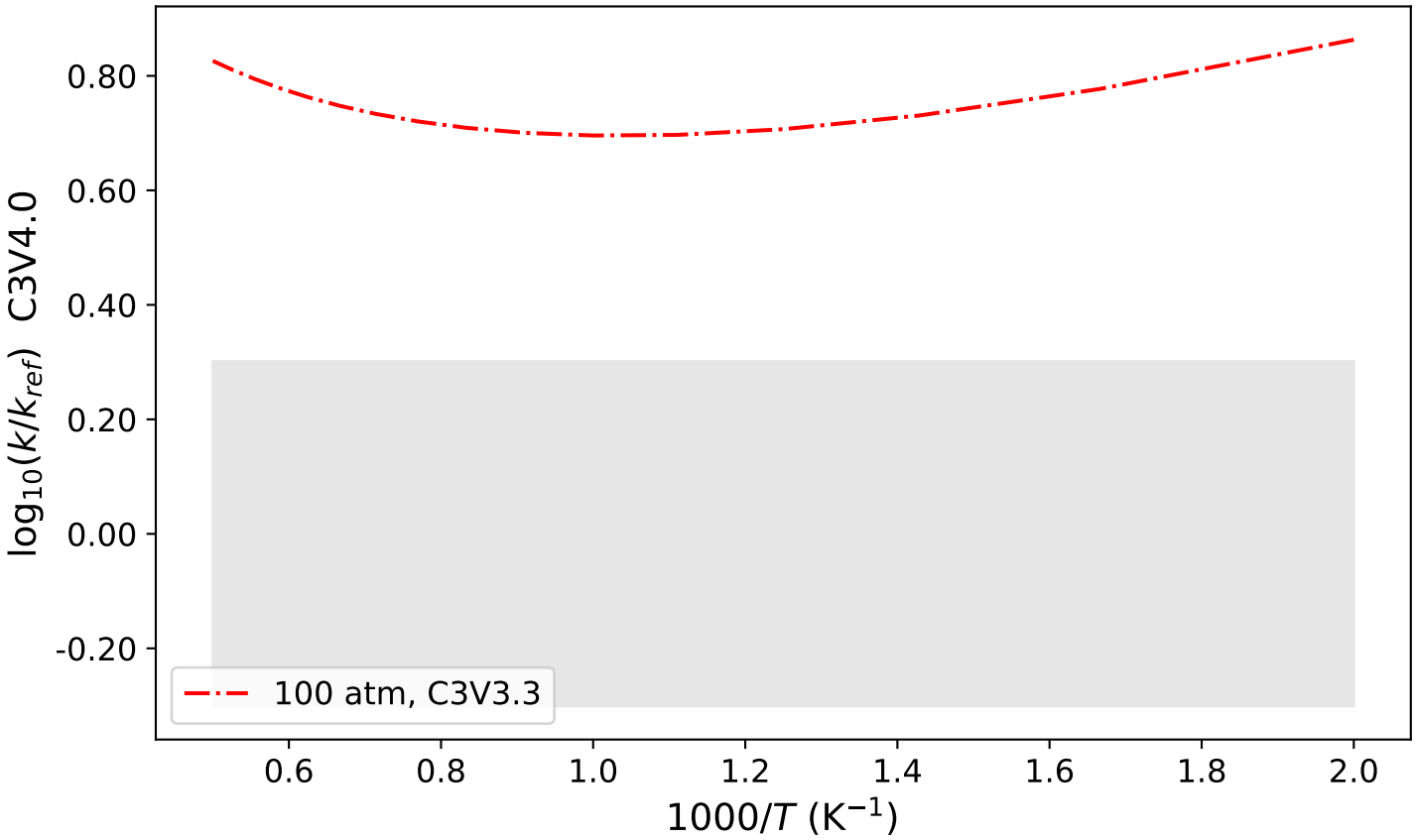
HO2 + C4H6 = H2O2 + C4H5-I



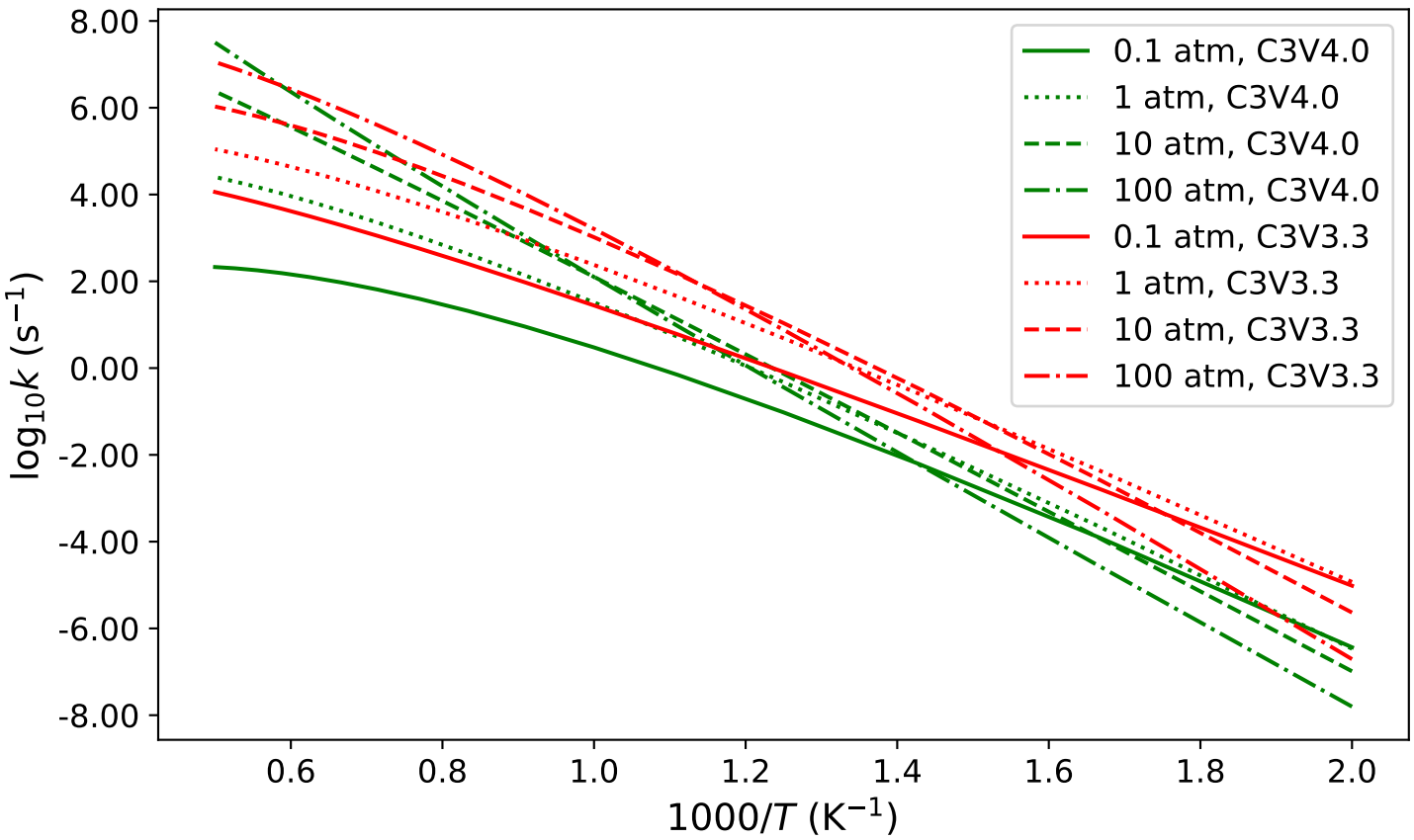



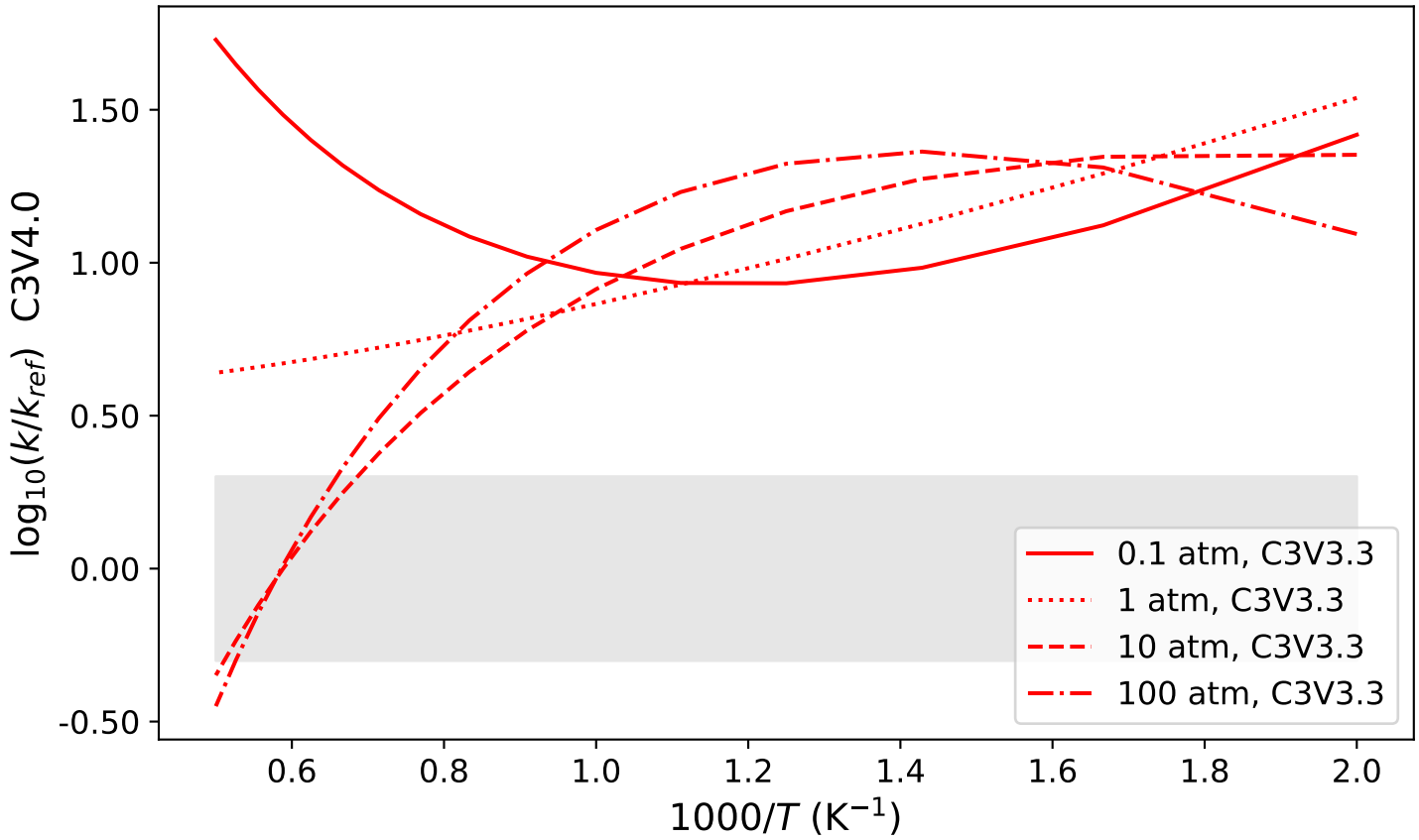
OH + C2H3CHO = H2O + C2H3CO



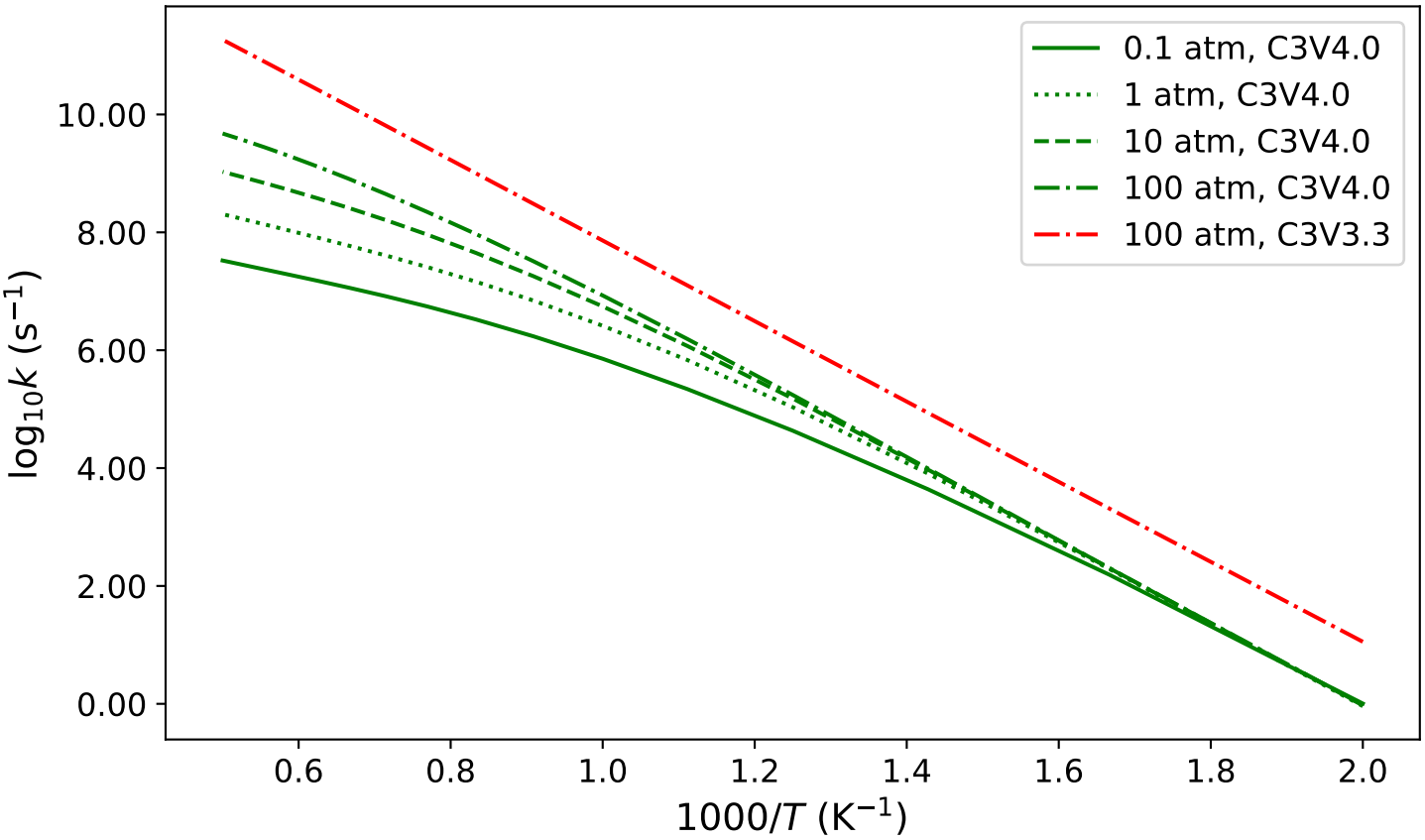



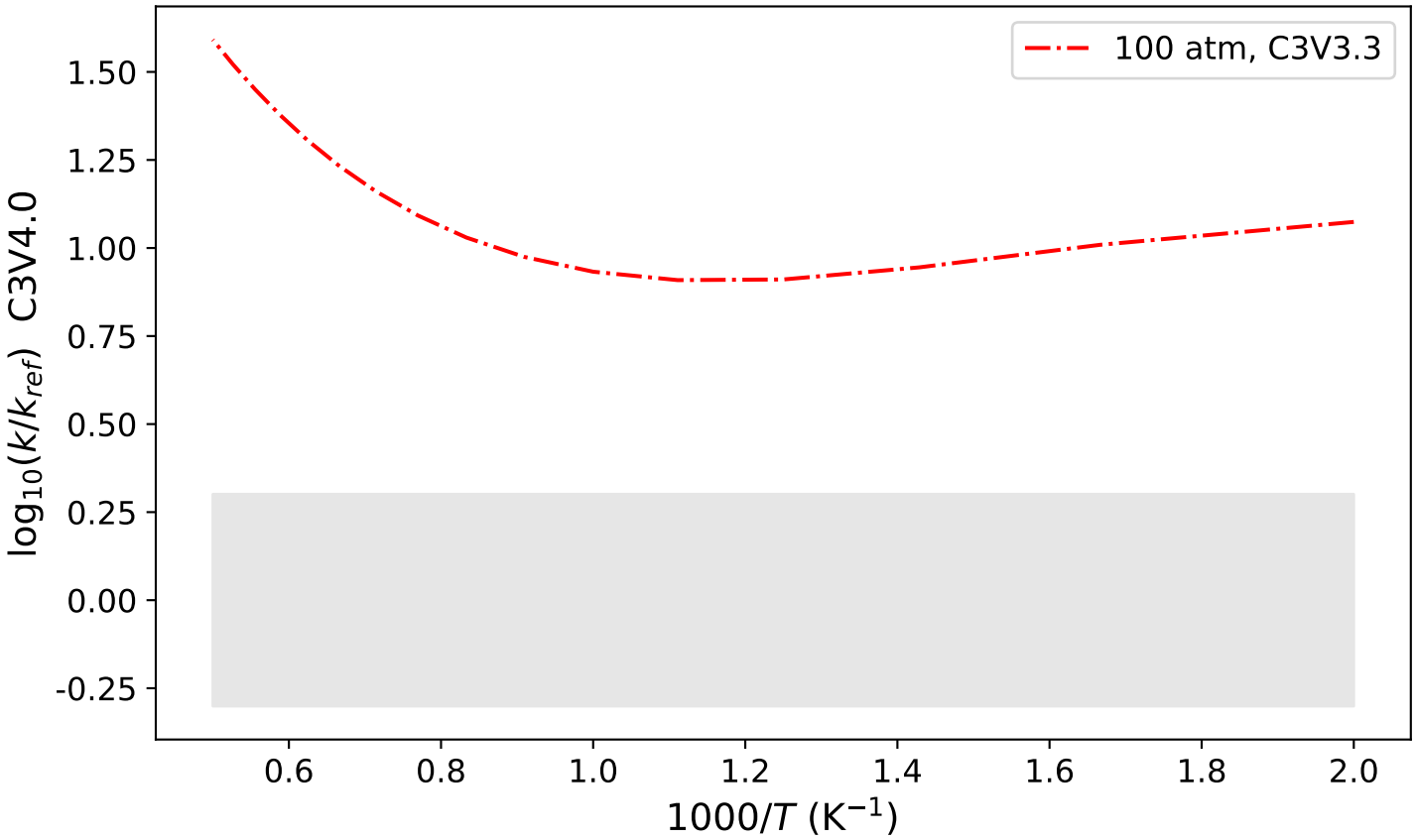
PC4H9 = H + C4H8-2



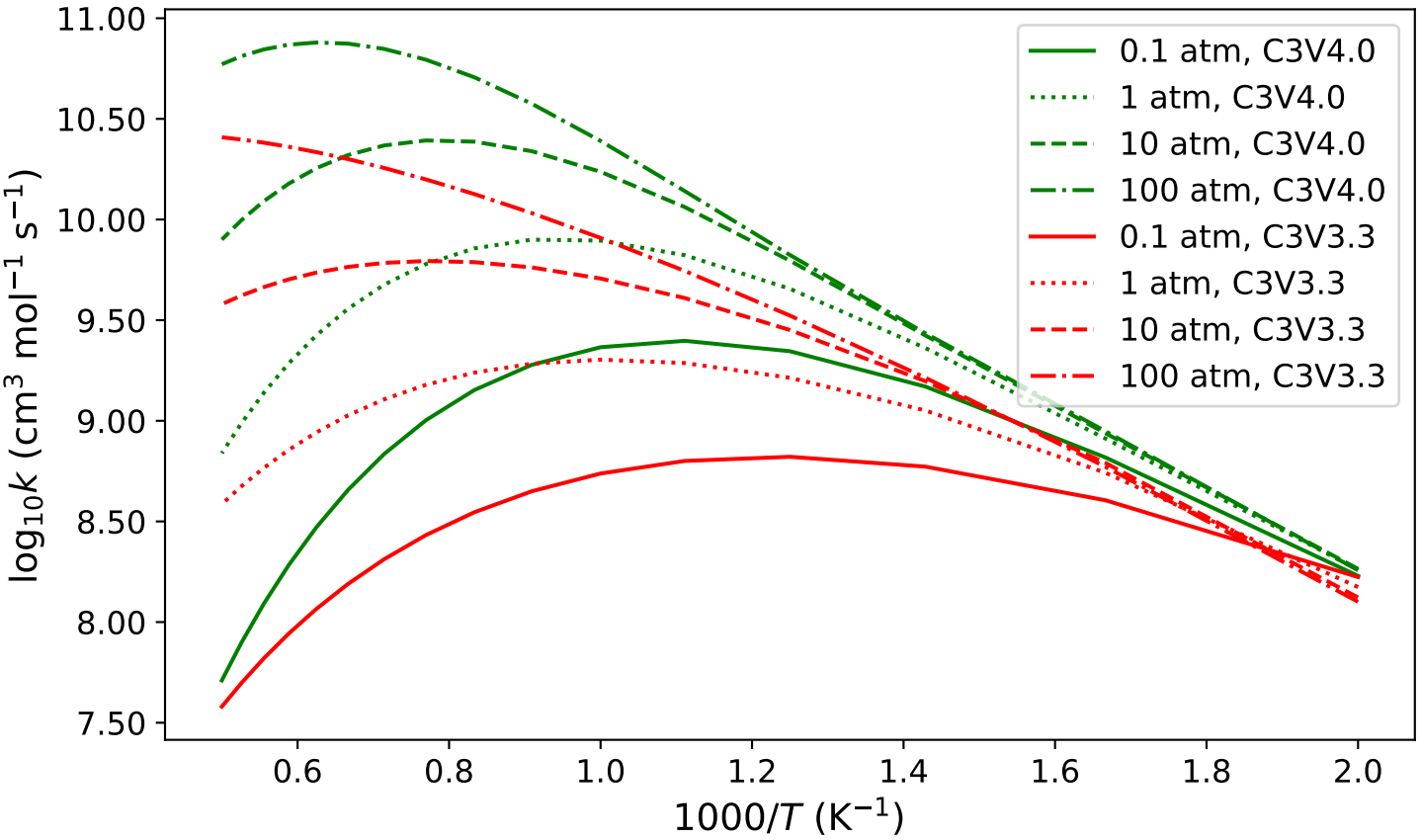



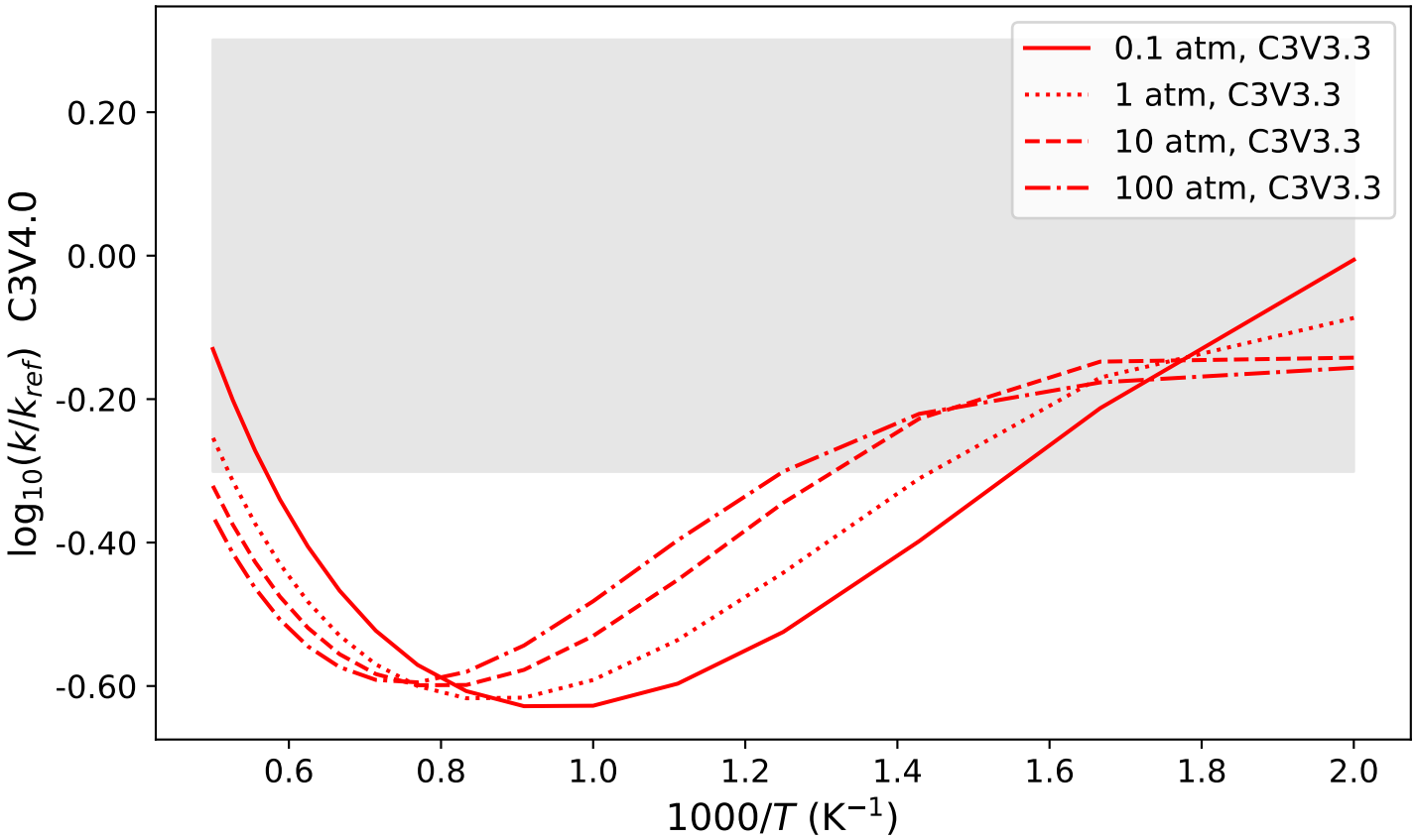
PC4H8OH-1 = C2H5 + C2H3OH



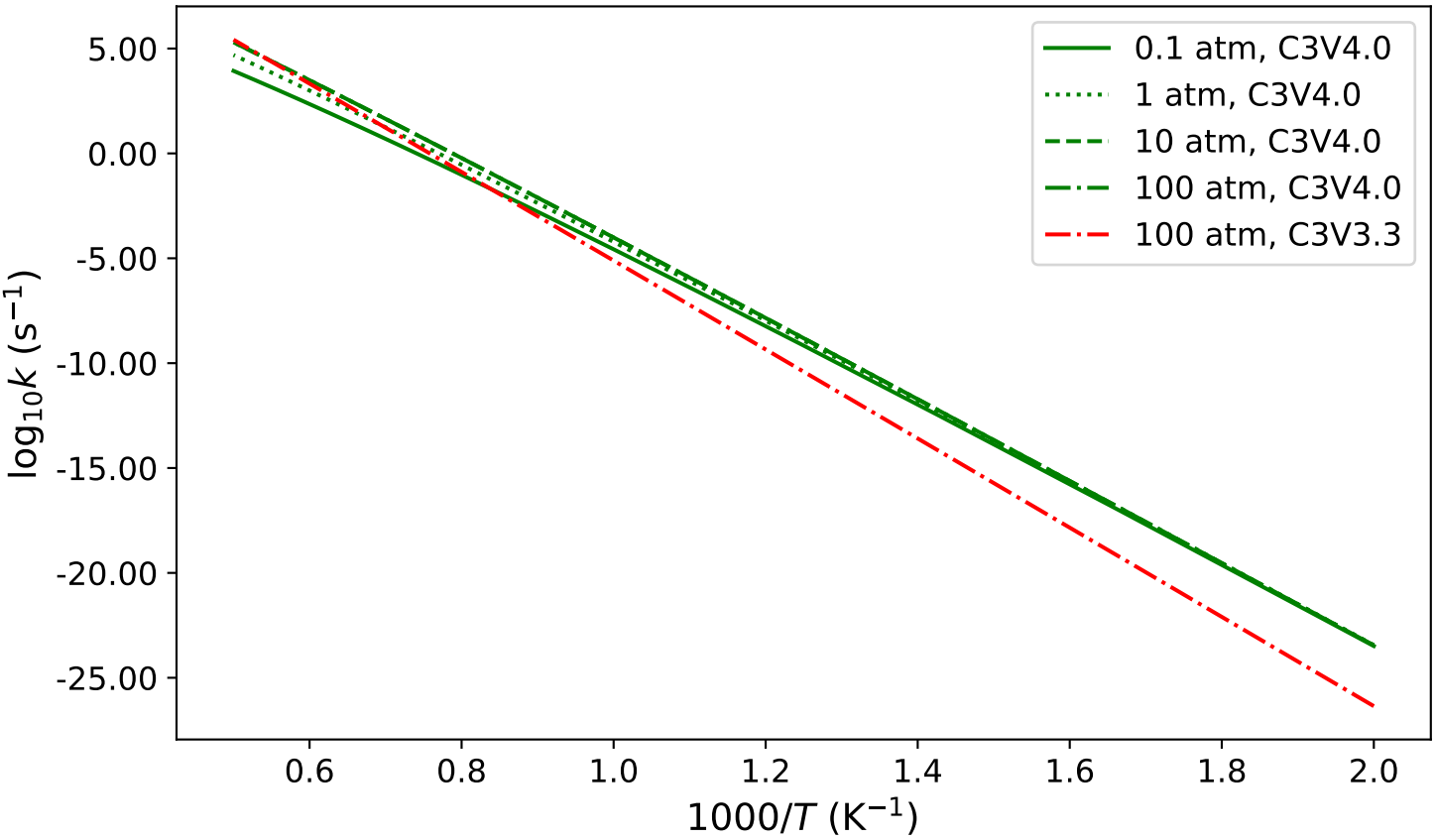



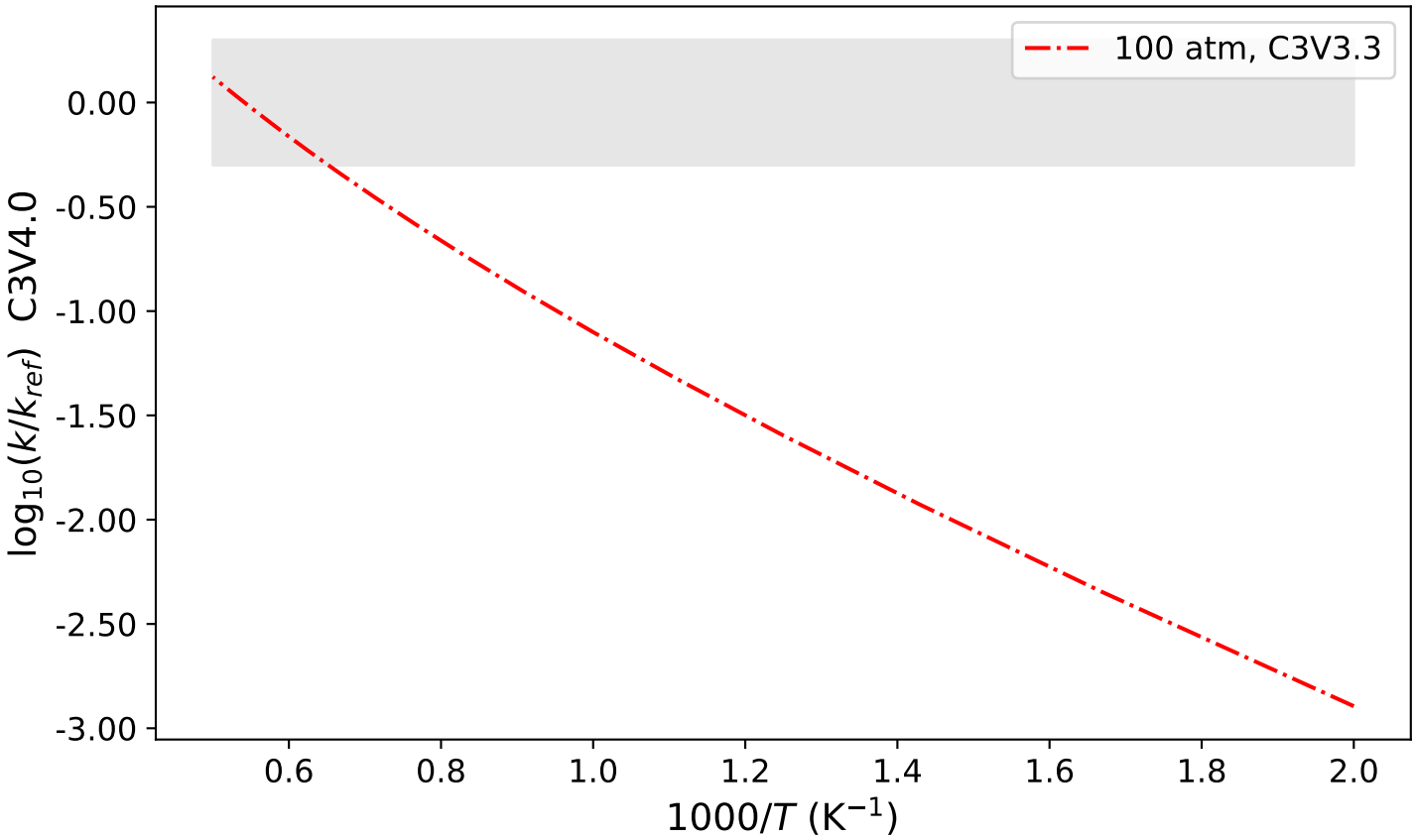
CH3 + C3H6 = SC4H9



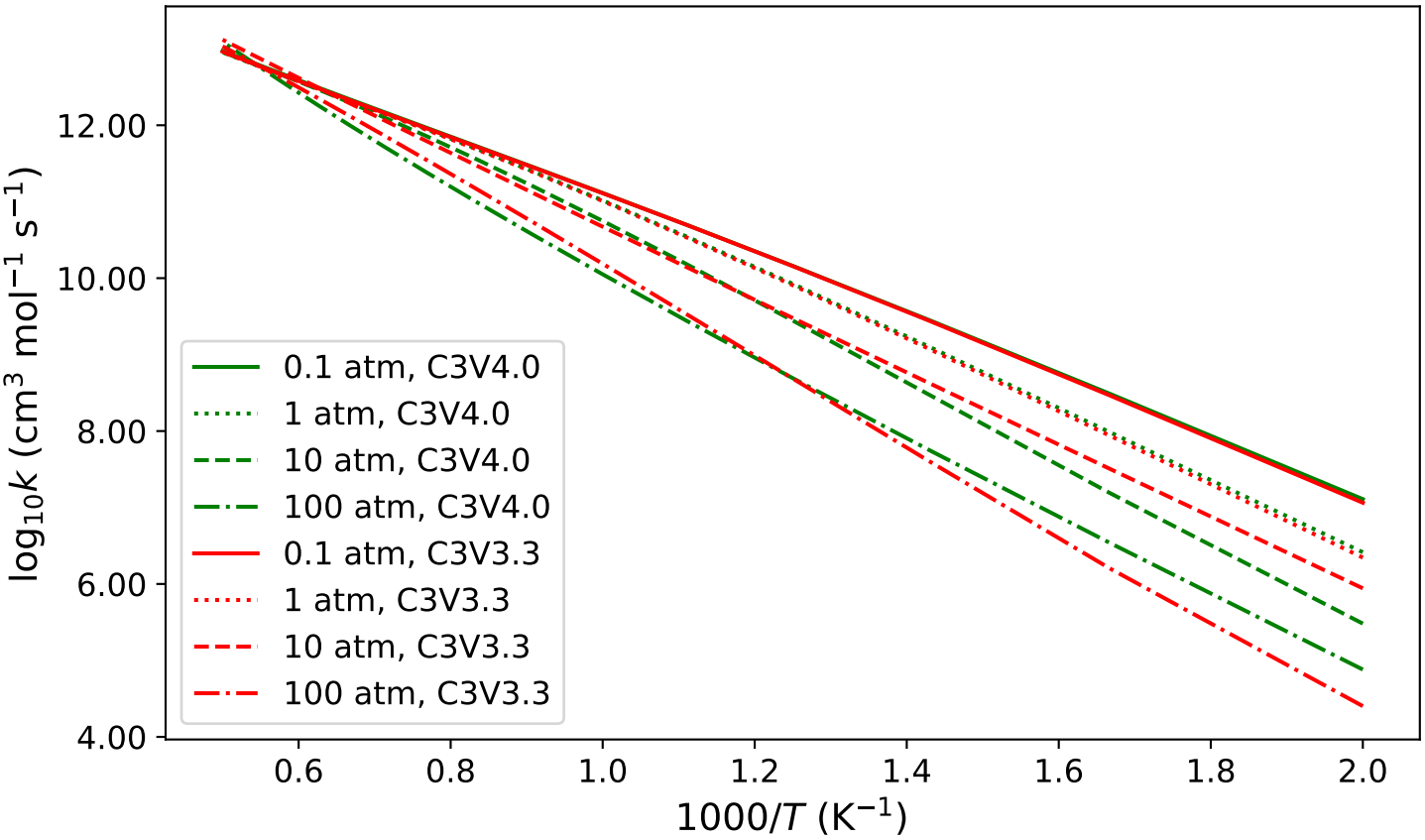



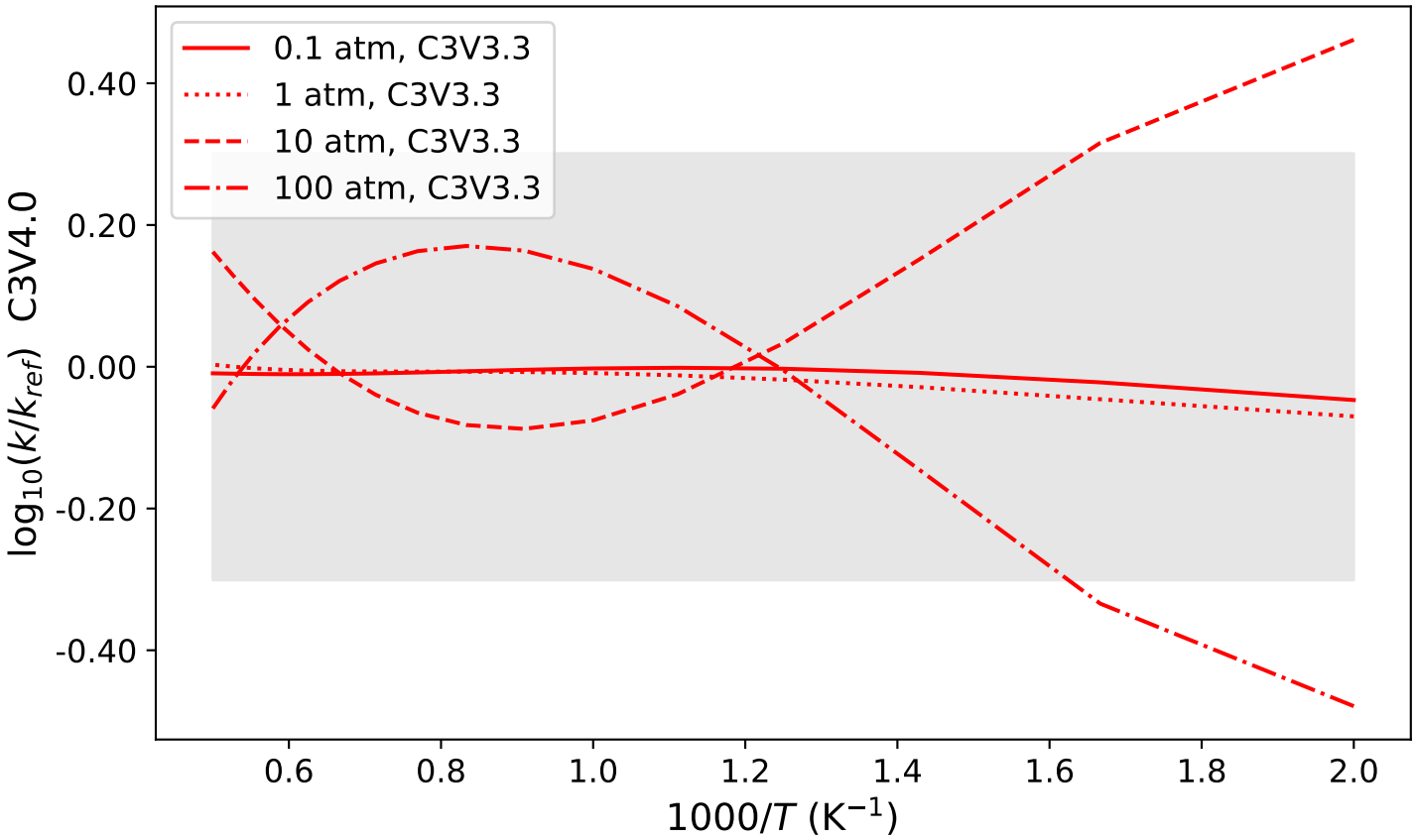
C4H6-1 = H + C4H5-3



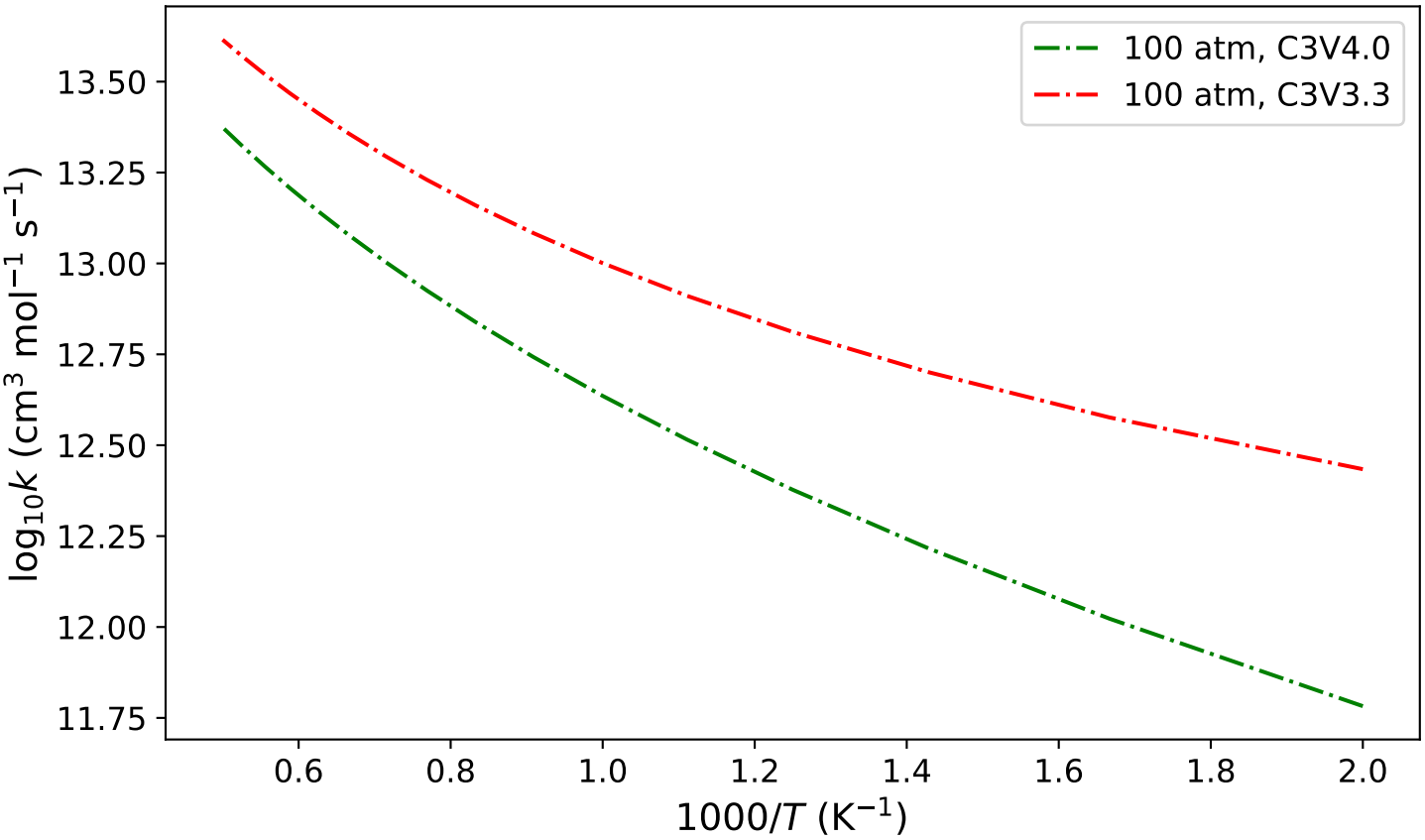



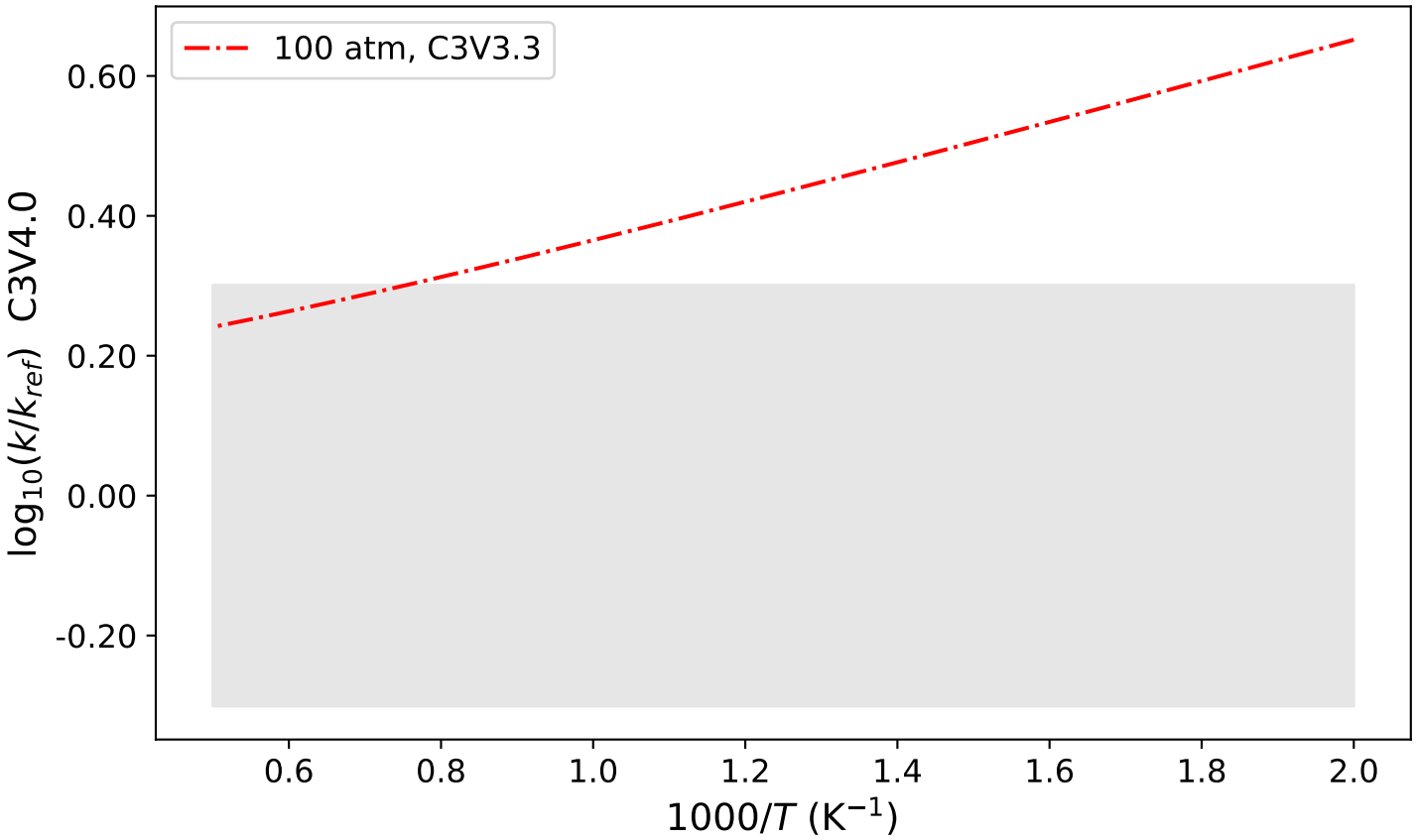
H + C4H6 = H + C4H612



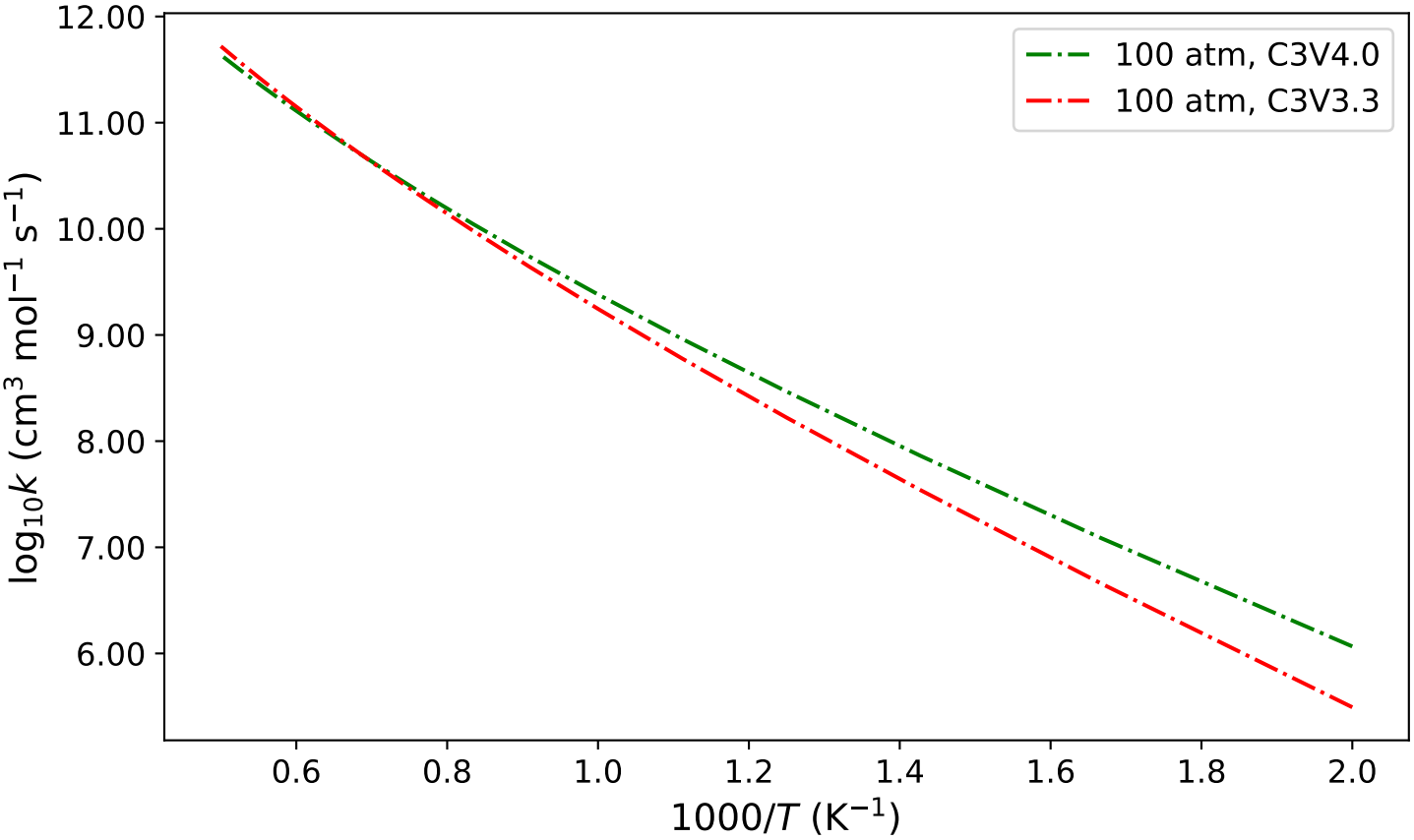



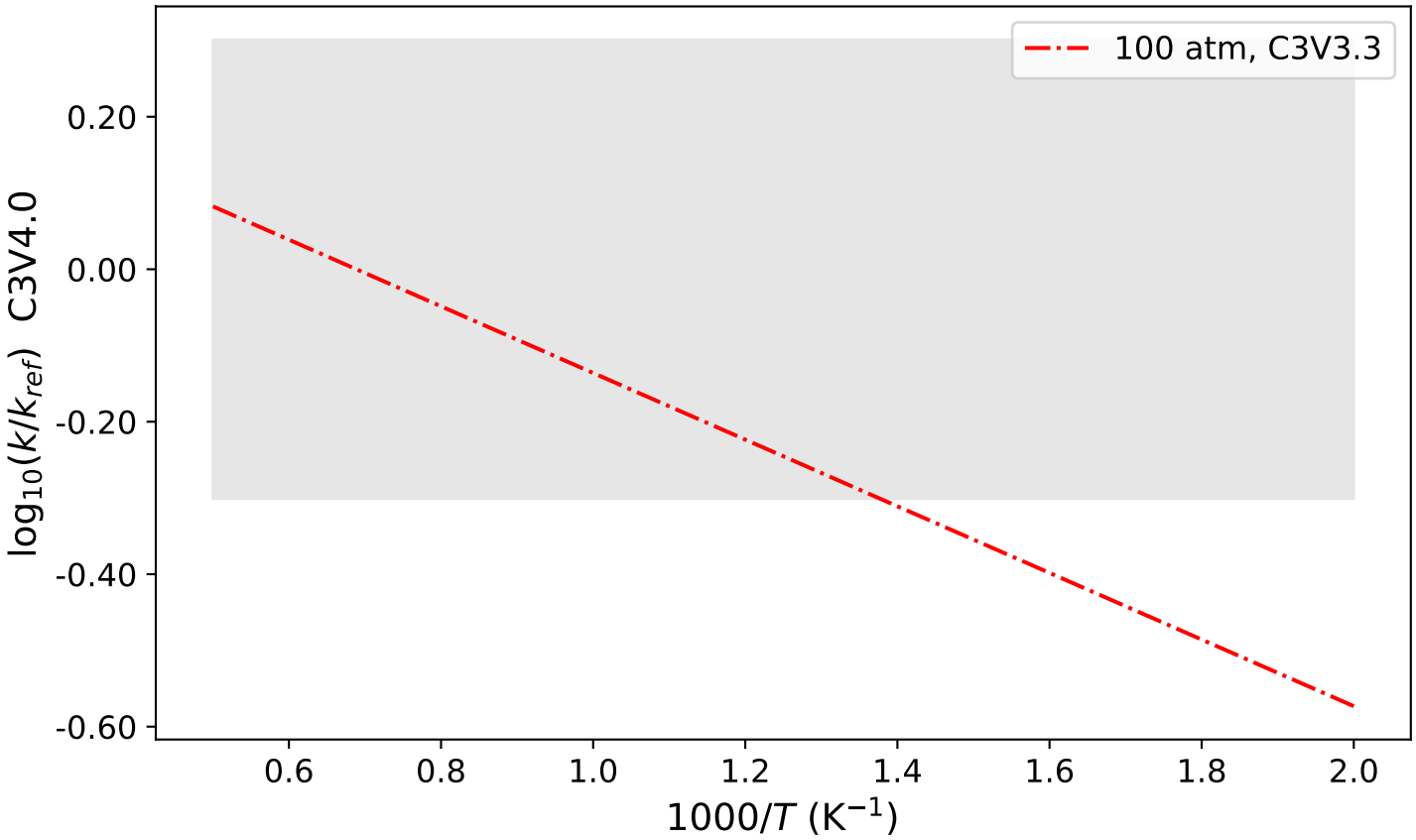
OH + C4H72-1OH = H2O + C4H64,2-1OH



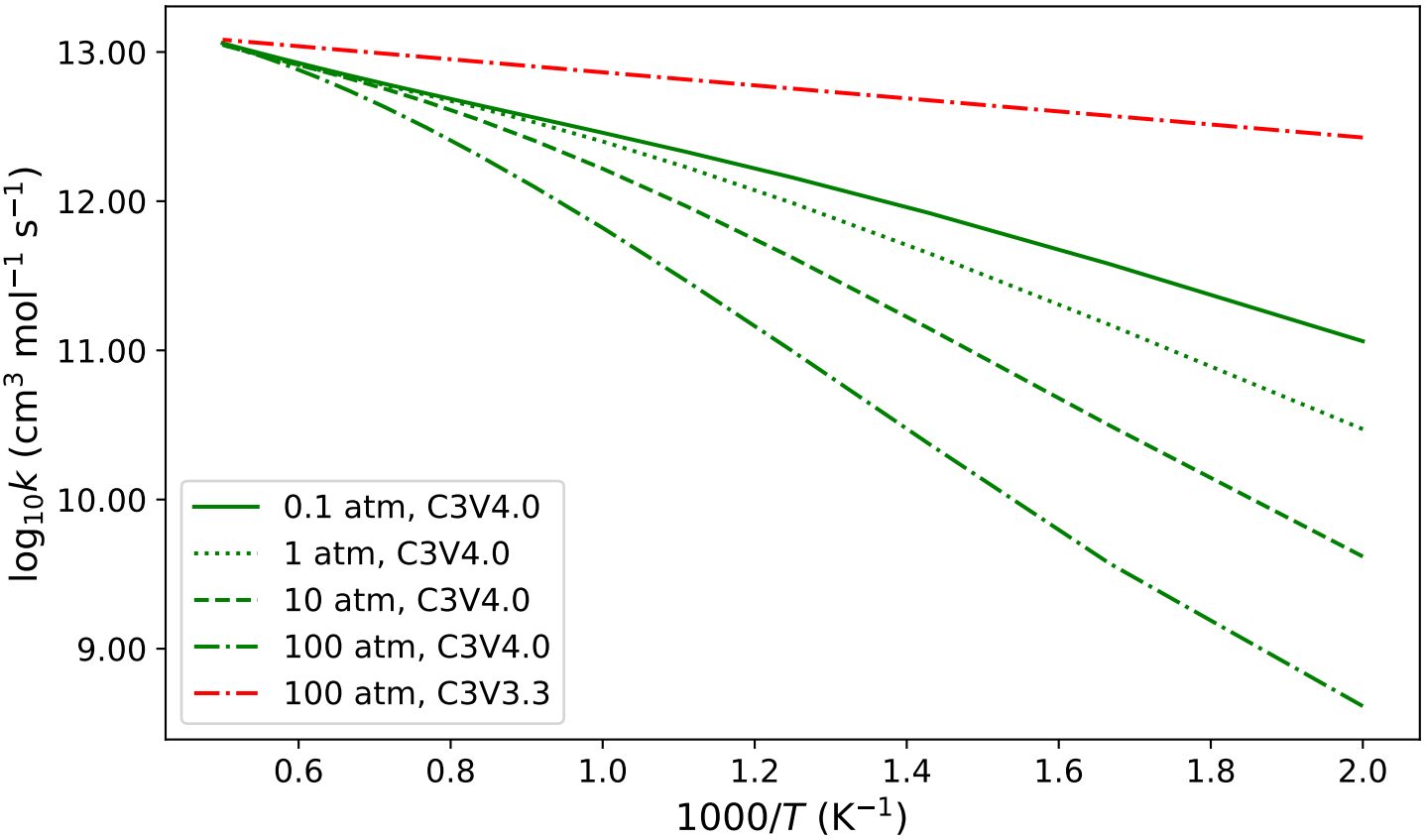



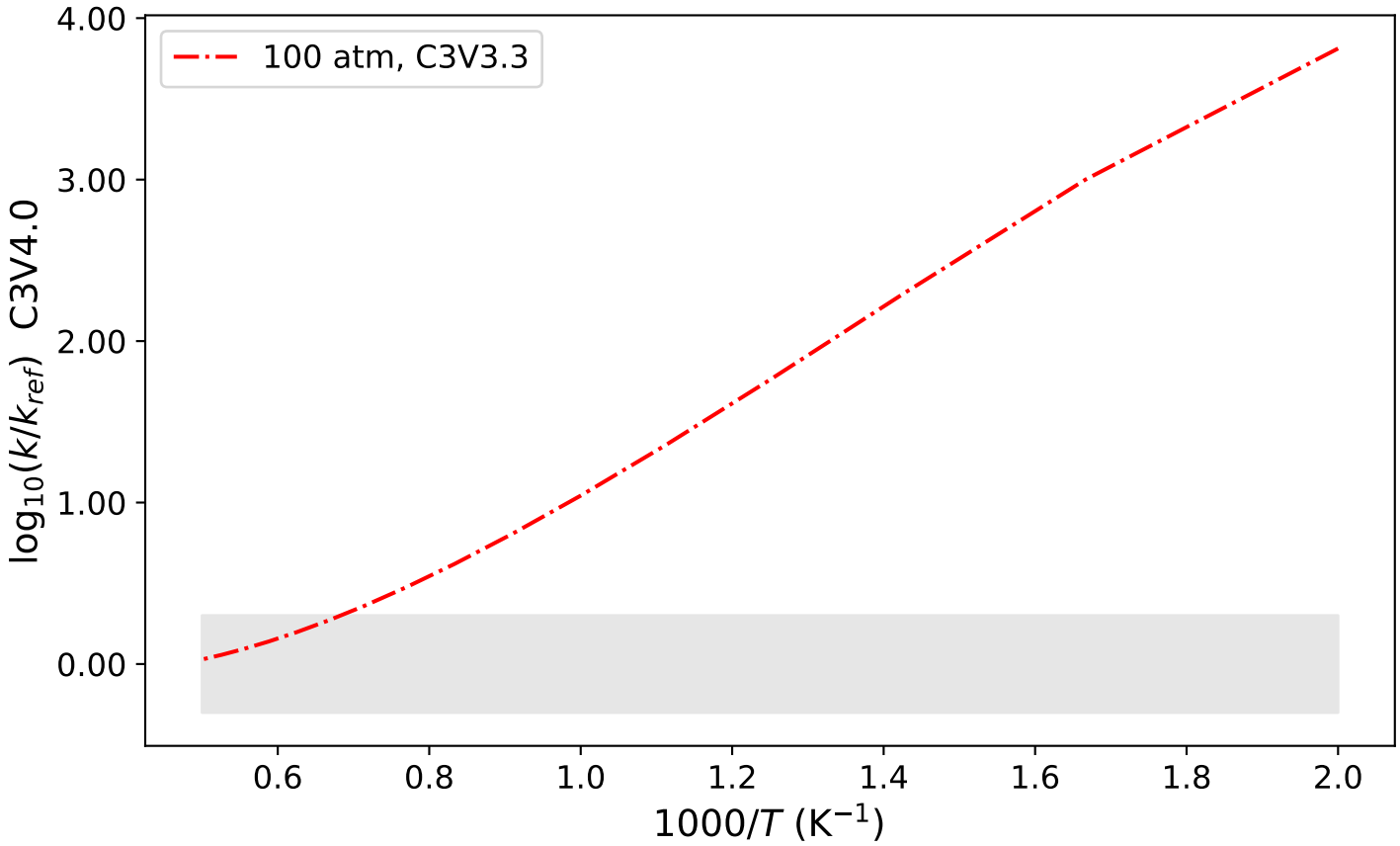
HO2 + C4H8-1 = H2O2 + C4H71-3



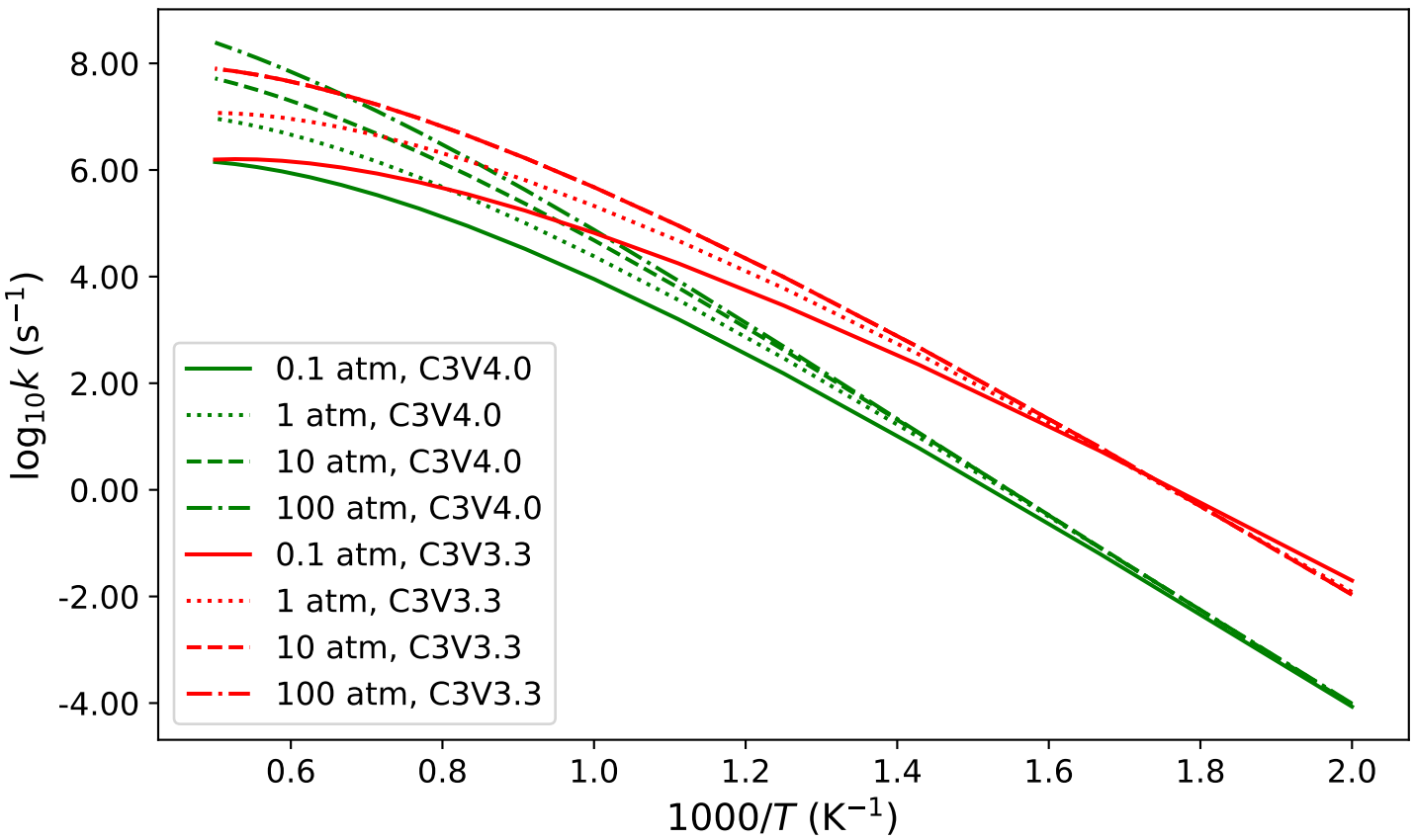



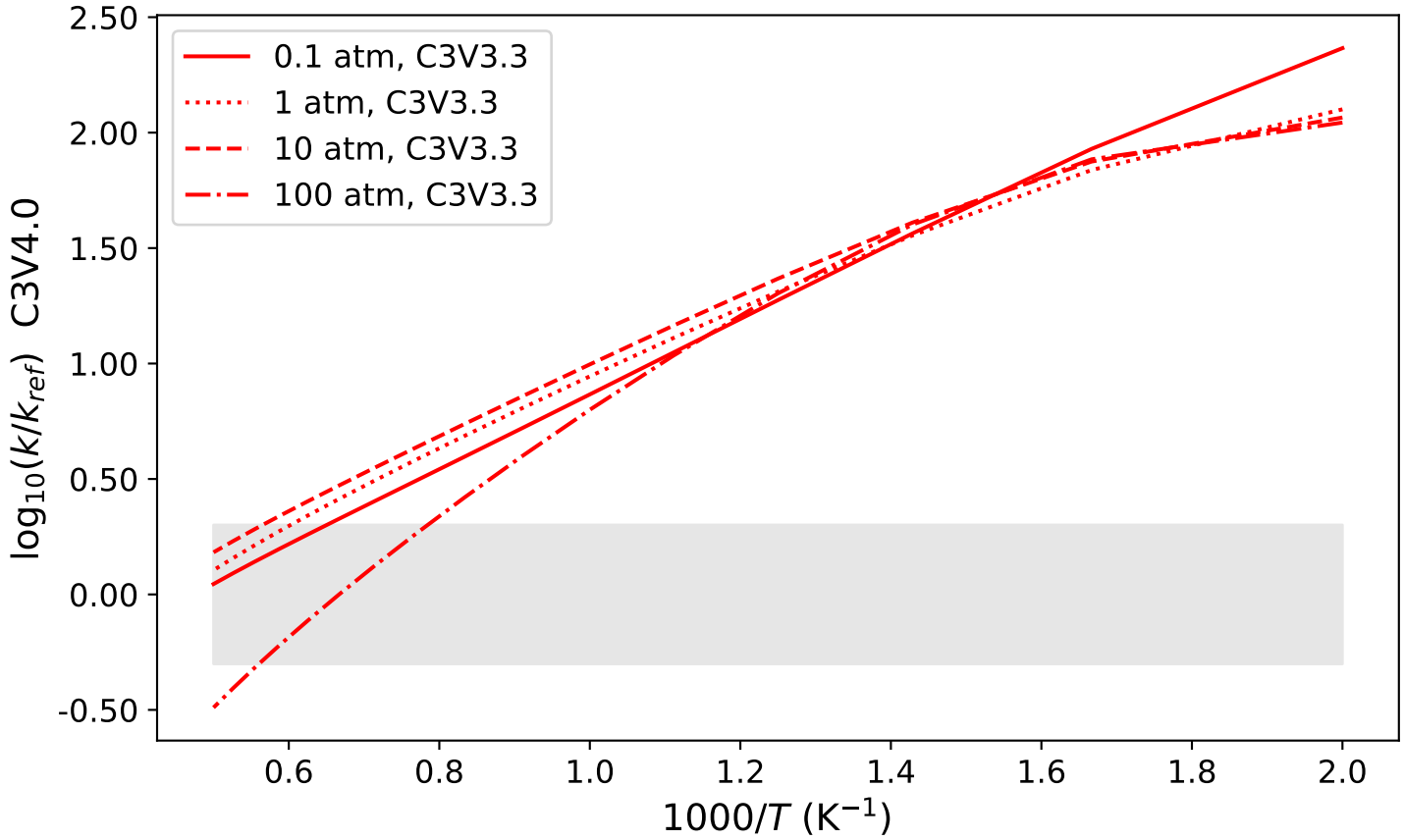
H + C4H612 = CH3 + C3H4-A



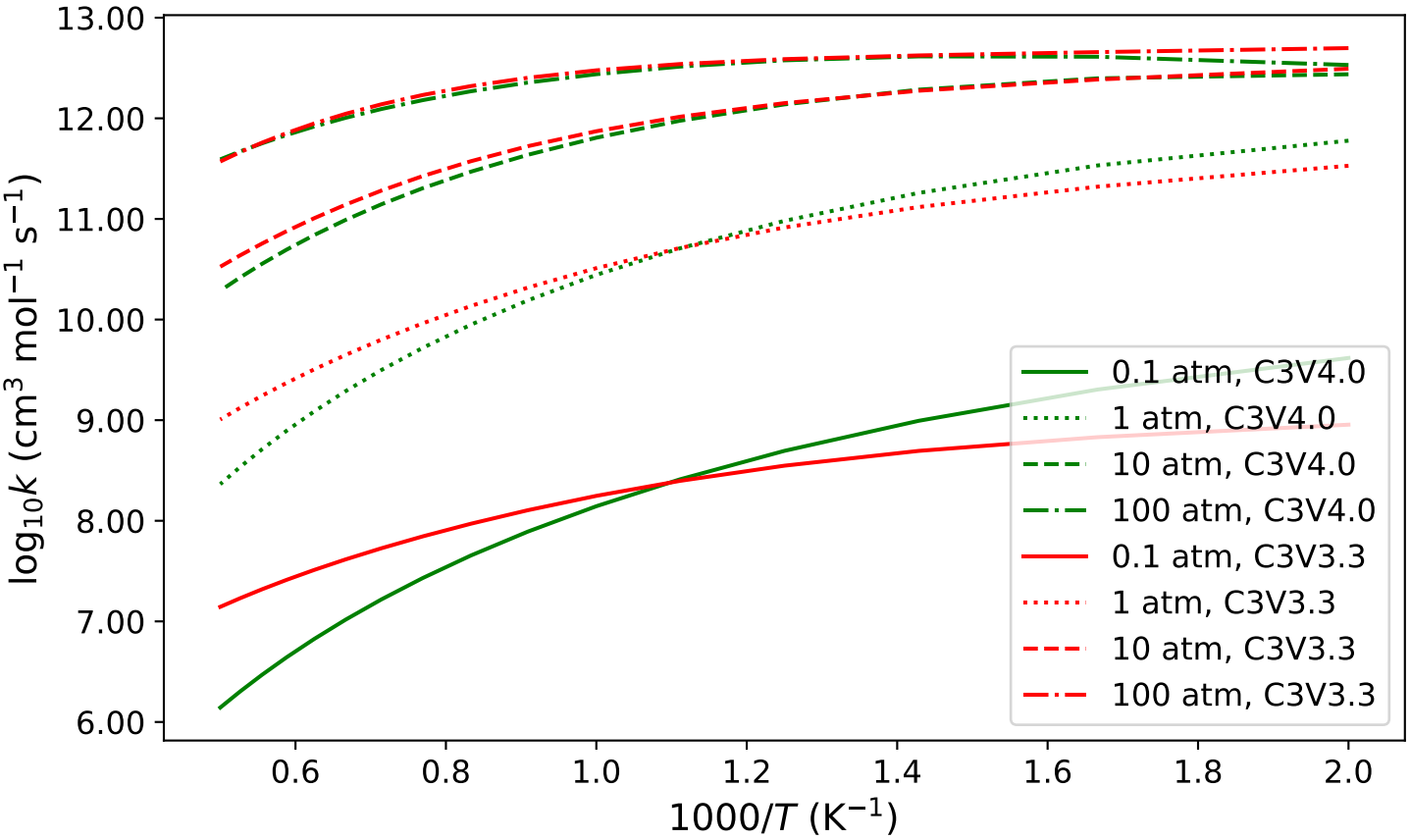



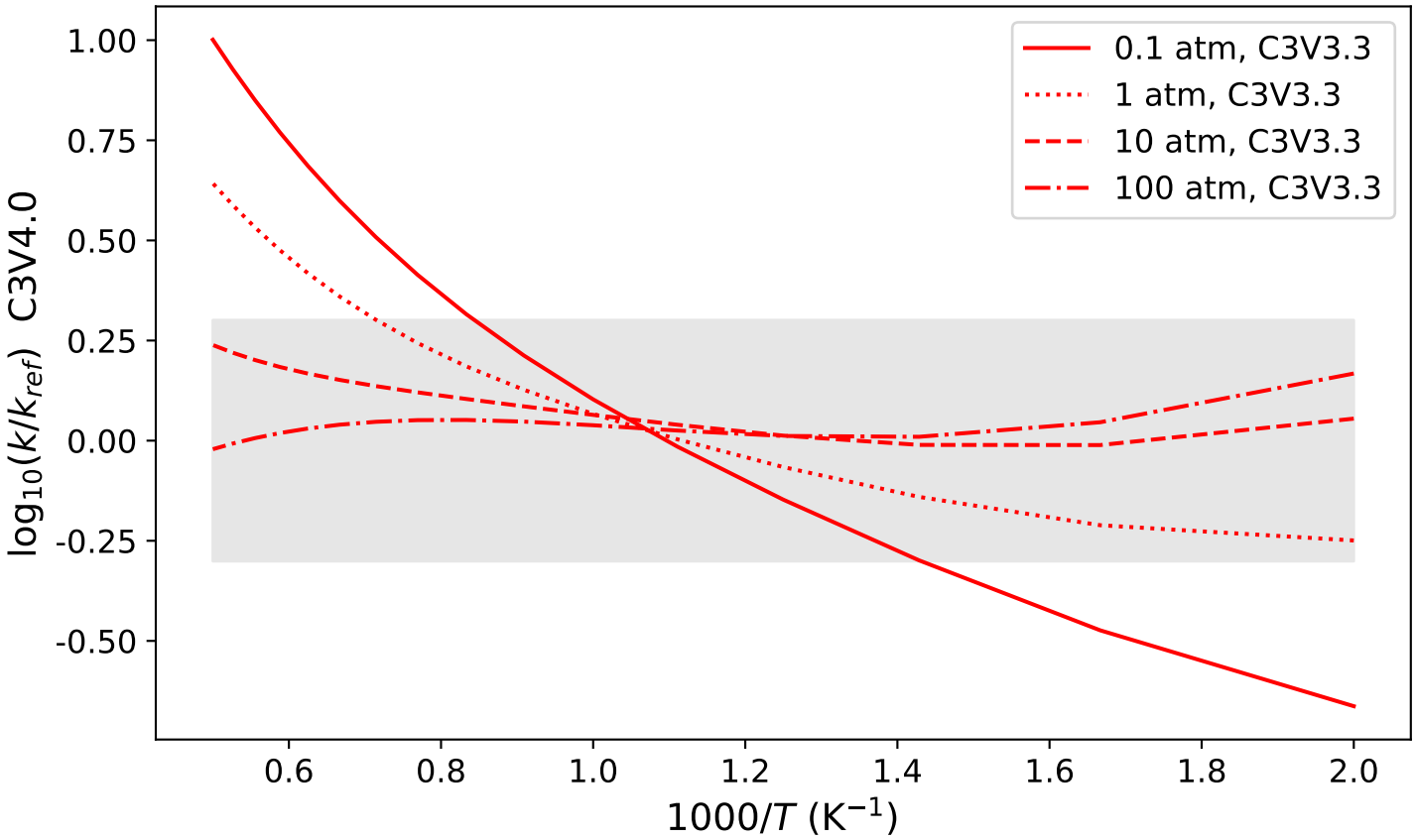
C4H5-N = H + C4H4


8.00
6.00
4.00
2.00
0.00
-2.00
-4.00
$\log_{10}k$ ($s^{-1}$)
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ ($K^{-1}$)
0.1 atm, C3V4.0
1 atm, C3V4.0
10 atm, C3V4.0
100 atm, C3V4.0
0.1 atm, C3V3.3
1 atm, C3V3.3
10 atm, C3V3.3
100 atm, C3V3.3

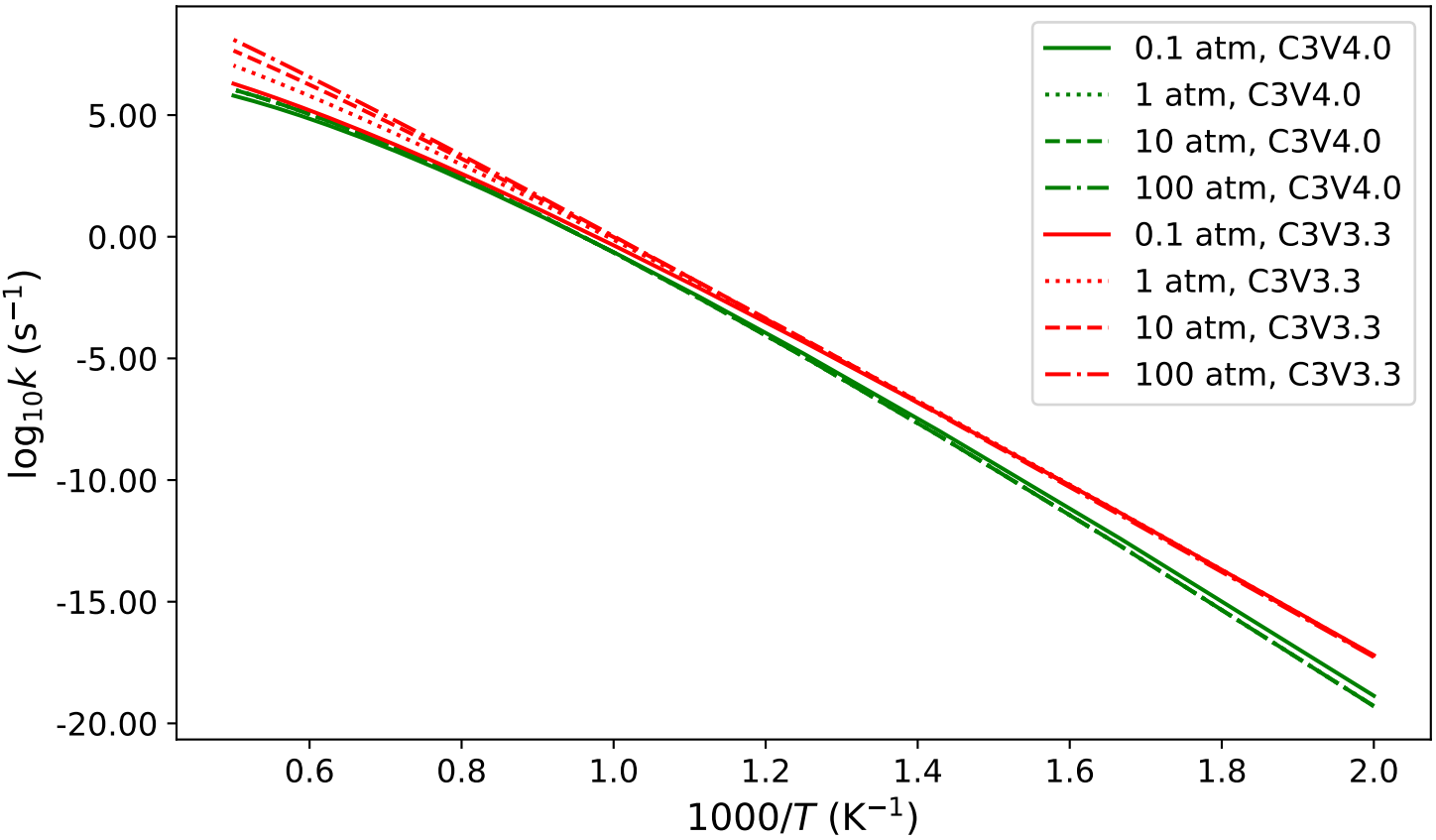

2.50
2.00
1.50
1.00
0.50
0.00
-0.50
$\log_{10}(k/k_{ref})$ C3V4.0
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ ($K^{-1}$)
0.1 atm, C3V3.3
1 atm, C3V3.3
10 atm, C3V3.3
100 atm, C3V3.3

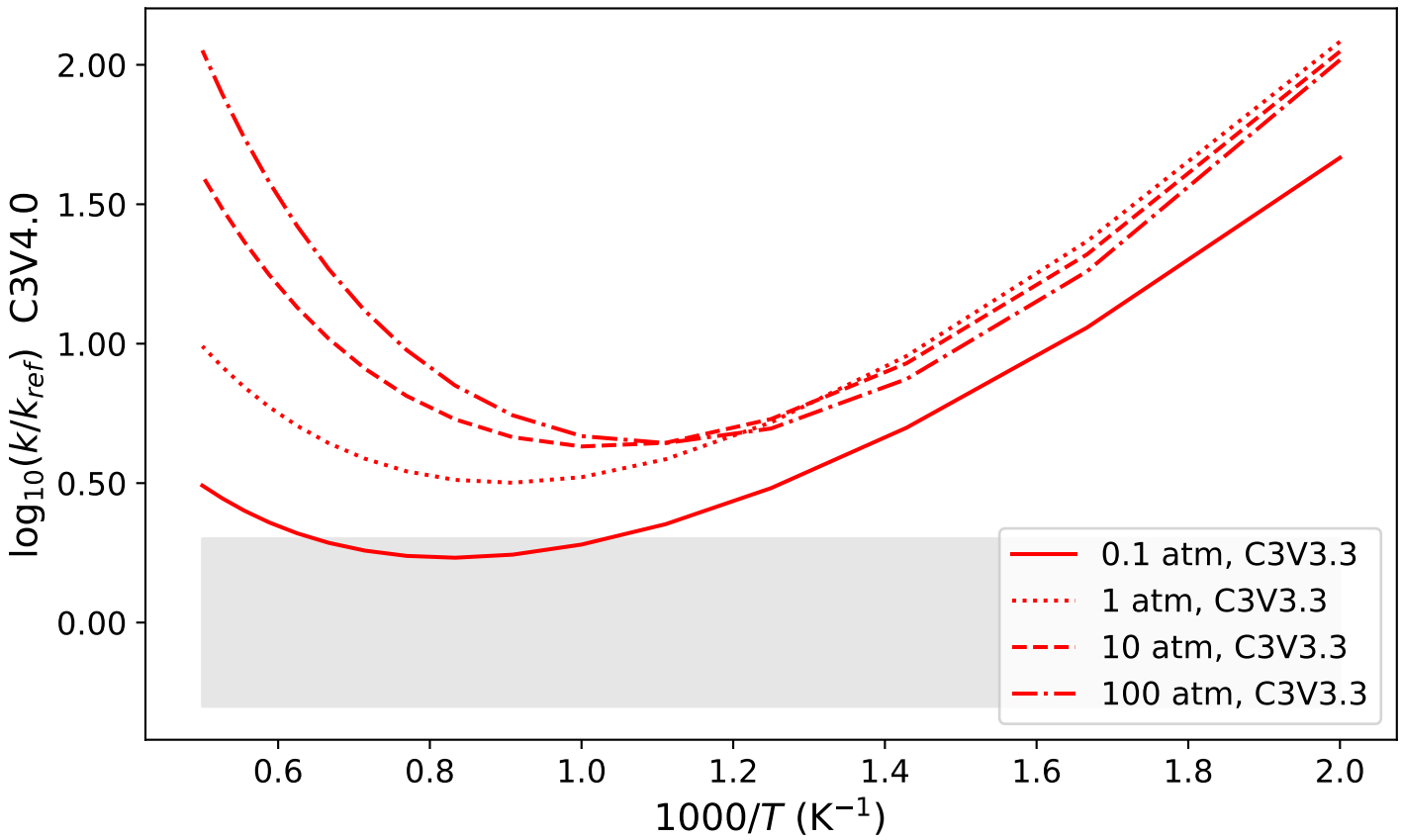
O2 + C4H71-1 = C4H71-1O2



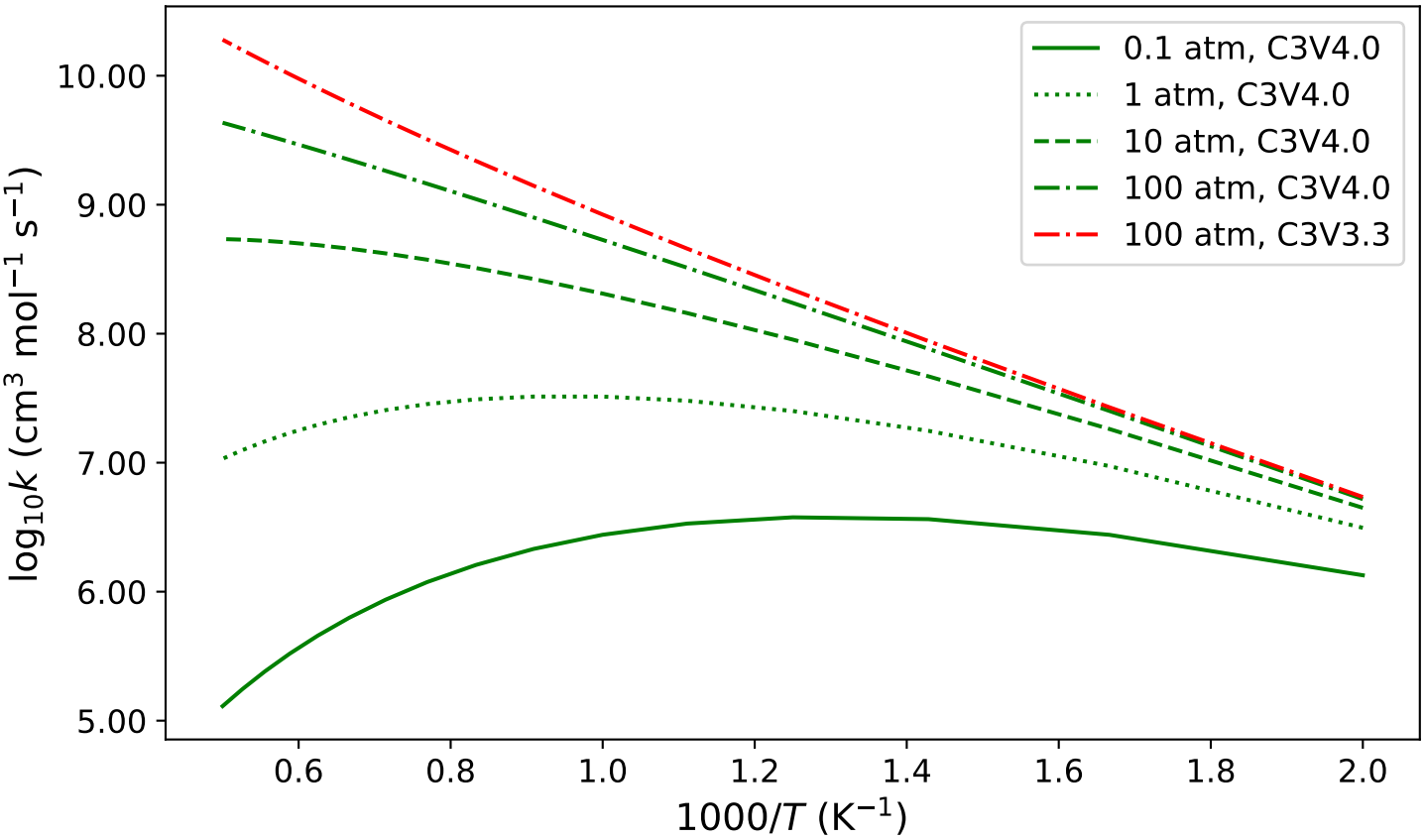



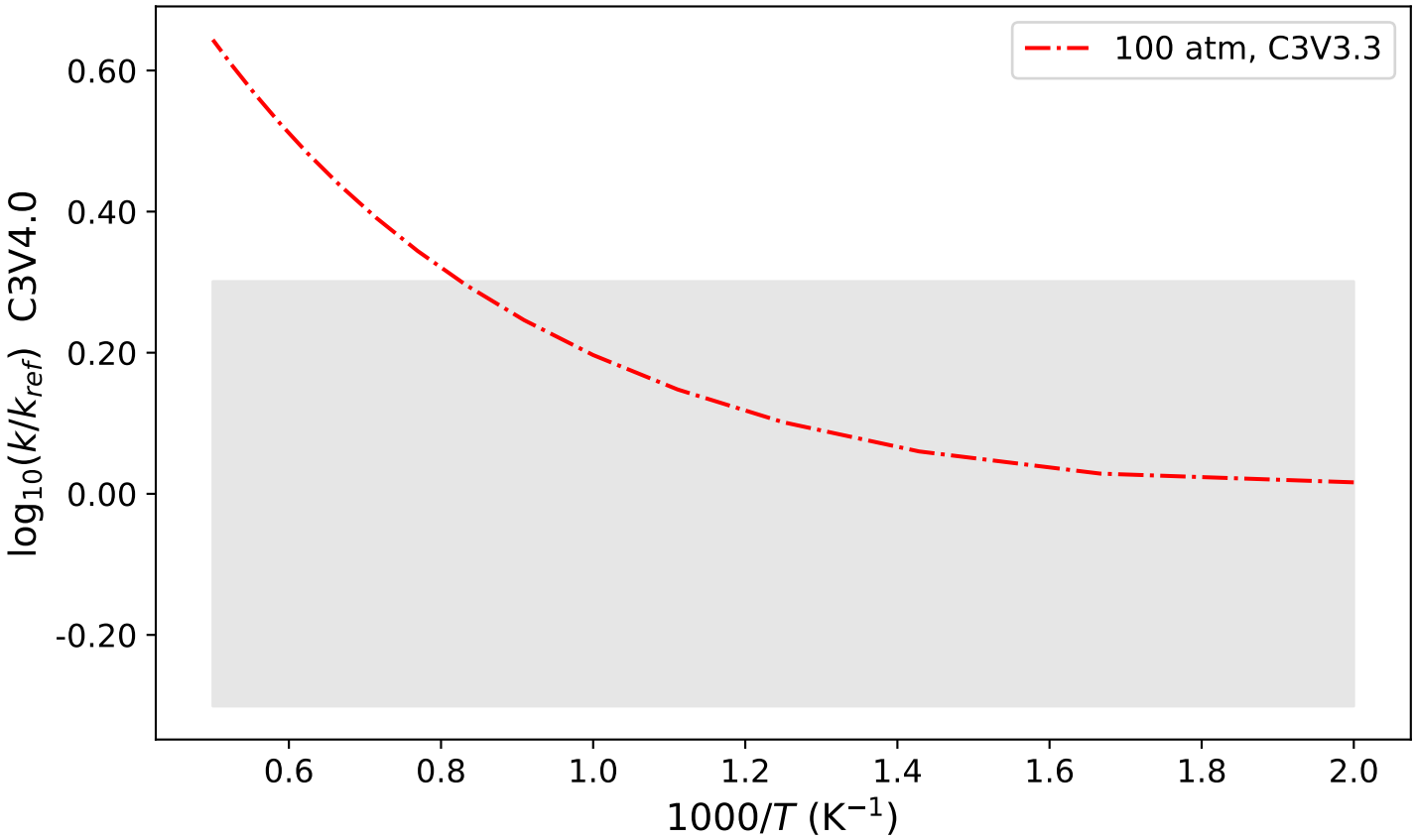
C4H6-1 = CH3 + C3H3



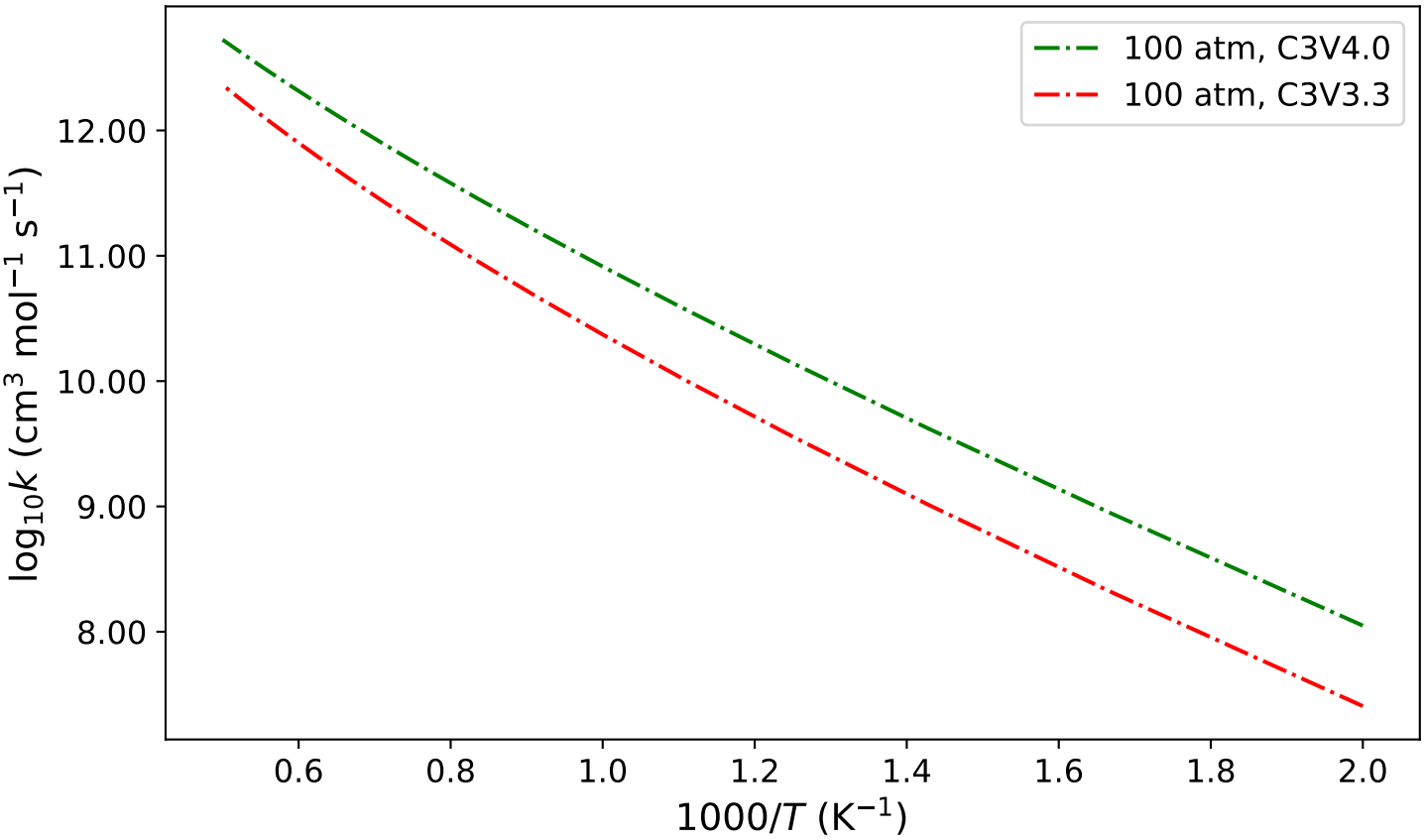



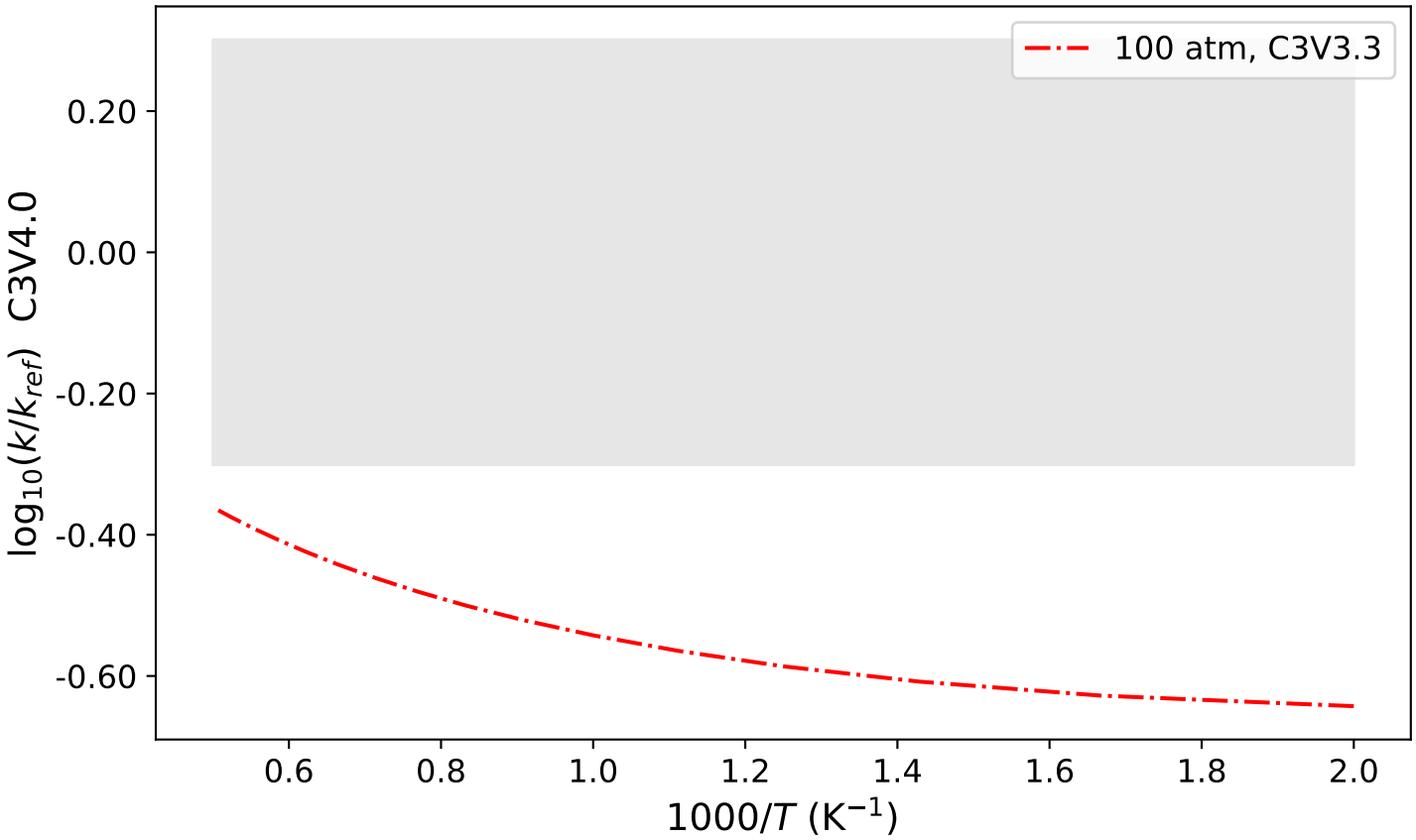
HO2 + C4H6 = C4H61-3OOH4



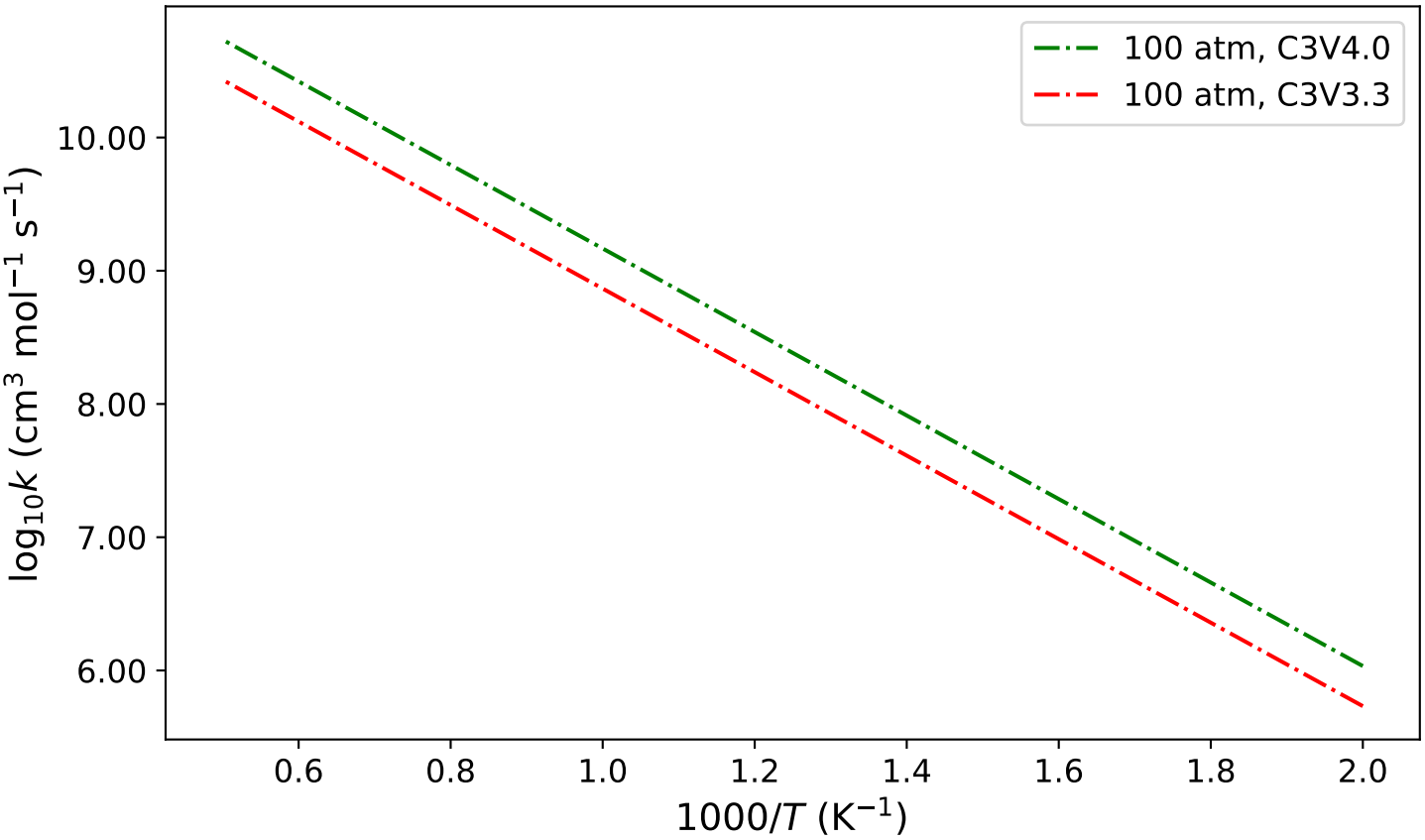



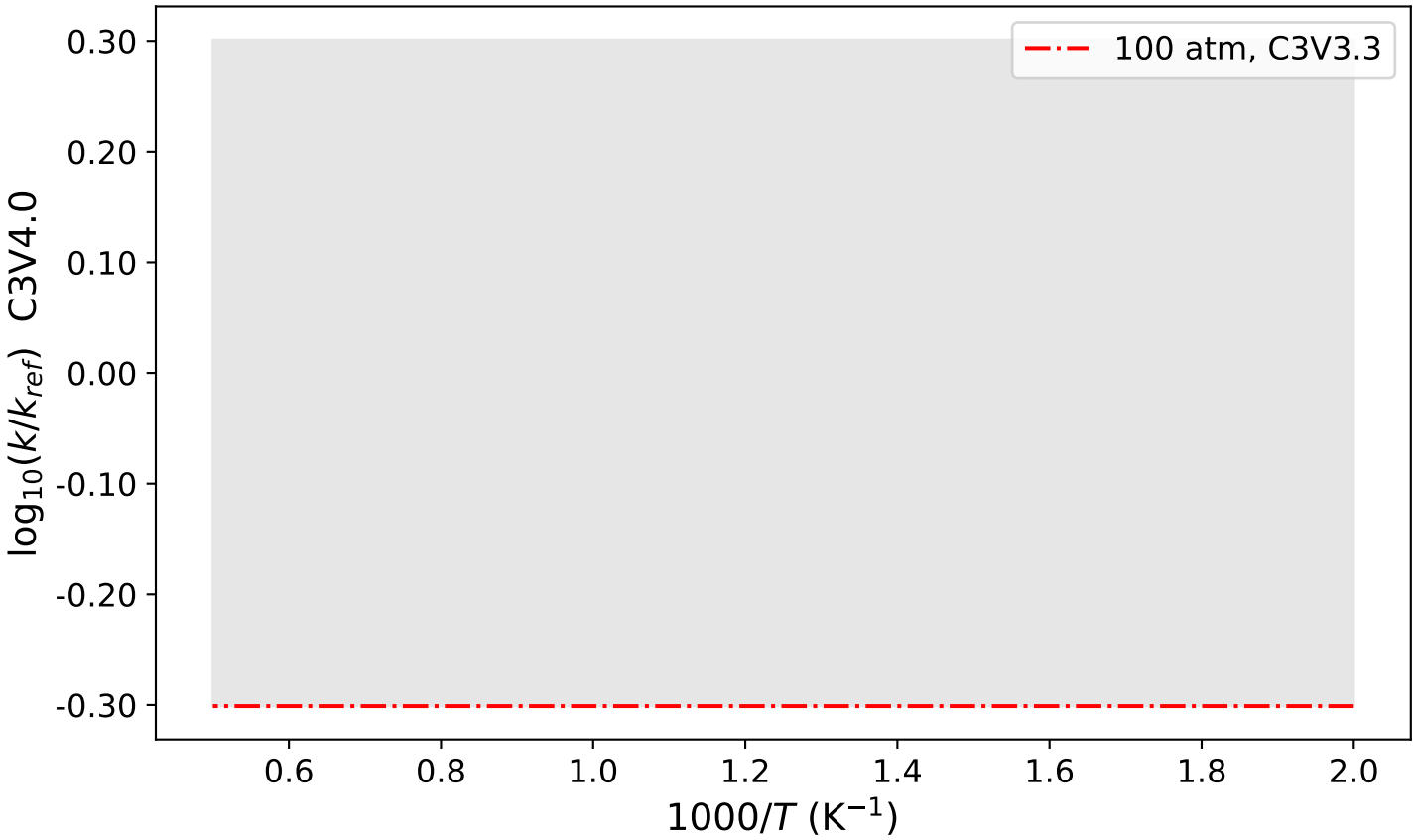
H + C3H6 = H2 + C3H5-T



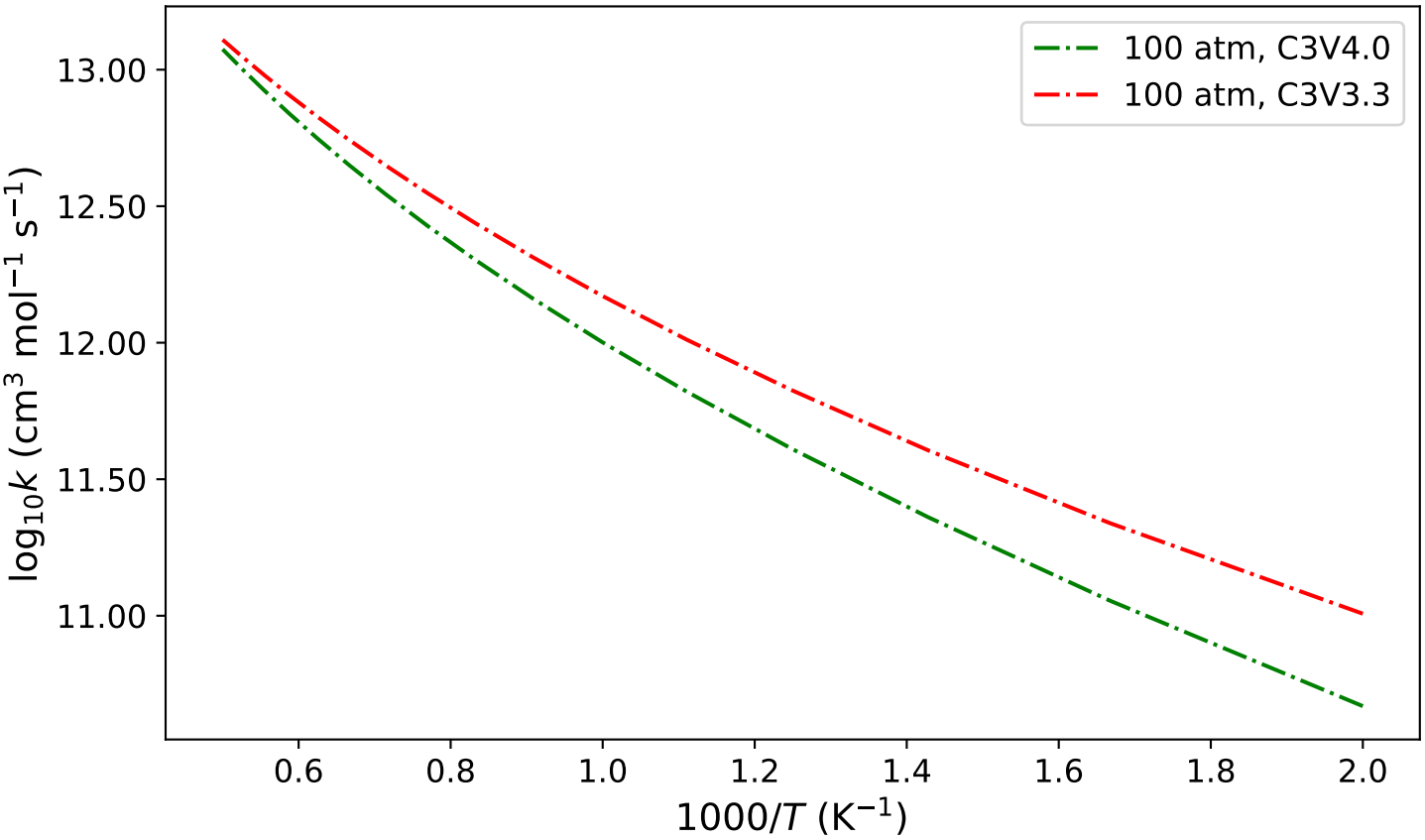



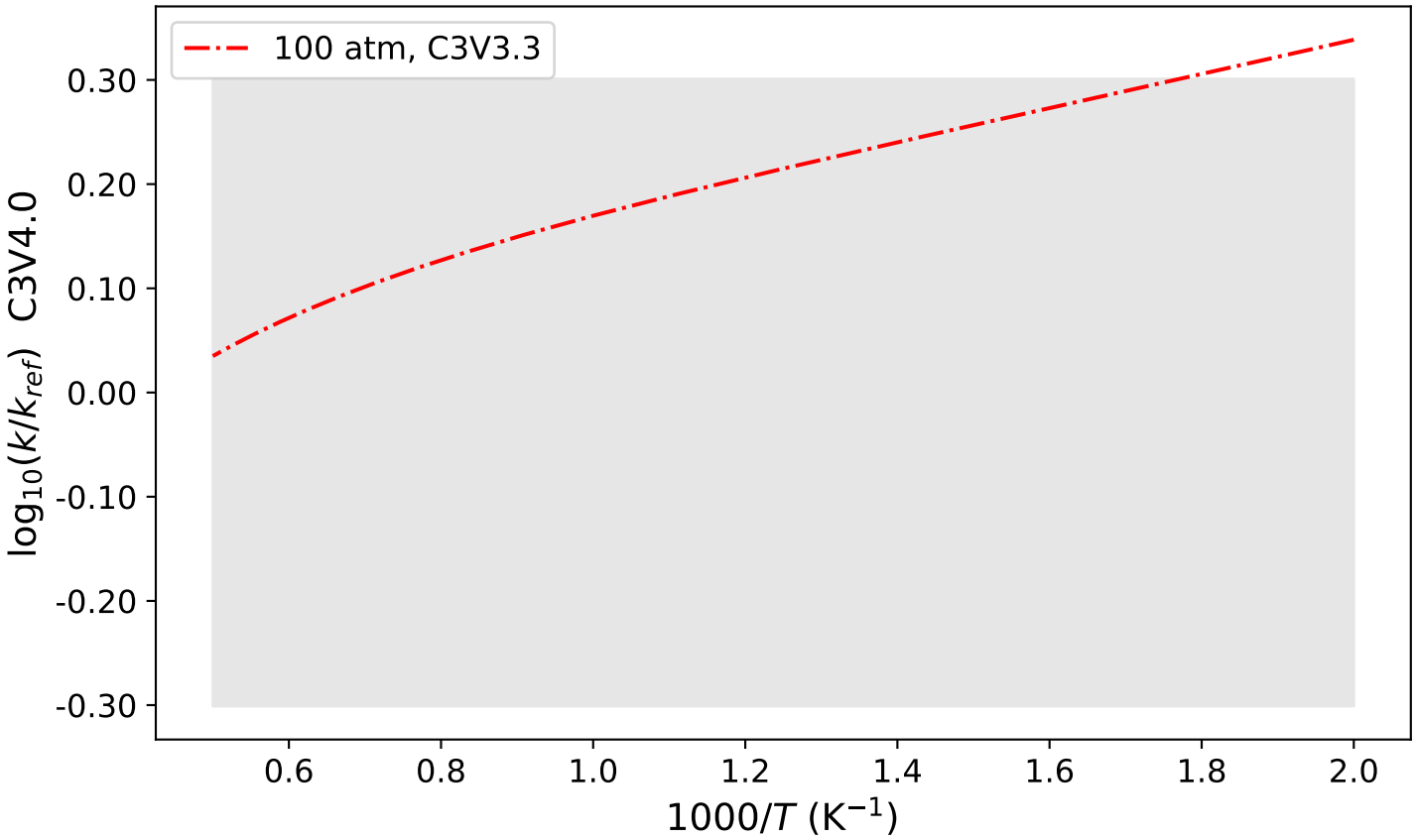
CH3O2 + C4H8-1 = CH3O + C4H8O1-2



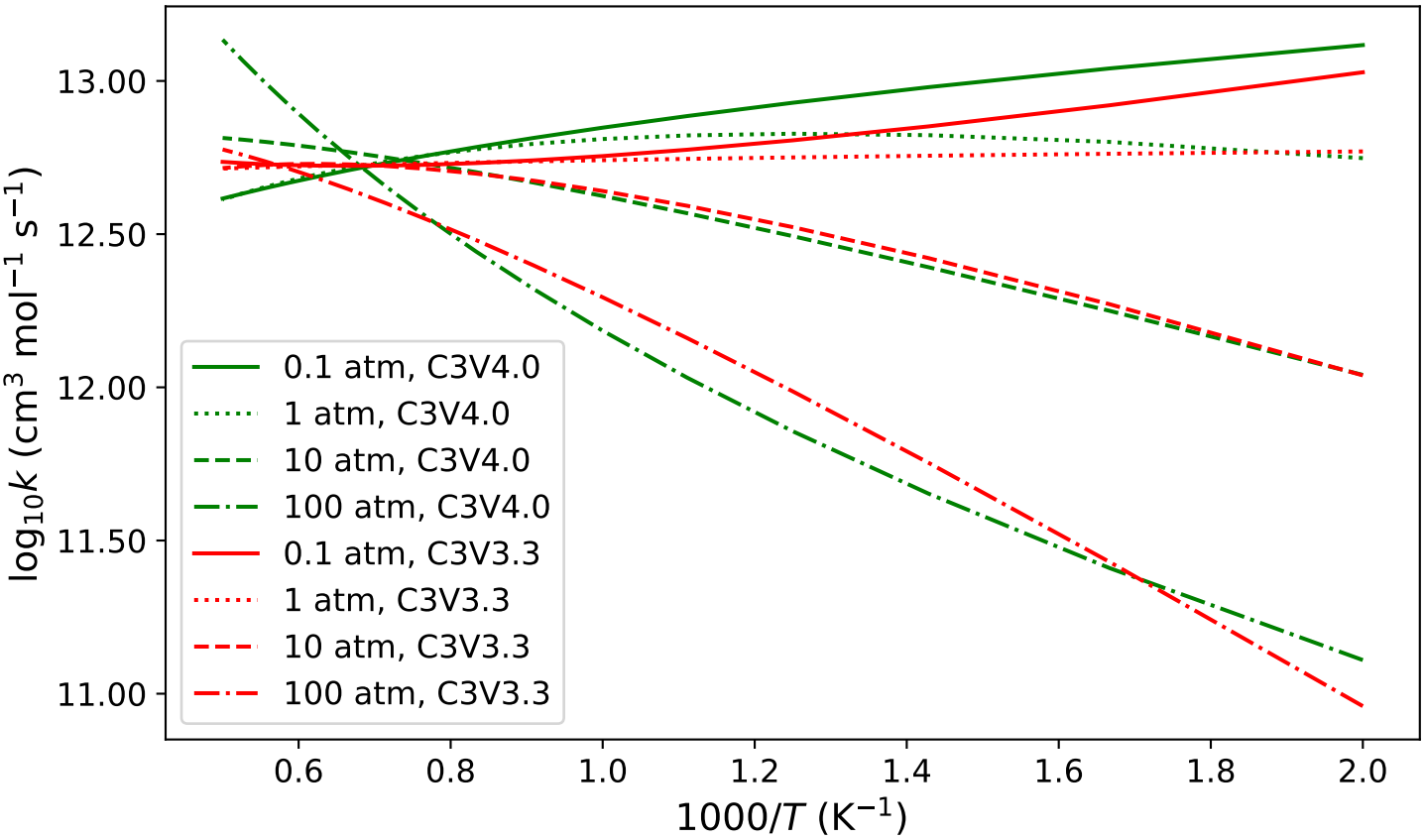



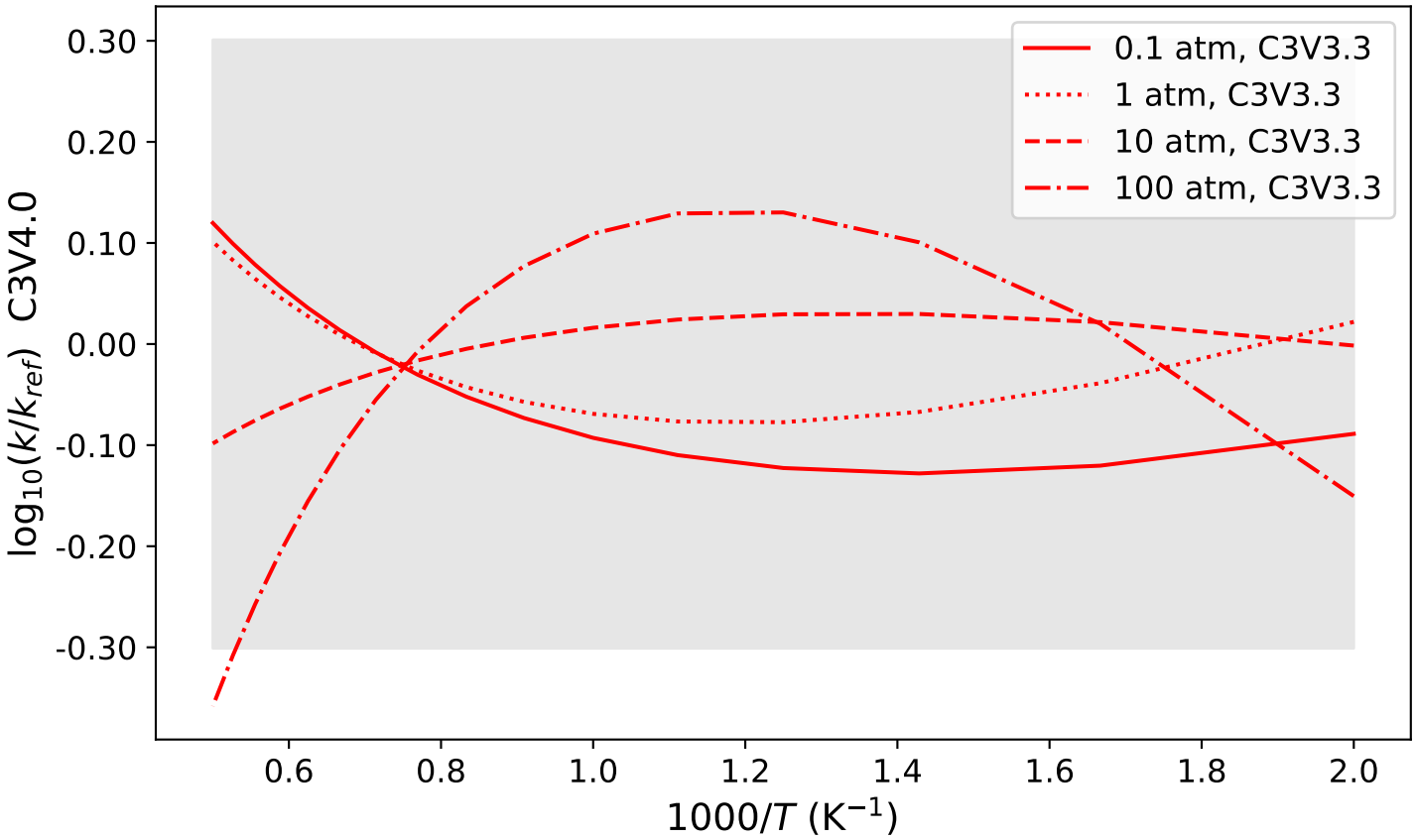
OH + C4H6 = H2O + C4H5-N



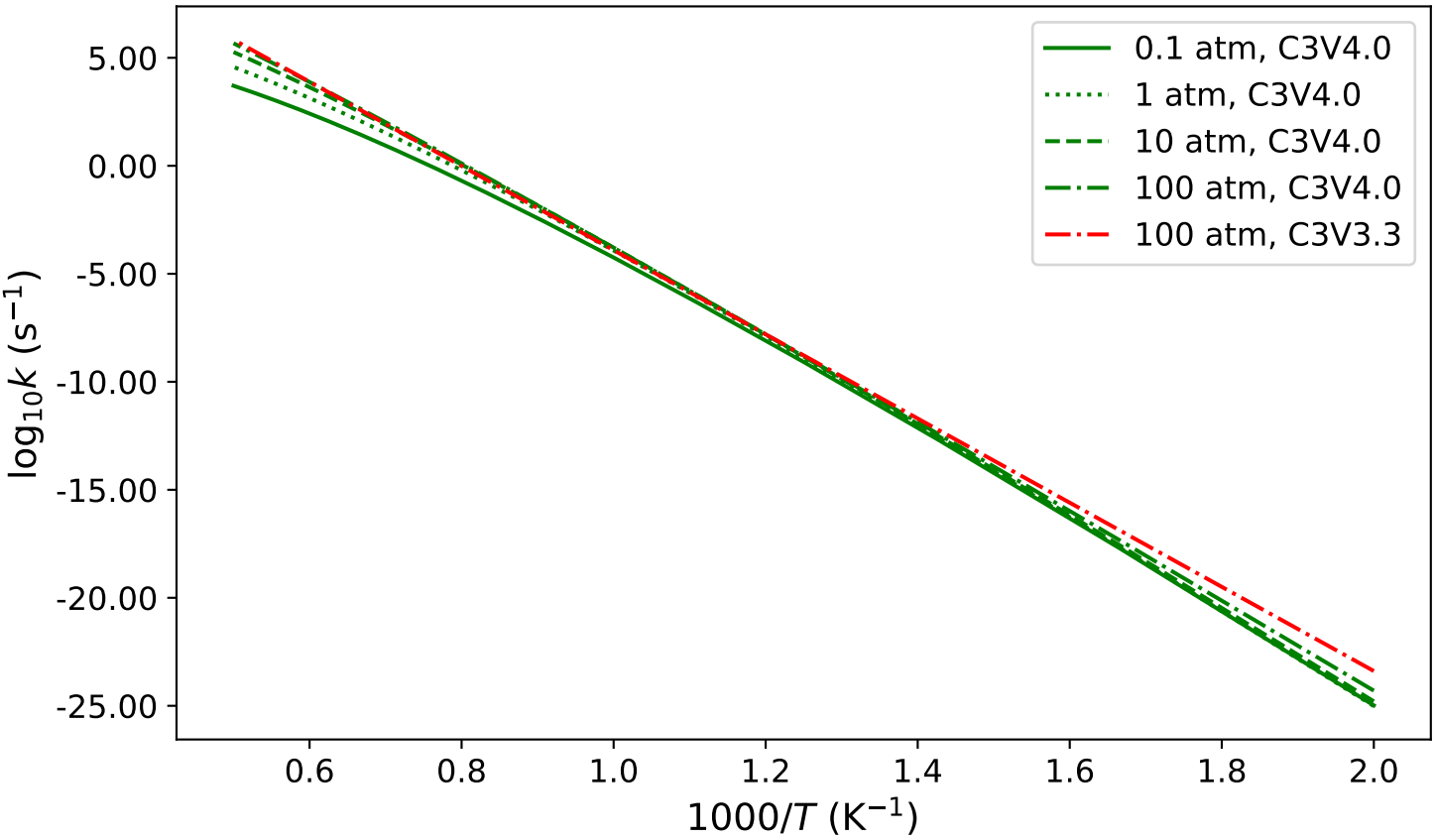



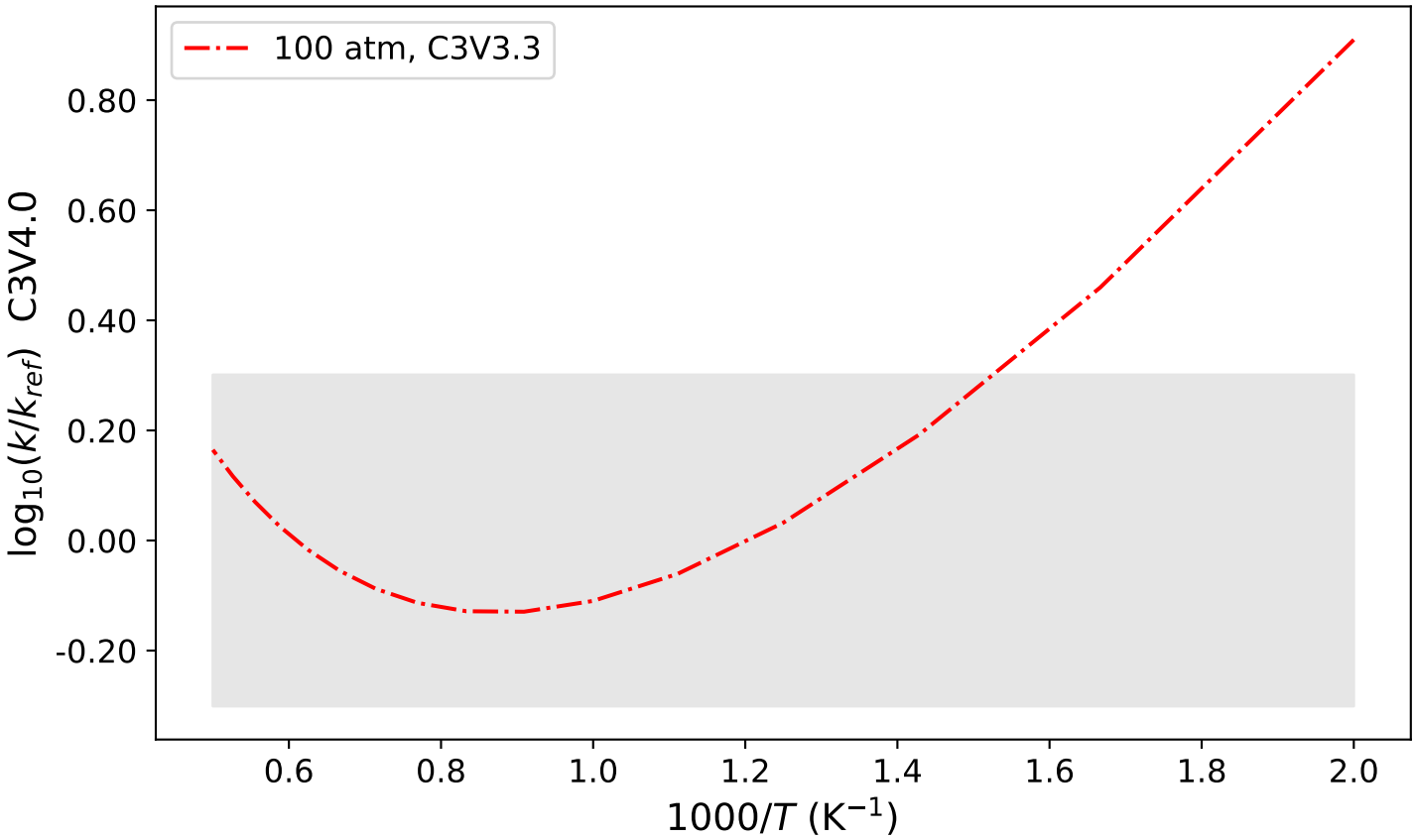
HO2 + IC4H7 = OH + IC4H7O



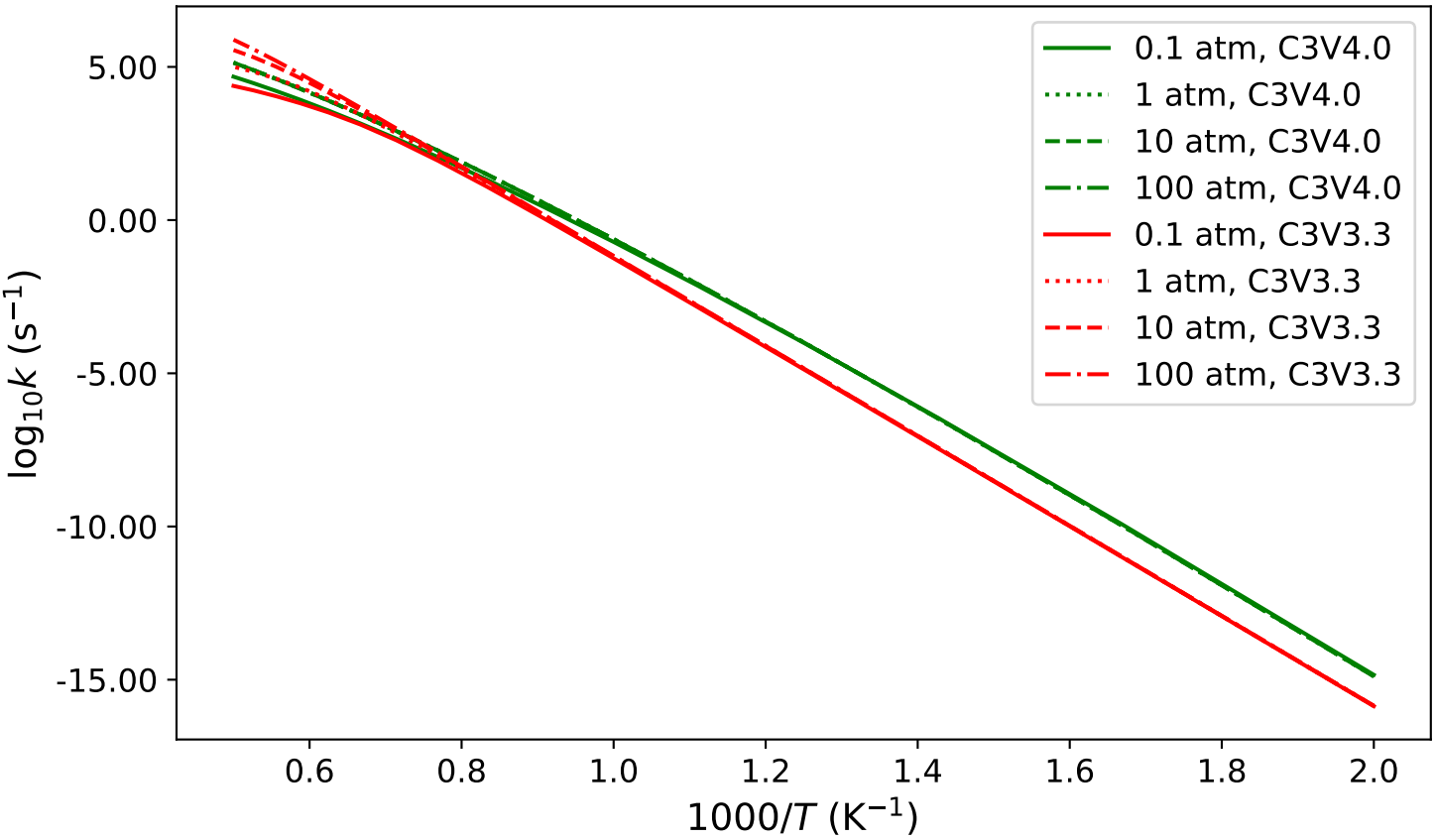



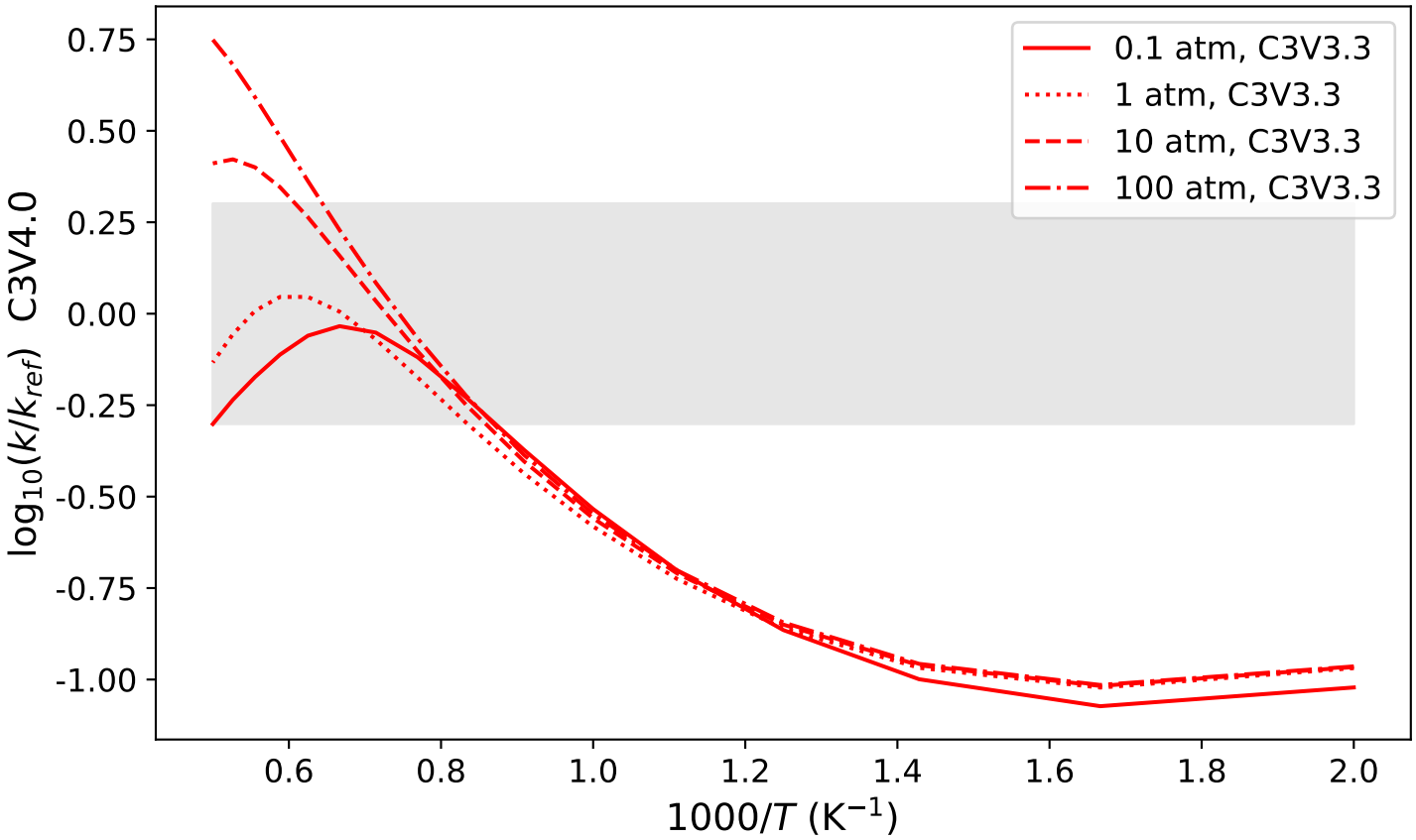
C4H6-2 = H + C4H5-2



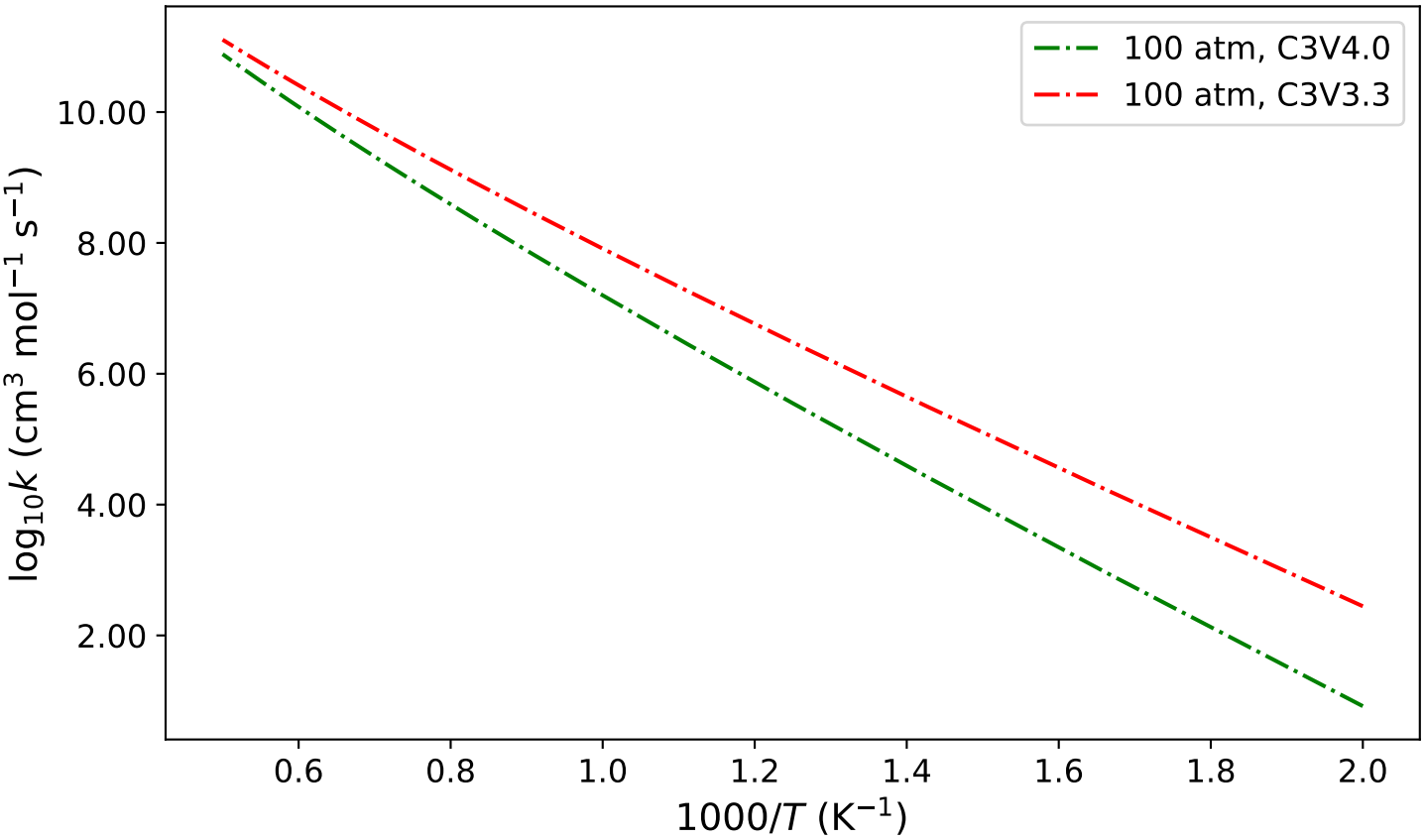



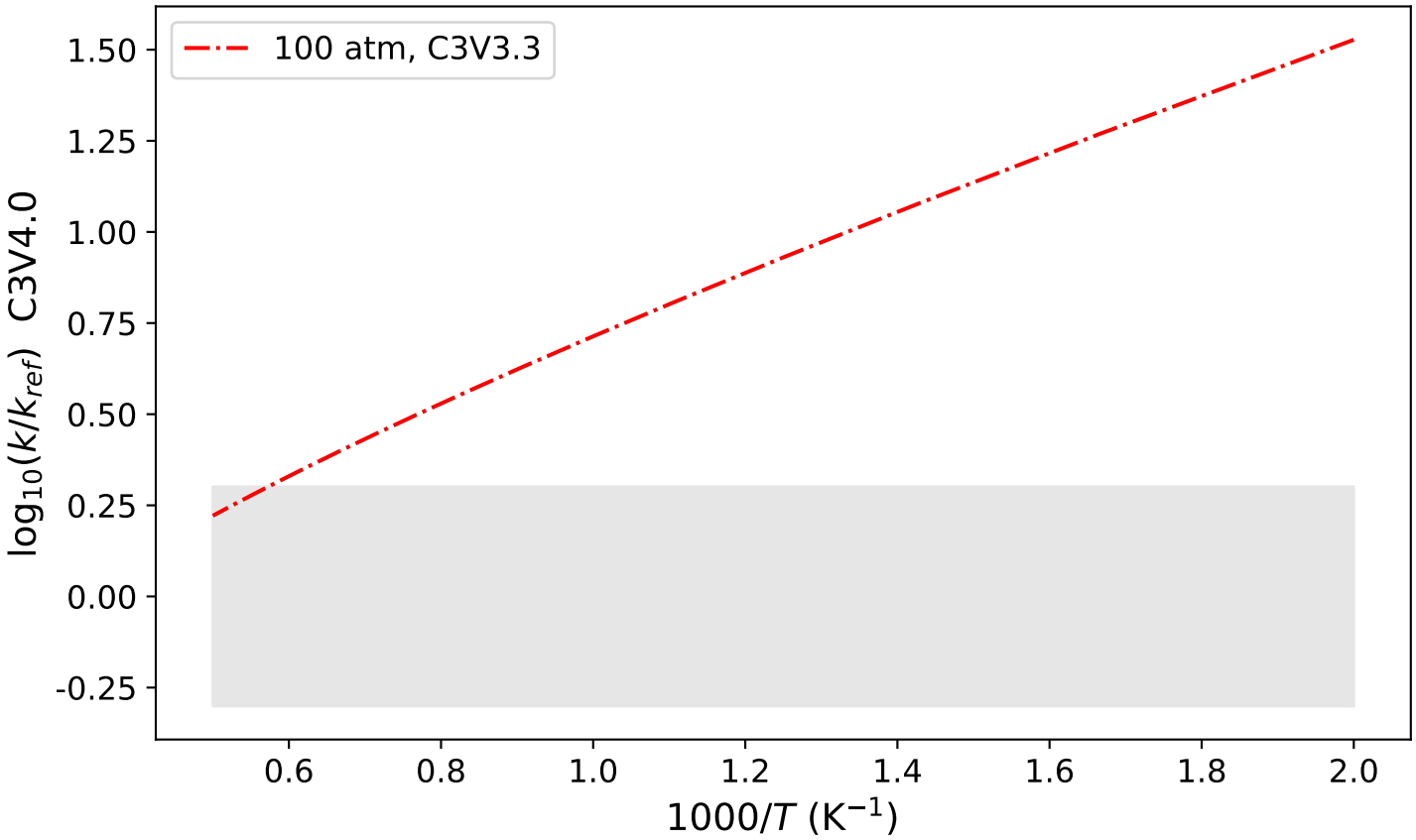
NC3H7OH = H2O + C3H6



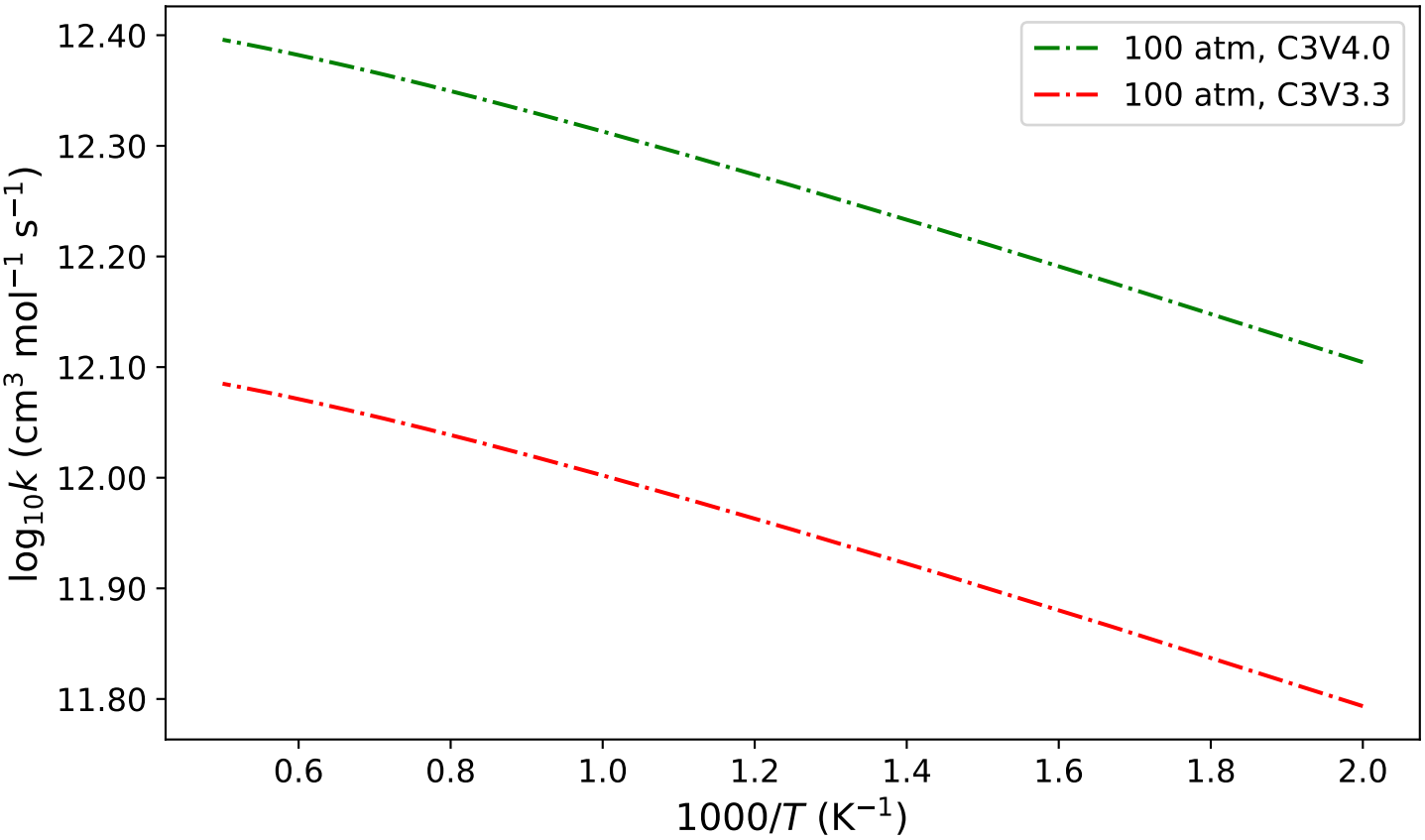



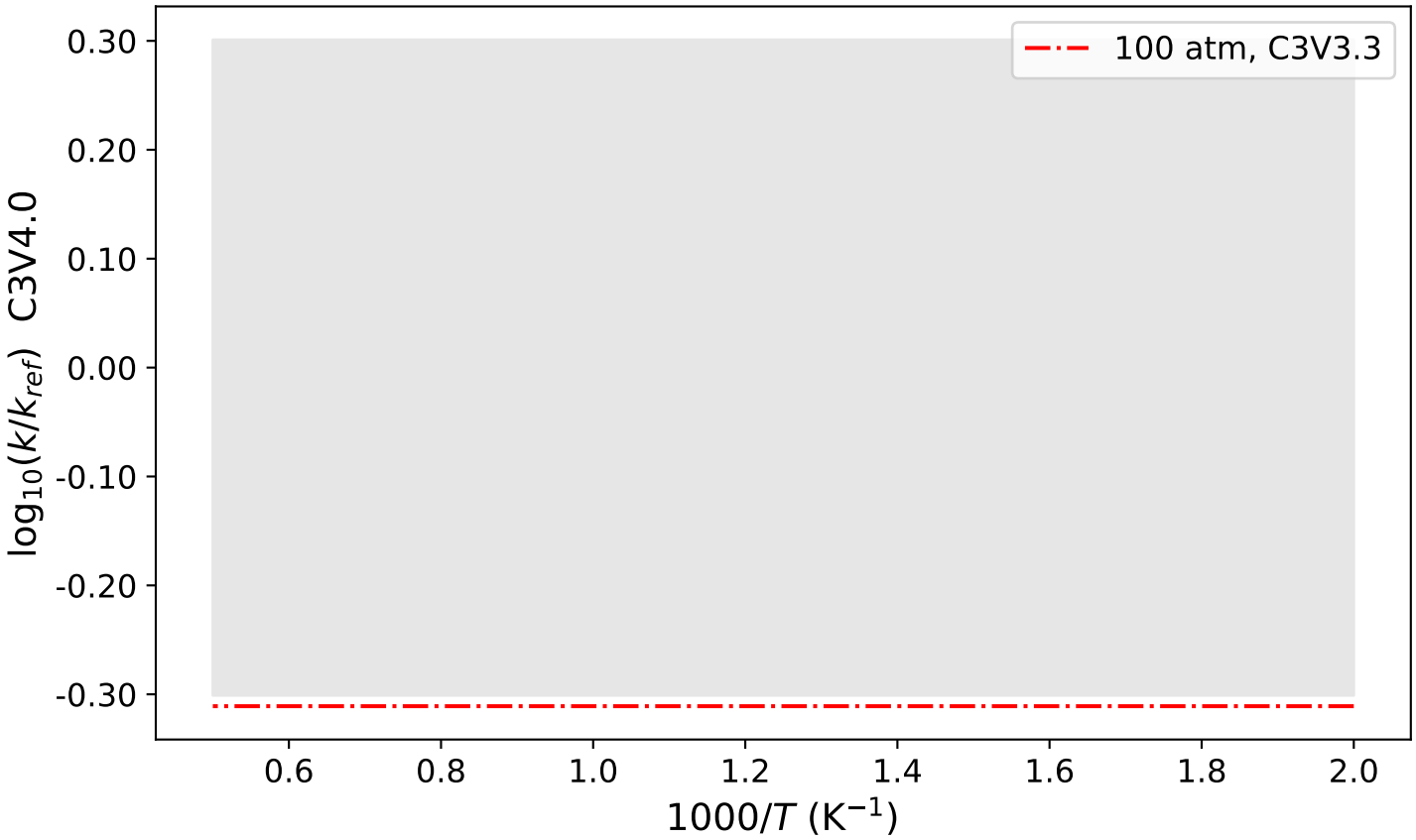
HO2 + C4H6 = H2O2 + C4H5-N



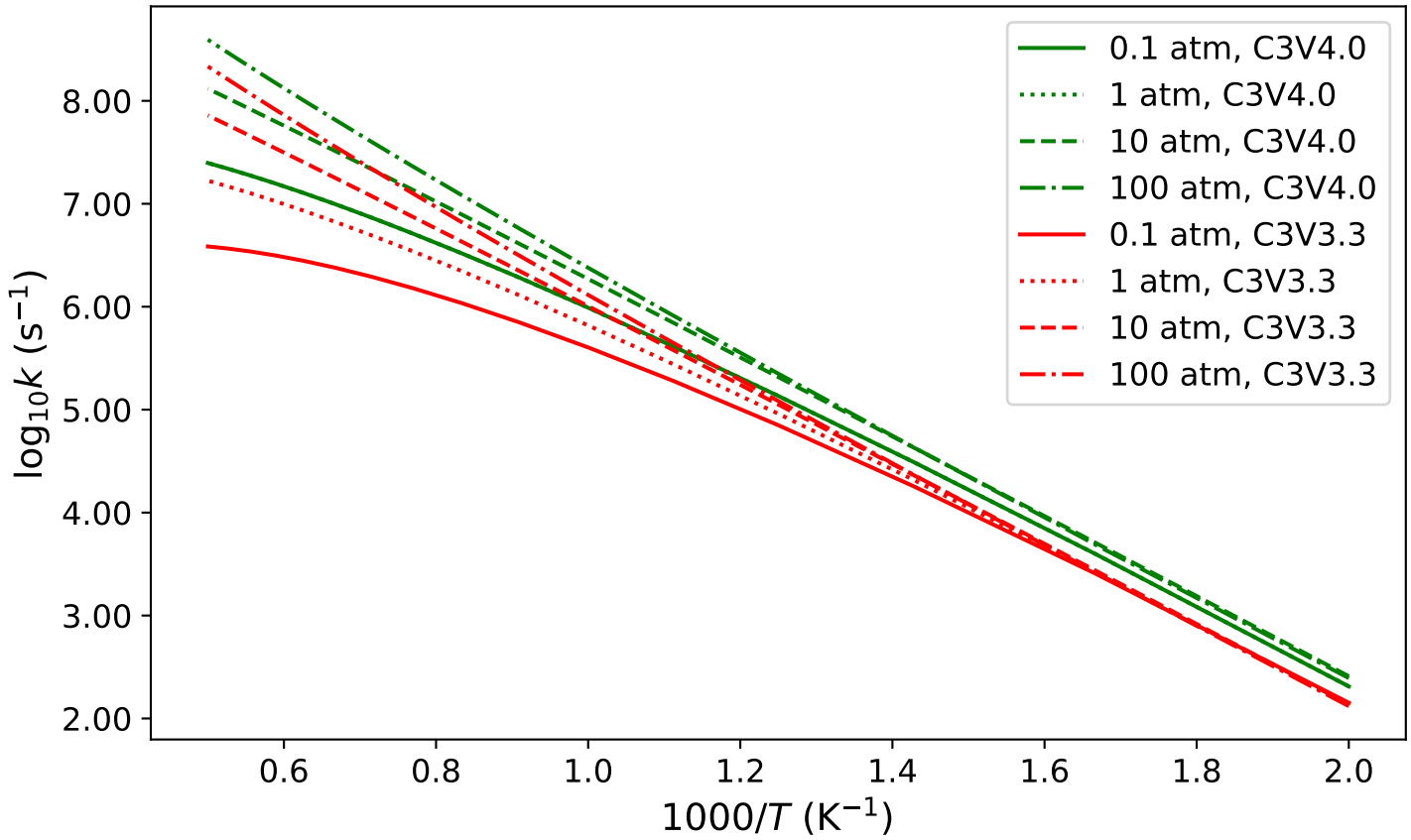



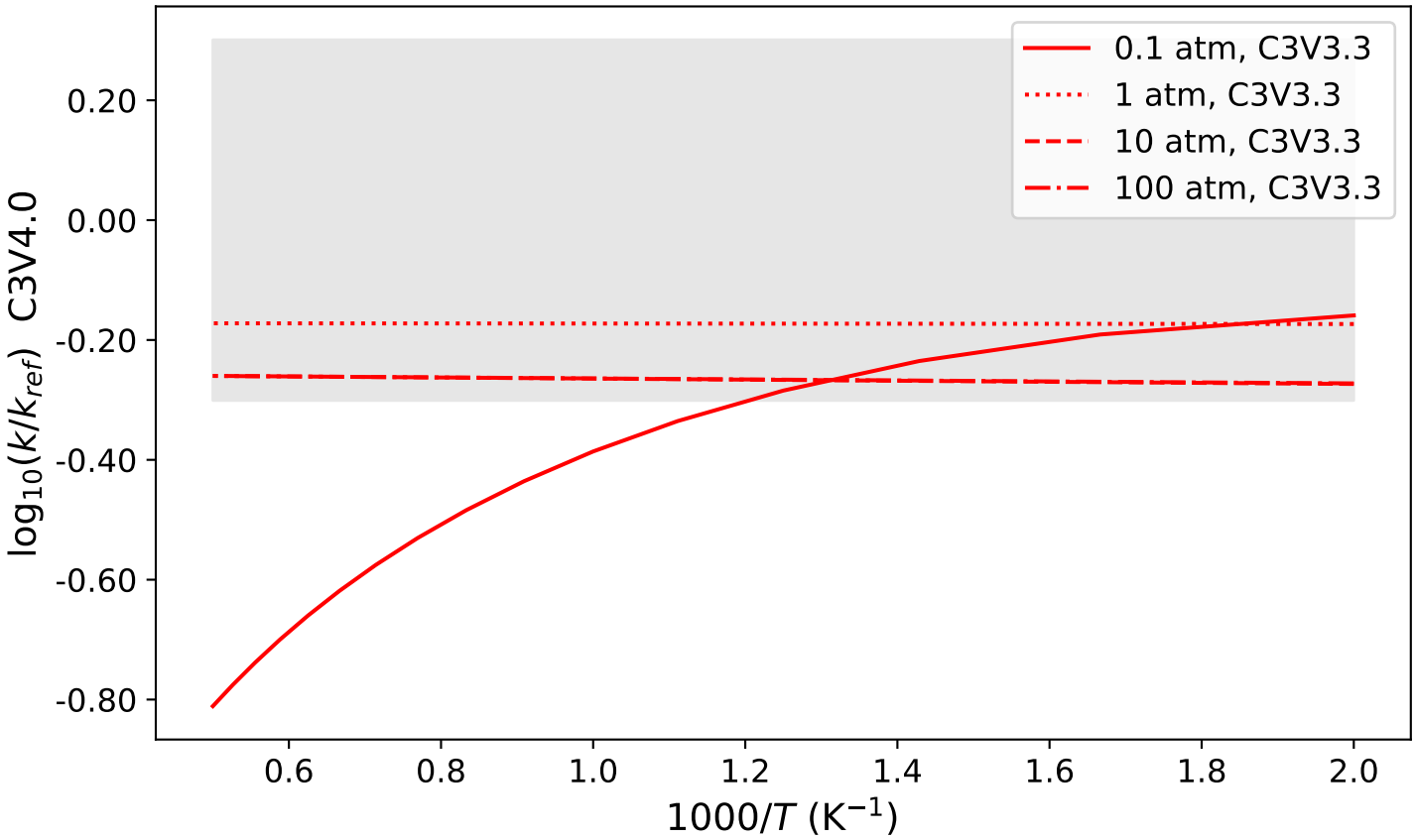
O2 + CH3CCO => H + CO + CH2CO2



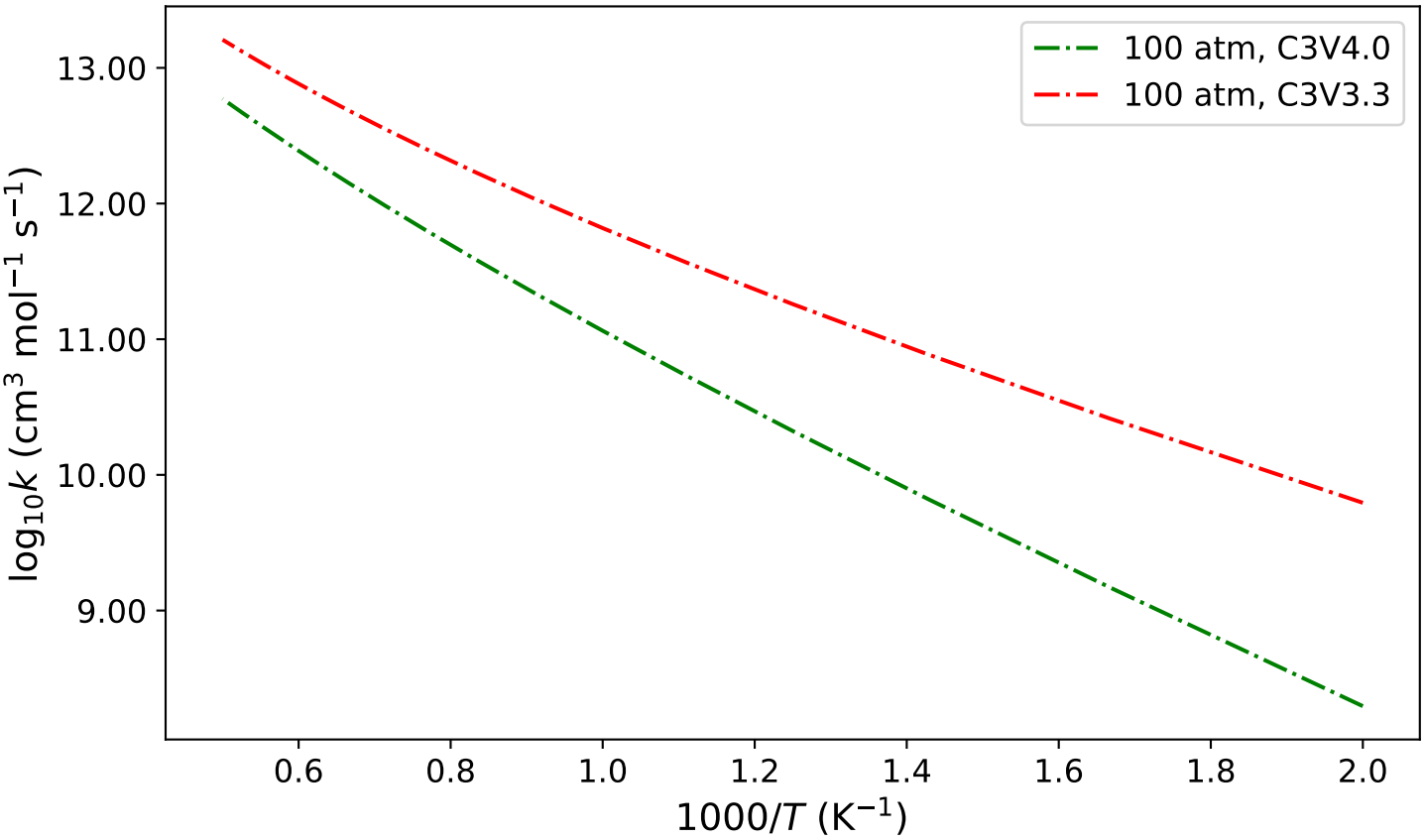



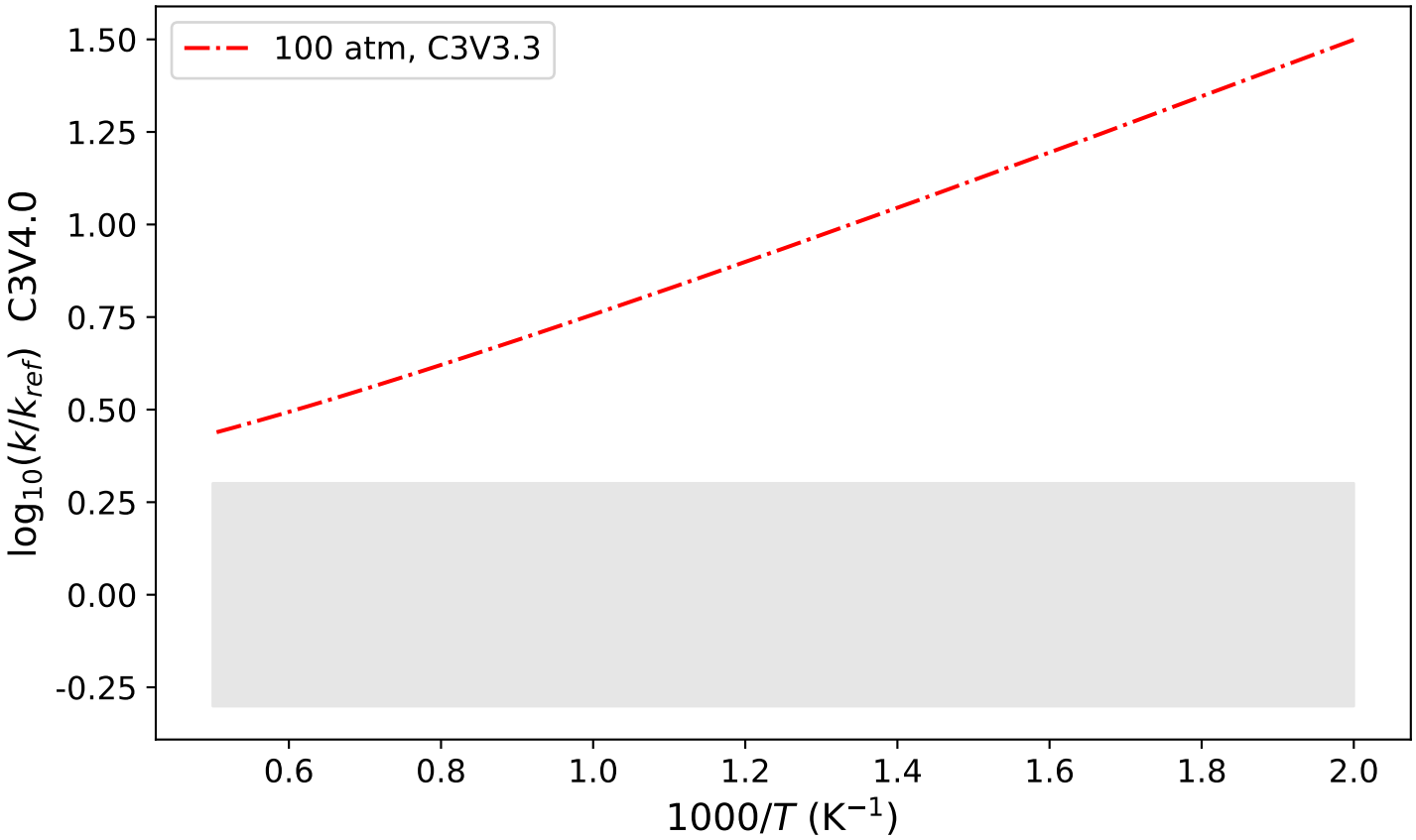
C3H6OOH1-3O2 = OH + C3KET13



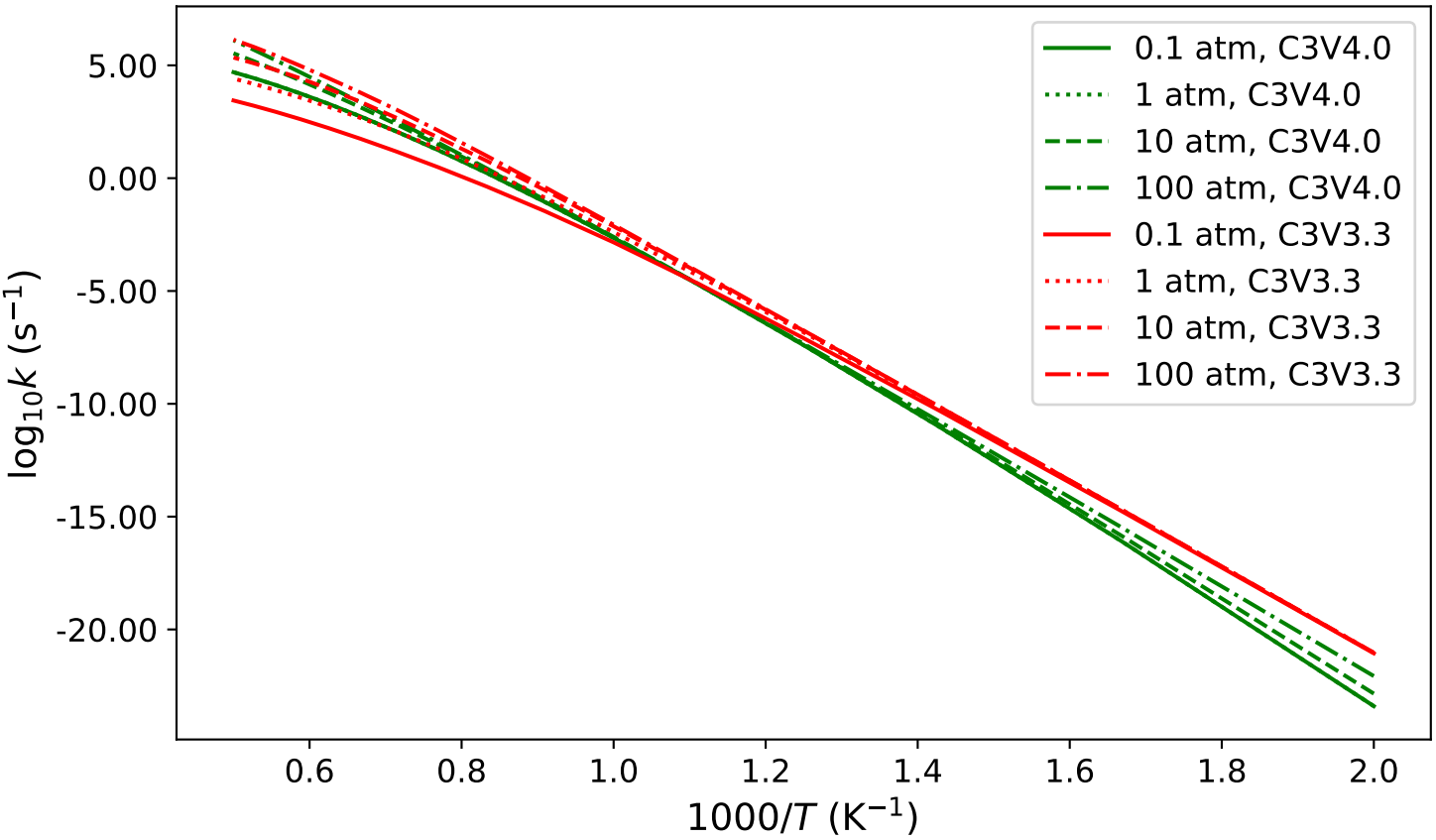



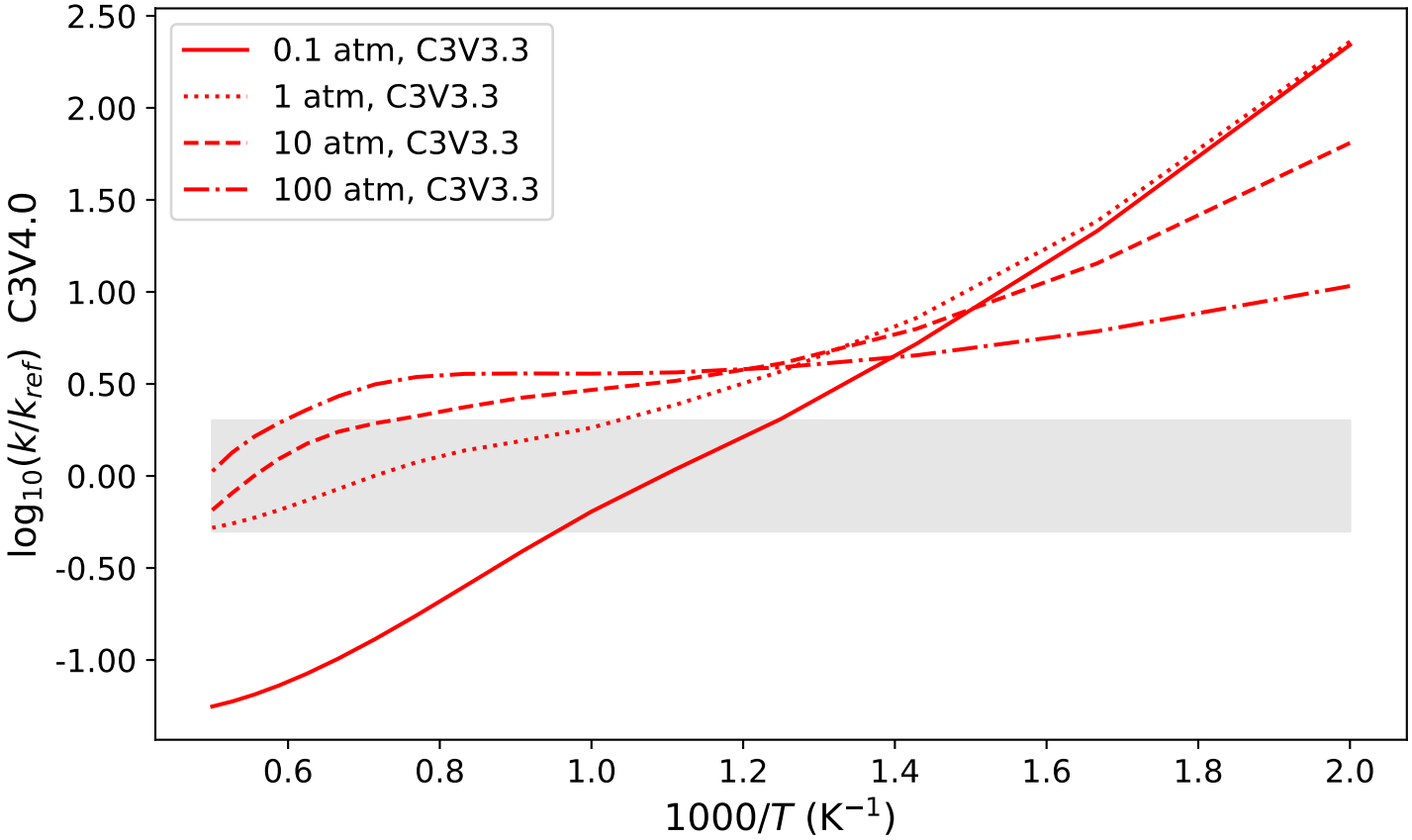
H + C3H5OH = H2 + CH2CCH2OH



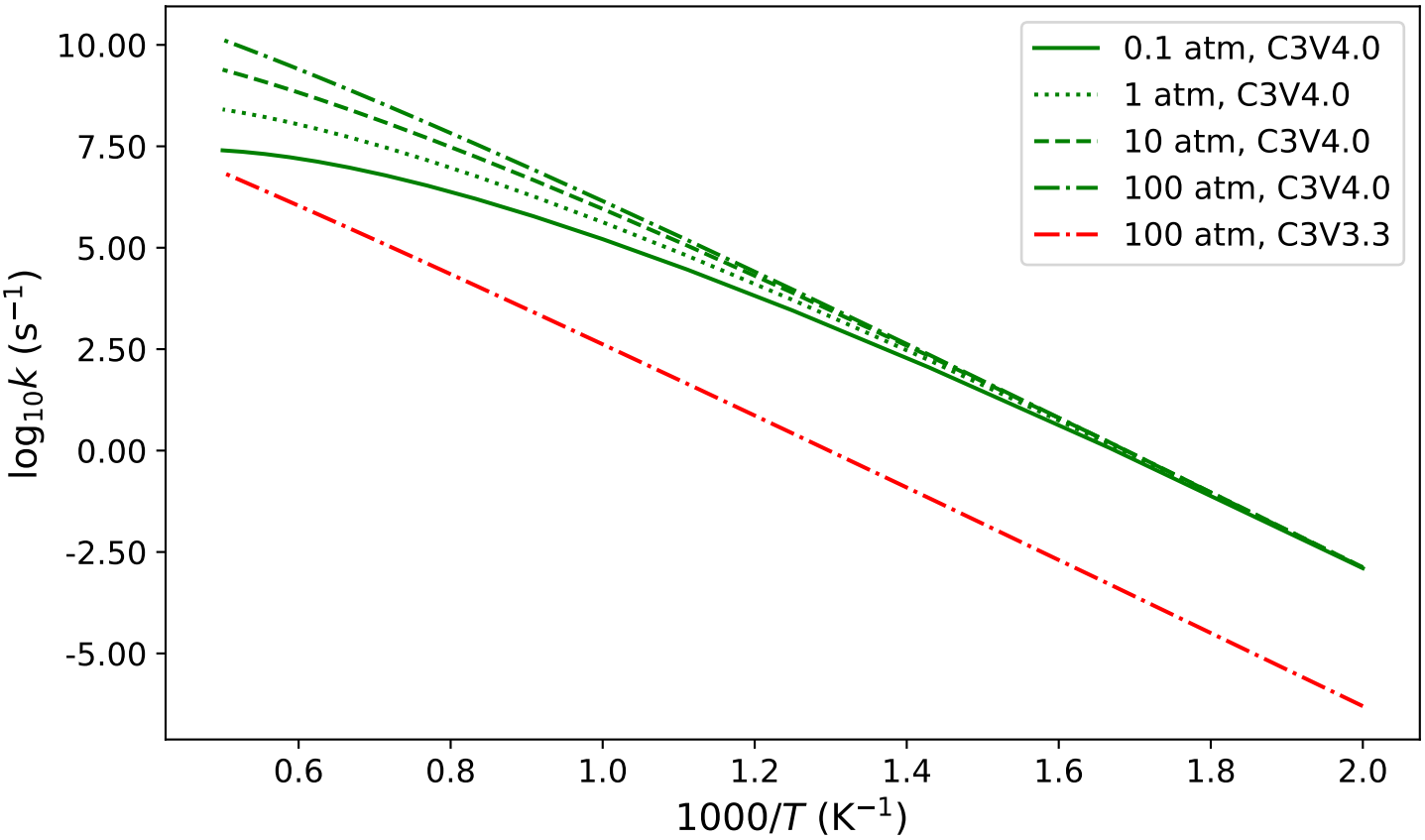



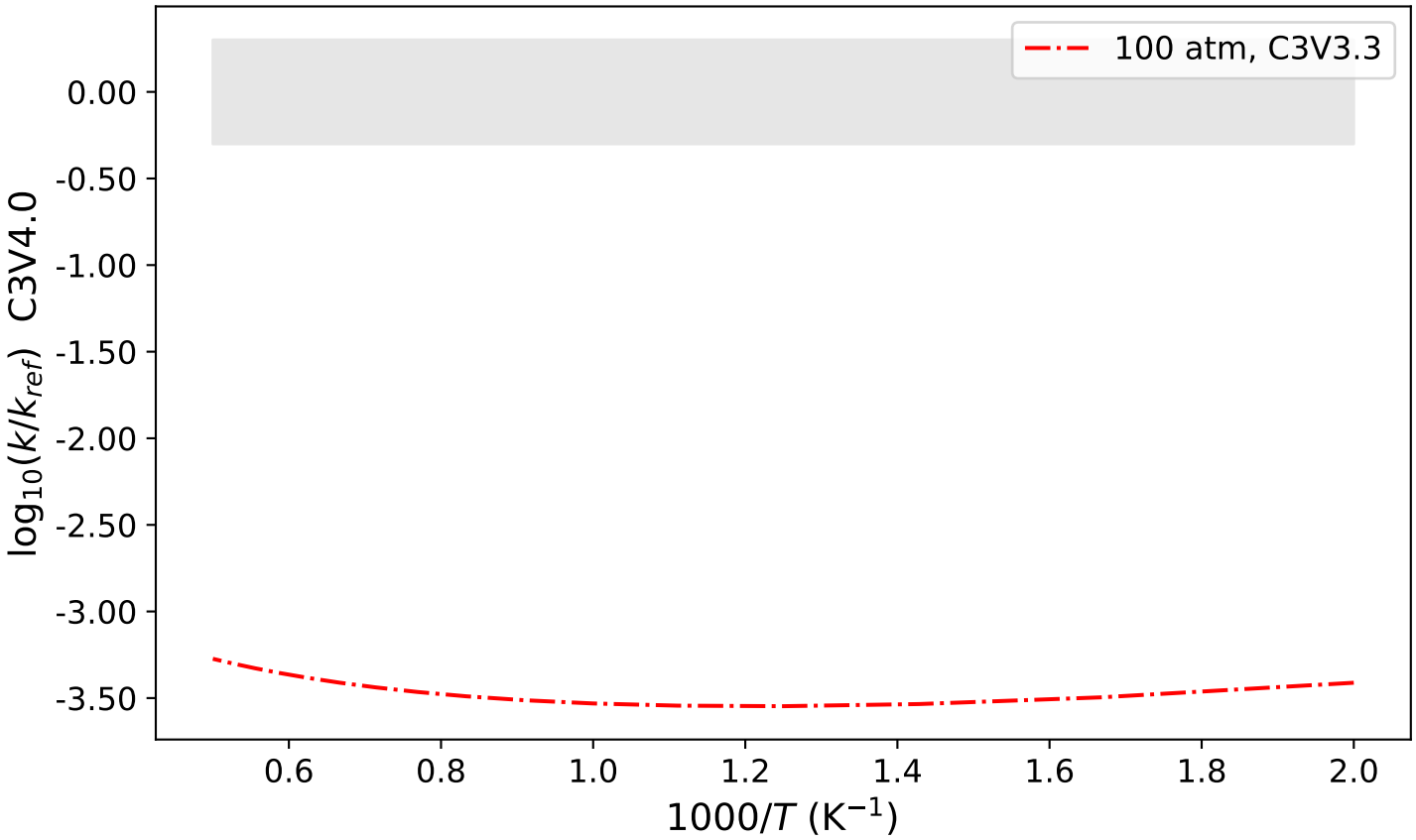
NC3H7OH = CH3 + PC2H4OH



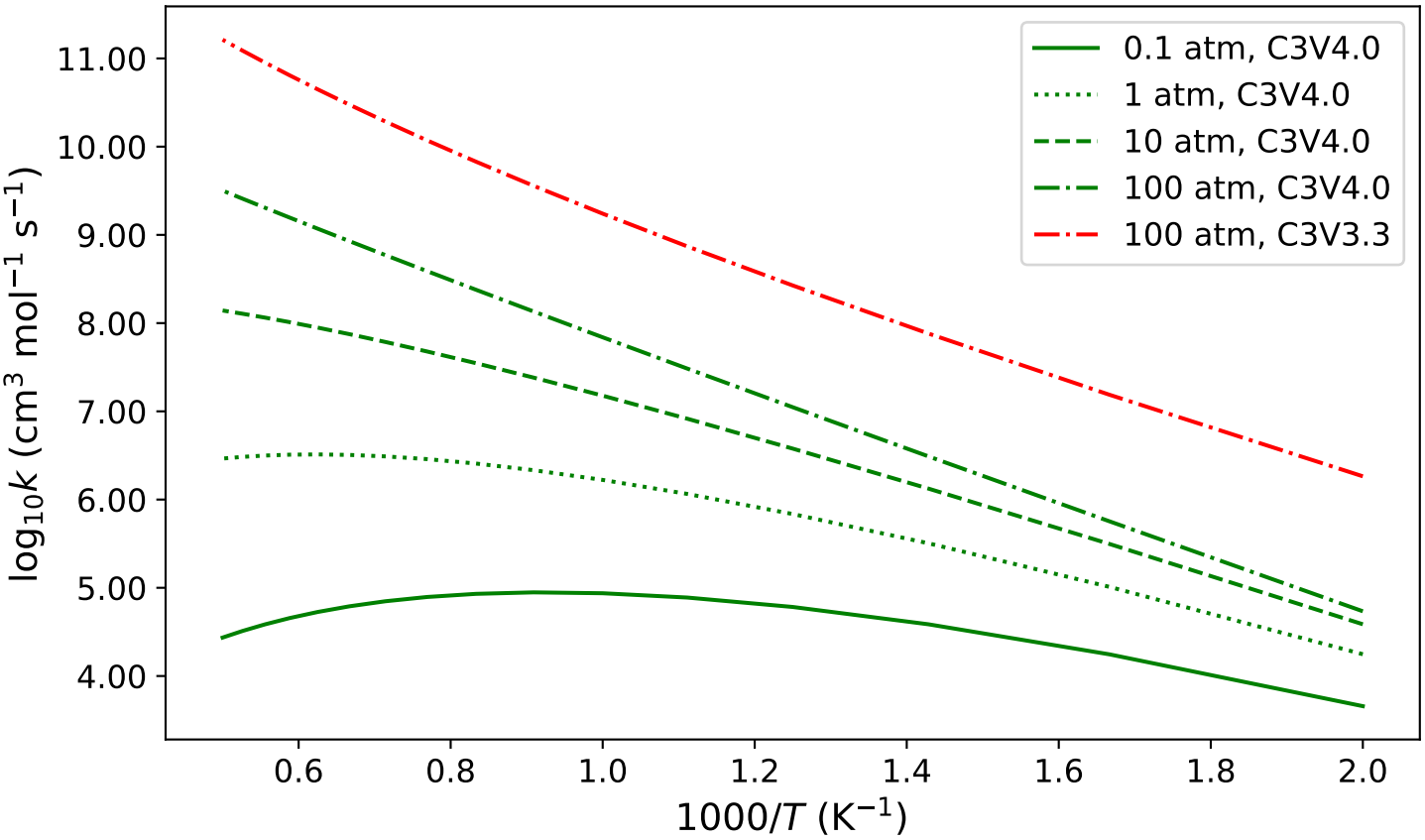



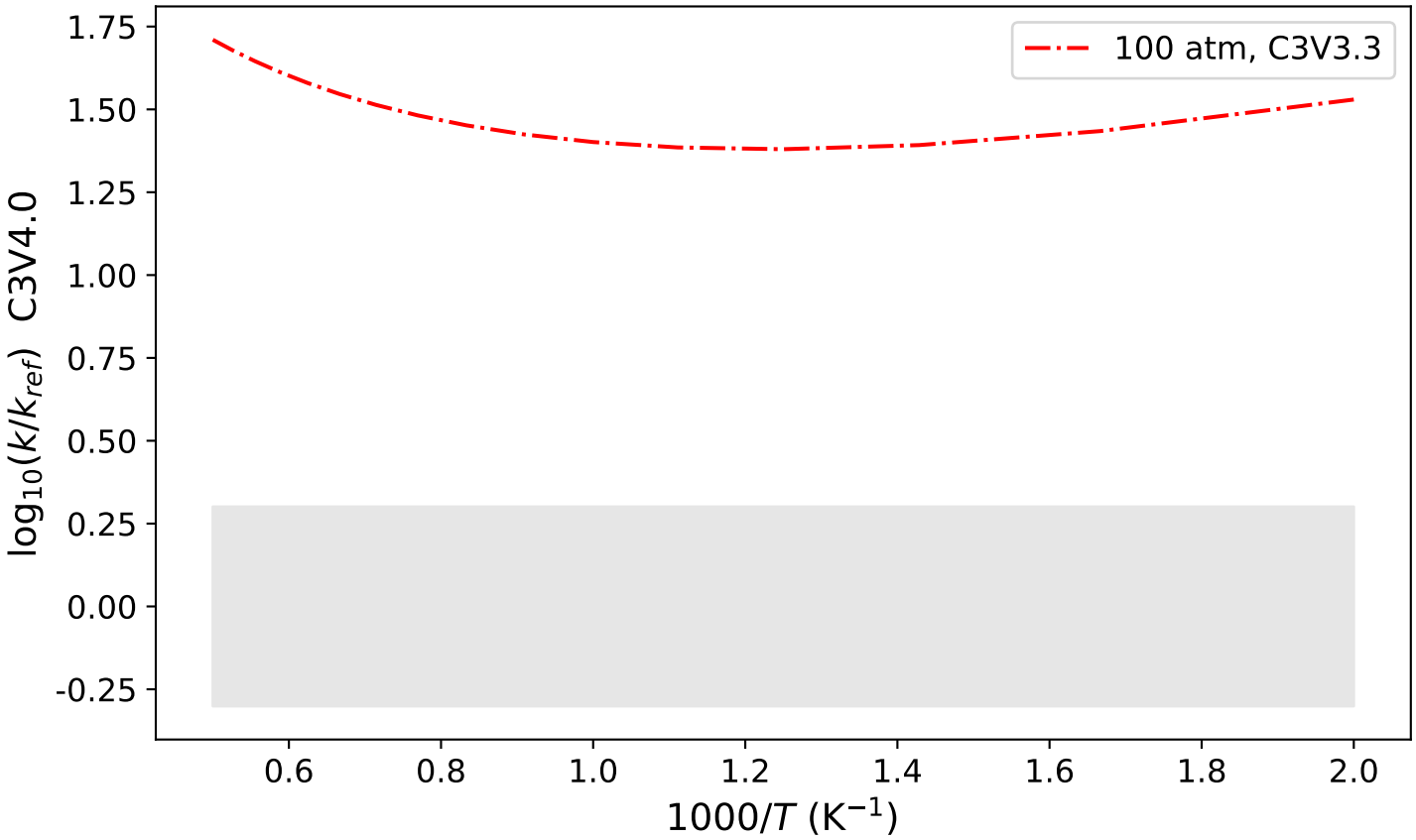
C4H6OHOOH1-4-3 = OH + C3H4CH2OH-3O



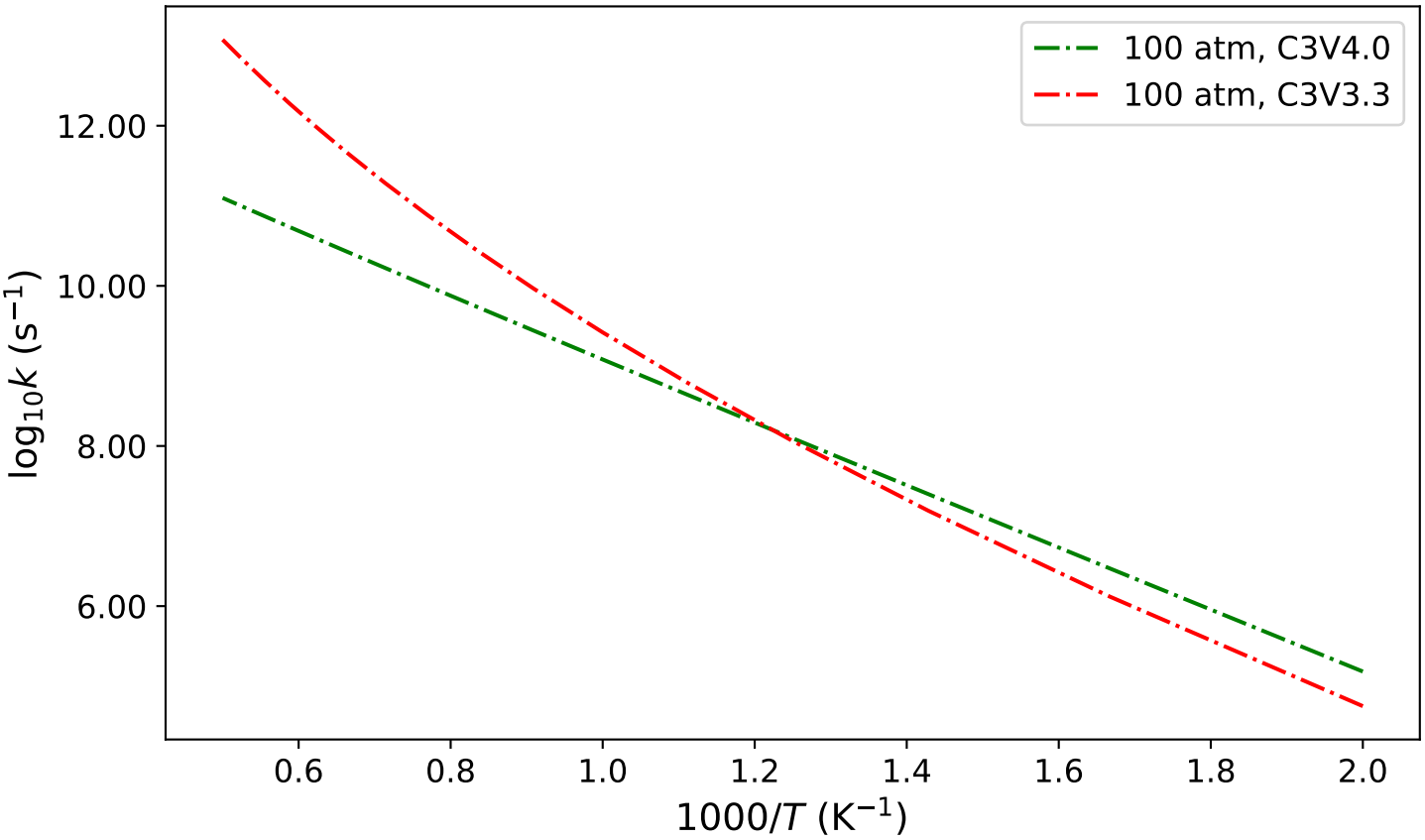



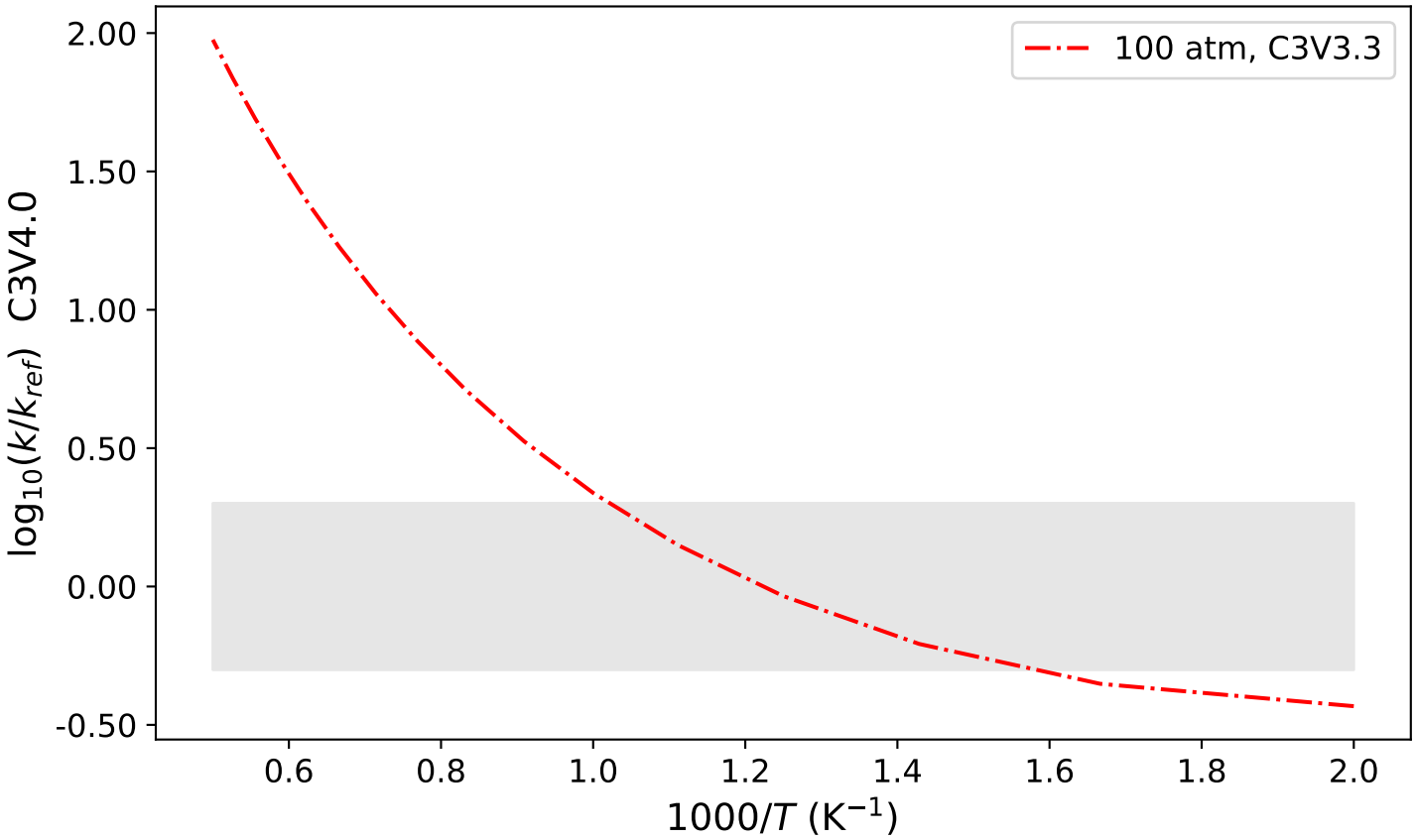
HO2 + C4H6 = C4H61OOH3-4



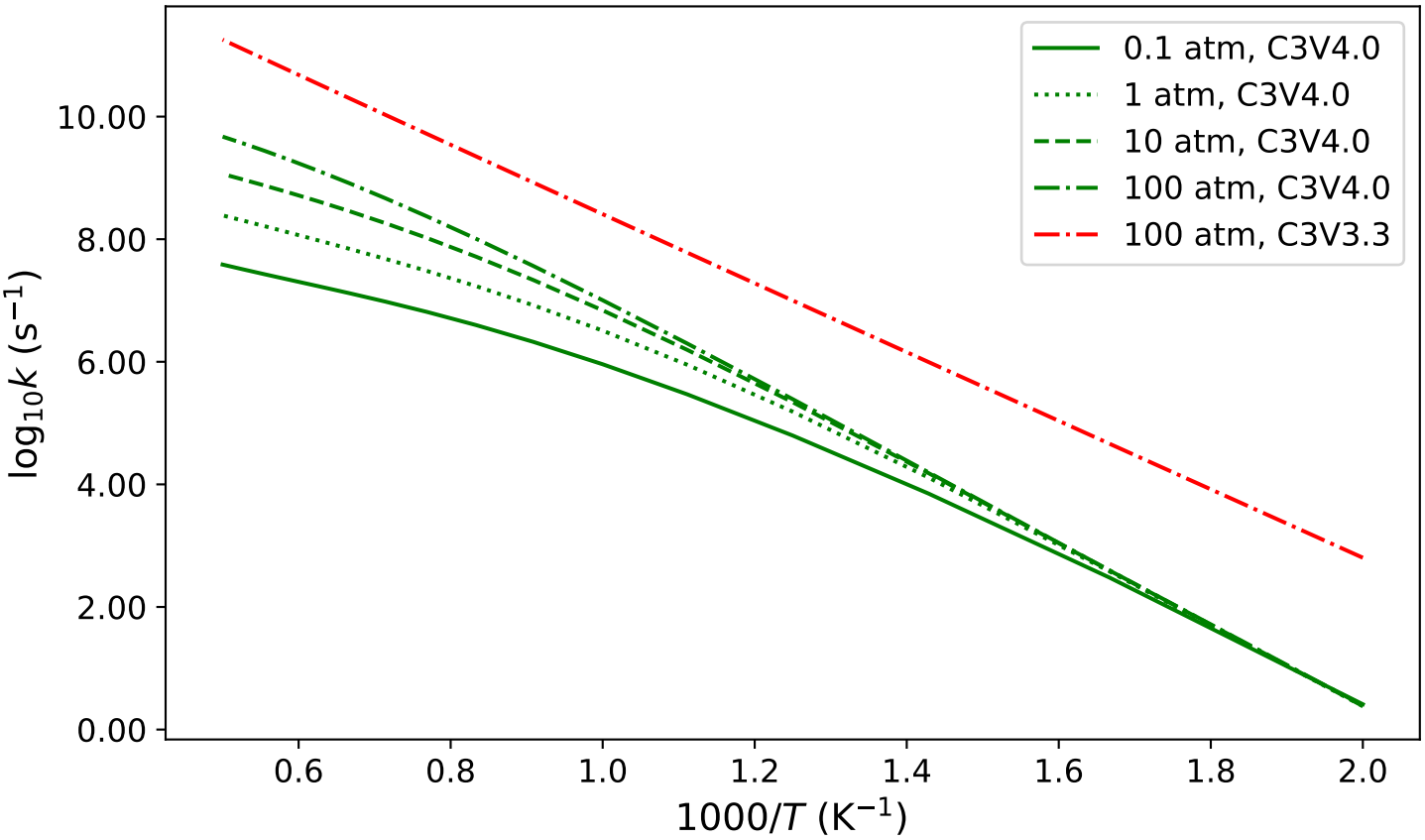



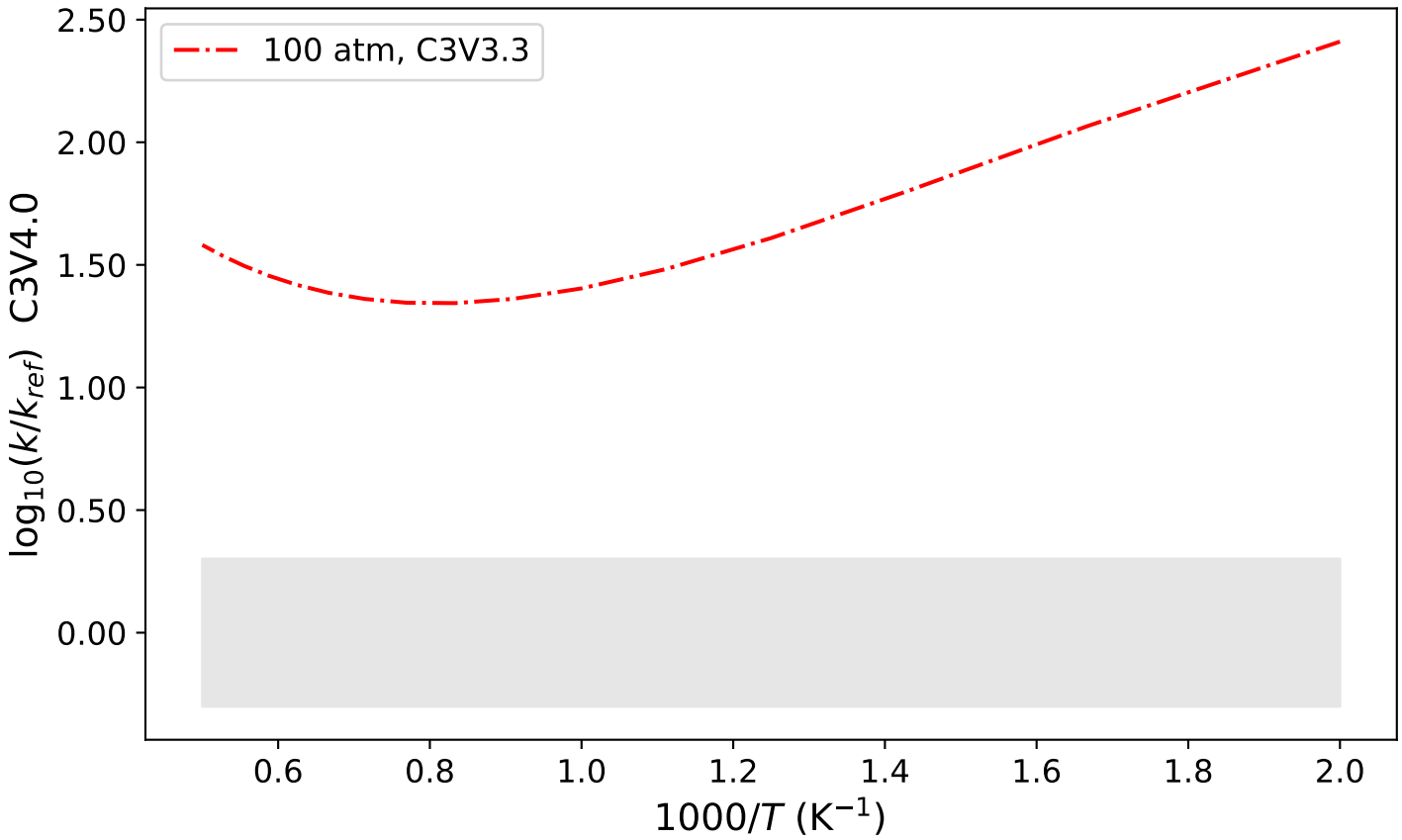
QHO2CH2CHCHO = HO2 + C2H3CHO



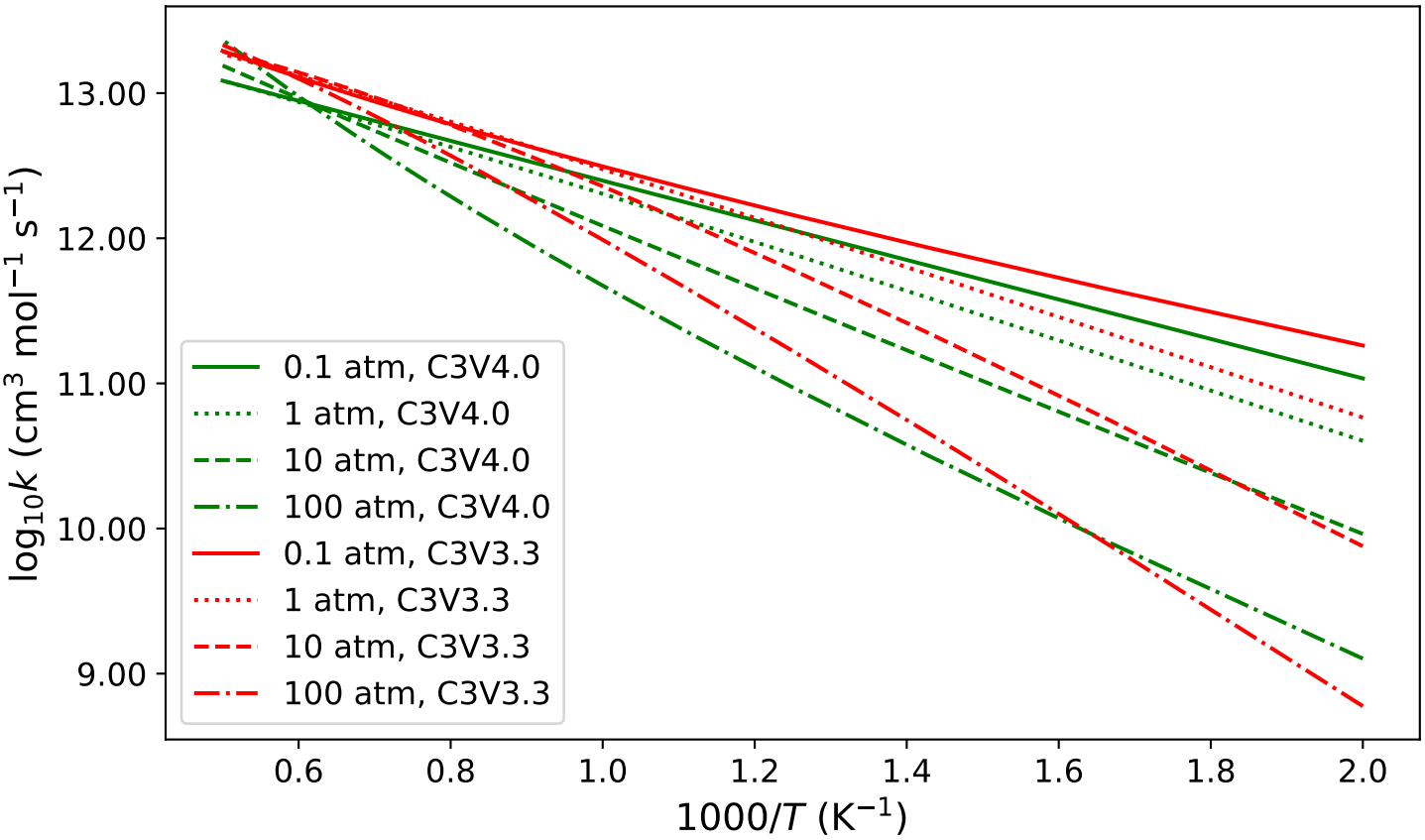



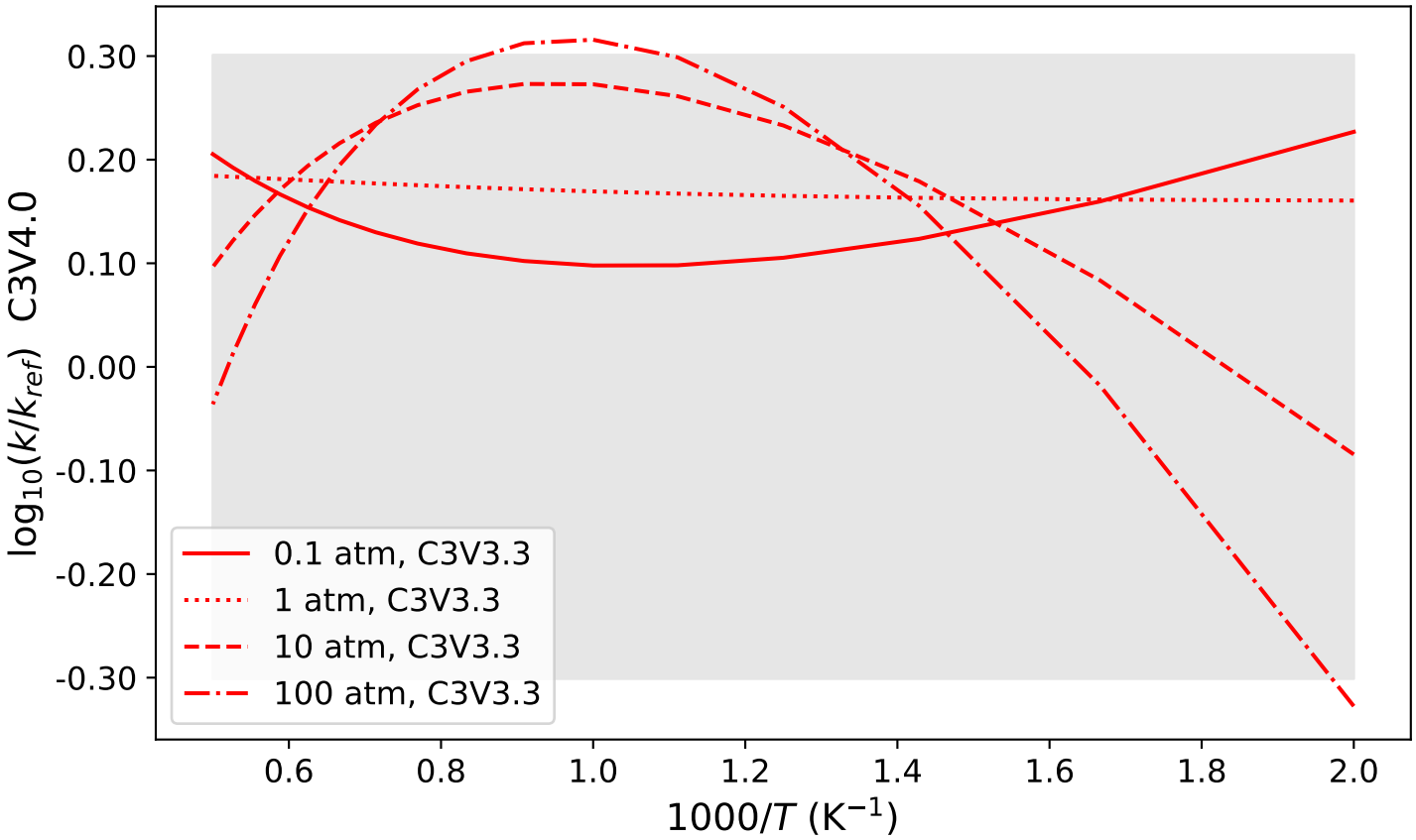
PC4H8OH-3 = C3H6 + CH2OH



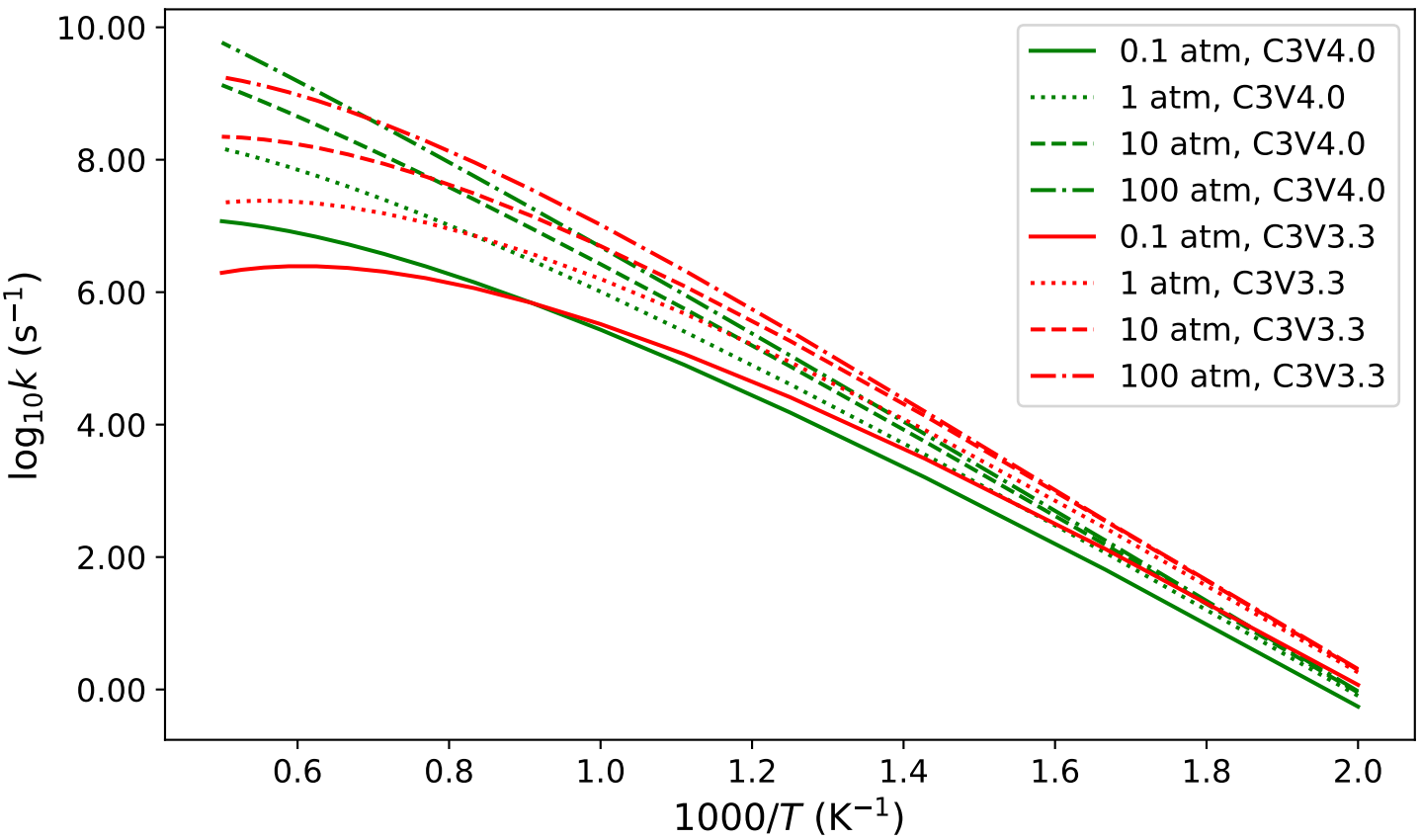



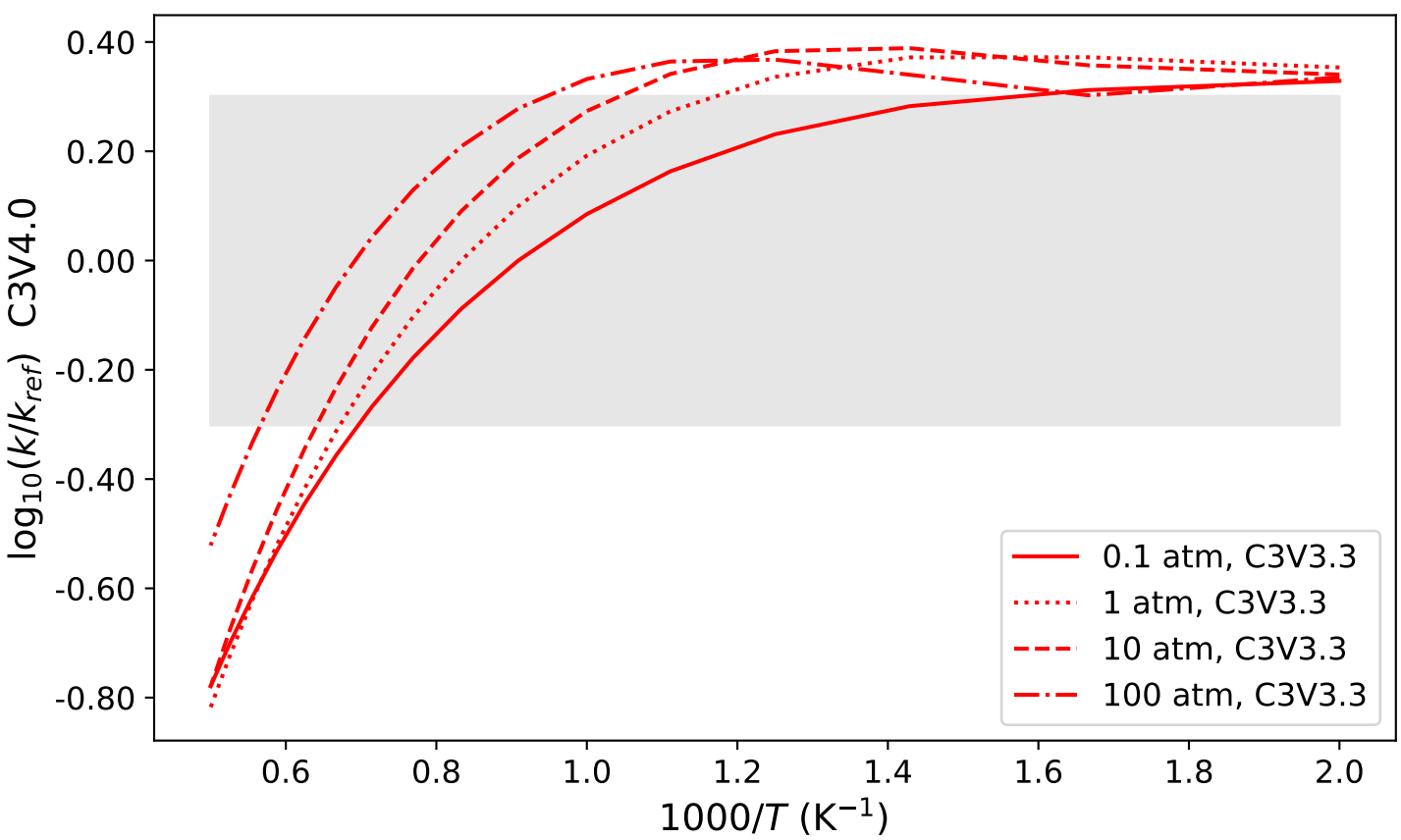
H + C4H8-1 = C2H5 + C2H4



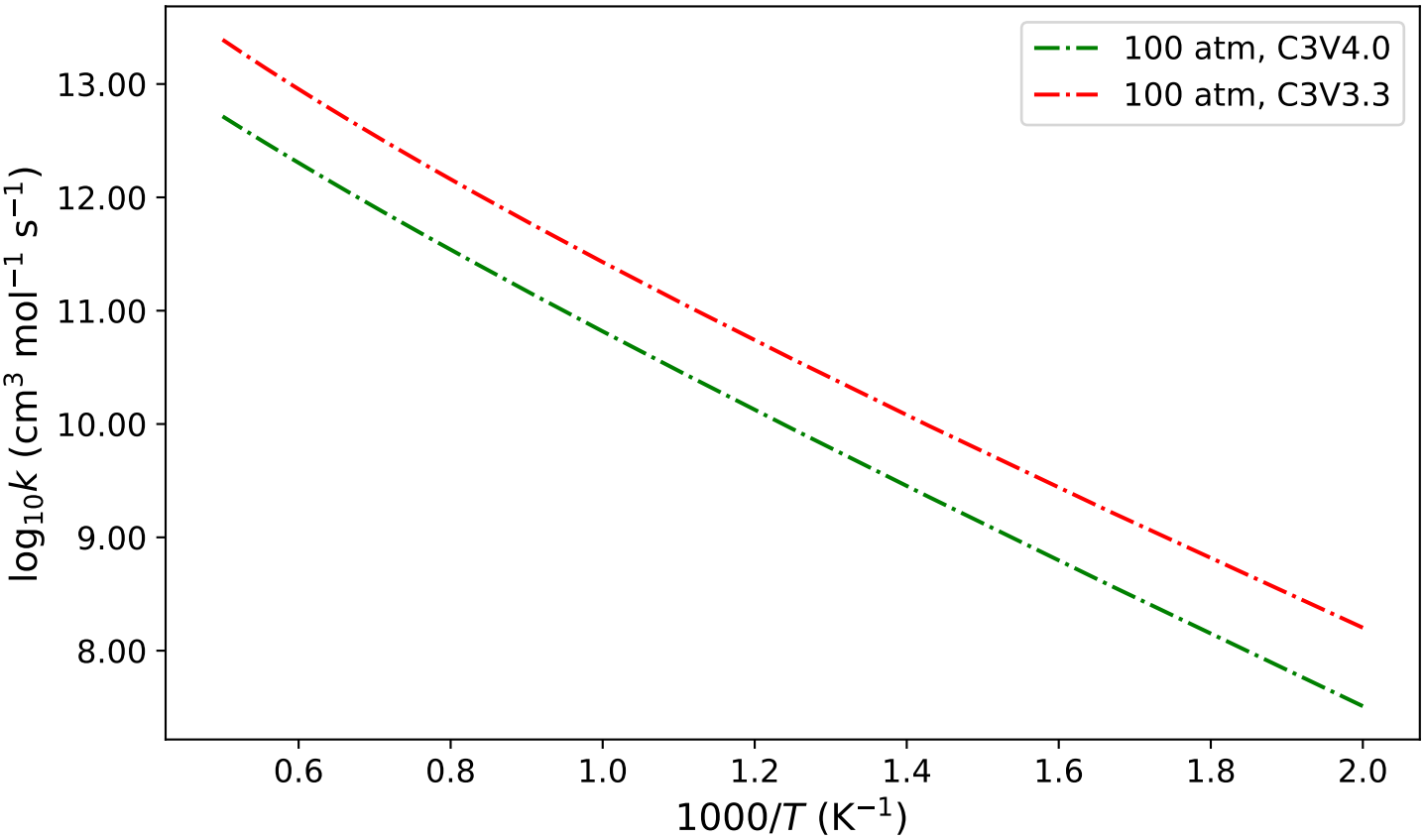



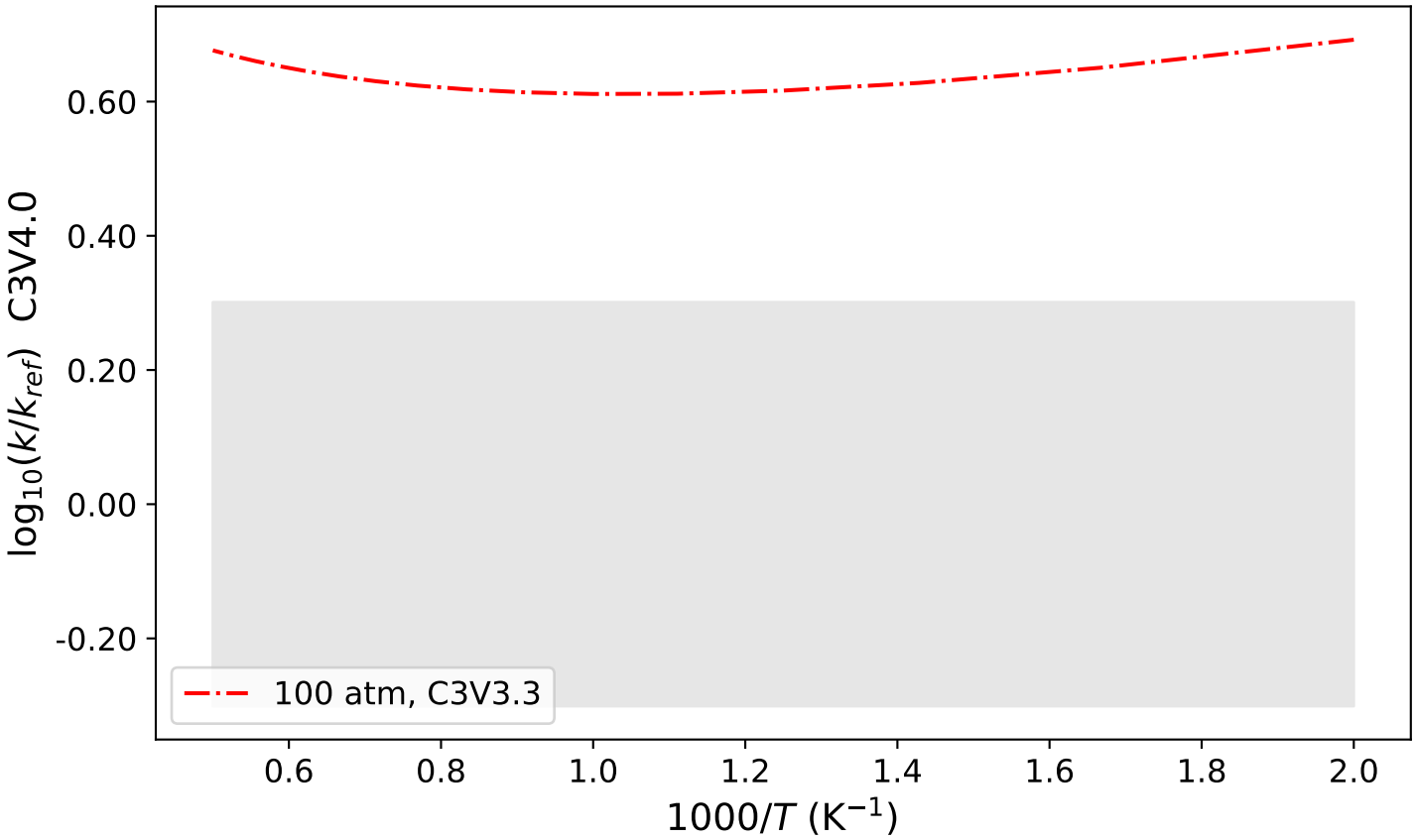
NC3H7 = CH3 + C2H4



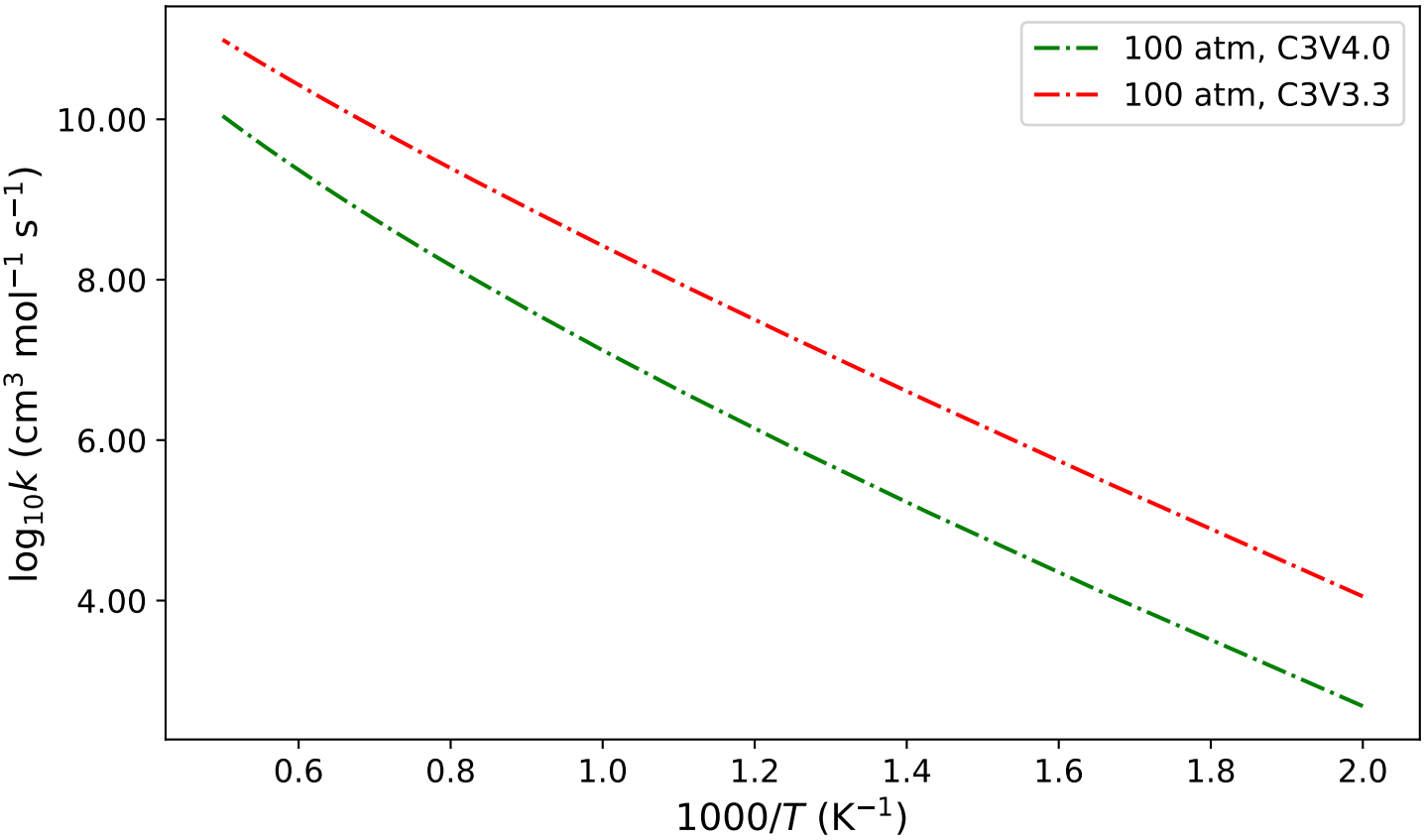



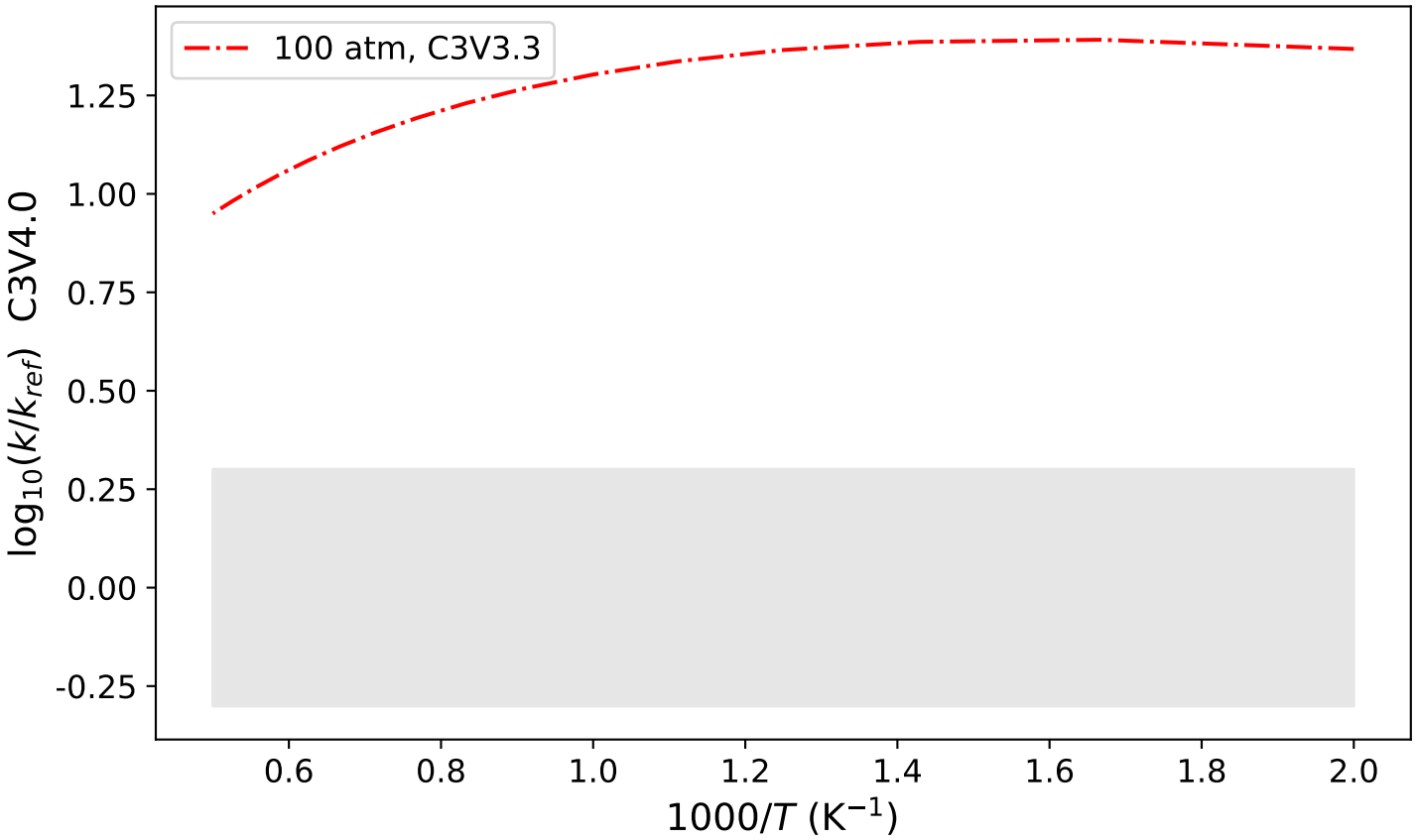
H + C4H6 = H2 + C4H5-I



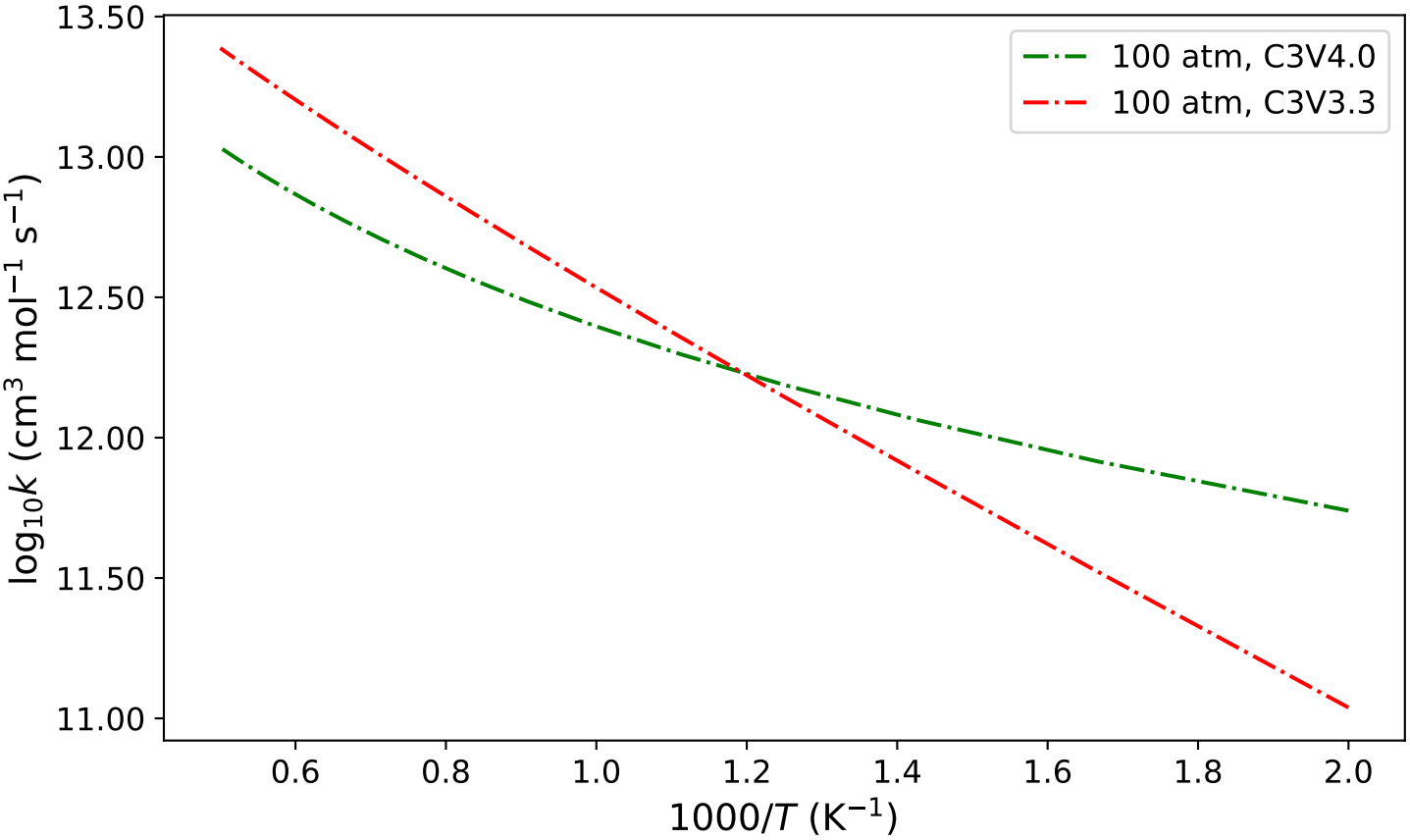



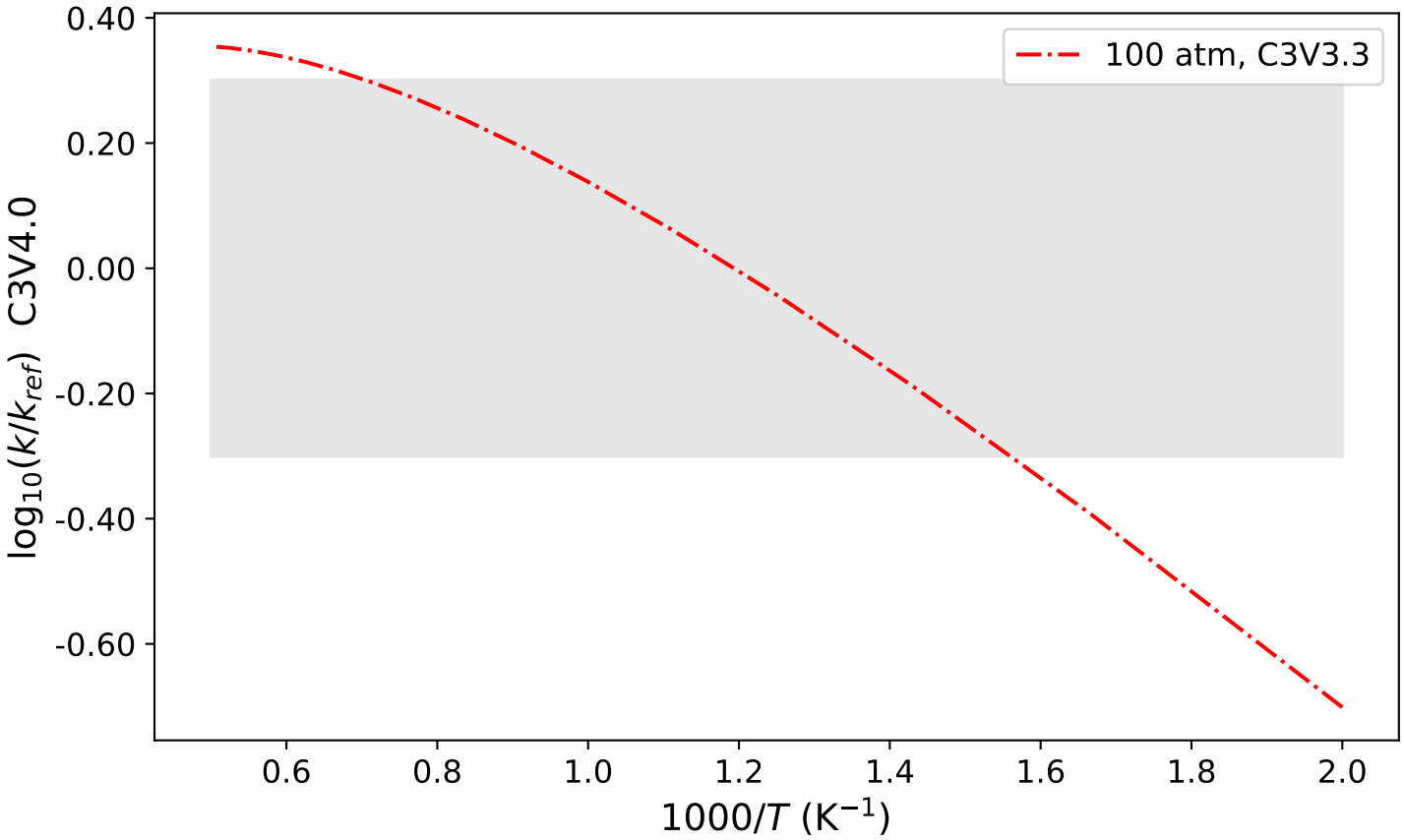
NC3H7OH + HO2 = H2O2 + C3H6OH1-3



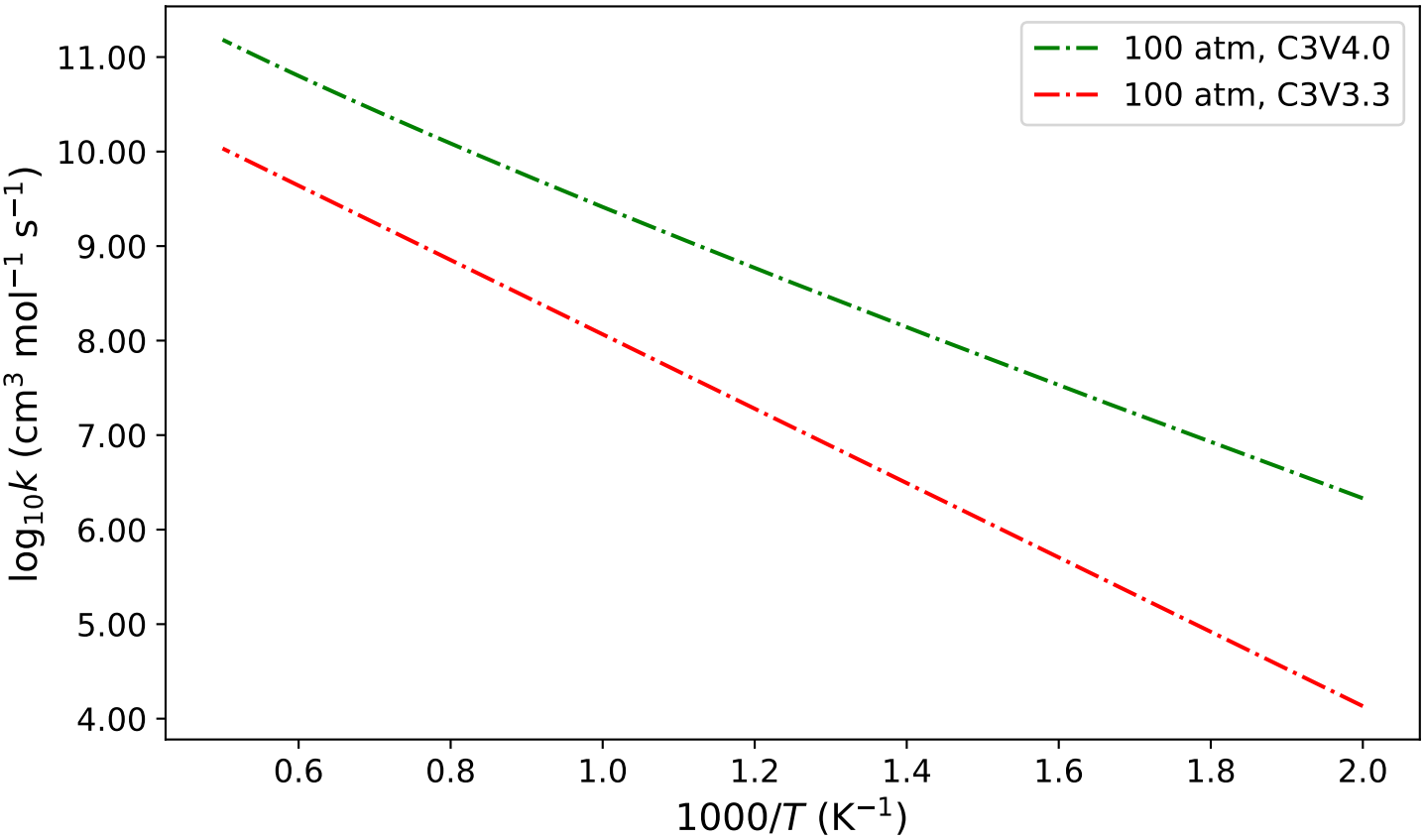



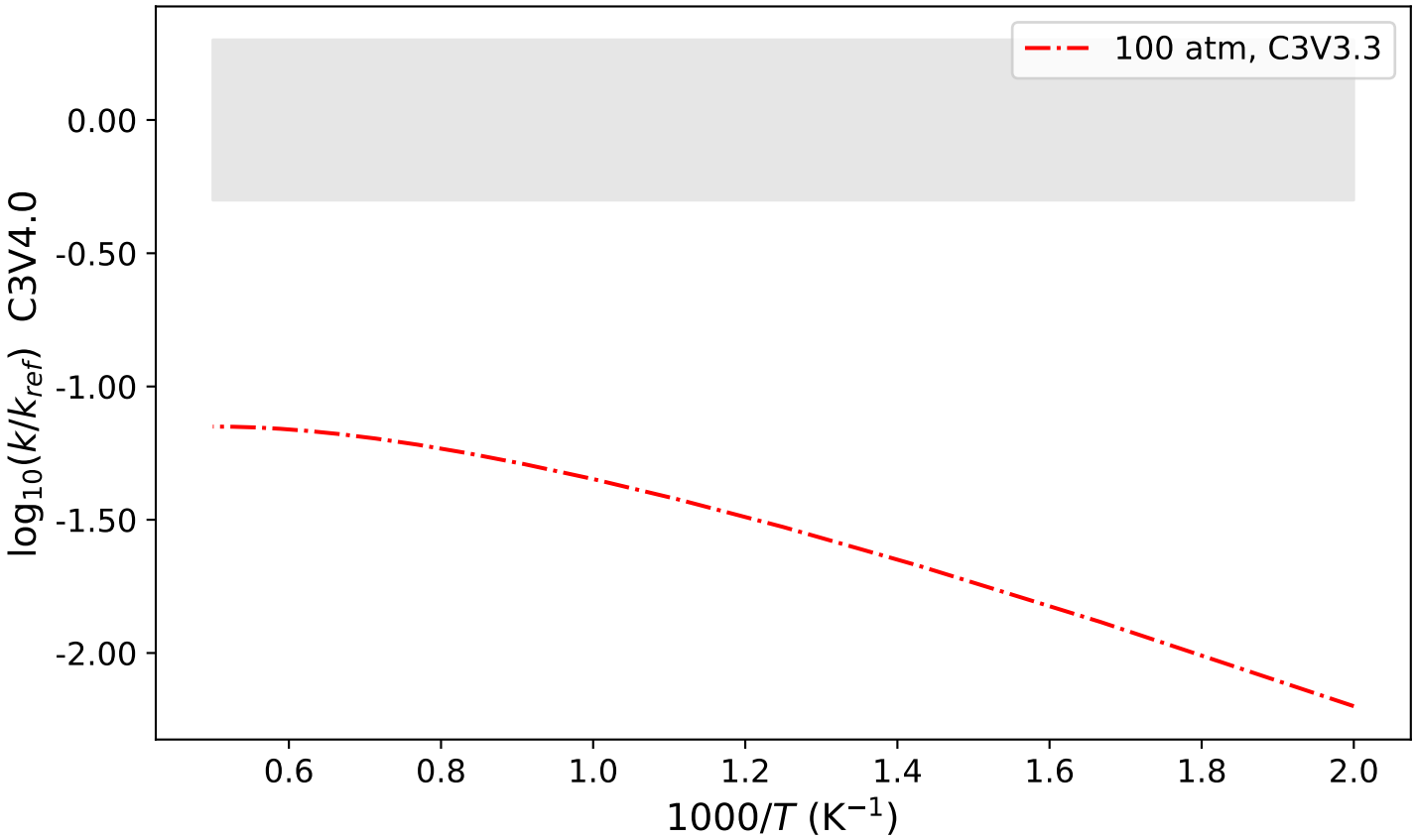
O + C3H4-A = OH + C3H3



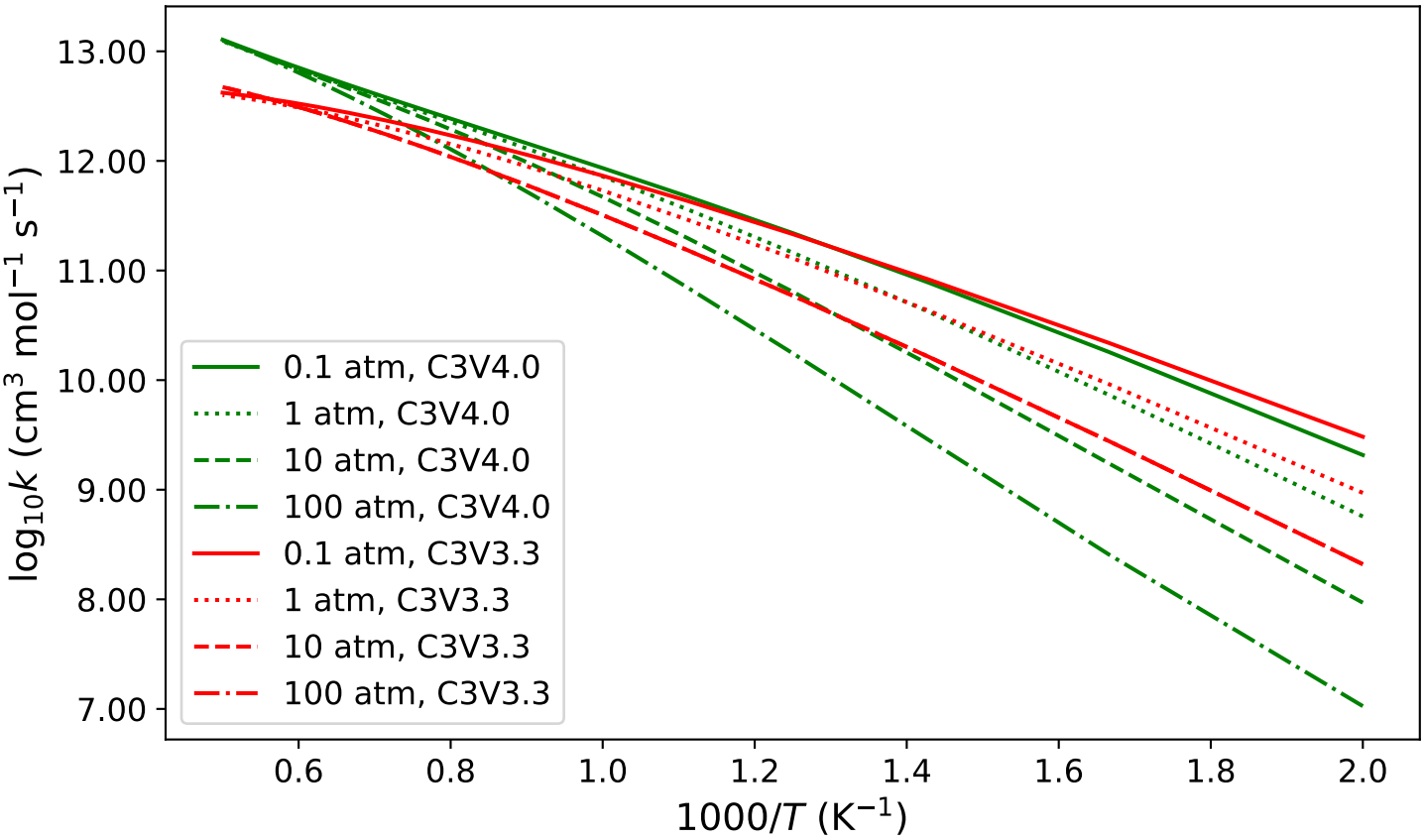



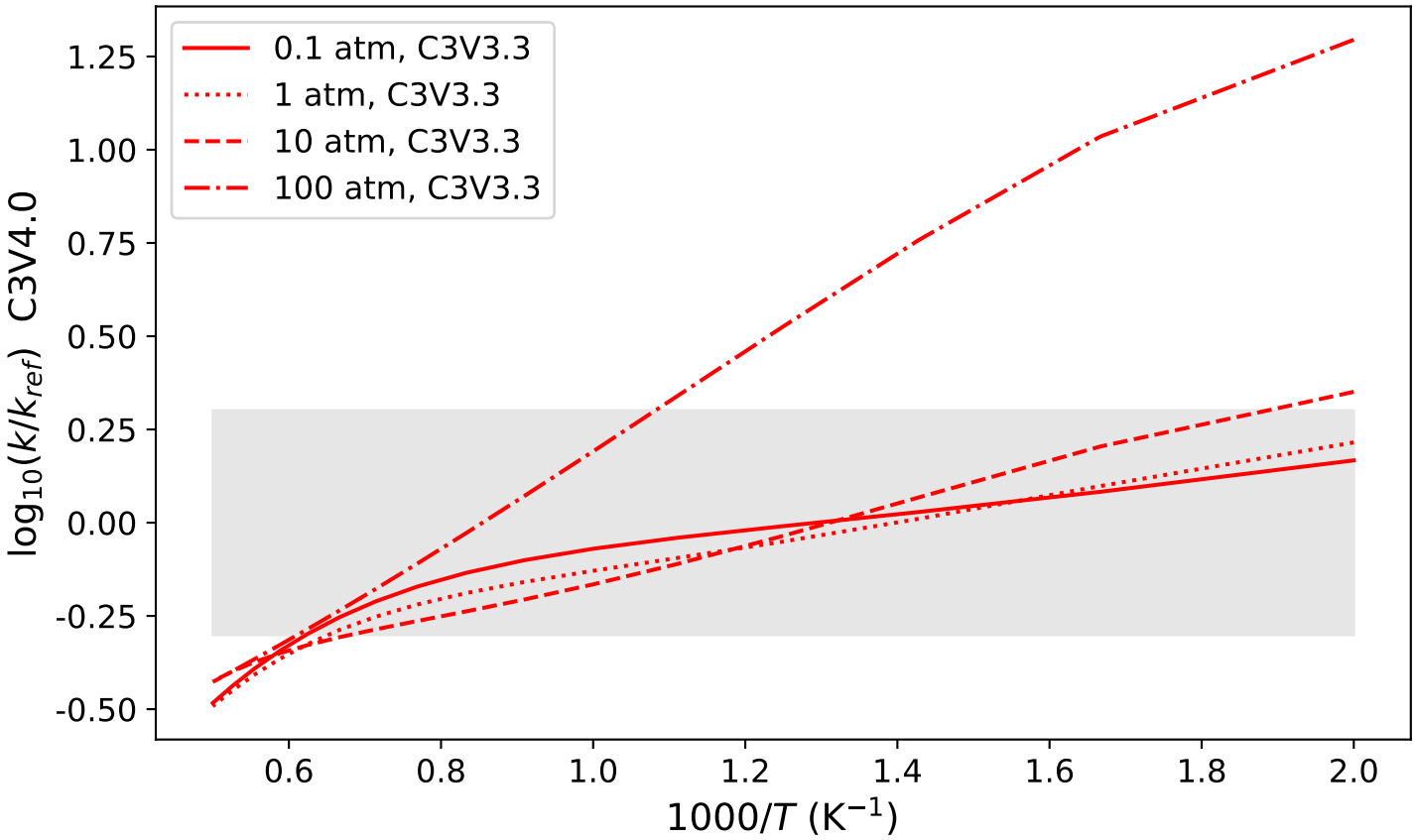
C2H3 + C4H10 = C2H4 + PC4H9



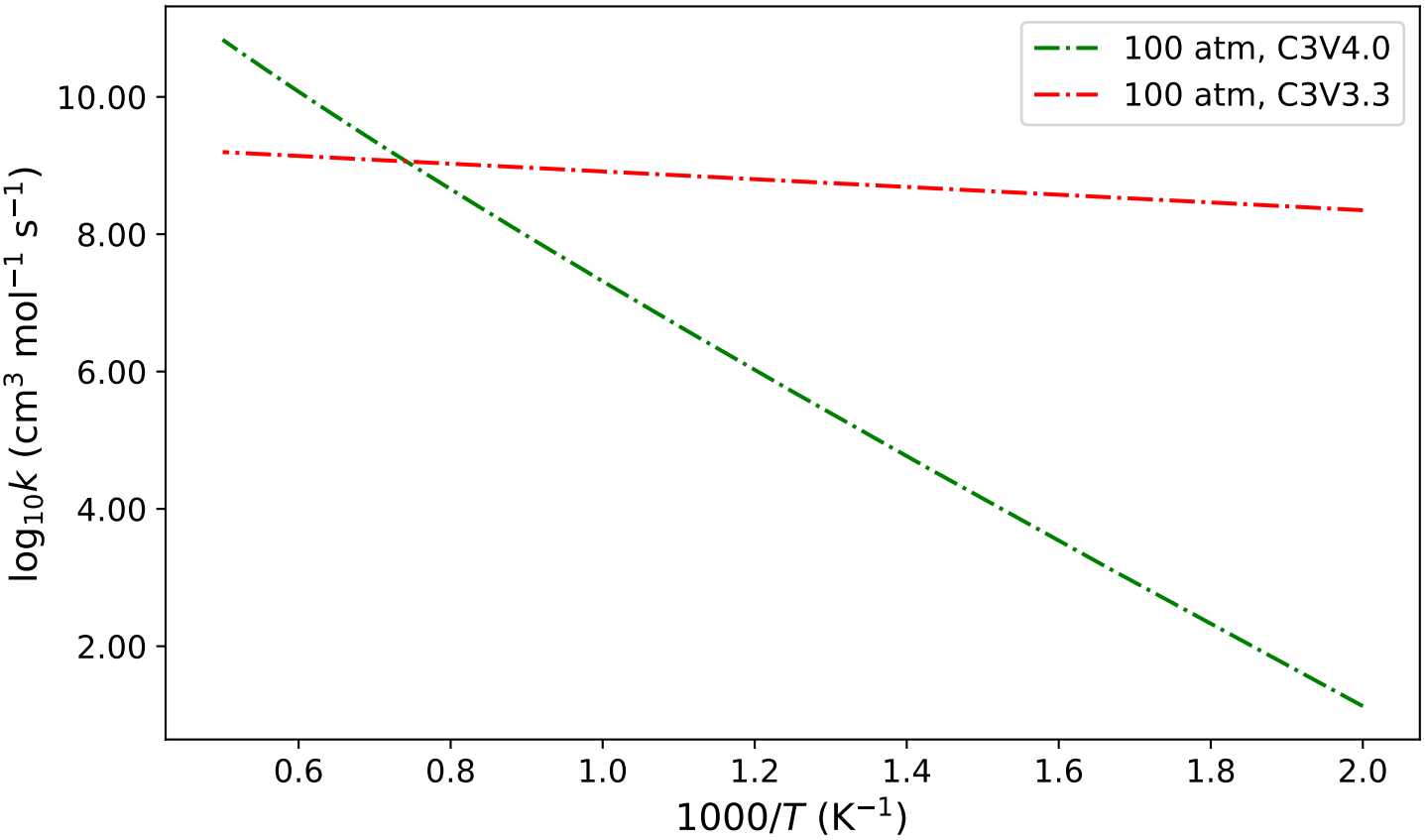



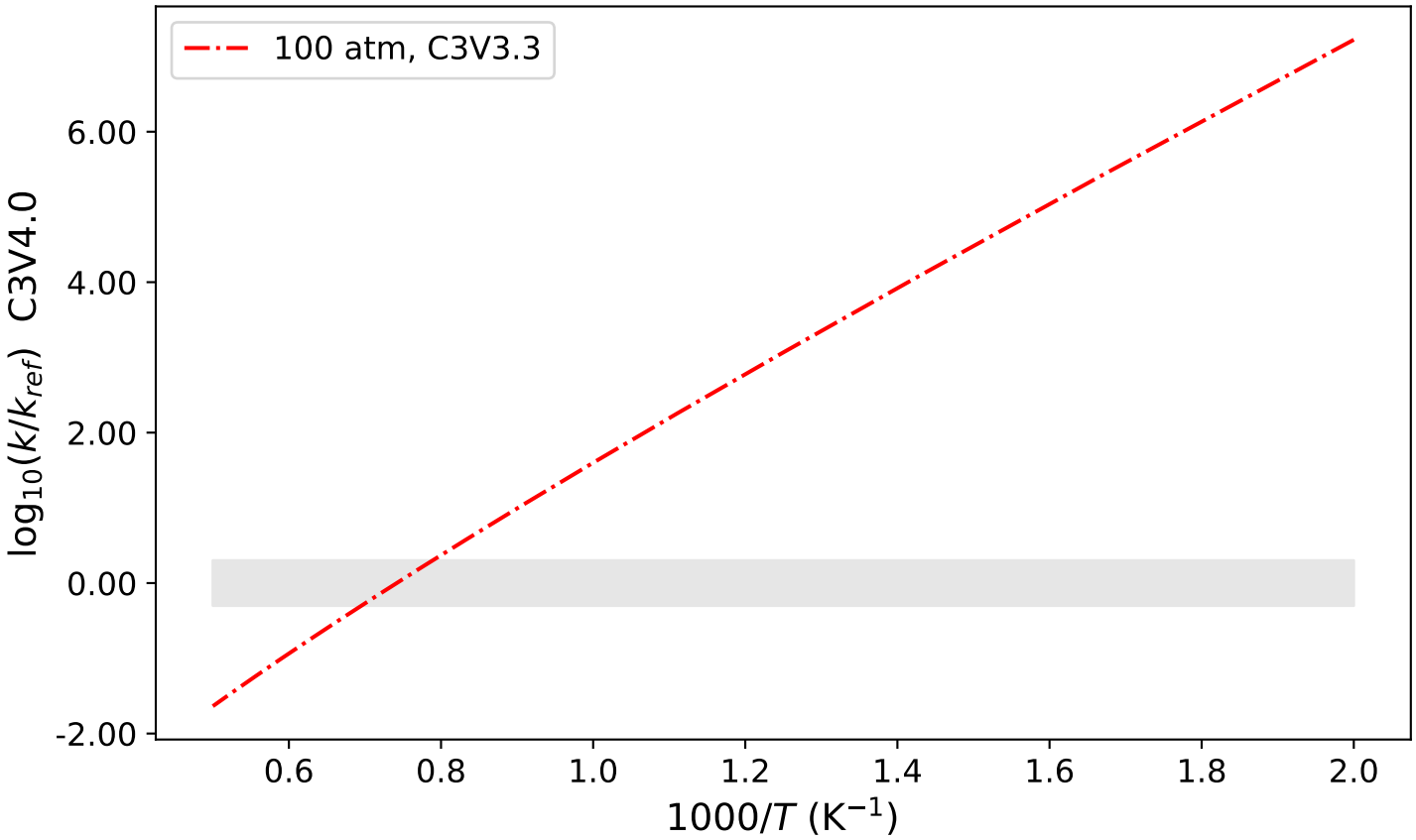
H + C4H4 = C2H3 + C2H2



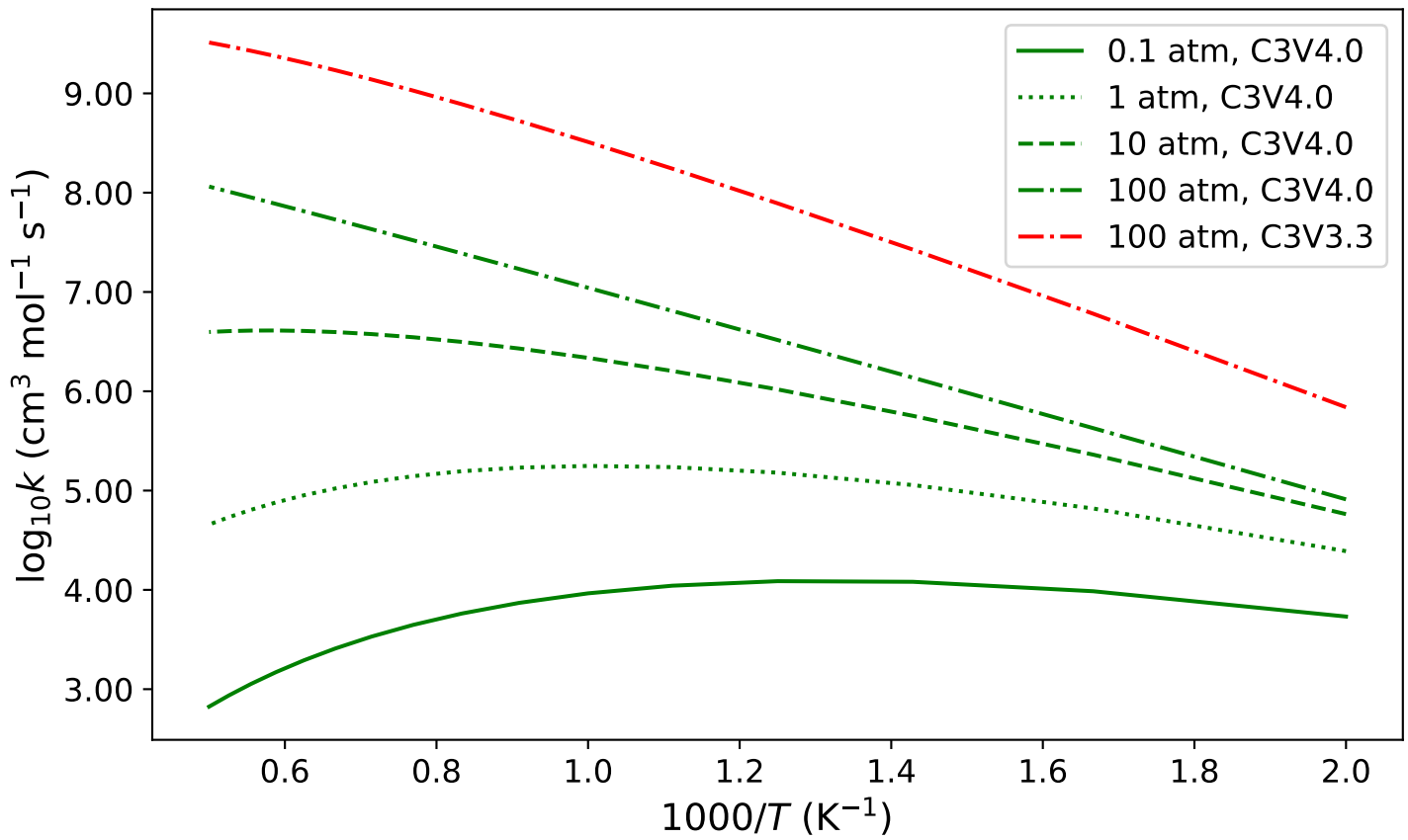



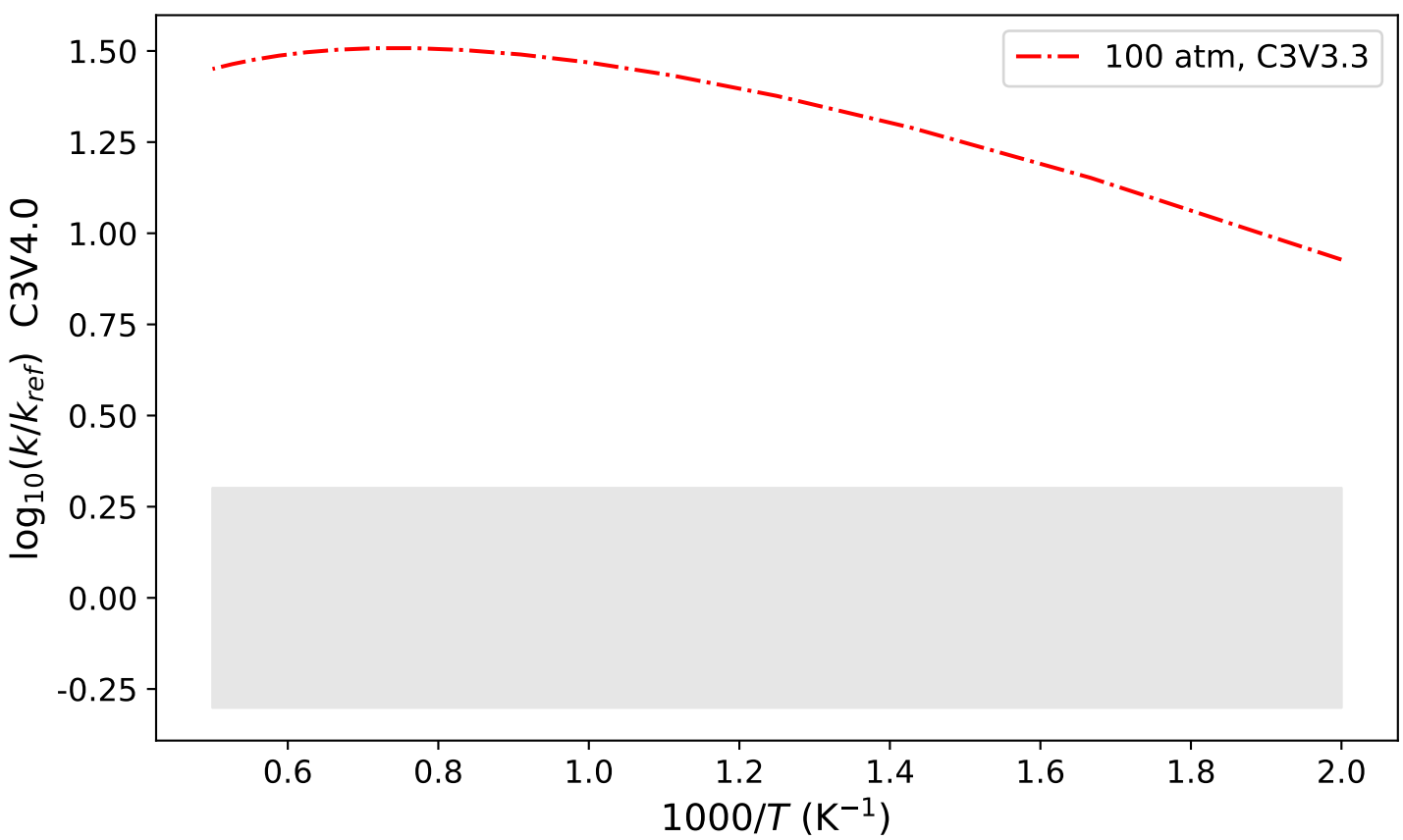
H2O2 + CH2CCH2OH = HO2 + C3H5OH



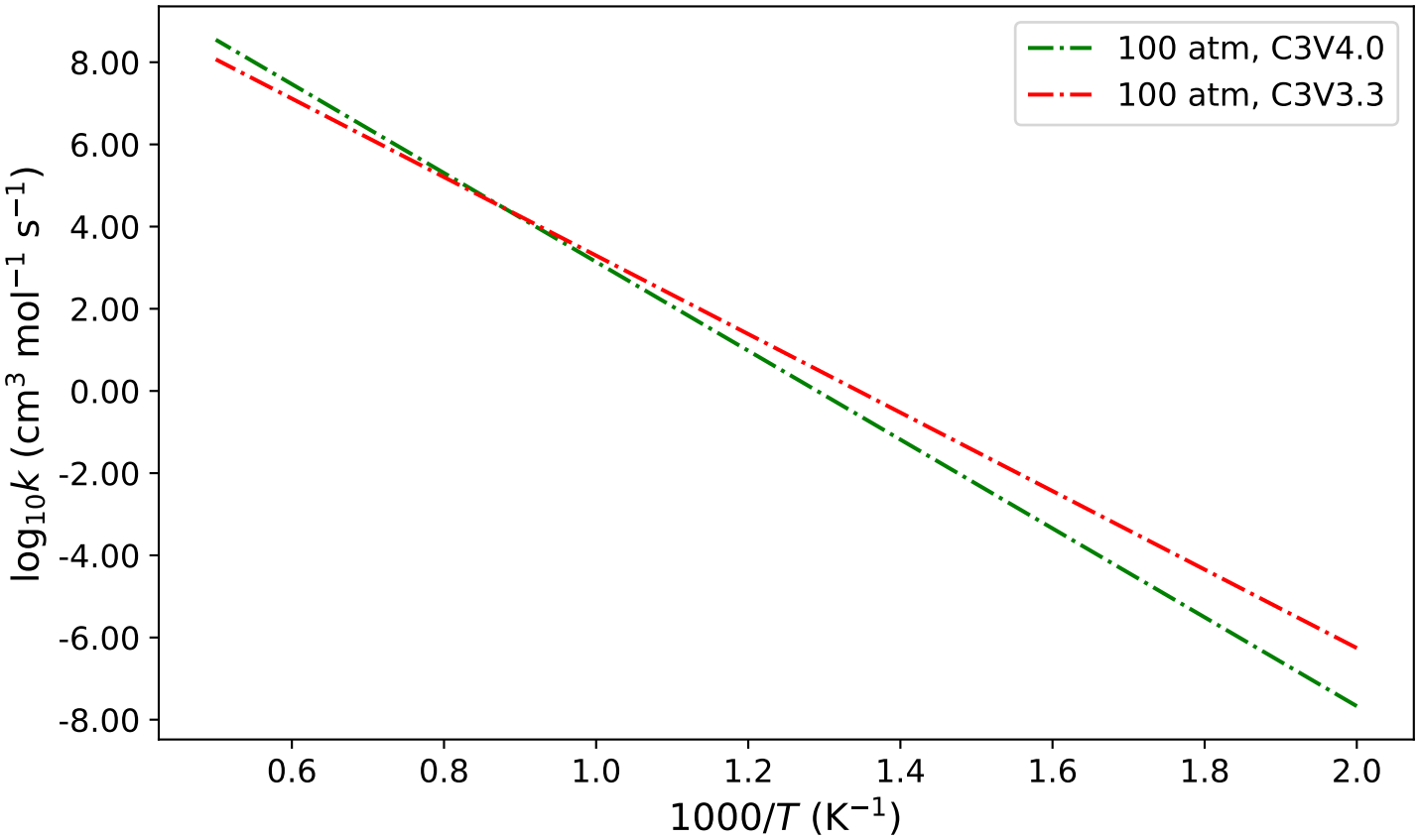



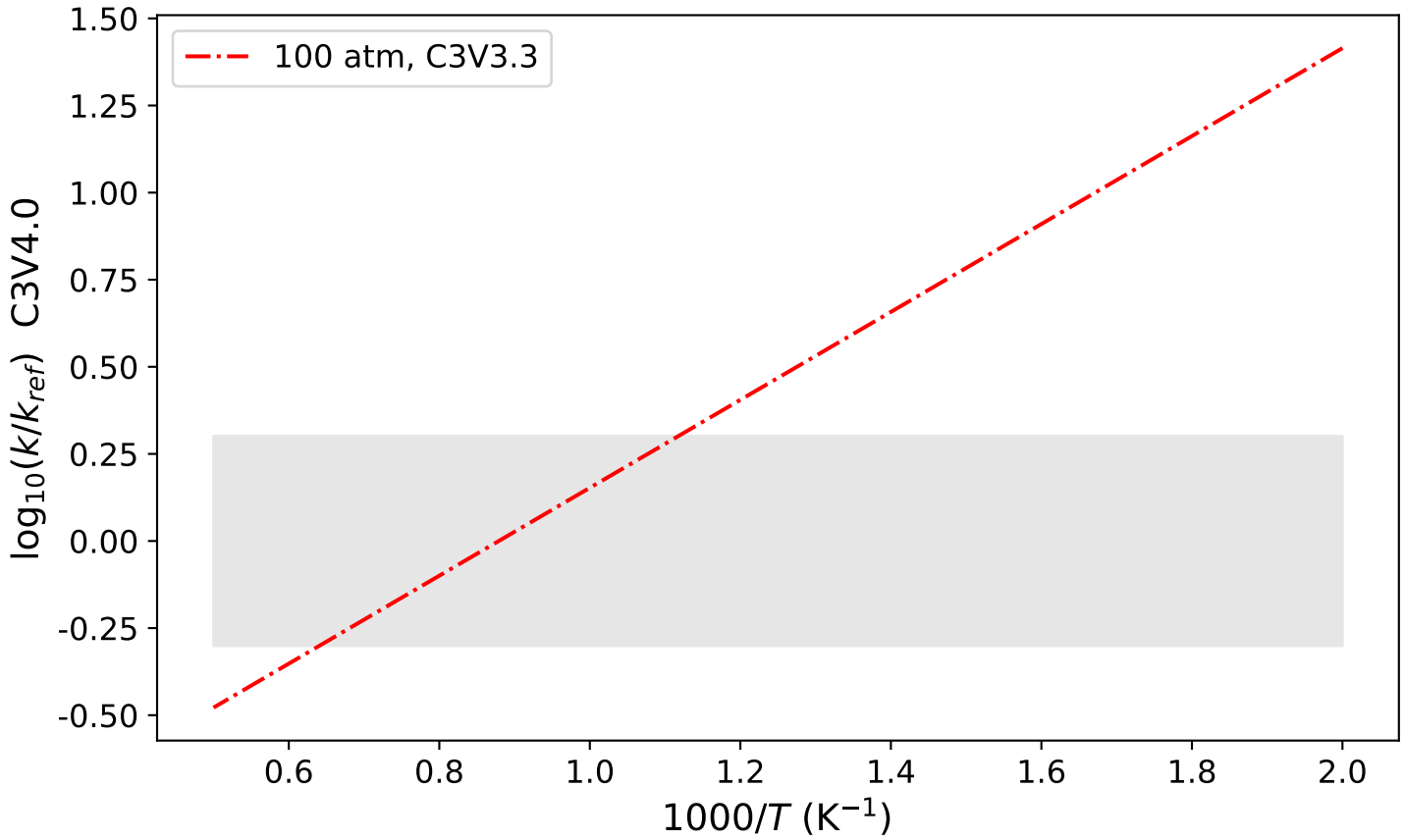
HO2 + C2H3CHO = CH3CH(OO)CHO



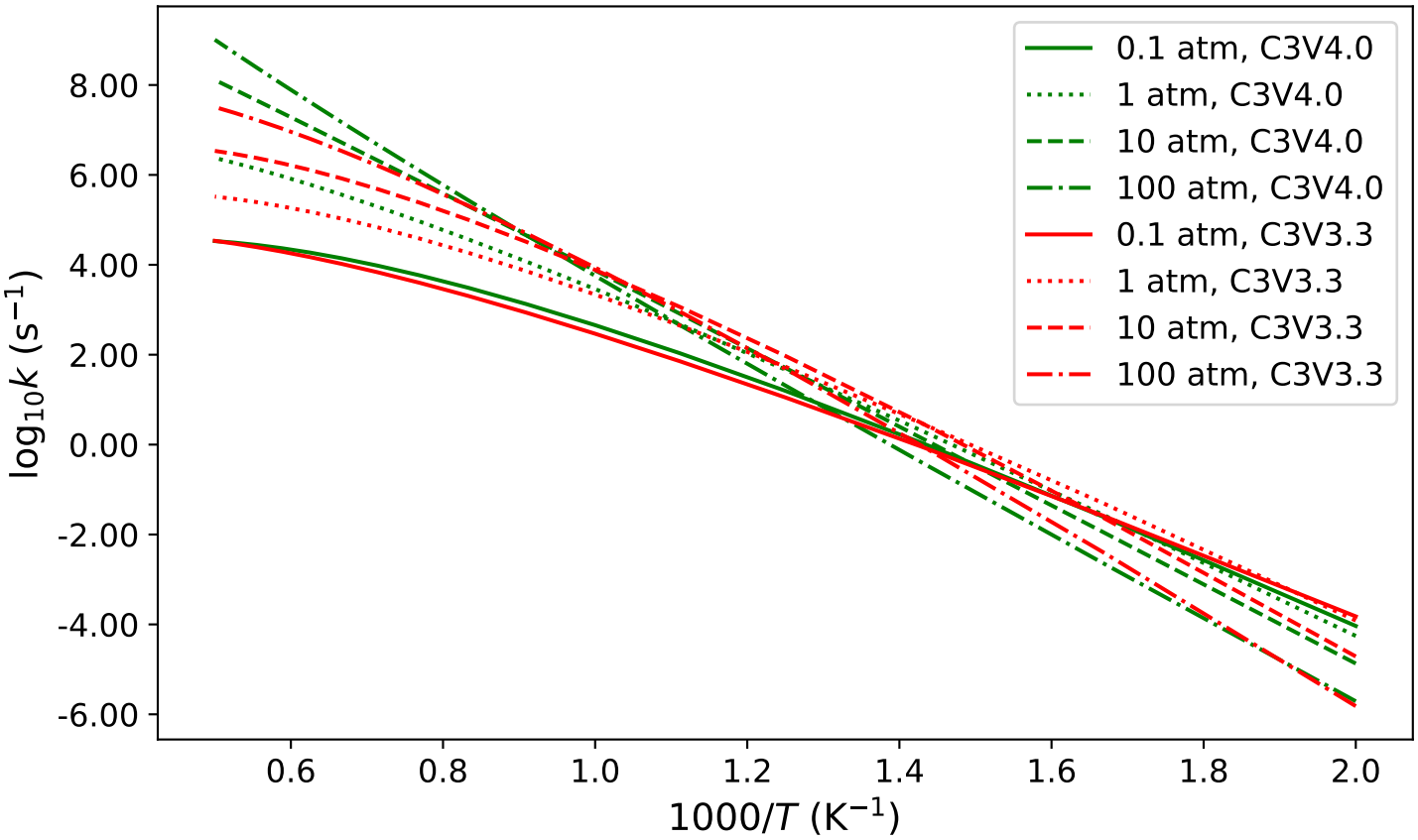



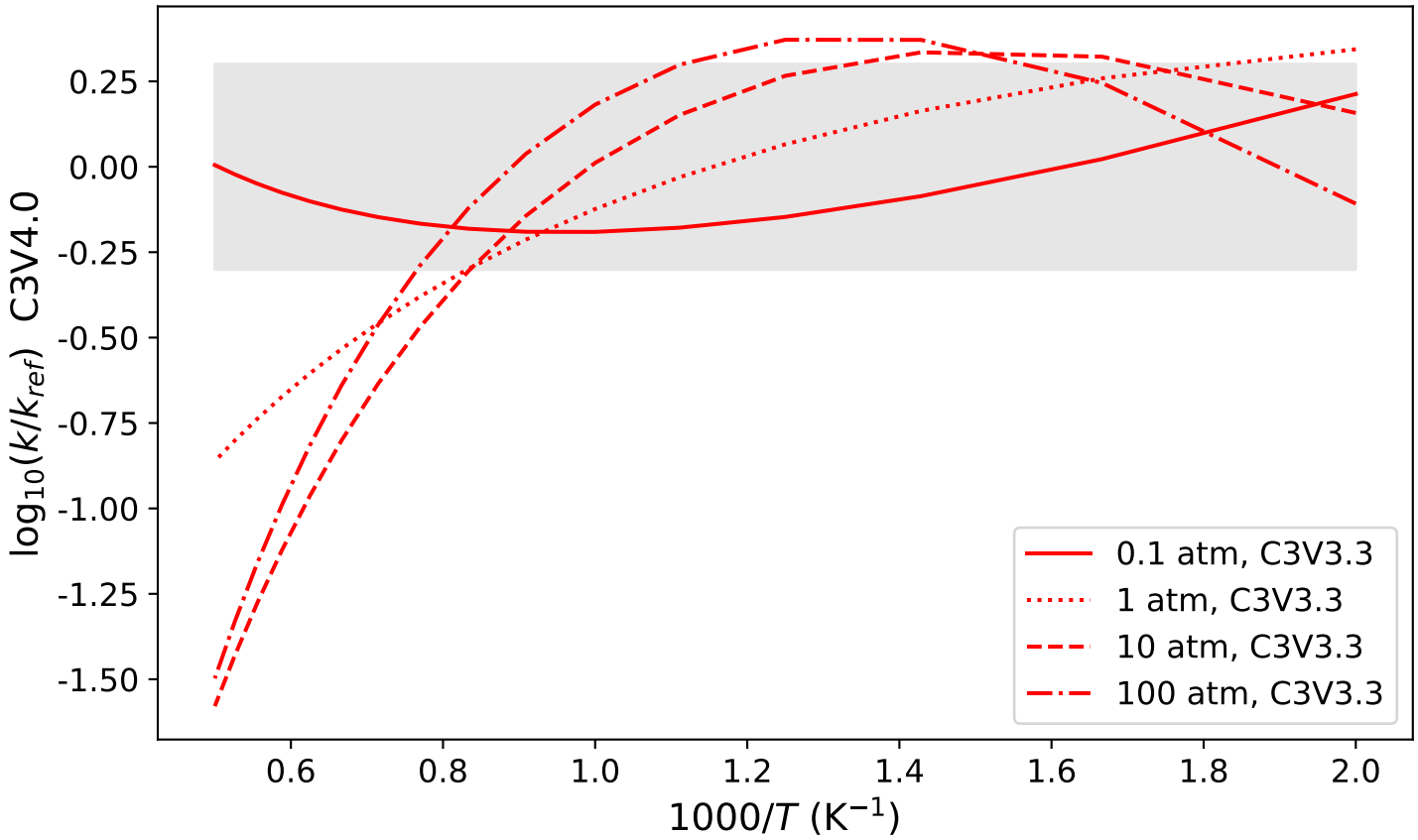
O2 + IC3H7OH = HO2 + TC3H6OH



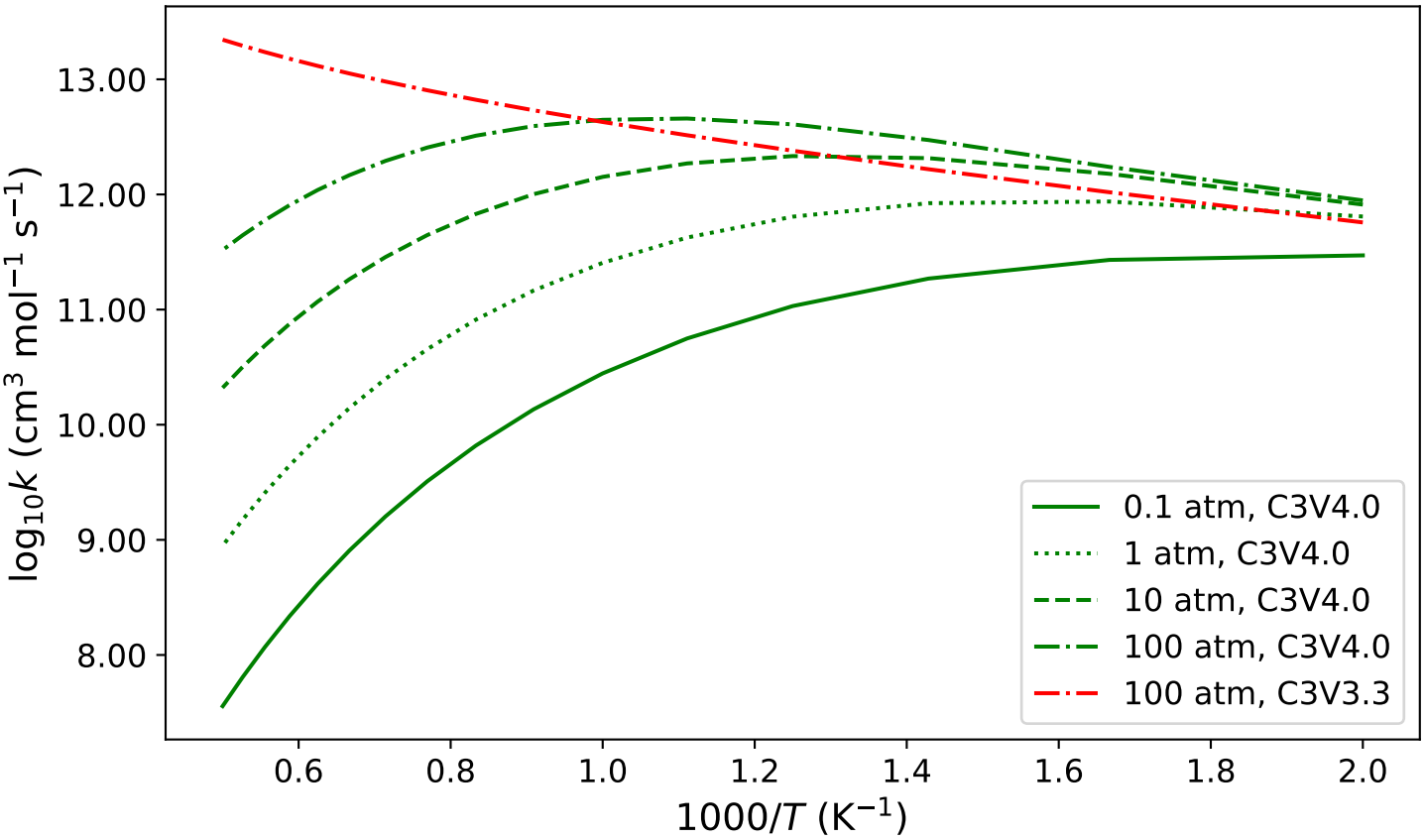



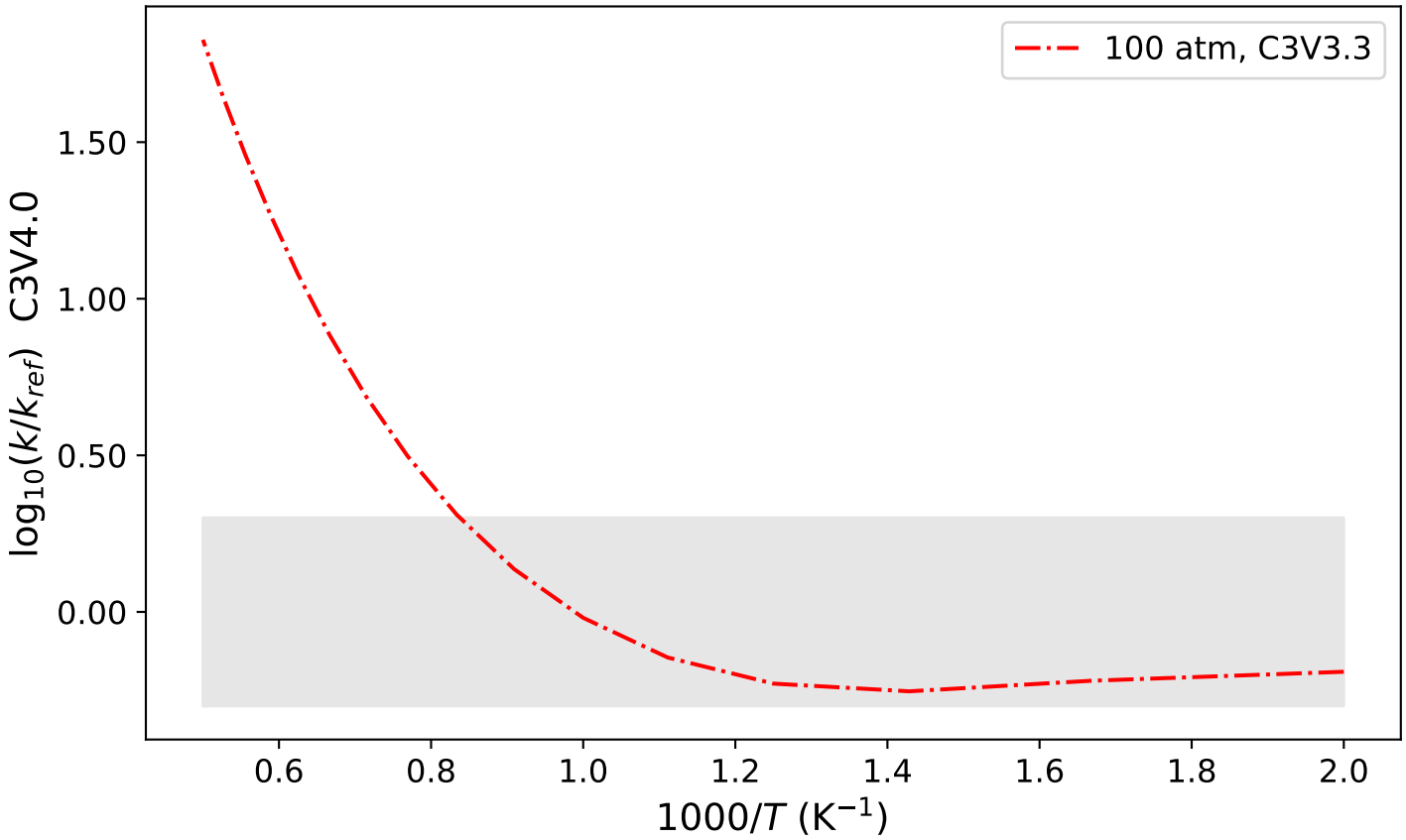
PC4H9 = CH3 + C3H6



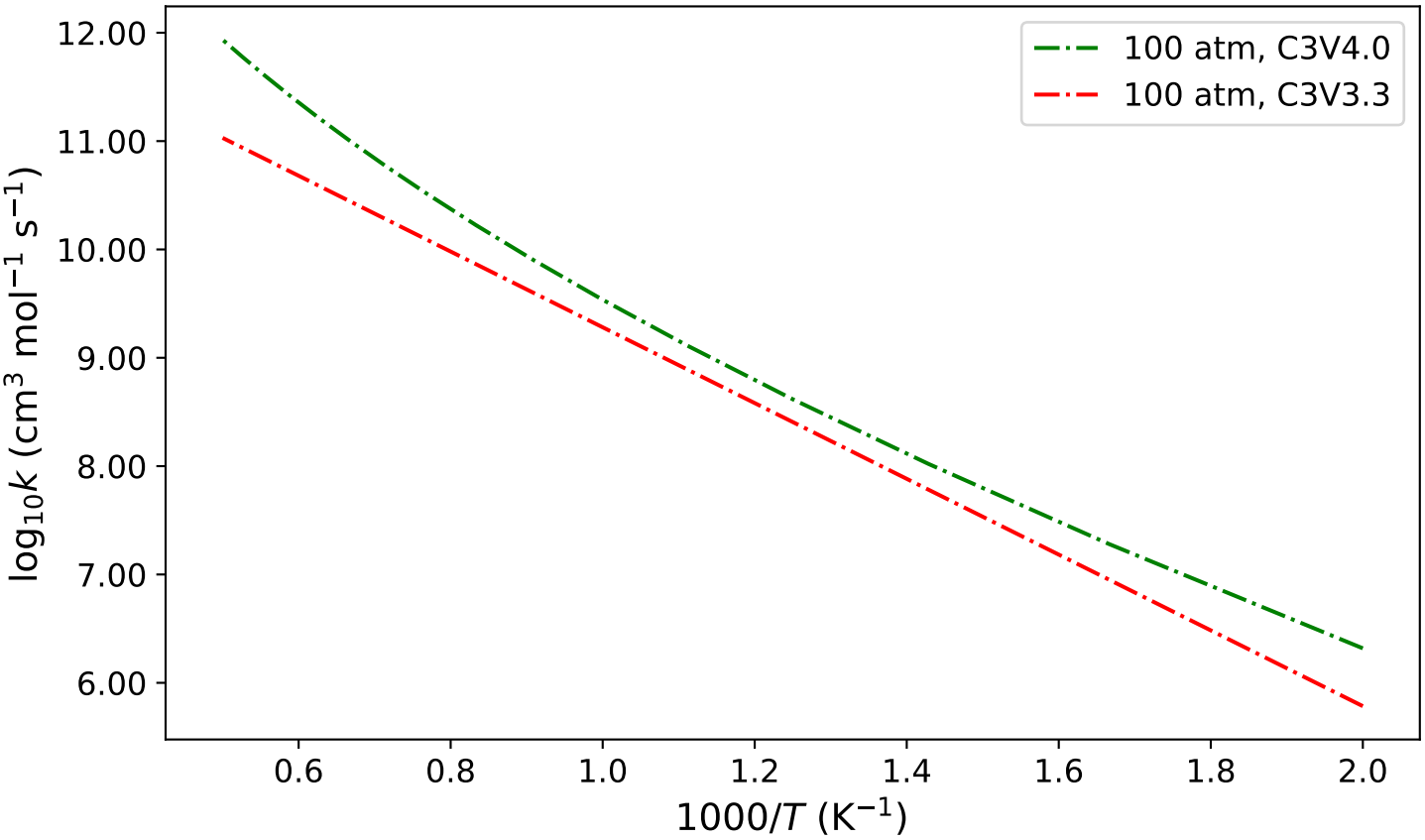



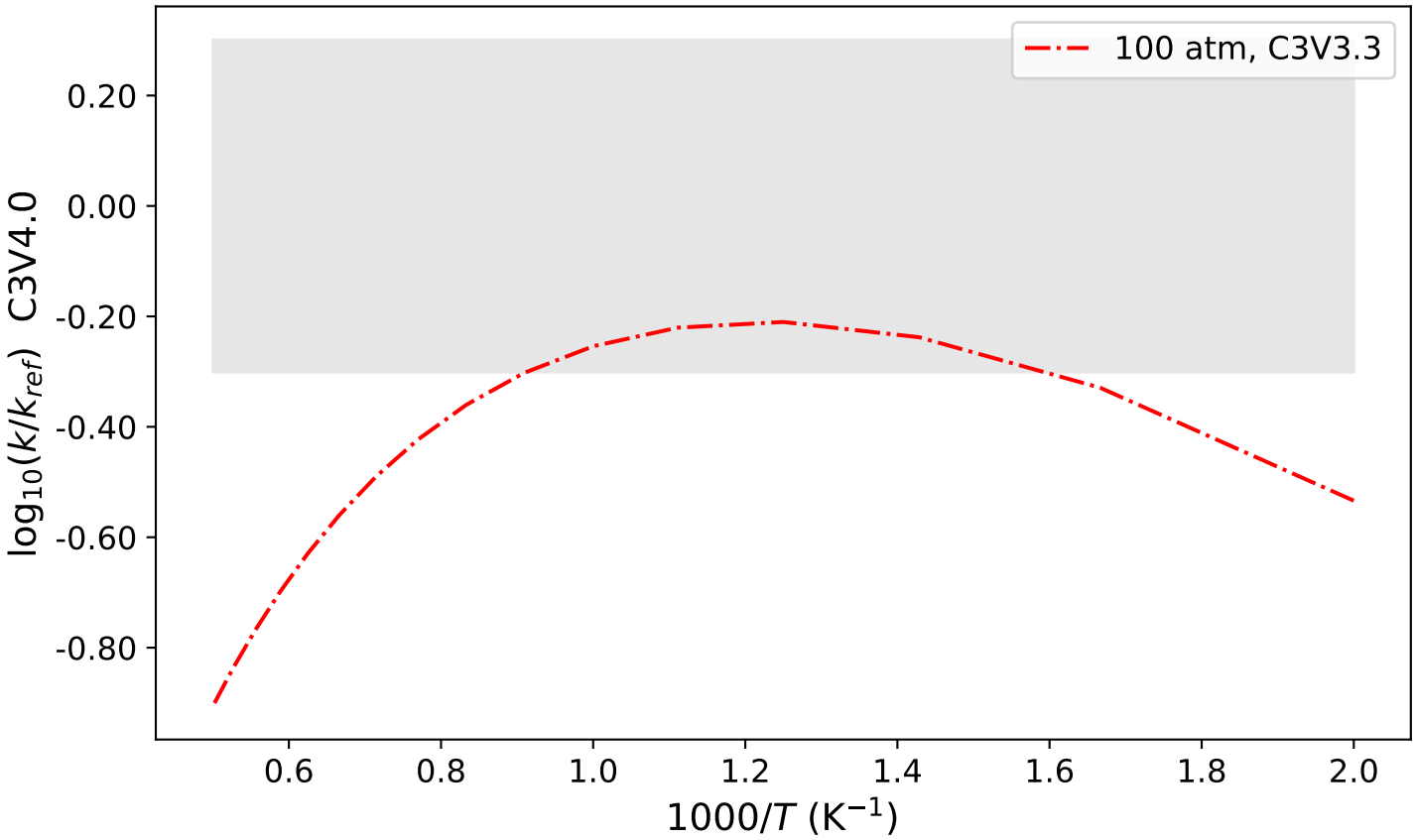
H + C4H612 = C4H72-2



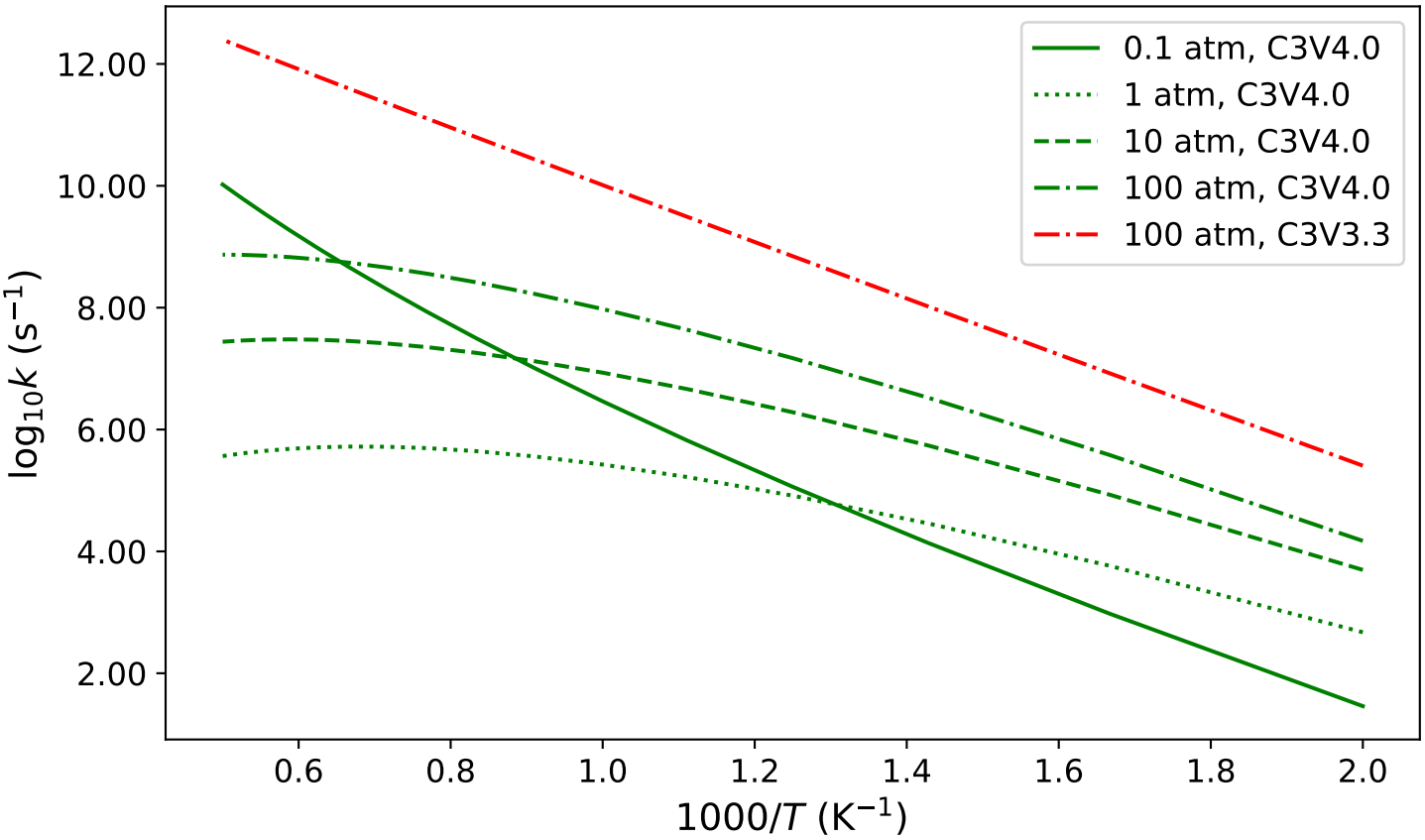



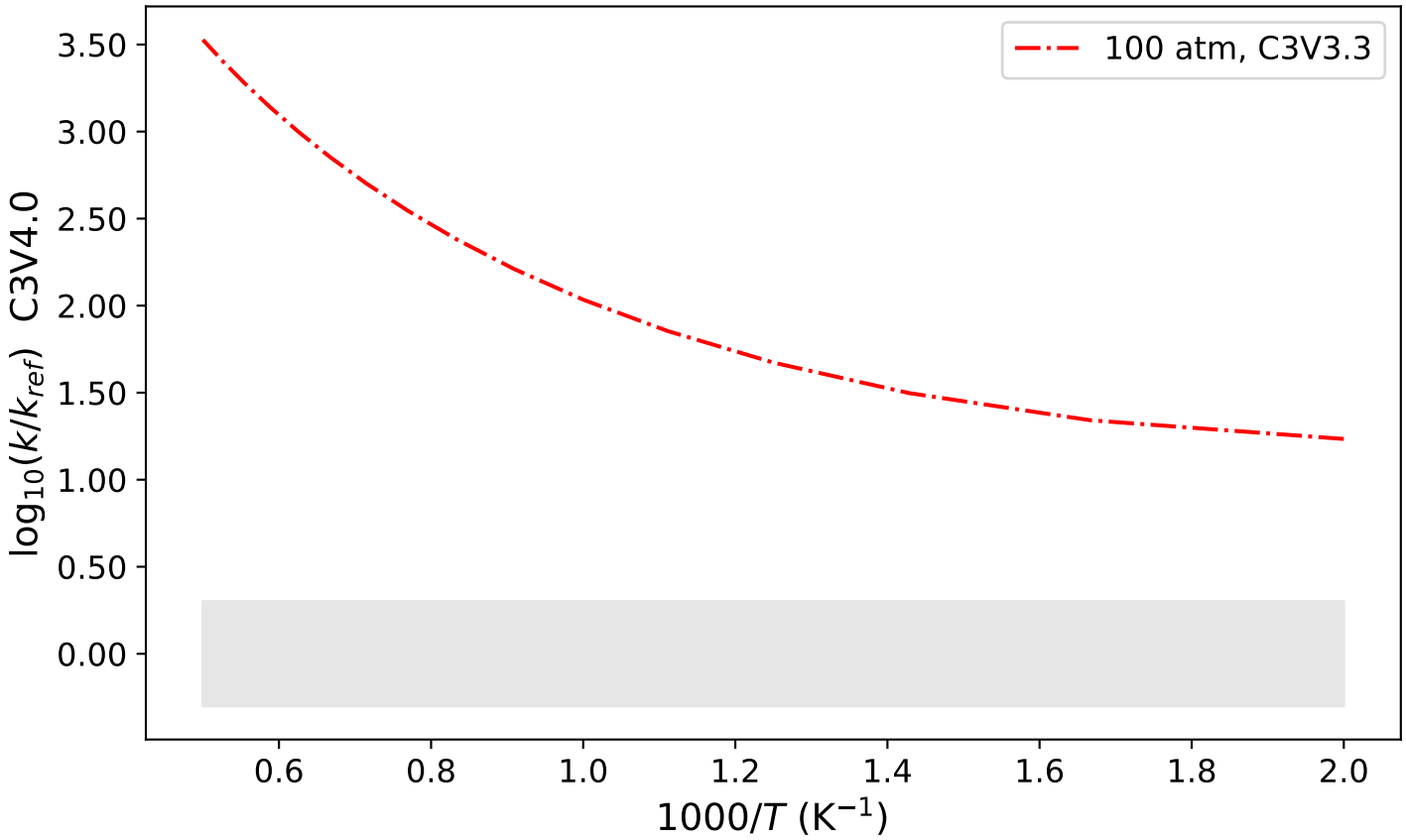
NC3H7OH + HO2 = H2O2 + C3H6OH1-1



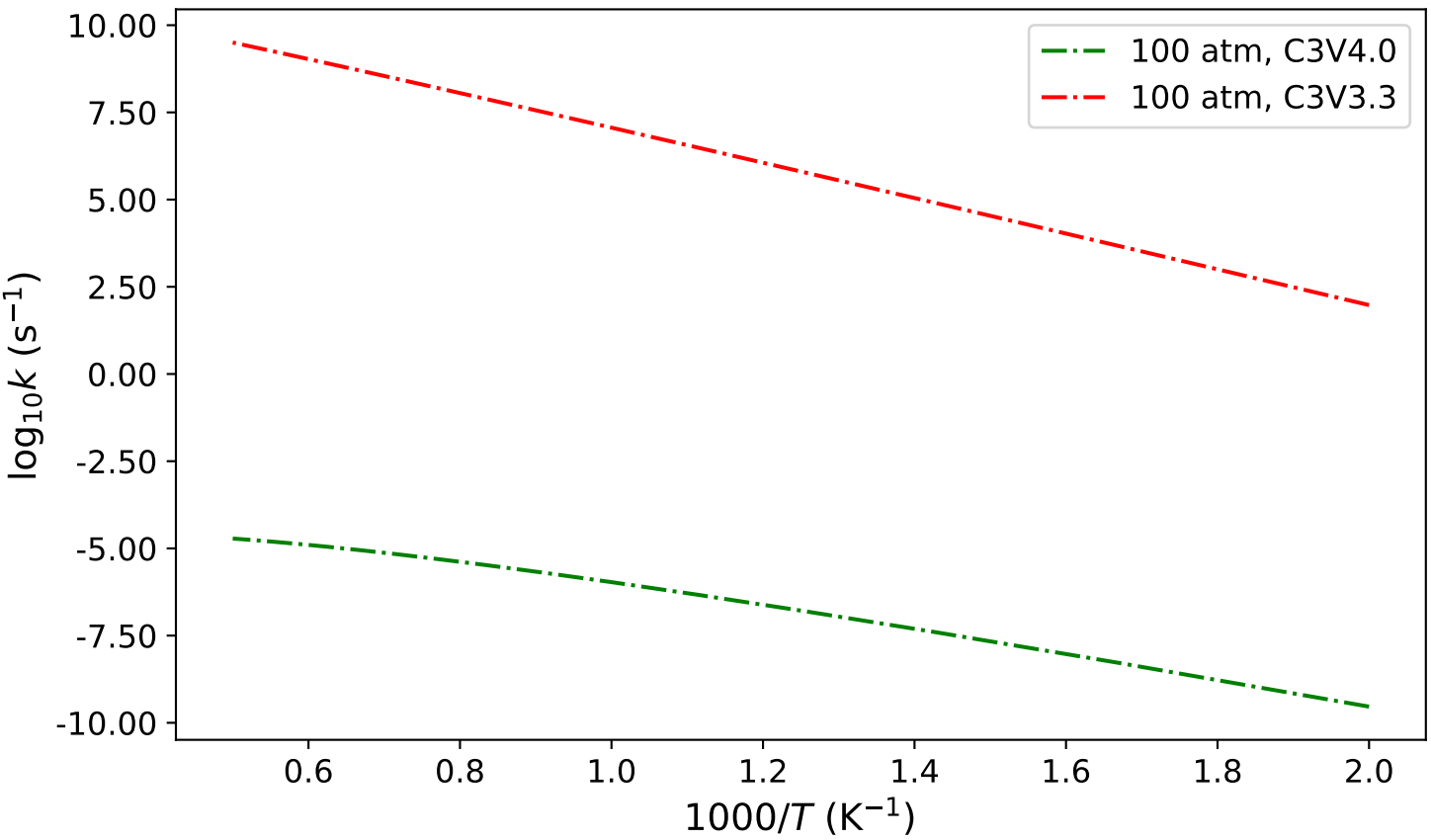



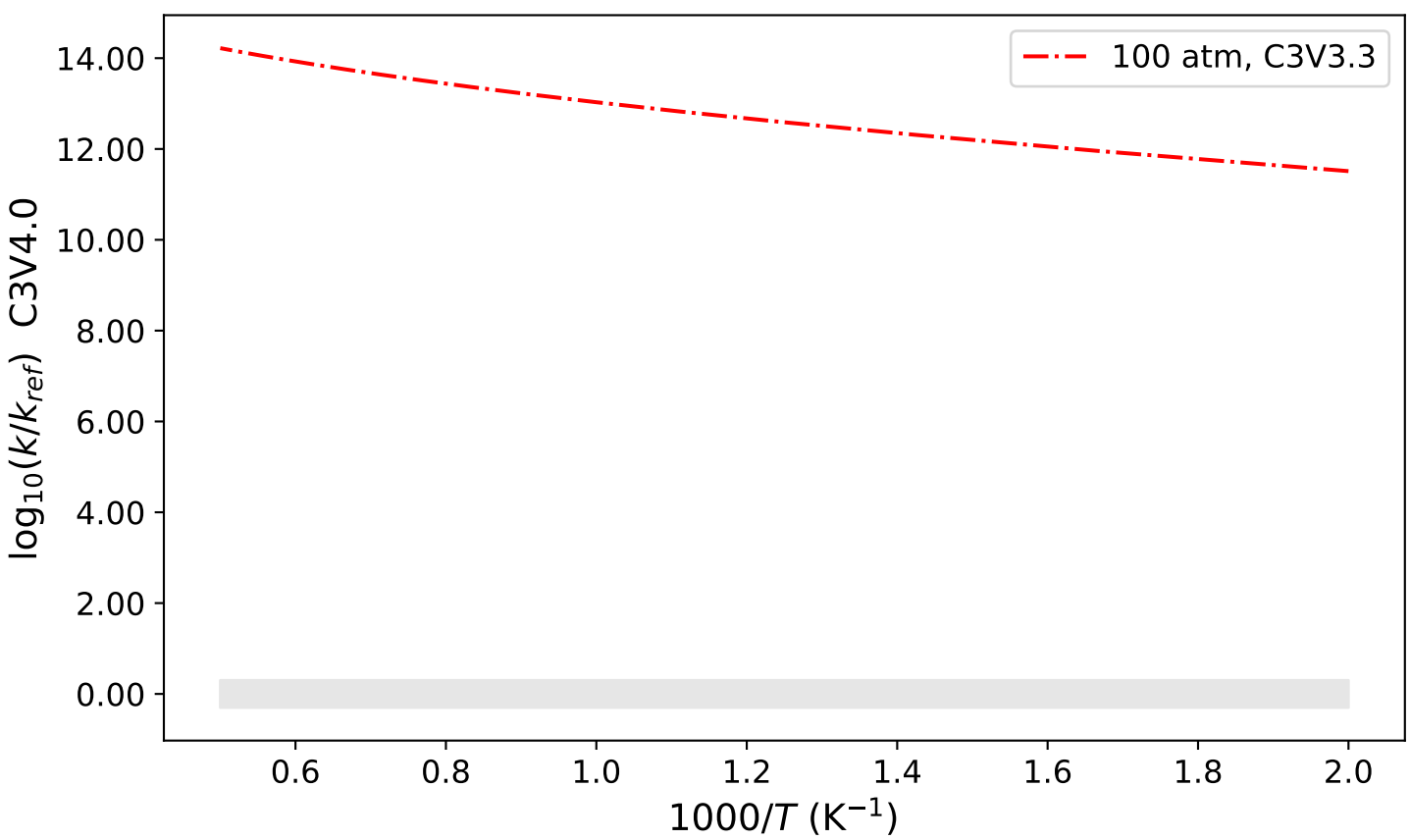
PC4H9O = H + NC3H7CHO



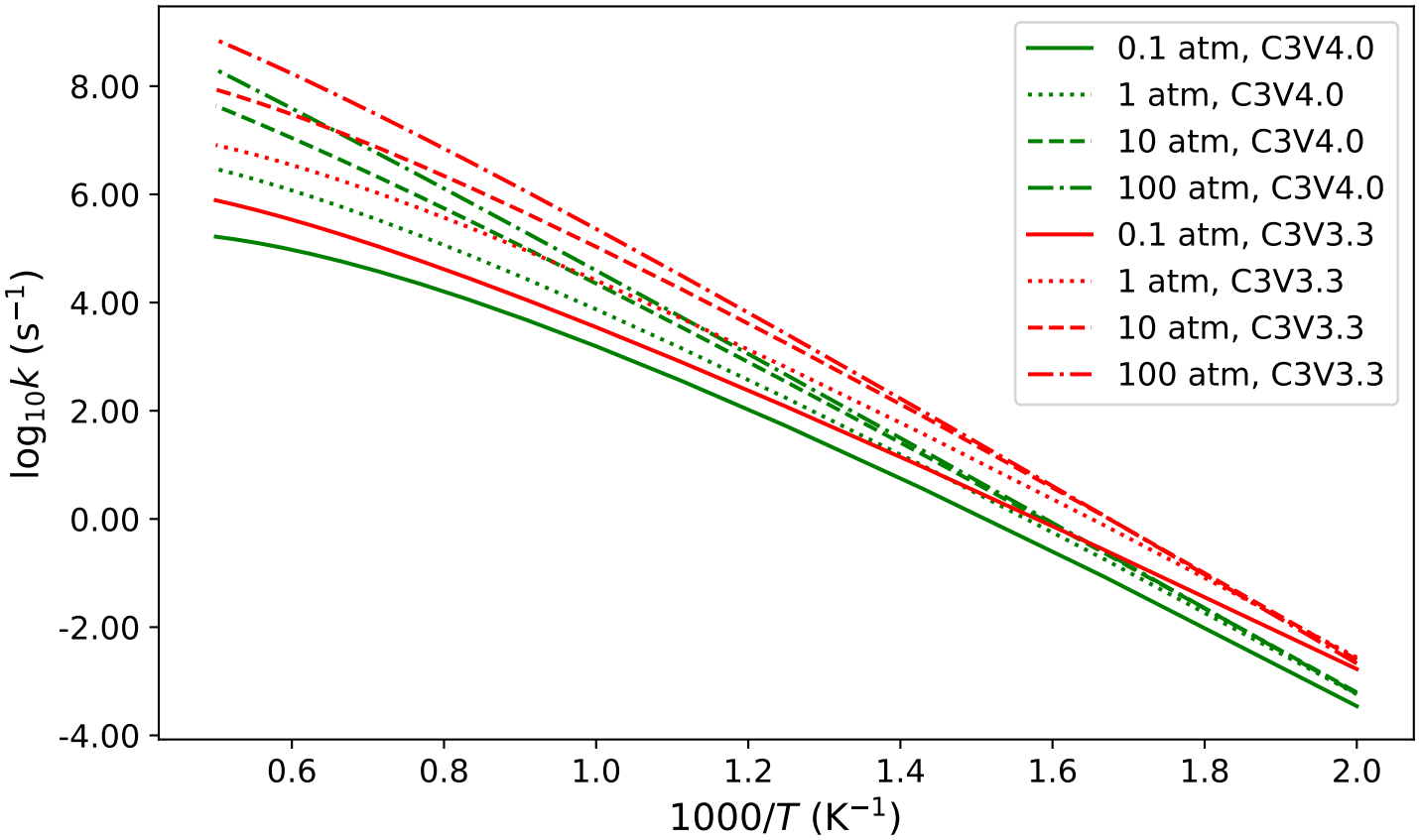



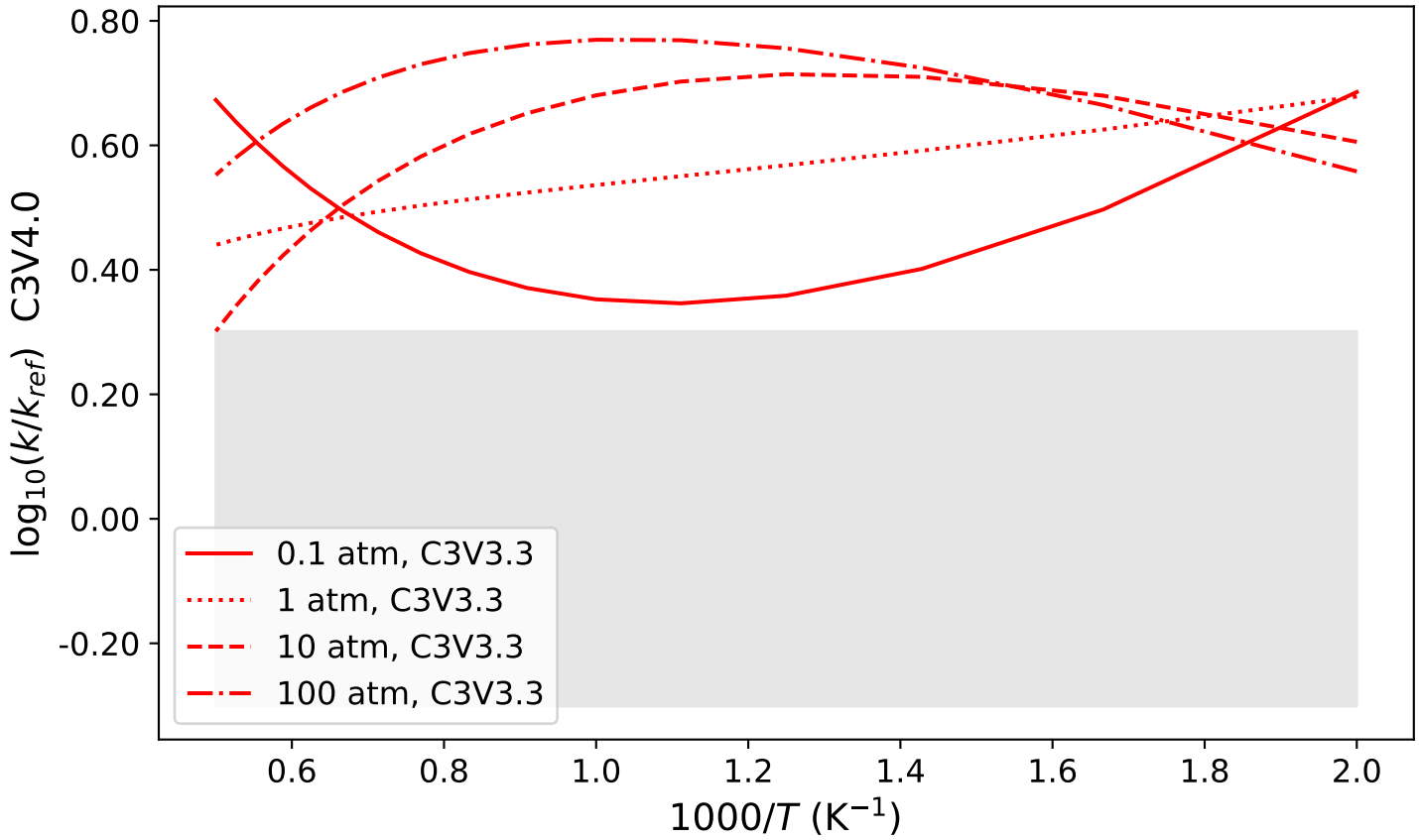
QCH3CH(OOH)CO = HO2 + CH3CHCO



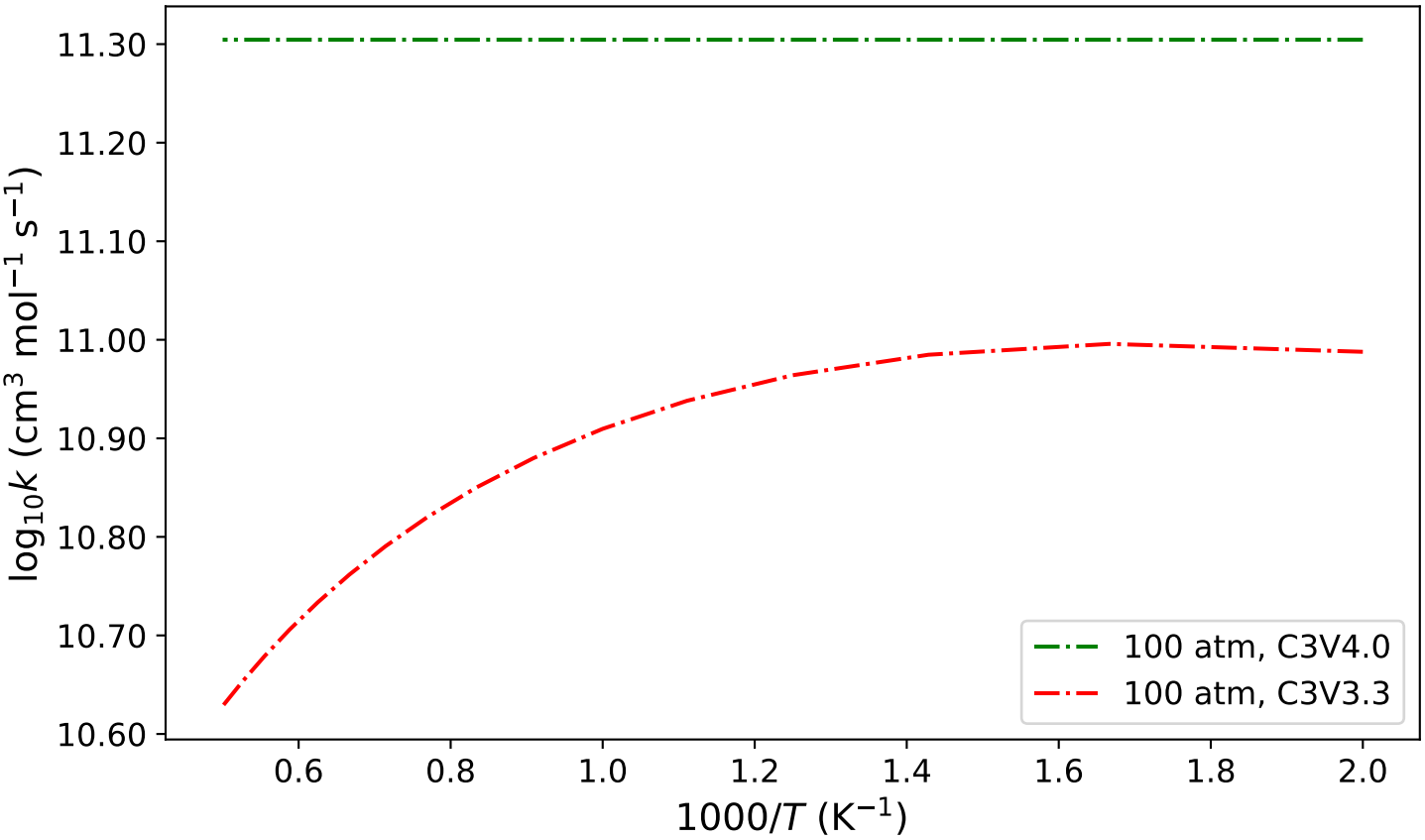



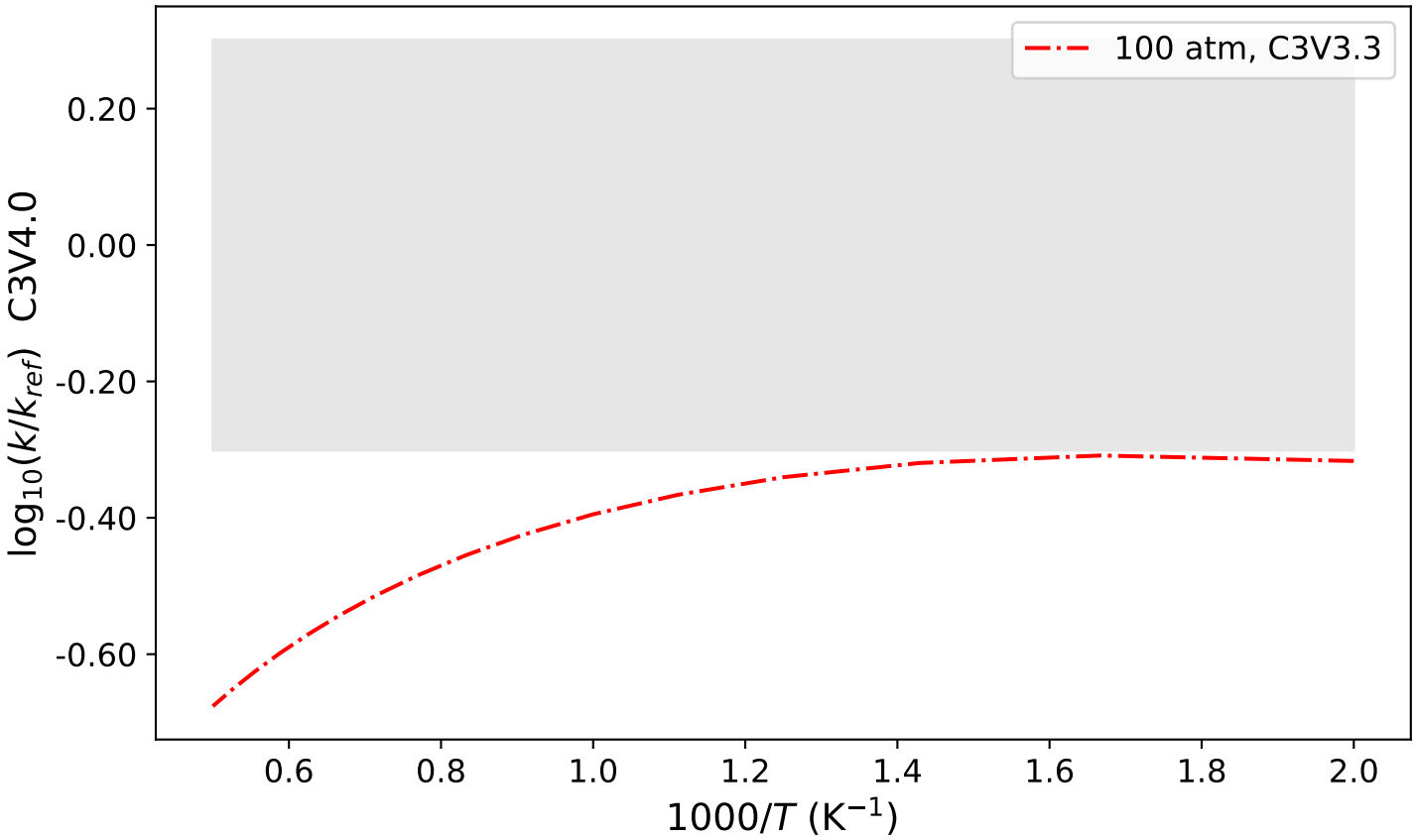
SC4H9 = H + C4H8-2



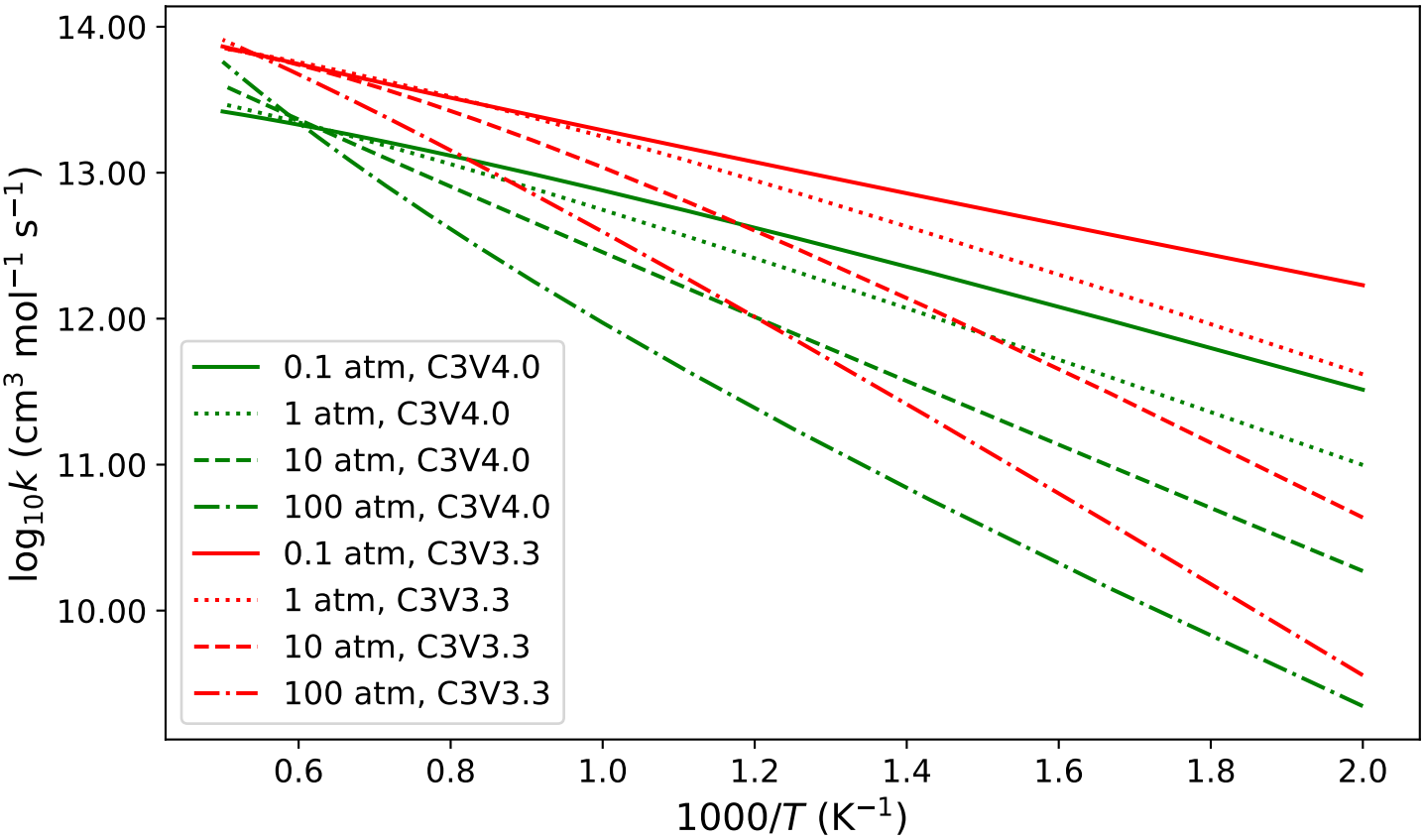



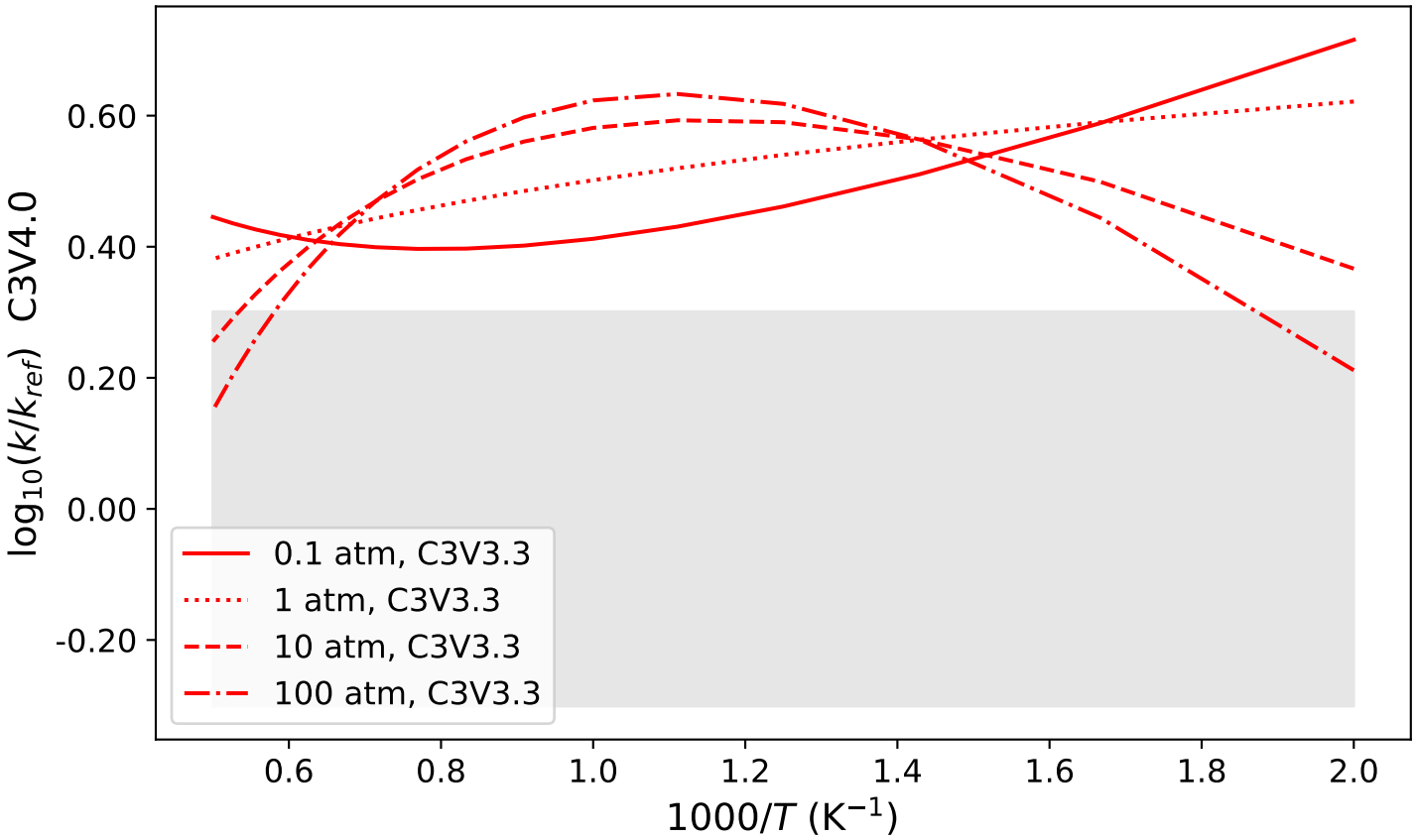
NC3H7O2 + CH3O2 => NC3H7O + CH3O + O2



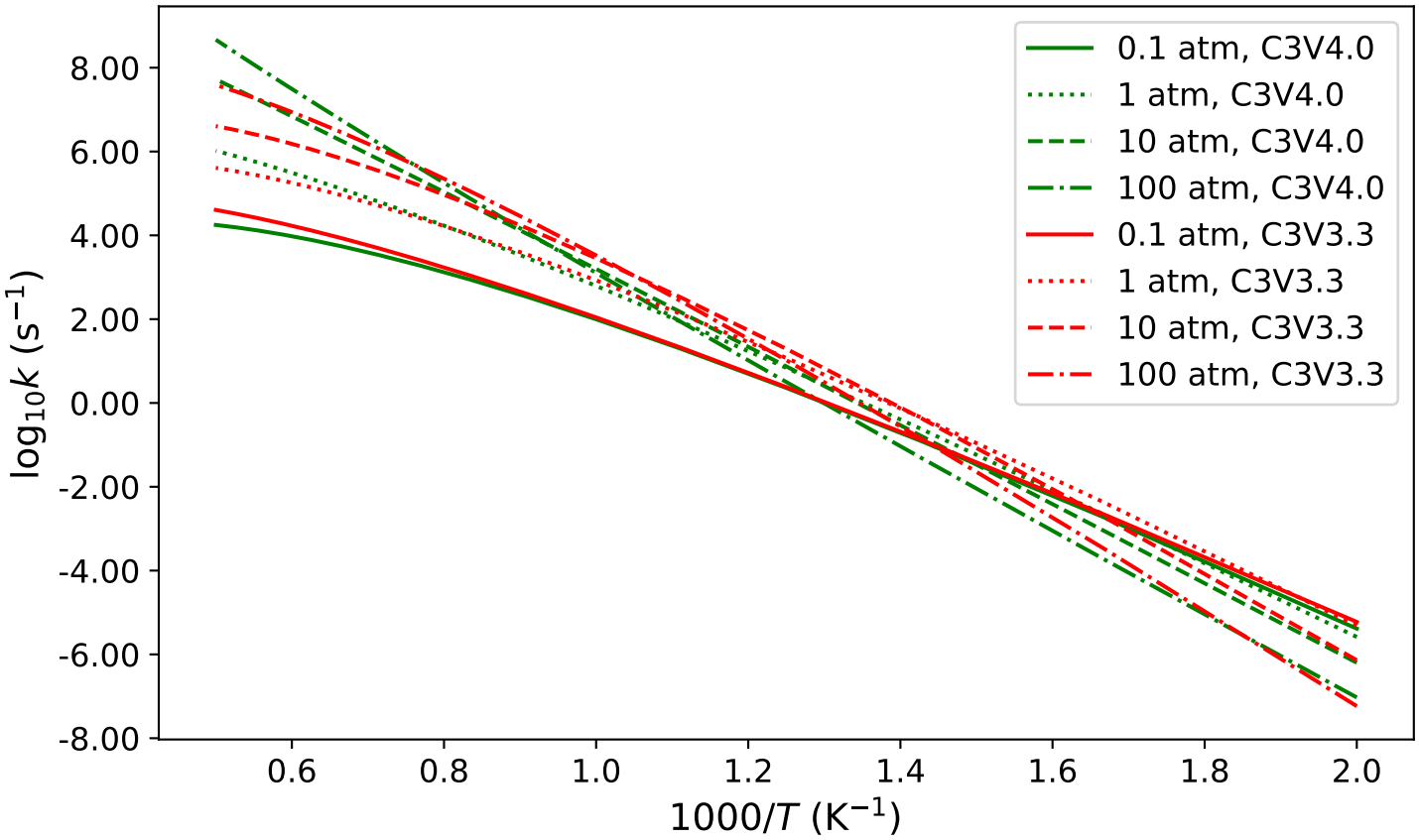



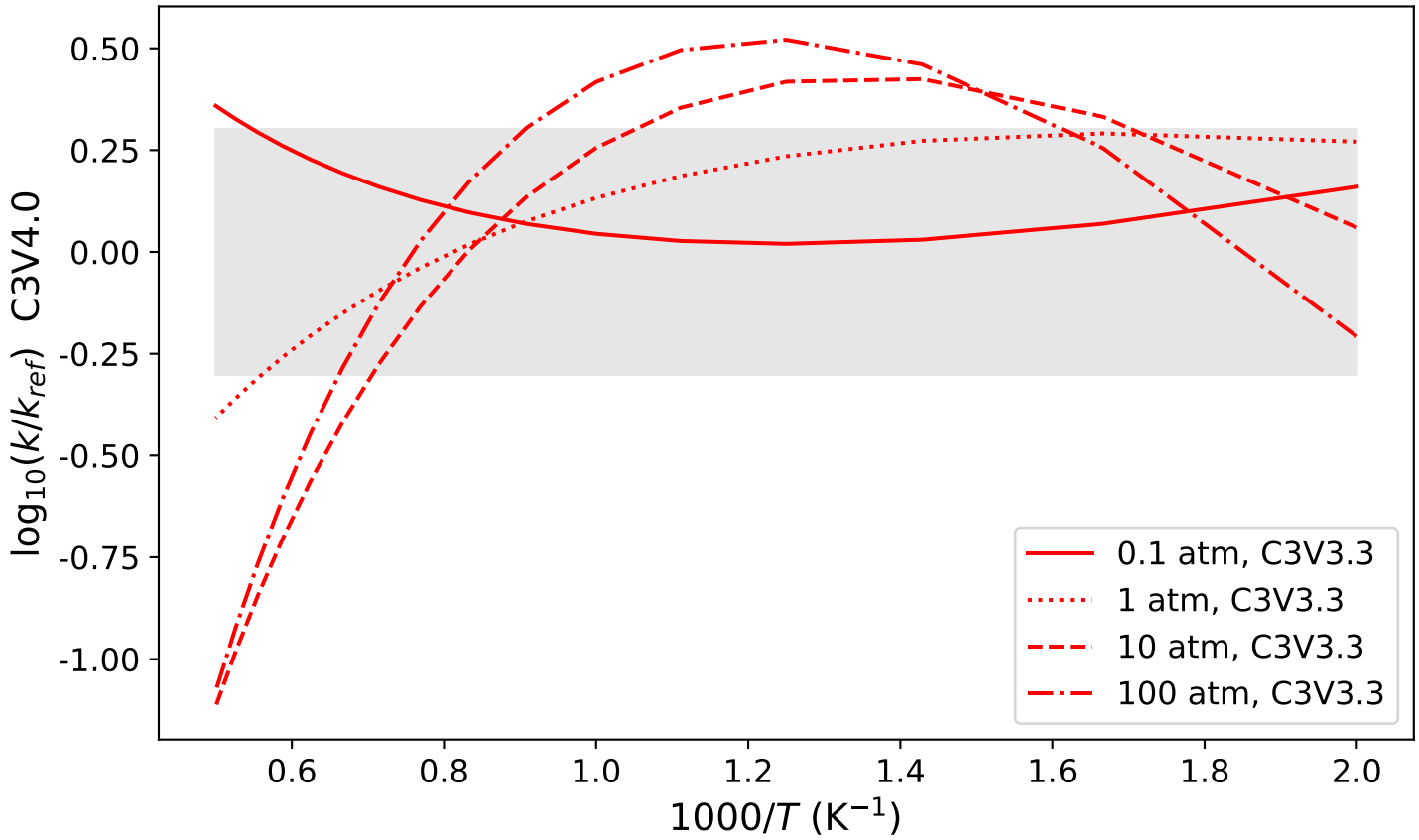
H + C4H8-2 = CH3 + C3H6



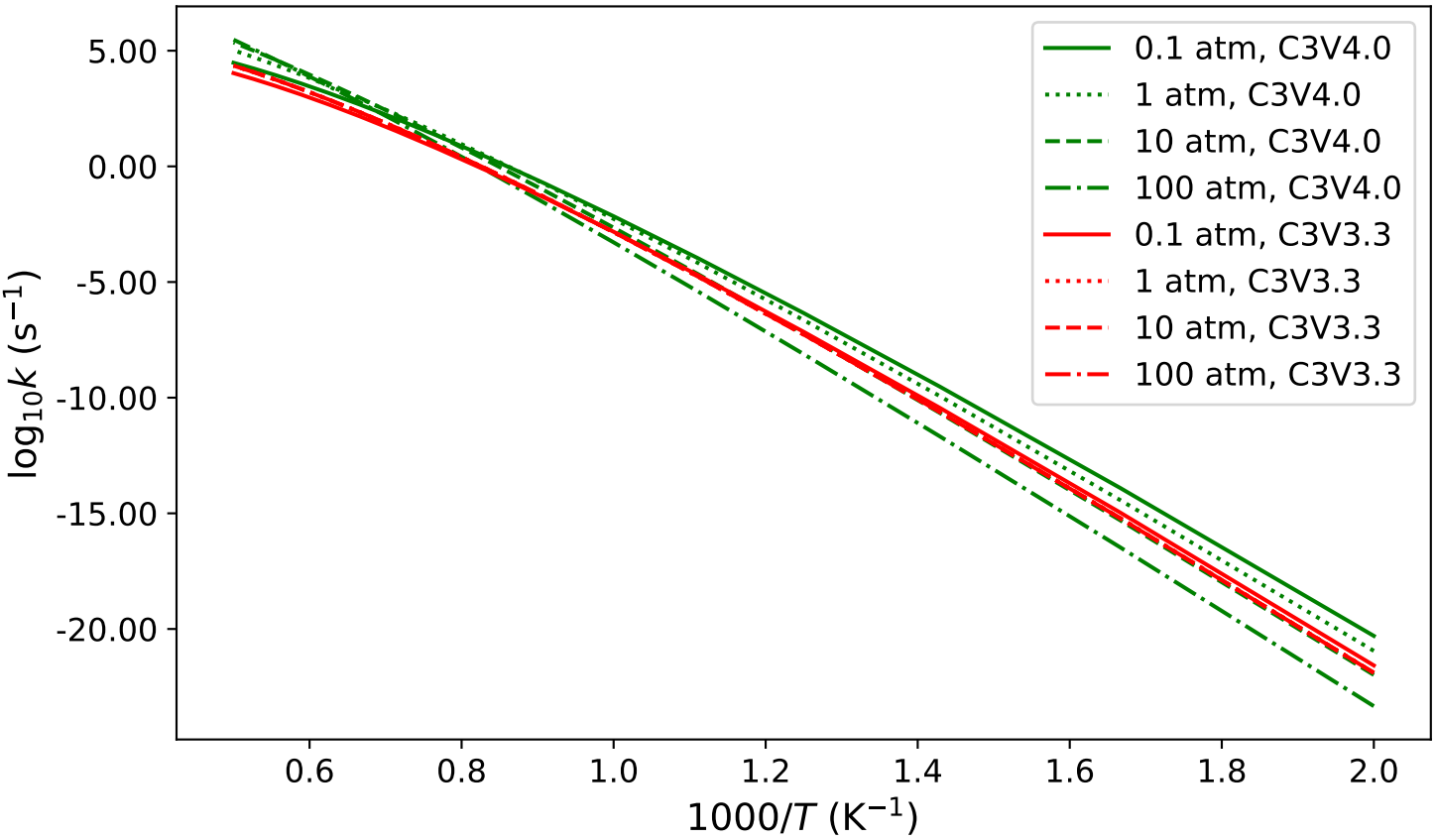



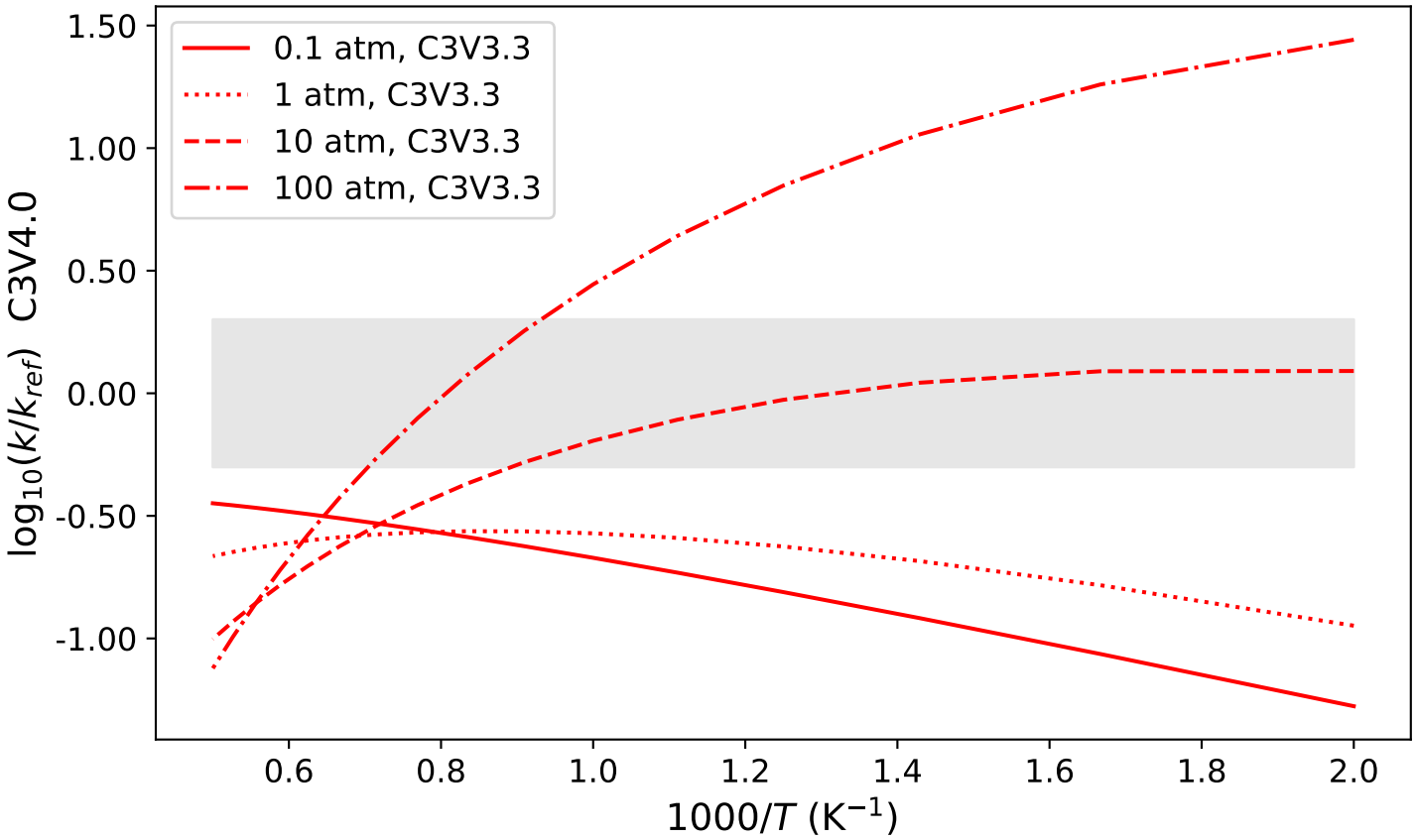
SC4H9 = C2H5 + C2H4



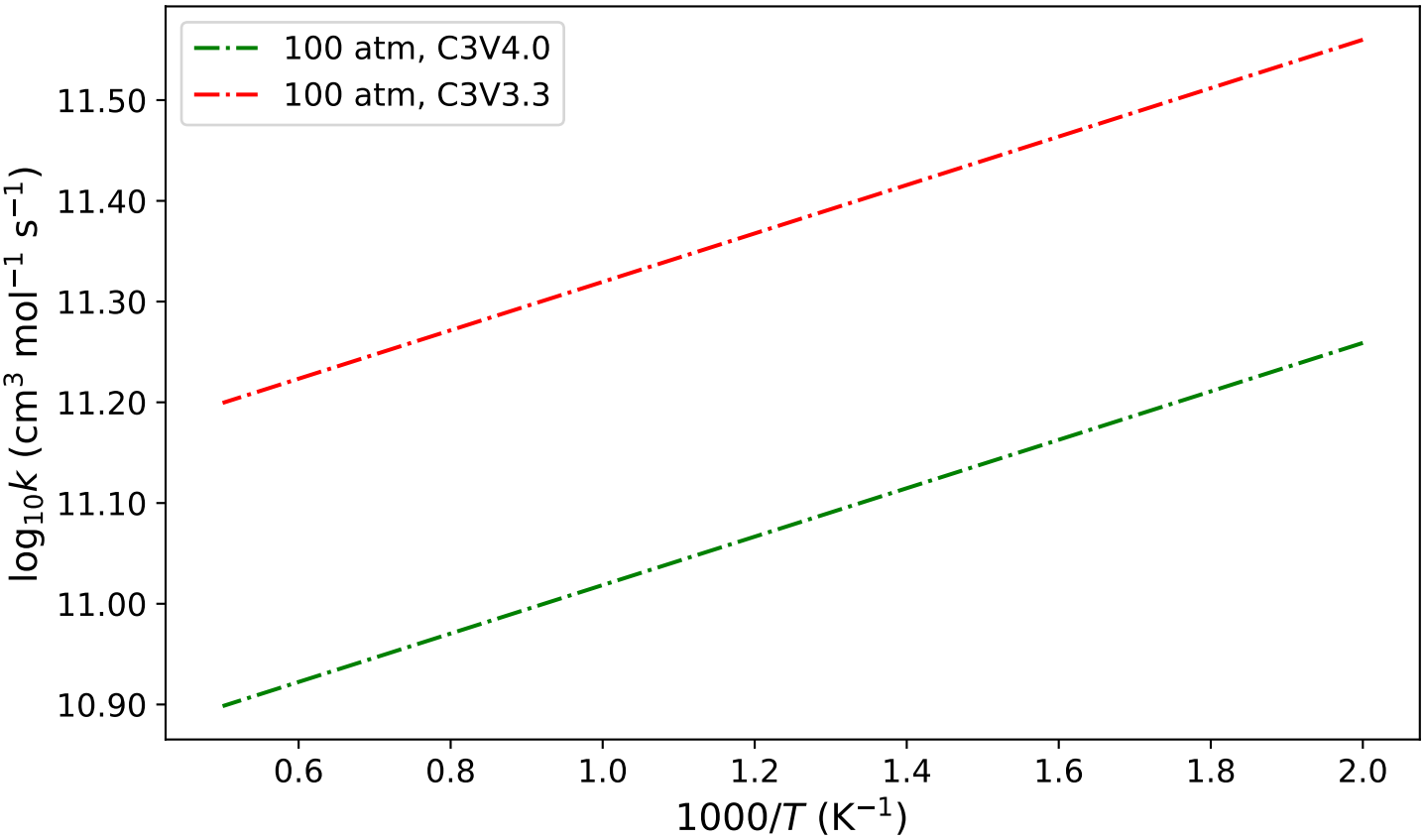



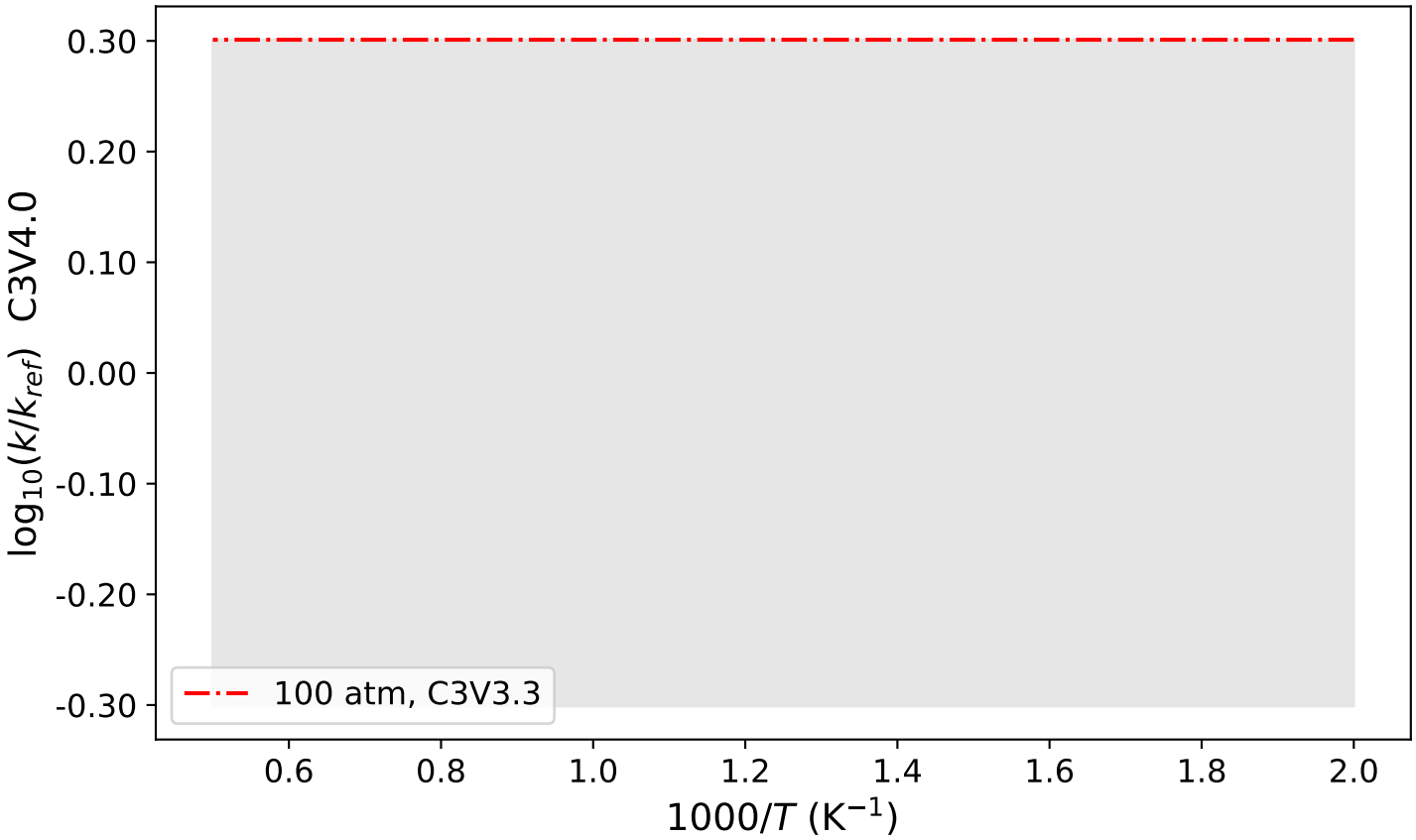
C4H6 = C4H6-2


0.1 atm, C3V4.0
1 atm, C3V4.0
10 atm, C3V4.0
100 atm, C3V4.0
0.1 atm, C3V3.3
1 atm, C3V3.3
10 atm, C3V3.3
100 atm, C3V3.3
5.00
0.00
-5.00
-10.00
-15.00
-20.00
$\log_{10}k$ (s$^{-1}$)
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ (K$^{-1}$)

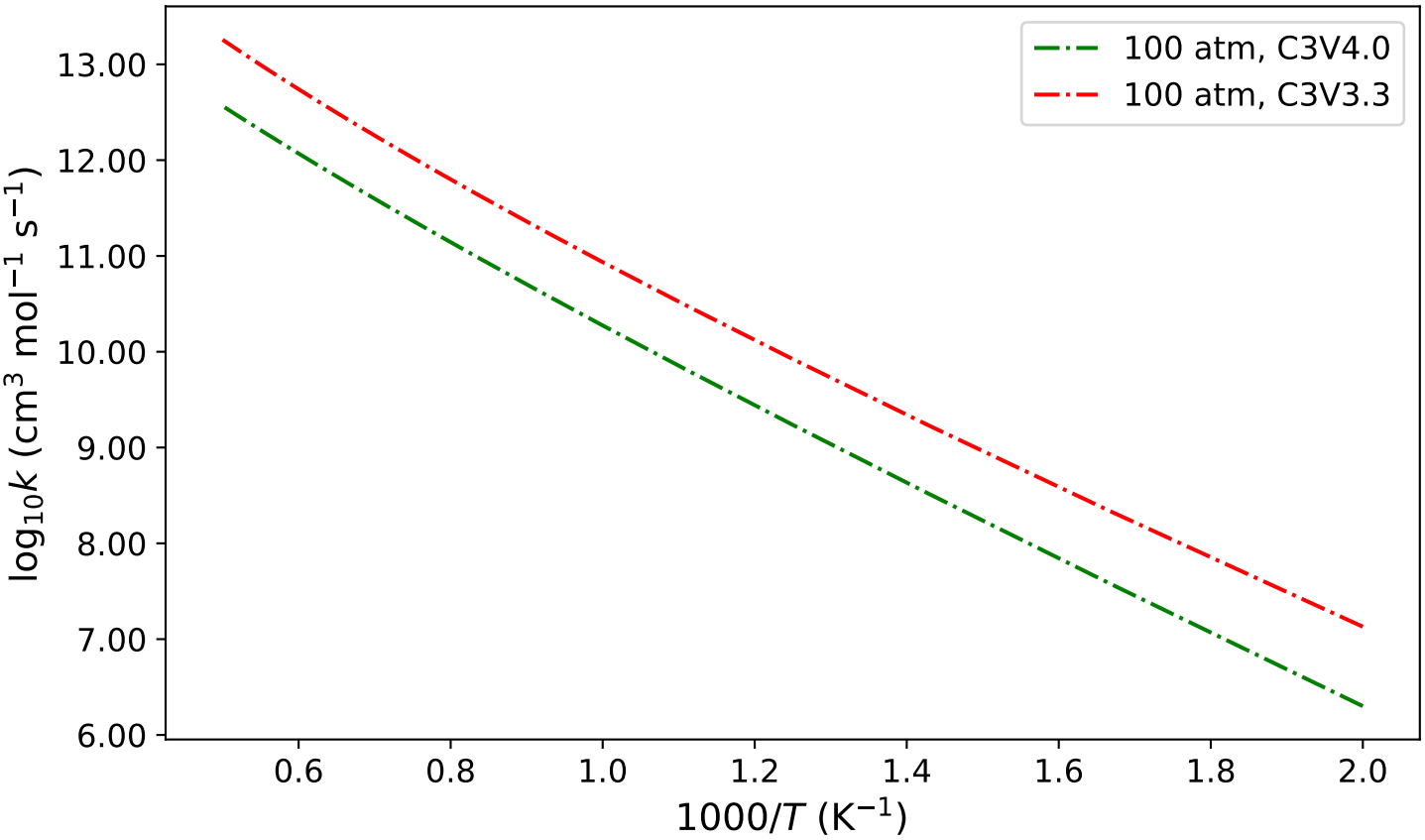

0.1 atm, C3V3.3
1 atm, C3V3.3
10 atm, C3V3.3
100 atm, C3V3.3
1.50
1.00
0.50
0.00
-0.50
-1.00
$\log_{10}(k/k_{ref})$ C3V4.0
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
1000/$T$ (K$^{-1}$)

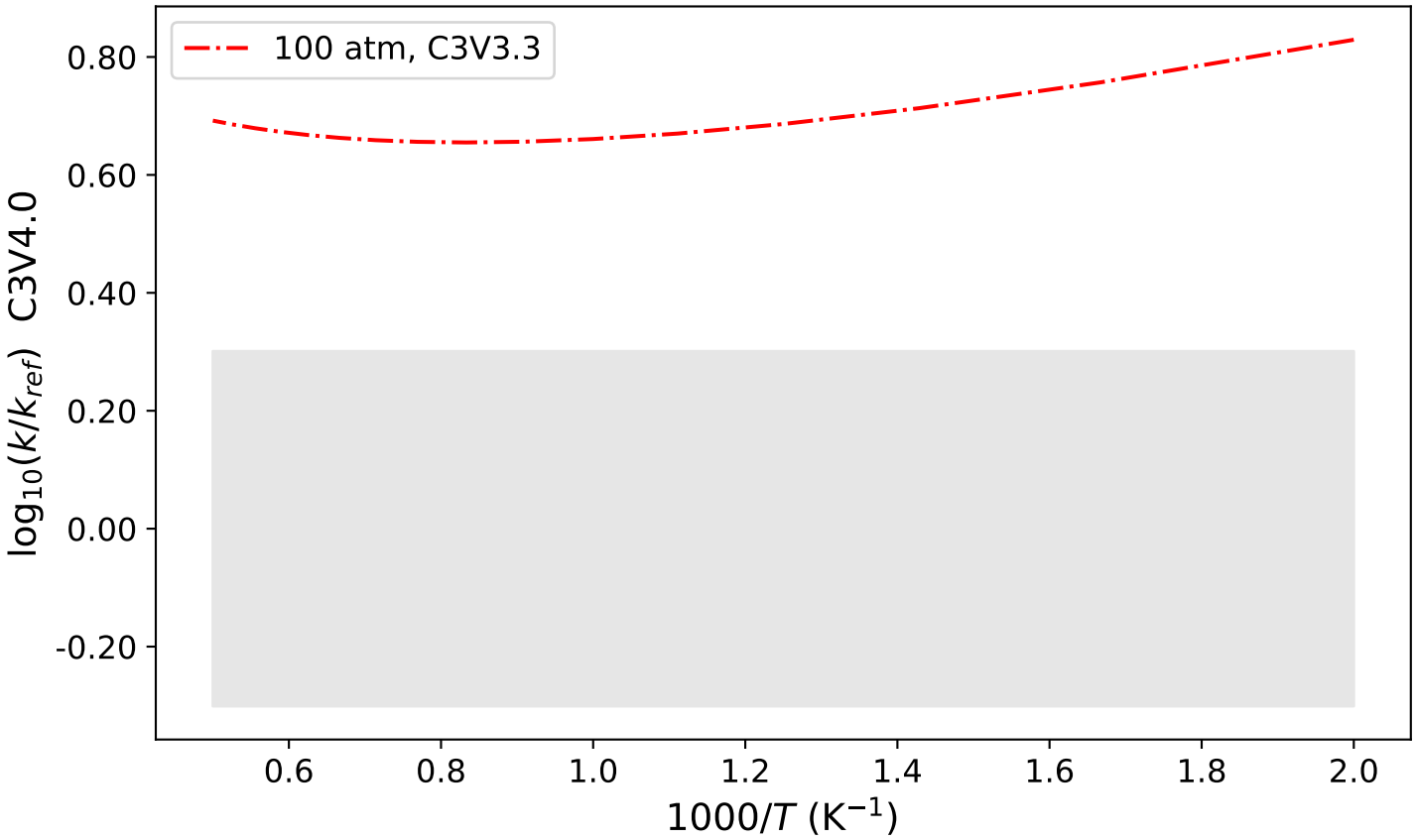
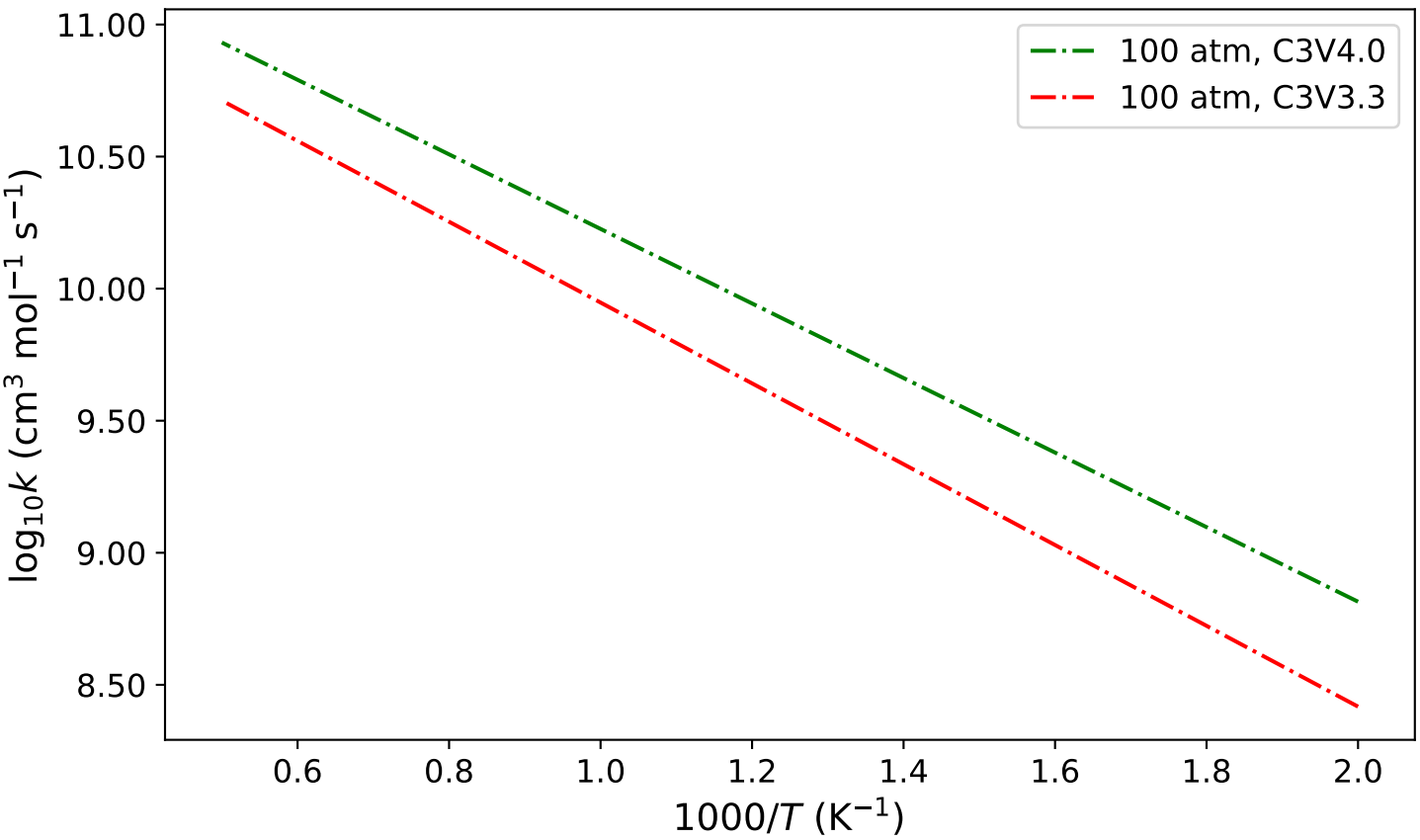
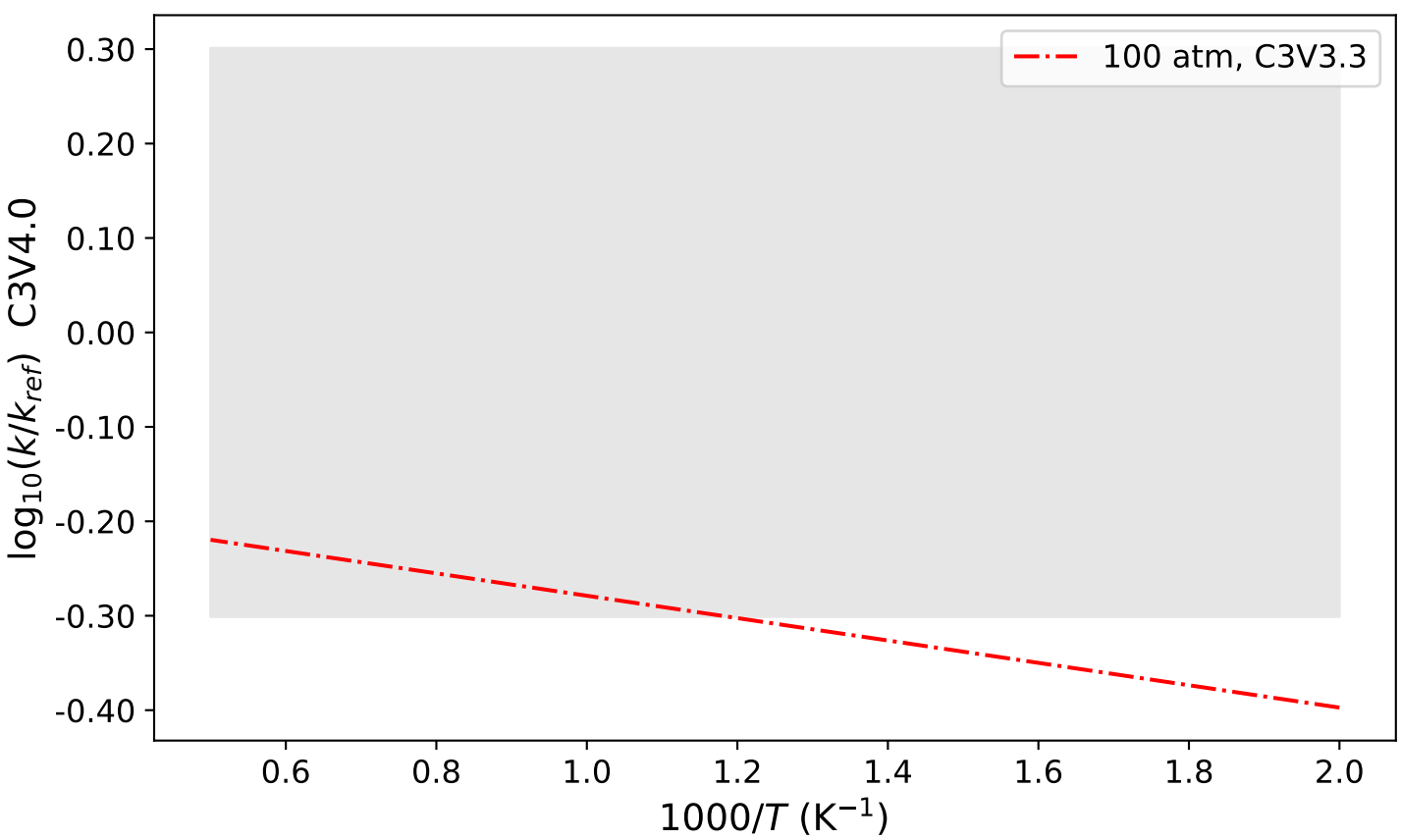
O2 + CH3COCH2 = CH3COCH2O2



H + C4H6 = H2 + C4H5-N






CH3O + C4H10 = CH3OH + PC4H9